Supergravity and the Superstring

A Spacetime-First Course, from the Worldsheet to Holography

Graduate lecture notes

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Abstract

These are notes for a one-semester graduate course in string theory and supergravity, written for readers who already know quantum field theory and general relativity. The worldsheet is treated as a means to an end. The bosonic string and the superstring are quantised covariantly, and only as far as is needed to reach the ten-dimensional spectrum quickly and honestly, after which the focus turns to spacetime and stays there. The core of the course is the low-energy effective actions and eleven-dimensional supergravity, the web of T-, S- and U-dualities with M-theory at its centre, D-branes and the BPS states that make the dualities calculable, and compactification on tori, orbifolds, K3 and Calabi-Yau manifolds, with the moduli fixed by flux and non-perturbative effects. Geometric engineering and a dedicated treatment of F-theory carry the geometry into four-dimensional model building, and the course closes with a holographic capstone: the microscopic counting of black-hole entropy and the AdS/CFT correspondence. Conformal field theory and scattering amplitudes appear as tools rather than as ends in themselves, the amplitudes including an honest account of where the perturbative expansion breaks down and what a string field theory would repair. Nine appendices collect the conventions and the spinor, supersymmetry, geometry and Lie-algebra background in self-contained form. Throughout, the written notes reach somewhat beyond what the lectures themselves would cover, with the optional material marked as such.

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Chapter 1

The Bosonic String: Covariant Quantisation

1.1 From point particle to string: the Nambu-Goto and Polyakov actions

The big picture

Goals for this chapter. By the end you should be able to:

- derive the classical equivalence of the Nambu-Goto and Polyakov actions, and read the worldsheet symmetries and stress tensor off the latter (this section);
- fix conformal gauge, solve the worldsheet wave equation, and impose the classical Virasoro constraints (Section 1.2);
- compute the free-boson central charge $c = D$ from the operator product expansion (Section 1.3);
- run the BRST-nilpotency argument that forces $D = 26$, and read off the massless spectrum (Section 1.4).

Each exercise below is tagged with the goal it exercises and a rough difficulty.

A relativistic point particle sweeps out a curve in spacetime, its *worldline*, and its history is a map $X^\mu(\tau)$ from a one-dimensional parameter space into D-dimensional spacetime. A string sweeps out a two-dimensional surface, the *worldsheet*. Its history is a map

$$X^\mu(\tau, \sigma), \qquad \mu = 0, 1, \ldots, D-1, \tag{1.1}$$

from a strip or cylinder into spacetime, with τ a timelike worldsheet coordinate and σ running along the string. Every construction in this chapter is built on that single upgrade, from one worldline parameter to two worldsheet parameters $\sigma^\alpha = (\tau, \sigma)$. We use worldsheet indices $\alpha, \beta \in \{0, 1\}$ as in Appendix A, and write $\dot{X}^\mu = \partial_\tau X^\mu$, $X'^\mu = \partial_\sigma X^\mu$, with $\partial_\alpha X \cdot \partial_\beta X \coloneqq \eta_{\mu\nu}\, \partial_\alpha X^\mu\, \partial_\beta X^\nu$.

The plan of this section is to write the classical action twice. The Nambu-Goto form is geometrically transparent (it measures the area of the worldsheet) but has a square root that resists quantisation. The Polyakov form trades the square root for an independent worldsheet metric and is the one we quantise in the rest of the chapter. Classically the two are equivalent. Establishing that equivalence, and reading off the symmetries and the stress tensor that the Polyakov action carries, is the classical backbone on which Sections 1.2 to 1.4 rest.

Two symbols carry the worldsheet geometry throughout, and it pays to fix them now. The letter $\gamma_{\alpha\beta}$, always with worldsheet indices α, β, denotes the *independent* (dynamical) metric of the

Polyakov formulation. The letter $h_{\alpha\beta}$ is reserved for the *induced* metric, the pullback of the spacetime metric to the surface. They coincide only on shell, and only up to a local rescaling; keeping them typographically distinct is what makes the equivalence proof below readable.

1.1.1 Warm-up: the relativistic particle, written twice

The action for a relativistic particle of mass m is proportional to the proper time along its worldline,

$$S_{\text{pp}} = -m \int \mathrm{d}\tau \, \sqrt{-\dot{X}^\mu \dot{X}_\mu} \,, \tag{1.2}$$

where the minus sign under the root is there because a timelike worldline has $\dot{X}^\mu \dot{X}_\mu < 0$ in the mostly-plus signature. This is reparametrisation invariant: under $\tau \to \tau'(\tau)$ the combination $\mathrm{d}\tau \sqrt{-\dot{X}^2}$ is unchanged, so no choice of clock along the worldline is preferred. That is the good news. The bad news is the square root. It is awkward to canonically quantise, and it makes the massless limit $m \to 0$ singular, killing the action outright.

Both problems are cured by introducing an auxiliary field on the worldline, the *einbein* $e(\tau)$, which one should think of as a worldline metric through $g_{\tau\tau} = -e^2$. Consider

$$S'_{\text{pp}} = \frac{1}{2} \int \mathrm{d}\tau \left(\frac{1}{e} \dot{X}^\mu \dot{X}_\mu - e\, m^2 \right). \tag{1.3}$$

There is no square root, and the massless limit is perfectly smooth: at $m = 0$ it becomes $\frac{1}{2}\int e^{-1} \dot{X}^2$, which still makes sense and describes null geodesics. The einbein carries no time derivative, so its equation of motion is algebraic,

$$-\frac{1}{e^2}\dot{X}^2 - m^2 = 0 \quad \Longrightarrow \quad e = \frac{\sqrt{-\dot{X}^2}}{m} \,. \tag{1.4}$$

Substituting this back into (1.3) collapses it to (1.2), so the two actions describe the same classical particle for $m \neq 0$. The einbein is pure gauge: under a reparametrisation it transforms so that $e\,\mathrm{d}\tau$ is invariant, and a single reparametrisation function is enough to set e to any value we like (for example $e = 1$, affine parametrisation). A one-dimensional metric has one component and one gauge symmetry to remove it, so there is nothing left over.

Everything the string does is a copy of this. Promote the worldline to a worldsheet, and promote the einbein to an independent worldsheet metric. What is new in two dimensions is that the metric now has three independent components while the reparametrisations are only two functions, so one redundancy is missing. Filling that gap is the role of Weyl symmetry, and it is the reason the two-dimensional case is special.

1.1.2 The Nambu-Goto action: area of the worldsheet

The direct generalisation of "length of the worldline" is "area of the worldsheet". The worldsheet inherits a metric from its embedding in spacetime, the *induced metric*

$$h_{\alpha\beta}(\sigma) \coloneqq \partial_\alpha X^\mu \, \partial_\beta X^\nu \, \eta_{\mu\nu} = \partial_\alpha X \cdot \partial_\beta X \,, \tag{1.5}$$

which is the pullback of the flat spacetime metric to the surface. For a physical (timelike) worldsheet its determinant $h \coloneqq \det h_{\alpha\beta}$ is negative, so the invariant area element is $\mathrm{d}^2\sigma \sqrt{-h}$. The Nambu-Goto action is the tension times this area,

$$S_{\text{NG}} = -T \int \mathrm{d}^2\sigma \, \sqrt{-h} = -\frac{1}{2\pi\alpha'} \int \mathrm{d}^2\sigma \, \sqrt{(\dot{X}\cdot X')^2 - \dot{X}^2 X'^2} \,, \tag{1.6}$$

using the string tension $T = 1/(2\pi\alpha')$ from Appendix A and $-h = (\dot{X}\cdot X')^2 - \dot{X}^2 X'^2$. The single dimensionful constant here is α', the *Regge slope* (equivalently the inverse tension up to

the 2π). It has dimensions of length squared and sets the scale of everything that follows: it defines the string length $\ell_s = \sqrt{\alpha'}$ and the string mass scale $M_s = 1/\ell_s$, and its name records the linear *Regge trajectories* $M^2 \propto J/\alpha'$ that relate the mass-squared M^2 and spin J of the string's excited states (Appendix A). The action is minus the tension times area for the same reason (1.2) is minus the mass times proper time: extremising it makes the string as "short" as its dynamics allow, and the sign is fixed by demanding a sensible Hamiltonian.

Geometrically this is the right answer, and one can quantise it, but the square root fights back. The canonical momenta $P_\mu = \partial\mathcal{L}/\partial\dot{X}^\mu$ are nonlinear in the velocities and obey constraints that follow from the reparametrisation invariance, and the Hamiltonian is a nonpolynomial function of the fields. None of this is fatal, but it is unpleasant, and it is the same unpleasantness the einbein removed for the particle. We do the same thing here.

1.1.3 The Polyakov action and classical equivalence

Introduce an independent worldsheet metric $\gamma_{\alpha\beta}(\sigma)$, a symmetric, non-degenerate, Lorentzian-signature tensor field on the worldsheet, with inverse $\gamma^{\alpha\beta}$ and determinant $\gamma = \det\gamma_{\alpha\beta} < 0$. It is a new dynamical variable on the same footing as X^μ, not the induced metric (1.5). The *Polyakov action* is

$$\boxed{S_{\rm P}[X,\gamma] = -\frac{1}{4\pi\alpha'}\int_\Sigma \mathrm{d}^2\sigma\,\sqrt{-\gamma}\,\gamma^{\alpha\beta}\,\partial_\alpha X^\mu\,\partial_\beta X^\nu\,\eta_{\mu\nu}\,.} \tag{1.7}$$

This is a theory of D scalar fields X^μ living on a two-dimensional curved worldsheet, coupled to worldsheet gravity $\gamma_{\alpha\beta}$. From the two-dimensional point of view it is a nonlinear sigma model with a flat target space; in Chapter 5 we curve the target and read the consistency conditions as spacetime equations of motion. The metric $\gamma_{\alpha\beta}$ appears in (1.7) without any derivatives of its own. It carries no kinetic term and propagates nothing. Its equation of motion is therefore algebraic, a constraint, and we can solve it and eliminate γ exactly.

Vary (1.7) with respect to $\gamma^{\alpha\beta}$, using $\delta\sqrt{-\gamma} = -\frac{1}{2}\sqrt{-\gamma}\,\gamma_{\alpha\beta}\,\delta\gamma^{\alpha\beta}$:

$$\delta_\gamma S_{\rm P} = -\frac{1}{4\pi\alpha'}\int \mathrm{d}^2\sigma\,\sqrt{-\gamma}\left(\partial_\alpha X\cdot\partial_\beta X - \tfrac{1}{2}\,\gamma_{\alpha\beta}\,\gamma^{\rho\sigma}\partial_\rho X\cdot\partial_\sigma X\right)\delta\gamma^{\alpha\beta}\,. \tag{1.8}$$

Setting the bracket to zero gives

$$\partial_\alpha X\cdot\partial_\beta X = \tfrac{1}{2}\,\gamma_{\alpha\beta}\,\gamma^{\rho\sigma}\partial_\rho X\cdot\partial_\sigma X\,. \tag{1.9}$$

The left-hand side is the induced metric $h_{\alpha\beta}$ of (1.5), so the equation says $\gamma_{\alpha\beta}$ is proportional to $h_{\alpha\beta}$ with a position-dependent proportionality factor. Take the determinant of both sides of (1.9): in two dimensions $\det(\lambda\,\gamma_{\alpha\beta}) = \lambda^2\gamma$, and the local factor cancels between numerator and denominator when we form $\sqrt{-h}$. Concretely, writing $\gamma_{\alpha\beta} = \Lambda(\sigma)\,h_{\alpha\beta}$ one finds $\sqrt{-\gamma}\,\gamma^{\alpha\beta} = \sqrt{-h}\,h^{\alpha\beta}$, because the two powers of Λ from $\sqrt{-\gamma}$ cancel the Λ^{-1} from each inverse-metric index. Substituting back,

$$S_{\rm P}\big|_{\text{on the constraint}} = -\frac{1}{4\pi\alpha'}\int \mathrm{d}^2\sigma\,\sqrt{-h}\,h^{\alpha\beta}h_{\alpha\beta} = -\frac{1}{2\pi\alpha'}\int \mathrm{d}^2\sigma\,\sqrt{-h} = S_{\rm NG}\,, \tag{1.10}$$

where $h^{\alpha\beta}h_{\alpha\beta} = 2$ in two dimensions. The Polyakov action, with its independent metric solved for and put back, is exactly the Nambu-Goto action. This is the classical equivalence promised in the chapter overview. The independent metric was a bookkeeping device that let us hide the square root; on shell it is fixed to be the induced metric (up to a Weyl factor we are about to understand), and the geometry of Section 1.1.2 returns.

Remark 1.1. The factor $\Lambda(\sigma)$ left undetermined in $\gamma_{\alpha\beta} = \Lambda\, h_{\alpha\beta}$ is not a defect of the derivation. The metric equation of motion (1.9) fixes $\gamma_{\alpha\beta}$ only up to an overall local rescaling, because the Polyakov action does not care about that rescaling. That indifference is Weyl symmetry, discussed next, and it is what makes the two-dimensional story richer than the one-dimensional einbein.

1.1.4 The symmetries

The Polyakov action carries three symmetries. Two are local (gauge) symmetries acting on the worldsheet, and one is a global symmetry acting on spacetime. Keeping them straight is essential, because the quantisation in Section 1.4 lives or dies by whether the local ones survive.

Worldsheet diffeomorphisms. The action is invariant under reparametrisations of the worldsheet, $\sigma^\alpha \to \sigma'^\alpha(\sigma)$, under which X^μ transforms as a set of scalars and $\gamma_{\alpha\beta}$ as a symmetric tensor. Infinitesimally, with $\delta\sigma^\alpha = \xi^\alpha(\sigma)$,

$$\delta X^\mu = \xi^\alpha \partial_\alpha X^\mu\,, \qquad \delta\gamma_{\alpha\beta} = \xi^\rho \partial_\rho \gamma_{\alpha\beta} + \partial_\alpha \xi^\rho\, \gamma_{\rho\beta} + \partial_\beta \xi^\rho\, \gamma_{\alpha\rho} = \nabla_\alpha \xi_\beta + \nabla_\beta \xi_\alpha\,, \tag{1.11}$$

with ∇ the worldsheet Levi-Civita connection built from $\gamma_{\alpha\beta}$. This is two arbitrary functions $\xi^\alpha(\sigma)$ of gauge freedom.

Weyl rescalings. The action is invariant under local rescalings of the worldsheet metric that leave X^μ untouched,

$$\gamma_{\alpha\beta}(\sigma) \to e^{2\omega(\sigma)}\, \gamma_{\alpha\beta}(\sigma)\,, \qquad X^\mu \to X^\mu\,. \tag{1.12}$$

This is one more arbitrary function $\omega(\sigma)$. Invariance holds because the combination appearing in (1.7) is Weyl inert in two dimensions:

$$\sqrt{-\gamma} \to e^{2\omega}\sqrt{-\gamma}\,, \qquad \gamma^{\alpha\beta} \to e^{-2\omega}\gamma^{\alpha\beta} \quad \Longrightarrow \quad \sqrt{-\gamma}\,\gamma^{\alpha\beta} \text{ is invariant.} \tag{1.13}$$

The cancellation is exact only in two dimensions: in d worldsheet dimensions $\sqrt{-\gamma} \to e^{d\omega}\sqrt{-\gamma}$, which balances the two inverse-metric factors only when $d = 2$. Weyl symmetry is the extra local symmetry that has no worldline analogue, and it is the redundancy that was missing from the counting in Section 1.1.1.

Two gauge functions from diffeomorphisms plus one from Weyl make three local functions, matching the three components of the symmetric 2×2 metric $\gamma_{\alpha\beta}$. Locally we can therefore use the gauge freedom to set $\gamma_{\alpha\beta}$ to whatever we please, for instance the flat metric $\eta_{\alpha\beta}$. That two real functions of reparametrisation always suffice to bring a Lorentzian two-metric to this conformally flat form is the existence of isothermal coordinates in two dimensions; we set this up in Section 1.2 and collect the CFT statement in Appendix F. Fixing $\gamma_{\alpha\beta} = \eta_{\alpha\beta}$ is conformal gauge, the starting point of Section 1.2. A worldsheet cosmological constant $\mu \int \mathrm{d}^2\sigma\, \sqrt{-\gamma}$ would spoil (1.13), so Weyl invariance forbids it: the vacuum energy of the worldsheet is pinned to zero by symmetry, classically.

Global spacetime Poincaré. Because the target metric is the constant $\eta_{\mu\nu}$, the action is invariant under

$$X^\mu \to \Lambda^\mu{}_\nu\, X^\nu + a^\mu\,, \qquad \Lambda \in \mathrm{SO}(1, D-1)\,, \tag{1.14}$$

with Λ, a constant. From the worldsheet viewpoint this is an internal global symmetry rotating the D scalar fields into each other; from the spacetime viewpoint it is the statement that the string propagates in flat Minkowski space with no preferred origin or frame. Its Noether currents are the worldsheet currents whose charges are the spacetime momentum p^μ and angular momentum $J^{\mu\nu}$ of the string.

Remark 1.2. One term is allowed by all three symmetries and we have quietly dropped it: the worldsheet Einstein-Hilbert term $\frac{\lambda}{4\pi}\int\sqrt{-\gamma}\,R^{(2)}$. In two dimensions $\int\sqrt{-\gamma}\,R^{(2)}=4\pi\chi$ is topological (Gauss-Bonnet), with χ the Euler characteristic of the worldsheet, so it does not affect the local dynamics. It contributes a constant $e^{-\lambda\chi}$ per worldsheet topology, which is how the string coupling $g_s=e^{\langle\phi\rangle}$ weights the genus expansion. We return to this in Chapter 4; here it plays no role.

1.1.5 The stress tensor: traceless and conserved

Since $\gamma_{\alpha\beta}$ is a metric, varying the action with respect to it defines a stress-energy tensor. We use the string normalisation

$$T_{\alpha\beta}\coloneqq-\frac{4\pi}{\sqrt{-\gamma}}\,\frac{\delta S_{\mathrm{P}}}{\delta\gamma^{\alpha\beta}}\,, \tag{1.15}$$

where the factor of 4π (rather than the general-relativity factor of 2) is a convention that makes the mode algebra of Section 1.3 come out clean. Reading it off (1.8),

$$T_{\alpha\beta}=\frac{1}{\alpha'}\Big(\partial_\alpha X\cdot\partial_\beta X-\tfrac{1}{2}\,\gamma_{\alpha\beta}\,\gamma^{\rho\sigma}\partial_\rho X\cdot\partial_\sigma X\Big)\,. \tag{1.16}$$

The metric equation of motion (1.9) is now simply

$$T_{\alpha\beta}=0\,. \tag{1.17}$$

These are constraints, not evolution equations, because $\gamma_{\alpha\beta}$ was non-dynamical. In conformal gauge they become the (classical) Virasoro constraints and organise the entire physical spectrum; that is the business of Sections 1.2 and 1.3. Two properties of $T_{\alpha\beta}$ follow directly from the two local symmetries, and both are worth seeing explicitly.

Tracelessness follows from Weyl invariance. Under (1.12) the inverse metric varies as $\delta_\omega\gamma^{\alpha\beta}=-2\omega\,\gamma^{\alpha\beta}$, so Weyl invariance of the action reads

$$0=\delta_\omega S_{\mathrm{P}}=\int\mathrm{d}^2\sigma\,\frac{\delta S_{\mathrm{P}}}{\delta\gamma^{\alpha\beta}}\,\delta_\omega\gamma^{\alpha\beta}=\frac{1}{2\pi}\int\mathrm{d}^2\sigma\,\sqrt{-\gamma}\,\omega\,\gamma^{\alpha\beta}T_{\alpha\beta}\,. \tag{1.18}$$

This must hold for every function $\omega(\sigma)$, so the trace vanishes,

$$T^{\alpha}{}_{\alpha}\equiv\gamma^{\alpha\beta}T_{\alpha\beta}=0\,. \tag{1.19}$$

This is an identity, true off shell and for any field configuration, not a consequence of any equation of motion. One checks it straight from (1.16) using $\gamma^{\alpha\beta}\gamma_{\alpha\beta}=2$: the two terms cancel identically. Tracelessness of the stress tensor is the hallmark of a classically conformal (scale-invariant) theory, and its fate under quantisation is the central drama of this chapter. The trace refuses to stay zero once we regularise the theory, and the coefficient of its failure is the central charge, the Weyl anomaly of Section 1.3.

Conservation follows from diffeomorphism invariance. Invariance of the action under (1.11), combined with the equation of motion for X^μ, forces $T_{\alpha\beta}$ to be covariantly conserved,

$$\nabla^\alpha T_{\alpha\beta}=0\,. \tag{1.20}$$

Unlike tracelessness this holds only on shell, that is, when X^μ solves its own equation of motion. Varying (1.7) with respect to X^μ gives that equation,

$$\frac{1}{\sqrt{-\gamma}}\,\partial_\alpha\Big(\sqrt{-\gamma}\,\gamma^{\alpha\beta}\partial_\beta X^\mu\Big)=\nabla^\alpha\nabla_\alpha X^\mu=0\,, \tag{1.21}$$

the covariant wave equation on the worldsheet. Verifying (1.20) is then a short computation: taking the divergence of (1.16), the term $\nabla^\alpha \partial_\alpha X \cdot \partial_\beta X$ vanishes by (1.21), and the remaining two terms cancel because $\nabla_\alpha \partial_\beta X = \nabla_\beta \partial_\alpha X$ (the connection is torsion-free).

The variation leading to (1.21) also produces a boundary term. For a closed string the worldsheet has no spatial boundary and it drops out; for an open string it does not, and its vanishing forces conditions on the endpoints. Those are the Neumann and Dirichlet boundary conditions, and we treat them carefully in Section 1.2, where they are planted as the seed of the D-branes of Chapter 6.

Careful

The stress tensor here is the *worldsheet* stress tensor, obtained by varying the two-dimensional metric $\gamma_{\alpha\beta}$. It is a completely different object from the spacetime stress tensor that sources Einstein's equations. When the two need to be distinguished later, the worldsheet one always carries worldsheet indices α, β and the spacetime one carries μ, ν.

1.1.6 Where this is going

We now have the classical theory in the form we want it: D free scalar fields X^μ on a worldsheet, dressed with a metric $\gamma_{\alpha\beta}$ that is pure gauge, and subject to the constraints $T_{\alpha\beta} = 0$. The three symmetries, worldsheet diffeomorphisms, Weyl rescalings, and spacetime Poincaré, are all we will use. The next step, in Section 1.2, is to spend the diffeomorphism and Weyl freedom to fix $\gamma_{\alpha\beta} = \eta_{\alpha\beta}$ (conformal gauge), whereupon the wave equation (1.21) becomes the flat two-dimensional wave equation and the constraints (1.17) become the Virasoro constraints. The subtlety we are walking towards is that Weyl invariance, a perfectly good classical redundancy, does not automatically survive quantisation. The obstruction is the central charge, and demanding that it cancel is what will single out $D = 26$ in Section 1.4. A light-cone gauge shortcut to the same spectrum exists, but we take the covariant route throughout, where the anomaly is visible and the logic is transparent.

Exercise 1.1 (The einbein in detail)**.** *(Warm-up. Goal: the einbein form of the point particle, the model for the Polyakov action.)* Starting from (1.3), show that eliminating e through its equation of motion reproduces (1.2) for $m \neq 0$. Then show that the reparametrisation $\tau \to \tau'(\tau)$ acts on the einbein as $e \to e\,\mathrm{d}\tau/\mathrm{d}\tau'$, and use this to argue that e can always be set to a constant. Finally, take $m = 0$ in (1.3) and show that the resulting equations of motion describe a null geodesic, whereas (1.2) is simply ill-defined there.

Exercise 1.2 (Why two dimensions)**.** *(Warm-up. Goal: the symmetry counting that makes $d = 2$ special and permits conformal gauge.)* Verify (1.13) and confirm that $\sqrt{-\gamma}\,\gamma^{\alpha\beta}$ is Weyl invariant only in $d = 2$ worldsheet dimensions. Count the components of a symmetric $d \times d$ metric against the d diffeomorphism parameters plus one Weyl parameter, and show that $d = 2$ is the unique dimension in which the local symmetries can gauge the metric away completely. What is left undetermined for $d = 1$ (the particle), and what is over-constrained for $d \geq 3$?

Exercise 1.3 (Tracelessness is an identity)**.** *(Standard. Goal: tracelessness from Weyl and conservation from diffeomorphisms, and which needs the equation of motion.)* Using (1.16) and $\gamma^{\alpha\beta}\gamma_{\alpha\beta} = 2$, show directly that $\gamma^{\alpha\beta}T_{\alpha\beta} = 0$ for any $X^\mu(\sigma)$, without using any equation of motion. Then take the divergence of (1.16) and show, using the wave equation (1.21) and the symmetry of second covariant derivatives, that $\nabla^\alpha T_{\alpha\beta} = 0$. Note carefully which of the two results needs the equation of motion and which does not, and connect each to the symmetry (Weyl or diffeomorphism) responsible for it.

1.2 Conformal gauge, mode expansions, and boundary conditions

The previous section left us with the Polyakov action,

$$S_{\mathrm{P}} \;=\; -\frac{1}{4\pi\alpha'}\int \mathrm{d}^2\sigma\,\sqrt{-\gamma}\,\gamma^{\alpha\beta}\,\partial_\alpha X^\mu\,\partial_\beta X^\nu\,\eta_{\mu\nu}, \tag{1.22}$$

a set of D scalar fields $X^\mu(\tau,\sigma)$ living on a two-dimensional worldsheet with its own intrinsic metric $\gamma_{\alpha\beta}(\tau,\sigma)$. We also have its stress tensor, obtained by varying $\gamma^{\alpha\beta}$,

$$T_{\alpha\beta} \;\propto\; \partial_\alpha X^\mu\,\partial_\beta X_\mu - \tfrac{1}{2}\,\gamma_{\alpha\beta}\,\gamma^{\rho\lambda}\,\partial_\rho X^\mu\,\partial_\lambda X_\mu, \tag{1.23}$$

which is traceless, $\gamma^{\alpha\beta}T_{\alpha\beta}=0$, and conserved on shell. The overall normalisation of Eq. (1.23) was fixed in the previous section; for everything that follows only the condition $T_{\alpha\beta}=0$ matters, and its normalisation drops out of that. The worldsheet indices α,β run over the two coordinates $(\sigma^0,\sigma^1)=(\tau,\sigma)$, as in Appendix A, and the spacetime metric is mostly plus.

The action Eq. (1.22) carries three local symmetries: worldsheet reparametrisations, two functions' worth, and Weyl rescalings $\gamma_{\alpha\beta}\to e^{2\omega(\tau,\sigma)}\gamma_{\alpha\beta}$, one function. As the previous section counted, a symmetric worldsheet metric has three independent components, so these local redundancies are just enough to erase it.

1.2.1 Reaching conformal gauge

In two dimensions, and only in two dimensions, the numbers match: the three components of $\gamma_{\alpha\beta}$ against two diffeomorphisms and one Weyl factor. The worldsheet metric therefore carries no local physics and can be gauged away entirely. A diffeomorphism brings $\gamma_{\alpha\beta}$ to the form $e^{2\omega}\eta_{\alpha\beta}$ (conformal, or isothermal, coordinates always exist locally for a two-dimensional metric; see Appendix F), and the Weyl symmetry then removes the conformal factor $e^{2\omega}$. What remains is the flat, or *conformal*, gauge

$$\gamma_{\alpha\beta} \;=\; \eta_{\alpha\beta} \;=\; \mathrm{diag}(-1,+1). \tag{1.24}$$

In this gauge the Polyakov action collapses to D free scalars on a flat worldsheet,

$$S \;=\; -\frac{1}{4\pi\alpha'}\int \mathrm{d}^2\sigma\,\eta^{\alpha\beta}\,\partial_\alpha X^\mu\,\partial_\beta X_\mu \;=\; \frac{1}{4\pi\alpha'}\int \mathrm{d}^2\sigma\left(\dot X^\mu \dot X_\mu - X'^\mu X'_\mu\right), \tag{1.25}$$

where $\dot X \equiv \partial_\tau X$ and $X' \equiv \partial_\sigma X$. The equation of motion is the free two-dimensional wave equation,

$$\left(\partial_\tau^2 - \partial_\sigma^2\right) X^\mu \;=\; 0. \tag{1.26}$$

This is the whole dynamical content of the string in conformal gauge: each spacetime coordinate of the string oscillates like a free relativistic field in $1+1$ dimensions.

Conformal gauge is local

The reduction $\gamma_{\alpha\beta}\to\eta_{\alpha\beta}$ is a *local* statement. On a worldsheet with nontrivial topology (a torus for the one-loop closed string, a surface with boundaries and handles at higher order) there are finitely many components of the metric that no diffeomorphism or Weyl rescaling can remove. These leftover parameters are the *moduli* of the surface, and integrating over them is what produces string scattering amplitudes in Chapter 4. For quantising a single free string they play no role, and we work on the infinite cylinder (or its conformal image, the plane).

A second subtlety is more serious and is the engine of this whole chapter. Weyl invariance was essential to reach Eq. (1.24). Classically it holds. Quantum mechanically it can fail: the trace of the stress tensor can pick up an anomaly proportional to the worldsheet curvature, with coefficient the *central charge* c. If $c \neq 0$ the gauge choice Eq. (1.24) is illegal at the quantum level. Computing c and demanding it cancel is the content of Sections 1.3 and 1.4, and it is what will force $D = 26$.

1.2.2 Residual conformal symmetry

Fixing the gauge Eq. (1.24) does not use up all of diff $\times$ Weyl. Some combinations of a diffeomorphism and a compensating Weyl rescaling map $\eta_{\alpha\beta}$ back to itself, and these are a genuine residual gauge freedom. An infinitesimal reparametrisation $\sigma^\alpha \to \sigma^\alpha + \xi^\alpha$ shifts the flat metric by $\delta\gamma_{\alpha\beta} = \partial_\alpha\xi_\beta + \partial_\beta\xi_\alpha$ (the same sign convention as the previous section); this can be undone by a Weyl transformation if and only if the shift is pure trace,

$$\partial_\alpha\xi_\beta + \partial_\beta\xi_\alpha \;=\; (\partial\cdot\xi)\,\eta_{\alpha\beta}, \tag{1.27}$$

the conformal Killing equation. It is cleanest in light-cone worldsheet coordinates

$$\sigma^\pm \;=\; \tau \pm \sigma, \qquad \partial_\pm \;=\; \tfrac{1}{2}(\partial_\tau \pm \partial_\sigma)\,, \tag{1.28}$$

in which Eq. (1.27) reduces to $\partial_-\xi^+ = 0$ and $\partial_+\xi^- = 0$, so

$$\xi^+ = \xi^+(\sigma^+), \qquad \xi^- = \xi^-(\sigma^-) \tag{1.29}$$

are arbitrary functions of a single light-cone coordinate each. The residual symmetry is therefore infinite-dimensional: independent reparametrisations of the two null directions. This is the two-dimensional conformal group, and its generators will close into two commuting copies of the Virasoro algebra in Section 1.3. Holding on to this residual freedom, instead of fixing it by hand, is what the covariant BRST treatment of Section 1.4 does. A light-cone gauge that fixes it by setting $X^+ \propto \tau$ also exists and trivialises what follows, at the cost of manifest Lorentz invariance; we do not take that route.

1.2.3 The Virasoro constraints

Gauge-fixing came at a price. By setting $\gamma_{\alpha\beta} = \eta_{\alpha\beta}$ before varying, we discarded the equation of motion for $\gamma_{\alpha\beta}$ itself. That equation was $T_{\alpha\beta} = 0$, and it does not follow from Eq. (1.26). It must be imposed by hand, as a constraint on the free-field solutions:

$$T_{\alpha\beta} \;=\; 0. \tag{1.30}$$

These are the classical *Virasoro constraints*. Tracelessness means only two of the three components are independent. In light-cone coordinates they are the two null components,

$$T_{++} \;=\; \partial_+X^\mu\,\partial_+X_\mu \;=\; 0, \qquad T_{--} \;=\; \partial_-X^\mu\,\partial_-X_\mu \;=\; 0, \tag{1.31}$$

while $T_{+-} = T_{-+} = 0$ is automatic by tracelessness. Equivalently, in terms of τ, σ derivatives,

$$\dot{X}^\mu\dot{X}_\mu + X'^\mu X'_\mu = 0, \qquad \dot{X}^\mu X'_\mu = 0. \tag{1.32}$$

Physically these say the velocity of the string and its extension are orthogonal and equal in magnitude: the induced worldsheet geometry has no preferred timelike direction picked out by the embedding. They are the shadow of the reparametrisation invariance we gauged away, and imposing them is what will project the naive Fock space down to physical string states.

1.2.4 The closed string: mode expansion

For the closed string σ is periodic, $\sigma \sim \sigma + 2\pi$, and X^μ obeys the wave equation with no boundary. The general periodic solution of Eq. (1.26) splits into a right-mover depending on $\sigma^- = \tau - \sigma$ and a left-mover depending on $\sigma^+ = \tau + \sigma$,

$$X^\mu(\tau,\sigma) \;=\; X^\mu_{\mathrm{R}}(\sigma^-) + X^\mu_{\mathrm{L}}(\sigma^+), \tag{1.33}$$

and each is expanded in Fourier modes on the circle,

$$X^\mu_{\mathrm{R}}(\sigma^-) = \frac{x^\mu}{2} + \frac{\alpha'}{2}\,p^\mu\,\sigma^- + i\sqrt{\frac{\alpha'}{2}}\sum_{n\neq 0}\frac{1}{n}\,\alpha^\mu_n\,e^{-in\sigma^-}, \tag{1.34}$$

$$X^\mu_{\mathrm{L}}(\sigma^+) = \frac{x^\mu}{2} + \frac{\alpha'}{2}\,p^\mu\,\sigma^+ + i\sqrt{\frac{\alpha'}{2}}\sum_{n\neq 0}\frac{1}{n}\,\tilde\alpha^\mu_n\,e^{-in\sigma^+}. \tag{1.35}$$

The centre-of-mass position x^μ and momentum p^μ are shared between the two sectors; the oscillators α^μ_n (right) and $\tilde\alpha^\mu_n$ (left) are independent. Adding the two pieces,

$$X^\mu(\tau,\sigma) = x^\mu + \alpha' p^\mu\,\tau + i\sqrt{\frac{\alpha'}{2}}\sum_{n\neq 0}\frac{1}{n}\left(\alpha^\mu_n\,e^{-in\sigma^-} + \tilde\alpha^\mu_n\,e^{-in\sigma^+}\right), \tag{1.36}$$

so the string as a whole moves as a free particle of momentum p^μ, dressed with left- and right-moving vibrations. Reality of X^μ forces $(\alpha^\mu_n)^* = \alpha^\mu_{-n}$ and $(\tilde\alpha^\mu_n)^* = \tilde\alpha^\mu_{-n}$. It is convenient to package the zero modes as oscillators too,

$$\alpha^\mu_0 \;=\; \tilde\alpha^\mu_0 \;=\; \sqrt{\frac{\alpha'}{2}}\,p^\mu, \tag{1.37}$$

which matches Appendix A: the zero mode *is* the spacetime momentum. Upon quantisation each α^μ_n becomes an operator with the algebra recorded in Appendix A,

$$[\alpha^\mu_m,\alpha^\nu_n] = m\,\delta_{m+n,0}\,\eta^{\mu\nu}, \qquad [\tilde\alpha^\mu_m,\tilde\alpha^\nu_n] = m\,\delta_{m+n,0}\,\eta^{\mu\nu}, \qquad [\alpha^\mu_m,\tilde\alpha^\nu_n] = 0, \tag{1.38}$$

so that, up to the normalisation $\sqrt{m}$, α^μ_{-m} for $m>0$ creates a quantum and α^μ_m destroys one. The timelike oscillators α^0_n carry the wrong-sign commutator (the $\eta^{00}=-1$), which is the origin of negative-norm states and the reason a no-ghost theorem is needed at all. That theorem is stated in Section 1.4 and proved in Appendix F.

1.2.5 Modes of the constraints and level matching

Feed the expansion Eq. (1.36) into the constraint Eq. (1.31). Using $\partial_- X^\mu_{\mathrm{R}} = \sqrt{\alpha'/2}\sum_n \alpha^\mu_n e^{-in\sigma^-}$, the constraint $T_{--}=0$ is a Fourier series whose coefficients are

$$L_m \;=\; \frac{1}{2}\sum_{n\in\mathbb{Z}}\alpha^\mu_{m-n}\,\alpha_{n\,\mu}, \qquad \tilde L_m \;=\; \frac{1}{2}\sum_{n\in\mathbb{Z}}\tilde\alpha^\mu_{m-n}\,\tilde\alpha_{n\,\mu}, \tag{1.39}$$

the classical Virasoro generators. The constraints Eq. (1.31) are then simply $L_m=0$ and $\tilde L_m=0$ for all $m\in\mathbb{Z}$. The zero modes are special. Separating the momentum piece and using Eq. (1.37),

$$L_0 = \frac{1}{2}\alpha_0^2 + \sum_{n\geq 1}\alpha^\mu_{-n}\alpha_{n\,\mu} = \frac{\alpha'}{4}\,p^\mu p_\mu + N, \qquad \tilde L_0 = \frac{\alpha'}{4}\,p^\mu p_\mu + \tilde N, \tag{1.40}$$

where $N = \sum_{n\geq 1} \alpha_{-n}\cdot\alpha_n$ and $\tilde{N} = \sum_{n\geq 1} \tilde{\alpha}_{-n}\cdot\tilde{\alpha}_n$ are the left- and right-moving level (occupation) numbers. With $M^2 = -p^\mu p_\mu$ in the mostly-plus convention, the constraints $L_0 = \tilde{L}_0 = 0$ read, classically,

$$\frac{\alpha'}{4} M^2 \;=\; N \;=\; \tilde{N}. \tag{1.41}$$

The quantum version acquires a normal-ordering constant on each side (the honest $\zeta(-1) = -\frac{1}{12}$ computation and the resulting $a = 1$ are done in Section 1.5); that shift is identical for left and right movers and so does not affect their difference.

The equality $N = \tilde{N}$ is *level matching*, and its origin is physical. A closed string has no marked point, so nothing distinguishes one value of σ from another: rigid rotations $\sigma \to \sigma + \text{const}$ are a symmetry. The generator of these rotations is the total worldsheet momentum in the σ direction,

$$P \;=\; \frac{1}{2\pi\alpha'} \int_0^{2\pi} \mathrm{d}\sigma\; \dot{X}^\mu X'_\mu \;\propto\; L_0 - \tilde{L}_0, \tag{1.42}$$

so demanding that physical states carry no such intrinsic momentum is the statement

$$L_0 - \tilde{L}_0 \;=\; 0 \quad\Longleftrightarrow\quad N = \tilde{N}. \tag{1.43}$$

Left and right movers must be excited to the same level. This single constraint survives quantisation unchanged and shapes the entire closed-string spectrum: it is why the massless closed sector is a symmetric two-index object rather than a vector, hence why the graviton appears there (Section 1.5).

1.2.6 The open string: boundary conditions

For the open string take $\sigma \in [0, \pi]$, a segment with two ends. The wave equation is unchanged, but now varying the gauge-fixed action Eq. (1.25) leaves a boundary term at each endpoint,

$$\delta S \;\supset\; -\frac{1}{2\pi\alpha'} \int \mathrm{d}\tau \,\Big[\, X'^\mu\, \delta X_\mu \,\Big]_{\sigma=0}^{\sigma=\pi}. \tag{1.44}$$

For a stationary action this must vanish separately at $\sigma = 0$ and $\sigma = \pi$, which leaves, for each spacetime direction μ and each endpoint, two consistent choices:

$$\text{Neumann:}\quad \partial_\sigma X^\mu\big|_{\text{end}} = 0, \qquad\qquad \text{Dirichlet:}\quad \delta X^\mu\big|_{\text{end}} = 0 \;\Rightarrow\; X^\mu\big|_{\text{end}} = \text{const}. \tag{1.45}$$

The two conditions are physically opposite. Neumann lets the endpoint move freely; setting $X'^\mu = 0$ says no worldsheet momentum flows off the end of the string, so the endpoint is dragged along by the string's tension and nothing is fixed. Dirichlet does the reverse: it nails the endpoint to a definite location $X^\mu = \text{const}$ in the μ direction, and momentum *can* leave the string there, absorbed by whatever the string is tied to.

Neumann conditions in every direction are the Poincaré-invariant choice, and they give the familiar freely-flapping open string. Its mode expansion is a single set of oscillators (the two ends reflect left- into right-movers, so α and $\tilde{\alpha}$ are identified), building standing waves,

$$X^\mu(\tau,\sigma) \;=\; x^\mu + 2\alpha'\, p^\mu\, \tau + i\sqrt{2\alpha'} \sum_{n\neq 0} \frac{1}{n}\, \alpha_n^\mu\, e^{-in\tau} \cos(n\sigma). \tag{1.46}$$

The endpoints at $\sigma = 0, \pi$ sit at antinodes and carry the centre-of-mass momentum p^μ.

Now impose Dirichlet at both ends of a given direction, pinning them to $X^\mu(\tau, 0) = x_0^\mu$ and $X^\mu(\tau, \pi) = x_\pi^\mu$. The cosines are replaced by sines, which vanish at both ends, and the linear-in-τ

momentum term is replaced by a linear-in-σ term that stretches the string between the two fixed points,

$$X^\mu(\tau,\sigma) \;=\; x_0^\mu + (x_\pi^\mu - x_0^\mu)\frac{\sigma}{\pi} + \sqrt{2\alpha'}\sum_{n\neq 0}\frac{1}{n}\,\alpha_n^\mu\, e^{-in\tau}\sin(n\sigma). \tag{1.47}$$

There is no continuous momentum in a Dirichlet direction, because the endpoints cannot move: translation invariance along μ is broken, as one expects when a boundary is stuck to a fixed wall.

The seed of D-branes

Suppose we impose Neumann conditions along the directions $X^0,\dots,X^p$ and Dirichlet conditions along $X^{p+1},\dots,X^{D-1}$. Then both endpoints of the open string are free to roam in $p+1$ of the dimensions but are pinned to fixed values in the remaining $D-p-1$. The endpoints are confined to a $(p+1)$-dimensional hypersurface in spacetime. That surface is a *Dirichlet p-brane*, or Dp-brane. At this stage it is only a boundary condition, a place where strings are allowed to end. What turns it into a dynamical, gravitating, charged object, and why T-duality forces its existence the moment you accept the open string at all, is the story of Chapter 6. Notice already that the Dirichlet expansion Eq. (1.47) describes a string stretched taut between two hyperplanes: its rest energy is the tension of Eq. (1.22) times the separation, a first glimpse of the BPS masses that will organise the whole duality web.

Exercise 1.4 (Residual conformal transformations)**.** Starting from the conformal Killing equation Eq. (1.27) on the flat worldsheet with $\gamma_{\alpha\beta}=\eta_{\alpha\beta}$, show that in light-cone coordinates it reduces to $\partial_-\xi^+=0$ and $\partial_+\xi^-=0$, so the residual gauge transformations are $\sigma^+\to f(\sigma^+)$, $\sigma^-\to g(\sigma^-)$ with f,g arbitrary. Verify that a global scaling $\sigma^\pm\to\lambda^{\pm1}\sigma^\pm$ and a rigid shift are special cases, and count how these relate to the finitely many transformations that survive as genuine (globally defined, invertible) symmetries on the plane.

Exercise 1.5 (Stretched strings and the tension)**.** Take a single Dirichlet direction with endpoints fixed at 0 and Δx, and set all oscillators in Eq. (1.47) to zero. Show that the resulting configuration is a straight static string of length Δx. Evaluate its energy directly from the Nambu-Goto action of the previous section and obtain

$$E \;=\; T\,\Delta x \;=\; \frac{\Delta x}{2\pi\alpha'}.$$

This is the ground-state mass of a string stretched between two parallel D-branes; anticipate the correction from the oscillator ground-state energy, and compare with the BPS mass formula you will meet in Chapter 6.

With the string reduced to free two-dimensional fields, the constraints identified, and the spectrum-labelling oscillators in hand, we have taken the classical theory as far as it goes. The next section quantises these fields as a worldsheet conformal field theory, computes the central charge that measures the Weyl anomaly, and thereby turns the innocuous-looking gauge choice Eq. (1.24) into the condition that fixes the dimension of spacetime.

1.3 The minimal worldsheet CFT: OPE, stress tensor, and the central charge

Conformal gauge did not spend all of the gauge freedom. After fixing the worldsheet metric to be flat, the previous section left a residual invariance: the two-dimensional conformal transformations, generated by the components of the worldsheet stress tensor T_{ab}, whose vanishing $T_{ab}=0$ is the constraint that a physical configuration must satisfy. To impose that constraint on the quantum theory we need the operator algebra of the worldsheet fields, and in particular the algebra obeyed by T_{ab} itself. That algebra is a two-dimensional conformal field theory (CFT).

We need very little of it. The whole of this section is three tools: the operator product expansion, the holomorphic stress tensor, and one number, the central charge c. The central charge is what makes the string interesting; it is simultaneously the coefficient in the Virasoro algebra, the coefficient of the Weyl anomaly, and (in the next section) the thing whose cancellation fixes $D = 26$. A fuller reference card for two-dimensional CFT, including the correlators and Ward identities we use later without re-deriving, lives in Appendix F.

1.3.1 Radial quantisation

Work in Euclidean worldsheet time $\tau_E = i\tau$ and pass to the complex coordinate of Appendix A,

$$z = e^{\tau_E - i\sigma}, \qquad \bar{z} = e^{\tau_E + i\sigma}, \qquad \sigma \sim \sigma + 2\pi. \tag{1.48}$$

This map takes the Euclidean cylinder, coordinatised by (τ_E, σ), onto the complex plane. A surface of constant worldsheet time τ_E becomes a circle of constant radius $|z| = e^{\tau_E}$; the infinite past $\tau_E \to -\infty$ is the single point $z = 0$, and the infinite future $\tau_E \to +\infty$ is the point at infinity. Translation in Euclidean time, $\tau_E \to \tau_E + a$, is the dilatation $z \to e^a z$. The worldsheet Hamiltonian, the generator of time translations, is therefore the generator of dilatations on the plane.

Two things follow, and both are conveniences we exploit constantly. First, ordinary time ordering of operators becomes *radial ordering*: the operator inserted at larger $|z|$ stands to the left,

$$\mathcal{R}\{A(z)\,B(w)\} = \begin{cases} A(z)\,B(w), & |z| > |w|, \\ B(w)\,A(z), & |z| < |w|, \end{cases} \tag{1.49}$$

for the bosonic operators of this chapter (fermionic ghosts in the next section carry an extra sign). Every operator product below is understood as radially ordered, and every equation between operators holds inside a correlation function. Second, a state is created by acting on the infinite past, that is, by an operator inserted at $z = 0$; this is the state–operator correspondence, which we use only lightly here and record properly in Appendix F.

1.3.2 The free-boson OPE

The matter fields $X^\mu(z, \bar{z})$ are D free scalars. Their equation of motion in conformal gauge is $\partial\bar{\partial}X^\mu = 0$, so each splits into a holomorphic (left-moving) and an antiholomorphic (right-moving) piece, and the basic two-point function is the Green's function of the two-dimensional Laplacian $\partial\bar{\partial}\,\ln|z|^2 = 2\pi\,\delta^2(z, \bar{z})$. Fixed by Appendix A, it reads

$$\langle X^\mu(z, \bar{z})\,X^\nu(w, \bar{w})\rangle = -\frac{\alpha'}{2}\,\eta^{\mu\nu}\,\ln|z - w|^2. \tag{1.50}$$

The logarithm means X^μ itself is not a well-behaved conformal field. The good local operators are its derivatives and its exponentials $:e^{ik\cdot X}:$ (the vertex operators used in the spectrum and in amplitudes). The workhorse is the holomorphic current $\partial X^\mu(z)$, of conformal weight 1. Differentiating Eq. (1.50) and keeping the holomorphic piece,

$$\partial X^\mu(z)\,\partial X^\nu(w) \;\sim\; -\frac{\alpha'}{2}\,\frac{\eta^{\mu\nu}}{(z - w)^2}\,, \tag{1.51}$$

with an identical antiholomorphic relation for $\bar{\partial}X^\mu(\bar{z})$, and no singularity in a holomorphic–antiholomorphic pair. This is our first operator product expansion (OPE). In general the OPE writes the radially-ordered product of two local operators, as their separation shrinks, as a Laurent series in $(z - w)$ whose coefficients are again local operators,

$$A(z)\,B(w) \;=\; \sum_n \frac{\{AB\}_n(w)}{(z - w)^n}\,, \tag{1.52}$$

and the symbol "$\sim$" keeps only the singular terms, the regular part being the normal-ordered product $:A(w)B(w):$ plus its derivatives. The singular terms carry all the algebraic content, because they are what produce poles when one integrates against a contour.

The OPE Eq. (1.51) is the canonical commutator in disguise. Inserting the mode expansion of Appendix A,

$$\partial X^\mu(z) = -i\sqrt{\tfrac{\alpha'}{2}} \sum_{n\in\mathbb{Z}} \frac{\alpha_n^\mu}{z^{n+1}}, \tag{1.53}$$

the double pole in Eq. (1.51) is equivalent, contour by contour, to the oscillator algebra $[\alpha_m^\mu, \alpha_n^\nu] = m\,\delta_{m+n,0}\,\eta^{\mu\nu}$ established in the previous section. The OPE is the same physics repackaged so that contour integrals do the work that mode sums did.

1.3.3 The stress tensor

Classically, the previous section found T_{ab} traceless and conserved. Tracelessness ($T_{z\bar z} = 0$) means the two surviving components are separately holomorphic and antiholomorphic; conservation then makes T_{zz} a function of z alone and $T_{\bar z\bar z}$ a function of $\bar z$ alone. Write the holomorphic component, with the normal ordering needed to make a product of coincident operators finite, as

$$T(z) \;\coloneqq\; T_{zz}(z) \;=\; -\frac{1}{\alpha'}\,{:}\partial X^\mu\,\partial X_\mu{:}(z), \qquad \bar T(\bar z) \;=\; -\frac{1}{\alpha'}\,{:}\bar\partial X^\mu\,\bar\partial X_\mu{:}(\bar z). \tag{1.54}$$

Normal ordering here means subtract the self-contraction $\langle \partial X^\mu \partial X_\mu\rangle$ predicted by Eq. (1.51), so that $T(z)$ has a finite expectation value. The overall sign differs from the Lorentzian component T_{--} of the previous section, which carried $+1/\alpha'$. That difference is only apparent: the holomorphic mode expansion Eq. (1.53) carries a factor $-i$, whose square supplies the compensating -1, so both conventions give the same zero mode $L_0 = \frac{\alpha'}{4}p^2 + N$ (see Eq. (1.59) below).

What does $T(z)$ do? It generates conformal transformations, and the OPE of T with any field encodes how that field transforms. Contract one ∂X inside $T(z)$ against $\partial X^\mu(w)$ using Eq. (1.51). The two factors in T contribute equally, giving a factor of two that cancels the $-1/\alpha'$ and the $-\alpha'/2$:

$$T(z)\,\partial X^\mu(w) \;\sim\; \frac{\partial X^\mu(w)}{(z-w)^2} + \frac{\partial^2 X^\mu(w)}{z-w}\,, \tag{1.55}$$

where the single pole comes from Taylor-expanding the surviving $\partial X^\mu(z)$ about w. A local operator $\mathcal{O}$ with the OPE $T(z)\mathcal{O}(w) \sim h\,\mathcal{O}(w)/(z-w)^2 + \partial\mathcal{O}(w)/(z-w)$ is called a *primary of weight* h, and Eq. (1.55) says ∂X^μ is a primary of weight $h = 1$. The double pole reads off the weight; the single pole is the statement that $\mathcal{O}$ shifts by its own derivative, that is, that it is dragged along by the transformation.

Exercise 1.6. *(Warm-up.)* (a) Verify Eq. (1.55) by contracting one ∂X in $T(z)$ against $\partial X^\mu(w)$ with Eq. (1.51), keeping the factor of two from the two equal contractions, and read off $h = 1$ from the double pole. (b) Using only the propagator Eq. (1.50), show that the vertex operator ${:}e^{ik\cdot X}{:}(w,\bar w)$ is primary with holomorphic weight $h = \alpha' k^2/4$. Contract each ∂X in $T(z)$ against the exponential; each contraction brings down a factor $-i\frac{\alpha'}{2}\,k^\mu/(z-w)$, and the two together assemble the coefficient $\alpha' k^2/4$ of the double pole.

1.3.4 The TT OPE and the central charge

Now feed T to itself. There are two kinds of Wick contraction between $T(z) = -\alpha'^{-1}{:}\partial X\cdot\partial X{:}(z)$ and $T(w)$: contracting both fields of $T(z)$ with both fields of $T(w)$ (a pure number, the double contraction), or contracting one pair and leaving the rest normal-ordered (the single contraction, which rebuilds T).

Take the double contraction first, since it carries the new physics. Each contraction supplies a factor $-\frac{\alpha'}{2}\,\eta/(z-w)^2$ from Eq. (1.51), and there are two ways to pair the fields. Tracking the index sums,

$$T(z)\,T(w)\Big|_{\text{double}} = \frac{1}{\alpha'^2}\,\eta_{\mu\nu}\eta_{\rho\sigma}\cdot 2\left(-\frac{\alpha'}{2}\right)^2\frac{\eta^{\mu\rho}\eta^{\nu\sigma}}{(z-w)^4} = \frac{1}{2}\,\frac{\eta_{\mu\nu}\eta^{\mu\nu}}{(z-w)^4} = \frac{D/2}{(z-w)^4}\,, \tag{1.56}$$

where $\eta_{\mu\nu}\eta^{\mu\nu} = D$ counts the D scalars. The single contraction reassembles T and its derivative in exactly the pattern of Eq. (1.55), now with a weight-2 operator, and produces $2T(w)/(z-w)^2 + \partial T(w)/(z-w)$. Together,

$$\boxed{T(z)\,T(w) \;\sim\; \frac{c/2}{(z-w)^4} + \frac{2\,T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}\,, \qquad c = D.} \tag{1.57}$$

The last two terms say that T would be a primary of weight 2 if that were the whole story; they are dictated by conformal invariance and are the same in any CFT. The first term is not dictated by symmetry. Its coefficient c, a pure c-number, is the *central charge*. For D free bosons $c = D$, one unit per scalar. The antiholomorphic copy gives $\bar{c} = D$ in the same way, and the two sectors do not talk to each other.

The central term is a genuinely quantum effect. It arises only from the double contraction, that is, only from the ordering of operators; classically T transforms as a weight-two tensor and there is no $(z-w)^{-4}$ piece. Geometrically, $c \neq 0$ means that under a finite conformal map $z \to f(z)$ the stress tensor fails to transform as a tensor and picks up a Schwarzian-derivative anomaly proportional to c, the price of quantising.

Exercise 1.7. *(Warm-up.)* Reproduce the central coefficient in Eq. (1.56) by counting Wick contractions. There are two ways to pair the two fields of $T(z)$ with the two fields of $T(w)$, and each pairing supplies two factors of $-\frac{\alpha'}{2}\eta/(z-w)^2$. Show the index contractions collapse to $\eta_{\mu\nu}\eta^{\mu\nu} = D$ and the prefactors to $D/2$. Then check that the single contraction, leaving one ∂X uncontracted in each T, rebuilds $2T(w)/(z-w)^2 + \partial T(w)/(z-w)$.

1.3.5 The Virasoro algebra

Expand T in modes as a weight-two field, $T(z) = \sum_{n\in\mathbb{Z}} L_n\, z^{-n-2}$, so that

$$L_n = \oint \frac{\mathrm{d}z}{2\pi i}\, z^{\,n+1}\, T(z). \tag{1.58}$$

Substituting Eq. (1.53) into Eq. (1.54) writes these modes as bilinears in the oscillators of the previous section,

$$L_n = \frac{1}{2}\sum_{m\in\mathbb{Z}} :\alpha_{n-m}\cdot\alpha_m:\,, \tag{1.59}$$

which is how the classical Virasoro constraints $T_{ab} = 0$ become operators. With $\alpha_0^\mu = \sqrt{\alpha'/2}\,p^\mu$, the zero mode $L_0 = \frac{\alpha'}{4}p^2 + \sum_{m\geq 1}\alpha_{-m}\cdot\alpha_m$ is the object that will give the mass-shell condition; the additive constant hiding in the normal ordering of Eq. (1.59) is the intercept a, fixed honestly in the spectrum section by the $\zeta(-1) = -\frac{1}{12}$ argument, and we leave it there.

The commutator of two modes follows from the OPE Eq. (1.57) by the standard contour manipulation: a commutator of contour integrals equals a single integral over a small circle of z about w, which picks up the residues of the OPE.

$$[L_m, L_n] = \oint \frac{\mathrm{d}w}{2\pi i}\, w^{\,n+1} \oint_w \frac{\mathrm{d}z}{2\pi i}\, z^{\,m+1}\, T(z)T(w). \tag{1.60}$$

The double and single poles of Eq. (1.57) produce, respectively, the linear term and the central term, and the result is the *Virasoro algebra*

$$[L_m, L_n] = (m-n)\, L_{m+n} + \frac{c}{12}\, m\,(m^2-1)\,\delta_{m+n,0}\,, \tag{1.61}$$

with $c = D$, and the identical algebra for the $\bar{L}_n$ built from $\bar{T}$. The central extension vanishes for $m = 0, \pm 1$: the generators $L_0, L_{\pm 1}$ still close on the anomaly-free $\mathrm{SL}(2,\mathbb{C})$ of global conformal transformations, which is why the vacuum can be $\mathrm{SL}(2,\mathbb{C})$-invariant even when $c \neq 0$. Everywhere else the c-term is an unavoidable quantum central charge, the operator shadow of the c-number in Eq. (1.57).

Exercise 1.8. *(Challenging; double-contour residue calculus.)* Derive the central term of Eq. (1.61) directly from the OPE. Insert Eq. (1.57) into Eq. (1.60) and do the inner z-contour by residues, using $\oint_w \frac{\mathrm{d}z}{2\pi i}\,\frac{z^{m+1}}{(z-w)^k} = \binom{m+1}{k-1} w^{m+2-k}$. Show that the $2T/(z-w)^2$ and $\partial T/(z-w)$ terms combine into $(m-n)L_{m+n}$, and that the $(z-w)^{-4}$ term gives $\frac{c}{12}(m^3-m)\delta_{m+n,0}$. Confirm the m^3-m structure is forced: it is the unique (up to normalisation and a shift absorbable into L_0) central extension of the Witt algebra. *Hint:* hold w fixed and do the inner z-integral first, so that the double-integral commutator collapses to a single residue about $z = w$; only then do the outer w-integral.

1.3.6 The central charge is the Weyl anomaly

The number $c = D$ we have been computing is the same number that obstructs Weyl invariance. Classically, the previous section derived $T^a{}_a = 0$ from Weyl symmetry of the Polyakov action. In the quantum theory this trace does not vanish once the worldsheet is curved. On a worldsheet with intrinsic Ricci scalar $R^{(2)}$, the expectation value of the trace is

$$\langle T^a{}_a \rangle = -\frac{c}{12}\, R^{(2)}\,, \tag{1.62}$$

the *Weyl (conformal) anomaly*. We assert the coefficient here rather than compute it; the derivation, and the precise curvature convention that fixes its sign, are in Appendix F. The point is that the very same c appears in Eq. (1.57), Eq. (1.61) and Eq. (1.62). A short-distance singularity in the TT OPE and a long-distance dependence on the worldsheet curvature are two faces of one anomaly.

This is the handle we need. The gauge-fixed Polyakov path integral is a physical theory only if it does not depend on the fiducial worldsheet metric we chose when going to conformal gauge, that is, only if the Weyl anomaly cancels. The matter fields X^μ contribute $c_{\text{matter}} = D$. They are not alone: the Faddeev-Popov determinant from fixing diffeomorphisms and Weyl symmetry is itself a conformal field theory, the bc ghost system of the next section, and it carries a central charge $c_{\text{gh}} = -26$. Central charges of decoupled sectors simply add, so

$$c_{\text{total}} = c_{\text{matter}} + c_{\text{gh}} = D - 26. \tag{1.63}$$

Demanding that the total Weyl anomaly vanish gives $D = 26$. The next section makes this statement airtight by showing that $c_{\text{total}} = 0$ is the condition for the BRST charge to be nilpotent, so that the gauge symmetry survives quantisation.

Remark 1.3. To be clear about what has and has not been shown: we *computed* $c = D$ for the matter sector from the free-boson OPE; we *quoted* the ghost value $c_{\text{gh}} = -26$ and the anomaly formula Eq. (1.62), deferring both to the next section and to Appendix F; and we have only *previewed* the deeper statement that in a curved spacetime background the condition for the Weyl anomaly to vanish is the spacetime equation of motion, so that Einstein's equations emerge as a consistency requirement of the worldsheet theory. That payoff waits for Chapter 5. For now one clean number, $c = D$, is all we carry into the derivation of the critical dimension.

1.4 Covariant quantisation, BRST, and the critical dimension $D = 26$

Conformal gauge did most of the work. It reduced the Polyakov path integral to D free bosons X^μ on a fixed worldsheet, with the residual conformal (Virasoro) symmetry and the matter central charge $c_m = D$ established in the previous section. Two questions remain, and they turn out to be one question wearing two masks. Was the gauge choice legitimate quantum mechanically, or does the conformal factor we threw away creep back as a Weyl anomaly? And the covariant Fock space contains timelike oscillators α^0_{-n}, whose commutator $[\alpha^0_m, \alpha^0_{-m}] = -m$ carries the wrong sign: are the negative-norm states they build actually in the spectrum? The apparatus that settles both at once is Faddeev-Popov gauge fixing promoted to a rigid fermionic symmetry, BRST. Its payoff is a Lorentz-covariant statement that spacetime must have $D = 26$ and that the normal-ordering constant is $a = 1$.

The big picture

Three conditions that look different are the same condition. The gauge-fixed path integral is independent of the conformal factor (no Weyl anomaly) iff the total central charge $c_{\rm tot} = c_m + c_{\rm gh}$ vanishes; the BRST charge squares to zero iff $c_{\rm tot} = 0$; and the no-ghost theorem holds in its critical form, with the timelike and longitudinal oscillators completely removed from the physical spectrum, when $c_{\rm tot} = 0$. Since $c_m = D$ and the ghosts contribute $c_{\rm gh} = -26$, all three read $D = 26$.

1.4.1 Faddeev-Popov gauge fixing and the bc ghosts

The Polyakov integral runs over metrics and embeddings divided by the volume of the local symmetry group,

$$Z = \int \frac{\mathcal{D}\gamma\, \mathcal{D}X}{V_{\rm diff\times Weyl}}\, e^{-S_P[X,\gamma]}. \tag{1.64}$$

Fixing the worldsheet metric $\gamma_{\alpha\beta}$ to a fiducial metric $\hat\gamma_{\alpha\beta}$ (flat, in conformal gauge; this rests on the existence of conformal coordinates in two dimensions, established in Appendix F) is a change of variables of the standard Faddeev-Popov type. Insert

$$1 = \Delta_{\rm FP}[\gamma] \int_{\rm diff\times Weyl} \mathcal{D}\zeta\; \delta\Big(\gamma - \hat\gamma^{\,\zeta}\Big)\,, \tag{1.65}$$

where $\hat\gamma^{\,\zeta}$ is the fiducial metric acted on by a diffeomorphism-plus-Weyl transformation ζ. The gauge orbit direction is spanned by the two infinitesimal generators

$$\delta_{\rm diff}\,\gamma_{\alpha\beta} = \nabla_\alpha\xi_\beta + \nabla_\beta\xi_\alpha, \qquad \delta_{\rm Weyl}\,\gamma_{\alpha\beta} = 2\,\omega\,\gamma_{\alpha\beta}. \tag{1.66}$$

Split the diffeomorphism variation into its trace and traceless parts, $\nabla_\alpha\xi_\beta + \nabla_\beta\xi_\alpha = (P_1\xi)_{\alpha\beta} + (\nabla\cdot\xi)\,\gamma_{\alpha\beta}$, where $(P_1\xi)_{\alpha\beta} = \nabla_\alpha\xi_\beta + \nabla_\beta\xi_\alpha - \gamma_{\alpha\beta}\,(\nabla\cdot\xi)$ is symmetric and traceless. The trace part combines with ω to shift the conformal factor, the piece a Weyl transformation is free to undo. The traceless part is the genuine content of the gauge slice, and the Jacobian $\Delta_{\rm FP} = \det{}'(P_1^\dagger P_1)^{1/2}$ is what the ghosts exponentiate. Writing the determinant as a Berezin integral introduces an anticommuting antighost $b_{\alpha\beta}$, conjugate to the traceless metric fluctuation and therefore symmetric and traceless, together with an anticommuting ghost c^α, a worldsheet vector carrying the diffeomorphism parameter. The determinant becomes a free path integral,

$$\Delta_{\rm FP} = \int \mathcal{D}b\,\mathcal{D}c\; e^{-S_{\rm gh}}, \qquad S_{\rm gh} = \frac{1}{2\pi}\int \mathrm{d}^2z\,\big(b\,\bar\partial c + \bar b\,\partial\bar c\big), \tag{1.67}$$

where in the complex frame of Appendix A the traceless antighost has the single holomorphic component $b \equiv b_{zz}$ and the vector ghost has $c \equiv c^z$ (plus their antiholomorphic partners $\bar b, \bar c$).

That $b_{\alpha\beta}$ is traceless, $b_\alpha{}^\alpha = 0$, is why the Weyl symmetry acquires no ghost of its own: the conformal factor is fixed algebraically, not by a propagating field. From here we work with one chirality; the antiholomorphic sector is an identical copy.

The action Eq. (1.67) is a first-order system with equations of motion $\bar\partial c = 0 = \bar\partial b$, so $b(z)$ and $c(z)$ are holomorphic. Reading off the propagator gives the operator products

$$b(z)\,c(w) \sim \frac{1}{z-w}, \qquad c(z)\,b(w) \sim \frac{1}{z-w}, \tag{1.68}$$

with $b(z)b(w)$ and $c(z)c(w)$ regular. Their conformal weights are inherited from their tensor structure: $b_{\alpha\beta}$ is a rank-two tensor of weight $h_b = 2$, and c^α is a vector of weight $h_c = -1$. The mode expansions consistent with these weights are

$$b(z) = \sum_{n\in\mathbb{Z}} \frac{b_n}{z^{n+2}}, \qquad c(z) = \sum_{n\in\mathbb{Z}} \frac{c_n}{z^{n-1}}, \qquad \{b_m, c_n\} = \delta_{m+n,0}, \tag{1.69}$$

all other anticommutators vanishing.

1.4.2 The ghost CFT and $c_{\rm gh} = -26$

The ghost stress tensor follows from varying Eq. (1.67) with respect to the metric before setting it flat. For a first-order bc system with b of weight λ and c of weight $1-\lambda$,

$$T^{\rm gh} = -\lambda\,{:}b\,\partial c{:} + (1-\lambda)\,{:}(\partial b)\,c{:}, \tag{1.70}$$

which for our case $\lambda = 2$ reads $T^{\rm gh} = -2\,{:}b\,\partial c{:} - {:}(\partial b)\,c{:}$. Its self-OPE has a quartic pole; feeding Eq. (1.68) into $T^{\rm gh}(z)\,T^{\rm gh}(w)$ and collecting terms (the contraction is the bookkeeping recorded in Appendix F) gives

$$T^{\rm gh}(z)\,T^{\rm gh}(w) \sim \frac{-13}{(z-w)^4} + \frac{2\,T^{\rm gh}(w)}{(z-w)^2} + \frac{\partial T^{\rm gh}(w)}{z-w}. \tag{1.71}$$

The quartic pole is $c_{\rm gh}/2$, so $c_{\rm gh} = -26$. The same number follows from the general first-order result, derived in Appendix F,

$$c_{bc}(\lambda) = 1 - 3(2\lambda-1)^2, \tag{1.72}$$

evaluated at $\lambda = 2$. The sign is a consequence of the anticommuting statistics: a commuting first-order system of the same weights would carry the opposite sign, and the value would fail to cancel the matter contribution. This one number is the whole reason the critical dimension is 26 and not some other integer. Nothing here is tuned by hand: the value is fixed by the tensor type ($\lambda = 2$) of the diffeomorphism ghost, which is fixed in turn by the fact that we are gauging a metric.

The ghost Virasoro modes $L_n^{\rm gh} = \oint \frac{\mathrm{d}z}{2\pi i}\, z^{n+1}\, T^{\rm gh}(z)$ satisfy the Virasoro algebra with central charge -26, and together with the matter modes L_n^m of the preceding section they build the total generators $L_n = L_n^m + L_n^{\rm gh}$.

1.4.3 The BRST charge

Gauge fixing broke the local symmetry, but a rigid fermionic remnant survives: the BRST transformation, under which X^μ shifts by a ghost-valued diffeomorphism, $\delta_B X^\mu \propto c\,\partial X^\mu$, and the antighost transforms into the total stress tensor, $\delta_B b \propto T = T^m + T^{\rm gh}$. The associated Noether current is

$$j_B = c\,T^m + \tfrac{1}{2}\,{:}c\,T^{\rm gh}{:} + \tfrac{3}{2}\,\partial^2 c, \tag{1.73}$$

and the BRST charge is its contour integral,

$$Q = \oint \frac{\mathrm{d}z}{2\pi i}\, j_B(z) = \sum_{n\in\mathbb{Z}} c_{-n}\, L_n^m + \sum_{m,n\in\mathbb{Z}} \frac{m-n}{2}\, {:}c_{-m}c_{-n}b_{m+n}{:} - a\, c_0. \tag{1.74}$$

The total-derivative improvement $\frac{3}{2}\,\partial^2 c$ is fixed by demanding that j_B be a weight-one primary, which is what allows Q to be well defined on the worldsheet, and this term ties the anomaly below to the combination $c_m - 26$. The constant a is the ghost normal-ordering constant, pinned down in Section 1.4.5.

Physically Q generates the gauge orbit fermionically: acting on a matter state it produces the Virasoro constraints weighted by ghosts, so that "Q-closed" is the operator statement of "satisfies the constraints". The point of the covariant formalism is that Q knows about the timelike and longitudinal modes and arranges their cancellation without our ever choosing a frame.

1.4.4 Nilpotency forces $D = 26$

Everything now hinges on one algebraic fact. BRST is a symmetry only if Q is nilpotent, $Q^2 = \frac{1}{2}\{Q, Q\} = 0$; otherwise a "gauge transformation" generated by Q fails to close and the cohomology below is not even defined. The obstruction is computed from the current OPE. With the improvement term in place,

$$j_B(z)\, j_B(w) \sim -\frac{c_m - 26}{2}\left[\frac{c\,\partial c(w)}{(z-w)^3} + \frac{\frac{1}{2}\,c\,\partial^2 c(w)}{(z-w)^2} + \frac{\frac{1}{6}\,c\,\partial^3 c(w)}{z-w}\right], \tag{1.75}$$

so that every singular term carries the same factor $c_m - 26$. Only the single pole contributes to the anticommutator of the charges, giving

$$\{Q, Q\} = \oint \frac{\mathrm{d}z}{2\pi i} \oint_z \frac{\mathrm{d}w}{2\pi i}\, j_B(z)\, j_B(w) = -\frac{c_m - 26}{12} \oint \frac{\mathrm{d}w}{2\pi i}\, c\,\partial^3 c(w). \tag{1.76}$$

The operator $\oint c\,\partial^3 c$ does not vanish, so

$$Q^2 = 0 \quad\Longleftrightarrow\quad c_m = 26 \quad\Longleftrightarrow\quad D = 26, \tag{1.77}$$

using $c_m = D$ for the D free bosons. Equivalently $c_{\mathrm{tot}} = c_m + c_{\mathrm{gh}} = D - 26$ must vanish: the matter central charge $+D$ is cancelled against the ghost -26. This is the fully covariant derivation of the critical dimension. No frame was chosen, Lorentz invariance was never sacrificed, and the number 26 came entirely from the weight-two tensor character of the metric ghost through Eq. (1.72). A light-cone gauge reaches the same 26 faster, by counting $D-2$ transverse oscillators directly, at the price of manifest Lorentz invariance; we took the covariant road because it keeps that invariance in plain sight and generalises without fuss to the superstring of Chapter 3.

Remark 1.4. The same $c_{\mathrm{tot}} = 0$ is the vanishing of the total Weyl anomaly. The gauge-fixed measure carries a factor $e^{-c_{\mathrm{tot}}\, S_L[\omega]/24\pi}$ in the discarded conformal factor ω, with S_L the Liouville action; the dependence on ω disappears when $c_{\mathrm{tot}} = 0$. When the flat target is replaced by a curved spacetime, this condition becomes the requirement that the sigma-model beta functions vanish, and those beta functions are the spacetime equations of motion. That is how gravity emerges as a consistency law rather than an input; we preview it here and pay it off in Chapter 5.

1.4.5 The intercept $a = 1$ from $\zeta(-1) = -\frac{1}{12}$

The mass-shell condition on physical states is $L_0^{\mathrm{tot}}\,|\psi\rangle = 0$, and on a matter state tensored with the ghost ground state this reduces to $(L_0^m - a)\,|\psi\rangle = 0$. The constant a is a zero-point energy, and computing it means confronting a divergent sum.

Take the matter piece first. Normal-ordering the quadratic term in L_0^m moves annihilators to the right and leaves a c-number,

$$L_0^m = \frac{\alpha'}{4}p^2 + \sum_{n=1}^{\infty} \alpha_{-n}\cdot\alpha_n + \frac{1}{2}\sum_{n=1}^{\infty} \eta_{\mu\nu}\,[\alpha_n^\mu, \alpha_{-n}^\nu], \qquad \eta_{\mu\nu}\,[\alpha_n^\mu, \alpha_{-n}^\nu] = n\,\eta_{\mu\nu}\eta^{\mu\nu} = n\,D. \tag{1.78}$$

The wrong-sign timelike commutator is automatically included in $\eta_{\mu\nu}\eta^{\mu\nu} = D$, so all D directions enter with the correct signs. The zero-point energy is therefore

$$E_0^m = \frac{D}{2}\sum_{n=1}^{\infty} n. \tag{1.79}$$

The sum diverges and has to be regulated. Insert a heat-kernel cutoff and expand,

$$\sum_{n=1}^{\infty} n\,e^{-\epsilon n} = -\frac{\mathrm{d}}{\mathrm{d}\epsilon}\frac{1}{e^\epsilon - 1} = \frac{1}{\epsilon^2} - \frac{1}{12} + \mathcal{O}\big(\epsilon^2\big)\,. \tag{1.80}$$

The $1/\epsilon^2$ piece is a local, cutoff-dependent divergence, proportional to the length of the string; it renormalises into the worldsheet cosmological constant and carries no physics. The universal finite part is the analytic continuation of $\zeta(s) = \sum_{n\geq 1} n^{-s}$ from its convergent domain $\mathrm{Re}\,s > 1$ to $s = -1$,

$$\sum_{n=1}^{\infty} n \;\longmapsto\; \zeta(-1) = -\frac{1}{12}. \tag{1.81}$$

The two regulators agree, which is the content of the claim that this finite part is physical and not an artefact. Hence $E_0^m = -D/24$.

The ghosts contribute a second zero-point energy, and it is not simply $-c_{\mathrm{gh}}/24$. The relevant ghost vacuum is the state annihilated by b_0, namely $|{\downarrow}\rangle = c_1|0\rangle = c(0)|0\rangle$; a short cylinder computation (a weight shift of this vacuum, carried out in Appendix F) gives

$$E_0^{\mathrm{gh}} = -1 + \frac{13}{12} = \frac{1}{12}. \tag{1.82}$$

Adding the two,

$$E_0 = E_0^m + E_0^{\mathrm{gh}} = -\frac{D}{24} + \frac{1}{12} = -\frac{D-2}{24}, \qquad \Longrightarrow \qquad a = \frac{D-2}{24} = 1 \;\text{ at } D = 26. \tag{1.83}$$

The bookkeeping reads as if only $D-2$ oscillator towers were present, which is the covariant echo of the transverse count: the ghost energy $+\frac{1}{12}$ is just what is needed to remove the timelike and longitudinal directions from the vacuum energy. The two determinations are consistent, as they must be, since $D = 26$ made Q nilpotent and only for a nilpotent Q is the reduction $L_0^{\mathrm{tot}} \to L_0^m - a$ meaningful.

1.4.6 Physical states as BRST cohomology

With $Q^2 = 0$ the physical spectrum is defined cohomologically. A state is physical if it is annihilated by Q (it satisfies the constraints), and two physical states are identified if they differ by a Q-exact state (a residual gauge transformation):

$$\mathcal{H}_{\mathrm{phys}} = \frac{\ker Q}{\mathrm{im}\,Q} = \{\;|\psi\rangle : Q\,|\psi\rangle = 0\;\}\Big/\{\;|\psi\rangle \sim |\psi\rangle + Q\,|\chi\rangle\;\}. \tag{1.84}$$

That Q-exact states have zero norm and decouple from all physical amplitudes is the reason Eq. (1.84) is well posed: gauge artefacts are the null directions. One works in the relative

cohomology, restricting to states with $b_0 |\psi\rangle = 0$, because $\{Q, b_0\} = L_0^{\text{tot}}$ makes b_0 the natural partner that enforces the mass-shell condition of Section 1.4.5.

The reason this reproduces old-fashioned covariant quantisation, where one imposes $L_n |\psi\rangle = 0$ for $n > 0$ and $(L_0 - a) |\psi\rangle = 0$ by hand, is that the cohomology at the relevant ghost number is carried by matter states obeying those conditions, dressed by the fixed ghost vacuum. The genuinely new content is that the negative-norm states are guaranteed to drop out.

Theorem 1.5 (No-ghost theorem, stated)**.** *For $D = 26$ and $a = 1$ the relative BRST cohomology Eq. (1.84) at ghost number one is a positive-definite Hilbert space, isomorphic to the Fock space generated by the $D - 2$ transverse oscillators acting on the momentum vacuum. In particular it contains no negative-norm states, and no states of null norm beyond the Q-exact ones already quotiented out.*

The proof constructs an explicit quartet mechanism pairing each unphysical matter oscillator with a ghost excitation, so that they cancel in cohomology; it is a self-contained piece of CFT and we defer it to Appendix F rather than interrupt the physics here. The dimension is genuinely critical in this statement. The spectrum can be made ghost-free for any $D \leq 26$, which is the content of the old-covariant no-ghost theorem; but for $D < 26$ the theory carries a Weyl anomaly and is non-critical, the conformal factor failing to decouple. Only at $D = 26$ is Q nilpotent, so only there is the BRST cohomology Eq. (1.84) defined at all, and only there do nilpotency, anomaly freedom and unitarity hold together. For $D > 26$ negative-norm states survive into the spectrum.

Careful

The absence of negative-norm states is not the absence of pathology. Theorem 1.5 guarantees a sensible inner product, but says nothing about the sign of M^2. The lowest state, at $N = 0$, has $M^2 = -a/\alpha' < 0$: a tachyon. It is a perfectly good, positive-norm state that happens to signal an unstable vacuum. Curing it is the business of the superstring in Chapter 3; enumerating the rest of the spectrum, including the massless graviton, Kalb-Ramond field and dilaton, is the business of the next section.

Exercise 1.9 (Moderate)**.** Using the general result Eq. (1.72), evaluate the central charge of an anticommuting first-order system whose fields have weights $(\frac{1}{2}, \frac{1}{2})$ and confirm it equals $+1$, the value for a complex free fermion. Explain in one line why replacing the metric ghost's weight $\lambda = 2$ by a different value would move the critical dimension. Finally, the superconformal ghost that fixes $D = 10$ in Chapter 3 is a *commuting* weight-$\frac{3}{2}$ system, so its central charge is given by the opposite-sign formula $c = -1 + 3(2\lambda - 1)^2$; evaluate this at $\lambda = \frac{3}{2}$ and check that it is $+11$ (*hint:* the sign in front of $3(2\lambda - 1)^2$ is set by the statistics of the ghosts, Fermi versus Bose).

Exercise 1.10 (Short)**.** The BRST charge is Hermitian, $Q^\dagger = Q$. Use this together with $Q^2 = 0$ to show that a Q-exact state $|s\rangle = Q |\chi\rangle$ is (i) itself Q-closed and (ii) orthogonal to every Q-closed state, so in particular $\langle s | s = 0$. Conclude that the physical inner product descends to BRST cohomology and that the identification $|\psi\rangle \sim |\psi\rangle + Q |\chi\rangle$ in Eq. (1.84) quotients out precisely the null directions. (At level one this is the statement that the longitudinal polarisation $\zeta_\mu \propto k_\mu$ of the would-be photon $\zeta \cdot \alpha_{-1}\, c_1 |0; k\rangle$ is Q-exact and decouples; Theorem 1.5 extends this to all levels.)

Exercise 1.11 (Harder)**.** Show from Eqs. (1.69) and (1.73) that $\{Q, b_0\} = L_0^{\text{tot}} = L_0^m + L_0^{\text{gh}}$. Deduce that any Q-closed state built on the ghost vacuum $|{\downarrow}\rangle$ satisfies the mass-shell condition $(L_0^m - a) |\psi\rangle = 0$ with the intercept of Eq. (1.83), and that a state failing the mass-shell condition can be Q-closed only if it is also Q-exact, hence trivial in cohomology. (*Hint:* the quickest route is the OPE $j_B(z)\, b(w)$, whose single pole is the total stress tensor $T^{\text{tot}}(w)$; the double contour then yields L_0^{tot}. Alternatively anticommute the modes in Eq. (1.74) against b_0 directly.)

1.5 The spectrum and its lessons

All the hard work is behind us. We have a consistent quantisation of the free string: the matter and ghost central charges cancel at $D = 26$, the normal-ordering constant is fixed to $a = 1$, and physical states are the cohomology of the BRST charge Q (Section 1.4). The no-ghost theorem tells us that this cohomology is a positive-norm space isomorphic to the transverse oscillator states, with the proof deferred to Appendix F. What remains is the payoff: turn the crank on the lowest few mass levels and see what spacetime fields the string is describing. Two of them will change the character of the whole course.

1.5.1 The mass formula

Physical states satisfy the Virasoro constraints derived in Sections 1.2 and 1.4: the positive modes annihilate them,

$$L_n \,|\psi\rangle = 0 \quad (n \geq 1), \qquad (L_0 - 1)\,|\psi\rangle = 0, \tag{1.85}$$

together with the tilded copies $\tilde{L}_n$, $\tilde{L}_0$ for the closed string. The constraint that carries the physics is the zero mode. Writing the matter Virasoro zero mode in terms of the oscillators of Appendix A,

$$L_0 = \frac{\alpha'}{4}\,p^2 + N, \qquad N \coloneqq \sum_{n=1}^{\infty} \alpha_{-n}\cdot\alpha_n, \tag{1.86}$$

where $\alpha_{-n}\cdot\alpha_n = \eta_{\mu\nu}\,\alpha_{-n}^{\mu}\alpha_n^{\nu}$ and N is the (holomorphic) *level operator*, an integer-valued number operator counting oscillator excitations weighted by their mode number. The intercept 1 in (1.85) is the normal-ordering constant $a = 1$, obtained in Section 1.4 by regulating $\frac{1}{2}\sum_{n\geq 1} n$ with $\zeta(-1) = -\frac{1}{12}$.

Using $\alpha_0^{\mu} = \sqrt{\alpha'/2}\,p^{\mu}$ and $M^2 = -p^2$ (mostly-plus signature), the closed-string mass-shell condition $(L_0 - 1)\,|\psi\rangle = 0$ becomes

$$M^2 = \frac{4}{\alpha'}\,(N - 1), \qquad N = \tilde{N}. \tag{1.87}$$

The second equation is *level matching*. The combination $L_0 - \tilde{L}_0$ generates rigid translations of the coordinate σ round the closed string, and since no point on the string is special, physical states must be invariant: $(L_0 - \tilde{L}_0)\,|\psi\rangle = 0$, i.e. $N = \tilde{N}$. Left and right movers must be excited to the same level.

For the open string there is a single set of oscillators and the zero mode is $\alpha_0^{\mu} = \sqrt{2\alpha'}\,p^{\mu}$ (Section 1.2), so $L_0 = \alpha' p^2 + N$ and

$$M^2 = \frac{1}{\alpha'}\,(N - 1). \tag{1.88}$$

Both spectra sit on Regge trajectories: straight lines of states whose M^2 rises linearly with the level N, the kinematic fingerprint of an extended string rather than a point particle. The slope is set by α', the Regge slope, which is the inverse string tension, $T = 1/(2\pi\alpha')$, and carries units of length squared. It fixes the string length $\ell_s = \sqrt{\alpha'}$ and the string mass scale $M_{\mathrm{s}} = 1/\ell_s$ (Appendix A); M_{s} sets the spacing between levels. At a given level N the closed-string M^2 is four times the open-string M^2.

1.5.2 The ground state: a tachyon

At $N = 0$ there are no oscillators; the state is the momentum eigenstate $|0; k\rangle$, a Lorentz scalar. Its mass is negative:

$$M^2_{\mathrm{closed}} = -\frac{4}{\alpha'}, \qquad M^2_{\mathrm{open}} = -\frac{1}{\alpha'}. \tag{1.89}$$

A field with $M^2 < 0$ is a tachyon. The name is a historical misnomer: nothing propagates faster than light. What $M^2 < 0$ signals is that the scalar field $T(x)$ associated with $|0;k\rangle$ has an effective potential with a *maximum*, not a minimum, at $T = 0$. The configuration we quantised around, empty 26-dimensional Minkowski space, is a hilltop. It is not a stable vacuum, and perturbation theory built on it is expanding about the wrong point.

Careful

The bosonic-string tachyon is a genuine sickness, not a bookkeeping artefact. It is a normalisable, gauge-invariant physical state (it survives the BRST cohomology), and its negative M^2 signals a true instability of the 26-dimensional vacuum. Unlike the massless fields below it, we have no reliable endpoint for the resulting rolling. This is one of the two reasons the bosonic string is not the final theory.

1.5.3 The massless closed string: graviton, B-field, dilaton

The first level that survives is $N = \tilde{N} = 1$, where (1.87) gives $M^2 = 0$. Level matching forbids $N \neq \tilde{N}$, so the unique massless state is built with one left and one right oscillator,

$$|e;k\rangle = e_{\mu\nu}\,\alpha^{\mu}_{-1}\,\tilde{\alpha}^{\nu}_{-1}\,|0;k\rangle\,, \qquad k^2 = 0, \tag{1.90}$$

with a polarisation tensor $e_{\mu\nu}$, an arbitrary $D \times D$ matrix. Not every $e_{\mu\nu}$ gives a physical state, and not every physical $e_{\mu\nu}$ is distinct. The remaining Virasoro constraints in (1.85) cut it down. Only L_1 and $\tilde{L}_1$ act nontrivially at this level, and $L_1\,\alpha^{\mu}_{-1}\,|0;k\rangle \propto \alpha^{\mu}_{0}\,|0;k\rangle \propto k^{\mu}\,|0;k\rangle$, so

$$k^{\mu}e_{\mu\nu} = 0, \qquad k^{\nu}e_{\mu\nu} = 0. \tag{1.91}$$

The polarisation is transverse on both indices. On top of this there is a redundancy: the null (BRST-exact) states $L_{-1}\,\tilde{\alpha}^{\nu}_{-1}\,|0;k\rangle$ and $\tilde{L}_{-1}\,\alpha^{\mu}_{-1}\,|0;k\rangle$ shift

$$e_{\mu\nu} \;\sim\; e_{\mu\nu} + k_{\mu}\,\zeta_{\nu} + \tilde{\zeta}_{\mu}\,k_{\nu}, \qquad k\cdot\zeta = k\cdot\tilde{\zeta} = 0. \tag{1.92}$$

These are physically null: adding them changes no inner product. So the physical content is a transverse tensor modulo (1.92), which is the data of a field carrying a linearised gauge invariance.

Decompose $e_{\mu\nu}$ into irreducible pieces under the little group of a massless particle in $D = 26$, namely SO(24) acting on the transverse directions:

- the symmetric traceless part, $g_{\mu\nu}$, whose gauge redundancy (1.92) is $\delta g_{\mu\nu} = k_{\mu}\zeta_{\nu} + k_{\nu}\zeta_{\mu}$, the momentum-space form of a linearised diffeomorphism. This is a **graviton**;
- the antisymmetric part, $B_{\mu\nu}$, the **Kalb-Ramond field**, with gauge transformation $\delta B_{\mu\nu} = k_{\mu}\zeta_{\nu} - k_{\nu}\zeta_{\mu}$, the momentum-space form of $\delta B = \mathrm{d}\lambda$;
- the trace, a single scalar ϕ, the **dilaton**.

These are the three massless fields of the closed bosonic string. They are the NS-NS sector fields of Appendix A, $g_{\mu\nu}, B_{\mu\nu}, \phi$, and they will reappear in every closed-string theory in this course. The counting checks out: the transverse tensor has $24^2 = 576$ components, split as

$$576 \;=\; \underbrace{299}_{\text{sym. traceless } g_{\mu\nu}} \;+\; \underbrace{276}_{\text{antisym. } B_{\mu\nu}} \;+\; \underbrace{1}_{\text{trace } \phi}\,, \tag{1.93}$$

using $\dim(\text{sym. traceless}) = \tfrac{1}{2}\cdot 24\cdot 25 - 1 = 299$ and $\dim(\text{antisym.}) = \tfrac{1}{2}\cdot 24\cdot 23 = 276$.

Level	State	$\alpha' M^2$	Spacetime content
$N = \tilde{N} = 0$	$\lvert 0; k\rangle$	-4	tachyon T (scalar)
$N = \tilde{N} = 1$	$e_{\mu\nu}\, \alpha^{\mu}_{-1} \tilde{\alpha}^{\nu}_{-1} \lvert 0; k\rangle$	0	$g_{\mu\nu} \oplus B_{\mu\nu} \oplus \phi$
$N = \tilde{N} = 2$	(see Exercise 1.13)	$+4$	massive SO(25) tensors

Table 1.1: The bottom of the closed bosonic string spectrum. All fields above the massless level have masses set by $M_{\rm s} = 1/\ell_s$ and decouple in the low-energy limit $\alpha' \to 0$.

Remark 1.6 (Disentangling the dilaton). The split (1.93) is a little subtler than a naive trace. A term proportional to $\eta_{\mu\nu}$ in $e_{\mu\nu}$ is not transverse, and the genuine physical scalar is the trace taken *after* projecting orthogonally to k. One introduces an auxiliary null vector $\bar{k}$ with $\bar{k} \cdot k = 1$ and defines the dilaton polarisation $e^{\phi}_{\mu\nu} \propto \eta_{\mu\nu} - k_\mu \bar{k}_\nu - \bar{k}_\mu k_\nu$, which is transverse and gauge-inequivalent to the graviton trace. The upshot is unchanged: one real propagating scalar.

1.5.4 The massless open string: a gauge boson and Chan-Paton factors

For the open string the first excited level is $N = 1$, massless by (1.88), with a single oscillator:

$$|e; k\rangle = e_\mu\, \alpha^{\mu}_{-1} \, |0; k\rangle \,, \qquad k^2 = 0. \tag{1.94}$$

The constraint $L_1 \, |e; k\rangle = 0$ gives transversality $k^\mu e_\mu = 0$, and the null state $L_{-1} \, |0; k\rangle$ supplies the redundancy $e_\mu \sim e_\mu + k_\mu$. A transverse vector modulo longitudinal shifts, with $D - 2 = 24$ physical polarisations: this is a massless U(1) gauge boson A_μ, with $e_\mu \sim e_\mu + k_\mu$ the momentum-space form of $\delta A_\mu = \partial_\mu \lambda$. The open string produces gauge theory as inevitably as the closed string produces gravity.

A single U(1) is not much of a gauge group. The string gives more for free. The two endpoints of an open string can carry extra discrete labels $i, j \in \{1, \ldots, N\}$, called *Chan-Paton factors*, one at each end. Every open-string state is then tensored with an $N \times N$ matrix of labels,

$$|k; a\rangle = \sum_{i,j=1}^{N} \lambda^{a}_{ij} \; |k; \, ij\rangle \,, \tag{1.95}$$

and the massless vector becomes a matrix A^a_μ of gauge bosons. These labels are spectators to the worldsheet dynamics but they are conserved at interactions, where strings join end to end and shared endpoint labels are summed. The resulting N^2 massless vectors fill out the adjoint of U(N) for oriented strings. (Unoriented strings project this to SO(N) or USp(N); we return to that in Chapter 6.) So the open string realises non-abelian gauge symmetry with the rank a free choice.

Remark 1.7 (Where the Chan-Paton labels live). In the pure quantisation of a single open string the labels i, j look like an arbitrary decoration. Their meaning is geometric. The boundary conditions on the string endpoints, Neumann in some directions and Dirichlet in others (planted in Section 1.2), define the hypersurfaces the endpoints are pinned to. The label i records *which* of N such surfaces an endpoint sits on. Those surfaces are D-branes, and U(N) is the gauge symmetry living on a stack of N coincident branes. That is the story of Chapter 6; the seed is already here.

1.5.5 The string coupling is the dilaton

The dilaton plays a role beyond that of one more massless scalar: its vacuum value fixes the strength of string interactions. The worldsheet action carries, in addition to the Polyakov term,

a coupling of ϕ to the worldsheet curvature (the Fradkin-Tseytlin term),

$$S_\phi = \frac{1}{4\pi}\int_\Sigma \mathrm{d}^2\sigma\,\sqrt{\gamma}\,\phi\,R^{(2)}, \tag{1.96}$$

where $R^{(2)}$ is the intrinsic Ricci scalar of the dynamical worldsheet metric $\gamma_{\alpha\beta}$. For a constant background $\phi = \langle\phi\rangle$, Gauss-Bonnet turns this integral into a topological invariant, the Euler characteristic $\chi(\Sigma) = 2 - 2g - b$ of a surface with g handles and b boundaries. A worldsheet of genus g is then weighted in the path integral by

$$e^{-S_\phi} = e^{-\chi\,\langle\phi\rangle} = \left(e^{\langle\phi\rangle}\right)^{2g-2+b}. \tag{1.97}$$

Each extra handle (one closed-string loop) costs a factor $e^{2\langle\phi\rangle}$, and each extra boundary (one open-string loop) a factor $e^{\langle\phi\rangle}$. This is a coupling-constant expansion, with the loop counting parameter

$$g_s = e^{\langle\phi\rangle}, \tag{1.98}$$

matching (A.5). The perturbative genus expansion this defines, weighted by g_s^{2g-2}, is the subject of Chapter 4. The point to carry forward is structural. The coupling g_s is fixed from within the theory: it is the expectation value of a dynamical field in the spectrum, not a dial we turn from outside. A theory has no dimensionless coupling to fix by hand; it has a modulus, and different values of $\langle\phi\rangle$ are different vacua of one theory. This is what makes strong-weak dualities (Chapter 7) even conceivable.

1.5.6 Two lessons

What the spectrum is telling us

Gravity is not optional. We quantised a relativistic string in flat space with no gravity put in, and a massless symmetric transverse-traceless field carrying linearised diffeomorphism invariance (1.92) fell out. A massless spin-2 with that gauge symmetry, once you make it interact consistently, is a graviton, and its only consistent low-energy theory is general relativity. The string did not assume gravity; it forced it.

The tachyon is fatal, and there are no fermions. The ground state (1.89) has $M^2 < 0$: the 26-dimensional vacuum is unstable. And every state we built used the bosonic oscillators α^μ_{-n} acting on a scalar, so every state is a spacetime tensor. There are no spacetime fermions anywhere in the spectrum. A theory of nature needs both a stable vacuum and electrons. The bosonic string has neither.

The first lesson is a promissory note. We saw the graviton *appear*; we have not yet seen Einstein's equations emerge as a *law*. That happens in Chapter 5. The requirement that the worldsheet theory remain conformal in a curved background $g_{\mu\nu}(X)$, that is, the vanishing of the sigma-model beta functions (equivalently, no Weyl anomaly), forces $g_{\mu\nu}$, $B_{\mu\nu}$ and ϕ to satisfy their spacetime equations of motion. To leading order in α' these are Einstein's equations coupled to $H = \mathrm{d}B$ and ϕ. Consistency of the string on a background *is* the field equation of that background. That is gravity as a law, and it is the reason this course exists.

The second lesson is a directive. Both diseases, the tachyon and the missing fermions, have a single cure: add worldsheet supersymmetry. Promoting each X^μ to a superfield introduces worldsheet fermions ψ^μ, whose spacetime-spinor ground states are the fermions we are missing, and the associated GSO projection removes the tachyon while keeping the theory consistent. The price is a new critical dimension, $D = 10$ instead of 26. That is the superstring, and it is where we go in Chapter 3, after we sharpen our CFT tools in Chapter 2.

Where this is going

We have quantised one string, covariantly, and read its spectrum: a tachyon we must remove, a graviton we cannot avoid, a Kalb-Ramond field and a dilaton that sets the coupling, and, on open strings, non-abelian gauge fields living on branes we have not yet named. The next chapter builds the conformal-field-theory machinery that makes all of this efficient (and that we leaned on here in stating the no-ghost theorem). The chapter after that fixes the string's two defects at once with supersymmetry. Everything downstream, effective actions, dualities, D-branes, compactification, is an elaboration of the fields we just found sitting at the massless level.

Exercise 1.12 (A null state decouples)**.** *(Short; on-ramp to the no-ghost theorem.)* Take the level-$(0,1)$ state $|\chi\rangle = \zeta_\nu\, \tilde{\alpha}^\nu_{-1}\,|0;k\rangle$ with $k^2 = 0$ and $k\cdot\zeta = 0$, and form the spurious state $L_{-1}\,|\chi\rangle$. Show that its polarisation is $e_{\mu\nu} \propto k_\mu \zeta_\nu$, reproducing the shift (1.92); that it satisfies the physical-state conditions (1.91); and that it has zero norm and zero overlap with any transverse physical state. This is the level-1 shadow of the no-ghost theorem: null (BRST-exact) states carry no physics and drop out of every inner product. *Hint:* the Virasoro algebra gives $[L_1, L_{-1}] = 2L_0$, and $L_0\,|\chi\rangle = 0$ on shell.

Exercise 1.13 (The first massive closed level)**.** *(Hard; multi-part.)* Enumerate the closed-string states at $N = \tilde{N} = 2$, built from $\{\alpha^\mu_{-2},\ \alpha^\mu_{-1}\alpha^\nu_{-1}\}$ on the left tensored with the same on the right. Show that $\alpha' M^2 = +4$, so the level is massive, and that the physical polarisations organise into representations of the *massive* little group SO(25) rather than SO(24). Why is it SO(25) and not SO(24) here? Count the states on one side and identify the symmetric two-tensor of SO(25) inside them. *Hint:* the little group of a massive particle in D dimensions is $\mathrm{SO}(D-1)$, since a massive momentum has a rest frame; a massless one only fixes $\mathrm{SO}(D-2)$.

Exercise 1.14 (Chan-Paton and U(N))**.** *(Moderate.)* The massless open-string vectors carry an $N\times N$ Chan-Paton matrix λ^a. Using that endpoint labels are conserved and summed when strings join, argue that the three-point coupling of the massless vectors is proportional to $\mathrm{Tr}(\lambda^a\lambda^b\lambda^c)$. Show that closure of the resulting algebra requires the λ^a to span a matrix algebra, and that for oriented strings this is $\mathfrak{u}(N)$ with N^2 generators. What extra condition on λ^a would project the gauge group to SO(N)?

Exercise 1.15 (The coupling is a field)**.** *(Short.)* Starting from (1.96), use Gauss-Bonnet, $\frac{1}{4\pi}\int_\Sigma \sqrt{\gamma}\,R^{(2)} = \chi(\Sigma)$ with $\chi = 2-2g-b$, to reproduce the genus weight (1.97). Deduce that a shift of the dilaton by a constant, $\langle\phi\rangle \to \langle\phi\rangle + c$, rescales the L-loop closed-string amplitude by $e^{(2L-2)c}$, and hence is equivalent to a redefinition of g_s. Explain in one sentence why this means the tree-level (sphere) normalisation of g_s is a convention, while ratios of amplitudes at different loop order are physical.

Chapter 2

Worldsheet CFT Tools and Vertex Operators

2.1 The state-operator map and radial quantisation

Chapter 1 handed us a Hilbert space: momentum eigenstates dressed by oscillators, cut down to the physical BRST cohomology, sitting on Regge trajectories. That is enough to read off a spectrum, but not enough to compute a scattering amplitude. A string amplitude is a correlation function on the worldsheet with one insertion for each external string, and those insertions are *local operators*, not abstract kets. So we need a dictionary that turns each asymptotic state into a definite local operator on the worldsheet. In a generic quantum field theory no clean dictionary exists: operators and states are different kinds of object, and many operators create the same state. In a two-dimensional CFT the dictionary is an isomorphism, the state-operator map, and building it is the organising job of this section.

This all takes place on the plane of radial quantisation from Section 1.3.1. We recall that setup, promote the offhand remark there ("a state is created by an operator at $z = 0$") into a precise statement, and set up the vocabulary of primaries, weights and descendants that sorts the worldsheet Hilbert space. The heavy CFT machinery was built in Chapter 1, and the careful convergence statements behind the map are catalogued in Appendix F; the map itself and the family structure it imposes are what this section adds.

The big picture

Radial quantisation puts constant-time slices on circles around the origin. The path integral over the disc inside such a circle, with a local operator inserted at the centre, prepares a state on the boundary. Shrinking the disc costs nothing in a scale-invariant theory, so the resulting state remembers only the operator at the point: states and local operators carry the same data. The operator carries two conformal weights $(h, \bar{h})$, whose sum $h + \bar{h}$ and difference $h - \bar{h}$ are its dilatation and rotation charges; primaries are the operators that transform simply, and descendants are generated from them by the raising Virasoro modes, the leading one being the derivative. Every string state we will eventually scatter is a primary of a definite weight, that is, a vertex operator.

2.1.1 Radial quantisation, recalled

Pass to Euclidean worldsheet time and the complex coordinate of Eq. (1.48), $z = e^{\tau_E - i\sigma}$. The cylinder becomes the plane: a slice of constant Euclidean time is a circle $|z| = e^{\tau_E}$, the infinite past $\tau_E \to -\infty$ collapses to the single point $z = 0$, and the infinite future opens out to

$z = \infty$. Time evolution is radial evolution. The generator of τ_E-translations on the cylinder is $L_0 + \tilde{L}_0 - (c + \bar{c})/24$, the dilatation generator shifted by the Casimir (vacuum) energy of the cylinder; on the plane the generator is the bare dilatation $L_0 + \tilde{L}_0$.

Now think about how a state is prepared, exactly as in ordinary canonical quantisation. There, field data on a spatial slice, together with the path integral over everything to its past, defines a wavefunctional, that is, a state. Here the "past" of the circle $|z| = r$ is the disc $|z| < r$ it bounds. The path integral over that disc, weighted by e^{-S} with the boundary values of X^μ held fixed, is a functional of those boundary values: a state on the circle. Run it with nothing inside and you get the vacuum $|0\rangle$; insert a local operator O at the centre $z = 0$ and you get some other state, written $|O\rangle$.

A scale-invariant theory has nothing in the disc path integral that cares about the radius r, so the state it prepares does not change as we shrink $r \to 0$; it is fixed entirely by whatever operator sits at the centre. Conversely, take any state on the circle and evolve it inward, accumulating it towards the origin. In a unitary theory with a discrete spectrum that ingoing evolution converges, and the limit is a local operator at $z = 0$, inverse to the first construction.

The vacuum is pinned down by the regularity of the stress tensor at the origin: $T(z)|0\rangle$ and $\bar{T}(\bar{z})|0\rangle$ carry no singular terms as $z \to 0$. In modes,

$$L_n|0\rangle = 0 \quad (n \geq -1), \qquad \tilde{L}_n|0\rangle = 0 \quad (n \geq -1), \tag{2.1}$$

one condition for every mode whose insertion would otherwise blow up. The three lowest, $n = -1, 0, 1$, are special: the Virasoro central term $\frac{c}{12}(n^3 - n)$ vanishes precisely there, so only $L_0, L_{\pm 1}$ (with their antiholomorphic partners) annihilate the vacuum as a genuine, anomaly-free symmetry. At the algebra level these span $\mathrm{SL}(2,\mathbb{C}) \times \overline{\mathrm{SL}(2,\mathbb{C})}$; the real form acting on the sphere $\mathbb{CP}^1$ is the Möbius group $\mathrm{PSL}(2,\mathbb{C})$, of complex dimension three, the global conformal group shared by in- and out-states. The remaining conditions $n \geq 2$ are not symmetry statements but the extra regularity that makes the ingoing limit well defined, so it is this regularity of the vacuum, not the global group alone, that makes the correspondence an isomorphism rather than a lossy projection.

2.1.2 States are local operators

Proposition 2.1 (State-operator isomorphism)**.** *In a unitary two-dimensional CFT with a discrete spectrum, radial quantisation on the plane gives a linear bijection between the Hilbert space $\mathcal{H}$ and the space of local operators,*

$$|O\rangle \;=\; \lim_{z,\bar{z}\to 0} O(z,\bar{z})\,|0\rangle, \tag{2.2}$$

under which the $\mathrm{SL}(2,\mathbb{C})$-invariant vacuum $|0\rangle$ is the image of the identity operator $\mathbf{1}$, and the inverse sends a state to the unique local operator that, inserted at the origin, prepares it.

We assert Theorem 2.1 at the level a spacetime-first course needs; the convergence of the ingoing evolution and the completeness behind it are in Appendix F. The forward direction, operator $\mapsto$ state, is the disc path integral of the previous subsection. The operators on the right are the normal-ordered composites of Chapter 1: normal ordering is what keeps $O(z,\bar{z})|0\rangle$ finite as the operator collides with the vacuum it acts on.

Two consequences matter for what follows. First, the map is linear and invertible, so counting states is the same as counting operators, and the partition function counts either. Second, it intertwines the conformal symmetry: acting on a state with a Virasoro mode equals acting on its operator. Here L_n is the contour integral Eq. (1.58) of T, and its contour can be deformed

onto the insertion because $L_n|0\rangle = 0$ for $n \geq -1$, so only the operator product of T with O contributes,

$$L_n\,|O\rangle \;=\; \oint_0 \frac{\mathrm{d}z}{2\pi i}\, z^{\,n+1}\, T(z)\, O(0,0)\,|0\rangle, \tag{2.3}$$

the contour a small circle around the origin. The rest of the section is contour manipulation of this one identity.

2.1.3 Reading the map on the free boson

Feed the abstract statement the free boson and it turns concrete. Take the holomorphic current $\partial X^\mu(z)$, with the mode expansion Eq. (1.53),

$$\partial X^\mu(z)\,|0\rangle = -i\sqrt{\tfrac{\alpha'}{2}} \sum_{n\in\mathbb{Z}} \alpha_n^\mu\, z^{-n-1}\,|0\rangle. \tag{2.4}$$

Regularity of $\partial X^\mu(z)|0\rangle$ at $z=0$ forces $\alpha_n^\mu|0\rangle = 0$ for $n \geq 0$, so only the negative modes survive and the sum runs over $n \leq -1$. The state-operator limit Eq. (2.2) keeps the single term with no power of z, the $n=-1$ term,

$$|\partial X^\mu\rangle \;=\; \lim_{z\to 0} \partial X^\mu(z)|0\rangle \;=\; -i\sqrt{\tfrac{\alpha'}{2}}\;\alpha_{-1}^\mu|0\rangle, \tag{2.5}$$

so the current at the origin is the first oscillator on the vacuum, up to the constant $-i\sqrt{\alpha'/2}$. Reading off higher powers of z in Eq. (2.4) gives the rest of the tower: the coefficient of z^{n-1} identifies $\alpha_{-n}^\mu|0\rangle$ with the operator $\frac{1}{(n-1)!}\partial^n X^\mu(0)$, again up to a constant, so the oscillators are simply the derivatives of X. A general Fock state, a product of oscillators on $|0\rangle$, is then a normal-ordered product of derivatives of X at the origin, and the oscillator Hilbert space of Chapter 1 and the space of local operators built from $\partial X, \partial^2 X, \ldots$ are one list written two ways.

The zero modes carry momentum. The state $|0;k\rangle$ of Chapter 1 is the image of the exponential operator,

$$|0;k\rangle \;=\; :e^{ik\cdot X}:(0,0)\,|0\rangle, \tag{2.6}$$

because $e^{ik\cdot X(0)}$ shifts the momentum eigenvalue by k^μ. This is the tachyon vertex operator; Section 2.2 dresses the polarised, momentum-carrying excited states into the graviton, Kalb-Ramond and dilaton vertex operators, and here it is the cleanest illustration of Eq. (2.2) at nonzero weight. That weight was computed in Exercise 1.6: $:e^{ik\cdot X}:$ is primary with $h = \bar{h} = \alpha' k^2/4$.

2.1.4 Primaries and their weights

Chapter 1 defined a primary O of weight $(h,\bar{h})$ by its operator product with the stress tensor,

$$T(z)\,O(w,\bar{w}) \sim \frac{h\,O(w,\bar{w})}{(z-w)^2} + \frac{\partial O(w,\bar{w})}{z-w}, \tag{2.7}$$

together with the antiholomorphic copy in which h becomes $\bar{h}$ and ∂ becomes $\bar{\partial}$. Feed Eq. (2.7) through Eq. (2.3): the contour around the insertion at w picks up the residue of z^{n+1} against the two poles,

$$L_n\,O(w,\bar{w})|0\rangle = h\,(n+1)\,w^{\,n}\,O(w,\bar{w})|0\rangle \;+\; w^{\,n+1}\,\partial O(w,\bar{w})|0\rangle, \tag{2.8}$$

and send $w \to 0$. For $n \geq 1$ both terms vanish. For $n=0$ only the first survives, giving h times the state. Collecting these with the antiholomorphic mirror,

$$L_0\,|O\rangle = h\,|O\rangle\,, \quad \tilde{L}_0\,|O\rangle = \bar{h}\,|O\rangle\,, \qquad L_n\,|O\rangle = \tilde{L}_n\,|O\rangle = 0 \;\; (n \geq 1). \tag{2.9}$$

A primary operator maps to a *highest-weight state*: an $L_0, \tilde{L}_0$ eigenstate annihilated by every lowering (positive) mode. The operator statement Eq. (2.7) and the state statement Eq. (2.9) are one fact carried across the map.

Definition 2.2 (Weight, dimension, spin)**.** The pair $(h, \bar{h})$ are the holomorphic and antiholomorphic *conformal weights*. Their sum $\Delta = h + \bar{h}$ is the *scaling dimension*, the eigenvalue of the dilatation generator $L_0 + \tilde{L}_0$; their difference $s = h - \bar{h}$ is the *spin*, the eigenvalue of the rotation generator $L_0 - \tilde{L}_0$. Since $L_0 + \tilde{L}_0$ generates dilatations, Δ is the plane worldsheet energy of $|O\rangle$.

This is the point of contact with the spectrum of Chapter 1. The L_0 eigenvalue in the mass formula there, $L_0 = \frac{\alpha'}{4}p^2 + N$ of Eq. (1.86), is exactly the holomorphic weight h of the corresponding operator, and N counts how far the operator has been pushed up its tower. The physical-state and level-matching conditions of Chapter 1 are conditions on h and $\bar{h}$; demanding total weight one on each side, $h = \bar{h} = 1$, is the on-shell condition on the external state, which Section 2.2 turns into the Weyl weight $(1, 1)$ of a good vertex operator.

2.1.5 Descendants and conformal families

The lowering modes annihilate a primary; the raising (negative) modes do not, and acting with them generates new states. Read Eq. (2.8) at $n = -1$: the first term drops through its factor $(n + 1)$, and the second gives

$$L_{-1}\,|O\rangle = |\partial O\rangle\,, \tag{2.10}$$

so L_{-1} is translation, and ∂O is a genuinely new local operator one unit higher in weight. More generally $L_{-n}\,|O\rangle$ is the state of an operator $(L_{-n}O)(0)$, built from O by a contour of T, of weight $h + n$. Repeated raising,

$$L_{-n_1}L_{-n_2}\cdots L_{-n_k}\,\tilde{L}_{-m_1}\cdots\,|O\rangle\,, \qquad n_i, m_j \geq 1, \tag{2.11}$$

builds the *descendants* of O. The primary together with all its descendants is the *conformal family* (Verma module) of O, graded by *level* $N = \sum_i n_i$ on the holomorphic side. The full worldsheet Hilbert space is the direct sum of such families, one per primary. That is what the map buys us: in place of an undifferentiated Fock space, a short list of primaries, each carrying a tower of raised states.

On the free boson the bookkeeping is transparent. The identity $\mathbf{1}$ has weight zero and its module is nearly empty, since $L_{-1}|0\rangle = |\partial\mathbf{1}\rangle = 0$, which is Eq. (2.1) again; its first nontrivial descendant is $L_{-2}|0\rangle = |T\rangle$, the stress tensor, at weight two. The current ∂X^μ is a separate primary of weight $(1, 0)$, and $\alpha^\mu_{-n}|0\rangle$ for $n \geq 2$ are its descendants $\partial^n X^\mu$. Each momentum sector is organised the same way around the primary $:e^{ik\cdot X}:$.

Remark 2.3 (Longitudinal is descendant, transverse is primary)*.* The family structure explains, from the worldsheet side, why the longitudinal polarisations of Chapter 1 were pure gauge. Acting on $|0; k\rangle$ with L_{-1} produces the longitudinal combination $k_\mu\alpha^\mu_{-1}\,|0; k\rangle$, a Virasoro descendant. The transverse combinations $e_\mu\alpha^\mu_{-1}\,|0; k\rangle$ with $e \cdot k = 0$ are not descendants of the primary $:e^{ik\cdot X}:$; they are new primaries. Descendants of the form $L_{-n}\,|\chi\rangle$ are exactly the spurious states of old covariant quantisation, null and decoupling (Exercise 1.10), while the physical polarisations are primaries, so the physical-state analysis becomes a statement about which operators are primary.

2.1.6 Out-states, and where this is going

An amplitude has incoming and outgoing legs, so we also need the conjugate. The out-state $\langle O|$ is prepared by an insertion at the point at infinity, reached from the origin by the inversion $z \to 1/z$. That map has $\mathrm{d}(1/z)/\mathrm{d}z = -z^{-2}$, and a weight-$(h, \bar{h})$ primary transforms with the $(h, \bar{h})$-th power of this Jacobian, which is where the prefactor comes from,

$$\langle O|\ =\ \lim_{z,\bar{z}\to\infty} z^{\,2h}\,\bar{z}^{\,2\bar{h}}\ \langle 0|\ O(z, \bar{z}), \tag{2.12}$$

the BPZ conjugate, whose conventions and inner product are collected in Appendix F. Chapter 4 glues an out-state at ∞ and an in-state at 0 and integrates the remaining insertions over the surface between them, which is the sense in which a scattering amplitude is a correlator of vertex operators. On a first reading this subsection is preview: the inner product it sets up is used in Chapter 4, not elsewhere in this chapter.

The map turns every asymptotic string state of Chapter 1 into a definite local operator, collected in Table 2.1; right-movers carry a tilde, $\tilde{\alpha}^{\mu}_{-n}$ and $\bar{\partial}X^{\mu}$, following Chapter 1. Section 2.2 builds the physical primaries explicitly, fixes their weight to $(1,1)$, and reads the on-shell and transversality conditions straight off; Section 2.3 then assembles the correlators reused in Chapters 3 and 4.

State	Local operator $O(0,0)$	Weight $(h,\bar{h})$
$\lvert 0\rangle$	$\mathbf{1}$	$(0,0)$
$\alpha^{\mu}_{-1}\lvert 0\rangle$	∂X^{μ}	$(1,0)$
$\tilde{\alpha}^{\mu}_{-1}\lvert 0\rangle$	$\bar{\partial}X^{\mu}$	$(0,1)$
$\lvert 0;k\rangle$	$:e^{ik\cdot X}:$	$(\frac{\alpha' k^2}{4}, \frac{\alpha' k^2}{4})$
$e_{\mu\nu}\,\alpha^{\mu}_{-1}\tilde{\alpha}^{\nu}_{-1}\,\lvert 0;k\rangle$	$:e_{\mu\nu}\,\partial X^{\mu}\,\bar{\partial}X^{\nu}\,e^{ik\cdot X}:$	$(1+\frac{\alpha' k^2}{4},\, 1+\frac{\alpha' k^2}{4})$

Table 2.1: The state-operator dictionary for the closed bosonic string. The last row is the massless vertex operator of Eq. (1.90); its weight becomes $(1,1)$ exactly on the mass shell $k^2=0$, where Section 2.2 takes over.

Exercise 2.1 (Spin from the weights)**.** *(Short; warm-up.)* Using Theorem 2.2, assign the scaling dimension $\Delta = h+\bar{h}$ and spin $s = h-\bar{h}$ to each operator in Table 2.1. Confirm that ∂X^{μ} carries $s=+1$ and $\bar{\partial}X^{\mu}$ carries $s=-1$, so a symmetric two-tensor built from one of each has $s=0$, as a graviton polarisation must. Which entries are rotational (spin-0) scalars, i.e. have $s=0$? Of those, only the identity is a genuine $\mathrm{SL}(2,\mathbb{C})$ (Möbius) scalar, since full conformal invariance forces $(h,\bar{h})=(0,0)$; check that the others have $s=0$ but nonzero Δ.

Exercise 2.2 (Highest weight from the OPE)**.** *(Standard.)* Fill in the contour step behind Eq. (2.9). Insert the primary OPE Eq. (2.7) into $L_n O(w)|0\rangle = \oint_w \frac{\mathrm{d}z}{2\pi i} z^{n+1} T(z)O(w)|0\rangle$ and take the residue at $z=w$ to reproduce Eq. (2.8). Then take $w\to 0$ and read off: $L_n\,|O\rangle = 0$ for $n\geq 1$, $L_0\,|O\rangle = h\,|O\rangle$, and $L_{-1}\,|O\rangle = |\partial O\rangle$. Explain in one line why the same computation on the vacuum ($O=\mathbf{1}$, $h=0$) reproduces $L_{-1}|0\rangle = 0$, hence part of Eq. (2.1).

Exercise 2.3 (Oscillators are derivatives)**.** *(Moderate.)* Starting from Eq. (2.4), expand $\partial X^{\mu}(z)|0\rangle$ in powers of z and match the coefficient of z^{n-1} to show

$$\alpha^{\mu}_{-n}\,|0\rangle \;=\; i\sqrt{\tfrac{2}{\alpha'}}\,\frac{1}{(n-1)!}\,\partial^{n}X^{\mu}(0)\,|0\rangle, \qquad n\geq 1.$$

Deduce that a general oscillator monomial $\prod_i \alpha^{\mu_i}_{-n_i}|0\rangle$ maps to a normal-ordered product of derivatives of X at the origin, and hence that the free-boson Fock space and the space of local operators in $\partial X, \partial^2 X, \ldots$ are isomorphic vector spaces, as Theorem 2.1 requires.

2.2 Vertex operators for the bosonic string

The state-operator map of Section 2.1 sends each asymptotic string state to a local operator inserted at a puncture. Here we make it concrete for the bosonic string and take the payoff: the three conditions that made a state physical, the mass shell, the transversality of the polarisation, and the decoupling of null states, collapse into a single statement about the conformal weight of its operator. A physical closed-string vertex operator is a weight-$(1,1)$ primary. The rest of this

section derives that slogan and reads it back as on-shell plus transversality, a result Chapters 3 and 4 use without pause.

The vertex-operator dictionary

An external string of momentum k^μ and polarisation data is emitted or absorbed by inserting a local operator on the worldsheet, integrated over where the insertion sits:

$$\text{state } |\psi; k\rangle \quad \longleftrightarrow \quad \mathcal{V}_\psi \;=\; g_\psi \int \mathrm{d}^2 z\, \mathcal{V}_\psi(z,\bar z),$$

with g_ψ a state-dependent normalisation fixed in Section 2.3, and $\mathcal{V}_\psi(z,\bar z)$ a normal-ordered polynomial in $\partial X, \bar\partial X$ times $:e^{ik\cdot X}:$. The insertion is a sensible, coordinate-independent object exactly when $\mathcal{V}_\psi$ is a conformal primary of weight $(h,\bar h) = (1,1)$. Two examples carry the section: the tachyon $:e^{ik\cdot X}:$ and the massless operator $e_{\mu\nu}{:}\partial X^\mu\, \bar\partial X^\nu\, e^{ik\cdot X}{:}$. Demanding weight $(1,1)$ on the first fixes the tachyon mass; on the second it forces $k^2 = 0$ and $k^\mu e_{\mu\nu} = 0$, the on-shell and transversality conditions of a massless graviton, B-field or dilaton.

2.2.1 Why weight $(1,1)$: the integrated insertion

Start from the requirement, not the answer. An amplitude is a number, and it cannot depend on the arbitrary complex coordinate z we used to label points on the worldsheet, nor on the fiducial metric we fixed when we chose conformal gauge in Section 1.2.1. An external state is inserted by integrating a local operator over the surface, so the integrated insertion must be invariant under the residual conformal reparametrisations $z \to z'(z)$, $\bar z \to \bar z'(\bar z)$ that survive that gauge choice (Section 1.2.2).

A primary $\mathcal{V}(z,\bar z)$ of weight $(h,\bar h)$ transforms, by definition, as a tensor of that weight,

$$\mathcal{V}(z,\bar z) \;=\; \left(\frac{\partial z'}{\partial z}\right)^{h} \left(\frac{\partial \bar z'}{\partial \bar z}\right)^{\bar h} \mathcal{V}'(z',\bar z')\,. \tag{2.13}$$

The integration measure carries the opposite weight,

$$\mathrm{d}^2 z \;=\; \left(\frac{\partial z'}{\partial z}\right)^{-1} \left(\frac{\partial \bar z'}{\partial \bar z}\right)^{-1} \mathrm{d}^2 z'\,, \tag{2.14}$$

so the product transforms with net weight $(h-1,\bar h-1)$,

$$\int \mathrm{d}^2 z\, \mathcal{V}(z,\bar z) \;=\; \int \mathrm{d}^2 z' \left(\frac{\partial z'}{\partial z}\right)^{h-1} \left(\frac{\partial \bar z'}{\partial \bar z}\right)^{\bar h-1} \mathcal{V}'(z',\bar z')\,. \tag{2.15}$$

This is invariant for every conformal map precisely when $h = \bar h = 1$.

The same rule wears a Weyl disguise. In the covariant frame the insertion is $\int \mathrm{d}^2\sigma\, \sqrt{\gamma}\, \mathcal{V}$ against the fiducial metric $\gamma_{\alpha\beta}$ of conformal gauge. Under a Weyl rescaling $\gamma_{\alpha\beta} \to e^{2\omega}\gamma_{\alpha\beta}$, the convention of Eq. (1.12), the measure factor scales as $\sqrt{\gamma} \to e^{2\omega}\sqrt{\gamma}$, so it has Weyl weight $+2$. A weight-$(h,\bar h)$ primary scales the opposite way, $\mathcal{V} \to e^{-(h+\bar h)\omega}\mathcal{V}$, Weyl weight $-(h+\bar h)$: this is the operator shadow of Eq. (2.13), since a residual conformal reparametrisation of z acts on the fixed fiducial metric exactly as a compensating Weyl rescaling, the identification set up in Section 1.2.2. The product $\sqrt{\gamma}\,\mathcal{V}$ is Weyl inert precisely when $h + \bar h = 2$, and adding $h = \bar h$ for a rotational scalar pins this to $(1,1)$. Conformal weight and Weyl weight are the same number here, so the conformal-invariance derivation and the Weyl-invariance derivation impose the identical condition.

One check is genuinely non-trivial, and it is where the exponential's Weyl anomaly bites. The Weyl variation of the renormalised insertion $\sqrt{\gamma}\,\mathcal{V}$ is proportional to $h + \bar h - 2$, so it vanishes at

$(1,1)$ on the nose: the marginality condition and the vanishing of the insertion's Weyl anomaly are one equation, not two. The general nonzero-weight version of this anomaly, and the central charge it descends from, is the subject of Section 1.3, and we take the $(1,1)$ statement as established there.

The condition $h = \bar{h} = 1$ has three faces, and later chapters lean on all three. It says the integrated insertion is conformally invariant. It says the operator is marginal, neither relevant nor irrelevant, so it can be added to the worldsheet action without breaking scale invariance. And, as we are about to see, it is the mass-shell-plus-transversality condition on the external particle. One equation carries all three readings.

2.2.2 The tachyon vertex operator

The closed-string ground state is the momentum eigenstate $|0;k\rangle$ with no oscillators. Under the state-operator map of Section 2.1 it corresponds to the simplest operator one can write with momentum k, the normal-ordered exponential

$$|0;k\rangle \quad \longleftrightarrow \quad :e^{ik\cdot X}:(z,\bar{z})\,. \tag{2.16}$$

Its conformal weight was computed in Exercise 1.6 straight from the propagator $\langle X^\mu(z,\bar{z})X^\nu(w,\bar{w})\rangle = -\frac{\alpha'}{2}\eta^{\mu\nu}\ln|z-w|^2$. Contracting each ∂X inside $T(z) = -\alpha'^{-1}{:}\partial X\cdot\partial X{:}$ against the exponential brings down a factor $-\frac{i\alpha'}{2}k^\mu/(z-w)$, and the two together assemble the double pole,

$$T(z)\,:e^{ik\cdot X}:(w,\bar{w}) \;\sim\; \frac{\alpha' k^2/4}{(z-w)^2}\,:e^{ik\cdot X}:(w,\bar{w}) + \frac{\partial\,:e^{ik\cdot X}:(w,\bar{w})}{z-w}\,, \tag{2.17}$$

with the antiholomorphic copy identical (the contraction is rehearsed in Exercise 2.4). So $:e^{ik\cdot X}:$ is a primary of weight

$$(h,\bar{h}) \;=\; \left(\frac{\alpha' k^2}{4},\,\frac{\alpha' k^2}{4}\right). \tag{2.18}$$

There is no cubic or higher pole: the exponential is primary for any k, because it carries no free Lorentz index for an extra contraction to hang on. Imposing the rule $(h,\bar{h}) = (1,1)$ gives $\alpha' k^2/4 = 1$, that is $k^2 = 4/\alpha'$, and with $M^2 = -k^2$ in the mostly-plus signature of Appendix A,

$$M^2 \;=\; -\frac{4}{\alpha'}\,. \tag{2.19}$$

This is the closed-string tachyon mass of Eq. (1.89), now recovered as the marginality condition on its vertex operator rather than from the Virasoro zero mode. The two derivations are the same physics: the weight $\frac{\alpha'}{4}k^2$ of the operator is L_0 evaluated on the momentum vacuum, and $(L_0 - 1)\,|\psi\rangle = 0$ is the statement $h = 1$.

Remark 2.4. The open-string tachyon works identically, with one change of arithmetic. Its vertex operator $:e^{ik\cdot X}:$ sits on the worldsheet boundary, where the Neumann conditions of Section 1.2 are enforced by the method of images (the doubling trick, developed in Chapter 6): reflecting the antiholomorphic field across the boundary through its holomorphic image doubles the propagator at coincident boundary points, so on the real boundary

$$\langle X^\mu(y_1)\,X^\nu(y_2)\rangle \;=\; -2\alpha'\,\eta^{\mu\nu}\,\ln|y_1 - y_2|\,, \tag{2.20}$$

twice the bulk value restricted to the boundary. Running the tachyon contraction with Eq. (2.20) gives the boundary weight $h = \alpha' k^2$ (a single holomorphic weight, since a boundary operator has no independent antiholomorphic partner), and marginality $h = 1$ gives $k^2 = 1/\alpha'$, hence $M^2 = -1/\alpha'$, the open-string value quoted alongside the closed one in Eq. (1.89). A boundary operator needs weight 1, not $(1,1)$, because it is integrated over the one-dimensional boundary.

2.2.3 The massless vertex operator: on-shell and transverse

Now for the operator that earns the section. The massless closed-string state of Eq. (1.90) carries one left- and one right-moving oscillator, $e_{\mu\nu}\,\alpha^{\mu}_{-1}\tilde{\alpha}^{\nu}_{-1}\,|0;k\rangle$. The oscillator α^{μ}_{-1} is the coefficient of z^0 in $\partial X^{\mu}(z)$, so under the state-operator map it becomes ∂X^{μ} at the insertion point, and likewise $\tilde{\alpha}^{\nu}_{-1} \to \bar{\partial} X^{\nu}$. Up to a normalisation fixed by Section 2.1, the dictionary reads

$$e_{\mu\nu}\,\alpha^{\mu}_{-1}\tilde{\alpha}^{\nu}_{-1}\,|0;k\rangle \quad \longleftrightarrow \quad e_{\mu\nu}\,{:}\partial X^{\mu}\,\bar{\partial} X^{\nu}\,e^{ik\cdot X}{:}(z,\bar{z})\,. \tag{2.21}$$

We want its conformal weight. Take the holomorphic side and consider $V^{\mu}(w) \coloneqq {:}\partial X^{\mu}\,e^{ik\cdot X}{:}(w)$, ignoring the spectator $\bar{\partial} X^{\nu}$ for the moment. Contracting $T(z)$ against $V^{\mu}(w)$ now has three kinds of term. Contracting one field of T with the explicit ∂X^{μ} reproduces the weight-1 double pole of Eq. (1.55); contracting both fields of T with the exponential produces the $\alpha' k^2/4$ of Eq. (2.17); and there is a new contribution in which one field of T contracts the ∂X^{μ} while the other contracts the exponential. That mixed double contraction has no analogue in the pure tachyon, and it produces a cubic pole. Carrying the factors through with the propagator of Appendix A (the termwise computation is Exercise 2.5),

$$T(z)\,V^{\mu}(w) \;\sim\; -\frac{i\alpha'}{2}\,\frac{k^{\mu}\,{:}e^{ik\cdot X}{:}(w)}{(z-w)^3} \;+\; \frac{\left(1+\frac{\alpha'}{4}k^2\right)V^{\mu}(w)}{(z-w)^2} \;+\; \frac{\partial V^{\mu}(w)}{z-w}\,. \tag{2.22}$$

The cubic pole is the obstruction to V^{μ} being primary. Its coefficient is proportional to k^{μ}, so it survives only because ∂X^{μ} carries a free index that k can point along. The antiholomorphic factor $\bar{\partial} X^{\nu}$ contributes the mirror image, a cubic pole in $(\bar{z}-\bar{w})$ proportional to k^{ν}. Assembling the full operator and contracting with the polarisation, the two cubic poles read

$$(k^{\mu}e_{\mu\nu}) \quad \text{and} \quad (e_{\mu\nu}k^{\nu}), \tag{2.23}$$

and $e_{\mu\nu}{:}\partial X^{\mu}\bar{\partial} X^{\nu}e^{ik\cdot X}{:}$ is a primary if and only if both vanish. That is transversality on each index,

$$k^{\mu}e_{\mu\nu} \;=\; 0\,, \qquad e_{\mu\nu}\,k^{\nu} \;=\; 0\,, \tag{2.24}$$

exactly the conditions Eq. (1.91) that the Virasoro constraints $L_1\,|\psi\rangle = \tilde{L}_1\,|\psi\rangle = 0$ imposed on the state. Once the operator is primary, Eq. (2.22) shows its weight is

$$(h,\bar{h}) \;=\; \left(1+\frac{\alpha' k^2}{4},\; 1+\frac{\alpha' k^2}{4}\right), \tag{2.25}$$

and the rule $(h,\bar{h}) = (1,1)$ forces $k^2 = 0$: the state is massless, $M^2 = 0$. The two requirements come from two different poles, transversality killing the cubic pole and masslessness setting the double pole to 1, and together they are the marginality of the operator.

Proposition 2.5 (Weight $(1,1)$ is the physical-state condition)**.** *The massless closed-string vertex operator $e_{\mu\nu}{:}\partial X^{\mu}\,\bar{\partial} X^{\nu}\,e^{ik\cdot X}{:}$ is a conformal primary of weight $(1,1)$ if and only if*

$$k^2 = 0 \qquad \textit{and} \qquad k^{\mu}e_{\mu\nu} = e_{\mu\nu}k^{\nu} = 0\,, \tag{2.26}$$

that is, if and only if the external particle is on-shell and its polarisation is transverse. These are the same conditions the Virasoro constraints impose on the corresponding state in Section 1.5. The residual gauge freedom $e_{\mu\nu} \sim e_{\mu\nu} + k_{\mu}\zeta_{\nu} + \tilde{\zeta}_{\mu}k_{\nu}$ of Eq. (1.92) corresponds to shifting the integrand by a total derivative, which drops out of $\int \mathrm{d}^2 z$.

The last clause closes the loop with the BRST-null states of Chapter 1. A spurious polarisation $e_{\mu\nu} = k_{\mu}\zeta_{\nu}$ gives the operator $\zeta_{\nu}{:}(k_{\mu}\partial X^{\mu})\,\bar{\partial} X^{\nu}\,e^{ik\cdot X}{:}$, and since $k_{\mu}{:}\partial X^{\mu}e^{ik\cdot X}{:} = -i\,\partial{:}e^{ik\cdot X}{:}$, this is $-i\zeta_{\nu}\,\partial\bar{\partial} X^{\nu}e^{ik\cdot X}{:}$ up to the equation of motion $\partial\bar{\partial} X = 0$. A total z-derivative integrates to zero over a closed worldsheet. The null states that Chapter 1 quotiented out of the Hilbert space are, on the operator side, precisely the total-derivative vertex operators that integrate to zero, so nothing is lost or double-counted in passing between the two languages.

Reading off the fields. With Eq. (2.26) in force, decompose the transverse polarisation $e_{\mu\nu}$ under the massless little group SO(24), exactly as for the state in Section 1.5.3. The symmetric traceless part is the graviton $g_{\mu\nu}$; its vertex operator is $g_{\mu\nu}{:}\partial X^{(\mu}\bar{\partial}X^{\nu)}e^{ik\cdot X}{:}$, and the residual shift $\delta g_{\mu\nu} = k_\mu\zeta_\nu + k_\nu\zeta_\mu$ is a linearised diffeomorphism. The antisymmetric part is the Kalb-Ramond field $B_{\mu\nu}$, with vertex operator $B_{\mu\nu}{:}\partial X^{[\mu}\bar{\partial}X^{\nu]}e^{ik\cdot X}{:}$ and gauge shift $\delta B_{\mu\nu} = k_\mu\zeta_\nu - k_\nu\zeta_\mu$. The trace is the dilaton. As in the remark around Eq. (1.93), the naive trace $\eta_{\mu\nu}$ is not transverse, so the honest dilaton polarisation uses an auxiliary null vector $\bar{k}$ with $\bar{k}\cdot k = 1$,

$$e^{\phi}_{\mu\nu} \;\propto\; \eta_{\mu\nu} - k_\mu\bar{k}_\nu - \bar{k}_\mu k_\nu\,, \tag{2.27}$$

which is transverse and gauge-inequivalent to the graviton trace. All three operators have the same weight $(1,1)$ at $k^2 = 0$; the decomposition sits entirely in the polarisation. This is the vertex-operator form of the statement that the massless closed bosonic string is $g_{\mu\nu} \oplus B_{\mu\nu} \oplus \phi$, and these are the operators Chapter 4 inserts to compute graviton, B-field and dilaton scattering.

2.2.4 Fixed and integrated insertions

The insertion in the big-picture box was written with an integral, $\int \mathrm{d}^2z\,\mathcal{V}$, the right object for a generic external leg. On surfaces with a residual global conformal symmetry some insertions are instead *fixed* at a chosen point. The obstruction to integrating everything is the conformal Killing group, the globally defined conformal transformations left unfixed by conformal gauge. On the sphere, the surface for closed-string tree amplitudes, this is the Möbius group $\mathrm{PSL}(2,\mathbb{C})$, three complex parameters; integrating a vertex operator over the whole sphere overcounts by the volume of this group, which is infinite. One fixes a few insertions to use up the parameters and integrates the rest. The counting across genus and boundaries is developed in Section 2.3.2; here we need only the form a fixed insertion takes.

A fixed insertion is not the bare primary $\mathcal{V}(z,\bar{z})$, which is not conformally invariant at a point. The invariant object dresses it with a ghost factor,

$$\text{integrated:}\quad \int \mathrm{d}^2z\;\mathcal{V}(z,\bar{z})\,, \qquad\qquad \text{fixed:}\quad c\bar{c}\,\mathcal{V}(z,\bar{z})\,, \tag{2.28}$$

where $c, \bar{c}$ are the reparametrisation ghosts of Section 1.4.1. The ghost c has weight -1, so $c\bar{c}\,\mathcal{V}$ has weight $(0,0)$ and sits at a definite point with no compensating integral, the ghost carrying the weight that the measure d^2z would otherwise have supplied. This is not optional bookkeeping. The fixed $c\bar{c}\,\mathcal{V}$ insertions saturate the c-ghost zero modes that would otherwise force the path integral to vanish, one complex $c\bar{c}$ for each complex conformal Killing vector. The sphere has three such vectors, so three vertex operators are fixed. On the disc the group is $\mathrm{PSL}(2,\mathbb{R})$, whose three real parameters are absorbed by fixing one interior insertion and one boundary insertion. The torus carries a single conformal Killing vector, the rigid translation, so one $c\bar{c}$ is still attached to soak up its complex c zero mode; there the group is compact and its volume finite, so fixing is not needed to cure a divergence, only to saturate that zero mode. This is the same bookkeeping as the later statement that the sphere has three Killing vectors and the torus has one. Getting the number right on each surface is part of the amplitude prescription of Chapter 4, where the open string runs on the disc with boundary operators of weight 1 integrated over the boundary circle and its three positions fixed as above.

Careful

Weight $(1,1)$ is necessary for a physical closed-string vertex operator, but on its own it is only the primary and marginal condition. The full physical-state statement is BRST invariance: the fixed operator $c\bar{c}\,\mathcal{V}$ must be annihilated by the BRST charge Q of Section 1.4.3. For a matter primary of weight $(1,1)$ these are equivalent, and Q-exact operators are exactly the total-derivative and pure-gauge insertions that drop out. When we add worldsheet fermions

and superghosts in Chapter 3 the weight bookkeeping shifts, the superstring vertex operators come in different "pictures", but the organising principle, a BRST-invariant local operator integrated against the right measure, is unchanged.

2.2.5 Where this leaves us

We now have the working half of the chapter's toolkit. Every asymptotic state of the bosonic string is a local operator, and the single requirement that its integrated insertion be a weight-$(1,1)$ primary reproduces, operator by operator, the mass-shell condition, the transversality of the polarisation, and the decoupling of null states that the covariant quantisation of Chapter 1 delivered state by state. What remains is to multiply these operators inside a correlation function, the business of Section 2.3: the handful of X^μ correlators and the Ward identities that turn a product of vertex operators into an amplitude, with the full CFT reference card collected in Appendix F.

Exercise 2.4 (The tachyon T-OPE)**.** *(Warm-up.)* Rehearse the double-pole mechanics before the harder mixed contraction of Exercise 2.5. Contract $T(z) = -\alpha'^{-1}{:}\partial X \cdot \partial X{:}(z)$ with ${:}e^{ik\cdot X}{:}(w,\bar w)$ using only the propagator of Appendix A: each ∂X in T contracts the exponential and brings down $-\frac{i\alpha'}{2}k^\mu/(z-w)$, and the two together build the double pole. Recover Eq. (2.17), read off $h = \alpha' k^2/4$, and confirm there is no cubic pole because the exponential carries no free index. Part (b) of Exercise 1.6 is the same computation for the holomorphic weight alone.

Exercise 2.5 (The cubic pole and transversality)**.** *(Standard.)* Verify Eq. (2.22) directly. Contract $T(z) = -\alpha'^{-1}{:}\partial X \cdot \partial X{:}(z)$ with $V^\mu(w) = {:}\partial X^\mu e^{ik\cdot X}{:}(w)$ using only the propagator of Appendix A, and organise the result by pole order. Show that (a) the single contraction with the explicit ∂X^μ gives the $(z-w)^{-2}$ term with coefficient 1; (b) the double contraction into the exponential gives the additional $\frac{\alpha'}{4}k^2$ in the double pole; and (c) the mixed double contraction, one field of T on ∂X^μ and one on the exponential, produces the cubic pole with coefficient $-\frac{i\alpha'}{2}k^\mu$. Conclude that contracting with $e_{\mu\nu}$ and demanding no cubic pole is transversality Eq. (2.24). *Hint:* the two identical fields in T supply the factor of 2 that turns $-\frac{\alpha'}{2}$ into the stated coefficient.

Exercise 2.6 (Null states are total derivatives)**.** *(Short.)* Take the spurious polarisation $e_{\mu\nu} = k_\mu \zeta_\nu$ with $k^2 = 0$ and $k\cdot\zeta = 0$. Using $k_\mu{:}\partial X^\mu e^{ik\cdot X}{:} = -i\,\partial{:}e^{ik\cdot X}{:}$ and the equation of motion $\partial\bar\partial X^\nu = 0$, show that the corresponding vertex operator is a total z-derivative, and hence that $\int \mathrm{d}^2 z\, e_{\mu\nu}{:}\partial X^\mu \bar\partial X^\nu e^{ik\cdot X}{:} = 0$. Identify this with the decoupling of the level-$(1,1)$ null state of Exercise 1.12, and explain in one line why the fixed insertion $c\bar c\,\mathcal{V}$ of the same spurious operator is instead Q-exact rather than literally zero.

Exercise 2.7 (The dilaton vertex operator)**.** *(Moderate.)* Show that the naive trace operator $\eta_{\mu\nu}{:}\partial X^\mu \bar\partial X^\nu e^{ik\cdot X}{:}$ is not transverse in the sense of Eq. (2.24) at $k^2 = 0$, so it is not by itself a $(1,1)$ primary. Introduce an auxiliary null vector $\bar k$ with $\bar k \cdot k = 1$ and verify that the polarisation Eq. (2.27) is transverse on both indices, gauge-inequivalent to the graviton trace under Eq. (1.92), and therefore a good weight-$(1,1)$ operator. This is the vertex operator whose insertion, via the Fradkin-Tseytlin coupling of Eq. (1.96), shifts the string coupling $g_s = e^{\langle\phi\rangle}$; comment on why a constant dilaton shift is a change of g_s rather than a genuine scattering process.

2.3 Correlators, Ward identities, and what we reuse

With the state-operator map (Section 2.1) and the vertex operators (Section 2.2) in hand, a string amplitude is a definite recipe: place vertex operators on a Riemann surface, take their correlation function in the worldsheet CFT, and integrate over the inequivalent positions of the

operators and shapes of the surface. Chapter 4 runs that recipe. This section collects the pieces of it that belong to the CFT rather than to the amplitude: the one family of correlators we actually evaluate, the residual symmetry that makes the position integral finite, and the Ward identities we invoke downstream without re-proving. Nothing here is pushed past the level a working knowledge needs. The full reference card, with the derivations, is Appendix F.

2.3.1 The tree-level correlator: exponentials on the sphere

Tree level is the sphere. In the genus expansion of Section 1.5.5, a closed-string surface with g handles is weighted by g_s^{2g-2}, so the leading contribution to any process comes from the genus-zero worldsheet, the Riemann sphere $\mathbb{CP}^1$, which is the plane of Section 1.3.1 with the point at infinity added. Every tree amplitude is built from correlators of vertex operators on $\mathbb{CP}^1$, and because the external tachyons and the massless states are all dressed by the exponential $:e^{ik\cdot X}:$, the master object is the correlator of a product of these exponentials.

It is a free-field Gaussian, so it collapses to a single identity. Normal ordering removes the self-contractions, and each distinct pair contracts once through the propagator Eq. (1.50). Separating the zero mode $X^\mu = x_0^\mu +$(oscillators) and doing the x_0 integral produces a momentum-conserving delta function,

$$\Big\langle \prod_{i=1}^{n} :e^{ik_i\cdot X}:(z_i,\bar z_i)\Big\rangle_{\mathbb{CP}^1} = \exp\Big(-\sum_{i<j} k_i\cdot k_j \,\langle X(z_i,\bar z_i)\, X(z_j,\bar z_j)\rangle\Big) \times (\text{zero mode}), \tag{2.29}$$

and inserting $\langle X(z_i)X(z_j)\rangle = -\frac{\alpha'}{2}\eta\,\ln|z_i-z_j|^2$ turns the exponent into a product of powers. Writing $z_{ij} \coloneqq z_i - z_j$,

$$\boxed{\Big\langle \prod_{i=1}^{n} :e^{ik_i\cdot X}:(z_i,\bar z_i)\Big\rangle_{\mathbb{CP}^1} = i\,C_{\mathbb{CP}^1}\,(2\pi)^D\,\delta^D\Big(\sum_i k_i\Big)\prod_{i<j}|z_{ij}|^{\,\alpha' k_i\cdot k_j}\,.} \tag{2.30}$$

This is the Koba-Nielsen factor. The delta function is the translation Ward identity: the action is invariant under $X^\mu \to X^\mu + a^\mu$, and the conserved charge is the total momentum, so a correlator vanishes unless $\sum_i k_i = 0$. The overall constant $C_{\mathbb{CP}^1}$ is a normalisation, not dynamics; the genus weighting fixes $C_{\mathbb{CP}^1} \propto g_s^{-2}$ (the Euler number of the sphere is $\chi = 2$, see Eq. (1.97)), and its precise value is pinned by unitarity in Appendix F. The interaction is carried entirely by the position dependence $\prod_{i<j}|z_{ij}|^{\alpha' k_i\cdot k_j}$.

Two remarks on scope. First, Eq. (2.30) is special to the sphere. On a surface of genus g the correlator is still Gaussian, but the propagator is replaced by the scalar Green's function on that surface, which is elementary only at $g = 0$; the higher-genus versions are assembled in Appendix F and used in Chapter 4. Second, states carrying oscillators (the graviton, the B-field, the dilaton) are dressed by ∂X^μ and $\bar\partial X^\nu$ alongside the exponential. Their correlators follow from Eq. (2.30) by Wick contraction: a single holomorphic current inserted into a sea of exponentials is replaced by a classical source,

$$\Big\langle \partial X^\mu(z)\prod_i :e^{ik_i\cdot X}:\Big\rangle = \Big(-\frac{i\alpha'}{2}\sum_i \frac{k_i^\mu}{z-z_i}\Big)\Big\langle \prod_i :e^{ik_i\cdot X}:\Big\rangle, \tag{2.31}$$

with additional currents contracting among themselves through the double pole $-\frac{\alpha'}{2}\eta^{\mu\nu}/(z-z')^2$ of Eq. (1.51). This one rule, the Koba-Nielsen factor differentiated, is the entire computational engine for massless tree amplitudes.

2.3.2 Three punctures the symmetry fixes: the conformal Killing group

The correlator Eq. (2.30) depends on where we put the operators, yet the amplitude cannot: the positions z_i are integration variables, not physical data, so we integrate over them. Done naively this overcounts, because a residual gauge symmetry survives conformal gauge, the same symmetry that let Section 2.2 fix three vertex operators. Here we develop that fixing in full.

Fixing the worldsheet metric to conformal gauge in Section 1.2.1 did not exhaust the gauge freedom. On the sphere a rigid subgroup survives, the globally defined conformal maps, the Möbius transformations

$$z \longmapsto \frac{az+b}{cz+d}, \qquad ad-bc=1, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathbb{C}). \tag{2.32}$$

These are the conformal Killing transformations of $\mathbb{CP}^1$, the automorphisms that move points without deforming the surface. At the level of the algebra they are two commuting copies of $\mathrm{SL}(2,\mathbb{C})$, generated by the holomorphic $L_{0,\pm1}$ and the antiholomorphic $\tilde{L}_{0,\pm1}$; the real transformations they generate on $\mathbb{CP}^1$ are the Möbius group, the conformal Killing group $\mathrm{PSL}(2,\mathbb{C}) = \mathrm{SL}(2,\mathbb{C})/\mathbb{Z}_2$, of complex dimension three. It is non-compact, so its volume is infinite. A correlator of vertex operators is invariant under Eq. (2.32) once the operators transform with their weights, so the position integral $\int \prod_i \mathrm{d}^2 z_i$ carries an infinite redundant factor, the volume of this group.

The cure is Faddeev-Popov again, one level up from Section 1.4.1. Divide by the group volume by using the three complex parameters of Eq. (2.32) to freeze three of the punctures at reference points, conventionally $z_1, z_2, z_3 = 0, 1, \infty$. The remaining $n-3$ positions are genuine moduli and are integrated. In the BRST language of Section 1.4.3 the same count appears as ghost zero modes: the sphere carries six real c-ghost zero modes, soaked up by attaching a $c\bar{c}$ factor to three vertex operators, which is exactly the fixed-versus-integrated distinction of Section 2.2.

The consequences are immediate and we use them constantly:

- The moduli space of the n-punctured sphere has complex dimension $n-3$. There is nothing to integrate for $n \leq 3$.
- Two- and three-point functions are pure kinematics, fixed by the symmetry up to a constant. The first amplitude with a genuine integral, and hence the first that can carry dynamical structure such as poles in a Mandelstam variable, is the four-point function, with one integrated position.
- This is the genus-zero face of a general story. The sphere has three conformal Killing vectors and, at $n=0$, no moduli; the torus has one Killing vector and one modulus, the shape parameter τ; a surface of genus $g \geq 2$ has no Killing vectors and $3g-3$ complex moduli. Once $2g-2+n>0$ the Killing vectors are exhausted and the moduli number $3g-3+n$. These counts and their ghost-number bookkeeping are tabulated in Appendix F and drive the moduli-space integral of Chapter 4.

2.3.3 The conformal Ward identity, and why low points are kinematics

The statement that low-point functions are fixed by symmetry is the content of the conformal Ward identity, which we quote and reuse rather than derive. For primaries $\mathcal{V}_i$ of holomorphic weight h_i, an insertion of the stress tensor is completely determined by the $T\mathcal{V}$ OPE of Eq. (1.55):

$$\Big\langle T(z) \prod_{i=1}^{n} \mathcal{V}_i(w_i,\bar{w}_i) \Big\rangle = \sum_{i=1}^{n} \Big(\frac{h_i}{(z-w_i)^2} + \frac{1}{z-w_i}\,\partial_{w_i} \Big) \Big\langle \prod_{i=1}^{n} \mathcal{V}_i(w_i,\bar{w}_i) \Big\rangle, \tag{2.33}$$

with the antiholomorphic copy for $\bar{T}$. The derivation is a contour deformation, the correlator version of the mode calculation in Section 1.3.5, and it lives in Appendix F. Two uses carry into the rest of the course.

The first is the fixing of low-point functions. Feeding Eq. (2.33) through the global generators $L_0, L_{\pm 1}$, the anomaly-free part of the algebra that exponentiates to the $\mathrm{PSL}(2,\mathbb{C})$ of Eq. (2.32), forces

$$\langle \mathcal{V}_1\, \mathcal{V}_2 \rangle \propto \frac{\delta_{h_1,h_2}}{z_{12}^{2h_1}}, \qquad \langle \mathcal{V}_1\, \mathcal{V}_2\, \mathcal{V}_3 \rangle \propto \prod_{i<j} z_{ij}^{-(h_i+h_j-h_k)}, \tag{2.34}$$

with k the third label in each factor, and analogously for the bars. No dynamics enters: the two-point function is a normalisation, and the three-point function is a single structure constant times a fixed kinematic prefactor. The Koba-Nielsen correlator Eq. (2.30) is consistent with Eq. (2.34), as it must be, and Exercise 2.8 checks the tachyon case explicitly.

The second use is the one we will lean on without comment. Integrating Eq. (2.33) against a conformal Killing vector, a global transformation, must annihilate the correlator, and this is exactly the requirement that an integrated vertex operator $\int \mathrm{d}^2 z\, \mathcal{V}$ be conformally invariant. The measure $\mathrm{d}^2 z$ carries weight $(-1,-1)$, so invariance demands $\mathcal{V}$ of weight $(1,1)$. That is the same $(1,1)$ condition derived from the state side in Section 2.2, where weight $(1,1)$ is the on-shell and transversality condition on the external state. The Ward identity is why it is also the condition for the amplitude to be well defined, independent of where the operator was inserted. A companion statement, that BRST-null insertions become total derivatives on moduli space and integrate to zero, is what makes the amplitude gauge invariant; it is the amplitude-level shadow of the null-state decoupling seen at level one in Exercise 1.12, and it is proved in Appendix F.

2.3.4 Modular invariance: the consistency condition that returns

One loop is the torus, and the torus carries a consistency condition with no analogue in point-particle field theory. This subsection is a preview: we flag the condition here and build the machinery that uses it in Chapter 3.

A torus is a parallelogram in the plane with opposite sides identified, fixed up to scale and rotation by a single complex modulus τ in the upper half plane, the $g = 1$ analogue of the three fixed punctures on the sphere. Different τ can describe the same torus: relabelling which cycle is which acts on τ by the modular group

$$\tau \longmapsto \frac{a\tau + b}{c\tau + d}, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathbb{Z}), \tag{2.35}$$

generated by $T : \tau \mapsto \tau + 1$ and $S : \tau \mapsto -1/\tau$. Integrating over inequivalent tori means integrating τ over a single fundamental domain of Eq. (2.35), not over the whole upper half plane, and for that to be consistent the one-loop integrand, the torus partition function $Z(\tau, \bar{\tau})$, must be invariant under the modular group. That invariance is a constraint on the spectrum, not an identity.

Two pieces of the constraint recur later. Invariance under T is the statement $L_0 - \tilde{L}_0 \in \mathbb{Z}$, level matching Eq. (1.43) promoted to a quantum consistency requirement. Invariance under S is the genuinely new content: it exchanges the short-distance ($\tau \to 0$) and long-distance regions, and the fundamental domain that survives excludes the dangerous $\tau \to 0$ corner, the origin of the string's improved ultraviolet behaviour. In the superstring the same requirement selects the GSO projection, which we develop in Chapter 3. The partition function, the fundamental domain, and their consequences for the consistent string spectra are assembled in Appendix F and Chapter 3.

The reusable toolkit

The pieces Chapters 3 and 4 stand on are now in place.

- **OPE catalogue.** The singular products $\partial X\,\partial X$, $T\,\partial X$, $T\,T$ of Section 1.3, extended to the ghost and (later) fermion systems in Appendix F.
- **State-operator map.** Every asymptotic state is a local vertex operator inserted at a puncture (Section 2.1).
- **Vertex operators.** The tachyon $:e^{ik\cdot X}:$ and the massless $:\partial X^\mu\,\bar\partial X^\nu\,e^{ik\cdot X}:$, physical exactly when they are weight $(1,1)$, that is, on shell and transverse (Section 2.2).
- **The tree correlator.** The Koba-Nielsen factor Eq. (2.30) and the current rule Eq. (2.31).
- **The finite-integral principle.** The conformal Killing group fixes three punctures; n-point trees integrate $n-3$ positions.
- **Ward identities.** Momentum conservation Eq. (2.30) and the conformal Ward identity Eq. (2.33), invoked but not re-derived.
- **Modular invariance** of the torus, the one-loop consistency condition, flagged for Chapter 3.

The full card is Appendix F.

Chapter 3 takes this machinery, adds worldsheet fermions and their vertex operators, and uses modular invariance to pin down the GSO projection. Chapter 4 takes the Koba-Nielsen correlator and the moduli count and produces the Veneziano amplitude and the genus expansion. From here the worldsheet recedes, and the rest of the course reads off the spacetime physics it has produced.

Exercise 2.8 (The three-tachyon amplitude is a constant)**.** *(Short.)* Take three on-shell tachyons, $k_i^2 = 4/\alpha'$ (weight $h_i = \alpha' k_i^2/4 = 1$), with $\sum_i k_i = 0$ from Eq. (2.30). (a) Using $2k_i\cdot k_j = (k_i+k_j)^2 - k_i^2 - k_j^2 = k_l^2 - k_i^2 - k_j^2$ with l the third label, show $\alpha' k_i\cdot k_j = -2$ for every pair, so the Koba-Nielsen factor is $\prod_{i<j}|z_{ij}|^{-2}$. (b) Check that this matches the fixed three-point form Eq. (2.34) with all $h_i = 1$. (c) Fix $z_1, z_2, z_3 = 0, 1, \infty$ and confirm the position dependence cancels the Möbius Jacobians, so that after dividing by the conformal Killing volume the amplitude is a pure number times $C_{\mathbb{CP}^1}$. There is no integral: three points is exactly the number the symmetry fixes.

Exercise 2.9 (Setting up Veneziano)**.** *(Leads into Chapter 4.)* Now take four tachyons. (a) From Section 2.3.2 the moduli space is $n-3 = 1$ complex dimensional, so one position is integrated. Fix $z_1, z_3, z_4 = 0, 1, \infty$ and call the integrated position $z_2 = z$. (b) Using Eq. (2.30) and dropping the $z_4 \to \infty$ factors (which are absorbed by the weight of the fixed operator there), show that the z-dependent integrand is

$$|z|^{\alpha' k_1\cdot k_2}\,|1-z|^{\alpha' k_2\cdot k_3},$$

so the closed-string four-tachyon amplitude is $\int \mathrm{d}^2 z$ of this. (c) Re-express the exponents through the Mandelstam invariants and the on-shell condition. Chapter 4 evaluates the integral: on the sphere it is the Virasoro-Shapiro amplitude, and its open-string analogue, with the punctures on a boundary and the conformal Killing group reduced to $\mathrm{PSL}(2,\mathbb{R})$, is Veneziano's Euler Beta function.

Chapter 3

The Superstring (RNS), GSO, and the Five Theories

3.1 Worldsheet supersymmetry and the RNS string

The bosonic string of Chapter 1 is a consistent quantum theory, and it is not the string we want. Two defects are fatal for physics. Its spectrum begins with a tachyon (Section 1.5.2), so the 26-dimensional vacuum we quantised around is unstable. And every state it produces is a spacetime boson: the graviton, the Kalb-Ramond field and the dilaton at the massless level, then an infinite tower of massive bosons above them. Nowhere in the Fock space built from the α^{μ}_{-n} is there a single spacetime fermion. A theory that aspires to contain quarks and leptons has to do better on both counts.

The cure is to enrich the worldsheet. We keep the D scalars X^{μ} and add to each of them a worldsheet fermion ψ^{μ}, a two-dimensional spinor that carries the same spacetime vector index. We then gauge the symmetry that rotates X^{μ} into ψ^{μ}, a local worldsheet supersymmetry, exactly as we gauged worldsheet diffeomorphisms in Chapter 1. The resulting theory is the *Ramond-Neveu-Schwarz* (RNS) string. Its worldsheet dynamics is a superconformal field theory, and the whole of the present chapter is a study of what that superconformal structure forces on spacetime.

There is more than one way to build a supersymmetric string. In the RNS description supersymmetry lives on the worldsheet and spacetime supersymmetry is not manifest; it emerges only after the projection of Section 3.3. The Green-Schwarz and pure-spinor formulations instead make the ten-dimensional spacetime supersymmetry manifest from the start, at the cost of a more intricate (and, for pure spinors, ghost-heavy) worldsheet. We use RNS throughout and name the alternatives only here, noting that all three give the same physical spectrum.

The big picture

Give each coordinate X^{μ} a worldsheet superpartner ψ^{μ}: a fermion on the worldsheet carrying a spacetime vector index. Gauge the supersymmetry that rotates them, so that the worldsheet theory becomes superconformal, with two generators, the stress tensor T and a supercurrent G. Each spacetime dimension now contributes central charge 1 from its boson and $\frac{1}{2}$ from its fermion, so the matter central charge is $c = \frac{3}{2}D$. Cancelling the ghosts of the enlarged gauge symmetry, which carry $c = -15$, forces $c = 15$, hence $D = 10 = 15/\frac{3}{2}$. The spacetime fermions this construction was built to deliver are still hidden at this stage; they appear in Section 3.2, from the two ways a worldsheet spinor can be continued around the string.

3.1.1 From worldsheet bosons to worldsheet superpartners

The field content is the point of departure, so fix it precisely. Alongside the embedding coordinates $X^\mu(\sigma^0,\sigma^1)$, which are worldsheet scalars and spacetime vectors, introduce

$$\psi^\mu(\sigma^0,\sigma^1), \qquad \mu = 0,1,\ldots,D-1, \tag{3.1}$$

a set of D Majorana fermions on the worldsheet. Each ψ^μ is a two-component real spinor under the worldsheet Lorentz group $\mathrm{SO}(1,1)$, and the collection $\{\psi^\mu\}$ transforms as a vector under the spacetime Lorentz group $\mathrm{SO}(1,D-1)$. This pairing of quantum numbers is the defining choice of the RNS string, and it repays a pause. The index μ that makes ψ^μ a spacetime vector is a flavour label from the worldsheet point of view; the spinor structure that makes ψ^μ a fermion is a worldsheet property with, so far, no spacetime meaning. Spacetime spinors are nowhere in sight. That they nonetheless emerge is the content of Section 3.2, and it is surprising: a spacetime spinor is assembled out of the worldsheet spinors ψ^μ by the way they close up around a periodic worldsheet coordinate, not written down by hand.

The two-dimensional worldsheet spinor splits into components of definite worldsheet chirality, which are the left- and right-moving pieces $\psi^\mu(z)$ and $\tilde\psi^\mu(\bar z)$. In the same way that X^μ decomposed into a holomorphic and an antiholomorphic half in Section 1.3, the fermion decomposes into a holomorphic ψ^μ of conformal weight $(\frac12,0)$ and an antiholomorphic $\tilde\psi^\mu$ of weight $(0,\frac12)$. We build the left-moving (holomorphic) sector explicitly and let the right-moving sector run in parallel; the closed string glues the two together in Section 3.2.

3.1.2 The super-Polyakov action and superconformal gauge

To gauge worldsheet supersymmetry we need its gauge fields. Worldsheet diffeomorphisms and Weyl rescalings were realised in Chapter 1 through the worldsheet metric $\gamma_{\alpha\beta}$, equivalently the zweibein $e_\alpha{}^a$. Local supersymmetry is the fermionic partner of a local translation, so its gauge field is the fermionic partner of the zweibein: a worldsheet gravitino χ_α, a vector-spinor. Together $(e_\alpha{}^a,\chi_\alpha)$ form the multiplet of two-dimensional supergravity, and coupling the matter (X^μ,ψ^μ) to it produces the *super-Polyakov action*

$$\begin{aligned} S = -\frac{1}{4\pi\alpha'}\int \mathrm{d}^2\sigma\, e\,\Big[\,&\gamma^{\alpha\beta}\,\partial_\alpha X^\mu\partial_\beta X_\mu + \alpha'\,\bar\psi^\mu\rho^\alpha\partial_\alpha\psi_\mu \\ &+ 2\alpha'\,\bar\chi_\alpha\rho^\beta\rho^\alpha\psi^\mu\,\partial_\beta X_\mu + \tfrac{\alpha'}{2}\,\bar\psi^\mu\psi_\mu\,\bar\chi_\alpha\rho^\beta\rho^\alpha\chi_\beta\,\Big], \end{aligned} \tag{3.2}$$

where $e=\det e_\alpha{}^a$, the ρ^α are worldsheet Dirac matrices obeying $\{\rho^\alpha,\rho^\beta\}=2\gamma^{\alpha\beta}$, and $\bar\psi=\psi^\dagger\rho^0$ is the Majorana conjugate. The first two terms are the free kinetic terms for the coordinates and their superpartners; the third couples the gravitino χ_α to the worldsheet *supercurrent* $\bar\chi_\alpha(\rho^\beta\rho^\alpha\psi^\mu\partial_\beta X_\mu)$, precisely as the metric couples to the stress tensor; the last term is a gravitino contact term fixed by local supersymmetry. We will not need the last two terms as written, because we are about to gauge χ_α away, and their coefficients are recorded here only to show the structure; the standard references carry the full derivation.

Beyond the three symmetries of the bosonic Polyakov action, worldsheet diffeomorphisms, Weyl rescalings and global spacetime Poincaré, the action Eq. (3.2) has two new local symmetries. There is a local supersymmetry, whose parameter is an anticommuting worldsheet spinor $\varepsilon(\sigma)$, under which $\delta X^\mu=\bar\varepsilon\psi^\mu$ and $\delta\psi^\mu=\rho^\alpha(\partial_\alpha X^\mu-\bar\chi_\alpha\psi^\mu)\varepsilon$, with matching variations of $e_\alpha{}^a$ and χ_α; and there is a local super-Weyl symmetry, the fermionic partner of the Weyl rescaling Eq. (1.12), which shifts the gravitino by $\delta\chi_\alpha=\rho_\alpha\eta$ for an arbitrary spinor $\eta(\sigma)$. The super-Weyl transformation is what makes the gravitino pure gauge in two dimensions: it has just enough components to be removed entirely.

Count the gauge freedom against the gauge fields, exactly as in Section 1.1.4. The gravitino χ_α is a two-by-two array of real components. Local supersymmetry supplies two (the two components of ε) and super-Weyl supplies the other two (the two components of η). So the entire gravitino can be gauged to zero, just as diffeomorphisms and Weyl set $\gamma_{\alpha\beta}$ to the flat metric. Fixing

$$e_\alpha{}^a = \delta_\alpha^a, \qquad \chi_\alpha = 0 \tag{3.3}$$

is *superconformal gauge*, the supersymmetric completion of the conformal gauge of Section 1.2. In this gauge the matter action collapses to D free bosons plus D free fermions. Passing to the complex coordinate $z = e^{\tau_E - i\sigma}$ of Appendix A and splitting into holomorphic and antiholomorphic halves, it reads

$$S = \frac{1}{2\pi}\int \mathrm{d}^2 z \left[\frac{2}{\alpha'}\,\partial X^\mu \bar\partial X_\mu + \psi^\mu \bar\partial \psi_\mu + \tilde\psi^\mu \partial \tilde\psi_\mu\right]. \tag{3.4}$$

The bosonic piece is the theory of Section 1.3 unchanged. The fermions are free, with equations of motion $\bar\partial\psi^\mu = 0$ and $\partial\tilde\psi^\mu = 0$, so $\psi^\mu = \psi^\mu(z)$ is holomorphic and $\tilde\psi^\mu = \tilde\psi^\mu(\bar z)$ antiholomorphic, as anticipated. Their propagators follow from Eq. (3.4) in the same way the $\partial X\,\partial X$ OPE followed in Section 1.3.2:

$$\psi^\mu(z)\,\psi^\nu(w) \sim \frac{\eta^{\mu\nu}}{z-w}, \qquad \tilde\psi^\mu(\bar z)\,\tilde\psi^\nu(\bar w) \sim \frac{\eta^{\mu\nu}}{\bar z - \bar w}, \tag{3.5}$$

with no singularity in a mixed holomorphic-antiholomorphic pair. The simple pole (weight $\frac{1}{2}$ each side, adding to the required 1) confirms that ψ^μ is a weight-$\frac{1}{2}$ field, and the symmetric $\eta^{\mu\nu}$ in the numerator, as against the $\partial X\,\partial X$ double pole, is the algebraic signature of a fermion: the two-point function is odd under exchange because $\psi^\mu(z)\psi^\nu(w) = -\psi^\nu(w)\psi^\mu(z)$ and $z - w$ is odd.

Exercise 3.1 (Checkpoint: the free fermion OPE; Warm-up, ∼10 min)**.** A quick check before the machinery accumulates. Treating $\psi^\mu(z)$ as a free holomorphic fermion of weight $\frac{1}{2}$ with the OPE Eq. (3.5): (a) confirm that the two-point function is odd under exchanging $z \leftrightarrow w$ together with $\mu \leftrightarrow \nu$, as an anticommuting field demands; (b) check that the weights of the two operators add to the pole order 1; (c) explain why a holomorphic $\psi^\mu(z)$ and an antiholomorphic $\tilde\psi^\nu(\bar w)$ have no singular OPE.

As in the bosonic case, gauge-fixing is not free. The Faddeev-Popov determinant for fixing diffeomorphisms and Weyl invariance is again the anticommuting bc ghost system of Section 1.4.1, with $c_{bc} = -26$. Fixing local supersymmetry and super-Weyl invariance produces a second ghost system, the *superconformal ghosts* (β, γ), the superpartners of (b, c). These are a *commuting* first-order system with weights $(\frac{3}{2}, -\frac{1}{2})$. The general first-order ghost formula gives central charge $c = 1 - 3(2\lambda - 1)^2$ for an anticommuting pair of weights $(\lambda, 1-\lambda)$ and $c = -1 + 3(2\lambda - 1)^2$ for a commuting one; at $\lambda = \frac{3}{2}$ the commuting case gives $c_{\beta\gamma} = -1 + 3(2 \cdot \frac{3}{2} - 1)^2 = -1 + 12 = +11$ (see Exercise 1.9). We will need this value in Section 3.1.5.

Remark 3.1 (Beyond the lectures: the superspace form)*.* The action Eq. (3.2) looks unmotivated as a list of terms; it is dictated by supersymmetry, and that is cleanest in superspace. Enlarge the worldsheet by two anticommuting coordinates $\theta, \bar\theta$ and package the matter into a single real superfield

$$Y^\mu(z, \bar z, \theta, \bar\theta) = X^\mu + \theta\,\psi^\mu + \bar\theta\,\tilde\psi^\mu + \theta\bar\theta\,F^\mu, \tag{3.6}$$

with F^μ an auxiliary field carrying no propagating degrees of freedom. With the superderivatives $D = \partial_\theta + \theta\partial$ and $\bar D = \partial_{\bar\theta} + \bar\theta\bar\partial$, the flat-space matter action is the single term

$$S = \frac{1}{4\pi\alpha'}\int \mathrm{d}^2 z\,\mathrm{d}\theta\,\mathrm{d}\bar\theta\;\bar D Y^\mu\, D Y_\mu, \tag{3.7}$$

and doing the θ integrals reproduces Eq. (3.4) after F^μ is eliminated by its algebraic equation of motion $F^\mu = 0$. The auxiliary F^μ is what makes the supersymmetry algebra close off-shell, without using the equations of motion; on-shell one may drop it, as we do in the component treatment below. The gauged version replaces the flat superspace by a curved one whose vielbein encodes $(e_\alpha{}^a, \chi_\alpha)$, and Eq. (3.2) is its component expansion.

3.1.3 Worldsheet supersymmetry and the supercurrent

With the gravitino gauged away, the local supersymmetry of Eq. (3.2) leaves behind a rigid (global) supersymmetry of the free action Eq. (3.4), in the same way that fixing the metric left behind the rigid conformal symmetry in Section 1.2.2. On the holomorphic fields it acts, with a constant anticommuting parameter ϵ, as

$$\delta X^\mu = -i\sqrt{\tfrac{\alpha'}{2}}\,\epsilon\,\psi^\mu, \qquad \delta\psi^\mu = i\sqrt{\tfrac{2}{\alpha'}}\,\epsilon\,\partial X^\mu, \tag{3.8}$$

with an independent copy generated by $\tilde\epsilon$ acting on X^μ and $\tilde\psi^\mu$ through $\bar\partial X^\mu$. The pattern is the essence of supersymmetry: the variation of a boson is a fermion, and the variation of that fermion is a derivative of the boson. Composing two of these transformations closes the algebra onto a translation. A short computation using Eq. (3.8) gives

$$[\,\delta_{\epsilon_1}, \delta_{\epsilon_2}\,]\,X^\mu = -2\,\epsilon_1\epsilon_2\,\partial X^\mu, \qquad [\,\delta_{\epsilon_1}, \delta_{\epsilon_2}\,]\,\psi^\mu = -2\,\epsilon_1\epsilon_2\,\partial\psi^\mu, \tag{3.9}$$

so composing two supersymmetries produces the holomorphic derivative ∂, the generator of left-moving translations. The variations $\delta_\epsilon = \epsilon\,Q$ carry the Grassmann parameter ϵ, so the commutator of two variations on the left encodes the anticommutator $\{Q, Q\}$ of the charges themselves. Squaring the supercharge therefore gives the Hamiltonian, the two-dimensional shadow of the spacetime relation $\{Q, Q\} \sim P$ that the whole subject is named after.

By Noether's theorem, the rigid symmetry Eq. (3.8) has a conserved current. Because the parameter is a worldsheet spinor, the current is a spinor-vector; its holomorphic component is a single weight-$\frac{3}{2}$ operator, the *supercurrent*

$$G(z) = i\sqrt{\tfrac{2}{\alpha'}}\,{:}\psi^\mu\,\partial X_\mu{:}(z), \qquad \tilde G(\bar z) = i\sqrt{\tfrac{2}{\alpha'}}\,{:}\tilde\psi^\mu\,\bar\partial X_\mu{:}(\bar z), \tag{3.10}$$

with normal ordering defined by subtracting the singular self-contraction, as in Section 1.3.3. The supercurrent stands to worldsheet supersymmetry as the stress tensor T stands to conformal symmetry: contour integrals of G against the fields generate Eq. (3.8). For instance,

$$\delta X^\mu(w) = \epsilon \oint_w \frac{\mathrm{d}z}{2\pi i}\, G(z)\, X^\mu(w), \tag{3.11}$$

and contracting ∂X in G against $X^\mu(w)$ with the propagator of Appendix A reproduces the first equation of Eq. (3.8). The holomorphic stress tensor gets a fermion contribution and now reads

$$T(z) = -\frac{1}{\alpha'}{:}\partial X^\mu\,\partial X_\mu{:} - \frac{1}{2}{:}\psi^\mu\,\partial\psi_\mu{:}, \tag{3.12}$$

the first term as in Eq. (1.54) and the second the standard weight-$\frac{1}{2}$ fermion stress tensor. The theory has one holomorphic supercurrent G and one antiholomorphic $\tilde G$, that is, one left-moving and one right-moving supersymmetry. In the standard counting this is worldsheet $\mathcal{N} = (1,1)$ supersymmetry. This $(1,1)$ counts worldsheet supercurrents, one from each moving sector; it is not the count of spacetime supercharges, which is a separate label introduced in Section 3.4 and differs between the two Type II theories even though both share this worldsheet $(1,1)$. When we build the heterotic string in Section 3.5 we will keep supersymmetry on only one side, $\mathcal{N} = (0,1)$, but for the Type II construction the symmetric $(1,1)$ theory is what we quantise.

3.1.4 The superconformal algebra

The operators T and G do not close among themselves as two independent symmetries; they mix, and the algebra they generate is the *superconformal* (or super-Virasoro) algebra. Everything about it is contained in three operator product expansions, which extend the single TT OPE of Eq. (1.57). Feeding T, G and each other through the Wick contractions of Eq. (3.5) and the $\partial X\,\partial X$ OPE gives

$$T(z)\,T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2\,T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}, \tag{3.13}$$

$$T(z)\,G(w) \sim \frac{\frac{3}{2}\,G(w)}{(z-w)^2} + \frac{\partial G(w)}{z-w}, \tag{3.14}$$

$$G(z)\,G(w) \sim \frac{2c/3}{(z-w)^3} + \frac{2\,T(w)}{z-w}. \tag{3.15}$$

The first line is the Virasoro OPE with a central charge c to be read off. The second says G is a Virasoro primary of weight $\frac{3}{2}$: the double pole carries the weight and the single pole is the transport term, exactly as in the reading of Eq. (1.55). The third line is the one that carries the new information. The single pole of G with itself is $2T$, which is the statement that two supersymmetries make a conformal transformation, the operator form of the closure Eq. (3.9). The most singular term, the triple pole, is a pure number proportional to the same central charge c. Computing that number is the quickest route to the critical dimension, so we do it now.

Example 3.2 (The central charge from $G\,G$)**.** Take the leading singularity of Eq. (3.15). Write $G(z) = i\sqrt{2/\alpha'}\,\psi^\mu \partial X_\mu(z)$ and contract it against $G(w)$ with both pairs of fields joined: the fermions through $\psi^\mu(z)\psi^\nu(w) \sim \eta^{\mu\nu}/(z-w)$ and the currents through $\partial X_\mu(z)\partial X_\nu(w) \sim -\frac{\alpha'}{2}\eta_{\mu\nu}/(z-w)^2$. The prefactor is $(i\sqrt{2/\alpha'})^2 = -2/\alpha'$, so, with both pairs contracted, the leading (triple) pole is

$$G(z)\,G(w)\Big|_{\text{triple pole}} = -\frac{2}{\alpha'}\cdot\frac{\eta^{\mu\nu}}{z-w}\cdot\left(-\frac{\alpha'}{2}\right)\frac{\eta_{\mu\nu}}{(z-w)^2} = \frac{\eta^{\mu\nu}\eta_{\mu\nu}}{(z-w)^3} = \frac{D}{(z-w)^3}. \tag{3.16}$$

Matching to the coefficient $2c/3$ in Eq. (3.15) gives $c = \frac{3}{2}D$. The counting is transparent: the $\eta^{\mu\nu}\eta_{\mu\nu} = D$ counts the D spacetime directions, one factor coming from the boson and one from the fermion in each G, and the α' cancels between the two propagators.

The same $c = \frac{3}{2}D$ follows from Eq. (3.13) by the Wick count of Eq. (1.57): the boson piece of T gives $c = D$ as before, and the $-\frac{1}{2}{:}\psi\partial\psi{:}$ piece adds $c = \frac{1}{2}$ per fermion, so $c = D + \frac{1}{2}D = \frac{3}{2}D$. Each spacetime dimension carries central charge

$$c_{\text{per dimension}} = \underbrace{1}_{X^\mu} + \underbrace{\tfrac{1}{2}}_{\psi^\mu} = \tfrac{3}{2}, \tag{3.17}$$

one unit from its coordinate and a half from its superpartner, and it is this number that every consistency condition in the chapter turns on.

Expanding T and G in modes turns the OPEs into an algebra of operators. As a weight-2 field $T(z) = \sum_{m\in\mathbb{Z}} L_m z^{-m-2}$, exactly as in Section 1.3.5; as a weight-$\frac{3}{2}$ field

$$G(z) = \sum_r G_r\, z^{-r-3/2}. \tag{3.18}$$

Here r runs over either the half-integers or the integers, and which one is not a matter of choice: it is fixed by how ψ^μ is continued around the closed string, a fermion being defined only up to a sign under $\sigma \to \sigma + 2\pi$. The half-integer moding is the Neveu-Schwarz sector and the integer moding is the Ramond sector, and the split between them is the whole of Section 3.2; here we simply record the algebra, which takes one form with $r, s \in \mathbb{Z} + \frac{1}{2}$ and another with $r, s \in \mathbb{Z}$.

Definition 3.3 ($\mathcal{N}=1$ super-Virasoro algebra)**.** The modes L_m ($m \in \mathbb{Z}$) and G_r obey

$$[\,L_m, L_n\,] = (m-n)\,L_{m+n} + \frac{c}{12}\,m(m^2-1)\,\delta_{m+n,0}, \tag{3.19}$$

$$[\,L_m, G_r\,] = \left(\frac{m}{2} - r\right) G_{m+r}, \tag{3.20}$$

$$\{\,G_r, G_s\,\} = 2\,L_{r+s} + \frac{c}{12}\,(4r^2-1)\,\delta_{r+s,0}. \tag{3.21}$$

The first line is the Virasoro algebra Eq. (1.61); the second says G_r lowers or raises L_0 like a weight-$\frac{3}{2}$ object; the third is the defining anticommutator, obtained from Eq. (3.15) by the contour argument of Section 1.3.5, with the same central charge c in all three.

Two features of Theorem 3.3 will do heavy work later. First, setting $m=0$ in Eq. (3.20) shows that G_r shifts the level L_0 by r, so the supercurrent modes move states up and down the mass spectrum in half-integer steps in the Neveu-Schwarz sector. Second, in the Ramond sector, where r is an integer, the zero mode G_0 exists, and Eq. (3.21) at $r=s=0$ reads

$$G_0^2 = L_0 - \frac{c}{24}. \tag{3.22}$$

This is a worldsheet Dirac equation: G_0 is a square root of the Hamiltonian. The L_0 appearing here is the plane-frame Virasoro zero mode; the physical, cylinder-frame number operator is $N = L_0 - c/24$, and in terms of it the relation is simply $G_0^2 = N$, with the additive $c/24$ absorbed. That shift-free form is the one Section 3.2 works with when it writes $\{G_0, G_0\} = 2L_0$ for L_0 the physical operator, and it is what forces the Ramond ground state to zero mass. The structure of G_0, built from the fermion zero modes ψ_0^μ, is what makes that ground state a spacetime spinor; we flag it here and pay it off in Section 3.2.

3.1.5 The critical dimension $D = 10$

The bosonic string fixed $D=26$ by demanding that the gauge symmetry survive quantisation: the total central charge, matter plus ghosts, had to vanish for the BRST charge to be nilpotent (Section 1.4.4). The superstring runs the identical argument with one more gauge symmetry and one more ghost system, and the number that comes out is 10.

The bookkeeping is short. The matter now carries

$$c_{\text{matter}} = \tfrac{3}{2}D, \tag{3.23}$$

from Eq. (3.17). Gauge-fixing the enlarged symmetry produces two ghost systems rather than one. Fixing diffeomorphisms and Weyl invariance gives the reparametrisation ghosts (b,c) with $c_{bc} = -26$, as in Chapter 1. Fixing local worldsheet supersymmetry and super-Weyl invariance gives the superconformal ghosts (β,γ) with $c_{\beta\gamma} = +11$. Their sum is

$$c_{\text{ghost}} = c_{bc} + c_{\beta\gamma} = -26 + 11 = -15. \tag{3.24}$$

The consistency condition is the same one as before. The super-BRST current is built from the matter and ghost T and G, and it is nilpotent, $Q^2=0$, precisely when the total superconformal anomaly vanishes, that is, when $c_{\text{matter}} + c_{\text{ghost}} = 0$. The computation is the supersymmetric copy of Eq. (1.76), with the $\beta\gamma$ sector added, and we quote its conclusion rather than repeat the contour algebra:

$$\boxed{\,c_{\text{matter}} + c_{\text{ghost}} = 0 \iff \tfrac{3}{2}D = 15 \iff D = 10.\,} \tag{3.25}$$

The critical central charge is $c=15$, and the way it factorises is the sentence to remember:

$$c = 15 = 10 \times \left(1 + \tfrac{1}{2}\right), \tag{3.26}$$

ten spacetime dimensions, each contributing one unit from its coordinate boson and a half from its worldsheet fermion. The tachyonic, fermion-free, 26-dimensional bosonic string has become a 10-dimensional theory whose full spectrum we are now equipped to build.

Remark 3.4 (Beyond the lectures: the light-cone count). There is a faster, less covariant route to the same 10, parallel to the transverse counting that reaches 26 in the bosonic case. In light-cone gauge one keeps only the $D-2$ transverse coordinates X^i and their partners ψ^i. Requiring that the quantised light-cone theory still realise the full spacetime Lorentz algebra, specifically that the anomaly in the commutator $[J^{i-}, J^{j-}]$ cancel, fixes $D-2=8$ transverse dimensions and simultaneously fixes the zero-point energies of the two sectors. The covariant central-charge argument of Eq. (3.25) keeps Lorentz invariance manifest and hides the transverse count; the light-cone argument trades manifest covariance for a direct handle on the ground-state energies, which we will need in Section 3.2 when we compute the mass of each sector's lowest state.

Remark 3.5 (Beyond the lectures: what the $\beta\gamma$ ghosts cost). The superconformal ghosts are more delicate than their bc cousins, and two of their features surface later. Being commuting fields of half-integer weight, they have inequivalent Neveu-Schwarz and Ramond vacua, and organising states by their $\beta\gamma$ ground state introduces the notion of *picture number* that vertex operators carry in Chapter 4. It is often convenient to trade the first-order $\beta\gamma$ system for a free boson plus a fermionic pair through bosonisation, $\beta \sim e^{-\phi}\partial\xi$, $\gamma \sim e^{\phi}\eta$, which is how the superghost central charge $+11$ is most cleanly reproduced and how the Ramond spin field is constructed. We record the machinery in Appendix F and use it sparingly.

3.1.6 Where this leaves us

We have taken the covariant engine of Chapter 1 and added a worldsheet fermion ψ^μ to each coordinate, gauged the supersymmetry that rotates them, and found a superconformal field theory with two generators, T and G, and central charge $c=\frac{3}{2}D$. Cancelling the ghosts of the enlarged gauge symmetry pins the spacetime dimension at $D=10$. None of this has yet produced a spacetime fermion or removed the tachyon; both are consequences of how the fermion ψ^μ is quantised, which turns on the one question we have deliberately postponed: is ψ^μ periodic or antiperiodic around the string? That question splits the theory into the Neveu-Schwarz and Ramond sectors of Section 3.2, delivers the spacetime spinors through the zero-mode relation Eq. (3.22), and sets up the projection of Section 3.3 that finally kills the tachyon and locks in spacetime supersymmetry.

Exercise 3.2 (The $G\,G$ central charge and $c=\frac{3}{2}D$; Standard, ∼30 min)**.** Fill in Theorem 3.2 in full. (a) Using the free-field OPEs Eq. (3.5) and $\partial X_\mu(z)\partial X_\nu(w) \sim -\frac{\alpha'}{2}\eta_{\mu\nu}/(z-w)^2$, compute all the singular terms of $G(z)G(w)$, not just the leading one, and confirm the single pole assembles $2T(w)$ with T given by Eq. (3.12). (b) Read off $c=\frac{3}{2}D$ from the triple pole. (c) Independently, evaluate the central term of the TT OPE Eq. (3.13) by adding the boson double-contraction of Eq. (1.56) to the fermion one, and check that the fermion contributes $\frac{1}{2}$ per flavour, so that $c=D+\frac{1}{2}D$ again.

Exercise 3.3 (The superconformal algebra from the OPEs; Moderate, ∼40 min)**.** Convert Eqs. (3.14) and (3.15) into the mode relations Eqs. (3.20) and (3.21) by the double-contour method of Exercise 1.8. (a) From Eq. (3.14) derive $[L_m, G_r]=(\frac{m}{2}-r)G_{m+r}$, and observe that the coefficient $\frac{m}{2}$ is the weight $\frac{3}{2}$ minus 1, the generic form for a weight-h primary. (b) From Eq. (3.15) derive $\{G_r, G_s\}=2L_{r+s}+\frac{c}{12}(4r^2-1)\delta_{r+s,0}$, taking care that the contour manipulation of two half-integer-weight fields produces an anticommutator rather than a commutator. (c) Set $r=s=0$ and recover the worldsheet Dirac relation Eq. (3.22); explain why G_0 exists only when the moding is integer.

Exercise 3.4 (The supercurrent generates supersymmetry; Standard, ∼20 min)**.** Verify

that G generates the transformations Eq. (3.8). (a) Using Eq. (3.11) and the XX propagator of Appendix A, show that $\delta X^\mu = -i\sqrt{\alpha'/2}\,\epsilon\,\psi^\mu$. (b) Show similarly that $\delta\psi^\mu(w) = \epsilon \oint_w \frac{\mathrm{d}z}{2\pi i} G(z)\psi^\mu(w)$ equals $i\sqrt{2/\alpha'}\,\epsilon\,\partial X^\mu(w)$. (c) Compose the two variations to obtain the closure Eq. (3.9), and identify the right-hand side as the action of $\oint G\,G \sim \oint 2T$, that is, a worldsheet translation.

3.2 NS and R sectors; spacetime fermions

In Section 3.1 we gave each coordinate X^μ a superpartner: a worldsheet Majorana fermion ψ^μ, a spacetime vector but a two-dimensional spinor, whose free action closes the superconformal algebra and fixes $D = 10$ through the central charge $c = 15$. The X^μ we already know how to quantise; that machinery was built in Chapter 1 and we reuse it wholesale. What is new sits in the fermion. Solving ψ^μ forces a choice that the boson never presented, because a coordinate must come back to itself around the string but a fermion, which enters the action only bilinearly, need only come back to itself up to a sign. That single sign splits the theory into two sectors. One of them carries every spacetime boson, the other every spacetime fermion. This section is the anatomy of that split.

3.2.1 Two boundary conditions

The worldsheet fermion obeys a first-order (Dirac) equation, so its solutions are purely left- or right-moving, $\psi^\mu = (\psi^\mu_+, \psi^\mu_-)$ with ψ^μ_+ holomorphic and ψ^μ_- antiholomorphic, exactly as X^μ split into X_L and X_R in Section 1.2.4. The precise action and its equation of motion live in Section 3.1; here we need only its variation, which for a string with a boundary leaves an endpoint term

$$\delta S_\psi \;\supset\; \frac{1}{2\pi}\int \mathrm{d}\tau\,\Big[\,\psi^\mu_+\,\delta\psi_{+\mu} - \psi^\mu_-\,\delta\psi_{-\mu}\,\Big]^{\sigma=\pi}_{\sigma=0}. \tag{3.27}$$

Take the open string, $\sigma \in [0,\pi]$. For Eq. (3.27) to vanish the two components must lock together at each end, $\psi^\mu_+ = \eta\,\psi^\mu_-$ with $\eta = \pm 1$. An overall sign is a relabelling of the field and carries no physics; the *relative* sign between the two ends does. Fix the residual freedom by setting $\psi^\mu_+(0) = \psi^\mu_-(0)$. Then the two inequivalent theories are

$$\psi^\mu_+(\pi) = +\,\psi^\mu_-(\pi): \qquad \text{Ramond (R)}, \tag{3.28}$$

$$\psi^\mu_+(\pi) = -\,\psi^\mu_-(\pi): \qquad \text{Neveu-Schwarz (NS)}. \tag{3.29}$$

It is cleaner to trade the strip with its two boundary conditions for a single field on a circle. Reflect $\sigma \to -\sigma$ and glue ψ_- to ψ_+, so that ψ^μ now lives on $\sigma \sim \sigma + 2\pi$. The endpoint data Eq. (3.27) become a periodicity,

$$\psi^\mu(\sigma{+}2\pi) \;=\; -\,\psi^\mu(\sigma) \quad \text{(NS, antiperiodic)}, \qquad \psi^\mu(\sigma{+}2\pi) \;=\; +\,\psi^\mu(\sigma) \quad \text{(R, periodic)}. \tag{3.30}$$

For the closed string there is no boundary to relate the movers, so ψ^μ_+ and ψ^μ_- are independent and each carries its own choice in Eq. (3.30). A closed string therefore has four sectors, NS-NS, NS-R, R-NS and R-R, one for each pair of left- and right-moving boundary conditions. The single-side projection that keeps a consistent set of modes is fixed in Section 3.3, and the four pairings are assembled into the closed string in Section 3.4. The physics of the split is already visible in a single set of modes, and that is what we quantise now.

Remark 3.6 (Why the boson had no such choice). X^μ is a map to spacetime, a genuine coordinate, so it must be single-valued: only the periodic condition is available and there is one X-sector. A fermion has no such interpretation. The action is invariant under the global $\mathbb{Z}_2$ that flips $\psi^\mu \to -\psi^\mu$, and fields are physical only up to that symmetry, so both signs in Eq. (3.30) define consistent theories. The two sectors are not a nuisance to be chosen between; a consistent string keeps both and sums over them, as the torus will insist in Section 3.3.

3.2.2 Mode expansions

Pass to the plane by $z = e^{\tau_E - i\sigma}$, the map of Appendix A, and work with the holomorphic (left-moving) component; the antiholomorphic one is identical with $z \to \bar{z}$. The free fermion has weight $\frac{1}{2}$, established in Section 3.1 through the operator product

$$\psi^\mu(z)\,\psi^\nu(w) \;\sim\; \frac{\eta^{\mu\nu}}{z-w}. \tag{3.31}$$

Trace the cylinder periodicity through the map. Under $z = e^{\tau_E - i\sigma}$ the shift $\sigma \to \sigma + 2\pi$ is $z \to e^{-2\pi i} z$, one full turn around the origin, and a weight-$\frac{1}{2}$ field transforms as $\psi \to (\mathrm{d}z'/\mathrm{d}z)^{1/2}\psi = e^{-i\pi}\psi = -\psi$ under that turn. So a field that is antiperiodic on the cylinder (NS) comes back to itself on the plane, single-valued, while a field that is periodic on the cylinder (R) picks up the extra sign and is double-valued, carrying a branch cut. The two moding patterns follow:

$$\text{NS:}\quad \psi^\mu(z) = \sum_{r \in \mathbb{Z}+\frac{1}{2}} \frac{\psi^\mu_r}{z^{\,r+1/2}}, \qquad r = \pm\tfrac{1}{2}, \pm\tfrac{3}{2}, \ldots \tag{3.32}$$

$$\text{R:}\quad \psi^\mu(z) = \sum_{n \in \mathbb{Z}} \frac{\psi^\mu_n}{z^{\,n+1/2}}, \qquad n = 0, \pm 1, \pm 2, \ldots \tag{3.33}$$

In the NS sector the exponents $r + \frac{1}{2}$ are integers, so $\psi^\mu(z)$ is single-valued; in the R sector they are half-integers and ψ^μ has a branch cut, the double-valuedness that is periodicity on the cylinder. The one structural difference between the sectors is already here: the R expansion contains a zero mode ψ^μ_0, the NS expansion does not.

Contour-integrating Eq. (3.31) turns the OPE into the canonical anticommutator, valid in both sectors with the appropriate range of indices,

$$\{\psi^\mu_r, \psi^\nu_s\} \;=\; \eta^{\mu\nu}\,\delta_{r+s,0}, \tag{3.34}$$

and the worldsheet Majorana (reality) condition makes the raising and lowering operators conjugate, $(\psi^\mu_r)^\dagger = \psi^\mu_{-r}$. Modes with $r > 0$ annihilate, modes with $r < 0$ create. The ground state of either sector is defined by

$$\psi^\mu_r\,|0\rangle \;=\; 0 \quad \text{for } r > 0, \tag{3.35}$$

together with the bosonic condition $\alpha^\mu_n|0\rangle = 0$ for $n > 0$ inherited from Chapter 1. In NS the smallest raising mode is $\psi^\mu_{-1/2}$ and Eq. (3.35) fixes the ground state uniquely up to normalisation. In R the zero modes are neither raising nor lowering: Eq. (3.35) says nothing about ψ^μ_0, and $\{\psi^\mu_0, \psi^\nu_0\} = \eta^{\mu\nu}$ from Eq. (3.34) means they cannot simply be set to zero. That single fact, unpacked in Section 3.2.4, is where spacetime fermions come from.

Exercise 3.5 (Contour to anticommutator (Warm-up, 10 min))**.** Invert Eqs. (3.32) and (3.33) to $\psi^\mu_r = \oint \frac{\mathrm{d}z}{2\pi i}\, z^{\,r-1/2}\,\psi^\mu(z)$, then evaluate $\{\psi^\mu_r, \psi^\nu_s\}$ as a double contour integral of the OPE Eq. (3.31) and recover Eq. (3.34). Check that the index ranges leave one zero mode ψ^μ_0 in R and none in NS, and that $(\psi^\mu_r)^\dagger = \psi^\mu_{-r}$ follows from reality of ψ^μ. Which of the two anticommutators, R or NS, is the one that will refuse to leave the ground state a singlet?

3.2.3 Zero-point energies and the two ground states

The mass of a state is read from L_0, and the constant piece of L_0, the intercept a, is a sum of vacuum energies over every oscillator tower. We computed the bosonic contribution in Section 1.4.5 by ζ-regularising $\sum_{n\geq 1} n = -\frac{1}{12}$. The fermion adds towers of its own, with two

differences: it contributes with the opposite sign, because normal-ordering an anticommutator costs a minus relative to a commutator, and its modes are half-integer in NS. Working in light-cone gauge, where only the $D-2=8$ transverse X^i and ψ^i survive and the bookkeeping is transparent, the zero-point energy of one transverse field is

$$E_0 \;=\; \pm\frac{1}{2}\sum_{\omega>0}\omega, \qquad \begin{cases} + \text{ boson} \\ - \text{ fermion} \end{cases} \tag{3.36}$$

with ω running over the positive mode numbers of that field. Two regularised sums do all the work,

$$\sum_{n=1}^{\infty} n \;=\; \zeta(-1) \;=\; -\frac{1}{12}, \qquad \sum_{n=0}^{\infty}\left(n+\tfrac{1}{2}\right) \;=\; \zeta\!\left(-1,\tfrac{1}{2}\right) \;=\; +\frac{1}{24}, \tag{3.37}$$

the second a Hurwitz zeta for the antiperiodic tower.

Example 3.7 (The two zero-point energies)**.** Assemble Eqs. (3.36) and (3.37) one field at a time. A transverse boson X^i is integer-moded, $\omega = 1,2,\ldots$, and carries the $+$ sign, so $E_0 = \frac{1}{2}\zeta(-1) = -\frac{1}{24}$. An NS fermion ψ^i is half-integer-moded, $\omega = \frac{1}{2},\frac{3}{2},\ldots$, and carries the $-$ sign, so $E_0 = -\frac{1}{2}\,\zeta(-1,\frac{1}{2}) = -\frac{1}{48}$. An R fermion is integer-moded like the boson but keeps the fermion sign, so $E_0 = -\frac{1}{2}\zeta(-1) = +\frac{1}{24}$, equal and opposite to the boson. The R fermion therefore cancels the boson term by term, whereas the NS fermion adds a further $-\frac{1}{48}$. Multiplying by the eight transverse directions reproduces Eqs. (3.38) and (3.39).

The intercept is $a = -E_0^{\text{tot}}$, so summing Eq. (3.36) over the eight transverse bosons (integer-moded, $-\frac{1}{24}$ each) and eight transverse fermions gives

$$\text{NS:}\quad E_0 = 8\Big[\underbrace{+\tfrac{1}{2}(-\tfrac{1}{12})}_{\text{boson}}\;\underbrace{-\tfrac{1}{2}(+\tfrac{1}{24})}_{\text{NS fermion}}\Big] = 8\Big(-\tfrac{1}{24}-\tfrac{1}{48}\Big) = -\tfrac{1}{2}, \qquad a_{\text{NS}} = \tfrac{1}{2}, \tag{3.38}$$

$$\text{R:}\quad E_0 = 8\Big[\underbrace{+\tfrac{1}{2}(-\tfrac{1}{12})}_{\text{boson}}\;\underbrace{-\tfrac{1}{2}(-\tfrac{1}{12})}_{\text{R fermion}}\Big] = 8(-\tfrac{1}{24}+\tfrac{1}{24}) = 0, \qquad a_{\text{R}} = 0. \tag{3.39}$$

The cancellation in the R line does not depend on the number eight. A periodic boson and a periodic fermion have identical mode numbers, and their vacuum energies differ only by the statistics sign in Eq. (3.36), so they cancel direction by direction. Section 3.2.4 shows that $a_{\text{R}} = 0$ is forced by supersymmetry on the worldsheet, not merely observed. The covariant computation, which restores the timelike and longitudinal towers and adds the commuting weight-$\frac{3}{2}$ superconformal $\beta\gamma$ ghosts alongside the bc ghosts of Section 1.4.5, gives the same two intercepts; we quote that agreement and leave the ghost arithmetic to Appendix F, as we did for the bosonic string.

With the intercepts in hand the open-superstring mass formula is $\alpha' M^2 = N - a$, where the level counts both kinds of oscillator,

$$N_{\text{NS}} = \sum_{n\geq 1}\alpha_{-n}\cdot\alpha_n + \sum_{r\geq 1/2} r\,\psi_{-r}\cdot\psi_r, \qquad N_{\text{R}} = \sum_{n\geq 1}\alpha_{-n}\cdot\alpha_n + \sum_{m\geq 1} m\,\psi_{-m}\cdot\psi_m. \tag{3.40}$$

Read off the bottom of each tower:

- **NS ground state.** $N=0$, so $\alpha' M^2 = -\frac{1}{2}$. The NS vacuum $|0\rangle_{\text{NS}}$ is a Lorentz scalar and a tachyon, the direct descendant of the bosonic tachyon of Section 1.5.2. Its removal is the first job of the GSO projection in Section 3.3.
- **First excited NS state.** The lowest raising mode gives $\psi^{\mu}_{-1/2}|0\rangle_{\text{NS}}$ at $N=\frac{1}{2}$, hence $\alpha' M^2 = 0$. This is a massless spacetime vector; its transverse polarisations fill the vector $\mathbf{8}_v$ of the massless little group SO(8). This is the boson that will pair with the Ramond fermion.

- **R ground state.** $N = 0$ with $a_{\mathrm{R}} = 0$ gives $\alpha' M^2 = 0$: massless. But this ground state is not one state, because of the zero modes. That is the subject of the next subsection.

3.2.4 The Ramond ground state is a spacetime spinor

Restrict Eq. (3.34) to the R zero modes:

$$\{\psi_0^\mu, \psi_0^\nu\} \;=\; \eta^{\mu\nu}. \tag{3.41}$$

Set $\Gamma^\mu \coloneqq \sqrt{2}\,\psi_0^\mu$ and this is exactly the Clifford algebra of Appendix A,

$$\{\Gamma^\mu, \Gamma^\nu\} \;=\; 2\,\eta^{\mu\nu}. \tag{3.42}$$

The zero modes commute with the number operator N_{R}, so they map the R ground state to itself: acting with ψ_0^μ never changes the mass level. The ground state cannot therefore be a single vector in the Hilbert space. It must be a whole multiplet carrying an irreducible representation of Eq. (3.42), that is, a module for the Dirac algebra in ten dimensions. In $D = 10$ the irreducible Clifford module has dimension $2^{D/2} = 2^5 = 32$: the Ramond ground state is a *32-component Dirac spinor of* $\mathrm{SO}(1,9)$. Every state built on it by the integer-moded oscillators is likewise a spacetime spinor. This is the origin of spacetime fermions in string theory, and remarkably little went into it. We added one worldsheet fermion per spacetime dimension, chose the periodic boundary condition, and the ground-state degeneracy had no option but to be a spinor, because Eq. (3.41) is the defining relation of the Γ-matrices. The Clifford construction, the counting $2^{D/2}$, and the reality and chirality conditions available in each dimension are the business of Appendix B. The one fact we lean on repeatedly is that $D = 10$, being $2 \bmod 8$, is exactly where a spinor can be simultaneously Majorana and Weyl: the 32 complex components of the Dirac spinor collapse to a Majorana-Weyl spinor of 16 real components, and the Dirac equation Eq. (3.45) halves that once more to the 8 physical states we are about to count.

There is a sharper worldsheet statement behind the masslessness. The R sector has an anticommuting zero mode of the supercurrent, G_0, Hermitian because $(\psi_r^\mu)^\dagger = \psi_{-r}^\mu$ makes $G_0^\dagger = G_0$. On the ground state, where the oscillator terms drop out, it reduces to

$$G_0 \;=\; \psi_0^\mu\,\alpha_{0\mu} + (\text{oscillators}) \;=\; \sqrt{2\alpha'}\,\psi_0^\mu\,p_\mu + (\text{oscillators}), \tag{3.43}$$

using the open-string zero mode $\alpha_0^\mu = \sqrt{2\alpha'}\,p^\mu$ of Section 1.5.1. The superconformal algebra of Section 3.1 does not set $\{G_0, G_0\} = 2L_0$ on the nose; it carries a central term, $\{G_0, G_0\} = 2L_0 - c/12$ from Eq. (3.21), equivalently $G_0^2 = L_0 - c/24$ from Eq. (3.22). The constant $c/24$ is precisely the Ramond zero-point energy that L_0 already carries, so $L_0 - c/24$ is the mass-shell operator with the zero-point stripped off. Squaring Eq. (3.43) on the ground state makes this concrete, using $(\psi_0^\mu p_\mu)^2 = \tfrac{1}{2}\{\psi_0^\mu, \psi_0^\nu\}p_\mu p_\nu = \tfrac{1}{2}p^2$,

$$G_0^2 \;=\; 2\alpha'\,(\psi_0^\mu p_\mu)^2 \;=\; \alpha'\,p^2 \;=\; -\,\alpha' M^2. \tag{3.44}$$

Because G_0 is Hermitian, $G_0^2 \geq 0$, and the physical-state condition $G_0\,|\mathrm{R}\rangle = 0$ forces the right-hand side to vanish, so $M^2 = 0$. Setting $N_{\mathrm{R}} = 0$ in $\alpha' M^2 = N_{\mathrm{R}} - a_{\mathrm{R}}$ then reads off $a_{\mathrm{R}} = 0$: the vanishing intercept found by hand in Eq. (3.39) is forced, because G_0 is a Hermitian square root of the mass-shell operator and the square of a square root carries no shift. The same constraint written out is the massless Dirac equation. With $\Gamma^\mu = \sqrt{2}\,\psi_0^\mu$ from Eq. (3.42) the coefficient in Eq. (3.43) is $\sqrt{2\alpha'}\,\psi_0^\mu p_\mu = \sqrt{\alpha'}\,\not{p}$, so

$$G_0\,|\mathrm{R}\rangle = 0 \quad\Longleftrightarrow\quad \not{p}\,|\mathrm{R}\rangle = 0, \tag{3.45}$$

a spacetime field equation emerging from a worldsheet constraint. The supercurrent evaluated on the Ramond vacuum is the Dirac operator, and a spacetime fermion field equation has appeared from a constraint on the string rather than being put in by hand.

The 32 is a covariant count, not the physical one

The covariant Clifford module includes the timelike zero mode ψ_0^0, whose norm is negative, $\{\psi_0^0, \psi_0^0\} = \eta^{00} = -1$, so a naive 32 contains unphysical directions exactly as the covariant boson Fock space contained negative-norm states in Chapter 1. The super-Virasoro constraints $G_r |\psi\rangle = 0$, with G_0 chief among them via Eq. (3.45), remove them. In light-cone gauge only the eight transverse zero modes ψ_0^i survive, giving the SO(8) Clifford algebra with module of dimension $2^4 = 16$. That 16 is reducible: the product $\bar{\Gamma} = \Gamma^1 \cdots \Gamma^8$ splits it into two Weyl spinors of opposite chirality,

$$\mathbf{16} = \mathbf{8}_s \oplus \mathbf{8}_c.$$

The physical Ramond ground state is these 16 states, and the GSO projection of Section 3.3 keeps just one chirality, eight physical fermionic states.

Line the two ground-state analyses up. After the GSO projection removes the NS tachyon, the lightest NS state is the massless vector $\mathbf{8}_v$ of Section 3.2.3, and the Ramond ground state is a massless spinor $\mathbf{8}_s$. Eight bosonic states, eight fermionic states, degenerate in mass. That equality is the first fingerprint of spacetime supersymmetry. The $8 = 8$ match is triality: $\mathbf{8}_v$, $\mathbf{8}_s$ and $\mathbf{8}_c$ are the three eight-dimensional representations of SO(8), permuted by its outer automorphism. Turning this numerical match into an exact projection, tying it to modular invariance of the torus, and reading off which theory results, is the work of Section 3.3. What this section has established is the raw material: a boson sector with a tachyon at the bottom and a massless vector above it, and a fermion sector whose very ground state is a spacetime spinor.

Remark 3.8 (Beyond the lectures: spin structures and why GSO is not optional). The NS/R split is the choice of boundary condition around the *spatial* circle of the string. On the torus that computes the one-loop vacuum energy there is a second circle, the Euclidean time direction, and ψ^μ needs a boundary condition around it too. Four spin structures result, and the partition function sums over them. Modular invariance, the requirement that the torus amplitude not depend on how one draws the two cycles (Section 2.3.4), fixes the relative signs in that sum uniquely. Those signs *are* the GSO projection: keeping both NS and R while summing with the modular-invariant weights is what a consistent string does, and it is not a choice imposed for elegance. The partition function and the modular arithmetic are carried out in Appendix F.

Remark 3.9 (Beyond the lectures: the spin field). In the operator language, the Ramond ground state is created from the NS vacuum by a *spin field* $S_\alpha(z)$, a twist operator that implements the branch cut of Eq. (3.33) by turning NS boundary conditions into R ones. Bosonising the ten ψ^μ into five chiral bosons H_a, the spin field is $S_\alpha = \exp(\frac{i}{2}\sum_a s_a H_a)$ with each $s_a = \pm 1$, the 2^5 sign choices reproducing the 32 components and the product $\prod_a s_a$ giving the chirality. Its conformal weight is $D/16 = 10/16 = 5/8$. This is the object one needs to write vertex operators for spacetime fermions, and it is where the amplitudes of Chapter 4 begin; we name it here and defer the construction.

Exercise 3.6 (The two intercepts (Moderate, 30 min))**.** Reproduce Eqs. (3.38) and (3.39) from Eqs. (3.36) and (3.37). Then redo the NS intercept covariantly: restore all ten of the X^μ and ψ^μ towers, add the bc reparametrisation ghosts of Section 1.4.5 together with the commuting weight-$\frac{3}{2}$ superconformal $\beta\gamma$ ghosts, and show the result is again $a_{\mathrm{NS}} = \frac{1}{2}$ at $D = 10$. (*Hint:* the superghost tower is bosonic and half-integer moded in the NS sector; its zero-point energy carries the boson sign in Eq. (3.36) with the antiperiodic sum of Eq. (3.37). Compare the central-charge bookkeeping you set up in Exercise 1.9.)

Exercise 3.7 (Building the Clifford module (Standard, 20 min))**.** Pair the ten zero modes into five fermionic creation/annihilation operators, $\psi_0^{\pm a} = \frac{1}{\sqrt{2}}(\psi_0^{2a} \pm i\,\psi_0^{2a+1})$ for $a = 0, \ldots, 4$,

and check from Eq. (3.41) that $\{\psi_0^{+a}, \psi_0^{-b}\} = \delta^{ab}$ with all other anticommutators vanishing. Conclude that acting with the five ψ_0^{-a} on a Clifford vacuum generates $2^5 = 32$ states, the Dirac spinor. Repeat with the eight transverse zero modes to get the $\mathbf{8}_s \oplus \mathbf{8}_c$ of Section 3.2.4, and identify the chirality operator that separates them.

Exercise 3.8 (The supercurrent is the Dirac operator (Moderate, 25 min))**.** Starting from $G_0^2 = L_0 - c/24$ (Eq. (3.22)), with $c/24$ the Ramond zero-point energy already carried by L_0, use Hermiticity of G_0 and the physical-state condition $G_0\,|\mathrm{R}\rangle = 0$ to argue that the ground state is massless, and hence that $a_{\mathrm{R}} = 0$ is forced rather than fitted. Then derive Eq. (3.45) explicitly from Eq. (3.43) using $\alpha_0^\mu = \sqrt{2\alpha'}\,p^\mu$ and $\Gamma^\mu = \sqrt{2}\,\psi_0^\mu$, tracking the factors of $\sqrt{2}$. Finally show that on a general R state $G_0\,|\psi\rangle = 0$ combines the Dirac operator with the fermionic oscillators, so the R tower is a stack of massive Dirac equations, one at each level.

Exercise 3.9 (The first levels, and level matching (Moderate, 30 min))**.** For the open superstring, list the states at the first massive NS level, $\alpha' M^2 = 1$, and organise them under the massive little group SO(9). Do the same for the first excited R level and check that its states again fill spinor representations of SO(9). For the closed superstring the mass formula per side reads $\frac{\alpha'}{4}M^2 = N - a = \tilde{N} - \tilde{a}$, the level-matching constraint of Section 1.2.5, so combining the two movers gives $\alpha' M^2 = 2\big[(N-a)+(\tilde{N}-\tilde{a})\big]$; note this differs from the open-string $\alpha' M^2 = N - a$ of Section 3.2.3 by the standard closed-string factor. Use it to explain why a massless closed-string state must pair a left ground state with a right ground state, and why choosing NS or R on each side then produces the four sectors of Section 3.2.1.

3.3 The GSO projection

At the end of Section 3.2 we had two towers of states and two problems. The Neveu-Schwarz tower is built on a ground state with $\alpha' M^2 = -\frac{1}{2}$, a tachyon, exactly as in the bosonic string of Chapter 1. The Ramond tower is built on a spacetime spinor, but it is the full Dirac spinor of $\mathrm{SO}(1,9)$, twice as large as anything a supersymmetric spectrum can use, and its energy levels do not line up with the NS ones: NS states sit at $\alpha' M^2 \in \{-\frac{1}{2}, 0, \frac{1}{2}, 1, \dots\}$, half-integer spaced, while R states sit at $\alpha' M^2 \in \{0, 1, 2, \dots\}$. A theory with a tachyon is unstable, and a theory whose bosons and fermions do not match cannot be spacetime supersymmetric. Both defects are cured at once by a single truncation of the Hilbert space.

That truncation is the projection Gliozzi, Scherk and Olive wrote down in 1976–77, the GSO projection. It keeps only the states invariant under a $\mathbb{Z}_2$ worldsheet symmetry, worldsheet fermion number. The tachyon is not invariant, so it goes. What is left has equal numbers of bosons and fermions at every mass level, which is the fingerprint of spacetime supersymmetry. This is the decisive step in building the superstring, so we take it slowly.

The GSO rule

Let $(-1)^F$ be worldsheet fermion number, defined below. The GSO projection keeps the states with

$$(-1)^F = +1, \qquad P_{\mathrm{GSO}} = \tfrac{1}{2}\big(1 + (-1)^F\big).$$

In the NS sector this keeps the states with an *odd* number of ψ oscillators. The ground state has none, so the **tachyon is removed**; the massless vector $\psi^\mu_{-1/2}|0\rangle$ survives. **In the R sector** $(-1)^F$ contains the ten-dimensional chirality matrix Γ_{11}, so the projection keeps **one spacetime chirality** of the ground-state spinor. The two payoffs: no tachyon, and a spectrum with matched Bose/Fermi content at every level, which is **spacetime supersymmetry**. The projection is not optional: it is what **modular invariance** of the one-loop partition function forces (Appendix F).

3.3.1 Worldsheet fermion number

The worldsheet fermions ψ^μ carry a $\mathbb{Z}_2$ symmetry: the free action of Section 3.1 is invariant under $\psi^\mu \to -\psi^\mu$ with X^μ untouched. The conserved charge that generates it is worldsheet fermion number F, counting how many fermionic oscillators an excited state carries. What matters for the projection is the operator $(-1)^F$, which is $+1$ on states with an even number of ψ's and -1 on states with an odd number. By construction $(-1)^F$ anticommutes with every fermion mode,

$$\{(-1)^F,\ \psi^\mu_r\} = 0, \tag{3.46}$$

and commutes with the bosonic modes α^μ_n. Acting on the oscillators alone the definition is unambiguous. The one genuine choice is the value assigned to each ground state, and that choice is the whole content of GSO.

The NS convention. In the NS sector the fermions are half-integer moded, $r \in \mathbb{Z} + \frac{1}{2}$, with $\{\psi^\mu_r, \psi^\nu_s\} = \eta^{\mu\nu}\delta_{r+s,0}$. Define

$$(-1)^F \ = \ -\,(-1)^{\sum_{r>0} \psi_{-r}\cdot\psi_r} \qquad \text{(NS sector)}. \tag{3.47}$$

The exponent counts the oscillators; the extra overall minus sign is the convention. It fixes the NS ground state to be *odd*,

$$(-1)^F\,|0\rangle_{\rm NS} = -|0\rangle_{\rm NS}, \tag{3.48}$$

and each application of ψ^μ_{-r} flips the sign. That single sign, dropped into (3.47) by hand, is what will kill the tachyon. Without it we would keep the ground state and throw away the vector, which is the wrong theory.

The R convention. In the R sector the fermions are integer moded, $n \in \mathbb{Z}$, and there are zero modes ψ^μ_0 obeying

$$\{\psi^\mu_0, \psi^\nu_0\} = \eta^{\mu\nu}, \tag{3.49}$$

which is the SO(1, 9) Clifford algebra rescaled, $\psi^\mu_0 = \Gamma^\mu/\sqrt{2}$ (Section 3.2 and Appendix B). The ground states therefore form a representation of that algebra: a Dirac spinor. Worldsheet fermion number cannot simply "count the zero modes", because the ψ^μ_0 neither raise nor lower the energy; they rotate the ground states into one another. The operator that plays the role of $(-1)^F$ on the ground-state spinor is the chirality matrix

$$\Gamma_{11} \ = \ \Gamma^0\Gamma^1\cdots\Gamma^9, \qquad (\Gamma_{11})^2 = 1, \qquad \{\Gamma_{11}, \Gamma^\mu\} = 0, \tag{3.50}$$

whose ± 1 eigenvalues are the two spacetime chiralities. Including the honest oscillator count for the non-zero modes, the full R-sector operator is

$$(-1)^F \ = \ \Gamma_{11}\,(-1)^{\sum_{n>0} \psi_{-n}\cdot\psi_n} \qquad \text{(R sector)}. \tag{3.51}$$

On the ground state, where no oscillators act, this is just Γ_{11}, so the projection $(-1)^F = +1$ selects a definite spacetime chirality and turns the bulky Dirac spinor into a Weyl spinor.

3.3.2 The Neveu-Schwarz sector: the tachyon goes

Recall the NS mass formula from Section 3.2, in light-cone form with the eight transverse fermions and bosons,

$$\alpha' M^2 = N_{\rm NS} - \tfrac{1}{2}, \qquad N_{\rm NS} = \sum_{n\geq 1} \alpha^i_{-n}\alpha^i_n + \sum_{r\geq 1/2} r\,\psi^i_{-r}\psi^i_r. \tag{3.52}$$

The $-\frac{1}{2}$ is the transverse zero-point energy, eight bosons at $-\frac{1}{24}$ each plus eight antiperiodic fermions at $-\frac{1}{48}$ each. Now apply the projector $P_{\rm GSO} = \frac{1}{2}(1 + (-1)^F)$ level by level.

- $N_{\rm NS}=0$, $\alpha' M^2=-\frac{1}{2}$: the ground state $|0\rangle_{\rm NS}$. By (3.48) it has $(-1)^F=-1$. **Projected out.** The tachyon is gone.
- $N_{\rm NS}=\frac{1}{2}$, $\alpha' M^2=0$: the states $\psi^i_{-1/2}|0\rangle_{\rm NS}$, $i=1,\dots,8$. One oscillator, so $(-1)^F=+1$. **Kept.** These eight states are the vector $\mathbf{8}_v$ of the transverse SO(8), a massless spacetime vector.
- $N_{\rm NS}=1$, $\alpha' M^2=+\frac{1}{2}$: the states $\alpha^i_{-1}|0\rangle_{\rm NS}$ (no fermions) and $\psi^i_{-1/2}\psi^j_{-1/2}|0\rangle_{\rm NS}$ (two fermions). Both are even, $(-1)^F=-1$. **The entire half-integer level is projected out.**

The pattern is now clear. The surviving NS states have an odd number of ψ oscillators, which pushes $N_{\rm NS}$ to the half-odd-integers $\frac{1}{2},\frac{3}{2},\frac{5}{2},\dots$, so that after the projection

$$\alpha' M^2 \in \{0,1,2,\dots\} \qquad \text{(NS, after GSO)}. \tag{3.53}$$

The tachyon has gone, and the surviving NS levels are now integer spaced, matching the R levels so that the two towers can line up.

3.3.3 The Ramond sector: one chirality

The R mass formula has no zero-point energy, the eight periodic fermions cancelling the eight bosons,

$$\alpha' M^2 = N_{\rm R}, \qquad N_{\rm R}=\sum_{n\geq 1}(\alpha^i_{-n}\alpha^i_n + n\,\psi^i_{-n}\psi^i_n). \tag{3.54}$$

The ground state is therefore massless, and it is degenerate: the zero modes ψ^μ_0 act within it. Splitting off the zero modes of the transverse fermions, the eight ψ^i_0 generate a $2^4=16$-dimensional spinor of the transverse SO(8), which is reducible into its two Weyl pieces,

$$\mathbf{16}=\mathbf{8}_s\oplus\mathbf{8}_c. \tag{3.55}$$

These are the two chiralities distinguished by Γ_{11}. The GSO projection $(-1)^F=+1$, which on the ground state is $\Gamma_{11}=+1$, keeps just one of them, say $\mathbf{8}_s$. (Choosing $\Gamma_{11}=-1$ keeps $\mathbf{8}_c$ instead; the two choices are physically distinct, and the relative choice between left- and right-movers is what separates Type IIA from Type IIB, Section 3.4.)

There is a clean spacetime reason to expect a Weyl spinor and not a Dirac one. The zero mode of the super-Virasoro current acts on the R ground state through its own zero modes,

$$G_0 \;=\; \alpha^\mu_0\,\psi_{0\,\mu} + (\text{oscillators}) \;=\; \sqrt{\alpha'}\,\not{p} + (\text{oscillators}), \tag{3.56}$$

where we used the single-side zero mode $\alpha^\mu_0=\sqrt{2\alpha'}\,p^\mu$ of Chapter 1 together with $\Gamma^\mu=\sqrt{2}\,\psi^\mu_0$, whose two square roots combine to $\sqrt{\alpha'}$. The physical-state condition $G_0\,|\psi\rangle=0$ is therefore the massless Dirac equation $\not{p}\,|\psi\rangle=0$ (Section 3.2). A massless Dirac equation together with a chirality projection is precisely the on-shell content of a Majorana-Weyl spinor, and $D=10$ is the dimension where Majorana and Weyl conditions are simultaneously consistent (Appendix B). The projection GSO performs on the worldsheet is the same halving that a Weyl condition performs in spacetime, and after it the massless R states number eight on-shell, the $\mathbf{8}_s$ of SO(8).

Careful

The two payoffs are separate choices dressed as one operator. The overall sign in (3.47) is fixed by demanding no tachyon: there is only one sensible choice. The sign in (3.51), the choice of which chirality to keep, is genuinely a fork in the road. Making it consistently on both movers of the closed string, or oppositely, gives two different ten-dimensional theories.

$\alpha' M^2$	sector	before GSO	after GSO
$-\frac{1}{2}$	NS	$\mathbf{1}$ (tachyon)	removed
0	NS	$\mathbf{8}_v$	$\mathbf{8}_v$
0	R	$\mathbf{16} = \mathbf{8}_s \oplus \mathbf{8}_c$	$\mathbf{8}_s$

Table 3.1: The bottom of the open RNS spectrum, in the light-cone SO(8). GSO deletes the tachyon and halves the Ramond ground state, leaving $\mathbf{8}_v$ bosons and $\mathbf{8}_s$ fermions: equal in number.

Do not read the single symbol P_{GSO} as forcing a unique answer in the R sector.

3.3.4 Matched levels: spacetime supersymmetry

Put the two projected towers side by side at the bottom. Table 3.1 shows the massless level and the (now absent) tachyon.

At the massless level the count is $8 = 8$: eight bosonic states $\mathbf{8}_v$ and eight fermionic states $\mathbf{8}_s$. The match is triality: the vector $\mathbf{8}_v$ and the two spinors $\mathbf{8}_s$, $\mathbf{8}_c$ are the three eight-dimensional representations of SO(8), permuted by the outer automorphisms of the algebra. A vector can equal a spinor in dimension only for this transverse group, which is to say only in $D = 10$. The critical dimension that Section 3.1 forced from the central charge, and the dimension in which the surviving spectrum can be supersymmetric, are one and the same: both are $D = 10$.

The equality does not stop at the massless level. Once the NS levels have been squeezed onto the integers by (3.53), the projected NS and R towers agree state for state at every $\alpha' M^2 = 0, 1, 2, \ldots$. The first massive level illustrates it.

Example 3.10 (The first massive level)**.** Take $\alpha' M^2 = 1$. In the NS sector this is $N_{\text{NS}} = \frac{3}{2}$, and the surviving (odd-fermion-number) states are

$$\psi^i_{-3/2}|0\rangle_{\text{NS}} \ (8), \qquad \alpha^i_{-1}\psi^j_{-1/2}|0\rangle_{\text{NS}} \ (64), \qquad \psi^i_{-1/2}\psi^j_{-1/2}\psi^k_{-1/2}|0\rangle_{\text{NS}} \ (56),$$

totalling $8 + 64 + 56 = 128$ bosonic states, which assemble into a massive representation of the little group SO(9). The projected R sector at $N_{\text{R}} = 1$ delivers exactly 128 fermionic states. The degeneracy $128_{\text{B}} = 128_{\text{F}}$ is spacetime supersymmetry at work one level up from massless.

What GSO buys, then, is a supersymmetric spectrum: the NS sector supplies spacetime bosons, the R sector supplies spacetime fermions, and after the projection they pair up level by level. We are stating this as a spectrum count, which is what the course needs. The supercharges that implement it, and the fact that they close on the ten-dimensional Poincaré algebra, follow from the worldsheet supercurrent; the covariant construction of the spacetime spinor supercharge is cleanest in the Green-Schwarz and pure-spinor formalisms, which we name here and do not develop, staying with RNS throughout.

3.3.5 Why the projection is forced: modular invariance

So far GSO looks like a hand-chosen truncation: we wrote a sign into (3.47) because it removed the tachyon and gave us supersymmetry. That is a good motivation but not a derivation. The derivation comes from a consistency condition on the closed string at one loop, and it leaves no freedom. We state the result and point to Appendix F for the calculation.

A one-loop closed-string vacuum amplitude is an integral over the moduli of the torus. The torus has two independent cycles, and a worldsheet fermion can be periodic (R) or antiperiodic

(NS) around each of them independently. That is four choices of boundary condition, the four *spin structures*:

$$\text{(NS,NS)}, \quad \text{(NS,R)}, \quad \text{(R,NS)}, \quad \text{(R,R)}. \tag{3.57}$$

The condition around the spatial cycle selects the sector, NS or R; the condition around the time cycle is precisely the insertion of $(-1)^F$ into the trace, because twisting the time-direction boundary condition by the $\mathbb{Z}_2$ symmetry is the same as weighting the trace by its generator. So the four spin structures are the four traces

$$\mathrm{Tr}_{\mathrm{NS}}\, q^{L_0 - c/24}, \quad \mathrm{Tr}_{\mathrm{NS}}\,(-1)^F q^{L_0 - c/24}, \quad \mathrm{Tr}_{\mathrm{R}}\, q^{L_0 - c/24}, \quad \mathrm{Tr}_{\mathrm{R}}\,(-1)^F q^{L_0 - c/24}, \tag{3.58}$$

with $q = e^{2\pi i \tau}$. The modular group $\mathrm{SL}(2,\mathbb{Z})$ acts on τ by relabelling the two cycles, and in doing so it *permutes the four spin structures*. A modular-invariant amplitude cannot use one spin structure alone; it must sum over all four with definite relative coefficients. Fixing those coefficients so that the sum is invariant, and reading off which states survive in the resulting trace, gives back exactly $P_{\mathrm{GSO}} = \frac{1}{2}(1 + (-1)^F)$ in each sector: GSO is the sum over spin structures, which is why it is mandatory rather than optional.

The eight transverse fermions make this concrete. Their contribution to the partition function in each spin structure is a fourth power of a Jacobi theta function over the Dedekind eta, and the modular-invariant combination is

$$Z_\psi(\tau) \;=\; \frac{1}{2\,\eta(\tau)^4}\Big[\,\theta_3(\tau)^4 - \theta_4(\tau)^4 - \theta_2(\tau)^4 - \theta_1(\tau)^4\,\Big], \tag{3.59}$$

where θ_3 is the (NS, no insertion) sector, θ_4 the (NS, $(-1)^F$) sector, θ_2 the (R, no insertion) sector, and θ_1 the (R, $(-1)^F$) sector. The relative signs are the ones modular invariance dictates, and they are the GSO signs. Two features of (3.59) carry the physics. First, $\theta_1 \equiv 0$: the last term vanishes identically, which is the statement that $\mathrm{Tr}_{\mathrm{R}}(-1)^F$ over the ground states counts $\mathbf{8}_s$ minus $\mathbf{8}_c$ with the chirality projection, and there is nothing left to weigh. Second, and more strikingly, Jacobi's identity

$$\theta_3(\tau)^4 = \theta_2(\tau)^4 + \theta_4(\tau)^4 \tag{3.60}$$

makes the whole bracket vanish, so $Z_\psi \equiv 0$. Jacobi called (3.60) an *aequatio identica satis abstrusa*, an "exceedingly obscure identity"; here it has a transparent meaning. Expanding both sides in powers of q, the coefficient at each level is the number of NS states minus the number of R states after GSO. The identity says those numbers are equal at every level. The vanishing of the one-loop partition function is spacetime supersymmetry written as an equation: an exact boson-fermion cancellation, level by level, all the way up the tower.

Remark 3.11 (Beyond the lectures). The sum over spin structures is a small gauge theory, and this aside is optional reading rather than examinable material. The four choices in (3.57) are the two-dimensional analogue of choosing a spin structure on a Riemann surface, and summing over them with modular-invariant phases is gauging the $\mathbb{Z}_2$ worldsheet fermion parity $(-1)^F$. Different consistent assignments of the relative phases are different, equally modular-invariant, theories. The GSO signs in (3.59) give the supersymmetric superstring. A different consistent choice, keeping the R sector without the chirality correlation to the NS sign, produces the non-supersymmetric *Type 0A* and *Type 0B* strings: modular invariant, no spacetime fermions in their simplest form, and still carrying a tachyon because the NS ground state survives. They are a useful reminder that "modular invariant" is a weaker condition than "supersymmetric"; GSO is the specific modular-invariant projection that also removes the tachyon and matches Bose to Fermi.

3.3.6 What we have and where it goes

The RNS string with the GSO projection is a consistent, tachyon-free, spacetime-supersymmetric string theory in ten dimensions. The bookkeeping is compact: keep NS states of odd worldsheet fermion number, keep one chirality of the R ground state, and the resulting spectrum has $\mathbf{8}_v$ bosons and $\mathbf{8}_s$ fermions at the massless level and matched multiplets above it. The projection is forced by modular invariance of the torus, and the supersymmetry of the spectrum is the vanishing of that torus amplitude.

Everything so far has been the single set of worldsheet fermions, appropriate to the open string or to one mover of the closed string. The single-side projection is what we have fixed here; the closed superstring carries two sets, left and right, each with its own GSO projection and its own R-sector chirality choice. Making the two chirality choices alike or opposite is the fork flagged in Section 3.3.3, and it is where the ten-dimensional theory splits into the non-chiral Type IIA and the chiral Type IIB. The four NS/R pairings and that left-right assembly are the subject of Section 3.4.

Exercise 3.10 (The empty $\alpha' M^2 = \frac{1}{2}$ NS level)**.** *(Standard.)* Verify the claim in Section 3.3.2 that GSO removes the whole $\alpha' M^2 = \frac{1}{2}$ NS level. List the states at $N_{\rm NS} = 1$, compute $(-1)^F$ for each from (3.47), and confirm that none has $(-1)^F = +1$. Then list the surviving states at $N_{\rm NS} = \frac{3}{2}$ and check that they number 128.

Exercise 3.11 (The chirality operator Γ_{11})**.** *(Standard.)* Using (3.49), show that the operator $\Gamma_{11} \propto \psi_0^0 \psi_0^1 \cdots \psi_0^9$ satisfies $(\Gamma_{11})^2 = 1$ and anticommutes with each ψ_0^μ, so that it splits the R ground-state Dirac spinor into two Γ_{11}-eigenspaces of equal dimension. Deduce the on-shell count of the surviving Ramond ground state after imposing both $G_0 \,|\psi\rangle = 0$ and the GSO chirality projection, and match it to $\mathbf{8}_s$.

Exercise 3.12 (Bose-Fermi cancellation from the theta functions)**.** *(Moderate.)* The identity (3.60) encodes level-by-level Bose-Fermi degeneracy. Using the product forms

$$\frac{\theta_3}{\eta} = q^{-1/24}\prod_{n\geq 1}(1+q^{n-1/2}), \qquad \frac{\theta_4}{\eta} = q^{-1/24}\prod_{n\geq 1}(1-q^{n-1/2}), \qquad \frac{\theta_2}{\eta} = \sqrt{2}\,q^{1/12}\prod_{n\geq 1}(1+q^n),$$

expand $Z_\psi(\tau)$ from (3.59) to the first two non-trivial orders in q and confirm that the bosonic (NS) and fermionic (R) coefficients cancel. Interpret the q^0 cancellation as $\mathbf{8}_v$ against $\mathbf{8}_s$.

3.4 Type IIA and Type IIB

The previous three sections did the hard work for a single set of oscillators. The RNS string in Section 3.1 gave worldsheet supersymmetry and the critical dimension $D = 10$; Section 3.2 split the fermion ψ^μ into an antiperiodic (NS) and a periodic (R) sector, with the R ground state a spacetime spinor; and Section 3.3 kept only the GSO-even states, killing the NS tachyon and leaving a Bose-Fermi degenerate spectrum. All of that was, in effect, the theory of *one* moving direction along the string. A closed string has two, the left- and right-movers, and they are independent. Assembling the closed superstring means taking a tensor product of a left-moving copy of the GSO-projected spectrum with a right-moving copy. There is one discrete choice to make in doing so, and it produces exactly two theories: Type IIA and Type IIB.

3.4.1 The building block: sixteen states per side

Fix conventions from Section 3.2. On each side the mass operator, in units where level matching reads $N_{\rm L} - a_{\rm L} = N_{\rm R} - a_{\rm R}$, is

$$\frac{\alpha'}{4} M^2 \;=\; N_{\rm R} - a_{\rm R} \;=\; N_{\rm L} - a_{\rm L}, \qquad a_{\rm NS} = \tfrac{1}{2}, \quad a_{\rm R} = 0, \tag{3.61}$$

with the intercepts derived in Section 3.2. The factor of 4 separates this closed-string normalisation from the open-string $\alpha' M^2 = N - a$ of that section: each mover carries $\frac{\alpha'}{4} M^2 = N - a$, and level matching sets the two movers equal, so the full closed-string mass is $\alpha' M^2 = 2\big[(N_{\mathrm{L}} - a_{\mathrm{L}}) + (N_{\mathrm{R}} - a_{\mathrm{R}})\big]$. A massless closed string therefore sits at the first GSO-surviving level on *both* sides at once: in an NS sector that is the single oscillator $\psi^{\mu}_{-1/2}$ acting on the vacuum ($N = \frac{1}{2}$), and in an R sector it is the ground state itself ($N = 0$). Higher levels are massive, at the string scale, and we do not track them here.

Massless states in ten dimensions are organised by the little group of a null momentum, $\mathrm{SO}(8) \subset \mathrm{SO}(1,9)$. The three inequivalent eight-dimensional irreducible representations of Spin(8) are the vector $\mathbf{8}_v$ and the two spinors $\mathbf{8}_s$, $\mathbf{8}_c$, related by the triality that makes Spin(8) special. After GSO, each side of the string contributes exactly one $\mathbf{8}_v$ and one spinor:

$$\underbrace{\psi^{i}_{-1/2}|0\rangle_{\mathrm{NS}}}_{\mathbf{8}_v} \oplus \underbrace{|0\rangle_{\mathrm{R}}}_{\mathbf{8}_s \text{ or } \mathbf{8}_c} \qquad (i = 1, \ldots, 8), \tag{3.62}$$

sixteen states in all. Which spinor appears, $\mathbf{8}_s$ or $\mathbf{8}_c$, is fixed by the sign chosen for the GSO projector in the R sector: the two choices keep the two opposite ten-dimensional chiralities of the Ramond ground state (Section 3.3). This sign is the one piece of freedom we exploit below.

The content (3.62), a transverse vector plus a chiral spinor, is precisely the on-shell field content of ten-dimensional super-Yang-Mills: a gauge boson A_i and a gaugino. This is the open-string face of the same GSO projection, which is why the massless closed superstring, built as a product of two such halves, lands on a supergravity multiplet with twice the supersymmetry.

3.4.2 Four sectors from left × right

Each side carries an NS and an R sector, so the closed string has four sectors, labelled by (left, right):

$$\text{NS-NS}, \qquad \text{R-R}, \qquad \text{NS-R}, \qquad \text{R-NS}. \tag{3.63}$$

An NS ground state is a worldsheet (and spacetime) boson; an R ground state is a spacetime fermion, because its zero modes furnish a Clifford algebra whose irreducible representation is a spinor (Section 3.2). A spacetime boson tensored with a spacetime boson is a boson, boson times fermion is a fermion. So NS-NS and R-R are spacetime *bosons*, while NS-R and R-NS are spacetime *fermions*. The massless spectrum is the product of two copies of (3.62), a $16 \times 16 = 256$-dimensional space, split as 128 bosonic and 128 fermionic on-shell degrees of freedom. Their equality is the massless shadow of spacetime supersymmetry, and it is forced by the same modular invariance that fixed GSO in the first place (Section 3.3, with the partition-function argument in Appendix F).

We now work out the four products in turn. The SO(8) tensor products we need are standard triality identities (collected in Appendix G); the ones that appear are

$$\mathbf{8}_v \otimes \mathbf{8}_v = \mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35}_v, \tag{3.64}$$

$$\mathbf{8}_v \otimes \mathbf{8}_s = \mathbf{8}_c \oplus \mathbf{56}_s, \qquad \mathbf{8}_v \otimes \mathbf{8}_c = \mathbf{8}_s \oplus \mathbf{56}_c, \tag{3.65}$$

$$\mathbf{8}_s \otimes \mathbf{8}_s = \mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35}_+, \tag{3.66}$$

$$\mathbf{8}_s \otimes \mathbf{8}_c = \mathbf{8}_v \oplus \mathbf{56}_v. \tag{3.67}$$

Each right-hand side sums to 64, as it must. The subscripts on the products record which of the two **35**'s (self-dual or anti-self-dual four-form) and which of the vector, s- or c-type **56**'s appear.

Example 3.12 (The vector square by hand)**.** The one product on this list that needs no triality is $\mathbf{8}_v \otimes \mathbf{8}_v$. A two-index tensor t_{ij} on the eight transverse directions decomposes into trace,

antisymmetric and symmetric-traceless parts,

$$t_{ij} \;=\; \underbrace{\tfrac{1}{8}\,\delta_{ij}\,t_{kk}}_{\mathbf{1}} \;+\; \underbrace{t_{[ij]}}_{\mathbf{28}} \;+\; \underbrace{\Big(t_{(ij)} - \tfrac{1}{8}\,\delta_{ij}\,t_{kk}\Big)}_{\mathbf{35}_v},$$

counting $1 + \binom{8}{2} + \left(\frac{8\cdot 9}{2} - 1\right) = 1 + 28 + 35 = 64$, which is (3.64). The spinor products cannot be reduced by symmetrisation alone, because they map between inequivalent Spin(8) representations; they are fixed by triality, worked out in Appendices B and G.

3.4.3 The universal NS-NS sector

The NS-NS states are $\psi^i_{-1/2}|0\rangle_{\mathrm{L}} \otimes \tilde{\psi}^j_{-1/2}|0\rangle_{\mathrm{R}}$, an $\mathbf{8}_v \otimes \mathbf{8}_v$. Decomposing a two-index object into its symmetric-traceless, antisymmetric and trace parts is exactly (3.64):

$$\underbrace{\mathbf{35}_v}_{g_{\mu\nu}} \;\oplus\; \underbrace{\mathbf{28}}_{B_{\mu\nu}} \;\oplus\; \underbrace{\mathbf{1}}_{\phi}. \tag{3.68}$$

The graviton $g_{\mu\nu}$ (a transverse-traceless symmetric tensor, 35 physical polarisations), the antisymmetric Kalb-Ramond field $B_{\mu\nu}$ (28), and the dilaton ϕ (1). This is the same NS-NS triplet the bosonic string produced in Chapter 1, and it appears identically in every closed superstring theory: IIA, IIB, Type I and both heterotic strings. The relative GSO sign we are about to vary lives entirely in the R sectors, so it never touches (3.68), and the metric, the B-field and the dilaton survive every choice we make below. That shared NS-NS core is why the low-energy actions of all five theories are built on one common gravity-dilaton-B kinetic term (Chapter 5).

3.4.4 Gravitini and dilatini: NS-R and R-NS

The mixed sectors carry a vector index from the NS side and a spinor index from the R side. Take NS-R: the state is $\psi^i_{-1/2}|0\rangle_{\mathrm{L}} \otimes |0\rangle_{\mathrm{R}}$, so it transforms as $\mathbf{8}_v$ times whichever spinor the right R sector supplies. If that spinor is $\mathbf{8}_s$, then (3.65) gives

$$\mathbf{8}_v \otimes \mathbf{8}_s = \underbrace{\mathbf{56}_s}_{\psi_\mu} \;\oplus\; \underbrace{\mathbf{8}_c}_{\lambda}. \tag{3.69}$$

The irreducible **56** is a vector-spinor with its gamma-trace removed, $\Gamma^\mu\psi_\mu = 0$: a massless *gravitino*, the gauge field of local supersymmetry and the superpartner of the graviton. The leftover **8** is the gamma-trace itself, a spin-$\frac{1}{2}$ *dilatino* λ, superpartner of the dilaton. R-NS gives a second gravitino and a second dilatino by the mirror computation, with the roles of the two sides swapped. Two gravitini means two independent local supersymmetries, so the closed superstring has $\mathcal{N} = 2$ supersymmetry in ten dimensions: thirty-two real supercharges, the maximal amount a consistent theory of gravity can carry (Appendix C). Each supercharge is a Majorana-Weyl spinor of SO(1, 9) with sixteen real components (Appendix B), and the two of them descend directly from the two R-sector ground states, one per side.

Throughout the Type II discussion $\mathcal{N} = (p, q)$ counts these spacetime supercharges by their ten-dimensional chirality, p of one handedness and q of the other. This is a different statement from the worldsheet $\mathcal{N} = (1, 1)$ supersymmetry of Section 3.1, which is the same for IIA and IIB; here the (p, q) split is exactly what the R-sector sign is about to decide.

The *chiralities* of those two supercharges are what we have not yet fixed, and they are the whole story.

3.4.5 The one choice: relative GSO chirality

In the R sector the GSO projector can be taken with either sign, keeping either $\mathbf{8}_s$ or $\mathbf{8}_c$ as the surviving ground state. On a single side this is a convention: $\mathbf{8}_s$ and $\mathbf{8}_c$ are exchanged by a spacetime parity (reflection of one transverse axis), so relabelling one for the other is just a choice of orientation. For the closed string there are two R sectors, one per side, and only the *relative* sign is physical. Two possibilities survive:

- **Opposite chiralities.** Left R gives $\mathbf{8}_s$, right R gives $\mathbf{8}_c$ (or vice versa). This is *Type IIA*. The two gravitini have opposite ten-dimensional chirality, the theory is non-chiral, and the supersymmetry is $\mathcal{N} = (1, 1)$: one left-handed and one right-handed supercharge.
- **Same chirality.** Both R sectors give $\mathbf{8}_s$. This is *Type IIB*. The two gravitini have the same chirality, the theory is chiral, and the supersymmetry is $\mathcal{N} = (2, 0)$: two supercharges of the same handedness.

That the same-chirality choice is a physically distinct theory, not a relabelling of the opposite-chirality one, is the content of $\mathbf{8}_s \neq \mathbf{8}_c$ as representations of SO(8): no field redefinition on the closed string maps IIB to IIA, because IIB has no parity symmetry to undo the alignment of the two gravitini. IIA does have one, since exchanging left and right movers acts as a spacetime parity, and that is why IIA is non-chiral. Aligned or anti-aligned R-sector chirality is the one bit of data that separates IIA from IIB.

Careful

The overall chirality convention (calling the aligned IIB spinors $\mathbf{8}_s$ rather than $\mathbf{8}_c$) is pure bookkeeping and flips under a spacetime reflection. What is invariant, and what defines the theory, is whether the two supercharges are aligned (IIB) or anti-aligned (IIA). Do not read physical chirality off the label of a single spinor; read it off the relative sign.

3.4.6 The R-R sector: p-form potentials from a bispinor

The R-R states carry a spinor index from each side, so the polarisation is a *bispinor* F_{ab} (both indices $\mathbf{8}_s$ in IIB) or $F_{a\dot{b}}$ (one $\mathbf{8}_s$, one $\mathbf{8}_c$ in IIA). A bispinor is expanded on the complete basis of antisymmetrised SO(8) gamma matrices,

$$F \;=\; \sum_p \frac{1}{p!}\, F_{i_1\cdots i_p}\, \gamma^{i_1\cdots i_p}, \qquad \gamma^{i_1\cdots i_p} \equiv \gamma^{[i_1}\cdots\gamma^{i_p]}, \tag{3.70}$$

and the coefficient $F_{i_1\cdots i_p}$ is an antisymmetric p-index tensor, that is, a p-form. Which ranks p appear is dictated by chirality. The product $\gamma^{i_1\cdots i_p}$ maps $\mathbf{8}_s \to \mathbf{8}_s$ for even p and $\mathbf{8}_s \to \mathbf{8}_c$ for odd p. So a same-chirality bispinor F_{ab} selects even forms and an opposite-chirality bispinor $F_{a\dot{b}}$ selects odd forms.

For Type IIB (same chirality) the surviving forms are $p = 0, 2, 4$, and the middle one obeys a self-duality condition inherited from $\gamma^{i_1 i_2 i_3 i_4}$ acting inside $\mathbf{8}_s$, namely $\gamma^{i_1\cdots i_4} = +\frac{1}{4!}\epsilon^{i_1\cdots i_8}\gamma_{i_5\cdots i_8}$. The result is $\mathbf{8}_s \otimes \mathbf{8}_s = \mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35}_+$, matching (3.66). These are the physical polarisations of the R-R potentials

$$\boxed{\text{IIB}: \quad C_0,\;\; C_2,\;\; C_4^+} \qquad \Longleftrightarrow \qquad \mathbf{1},\; \mathbf{28},\; \mathbf{35}_+, \tag{3.71}$$

a scalar (the R-R axion), a two-form, and a four-form whose field strength $F_5 = \mathrm{d}C_4$ is self-dual (hence the 35, exactly half of $\binom{8}{4} = 70$).

For Type IIA (opposite chirality) the surviving forms are $p = 1, 3$, giving $\mathbf{8}_s \otimes \mathbf{8}_c = \mathbf{8}_v \oplus \mathbf{56}_v$ (3.67), the potentials

$$\boxed{\text{IIA}: \quad C_1,\;\; C_3} \qquad \Longleftrightarrow \qquad \mathbf{8}_v,\; \mathbf{56}_v, \tag{3.72}$$

Sector	SO(8) content	Fields	d.o.f.
NS-NS	$\mathbf{35}_v \oplus \mathbf{28} \oplus \mathbf{1}$	$g_{\mu\nu},\ B_{\mu\nu},\ \phi$	64
R-R	$\mathbf{8}_v \oplus \mathbf{56}_v$	$C_1,\ C_3$	64
NS-R	$\mathbf{56}_c \oplus \mathbf{8}_s$	$\psi_\mu^{(-)},\ \lambda^{(-)}$	64
R-NS	$\mathbf{56}_s \oplus \mathbf{8}_c$	$\psi_\mu^{(+)},\ \lambda^{(+)}$	64
Total		$128_B + 128_F,\ \mathcal{N} = (1,1)$	

Table 3.2: Massless spectrum of Type IIA. The two gravitini $\psi_\mu^{(\pm)}$ have opposite ten-dimensional chirality, so the theory is non-chiral. Exchanging left and right movers is a symmetry acting as spacetime parity.

Sector	SO(8) content	Fields	d.o.f.
NS-NS	$\mathbf{35}_v \oplus \mathbf{28} \oplus \mathbf{1}$	$g_{\mu\nu},\ B_{\mu\nu},\ \phi$	64
R-R	$\mathbf{35}_+ \oplus \mathbf{28} \oplus \mathbf{1}$	$C_4^+,\ C_2,\ C_0$	64
NS-R	$\mathbf{56}_s \oplus \mathbf{8}_c$	$\psi_\mu^{(1)},\ \lambda^{(1)}$	64
R-NS	$\mathbf{56}_s \oplus \mathbf{8}_c$	$\psi_\mu^{(2)},\ \lambda^{(2)}$	64
Total		$128_B + 128_F,\ \mathcal{N} = (2,0)$	

Table 3.3: Massless spectrum of Type IIB. The two gravitini $\psi_\mu^{(1,2)}$ have the same chirality, so the theory is chiral. There is no perturbative parity symmetry.

a one-form and a three-form. The field strengths are $F_{p+1} = \mathrm{d}C_p$, so IIB carries odd field strengths F_1, F_3, F_5 and IIA even field strengths F_2, F_4; the democratic completion of Appendix A adds the Hodge duals (C_5, C_7 in IIA and C_6, C_8 in IIB), with self-duality imposed at the level of the equations of motion so nothing is double counted.

Two features of this sector deserve to be stated plainly. First, no perturbative string state is charged under a R-R potential. The vertex operator built from (3.70) couples to the field strength, not to C_p directly, so a fundamental string is R-R neutral. The objects that do carry R-R charge are the D-branes, and a Dp-brane is the electric source of C_{p+1}: the even/odd split of (3.71) and (3.72) is exactly why IIA has even-dimensional D-branes (D0, D2, D4, D6, D8) and IIB odd-dimensional ones (D1, D3, D5, D7, and the instantonic D(-1)). We develop this in Chapter 6. Second, the self-dual four-form of IIB has no covariant Lagrangian with a standard kinetic term, which is one of the technical headaches of writing down IIB supergravity and one reason the effective action of Chapter 5 treats it with the democratic formalism.

Remark 3.13 (Beyond the lectures: field strengths versus potentials). Whether one reads the bispinor (3.70) as the potential C_p or the field strength F_{p+1} is a matter of which physical object one identifies with the on-shell polarisation. A transverse, on-shell $(p+1)$-form field strength carries the same $\binom{8}{p}$ physical components as its p-form potential and sits in the same SO(8) representation, so the counting is identical either way. The vertex-operator computation naturally produces the gauge-invariant field strength; we quote potentials here because those are the fields that appear in the spacetime action and that D-branes couple to.

3.4.7 The two spectra

Collecting the four sectors gives the full massless content. Tables 3.2 and 3.3 list them side by side. In both theories the bosons total 128 and the fermions total 128, as supersymmetry requires; the difference is confined to the R-R and mixed sectors, where the chirality choice acts.

These are the field contents of the two maximal ten-dimensional supergravities, $\mathcal{N} = (1,1)$ (IIA) and $\mathcal{N} = (2,0)$ (IIB). Their actions, symmetries and equations of motion are the subject of Chapter 5; here we have derived the multiplets from the worldsheet, which is all the massless data those actions encode.

3.4.8 Chirality, anomalies, and where the two theories go next

Type IIB is chiral, and a chiral theory of gravity in ten dimensions is not obviously consistent: chiral gravitini, chiral dilatini and a self-dual tensor each contribute to the gravitational anomaly. That these contributions cancel among the IIB fields is a genuine feature of the spectrum, first checked by Alvarez-Gaumé and Witten. The same anomaly-cancellation logic, extended to the gauge sector, is what pins down the two heterotic gauge groups in Section 3.5; the mechanism and the arithmetic are in Appendix H and Chapter 5. Type IIA is non-chiral and free of gravitational anomalies for the trivial reason that parity is a symmetry.

The non-chirality of IIA has a deeper origin, which is the best single piece of evidence that string theory is not the whole story. The IIA massless spectrum is exactly what one gets by reducing eleven-dimensional supergravity on a circle: the eleven-dimensional metric gives $g_{\mu\nu}$, the dilaton and the Kaluza-Klein photon C_1; the eleven-dimensional three-form gives C_3 and B_2. The R-R one-form is the KK photon of a hidden eleventh dimension whose radius grows with the string coupling. This is the doorway to M-theory, and we walk through it in Chapter 7. IIB, by contrast, has no such eleven-dimensional lift; instead it has a non-perturbative $\mathrm{SL}(2,\mathbb{Z})$ self-duality, described next.

Remark 3.14 (Beyond the lectures: $\mathrm{SL}(2,\mathbb{Z})$ of IIB). Type IIB has a hidden $\mathrm{SL}(2,\mathbb{Z})$ symmetry acting on its scalars and two-forms. The dilaton and the R-R axion combine into the complex axio-dilaton

$$\tau \;=\; C_0 + i\, e^{-\phi}, \tag{3.73}$$

on which $\mathrm{SL}(2,\mathbb{Z})$ acts by fractional linear transformations $\tau \mapsto (a\tau + b)/(c\tau + d)$, while the NS-NS and R-R two-forms (B_2, C_2) transform as a doublet and C_4 is invariant. The S-generator $\tau \mapsto -1/\tau$ sends $g_s \to 1/g_s$ at $C_0 = 0$: it is a strong-weak duality that maps the fundamental string to the D1-brane. This symmetry is the seed of F-theory, where τ is geometrised as the complex structure of an auxiliary two-torus fibred over spacetime (Chapter 11). None of it is visible in perturbation theory; it is a statement about the full non-perturbative IIB.

The two Type II theories exhaust the closed oriented superstrings. Type IIB has a further symmetry the others lack: it is invariant under the worldsheet parity Ω that exchanges left and right movers, precisely because its two sides carry the same chirality. Gauging Ω projects IIB to an unoriented theory, and the resulting inconsistencies (tadpoles) are cancelled only by adding an open-string sector whose Chan-Paton factors are pinned to gauge group SO(32). That construction yields Type I, and it is the subject of Section 3.5, alongside the two heterotic strings. The complete catalogue of all five is assembled in Section 3.6.

Exercise 3.13 (Counting the Type II multiplet)**.** *(Standard, ∼20 min.)* Verify that each of the four sectors contributes 64 on-shell degrees of freedom, using $\binom{8}{p}$ for a transverse p-form and $\frac{(D-2)(D-1)}{2} - 1$ for the graviton with $D = 10$. Confirm that the self-dual four-form contributes $\frac{1}{2}\binom{8}{4} = 35$ and the gravitino $(D-3) \times$ (spinor dimension) $= 7 \times 8 = 56$. Conclude that both theories have 128 bosonic and 128 fermionic massless states, and identify this 256 with the tensor square $16 \otimes 16$ of (3.62).

Exercise 3.14 (IIA from eleven dimensions)**.** *(Moderate, ∼30 min.)* The bosonic fields of eleven-dimensional supergravity are a metric g_{MN} and a three-form A_3. Reduce on a circle, splitting $M = (\mu, 11)$, and show that the massless bosonic content organises into $g_{\mu\nu}$, a vector

$g_{\mu\,11} = C_1$, a scalar $g_{11\,11} \sim e^{\phi}$, and from A_3 a three-form C_3 and a two-form $A_{\mu\nu\,11} = B_2$. Match this to Table 3.2 and count that the bosonic degrees of freedom agree ($44 + 84$ in eleven dimensions against $64 + 64$ in ten). Which IIA field is the KK photon whose charge is carried by the D0-brane?

Exercise 3.15 (Why IIB has no parity)**.** *(Moderate, ∼25 min.)* A ten-dimensional parity reverses one spatial coordinate and exchanges $\mathbf{8}_s \leftrightarrow \mathbf{8}_c$. Show that, acting on the IIA spectrum of Table 3.2, this maps the theory to itself (it exchanges the two gravitini and sends C_1, C_3 to themselves up to sign). Show that on the IIB spectrum of Table 3.3 the same operation would flip the common chirality of both gravitini and reverse the duality of C_4^+, taking IIB to a would-be "$\overline{\text{IIB}}$" that is the same theory only after also reversing orientation. Explain in one sentence why this makes IIB chiral and IIA non-chiral.

3.5 Type I and the heterotic strings

The big picture

Section 3.4 built the two Type II theories from a closed string whose left and right movers are *both* superstrings, with the relative GSO chirality as the only free choice. Three consistent theories remain, and each one relaxes one of those assumptions. Keep the closed string but gauge worldsheet parity: the oriented Type IIB string becomes the unoriented Type I string, and the charge this leaves behind must be soaked up by open strings, which arrive carrying an SO(32) gauge symmetry. Or give up the idea that the two sides are the same kind of string. Pair a right-moving superstring with a left-moving bosonic string. The two want to live in 10 and 26 dimensions; sharing ten spacetime coordinates leaves 16 purely left-moving internal degrees of freedom, and modular invariance allows their momenta to fill exactly two lattices. Those two lattices deliver the gauge groups $\mathrm{E}_8 \times \mathrm{E}_8$ and $\mathrm{Spin}(32)/\mathbb{Z}_2$.

Both constructions are projections or amputations of machinery you already have. Nothing new needs to be quantised. Type I reuses the Type IIB spectrum of Section 3.4 and throws away half of it; the heterotic string reuses the right-moving superstring of Sections 3.2 and 3.3 and the left-moving bosonic string of Chapter 1, and bolts them together. What is genuinely new is the arithmetic of even self-dual lattices, and that we quote.

3.5.1 Type I: gauging worldsheet parity

The Type II strings are theories of *oriented* closed strings: the worldsheet carries an arrow, and reversing it gives a physically different configuration. There is an obvious $\mathbb{Z}_2$ symmetry that reverses the arrow.

Definition 3.15 (Worldsheet parity and the orientifold)**.** Worldsheet parity Ω is the operation

$$\Omega : \quad \sigma \longmapsto 2\pi - \sigma \tag{3.74}$$

on the closed string, which reverses the orientation of the string and hence exchanges left- and right-moving oscillators, $\alpha_n^\mu \leftrightarrow \tilde{\alpha}_n^\mu$ and $\psi_r^\mu \leftrightarrow \tilde{\psi}_r^\mu$ (up to a fixed sign in each sector). It satisfies $\Omega^2 = 1$. *Gauging* Ω, that is keeping only Ω-even states and summing over all worldsheet topologies including the unoriented ones, is called an *orientifold* projection.

A discrete symmetry can be gauged only if it is actually a symmetry, and this is the point that selects Type IIB rather than Type IIA. Under Ω the two sides of the string are swapped, so the theory is invariant only if its left and right sectors are built identically. In Section 3.4 the two supercharges of Type IIB were found to have the *same* spacetime chirality, precisely because its two R sectors carry the same GSO projection: Type IIB is left-right symmetric, so Ω is a

symmetry of it. Type IIA has opposite chiralities on the two sides, so a bare Ω is not a symmetry there and the orientifold must be built on Type IIB.

Now apply the projection to the massless closed spectrum of Section 3.4. In the NS-NS sector the graviton and B-field sit in

$$\psi^{\mu}_{-1/2}\,\tilde{\psi}^{\nu}_{-1/2}|0\rangle_{\rm NS} \;\sim\; \psi^{(\mu}_{-1/2}\tilde{\psi}^{\nu)}_{-1/2}|0\rangle \;\oplus\; \psi^{[\mu}_{-1/2}\tilde{\psi}^{\nu]}_{-1/2}|0\rangle, \tag{3.75}$$

the symmetric traceless part being the graviton $g_{\mu\nu}$, the trace the dilaton ϕ, and the antisymmetric part the two-form $B_{\mu\nu}$. Because Ω swaps the two oscillators, it acts as $+1$ on the symmetric combination and -1 on the antisymmetric one. The graviton and dilaton survive while $B_{\mu\nu}$ is projected out, so perturbative Type I has no B-field.

The R-R sector of Type IIB held the forms C_0, C_2 and the self-dual C_4^+. Under Ω these have definite parities, and the survivor is the two-form:

$$\text{Type I closed, bosonic:} \qquad g_{\mu\nu}, \quad \phi, \quad C_2. \tag{3.76}$$

The NS-R and R-NS gravitini of Type IIB are exchanged by Ω, and only their sum survives, leaving one Majorana-Weyl gravitino and one dilatino. Ten-dimensional $\mathcal{N}=(2,0)$ supersymmetry, thirty-two supercharges, is cut in half to $\mathcal{N}=(1,0)$, sixteen supercharges. The fields in (3.76) together with the surviving fermions are exactly the ten-dimensional $\mathcal{N}=1$ supergravity multiplet.

If that were the whole story, Type I would be an unoriented closed string and nothing more. It is not consistent on its own. Gauging Ω forces unoriented worldsheets into the sum, and the leading unoriented surfaces (the Klein bottle at one loop, the crosscap $\mathbb{RP}^2$ at tree level) produce a divergence: a non-vanishing tadpole for the unphysical ten-form R-R potential C_{10}. In brane language, gauging Ω fills spacetime with an orientifold nine-plane, an O9-plane, carrying -32 units of R-R nine-brane charge. Gauss's law on the compact transverse space (there is none here, so the total charge must simply vanish) is violated unless something supplies $+32$ units, and that something is a stack of 32 space-filling D9-branes.

The open strings stretched on those D9-branes are what bring in the gauge symmetry. Their endpoints carry Chan-Paton labels $i=1,\dots,32$, and the Ω projection descends to the Chan-Paton matrices, keeping the antisymmetric ones. Thirty-two labels with an antisymmetric reality condition give the adjoint of SO(32):

$$\text{Type I open, massless:} \qquad A_{\mu}^{[ij]}\ \text{(NS)} \;\oplus\; \lambda^{[ij]}\ \text{(R)}, \qquad [ij] \in \text{ adjoint of SO(32)}, \tag{3.77}$$

a ten-dimensional $\mathcal{N}=1$ vector multiplet, gauge boson plus gaugino, in the 496-dimensional adjoint. The sign of the orientifold charge is what picks SO(32) rather than the other option USp(32); the tadpole computation that fixes both the number 32 and the group is done in Chapter 6, and we quote its result here.

Two different things called Chan-Paton

The SO(32) here is a gauge symmetry carried by open-string endpoints on a stack of D9-branes, and its rank is fixed by a cancellation, not chosen. Contrast the U(N) of a stack of N D-branes met in passing in Chapter 1, where N is free. In Type I the number is pinned at 32 by consistency of the vacuum, which is why SO(32) is not one gauge group among many but *the* gauge group of the theory.

Type I is therefore a theory of unoriented open and closed strings: $\mathcal{N}=1$ supergravity from the closed sector, coupled to SO(32) super-Yang-Mills from the open sector. Gravity comes from the sphere and the gauge fields from the disk, so the two sectors sit at different powers of $g_s = e^{\langle\phi\rangle}$. That mismatch is a fingerprint that reappears in Chapter 7: Type I is the strong-coupling limit of the SO(32) heterotic string, the two related by $g_s \to 1/g_s$.

3.5.2 The heterotic idea: two strings in one

On a closed string the left- and right-moving sectors are decoupled. They share only the zero mode, the centre-of-mass position and momentum, tied together by level matching. Everything else, the oscillators and their vacua, factorises. The Type II construction made both sides superstrings out of habit, not necessity. The heterotic string (from *heterosis*, hybrid vigour) takes the two sides to be different theories:

- the **right-movers** are a full RNS superstring, ten bosons X^μ and ten worldsheet fermions ψ^μ, with NS and R sectors and the GSO projection of Section 3.3;
- the **left-movers** are a plain bosonic string, of the kind quantised in Chapter 1.

The only constraint linking the two is that they share the same ten spacetime coordinates $X^\mu(z,\bar{z})$, so the theory has a single ten-dimensional Poincaré symmetry.

The tension in this marriage is arithmetic. The right-moving superconformal theory is critical at $c_R = 15$, which as in Section 3.1 means ten dimensions. The left-moving bosonic theory is critical at $c_L = 26$. Counting the shared spacetime bosons on each side,

$$c_R \;=\; \underbrace{10}_{X^\mu} \;+\; \underbrace{10\times\tfrac{1}{2}}_{\psi^\mu} \;=\; 15, \qquad c_L \;=\; \underbrace{10}_{X^\mu} \;+\; \underbrace{16}_{X^I_{\mathrm{L}}} \;=\; 26. \tag{3.78}$$

The right-movers close their books with the ten fermions, worth $c = 5$. The left-movers are 16 short. Those sixteen units of central charge must come from sixteen extra left-moving bosons $X^I_{\mathrm{L}}(z)$, $I = 1,\dots,16$, that carry no spacetime index. They are purely internal and purely left-moving (chiral), and the whole content of the heterotic construction is what they are allowed to be.

Remark 3.16 (Beyond the lectures: supersymmetry is not manifest here). The right-moving superstring gives this theory ten-dimensional spacetime supersymmetry after the GSO projection, but, exactly as in Section 3.3, that supersymmetry is invisible term by term in the RNS variables and appears only once the projection is imposed. Formulations that make the ten-dimensional supersymmetry manifest from the outset exist, the Green-Schwarz and the pure-spinor (Berkovits) formalisms, and we do not use them.

3.5.3 The internal lattice and modular invariance

Sixteen purely left-moving bosons cannot roam a non-compact $\mathbb{R}^{16}$: a chiral boson with continuous momentum is inconsistent. They must be compactified, so that X^I_{L} lives on a torus $\mathbb{R}^{16}/(2\pi\Lambda)$ and the internal momentum $p = (p^I)$ takes values in a sixteen-dimensional lattice Λ. Which lattice is not free. It is fixed almost completely by demanding that the one-loop vacuum amplitude, the torus partition function, be modular invariant. That is the same consistency requirement that pinned the GSO projection in Section 3.3, now applied to the internal sector; the modular machinery lives in Appendix F.

The internal momenta enter the partition function only through the lattice sum

$$\Theta_\Lambda(\tau) \;=\; \sum_{p\in\Lambda} q^{\,p^2/2}, \qquad q = e^{2\pi i\tau}, \tag{3.79}$$

divided by powers of the Dedekind η from the oscillators. Modular invariance rests on the two generators of the modular group:

- Under $T:\tau\to\tau+1$, each term picks up $e^{i\pi p^2}$. Together with the oscillator and ghost contributions, invariance forces $p^2 \in 2\mathbb{Z}$ for every $p\in\Lambda$. The lattice must be **even**. Physically this is level matching: the left-moving internal contribution to $L_0 - \tilde{L}_0$ must be an integer.

- Under $S : \tau \to -1/\tau$, Poisson resummation turns Θ_Λ into Θ_{Λ^*}, the sum over the *dual* lattice Λ^*, times a volume factor. Invariance forces $\Lambda^* = \Lambda$. The lattice must be **self-dual** (unimodular).

So the sixteen internal momenta fill an even self-dual Euclidean lattice of rank 16.

Here we cash in the arithmetic quoted from Appendix G. Even self-dual Euclidean lattices exist only when the rank is a multiple of 8. In rank 8 there is exactly one, the root lattice Γ_8 of E_8. In rank 16 there are exactly *two*:

$$\Gamma_8 \oplus \Gamma_8 \qquad \text{and} \qquad \Gamma_{16} \equiv D_{16}^{+}. \tag{3.80}$$

The first is two copies of the E_8 lattice. The second is the D_{16} root lattice (integer vectors with even coordinate sum) enlarged by one spinor conjugacy class, the vectors $(\pm\frac{1}{2})^{16}$ with an even number of minus signs. There are no others, so the internal sector allows exactly two heterotic strings. (In rank 24 the count jumps to the twenty-four Niemeier lattices, one of which is the Leech lattice, but that is another course.)

3.5.4 The spectrum and the gauge group

Assemble the mass formula from the two sides. With the internal momentum $p \in \Lambda$ and oscillator numbers N_R, N_L,

$$\frac{\alpha'}{4} M_R^2 \;=\; N_R - a_R, \qquad \frac{\alpha'}{4} M_L^2 \;=\; N_L + \frac{p^2}{2} - 1, \tag{3.81}$$

where each side is normalised as for a single mover, $\frac{\alpha'}{4}M^2 = N - a$, and the intercept a is minus the sum of the transverse zero-point energies. On the right, $a_R = \frac{1}{2}$ (NS) and $a_R = 0$ (R) are the superstring intercepts of Section 3.2. On the left, the $8 + 16 = 24$ transverse bosons each contribute $-\frac{1}{24}$ to that sum, so $a_L = -(24 \times (-\frac{1}{24})) = +1$, the ordinary bosonic intercept written as the -1 in (3.81). Level matching sets $M_L^2 = M_R^2$.

The right-moving massless states are the GSO survivors of Section 3.3: the vector $\psi^\mu_{-1/2}|0\rangle_{\mathrm{NS}}$ and the ground-state spinor of the R sector, together the $\mathbf{8}_v \oplus \mathbf{8}_s$ of the transverse SO(8). This is a single ten-dimensional $\mathcal{N} = 1$ vector multiplet on the right, and it multiplies everything.

The left-moving massless states solve $N_L + \frac{1}{2}p^2 = 1$:

$$\text{(a)} \quad N_L = 1,\ p = 0: \quad \alpha^\mu_{-1}|0\rangle,\ \ \alpha^I_{-1}|0\rangle; \qquad \text{(b)} \quad N_L = 0,\ p^2 = 2: \quad |p\rangle\,. \tag{3.82}$$

Case (a) with the spacetime oscillator α^μ_{-1} carries a vector index. Tensored with the right-moving $\mathbf{8}_v \oplus \mathbf{8}_s$ it builds the ten-dimensional $\mathcal{N} = 1$ supergravity multiplet: graviton $g_{\mu\nu}$, two-form $B_{\mu\nu}$, dilaton ϕ, gravitino and dilatino. That much is the same for both lattices and is the gravitational backbone of the theory.

The rest of case (a), the sixteen internal oscillators $\alpha^I_{-1}|0\rangle$, and all of case (b), the momentum states $|p\rangle$ with $p^2 = 2$, carry an *internal* index and no spacetime index. Tensored with the right-moving vector multiplet they are spacetime gauge bosons and their gauginos. The sixteen $\alpha^I_{-1}|0\rangle$ are the Cartan generators, a $\mathrm{U}(1)^{16}$. Each $|p\rangle$ with $p^2 = 2$ is a charged gauge boson, a root. The gauge algebra is a level-one current algebra (see Appendix G) whose roots are exactly the norm-two vectors of Λ. Counting them,

$$\dim G \;=\; \underbrace{16}_{\text{Cartan}} + \#\{p \in \Lambda : p^2 = 2\} \;=\; 16 + 480 \;=\; 496, \tag{3.83}$$

for both lattices. For $\Gamma_8 \oplus \Gamma_8$ the 480 norm-two vectors are the 240 roots of each E_8, giving $G = \mathrm{E}_8 \times \mathrm{E}_8$. For D_{16}^{+} they are the 480 roots of D_{16}, giving the algebra $\mathfrak{so}(32)$; the added

spinor class contributes vectors of norm 4, which are massive, so it changes the *global* form of the group without adding massless gauge bosons. The gauge group is $\mathrm{Spin}(32)/\mathbb{Z}_2$, not $\mathrm{SO}(32)$: the charge lattice D_{16}^{+} carries the adjoint (norm-two, massless) and a spinor conjugacy class (norm-four, hence appearing only at massive levels) but no vector of $\mathrm{SO}(32)$, and it is the absence of the vector representation that fixes the global form as $\mathrm{Spin}(32)/\mathbb{Z}_2$. We leave the global-form bookkeeping to Appendix G.

Example 3.17 (The 240 roots of E_8)**.** The norm-two vectors of the E_8 lattice Γ_8 come in two families. The integer vectors of its D_8 sublattice with $p^2 = 2$ are $(\pm 1, \pm 1, 0, \ldots, 0)$ up to permutation: choosing the two nonzero slots and their two signs gives $\binom{8}{2} \times 2^2 = 112$. The spinor class adds the half-integer vectors $(\pm\frac{1}{2})^8$ with an even number of minus signs, each of norm $8 \times \frac{1}{4} = 2$; the even-sign condition leaves $2^8/2 = 128$ of them. Together $112 + 128 = 240$, the roots of E_8. Two copies give 480 norm-two vectors, so with the sixteen Cartan directions $\dim(\mathrm{E}_8 \times \mathrm{E}_8) = 480 + 16 = 496$. The parallel count for D_{16}^{+}, left to Exercise 3.18, reaches the same 496 from the 480 roots of D_{16} alone, the half-integer spinor class this time having norm $16 \times \frac{1}{4} = 4$ and so contributing nothing massless.

Both groups have dimension 496 and rank 16. The full massless content of either heterotic string is then

$$\big[\,\mathcal{N} = 1 \text{ supergravity}\,\big] \;\oplus\; \big[\,\mathcal{N} = 1 \text{ super-Yang-Mills, gauge group } G\,\big], \tag{3.84}$$

with $G = \mathrm{E}_8 \times \mathrm{E}_8$ or $\mathrm{Spin}(32)/\mathbb{Z}_2$, and sixteen supercharges. The gauge fields here descend from the *closed* string, so gauge and gravitational couplings share the same power of g_s, in contrast to Type I. This is why the two $\mathrm{SO}(32)$-type theories, heterotic and Type I, look so different perturbatively yet turn out to be the same theory at strong and weak coupling.

Remark 3.18 (Beyond the lectures: the fermionic construction). Sixteen chiral bosons on Λ are one presentation. An equivalent one replaces them, through two-dimensional boson-fermion equivalence, by 32 left-moving Majorana-Weyl worldsheet fermions λ^A, $A = 1, \ldots, 32$, with $32 \times \frac{1}{2} = 16$ units of central charge. Modular invariance now becomes a choice of boundary conditions: making all 32 fermions share one condition (one spin structure) realises $\mathrm{Spin}(32)/\mathbb{Z}_2$, while splitting them $16 + 16$ into two independently-projected groups realises $\mathrm{SO}(16) \times \mathrm{SO}(16)$, which enhances to $\mathrm{E}_8 \times \mathrm{E}_8$ once the spinor conjugacy classes are added. The two consistent projections reproduce the two lattices of (3.80), from the other side of the bosonisation.

3.5.5 Why exactly these two gauge groups

We have now met 496 three times: as $\dim \mathrm{SO}(32)$ in Type I, and as $\dim(\mathrm{E}_8 \times \mathrm{E}_8) = \dim \mathrm{Spin}(32)/\mathbb{Z}_2$ in the heterotic strings. The same number keeps surfacing for a spacetime reason that the worldsheet cannot see directly.

A ten-dimensional $\mathcal{N} = 1$ theory with a chiral gravitino, chiral gauginos and chiral fermions is generically anomalous. The hexagon diagram produces a twelve-form anomaly polynomial I_{12} that, for a generic gauge group, cannot be cancelled by any local counterterm. Green and Schwarz found that for special groups I_{12} *factorises*,

$$I_{12} \;=\; X_4 \wedge X_8, \tag{3.85}$$

and the residual anomaly is then cancelled by a one-loop counterterm involving the two-form (B in the heterotic case, C_2 in Type I) with a modified Bianchi identity. The factorisation (3.85) demands both a group of dimension 496 and a set of trace identities on the generators. Exactly four gauge algebras pass: $\mathfrak{so}(32)$, $\mathfrak{e}_8 \oplus \mathfrak{e}_8$, $\mathfrak{e}_8 \oplus \mathrm{U}(1)^{248}$ and $\mathrm{U}(1)^{496}$. Only the first two are non-abelian and admit a weakly-coupled string realisation, and they are precisely the two groups the lattices delivered. The anomaly polynomial, its factorisation and the counterterm are worked out in Appendix H and used in Chapter 5; we forward-reference them here as the deeper reason

the list is so short. (This is the same Green and Schwarz as the alternative worldsheet formalism named above, a different piece of their work; the community calls this one the Green-Schwarz mechanism.)

The convergence deserves to be stated plainly. Modular invariance is a consistency condition on the worldsheet, on a genus-one Riemann surface. Anomaly freedom is a consistency condition in ten-dimensional spacetime, on chiral fermion loops. They know nothing of each other, and yet the worldsheet condition (even self-dual rank-sixteen lattices) and the spacetime condition (Green-Schwarz factorisation) single out the same two non-abelian groups. When a theory is consistent, its independent consistency conditions must agree, and the heterotic string is among the sharpest examples of that in the whole subject. Historically the agreement ran the other way: the 1984 discovery that SO(32) and $\mathrm{E}_8 \times \mathrm{E}_8$ cancel anomalies came first, and the heterotic construction that realises them followed within months.

Remark 3.19 (Beyond the lectures: the two heterotic strings are one in nine dimensions)*.* In exactly ten dimensions $\mathrm{E}_8 \times \mathrm{E}_8$ and $\mathrm{Spin}(32)/\mathbb{Z}_2$ are distinct theories. Compactify either on a circle and switch on Wilson lines, and the moduli of the two coincide: both are governed by the same Narain lattice $\Gamma_{17,1}$, and one deforms continuously into the other. The two ten-dimensional strings are boundary points of a single nine-dimensional moduli space. This T-duality, and the Narain lattice behind it, are developed in Chapter 7 and Appendix G; the strong-coupling story, where $\mathrm{E}_8 \times \mathrm{E}_8$ grows an eleventh dimension into an interval (Hořava-Witten), is also in Chapter 7.

3.5.6 Where this leaves us

We now have all five. Type IIA and Type IIB from the relative GSO chirality of a doubled superstring; Type I from the Ω orientifold of Type IIB with its 32 D9-branes and SO(32); and the two heterotic strings, $\mathrm{E}_8 \times \mathrm{E}_8$ and $\mathrm{Spin}(32)/\mathbb{Z}_2$, from a bosonic left-mover on the two even self-dual rank-sixteen lattices. Each carries $\mathcal{N} = 1$ or $\mathcal{N} = 2$ supersymmetry, no tachyon, and a definite gauge and chirality structure. Section 3.6 lays the five side by side in a single table and reads off the pattern: sixteen supercharges for Type I and the heterotic strings, thirty-two for the Type II pair, and a gauge group only where open strings or an internal lattice supply one. The low-energy limits of these five spectra are the ten-dimensional supergravities assembled in Chapter 5, and the dualities hinted at above (Type I with heterotic SO(32), the two heterotic strings, Type IIA with M-theory) are the threads of Chapter 7.

Exercise 3.16 (The intercepts from central charge)**.** *(Warm-up, ∼15 min. Goal: reconcile the two zero-point energies in* (3.81)*.)* Using $\zeta(-1) = -\frac{1}{12}$ as in Chapter 1, and the rule that each transverse boson contributes $E_0 = -\frac{1}{24}$, a transverse periodic (R) fermion $E_0 = +\frac{1}{24}$ and an antiperiodic (NS) fermion $E_0 = -\frac{1}{48}$ per real component, verify that the 8 transverse X^μ plus 8 transverse ψ^μ give $a_R = \frac{1}{2}$ (NS) and $a_R = 0$ (R). Then verify that the 8 transverse X^μ plus 16 internal X^I_{L} give $a_L = +1$. Throughout, the intercept is $a = -\sum E_0$ in the common normalisation $\frac{\alpha'}{4}M^2 = N - a$. Explain in one line why the left side reproduces the bosonic value while the right side does not.

Exercise 3.17 (The orientifold projection on the NS-NS sector)**.** *(Standard, ∼25 min. Goal: see $B_{\mu\nu}$ leave and $g_{\mu\nu}, \phi$ stay.)* Take the massless NS-NS state $\psi^\mu_{-1/2}\tilde{\psi}^\nu_{-1/2}|0\rangle$ of Type IIB and let Ω act by exchanging the barred and unbarred oscillators with the sector sign appropriate to Section 3.4. Decompose into symmetric-traceless, trace and antisymmetric parts, and show that Ω acts as $+1$ on the graviton and dilaton and -1 on $B_{\mu\nu}$. Repeat for the R-R sector at the level of parities and confirm that of C_0, C_2, C_4^+ only C_2 survives, reproducing (3.76).

Exercise 3.18 (Counting the roots: 480 twice)**.** *(Standard, ∼25 min. Goal: the* 496 *of* (3.83) *from lattice arithmetic.)* (a) The D_{16} root lattice is the set of integer vectors in $\mathbb{R}^{16}$ with even

coordinate sum. Show that its vectors of norm-squared 2 are exactly $(\pm1, \pm1, 0, \ldots, 0)$ up to permutation, and count them: $\binom{16}{2} \times 2^2 = 480$. Add the sixteen Cartan generators to get $\dim \mathrm{SO}(32) = 496$. (b) Show that the extra spinor class $(\pm\frac{1}{2})^{16}$ (even number of minus signs) that builds D_{16}^+ has norm-squared 4, hence contributes no massless gauge boson. (c) Using the count of Theorem 3.17 that Γ_8 has 240 norm-two vectors, conclude that $\Gamma_8 \oplus \Gamma_8$ also gives $480 + 16 = 496$.

3.6 The five theories: a catalogue

The constructions of the previous sections all end in the same place: a modular-invariant, tachyon-free, spacetime-supersymmetric string theory in ten dimensions. There are exactly five of them. This section is the reference card. It sets the endpoints of Sections 3.4 and 3.5 side by side, reads the physics off the resulting table, and points to where each theory is taken up again as a low-energy effective action (Chapter 5) and as one corner of a single unified framework (Chapter 7). Nothing here is new. It is bookkeeping, but bookkeeping worth doing carefully, because the pattern of what agrees and what differs across the five rows is the entire content of string duality.

The five theories at a glance

Two have maximal supersymmetry (32 supercharges) and no gauge group: **Type IIA**, which is non-chiral, and **Type IIB**, which is chiral. They differ only in the relative GSO chirality of the left- and right-moving Ramond sectors. Three have half-maximal supersymmetry (16 supercharges), are chiral, and carry a rank-16 gauge group of dimension 496: **Type I** with SO(32), and the two **heterotic** strings with $\mathrm{E}_8 \times \mathrm{E}_8$ and $\mathrm{Spin}(32)/\mathbb{Z}_2$. Every one of them contains a massless graviton, and every one is free of the tachyon that plagued the bosonic string. That is the whole output of adding ψ^μ and projecting with GSO.

3.6.1 The master table

Table 3.4 is the catalogue. Each row is one consistent superstring theory; the columns record the amount of spacetime supersymmetry, whether the theory is chiral, the gauge group, and the massless bosonic and fermionic content organised by worldsheet sector. The universal graviton multiplet aside, everything in the table was derived in Section 3.4 (the Type II pair) and Section 3.5 (Type I and the heterotic strings); here it is only collected.

Three features of the table organise everything that follows.

The universal sector. Every closed string contains a massless graviton $g_{\mu\nu}$, an antisymmetric two-form, and a scalar dilaton ϕ. These come from the NS-NS sector, from the product of two vectors $\mathbf{8}_v \otimes \mathbf{8}_v = \mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35}$ under the transverse little group SO(8), split as dilaton, B-field, graviton (Eq. (3.64)). This is the same massless triple $g \oplus B \oplus \phi$ that appeared at level one of the closed bosonic string in Chapter 1; the superstring inherits it unchanged. The dilaton is not a spectator. Its vacuum value sets the string coupling $g_s = e^{\langle\phi\rangle}$ of Chapter 1, so every one of these theories has a tunable coupling rather than a dimensionless constant put in by hand. Type I is the one apparent exception in the table, where the antisymmetric two-form is Ramond-Ramond (C_2) rather than NS-NS; the reason appears below.

The Type II pair. Type IIA and Type IIB both have 32 supercharges, the maximum a consistent theory of gravity in ten dimensions can carry. They share the NS-NS sector above and share the fermionic content of two gravitini and two dilatini. The single difference is the relative chirality of the two Ramond ground states, fixed by the relative GSO projection of Section 3.3.

Theory	SUSY	Chiral?	Gauge group	NS-NS	R-R	Fermions
Type IIA	$\mathcal{N} = (1,1)$, $32\,Q$	no	none	$g_{\mu\nu}, B_{\mu\nu}, \phi$	C_1, C_3	$2\times$ gravitino (opposite chirality), $2\times$ dilatino
Type IIB	$\mathcal{N} = (2,0)$, $32\,Q$	yes	none	$g_{\mu\nu}, B_{\mu\nu}, \phi$	C_0, C_2, C_4^+	$2\times$ gravitino (same chirality), $2\times$ dilatino
Type I	$\mathcal{N} = 1$, $16\,Q$	yes	SO(32)	$g_{\mu\nu}, \phi$	C_2	gravitino, dilatino, gaugini (adjoint)
Heterotic	$\mathcal{N} = 1$, $16\,Q$	yes	$\mathrm{E}_8 \times \mathrm{E}_8$	$g_{\mu\nu}, B_{\mu\nu}, \phi$	none	gravitino, dilatino, gaugini (adjoint)
Heterotic	$\mathcal{N} = 1$, $16\,Q$	yes	$\mathrm{Spin}(32)/\mathbb{Z}_2$	$g_{\mu\nu}, B_{\mu\nu}, \phi$	none	gravitino, dilatino, gaugini (adjoint)

Table 3.4: The five consistent superstring theories in $D = 10$. Here Q counts real supercharges, and "chiral" refers to the spacetime chirality of the supercharges. C_p denotes a Ramond-Ramond p-form potential (field strength $F_{p+1} = \mathrm{d}C_p$ in the conventions of Appendix A); C_4^+ has self-dual field strength. The gaugini fill out the adjoint of the gauge group. Type I is a theory of unoriented open and closed strings; the other four are oriented closed strings.

Choosing the two Ramond sectors with *opposite* chirality gives IIA, whose two gravitini have opposite chirality: the theory is non-chiral, and its supersymmetry is labelled $\mathcal{N} = (1,1)$. Choosing them with the *same* chirality gives IIB, whose two gravitini have the same chirality: the theory is chiral, $\mathcal{N} = (2,0)$. That one sign propagates into the Ramond-Ramond spectrum, which is the product of the two Ramond ground states. Using SO(8) triality (Eq. (3.66)),

$$\text{IIA:}\quad \mathbf{8}_s \otimes \mathbf{8}_c = [1] \oplus [3] \;=\; \mathbf{8}_v \oplus \mathbf{56}_v \;\longrightarrow\; C_1,\, C_3, \tag{3.86}$$

$$\text{IIB:}\quad \mathbf{8}_s \otimes \mathbf{8}_s = [0] \oplus [2] \oplus [4]_+ \;=\; \mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35}_+ \;\longrightarrow\; C_0,\, C_2,\, C_4^+, \tag{3.87}$$

where $[n]$ is a rank-n antisymmetric tensor and the subscript on $[4]_+$ records that the corresponding five-form field strength is self-dual. IIA carries Ramond-Ramond potentials of odd rank, IIB of even rank. This even-odd split has direct physical content: a C_{p+1} potential couples electrically to a Dp-brane, so IIA supports even-dimensional and IIB odd-dimensional D-branes, a fact Chapter 6 turns into T-duality.

The half-maximal trio. Type I and the two heterotic strings each have 16 supercharges, half the maximum, and each is chiral. Their massless sector is a ten-dimensional $\mathcal{N} = 1$ supergravity multiplet (graviton, one gravitino, dilatino, dilaton, and a two-form) coupled to an $\mathcal{N} = 1$ super-Yang-Mills multiplet (gauge bosons and gaugini) in the adjoint of a gauge group. What distinguishes the three is the origin of that gauge group and its precise identity. In Type I the gauge symmetry lives on the open-string endpoints: 32 D9-branes together with the orientifold plane carry Chan-Paton labels filling out SO(32), as in Section 3.5. In the heterotic strings there are no open strings; the gauge symmetry is internal, carried by sixteen left-moving bosons compactified on an even self-dual lattice, and the two admissible lattices give $\mathrm{E}_8 \times \mathrm{E}_8$ and $\mathrm{Spin}(32)/\mathbb{Z}_2$. All three gauge groups have rank 16 and dimension 496.

Same algebra, different group

Type I and the $\mathrm{Spin}(32)/\mathbb{Z}_2$ heterotic string share the Lie algebra $\mathfrak{so}(32)$, but not the global form of the gauge group. The open-string Chan-Paton construction produces SO(32), whereas the heterotic lattice Γ_{16} carries one spinor conjugacy class of $\mathfrak{so}(32)$ and produces the quotient $\mathrm{Spin}(32)/\mathbb{Z}_2$, which is neither SO(32) nor its full cover Spin(32). At the level of the massless spectrum and the perturbative interactions the distinction is invisible, but it

matters for which line operators and solitons the theory admits, and it is one of the discrete data that must match when Type I and the $\mathrm{Spin}(32)/\mathbb{Z}_2$ heterotic string are related by strong-weak duality in Chapter 7.

3.6.2 Reading the counting

The massless spectra in Table 3.4 are constrained by supersymmetry: bosons and fermions must match state by state. Counting them is a good check and a good way to see the multiplet structure.

Example 3.20 (Bose-Fermi matching in Type II)**.** Work in light-cone gauge, where the massless little group is SO(8) and each of the four closed-string sectors is a tensor product of an $\mathbf{8}_v$ (from an NS side) or an $\mathbf{8}_{s,c}$ (from an R side); the products are those of Eqs. (3.64) to (3.66). Take IIB, with both Ramond ground states in $\mathbf{8}_s$. The bosons are

$$\underbrace{\mathbf{8}_v \otimes \mathbf{8}_v}_{\text{NS-NS}} \oplus \underbrace{\mathbf{8}_s \otimes \mathbf{8}_s}_{\text{R-R}} = (\mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35}) \oplus (\mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35}_+), \tag{3.88}$$

which is $64 + 64 = 128$ bosonic states. The fermions come from the mixed sectors,

$$\underbrace{\mathbf{8}_v \otimes \mathbf{8}_s}_{\text{NS-R}} \oplus \underbrace{\mathbf{8}_s \otimes \mathbf{8}_v}_{\text{R-NS}} = (\mathbf{8}_c \oplus \mathbf{56}_s) \oplus (\mathbf{8}_c \oplus \mathbf{56}_s), \tag{3.89}$$

which is $64 + 64 = 128$ fermionic states: two gravitini ($\mathbf{56}_s$) and two dilatini ($\mathbf{8}_c$). The two gravitini share one SO(8) chirality and the two dilatini the opposite one; what makes the theory chiral is that both supercharges carry the same chirality, the $\mathcal{N} = (2, 0)$ entry in the table. Bosons and fermions balance, $128 = 128$. The IIA count is identical after replacing one $\mathbf{8}_s$ by $\mathbf{8}_c$, which flips the chirality of one gravitino and one dilatino and turns the R-R product (3.87) into (3.86). The total, $128 + 128 = 256 = 2^8$, is the dimension of the maximal ten-dimensional supergravity multiplet.

For the $\mathcal{N} = 1$ theories the same arithmetic splits into two multiplets. The supergravity multiplet has 64 bosonic and 64 fermionic states: bosons $g_{\mu\nu}$ (35) $\oplus$ $B_{\mu\nu}$ (28) $\oplus$ ϕ (1), fermions gravitino (56)$\oplus$ dilatino (8). Each super-Yang-Mills multiplet adds a gauge boson (8) and a gaugino (8), and there are 496 of them, one per generator. In Type I the closed-string sector is what survives the orientifold projection Ω of IIB: Ω keeps the symmetric NS-NS combination, so g and ϕ survive while $B_{\mu\nu}$, antisymmetric under exchange of the two sides, is removed (Exercise 3.17); and it keeps the antisymmetric part of the R-R product $\mathbf{8}_s \otimes \mathbf{8}_s$, so of the three potentials in (3.87) only C_2 remains. That the surviving two-form of Type I is Ramond-Ramond becomes central in Chapter 7: the fundamental heterotic string couples to an NS-NS B-field, so under the duality relating Type I to the $\mathrm{Spin}(32)/\mathbb{Z}_2$ heterotic string the object charged under the Type I C_2 is not the fundamental string but the D1-brane.

3.6.3 Why exactly 496

The recurrence of 496 across the three chiral, half-maximal theories comes from anomaly cancellation, and it is one of the sharpest consistency conditions in the subject. A chiral theory of gravity coupled to gauge fields and antisymmetric tensors generically suffers gauge and gravitational anomalies: the one-loop effective action fails to be invariant, and the theory is inconsistent. Green and Schwarz found that for ten-dimensional $\mathcal{N} = 1$ supergravity coupled to super-Yang-Mills the anomaly does not vanish for a generic group, but it factorises, and so can be cancelled by a local counterterm, precisely when the gauge group has dimension 496 and its trace invariants satisfy a set of identities. The full statement and its derivation are deferred to Chapter 5 and Appendix H; here we quote the outcome.

Proposition 3.21 (Green-Schwarz constraint, quoted)**.** *The anomaly of ten-dimensional chiral $\mathcal{N}=1$ supergravity coupled to super-Yang-Mills factorises for exactly four Lie algebras, all of dimension* 496*: $\mathfrak{so}(32)$, $\mathfrak{e}_8 \oplus \mathfrak{e}_8$, $\mathfrak{e}_8 \oplus \mathfrak{u}(1)^{248}$, and $\mathfrak{u}(1)^{496}$. Of these, only the first two are realised by a consistent string theory. The derivation is in Appendix H.*

The dimensions check at once: $\dim \mathrm{SO}(32) = \frac{1}{2}\cdot 32\cdot 31 = 496$ and $\dim(\mathrm{E}_8 \times \mathrm{E}_8) = 248+248 = 496$. String theory then does two things simultaneously. It supplies the two-form and the specific coupling needed to write the Green-Schwarz counterterm, namely the B-field of the heterotic string or the C_2 of Type I, so the counterterm is a field the theory already contains rather than an addition by hand. And it selects, out of the four algebras allowed by the anomaly, exactly the two that arise from an even self-dual lattice or from a consistent orientifold. The purely abelian and mixed options in Theorem 3.21 have no such realisation: there is no even self-dual lattice in sixteen dimensions delivering $\mathfrak{u}(1)^{496}$ or $\mathfrak{e}_8 \oplus \mathfrak{u}(1)^{248}$. The classification of those lattices, which returns $\Gamma_8 \oplus \Gamma_8$ and Γ_{16} and nothing else, is used in Section 3.5 and proved in Appendix G. Two independent consistency conditions converge on the same pair of gauge groups: worldsheet modular invariance through the lattice, and spacetime anomaly cancellation through Green-Schwarz. That two such different requirements select the same answer is a strong hint that they probe one underlying structure, a theme Chapter 7 takes up.

Remark 3.22 (Beyond the lectures: strings without spacetime supersymmetry)*.* The five theories in Table 3.4 are the consistent superstrings *with* unbroken spacetime supersymmetry. Relaxing that requirement while keeping modular invariance enlarges the list. Choosing the GSO projection differently in the two Type II sectors produces the Type 0A and Type 0B strings, purely bosonic in spacetime, with a doubled R-R sector and a surviving tachyon, so they do not describe a stable ten-dimensional vacuum. On the heterotic side there is the tachyon-free but non-supersymmetric $\mathrm{SO}(16) \times \mathrm{SO}(16)$ string, obtained by a different chiral GSO-type projection. These are genuine string theories, studied for cosmology and as endpoints of tachyon condensation, but they sit outside the supersymmetric catalogue this chapter is built around and we will not need them. The point to carry forward is that "the five" is shorthand for "the five that are ten-dimensional, tachyon-free, and spacetime supersymmetric"; GSO is a choice, and other consistent choices exist.

3.6.4 Where this is going

The table is an endpoint for this chapter and a starting point for the rest of the course. Two threads lead out of it.

The first is the low-energy limit. Send $\alpha' \to 0$, so that the massive string states of mass $\sim M_{\mathrm{s}}$ decouple and only the massless fields of Table 3.4 survive. What remains is a two-derivative field theory, fixed almost completely by the supersymmetry and field content of each row. For the Type II strings these are the maximal ten-dimensional supergravities: Type IIA supergravity, non-chiral, which is also what one obtains by reducing eleven-dimensional supergravity on a circle, and Type IIB supergravity, chiral, with its self-dual five-form field strength. For Type I and the heterotic strings the limit is ten-dimensional $\mathcal{N}=1$ supergravity coupled to $\mathcal{N}=1$ super-Yang-Mills with the appropriate gauge group. Chapter 5 writes these actions down, constructs the Green-Schwarz counterterm quoted in Theorem 3.21, and brings eleven-dimensional supergravity into the story.

The second thread is that the five rows are not five separate theories. The agreements in the table, the identical NS-NS sectors, the repeated 496, the two routes to SO(32), are the visible edge of a web of dualities that identifies all five, together with eleven-dimensional supergravity, as limits of one underlying framework. Compactify IIA and IIB each on a circle and they become the same theory, related by T-duality; the two heterotic strings are likewise T-dual on a circle. Send the coupling to infinity and new descriptions appear: strongly coupled IIA grows

an eleventh dimension and becomes M-theory, whose low-energy limit is eleven-dimensional supergravity, and strongly coupled $\mathrm{E}_8 \times \mathrm{E}_8$ heterotic becomes M-theory on an interval. Type I is the strong-coupling dual of the $\mathrm{Spin}(32)/\mathbb{Z}_2$ heterotic string, which is why the D1-brane of one maps to the fundamental string of the other. Chapter 7 assembles these statements into the duality web and its eleven-dimensional apex. Read that way, Table 3.4 is less a list of five theories than a first look at one theory seen from five perturbative corners.

Exercise 3.19 (D-branes from the Ramond-Ramond spectrum)**.** *(Short.)* A Ramond-Ramond potential C_{p+1} couples electrically to a Dp-brane and magnetically to a D$(6-p)$-brane. Using the R-R content of Eqs. (3.86) and (3.87), list the D-brane dimensions predicted for Type IIA and for Type IIB, and confirm that IIA carries even-dimensional and IIB odd-dimensional D-branes. Explain why the self-dual C_4^+ of IIB yields a single self-dual object, the D3-brane, that is its own magnetic dual. (This anticipates Chapter 6; you need only the electric-magnetic rule stated here.)

Exercise 3.20 (The Type I closed sector as an $\mathcal{N}=1$ multiplet)**.** *(Standard.)* Taking as given that the orientifold projection Ω removes the NS-NS two-form (Exercise 3.17), determine which of the three R-R potentials in (3.87) survives Ω, and assemble the surviving closed-string bosons and their superpartners into a ten-dimensional $\mathcal{N}=1$ supergravity multiplet. Check that the multiplet has 64 bosonic and 64 fermionic states. In one sentence, say why exactly one of the two IIB gravitini survives the projection, so that Type I has $\mathcal{N}=1$ rather than $\mathcal{N}=2$.

Exercise 3.21 (Counting the heterotic spectrum, and the missing fifth group)**.** *(Moderate.)* Compute the total number of massless states of the $\mathrm{E}_8 \times \mathrm{E}_8$ heterotic string: add the $64+64$ of the supergravity multiplet to the $496 \times (8+8)$ of the gauge sector. Then use Theorem 3.21 together with the lattice classification of Appendix G to explain why the anomaly-allowed algebras $\mathfrak{u}(1)^{496}$ and $\mathfrak{e}_8 \oplus \mathfrak{u}(1)^{248}$ do *not* appear as further rows of Table 3.4: there is no even self-dual rank-16 lattice, and no consistent orientifold, that realises them. What does this say about the relation between worldsheet consistency and spacetime anomaly freedom?

Chapter 4

Amplitudes as a Cartoon, and the Limits of First Quantisation

4.1 The recipe: sum over surfaces

A scattering amplitude in quantum field theory is a sum of Feynman diagrams. You draw every graph with the right external legs, assign a propagator to each line and a coupling to each vertex, integrate the loop momenta, and add the results with their symmetry factors. The vertices are input: they come from the interaction terms you wrote in the Lagrangian, and the couplings are numbers you fix by measurement. String perturbation theory follows the same logic with one structural change. The graph is replaced by a smooth two-dimensional surface, the worldsheet, and everything else follows from the geometry of that surface. There are no vertices to choose and no couplings to assign by hand.

This section assembles the prescription. The individual ingredients were built in Chapter 2: the external states became vertex operators through the state-operator map (Sections 2.1 and 2.2), the tree correlator became the Koba-Nielsen factor (Eq. (2.30)), and the residual conformal symmetry was shown to fix three operator positions on the sphere (Section 2.3.2). Here we put the pieces together into the perturbation series, read off its structure, and are honest about what the cartoon leaves out. We do not construct the path-integral measure or its Faddeev-Popov determinant; those live in Appendix F, and the point of this chapter is the physics they support, not their machinery.

4.1.1 From worldlines to worldsheets

Start with the first-quantised description of a single relativistic particle, the worldline formalism that some of you will have met as the Schwinger proper-time representation of a propagator. A particle traces a worldline through spacetime, and its propagator is a sum over all such paths weighted by $e^{-m\,(\text{length})}$. That path integral gives you one propagator. To build an amplitude you still have to glue propagators together at interaction vertices, and the vertices, together with their couplings, are extra data imposed from outside the worldline. The S-matrix is then a sum over graphs: which lines meet at which vertices, and how many loops.

Replace the particle by a string and the worldline fattens into a two-dimensional worldsheet. A free closed string sweeps out a tube. Two strings that join, or one string that splits, sweep out a surface shaped like a pair of trousers: one tube coming in at the waist, two going out at the legs. Nothing singular happens where the split occurs. Run your finger along the trousers and there is no point you can call "the interaction", no place where the geometry fails to be smooth. Contrast the Feynman graph, where the interaction sits at a sharply localised vertex.

The string interaction is delocalised over the whole surface, and where a QFT observer in one Lorentz frame says the string split "here" and in another frame "there", those are the same smooth worldsheet cut at different times. That frame-independence is worldsheet conformal invariance, and it is the reason the coupling cannot be a free vertex: the three-string coupling is fixed by the same conformal data that fixes everything else, and its overall strength is the one number g_s, the exponential of the dilaton expectation value (Section 1.5.5).

The payoff is a dramatic economy of diagrams. In field theory a four-point tree amplitude already needs several graphs, one for each channel (s, t, u) through which a pair of particles can fuse and re-split. In string theory those channels are not different diagrams. They are different regions of one and the same surface, the four-punctured sphere, reached by sliding the punctures around. A single worldsheet topology carries what field theory splits across many graphs. This is the geometric root of the s-t duality that we will see explicitly in the Veneziano amplitude of Section 4.2; here the sum over diagrams has already collapsed to a sum over surfaces, one representative per topology.

4.1.2 The genus expansion is the loop expansion

Classify the surfaces. For oriented closed strings a connected closed worldsheet is a compact orientable surface, and topology sorts these completely by a single integer, the genus g, the number of handles. The sphere has $g = 0$, the torus $g = 1$, and so on. The invariant that controls the weighting is the Euler characteristic, which for a genus-g surface carrying n punctures (one for each external string) is

$$\chi = 2 - 2g - n. \tag{4.1}$$

The dilaton coupling of Section 1.5.5 weights each topology by $g_s^{-\chi}$: the worldsheet Einstein-Hilbert term is topological in two dimensions, its integral is χ by Gauss-Bonnet, and it exponentiates to a power of the coupling. Written out through Eq. (1.97), a closed surface of genus g therefore comes with

$$g_s^{2g-2} \tag{4.2}$$

from the handles alone, and each of the n external strings contributes one further factor of g_s, so an n-point amplitude at genus g scales as g_s^{2g-2+n}. Adding a handle costs g_s^2. That is the string version of adding a loop: a handle is a closed string emitted and reabsorbed, exactly one closed-string loop. The genus expansion is the loop expansion, with g_s^2 the loop-counting parameter.

Figure 4.1 shows the first few terms for a four-point process, so $n = 4$ throughout and each topology carries the full weight g_s^{2g-2+n}. The sphere is tree level, g_s^2; the torus is one loop, g_s^4; the genus-two surface is two loops, g_s^6. Each step up in genus multiplies by g_s^2, the price of one handle, one closed-string loop. There is one surface at each order, not a proliferating set of graphs.

One caution before we write the formula. The coupling g_s is not a dial we are free to turn: it is the expectation value of a field in the spectrum, the dilaton, so the genus expansion is an expansion about a particular vacuum (Section 1.5.5). Calling it "perturbation theory" commits us to small g_s and to a fixed background, and both of those commitments are exactly what the later parts of this chapter interrogate. For now we take them as given.

Remark 4.1 (Open and unoriented strings). For open strings and for unoriented strings the surfaces also carry boundaries and crosscaps, and the Euler characteristic generalises to $\chi = 2 - 2g - b - c$ with b boundaries and c crosscaps (Eq. (1.97)). The disc, $g = 0$, $b = 1$, has $\chi = 1$ and weight g_s^{-1}: it is the open-string tree. The annulus, $\chi = 0$, weight g_s^0, is the open-string one-loop diagram, and reading the same surface in the other channel exhibits it as a closed string exchanged between two boundaries, the first sighting of open-closed duality. We keep

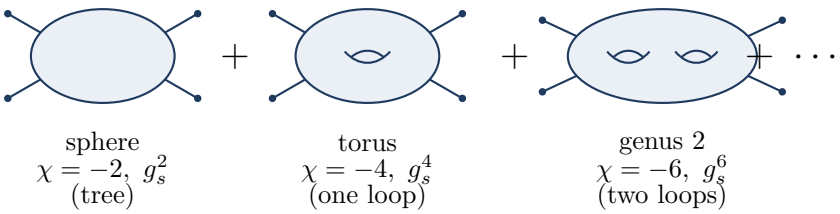


Figure 4.1: The genus expansion of a four-point closed-string amplitude. Each term is a single worldsheet topology with the four external states attached at punctures, so $\chi = 2 - 2g - n$ (Eq. (4.1)) with $n = 4$, and the weight is $g_s^{-\chi} = g_s^{2g-2+n}$: a factor g_s^{2g-2} from the g handles and one g_s for each external state. Adding a handle adds one closed-string loop and a factor g_s^2. Field theory would draw many graphs at each order; string theory draws one surface.

the main development to oriented closed strings and pick these up when they are needed for D-branes in Chapter 6, with one exception: the tree-level disc, which the next section runs the Veneziano amplitude on and whose conformal Killing group we set up in Section 4.1.4.

4.1.3 The master formula

Everything now condenses to one prescription. The n-point amplitude for oriented closed strings is

$$\boxed{\mathcal{A}(1,\ldots,n) \;=\; \sum_{g=0}^{\infty} g_s^{2g-2+n} \int_{\mathcal{M}_{g,n}} \mathrm{d}\mu_{g,n} \left\langle \mathcal{V}_1\, \mathcal{V}_2 \cdots \mathcal{V}_n \right\rangle_{\Sigma_g}} \tag{4.3}$$

and the whole of first-quantised string perturbation theory is contained in reading this equation correctly. Take it piece by piece.

The external states enter as vertex operators $\mathcal{V}_i$, one per leg, inserted at punctures on the surface. A momentum eigenstate of the string maps to a local operator by the state-operator correspondence (Section 2.1); a physical external state is a weight-$(1,1)$ primary (Section 2.2), which is the on-shell and transversality condition made local. For the tachyon $\mathcal{V} = {:}e^{ik\cdot X}{:}$, for the massless states $\mathcal{V} = {:}\partial X^\mu \bar{\partial} X^\nu\, e^{ik\cdot X}{:}$ contracted with a polarisation. The angle brackets are the correlation function of these operators in the worldsheet CFT on the surface Σ_g; at genus zero it is the Koba-Nielsen factor Eq. (2.30), and its higher-genus versions are assembled in Appendix F.

The symbol $\mathcal{M}_{g,n}$ is the moduli space of the genus-g surface with n marked points: the space of inequivalent shapes the worldsheet can take, and inequivalent places the operators can sit, after all the redundancy of reparametrisations and Weyl rescalings has been removed. We integrate over it with a measure $\mathrm{d}\mu_{g,n}$. That measure is where the Faddeev-Popov determinant and the ghost path integral live, and constructing it is the grinding we set aside; Appendix F does it. What matters for the physics is that the integral is over a finite-dimensional space of genuine shape parameters, not over field configurations.

The powers of g_s are the bookkeeping of Section 4.1.2: g_s^{2g-2} for the handles of the surface and one factor of g_s for each external vertex operator. When you compare the same process at successive loop orders, the external factors are common and the relative weight is g_s^2 per handle, as it must be for a loop expansion.

The sum over g is put in by hand. This is the defining feature of the first-quantised recipe, and the reason for the word "first". We quantised a single string, one worldsheet at a time, and then declared that the amplitude is the sum over topologies of one string's history. Nothing in the quantisation of one worldsheet told us to add the others; there is no operator in this framework that creates or destroys strings, no field whose Feynman rules would generate Eq. (4.3) for us.

The prescription is a rule about which surfaces to draw, exactly as the sum over Feynman graphs is a rule in the worldline formalism of ordinary QFT. That this rule is imposed rather than derived is the seam that Sections 4.4 and 4.5 pull on.

4.1.4 Moduli space, and the three punctures the symmetry fixes

The one part of Eq. (4.3) worth unpacking here, because it has no analogue in the Feynman rules and because it controls the whole structure of the integral, is the moduli space. Its dimension is fixed by counting, and the count came out of Section 2.3.2; we restate the result and its meaning, and send you there and to Appendix F for the derivation with its ghost bookkeeping.

A complex-dimension count balances the shape deformations of the surface against the symmetries that act trivially on it. Deformations of the metric that are not undone by a diffeomorphism or a Weyl rescaling are moduli; infinitesimal symmetries that fix the surface, the conformal Killing vectors, subtract. For a genus-g surface with n punctures the net count is

$$\dim_{\mathbb{C}} \mathcal{M}_{g,n} \;=\; 3g - 3 + n \tag{4.4}$$

once $2g - 2 + n > 0$, the regime in which the conformal Killing vectors are exhausted. The three cases that organise the course sit inside this one formula.

At genus zero the surface is the Riemann sphere $\mathbb{CP}^1$. It has an unusually large symmetry: the globally defined conformal maps are the Möbius transformations $z \mapsto (az + b)/(cz + d)$ of Eq. (2.32), the group $\mathrm{PSL}(2, \mathbb{C})$ of complex dimension three. This is the conformal Killing group of the sphere. It is non-compact, so its volume is infinite, and a naive integral over all n puncture positions would carry that infinite factor as pure redundancy. The cure is to spend the three complex parameters of the group fixing three of the punctures at reference points, conventionally 0, 1, and ∞, and to integrate only the remaining $n - 3$. In the vertex-operator language this is the fixed-versus-integrated distinction of Section 2.2: three operators are "fixed", carrying a $c\bar{c}$ ghost factor that soaks up the six real ghost zero modes of the sphere, and the rest are "integrated", $\int \mathrm{d}^2 z\, \mathcal{V}$. For genus zero the measure in Eq. (4.3) is then nothing fancier than

$$\int_{\mathcal{M}_{0,n}} \mathrm{d}\mu_{0,n} \;=\; \int \prod_{i=4}^{n} \mathrm{d}^2 z_i, \tag{4.5}$$

with z_1, z_2, z_3 frozen. The disc, the surface for open-string trees, works the same way with the smaller group $\mathrm{PSL}(2, \mathbb{R})$: its three real parameters fix three boundary punctures at reference points on the real axis, so the four-point disc keeps one real modulus running over an interval. That disc is the single exception to the deferral of open strings in Theorem 4.1, since the Veneziano amplitude of Section 4.2 runs on it. The nontrivial part of $\mathrm{d}\mu_{g,n}$, the part that needs the b-ghosts folded against the shape deformations, appears only once there are handle moduli to integrate, that is, from genus one upwards.

Setting $g = 0$ in Eq. (4.4) makes the pattern of low-point amplitudes immediate. For $n \leq 3$ there is nothing to integrate: two- and three-point functions are pure kinematics, fixed by the symmetry up to a constant, so they cannot carry the poles and cuts of a genuine scattering process. The first amplitude with a genuine integral is the four-point function. On the sphere $\dim_{\mathbb{C}} \mathcal{M}_{0,4} = 1$, a single complex modulus integrated over the plane, and that integral is the closed-string Virasoro-Shapiro amplitude. The open-string version puts the four punctures on the boundary of the disc, where $\mathrm{PSL}(2, \mathbb{R})$ leaves one *real* modulus running over an interval, and that integral is the Veneziano amplitude that Section 4.2 evaluates; Exercise 2.9 already set the closed-string version up.

At genus one the surface is the torus, with one complex modulus τ, the shape of the parallelogram, and one leftover conformal Killing vector, the rigid translation. That translation is spent fixing

the single puncture, so a one-point function on the torus integrates over τ alone: $\dim_{\mathbb{C}} \mathcal{M}_{1,1} = 1$, with the vertex operator held at a reference point. From genus two the conformal Killing vectors are gone entirely and the moduli number a plain $3g - 3$. These counts, and the ghost-number arithmetic that makes them precise, are tabulated in Appendix F.

There is one more feature of $\mathcal{M}_{g,n}$ that will matter throughout the chapter: it has a boundary, and the boundary is where the surface degenerates. A handle can pinch to zero thickness, or two punctures can collide, and in either limit the surface develops a long thin neck. A long neck is a string propagating a long proper distance, so these degeneration limits are where an internal string goes on shell and the amplitude develops a pole or a cut. The boundary of moduli space is the infrared, the region of physical intermediate states, not a short-distance region to be regulated. We return to this reading in Section 4.4, because it is exactly the place where the first-quantised measure has to be handled with care.

Remark 4.2 (Beyond the lectures: the measure and super-moduli). The measure $\mathrm{d}\mu_{g,n}$ is the image of the Polyakov path integral after gauge fixing. Concretely each of the $3g-3+n$ complex moduli is paired with an insertion of the antighost b folded against the corresponding Beltrami differential, $\prod_k (b, \mu_k)$, and the integral of these against the matter-plus-ghost correlator is a top form on $\mathcal{M}_{g,n}$. At genus one this reproduces the familiar $\mathrm{d}^2\tau/(\operatorname{Im}\tau)^2$ up to the partition function; the general construction, including holomorphic factorisation, is standard and lives in Appendix F. For the superstring the story is genuinely harder: the worldsheet has fermions and local supersymmetry, so one integrates over a super-moduli space, and the picture-changing operators of the RNS formalism are a device for handling its odd directions. This is smooth at low genus but not globally: Donagi and Witten showed that super-moduli space fails to be split, that is, it is not a fermionic bundle over the ordinary moduli space, from genus five onwards, so the naive recipe of reducing the superstring measure to an integral over bosonic moduli with picture-changing insertions breaks down there. None of this is needed for the cartoon, and we will not use it; it is flagged so that "we skipped the measure" does not read as "the measure is trivial".

4.1.5 What the recipe buys, and what it costs

Two structural consequences of Eq. (4.3) deserve to be stated plainly, one a genuine triumph and one the seam we have already flagged.

The triumph is the ultraviolet behaviour. In field theory a loop is an integral over the momentum of a virtual particle running to infinity, and that is where divergences are born; you regulate and renormalise them. In Eq. (4.3) there is no such integral. A handle is a shape, catalogued by the finite-dimensional, compactified moduli space $\mathcal{M}_{g,n}$, and nowhere in that catalogue does a momentum run off to infinity. There is no separate short-distance region to integrate over, because a string of finite size has no arbitrarily short-distance structure to probe. On the torus this is sharp: the dangerous limit would be $\tau \to 0$, a degenerate parallelogram, and modular invariance (Section 2.3.4) identifies that corner with the well-behaved $\tau \to i\infty$ limit and excises it from the fundamental domain. What field theory would call an ultraviolet divergence is, on the worldsheet, either absent or re-expressed as an infrared feature at the boundary of moduli space, a physical intermediate state. This is the origin of the string's ultraviolet softness, and the Veneziano amplitude of Section 4.2 will display it as an exponential fall-off at high energy.

The cost is the one already named: Eq. (4.3) is a first-quantised, on-shell, asymptotic recipe. First-quantised, because it sums over the histories of one string with the sum over topologies put in by hand. On-shell, because the external $\mathcal{V}_i$ are weight-$(1,1)$ primaries, and only on-shell physical states are weight $(1,1)$: the formula computes S-matrix elements between asymptotic states and offers no off-shell continuation, no analogue of an off-shell Green's function you could cut open or use as a building block. Asymptotic, because the genus sum is not a convergent series;

the number of surfaces grows fast enough with g that the series has zero radius of convergence, a point Section 4.4 makes quantitative. And the recipe is silent about physics that is not organised by the genus expansion at all: shifts of the vacuum when a tadpole fails to cancel, the renormalisation of masses that a first-quantised on-shell formula cannot even phrase, and effects of order e^{-1/g_s} that no finite number of terms in Eq. (4.3) can see. Cataloguing these is the business of Section 4.4, and the framework that repairs them, string field theory, is the business of Section 4.5.

The recipe in one box

A string amplitude is a sum over worldsheet topologies. Each topology contributes once, weighted by g_s^{2g-2+n}: a factor g_s^{2g-2} from the g handles (one closed-string loop each) and one g_s per external state. The contribution is an integral over the moduli space $\mathcal{M}_{g,n}$ of the surface, of dimension $3g-3+n$, of a correlation function of vertex operators in the worldsheet CFT. On the sphere the conformal Killing group $\mathrm{PSL}(2,\mathbb{C})$ fixes three punctures at $0, 1, \infty$ and the remaining $n-3$ are integrated. The finite, compact moduli space is why there is no ultraviolet region and no ultraviolet divergence. The price is that the recipe is first-quantised, on-shell, and an asymptotic series: honest limits set out in Section 4.4, and repaired in Section 4.5.

The next section runs the recipe once, on the four-tachyon amplitude, and extracts the physics that made string theory worth taking seriously in the first place: linear Regge trajectories, s-t duality, and the soft ultraviolet.

Exercise 4.1 (Counting couplings and loops)**.** *(Short.)* Use Eqs. (4.1) and (4.2). (a) For oriented closed strings, confirm that the four-point amplitude scales as g_s^2 at genus zero, g_s^4 at genus one, and that the ratio of successive orders is g_s^2, independent of n. (b) For open strings the surface carries boundaries, $\chi = 2-2g-b$ (Theorem 4.1), the surface is weighted by $g_s^{-\chi}$, and each external open-string state brings one factor of the open-string coupling $g_o = g_s^{1/2}$ in place of the closed-string g_s. Show that the disc four-point ($g=0$, $b=1$) scales as $g_s^{-1}g_o^4 = g_o^2 = g_s$, and the annulus four-point ($g=0$, $b=2$) as $g_s^0 g_o^4 = g_o^4 = g_s^2$, a relative factor g_s. Conclude that each extra boundary, like each crosscap, costs one power of g_s, half the g_s^2 that a handle costs, so an open-string loop is half a closed-string loop in the g_s-counting and it takes a pair of boundaries to match one handle. (c) A theory with both open and closed strings can attach either kind of external state; explain in one line why, since $g_s = e^{\langle\phi\rangle}$, a constant shift $\langle\phi\rangle \to \langle\phi\rangle + c$ multiplies an n-point L-loop amplitude by $e^{(2L-2+n)c}$, and why the loop-counting part of this, $e^{(2L-2)c}$, is a redefinition of g_s rather than a physical operation, the e^{nc} from the external legs being absorbed into the normalisation of the external states.

Exercise 4.2 (Where the first integral appears)**.** *(Short.)* From Eq. (4.4): (a) evaluate $\dim_{\mathbb{C}} \mathcal{M}_{g,n}$ for $(g,n) = (0,3), (0,4), (0,5), (1,1), (2,0)$, and identify in each case how many positions or shape parameters are integrated. Treat $(1,0)$ separately: the formula returns $3g-3+n = 0$, moduli minus conformal Killing vectors $(1-1)$, yet the vacuum torus still carries one genuine modulus, τ, the translation conformal Killing vector being exactly the -1 that the count removes, so the vacuum torus integrates one shape parameter even though the net count vanishes; this is the case that the condition $2g-2+n>0$ excludes. (b) Explain why the genus-zero three-point function and the genus-one zero-point function (the vacuum torus amplitude) sit at opposite ends of the story even though both have a small moduli space: one is fixed entirely by the conformal Killing group, the other is a nonzero integral over the single modulus τ. (c) A student proposes computing a genus-zero two-point function to read off a propagator. Using the conformal Killing group count, explain why Eq. (4.3) gives them only a kinematic constant, and why an honest off-shell propagator is not something the on-shell recipe can produce, pointing ahead to Section 4.5.

4.2 The Veneziano amplitude

Two- and three-point functions are pure kinematics, on the sphere and on the disc alike. The conformal Killing group fixes them up to constants (Section 4.1.4): the two-point function is a normalisation, the three-point function is one coupling times a rigid prefactor, and neither has a modulus to integrate. The counting is the same for the closed string on the sphere, whose residual symmetry is $\mathrm{PSL}(2,\mathbb{C})$, and for the open string on the disc, whose residual symmetry is $\mathrm{PSL}(2,\mathbb{R})$; it is the disc we now work. The four-point function is the first with a genuine integral, and so the first that can grow poles in a Mandelstam variable, resonate on the tower of string states, and develop a nontrivial high-energy limit. We work it once, for the simplest external states the string has, its tachyons, and read the entire physics of tree-level string scattering off a single special function.

That function is an Euler beta function, and its appearance here is where string theory started. Veneziano wrote it down in 1968 as a guess: a closed-form four-point amplitude engineered by hand to be crossing symmetric and to carry the resonance-plus-Regge structure that strong-interaction phenomenology demanded, at a time when nobody knew it came from anything. The worldsheet arrived afterwards, when Nambu, Nielsen and Susskind recognised that Veneziano's formula was the scattering amplitude of a relativistic string. The dual-model bootstrap of the late 1960s is the ancestor of everything in these notes, and the amplitude below is the thing it was bootstrapping.

What one amplitude teaches

The four-tachyon disc amplitude collapses to a beta function,

$$A(s,t) \;\propto\; B(-\alpha(s),-\alpha(t)) \;=\; \frac{\Gamma(-\alpha(s))\,\Gamma(-\alpha(t))}{\Gamma(-\alpha(s)-\alpha(t))}, \qquad \alpha(x)=1+\alpha' x.$$

Three facts fall straight out of the gamma functions, and each is a pillar of the subject. The **poles** sit at $\alpha(s)=0,1,2,\ldots$, exactly the masses of the free open string, with residues of growing spin: the string's *linear Regge trajectories*, $J=\alpha(M^2)=1+\alpha' M^2$, seen now as resonances in a scattering process. The beta function is **symmetric**, $B(a,b)=B(b,a)$, so one term already carries the full tower of s-channel poles *and* the full tower of t-channel poles: *worldsheet duality*, with no separate diagrams to add. And at high energy and fixed angle the amplitude is **exponentially soft**, $|A|\sim e^{-\alpha' E^2 f(\theta)}$, far below any power law, which is where the ultraviolet finiteness of string perturbation theory becomes visible already in a tree graph.

4.2.1 Four open tachyons on the disc

Tree level for open strings is the disc, whose boundary is the worldsheet locus where the endpoints live (Section 4.1). We scatter four open-string tachyons, the ground states of Section 1.5.2, each represented by the boundary vertex operator $:e^{ik_i\cdot X}:$ of Section 2.2.2 inserted on the real axis at a point y_i. The choice of the open string is deliberate: it produces the Veneziano amplitude in its original form, a single beta function over one real modulus. The closed string on the sphere gives the analogous Virasoro-Shapiro amplitude, a product of three gamma ratios over one complex modulus, and we quote it in the remark at the end.

The whole dynamical content is the boundary Koba-Nielsen correlator. It is the disc version of Eq. (2.30), built from the boundary propagator $\langle X^\mu(y)X^\nu(y')\rangle = -2\alpha'\,\eta^{\mu\nu}\ln|y-y'|$ of Eq. (2.20), whose doubled coefficient (Neumann conditions enforced by the method of images) is the only

change from the bulk. A free-field Gaussian contraction gives

$$\Big\langle \prod_{i=1}^{4} :e^{ik_i\cdot X}:(y_i) \Big\rangle_{D_2} \;=\; i\,C_{D_2}\,(2\pi)^D\,\delta^D\Big(\sum_i k_i\Big)\prod_{i<j}|y_{ij}|^{\,2\alpha' k_i\cdot k_j}, \qquad y_{ij}\coloneqq y_i-y_j, \tag{4.6}$$

with $D=26$ and $C_{D_2}\propto g_o^{-2}$ the disc normalisation, one factor for the disc Euler number $\chi(D_2)=1$ (Section 1.5.5). The delta function is momentum conservation, the boundary Ward identity; the position dependence carries the interaction.

The positions y_i are not physical data. They are integrated, but only after dividing by the residual gauge symmetry that survives conformal gauge, the conformal Killing group of the disc, $\mathrm{PSL}(2,\mathbb{R})$, with three real parameters (Section 4.1.4). Three of the four insertions are fixed to reference points and the fourth is integrated. Take the ordering $y_1<y_2<y_3<y_4$ along the boundary and fix

$$y_1=0, \qquad y_3=1, \qquad y_4\to\infty, \tag{4.7}$$

leaving $y_2\equiv x$ to run over the interval $(0,1)$ between its fixed neighbours. The Jacobian of the fixing is the c-ghost correlator $\langle c(y_1)c(y_3)c(y_4)\rangle = y_{13}\,y_{14}\,y_{34}$, and its derivation belongs to Appendix F. It combines with the $y_4\to\infty$ tail of Eq. (4.6) so that the limit is finite: every factor carrying y_4 assembles by momentum conservation into $|y_4|^{\,2\alpha' k_4\cdot\sum_{i\neq4}k_i}=|y_4|^{-2\alpha' k_4^2}=|y_4|^{-2}$, which the ghost Jacobian $|y_4|^2$ cancels on the mass shell. What survives is an integral over the one real modulus x, with only the two factors $|y_{12}|=x$ and $|y_{23}|=1-x$ left ($|y_{13}|=1$):

$$A(1,2,3,4) \;=\; i\,g_o^4\,C_{D_2}\,(2\pi)^D\,\delta^D\Big(\sum_i k_i\Big)\int_0^1 \mathrm{d}x\; x^{\,2\alpha' k_1\cdot k_2}\,(1-x)^{\,2\alpha' k_2\cdot k_3}, \tag{4.8}$$

one power of the open-string coupling g_o for each of the four vertices. Against the disc normalisation $C_{D_2}\propto g_o^{-2}$ the amplitude scales as g_o^2, the disc face of the genus counting Eq. (1.97). The moduli space of the four-punctured disc is this single interval, and the amplitude is a definite integral over it.

4.2.2 Kinematics, and the integral is a beta function

Fix the kinematics before doing the integral. With all four momenta taken incoming, momentum conservation is $\sum_i k_i=0$, and each external tachyon is on shell at the mass of Eq. (1.89),

$$M^2=-\frac{1}{\alpha'}, \qquad \text{so} \qquad k_i^2=-M^2=\frac{1}{\alpha'}. \tag{4.9}$$

The Mandelstam invariants, in the mostly-plus signature of Appendix A, are

$$s=-(k_1+k_2)^2, \qquad t=-(k_2+k_3)^2, \qquad u=-(k_1+k_3)^2, \tag{4.10}$$

so that s measures the invariant mass flowing through the adjacent (12) pair, t through the (23) pair. Momentum conservation and Eq. (4.9) give the familiar constraint

$$s+t+u=\sum_i M_i^2=-\frac{4}{\alpha'}. \tag{4.11}$$

Expanding one dot product, $s=-(k_1^2+2k_1\cdot k_2+k_2^2)=-\frac{2}{\alpha'}-2k_1\cdot k_2$, and the same for the (23) pair, converts the exponents in Eq. (4.8) into linear functions of s and t:

$$2\alpha' k_1\cdot k_2=-\alpha' s-2=-\alpha(s)-1, \qquad 2\alpha' k_2\cdot k_3=-\alpha' t-2=-\alpha(t)-1, \tag{4.12}$$

where we have collected the natural combination into the *Regge trajectory function*

$$\alpha(x)\;\coloneqq\;1+\alpha' x. \tag{4.13}$$

Its slope is α': this is the object after which α' was named the Regge slope back in Section 1.5.1, and it is about to earn the name.

Substituting Eq. (4.12) into Eq. (4.8), the modulus integral is the Euler beta function on the nose,

$$\int_0^1 \mathrm{d}x\, x^{-\alpha(s)-1}\,(1-x)^{-\alpha(t)-1} \;=\; B(-\alpha(s),-\alpha(t)) \;=\; \frac{\Gamma(-\alpha(s))\,\Gamma(-\alpha(t))}{\Gamma(-\alpha(s)-\alpha(t))}. \tag{4.14}$$

The integral converges only where both exponents exceed -1, that is $\alpha(s)<0$ and $\alpha(t)<0$, deep below every threshold; everywhere else the amplitude is defined by the gamma-function expression on the right, its unique analytic continuation. Stripping the kinematical prefactor, the colour-ordered four-tachyon amplitude is

$$\boxed{A(s,t) \;=\; B(-\alpha(s),-\alpha(t)) \;=\; \frac{\Gamma(-\alpha(s))\,\Gamma(-\alpha(t))}{\Gamma(-\alpha(s)-\alpha(t))}\,, \qquad \alpha(x)=1+\alpha' x\,.} \tag{4.15}$$

This is the Veneziano amplitude. The three inequivalent cyclic orderings of the four legs on the boundary contribute

$$A_4 \;\propto\; B(-\alpha(s),-\alpha(t)) + B(-\alpha(t),-\alpha(u)) + B(-\alpha(u),-\alpha(s)), \tag{4.16}$$

each carried by its own Chan-Paton trace $\mathrm{Tr}(\lambda^{a_1}\lambda^{a_2}\lambda^{a_3}\lambda^{a_4})$ when the endpoints wear the labels of Eq. (1.95). One beta function does the work; the sum only symmetrises over which legs sit adjacent. We now read Eq. (4.15) three ways.

4.2.3 Linear Regge trajectories

The gamma function $\Gamma(z)$ has simple poles at $z=0,-1,-2,\ldots$, so $\Gamma(-\alpha(s))$ in Eq. (4.15) makes $A(s,t)$ blow up whenever

$$\alpha(s)=n, \quad n=0,1,2,\ldots \qquad\Longleftrightarrow\qquad s=\frac{n-1}{\alpha'}=M_n^2, \tag{4.17}$$

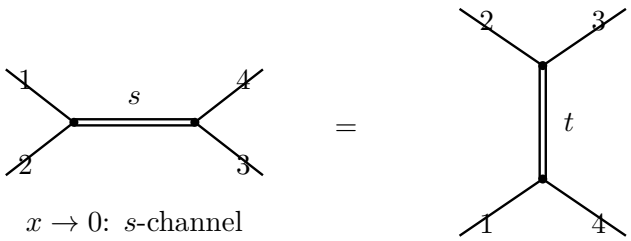
and $M_n^2=(n-1)/\alpha'$ is exactly the open-string mass formula $M^2=(N-1)/\alpha'$ of Eq. (1.88) at level $N=n$. The poles of the scattering amplitude sit precisely on the free spectrum. The lowest, at $\alpha(s)=0$, is the tachyon itself ($s=-1/\alpha'$), a pole the amplitude inherits from the vacuum instability of Section 1.5.2; the next, at $\alpha(s)=1$, is the massless gauge boson ($s=0$); then the whole tower. This is unitarity made concrete. A tree amplitude develops a pole at $s=M^2$ when an on-shell intermediate string of mass M can be exchanged in the s-channel, and the residue is the product of the two couplings where it is emitted and absorbed.

What is being exchanged at each pole has a definite spin, and that is where the Regge structure lives. Near $\alpha(s)=n$ the residue is controlled by the other two gamma functions,

$$A(s,t) \;\xrightarrow{\alpha(s)\to n}\; \frac{1}{n-\alpha(s)}\,\frac{(-1)^n}{n!}\,\frac{\Gamma(-\alpha(t))}{\Gamma(-n-\alpha(t))} \;=\; \frac{1}{n-\alpha(s)}\,\frac{(-1)^n}{n!}\,P_n(\alpha(t)), \tag{4.18}$$

where the sign follows from $\mathrm{Res}\,\Gamma(z)\big|_{z=-n}=(-1)^n/n!$ evaluated at $z=-\alpha(s)$, and $P_n(\alpha(t))=(-\alpha(t)-1)(-\alpha(t)-2)\cdots(-\alpha(t)-n)$ is a polynomial of degree n in $\alpha(t)$, hence of degree n in t. At fixed s the invariant t is a linear function of the s-channel scattering angle $\cos\theta$, so a residue polynomial of degree n in t is a sum of Legendre partial waves up to spin n. The state at level n therefore carries spin up to

$$J_{\max}(n)=n=\alpha(s)\big|_{s=M_n^2}=1+\alpha' M_n^2. \tag{4.19}$$



$x \to 0$: s-channel $\qquad$ $x \to 1$: t-channel

Figure 4.2: Worldsheet duality. The four-punctured disc has a single real modulus $x \in (0,1)$. As $x \to 0$ the surface pinches into two discs joined by a long strip, and the amplitude factorises on s-channel string states; as $x \to 1$ it pinches the other way and factorises on t-channel states. One integral over x interpolates between them, so the two "channels" are the two ends of a single integration region, not separate diagrams. The double line is the string propagating along the degenerating strip.

Plot spin against mass-squared and the highest-spin state at each level lies on a straight line, the *leading Regge trajectory* $J = 1 + \alpha' M^2$, of slope α' and intercept 1. The linearity is the kinematic fingerprint of an extended relativistic object: a rotating string of tension $T = 1/(2\pi\alpha')$ has $J \propto M^2$, whereas a point particle has no such tower. The Regge slope α' that set the level spacing in Section 1.5.1 is the same α' that here is literally the slope of $J(M^2)$.

The trajectory also governs the amplitude away from its poles. In the Regge limit, $s \to \infty$ at fixed t, the ratio of gammas in Eq. (4.15) has the asymptotics $\Gamma(-\alpha(s))/\Gamma(-\alpha(s)-\alpha(t)) \sim (-\alpha(s))^{\alpha(t)}$, so

$$A(s,t) \xrightarrow{s\to\infty,\ t \text{ fixed}} \Gamma(-\alpha(t))\,(\alpha' s)^{\alpha(t)} \times (\text{phase}). \tag{4.20}$$

The amplitude grows as a *moving* power $s^{\alpha(t)}$, not the fixed s^J that a single spin-J exchange would give in field theory. The exponent slides along the trajectory as t varies. This is Regge behaviour in its original phenomenological sense, and it is the whole trajectory, not any one particle, doing it to the high-energy amplitude.

4.2.4 Duality: one term, two channels

The beta function is symmetric,

$$B(-\alpha(s), -\alpha(t)) = B(-\alpha(t), -\alpha(s)), \tag{4.21}$$

which is a stronger statement than it looks. The single term Eq. (4.15) has poles from $\Gamma(-\alpha(s))$, the entire s-channel tower, and poles from $\Gamma(-\alpha(t))$, the entire t-channel tower, both present at once. It is simultaneously a sum over s-channel resonances and a sum over t-channel resonances, and the two sums are the same function.

Compare the field-theory habit. There, the s-channel exchange diagram and the t-channel exchange diagram are distinct Feynman graphs, computed separately and added. Here, if you wrote the amplitude once as its sum of s-channel poles and again as its sum of t-channel poles, and then added the two, you would double count: each expression is already the complete amplitude. This is Dolen-Horn-Schmid duality, the observation that resonances and Regge exchange are not independent contributions but two expansions of one object. Veneziano's formula was built to realise it.

The worldsheet says why. The four-punctured disc has one real modulus, the position x of Eq. (4.8). As $x \to 0$ the two operators at $y_1 = 0$ and $y_2 = x$ collide; on the surface this is a long thin strip growing out of the disc, a string of definite mass propagating between the (12) pair

and the (34) pair, and the amplitude factorises on s-channel states (Fig. 4.2). The endpoint $x = 0$ of the integral is where $x^{-\alpha(s)-1}$ produces, term by term in its Taylor expansion, the poles at $\alpha(s) = 0, 1, 2, \ldots$. As $x \to 1$ the operators at y_2 and y_3 collide, the strip grows the other way, and $(1-x)^{-\alpha(t)-1}$ generates the t-channel poles. One modulus, integrated once over $(0,1)$, passes continuously from the s-channel degeneration at one end to the t-channel degeneration at the other. The two channels are two limits of the same worldsheet, which is why summing over both would count the same surface twice. Duality here is forced by the topology of the disc, not by any accident of the gamma functions.

4.2.5 Exponential softness, and where the ultraviolet went

The last reading is the high-energy behaviour at fixed scattering angle, the hard-scattering limit: send $s, t, u \to \infty$ together with their ratios held fixed. It is cleanest to sit in the regime where the integral Eq. (4.14) genuinely converges, all Mandelstams deep below threshold, so that $a \coloneqq -\alpha(s)$ and $b \coloneqq -\alpha(t)$ are large and positive. The integrand $x^{a-1}(1-x)^{b-1}$ is then sharply peaked, with a saddle point at $x_* = a/(a+b)$. Evaluating the exponent there, or equivalently applying Stirling's formula $\ln \Gamma(z) \simeq z \ln z - z$ to Eq. (4.14),

$$\ln A(s,t) \;\simeq\; a \ln a + b \ln b - (a+b)\ln(a+b), \qquad a = -\alpha(s), \;\; b = -\alpha(t). \tag{4.22}$$

At the symmetric point $a = b$ this is $2a \ln a - 2a \ln 2a = -2a \ln 2 < 0$, and it stays negative and grows like the energy squared as $a, b \to \infty$. With $\alpha(s) \simeq \alpha' s$ at high energy, the magnitude falls as

$$|A| \;\sim\; \exp\!\big[-\alpha'\, E^2\, f(\theta)\big], \qquad f(\theta) > 0, \tag{4.23}$$

where $E^2 \sim s$ is the centre-of-mass energy squared and f a positive function of the fixed angle. The amplitude is Gaussian-soft: it decays faster than any power of the energy.

Set this against local quantum field theory, where a fixed-angle amplitude falls only as a power of E, set by the dimension of the interaction, and where individual loop integrals diverge in the ultraviolet and must be renormalised. The string does neither. The exponential in Eq. (4.23) is the short-distance smearing of an extended object: a string of size $\ell_s = \sqrt{\alpha'}$ cannot be probed below ℓ_s, a would-be point interaction is spread over the worldsheet, and the high-momentum region that produces ultraviolet divergences in field theory is simply absent. Already at tree level that softness is on display as the decay of a single beta function. At loop level the same finiteness returns as the modular invariance of Section 2.3.4, whose fundamental domain excises the short-distance corner $\tau \to 0$ of the torus. It is one physics throughout: there is no ultraviolet region on the string worldsheet in which to diverge.

Remark 4.3 (Beyond the lectures: the closed string and the Gross-Mende saddle). The closed-string four-tachyon amplitude runs on the sphere. Fixing the $\mathrm{PSL}(2,\mathbb{C})$ of Section 4.1.4 leaves one *complex* modulus, integrated over the whole plane, and the bulk Koba-Nielsen factor Eq. (2.30) turns the integral into a complex beta function. The result is the Virasoro-Shapiro amplitude,

$$A_{\mathrm{VS}}(s,t,u) \;\propto\; \frac{\Gamma\!\left(-1-\frac{\alpha' s}{4}\right)\Gamma\!\left(-1-\frac{\alpha' t}{4}\right)\Gamma\!\left(-1-\frac{\alpha' u}{4}\right)}{\Gamma\!\left(2+\frac{\alpha' s}{4}\right)\Gamma\!\left(2+\frac{\alpha' t}{4}\right)\Gamma\!\left(2+\frac{\alpha' u}{4}\right)}, \tag{4.24}$$

fully symmetric in s, t, u (a closed string carries no boundary ordering to break the symmetry), with poles at $\frac{\alpha'}{4} s = n-1$, that is $M^2 = 4(n-1)/\alpha'$, the closed-string spectrum of Eq. (1.87). Its leading trajectory has intercept 2: the graviton, spin 2 at $M^2 = 0$, sits at the bottom of it, the closed-string counterpart of the gauge boson on the open trajectory. The high-energy behaviour is again exponentially soft. Gross and Mende showed that the fixed-angle amplitude, at every order in the genus expansion, is dominated by a single saddle-point worldsheet, with leading behaviour $\exp[-\frac{\alpha'}{2}(s \ln \alpha' s + t \ln \alpha' t + u \ln \alpha' u)]$ up to phases; the coefficient of the exponent

is the area of a minimal surface, and the saddle is a stringy analogue of an instanton. That the saddle survives to all genus while the sum over genus does not converge is a preview of Section 4.4.

Remark 4.4 (Beyond the lectures: how little freedom there was). The rigidity here is easy to underestimate. Demand that a four-point function (i) have only simple poles at a linearly rising set of masses, (ii) have polynomial-bounded residues, so that the spins exchanged at each level are finite, and (iii) be crossing symmetric. The beta function is essentially the unique solution, and the string spectrum $M^2 \propto (N-1)$ with the tachyon at the bottom is read backwards from the pole positions rather than put in. This is the concrete sense in which the amplitude came first and the string second, and it is an early instance of the modern bootstrap: consistency conditions on amplitudes, taken seriously, can pin the dynamics.

Exercise 4.3 (From Mandelstams to poles). *(Standard.)* Using $\sum_i k_i = 0$ and the on-shell condition Eq. (4.9), verify the exponent identities Eq. (4.12) and the sum rule Eq. (4.11). Then locate the first three s-channel poles of $A(s,t)$ from Eq. (4.17), and confirm that they sit at the tachyon ($n=0$), the massless gauge boson ($n=1$) and the first massive level ($n=2$) of the open-string spectrum Eq. (1.88). Which pole is the amplitude's memory of the vacuum instability of Section 1.5.2?

Exercise 4.4 (Residues and spin). *(Standard.)* Expand Eq. (4.15) about $\alpha(s)=1$ and show the residue is linear in t, the partial-wave content of a single spin-1 exchange: the gauge boson. Repeat about $\alpha(s)=0$ and show the residue is constant in t, a spin-0 (tachyon) exchange. In general, from Eq. (4.18), argue that the residue at $\alpha(s)=n$ is a degree-n polynomial in t and so contains spins up to n, giving the leading trajectory Eq. (4.19). *Hint:* t is linear in $\cos\theta$ at fixed s, and a degree-n polynomial in $\cos\theta$ is a sum of Legendre polynomials $P_0, \dots, P_n$.

Exercise 4.5 (Duality by partial fractions). *(Moderate.)* Write $A(s,t) = \sum_{n\geq 0} \dfrac{R_n(t)}{n-\alpha(s)}$ as a sum over its s-channel poles, reading $R_n(t)$ off Eq. (4.18). Argue that this single sum, resummed, already reproduces the t-channel poles of A, so that adding a separate t-channel sum would double count. Match the two ends $x \to 0$ and $x \to 1$ of the modulus integral Eq. (4.8) to the two expansions, and hence to the two degenerations of Fig. 4.2.

Exercise 4.6 (Softness from Stirling). *(Standard.)* Take the beta-function form Eq. (4.14) in the convergent regime, with $a = -\alpha(s)$ and $b = -\alpha(t)$ large and positive, and apply Stirling's formula $\ln\Gamma(z) \simeq z\ln z - z$ to derive the saddle exponent Eq. (4.22). Specialise to the symmetric point $a=b$, show that $\ln|A| \simeq -2a\ln 2$, and, with $\alpha(s) \simeq \alpha' s$ at high energy, recover the Gaussian-soft falloff Eq. (4.23). Compare with the power-law fixed-angle behaviour a renormalisable field theory would give, and say in one line where the string's extra suppression comes from.

Exercise 4.7 (The Virasoro-Shapiro integral). *(Beyond the lectures.)* Set up the closed-string four-tachyon amplitude on $\mathbb{CP}^1$. Fix $z_1, z_3, z_4 = 0, 1, \infty$ with the $\mathrm{PSL}(2,\mathbb{C})$ of Section 4.1.4, use the bulk Koba-Nielsen factor Eq. (2.30), and reduce the amplitude to the complex integral

$$\int_{\mathbb{C}} \mathrm{d}^2 z\, |z|^{\alpha' k_1\cdot k_2}\, |1-z|^{\alpha' k_2\cdot k_3}.$$

Quoting the complex-beta identity from Appendix F, obtain Eq. (4.24), and read off the closed-string mass spectrum and the intercept-2 (graviton) leading trajectory.

4.3 The $\alpha' \to 0$ limit: spacetime emerges

Every amplitude in this chapter depends on the external kinematics only through the dimensionless products $\alpha' s$, $\alpha' t$, $\alpha' u$. The worldsheet allows nothing else. On it α' is the only

dimensionful constant, and a scattering amplitude is a pure number, so an external momentum can enter only through the combination $\alpha' k^2$. The low-energy regime is the one in which every external momentum is small in string units,

$$\alpha' k^2 \ll 1 \qquad \Longleftrightarrow \qquad E \ll M_{\rm s} = 1/\ell_s, \tag{4.25}$$

and we reach it cleanly by holding the physical momenta fixed and sending $\alpha' \to 0$. Something survives that limit. What survives is an ordinary quantum field theory of the massless string states, and this section is about computing which one. For the open string the answer is Yang-Mills; for the closed string it is Einstein gravity, packaged with the B-field and the dilaton. Both answers are read straight off the three-point functions, where the couplings live and where, because three punctures carry no moduli (Section 4.1), no integral stands between the worldsheet correlator and the spacetime vertex.

This is the first of two places in the course where gravity falls out of the sum over surfaces. Here it appears as a *state*: the graviton is one massless excitation among the string's vibrations, and its on-shell scattering is pinned, by the worldsheet, to exactly what general relativity predicts. In Chapter 5 it will return as a *law*, when Weyl invariance of the nonlinear sigma model forces the Einstein equation on the background the string propagates in. On-shell here, off-shell there. The distinction is worth holding onto, because it is why the same physics earns two independent derivations, and why neither on its own is the whole story.

4.3.1 Decoupling the tower

Start from the four-point function of Section 4.2, the Veneziano amplitude

$$A(s,t) \;=\; B\big(-\alpha(s), -\alpha(t)\big) \;=\; \frac{\Gamma(-\alpha(s))\,\Gamma(-\alpha(t))}{\Gamma(-\alpha(s)-\alpha(t))}, \qquad \alpha(x) = 1 + \alpha' x, \tag{4.26}$$

with the linear trajectory $\alpha(x)$ of Eq. (4.13). Its poles sit where $\alpha(s) = n$ for $n = 0, 1, 2, \ldots$, that is at $\alpha' s = n - 1$, one pole for every state in the open-string tower, $M_n^2 = (n-1)/\alpha'$. Send $\alpha' \to 0$ at fixed s. Every pole above the massless level, $n \geq 2$, runs off to $s = (n-1)/\alpha' \to \infty$: the massive string states become infinitely heavy and decouple, one after another, exactly as heavy fields decouple from low-energy physics in any Wilsonian effective theory. What is left at finite s is the exchange of the lightest states. In the bosonic string that set still includes the $n = 0$ tachyon, an embarrassment we will have to name; in a GSO-projected superstring (Chapter 3) the tachyon is gone and the lightest exchanged state is massless. The residue at the surviving massless pole factorises, by unitarity, into a product of two three-point couplings. That is the structural reason the three-point data fix the leading low-energy amplitude completely: everything else is either a heavy state that has decoupled or a contact term suppressed by α'.

Organise the survivors as a Taylor expansion in α'. Each amplitude becomes

$$A \;=\; A^{(0)} \;+\; \alpha'\, A^{(1)} \;+\; \alpha'^2\, A^{(2)} \;+\; \cdots, \tag{4.27}$$

where $A^{(0)}$ is a two-derivative field-theory amplitude of the massless fields and the higher terms are matrix elements of higher-derivative operators, the α'-corrections to the effective action. The rest of this section identifies $A^{(0)}$ in each sector and names the first correction $A^{(1)}$.

One bookkeeping point before the calculation. The vertex-operator normalisations, call them g_o for the open string and g_c for the closed, are not the physical couplings, and by dimensional analysis they carry α'-dependent factors. The finite quantities that survive (4.25) are the Yang-Mills coupling $g_{\rm YM}$ and the gravitational coupling κ; the map from (g_o, g_c) to $(g_{\rm YM}, \kappa)$ is fixed by unitarity, that is by demanding that the residues on the massless poles factorise correctly. We quote that dictionary where it is needed and collect the gravitational value in Appendix A. Throughout, ε denotes the state's polarisation, the object written e in the vertex-operator table of Chapter 2.

Exercise 4.8 (Decoupling and the α' counting)**.** *(Warm-up. Goal: watch the tower decouple and the low-energy expansion appear.)* Take the Veneziano amplitude (4.26) with the open-string trajectory $\alpha(s) = 1 + \alpha' s$, so the poles sit at $\alpha' s = n - 1$. Show that at fixed s and $\alpha' \to 0$ every pole with $n \geq 2$ leaves the finite-s region, and that the two lightest are the tachyon ($n = 0$) and the massless gauge boson ($n = 1$). Expand $A(s,t)$ to first nontrivial order in α' about the massless pole and identify the leading term as a massless exchange plus a contact term. Comment on why the $n = 0$ tachyon makes this expansion untrustworthy in the bosonic string, and what the GSO projection of Chapter 3 does about it.

4.3.2 The open string gives Yang-Mills

The massless open-string state is a gauge boson. Its vertex operator follows from the state-operator map of Chapter 2, now applied on the boundary of the disc rather than in the bulk of the sphere: it is the tachyon exponential dressed with the tangential derivative that raises the boundary weight from $\alpha' k^2$ to $\alpha' k^2 + 1$,

$$V \;=\; g_o\, \lambda^a \int \mathrm{d}x\, :\varepsilon_\mu\, \dot{X}^\mu\, e^{ik\cdot X}:(x), \qquad k^2 = 0, \quad \varepsilon \cdot k = 0, \tag{4.28}$$

integrated along the boundary. Here $\dot{X} = \partial_x X$ is the derivative tangent to the boundary, λ^a is the Chan-Paton generator that carries the colour (Section 1.5.4, with the matrix labels of Eq. (1.95)), and the two on-shell conditions are the content of weight one. The exponential $:e^{ik\cdot X}:$ contributes $\alpha' k^2$ to the boundary weight and $\dot{X}$ contributes one, so weight one forces $k^2 = 0$, while transversality $\varepsilon \cdot k = 0$ removes the longitudinal state, which is null and pure gauge.

Three gauge bosons put three punctures on the boundary of the disc, and the residual $\mathrm{PSL}(2,\mathbb{R})$ conformal symmetry fixes all three positions with no leftover modulus (Section 4.1). We may therefore drop the position integrals, fix x_1, x_2, x_3, and dress the insertions with the c-ghost factors that soak up the three conformal Killing zero modes. The amplitude collapses to a single boundary correlator,

$$\Big\langle \textstyle\prod_{i=1}^{3}\, \varepsilon_i \cdot \dot{X}(x_i)\; e^{ik_i\cdot X(x_i)} \Big\rangle, \tag{4.29}$$

evaluated with the boundary-to-boundary propagator $\langle X^\mu(x) X^\nu(y)\rangle = -2\alpha'\,\eta^{\mu\nu}\ln|x-y|$ of Section 4.2. Two facts make the evaluation short. First, for three massless particles every Mandelstam invariant vanishes, since $2k_i \cdot k_j = (k_i + k_j)^2 = k_l^2 = 0$, so the Koba-Nielsen factor $\prod |x_{ij}|^{2\alpha' k_i\cdot k_j}$ is unity and no position-dependent exponent survives. Second, the three tangential derivatives must be removed by Wick contractions, and there are only two ways to do it:

- one pair $\varepsilon_i \cdot \dot{X}$ with $\varepsilon_j \cdot \dot{X}$ contracts among themselves to $\alpha'\, \varepsilon_i \cdot \varepsilon_j / x_{ij}^2$, while the third $\varepsilon_l \cdot \dot{X}$ contracts with the exponentials to give $\alpha'\, \varepsilon_l \cdot k_j / x_{lj}$ summed over the other two insertions $j \neq l$: two powers of α', one of momentum;
- all three $\varepsilon_i \cdot \dot{X}$ contract with exponentials, giving $\alpha'^3 \prod_i \varepsilon_i \cdot k$: three powers of α', three of momentum.

The second pattern is down by one factor of $\alpha' k^2$ relative to the first, so it is the leading α'-correction rather than part of the leading amplitude. Collecting the position factors against the c-ghost correlator, which supplies $\prod |x_{ij}|$ and renders the result independent of where we fixed the punctures, as $\mathrm{PSL}(2,\mathbb{R})$ invariance demands, the amplitude for one cyclic ordering is

$$A_3 \;=\; i\, g_o\, (2\pi)^D \delta^D\Big(\textstyle\sum_i k_i\Big)\; \mathrm{Tr}(\lambda^{a_1}\lambda^{a_2}\lambda^{a_3})\, \mathcal{N}_{123} \;+\; (\text{reversed ordering}), \tag{4.30}$$

with the kinematic numerator

$$\mathcal{N}_{123} \;=\; \varepsilon_1{\cdot}\varepsilon_2\, \varepsilon_3{\cdot}k_{12} + \varepsilon_2{\cdot}\varepsilon_3\, \varepsilon_1{\cdot}k_{23} + \varepsilon_3{\cdot}\varepsilon_1\, \varepsilon_2{\cdot}k_{31} + \frac{\alpha'}{2}\, \varepsilon_1{\cdot}k_{23}\; \varepsilon_2{\cdot}k_{31}\; \varepsilon_3{\cdot}k_{12}, \qquad k_{ij} \coloneqq k_i - k_j. \tag{4.31}$$

The first three terms are the two-derivative piece; the last, cubic in momenta, is the $\mathcal{O}(\alpha')$ correction the contraction counting promised.

Summing the two boundary orderings turns the colour factor into a commutator on the antisymmetric part and an anticommutator on the symmetric part: $\mathrm{Tr}\,(\lambda^{a_1}[\lambda^{a_2},\lambda^{a_3}]) \propto f^{a_1a_2a_3}$ and $\mathrm{Tr}\,(\lambda^{a_1}\{\lambda^{a_2},\lambda^{a_3}\}) \propto d^{a_1a_2a_3}$. The antisymmetric structure constant multiplies the two-derivative numerator; the symmetric d-symbol multiplies the cubic term. On the two-derivative piece the on-shell conditions simplify k_{ij}: since $\varepsilon_3\cdot k_3 = 0$ and $\sum_i k_i = 0$ give $\varepsilon_3\cdot k_{12} = \varepsilon_3\cdot(k_1-k_2) = 2\,\varepsilon_3\cdot k_1$, and cyclically for the rest, the antisymmetric part reduces to

$$A_3^{(0)} \;\propto\; g_o\, f^{a_1a_2a_3}\Big[\,\varepsilon_1{\cdot}\varepsilon_2\;\varepsilon_3{\cdot}k_1+\varepsilon_2{\cdot}\varepsilon_3\;\varepsilon_1{\cdot}k_2+\varepsilon_3{\cdot}\varepsilon_1\;\varepsilon_2{\cdot}k_3\,\Big]. \tag{4.32}$$

Now the field theory. Non-abelian Yang-Mills with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + g_{\mathrm{YM}} f^{abc} A^b_\mu A^c_\nu$ and $\mathcal{L} = -\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$ has the cubic term $-g_{\mathrm{YM}} f^{abc}(\partial_\mu A^a_\nu)A^{b\,\mu}A^{c\,\nu}$. Its momentum-space vertex, contracted with three transverse polarisations and reduced on shell, is

$$V_3^{\mathrm{YM}} \;=\; -2i\, g_{\mathrm{YM}}\, f^{a_1a_2a_3}\Big[\,\varepsilon_1{\cdot}\varepsilon_2\;\varepsilon_3{\cdot}k_1+\varepsilon_2{\cdot}\varepsilon_3\;\varepsilon_1{\cdot}k_2+\varepsilon_3{\cdot}\varepsilon_1\;\varepsilon_2{\cdot}k_3\,\Big], \tag{4.33}$$

identical in structure to (4.32), the same on-shell reduction from k_{ij} to $2k_i$ having been used on both sides. The disc has handed us the three-gluon vertex of Yang-Mills, with the string normalisation g_o mapping onto g_{YM}; the precise dictionary, including the power of α' that makes g_{YM} dimensionless in the relevant dimension, is fixed by unitarity when the open-string gauge theory on D-branes is assembled in Chapter 6.

The $\mathcal{O}(\alpha')$ term in (4.31) is not junk to be discarded. The symmetric d^{abc} cubic vertex is the three-point matrix element of a $\mathrm{Tr}\,F^3$ operator, the leading higher-derivative correction to Yang-Mills in the open bosonic string. In the superstring this cube is absent and the first correction is a $\mathrm{Tr}\,F^4$; resumming the whole abelian open-string tree series then turns $-\frac{1}{4}F^2$ into the Born-Infeld Lagrangian $\sqrt{-\det(\eta_{\mu\nu}+2\pi\alpha' F_{\mu\nu})}$, whose non-abelian and curved-space completion is the leading part of the D-brane worldvolume action of Chapter 6. The low-energy limit hands us Yang-Mills together with the first entries of its α'-expansion, both from the same disc correlator.

Exercise 4.9 (The gauge-boson numerator from the boundary correlator)**.** *(Standard. Goal: derive* (4.31) *from the two contraction patterns.)* From the boundary propagator $\langle X^\mu(x)X^\nu(y)\rangle = -2\alpha'\,\eta^{\mu\nu}\ln|x-y|$, deduce the two building blocks $\langle \dot X^\mu(x_i)\,\dot X^\nu(x_j)\rangle = -2\alpha'\,\eta^{\mu\nu}/x_{ij}^2$ and $\langle \dot X^\mu(x_i)\,e^{ik_j\cdot X(x_j)}\rangle = -2i\alpha'\,k_j^\mu/x_{ij}$ (the latter meaning the contraction of $\dot X$ with that exponential inside the correlator). Apply them to (4.29). Show that the pattern in which one $\varepsilon\cdot\dot X$ pair self-contracts feeds the three two-derivative terms of $\mathcal{N}_{123}$, and that the pattern in which all three contract with exponentials feeds the cubic term, with the relative $\frac{\alpha'}{2}$ fixed by the momentum factors. Use $2k_i\cdot k_j = 0$ to set the Koba-Nielsen exponent to zero throughout.

Exercise 4.10 (The Yang-Mills vertex from the Lagrangian)**.** *(Standard. Goal: reproduce* (4.33) *and see the on-shell reduction.)* Expand $\mathcal{L} = -\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$ to cubic order in A and read off the three-gluon vertex $V^{abc}_{\mu\nu\rho}(k_1,k_2,k_3) = -g_{\mathrm{YM}} f^{abc}\big[\eta_{\mu\nu}(k_1-k_2)_\rho+\eta_{\nu\rho}(k_2-k_3)_\mu+\eta_{\rho\mu}(k_3-k_1)_\nu\big]$, all momenta incoming. Contract with $\varepsilon_1^\mu\varepsilon_2^\nu\varepsilon_3^\rho$, impose $\varepsilon_i\cdot k_i = 0$ and $\sum_i k_i = 0$, and obtain (4.33). Confirm that the same steps carry the string numerator (4.31) to (4.32), so the two agree term by term at leading order in α'.

4.3.3 The closed string gives Einstein

The massless closed-string state is a two-index tensor built from one left-moving and one right-moving oscillator (Chapter 2). Its vertex operator sits at an interior point of the sphere and is

integrated over it,

$$V \;=\; g_c\,\varepsilon_{\mu\nu}\int \mathrm{d}^2z\,:\partial X^\mu\,\bar\partial X^\nu\,e^{ik\cdot X}:(z,\bar z), \qquad k^2=0, \quad k^\mu\varepsilon_{\mu\nu}=0, \tag{4.34}$$

of weight $(1,1)$, hence $k^2=0$, with transversality on each index. The same exponential that carried boundary weight $\alpha' k^2$ in (4.28) carries only $\alpha' k^2/4$ on each index here; the factor of four is a factor of two from the boundary propagator being doubled to $-2\alpha'\ln|y-y'|$ by the method of images, times another from the bulk weight splitting across the left- and right-movers. The polarisation tensor decomposes into the pieces already met in the massless spectrum: its symmetric traceless part is the graviton, its antisymmetric part the Kalb-Ramond field $B_{\mu\nu}$, and its trace the dilaton. Keep the symmetric traceless piece for graviton scattering; the rest ride along for free, and we return to them.

Three gravitons put three punctures on the sphere, and $\mathrm{PSL}(2,\mathbb{C})$ fixes all three with no modulus (Section 4.1), the same three-point simplification as on the disc. The structural fact that makes the closed answer follow from the open one is that the free-field correlator on the sphere splits into a holomorphic and an antiholomorphic half, because ∂X contracts only with ∂X and $\bar\partial X$ only with $\bar\partial X$. On shell the polarisation splits the same way, $\varepsilon_{\mu\nu}=\varepsilon^{\mathrm{L}}_\mu\,\varepsilon^{\mathrm{R}}_\nu$, and the three-graviton correlator becomes a product of two copies of the open-string numerator (4.31),

$$A_3^{\text{closed}} \;=\; i\,\kappa\,(2\pi)^D\delta^D\Big(\textstyle\sum_i k_i\Big)\,\mathcal{N}^{\mathrm{L}}_{123}\,\mathcal{N}^{\mathrm{R}}_{123}, \qquad \mathcal{N}^{\mathrm{L}}\text{ from }\varepsilon^{\mathrm{L}}_i, \quad \mathcal{N}^{\mathrm{R}}\text{ from }\varepsilon^{\mathrm{R}}_i, \tag{4.35}$$

with κ absorbing the g_c normalisation and the sphere weighting. This is the three-point instance of the slogan "gravity is the square of gauge theory": the holomorphic factorisation of the closed string is the worldsheet origin of the KLT relations, and at three points it says the graviton amplitude is literally the Yang-Mills numerator squared. At leading order in α' the product $\mathcal{N}^{\mathrm{L}}\mathcal{N}^{\mathrm{R}}$ is the cubic graviton vertex you get by expanding the Einstein-Hilbert action to third order in $h_{\mu\nu}=g_{\mu\nu}-\eta_{\mu\nu}$. That vertex is a polynomial with on the order of a hundred index contractions, so we do not write it out; the assertion is that the direct spacetime computation agrees with (4.35) term by term, a check first carried through in full by Sannan, and the squaring makes the agreement structural rather than a coincidence to be marvelled at. The worldsheet has produced the graviton self-coupling of general relativity.

Matching the overall normalisation of (4.35) to the Einstein-Hilbert action $\frac{1}{2\kappa^2}\int\sqrt{-g}\,R$ fixes κ in terms of g_c and α'; combined with the Euler-number weighting that ties g_c to the string coupling g_s (Section 4.1), this reproduces the ten-dimensional value

$$2\kappa_{10}^2 \;=\; (2\pi)^7\,\alpha'^4\,g_s^2 \tag{4.36}$$

recorded in Eq. (A.6). That number is written for the ten-dimensional superstring, with the tachyon projected out; the algebra above ran in the bosonic $D=26$ setting, but the vertex structure is dimension-agnostic and only the numerical normalisation is specific to the superstring. The other polarisations of $\varepsilon_{\mu\nu}$ sit on the same footing, so the low-energy closed string is not pure gravity. The graviton arrives bundled with the B-field and the dilaton, and their three-point couplings are the cubic vertices of the universal NS-NS action

$$S_{\mathrm{NS}} \;=\; \frac{1}{2\kappa^2}\int \mathrm{d}^Dx\,\sqrt{-g}\,e^{-2\phi}\Big(R+4(\partial\phi)^2-\tfrac{1}{12}|H|^2\Big), \tag{4.37}$$

with $H=\mathrm{d}B$ the field strength of the Kalb-Ramond field and $|H|^2=H_{\mu\nu\rho}H^{\mu\nu\rho}$. Its full off-shell form is derived, by the completely different route of σ-model Weyl invariance, in Chapter 5. Reading the cubic couplings off the S-matrix here and deriving the whole action from the vanishing of the beta functions there are two views of one object, and the check that they match is the subject of Chapter 5.

The first α'-correction to (4.37) is already visible at four points. The four-graviton amplitude on the sphere is the Virasoro-Shapiro amplitude, which for massless external states takes the form

$$A_4 \;=\; \mathcal{C}\,\kappa^2\,\mathcal{K}\,\frac{\Gamma(-\frac{\alpha'}{4}s)\,\Gamma(-\frac{\alpha'}{4}t)\,\Gamma(-\frac{\alpha'}{4}u)}{\Gamma(1+\frac{\alpha'}{4}s)\,\Gamma(1+\frac{\alpha'}{4}t)\,\Gamma(1+\frac{\alpha'}{4}u)}, \qquad s+t+u=0, \tag{4.38}$$

the massless cousin of Eq. (4.24), with $\mathcal{K}$ the degree-eight kinematic factor that carries the four-graviton polarisation structure (the $t_8t_8R^4$ tensor). Using the identity

$$\frac{\Gamma(-x)\Gamma(-y)\Gamma(-z)}{\Gamma(1+x)\Gamma(1+y)\Gamma(1+z)} = -\frac{1}{xyz}\,\exp\Big[\,2\sum_{n\geq 1}\frac{\zeta(2n+1)}{2n+1}\big(x^{2n+1}+y^{2n+1}+z^{2n+1}\big)\Big], \tag{4.39}$$

valid on the surface $x+y+z=0$, with $x=\frac{\alpha'}{4}s$ and cyclically, the low-energy expansion of (4.38) organises itself. The prefactor $-1/(xyz) = -(4/\alpha')^3/(stu)$ is the massless graviton exchanged in the three channels, precisely the field-theory answer built from (4.37). The leading term in the exponential is of relative order $\zeta(3)\,\alpha'^3\,stu$, and it is the R^4 operator. For the type II superstring α'^3R^4 is genuinely the first correction to Einstein gravity, the candidate four- and six-derivative terms (R^2 at order α', R^3 at order α'^2) being absent or removable by field redefinition; for the bosonic and heterotic strings a curvature-squared term already appears at order α'. We assert (4.38) and (4.39) rather than derive them: (4.38) is a sphere four-point computation of exactly the type set up in Section 4.2 and Appendix F, and (4.39) is the exercise below.

4.3.4 What a limit of the S-matrix can and cannot give you

We have fixed on-shell three-point amplitudes of the massless fields and named their first α'-corrections. That is real, coupling-level physics, and it determines the two-derivative action (4.37) up to the ambiguities any S-matrix leaves. Be honest about those ambiguities. On-shell amplitudes are invariant under field redefinitions and blind to operators that vanish on the equations of motion, so scattering data fix the effective action only up to such terms. An S-matrix is not an action: it hands you the couplings evaluated on physical external states, not the off-shell functional whose variation is the field equation. Two Lagrangians related by $g_{\mu\nu} \to g_{\mu\nu} + a\,R_{\mu\nu} + \cdots$ have the same S-matrix and are, from this vantage, the same theory.

That is precisely why the effective action deserves a second, independent derivation, and gets one in Chapter 5, where demanding that the worldsheet sigma model stay Weyl invariant produces the spacetime field equations off shell, as genuine equations of motion rather than as a fit to scattering data. It is also the first appearance of a limitation we take up properly in Section 4.4: the first-quantised recipe computes on-shell amplitudes and nothing else, and recovering the off-shell content of the theory is one of the things string field theory (Section 4.5) is built to do.

Two ways gravity emerges

Gravity comes out of the worldsheet twice, and the two are logically different.

- **As a state (this section).** The graviton is a massless string excitation. Its on-shell scattering, computed from the sum over surfaces in the $\alpha' \to 0$ limit, is the Einstein-Hilbert three-point (and higher) vertex, with α'-corrections starting at R^4 for the type II string. This fixes the S-matrix of the massless sector, and through it the two-derivative action up to field redefinition.
- **As a law (Chapter 5).** Consistency of the string in a curved background, that is Weyl invariance of the sigma model, forces the background to satisfy $R_{\mu\nu}+2\nabla_\mu\nabla_\nu\phi+\cdots=0$. This is the Einstein equation itself, holding off shell as an equation of motion, not merely its on-shell scattering shadow.

The two must agree where they overlap, and they do: the beta-function equations of

Chapter 5 are the equations of motion of the same action (4.37) whose cubic vertices we read off here.

Exercise 4.11 (Graviton polarisation as a square)**.** *(Standard. Goal: make the "gravity = gauge^2" factorisation concrete.)* Take a graviton of momentum k ($k^2 = 0$) and write $\varepsilon_{\mu\nu} = \varepsilon^{\mathrm{L}}_{\mu}\varepsilon^{\mathrm{R}}_{\nu}$ with $\varepsilon^{\mathrm{L}} \cdot k = \varepsilon^{\mathrm{R}} \cdot k = 0$. Show that the symmetric traceless, antisymmetric, and trace parts of $\varepsilon_{\mu\nu}$ reproduce the graviton, B-field, and dilaton, and that transversality $k^{\mu}\varepsilon_{\mu\nu} = 0$ follows from the two vector conditions. Using (4.35), verify that the three-graviton amplitude is symmetric under permutations of the three particles, and count powers of momentum to confirm it is a two-derivative vertex, as the Einstein-Hilbert action requires.

Exercise 4.12 (The first α'-correction)**.** *(Harder. Goal: extract the R^4 coefficient from the Virasoro-Shapiro amplitude.)* Verify the identity (4.39) by expanding $\ln\Gamma(1+x) = -\gamma x + \sum_{k\geq 2}(-1)^k \frac{\zeta(k)}{k} x^k$, writing $\Gamma(-x) = \Gamma(1-x)/(-x)$, and using $x + y + z = 0$ to kill the Euler-γ and even-power terms. With $x = \frac{\alpha'}{4}s$ and cyclic, expand (4.38) to the first two orders: show that the leading term is $-(4/\alpha')^3\,\mathcal{K}/(stu)$, the massless exchange, and that the first correction is the R^4 contact term with coefficient proportional to $\zeta(3)\,\alpha'^3$. Explain why $x^3 + y^3 + z^3 = 3xyz$ when $x + y + z = 0$, and why that is what makes the correction start at cubic order in the Mandelstam invariants.

The two low-energy limits are now on the table: the open string has handed us Yang-Mills and the closed string Einstein gravity, each with the leading term of its α'-expansion attached. Both were read off three-point functions that carry no moduli, so no worldsheet integral stood in the way. What we have not produced is an action we can trust off shell, and we have been careful to say so. The next section makes the boundaries of the first-quantised recipe explicit, and the one after that shows what string field theory does to move past them.

4.4 Where first quantisation breaks down

The recipe of Section 4.1 is an efficient machine. Fix the external momenta on shell, fix the genus, and it returns a number: an integral over the moduli space of the surface of a correlator of vertex operators. Iterate in genus and the numbers assemble into a power series in g_s. Worked once on the sphere the machine gave Veneziano (Section 4.2); expanded at low energy it handed us the Yang-Mills and Einstein vertices (Section 4.3). It is tempting to mistake the machine for the theory. It is not: it is a procedure for generating the terms of a series, and a series is not a function.

This section is the honest accounting: five things the sum over surfaces cannot do. They are not mistakes waiting to be fixed by a cleverer calculation. Each is a structural limit of computing a scattering matrix, order by order in g_s, about a background that was chosen once and for all when the worldsheet action was written down. Setting them out is also the motivation for what comes next, because each is repaired by passing to a second-quantised description, a field theory whose field is the string.

Five silences of the sum over surfaces

The first-quantised prescription is silent, or wrong, in five places.

1. **The genus expansion is asymptotic.** The series $\sum_g A_g\, g_s^{2g-2}$ has zero radius of convergence. It defines the amplitude order by order and no further.
2. **External states are on shell only.** A vertex operator must be a weight-$(1,1)$ primary, which is the mass-shell condition. There is no worldsheet object that continues an external leg off shell, hence no Green's functions and no effective potential.

3. **Tadpoles force a shift of the background.** A nonzero massless one-point function signals that the chosen vacuum does not solve its own equations of motion, and produces an infrared divergence the formalism cannot cure from within.
4. **Mass renormalisation needs contour deformation.** Once masses run with g_s, the physical pole no longer sits at the tree-level mass, and the amplitude is only defined by deforming the integration contour in moduli space.
5. **Effects of order e^{-C/g_s} are invisible.** No sum over smooth surfaces of any genus can see a D-instanton; these effects have vanishing Taylor coefficients to all orders in g_s.

The cure for all five is the string field theory of Section 4.5.

4.4.1 The genus expansion is asymptotic

Start with a genuine triumph, so that the failure lands in the right place. Each individual genus-g amplitude is ultraviolet finite. Modular invariance, the statement that the integral runs over the moduli space of the surface and not over its Teichmüller cover, cuts off the region that in field theory would be the ultraviolet, and the exponential softness of the Veneziano amplitude (Section 4.2.5) is the tree-level face of the same fact. The string has no ultraviolet divergences to renormalise. That is not in question here.

What fails is the sum over g. Write a fixed amplitude, or the vacuum free energy, as

$$A(g_s) \;\sim\; \sum_{g\geq 0} A_g\, g_s^{2g-2}, \tag{4.40}$$

the genus counting of Eqs. (1.97) and (1.98). The coefficients A_g do not stay bounded; they grow factorially with the genus,

$$A_g \;\sim\; (2g)!\, C^{-2g} \qquad (g\to\infty), \tag{4.41}$$

for some positive constant C set by the theory. This is asserted, not derived here. It was established for the $c<1$ strings by their exact matrix-model solutions (Gross and Periwal), argued to hold for the critical string by Shenker, and it can be traced to the growth of the volume of moduli space itself: the Weil-Petersson volume of $\mathcal{M}_g$ grows like $(2g)!$ at large g (Mirzakhani and others, up to exponential and power-law factors). The factorial is geometric in origin, a proliferation of inequivalent surfaces, and has nothing to do with ultraviolet subdivergences. It is worse than the $g!$ of a field-theory loop expansion, and it is there even though every term it multiplies is finite.

A series whose coefficients grow like $(2g)!$ has zero radius of convergence. It is asymptotic in the sense of Poincaré: for small g_s the partial sums first approach the true answer and then, past an optimal order, run away. The best one can do is truncate where the terms stop shrinking. That optimal term sets the intrinsic ambiguity of the whole series. With A_g as in Eq. (4.41), the term $A_g\, g_s^{2g}$ is smallest near $2g_\star \simeq C/g_s$, and its size there is

$$\min_g \left|A_g\, g_s^{2g}\right| \;\sim\; e^{-C/g_s}. \tag{4.42}$$

Two functions with the same asymptotic expansion (4.40) can differ by anything of order e^{-C/g_s}, because such a difference has vanishing Taylor coefficients at $g_s=0$ and contributes nothing to any A_g. The perturbative series fixes the amplitude only up to a term the size of (4.42). That term is physical: its exponent C/g_s is the action of a D-instanton, and Section 4.4.5 returns to it. For now the lesson is narrow and firm: the sum over surfaces defines the theory order by order in g_s and not beyond.

4.4.2 On shell only: there is no off-shell continuation

An external string enters an amplitude as a vertex operator, and a vertex operator is admissible only if it is a weight-$(1,1)$ primary (Section 2.2), equivalently a BRST-closed insertion built on the cohomology of Section 1.4.6. That single requirement is the on-shell condition. For the (closed-string) tachyon the weight is $(h,\bar{h}) = (\alpha' k^2/4,\, \alpha' k^2/4)$ from Eq. (2.18), so marginality fixes k^2; for the massless operator it forces $k^2 = 0$ together with transversality (Theorem 2.5). The open-string tachyon of Section 4.2 sits on the boundary and instead carries weight $\alpha' k^2$; the factor of four between the two is the ratio of the boundary propagator coefficient $-2\alpha'$ to the bulk $-\alpha'/2$, and it is real, not a typo. Momentum enters the worldsheet only through $:e^{ik\cdot X}:$, and the weight of that exponential is rigidly tied to k^2. You cannot dial k^2 off its on-shell value and keep a sensible insertion.

Try it and watch what breaks. Give an external leg an off-shell momentum, so that its operator has weight $(h,\bar{h}) \neq (1,1)$. The integrated insertion $\int \mathrm{d}^2 z\, \mathcal{V}(z,\bar{z})$ is then not conformally invariant: under a rescaling $z \to \lambda z$ it is multiplied by $|\lambda|^{2-2h} \neq 1$, so the number it computes depends on the coordinate one happened to use on the worldsheet. That is not the coordinate-independent quantity an amplitude must be, and on the BRST side it is no longer annihilated by Q, so it does not descend to a well-defined object on the cohomology at all. The worldsheet has no room for an off-shell external state.

Behind the technical failure is a fact about what first quantisation computes. It computes an S-matrix element, $\langle \mathrm{out} \,|\, \mathrm{in} \rangle$ between asymptotic on-shell states, and an S-matrix is on shell by construction. There is no worldsheet correlator that plays the role of an off-shell Green's function $\langle \Phi(x_1) \cdots \Phi(x_n) \rangle$ of local spacetime fields, because the formalism never introduces those local fields; it introduces their asymptotic quanta. Several things one wants in field theory are therefore simply absent. There is no effective potential $V(\Phi)$ to extremise, so the formalism cannot be asked where its vacuum sits. There is no one-particle-irreducible effective action, no off-shell current, no propagator to amputate. The spacetime fields $\Phi(x)$ whose action we assemble in Chapter 5 are reconstructed by matching their on-shell amplitudes to those the worldsheet delivers; they are inferred from the S-matrix, not read off a worldsheet correlator. Section 4.3.4 made the same point from the side of the effective action, where the residual field-redefinition ambiguity is its shadow.

Remark 4.5 (Beyond the lectures: off shell is a choice, not a formula). One can define off-shell string amplitudes in first quantisation, but not uniquely. Extending a vertex operator off shell requires choosing a local coordinate around each puncture, and the answer depends on that choice. On shell the dependence drops out, since a weight-$(1,1)$ primary transforms without an anomalous factor, which is exactly why the on-shell S-matrix is unambiguous. Off shell the coordinate choice is genuine data. Fixing it consistently across all surfaces, so that the pieces sew together into a single amplitude, is precisely the combinatorial content of string field theory: the choice of local coordinates is the choice of interaction vertices, and different consistent choices are related by field redefinitions. This is the viewpoint developed by Sen and Zwiebach, and it is the sense in which Section 4.5 supplies the missing off-shell continuation.

4.4.3 Tadpoles and the shift of the background

The remaining failures live at the boundary of moduli space, where a surface degenerates. Write down once the object that controls that boundary, because it does the work for both this subsection and the next.

When a Riemann surface develops a long thin neck, the neck is a cylinder of some length ℓ, and sewing across it inserts the closed-string propagator. In the operator language of Chapter 1 a cylinder of length ℓ acts as $e^{-\ell(L_0+\bar{L}_0)}$, dressed by the ghost zero modes $b_0\bar{b}_0$ from the measure,

and integrating over the length gives

$$\int_0^\infty \mathrm{d}\ell\; e^{-\ell(L_0+\bar{L}_0)} \;=\; \frac{1}{L_0+\bar{L}_0}. \tag{4.43}$$

On an intermediate state of mass M and momentum p flowing through the neck, $L_0+\bar{L}_0 = \frac{\alpha'}{2}(p^2+M^2)$, so (4.43) is the propagator $2/\alpha'(p^2+M^2)$. The neck length ℓ is a Schwinger proper time, and the degeneration limit $\ell\to\infty$ is the region of moduli space where that propagator goes on shell. The integral (4.43) converges at $\ell\to\infty$ only when $p^2+M^2>0$, that is only when the internal line is off its mass shell. Everything below is a statement about what happens when it is not.

Now the tadpole. A one-point function of a massless field on a surface of nonzero genus, or on the disc and crosscap for open and unoriented strings, need not vanish:

$$\big\langle\, \mathcal{V}_\phi \,\big\rangle_\Sigma \;\neq\; 0. \tag{4.44}$$

The dilaton is the usual culprit, but any massless scalar can source one. Read (4.44) as a spacetime statement and it is alarming: the one-point function of a field is the gradient of the effective action, $\langle\mathcal{V}_\phi\rangle \propto \partial S_{\text{eff}}/\partial\phi$ evaluated at the background one expanded around. A nonzero tadpole says $\partial S_{\text{eff}}/\partial\phi\neq 0$: the chosen vacuum is not a stationary point of the action, so it does not solve the theory.

The worldsheet reports the problem as a divergence. Attach the tadpole (4.44) to any other amplitude through a neck. The massless mode then runs through that neck at zero momentum, $p=0$ and $M=0$, the exponent in (4.43) vanishes, and

$$\int_0^\infty \mathrm{d}\ell\; e^{-\ell\cdot 0} \;=\; \infty. \tag{4.45}$$

This is an infrared divergence, from the far boundary of moduli space where the surface splits off a long tube carrying a massless state to zero momentum. It is not a UV problem and no worldsheet counterterm removes it. Its meaning is that the perturbation expansion was set up around the wrong point. First quantisation fixes the target-space background once, in the sigma-model action, and then has no dynamical field with which to move it; the divergence (4.45) is the formalism protesting that it has been asked to expand around a non-solution.

The classic cure is the Fischler-Susskind mechanism: shift the background order by order in g_s, adjusting the metric, dilaton and other massless fields, so that the tadpole (4.44) is cancelled at each order and the corrected configuration does solve the corrected equations of motion. It works, and it is how one computes in practice, but it is grafted on by hand. In supersymmetric type II strings in flat space the NS-NS tadpoles cancel by supersymmetry and the question is moot at leading order; in non-supersymmetric strings, in brane-antibrane systems, and in orientifolds with uncancelled disc and crosscap tadpoles, the vacuum genuinely moves, and the first-quantised recipe has no built-in way to follow it. A second-quantised action does: shifting the vacuum means solving the string field equation of motion $Q\Psi+\Psi*\Psi=0$ for a new Ψ, the business of Section 4.5, where this reappears as tachyon condensation and the systematic relocation of the vacuum.

4.4.4 Mass renormalisation and contour deformation

Masses run. The physical mass of a string state is its tree-level value plus loop corrections,

$$M^2 \;=\; M_0^2 + g_s^2\,\delta M^2 + \mathcal{O}\big(g_s^4\big)\,, \tag{4.46}$$

and the physical pole of the propagator sits at the corrected value, not at M_0^2. For BPS states the correction is protected and vanishes, which is one reason BPS quantities are computable; for

a generic massive string state it does not vanish, and (4.46) is a real shift. The first-quantised recipe has already committed to M_0^2: the external vertex operator is built to be marginal at the tree-level mass shell, $k^2 = -M_0^2$, because that is what weight $(1,1)$ demands of it.

That commitment collides with the neck integral. A self-energy correction on an external leg is a sub-surface joined to the rest by a neck, and the state running through that neck is the external state itself, sitting at $p^2 = -M_0^2$. The exponent in (4.43) is then $\frac{\alpha'}{2}(p^2 + M_0^2) = 0$, exactly where the loop correction wants to move the pole, and the moduli integral sits on the edge of convergence,

$$\int_0^\infty \mathrm{d}\ell \; e^{-\ell \, \frac{\alpha'}{2}(p^2+M_0^2)}\Big|_{p^2=-M_0^2} \;=\; \int_0^\infty \mathrm{d}\ell \;=\; \infty. \tag{4.47}$$

The naive prescription, integrate the real modulus ℓ from 0 to ∞, is simply not defined here. The physics it fails to represent is elementary: the true pole has moved to $p^2 = -M_0^2 - g_s^2 \delta M^2$, and a Dyson resummation of self-energies would put it there. The resolution, due to Sen, is to define the amplitude by deforming the integration contour of the plumbing parameter into the complex plane, away from the pinch, so that (4.47) becomes a convergent integral that continues analytically to $1/(p^2 + M^2)$ with the corrected mass (4.46) in the denominator. The loop amplitude is properly defined by integrating over a deformed contour in the complexified moduli space rather than over the moduli space itself, with the deformation dictated by where the renormalised poles sit.

This adds a genuine rule to the recipe rather than a subtlety of evaluation. The external mass shell one is told to use is the tree-level one, and correcting it needs information, the shifted pole, that lives at the next order in g_s, together with a prescription, contour deformation, that the sum-over-surfaces rules do not contain. In a field theory of strings none of this appears as a separate rule: the propagator is $b_0 \bar{b}_0/(L_0 + \bar{L}_0)$, self-energy insertions resum geometrically, and the pole moves to its physical location by the same Dyson logic as in any quantum field theory. Section 4.5 is where that propagator lives.

4.4.5 Non-perturbative effects of order e^{-1/g_s}

The sum over surfaces is an expansion about $g_s = 0$. Anything that vanishes faster than every power of g_s as $g_s \to 0$ is invisible to it. The cleanest examples are the exponentials

$$e^{-C/g_s} \qquad \text{and} \qquad e^{-C/g_s^2}, \tag{4.48}$$

whose Taylor coefficients at $g_s = 0$ all vanish, so no finite genus, and no infinite sum over genus, produces them. These are not hypothetical. The first is the weight of a D-instanton, a Euclidean D-brane whose action scales as the inverse coupling, $S \sim 1/g_s$; the second is the weight of a Euclidean NS5-brane, whose tension scales as $1/g_s^2$. The exponent C/g_s is exactly the scale (4.42) at which the genus series stopped making sense, and that is no coincidence. A series with coefficients growing like $(2g)!$ is the perturbative shadow of a saddle at action C/g_s; the factorial divergence of Section 4.4.1 and the D-instanton of (4.48) are two views of one object. This connection, guessed by Shenker before D-branes were understood and later pinned on the D-instanton by Polchinski, is why the asymptotic character of the genus expansion is a signpost rather than a technicality.

The point that matters for the rest of the course is that these effects are physical and often decisive. Euclidean D-branes and worldsheet instantons wrapping holomorphic cycles generate the superpotentials that stabilise moduli (Chapter 9); the same e^{-1/g_s} objects are counted in the microscopic degeneracies that reproduce black-hole entropy (Chapter 12); and the strong-weak dualities of Chapter 7 exchange them with perturbative states, so what is non-perturbative in

one frame is a fundamental string in another. A formalism that expands smooth worldsheets in g_s cannot reach any of this. It needs new saddle points, worldsheets whose boundaries end on Euclidean branes, that are absent from the perturbative expansion. First quantisation does not forbid them; it simply cannot see them.

One fact, five masks

The five failures are one fact viewed from five sides. First quantisation computes an on-shell S-matrix, order by order in g_s, about a background chosen in advance. An S-matrix in a fixed background is a shadow of a theory, not the theory: it says nothing off shell, it cannot move its own vacuum, it puts external legs on the wrong mass shell once loops run, its perturbative series is merely asymptotic, and it is blind to any effect not analytic at $g_s = 0$. What is missing is exactly what a field theory supplies for point particles: a spacetime action, a field whose quanta are the objects being scattered, and from it off-shell Green's functions, a variational principle that can relocate the vacuum, propagators whose poles resum to physical masses, and, in principle, the non-perturbative saddles. Writing that action for the string is the business of Section 4.5.

Exercise 4.13 (Naming the failure)**.** *(Short.)* The neck integral (4.43) appears three times in this section, each time diagnosing a different item on the five-silences list. Without calculating, say which failure is signalled by each degenerate limit, and give the one-line physical picture behind it: (i) the tadpole divergence (4.45), a massless mode at zero momentum; (ii) the mass-renormalisation divergence (4.47), an external leg on its tree-level shell; (iii) a generic internal line held off shell, $p^2 + M^2 > 0$. Which of the three gives a finite integral, and why is that the one that behaves?

Exercise 4.14 (Optimal truncation and the D-instanton scale)**.** *(Standard.)* Take the genus coefficients to grow as $A_g = (2g)!\, C^{-2g}$ at large g, and consider the terms $T_g = A_g\, g_s^{2g}$ of the series (4.40). Using Stirling's formula $\ln(2g)! \approx 2g\ln(2g) - 2g$, show that $\ln T_g$ is stationary in g near $2g_\star \simeq C/g_s$, and that at this optimal order the term has size $T_{g_\star} \sim e^{-C/g_s}$. Conclude that the best accuracy the truncated genus expansion can reach is set by a term of exactly the size of a D-instanton contribution, and say in one sentence why this is consistent with the series having zero radius of convergence.

Exercise 4.15 (The neck integral decides everything)**.** *(Standard.)* Consider the degeneration integral (4.43) for a single intermediate state of mass M and momentum p, so that the exponent is $x = \frac{\alpha'}{2}(p^2 + M^2)$.

(a) Show that $\int_0^\infty \mathrm{d}\ell\, e^{-\ell x}$ converges precisely when $x > 0$, equals $1/x$ there, and so reproduces the propagator $2/\alpha'(p^2 + M^2)$. Interpret "$x > 0$" as the statement that the internal line is off its mass shell.

(b) Set $M = 0$ and $p = 0$, a massless mode at zero momentum, as sourced by a tadpole. Show that the integral diverges, and identify this with the infrared divergence (4.45).

(c) Set $M^2 = M_0^2$ and $p^2 = -M_0^2$, an external leg on its tree-level shell receiving a self-energy correction. Show again that $x = 0$, and explain why defining the integral now requires continuing ℓ into the complex plane so that the pole lands at the corrected mass $M_0^2 + g_s^2\delta M^2$ rather than at M_0^2.

Exercise 4.16 (Off shell is a choice)**.** *(Moderate.)* The massless closed-string operator $V = \varepsilon_{\mu\nu}{:}\partial X^\mu \bar\partial X^\nu e^{ik\cdot X}{:}$ has weight $(1 + \alpha' k^2/4,\, 1 + \alpha' k^2/4)$. On shell, $k^2 = 0$, its integral $\int \mathrm{d}^2 z\, V$ is invariant under $z \to \lambda z$. Take $k^2 \neq 0$ (off shell) and show that $\int \mathrm{d}^2 z\, V$ is then multiplied by $|\lambda|^{-\alpha' k^2/2}$, so the insertion depends on the worldsheet coordinate. Argue that removing this

dependence amounts to choosing, by hand, a local coordinate at the puncture, and that this is the choice a string field theory must fix once and for all when it defines its interaction vertices.

4.5 A cartoon of string field theory

The list of failures in Section 4.4 has a common root. The recipe of Section 4.1 quantises *one* string and then puts multi-string physics in by hand, as a sum over worldsheet topologies with on-shell states glued into the punctures. That is a first-quantised prescription, in the exact sense that the worldline path integral is a first-quantised prescription for a particle: it computes on-shell S-matrix elements and nothing else. The cure, in both cases, is second quantisation. Introduce a field whose quanta are the objects in question, write an action for it, and read the perturbation series off the Feynman rules. For a particle that field is $\phi(x)$ and the theory is ordinary quantum field theory. For a string the field is a field of a string, and the theory is string field theory (SFT).

This section is a cartoon of that construction: enough structure to see what the theory *is* and what it fixes, and no more. There will be no gauge-fixed measure, no ghost determinants, no covering of moduli space beyond a statement of the theorem, and no computation carried past the point where the structure is visible. The reward is conceptual. It is also the most abstract stretch of the chapter, and a natural second-week or optional read: nothing later in the course needs its machinery, only the conclusions gathered at the end. Every entry on the breakdown list of Section 4.4 becomes, in SFT, an ordinary question about a field theory that happens to have infinitely many fields.

From a sum over surfaces to a field of strings

First quantisation gives you a machine that eats on-shell external states and returns numbers, order by order in g_s. It never defines an off-shell Green's function, a potential to sit at the bottom of, or a propagator whose pole can move. String field theory supplies exactly those. Its dynamical variable is a single object Ψ, a point in the first-quantised Hilbert space, that packages the entire spectrum: the tachyon, the graviton, the gauge field, and the whole infinite Regge tower, each as an ordinary spacetime field. Its free equation of motion is $Q\Psi = 0$ with Q the worldsheet BRST charge of Chapter 1, so the free spectrum is the BRST cohomology and nothing new happens there. The content is in the interaction, and the interaction is geometry: strings join by gluing.

4.5.1 The worldline warm-up

Recall how second quantisation undoes the first-quantised particle. The worldline path integral for a scalar of mass m gives the propagator as an integral over the length T of the worldline,

$$G(x_f, x_i) = \int_0^\infty \mathrm{d}T \ \langle x_f |\ e^{-T(p^2+m^2)}\ | x_i \rangle \,, \tag{4.49}$$

and higher-point functions come from summing worldline graphs, with cubic vertices inserted by hand wherever three lines meet. Field theory reorganises this. The same propagator is $(p^2+m^2)^{-1}$, the inverse of a *local* kinetic operator; the vertices come from a *local* term $\frac{g}{3!}\phi^3$ in an action; and off-shell Green's functions, which Eq. (4.49) never defines because it wants x_i, x_f but not a continuation in p^2 away from the mass shell, are simply the correlators of the field ϕ. The kinetic operator p^2+m^2 is the object whose vanishing is the mass-shell condition.

String field theory is this move applied to the worldsheet. The mass-shell condition for a string is not a single p^2+m^2 but the whole Virasoro/BRST content of Chapter 1, so the kinetic operator will be the BRST charge Q itself, and the local cubic vertex will be a specific gluing of strings.

What makes the string case rich, and what took from the discovery of the dual model to Witten's 1986 construction to get right, is that "local" for a string means local on the worldsheet, and there is a nontrivial geometric choice in how three strings should meet.

4.5.2 The string field

A string field is a state in the first-quantised Hilbert space of Chapters 1 and 2, ghosts included. Concretely, expand it in the oscillator basis with arbitrary spacetime fields as coefficients. For the open bosonic string, at the lowest two levels,

$$|\Psi\rangle = \int \frac{\mathrm{d}^{26}k}{(2\pi)^{26}} \Big[t(k)\, c_1\, |0;k\rangle \;+\; A_\mu(k)\, \alpha_{-1}^{\mu} c_1\, |0;k\rangle \;+\; (\text{massive}) \Big], \tag{4.50}$$

so the component fields $t(k)$, $A_\mu(k)$, and their infinitely many massive partners are the tachyon, the gauge potential and the Regge tower, now treated as independent off-shell spacetime fields. Equivalently Ψ is a functional $\Psi[X(\sigma), c(\sigma), b(\sigma)]$ of the string's embedding and ghosts, which is the sense in which it is one field carrying a continuum of oscillator labels: a field *of a string*. No mass-shell condition is imposed on the components; that is the whole point of going off-shell.

Two gradings are fixed once and for all. The classical open-string field sits at ghost number one, the value at which the physical cohomology of Section 1.4.6 lives, so $c_1\,|0;k\rangle$ and $\alpha_{-1}^{\mu} c_1\,|0;k\rangle$ in Eq. (4.50) are the ghost-number-one representatives already met as the tachyon and photon vertex operators. Ghost number one is odd, so Ψ is Grassmann odd. Both facts matter below: the pairing that defines the action is nonzero only at total ghost number three, and signs in the gauge algebra track the Grassmann parity.

4.5.3 The free action: BRST cohomology as a spectrum

The pairing is the BPZ inner product $\langle\, \cdot\, , \cdot\, \rangle$ set up in Section 2.1.6 and collected in Appendix F: glue an out-state at infinity to an in-state at the origin on the disc and read off the correlator. On the disc the ghost-number anomaly makes $\langle A, B\rangle$ vanish unless the ghost numbers of A and B add to three. The free string field action is

$$S_2[\Psi] = \tfrac{1}{2}\, \langle \Psi, Q\Psi \rangle, \tag{4.51}$$

with Q the BRST charge (1.74). Ghost number counts: Ψ carries one, Q raises it by one, so the integrand sits at three and the action is nonzero. The equation of motion and the linearised gauge symmetry are

$$Q\Psi = 0, \qquad \delta\Psi = Q\Lambda, \tag{4.52}$$

where Λ is a ghost-number-zero (Grassmann even) string field. This is precisely the definition of the physical spectrum from Section 1.4.6: solutions modulo gauge are $\ker Q / \operatorname{im} Q$, the BRST cohomology. The free SFT therefore contains no new information beyond Chapter 1; it re-expresses the on-shell spectrum as the classical solutions of a field theory. What it adds is viewpoint. The decoupling of null states, which Theorem 1.5 guaranteed by hand, is now a gauge redundancy $\delta\Psi = Q\Lambda$ of an action, exactly as the decoupling of longitudinal photons is the gauge redundancy of Maxwell theory.

Exercise 4.17 (Gauge invariance of the free action)**.** *(Short.)* The BPZ pairing makes Q formally self-adjoint up to a sign, $\langle QA, B\rangle = -(-1)^{|A|}\langle A, QB\rangle$, where $|A|$ is the Grassmann parity. Using this and $Q^2 = 0$, show that Eq. (4.51) is invariant under $\delta\Psi = Q\Lambda$. Show also that varying S_2 gives the equation of motion $Q\Psi = 0$, so that on-shell field configurations are Q-closed and gauge orbits are Q-exact. One line each.

4.5.4 The star product and Witten's cubic action

The interaction is where strings differ from particles. A particle vertex is a point where three worldlines meet; there is no choice to make. Three strings must meet along a curve, and Witten's construction makes a specific choice: split each open string at its midpoint into a left half and a right half, and glue the right half of the first string to the left half of the second. The result is a third string, the left half of the first joined to the right half of the second, with the two inner halves integrated over. This defines an associative product

$$* : \quad (A, B) \longmapsto A * B, \tag{4.53}$$

the star product, on the space of string fields. It preserves ghost number, $\mathrm{gh}(A * B) = \mathrm{gh}(A) + \mathrm{gh}(B)$, and it is noncommutative, because gluing A to B is a different worldsheet from gluing B to A. The common midpoint of all three strings is a distinguished point, and the singular way operators behave as they approach it is the source of most of the technical difficulty in the subject; a cartoon steps around it.

With $*$ in hand, Witten's cubic open string field theory is

$$S[\Psi] = \tfrac{1}{2}\,\langle \Psi, Q\Psi \rangle \;+\; \tfrac{1}{3}\,\langle \Psi, \Psi * \Psi \rangle, \tag{4.54}$$

where the string coupling has been scaled into Ψ and can be reinstated by $\Psi \to \Psi/g_s$, which pulls an overall $1/g_s^2$ in front. The cubic term is nonzero for the same ghost-number reason as the quadratic one: three fields of ghost number one saturate the anomaly at three. It is convenient to write the pairing as an integration, $\langle A, B \rangle = \int A * B$, where $\int$ folds a single string's left half onto its right half and returns a number; the two forms are the same object.

What makes Eq. (4.54) consistent is a short list of algebraic axioms, all provable from the worldsheet definitions and all that a cartoon needs:

$$\text{(i)} \quad Q^2 = 0, \qquad \text{(nilpotency, hence } D = 26\text{)} \tag{4.55}$$

$$\text{(ii)} \quad Q(A * B) = (QA) * B + (-1)^{|A|} A * (QB), \qquad (Q \text{ is a derivation of } *) \tag{4.56}$$

$$\text{(iii)} \quad (A * B) * C = A * (B * C), \qquad \text{(associativity)} \tag{4.57}$$

$$\text{(iv)} \quad \int QA = 0, \quad \int A * B = (-1)^{|A||B|} \int B * A. \qquad (Q\text{-invariance and cyclicity}) \tag{4.58}$$

The data (Ψ-space, Q, $*$, $\int$) is a differential graded algebra with a cyclic, Q-invariant trace: Q is the differential, $*$ the associative product, $\int$ the trace. That is the entire algebraic skeleton of open SFT. It should look familiar. With the dictionary $\Psi \leftrightarrow A$, $Q \leftrightarrow \mathrm{d}$, $* \leftrightarrow \wedge$, $\int \leftrightarrow \int_M \mathrm{Tr}$, the action (4.54) is the Chern-Simons action

$$S_{\mathrm{CS}} = \int_M \mathrm{Tr}\big(\tfrac{1}{2}\, A \wedge \mathrm{d}A + \tfrac{1}{3}\, A \wedge A \wedge A\big), \tag{4.59}$$

with the string field playing the role of the connection and Q the role of the exterior derivative. Ghost number one is the analogue of form degree one, and Q raises it just as d raises form degree. String field theory is Chern-Simons theory of a differential graded algebra. The two actions are the same expression evaluated on different algebras, so the correspondence is structural rather than a loose analogy.

4.5.5 Gauge symmetry and classical solutions

The Chern-Simons form of Eq. (4.54) fixes the full nonlinear gauge symmetry by inspection. For a Grassmann-even, ghost-number-zero parameter Λ,

$$\delta\Psi = Q\Lambda + \Psi * \Lambda - \Lambda * \Psi, \tag{4.60}$$

the string-field version of $\delta A = \mathrm{d}\Lambda + [A, \Lambda]$, with the star commutator replacing the matrix one. Invariance of Eq. (4.54) under Eq. (4.60) follows from axioms (i)–(iv) and nothing else (see Exercise 4.18). At linear order it reduces to the free gauge symmetry $\delta\Psi = Q\Lambda$ of Eq. (4.52), so the massless component A_μ inherits the ordinary spacetime gauge transformation $\delta A_\mu = \partial_\mu \lambda$ from the leading term, and the massive components inherit their own Stueckelberg-like shifts. The full equation of motion is

$$Q\Psi + \Psi * \Psi = 0, \tag{4.61}$$

a Maurer-Cartan ("flatness") equation. Its perturbative solution $\Psi = 0$ is the D-brane background one started on; its *other* solutions are other backgrounds, reached without leaving the theory. That last sentence is the germ of everything the second-quantised formulation buys, and the next subsection cashes it in.

Exercise 4.18 (Equation of motion and full invariance)**.** *(Moderate.)* Vary Eq. (4.54) and use cyclicity (4.58) to show that the factor of $\frac{1}{3}$ in the cubic term produces the undivided $\Psi * \Psi$ in the equation of motion (4.61); the three cyclic placements of $\delta\Psi$ each contribute equally. Then show $\delta S = 0$ under Eq. (4.60) using the derivation property (4.56), associativity (4.57) and cyclicity (4.58). *Hint:* it is the same computation that proves Chern-Simons gauge invariance; group the variation into a Q-exact piece that integrates to zero and a triple product whose cyclic images cancel in pairs.

4.5.6 What the second-quantised theory buys

Take the breakdown list of Section 4.4 in turn.

Off-shell amplitudes. Gauge-fix Eq. (4.54) in Siegel gauge, $b_0\Psi = 0$. The kinetic operator becomes L_0 (since $\{Q, b_0\} = L_0$, the mass-shell operator of Chapter 1), so the propagator is

$$\frac{b_0}{L_0} = b_0 \int_0^\infty \mathrm{d}s \; e^{-sL_0}, \tag{4.62}$$

a flat strip of worldsheet of width s, the Schwinger parameter. The cubic vertex is the Witten gluing of three half-strings at their midpoints. Feynman diagrams built from these two pieces generate worldsheets, and the theorem of Giddings, Martinec and Witten is that they cover the moduli space of bordered Riemann surfaces *exactly once*, with no region missed and none double-counted. The upshot is that the SFT S-matrix equals the first-quantised amplitudes of Sections 4.1 and 4.2 for on-shell external states, and now the external legs may be continued off-shell like any amputated field-theory Green's function. The "on-shell only" limitation is gone by construction, because Q-closure was never imposed on the components in Eq. (4.50).

Vacuum shifts and tachyon condensation. The open bosonic string tachyon $t(k)$ in Eq. (4.50) has $M^2 < 0$: the D-brane one quantised on is unstable, and first quantisation, which knows only how to scatter on-shell states about a fixed background, has no language for where the field then rolls. String field theory does, because Eq. (4.61) is a genuine equation with genuine solutions. There is a nontrivial Ψ_* solving $Q\Psi_* + \Psi_* * \Psi_* = 0$, the tachyon vacuum, and expanding $\Psi = \Psi_* + \psi$ shifts the kinetic operator from Q to a new nilpotent operator $Q_*\psi = Q\psi + \Psi_* * \psi - (-1)^{|\psi|}\psi * \Psi_*$. Sen's conjectures, that the energy of Ψ_* relative to the perturbative vacuum equals minus the D-brane tension (the brane has decayed to the closed-string vacuum) and that Q_* has trivial cohomology (no open-string excitations survive), were confirmed first numerically in level truncation and then analytically by Schnabl's exact solution. This is the direct answer to the "tadpoles and background shifts" entry of Section 4.4: SFT relocates the vacuum systematically rather than fighting a tadpole diagram by diagram.

Mass renormalisation. A massive string state receives loop corrections to its mass. In first quantisation the physical mass is the pole of a propagator one has not defined, so the shift shows up indirectly, as a divergence from the boundary of moduli space where a handle degenerates, cured only by an ad hoc contour deformation. In SFT the propagator (4.62) is an honest object; the full two-point function has a pole; and the pole moves under loops exactly as in any field theory, with the analytic continuation off the tree-level mass supplied automatically by the off-shell structure. The "mass renormalisation and contour deformation" entry of Section 4.4 becomes textbook renormalised perturbation theory. Making this precise for the physical S-matrix, including the correct $i\epsilon$ and the proof of unitarity, is the content of Sen's work on the subject.

What SFT does *not* do is sum the genus expansion. The g_s series stays asymptotic, and effects of order e^{-1/g_s} (Section 4.4) remain non-perturbative. The gain there is subtler and recent: because SFT renders every individual amplitude finite and unambiguous, it also fixes the otherwise ambiguous *normalisations* of D-instanton contributions, where naive first-quantised moduli integrals give ill-defined zero-mode integrals. SFT leaves the non-perturbative physics in place while making its coefficients well posed.

4.5.7 Closed string field theory

The closed string carries gravity (Section 1.5.3), so its field theory is the one that matters for this course, and it is markedly harder to build. The closed string field is a ghost-number-two state, subject to the level-matching constraints $(b_0 - \bar{b}_0)\Psi = 0$ and $(L_0 - \bar{L}_0)\Psi = 0$, and its kinetic term carries an extra insertion, $S_2 = \frac{1}{2}\langle \Psi, c_0^- Q\Psi\rangle$ with $c_0^- = \frac{1}{2}(c_0 - \bar{c}_0)$. The real change is in the interactions. There is no closed-string analogue of the midpoint gluing that let a single cubic vertex cover moduli space for the open string, so Zwiebach's construction requires an infinite tower of elementary vertices $\{\Psi, \dots, \Psi\}_{g,n}$, one for every number of legs n and every genus g, shaped so that vertices and propagators together still tile the moduli space of closed Riemann surfaces exactly once. Taken to cubic order and to $\alpha' \to 0$, the classical closed SFT action reproduces the Einstein and B-field vertices found from amplitudes in Section 4.3; closed SFT is the off-shell completion whose leading low-energy content is the supergravity effective action assembled in Chapter 5.

Remark 4.6 (Beyond the lectures: the homotopy algebra of closed strings). The infinite vertex tower has an algebraic name. Witten's open string gave an associative product; the closed-string multi-string brackets $\{\cdot, \dots, \cdot\}$, with Q as the differential, are only homotopy-associative, in the sense that they satisfy the relations of an L_∞ (strong-homotopy Lie) algebra, in which the Jacobi identity for the two-bracket holds only up to a Q-exact correction generated by the three-bracket, and so on up the tower. Including loops, the whole quantum theory is organised by the Batalin-Vilkovisky formalism: the quantum master equation $\frac{1}{2}\{S, S\} + \hbar\,\Delta S = 0$ encodes gauge invariance and the sum over genus at once, with the classical L_∞ structure as its tree-level shadow. The open-string counterpart, where associativity is relaxed in the same way, is an A_∞ algebra; both structures return when the superstring is quantised, in the remark below.

Remark 4.7 (Beyond the lectures: background independence). The remark under Eq. (4.61) has teeth. Two nearby string backgrounds are two solutions Ψ_* of one and the same SFT action, and expanding about either gives the string in that background. Sen and Zwiebach proved this explicitly for closed SFT to the relevant order: a marginal deformation of the worldsheet CFT (a change of background) is a solution of the spacetime equation of motion (4.61), and the action reorganises into the SFT built on the new background. So the framework is background independent in a controlled, if perturbative, sense: the sum over surfaces of Section 4.1 is an expansion of a single functional about one of its critical points, and the choice of background, fixed and external in first quantisation, is here a choice of solution. This is as close as the

perturbative formalism comes to the non-perturbative dream of a single action with all string vacua as classical solutions.

Remark 4.8 (Beyond the lectures: the superstring costs more). The superstring version is not a cosmetic upgrade. The worldsheet now carries the $\beta\gamma$ superghosts and their picture number, and a naive cubic action suffers from contact divergences at coincident picture-changing insertions. The modern resolution, due to Sen, Erler, Konopka and others, keeps the Neveu-Schwarz sector at a fixed picture and treats the Ramond sector with an auxiliary string field (or, equivalently, a "vertical integration" prescription in the supermoduli directions), giving an action with the same A_∞ (open) or L_∞ (closed) skeleton as the bosonic case. The bosonic cartoon of this section survives intact as the algebra; the analytic cost is entirely in constructing vertices compatible with picture. This is live research, and the reviews by Erler are the place to start.

Where this leaves us

The perturbative string of Sections 4.1 and 4.2 is a first-quantised recipe, exact on-shell and mute off it. String field theory is the second-quantised completion: a single field Ψ, a kinetic term built from the BRST charge Q of Chapter 1, a cubic vertex that is nothing but the geometric gluing of strings, and a Chern-Simons gauge symmetry (4.60) that packages null-state decoupling and spacetime gauge invariance together. It returns the same S-matrix, now off-shell; it relocates the vacuum when the perturbative one is unstable; and it renormalises masses as a field theory should. What it hands to the rest of the course is a licence to work with the low-energy *fields* rather than the worldsheet. The graviton, B-field and dilaton are components of a closed string field whose classical dynamics, at leading order in α', is a spacetime action. Building and using that action is the business of Chapter 5, and from there on the string recedes and the supergravity takes over.

Chapter 5

Low-Energy Effective Actions: 10D and 11D Supergravity

5.1 The sigma model and the beta-function equations

In Chapter 1 the graviton fell out of the string spectrum as a massless state: a symmetric transverse-traceless polarisation carrying the gauge invariance of a linearised diffeomorphism (Section 1.5.3). That was existence, not dynamics. We saw the field; we did not see the equation it obeys. This section pays off the promissory note left at the end of Section 1.5.6. Put the string in a spacetime that is curved, switch on the other two massless fields it carries, and demand that the worldsheet theory still make sense as a quantum theory. The demand is quantum Weyl invariance, and its price is that the background fields solve their own equations of motion. Gravity comes out as a consistency condition of the worldsheet.

The big picture

Weyl invariance was the redundancy that let us fix conformal gauge in Section 1.2.1. In a curved background the couplings of the worldsheet theory *are* the spacetime fields $G_{\mu\nu}(X)$, $B_{\mu\nu}(X)$, $\phi(X)$, and as couplings they run. Their beta functions measure the failure of Weyl invariance. Restoring the symmetry means

$$\beta^G_{\mu\nu} = \beta^B_{\mu\nu} = \beta^\phi = 0,$$

and to leading order in α' these three equations are Einstein's equation, the B-field equation, and the dilaton equation. The spacetime effective action of Section 5.2 is the functional whose stationary points are exactly this locus.

5.1.1 The string in a background

A nonzero classical field is a coherent state of its quanta. In electrodynamics a background $A_\mu(x)$ is a condensate of photons; a background metric $G_{\mu\nu}(X) = \eta_{\mu\nu} + \chi_{\mu\nu}(X)$ is, in the same sense, a coherent state of gravitons. Made precise, this is the statement that exponentiating the graviton vertex operator of Section 1.5.3 inside the flat-space path integral and resumming reproduces the worldsheet action with $\eta_{\mu\nu}$ replaced by $G_{\mu\nu}$; the same holds for the Kalb-Ramond field $B_{\mu\nu}$ and the dilaton ϕ. We take that resummation as given here, an asserted standard result rather than one we derive: it is the coherent-state counterpart of the amplitude expansion of Chapter 4, built on the same vertex operators. So the natural arena for the massless closed-string fields is

not a fixed flat target but the theory of a string moving in an arbitrary (G, B, ϕ) background,

$$S_\sigma = -\frac{1}{4\pi\alpha'}\int_\Sigma \mathrm{d}^2\sigma\,\Big(\sqrt{-\gamma}\,\gamma^{\alpha\beta}\,G_{\mu\nu}(X)+\epsilon^{\alpha\beta}\,B_{\mu\nu}(X)\Big)\partial_\alpha X^\mu\,\partial_\beta X^\nu \;+\; \frac{1}{4\pi}\int_\Sigma \mathrm{d}^2\sigma\,\sqrt{-\gamma}\,\phi(X)\,R^{(2)}. \tag{5.1}$$

The first term is the Polyakov action (1.7) with the flat metric promoted to the curved $G_{\mu\nu}(X)$. The second uses the worldsheet antisymmetric symbol $\epsilon^{\alpha\beta}$ (with $\epsilon^{01}=+1$), a density that needs no worldsheet metric to contract its indices; this is how the string couples to B, and it is why the fundamental string is an electric source of B (Section 5.6). The last term is the Fradkin-Tseytlin coupling (1.96), now with a position-dependent $\phi(X)$; $R^{(2)}$ is the intrinsic curvature of the worldsheet metric.

This is a two-dimensional *nonlinear sigma model*: D scalar fields X^μ with a field-dependent target metric. The point to hold onto is that $G_{\mu\nu}(X)$, $B_{\mu\nu}(X)$ and $\phi(X)$ are not three coupling constants but three infinite towers of them. Expanding a background about a point x_0,

$$G_{\mu\nu}(X) = G_{\mu\nu}(x_0) + \partial_\rho G_{\mu\nu}(x_0)\,(X-x_0)^\rho + \tfrac{1}{2}\,\partial_\rho\partial_\sigma G_{\mu\nu}(x_0)\,(X-x_0)^\rho(X-x_0)^\sigma + \cdots, \tag{5.2}$$

each Taylor coefficient is an independent coupling of an interacting theory. Chapter 1's free bosons were the special case of a flat, constant target, where every coefficient beyond $\eta_{\mu\nu}$ vanishes.

What organises this infinite set is that α' is dimensionful. If the background varies over a characteristic curvature radius L, the dimensionless expansion parameter of the sigma model is

$$\frac{\alpha'}{L^2} = \Big(\frac{\ell_s}{L}\Big)^2, \tag{5.3}$$

the string length $\ell_s = \sqrt{\alpha'}$ measured against the scale over which spacetime bends. Gentle curvature on the string scale is weak worldsheet coupling. One loop in the two-dimensional theory brings one factor of α', and each extra derivative on the background comes with a factor of ℓ_s. This is the α' expansion, a low-energy (long-wavelength) expansion of spacetime physics, valid when the curvature radius is large compared with ℓ_s. The Fradkin-Tseytlin term carries an explicit $1/4\pi$ with no $1/\alpha'$, so ϕ sits one order higher in α' than G and B; that bookkeeping is exactly why the dilaton first appears in β^G alongside a single loop of the metric coupling.

The sigma model is not renormalisable in the naive power-counting sense, but it is renormalisable order by order in the α' expansion: the ultraviolet divergences of the two-dimensional theory are local and covariant in the target, so they can be absorbed by redefining the couplings G, B, ϕ themselves. The counterterms are geometric tensors built from the background, so the beta functions that govern them are themselves spacetime tensors, and the equations they yield are covariant spacetime field equations.

5.1.2 Weyl invariance is the vanishing of the beta functions

The two-derivative part of (5.1), the (G, B) terms, is classically Weyl invariant: in two dimensions $\sqrt{-\gamma}\,\gamma^{\alpha\beta}$ is inert under $\gamma_{\alpha\beta} \to e^{2\omega}\gamma_{\alpha\beta}$, and the B term never saw the metric to begin with. The dilaton term is not Weyl invariant even classically, since $\sqrt{-\gamma}\,R^{(2)}$ shifts by a total derivative under a Weyl rescaling. Its non-invariance is suppressed by one power of α' relative to the metric term, which is precisely the order at which the quantum anomaly of the metric coupling shows up, so the classical dilaton variation and the one-loop matter anomaly combine into a single object.

That object is the trace of the worldsheet stress tensor. A coupling that runs is one whose value depends on the scale at which it is probed, so a running coupling is scale invariance lost; in two dimensions the loss of scale invariance is measured by the trace $T^\alpha{}_\alpha$. Classically $T^\alpha{}_\alpha = 0$

was the hallmark of that symmetry (Exercise 1.3). Quantum-mechanically the trace fails to vanish by an amount fixed, one coupling at a time, by the corresponding beta function. In the normalisation of Appendix F, and on the Euclidean worldsheet reached by the continuation $z = e^{\tau_E - i\sigma}$ of Appendix A (on which $\sqrt{\gamma}$ carries no minus sign), it reads

$$T^{\alpha}{}_{\alpha} = -\frac{1}{2\alpha'}\,\beta^{G}_{\mu\nu}\,\gamma^{\alpha\beta}\,\partial_\alpha X^\mu \partial_\beta X^\nu \;-\; \frac{i}{2\alpha'}\,\beta^{B}_{\mu\nu}\,\frac{\epsilon^{\alpha\beta}}{\sqrt{\gamma}}\,\partial_\alpha X^\mu \partial_\beta X^\nu \;-\; \tfrac{1}{2}\,\beta^{\phi}\,R^{(2)}. \tag{5.4}$$

Each beta function multiplies exactly the worldsheet operator that its coupling multiplies in the action: β^G against the symmetric $\partial X \partial X$, β^B against the antisymmetric one, β^ϕ against $R^{(2)}$. The factor of i on the B term is an artefact of that Euclidean continuation and is absent on the Lorentzian worldsheet; the derivation of (5.4) is the normal-coordinate computation collected in Appendix F. Weyl invariance of the quantum theory is the statement that this trace vanishes as an operator. Because the three operators on the right are independent, that happens if and only if

$$\beta^{G}_{\mu\nu}(x) = 0, \qquad \beta^{B}_{\mu\nu}(x) = 0, \qquad \beta^{\phi}(x) = 0 \tag{5.5}$$

at every spacetime point x. These are not the usual numerical beta-function conditions of a QFT with finitely many couplings; they are functional conditions, one equation per point of spacetime for each field. A fixed point of the worldsheet renormalisation group is a conformal field theory, hence a consistent string background, hence a solution of (5.5).

5.1.3 The one-loop beta functions

Computing the beta functions is a background-field calculation: expand X^μ about a classical solution in Riemann normal coordinates, integrate out the quantum fluctuation to one loop, and read off the logarithmically divergent, and hence scale-dependent, corrections to G, B and ϕ. One worldsheet loop is one power of α'. The computation is standard and self-contained, and reproducing it here would be a detour into two-dimensional perturbation theory; the normal-coordinate machinery is collected as a cheat-sheet in Appendix F. The result, to leading order in α', is

$$\beta^{G}_{\mu\nu} = \alpha'\left(R_{\mu\nu} + 2\,\nabla_\mu\nabla_\nu\phi - \tfrac{1}{4}\,H_{\mu\lambda\sigma}H_{\nu}{}^{\lambda\sigma}\right) + \mathcal{O}\!\left(\alpha'^2\right), \tag{5.6}$$

$$\beta^{B}_{\mu\nu} = \alpha'\left(-\tfrac{1}{2}\,\nabla^\lambda H_{\lambda\mu\nu} + \nabla^\lambda\phi\,H_{\lambda\mu\nu}\right) + \mathcal{O}\!\left(\alpha'^2\right), \tag{5.7}$$

$$\beta^{\phi} = \frac{D-26}{6} - \frac{\alpha'}{2}\,\nabla_\mu\nabla^\mu\phi + \alpha'\,\nabla_\mu\phi\,\nabla^\mu\phi - \frac{\alpha'}{24}\,H_{\mu\nu\rho}H^{\mu\nu\rho} + \mathcal{O}\!\left(\alpha'^2\right), \tag{5.8}$$

where $R_{\mu\nu}$ is the spacetime Ricci tensor in the conventions of Appendix A (a round sphere has $R > 0$), ∇ is its Levi-Civita connection (on a scalar $\nabla_\mu\phi = \partial_\mu\phi$, and we abbreviate $(\nabla\phi)^2 := \nabla_\mu\phi\,\nabla^\mu\phi$), and

$$H = \mathrm{d}B, \qquad H_{\mu\nu\rho} = 3\,\partial_{[\mu}B_{\nu\rho]} = \partial_\mu B_{\nu\rho} + \partial_\nu B_{\rho\mu} + \partial_\rho B_{\mu\nu} \tag{5.9}$$

is the gauge-invariant field strength, with the Bianchi identity $\mathrm{d}H = 0$ built in. The two-derivative terms displayed here are universal; the $\mathcal{O}(\alpha'^2)$ pieces (quadratic in curvature) depend on the renormalisation scheme, a point we return to below.

5.1.4 Gravity as a law

Read the three equations (5.5) in turn.

The metric condition $\beta^G_{\mu\nu} = 0$ is

$$R_{\mu\nu} + 2\,\nabla_\mu\nabla_\nu\phi - \tfrac{1}{4}\,H_{\mu\lambda\sigma}H_{\nu}{}^{\lambda\sigma} = 0. \tag{5.10}$$

This is Einstein's equation. The dilaton gradient and the three-form flux appear on the right-hand side as the matter that curves spacetime. Setting $B = 0$ and ϕ constant collapses it to $R_{\mu\nu} = 0$, the vacuum Einstein equation of pure general relativity, recovered here from nothing more than the requirement that the string propagate consistently.

The B-field condition $\beta^B_{\mu\nu} = 0$ is $\nabla^\lambda H_{\lambda\mu\nu} = 2\,(\nabla^\lambda\phi)H_{\lambda\mu\nu}$, which is cleanest in form language,

$$\mathrm{d}\Big(e^{-2\phi} \star H\Big) = 0. \tag{5.11}$$

This is a Maxwell equation for the two-form potential, with the dilaton playing the role of a position-dependent gauge coupling $e^{-2\phi}$ in front. Its partner is the Bianchi identity $\mathrm{d}H = 0$ from (5.9).

The dilaton condition $\beta^\phi = 0$ carries two messages at once. The constant piece $(D-26)/6$ is proportional to the central-charge deficit $c - 26$ (with $c = D$ at leading order, one unit per free boson); requiring it to vanish when the derivative terms switch off fixes the critical dimension $D = 26$ of the bosonic string, exactly the number that BRST nilpotency delivered in Section 1.4.4. The derivative terms are the wave equation for the dilaton itself.

These three equations are not independent, and their dependence is the structural fact that makes the rest of this chapter possible. There is a Bianchi-like identity relating the divergence of β^G and the gradient of β^ϕ (the Curci-Paffuti relation), so on the common zero locus (5.5) the dilaton beta function is forced to be a spacetime *constant*, equal to $(c-26)/6$ with c the exact central charge of the background CFT. The same identity guarantees that (5.6)–(5.8) are the gradient of a single spacetime functional. That functional is the low-energy effective action. Constructing it, and checking that its Einstein equation is precisely (5.10), is the business of Section 5.2.

String frame

The equations (5.10)–(5.11) are written in *string frame*, the frame in which the worldsheet metric $G_{\mu\nu}$ appears directly. The dilaton multiplies the curvature through the overall $e^{-2\phi}$ that the effective action will carry, so $\beta^G_{\mu\nu} = 0$ is not literally $R_{\mu\nu} = \kappa^2 T_{\mu\nu}$ with a canonical Einstein-Hilbert term. Passing to Einstein frame, where the graviton kinetic term is canonical, is a Weyl rescaling $G_{\mu\nu} \to e^{-4\phi/(D-2)}G_{\mu\nu}$ (equivalently $e^{-\phi/2}$ at $D = 10$), carried out in Section 5.2. Comparisons of masses and tensions must always state the frame.

Example 5.1 (The graviton reappears as a wave)**.** Take the vacuum $B = 0$, $\phi = \phi_0$ constant, and write $G_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$. Then $\beta^B \equiv 0$, the derivative terms in β^ϕ vanish and its constant fixes $D = 26$, and $\beta^G_{\mu\nu} = 0$ reduces to $R_{\mu\nu} = 0$. Linearising the Ricci tensor,

$$R^{(1)}_{\mu\nu} = -\tfrac{1}{2}\left(\partial^2 h_{\mu\nu} + \partial_\mu\partial_\nu h - \partial_\mu\partial^\lambda h_{\lambda\nu} - \partial_\nu\partial^\lambda h_{\lambda\mu}\right), \qquad h \equiv \eta^{\mu\nu}h_{\mu\nu}, \tag{5.12}$$

with $\partial^2 = \partial_\mu\partial^\mu$. In de Donder gauge $\partial^\lambda \bar{h}_{\lambda\mu} = 0$ (where $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$), the last two terms drop and $R_{\mu\nu} = 0$ becomes $\partial^2 h_{\mu\nu} = 0$: a massless wave. The residual gauge freedom $\delta h_{\mu\nu} = \partial_\mu\xi_\nu + \partial_\nu\xi_\mu$ is the linearised diffeomorphism. Compare Section 1.5.3: the field whose equation of motion we have now derived is the same massless graviton whose *state* the string produced there, the state and the law being two descriptions of one object.

Remark 5.2 (Beyond the lectures: β^ϕ as a c-function). That β^ϕ is constant on-shell is the sigma-model face of Zamolodchikov's c-theorem. Away from a fixed point β^ϕ tracks a monotonic quantity that counts worldsheet degrees of freedom; at a fixed point it equals the central charge c, and the derivative structure of (5.8) is exactly what a gradient flow requires. For the superstring the arithmetic shifts but the logic does not: the worldsheet carries D bosons and their fermionic partners, the critical central charge is reached at $D = 10$ (Chapter 3), and the constant in β^ϕ

vanishes there. The two-derivative equations (5.10)–(5.11) for the common NS-NS fields are unchanged; only the higher-α' tail differs between the bosonic and supersymmetric strings. The deeper statement, that β^G, β^B and β^ϕ descend from a single local spacetime functional to all orders in α' (the Weyl consistency conditions of Osborn), is what guarantees the effective action exists; we take it up where that action is built, in Section 5.2.

Remark 5.3 (Beyond the lectures: scheme dependence at order α'^2). The $\mathcal{O}(\alpha'^2)$ terms in (5.6)–(5.8) are not physical on their own. A local field redefinition of the target metric, $G_{\mu\nu} \to G_{\mu\nu} + \alpha'\,(a\,R_{\mu\nu} + b\,G_{\mu\nu}R + \cdots)$, is a change of renormalisation scheme on the worldsheet; it reshuffles the two-loop beta functions without touching the leading terms. This is why the first α' correction is sometimes written with $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ and sometimes with other curvature invariants: they are related by field redefinition. What is unambiguous is the on-shell content, the S-matrix of the massless fields, which is why the honest way to fix the higher corrections is to match to string scattering amplitudes (Chapter 4). The leading equations (5.10)–(5.11) are the same in every scheme.

Exercise 5.1 (The B-field equation as a Maxwell law)**.** Starting from $\beta^B_{\mu\nu} = 0$ in (5.7), show that

$$\nabla^\lambda\Big(e^{-2\phi}H_{\lambda\mu\nu}\Big) = 0,$$

and hence, in form language, $\mathrm{d}\Big(e^{-2\phi}\star H\Big) = 0$, the equation (5.11). Read this as a Maxwell equation for the two-form potential with $e^{-2\phi}$ a position-dependent coupling, and identify its companion Bianchi identity. *(Hint: expand the covariant divergence and factor out $e^{-2\phi}$.)* The conserved charge these equations assign, so that the fundamental string is an electric source for B and its magnetic dual the NS5-brane sources $\star H$, is worked from the action side in Exercise 5.4 and used in Section 5.6.

Exercise 5.2 (The state and the law, from the vertex operator)**.** The graviton vertex operator is $V = e_{\mu\nu}\,{:}\partial X^\mu\,\bar\partial X^\nu\,e^{ik\cdot X}{:}$ with polarisation $e_{\mu\nu}$. Using the free-boson weights of Section 1.3 (each exponential contributes holomorphic weight $\alpha' k^2/4$, each ∂X weight 1), show that V is a marginal operator, of weight $(1,1)$, precisely when $k^2 = 0$, and that requiring it to be primary (no descendant $\partial(\cdots)$ piece) forces the transversality $k^\mu e_{\mu\nu} = 0$. Compare with the linearised form of (5.10) about flat space with constant ϕ and $B = 0$: marginality of the vertex operator is the same condition as the leading vanishing of β^G. A physical string state is a marginal deformation of the worldsheet CFT, and a marginal deformation is a solution of the linearised spacetime equations.

With the beta functions in hand we have turned the Chapter 1 slogan into a theorem: consistency of the string on a background is the field equation of that background. What remains is to package (5.10)–(5.11) as the variation of a spacetime action, read off the correct dilaton prefactor and normalisation, and move from string frame to Einstein frame. That action, universal to every closed string through its NS-NS sector, is the subject of Section 5.2.

5.2 The universal NS-NS action

The previous section did the hard worldsheet work. Demanding that a string propagate consistently in a curved background $(g_{\mu\nu}, B_{\mu\nu}, \phi)$, that is, demanding that the two-dimensional sigma model stay Weyl invariant at the quantum level, produced three conditions: the vanishing of the beta functions β^G, β^B and β^ϕ. Those conditions are differential equations on the spacetime fields. This section reads them backwards. We show that they are the Euler-Lagrange equations of a single spacetime action, and we write that action down.

The reason to build it is the payoff promised at the end of Chapter 1. The graviton appeared there as a state in the closed-string spectrum; its field equation now appears as the condition

that the background not spoil conformal invariance. Packaging that condition as $\delta S = 0$ for an explicit S turns it into Einstein's equations, coupled to B and ϕ, written in the form a relativist would recognise.

The action we are about to build involves only the fields $g_{\mu\nu}$, $B_{\mu\nu}$ and ϕ. These three make up the massless Neveu-Schwarz-Neveu-Schwarz (NS-NS) sector, common to the closed-string spectrum of every one of the five ten-dimensional superstrings and of the bosonic string as well. Type IIA, type IIB, type I and the two heterotic theories differ in their Ramond-Ramond content, their gauge sectors and their fermions, but they all contain this same graviton, Kalb-Ramond field and dilaton, with the same two-derivative dynamics. That is why the action of this section is *universal*: it is the piece of every supergravity that comes straight from the beta functions, before any theory-specific field is switched on. The theory-specific pieces are collected in Appendix I and quoted in Sections 5.4 and 5.5.

5.2.1 Reconstructing the action from the beta functions

We quote the one-loop beta functions from Section 5.1, in the conventions of Appendix A. Writing $H = \mathrm{d}B$ for the three-form field strength, $H^2 \coloneqq H_{\mu\nu\lambda}H^{\mu\nu\lambda}$, and $(\nabla\phi)^2 \coloneqq \nabla_\mu\phi\,\nabla^\mu\phi$ (on a scalar $\nabla_\mu\phi = \partial_\mu\phi$, so the distinction between covariant and partial derivative is immaterial for the dilaton gradient; we write ∇ throughout), they are

$$\beta^G_{\mu\nu} = \alpha'\Big(R_{\mu\nu} + 2\nabla_\mu\nabla_\nu\phi - \tfrac{1}{4}H_{\mu\lambda\sigma}H_\nu{}^{\lambda\sigma}\Big) + \mathcal{O}\Big(\alpha'^2\Big)\,, \tag{5.13}$$

$$\beta^B_{\mu\nu} = \alpha'\Big(-\tfrac{1}{2}\nabla^\lambda H_{\lambda\mu\nu} + \nabla^\lambda\phi\, H_{\lambda\mu\nu}\Big) + \mathcal{O}\Big(\alpha'^2\Big)\,, \tag{5.14}$$

$$\beta^\phi = \frac{D - D_c}{6} - \frac{\alpha'}{2}\Big(\nabla^2\phi - 2(\nabla\phi)^2 + \tfrac{1}{12}H^2\Big) + \mathcal{O}\Big(\alpha'^2\Big)\,, \tag{5.15}$$

with $\nabla^2 \coloneqq \nabla_\mu\nabla^\mu$ the covariant Laplacian and D_c the critical dimension (26 for the bosonic string, 10 for the superstring). Conformal invariance requires all three to vanish. For the superstrings of this chapter $D = D_c = 10$, so the constant in (5.15) drops out; we keep it visible for a moment because it becomes a potential for the dilaton, then set it to zero.

The claim is that the equations $\beta^G_{\mu\nu} = \beta^B_{\mu\nu} = \beta^\phi = 0$ are exactly the field equations of

$$\boxed{S_{\mathrm{NS}} = \frac{1}{2\kappa_{10}^2}\int \mathrm{d}^{10}x\,\sqrt{-g}\;e^{-2\phi}\,\Big(R + 4\,\nabla_\mu\phi\,\nabla^\mu\phi - \tfrac{1}{12}\,H_{\mu\nu\lambda}H^{\mu\nu\lambda}\Big)\,.} \tag{5.16}$$

The constant $2\kappa_{10}^2$ is fixed in Section 5.2.2; it does not affect the equations of motion and can be ignored while we vary. In $D \neq D_c$ dimensions one adds a term $-\frac{2(D-D_c)}{3\alpha'}$ inside the brackets, a cosmological-constant-like potential; at criticality it vanishes and (5.16) is the whole story at two derivatives.

Two features of (5.16) matter before we verify it. The overall factor $e^{-2\phi}$ multiplies *every* term, including the Einstein-Hilbert term R; a dilaton-dependent coefficient in front of R is the signature of a scalar-tensor theory rather than ordinary general relativity, and it is what makes the variation below less routine than it looks. The dilaton kinetic term appears with a $+4$ and the "wrong" sign relative to a canonical scalar; this is not a ghost, and Section 5.2.3 shows why once the metric is rescaled.

The B-field equation. Start with the easiest variation. Only the last term of (5.16) contains B, through $H = \mathrm{d}B$. Varying B and integrating by parts gives

$$\nabla^\mu\Big(e^{-2\phi}H_{\mu\nu\lambda}\Big) = 0 \qquad\Longleftrightarrow\qquad \mathrm{d}\Big(e^{-2\phi}\star H\Big) = 0. \tag{5.17}$$

Expanding the covariant derivative, $\nabla^\mu H_{\mu\nu\lambda} - 2\,\nabla^\mu\phi\, H_{\mu\nu\lambda} = 0$, which is $-2/\alpha'$ times $\beta^B_{\mu\nu}$ from (5.14). So the B equation of motion is precisely $\beta^B_{\mu\nu} = 0$. The dilaton has turned the Bianchi-partner equation $\mathrm{d}\star H = 0$ of ordinary p-form electrodynamics into $\mathrm{d}(e^{-2\phi}\star H) = 0$, with $e^{-2\phi}$ playing the role of a position-dependent coupling for the Kalb-Ramond field, exactly as it does for the metric.

The graviton equation. Vary (5.16) with respect to $g^{\mu\nu}$. The delicate term is $\sqrt{-g}\,e^{-2\phi}R$. In ordinary general relativity the piece $\sqrt{-g}\,g^{\mu\nu}\delta R_{\mu\nu}$ is a total derivative and is discarded; here it is multiplied by the non-constant factor $e^{-2\phi}$ and survives. This is the standard scalar-tensor (Brans-Dicke) situation. Varying $\int\sqrt{-g}\,f(\phi)\,R$ with the curvature-variation identities of Appendix D yields

$$\frac{1}{\sqrt{-g}}\,\frac{\delta}{\delta g^{\mu\nu}}\int\sqrt{-g}\,f\,R = f\Big(R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R\Big) + (g_{\mu\nu}\nabla^2 - \nabla_\mu\nabla_\nu)f, \tag{5.18}$$

where the two derivatives on the right come from twice integrating the surviving total derivative by parts against $f = e^{-2\phi}$; Exercise 5.3 asks you to fill in the steps. Using $\nabla_\mu\nabla_\nu e^{-2\phi} = e^{-2\phi}(-2\nabla_\mu\nabla_\nu\phi + 4\nabla_\mu\phi\,\nabla_\nu\phi)$ and $\nabla^2 e^{-2\phi} = e^{-2\phi}(-2\nabla^2\phi + 4(\nabla\phi)^2)$, the Einstein-Hilbert variation contributes, per factor $e^{-2\phi}$,

$$R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R + 2\nabla_\mu\nabla_\nu\phi - 4\nabla_\mu\phi\,\nabla_\nu\phi + g_{\mu\nu}\big(-2\nabla^2\phi + 4(\nabla\phi)^2\big).$$

The dilaton kinetic term $4(\nabla\phi)^2 = 4\,g^{\rho\sigma}\nabla_\rho\phi\,\nabla_\sigma\phi$ contributes $4\nabla_\mu\phi\,\nabla_\nu\phi - 2g_{\mu\nu}(\nabla\phi)^2$, and the field-strength term $-\frac{1}{12}H^2$ contributes $-\frac{1}{4}H_{\mu\lambda\sigma}H_\nu{}^{\lambda\sigma} + \frac{1}{24}g_{\mu\nu}H^2$. The $\nabla_\mu\phi\,\nabla_\nu\phi$ pieces cancel between the first and second contributions, and collecting the rest gives the graviton equation

$$R_{\mu\nu} + 2\nabla_\mu\nabla_\nu\phi - \tfrac{1}{4}H_{\mu\lambda\sigma}H_\nu{}^{\lambda\sigma} = \tfrac{1}{2}\,g_{\mu\nu}\Big(R + 4\nabla^2\phi - 4(\nabla\phi)^2 - \tfrac{1}{12}H^2\Big). \tag{5.19}$$

The left-hand side is exactly $\beta^G_{\mu\nu}/\alpha'$. The right-hand side is $\frac{1}{2}g_{\mu\nu}$ times a scalar we are about to meet as the dilaton equation.

The dilaton equation. Vary (5.16) with respect to ϕ. The -2 from differentiating $e^{-2\phi}$ hits every term, and the $+4(\nabla\phi)^2$ term also varies through its explicit ϕ; integrating the resulting $\nabla^\mu\phi\,\nabla_\mu\delta\phi$ by parts against $e^{-2\phi}$ produces the Laplacian. The result is

$$R + 4\nabla^2\phi - 4(\nabla\phi)^2 - \tfrac{1}{12}H^2 = 0. \tag{5.20}$$

This is precisely the scalar sitting in the bracket on the right of (5.19).

Putting the three together. The dilaton equation (5.20) sets the right-hand side of the graviton equation (5.19) to zero. What is left is

$$R_{\mu\nu} + 2\nabla_\mu\nabla_\nu\phi - \tfrac{1}{4}H_{\mu\lambda\sigma}H_\nu{}^{\lambda\sigma} = 0, \tag{5.21}$$

which is $\beta^G_{\mu\nu} = 0$. To recover $\beta^\phi = 0$, take the trace of (5.21) with $g^{\mu\nu}$, giving $R + 2\nabla^2\phi - \frac{1}{4}H^2 = 0$, and subtract it from the dilaton equation (5.20). The Ricci scalar drops out and one is left with

$$\nabla^2\phi - 2(\nabla\phi)^2 + \tfrac{1}{12}H^2 = 0, \tag{5.22}$$

which is $-2/\alpha'$ times the critical β^ϕ of (5.15). So the three Weyl-anomaly conditions and the three field equations of (5.16) are the same set of equations, written twice.

This is the concrete content of "gravity as a law" at leading order. Gravity is not put into string theory by hand; it is forced out of it, as the condition that a string not notice the two-dimensional metric it was quantised with. The action (5.16) is the generating functional for that condition, and Einstein's equations together with the B-field and dilaton equations are what conformal invariance requires of a nontrivial background.

Remark 5.4 (Beyond the lectures: why an action exists at all). We reconstructed (5.16) by guessing and checking. That an action exists is not automatic: three independent functions $\beta^G, \beta^B, \beta^\phi$ need not be the gradient of a single functional. The obstruction is controlled by a Wess-Zumino-type consistency condition on the Weyl anomaly, the Curci-Paffuti identity, which forces $\nabla_\mu\beta^\phi$ to be a specific linear combination of β^G and β^B (schematically $\nabla_\mu\beta^\phi \sim \nabla^\nu\beta^G_{\mu\nu} + \dots$). Its practical consequence is that β^ϕ is spacetime-constant whenever β^G and β^B vanish, which is why setting the graviton and B equations to zero already implies most of the dilaton equation, and why a single S can encode all three. The same structure underlies Zamolodchikov's c-theorem: the quantity β^ϕ evaluated on solutions is (up to normalisation) the central charge, and the beta-function flow is gradient flow for it. This is the natural home for the statement; Section 5.1 points back to it. We will not need the machinery again, but it is the reason the manipulation above was guaranteed to succeed.

Remark 5.5 (Beyond the lectures: higher orders and scheme dependence). Equations (5.13)–(5.15) are the first term in an expansion in α', equivalently in worldsheet loops. The next term feeds curvature-squared corrections into the action. For the bosonic string the leading correction is $\frac{\alpha'}{4}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ at order α' beyond (5.16); for the superstring the first correction to the NS-NS action is pushed to order α'^3, a quartic Riemann invariant R^4 whose coefficient is fixed by four-graviton scattering. Neither the beta functions nor the action are unique beyond leading order: a field redefinition $g_{\mu\nu} \to g_{\mu\nu} + \alpha'\, a\, R_{\mu\nu} + \dots$ (a change of renormalisation scheme on the worldsheet) reshuffles the higher terms, and only on-shell quantities such as the S-matrix are scheme-independent. This is the one home for the scheme-dependence point; the higher-loop beta functions of Section 5.1 carry the same freedom. The two-derivative action (5.16) is the part everyone agrees on, and it is all we use in this chapter.

Example 5.6 (The linear-dilaton background)**.** Take flat space $g_{\mu\nu} = \eta_{\mu\nu}$, no flux $H = 0$, and a dilaton linear in the coordinates, $\phi = V_\mu x^\mu$ with V_μ constant. Then $R_{\mu\nu} = 0$ and $\nabla_\mu\nabla_\nu\phi = 0$, so (5.21) and the B equation hold identically. The dilaton equation reduces to the constant $\beta^\phi = 0$, which fixes the gradient:

$$\frac{D - D_c}{6} + \alpha'\, V_\mu V^\mu = 0 \qquad \Longrightarrow \qquad V_\mu V^\mu = \frac{D_c - D}{6\alpha'}.$$

For a subcritical string ($D < D_c$) this has real spacelike solutions: the string lives happily in $D < 26$ or $D < 10$ dimensions provided the dilaton runs linearly, its slope tuned to make up the central-charge deficit. This is the *linear-dilaton* vacuum, and it is one of the very few backgrounds that solves the beta functions *exactly*, to all orders in α', because the worldsheet theory is still free (the dilaton coupling is a total-derivative improvement of the stress tensor). At criticality $D = D_c$ the equation forces $V^2 = 0$: only a constant or null-gradient dilaton survives, which is the case relevant to the ten-dimensional superstrings.

Exercise 5.3. The graviton equation rests on the scalar-tensor identity (5.18), the one asserted step in the "gravity as a law" derivation. Fill it in. Vary $\int \mathrm{d}^D x\, \sqrt{-g}\, f(\phi)\, R$ with respect to $g^{\mu\nu}$, treating f as a fixed function of an independent field. The algebraic terms give $f\big(R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R\big)$; the piece $\sqrt{-g}\, f\, g^{\mu\nu}\delta R_{\mu\nu}$, which is a pure boundary term when f is constant, now survives because f is not. Using $g^{\mu\nu}\delta R_{\mu\nu} = \nabla_\mu(g^{\rho\sigma}\delta\Gamma^\mu_{\rho\sigma} - g^{\mu\sigma}\delta\Gamma^\rho_{\rho\sigma})$, integrate by parts twice to move both derivatives onto f, and obtain the $(g_{\mu\nu}\nabla^2 - \nabla_\mu\nabla_\nu)f$ term. Specialise to $f = e^{-2\phi}$ and confirm the Einstein-Hilbert contribution used above.

Exercise 5.4. The B-field equation of motion, its Bianchi partner $\mathrm{d}H = 0$, and their reading as the electric and magnetic Maxwell statements for the Kalb-Ramond field are worked out in Exercise 5.1; do not re-derive them here. Instead pursue the charge content of $\mathrm{d}(e^{-2\phi} \star H) = 0$. First complete the action-side calculation: starting from the last term of (5.16) with $H = \mathrm{d}B$, vary B, integrate by parts, and confirm (5.17). The seven-form $e^{-2\phi} \star H$ is closed on shell, so

its integral over a closed seven-cycle Σ_7 is conserved; read $\nabla^\mu(e^{-2\phi}H_{\mu\nu\lambda}) = 0$ as the statement that this current is covariantly conserved. Then show that $\int_{\Sigma_7} e^{-2\phi} \star H$ counts the number of fundamental strings threading Σ_7, the charge that couples electrically to B, and explain why $\mathrm{d}H = 0$ is instead the kinematic, magnetic statement.

5.2.2 The dilaton in front: string frame and the sphere

The factor $e^{-2\phi}$ multiplying the whole Lagrangian in (5.16) records that this action is the *tree-level* effective action of the closed string, generated by worldsheets of spherical topology.

Recall from Chapter 1 the Fradkin-Tseytlin coupling of the dilaton to the worldsheet curvature, $S_\phi = \frac{1}{4\pi}\int \sqrt{\gamma}\,\phi\, R^{(2)}$ of (1.96). For a constant dilaton this integrates by Gauss-Bonnet to ϕ times the Euler number $\chi = 2 - 2g$ of a genus-g closed surface, so a worldsheet of genus g is weighted by $e^{-\chi\langle\phi\rangle} = g_s^{2g-2}$, as in (1.97). The sphere has $g = 0$, $\chi = 2$, and weight $g_s^{-2} = e^{-2\langle\phi\rangle}$. Every term in (5.16) carries the same $e^{-2\phi}$ because they all descend from the same sphere: the graviton, the B-field and the dilaton are all NS-NS massless closed strings, and their tree-level two-point and three-point functions on the sphere are what the two-derivative action reproduces. Promoting the constant $\langle\phi\rangle$ to the field $\phi(x)$ turns the topological weight g_s^{-2} into the local factor $e^{-2\phi(x)}$ standing in front of the Lagrangian. A metric written in the frame where the closed-string tree-level action takes the form (5.16), with $e^{-2\phi}$ common to all terms, is called the *string frame* (or sigma-model frame): it is the metric that appears directly in the worldsheet action, the metric a string "sees".

This tells us where the loop corrections live. Adding a handle to the worldsheet costs $g_s^2 = e^{2\langle\phi\rangle}$, so the one-loop (torus) correction to the effective action comes with two fewer powers of $e^{-\phi}$, that is, with $e^0 = 1$ out front, and the ℓ-loop term with $e^{(2\ell-2)\phi}$. The single power $e^{-2\phi}$ in (5.16) is therefore a sharp statement: this is the leading term in the closed-string coupling expansion, and nothing in it has been corrected by string loops. The Ramond-Ramond fields of Section 5.4 will enter with a different power of e^{ϕ}, precisely because their vertex operators dress the dilaton differently; that mismatch is what makes the RR sector look strange in the string frame and clean in the Einstein frame.

The overall constant. The prefactor $1/2\kappa_{10}^2$ sets the units of the action and, after we fix the frame, the value of Newton's constant. Its relation to the string data is (A.6) of Appendix A,

$$2\kappa_{10}^2 = (2\pi)^7\,\alpha'^4\,g_s^2 = (2\pi)^7\,\ell_s^8\,g_s^2. \tag{5.23}$$

Two of the three factors follow from cheap arguments. In $D = 10$ the Einstein-Hilbert term requires $[2\kappa_{10}^2] = (\text{length})^8$, since $\int \mathrm{d}^{10}x\, R$ has dimensions $(\text{length})^{10}\cdot(\text{length})^{-2}$ and the action is dimensionless; the only length in the game is $\ell_s = \sqrt{\alpha'}$, so $2\kappa_{10}^2 \propto \alpha'^4$ on dimensional grounds alone. The factor g_s^2 is the sphere weight: whether one writes the prefactor as $1/2\kappa^2$ with the full dilaton left in $e^{-2\phi}$, or evaluates $e^{-2\langle\phi\rangle} = g_s^{-2}$ and absorbs it into the constant, the tree-level graviton kinetic term is proportional to $g_s^{-2}\times(\text{const})$, and matching the two conventions puts the g_s^2 where (5.23) shows it. The pure number $(2\pi)^7$ is the only part that requires real work: it comes from the properly normalised three-graviton amplitude on the sphere, computed once and for all in the amplitude machinery of Chapter 4. We quote it.

Where the g_s lives

Conventions differ on whether g_s sits inside κ or inside the dilaton. Two choices are common. (i) Keep the *full* dilaton in $e^{-2\phi}$ and use a g_s-free constant $2\kappa_0^2 = (2\pi)^7\alpha'^4$; then the *physical* gravitational coupling read off at the vacuum is $2\kappa_{10}^2 = 2\kappa_0^2\,g_s^2$, because $e^{-2\langle\phi\rangle} = g_s^{-2}$ multiplies it. (ii) Split $\phi = \langle\phi\rangle + \tilde{\phi}$, absorb $e^{-2\langle\phi\rangle} = g_s^{-2}$ into the constant so that the prefactor is $1/2\kappa_{10}^2$ with $2\kappa_{10}^2 = (2\pi)^7\alpha'^4 g_s^2$ as in (5.23), and let only the fluctuation $\tilde{\phi}$

appear in $e^{-2\tilde{\phi}}$. We follow Appendix A, which quotes the physical coupling (5.23). The two conventions describe the same physics; the danger is mixing them and double-counting a g_s^2. When comparing formulae across textbooks, always check which dilaton is sitting in the exponential.

Because $g_s = e^{\langle\phi\rangle}$ is the vacuum value of a field and not a fixed number, so is Newton's constant. From (5.23), $G_{10} \propto \kappa_{10}^2 \propto g_s^2\,\alpha'^4$: gravity is weak precisely when the string is weakly coupled, and it switches off entirely as $g_s \to 0$. Because the strength of gravity is a modulus rather than a constant of nature, the strong-weak dualities of Chapter 7 can reorganise the theory so drastically.

5.2.3 String frame versus Einstein frame

The string frame is natural from the worldsheet, but it is awkward for gravitational physics. The coefficient of R in (5.16) is $e^{-2\phi}$, a field, so the "Newton constant" varies from point to point and the graviton kinetic term is not canonically normalised. To read off masses, compare with the ADM formulae, or set up the BPS bound of Section 5.6, one wants ordinary Einstein gravity with a fixed κ. That is achieved by a field-dependent Weyl rescaling of the metric.

Write

$$g_{\mu\nu} = e^{2\omega(\phi)}\,\tilde{g}_{\mu\nu}, \tag{5.24}$$

and choose ω so that the $e^{-2\phi}$ in front of R is cancelled. Under (5.24) the Ricci scalar transforms (in D dimensions, in the curvature conventions of Appendix A) as

$$\sqrt{-g}\,R = \sqrt{-\tilde{g}}\,e^{(D-2)\omega}\Big(\tilde{R} - 2(D-1)\tilde{\nabla}^2\omega - (D-1)(D-2)\,\tilde{\nabla}_\mu\omega\,\tilde{\nabla}^\mu\omega\Big), \tag{5.25}$$

so $\sqrt{-g}\,e^{-2\phi}R = \sqrt{-\tilde{g}}\,e^{(D-2)\omega-2\phi}(\tilde{R} + \dots)$. The prefactor becomes unity when

$$\omega = \frac{2\phi}{D-2}, \qquad \text{i.e. in } D=10: \quad \omega = \frac{\phi}{4}, \quad g_{\mu\nu} = e^{\phi/2}\,\tilde{g}_{\mu\nu}. \tag{5.26}$$

The metric $\tilde{g}_{\mu\nu}$ is the *Einstein-frame* metric; the string-frame metric is $e^{\phi/2}$ times heavier. Carrying the rescaling through every term of (5.16) is a short computation (Exercise 5.5). The $\tilde{\nabla}^2\omega$ term in (5.25) is a total derivative and drops. What remains, in $D = 10$, is

$$\boxed{S_{\rm NS} = \frac{1}{2\kappa_{10}^2}\int \mathrm{d}^{10}x\,\sqrt{-\tilde{g}}\,\Big(\tilde{R} - \tfrac{1}{2}\,\tilde{\nabla}_\mu\phi\,\tilde{\nabla}^\mu\phi - \tfrac{1}{12}\,e^{-\phi}\,\tilde{H}_{\mu\nu\lambda}\tilde{H}^{\mu\nu\lambda}\Big)\,,} \tag{5.27}$$

where all contractions and the Laplacian now use $\tilde{g}$. Three things changed. The gravitational term is canonical Einstein-Hilbert, $\sqrt{-\tilde{g}}\,\tilde{R}$ with a fixed coefficient $1/2\kappa_{10}^2$. The dilaton kinetic term has become $-\frac{1}{2}(\tilde{\nabla}\phi)^2$, a properly normalised, healthy scalar with the correct sign in mostly-plus signature: the alarming $+4(\nabla\phi)^2$ of the string frame was an artefact of the $e^{-2\phi}R$ mixing, and the rescaling exposes the true kinetic term. The B-field kinetic term picked up a factor $e^{-\phi}$, so the dilaton now couples to the Kalb-Ramond field as $e^{-\phi}\tilde{H}^2$, and the string coupling controls the strength of the H-flux in Einstein frame.

Which frame to use. The two frames describe identical physics; the metric is a field redefinition apart. The choice is one of convenience.

- *String frame* is the one to use for anything that refers back to the worldsheet: the tension of a fundamental string is $1/2\pi\alpha'$ measured with $g_{\mu\nu}$, independent of g_s, and distances that a probe string measures are string-frame distances. Brane tensions and the sigma-model expansion are cleanest here.

- *Einstein frame* is the one to use for gravitational and thermodynamic statements: ADM mass, the Schwarzschild-like fall-off of a metric, black-hole entropy, and the BPS bound $M \geq |Z|$ all presume canonically normalised gravity with a ϕ-independent κ. Moduli-space geometry and the kinetic terms of four-dimensional effective theories are read off in Einstein frame.

A tension or a mass computed in one frame differs from the other by powers of e^{ϕ}; the g_s-scalings of brane tensions quoted in Section 5.6 and Chapter 6 (F1 $\sim g_s^0$, Dp $\sim g_s^{-1}$, NS5 $\sim g_s^{-2}$) are frame-sensitive statements, and one must say which metric is being used. When in doubt, convert to Einstein frame, where "mass" means what a relativist means by it.

Exercise 5.5. Carry out the Weyl rescaling (5.24)–(5.26) in general dimension D. Using (5.25), show that the choice $\omega = 2\phi/(D-2)$ removes the $e^{-2\phi}$ in front of $\tilde{R}$, and that the dilaton kinetic term in Einstein frame is

$$-\frac{4}{D-2}\,\tilde{\nabla}_\mu\phi\,\tilde{\nabla}^\mu\phi,$$

which is $-\frac{1}{2}(\tilde{\nabla}\phi)^2$ at $D = 10$. Show that the B-field term acquires a factor $e^{-8\phi/(D-2)}$, equal to $e^{-\phi}$ at $D = 10$. Finally, redo the computation keeping the cosmological term $-\frac{2(D-D_c)}{3\alpha'}$ inside the brackets of (5.16), and show that, because ϕ is not rescaled by (5.24), in Einstein frame it becomes a runaway exponential dilaton potential $\propto (D-D_c)\,e^{4\phi/(D-2)}$, with no stationary point: the non-critical string has no static maximally symmetric vacuum, the content of Theorem 5.6 seen from the target-space action.

5.2.4 Universality, and what the NS-NS action leaves out

Everything in (5.16) followed from the massless NS-NS closed-string spectrum and the requirement of Weyl invariance, with no reference to which superstring we started from. The graviton, the two-form and the dilaton are shared by IIA, IIB, type I, the two heterotics and the bosonic string, and their two-derivative dynamics is identical in all of them, so (5.16) is the common core of every ten-dimensional supergravity. This is why the full supergravities can be assembled economically: one writes the universal NS-NS action once, then adds the sector that distinguishes the theory.

What is added, and where it is derived, sets up the rest of the chapter. The type II theories bolt on Ramond-Ramond p-form fields with their own kinetic and Chern-Simons terms; that is Section 5.4, where the different dilaton dressing of the RR sector (foreshadowed above) makes the Einstein frame the natural home for those fields. Type I and the heterotic strings add a ten-dimensional Yang-Mills sector and modify the field strength to $H = \mathrm{d}B - \frac{\alpha'}{4}(\omega_{\mathrm{YM}} - \omega_{\mathrm{L}})$, whose Bianchi identity $\mathrm{d}H \neq 0$ carries the Green-Schwarz anomaly-cancellation story of Section 5.5. We take the genuinely new piece first: eleven-dimensional supergravity in Section 5.3, which cannot be read off from any worldsheet, and only then return to bolt the RR and Yang-Mills sectors onto the universal core. All four theory-specific actions are collected in one convention in Appendix I and quoted rather than rebuilt. The universal piece (5.16) we have earned: it is the beta functions of Section 5.1 written as an action.

5.3 Eleven-dimensional supergravity

The superstring lives in ten dimensions, so it is fair to ask what an eleventh dimension is doing in a chapter on string effective actions. The answer, which we make precise by the end of the section and cash out in full in Chapter 7, is that the strongly coupled type IIA string grows an extra dimension. Its low-energy limit is not a ten-dimensional theory at all but an eleven-dimensional one, and that eleven-dimensional theory is the unique supergravity in the highest spacetime dimension that admits one. We build it here for three reasons. It is the simplest supergravity

in the course: a single multiplet, with no choice of matter to make. It is the most economical route to type IIA, which drops out when we put the theory on a circle. And it is the low-energy shadow of M-theory, the object at the centre of the duality web.

Why eleven?

The five ten-dimensional superstring theories are not the end of the story. Type IIA at strong coupling opens up an eleventh dimension whose size grows with g_s, and the low-energy limit is the unique supergravity that lives in eleven dimensions. This section builds that theory (the metric, a three-form C_3, a gravitino, and the Chern-Simons term $C_3 \wedge F_4 \wedge F_4$), identifies its M2 and M5 brane sources, and reduces it on a circle to recover type IIA with g_s traded for the radius R_{11}. It is the shortest path into M-theory (Chapter 7) and the organising centre of the duality web.

5.3.1 Why eleven, and why the theory is unique

The number eleven is not an input. It is forced by asking how much supersymmetry a theory of gravity can carry. A supercharge is a spinor, and each one shifts the spin of a state by half a unit. Pile up too many and the massless multiplet you build is forced to contain a field of spin greater than two, for which no consistent interacting theory coupled to gravity is known. Counting this carefully (the representation theory is in Appendix C) gives a ceiling of 32 real supercharges.

One Majorana supercharge in eleven dimensions has exactly 32 real components, so eleven dimensions saturates the bound with a single spinor: $\mathcal{N} = 1$. Reduce those 32 supercharges to four dimensions and they become the eight Majorana supercharges of $\mathcal{N} = 8$ supergravity, the maximal four-dimensional theory, whose massless multiplet runs over helicities from -2 to $+2$ with the graviton at the top. Add one supercharge beyond 32 and the top helicity exceeds two. Eleven is the largest dimension in which the smallest spinor still fits under the ceiling: a minimal spinor has 32 real components in $D = 11$ and 64 in $D = 12$.

Once the supercharge count is fixed, everything else follows. The massless multiplet is unique, the field content is forced by filling it out, and the interacting theory, meaning the full action with all its couplings, is fixed uniquely by local supersymmetry. This is the theorem of Cremmer, Julia and Scherk (1978): there is exactly one supergravity in eleven dimensions. We quote the field content from Appendix I and assemble the bosonic action on the board below; the fermion completion that pins down every coefficient is stated in Appendix C.

5.3.2 The field content

The multiplet has three fields: the metric g_{MN}, an antisymmetric three-form gauge potential C_{MNP}, and a single Majorana gravitino ψ_M (a vector-spinor). Here $M, N = 0, 1, \ldots, 10$. Massless states in eleven dimensions organise into representations of the little group Spin(9), and counting on-shell degrees of freedom this way gives the tally in the table.

Field	Spin(9) content	on-shell dof
g_{MN} (graviton)	symmetric traceless	44
C_{MNP} (three-form)	antisymmetric rank 3	84
ψ_M (gravitino)	Γ-traceless vector-spinor	128
bosons / fermions		$128 + 128$

The graviton contributes the symmetric traceless tensor of Spin(9), of dimension $\frac{1}{2} \cdot 9 \cdot 10 - 1 = 44$. The three-form contributes $\binom{9}{3} = 84$. Their sum, 128, matches the on-shell count of the gravitino to the number: a vector-spinor of Spin(9) has $9 \times 16 = 144$ components, and the irreducibility

(gamma-tracelessness) constraint removes 16, leaving 128. Bose and Fermi degrees of freedom balance, as any supersymmetric theory demands.

One feature of the list drives the rest of the section: there is no scalar. Eleven-dimensional supergravity has no dilaton, hence no dimensionless coupling constant and no tunable parameter at all. The only scale is the eleven-dimensional Planck length ℓ_p. When we reduce to ten dimensions the type IIA string coupling g_s has to come from somewhere, and the answer is that it is geometrised: it becomes the size of the circle.

5.3.3 The action

The bosonic sector is an Einstein-Hilbert term, a Maxwell-type kinetic term for the four-form field strength

$$F_4 = \mathrm{d}C_3, \qquad \mathrm{d}F_4 = 0 \ \ \text{(Bianchi)}, \tag{5.28}$$

and one term with no metric in it at all, a Chern-Simons coupling built purely from wedge products. In the conventions of Appendix A,

$$\boxed{S_{11} = \frac{1}{2\kappa_{11}^2}\int \mathrm{d}^{11}x\,\sqrt{-g}\left(R - \tfrac{1}{2}\,|F_4|^2\right) \;-\; \frac{1}{12\kappa_{11}^2}\int C_3 \wedge F_4 \wedge F_4\,,} \tag{5.29}$$

with $|F_4|^2 \coloneqq \frac{1}{4!}\,F_{MNPQ}F^{MNPQ}$, so that $-\frac{1}{2}|F_4|^2\sqrt{-g}\,\mathrm{d}^{11}x = -\frac{1}{2}\,F_4 \wedge \star F_4$. The single dimensionful constant is Newton's constant $2\kappa_{11}^2 = (2\pi)^8\ell_p^9$, quoted from Appendix I.

The Chern-Simons term is the interesting one. It is topological in the sense of being independent of the metric, and it cannot be guessed from bosonic reasoning alone. Its presence, and its coefficient relative to the Einstein and kinetic terms, are forced by closure of the local supersymmetry algebra: without it, the variation of the action under a supersymmetry transformation fails to cancel. This is the precise sense in which the theory is unique. The fermion terms and the derivation of the coefficient live in Appendix C; here we take the $C_3 \wedge F_4 \wedge F_4$ term as given and read off its consequences.

Varying Eq. (5.29) with respect to C_3 (using $\delta\int C_3 \wedge F_4 \wedge F_4 = 3\int \delta C_3 \wedge F_4 \wedge F_4$ after integration by parts, since $F_4 = \mathrm{d}C_3$) gives the gauge-field equation of motion

$$\mathrm{d}\star F_4 + \tfrac{1}{2}\,F_4 \wedge F_4 = 0. \tag{5.30}$$

The nonlinear second term is the fingerprint of the Chern-Simons coupling: the four-form sources itself. It is what makes Eq. (5.30) more than a free Maxwell equation, and, as we will see in Section 5.4, it descends to the Chern-Simons term of type IIA.

5.3.4 The M2 and M5 branes

A three-form gauge potential couples naturally to a three-dimensional worldvolume, in exactly the way a Maxwell one-form couples to the one-dimensional worldline of a point charge. A $(p+1)$-form worldvolume integral $\int C_{p+1}$ needs a $(p+1)$-dimensional history, that is, a p-brane. For C_3 the electric source is therefore a 2-brane, the *M2-brane* or membrane, with worldvolume coupling

$$S_{\mathrm{M2}} \supset T_{\mathrm{M2}}\int_{\mathcal{W}_3} C_3, \tag{5.31}$$

alongside a Nambu-Goto area term. The membrane carries electric C_3-charge, $\int_{S^7}\star F_4 \propto Q_{\mathrm{M2}}$, measured on a seven-sphere surrounding it in the eight transverse dimensions.

The magnetic dual is found by trading F_4 for its Hodge dual. Define the seven-form

$$F_7 \coloneqq \star F_4, \qquad \mathrm{d}F_7 = -\tfrac{1}{2}\,F_4 \wedge F_4, \tag{5.32}$$

where the right-hand side is just Eq. (5.30) rewritten. Locally $F_7 = \mathrm{d}C_6 - \frac{1}{2}\,C_3 \wedge F_4$ for a six-form potential C_6, whose electric source is a 5-brane, the *M5-brane*. What is electric under C_3 is magnetic under C_6 and vice versa, so the M2 and M5 are electric-magnetic duals. Their tensions, quoted from Appendix I, are

$$T_{\mathrm{M2}} = \frac{1}{(2\pi)^2\,\ell_p^3}, \qquad T_{\mathrm{M5}} = \frac{1}{(2\pi)^5\,\ell_p^6}, \tag{5.33}$$

and they are not independent. Charge quantisation in the Dirac-Nepomechie-Teitelboim sense ties the electric and magnetic charges together; in these normalisations $2\kappa_{11}^2\,T_{\mathrm{M2}}\,T_{\mathrm{M5}} = 2\pi$, the minimal Dirac unit (Exercise 5.8).

Both branes are half-BPS. Each is a solution of the field equations that leaves invariant exactly half of the 32 supersymmetries, 16 Killing spinors, which is why its tension equals its charge and why the two never repel: parallel M2-branes, or parallel M5-branes, feel no net force. We set up that statement properly in Section 5.6; the harmonic-function solutions and the full brane catalogue are collected in Appendix I.

Remark 5.7 (Beyond the lectures). The M5-brane is subtler than the M2. Its worldvolume theory contains a two-form gauge field b_2 whose three-form field strength is self-dual, $h_3 = \star_6 h_3$. A self-dual field has no simple Lorentz-covariant Lagrangian (the naive kinetic term vanishes), which is the same obstruction that afflicts the type IIB self-dual five-form in Section 5.4. This chiral tensor multiplet on the M5 is the six-dimensional $(2,0)$ theory, one of the exotic superconformal theories with no Lagrangian description at all; it reappears in Chapter 7 and again in the black-hole entropy counting of Chapter 12.

5.3.5 Supersymmetry and Killing spinors

The one piece of the fermion sector we need explicitly is the supersymmetry variation of the gravitino, because it is the object that decides how much supersymmetry a given background preserves. To leading order in fermions it reads

$$\delta\psi_M = \nabla_M\varepsilon + \frac{1}{288}\Big(\Gamma_M{}^{NPQR} - 8\,\delta_M^N\,\Gamma^{PQR}\Big)F_{NPQR}\,\varepsilon, \qquad \nabla_M\varepsilon = \Big(\partial_M + \tfrac{1}{4}\,\omega_{Mab}\Gamma^{ab}\Big)\varepsilon, \tag{5.34}$$

where ε is a 32-component Majorana spinor, ω_{Mab} the spin connection, and the coefficient $1/288$ is fixed together with the Chern-Simons coefficient of Eq. (5.29). We quote Eq. (5.34) from Appendix C rather than deriving it.

For a purely bosonic background the fermions vanish, and the only nontrivial supersymmetry condition is that the gravitino variation itself vanish. A spinor ε solving

$$\delta\psi_M = 0 \qquad \text{(the Killing spinor equation)} \tag{5.35}$$

generates a supersymmetry preserved by the background. In flat space, $F_4 = 0$ and $\omega = 0$, Eq. (5.35) reduces to $\partial_M\varepsilon = 0$, which has 32 constant solutions: the Minkowski vacuum is maximally supersymmetric. Switch on the M2 or M5 background and Eq. (5.35) projects the 32 down to 16, the half-BPS count quoted above. The fraction of supersymmetry a background preserves is precisely the number of independent solutions of Eq. (5.35), which is the cleanest invariant we have of a supergravity solution.

5.3.6 Reduction on a circle: type IIA

Now the payoff. Compactify the eleventh direction on a circle, $x^{11} \simeq x^{11} + 2\pi R_{11}$, and ask what a ten-dimensional observer sees. The eleven-dimensional metric splits into a ten-dimensional

metric, a vector (the Kaluza-Klein photon), and a scalar (the size of the circle). The correct ansatz, written so that the ten-dimensional fields come out in string frame, is

$$\mathrm{d}s_{11}^2 = e^{-2\phi/3}\, g_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu + e^{4\phi/3}(\mathrm{d}x^{11} + C_\mu\,\mathrm{d}x^\mu)^2, \qquad \mu = 0,\ldots,9. \tag{5.36}$$

Read off the three ten-dimensional fields hiding in g_{MN}. The block $g_{\mu\nu}$ is the ten-dimensional graviton. The off-diagonal component $C_\mu = g_{\mu,11}/g_{11,11}$ is a one-form: the Kaluza-Klein photon. And the size of the circle, $g_{11,11} = e^{4\phi/3}$, is a scalar, which we have named ϕ because it is going to be the type IIA dilaton. The Weyl factor $e^{-2\phi/3}$ in front of $g_{\mu\nu}$ is chosen precisely so that reducing the Einstein-Hilbert term lands on the string-frame action of Section 5.2, with its $e^{-2\phi}R$ prefactor and the tell-tale $+4(\nabla\phi)^2$ (Exercise 5.7).

The three-form splits by whether or not its legs point along the circle:

$$C_{\mu\nu\rho} \longrightarrow C_3 \ \text{(RR three-form)}, \qquad C_{\mu\nu,11} \longrightarrow B_{\mu\nu} \ \text{(NS-NS two-form)}. \tag{5.37}$$

Collecting the harvest: from g_{MN} we get the graviton $g_{\mu\nu}$, the dilaton ϕ, and a one-form C_1; from C_3 we get a two-form B_2 and a three-form C_3. Sort them by how they transform under the dilaton and one finds exactly the type IIA field content (the IIA action itself is assembled in Section 5.4; here we need only its field content): the NS-NS sector $(g_{\mu\nu}, B_{\mu\nu}, \phi)$, and the Ramond-Ramond sector with a one-form C_1 and a three-form C_3. The RR one-form of IIA *is* the Kaluza-Klein photon of the M-theory circle. The single Majorana gravitino likewise splits into two ten-dimensional gravitinos of opposite chirality, which is the non-chiral $(1,1)$ spinor content of IIA. The Chern-Simons term of Eq. (5.29) reduces to the $B \wedge F_4 \wedge F_4$ Chern-Simons term of IIA (Section 5.4). Nothing is left over and nothing is missing; the reduction is exact.

The coupling is the radius. Momentum around the circle is quantised, $p_{11} = n/R_{11}$, so the ten-dimensional observer sees a tower of Kaluza-Klein states of mass $|n|/R_{11}$, each carrying n units of charge under the photon C_1. Since C_1 is the RR one-form, the objects charged under it are the D0-branes of type IIA, whose mass is $1/(g_s\ell_s)$ (quoted from Chapter 6). Matching the two towers, D0-branes with Kaluza-Klein momenta, fixes the physical radius of the circle:

$$R_{11} = g_s\,\ell_s, \qquad \ell_p = g_s^{1/3}\,\ell_s, \qquad \text{equivalently} \quad R_{11} = g_s^{2/3}\,\ell_p, \tag{5.38}$$

with the first relation the physical statement and the others following from the reduction of Newton's constant, $2\kappa_{10}^2 = 2\kappa_{11}^2/(2\pi R_{11})$; the numerical chain is spelled out in Appendix I. The content of Eq. (5.38) is the headline of Chapter 7. At weak coupling $g_s \to 0$ the circle shrinks below the string length and the eleventh dimension is invisible: type IIA looks ten-dimensional. As g_s grows the circle grows with it, the D0-brane tower becomes light, and at strong coupling the eleventh dimension decompactifies. Strongly coupled type IIA is M-theory on a large circle, and eleven-dimensional supergravity is what governs it at low energies.

The branes track this too. Every M-theory brane becomes an IIA brane according to whether it wraps the circle, and the dictionary is a preview of the web to come.

IIA object	M-theory origin
D0-brane	Kaluza-Klein momentum on S^1
fundamental string F1	M2 wrapped on S^1
D2-brane	M2 transverse to S^1
D4-brane	M5 wrapped on S^1
NS5-brane	M5 transverse to S^1
D6-brane	Kaluza-Klein monopole (Taub-NUT)

The fundamental string of IIA is a wrapped membrane: shrink the circle and the tension $T_{\mathrm{M2}} \cdot 2\pi R_{11}$ of the wrapped M2 becomes the string tension $1/(2\pi\alpha')$. The details, including the one IIA brane with no clean M-theory origin (the D8), belong to Chapter 7. For our purposes the entire IIA brane spectrum is accounted for by the M2, the M5, and the geometry of the circle.

With the eleven-dimensional theory in hand and its reduction to IIA understood, the two ten-dimensional type II actions are the natural next stop. We assemble them, in the same conventions, in Section 5.4.

Exercise 5.6 (Degree-of-freedom count)**.** Working with the little group Spin(9), confirm the on-shell counts in the field-content table. Show that the symmetric traceless tensor gives 44, the antisymmetric three-index tensor gives $\binom{9}{3} = 84$, and a gamma-traceless vector-spinor gives $9 \times 16 - 16 = 128$. Conclude that bosonic and fermionic degrees of freedom match at 128 each, as supersymmetry requires. (Difficulty: warm-up.)

Exercise 5.7 (Circle reduction of the Einstein-Hilbert term)**.** Insert the ansatz Eq. (5.36) into $\frac{1}{2\kappa_{11}^2}\int \mathrm{d}^{11}x\,\sqrt{-g_{11}}\,R_{11}$, integrate over x^{11} to produce a factor $2\pi R_{11}$, and verify that the Weyl factor $e^{-2\phi/3}$ delivers the string-frame result

$$\frac{1}{2\kappa_{10}^2}\int \mathrm{d}^{10}x\,\sqrt{-g}\,e^{-2\phi}\Big(R + 4(\nabla\phi)^2\Big) + \ldots, \qquad \frac{1}{2\kappa_{10}^2} = \frac{2\pi R_{11}}{2\kappa_{11}^2},$$

matching the dilaton gravity of Section 5.2. Track the field strength $\mathrm{d}C_1$ and $\mathrm{d}B_2$ terms as well and identify which IIA kinetic terms they become. (Difficulty: the standard Kaluza-Klein computation; do it once in your life.)

Exercise 5.8 (Dirac quantisation for M2 and M5)**.** Using $2\kappa_{11}^2 = (2\pi)^8\ell_p^9$ and the tensions Eq. (5.33), verify $2\kappa_{11}^2\,T_{\mathrm{M2}}\,T_{\mathrm{M5}} = 2\pi$. Interpret this as the statement that the electric charge of one membrane and the magnetic charge of one fivebrane saturate the minimal Dirac unit, so no fractional M-brane charge is allowed. (Difficulty: short.)

5.4 Type IIA and IIB supergravity

The universal NS-NS action of Section 5.2 is the piece every ten-dimensional string theory shares. It is not, by itself, a supergravity: it has no fermions and no supersymmetry made manifest. To close a supermultiplet we have to add the Ramond-Ramond fields and the gravitini, and there are exactly two ways to do this at the maximal level of supersymmetry. Those two completions are type IIA and type IIB supergravity, the low-energy limits of the two type II superstrings whose spectra you met in Chapter 3.

Both theories carry 32 real supercharges, the most a ten-dimensional theory of gravity can have. Two spacetime dimensions higher this is the 32 of eleven-dimensional supergravity (Section 5.3); one cannot do better without fields of spin greater than two. Maximal supersymmetry is a straitjacket: once the field content is fixed, the two-derivative action is determined up to field redefinitions, with no free couplings beyond κ_{10} and the normalisations of the gauge fields, so there is nothing to fit. We assemble the field content, then quote the action that supersymmetry forces on it, exactly as collected in Appendix I.

The whole content of the distinction is the Ramond-Ramond sector and the chirality of the fermions:

- **Type IIA** is non-chiral. Its two gravitini have *opposite* ten-dimensional chirality, so it is a $\mathcal{N} = (1,1)$ theory. Its Ramond-Ramond potentials are odd forms carrying *even* field strengths: a one-form C_1 and a three-form C_3. IIA is what you get by reducing eleven-dimensional supergravity on a circle (Section 5.3).

- **Type IIB** is chiral. Its two gravitini have the *same* chirality, a $\mathcal{N} = (2,0)$ theory. Its Ramond-Ramond potentials are even forms: a scalar C_0, a two-form C_2, and a four-form C_4 whose field strength is self-dual. IIB has a classical $\mathrm{SL}(2,\mathbb{R})$ symmetry that becomes the $\mathrm{SL}(2,\mathbb{Z})$ S-duality of Chapter 7.

Everything else follows from these two lines.

5.4.1 Field content: two ways to fill the Ramond-Ramond sector

The massless closed-string spectrum of a type II theory is the tensor product of a left-moving and a right-moving ground sector, each of which is either Neveu-Schwarz or Ramond after the GSO projection (Chapter 3). Schematically,

$$(\mathrm{NS} \oplus \mathrm{R})_{\mathrm{L}} \otimes (\mathrm{NS} \oplus \mathrm{R})_{\mathrm{R}}. \tag{5.39}$$

The NS⊗NS product is the same in both theories and gives the metric $g_{\mu\nu}$, the Kalb-Ramond field $B_{\mu\nu}$ and the dilaton ϕ: the NS-NS sector. The two mixed products NS⊗R and R⊗NS are spacetime fermions (the gravitini and dilatini). The R⊗R product is a bosonic sector built from a pair of spinor ground states, and this is where the two theories part company.

A Ramond ground state in ten dimensions is a Majorana-Weyl spinor of Spin(8), the transverse little group, in one of the two chiralities $\mathbf{8}_s$ or $\mathbf{8}_c$. Tensoring two of them and decomposing into antisymmetric tensors turns a bispinor into a sum of p-forms. The GSO projection fixes each factor's chirality, and the *relative* choice is precisely the IIA/IIB fork:

$$\text{IIA}: \quad \mathbf{8}_s \otimes \mathbf{8}_c \;=\; \mathbf{8}_v \oplus \mathbf{56}_v \;=\; [1] \oplus [3], \tag{5.40}$$

$$\text{IIB}: \quad \mathbf{8}_s \otimes \mathbf{8}_s \;=\; [0] \oplus [2] \oplus [4]_+ \;=\; \mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35}_+. \tag{5.41}$$

Here $[p]$ denotes a p-form of Spin(8). The opposite-chirality product (5.40) gives a one-form and a three-form, which we read as the field strengths F_2 and F_4 of potentials C_1 and C_3. The same-chirality product (5.41) gives a scalar, a two-form and a *self-dual* four-form $\mathbf{35}_+$, the field strengths F_1, F_3 and the self-dual F_5 of C_0, C_2, C_4. The self-duality of F_5 is not something we impose for convenience: it is already there in the spectrum, forced by taking two spinors of the same chirality. What we struggle with later is only the writing of a Lorentz-invariant Lagrangian for a field that is self-dual.

Table 5.1 collects the bosons. Each theory has 64 NS-NS and 64 Ramond-Ramond on-shell degrees of freedom, so 128 bosonic states, matched by 128 fermionic states (two gravitini at 56 each and two dilatini at 8 each). That $128 + 128$ is the maximal 10D supergravity multiplet.

Two remarks before we write the actions. First, each RR potential C_p has an electric-magnetic dual C_{8-p}: C_1 pairs with C_7, C_3 with C_5 in IIA, and C_0 with C_8, C_2 with C_6, C_4 with itself in IIB. We keep only the lower half in the action and let the equations of motion supply the rest; the full democratic set is what makes the D-brane couplings transparent in Chapter 6. The upshot for branes is the familiar even/odd rule: a Dp-brane is the electric source of C_{p+1}, so IIA carries D0, D2, D4, D6, D8 and IIB carries D(-1), D1, D3, D5, D7, D9. Second, the RR kinetic terms below carry *no* factor of $e^{-2\phi}$ in the string frame, unlike the NS-NS terms. That asymmetry is physical: RR fields are not sourced by fundamental strings at any order, and the branes that do source them have tension $\sim 1/g_s$ rather than the $\sim 1/g_s^2$ of an NS-NS soliton. We return to those scalings in Section 5.6.

Throughout we use the norm

$$|F_p|^2 \;\coloneqq\; \frac{1}{p!}\, F_{\mu_1\cdots\mu_p} F^{\mu_1\cdots\mu_p}, \qquad \text{so that} \qquad F_p \wedge \star F_p \;=\; |F_p|^2 \sqrt{-g}\, \mathrm{d}^{10}x, \tag{5.42}$$

matching Appendix A. In this notation the NS-NS three-form term of Section 5.2 is $-\frac{1}{2}|H_3|^2 = -\frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho}$, and $H_3 = \mathrm{d}B$.

field	Spin(8) irrep	on-shell dof	sector / theory
$g_{\mu\nu}$	$\mathbf{35}_v$	35	NS-NS (both)
$B_{\mu\nu}$	$\mathbf{28}$	28	NS-NS (both)
ϕ	$\mathbf{1}$	1	NS-NS (both)
C_1	$\mathbf{8}_v$	8	RR (IIA)
C_3	$\mathbf{56}_v$	56	RR (IIA)
C_0	$\mathbf{1}$	1	RR (IIB)
C_2	$\mathbf{28}$	28	RR (IIB)
C_4 (self-dual F_5)	$\mathbf{35}_+$	35	RR (IIB)

Table 5.1: Bosonic field content of the two maximal ten-dimensional supergravities, organised by the transverse little group Spin(8). The NS-NS sector (64 states) is common; the Ramond-Ramond sector (64 states each) distinguishes them. Counting the RR degrees of freedom on shell means keeping the little-group representations, with the four-form entry already halved by self-duality.

5.4.2 The type IIA action

The bosonic action of type IIA supergravity, in the string frame and to two derivatives, is

$$\begin{aligned} S_{\text{IIA}} \;=\; \frac{1}{2\kappa_{10}^2}\int \mathrm{d}^{10}x\,\sqrt{-g}\,\Big[\, e^{-2\phi}\Big(R + 4\,\nabla_\mu\phi\,\nabla^\mu\phi - \tfrac{1}{2}|H_3|^2\Big) \\ - \tfrac{1}{2}|F_2|^2 - \tfrac{1}{2}\big|\tilde{F}_4\big|^2\,\Big] \;-\; \frac{1}{4\kappa_{10}^2}\int B_2\wedge F_4\wedge F_4, \end{aligned} \tag{5.43}$$

with the field strengths

$$F_2 = \mathrm{d}C_1, \qquad F_4 = \mathrm{d}C_3, \qquad \tilde{F}_4 = \mathrm{d}C_3 - H_3\wedge C_1, \qquad H_3 = \mathrm{d}B_2. \tag{5.44}$$

The first bracket is the NS-NS action derived in Section 5.2; the second bracket is the Ramond-Ramond kinetic energy; the last term, with no metric and no $\sqrt{-g}$, is a topological Chern-Simons (or Wess-Zumino) term. The modification $\tilde{F}_4 = F_4 - H_3\wedge C_1$ is not optional. It makes $\tilde{F}_4$ invariant under the RR gauge transformations $C_1 \to C_1 + \mathrm{d}\lambda_0$ and $C_3 \to C_3 + \mathrm{d}\lambda_2 - H_3\wedge\lambda_0$; invariance under the Kalb-Ramond shift $B_2 \to B_2 + \mathrm{d}\lambda_1$ comes for free, since $\tilde{F}_4$ sees B_2 only through $H_3 = \mathrm{d}B_2$. It is also the form-degree-four piece of the polyform relation $F = \mathrm{d}C - H\wedge C$ quoted in Appendix A. The pattern, a bare exterior derivative corrected by a wedge with H_3, recurs in every RR field strength, and it is the reason the Bianchi identities read $\mathrm{d}\tilde{F}_4 = H_3\wedge F_2$ rather than $\mathrm{d}\tilde{F}_4 = 0$. The next example verifies both facts.

Example 5.8 (Twisted field strength: Bianchi and gauge invariance)**.** The corrected four-form $\tilde{F}_4 = \mathrm{d}C_3 - H_3\wedge C_1$ is the first piece of genuinely new RR machinery, so we check its two defining properties by hand.

Bianchi identity. Using $\mathrm{d}^2 = 0$ and $\mathrm{d}H_3 = 0$,

$$\mathrm{d}\tilde{F}_4 \;=\; -\mathrm{d}(H_3\wedge C_1) \;=\; -(\mathrm{d}H_3\wedge C_1 - H_3\wedge \mathrm{d}C_1) \;=\; H_3\wedge F_2, \tag{5.45}$$

where the internal minus sign is the $(-1)^3$ from carrying d past the three-form H_3, and $\mathrm{d}C_1 = F_2$. The naive $\mathrm{d}F_4 = 0$ is replaced by a source term built from the other fields, exactly the Bianchi quoted above.

Gauge invariance. With λ_0 a function and λ_2 a two-form, the RR shifts of (5.44) give

$$\delta\tilde{F}_4 \;=\; \mathrm{d}(\mathrm{d}\lambda_2 - H_3\wedge\lambda_0) - H_3\wedge\mathrm{d}\lambda_0 \;=\; H_3\wedge\mathrm{d}\lambda_0 - H_3\wedge\mathrm{d}\lambda_0 \;=\; 0, \tag{5.46}$$

using $\mathrm{d}(H_3 \wedge \lambda_0) = -H_3 \wedge \mathrm{d}\lambda_0$. Under $B_2 \to B_2 + \mathrm{d}\lambda_1$ the three-form $H_3 = \mathrm{d}B_2$ is unchanged, so $\tilde{F}_4$ is invariant with no compensating shift of C_3. The same two calculations, with the degrees relabelled, dispatch $\tilde{F}_3$ and $\tilde{F}_5$ in the type IIB field strengths (5.49).

The Chern-Simons term deserves a second look. It is cubic in the gauge fields, it involves no metric, and it is there because supersymmetry demands it: drop it and the variation of (5.43) no longer closes into the supersymmetry algebra. On dimensional reduction it is the seed of many Chern-Simons couplings in lower dimensions, and it is itself the direct descendant of the $C_3 \wedge F_4 \wedge F_4$ term of eleven-dimensional supergravity, from which (5.43) as a whole descends (Section 5.3). Like every coefficient in the action, its normalisation is fixed by supersymmetry.

Remark 5.9 (Beyond the lectures: massive IIA). There is a one-parameter deformation of (5.43), *massive* (Romans) IIA, in which the RR sector acquires a zero-form field strength $F_0 = m$, a cosmological-constant-like mass parameter. One adds a term $-\frac{1}{2}m^2$ to the potential, promotes $F_2 \to F_2 + mB_2$ and $\tilde{F}_4 \to \tilde{F}_4 + \frac{1}{2}mB_2 \wedge B_2$, and finds that B_2 eats C_1 and becomes massive. The parameter m is quantised and is the RR flux threading spacetime that a D8-brane sources. Massive IIA has no known lift to eleven dimensions at the level of a Lagrangian, which makes it a persistent curiosity in the duality web. We will not use it, but you should know the standard $m = 0$ theory above is a special point of a slightly larger family.

5.4.3 The type IIB action and the self-dual five-form

Type IIB is chiral, and its four-form potential C_4 has a self-dual five-form field strength. That single fact obstructs writing an ordinary covariant action, and it pays to see exactly why before quoting the standard workaround.

Suppose we tried the usual Maxwell-type kinetic term $-\frac{1}{2}|F_5|^2$ for a five-form $F_5 = \mathrm{d}C_4$, and then asked the field to be self-dual, $F_5 = \star F_5$. The kinetic density is $F_5 \wedge \star F_5$. But for any five-form in ten dimensions,

$$F_5 \wedge F_5 \;=\; (-1)^{5\cdot 5}\, F_5 \wedge F_5 \;=\; -\,F_5 \wedge F_5 \;=\; 0, \tag{5.47}$$

so on a self-dual field, where $\star F_5 = F_5$, the Lagrangian $F_5 \wedge \star F_5 = F_5 \wedge F_5$ vanishes identically. A self-dual field strength has no covariant, Lorentz-invariant, polynomial action of the naive kind: the object that would be its energy is a total antisymmetry away from zero. This is a general feature of self-dual $(2k{+}1)$-forms in $4k{+}2$ dimensions, familiar already from the chiral boson in two dimensions and the self-dual tensor of the six-dimensional $(2,0)$ theory.

The standard resolution is a *pseudo-action*. One writes an action that produces the correct *second*-order equation of motion, and then imposes the first-order self-duality relation *by hand* on its solutions. The bosonic type IIB pseudo-action, in the string frame, is

$$\begin{aligned} S_{\mathrm{IIB}} \;=\; \frac{1}{2\kappa_{10}^2}\int \mathrm{d}^{10}x\,\sqrt{-g}\,\Big[\, e^{-2\phi}\Big(R + 4\,\nabla_\mu\phi\,\nabla^\mu\phi - \tfrac{1}{2}|H_3|^2\Big) \\ - \tfrac{1}{2}|F_1|^2 - \tfrac{1}{2}\big|\tilde{F}_3\big|^2 - \tfrac{1}{4}\big|\tilde{F}_5\big|^2\,\Big] \;-\; \frac{1}{4\kappa_{10}^2}\int C_4 \wedge H_3 \wedge F_3, \end{aligned} \tag{5.48}$$

with

$$F_1 = \mathrm{d}C_0, \quad F_3 = \mathrm{d}C_2, \quad \tilde{F}_3 = F_3 - C_0\,H_3, \quad \tilde{F}_5 = \mathrm{d}C_4 - \tfrac{1}{2}C_2 \wedge H_3 + \tfrac{1}{2}B_2 \wedge F_3, \tag{5.49}$$

supplemented by the self-duality constraint

$$\boxed{\;\tilde{F}_5 \;=\; \star\,\tilde{F}_5\;} \tag{5.50}$$

imposed on the equations of motion *after* varying, never before. Two signatures of the pseudo-action are visible in (5.48). The five-form term carries $-\frac{1}{4}$ rather than the $-\frac{1}{2}$ of an ordinary

p-form: the halving compensates for the fact that self-duality would otherwise make the electric and magnetic descriptions count the same physical field twice. And the Euler-Lagrange equation for C_4, namely $\mathrm{d}\star\tilde{F}_5 = H_3 \wedge F_3$, only becomes first order once (5.50) is used to trade $\star\tilde{F}_5$ for $\tilde{F}_5$; the Bianchi identity $\mathrm{d}\tilde{F}_5 = H_3 \wedge F_3$ then reproduces the same relation, which is the consistency check that the constraint may be imposed at all.

The pseudo-action is not an ordinary action

You must vary (5.48) with $\tilde{F}_5$ treated as unconstrained, obtain the equations of motion, and only then set $\tilde{F}_5 = \star\tilde{F}_5$. Imposing self-duality inside the functional integral or before varying gives the wrong (indeed vanishing) five-form energy, by (5.47). The action (5.48) is a bookkeeping device that packages the correct second-order dynamics; the physics is the second-order equations together with the constraint.

That self-duality can be imposed at all in Lorentzian signature is a signature-dependent fact. On a five-form in ten dimensions the Hodge star squares to

$$\star\star\, \tilde{F}_5 = -(-1)^{5\cdot 5}\, \tilde{F}_5 = +\tilde{F}_5, \tag{5.51}$$

using the Lorentzian rule of Appendix A. Because $\star^2 = +1$ here, the star has real eigenvalues ± 1 and a real self-dual condition exists. In Euclidean signature $\star^2 = -1$ on the same form and one would be forced into complex or doubled fields. The counting works out: an unconstrained four-form in the $\mathbf{35}_+ \oplus \mathbf{35}_-$ of Spin(8) carries 70 on-shell states, and (5.50) projects onto the $\mathbf{35}_+$, the 35 states listed in Table 5.1.

Remark 5.10 (Beyond the lectures: honest actions for the five-form). The pseudo-action is a convenience, not a limitation of the physics. There are fully covariant Lagrangians for self-dual fields at the cost of extra structure. The Pasti-Sorokin-Tonin (PST) formulation introduces an auxiliary scalar and a compensating gauge symmetry that removes it, yielding (5.50) as a genuine field equation. One can also keep both C_4 and its dual on the same footing in the democratic formulation of Appendix A, doubling the RR potentials and imposing $F_p = (-1)^{\lfloor p/2 \rfloor} \star F_{10-p}$ as constraints; the pseudo-action is what you get after solving half of them. All three routes agree on the equations of motion, which is all that supergravity is really committed to.

5.4.4 $\mathrm{SL}(2,\mathbb{R})$ and the axio-dilaton

Type IIB has a global symmetry that IIA lacks. It acts on the dilaton and the RR scalar together, and it is the classical shadow of the S-duality that will reorganise the entire ten-dimensional landscape in Chapter 7. The symmetry is cleanest in the *Einstein frame*, obtained by the Weyl rescaling

$$g^{\mathrm{E}}_{\mu\nu} = e^{-\phi/2}\, g_{\mu\nu}, \tag{5.52}$$

which strips the $e^{-2\phi}$ off the Einstein-Hilbert term (in ten dimensions the exponent is $2\phi/(D-2) = \phi/4$ per index, hence $\phi/2$ on the metric). Assemble the two scalars into the *axio-dilaton*

$$\tau \;:=\; C_0 + i\, e^{-\phi}, \qquad \mathrm{Im}\,\tau = e^{-\phi} = 1/g_s\big|_{\langle\phi\rangle}, \tag{5.53}$$

which lives on the upper half-plane $\mathrm{Im}\,\tau > 0$, and the two two-forms into a doublet

$$\mathcal{B} = \begin{pmatrix} C_2 \\ B_2 \end{pmatrix}. \tag{5.54}$$

Then the Einstein-frame action can be written in a manifestly $\mathrm{SL}(2,\mathbb{R})$-invariant form,

$$S_{\mathrm{IIB}} = \frac{1}{2\kappa_{10}^2}\int \mathrm{d}^{10}x\,\sqrt{-g_{\mathrm{E}}}\left(R_{\mathrm{E}} - \frac{\nabla_\mu\bar{\tau}\,\nabla^\mu\tau}{2(\mathrm{Im}\,\tau)^2} - \tfrac{1}{2}|G_3|^2 - \tfrac{1}{4}|\tilde{F}_5|^2\right) - \frac{i}{8\kappa_{10}^2}\int C_4 \wedge G_3 \wedge \bar{G}_3, \tag{5.55}$$

where the complex three-form field strength is

$$G_3 \;=\; \frac{F_3 - \tau H_3}{\sqrt{\operatorname{Im}\tau}}. \tag{5.56}$$

The Chern-Simons term is metric-free, so it is the same object as in the string frame (5.48): a short computation gives $G_3 \wedge \bar{G}_3 = 2i\, F_3 \wedge H_3$, in which the factors of $\operatorname{Im}\tau$ from the two square roots cancel, so the term is already independent of τ and no $1/\operatorname{Im}\tau$ appears. Under a matrix $\Lambda = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{R})$, with $ad - bc = 1$, the fields transform as

$$\tau \;\longmapsto\; \frac{a\tau + b}{c\tau + d}, \qquad \mathcal{B} \;\longmapsto\; \Lambda\,\mathcal{B}, \qquad g^{\mathrm{E}}_{\mu\nu} \text{ and } C_4 \text{ invariant.} \tag{5.57}$$

The Einstein-frame metric is inert, which is the whole point of passing to it: only in that frame does the duality act as a symmetry of a fixed geometry rather than mixing into the metric. The doublet transformation sends (F_3, H_3) into linear combinations of themselves, and G_3 picks up only a phase, $G_3 \to \frac{c\bar{\tau}+d}{|c\tau+d|}\, G_3$, so $|G_3|^2$ and $G_3 \wedge \bar{G}_3$ are both invariant. The scalar kinetic term in (5.55) is the $\mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ coset sigma model: τ is a coordinate on the hyperbolic upper half-plane and $\nabla\bar{\tau}\,\nabla\tau/(\operatorname{Im}\tau)^2$ is its Poincaré metric, on which $\mathrm{SL}(2,\mathbb{R})$ acts by isometries.

This is a classical statement about the two-derivative action, and charge quantisation breaks it. The fundamental string and the D-string carry integer charges under B_2 and C_2, so the doublet $\mathcal{B}$ can only be rotated by integer matrices, and $\mathrm{SL}(2,\mathbb{R})$ is reduced to its arithmetic subgroup $\mathrm{SL}(2,\mathbb{Z})$. That discrete remnant is the exact non-perturbative S-duality of type IIB string theory, developed in Chapter 7.

Example 5.11 (S-duality inverts the string coupling)**.** The element $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{Z})$ acts by $\tau \mapsto -1/\tau$. Set $C_0 = 0$, so $\tau = ie^{-\phi}$ and $-1/\tau = ie^{\phi}$; the dilaton flips sign, $e^{-\phi} \to e^{+\phi}$, that is $g_s \to 1/g_s$. This is a genuine strong-weak duality. On the doublet S sends $C_2 \to -B_2$ and $B_2 \to C_2$, exchanging the fundamental string (electric under B_2) with the D-string (electric under C_2). A theory at coupling $g_s \gg 1$, where the D-string is light and the fundamental string is not the natural degree of freedom, is identical to a weakly coupled IIB theory in which the roles are swapped.

Remark 5.12 (Beyond the lectures: (p,q)-strings and the road to F-theory). Acting with $\mathrm{SL}(2,\mathbb{Z})$ on the fundamental string generates a whole lattice of $\frac{1}{2}$-BPS strings labelled by a coprime pair of charges (p,q): p units of B_2 charge and q of C_2 charge, a bound state of p fundamental strings and q D-strings, with tension $\propto |p - q\tau|/\sqrt{\operatorname{Im}\tau}$. The same $\mathrm{SL}(2,\mathbb{Z})$ acts on the seven-branes that source τ magnetically: because τ is a genuine complex modulus and seven-branes make it multivalued (a D7-brane shifts C_0 by one on encircling it), the geometry of type IIB with varying τ is precisely the geometry of an elliptic curve fibred over spacetime. Promoting τ to the modulus of a physical two-torus is the definition of F-theory (Chapter 11). The classical $\mathrm{SL}(2,\mathbb{R})$ of (5.57) is the flat first sign of all of this.

5.4.5 How the two theories fit together

The IIA and IIB actions are not two isolated facts to memorise. They are two faces of one structure, and the two links that connect them are the backbone of Chapters 6 and 7.

The first link is upward, to eleven dimensions. Type IIA supergravity (5.43) is exactly what falls out of reducing eleven-dimensional supergravity on a circle: the eleven-dimensional metric gives the ten-dimensional metric, the dilaton and the RR one-form C_1 (the Kaluza-Klein photon), while the eleven-dimensional three-form C_3 gives the RR three-form and, through one leg on the circle, the NS-NS two-form B_2. The string coupling is set by the size of the circle through $g_s = R_{11}/\ell_s$ (Section 5.3), so a strongly coupled IIA string grows an eleventh dimension as

$R_{11} \to \infty$. We build this reduction in Section 5.3; the Chern-Simons term of (5.43) is the direct descendant of the eleven-dimensional $C_3 \wedge F_4 \wedge F_4$.

The second link is sideways, between IIA and IIB. Compactify either theory on a circle and T-dualise (Chapter 6): IIA on a circle of radius R is the same theory as IIB on a circle of radius α'/R. T-duality is why the RR potentials come in even and odd sets that differ by one form degree, and why a Dp-brane wrapping the circle turns into a D$(p\mp1)$-brane. The single family of maximal ten-dimensional supergravities, presented above in one convention and collected in Appendix I, is therefore best pictured as one theory in eleven dimensions viewed through two different circles. The BPS states that make these statements sharp, the fundamental string as an electric source of B_2, the NS5-brane as its magnetic dual, and the D-branes charged under the RR fields, are the subject of Section 5.6.

Exercise 5.9 (Counting the maximal multiplet)**.** Using the transverse little group Spin(8), verify that the bosonic and fermionic on-shell states of each type II theory number 128.

1. Show that the graviton $g_{\mu\nu}$ carries 35 states (the symmetric traceless tensor of Spin(8)), $B_{\mu\nu}$ carries 28 and ϕ carries 1, so the NS-NS sector has 64.
2. For IIA, show C_1 and C_3 add $8 + 56 = 64$. For IIB, show C_0, C_2 and the *self-dual* C_4 add $1 + 28 + 35 = 64$, where self-duality halves the naive $\binom{8}{4} = 70$.
3. A ten-dimensional Majorana-Weyl gravitino carries 56 on-shell states and a Majorana-Weyl dilatino carries 8. Confirm that two of each give 128 fermionic states, and that this matches the boson count in both theories.

Exercise 5.10 (Why the five-form needs a pseudo-action)**.** Let F_5 be a real five-form in ten Lorentzian dimensions.

1. Using the graded commutativity of the wedge product, prove $F_5 \wedge F_5 = 0$ identically, and hence that $F_5 \wedge \star F_5 = 0$ whenever $F_5 = \star F_5$.
2. Using the Lorentzian rule $\star\star\,\omega_p = -(-1)^{p(D-p)}\omega_p$ from Appendix A, show $\star^2 = +1$ on five-forms in $D = 10$, so a real self-dual condition is consistent. What goes wrong in Euclidean signature?
3. Argue from parts (i)-(ii) that no covariant polynomial action of the standard Maxwell type can yield $F_5 = \star F_5$ as an Euler-Lagrange equation, motivating the pseudo-action (5.48) with self-duality imposed afterwards.

Exercise 5.11 (The axio-dilaton coset)**.** Take the Einstein-frame scalar sector of type IIB, the term $-\dfrac{\nabla_\mu \bar\tau\, \nabla^\mu \tau}{2(\operatorname{Im}\tau)^2}$ with $\tau = C_0 + ie^{-\phi}$.

1. Show that this kinetic term is invariant under $\tau \mapsto (a\tau + b)/(c\tau + d)$ for any real $\Lambda = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with $ad - bc = 1$. (Hint: compute how $\operatorname{Im}\tau$ and $\mathrm{d}\tau$ transform under a Möbius map.)
2. Identify the two obvious symmetries hidden inside $\mathrm{SL}(2,\mathbb{R})$: the shift $C_0 \to C_0 + b$ (integer b survives quantum mechanically, as C_0 has period 1) and the scaling generated by $\left(\begin{smallmatrix} a & 0 \\ 0 & a^{-1} \end{smallmatrix}\right)$. Which combination gives the strong-weak inversion $g_s \to 1/g_s$?
3. Verify that $G_3 = (F_3 - \tau H_3)/\sqrt{\operatorname{Im}\tau}$ transforms by a pure phase, so that $|G_3|^2$ in (5.55) is $\mathrm{SL}(2,\mathbb{R})$-invariant.

5.5 Type I and heterotic supergravity, and Green-Schwarz

The two theories of the previous section, IIA and IIB, exhaust the maximal supergravities in ten dimensions: they carry $\mathcal{N}=2$ supersymmetry, thirty-two real supercharges, and their massless content is fixed entirely by the graviton multiplet. There is no room in them for an independent Yang-Mills sector, because with thirty-two supercharges the smallest massless multiplet is already the supergravity multiplet itself; a separate vector multiplet is not among the representations. Gauge fields in ten dimensions therefore require the minimal amount of supersymmetry, $\mathcal{N}=1$: sixteen real supercharges, a single Majorana-Weyl supersymmetry parameter. This section is about the $\mathcal{N}=1$ supergravities and the string theories that realise them.

Three of the five superstrings land here: Type I (unoriented open plus closed strings) and the two heterotic strings, with gauge groups $\mathrm{SO}(32)$ and $\mathrm{E}_8\times\mathrm{E}_8$. Their worldsheet constructions belong to Chapter 3; here we need only the common low-energy theory. All three reduce, below the string scale, to the same two-derivative Lagrangian: $\mathcal{N}=1$ supergravity in ten dimensions coupled to super-Yang-Mills. The theories differ in the gauge group they carry and in which two-form of the spectrum plays the role of B, but the structure is one and the same. That structure comes with a price. The fermions are chiral, so the theory is anomalous unless the gauge group is chosen with surgical precision, which is the Green-Schwarz story and the reason $\mathrm{SO}(32)$ and $\mathrm{E}_8\times\mathrm{E}_8$ are singled out.

5.5.1 The $\mathcal{N}=1$ supergravity–Yang-Mills system

The massless fields split into two supermultiplets. The *gravity multiplet* is the NS-NS content of Section 5.2 together with its fermionic partners,

$$(g_{MN},\ B_{MN},\ \phi;\ \ \psi_M,\ \lambda), \tag{5.58}$$

a graviton, a two-form, a dilaton, a Majorana-Weyl gravitino ψ_M and a Majorana-Weyl dilatino λ of the opposite chirality. On shell this is 64 bosonic and 64 fermionic states. The *vector multiplet* is one copy, for each generator of the gauge algebra, of

$$(A^a_M,\ \chi^a), \qquad a=1,\dots,\dim G, \tag{5.59}$$

a gauge field and a Majorana-Weyl gaugino χ^a in the adjoint, $8+8$ states each. The gaugino has the same chirality as the gravitino. Here $M,N=0,\dots,9$ are curved ten-dimensional indices, and the reality and chirality carried by the spinors are the Majorana-Weyl conditions that $D=10$ uniquely allows (Appendix B).

Everything chiral in (5.58)–(5.59) feeds the anomaly. Three sets of chiral fermions run in loops: the gravitino, the dilatino, and $\dim G$ gauginos. In ten dimensions a chiral fermion coupled to gravity and to gauge fields produces a hexagon anomaly (six external legs), captured by a twelve-form anomaly polynomial. Left alone, that polynomial does not vanish, and a theory with an uncancelled gauge or gravitational anomaly is inconsistent: diffeomorphism and gauge invariance fail at one loop, unphysical polarisations no longer decouple, and unitarity is lost. Consistency of the quantum theory is therefore a constraint on the field content, and it will turn out to fix $\dim G$ and then the group itself.

Remark 5.13 (Which of the five need saving). Among the type II theories, IIA is non-chiral and automatically anomaly-free, while IIB is chiral but its anomaly cancels on its own: the two gravitini, the two dilatini and the self-dual four-form conspire so that the twelve-form vanishes identically (Alvarez-Gaumé and Witten). Only the $\mathcal{N}=1$ theories, Type I and the two heterotics, carry a residual anomaly, and only they need the Green-Schwarz mechanism. That the mechanism works at all, and works for exactly the two gauge groups the worldsheet independently produces, is the coincidence this section is built around.

5.5.2 The bosonic action

We quote the two-derivative bosonic action; the supersymmetric completion, the fermion terms, and the precise trace and coupling normalisations are collected in Appendix I, which is the source of truth for signs and factors. The three-form field strength that appears below is not the naive $\mathrm{d}B_2$ but the modified combination

$$H_3 = \mathrm{d}B_2 - \tfrac{\alpha'}{4}(\omega_3^{\mathrm{YM}} - \omega_3^{\mathrm{L}}),$$

with ω_3^{YM} and ω_3^{L} the gauge and Lorentz Chern-Simons three-forms defined in Section 5.5.3. With that understood, in string frame the heterotic action is the NS-NS action of Section 5.2 with a Yang-Mills sector bolted on under the same dilaton weighting,

$$S_{\mathrm{het}} = \frac{1}{2\kappa_{10}^2}\int \mathrm{d}^{10}x\,\sqrt{-g}\,e^{-2\phi}\left[R + 4\,(\nabla\phi)^2 - \frac{1}{2}\,|H_3|^2 - \frac{\alpha'}{4}\,\mathrm{tr}|F_2|^2\right], \tag{5.60}$$

with $|H_3|^2 = \frac{1}{3!}H_{MNP}H^{MNP}$ and $|F_2|^2 = \frac{1}{2!}F_{MN}F^{MN}$ in the form norm of Appendix A, so that $-\frac{1}{2}|H_3|^2 = -\frac{1}{12}H_{MNP}H^{MNP}$ reproduces the coefficient of Section 5.2. Here H_3 is the modified three-form field strength quoted above and defined in full in (5.62), and $F = \mathrm{d}A + A\wedge A$ is the field strength of the gauge connection $A = A^a T_a$. The single common factor $e^{-2\phi}$ shows that the whole action descends from the sphere, with the disk nowhere in sight: this is a theory of closed strings, and the gauge fields live in the closed-string spectrum. The gauge kinetic term enters at one power of α' relative to gravity, the heterotic α' counting.

Type I has a similar-looking action assembled from different ingredients, and the differences are physical. Type I is the orientifold of IIB (Chapter 3): the orientation projection removes the NS-NS B-field and the RR scalar and four-form, keeps the RR two-form C_2, and adds an SO(32) open-string sector from 32 D9-branes cancelling the charge of an O9-plane. The two-form carrying the field strength is now the *RR* C_2, and the gauge fields come from the disk, so they enter at order $e^{-\phi}$ rather than $e^{-2\phi}$,

$$S_{\mathrm{I}} = \frac{1}{2\kappa_{10}^2}\int \mathrm{d}^{10}x\,\sqrt{-g}\left[e^{-2\phi}(R + 4(\nabla\phi)^2) - \frac{1}{2}\,|\tilde{F}_3|^2 - \frac{\alpha'}{4}\,e^{-\phi}\,\mathrm{tr}|F_2|^2\right], \tag{5.61}$$

where $\tilde{F}_3 = \mathrm{d}C_2 - \frac{\alpha'}{4}(\omega_3^{\mathrm{YM}} - \omega_3^{\mathrm{L}})$ is the RR three-form field strength, carrying the same Chern-Simons modification (5.63) and no dilaton prefactor on its kinetic term, as befits a RR field. The relative dilaton weights of the two sectors, $e^{-2\phi}$ for closed and $e^{-\phi}$ for open, are exactly what a genus expansion predicts.

These two actions are not independent theories at the two-derivative level. Passing to Einstein frame and sending $\phi \to -\phi$ maps (5.60) for SO(32) into (5.61), exchanging weak and strong coupling and trading the NS-NS B_2 for the RR C_2. This is the Type I / heterotic-SO(32) S-duality; we take it up properly, as a strong-weak duality tested on BPS states, in Chapter 7.

5.5.3 The modified Bianchi identity

The field strength in (5.60) is not $\mathrm{d}B$ alone. In the heterotic and Type I theories the three-form is

$$\boxed{H_3 = \mathrm{d}B_2 - \frac{\alpha'}{4}(\omega_3^{\mathrm{YM}} - \omega_3^{\mathrm{L}})\,,} \tag{5.62}$$

where ω_3^{YM} and ω_3^{L} are the Chern-Simons three-forms of the gauge connection and of the Lorentz (spin) connection,

$$\omega_3^{\mathrm{YM}} = \mathrm{tr}\Big(A\wedge\mathrm{d}A + \tfrac{2}{3}\,A\wedge A\wedge A\Big), \qquad \omega_3^{\mathrm{L}} = \mathrm{tr}\Big(\Omega\wedge\mathrm{d}\Omega + \tfrac{2}{3}\,\Omega\wedge\Omega\wedge\Omega\Big), \tag{5.63}$$

with Ω the matrix-valued spin-connection one-form and $\mathcal{R} = \mathrm{d}\Omega + \Omega \wedge \Omega$ the Riemann curvature two-form built from it. (Throughout the Chern-Simons and anomaly formulae, $\mathcal{R}$ denotes this curvature two-form, distinct from the Ricci scalar R that sits in the actions above.) The exterior derivatives of the two Chern-Simons forms are the Pontryagin four-forms,

$$\mathrm{d}\omega_3^{\mathrm{YM}} = \mathrm{tr}(F \wedge F), \qquad \mathrm{d}\omega_3^{\mathrm{L}} = \mathrm{tr}(\mathcal{R} \wedge \mathcal{R}). \tag{5.64}$$

Take d of (5.62) and use (5.64): the exact piece $\mathrm{d}(\mathrm{d}B)$ drops out and H is no longer closed. It obeys the *modified Bianchi identity*

$$\boxed{\mathrm{d}H_3 = \frac{\alpha'}{4}\left(\mathrm{tr}\,\mathcal{R} \wedge \mathcal{R} - \mathrm{tr}\,F \wedge F\right),} \tag{5.65}$$

with $\mathcal{R}$ the Riemann curvature two-form of (5.63), not the Ricci scalar. Here $\mathrm{tr}\,\mathcal{R} \wedge \mathcal{R}$ traces over the tangent-space (Lorentz) indices of the curvature two-form, and $\mathrm{tr}\,F \wedge F$ uses a reference-normalised trace over the gauge algebra, fixed in Appendix G so that the two four-forms enter on the same footing. We abbreviate $\mathrm{tr}\,\mathcal{R}^2 \coloneqq \mathrm{tr}\,\mathcal{R} \wedge \mathcal{R}$ and $\mathrm{tr}\,F^2 \coloneqq \mathrm{tr}\,F \wedge F$ from here on.

The content of (5.65) is that gauge and gravitational instanton density sources the H-flux. On any closed four-cycle C_4 the periods of $\mathrm{tr}\,F^2$ and $\mathrm{tr}\,\mathcal{R}^2$ are integers (instanton numbers, in the normalisation of Appendix H), and integrating (5.65) with $\int_{C_4} \mathrm{d}H_3 = 0$ forces the two integers to agree,

$$\int_{C_4} \mathrm{tr}\,F^2 \;=\; \int_{C_4} \mathrm{tr}\,\mathcal{R}^2. \tag{5.66}$$

This tadpole-like condition is what fixes the number of gauge-bundle instantons that must sit on an internal manifold: the "24 instantons on K3" of heterotic compactification are nothing but (5.66) evaluated on K3, whose tangent bundle carries $\int_{\mathrm{K3}} \mathrm{tr}\,\mathcal{R}^2 = 24$ in these units (Chapter 8). At the two-derivative level ω_3^{YM} is required for gauge invariance of the action, while ω_3^{L} is a genuine α' correction; both are locked in by the anomaly argument we turn to now.

5.5.4 Green-Schwarz anomaly cancellation and the 496 miracle

The modified Bianchi (5.65) is the visible tip of the anomaly cancellation. The full statement is a claim about the twelve-form anomaly polynomial I_{12} built from the chiral fermions of Section 5.5.1. By the Wess-Zumino descent formalism (Appendix H), the one-loop gauge and gravitational anomaly of the effective action is encoded in I_{12} through $I_{12} = \mathrm{d}\, I_{11}^{(0)}$, $\delta I_{11}^{(0)} = \mathrm{d}\, I_{10}^{(1)}$, with the anomaly equal to $\int I_{10}^{(1)}$. Summing the gravitino $(+)$, the dilatino $(-)$ and $\dim G$ gauginos $(+)$ gives a polynomial in $\mathrm{tr}\,\mathcal{R}^2, \mathrm{tr}\,\mathcal{R}^4, \mathrm{tr}\,\mathcal{R}^6$ and $\mathrm{tr}\,F^2, \mathrm{tr}\,F^4, \mathrm{tr}\,F^6$. Generically it contains *irreducible* pieces, terms like $\mathrm{tr}\,\mathcal{R}^6$ and $\mathrm{tr}\,F^6$ that cannot be written as wedge products of lower forms. No local counterterm removes such terms, so any that survive leave the theory inconsistent.

Green and Schwarz observed that for special gauge groups the whole polynomial *factorises*,

$$I_{12} \;\propto\; X_4 \wedge X_8, \tag{5.67}$$

into a product of a four-form and an eight-form, with

$$X_4 = \mathrm{tr}\,\mathcal{R}^2 - \mathrm{tr}\,F^2, \qquad X_8 = \alpha\ \mathrm{tr}\,\mathcal{R}^4 + \beta\,(\mathrm{tr}\,\mathcal{R}^2)^2 + \gamma\ \mathrm{tr}\,\mathcal{R}^2\ \mathrm{tr}\,F^2 + \delta\ \mathrm{tr}\,F^4 + \varepsilon\,(\mathrm{tr}\,F^2)^2, \tag{5.68}$$

where X_8 is a specific quartic in the two curvatures whose numerical coefficients $\alpha, \ldots, \varepsilon$ are fixed by the descent and listed in Appendix H. The four-form X_4 is exactly the combination on the right of the modified Bianchi (5.65), and that identification is the mechanism, not a

coincidence. Once I_{12} factorises, the anomaly is cancelled by adding a local counterterm to the action,

$$S_{\mathrm{GS}} \sim \int B_2 \wedge X_8, \tag{5.69}$$

together with the anomalous transformation of B_2 implied by (5.62). Under a gauge or Lorentz transformation the Chern-Simons forms shift by exact pieces, so B_2 must shift to keep H_3 invariant; that shift, fed through (5.69), produces a variation cancelling the descended hexagon anomaly $\int I_{10}^{(1)}$. Diagrammatically, a tree-level exchange of B_2 between the Green-Schwarz vertex and the anomalous coupling cancels the one-loop hexagon. The gravitational Chern-Simons term ω_3^{L} in H_3 is present for exactly the same reason: to supply the $\operatorname{tr}\mathcal{R}^2$ half of X_4.

Factorisation imposes two conditions.

Dimension. The irreducible $\operatorname{tr}\mathcal{R}^6$ term cancels among gravitino, dilatino and gauginos only if the number of gauginos is

$$\dim G = 496. \tag{5.70}$$

The gravitino and dilatino leave a fixed $\operatorname{tr}\mathcal{R}^6$ residue; each gaugino cancels a fixed amount of it; the books balance at 496 and nowhere else.

Trace reduction. The irreducible $\operatorname{tr} F^6$ term must be expressible through lower Casimirs. This is the deeper of the two conditions, and beyond what the lectures strictly need: the algebra must satisfy

$$\operatorname{Tr}_{\mathrm{adj}} F^6 = a\,(\operatorname{Tr}_{\mathrm{adj}} F^2)(\operatorname{Tr}_{\mathrm{adj}} F^4) + b\,(\operatorname{Tr}_{\mathrm{adj}} F^2)^3, \tag{5.71}$$

with particular coefficients a, b, fixed in Appendix H, that make I_{12} collapse to the product (5.67). In invariant-theory language, the independent sixth-order Casimir of the algebra must vanish.

The Lie algebras of dimension 496 that satisfy (5.71) are $\mathfrak{so}(32)$, $\mathfrak{e}_8 \oplus \mathfrak{e}_8$, and two abelian-contaminated cases, $\mathfrak{e}_8 \oplus \mathfrak{u}(1)^{248}$ and $\mathfrak{u}(1)^{496}$. The last two do not arise from any consistent chiral string, so string theory delivers exactly two gauge groups,

$$\dim \mathrm{SO}(32) = \frac{32 \cdot 31}{2} = 496, \qquad \dim\,(\mathrm{E}_8 \times \mathrm{E}_8) = 2 \cdot 248 = 496. \tag{5.72}$$

This is the 496 *miracle*. Two entirely different groups, an orthogonal group and a product of the largest exceptional group with itself, have the same dimension and both pass the trace test, and both are realised: Type I and heterotic with SO(32), heterotic with $\mathrm{E}_8 \times \mathrm{E}_8$. The full anomaly polynomial, the coefficients in X_8, and the descent computation live in Appendix H.

The trace "tr" is group-dependent

For SO(32) the reference trace tr in (5.65) and (5.68) is the trace in the vector (fundamental) **32**. The group E_8 has no representation smaller than the adjoint **248**, so there is no fundamental to trace over; one defines $\operatorname{tr} F^2 := \frac{1}{30}\operatorname{Tr}_{\mathrm{adj}} F^2$, and the quartic Casimir of E_8 is itself not independent, $\operatorname{Tr}_{\mathrm{adj}} F^4 = \frac{1}{100}(\operatorname{Tr}_{\mathrm{adj}} F^2)^2$, which is what lets the $\mathrm{E}_8 \times \mathrm{E}_8$ polynomial factorise. These normalisations are collected in Appendix G; do not mix them across groups.

Remark 5.14 (Two roads to the same two groups). Green-Schwarz is a spacetime argument: cancel the hexagon anomaly of the ten-dimensional field theory. The worldsheet gives an independent one. The heterotic string carries sixteen internal left-moving bosons on an even self-dual Euclidean lattice, and modular invariance of the one-loop partition function requires that lattice to be even and self-dual of rank sixteen. There are exactly two such lattices, Γ_{16} and $\Gamma_8 \oplus \Gamma_8$, giving gauge groups $\mathrm{Spin}(32)/\mathbb{Z}_2$ and $\mathrm{E}_8 \times \mathrm{E}_8$ (Chapter 3, classification in Appendix G). The two selections agree. The gauge group of the heterotic SO(32) string is really $\mathrm{Spin}(32)/\mathbb{Z}_2$,

a specific global quotient sharing the algebra $\mathfrak{so}(32)$; the anomaly polynomial sees only the algebra, so (5.72) cannot tell SO(32) from $\mathrm{Spin}(32)/\mathbb{Z}_2$, but the string spectrum can.

Remark 5.15 (Beyond the lectures: anomaly inflow and the missing groups). Two further points, both past what the lectures need. First, the Green-Schwarz cancellation has a modern reading as *anomaly inflow*: the coupling (5.69) makes the two-form an object whose flux carries the anomaly in from the "bulk" of field space, and the same $B \wedge X_8$ term is what assigns the correct anomalous couplings to the NS5-brane and to the various D-branes that source these forms (Chapter 6). Second, the two ruled-out solutions $\mathfrak{u}(1)^{496}$ and $\mathfrak{e}_8 \oplus \mathfrak{u}(1)^{248}$ are perfectly consistent as anomaly-free field theories but have no string realisation, and there is by now good evidence, from the swampland programme, that they admit no consistent completion into quantum gravity at all, so string consistency is the stronger requirement.

Why exactly these two groups

A ten-dimensional $\mathcal{N} = 1$ supergravity is chiral, hence anomalous, hence inconsistent as it stands. The anomaly is a twelve-form I_{12}. It can be cancelled by a local counterterm $\int B \wedge X_8$ only if I_{12} factorises as $X_4 \wedge X_8$, and factorisation demands $\dim G = 496$ together with a trace identity killing the independent sixth-order Casimir. Precisely two compact groups pass, SO(32) and $\mathrm{E}_8 \times \mathrm{E}_8$, and both are handed to us by the heterotic worldsheet and by the Type I tadpole condition. The modified Bianchi $\mathrm{d}H = \frac{\alpha'}{4}(\operatorname{tr}\mathcal{R}^2 - \operatorname{tr}F^2)$ is the low-energy shadow of the whole story: the four-form on its right-hand side is X_4 itself.

Where this is going

We now hold all five ten-dimensional supergravities and the eleven-dimensional one of Section 5.3: two maximal (IIA, IIB), three minimal (Type I, heterotic SO(32), heterotic $\mathrm{E}_8 \times \mathrm{E}_8$), and the parent above them. The relations glimpsed here, Type I as the S-dual of heterotic SO(32), and the hint that $\mathrm{E}_8 \times \mathrm{E}_8$ at strong coupling grows an eleventh dimension, are the first threads of the duality web of Chapter 7. What makes those relations testable is that each theory has a protected spectrum of states saturating a mass bound. The next section sets up that bound, $M \geq |Z|$, from the supersymmetry algebra and its central charges.

Exercise 5.12 (The 496 arithmetic). *(Short; goal: the miracle is real.)* Verify $\dim \mathrm{SO}(N) = \frac{1}{2}N(N-1)$ and hence $\dim \mathrm{SO}(32) = 496$, and $\dim \mathrm{E}_8 = 248$ so $\dim(\mathrm{E}_8 \times \mathrm{E}_8) = 496$. Check that the two abelian solutions $\mathrm{E}_8 \times \mathrm{U}(1)^{248}$ and $\mathrm{U}(1)^{496}$ also have dimension 496. In one sentence, name the physical requirement, beyond the field-theory anomaly count, that excludes them.

Exercise 5.13 (Chern-Simons and the anomalous shift of B). *(Moderate; goal: how B absorbs the anomaly.)* Using $F = \mathrm{d}A + A \wedge A$ and cyclicity of the trace, show that $\mathrm{d}\,\omega_3^{\mathrm{YM}} = \operatorname{tr}(F \wedge F)$ for the Chern-Simons form (5.63). Under an infinitesimal gauge transformation $\delta A = \mathrm{d}\lambda + [A, \lambda]$, show that ω_3^{YM} shifts by an exact form, $\delta\omega_3^{\mathrm{YM}} = \mathrm{d}\operatorname{tr}(\lambda\,\mathrm{d}A)$. Deduce that invariance of H_3 in (5.62) forces $\delta B_2 = \frac{\alpha'}{4}\operatorname{tr}(\lambda\,\mathrm{d}A)$, so B_2 is not gauge-invariant. *Hint:* $\delta F = [F, \lambda]$, and $\operatorname{tr}(F[F, \lambda]) = 0$.

Exercise 5.14 (The Bianchi as a Gauss law). *(Moderate; goal: flux quantisation from the modified Bianchi.)* Take the heterotic string on a background containing a compact four-cycle C_4 without boundary. Integrate (5.65) over C_4 and use $\int_{C_4} \mathrm{d}H_3 = 0$ to derive (5.66). Reading each side as an integer instanton number, argue that the gauge instanton number on C_4 is fixed by the geometry alone (the second Chern class of the tangent bundle), and evaluate it for $C_4 = \mathrm{K3}$. This is the seed of the tadpole conditions of Chapter 8. *Hint:* a total exterior derivative integrates to zero on a closed manifold, so the factor $\frac{\alpha'}{4}$ cancels between the two sides.

5.6 Central charges and the BPS bound

We have now assembled the bosonic content of every ten- and eleven-dimensional supergravity in this course, and each one propagates a collection of antisymmetric tensor gauge fields: the NS-NS two-form B_2, the RR potentials C_{p+1} of Section 5.4, and the eleven-dimensional three-form C_3 of Section 5.3. A gauge field with no charged source is pure redundancy. Maxwell theory is interesting because there are electrons; B_2 is interesting because something couples to it. The objects that carry these charges are extended: a p-form potential is sourced by a p-dimensional worldvolume, a *brane*. This section identifies those branes and works out what supersymmetry does to them: it ties their masses to their charges by an exact inequality, the BPS bound, and pins a distinguished subset of them to the boundary of that inequality, where their properties become rigid enough to survive all the way to strong coupling. That rigidity is the single most useful fact in the second half of the course.

5.6.1 The superalgebra carries brane charges

The super-Poincaré algebra you met in a first course on supersymmetry has, as its defining relation, the anticommutator of two supercharges closing on the momentum, schematically $\{Q, Q\} \sim P$. In four dimensions with extended supersymmetry a scalar central charge can appear on the right, $\{Q, Q\} \sim P + Z$, and it is central in the strict sense: it commutes with everything in the algebra. In ten and eleven dimensions the story is richer. The right-hand side can carry not just a scalar but a whole tower of antisymmetric tensor charges, one for each rank p allowed by the spinor representation. With real (Majorana) supercharges Q_α the most general closure consistent with Lorentz symmetry is

$$\{Q_\alpha, Q_\beta\} \;=\; (C\Gamma^{M})_{\alpha\beta}\, P_M \;+\; \sum_{p} \frac{1}{p!}\, (C\Gamma^{M_1\cdots M_p})_{\alpha\beta}\, Z_{M_1\cdots M_p}, \tag{5.73}$$

where C is the charge-conjugation matrix and the sum runs over exactly those ranks p for which $C\Gamma^{M_1\cdots M_p}$ is symmetric in $\alpha\beta$, so that it can appear on the left-hand side. The $Z_{M_1\cdots M_p}$ are the p-form *central charges*. They commute with the supercharges and with P_M, but they are not Lorentz scalars, so "central" is a mild abuse of language inherited from four dimensions. The precise statement, and the full representation theory, live in Appendix C; here we need only their physical meaning, established by de Azcárraga, Gauntlett, Izquierdo and Townsend: *each $Z_{M_1\cdots M_p}$ is the charge of a p-brane.*

The cleanest place to see the counting close is eleven dimensions, where the supercharge is a single 32-component Majorana spinor and (5.73) is a symmetric 32×32 matrix equation. The left-hand side has $\tfrac{1}{2}\cdot 32 \cdot 33 = 528$ independent components. On the right, the symmetric gamma structures in $D = 11$ are exactly $C\Gamma^{M}$, $C\Gamma^{MN}$ and $C\Gamma^{MNPQR}$, so

$$\{Q_\alpha, Q_\beta\} = (C\Gamma^{M})_{\alpha\beta}\, P_M + \tfrac{1}{2}(C\Gamma^{MN})_{\alpha\beta}\, Z_{MN} + \tfrac{1}{5!}(C\Gamma^{MNPQR})_{\alpha\beta}\, Z_{MNPQR}, \tag{5.74}$$

and the bookkeeping is

$$528 \;=\; \underbrace{11}_{P_M} \;+\; \underbrace{55}_{Z_{MN}} \;+\; \underbrace{462}_{Z_{MNPQR}} \quad . \tag{5.75}$$

Every symmetric bilinear in the supercharges is accounted for. The two-form charge Z_{MN} is the charge of the **M2-brane** and the five-form charge Z_{MNPQR} is the charge of the **M5-brane**; together with the momentum they exhaust the right-hand side. This is why (5.74) is called the "M-theory algebra": its central charges fix the brane content of the eleven-dimensional theory before any field equation is solved.

Reducing to ten dimensions splits these tensor charges into pieces of various rank, and the type II algebras (non-chiral $\mathcal{N} = (1,1)$ for IIA, chiral $\mathcal{N} = (2,0)$ for IIB) carry a p-form charge for

every brane in the theory: the string and fivebrane of the NS-NS sector, and the full ladder of D-branes in the RR sector. We collect the dimension-by-dimension list in Appendix I; the physics we need follows from the structure of (5.73) alone.

Remark 5.16 (Beyond the lectures: the charges are surface terms). If you compute $\{Q, Q\}$ naively from the supercurrent of a field theory in infinite flat space, the p-form pieces integrate to total derivatives and appear to vanish. That is the classic "theorem" that the only central charge is the trivial one. The loophole is that the total derivatives are boundary integrals over the sphere at spatial infinity, and they are nonzero precisely when a brane sits inside and sources flux through that sphere. So $Z_{M_1 \cdots M_p}$ is literally a flux at infinity: for the fundamental string it is $\int_{S^7} \star(e^{-2\phi} H_3)$, for the NS5-brane it is $\int_{S^3} H_3$, and so on. The central charge and the Gauss-law charge of the brane are the same number, computed by the same surface integral.

5.6.2 The BPS bound

The algebra (5.73) constrains the spectrum through one elementary fact: an anticommutator of a Hermitian operator with itself is positive. Take the supercharges in a Majorana basis where $Q_\alpha^\dagger = Q_\alpha$, and act on any one-particle state $|\psi\rangle$. For an arbitrary real vector u^α set $\hat{Q} = u^\alpha Q_\alpha$, which is again Hermitian, so

$$u^\alpha u^\beta \, \langle \psi | \{Q_\alpha, Q_\beta\} | \psi \rangle \;=\; 2 \, \langle \psi | \hat{Q}^2 | \psi \rangle \;=\; 2 \, \| \hat{Q} \, |\psi\rangle \|^2 \;\geq\; 0. \tag{5.76}$$

The matrix $\mathcal{M}_{\alpha\beta} \coloneqq \langle \psi | \{Q_\alpha, Q_\beta\} | \psi \rangle$ is therefore positive semi-definite: all of its eigenvalues are non-negative. Now evaluate the right-hand side of (5.73) on a massive state in its rest frame, $P_M = (M, \mathbf{0})$. The momentum term becomes $M \, (C\Gamma^0)_{\alpha\beta}$, which in the Majorana basis is proportional to the identity on spinor space, while the central-charge term becomes a fixed traceless matrix $\mathcal{Z}_{\alpha\beta}$ built from the values of the Z's carried by the state. Schematically,

$$\mathcal{M} \;=\; M \, \mathbf{1} + \mathcal{Z}. \tag{5.77}$$

The relevant gamma structures square to the identity and are traceless, so $\mathcal{Z}$ has eigenvalues that come in pairs $\pm |Z|$, where $|Z|$ is a definite norm of the central charge (the precise combination, with the projectors that select it, is worked out in Appendix C). The eigenvalues of $\mathcal{M}$ are then $M \pm |Z|$, and positivity of the smallest one gives the **BPS bound**

$$\boxed{M \;\geq\; |Z| \, .} \tag{5.78}$$

The mass of any state is bounded below by the charge it carries. For a brane the "mass" is a tension times an infinite worldvolume; dividing (5.78) by that volume turns it into a bound on the tension by the charge density, $T \geq |q|$. Named for Bogomol'nyi, Prasad and Sommerfield, who found the same inequality for monopoles, the bound is a pure consequence of the supersymmetry algebra and holds state by state, quantum mechanically.

The interesting states are the ones that *saturate* it. Equality $M = |Z|$ happens precisely when $\mathcal{M}$ has a zero eigenvalue, and by (5.76) a zero eigenvalue means there is a combination $\hat{Q}$ with

$$\hat{Q} \, |\psi\rangle = 0. \tag{5.79}$$

Some supercharges annihilate the state. A state that saturates the bound is invariant under a subalgebra of the supersymmetry: it is a **BPS state**, and it *preserves* that fraction of the supersymmetry. States above the bound, $M > |Z|$, are annihilated by no supercharge; they break all of it.

This algebraic distinction has an immediate consequence for the size of multiplets. With $\mathcal{N}$ real supercharges, half act as raising operators and half as lowering operators on a Clifford vacuum,

so a generic massive (*long*) multiplet has $2^{\mathcal{N}/2}$ states. In $D = 10$ and $D = 11$ we have $\mathcal{N} = 32$, hence 2^{16} states in a long multiplet. When $\nu\mathcal{N}$ of the supercharges annihilate the state, they build nothing, and only the remaining $(1-\nu)\mathcal{N}$ act. The multiplet collapses to

$$\dim(\nu\text{-BPS multiplet}) \;=\; 2^{\,(1-\nu)\mathcal{N}/2}, \tag{5.80}$$

a *short* multiplet. A $\frac{1}{2}$-BPS state ($\nu = \frac{1}{2}$, sixteen preserved supercharges) sits in a multiplet of $2^8 = 256$ states, precisely the $128 + 128$ of the massless supergravity multiplet: the graviton multiplet itself is a short multiplet. A $\frac{1}{4}$-BPS state has 2^{12} states, and so on. The gap between 2^8 and 2^{16} is what matters here, because a short multiplet cannot be deformed into a long one without $2^{16} - 2^8$ states appearing discontinuously, and states cannot materialise as you vary a parameter smoothly. The shortness of a BPS multiplet is therefore topologically protected. We return to why that makes BPS masses exact in Section 5.6.4; the representation theory that fixes which fractions occur in which dimension is in Appendix C.

Remark 5.17 (The no-force condition). Saturating $M = |Z|$ has a mechanical avatar. Two parallel identical BPS branes exert no net force on one another: the gravitational and dilatonic attraction is cancelled exactly by the repulsion of the tensor gauge field to which they are both charged. This is why static multi-brane configurations exist as solutions of supergravity with the branes at arbitrary separation, and it is the field-theory face of the algebraic statement that the same supercharges annihilate any number of coincident BPS branes. You will meet these as the harmonic-function "extremal" p-brane solutions of Appendix I and Chapter 6.

5.6.3 The BPS brane spectrum

Now put charges to objects. A p-form potential C_p (with field strength $F_{p+1} = \mathrm{d}C_p$) couples electrically to an object whose p-dimensional worldvolume it can integrate over, $\int_{\text{worldvol}} C_p$; that object is a $(p-1)$-brane. Its magnetic dual is defined by the dual potential $\widetilde{C}_{D-p-2}$ obtained from $\star F_{p+1}$, and is a $(D-p-3)$-brane. In $D = 10$ this pairs a $(p-1)$-brane with a $(7-p)$-brane. Run this through the fields we have built.

The NS-NS sector: F1 and NS5. The Kalb-Ramond field B_2 couples electrically to a one-dimensional object, the **fundamental string** F1. This is not a new ingredient; it is the string itself, whose worldsheet action already contains the term $\int B_2$ from Chapter 1. The string is the electric source of B_2, and its tension is the one we started with,

$$T_{\mathrm{F1}} \;=\; \frac{1}{2\pi\alpha'} \;=\; \frac{1}{2\pi\ell_s^2} \;\sim\; g_s^0. \tag{5.81}$$

It carries no power of the string coupling: the fundamental string is the object whose perturbation theory *defines* g_s. The magnetic dual of B_2 in ten dimensions is a five-brane, the **NS5-brane**, the source of $H_3 = \mathrm{d}B$ flux through a surrounding S^3. It is a soliton of the supergravity fields, and like every soliton its tension scales inversely with a power of the coupling, here two powers:

$$T_{\mathrm{NS5}} \;\sim\; \frac{1}{g_s^2\,\ell_s^6}. \tag{5.82}$$

The RR sector: Dp-branes. Each RR potential C_{p+1} couples electrically to a p-dimensional object, the **Dp-brane**, with p even in IIA and p odd in IIB. These are the branes on which open strings end; their microscopic construction, worldvolume gauge theory and RR charge are the business of Chapter 6. What we need here is the tension, which sits one power of the coupling below the NS5:

$$T_{\mathrm{Dp}} \;\sim\; \frac{1}{g_s\,\ell_s^{p+1}}. \tag{5.83}$$

The single power $1/g_s$ is the signature of a D-brane. It is heavier than a fundamental string at weak coupling but lighter than an NS5-brane, and it interpolates between the two under duality.

Eleven dimensions: M2 and M5. The M2 and M5 read off from (5.74) are the electric and magnetic sources of the eleven-dimensional C_3. There is no dilaton in eleven dimensions and hence no g_s; their tensions are set purely by the eleven-dimensional Planck length, $T_{\rm M2} \sim \ell_p^{-3}$ and $T_{\rm M5} \sim \ell_p^{-6}$. The ten-dimensional coupling dependence of (5.81)–(5.83) is what emerges when these are reduced on the M-theory circle, as in Section 5.3.

The parametric tensions are collected in Table 5.2; the exact numerical coefficients (all the factors of 2π) are in Appendix I, and are fixed by the Dirac quantisation of the corresponding charges together with $2\kappa_{10}^2 = (2\pi)^7 \alpha'^4 g_s^2$.

Brane	Couples to	Source of	Worldvol. dim.	Tension scaling
F1	B_2	electric (B_2)	2	$1/(2\pi\ell_s^2) \sim g_s^0$
NS5	$\widetilde{B}_6$	magnetic (B_2)	6	$1/(g_s^2\ell_s^6)$
Dp	C_{p+1}	electric (C_{p+1})	$p+1$	$1/(g_s\,\ell_s^{p+1})$
M2	C_3	electric (C_3)	3	$1/\ell_p^3$
M5	$\widetilde{C}_6$	magnetic (C_3)	6	$1/\ell_p^6$

Table 5.2: The BPS brane spectrum whose charges are the central charges of (5.73). In IIA the Dp have $p = 0, 2, 4, 6, 8$; in IIB, $p = 1, 3, 5, 7, 9$. Exact coefficients are in Appendix I.

Look at the three ten-dimensional scalings together:

$$T_{\rm F1} \sim g_s^0, \qquad T_{\rm Dp} \sim g_s^{-1}, \qquad T_{\rm NS5} \sim g_s^{-2}. \tag{5.84}$$

At weak coupling, $g_s \to 0$, the D-branes and the NS5 are heavy, becoming infinitely so relative to the string: they are non-perturbative, invisible in string perturbation theory, which is exactly why the perturbative spectrum of Chapter 3 contained only strings. Turn the coupling up and the ordering inverts. A D0-brane, with $M_{\rm D0} \sim 1/(g_s\ell_s)$, becomes light as $g_s \to \infty$, and a whole tower of n D0-branes with masses $M_n \sim n/(g_s\ell_s)$ becomes light together. A continuum of states becoming light as a parameter is sent to a limit is the unmistakable signature of a decompactifying dimension: the tower is the Kaluza-Klein spectrum of an eleventh circle of radius $R_{11} = g_s\ell_s$, opening up as $g_s \to \infty$. Read at strong coupling, the tension scalings already contain the growth of the eleventh dimension, and Chapter 7 follows that thread.

5.6.4 Why BPS states are protected: the golden thread

Everything above would be a pleasant piece of representation theory were it not for one property that makes BPS states the load-bearing objects of the whole subject. Their masses are *exact*.

Consider a $\frac{1}{2}$-BPS state at weak coupling, sitting in its 256-state short multiplet. Now imagine dialling g_s up towards strong coupling. The mass $M(g_s)$ and the charge $|Z|$ both depend on the coupling, but the charge is quantised (Dirac quantisation of the flux integral of the previous remark) and cannot vary continuously, so $|Z|$ is locked. Could $M(g_s)$ drift above $|Z|$? Only if the short multiplet turned into a long one, and that would require $2^{16} - 2^8$ extra states to appear discontinuously. States do not do that under a smooth deformation. The multiplet is short at every coupling, the bound is saturated at every coupling, and therefore

$$M(g_s) \;=\; |Z| \qquad \text{for all } g_s. \tag{5.85}$$

The BPS mass formula is a non-renormalisation theorem. We derived it in a weakly coupled corner where we could count states, and it holds where we cannot compute at all. (Strictly, what is protected is a supersymmetric index, an alternating count of BPS multiplets; two short multiplets can occasionally pair up into a long one and lift off the bound, but only in pairs, so

the index is invariant. This subtlety is developed in Appendix C and matters for the entropy counting of Chapter 12.)

Why we care about BPS states

A BPS state is a quantity you can compute at weak coupling and trust at strong coupling. Two enterprises in this course rest entirely on that.

Dualities become provable. A strong-weak duality (Chapter 7) claims that one theory at coupling g_s is the same as another at coupling $1/g_s$. You cannot check this by comparing perturbation series, because when one side is weak the other is strong. But both sides have BPS spectra with exact masses, and a duality must map BPS states to BPS states with matching $M = |Z|$. Those you *can* compare across the duality. The D0-brane tower matching the Kaluza-Klein tower of the eleventh dimension, the F1-NS5 pair of IIB mapping to the D1-D5 pair under S-duality: these are BPS checks, and they are the evidence that the duality web is real.

Black-hole microstates become countable. A bound state of branes with the right charges gravitates, and at strong coupling it is a black hole with a Bekenstein-Hawking entropy $S = A/4G$. At weak coupling the same charges are a gas of open strings on a brane stack in flat space, whose degeneracy you can count by hand. Because the states are BPS, the count is protected and the two answers must agree. That agreement is the Strominger-Vafa derivation of black-hole entropy from microstates (Chapter 12), and it is possible only because the bound (5.78) freezes the counting against the enormous change in coupling between the two pictures.

This is the golden thread. The BPS bound turns a subset of the brane spectrum into rigid, coupling-independent data, and rigid data is what lets you reason across regimes where nothing else survives. The rest of the course is, to a large extent, the exploitation of (5.85).

Where this leaves us

We built the ten- and eleven-dimensional supergravities from the demand that the worldsheet stay conformal, read their gauge fields, and found the sources those gauge fields require: the fundamental string and the NS5-brane in the NS-NS sector, the Dp-branes in the RR sector, the M2 and M5 upstairs. Supersymmetry bounded their tensions by their charges and saturated the bound for the BPS ones, protecting their masses for good. The next chapter constructs the D-branes microscopically and turns T-duality loose on them; the chapter after that assembles the duality web whose girders are the BPS matchings we have just motivated.

Exercise 5.15 (The bound from a single central charge)**.** *(Short.)* Model the essential algebra with four Hermitian supercharges assembled into two complex combinations $a_i = \frac{1}{2}(Q_i + i\,Q_{i+2})$, $i = 1, 2$, obeying $\{a_i, a_j^\dagger\} = (M + \sigma\, Z)_{ij}$ with $\sigma = \mathrm{diag}(+1, -1)$ and Z real. By demanding that $\{a_i, a_i^\dagger\} = \|a_i^\dagger\,|\psi\rangle\|^2 + \|a_i\,|\psi\rangle\|^2 \geq 0$ for each i, derive $M \geq |Z|$. Show that at $M = |Z|$ one of the two oscillators annihilates the state, so half the supercharges are preserved, and the multiplet built on $|\psi\rangle$ has half the dimension of the generic one. *Moral:* the ten-dimensional derivation is this argument with 32 supercharges and a p-form Z in place of the scalar.

Exercise 5.16 (Dirac quantisation of the F1 and NS5 charges)**.** *(Moderate.)* The fundamental string is the electric source of B_2, with charge $Q_\mathrm{e} = \int_{S^7} \star(e^{-2\phi} H_3)$, and the NS5-brane is its magnetic source, with $Q_\mathrm{m} = \int_{S^3} H_3$. Using $H_3 = \mathrm{d}B$ and the equation of motion for B from the NS-NS action of Section 5.2, argue that Q_e and Q_m are conserved (one by an equation of motion, one by a Bianchi identity), and that the linking of the S^3 and S^7 around a string-fivebrane pair forces a Dirac quantisation condition $Q_\mathrm{e} Q_\mathrm{m} \in 2\pi\mathbb{Z}$. Explain in one line why this is what makes $|Z|$ in (5.85) a discrete, hence non-renormalised, quantity.

Exercise 5.17 (The eleventh dimension from a brane tension)**.** *(Moderate; foreshadows Chapter 7.)* Take the eleven-dimensional M2-brane tension $T_{\mathrm{M2}} = 1/((2\pi)^2 \ell_p^3)$ and reduce on a circle of radius R_{11}. Wrapping the M2 on the circle gives a string of tension $T = 2\pi R_{11}\, T_{\mathrm{M2}}$; not wrapping it gives a membrane. Impose that the wrapped M2 is the fundamental string, $T = 1/(2\pi \ell_s^2)$, and that the unwrapped M2 is the D2-brane, $T_{\mathrm{D2}} = 1/((2\pi)^2 g_s \ell_s^3)$. Solve the two conditions for R_{11} and ℓ_p in terms of g_s and ℓ_s, and recover $R_{11} = g_s \ell_s$ and $\ell_p = g_s^{1/3} \ell_s$. Deduce that the D0-brane mass $1/(g_s \ell_s)$ is exactly one unit of Kaluza-Klein momentum $1/R_{11}$, and hence that the D0 tower of Section 5.6.3 is the eleventh dimension seen from ten.

Chapter 6

T-Duality, D-Branes, Orientifolds, and BPS States

6.1 T-duality of the closed string

A point particle on a circle knows one thing about the circle: its size. The quantised momentum $p = n/R$ climbs as R shrinks, and in the limit $R \to 0$ every excited Kaluza-Klein mode runs off to infinite energy and decouples, leaving the theory in one fewer dimension. A string does something a particle cannot. Because it is extended, it can be wrapped around the circle, and a wrapped configuration cannot unwind without cutting the string. That wrapping is a new, conserved, integer-valued charge, the *winding* w, and its energy grows as the circle grows. Momentum and winding respond to the radius in opposite ways, and the string spectrum turns out to depend on R only through the pair (n, w) in a way that is completely symmetric under swapping them while sending $R \to \alpha'/R$. This is *T-duality*. It is an exact equivalence between string theory on a circle of radius R and string theory on a circle of radius α'/R, order by order in perturbation theory, and it is the simplest and cleanest of the dualities this course is built around. We develop it for the bosonic string; the momentum-winding structure below carries over verbatim to the superstring, where only the zero-point energy in the mass formula changes.

Why start here

T-duality is the template. Every later duality (mirror symmetry, the IIA/IIB and heterotic pairs, the strong-weak S-duality of Chapter 7) is a statement that two apparently different string backgrounds are the same physics in different variables. T-duality is the one case where we can prove the equivalence outright, by matching spectra state for state, with no strong-coupling input. It also does the concrete work this chapter needs: applied to open strings in Section 6.2 it turns a freely moving endpoint into a pinned one, which is what forces D-branes into existence.

The closed-string version comes first because the same worldsheet identity drives everything after it. What it unlocks (D-branes, orientifolds, BPS states) is real physics, not bookkeeping, and the chapter runs to more than a single week. Treat Section 6.1 through the Buscher rules, Section 6.2 through the D-brane construction, and the BPS spectrum of Section 6.3 as the core; the orientifold tadpoles of Section 6.4, the Dirac-Born-Infeld and Wess-Zumino deep dives, and the BPS thread of Section 6.5 are extension reading.

What you should be able to do. By the end of this section you can:

- write the compact boson as $X = X_{\mathrm{L}} + X_{\mathrm{R}}$ and read off the split momenta $p_{\mathrm{L}}, p_{\mathrm{R}}$

(Exercise 6.1);

- derive the circle mass formula and the deformed level-matching $N - \tilde{N} = nw$, and read the $R \to \alpha'/R$, $n \leftrightarrow w$ symmetry straight off it (Exercise 6.2);
- locate the self-dual radius and explain the $\mathrm{SU}(2)_\mathrm{L} \times \mathrm{SU}(2)_\mathrm{R}$ enhancement as a Higgs mechanism run backwards (Exercise 6.3);
- apply the Buscher rules to dualise a background with a compact isometry, and watch momentum traded for winding charge (Exercise 6.4).

6.1.1 The closed string on a circle

Take the flat closed-string background of Chapter 1 and curl up one spatial direction. Single out one coordinate, call it X, and make it periodic,

$$X \;\sim\; X + 2\pi R, \tag{6.1}$$

so that X parametrises a circle of radius R while the remaining X^μ ($\mu = 0, \dots, D-2$) stay non-compact. Nothing changes locally: the worldsheet theory is still free bosons obeying the wave equation of Section 1.2. What changes is global, and it shows up in exactly two places.

The first is familiar from Kaluza-Klein reduction. A centre-of-mass wavefunction e^{ipX} must be single-valued under (6.1), which quantises the momentum along the circle,

$$p \;=\; \frac{n}{R}, \qquad n \in \mathbb{Z}. \tag{6.2}$$

The integer n is the Kaluza-Klein momentum number. The second effect has no particle analogue. The field $X(\tau, \sigma)$ need not be strictly periodic in the worldsheet coordinate $\sigma \sim \sigma + 2\pi$; it need only return to a point identified with its start, so it may wind around the target circle an integer number of times before closing,

$$X(\tau, \sigma + 2\pi) \;=\; X(\tau, \sigma) + 2\pi w R, \qquad w \in \mathbb{Z}. \tag{6.3}$$

The integer w is the winding number. A configuration with $w \neq 0$ is a string stretched taut w times around the circle; it cannot relax to $w = 0$ without breaking, so w is a topological charge, conserved for the same reason n is.

Both effects live in the zero modes. The oscillators are untouched, because the local dynamics is unchanged, so the mode expansion of (1.36) is modified only in its centre-of-mass piece. Momentum (6.2) sits in the term linear in τ, and winding (6.3) demands a term linear in σ:

$$X(\tau, \sigma) \;=\; x + \alpha' \frac{n}{R}\,\tau + wR\,\sigma + i\sqrt{\frac{\alpha'}{2}} \sum_{k \neq 0} \frac{1}{k} \Big(\alpha_k\, e^{-ik\sigma^-} + \tilde{\alpha}_k\, e^{-ik\sigma^+} \Big), \tag{6.4}$$

with $\sigma^\pm = \tau \pm \sigma$ as before. The $wR\,\sigma$ term is the whole story: under $\sigma \to \sigma + 2\pi$ it reproduces (6.3), and it is precisely the term a non-compact coordinate forbids, where single-valuedness kills it. The non-compact coordinates X^μ keep the expansion (1.36) unchanged.

It pays to split (6.4) into its left- and right-moving halves $X = X_\mathrm{L}(\sigma^+) + X_\mathrm{R}(\sigma^-)$ and read off the zero modes of each. Matching the τ and σ coefficients gives left- and right-moving momenta

$$p_\mathrm{L} \;=\; \frac{n}{R} + \frac{wR}{\alpha'}, \qquad p_\mathrm{R} \;=\; \frac{n}{R} - \frac{wR}{\alpha'}, \tag{6.5}$$

which enter the mode expansion exactly as the ordinary momentum did in Appendix A: the compact zero-mode oscillators are

$$\alpha_0 \;=\; \sqrt{\frac{\alpha'}{2}}\, p_\mathrm{R}, \qquad \tilde{\alpha}_0 \;=\; \sqrt{\frac{\alpha'}{2}}\, p_\mathrm{L}. \tag{6.6}$$

For the non-compact string $p_{\rm L} = p_{\rm R} = p$ and (6.6) collapses to the single relation (1.37). The compact circle is what splits them apart: momentum contributes symmetrically to $p_{\rm L}$ and $p_{\rm R}$, winding antisymmetrically. Two integers, n and w, now label the zero-mode sector, where the non-compact string had only the continuous momentum p.

The mass formula. The physics is in the Virasoro zero-mode constraints $L_0 = \tilde{L}_0 = a$ of Section 1.5.1, with the bosonic intercept $a = 1$. Splitting the momentum into its non-compact part p^μ (with $M^2 = -p^\mu p_\mu$ the mass measured in the $D-1$ large dimensions) and its compact part,

$$L_0 = -\frac{\alpha'}{4}M^2 + \frac{\alpha'}{4}\,p_{\rm R}^2 + N, \qquad \tilde{L}_0 = -\frac{\alpha'}{4}M^2 + \frac{\alpha'}{4}\,p_{\rm L}^2 + \tilde{N}, \tag{6.7}$$

with $N, \tilde{N}$ the right- and left-moving level operators of Section 1.2.5. Setting each to 1 and adding gives the bosonic-string mass formula on a circle,

$$\boxed{M^2 \;=\; \frac{n^2}{R^2} + \frac{w^2R^2}{\alpha'^2} + \frac{2}{\alpha'}(N + \tilde{N} - 2)} \tag{6.8}$$

where I used $\frac{1}{2}(p_{\rm L}^2 + p_{\rm R}^2) = n^2/R^2 + w^2R^2/\alpha'^2$. Each of the three pieces has a clean reading. A Kaluza-Klein mode costs energy n/R, heavier on a smaller circle. A winding mode costs wR/α', the string tension $T = 1/(2\pi\alpha')$ times the length $2\pi wR$ it is stretched over, heavier on a larger circle. The oscillators contribute as in flat space.

Subtracting the two constraints in (6.7) instead of adding them gives the level-matching condition, and here it carries a surprise. The naive $N = \tilde{N}$ of the non-compact string (1.43) is deformed by the zero modes:

$$N - \tilde{N} \;=\; \frac{\alpha'}{4}(p_{\rm L}^2 - p_{\rm R}^2) \;=\; n\,w. \tag{6.9}$$

A state carrying both momentum and winding must carry a compensating oscillator imbalance. The origin is the same rigid σ-translation invariance as before; what is new is that a string with $n \neq 0$ and $w \neq 0$ already carries worldsheet momentum in its zero modes, so the oscillators must make up the difference.

Exercise 6.1 (Reading off the split momenta)**.** *(Short.)* Starting from (6.4), split $X = X_{\rm L}(\sigma^+) + X_{\rm R}(\sigma^-)$ and match the coefficients of τ and of σ to derive the left- and right-moving momenta (6.5). Check that $p_{\rm L} = p_{\rm R}$ once the winding w is switched off, so that (6.6) collapses to the non-compact relation (1.37).

Exercise 6.2 (The circle spectrum from scratch)**.** *(Hard.)* Feed the zero modes (6.6) into L_0 and $\tilde{L}_0$ and reproduce the mass formula (6.8) and the level-matching (6.9), being careful that M^2 is the mass in the non-compact directions only. Then verify that in the decompactification limit $R \to \infty$ at fixed n, the winding tower disappears and the momentum sum $\sum_n$ turns into a continuous integral $R\int \mathrm{d}p$, recovering the flat-space spectrum of Section 1.5 with one extra non-compact dimension.

6.1.2 The duality: $R \to \alpha'/R$ and $n \leftrightarrow w$

Look at the mass formula (6.8) and perform the substitution

$$R \;\longrightarrow\; \tilde{R} = \frac{\alpha'}{R}, \qquad n \;\longleftrightarrow\; w. \tag{6.10}$$

The two zero-mode terms swap into each other: $n^2/R^2 \to w^2/\tilde{R}^2 = w^2R^2/\alpha'^2$ and $w^2R^2/\alpha'^2 \to n^2\tilde{R}^2/\alpha'^2 = n^2/R^2$. The oscillator term is untouched, and level matching (6.9) is invariant because nw is. The entire spectrum is left exactly where it was: every state of the theory at

radius R has a partner of identical mass at radius $\tilde{R} = \alpha'/R$, obtained by trading its momentum number for its winding number, an exact match level by level.

The match lifts to an isomorphism of the full conformal field theories. Under (6.10) the left- and right-moving momenta (6.5) transform as

$$p_{\mathrm{L}} \longrightarrow p_{\mathrm{L}}, \qquad p_{\mathrm{R}} \longrightarrow -p_{\mathrm{R}}. \tag{6.11}$$

T-duality is a one-sided reflection: it flips the sign of everything right-moving and leaves everything left-moving alone. Extending the reflection to the oscillators, $\alpha_k \to -\alpha_k$ and $\tilde{\alpha}_k \to \tilde{\alpha}_k$, defines a canonical transformation

$$X_{\mathrm{L}} \longrightarrow X_{\mathrm{L}}, \qquad X_{\mathrm{R}} \longrightarrow -X_{\mathrm{R}}, \tag{6.12}$$

which preserves the worldsheet action, the stress tensor, and the oscillator algebra of Appendix A. It is a symmetry of the free-boson CFT that acts trivially on left-movers and as a parity on right-movers. The two descriptions are literally the same theory, written in two coordinate systems related by (6.12).

The reflection (6.12) produces a dual coordinate worth naming. Define

$$\tilde{X} \;\coloneqq\; X_{\mathrm{L}} - X_{\mathrm{R}}, \tag{6.13}$$

the coordinate the reflection builds out of X. The reflection exchanges the two, $X \leftrightarrow \tilde{X}$: flipping the sign of X_{R} turns $X = X_{\mathrm{L}} + X_{\mathrm{R}}$ into $X_{\mathrm{L}} - X_{\mathrm{R}} = \tilde{X}$ and back, and the one thing it leaves fixed is X_{L}. Comparing zero modes, $\tilde{X}$ has momentum w around a circle of radius $\tilde{R} = \alpha'/R$ and winding n, so the roles of (6.2) and (6.3) are exchanged. For the worldsheet derivatives, use that X_{L} and X_{R} depend only on $\sigma^{\pm} = \tau \pm \sigma$, so $\partial_\tau X_{\mathrm{L}} = \partial_\sigma X_{\mathrm{L}} = X_{\mathrm{L}}'$ while $\partial_\tau X_{\mathrm{R}} = -\partial_\sigma X_{\mathrm{R}} = X_{\mathrm{R}}'$; then

$$\partial_\tau \tilde{X} = X_{\mathrm{L}}' - X_{\mathrm{R}}' = \partial_\sigma X, \qquad \partial_\sigma \tilde{X} = X_{\mathrm{L}}' + X_{\mathrm{R}}' = \partial_\tau X, \tag{6.14}$$

so a gradient of X along the string becomes a time derivative of $\tilde{X}$, and vice versa. Equation (6.14) is the operative heart of T-duality. When we hit the open string with it in Section 6.2, the two derivatives exchange, which turns the Neumann condition $\partial_\sigma X = 0$ into the Dirichlet condition $\partial_\tau \tilde{X} = 0$. Hold onto it.

No sub-string-scale geometry. The invariance (6.10) says that the circle of radius R and the circle of radius α'/R are indistinguishable to a string. Push this. As $R \to 0$ the Kaluza-Klein tower (6.2) becomes infinitely heavy and decouples, exactly as for a particle, but the winding tower (6.3) becomes light: a string wrapped on a tiny circle costs almost nothing. The nearly-massless winding states form a continuum in the limit, and that continuum is precisely the momentum continuum of a *new* non-compact direction of radius $\tilde{R} = \alpha'/R \to \infty$. A string cannot tell a very small circle from a very large one; the geometry it sees bottoms out. The fixed point of (6.10) is

$$R_{\mathrm{sd}} \;=\; \sqrt{\alpha'} \;=\; \ell_s, \tag{6.15}$$

the *self-dual radius*, where the theory maps to itself, and the shortest distance the string resolves on the circle. Here is the first appearance of a lesson the course returns to: string geometry is not Riemannian geometry, and lengths below ℓ_s are reparametrised into lengths above it rather than probed directly.

T-duality is exact, but read the fine print

The equivalence holds order by order in the g_s expansion, not merely at tree level, and it holds exactly in α': it is an isomorphism of the exact worldsheet CFT rather than a

low-energy approximation. Two caveats. First, the string coupling is *not* inert: matching the lower-dimensional Newton constant forces the dilaton to shift, $g_s \to g_s\,\ell_s/R$, which is the Buscher dilaton rule of Section 6.1.4 and is invisible in the free spectrum above only because it multiplies the whole partition function. Second, T-duality is a statement about a background with a compact isometry; on a circle it is exact and unbroken, but a generic curved background need not have one, and when the isometry has fixed points the naive rules degenerate.

6.1.3 The self-dual radius and enhanced gauge symmetry

Something special happens at $R = \ell_s$. To see it, ask which states in (6.8) are massless, and in particular which are massless spacetime vectors, since a massless vector in the non-compact dimensions is a gauge boson.

At generic radius the answer is dull. A massless vector needs one oscillator carrying a non-compact index and enough level to reach $M^2 = 0$. With $n = w = 0$ (hence $p_{\rm L} = p_{\rm R} = 0$) the states

$$\alpha^{\mu}_{-1}\,\tilde{\alpha}_{-1}\,|0\rangle\,, \qquad \alpha_{-1}\,\tilde{\alpha}^{\mu}_{-1}\,|0\rangle \tag{6.16}$$

are massless, where the un-indexed oscillator is the internal (compact) one. These are the two Kaluza-Klein gauge bosons: the graviphoton from $g_{\mu,\text{circle}}$ and the winding photon from $B_{\mu,\text{circle}}$, familiar from the reduction of the metric and B-field of Section 1.5.3. The generic gauge symmetry of the circle compactification is $\mathrm{U}(1)_{\rm L} \times \mathrm{U}(1)_{\rm R}$, one factor for each internal current $\partial X_{\rm L}$ and $\bar\partial X_{\rm R}$.

At the self-dual radius (6.15) the momenta (6.5) become $p_{\rm L} = (n+w)/\ell_s$ and $p_{\rm R} = (n-w)/\ell_s$, so

$$\frac{\alpha'}{4}\,p_{\rm L}^2 = \frac{(n+w)^2}{4}, \qquad \frac{\alpha'}{4}\,p_{\rm R}^2 = \frac{(n-w)^2}{4}. \tag{6.17}$$

These are the conformal weights of the zero-mode vertex operator (the momentum-weight relation $h = \frac{\alpha'}{4}p^2$ of the state-operator map, Section 2.1), and at the self-dual radius they can equal 1 with no oscillators at all. Consider a vector polarised by the left-moving oscillator $\tilde\alpha^{\mu}_{-1}$ (so $\tilde N = 1$). Masslessness through (6.7) forces $p_{\rm L} = 0$ and $\frac{\alpha'}{4}p_{\rm R}^2 = 1$, that is $n + w = 0$ and $(n-w)^2 = 4$. The solutions are

$$(n,w) = (1,-1) \quad \text{and} \quad (n,w) = (-1,1), \tag{6.18}$$

two new massless vectors that exist only at $R = \ell_s$. Together with the generic one from (6.16) they make three. Their internal parts are the three right-moving currents

$$J^3(\bar z) \;\sim\; \bar\partial X_{\rm R}, \qquad J^{\pm}(\bar z) \;\sim\; {:}e^{\pm 2iX_{\rm R}/\ell_s}{:}, \tag{6.19}$$

each of holomorphic (here antiholomorphic) weight 1 by (6.17) with $(n-w) = \pm 2$. Three weight-one currents whose charges are the momentum/winding lattice points ± 2 close into an $\mathfrak{su}(2)$ current algebra at level one; the derivation of the operator products is a standard vertex-operator computation, collected in Appendix G. Repeating the analysis for a vector polarised by the right-moving α^{μ}_{-1} produces the mirror triple of left-moving currents, with $(n,w) = (1,1), (-1,-1)$.

The chirality label needs care. The subscript on each SU(2) names the moving-sector of the internal current that generates it, while the spacetime polarisation rides the oscillator of the *opposite* chirality. The left-moving $\tilde\alpha^{\mu}_{-1}$ carries a right-moving internal dressing and so joins $\mathrm{SU}(2)_{\rm R}$; the right-moving α^{μ}_{-1} joins $\mathrm{SU}(2)_{\rm L}$. The two triples read:

Polarisation oscillator	Extra states (n, w)	Enhanced factor
$\tilde{\alpha}^{\mu}_{-1}$ (left-moving)	$(1,-1),\ (-1,1)$	$\mathrm{SU}(2)_{\mathrm{R}}$
α^{μ}_{-1} (right-moving)	$(1,1),\ (-1,-1)$	$\mathrm{SU}(2)_{\mathrm{L}}$

Each factor pairs its two generic gauge bosons (6.16) with the two extra states of its own chirality, and the gauge symmetry is enhanced,

$$\mathrm{U}(1)_{\mathrm{L}} \times \mathrm{U}(1)_{\mathrm{R}} \xrightarrow{R=\ell_s} \mathrm{SU}(2)_{\mathrm{L}} \times \mathrm{SU}(2)_{\mathrm{R}}, \tag{6.20}$$

from two gauge bosons to six.

Moving off the self-dual radius switches the enhancement off, and the mechanism is the ordinary Higgs mechanism run backwards. The radius modulus is the vacuum expectation value of the massless internal scalar $\alpha_{-1}\tilde{\alpha}_{-1}\,|0\rangle$, the state $g_{\mathrm{circle,circle}}$ that sets R. At $R = \ell_s$ this scalar sits in the adjoint of the enhanced group; turning it on away from ℓ_s is a Higgs field that gives mass to four of the six vectors and breaks $\mathrm{SU}(2)^2$ back to $\mathrm{U}(1)^2$. The four extra vectors (6.18) and their mirrors are massless only at the symmetric point and grow a mass $\propto |R - \ell_s|$ on either side of it. They become massive by eating scalars, in the ordinary way, which is what certifies the enhancement as a genuine gauge symmetry.

Remark 6.1 (Beyond the lectures: the Narain lattice). The single circle is the $d = 1$ case of toroidal compactification on T^d. There the $2d$ left- and right-moving momenta $(p_{\mathrm{L}}, p_{\mathrm{R}})$ live on an even self-dual lattice of signature (d, d), the *Narain lattice* $\Gamma_{d,d}$, and level matching is the statement that the lattice is even, $p_{\mathrm{L}}^2 - p_{\mathrm{R}}^2 \in 2\mathbb{Z}$. The moduli are the constant background metric G_{mn} and B-field B_{mn} on the torus, parametrising the coset $\mathrm{O}(d,d)/(\mathrm{O}(d) \times \mathrm{O}(d))$, and the T-duality group is the arithmetic subgroup $\mathrm{O}(d, d; \mathbb{Z})$ that maps the lattice to itself. The single-circle map (6.10) is the generator of $\mathrm{O}(1, 1; \mathbb{Z}) = \mathbb{Z}_2$. Enhanced gauge symmetry occurs on the sublocus where extra lattice points reach $(p_{\mathrm{L}}^2, p_{\mathrm{R}}^2) = (2, 0)$ or $(0, 2)$, and the possible enhanced groups are exactly the simply-laced (ADE) ones, classified in Appendix G. This is the same lattice technology that, applied to the sixteen chiral bosons of the heterotic string, produces its $\mathrm{E}_8 \times \mathrm{E}_8$ and $\mathrm{Spin}(32)/\mathbb{Z}_2$ gauge groups (Chapter 3); T-duality on the heterotic torus is what continuously connects the two, one face of the HE/HO equivalence taken up in Chapter 7.

Exercise 6.3 (Counting the enhanced spectrum). *(Moderate.)* At $R = \ell_s$, list every state with $N + \tilde{N} \leq 1$ that is massless by (6.8), keeping track of (n, w) and of which oscillators are excited. Show that the massless vectors organise into the adjoint $(\mathbf{3}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{3})$ of $\mathrm{SU}(2)_{\mathrm{L}} \times \mathrm{SU}(2)_{\mathrm{R}}$, and find the massless scalars with $N = \tilde{N} = 0$, namely $(n, w) = (\pm 2, 0), (0, \pm 2)$. Verify that these scalars, together with the radius modulus, fill out the representation eaten in the Higgsing (6.20). Confirm that away from $R = \ell_s$ none of the extra states (6.18) is massless.

6.1.4 Buscher rules: T-duality in a general background

The circle was flat and the B-field was off. String backgrounds are more general: the sigma model of Chapter 5 carries an arbitrary metric $G_{\mu\nu}(X)$, an antisymmetric $B_{\mu\nu}(X)$, and a dilaton $\phi(X)$. T-duality survives whenever the background has a compact isometry, a direction one can shift without changing G, B or ϕ. Choose adapted coordinates $X^{\mu} = (\theta, X^{i})$ in which the isometry is the shift $\theta \to \theta + \text{const}$ with $\theta \sim \theta + 2\pi$, so that none of the background fields depends on θ. Dualising along θ produces a new background $(\tilde{G}, \tilde{B}, \tilde{\phi})$, and the map between them is the set of *Buscher rules*:

$$\tilde{G}_{\theta\theta} = \frac{1}{G_{\theta\theta}}, \tag{6.21a}$$

$$\tilde{G}_{\theta i} = \frac{B_{\theta i}}{G_{\theta\theta}}, \qquad \tilde{B}_{\theta i} = \frac{G_{\theta i}}{G_{\theta\theta}}, \tag{6.21b}$$

$$\tilde{G}_{ij} = G_{ij} - \frac{G_{\theta i}G_{\theta j} - B_{\theta i}B_{\theta j}}{G_{\theta\theta}}, \qquad \tilde{B}_{ij} = B_{ij} - \frac{G_{\theta i}B_{\theta j} - B_{\theta i}G_{\theta j}}{G_{\theta\theta}}, \tag{6.21c}$$

$$\tilde{\phi} = \phi - \tfrac{1}{2}\ln G_{\theta\theta}. \tag{6.21d}$$

Here the metric and B-field components are dimensionless, measured in units of α', so that a flat circle of radius R is the constant background $G_{\theta\theta} = R^2/\alpha'$.

First, the mixed rule (6.21b) exchanges the off-diagonal metric with the off-diagonal B-field. An off-diagonal $G_{\theta i}$ is a Kaluza-Klein graviphoton, whose charge is momentum along the circle; an off-diagonal $B_{\theta i}$ is the winding photon. Their swap is the general-background face of $n \leftrightarrow w$: momentum and winding are exchanged, and so are the gauge fields that count them. In the flat, $B = 0$ case only (6.21a) survives, and it reads $\tilde{G}_{\theta\theta} = \alpha'/R^2 = \tilde{R}^2/\alpha'$, that is $\tilde{R} = \alpha'/R$, recovering (6.10).

Second, the dilaton shift (6.21d) is a genuinely stringy, one-loop effect, invisible in the classical sigma model. It comes from the path-integral measure: gauging the isometry and integrating out the gauge field produces a Jacobian whose regularisation shifts ϕ by $-\frac{1}{2}\ln G_{\theta\theta}$. Its job is to keep the theory physically the same. Under $R \to \alpha'/R$ it gives $g_s \to g_s\,\ell_s/R$, and one checks that the combination $g_s^2/(2\pi R)$ that sets the lower-dimensional Newton constant $G_{D-1} \sim G_D/(2\pi R)$ is invariant. Without the dilaton shift, T-duality would change Newton's constant and could not be an equivalence.

Remark 6.2 (Beyond the lectures: where the rules come from). The clean derivation gauges the shift isometry. Promote $\theta \to \theta + \lambda(\sigma^\alpha)$ to a local symmetry by introducing a worldsheet gauge field A_α, add a Lagrange multiplier $\tilde{\theta}$ enforcing $F_{\alpha\beta} = 0$ (so that A is pure gauge and nothing has changed), and then integrate out. Integrating out $\tilde{\theta}$ first returns the original model; integrating out A_α first, which is Gaussian, returns the dual model with the fields (6.21), and the Gaussian determinant is the source of the dilaton shift (6.21d). The construction is due to Buscher; we quote the result rather than run the worldsheet path integral here, and the Weyl-invariance conditions that make $(\tilde{G}, \tilde{B}, \tilde{\phi})$ a consistent string background are the beta-function equations of Chapter 5. The same gauging, carried out for the Ramond-Ramond sector, is what maps the IIA and IIB RR potentials into each other and shifts every p-form degree by one, which is why the branes of Section 6.2 change dimension under T-duality.

Example 6.3 (A twisted circle becomes a B-field)**.** Take a circle fibred over a base so that the metric has a constant off-diagonal piece, $\mathrm{d}s^2 = R^2(\mathrm{d}\theta + a\,\mathrm{d}x)^2 + \mathrm{d}x^2$ with a a constant and $B = 0$. In string units $G_{\theta\theta} = R^2/\alpha'$, $G_{\theta x} = aR^2/\alpha'$, and $G_{xx} = (1 + a^2R^2)/\alpha'$. Applying (6.21): the dual metric is diagonal, $\tilde{G}_{\theta\theta} = \alpha'/R^2$ and $\tilde{G}_{xx} = G_{xx} - G_{\theta x}^2/G_{\theta\theta} = 1/\alpha'$, but a B-field switches on, $\tilde{B}_{\theta x} = G_{\theta x}/G_{\theta\theta} = a$. Momentum along the twisted fibre has become winding charge under a B-field: a purely geometric background and a background with flux are the same string theory. This is the baby version of the geometry/flux exchanges that reorganise the flux compactifications of Chapter 9.

Exercise 6.4 (Dualising a general background)**.** *(Moderate.)* (a) Set $B = 0$ and take the flat circle $G_{\theta\theta} = R^2/\alpha'$ with all other components trivial. Show that only (6.21a) acts, returning $\tilde{G}_{\theta\theta} = \alpha'/R^2$, that is $\tilde{R} = \alpha'/R$, and confirm from (6.21d) that $g_s \to g_s\,\ell_s/R$. (b) Now switch on a constant off-diagonal $B_{\theta i}$ with $G_{\theta i} = 0$. Apply (6.21b) and read off that the winding photon $B_{\theta i}$ has turned into the graviphoton $\tilde{G}_{\theta i}$, so that the momentum and winding gauge fields are exchanged exactly as $n \leftrightarrow w$ demands.

Where this is going

The closed string handed us T-duality as an exact equivalence: $R \leftrightarrow \alpha'/R$, $n \leftrightarrow w$, implemented on the worldsheet by the one-sided reflection $X_{\mathrm{R}} \to -X_{\mathrm{R}}$, and extended to any background

with a compact isometry by the Buscher rules (6.21). The single most useful thing to carry forward is the worldsheet identity (6.14), $\partial_\tau \tilde{X} = \partial_\sigma X$. For a closed string it merely relabels momentum as winding. For an open string it does something far more drastic: it exchanges the two kinds of endpoint condition of Section 1.2.6, Neumann for Dirichlet. A theory with freely moving string ends turns, under the very same duality we have just proved exact, into a theory whose string ends are pinned to a hypersurface. That hypersurface is a D-brane, and building it is the business of Section 6.2.

6.2 D-branes from T-dualising open strings

Closed-string T-duality, as we set it up in Section 6.1, takes a theory of closed strings on a circle of radius R and returns another theory of closed strings on a circle of radius α'/R, trading momentum for winding. Both Type II strings and both heterotic strings are theories of closed strings, so on a circle T-duality is either a symmetry of one theory or an exact relation between two of them. Feed the same operation an *open* string and something new comes out.

An open string has two ends. In a Neumann direction those ends move freely and the string carries centre-of-mass momentum, but it cannot wind: a segment with loose ends can always be pulled straight, so there is no conserved winding number to speak of. The open string sees only half of the momentum-winding structure that made closed-string T-duality tick, and T-duality exchanges the two halves. In the dual frame the open string must therefore acquire the half it was missing. Its momentum turns into winding, and its free Neumann endpoints turn into endpoints pinned to a hyperplane. That hyperplane is a D-brane. You cannot keep both open strings and T-duality without also admitting D-branes into the theory.

Why T-duality forces D-branes

The boundary conditions of Section 1.2.6 offered Neumann and Dirichlet as two equally allowed choices, and Neumann looked like the physical one because it preserves translation invariance. T-duality removes the choice. Applied along a compact Neumann direction it produces, term by term in the mode expansion, a Dirichlet direction: the endpoints stop at a fixed value of the dual coordinate. Poincaré invariance survives the move, relocated from the original circle to the $(p+1)$-dimensional hyperplane the endpoints now live on. That hyperplane is a dynamical object, the Dirichlet p-brane, with its own worldvolume fields (a gauge field and a set of transverse scalars whose values are its position) and its own action (Dirac-Born-Infeld plus a Wess-Zumino coupling to the Ramond-Ramond forms). Everything the rest of this chapter says about D-brane charge, tension and supersymmetry rests on the elementary mode computation of Section 6.2.2.

Learning objectives. By the end of this section you should be able to:

- run the pinning computation: T-dualise a compact Neumann open string and show its endpoints become Dirichlet, fixed on a hyperplane (Exercise 6.5);
- read off how a single T-duality changes a brane, D$p \to$ D$(p \mp 1)$, and shuttles between IIA and IIB (Table 6.1);
- identify a diagonal Wilson line with a set of brane positions, and recover the stretched-string Higgs mechanism from brane separations;
- name the worldvolume field content of a Dp-brane and recognise it as the dimensional reduction of ten-dimensional super-Yang-Mills;
- expand the Dirac-Born-Infeld action to Maxwell-plus-scalars and read off g_{YM} (Exercise 6.6),

and write the Wess-Zumino coupling that dissolves lower branes into worldvolume flux (Exercise 6.7).

The pinning computation of Section 6.2.2, together with its consequences through the worldvolume field content of Section 6.2.5, is the core of the section. The Dirac-Born-Infeld and Wess-Zumino actions (Sections 6.2.6 and 6.2.7) are written up more fully than a first pass needs; the two results worth keeping from them are flagged as we reach them.

6.2.1 The open string on a circle

Everything below is the ten-dimensional Type II superstring, with the mechanics identical for the bosonic string. The pinning argument uses only the momentum-winding structure of Section 6.1, not the intercept, so the switch from the bosonic circle mass formula (6.8) to the superstring changes only the zero-point energy, not a step of the computation.

Take the open string and compactify one Neumann direction. Call it $X \equiv X^9$, periodic with $X \sim X + 2\pi R$, and leave $X^0, \dots, X^8$ non-compact. All directions carry Neumann conditions, so the endpoints are free and the string is the boundary of a space-filling brane; we return to that identification once the dual has told us what kind of brane it is.

Compactification changes the Neumann mode expansion (1.46) in one way: the centre-of-mass momentum along X is quantised, $p = n/R$ with $n \in \mathbb{Z}$, so that e^{ipX} is single-valued on the circle. There is no winding quantum number, because the cosine mode expansion has antinodes at the ends rather than a term linear in σ. Writing $\sigma^{\pm} = \tau \pm \sigma$ with $\sigma \in [0, \pi]$, split the field into left- and right-moving halves, $X = X_{\mathrm{L}}(\sigma^+) + X_{\mathrm{R}}(\sigma^-)$, with

$$X_{\mathrm{L}}(\sigma^+) = \tfrac{1}{2}x + \alpha' \frac{n}{R}\,\sigma^+ + \tfrac{i}{2}\sqrt{2\alpha'} \sum_{m\neq 0} \frac{\alpha_m}{m}\, e^{-im\sigma^+}, \tag{6.22}$$

$$X_{\mathrm{R}}(\sigma^-) = \tfrac{1}{2}x + \alpha' \frac{n}{R}\,\sigma^- + \tfrac{i}{2}\sqrt{2\alpha'} \sum_{m\neq 0} \frac{\alpha_m}{m}\, e^{-im\sigma^-}. \tag{6.23}$$

The two halves carry the *same* oscillators α_m. That identification is the Neumann boundary condition, which reflects a left-mover into a right-mover at each end. Adding the two lines reproduces (1.46) with $p = n/R$, the two momentum terms summing to the expected $2\alpha'(n/R)\tau$, and the endpoints sit at antinodes carrying momentum n/R.

6.2.2 T-duality pins the endpoints

T-duality flips the sign of the right-movers, exactly as for the closed string in Section 6.1. Define the dual coordinate

$$\tilde{X}(\tau, \sigma) \coloneqq X_{\mathrm{L}}(\sigma^+) - X_{\mathrm{R}}(\sigma^-). \tag{6.24}$$

Subtract (6.23) from (6.22). The constant $\frac{1}{2}x$ cancels and is replaced by an undetermined additive constant $\tilde{x}$, the zero mode of the dual field. The momentum terms give $\alpha'(n/R)(\sigma^+ - \sigma^-) = 2\alpha'(n/R)\sigma$, and the oscillator sum turns cosines into sines through $e^{-im\sigma^+} - e^{-im\sigma^-} = -2i\, e^{-im\tau} \sin(m\sigma)$. Collecting the pieces,

$$\tilde{X}(\tau, \sigma) \;=\; \tilde{x} \;+\; 2\alpha' \frac{n}{R}\,\sigma \;+\; \sqrt{2\alpha'} \sum_{m\neq 0} \frac{\alpha_m}{m}\, e^{-im\tau} \sin(m\sigma). \tag{6.25}$$

Compare this with the Dirichlet expansion (1.47). The oscillators multiply $\sin(m\sigma)$, which vanishes at both ends, and there is no term linear in τ, so the dual coordinate carries no momentum. The endpoints are stuck. Read off their positions:

$$\tilde{X}\big|_{\sigma=0} = \tilde{x}, \qquad \tilde{X}\big|_{\sigma=\pi} = \tilde{x} + 2\pi n\, \tilde{R}, \qquad \tilde{R} \coloneqq \frac{\alpha'}{R}. \tag{6.26}$$

Both ends sit at the same value of the dual coordinate: the two endpoint positions differ by $2\pi n\,\tilde{R}$, which is n complete circuits of the dual circle of radius $\tilde{R} = \alpha'/R$. The string winds n times around that circle and comes back to where it started. What was momentum n/R of the original open string is winding number n of the dual open string, so the missing half of the momentum-winding structure has reappeared, precisely as T-duality demands.

A Neumann direction has become a Dirichlet direction. The endpoints, free to roam the original circle, are confined in the dual frame to the hyperplane $\tilde{X} = \tilde{x}$. That hyperplane, the nine-dimensional worldvolume spanned by time together with the eight directions transverse to $\tilde{X}$, is where the open strings must end. This is the D-brane promised by the boundary-condition analysis of Section 1.2.6, now derived rather than posited: T-duality has manufactured it out of an ordinary Neumann string.

T-duality acts on the worldsheet, not on spacetime

Nothing in (6.24) moved a point of spacetime. T-duality is a relabelling of the worldsheet degrees of freedom, $X_{\mathrm{R}} \to -X_{\mathrm{R}}$, that reorganises the same physics into a description with a different geometric circle and a different-looking brane. The original theory and its dual are the *same* quantum theory. When we say "a D-brane appears", we mean that the natural spacetime picture of that one theory, read in the dual variables, contains a hyperplane on which strings end. The brane is as real as the string it is dual to, which is the point of the whole construction.

6.2.3 Which brane, and how the type flips

We started with all directions Neumann, an object filling all ten dimensions, and T-dualising one direction cut that down to a hyperplane transverse to $\tilde{X}$: the object lost a dimension. The rule generalises directly. Take a Dp-brane, meaning Neumann conditions along $X^0, \ldots, X^p$ and Dirichlet conditions along $X^{p+1}, \ldots, X^9$, so the endpoints live on a $(p+1)$-dimensional worldvolume. T-dualise one compact direction, and the mode computation of Section 6.2.2 runs the same way, exchanging the boundary condition in that one direction:

- **T-dualise a direction along the brane** (a Neumann direction, $1 \le i \le p$). Neumann becomes Dirichlet, the worldvolume shrinks by one, and D$p \to$ D$(p{-}1)$. The brane that used to wrap the circle now sits at a point on the dual circle.
- **T-dualise a direction transverse to the brane** (a Dirichlet direction, $p{+}1 \le i \le 9$). Dirichlet becomes Neumann, the worldvolume grows by one, and D$p \to$ D$(p{+}1)$. The brane that sat at a point now wraps the dual circle.

Either way the brane dimension changes by one, so the rank of the Ramond-Ramond form it couples to changes by one, and the theory jumps between the even and odd D-brane ladders. This is the brane-side face of the statement that T-duality on a circle relates IIA and IIB: IIA on radius R is IIB on radius α'/R, IIA carries the even-dimensional branes D0, D2, D4, D6, D8 and IIB the odd ones D(-1), D1, D3, D5, D7, D9 (Section 3.4.6), and a single T-duality shuttles between the two lists one rung at a time. The Ramond-Ramond potentials of the two theories differ by one form degree for the same reason: T-duality on the worldsheet fermions rotates a C_p into a $C_{p\pm1}$, and the branes follow their electric sources.

The unoriented and heterotic strings complete the picture. T-dualising Type I (Section 3.5.1) along a circle turns the space-filling orientifold nine-plane into a pair of orientifold eight-planes sitting at the ends of the dual interval, with the 32 D9-branes becoming D8-branes free to move between them; the resulting theory is Type I′, and the O-planes it contains are the subject of Section 6.4. The two heterotic strings give a closed-string analogue with no branes involved: heterotic $\mathrm{E}_8 \times \mathrm{E}_8$ on a circle, with a Wilson line, is T-dual to heterotic $\mathrm{Spin}(32)/\mathbb{Z}_2$ on the dual

Direction T-dualised	Effect on the brane	Type
along the worldvolume (Neumann)	$\mathrm{D}p \to \mathrm{D}(p-1)$	IIA $\leftrightarrow$ IIB
transverse to it (Dirichlet)	$\mathrm{D}p \to \mathrm{D}(p+1)$	IIA $\leftrightarrow$ IIB

Table 6.1: One T-duality changes the D-brane dimension by one and swaps the two Type II theories. Iterating builds the whole D-brane spectrum of Table 5.2 from any single member of it.

circle, the gauge groups exchanged through the arithmetic of the Narain lattice rather than through any endpoint. The lattice machinery behind that exchange is collected in Appendix G, and the equivalence of the two heterotic strings, with its place in the duality web, is established in Chapter 7; we only flag it here.

6.2.4 Wilson lines are brane positions

So far the dual brane sat at a single location $\tilde{x}$, left undetermined by the mode expansion. Its value is physical, and it is controlled by a background field on the original brane. Recall from Section 1.5.4 that open-string endpoints carry Chan-Paton labels $i = 1, \dots, N$, promoting the massless vector to a $\mathrm{U}(N)$ gauge field A_μ. Switch on a constant component of that gauge field along the compact direction,

$$A_9 = \frac{1}{2\pi R}\,\mathrm{diag}(\theta_1, \dots, \theta_N), \tag{6.27}$$

a flat connection. Its field strength vanishes, so it is pure gauge locally, but its holonomy around the circle, $\exp\!\left(i \oint A_9\, \mathrm{d}X\right) = \mathrm{diag}(e^{i\theta_1}, \dots, e^{i\theta_N})$, cannot be removed, because the gauge transformation that would undo it is not periodic. This is a Wilson line. An open string with Chan-Paton labels (i, j) has one endpoint of charge $+1$ under the i-th U(1) and the other of charge -1 under the j-th, so the Wilson line shifts its quantised momentum to

$$p = \frac{2\pi n + \theta_i - \theta_j}{2\pi R}, \qquad n \in \mathbb{Z}. \tag{6.28}$$

Run this fractional momentum through the T-duality of Section 6.2.2. The endpoint carrying label i lands not at a common $\tilde{x}$ but at

$$\tilde{X}_i = \theta_i\, \tilde{R} = 2\pi\alpha'\, A_{9,ii}, \tag{6.29}$$

an angle θ_i around the dual circle. A stack of N coincident branes with a diagonal Wilson line becomes N D-branes sitting at N distinct points $\tilde{X}_1, \dots, \tilde{X}_N$ on the dual circle. The gauge field that pointed along the compactified direction has become the transverse position of the branes: the internal component A_9 turned into a coordinate.

This is the origin of the transverse scalars. Let the Wilson line fluctuate, $A_9 \to A_9(\xi)$ depending on the worldvolume coordinates ξ^a of the brane; the dual statement is a brane whose transverse position varies, $\tilde{X}(\xi) = 2\pi\alpha'\, A_9(\xi)$. A would-be gauge-field component along a T-dualised direction becomes, in the dual frame, a dynamical scalar field on the worldvolume whose value records where the brane is. We give it a name and a canonical normalisation below.

The separations $\tilde{X}_i - \tilde{X}_j$ are visible in the spectrum. A string stretched from brane i to brane j has minimum length $|\tilde{X}_i - \tilde{X}_j|$, and by the tension formula its ground state has mass

$$M_{ij} = \frac{|\tilde{X}_i - \tilde{X}_j|}{2\pi\alpha'} = \frac{|\theta_i - \theta_j|}{2\pi R} = |A_{9,ii} - A_{9,jj}|, \tag{6.30}$$

which is exactly the shifted momentum (6.28) at $n = 0$, as it must be, since T-duality does not change the spectrum. When all N branes coincide, every $\tilde{X}_i$ is equal, all N^2 stretched strings

become massless, and the gauge symmetry is the full $\mathrm{U}(N)$ carried by the coincident stack. Pull one brane off, and the strings ending on it acquire the mass (6.30): those gauge bosons become massive and $\mathrm{U}(N)$ breaks to $\mathrm{U}(N-1)\times\mathrm{U}(1)$. This is the Higgs mechanism, read geometrically. The Higgs field is the transverse position, its expectation value is the brane separation, and the mass of the W-boson is the tension times the length of the string that stretches across the gap. Brane positions are moduli, flat directions of the worldvolume potential, and moving along them Higgses the gauge theory.

Exercise 6.5 (The pinning computation)**.** *(Moderate.)* Fill in Section 6.2.2 in full. Starting from (6.22) and (6.23), verify that $X = X_{\mathrm{L}} + X_{\mathrm{R}}$ reproduces the Neumann expansion (1.46) with $p = n/R$, and that $\tilde{X} = X_{\mathrm{L}} - X_{\mathrm{R}}$ gives (6.25). Confirm that $\partial_\sigma X = 0$ at the endpoints (Neumann for X) is equivalent to $\partial_\tau \tilde{X} = 0$ at the endpoints (Dirichlet for $\tilde{X}$), so that the two boundary conditions are exchanged term by term. Finally, show that the mass of a state built on this string is unchanged by T-duality, with the momentum contribution $(n/R)^2$ to M^2 reappearing as the winding contribution $(n\tilde{R}/\alpha')^2$.

6.2.5 The fields that live on the brane

The T-duality argument has already handed us the field content of a single Dp-brane. Start from the massless open-string states in ten dimensions: a $\mathrm{U}(1)$ gauge field A_M, $M = 0,\dots,9$, and, for the superstring, its gaugino superpartner, together the ten-dimensional $\mathcal{N}=1$ Maxwell multiplet living on the space-filling D9-brane. T-dualise $9-p$ directions to reach a Dp-brane. The gauge field splits according to Section 6.2.4: the $p+1$ components A_a along the surviving Neumann directions stay a gauge field on the worldvolume, while the $9-p$ components A_i along the T-dualised directions become transverse scalars. Naming the transverse coordinates X^i and normalising so that the scalar has mass dimension one in the worldvolume theory,

$$\Phi^i \coloneqq \frac{X^i}{2\pi\alpha'}, \qquad i = p{+}1,\dots,9, \tag{6.31}$$

the worldvolume of a Dp-brane carries a $\mathrm{U}(1)$ gauge field A_a ($a = 0,\dots,p$), a set of $9-p$ real scalars Φ^i recording the position of the brane in the transverse space, and the fermions completing the supermultiplet. This is the dimensional reduction of ten-dimensional $\mathcal{N}=1$ super-Yang-Mills from ten dimensions to the $p+1$ worldvolume dimensions: the ten-dimensional gauge field reduces to A_a plus $9-p$ scalars, and the ten-dimensional gaugino to the worldvolume fermions.

For N coincident branes every field is promoted to an $N\times N$ Hermitian matrix in the adjoint of $\mathrm{U}(N)$, and the scalars Φ^i become matrices whose simultaneous eigenvalues are the N brane positions. Writing out the bosonic part, the worldvolume theory is

$$S_{\mathrm{SYM}} = -\frac{1}{g_{\mathrm{YM}}^2}\int \mathrm{d}^{p+1}\xi\ \mathrm{Tr}\Big(\tfrac{1}{4}\,F_{ab}F^{ab} + \tfrac{1}{2}\,D_a\Phi^i\,D^a\Phi_i - \tfrac{1}{4}\,[\Phi^i,\Phi^j][\Phi_i,\Phi_j]\Big) + (\text{fermions}), \tag{6.32}$$

with $F_{ab} = \partial_a A_b - \partial_b A_a - i[A_a, A_b]$, $D_a\Phi^i = \partial_a\Phi^i - i[A_a,\Phi^i]$, and g_{YM} the coupling (6.37) read off below. The scalars sit in a commutator-squared potential. Its flat directions, where all $[\Phi^i,\Phi^j]=0$ so the Φ^i are simultaneously diagonalisable, are labelled by the N eigenvalue-tuples, which are the N brane positions of Section 6.2.4; the off-diagonal entries are the stretched strings of (6.30). This is the $\mathrm{U}(N)$ super-Yang-Mills whose Coulomb branch labels the separated branes, and whose role as the microscopic origin of gauge theory from geometry is taken up in Section 6.5. Here we have established what fields are present and where they come from.

Remark 6.4 (Beyond the lectures: non-commuting positions)*.* When the scalars Φ^i are $N\times N$ matrices, their simultaneous eigenvalues are the brane positions, but the off-diagonal entries carry information too: they are the stretched strings of (6.30). Configurations in which the Φ^i

do not commute, $[\Phi^i, \Phi^j] \neq 0$, cost energy through the potential in (6.32) and have no classical geometric interpretation as N points; they are the seed of genuinely stringy geometry. The Myers effect, in which a stack of Dp-branes in a background Ramond-Ramond flux polarises into a higher-dimensional fuzzy brane, and the matrix-model proposals for M-theory both live in this non-commuting corner. The action that governs it is the non-abelian Dirac-Born-Infeld action with a symmetrised-trace prescription, which we quote where a specific computation needs it and not before.

6.2.6 The Dirac-Born-Infeld action

The two subsections that follow give the brane its action. They are a first orientation: the load-bearing results are the tension (6.35) and the minimal Ramond-Ramond coupling (6.40), and the curvature refinements that dress them can wait for a second reading. The bosonic action has two pieces: a Dirac-Born-Infeld term that measures the worldvolume and couples it to the Neveu-Schwarz sector, and a Wess-Zumino term that couples it to the Ramond-Ramond forms. Neither is derived here from first principles; both are fixed by open-string scattering amplitudes and by the requirement that T-duality act on the action as it acts on the branes. We state them and read off their content, and point to where each ingredient is pinned down.

The Dirac-Born-Infeld (DBI) action is

$$S_{\mathrm{DBI}} = -\,\tau_p \int_{\mathcal{W}} \mathrm{d}^{p+1}\xi \; e^{-(\phi-\phi_0)} \sqrt{-\det\bigl(P[g+B]_{ab} + 2\pi\alpha'\, F_{ab}\bigr)}\,, \tag{6.33}$$

where $\mathcal{W}$ is the $(p+1)$-dimensional worldvolume with coordinates ξ^a, τ_p is the physical tension (6.35), $\phi_0 = \langle\phi\rangle$ is the asymptotic dilaton with $g_s = e^{\phi_0}$, $F_{ab} = \partial_a A_b - \partial_b A_a$ is the worldvolume gauge field strength, and $P[\,\cdot\,]$ denotes the pullback of a spacetime tensor to the worldvolume,

$$P[g+B]_{ab} = (\partial_a X^\mu)(\partial_b X^\nu)(g_{\mu\nu} + B_{\mu\nu})(X). \tag{6.34}$$

The action packs three physical statements. The geometry first: with the gauge field and B switched off and $\phi = \phi_0$, (6.33) reduces to $-\tau_p \int \sqrt{-\det P[g]}$, the volume of the worldvolume swept out in spacetime, which is the higher-dimensional Nambu-Goto action. A D-brane, like a string, minimises its worldvolume, and its rest energy is a tension times a volume. Then the coupling: the exponential is the dilaton measured from its vacuum value, the string-loop origin of the $e^{-\phi}$ that customarily sits out front, and because $\tau_p \propto e^{-\phi_0} = g_s^{-1}$ the D-brane tension carries one inverse power of the string coupling, the hallmark g_s^{-1} recorded as the scaling $T_{\mathrm{D}p} \sim g_s^{-1}$ in Eq. (5.83). Explicitly,

$$\boxed{\tau_p = \frac{1}{(2\pi)^p\, g_s\, \ell_s^{p+1}}}\,, \qquad g_s\,\tau_p = \frac{1}{(2\pi)^p\, \ell_s^{p+1}}\,, \tag{6.35}$$

the coupling-independent combination $g_s\tau_p$ on the right being the one that reappears as the brane's Ramond-Ramond charge in Section 6.2.7. The value of τ_p, and the fact that it equals that charge so that the brane saturates the BPS bound (5.78), is the Polchinski computation of Section 6.3; here it is quoted. The third statement is the field combination: B and F enter only through the T-duality-invariant $B_{ab} + 2\pi\alpha' F_{ab}$, the same gauge-invariant field strength that appears on the worldsheet boundary. That combination is what makes the DBI action transform correctly under T-duality, and it is why a worldvolume flux is indistinguishable from a B-field background.

To see the familiar low-energy theory buried inside (6.33), work in static gauge, $\xi^a = X^a$ for $a = 0, \dots, p$, take flat space $g = \eta$ with $B = 0$ and $\phi = \phi_0$, and use the scalars (6.31). The

pullback becomes $P[\eta]_{ab} = \eta_{ab} + (2\pi\alpha')^2\,\partial_a\Phi^i\,\partial_b\Phi_i$, and expanding the square root for slowly varying fields,

$$S_{\mathrm{DBI}} \simeq -\,\tau_p \int \mathrm{d}^{p+1}\xi \left[\, 1 + \frac{(2\pi\alpha')^2}{4}\,F_{ab}F^{ab} + \frac{(2\pi\alpha')^2}{2}\,\partial_a\Phi^i\,\partial^a\Phi_i + \dots \right]. \tag{6.36}$$

The constant term is the tension times the worldvolume, the rest energy of the brane. The quadratic terms are a Maxwell action for A_a and a canonical kinetic term for the transverse scalars Φ^i: the worldvolume field theory is Yang-Mills coupled to adjoint scalars, exactly the dimensional reduction identified in Section 6.2.5. Reading the coefficient of F^2 gives the worldvolume gauge coupling

$$\frac{1}{g_{\mathrm{YM}}^2} = (2\pi\alpha')^2\,\tau_p \quad\Longrightarrow\quad g_{\mathrm{YM}}^2 = (2\pi)^{p-2}\,g_s\,\ell_s^{p-3}. \tag{6.37}$$

For $p = 3$ the powers of ℓ_s cancel and $g_{\mathrm{YM}}^2 = 2\pi g_s$ is dimensionless, which is why the D3-brane carries an honest four-dimensional gauge theory, a fact the AdS/CFT correspondence of Chapter 13 rests on. The higher terms in (6.36), the ones the Born-Infeld square root resums, are an infinite series of α' corrections encoding the full non-linear response of the brane to strong worldvolume fields.

Exercise 6.6 (Born-Infeld from the square root)**.** *(Moderate.)* Take (6.33) in flat space with $B = 0$, $\phi = \phi_0$ and constant Φ^i, so only F_{ab} survives. Using $\det(\mathbf{1} + M) = \exp \mathrm{Tr}\ln(\mathbf{1} + M)$ and the antisymmetry of F, expand $\sqrt{-\det(\eta_{ab} + 2\pi\alpha' F_{ab})}$ to fourth order in F and confirm the Maxwell term in (6.36). Then specialise to $p = 1$ (a D-string) with a single electric field $F_{01} = E$, and show that the Lagrangian density is $-\tau_1\sqrt{1 - (2\pi\alpha' E)^2}$, so that the electric field cannot exceed a critical value $E_{\mathrm{crit}} = 1/(2\pi\alpha')$. Interpret E_{crit} as the field at which the energy stored in the string equals its tension, the point at which fundamental strings are pulled out of the brane.

6.2.7 The Wess-Zumino coupling and induced charges

The DBI action knows nothing about Ramond-Ramond fields. The coupling that makes a Dp-brane the electric source of C_{p+1}, responsible for its charge and hence for everything BPS about it, is the Wess-Zumino (or Chern-Simons) term

$$S_{\mathrm{WZ}} = \mu_p^{(\mathrm{WZ})} \int_{\mathcal{W}} P\Big[\, C \wedge e^{B} \,\Big] \wedge e^{2\pi\alpha' F} \wedge \sqrt{\hat{A}(\mathcal{R})}\,, \qquad C \coloneqq \bigoplus_q C_q, \tag{6.38}$$

where C is the formal sum of all Ramond-Ramond potentials, the wedge products are pulled back to the worldvolume, and the integral extracts the $(p+1)$-form piece of the integrand, the only piece that can be integrated over $\mathcal{W}$. The charge $\mu_p^{(\mathrm{WZ})}$ is the coupling-independent combination of (6.35),

$$\mu_p^{(\mathrm{WZ})} = g_s\,\tau_p = \frac{1}{(2\pi)^p\,\ell_s^{p+1}}, \tag{6.39}$$

which carries no power of g_s because the string-frame Ramond-Ramond field, unlike the Neveu-Schwarz fields, comes with no $e^{-2\phi}$. In the gravitational frame of Section 6.3 the brane's physical charge equals its physical tension, $\mu_p = \tau_p$, which is the worldvolume statement of $M = |Z|$; the frame bookkeeping relating $\mu_p^{(\mathrm{WZ})}$ to the physical μ_p is the warning box there. The magnitude of the charge is fixed by Dirac quantisation together with the DBI tension, and is established in Section 6.3.

The last factor, $\sqrt{\hat{A}(\mathcal{R})}$, is a refinement for branes wrapping curved cycles and may be skipped on a first pass. It is the square root of the A-roof genus built from the curvature two-form $\mathcal{R}$ of the worldvolume (more precisely a ratio of A-roof genera of the tangent and normal bundles), a

sum of $4k$-forms starting at 1, and its curvature terms contribute only when the brane wraps a curved cycle. The characteristic classes it is built from are collected in Appendix D.

The leading term of (6.38) is the minimal electric coupling,

$$\mu_p^{(\mathrm{WZ})} \int_{\mathcal{W}} P[C_{p+1}]\,, \tag{6.40}$$

the higher-dimensional analogue of the point particle's $\int A$ and the string's $\int B$: a $(p+1)$-dimensional worldvolume couples to a $(p+1)$-form potential, so the Dp-brane carries $\mu_p^{(\mathrm{WZ})}$ units of C_{p+1} charge. The rest of the expansion is where the structure lives. Multiplying out the exponentials produces lower-degree Ramond-Ramond forms wedged with powers of $B + 2\pi\alpha' F$,

$$S_{\mathrm{WZ}} = \mu_p^{(\mathrm{WZ})} \int_{\mathcal{W}} \Big[\, C_{p+1} + C_{p-1} \wedge (B + 2\pi\alpha' F) + \tfrac{1}{2}\, C_{p-3} \wedge (B + 2\pi\alpha' F)^2 + \dots \,\Big]. \tag{6.41}$$

Each term couples the brane to a *lower* Ramond-Ramond potential, weighted by worldvolume flux. A Dp-brane carrying k units of magnetic flux $\int F$ on a two-cycle in its worldvolume also carries the charge of a D$(p-2)$-brane: it is a bound state of a Dp-brane and some D$(p-2)$-branes dissolved into it as flux. This is why a single T-duality can turn one brane into another (Section 6.2.3), and why brane charges must always be counted together with the flux they carry. The curvature terms from $\sqrt{\hat{A}}$ do the same job geometrically: a D-brane wrapping a topologically non-trivial cycle picks up induced lower-brane charges proportional to the Pontryagin numbers of the cycle, even with no worldvolume flux. These induced charges are not bookkeeping to be waved away: the tadpole-cancellation conditions of Chapters 9 to 11 sum over exactly them, so a miscounted A-roof term is a miscounted tadpole. The curvature couplings and the induced charges are worked out in Appendix D.

Remark 6.5 (Beyond the lectures: anomaly inflow fixes the curvature terms). The $\sqrt{\hat{A}}$ factor is forced, not an optional refinement. A D-brane on which chiral fermions live can suffer a gauge or gravitational anomaly localised on its worldvolume, and consistency demands that the anomaly be cancelled by inflow from the bulk. The bulk mechanism is precisely the variation of the Wess-Zumino coupling (6.38) in the presence of the bulk Ramond-Ramond fields, and matching the inflow to the worldvolume anomaly fixes every term in $\sqrt{\hat{A}(\mathcal{R}_T)/\hat{A}(\mathcal{R}_N)}$, tangent bundle over normal bundle. The same $B \wedge X_8$ structure that made the Green-Schwarz mechanism work in Chapter 5 reappears here as the brane's anomalous coupling; the two are the same calculation seen from opposite sides of the brane.

Two results from these last two subsections carry forward: the physical tension (6.35), $\tau_p \sim g_s^{-1}$, and the minimal Ramond-Ramond coupling (6.40) that makes the brane the electric source of C_{p+1}. The A-roof and anomaly-inflow material is depth for later chapters and changes neither headline.

Where this leaves us

T-duality, applied to the open string, does not leave the choice of boundary conditions intact: it converts Neumann into Dirichlet and stands up a dynamical hyperplane where the string ends. That hyperplane is the Dp-brane, carrying a U(1) (or U(N)) gauge field and $9-p$ transverse scalars that report its position, and governed by the Dirac-Born-Infeld action together with the Wess-Zumino coupling to the Ramond-Ramond forms. We have read off its tension, seen the Yang-Mills theory sitting inside the DBI square root, and identified the minimal Ramond-Ramond coupling that makes it charged. What we have not yet done is compute that charge and prove that it locks the tension to it. That computation, Polchinski's, is the next section: it turns the D-brane from a T-duality artefact into a BPS object of definite charge, and threads it onto the BPS thread of Section 5.6.

Exercise 6.7 (Dissolving a brane into flux)**.** *(Moderate; this bound state is the workhorse of the microstate counting in Chapter 12.)* Consider a D4-brane wrapping a four-torus T^4 inside its worldvolume, with a constant magnetic flux F on T^4 carrying integer instanton number k. Using the Wess-Zumino expansion (6.41) with $B = 0$, show that this configuration carries $\mu_4^{(\mathrm{WZ})}$ units of C_5 charge and, from the $C_1 \wedge (2\pi\alpha' F)^2$ term, an induced C_1 charge proportional to k; fix the proportionality constant from your flux-quantisation convention, and identify the induced charge as that of k D0-branes bound to the D4. Explain in one sentence why this makes "a D4 with k units of flux" and "a D4-D0 bound state" two names for the same object.

6.3 D-branes carry Ramond-Ramond charge

In Section 6.2 the Dp-brane appeared as a place: the hyperplane on which open strings are forced to end once T-duality turns a Neumann boundary condition into a Dirichlet one. That is a kinematic statement. It says nothing about whether the hyperplane is a dynamical object, let alone one that gravitates or carries charge. Chapter 5 came at the same object from the opposite side: every Type II supergravity propagates a ladder of Ramond-Ramond potentials C_{p+1}, and a gauge field with no charged source is redundancy. Something must couple to C_{p+1}, and Section 5.6 named it the Dp-brane and quoted, without proof, a tension of order $g_s^{-1}\ell_s^{-(p+1)}$.

Polchinski's 1995 argument welds these two pictures into one. The very same hyperplane that open strings end on is the elementary electric source of C_{p+1}, and its tension and charge can be computed exactly from a single one-loop open-string diagram. The charge comes out non-zero, it saturates the Dirac quantisation condition at the minimal quantum, and it equals the tension: the D-brane is BPS. This section is that computation, and it is where the D-brane stops being a boundary condition and becomes a state.

This section has one computation at its centre, and four things you should be able to do with it:

- read the cylinder in both channels, and say why one loop of open string equals a tree-level closed-string exchange (Section 6.3.1);
- run the Jacobi-identity cancellation of the amplitude and recognise it as the no-force condition between parallel branes (Exercise 6.10);
- collapse the quoted amplitude to the D-brane tension τ_p and check that it lands on the minimal Dirac quantum (Theorem 6.6 and Exercise 6.8);
- state why $\tau_p = \mu_p$ is the BPS bound saturated, and why saturation freezes the tension against the coupling (Section 6.3.5).

This is core material for the week: the tension it fixes is quoted throughout Sections 6.2 and 6.5 and is the input to both the duality web and the black-hole microstate count.

6.3.1 The one-loop vacuum diagram is a closed-string tree

Put two parallel Dp-branes a transverse distance y apart and ask for the force between them. In field theory you would exchange gravitons, dilatons and C_{p+1} quanta and add up the potentials. In string theory the leading interaction is the cylinder: the vacuum diagram of an open string with one end on each brane, swept once around a loop. It has the topology of an annulus, a cylinder with two boundaries, one on each brane.

That single surface admits two readings, and the equality of the two is the whole engine of the argument. Slice the cylinder the short way and you see an open string, born on one brane and ending on the other, running around a loop of Euclidean worldsheet time: this is a *one-loop open-string vacuum amplitude*, and it is weighted by g_s^0 (an annulus has Euler number zero).

Slice it the long way and you see a *closed* string emitted from one brane, propagating across the gap, and absorbed by the other: this is a *tree-level closed-string exchange*. The brane emits a closed string. Whatever closed-string fields the brane couples to, the cylinder sees them all, and it sees them at the order in g_s at which they first appear.

$$\underbrace{\text{one loop of open string}}_{\text{stretched, } g_s^0} \;=\; \underbrace{\text{tree exchange of closed string}}_{\text{graviton, dilaton, } C_{p+1}} \tag{6.42}$$

The open string that lives on the brane knows only about the massless open-string modes, the worldvolume gauge field and transverse scalars of Section 6.2. Yet the closed-string reading of the same amplitude contains gravity and the RR sector. Channel duality forces the brane to be a source for closed strings, and reading off which closed strings, and with what strength, is how we measure its tension and its charge.

6.3.2 The cylinder amplitude and the no-force condition

The open-string vacuum amplitude is a trace over the stretched-string Hilbert space, with the GSO projection inserted. Writing t for the modulus of the annulus (the circumference of the loop in units of its width), the superstring result is

$$\mathcal{A} \;=\; V_{p+1}\int_0^\infty \frac{\mathrm{d}t}{2t}\,(8\pi^2\alpha' t)^{-\frac{p+1}{2}}\,e^{-\,y^2 t/2\pi\alpha'}\;\frac{\vartheta_3(0|it)^4-\vartheta_4(0|it)^4-\vartheta_2(0|it)^4}{\eta(it)^{12}}. \tag{6.43}$$

Read the integrand factor by factor. The worldvolume V_{p+1} is there because the branes are infinite and the amplitude is really an energy per unit worldvolume. The power $(8\pi^2\alpha' t)^{-(p+1)/2}$ is the Gaussian integral over the $p+1$ non-compact momenta the string can carry along the branes. The exponential $e^{-y^2 t/2\pi\alpha'}$ is the classical action of an open string stretched across the gap y: it is the only place the separation enters, and it is what makes the force fall off with distance. The ratio of Jacobi theta functions is the oscillator partition function of the eight transverse worldsheet directions, summed over spin structures with the signs the GSO projection assigns: ϑ_3^4 from the NS sector, $-\vartheta_4^4$ from the NS sector with $(-1)^F$ inserted, and $-\vartheta_2^4$ from the R sector, which counts the spacetime fermions. These $\vartheta_i(z|\tau)$, taken at $z=0$ and $\tau=it$, and the Dedekind eta $\eta(\tau)$ are the modular forms of Section 3.3.5 (written there as $\theta_i(\tau)\equiv\vartheta_i(0|\tau)$); their definitions, the $t\to 1/t$ transformation used in Section 6.3.3, and the identity below are set out there and in Appendix F.

Now use Jacobi's identity, the *aequatio identica satis abstrusa*,

$$\vartheta_3(0|it)^4 \;=\; \vartheta_4(0|it)^4+\vartheta_2(0|it)^4, \qquad \text{so that} \qquad \vartheta_3^4-\vartheta_4^4-\vartheta_2^4 \;\equiv\; 0, \tag{6.44}$$

already met as (3.60), where it made the one-loop vacuum energy of the superstring vanish. The integrand of (6.43) vanishes identically, so $\mathcal{A}=0$. Two parallel Dp-branes feel no force at any separation. This is the string incarnation of the no-force property of Section 5.6: it is why a stack of coincident branes is a solution at all, and it holds because the spacetime bosons the branes exchange (the NS sector, $\vartheta_3^4-\vartheta_4^4$) cancel against the spacetime fermions and, in the closed-string reading, against the RR exchange.

The cancellation is exactly what we want to unpick. A vanishing total is the difference of two non-zero pieces, and separating them measures the two charges of the brane.

6.3.3 Tension and charge from the massless exchange

Before resolving the amplitude, fix the two symbols it will produce. We write τ_p for the physical tension, the coupling-independent DBI prefactor $T_{\mathrm{D}p}$ of Eq. (6.35) carrying its dilaton factor, $\tau_p = T_{\mathrm{D}p}/g_s$, and μ_p for the RR charge measured in the same gravitational frame; the warning below reconciles this μ_p with the coupling-independent charge of Section 6.2.

To resolve the amplitude into closed-string exchanges, pass to the long channel by the modular transformation $t \to 1/t$, which turns the loop of open string into a closed string propagating for proper time $\ell \propto 1/t$. The eta and theta functions go over to their closed-channel form (the $t \to 1/t$ transform sends $\vartheta_2 \leftrightarrow \vartheta_4$ and fixes ϑ_3), and the exponent $-y^2 t/2\pi\alpha'$ becomes $-y^2/(2\pi\alpha'\ell)$, the action of a closed string reaching across the gap. At large separation the integral is dominated by long, thin cylinders, $\ell \to \infty$, where only the massless closed strings survive. Sorting that exchange by closed-string sector, the NS-NS channel (graviton and dilaton) supplies the attraction and the RR channel (a single C_{p+1} quantum) the repulsion; the precise dictionary from the three open-channel spin structures to these sectors is Polchinski's spin-structure bookkeeping, whose result we quote. Keeping the massless poles,

$$\mathcal{A} \xrightarrow{y\to\infty} V_{p+1}\,(\tau_p^2 - \mu_p^2)\,(2\kappa_{10}^2)\,G_{9-p}(y), \qquad G_{9-p}(y) = \frac{\Gamma\!\left(\frac{7-p}{2}\right)}{4\,\pi^{(9-p)/2}}\,\frac{1}{y^{\,7-p}}, \tag{6.45}$$

where G_{9-p} is the massless scalar propagator in the $9-p$ dimensions transverse to the brane, falling off as the Coulomb tail of a source surrounded by an S^{8-p}. Read this against field theory. Two objects of tension τ_p attract by graviton and dilaton exchange, contributing $+\tau_p^2$; two objects of RR charge μ_p repel by C_{p+1} exchange, contributing $-\mu_p^2$; both couplings are normalised by the same $2\kappa_{10}^2$ that sits in front of the RR kinetic term of Eq. (5.43). The string amplitude (6.43) vanishes, so

$$\boxed{\tau_p \;=\; \mu_p} \tag{6.46}$$

The tension equals the RR charge. This is the BPS saturation $M = |Z|$ of Eq. (5.78) written for a brane: gravitational attraction and RR repulsion cancel because tension and charge are one number, and the no-force result of Section 6.3.2 is that cancellation seen in the total amplitude. We return to this reading in Section 6.3.5.

The overall coefficient in (6.45) is not zero, even though the difference in brackets is, and computing the NS-NS piece on its own fixes the magnitude. The result, which we quote from Polchinski's modular bookkeeping rather than reproduce line by line, is

$$\tau_p^2 \;=\; \frac{\pi}{\kappa_{10}^2}\,\left(4\pi^2\alpha'\right)^{3-p}. \tag{6.47}$$

Feeding in $2\kappa_{10}^2 = (2\pi)^7\alpha'^4 g_s^2$ from Eq. (A.6) and $\ell_s = \sqrt{\alpha'}$ collapses this to the clean form

$$\boxed{\tau_p \;=\; \frac{1}{(2\pi)^p\, g_s\, \ell_s^{\,p+1}}} \tag{6.48}$$

Example 6.6 (The tension collapse, and the clean D3 case)**.** The step from (6.47) to (6.48) is pure algebra, worth doing once by hand. Trade κ_{10} for the convention value, so that $\pi/\kappa_{10}^2 = 2\pi/(2\kappa_{10}^2) = 2\pi/\left[(2\pi)^7\alpha'^4 g_s^2\right]$, and write $4\pi^2 = (2\pi)^2$ in the remaining factor:

$$\tau_p^2 \;=\; \frac{2\pi}{(2\pi)^7\alpha'^4 g_s^2}\left[(2\pi)^2\alpha'\right]^{3-p} \;=\; \frac{(2\pi)^{1+2(3-p)}}{(2\pi)^7\, g_s^2}\,\alpha'^{\,(3-p)-4} \;=\; \frac{1}{(2\pi)^{2p}\, g_s^2}\,\alpha'^{-(p+1)}.$$

Taking the square root and using $\ell_s = \sqrt{\alpha'}$ gives $\tau_p = (2\pi)^{-p} g_s^{-1} \ell_s^{-(p+1)}$, the boxed tension. The power of α' is fixed by dimensions alone; the powers of 2π come from the careful normalisation of (6.43). The tidiest case is $p = 3$, where $3 - p = 0$: the α'-dependent factor is 1, so $\tau_3^2 = \pi/\kappa_{10}^2$ outright and $\tau_3 = 1/[(2\pi)^3 g_s \ell_s^4]$. That is the tension behind the dimensionless four-dimensional coupling $g_{\rm YM}^2 = 2\pi g_s$ of Eq. (6.37).

This is the tension that Section 5.6 asked us to take on trust, now derived.

The dimension is right: a $(p+1)$-dimensional worldvolume has an energy per unit volume of mass dimension $p+1$, and $\ell_s^{-(p+1)}$ supplies it. The factor $(2\pi)^{-p}$ is the honest normalisation that makes the Dirac condition below come out with the minimal quantum. And the coupling dependence is the headline: $\tau_p \propto 1/g_s$. A fundamental string has $T_{\rm F1} \sim g_s^0$, because its perturbation theory *defines* g_s; a D-brane is heavier by one inverse power of the coupling. As $g_s \to 0$ it becomes infinitely heavy relative to the string, which is why it is invisible in the perturbative spectrum of Chapter 3: a non-perturbative, solitonic object, a lump of energy that string perturbation theory never produces as a state. The $1/g_s$, sitting between the g_s^0 of the fundamental string and the g_s^{-2} of the NS5-brane in Eq. (5.84), is the fingerprint of a D-brane.

Which "charge" is μ_p

The equality $\tau_p = \mu_p$ in (6.46) holds when both are measured in the same gravitational normalisation, the one in which the graviton and RR kinetic terms share the prefactor $1/2\kappa_{10}^2$. In that convention $\mu_p = \tau_p \sim 1/g_s$. If instead you read the charge off as the coefficient of $\int_{\mathcal{W}} C_{p+1}$ in the string-frame worldvolume action of Section 6.2, you get $\mu_p^{(\rm WZ)} = g_s\, \tau_p = (2\pi)^{-p} \ell_s^{-(p+1)} = T_{\rm Dp}$, the coupling-independent prefactor of Eq. (6.35) and the quantity written μ_p in Section 6.2. The two differ by the g_s hiding in the string-frame RR field, which unlike the NS-NS fields carries no $e^{-2\phi}$. This is pure bookkeeping, but it is the single most common place to drop a factor of g_s, so fix a frame and keep it.

6.3.4 Dirac quantisation of Ramond-Ramond charge

A Dp-brane sources $F_{p+2} = {\rm d}C_{p+1}$. Its electric charge is the flux through the sphere S^{8-p} that links the brane in the transverse space,

$$\mu_p \;\propto\; \int_{S^{8-p}} \star F_{p+2}, \tag{6.49}$$

conserved by the RR equation of motion. The magnetic dual is the object that the same field strength encircles: a brane measured by $\int_{S^{p+2}} F_{p+2}$, whose worldvolume is $7-p$ dimensional. That is a D$(6-p)$-brane, coupling electrically to the dual potential C_{7-p} with $F_{8-p} = \star F_{p+2}$. The RR ladder pairs up electrically and magnetically as

$$\text{D0-D6}, \qquad \text{D1-D5}, \qquad \text{D2-D4}, \qquad \text{D3-D3}, \tag{6.50}$$

each pair summing to $p + (6-p) = 6$, and each pair sitting inside a single Type II theory (p even in IIA, odd in IIB). The D3-brane is its own magnetic dual, the geometric face of the self-dual five-form of Eq. (5.50).

An electric and a magnetic source of the same gauge field cannot carry arbitrary charges. Wrapping the electric brane's Wilson-line phase around the magnetic brane's flux must be single-valued, and for p-form charges in ten dimensions this is the Dirac-Nepomechie-Teitelboim condition. In the normalisation of the RR action it reads

$$\boxed{2\kappa_{10}^2\, \mu_p\, \mu_{6-p} \;=\; 2\pi n, \qquad n \in \mathbb{Z}.} \tag{6.51}$$

The tension (6.48) that Polchinski computed is exactly the value that puts a single D-brane at the minimal quantum $n = 1$: substituting $\mu_p = \tau_p$ gives $2\kappa_{10}^2\,\tau_p\,\tau_{6-p} = 2\pi$ on the nose (Exercise 6.8). A separate one-loop computation and an integrality condition that know nothing about each other agree perfectly, and that agreement is one of the sharpest internal consistency checks in the subject. Had the D-brane charge come out irrational, or a non-integer multiple of the Dirac unit, the theory would have been inconsistent; instead it lands exactly on $n = 1$.

Remark 6.7 (Beyond the lectures: the boundary state). The closed-string reading of the cylinder can be made precise with a *boundary state* $|B_p\rangle$, a particular coherent state in the closed-string Hilbert space that encodes the Dirichlet and Neumann conditions of the brane as gluing conditions on the oscillators, $(\alpha_n + \tilde{\alpha}_{-n})\,|B_p\rangle = 0$ along the brane and $(\alpha_n - \tilde{\alpha}_{-n})\,|B_p\rangle = 0$ transverse to it, together with their fermionic partners. The cylinder is then literally the closed-string propagator sandwiched between two boundary states, $\mathcal{A} = \langle B_p|\ \Delta\ |B_p\rangle$ with $\Delta = (L_0 + \tilde{L}_0)^{-1}$. The boundary state has an NS-NS part and, crucially, an R-R part; the RR component is where the charge μ_p lives, and its very existence, that a Dirichlet boundary can carry an R-R quantum number, is the technical statement of "D-branes carry RR charge". The $p = -1$ member, the D-instanton, is a boundary state localised in Euclidean time as well as space; it sources contributions of order $e^{-2\pi/g_s}$ to amplitudes, weighted by the D-instanton action $S = 2\pi/g_s$, the non-perturbative effects flagged in Chapter 4.

6.3.5 D-branes are BPS

Collect what the cylinder has told us. The Dp-brane sources the graviton, the dilaton and C_{p+1}; its tension is $\tau_p = (2\pi)^{-p} g_s^{-1} \ell_s^{-(p+1)}$; its RR charge equals that tension; and it carries the minimal Dirac quantum. Equality of tension and charge, $\tau_p = \mu_p$, is the tension-density form of the BPS bound $M = |Z|$ from Eq. (5.78). The D-brane saturates the bound $\tau \geq |q|$, and a saturating object is annihilated by half of the supercharges. A single flat Dp-brane is $\frac{1}{2}$-BPS, preserving the sixteen supersymmetries selected by the projector $\frac{1}{2}(1 + \Gamma_{\text{brane}})$ of Eq. (C.42), whose R-R phase is exactly the datum the boundary state carries.

This closes the loop the chapter opened. T-duality produced the D-brane as a hyperplane; the cylinder shows the hyperplane is a BPS state carrying protected RR charge; and BPS means the tension (6.48), computed here at weak coupling from a one-loop diagram, is exact. It does not receive corrections as g_s grows, for the reasons laid out in Section 5.6.4: the charge is quantised by (6.51) and cannot drift, and the short multiplet cannot lengthen. That is what licenses using D-branes at strong coupling, where the duality web of Chapter 7 and the microstate counts of Chapter 12 both live. The same equality $\tau_p = \mu_p$ returns as the extremality condition of the supergravity p-brane solution sourced by a harmonic function: the harmonic-function extremal branes are collected in Appendix I, and the near-horizon geometry that a stack of D3-branes grows into at large $g_s N$ is the starting point of Chapter 13.

Exercise 6.8 (The tension is the minimal Dirac quantum). *(Short.)* Using $\tau_p = (2\pi)^{-p} g_s^{-1} \ell_s^{-(p+1)}$ with $\ell_s = \sqrt{\alpha'}$, and $2\kappa_{10}^2 = (2\pi)^7 \alpha'^4 g_s^2$, verify that

$$2\kappa_{10}^2\,\tau_p\,\tau_{6-p} \;=\; 2\pi,$$

independent of p and of g_s. Conclude that a single D-brane realises the $n = 1$ case of (6.51). Which value of p gives an electrically and magnetically self-dual brane, and why is that the one that could have obstructed writing a covariant action in Section 5.4?

Exercise 6.9 (T-duality moves the tension consistently). *(Moderate; uses Sections 6.1 and 6.2.)* T-duality along a circle of radius R that a brane is transverse to turns a Dp into a D$(p{+}1)$ wrapping the dual circle of radius $R' = \alpha'/R$, and shifts the coupling as $g_s \to g_s' = g_s\,\ell_s/R$.

Show that the effective $(p+1)$-dimensional tension of the wrapped D$(p{+}1)$,

$$(2\pi R')\,\tau_{p+1}(g_s', R') \;=\; \tau_p(g_s),$$

reproduces the original Dp tension exactly. Interpret the factor $2\pi R'$, and explain in one sentence why this had to work if T-duality is a symmetry rather than an approximation.

Exercise 6.10 (Why the fundamental string is different)**.** *(Short, conceptual.)* Two parallel fundamental strings do feel a net static force, whereas two parallel D-branes do not. In terms of the sectors summed in (6.43), identify which exchange the D-brane has that a neutral object lacks, and state which physical charge the cancellation (6.44) balances against gravity plus the dilaton. What would go wrong with the no-force condition if μ_p were even slightly different from τ_p?

6.4 Orientifolds and O-planes

The previous sections took an oriented string and let T-duality act on it. Neumann boundary conditions turned Dirichlet, and a place where a string is merely allowed to end became a dynamical object that gravitates and carries Ramond-Ramond charge, a D-brane (Sections 6.2 and 6.3). This section gauges a discrete symmetry that the oriented string has been carrying all along: the worldsheet parity Ω that reverses the string's orientation. The theory that survives the projection is *unoriented*, and it contains a new kind of extended object living at the fixed loci of the projection, an *orientifold plane*. An O-plane is not a brane. Nothing lives on it, it has no fluctuations, and no open string ends on it. It nonetheless gravitates and it sources a Ramond-Ramond field, and both its tension and its charge are *negative*.

That negative charge is the point. On a compact space the flux sourced by all the branes has nowhere to escape, so the total Ramond-Ramond charge is forced to vanish (Gauss's law with no boundary). Positive-charge D-branes cannot cancel among themselves; something with the opposite sign is needed, and the O-plane supplies it. The resulting equality, "total Ramond-Ramond charge $= 0$", is the *tadpole condition*. It is the single most important consistency requirement on the compactifications of Chapter 9, Chapter 10 and Chapter 11, and Chapter 3 already leaned on its ten-dimensional instance to assert that Type I has gauge group SO(32). Here we build the machinery and pay off that promise. The T-duality and D-brane spine of Sections 6.1 to 6.3 stands on its own; this section and the BPS thread of Section 6.5 are the chapter's second pass, and the tadpole condition assembled here is a genuine new constraint, not bookkeeping.

What you should be able to do. By the end of this section you should be able to:

- gauge worldsheet parity Ω: say how it acts on the closed-string fields and on the Chan-Paton factors (6.53), and read off SO(N) versus USp(N) (Exercise 6.13);
- locate an Op-plane as the fixed locus of $\Omega\,\mathcal{R}$, and count the 2^{9-p} planes a T-duality on T^{9-p} produces (Exercise 6.11);
- quote the O-plane tension and charge $\tau_{\mathrm{O}p^{\mp}} = \mp\,2^{\,p-4}\tau_p$, and explain why a non-dynamical object may carry negative and even fractional Ramond-Ramond charge;
- state the Ramond-Ramond tadpole condition as a Gauss law and obtain it from the C_{10} equation of motion (Exercise 6.12);
- run the Type I count $\mathrm{O9}^- + 32\,\mathrm{D9} = \mathrm{SO}(32)$ and say why the vacuum, not a modelling choice, fixes $N = 32$.

6.4.1 Gauging worldsheet parity

Worldsheet parity was defined in Theorem 3.15. On the closed string it is the reflection

$$\Omega : \quad \sigma \longmapsto 2\pi - \sigma, \tag{6.52}$$

which reverses the arrow on the worldsheet and so exchanges left- and right-moving oscillators, $\alpha_n^\mu \leftrightarrow \tilde{\alpha}_n^\mu$ and $\psi_r^\mu \leftrightarrow \tilde{\psi}_r^\mu$ (up to a fixed sign in each sector), with $\Omega^2 = 1$. It is a genuine symmetry only when the two movers are built identically, which in the superstring singles out the left-right symmetric Type IIB (Section 3.4). A symmetry with $\Omega^2 = 1$ generates a $\mathbb{Z}_2$, and *gauging* that $\mathbb{Z}_2$ means two things at once: keep only the Ω-even states in the spectrum, and sum over *all* worldsheet topologies, including the non-orientable ones (the Klein bottle, the Möbius strip, the crosscap $\mathbb{RP}^2$) that an oriented string theory would never have included.

Keeping the even states is the easy half. On the closed string it removes the Kalb-Ramond field $B_{\mu\nu}$, keeps the graviton and dilaton, and in the Ramond-Ramond sector keeps exactly those potentials C_p that are Ω-even; for Type IIB the survivor is C_2, which is why Type I carries a Ramond-Ramond two-form and not the B-field (Eq. (3.76)). On the open string Ω swaps the two endpoints, so it acts on the Chan-Paton matrix λ as a transpose,

$$\Omega : \quad \lambda \longmapsto -\gamma\, \lambda^{\mathsf{T}} \gamma^{-1}, \tag{6.53}$$

for a fixed matrix γ obeying $\gamma\gamma^{\mathsf{T}} = \pm\mathbf{1}$. Projecting onto Ω-invariant states keeps the matrices with $\gamma\lambda^{\mathsf{T}}\gamma^{-1} = -\lambda$, and the $\mathrm{U}(N)$ of the oriented string (Section 1.5.4) collapses to a real subgroup: to $\mathrm{SO}(N)$ when γ is symmetric and to $\mathrm{USp}(N)$ when γ is antisymmetric. Which of the two occurs is fixed by the sign of the crosscap contribution to the vacuum amplitude, the same sign that will set the orientifold charge below.

The summing-over-topologies half is where the physics of this section hides. The unoriented surfaces are what force the O-planes into the theory and what generate the tadpole. Before computing anything it helps to see *where* an O-plane sits, and for that the cleanest route is T-duality.

6.4.2 O-planes are fixed planes of a parity

Gauging Ω on ten-dimensional Type IIB fills all of spacetime with the orientifold: the projection acts trivially on the coordinates X^μ, so the "fixed locus" is everything. The object that fills spacetime is an orientifold nine-plane, an O9-plane. To see a lower-dimensional O-plane, and to see that it really is localised, combine Ω with T-duality.

Recall from Section 6.1 that T-duality along X^9 acts on the mode decomposition as $X^9 = X^9_{\mathrm{L}} + X^9_{\mathrm{R}} \mapsto X'^9 = X^9_{\mathrm{L}} - X^9_{\mathrm{R}}$: it flips the sign of the right-movers. Worldsheet parity exchanges left and right, $X^9_{\mathrm{L}} \leftrightarrow X^9_{\mathrm{R}}$. Compose the two and act on the dual coordinate,

$$\Omega : \; X'^9 = X^9_{\mathrm{L}} - X^9_{\mathrm{R}} \longmapsto X^9_{\mathrm{R}} - X^9_{\mathrm{L}} = -X'^9. \tag{6.54}$$

In the T-dual frame the pure parity Ω has become $\Omega\,\mathcal{R}_9$, worldsheet parity *times* a reflection $\mathcal{R}_9 : X'^9 \mapsto -X'^9$ of the dualised coordinate. This is the general rule: T-duality on a set of directions turns the orientifold generator Ω into $\Omega\,\mathcal{R}$, where $\mathcal{R}$ reflects exactly those directions.

Now the O-plane is visible. On the dual circle $X'^9 \sim X'^9 + 2\pi R'$ the reflection $\mathcal{R}_9$ has two fixed points, $X'^9 = 0$ and $X'^9 = \pi R'$, and the quotient is the line interval $[0, \pi R']$. The parity is a symmetry of the physics everywhere, but the two fixed points are special: they are the places left invariant by the orientifold action, and an *orientifold eight-plane*, an O8-plane, sits at each. Type IIA on this interval is the theory called Type I′ (or Type IA). The general statement follows by iterating: an orientifold that combines Ω with a reflection of $9-p$ coordinates has Op-plane

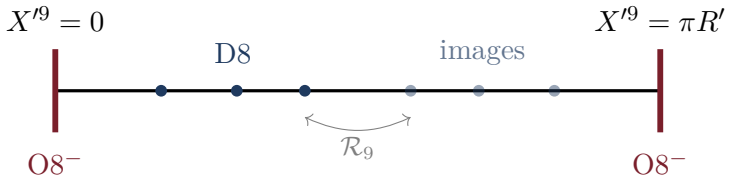


Figure 6.1: Type I′: Type IIA on the interval $[0, \pi R'] = S^1/\mathbb{Z}_2$, the T-dual of Type I on a circle. Each end is an O8-plane (the fixed points of the reflection $\mathcal{R}_9$). The 32 D9-branes of Type I become D8-branes on the covering circle; the reflection pairs each D8-brane with a mirror image, so 16 of them move independently and their positions are the Wilson lines of the original SO(32).

fixed loci wherever the reflection has a fixed point. On a torus T^{9-p} built from those directions the reflection $\vec{y} \mapsto -\vec{y}$ has 2^{9-p} fixed points, so the orientifold of Section 6.4.1, T-dualised on T^{9-p}, produces 2^{9-p} Op-planes.

Definition 6.8 (Orientifold and O-plane)**.** An *orientifold* is the quotient of an oriented string theory by a $\mathbb{Z}_2$ generated by $\Omega\,\sigma$, where Ω is worldsheet parity and σ is an involution of spacetime (a reflection of some coordinates, or more generally an isometric, holomorphic involution of the internal manifold in Chapter 8). Its *orientifold p-plane* is the fixed-point locus of σ, a $(p+1)$-dimensional hypersurface. The O-plane is not a dynamical object: it carries no worldvolume fields, has no moduli, and no open string ends on it. It is the geometric shadow of a gauge identification.

The contrast with a D-brane is what keeps the negative tension consistent. A D-brane is dynamical: it can move (its transverse position is a scalar field on its worldvolume), it carries a gauge field, and open strings end on it. An O-plane has none of that. It is the set of points the orientifold projection holds fixed, a special locus of spacetime in the same sense that the fixed point of a $\mathbb{Z}_2$ orbifold is, not a particle sitting there. With no localised degrees of freedom on the O-plane, there is nothing whose kinetic term could acquire the wrong sign, and it can carry negative tension without producing a ghost or an instability.

6.4.3 Tension and charge of an O-plane

An O-plane parallel to a stack of Dp-branes preserves the same sixteen supercharges the branes do. This is the T-dual statement of the fact that gauging Ω cut the thirty-two supersymmetries of Type IIB to the sixteen of Type I (Section 3.5), and it makes the Op-plane a BPS object in the sense of Section 5.6: its tension is locked to its charge. The lock works exactly as it did for the D-brane, with one sign flipped. Writing μ_p for the Ramond-Ramond charge of a single Dp-brane and $\tau_p = \mu_p$ for its tension in the BPS units of Section 6.3, the Op-plane carries

$$Q_{\mathrm{O}p^{\mp}} \;=\; \mp 2^{p-4}\,\mu_p, \qquad \tau_{\mathrm{O}p^{\mp}} \;=\; \mp 2^{p-4}\,\tau_p, \tag{6.55}$$

so that $|\tau| = |Q|$ holds for the O-plane just as for the brane. The two signs label the two kinds of O-plane. The Op^--plane has *negative* tension and charge, and this is the one the ordinary Ω projection produces; it forces the antisymmetric Chan-Paton condition in Eq. (6.53) and hence an SO gauge group. The Op^+-plane has the opposite sign and gives a USp group.

The coefficient 2^{p-4} is fixed by a one-loop string computation, sketched in Section 6.4.5; a quick self-consistency check makes it plausible already. T-duality preserves total Ramond-Ramond charge, and it turns the single O9$^-$ into 2^{9-p} Op^--planes on T^{9-p}. If the O9$^-$ carries $-32\,\mu_9$ then each Op^- must carry $-32\,\mu_p/2^{9-p} = -2^{p-4}\mu_p$, which is Eq. (6.55).

Example 6.9 (O7-planes on a two-torus)**.** Take the orientifold of Section 6.4.1 T-dualised on a two-torus, so $k = 9 - p = 2$ and $p = 7$. The reflection $(y^8, y^9) \mapsto (-y^8, -y^9)$ has $2^2 = 4$

fixed points, at $(0,0)$, $(0,\pi R')$, $(\pi R',0)$ and $(\pi R',\pi R')$, so the quotient carries four O7$^-$-planes. T-duality preserves the total Ramond-Ramond charge, $-32\,\mu_9$ on the O9$^-$, and distributes it equally: each O7$^-$ carries $-32\,\mu_7/4 = -8\,\mu_7 = -2^{7-4}\mu_7$, which is Eq. (6.55) at $p=7$. The 32 D9-branes become 32 D7-branes on the covering torus (sixteen mobile branes and their images), and the total charge cancels, $32\,\mu_7 - 4\times 8\,\mu_7 = 0$. Exercise 6.11 runs the same count for general p.

Two conventions for the O-plane charge, differing by a factor of two

The literature usually quotes $Q_{\mathrm{O}p^-} = -2^{p-5}\mu_p$ and, famously, $Q_{\mathrm{O3}^-} = -\frac{1}{4}\,\mu_3$. That is Eq. (6.55) written in a different unit. There, a "physical" Dp-brane means a brane *together with* its orientifold mirror image, a pair carrying $2\mu_p$ on the covering space, and the O-plane charge is measured against that pair. Our $-2^{p-4}\mu_p$ is $-2^{p-5}\times(2\mu_p)$, so the two agree; only the counting unit differs, and O3$^-$ reads $-\frac{1}{4}$ of a brane-plus-image. For the space-filling O9 there is no transverse space and hence no image, so the two units coincide: there are genuinely 32 D9-branes and $Q_{\mathrm{O9}^-} = -32\,\mu_9$. We use the covering-space unit Eq. (6.55) throughout so that the O9 number comes out right without bookkeeping.

The charge in Eq. (6.55) can be fractional: in our units $Q_{\mathrm{O3}^-} = -\frac{1}{2}\mu_3$. A dynamical object with fractional Ramond-Ramond charge would violate the Dirac quantisation of Section 6.3, and the O-plane escapes only because it is not dynamical. Dirac's argument constrains the charges that can be transported around a dual charge on a closed loop, and a fixed locus cannot be moved.

The balance with a brane is a genuine equilibrium. An Op^--plane and a parallel Dp-brane exert no net force on each other. The gravitational and dilaton exchange, proportional to the product $\tau_{\mathrm{O}p}\,\tau_p < 0$, is repulsive; the Ramond-Ramond exchange, proportional to $Q_{\mathrm{O}p}\,\mu_p < 0$, is attractive; and the two cancel to the same order they cancel between two D-branes, so an Op^--plane and a parallel Dp-brane sit in static BPS balance.

6.4.4 The Gauss law: the tadpole condition

Why must the charges cancel at all? The cleanest way to see it uses the top-form potential. A D9-brane and an O9-plane both couple electrically to the Ramond-Ramond ten-form C_{10}, through $\mu_9\int C_{10}$ and the corresponding O-plane coupling. In ten dimensions C_{10} is a peculiar field: its field strength $F_{11} = \mathrm{d}C_{10}$ is an eleven-form and vanishes identically, so C_{10} has no propagating degree of freedom. It is a Lagrange multiplier. Its equation of motion is obtained by varying the total action with respect to it, and since it appears only linearly, multiplying the sum of all the charge densities that source it, that equation is

$$\frac{\delta S}{\delta C_{10}} = 0 \quad\Longrightarrow\quad \sum_{\mathrm{D9}} \mu_9 \;+\; Q_{\mathrm{O9}} \;=\; 0. \tag{6.56}$$

There is no field to carry away any excess; the constraint is that the net source vanish. This is Gauss's law in its starkest form. For a lower Op-plane the same logic runs on the compact space transverse to the branes: the flux of $\star F_{p+2}$ through a surface surrounding all the sources equals the enclosed charge, and on a compact transverse manifold that surface can be deformed to nothing, so the enclosed charge must be zero,

$$\sum_a Q_{\mathrm{D}p,a} \;+\; \sum_i Q_{\mathrm{O}p,i} \;=\; 0 \qquad \text{(RR tadpole condition)}. \tag{6.57}$$

Read from the worldsheet, an uncancelled charge is a *tadpole*: a nonzero one-point function of the massless C_{p+1} on the disk and the crosscap at zero momentum. Such a one-point function is an amplitude for the vacuum to emit a single quantum of a massless field into empty space, and

it signals that the configuration one expanded around does not solve the equations of motion. In the low-energy language it is exactly the failure of Eq. (6.56): the source term for C_{p+1} has no solution unless it integrates to zero. The divergence it produces (the massless closed string propagates to zero momentum, where its propagator $1/k^2$ blows up) is the string-theoretic symptom, first diagnosed by Fischler and Susskind, and cancelling it is not optional. A string background with an uncancelled Ramond-Ramond tadpole is inconsistent.

Two refinements matter for the later chapters and are established in the sections that own them, so we only flag them here. The Wess-Zumino couplings of Section 6.3, of the schematic form $\int C \wedge e^{B+F} \wedge \sqrt{\hat{A}(R)}$, mean that a curved or flux-carrying brane also carries *induced* charge under the lower Ramond-Ramond potentials, and O-planes carry an analogous curvature coupling built from the Hirzebruch $\hat{L}$-genus rather than $\hat{A}$. So the honest tadpole budget of Chapter 9 and Chapter 10 sums induced D-brane charges, O-plane charges, and background H- and F-fluxes, all against the curvature of the compactification manifold. The general form Eq. (6.57), with each term read off from the appropriate Wess-Zumino coupling, is the constraint those chapters impose on every model.

6.4.5 Type I: $\mathrm{O9} + 32\,\mathbf{D9} = \mathrm{SO}(32)$

Everything comes together in the theory that started the story. Gauging Ω on Type IIB fills spacetime with a single $\mathrm{O9}^-$-plane. By Eq. (6.55) it carries

$$Q_{\mathrm{O9}^-} \;=\; -\,2^{\,9-4}\,\mu_9 \;=\; -\,32\,\mu_9, \tag{6.58}$$

and it is space-filling, so its charge cannot leak out of any boundary. The tadpole condition Eq. (6.56) then demands $+32\,\mu_9$ from somewhere, and the only objects that carry C_{10} charge are D9-branes. Consistency forces exactly

$$N \;=\; 32 \qquad \text{space-filling D9-branes.} \tag{6.59}$$

Not fewer (the tadpole would not cancel) and not more (likewise). The rank is fixed by the vacuum, not chosen as it was for the free $\mathrm{U}(N)$ of an oriented D-brane stack in Chapter 1. The open strings on those 32 branes carry Chan-Paton labels $i, j \in \{1, \ldots, 32\}$, the Ω projection Eq. (6.53) with the $\mathrm{O9}^-$ sign keeps the antisymmetric matrices, and 32 labels with an antisymmetric reality condition fill the adjoint of $\mathrm{SO}(32)$. This is the ten-dimensional $\mathcal{N} = 1$ vector multiplet of Type I, and it reproduces exactly what Section 3.5 quoted.

The number 32 and the sign that selects SO over USp both come from the one-loop vacuum amplitude, and it is worth seeing the structure even though the full theta-function bookkeeping belongs to Polchinski's text. At one loop the unoriented theory has three diagrams beyond the oriented torus: the cylinder, with two boundaries; the Möbius strip, with one boundary and one crosscap; and the Klein bottle, with two crosscaps. Each has a limit in which the surface degenerates into a long thin tube, and in that limit the amplitude factorises into a closed string emitted from one end and absorbed at the other. A boundary emits with amplitude proportional to the number of D-branes it carries, n; a crosscap emits with a fixed amplitude c. Summing the three diagrams organises the emission into a perfect square,

$$\mathcal{A}_{\mathrm{tube}} \;\sim\; \big(\, n\,|\mathrm{D9}\rangle + |\mathrm{C}\rangle \,\big)^2 \;\longrightarrow\; (n\,\mu_9 + Q_{\mathrm{O9}})^2 \;=\; (n-32)^2\,\mu_9^2 \tag{6.60}$$

in the massless Ramond-Ramond channel, where $|\mathrm{D9}\rangle$ and $|\mathrm{C}\rangle$ are the boundary and crosscap states and the crosscap-to-boundary ratio in this channel evaluates to $32 = 2^5$. The tadpole is the coefficient of the divergence, the square itself, and it vanishes only at $n = 32$. The corresponding NS-NS square is the dilaton and graviton tadpole; it cancels at the same $n = 32$, which is why Type I is supersymmetric rather than merely charge-neutral. The sign inside the Möbius cross-term, set by the choice $\mathrm{O9}^-$ rather than $\mathrm{O9}^+$, is the sign that lands on the antisymmetric Chan-Paton projection and hence on $\mathrm{SO}(32)$.

Remark 6.10 (The SO(32) that is really $\mathrm{Spin}(32)/\mathbb{Z}_2$). The tadpole fixes the algebra $\mathfrak{so}(32)$ and the antisymmetric projection, but not the global form of the group. The perturbative open-string spectrum realises tensors of SO(32), yet the full non-perturbative Type I string also contains states in the spinor conjugacy class, carried by a D-particle bound state, and their presence promotes the group to $\mathrm{Spin}(32)/\mathbb{Z}_2$, exactly the lattice statement made for the heterotic string in Section 3.5. The two theories are S-dual (Section 5.5, Chapter 7), so they had better share a gauge group down to its global form, and they do.

6.4.6 Type I′ and the geometry of the gauge group

T-dualising Type I on a circle turns the picture of Fig. 6.1 into a laboratory for the tadpole condition. The 32 D9-branes become 32 D8-branes on the covering circle, and $\Omega\mathcal{R}_9$ pairs a D8-brane at position x with a mirror at $-x$. Generic positions give 16 D8-branes free to move, each dragging an image, so the gauge symmetry is broken to $\mathrm{U}(1)^{16}$, the Cartan of SO(32); the D8-brane positions are precisely the sixteen Wilson lines of the ten-dimensional SO(32) wrapped on the circle. Move k of the D8-branes (and their images) onto one O8-plane and the coincident stack enhances the gauge group there to $\mathrm{SO}(2k)$; split branes off and it breaks again. The compactified gauge group is whatever the D8-brane positions allow, and the tadpole condition Eq. (6.57) keeps the total D8 charge fixed at $-\sum_i Q_{\mathrm{O8},i} = +32\,\mu_8$ throughout: two O8$^-$-planes at $-2^{8-4}\mu_8 = -16\,\mu_8$ each. This is the "16 D9-branes and their images plus an O9" counting of the chapter overview, made geometric: the 16 movable branes are the physical degrees of freedom, the 32 Chan-Paton labels and the -32 of the O9 are the covering-space bookkeeping, and the two descriptions are the same theory viewed through the circle.

Remark 6.11 (Beyond the lectures: the zoo of O-planes). The two signs in Eq. (6.55) are not the whole story. Two $\mathbb{Z}_2$ discrete-torsion labels, for the B-field and the Ramond-Ramond potential, together with the option of a stuck half-D-brane, refine each O$p^{\pm}$ into a $\widetilde{\mathrm{O}p}^{\pm}$ variant. For $p = 3$ this produces four O3-planes, $\mathrm{O3}^-, \widetilde{\mathrm{O3}}^-, \mathrm{O3}^+, \widetilde{\mathrm{O3}}^+$, with charges $-\frac{1}{4}, +\frac{1}{4}, +\frac{1}{4}, +\frac{1}{4}$ (physical units) and gauge groups $\mathrm{SO}(2N), \mathrm{SO}(2N{+}1), \mathrm{USp}(2N), \mathrm{USp}(2N)$ respectively; only the O3$^-$ carries negative charge. The $\mathrm{SL}(2,\mathbb{Z})$ duality of Chapter 7 fixes O3$^-$ and $\widetilde{\mathrm{O3}}^+$ and exchanges O3$^+$ with $\widetilde{\mathrm{O3}}^-$, so the charge assignment is duality-consistent. These four are the microscopic reason the $\mathcal{N} = 4$ super-Yang-Mills theories with gauge groups $\mathrm{SO}(2N), \mathrm{SO}(2N+1), \mathrm{USp}(2N)$ arise on D3-branes at orientifolds; the full classification is a statement about $\mathbb{Z}_2$-valued fluxes (B-field and Ramond-Ramond torsion) and is cleanest in K-theory. The USp(32) theory built from an O9$^+$-plane and 32 anti-D9-branes (Sugimoto's model) cancels the Ramond-Ramond tadpole Eq. (6.56) but not the NS-NS one, leaving a runaway dilaton potential and broken supersymmetry: Ramond-Ramond tadpole cancellation is necessary for consistency but does not by itself guarantee a stable vacuum.

Exercise 6.11 (The O-plane charge from T-duality)**.** Take the O9$^-$ charge Eq. (6.58) as given and T-dualise on a k-torus T^k, $k = 9 - p$. (a) Argue that the reflection $\vec{y} \mapsto -\vec{y}$ on T^k has 2^k fixed points, so the orientifold has 2^k Op-planes. (b) Using that T-duality preserves the total Ramond-Ramond charge and that the 32 D9-branes become 32 Dp-branes on the covering torus, show that each Op^--plane must carry $-2^{p-4}\mu_p$, reproducing Eq. (6.55). (c) Convert to physical (brane-plus-image) units and recover the textbook -2^{p-5} and the value $-\frac{1}{4}$ for the O3$^-$-plane. *(Short.)*

Exercise 6.12 (The ten-form is a Gauss constraint)**.** The Ramond-Ramond ten-form C_{10} couples to D9-branes and the O9-plane but has no kinetic term, since its field strength $\mathrm{d}C_{10}$ is an eleven-form in ten dimensions. Write the part of the action linear in C_{10}, $S \supset \int C_{10} \wedge \rho$, with ρ the total (D9 plus O9) charge density, treated as a spacetime-filling zero-form times the volume form. Vary with respect to C_{10} and show that its equation of motion is $\rho = 0$, that is, Eq. (6.56). Explain in one line why the same argument for a propagating field such as C_2 gives

a differential equation with sources rather than a pure constraint. *(Short.)*

Exercise 6.13 (SO versus USp from the Möbius sign)**.** The Chan-Paton projection Eq. (6.53) keeps antisymmetric λ for symmetric γ (giving SO(N)) and symmetric λ for antisymmetric γ (giving USp(N)). Show that consistency of $\Omega^2 = 1$ acting on the open-string oscillator ground state forces $\gamma\gamma^{\mathsf{T}} = \pm\mathbf{1}$, and that the two signs are the two O-plane types O$p^{\mp}$. Count the dimensions: SO(32) has 496 generators while USp(32) has 528, so the size of the gauge group does not by itself pick one out. Argue instead from the sign of the Möbius cross-term in Eq. (6.60), equivalently from cancelling the NS-NS dilaton and graviton tadpole and not merely the Ramond-Ramond one, that the supersymmetric ten-dimensional Type I string selects the O9$^-$ projection and hence SO(32); the O9$^+$ choice gives USp(32) with supersymmetry broken, the Sugimoto model of Theorem 6.11. *(Moderate.)*

Orientifold planes complete the cast of BPS objects the chapter set out to assemble: the fundamental string, the D-branes, and now the negative-charge planes that let a compact configuration balance its books. What they buy is leverage. Every intersecting-brane and magnetised-brane model of Chapter 10, every flux vacuum of Chapter 9, and the F-theory constructions of Chapter 11 live or die by the tadpole condition Eq. (6.57), with the O-plane charge on one side of the ledger and the branes and fluxes on the other. The next section threads the BPS branes of this chapter onto the central-charge spine of Section 5.6 and reads off the spectrum that survives to strong coupling.

6.5 The BPS brane spectrum and the BPS thread

The big picture

This chapter built the Dp-branes twice: as the places where open strings end once T-duality turns a Neumann direction Dirichlet (Section 6.2), and as dynamical carriers of Ramond-Ramond charge with a tension read off a cylinder diagram (Section 6.3). Step back now and set them beside the two objects that were in the theory before any D-brane existed: the fundamental string F1, the electric source of the NS-NS two-form, and its magnetic dual the NS5-brane. These three families, F1, Dp and NS5, are the entire BPS brane spectrum of the Type II string, and their tensions scale with the string coupling as g_s^0, g_s^{-1} and g_s^{-2}, one representative from each power. Supersymmetry ties each tension to a conserved charge and freezes the relation against changes in g_s. That freezing, the BPS thread, is what makes the strong-coupling dualities of Chapter 7 testable and the black-hole microstate count of Chapter 12 possible.

This closing section gathers the chapter's threads rather than opening a new computation. On a single-week pass it can be read after the core material of Sections 6.1 to 6.3, and it is where the payoff for the next two chapters is assembled.

Learning objectives. By the end of this section you should be able to:

- place F1, Dp and NS5 in the BPS spectrum and read off the g_s^0, g_s^{-1}, g_s^{-2} scaling of their tensions from the leading worldsheet topology (Table 6.2);
- show that a wound fundamental string saturates the BPS bound, and that T-duality exchanges it with a pure momentum state (Exercise 6.14);
- distinguish a marginal (threshold) bound state from a non-threshold one, and derive the (p, q)-string tension from central charges that add in quadrature (Exercise 6.15);

- read the induced RR charges off the Wess-Zumino coupling, for instance the k D0 charges carried by a D4 of instanton number k (Exercise 6.16);
- state why BPS protection lets a weak-coupling count survive to strong coupling, and how that underwrites both the duality web and the black-hole microstate count.

The BPS bound itself, the collapse of a saturating state into a short multiplet, and the argument that shortness is protected against continuous deformation were derived in Section 5.6 and, in more detail, in Appendix C. We do not repeat those derivations. What this section adds is the assembly of the spectrum with the D-branes now constructed rather than merely postulated, a worldsheet explanation for why the three powers of g_s come out as they do, and the thread run forward to the two chapters that rest on it.

6.5.1 The spectrum, object by object

A $(p+1)$-form potential is sourced electrically by a p-brane, whose $(p+1)$-dimensional worldvolume it can be integrated over, and magnetically by the object charged under the dual potential. In ten dimensions the dual of a $(p+1)$-form is a $(7-p)$-form, so the magnetic partner of an electric p-brane is a $(6-p)$-brane. Running this through the gauge fields catalogued in Chapter 5 gives three sectors.

F1: the string is its own BPS state. The fundamental string is one we have already quantised (Chapters 1 and 3); what is new is reading it as the electric source of the NS-NS field B_2. The coupling is the term $\int_\Sigma B_2$ already sitting in the worldsheet action, and the tension is the one we started with,

$$T_{\mathrm{F1}} \;=\; \frac{1}{2\pi\alpha'} \;=\; \frac{1}{2\pi\ell_s^2} \;\sim\; g_s^0. \tag{6.61}$$

No power of the coupling appears, because the fundamental string is the object whose loop expansion *defines* g_s.

That the string is BPS can be seen without leaving the free spectrum. Put the Type II superstring on a circle of radius R in the direction X, as in Section 6.1, with integer momentum n and winding w. The left- and right-moving momenta are

$$p_{\mathrm{L}} = \frac{n}{R} + \frac{wR}{\alpha'}, \qquad p_{\mathrm{R}} = \frac{n}{R} - \frac{wR}{\alpha'}, \tag{6.62}$$

and the mass in the nine noncompact directions follows from the two Virasoro constraints. This is the superstring version of the circle mass-shell of Section 6.1: the bosonic intercept that produced the -2 in (6.8) is cancelled here between the worldsheet bosons and fermions on each side, leaving

$$\tfrac{1}{4}\alpha' M^2 \;=\; N + \tfrac{1}{4}\alpha' p_{\mathrm{R}}^2 \;=\; \tilde{N} + \tfrac{1}{4}\alpha' p_{\mathrm{L}}^2, \qquad N - \tilde{N} = nw, \tag{6.63}$$

where $N, \tilde{N}$ are the right- and left-moving oscillator levels (bosonic plus fermionic) of Section 6.1 and the last relation is level matching. The right-moving worldsheet fields carry half the spacetime supersymmetry, so a state that is a ground state on the right, $N=0$, is annihilated by sixteen of the thirty-two supercharges: it is $\tfrac{1}{2}$-BPS. Its mass saturates $M = |p_{\mathrm{R}}|$. The simplest case is a static string of pure winding, $n=0$ and $N=\tilde{N}=0$, for which

$$M \;=\; |p_{\mathrm{R}}| \;=\; \frac{|w|R}{\alpha'} \;=\; |w|\,(2\pi R)\,T_{\mathrm{F1}}. \tag{6.64}$$

The mass is the tension times the length of the wound string. The winding number is precisely the electric B_2 charge, so Eq. (6.64) is the brane version of the BPS bound $T \geq |q|$, saturated. The whole momentum-winding lattice of Section 6.1 is shot through with such states, and T-duality, which swaps $n \leftrightarrow w$ and $R \leftrightarrow \alpha'/R$, maps this tower to itself.

Dp: the RR sources, assembled. Each RR potential C_{p+1} couples electrically to a Dp-brane, with p even in IIA and odd in IIB. Section 6.3 constructed these as the endpoints of open strings, computed the force between two of them from the cylinder, and extracted the physical tension τ_p, the energy density $T_{\mathrm{D}p}/g_s$ of the DBI action,

$$\tau_p \;=\; \frac{1}{(2\pi)^p\, g_s\, \ell_s^{p+1}} \;\sim\; g_s^{-1}, \tag{6.65}$$

together with an equal RR charge, the Polchinski result. The single power $1/g_s$ is the signature of a D-brane: heavier than a fundamental string at weak coupling, lighter than the NS5, and interpolating between them under duality. A single Dp-brane preserves half the supersymmetry, as the projector argument below makes precise, so it too is $\frac{1}{2}$-BPS. The RR ladder runs from the spacetime-filling $p = 9$ down to a $p = -1$ instanton in IIB (and $p = 0, \ldots, 8$ in IIA); the D9 and its negative-charge orientifold partner are what Section 6.4 balances against each other in the tadpole condition, and the D-instanton is the source of the $e^{-2\pi/g_s}$ effects flagged in Chapter 4.

NS5: the magnetic soliton. The magnetic dual of B_2 in ten dimensions is a five-brane, the NS5, the source of $H_3 = \mathrm{d}B$ flux through a surrounding S^3. No open string ends on it. It is a genuine soliton of the NS-NS fields, and its tension carries two powers of the coupling,

$$T_{\mathrm{NS5}} \;\sim\; \frac{1}{g_s^2\, \ell_s^6}. \tag{6.66}$$

Its worldvolume theory is correspondingly harder to name: the IIA NS5 carries the same six-dimensional $(2,0)$ tensor multiplet as the M5-brane, the IIB NS5 a $(1,1)$ vector multiplet, and in the limit that decouples gravity these become the little string theories, six-dimensional non-gravitational theories with a string scale but no coupling to send to zero. We will not need their dynamics, only their tension and charge.

6.5.2 The three powers of the coupling

The pattern g_s^0, g_s^{-1}, g_s^{-2} comes from one fact: the topology of the leading worldsheet that couples to each object. Recall from Chapter 4 that a worldsheet of Euler characteristic χ enters a closed-string amplitude weighted by $g_s^{-\chi}$. Then:

- The fundamental string *is* the worldsheet. Its tension is the input scale that the g_s-expansion is built on top of, so it carries g_s^0 by definition; there is nothing for a worldsheet to end on.
- A D-brane is where a worldsheet can end. The lowest-order coupling of a closed string to a Dp-brane is a disk with its boundary on the brane, $\chi = 1$, weight g_s^{-1}. The tension is the coefficient of that disk one-point function; equivalently, the cylinder of Section 6.3, which computes the force between two branes, factorises on the closed-string channel into two such disk sources, each a factor g_s^{-1}, fixing $\tau_p \sim g_s^{-1}$. The same counting shows up in the D-instanton action $S = 2\pi/g_s$: a disk, $\chi = 1$, hence the exponent $\propto 1/g_s$.
- The NS5 is a soliton of the NS-NS action, and the entire NS-NS sector sits in front of $e^{-2\phi} = g_s^{-2}$ in string frame. A classical solution of an action of the form $g_s^{-2}\int(\partial\phi)^2 + \cdots$ has energy of order g_s^{-2}: this is the sphere, $\chi = 2$. Solitons cost $1/g_s^2$, the same g_s-power a gravitational lump would.

So the coupling dependence is a topological label: $g_s^{-\chi}$ with $\chi = 0, 1, 2$ for the string, the D-brane and the closed-string soliton. The tensions are collected in Table 6.2.

Object	Worldvol.	Charge	Mag. dual	Tension	Leading worldsheet
F1	2	electric B_2	NS5	$\dfrac{1}{2\pi\ell_s^2} \sim g_s^0$	the string, $\chi = 0$
Dp	$p+1$	electric C_{p+1}	D$(6-p)$	$\dfrac{1}{(2\pi)^p g_s \ell_s^{p+1}} \sim g_s^{-1}$	disk, $\chi = 1$
NS5	6	magnetic B_2	F1	$\dfrac{1}{(2\pi)^5 g_s^2 \ell_s^6} \sim g_s^{-2}$	sphere, $\chi = 2$

Table 6.2: The BPS brane spectrum of the Type II string. In IIA the D-branes have $p = 0, 2, 4, 6, 8$; in IIB, $p = -1, 1, 3, 5, 7, 9$. Each electric brane is paired with its magnetic dual under Hodge duality of the sourcing field. The final column records the Euler characteristic χ of the leading worldsheet, whose weight $g_s^{-\chi}$ fixes the power of the coupling. Exact coefficients are collected in Appendix I.

Careful

Two normalisations of "tension" circulate, and they differ by a factor of g_s. The physical energy per unit worldvolume of a Dp-brane, τ_p in Eq. (6.65), carries the factor $1/g_s$ through the $e^{-\phi}$ in front of the DBI action. The charge μ_p that multiplies the RR potential in the Wess-Zumino term $\mu_p \int C_{p+1}$ does *not* carry that factor. BPS saturation is the statement that the two are equal once the dilaton factor of the DBI action is restored, and it is what the cylinder computation of Section 6.3 verifies. In Table 6.2 and everywhere below, "tension" means the physical energy density τ_p.

Electric and magnetic charges are not independent: they are tied by Dirac quantisation. The Dp (electric under C_{p+1}) and the D$(6-p)$ (magnetic under the same field) obey a quantisation condition that fixes the product of their charges to an integer multiple of 2π, $\mu_p\, \mu_{6-p} \in 2\pi\mathbb{Z}$ (Section 6.3). The pairs are D0-D6, D1-D5, D2-D4, with the D3 self-dual in IIB; in the NS-NS sector the same statement pairs F1 with NS5, as in Exercise 5.16. Quantised charge is what makes the BPS thread hold: a central charge assembled from integers cannot drift continuously, so the mass it bounds cannot either.

6.5.3 Charges, projectors and bound states

The physical content of "BPS" for a brane is that its charge is a central charge in the supersymmetry algebra. Section 5.6 showed that the ten- and eleven-dimensional algebras close on a tower of p-form charges $Z_{M_1 \cdots M_p}$, one for every brane, and that repeating the positivity argument for a static extended object turns $M \geq |Z|$ into a bound on the tension by the charge density, $T \geq |z|$. For a single flat brane the unbroken supersymmetries are those preserved by a projector built from the worldvolume gamma matrices,

$$\mathcal{P} = \tfrac{1}{2}(\mathbf{1} + \Gamma_{\text{brane}}), \qquad \Gamma_{\text{brane}} \propto \Gamma^0 \Gamma^1 \cdots \Gamma^p, \qquad \Gamma_{\text{brane}}^2 = \mathbf{1}, \tag{6.67}$$

so that half the components survive, $\Gamma_{\text{brane}} \epsilon = \epsilon$, and every elementary brane is $\tfrac{1}{2}$-BPS. The derivation, including the RR phase that distinguishes a D-brane from an anti-D-brane, is in Appendix C.3.4. Two consequences drive the rest of the course.

Coincident branes and marginal binding. Stack N parallel Dp-branes. Each contributes the same projector Eq. (6.67), so the stack preserves exactly the supersymmetry a single brane does: it is still $\tfrac{1}{2}$-BPS, with mass $M = N\tau_p \times$ volume. The mass is additive and the binding energy vanishes, so the branes can be separated at no cost. This is the no-force condition of Section 5.6: the gravitational and dilatonic attraction is cancelled by the RR repulsion between

like charges. The flat directions along which the branes separate are the expectation values of the transverse scalars, and the theory living on the stack is the U(N) super-Yang-Mills of Section 6.2, its Coulomb branch labelling the brane positions. At large g_sN the stack backreacts on spacetime, and the same tension and charge become the parameters of the extremal p-brane solution of Chapter 5: a supergravity geometry built from a single harmonic function $H(y)$ in the $9-p$ transverse coordinates, tending to 1 at infinity with a $1/|y|^{7-p}$ Coulomb tail whose coefficient is proportional to g_sN, and whose extremality is the BPS equality $\tau_p = \mu_p$. That black brane is the starting point of Chapter 12. A marginal, or threshold, bound state of this kind sits exactly at $M = \sum_i M_i$, and a short multiplet can be built there.

Non-threshold bound states and induced charges. Now bind objects of different type. A fundamental string can dissolve into a D-brane as worldvolume electric flux, and a lower D-brane can dissolve into a higher one as magnetic flux, because the Wess-Zumino coupling of Section 6.3,

$$S_{\mathrm{WZ}} = \mu_p \int_{W_{p+1}} C \wedge e^{\,2\pi\alpha' F + B} \wedge \sqrt{\hat{A}}, \tag{6.68}$$

makes a Dp-brane carrying flux couple to C_{p-1}, C_{p-3}, and so on. The brane therefore carries a whole vector of RR charges, the coefficients in $\mu_p\, e^{2\pi\alpha' F+B}\sqrt{\hat{A}}$, not a single one. When the additional charge points in a new direction of the charge lattice, the central charges add in quadrature rather than linearly, and the bound-state mass sits strictly below the sum of the constituents:

$$M \;=\; |Z| \;=\; \sqrt{|Z_1|^2 + |Z_2|^2} \;<\; |Z_1| + |Z_2|. \tag{6.69}$$

This is a non-threshold bound state: it has genuine binding energy and cannot be pulled apart into its constituents, in contrast to the marginal stack above. The prototype is the (p,q) string of IIB, a bound state of p fundamental strings and q D1-branes with $\gcd(p,q)=1$ and tension

$$T_{(p,q)} \;=\; T_{\mathrm{F1}} \sqrt{p^2 + \frac{q^2}{g_s^2}}\,, \tag{6.70}$$

the two central charges being the orthogonal B_2 and C_2 charges. Exercise 6.15 works this out from Eq. (6.69). The fact that $T_{(p,q)}$ is invariant, up to an overall g_s rescaling (that is, in Einstein frame), under $g_s \to 1/g_s$ together with $(p,q) \to (q,p)$ is a first sight of the S-duality of Chapter 7.

Remark 6.12 (Beyond the lectures: charge lives in K-theory). The charge vector $\mu_p\, e^{2\pi\alpha' F+B}\sqrt{\hat{A}}$ looks like a differential form, and for most purposes treating D-brane charge as an element of the cohomology $H^{\mathrm{even/odd}}(X)$ is enough. That is not the whole story. The correct home for D-brane charge is the topological K-theory of spacetime, $K^0(X)$ for IIB and $K^1(X)$ for IIA (Minasian and Moore 1997; Witten 1998). K-theory sees information that cohomology misses: the torsion charges of branes stabilised only by a discrete $\mathbb{Z}_k$, and the pattern by which a brane-antibrane pair with a tachyon condenses to a lower brane. The map from a K-theory class to its cohomological image is exactly the modified charge $\mathrm{ch}(F)\sqrt{\hat{A}}$ appearing in Eq. (6.68), so the Wess-Zumino term is the shadow in cohomology of a statement that is really about K-theory.

For several branes the surviving spinors must satisfy one projection Eq. (6.67) per brane. Each independent, mutually commuting projector halves the count again: two give $\tfrac{1}{4}$-BPS, three give $\tfrac{1}{8}$-BPS. The compatibility of the projectors is what separates a supersymmetric bound state from an ordinary pile of branes that would attract, radiate and collapse. The configuration that reproduces the black-hole entropy of Chapter 12, the D1-D5-P system, is exactly a $\tfrac{1}{8}$-BPS intersection of this kind, three commuting projectors leaving four of the thirty-two supercharges unbroken.

6.5.4 The thread: what the bound buys

Assemble the logic once more, now with the branes in hand. A BPS brane sits in a short multiplet; the multiplet dimension is an integer; an integer cannot change under a continuous deformation of the coupling; the only way out is for short multiplets to recombine in pairs into a long one, which needs a coincidence of charges at a wall of marginal stability. Away from those walls the relation $M = |Z|$, and the protected index that counts such states, are exact. The derivation is in Section 5.6.4 and Appendix C.3.3; the point to carry away is that a quantity computed in the weakly coupled description survives to strong coupling untouched. Two enterprises use this, and both need the branes of this chapter to be concrete.

The duality web (Chapter 7). A strong-weak duality relates a theory at g_s to another at $1/g_s$, so it cannot be checked by comparing perturbation series. It *can* be checked on BPS states, whose masses are exact on both sides, because a duality must map BPS to BPS with matching $|Z|$. The spectrum of this section is the raw material. Under the strong-coupling limit of IIA, the eleven-dimensional origin of each Type II brane becomes visible:

M-theory origin	Type IIA object
M2 wrapped on the circle	F1
M2 transverse to the circle	D2
M5 wrapped on the circle	D4
M5 transverse to the circle	NS5
momentum along the circle	D0
circle fibred as a KK monopole	D6

The D0 tower, with masses $n/(g_s\ell_s)$ growing dense as $g_s \to \infty$, is the Kaluza-Klein spectrum of the eleventh circle of radius $R_{11} = g_s\ell_s$ opening up (Exercise 5.17). In IIB the same bookkeeping is organised by S-duality: F1 and D1 form a doublet, NS5 and D5 form a doublet, and the D3 is self-dual, the (p, q) strings of Eq. (6.70) filling out the orbit. Every one of these matchings is a BPS statement, and each is a nontrivial check that the objects really do lift to eleven dimensions.

The black hole (Chapter 12). Take a $\frac{1}{8}$-BPS D1-D5-P bound state with charges large enough that at strong coupling it gravitates into a black hole with a finite horizon and a Bekenstein-Hawking entropy $S = A/4G$. At weak coupling the same charges are a gas of open strings on a brane intersection in flat space, whose degeneracy is a counting problem in the U(N) theory of Section 6.2. Because the states are BPS, the count is protected across the enormous change in g_s between the two pictures, so the two numbers must agree. They do, to leading order and beyond, and that agreement, the Strominger-Vafa result, is a genuine microscopic derivation of black-hole entropy. It is possible only because Eq. (6.64) and its cousins are frozen against the coupling.

Where this leaves us

T-duality forced Neumann boundary conditions to become Dirichlet, which conjured D-branes; the D-branes turned out to carry RR charge and to be $\frac{1}{2}$-BPS; and the BPS bound turned that fact into a piece of exact, coupling-independent data. Setting the D-branes beside the fundamental string and the NS5-brane completes the spectrum of stable extended objects in the Type II string, one for each power of g_s from zero to two. The next chapter treats that spectrum as the input to a network of dualities, following the D0 tower up to the eleventh dimension and reading the whole Type II and heterotic landscape as different limits of a single theory. The rigidity established here is what lets a statement proved in one corner of that network be trusted in another.

Exercise 6.14 (Perturbative BPS states of the wound string)**.** *(Moderate.)* Start from the circle mass-shell Eq. (6.63) for the Type II superstring, with $p_{\mathrm{L,R}}$ as in Eq. (6.62).

1. Show that the $\frac{1}{2}$-BPS condition $N = 0$ forces $nw \leq 0$ through level matching, and gives $M = |p_{\mathrm{R}}| = |n/R - wR/\alpha'|$.
2. The two central charges carried by the state are the momentum $Z_n = n/R$ and the winding $Z_w = wR/\alpha'$. Show $M = |p_{\mathrm{R}}| = |Z_n - Z_w|$, and that the BPS branch $nw \leq 0$ is precisely the one on which this collapses to the threshold sum $|Z_n| + |Z_w|$. Contrast this with the D-brane bound state of Exercise 6.15, where the distinct NS-NS and RR charges instead add in quadrature.
3. Apply a T-duality $R \to \alpha'/R$, $n \leftrightarrow w$ and verify that the BPS mass Eq. (6.64) is invariant, so that a purely wound string in one frame is a purely momentum state in the dual frame, both BPS.

Exercise 6.15 (The (p, q) string as a non-threshold bound state)**.** *(Moderate.)* A (p, q) string carries p units of F1 charge and q units of D1 charge. Because B_2 and C_2 are distinct fields, the two central charges are orthogonal, $Z_{\mathrm{F1}} = p\,T_{\mathrm{F1}}$ and $Z_{\mathrm{D1}} = q\,\tau_1$ with $\tau_1 = T_{\mathrm{F1}}/g_s$.

1. Use $M = |Z| = \sqrt{|Z_{\mathrm{F1}}|^2 + |Z_{\mathrm{D1}}|^2}$ to derive the tension Eq. (6.70), and show the binding energy $(pT_{\mathrm{F1}} + q\tau_1) - T_{(p,q)}$ is strictly positive unless p or q vanishes.
2. Show $T_{(p,q)}$ is invariant under $g_s \to 1/g_s$ combined with $(p, q) \to (q, p)$ up to an overall rescaling by g_s, and interpret this as the exchange of the F1 with the D1. Why does the condition $\gcd(p, q) = 1$ select the genuinely bound states rather than superpositions?

Exercise 6.16 (Induced charge from the Wess-Zumino term)**.** *(Short.)* Take a D4-brane in IIA carrying a worldvolume gauge field with instanton number $k = \frac{1}{8\pi^2}\int_{RR^4} F \wedge F$ in its four spatial directions. The relevant piece of Eq. (6.68) is $\mu_4 \int C_1 \wedge \frac{(2\pi\alpha')^2}{2} F \wedge F$. Using the elementary RR charges $\mu_p = (2\pi)^{-p}\ell_s^{-(p+1)}$, show that the coefficient of $\int C_1$ is exactly $k\,\mu_0$: the D4 with k units of instanton flux carries the RR charge of k D0-branes. This is the microscopic version of the statement that a small instanton in the five-dimensional gauge theory on the D4 *is* a D0-brane bound to it.

Chapter 7

The Web of Dualities: S, U, and M-Theory

7.1 S-duality of type IIB

Everything in the previous chapters was, in one way or another, perturbative. The genus expansion of Chapter 4 is a power series in g_s; the effective actions of Chapter 5 are its leading term; T-duality and the D-branes of Chapter 6 are exact but they never move the string coupling. We now cross that line. Type IIB string theory has an exact symmetry that inverts the coupling, $g_s \to 1/g_s$, and turns the fundamental string into a solitonic D-string. It reorganises the entire perturbation series, and it is the first duality in these notes that is not visible at any finite order in g_s.

The big picture

In Section 5.4.4 the two IIB scalars combined into the axio-dilaton $\tau = C_0 + ie^{-\phi}$, and the Einstein-frame action turned out to be invariant under a continuous $\mathrm{SL}(2,\mathbb{R})$ acting on τ by fractional linear transformations. That was a fact about a classical two-derivative field theory. In the full quantum string the charges carried by the fundamental string and the D-string are integers, and only the integer subgroup $\mathrm{SL}(2,\mathbb{Z})$ maps the charge lattice to itself. The conjecture, now supported by a great deal of evidence, is that this $\mathrm{SL}(2,\mathbb{Z})$ is an *exact* symmetry of type IIB. Its generator $S : \tau \to -1/\tau$ sends $g_s \to 1/g_s$: strong coupling is weak coupling in disguise. The elementary string and the D-string are two members of a single $\mathrm{SL}(2,\mathbb{Z})$ multiplet of (p,q)-strings, and which one you call "fundamental" is a choice of duality frame.

7.1.1 From a classical symmetry to a quantum duality

Recall the setup of Section 5.4.4, which we take as given rather than rederive. The RR scalar C_0 and the dilaton ϕ assemble into

$$\tau \;=\; C_0 + i\,e^{-\phi}, \qquad \operatorname{Im}\tau = e^{-\phi} = \frac{1}{g_s}\Big|_{\langle\phi\rangle}, \tag{7.1}$$

a point in the upper half-plane $\mathbb{H} = \{\operatorname{Im}\tau > 0\}$, and the NS-NS and RR two-forms assemble into a doublet $\mathcal{B} = (C_2, B_2)^{\mathsf{T}}$. In the Einstein frame, where the metric $g^{\mathrm{E}}_{\mu\nu} = e^{-\phi/2} g_{\mu\nu}$ carries a canonical Einstein-Hilbert term and does not mix with the scalars, the bosonic action (5.55) is

invariant under

$$\Lambda = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathbb{R}), \qquad \tau \longmapsto \frac{a\tau + b}{c\tau + d}, \qquad \mathcal{B} \longmapsto \Lambda\,\mathcal{B}, \tag{7.2}$$

with $ad - bc = 1$, the Einstein metric and the four-form C_4 (whose five-form field strength F_5 is self-dual) left inert. The scalar kinetic term $\partial\bar{\tau}\,\partial\tau/(\mathrm{Im}\,\tau)^2$ is the Poincaré metric on $\mathbb{H} = \mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$, and $\mathrm{SL}(2,\mathbb{R})$ acts on $\mathbb{H}$ by isometries.

A continuous symmetry acting on the coupling should make you suspicious. If $\mathrm{SL}(2,\mathbb{R})$ survived in the quantum theory, then any value of g_s could be dialled continuously to any other, and there would be no such thing as a distinguished weakly coupled description. What actually happens is that the quantum theory keeps only a discrete remnant, and the mechanism is the same one that quantises electric charge.

The two-form fields B_2 and C_2 have charged sources: the fundamental string F1 is the electric source of B_2, the D-string D1 the electric source of C_2 (Section 5.6). A string can carry p units of B_2 charge and q units of C_2 charge, and Dirac quantisation against the magnetic duals (the NS5 and D5) forces $(p,q) \in \mathbb{Z}^2$. The set of allowed charges is therefore a lattice $\mathbb{Z}^2 \subset \mathbb{R}^2$. A symmetry of the theory must permute the physical spectrum, so it must map this lattice to itself. The doublet $\mathcal{B} \mapsto \Lambda\mathcal{B}$ acts on charges by the same Λ, and the largest subgroup of $\mathrm{SL}(2,\mathbb{R})$ preserving $\mathbb{Z}^2$ is the modular group

$$\mathrm{SL}(2,\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} :\ a,b,c,d \in \mathbb{Z},\ ad - bc = 1 \right\}. \tag{7.3}$$

The continuous classical symmetry is broken to its arithmetic subgroup by charge quantisation, exactly as the continuous shift symmetry of a periodic scalar is broken to integer shifts once you know the periods.

That $\mathrm{SL}(2,\mathbb{Z})$, and not merely its manifest perturbative part, is a symmetry of the full string theory is a conjecture, proposed by Hull and Townsend and by Schwarz and Sen around 1994–95 as part of the larger web this chapter assembles. It is not a theorem. What makes it credible is that it acts correctly on the objects it can be tested against, above all the BPS spectrum, and we take up that evidence in Section 7.1.5 and systematically in Section 7.4.

7.1.2 The modular group and its two generators

$\mathrm{SL}(2,\mathbb{Z})$ is generated by two elements,

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} :\ \tau \mapsto \tau + 1, \qquad S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} :\ \tau \mapsto -\frac{1}{\tau}, \tag{7.4}$$

subject to $S^2 = (ST)^3 = -\mathbf{1}$, so that the group acting faithfully on $\mathbb{H}$ is $\mathrm{PSL}(2,\mathbb{Z}) = \mathrm{SL}(2,\mathbb{Z})/\{\pm\mathbf{1}\}$. The two generators could not be more different in character.

The generator T shifts $\tau \to \tau + 1$, that is $C_0 \to C_0 + 1$ at fixed dilaton. The RR scalar is an axion: its continuous shift symmetry is broken only by objects with integer C_0 charge (D-instantons, and the D7-branes around which C_0 winds), so C_0 is periodic with period one, $C_0 \cong C_0 + 1$. This periodicity is a symmetry order by order in g_s, because it never touches the dilaton: it is entirely perturbative, the string-theory cousin of the θ-angle periodicity of a gauge theory.

The generator S sends $\tau \to -1/\tau$. Set $C_0 = 0$ so that $\tau = i/g_s$; then $-1/\tau = i g_s$, and

$$S: \quad g_s \longmapsto \frac{1}{g_s}. \tag{7.5}$$

This is a strong-weak duality. Nothing in perturbation theory suggests it: the genus expansion is an asymptotic series about $g_s = 0$, and g_s and $1/g_s$ sit at opposite ends of it. In the language of the upper half-plane, string perturbation theory is an expansion about the cusp $\mathrm{Im}\,\tau \to \infty$, and $\mathrm{SL}(2,\mathbb{Z})$ maps that cusp to the rational points on the real axis and folds the interior of $\mathbb{H}$ down to a single fundamental domain. Weakly coupled IIB explores a neighbourhood of one cusp and never sees that the interior has been identified.

Because τ is a modulus (the vacuum expectation value of a field, not an input parameter) $\mathrm{SL}(2,\mathbb{Z})$ is a *gauge identification* on the space of vacua: τ and $\Lambda\tau$ label the same physics, in the same sense that two large gauge transformations, or two points of Teichmüller space related by the mapping class group, describe the same configuration. There is no separate "S-dual theory". IIB is self-dual, and the redundancy is $\mathrm{SL}(2,\mathbb{Z})$.

Weak coupling is not preferred, it is chosen

It is tempting to say IIB "is" weakly coupled and S-duality is a trick for reaching strong coupling. That reads the arrow backwards. There is one theory and one modulus τ; a duality frame is a choice of which cusp of the fundamental domain you expand around and which BPS object you call the fundamental string. At generic τ, of order one g_s, no frame is weakly coupled and none is preferred. The perturbative string is the description that happens to be useful near a cusp.

7.1.3 The (p,q)-string multiplet

If S exchanges the fundamental string and the D-string, the two cannot be independent objects: they are two points in one $\mathrm{SL}(2,\mathbb{Z})$ orbit. Label a string by its charge vector

$$\begin{pmatrix} p \\ q \end{pmatrix}, \qquad p = \text{units of } B_2 \text{ charge (F1 number)}, \quad q = \text{units of } C_2 \text{ charge (D1 number)}, \tag{7.6}$$

so that $(1,0)$ is the fundamental string and $(0,1)$ is the D-string. Under the generators the charge transforms as $\binom{p}{q} \mapsto \Lambda\binom{p}{q}$, chosen so that the coupling of the string to $pB_2 + qC_2$ is duality invariant:

$$S: \begin{pmatrix} p \\ q \end{pmatrix} \mapsto \begin{pmatrix} -q \\ p \end{pmatrix}, \qquad T: \begin{pmatrix} p \\ q \end{pmatrix} \mapsto \begin{pmatrix} p+q \\ q \end{pmatrix}. \tag{7.7}$$

S sends the fundamental string $(1,0)$ to the D-string $(0,1)$ and back (with a reversed orientation), which is exactly the exchange found in the example of Section 5.4.4. T leaves $(1,0)$ alone but sends $(0,1) \to (1,1)$: switching on one unit of the RR axion binds a fundamental string to every D-string. That last statement is the string-theoretic Witten effect, and it is why C_0 and B_2-charge cannot be discussed separately.

The tension. A (p,q)-string is $\frac{1}{2}$-BPS, so its tension is fixed by its charge. When the supersymmetry algebra is extended by brane charges, the tension of a $\frac{1}{2}$-BPS p-brane is bounded below by a p-form central charge, and a string is the $p = 1$ case of that bound (Appendix C); saturating it gives the relation $T = |Z|$ of Section 5.6. The central charge of a string charged (p,q) under the doublet (B_2, C_2) is the norm of the charge vector in an $\mathrm{SL}(2,\mathbb{Z})$-covariant metric $\mathcal{M}(\tau)$ on the charge lattice, built from the same scalar-coset data as the two-form kinetic term. Following the pattern of Section 5.6 and Appendix C,

$$T^2_{(p,q)} \;=\; \frac{1}{(2\pi\alpha')^2}\,(p \;\; q)\,\mathcal{M}(\tau)\begin{pmatrix} p \\ q \end{pmatrix}, \qquad \mathcal{M}(\tau) = \frac{1}{\mathrm{Im}\,\tau}\begin{pmatrix} 1 & -\mathrm{Re}\,\tau \\ -\mathrm{Re}\,\tau & |\tau|^2 \end{pmatrix}, \tag{7.8}$$

measured in the Einstein frame, where $\mathrm{SL}(2,\mathbb{Z})$ fixes the metric and tensions may be compared directly. Multiplying out the quadratic form gives the clean result

$$\boxed{T_{(p,q)} \;=\; \frac{1}{2\pi\alpha'}\,\frac{|p-q\tau|}{\sqrt{\mathrm{Im}\,\tau}} \;=\; \frac{1}{2\pi\alpha'}\sqrt{(p-qC_0)^2\,g_s+\frac{q^2}{g_s}}\;,} \tag{7.9}$$

using (7.1) (and $\alpha' = \ell_s^2$, so the prefactor $1/(2\pi\alpha')$ is the same as $1/(2\pi\ell_s^2)$). This is the formula quoted without proof in Section 5.4.4. Three checks fix its meaning. At $C_0 = 0$ the fundamental string $(1,0)$ has $T = \frac{1}{2\pi\alpha'}\sqrt{g_s}$ and the D-string $(0,1)$ has $T = \frac{1}{2\pi\alpha'}\sqrt{1/g_s}$; these are the string-frame tensions $\frac{1}{2\pi\alpha'}$ and $\frac{1}{2\pi\alpha' g_s}$ of Eqs. (5.81) and (5.83) multiplied by the Weyl factor $g_s^{1/2}$ that converts a string tension from string frame to Einstein frame. That factor has a one-line origin: since $g_{\mu\nu} = e^{\phi/2} g^{\mathrm{E}}_{\mu\nu}$, the two-dimensional worldsheet volume element $\sqrt{-\det g}$ picks up $e^{\phi/2}$, equal to $g_s^{1/2}$ at the dilaton vev (the frame bookkeeping is collected in Appendix I). Under $g_s \to 1/g_s$ the two swap, as S-duality requires. And the whole expression is manifestly invariant under (7.7) together with (7.4), since $|p-q\tau|/\sqrt{\mathrm{Im}\,\tau}$ picks up the same $|c\tau+d|$ from numerator and denominator and it cancels.

Why the charges must be coprime. Not every (p,q) is a single string. The tension (7.9) is the norm $\|\vec{Q}\| \equiv \sqrt{\vec{Q}^{\mathsf{T}}\mathcal{M}\vec{Q}}$ of the charge vector $\vec{Q} = (p,q)$ in a positive-definite metric (positive because $\mathrm{Im}\,\tau > 0$), and a norm is subadditive,

$$\|\vec{Q}_1 + \vec{Q}_2\| \;\le\; \|\vec{Q}_1\| + \|\vec{Q}_2\|, \tag{7.10}$$

with equality if and only if $\vec{Q}_1$ and $\vec{Q}_2$ are parallel. Suppose $\gcd(p,q) = d$, so $\vec{Q} = d\,\vec{Q}_0$ with $\vec{Q}_0$ primitive. Then $T_{(p,q)} = d\,T_{\vec{Q}_0}$ exactly: the state has precisely the tension of d separated copies of the primitive $(p/d, q/d)$-string, sits at threshold, and can fragment at no cost in energy. It is not a bound state. If instead $\gcd(p,q) = 1$, the vector is primitive, no decomposition into charges carrying the same total is parallel, and (7.10) is strict: the (p,q)-string is strictly lighter than any collection of strings with the same charge, hence a genuine, stable bound state. The physical $\frac{1}{2}$-BPS strings of IIB are therefore labelled by *coprime* pairs (p,q), one orbit of $\mathrm{SL}(2,\mathbb{Z})$ acting on $(1,0)$.

Exercise 7.1 (Stability and threshold from the triangle inequality)**.** Take $C_0 = 0$ and $g_s = 1$, so $\mathcal{M} = \mathbf{1}$ and $T_{(p,q)} \propto \sqrt{p^2+q^2}$.

1. Show that $(2,0)$ has exactly twice the tension of $(1,0)$, so a $(2,0)$ "string" is two coincident fundamental strings at threshold, with no binding.
2. Show that the coprime state $(1,1)$ has tension $\sqrt{2}/(2\pi\alpha')$, strictly below the $2/(2\pi\alpha')$ of an unbound $(1,0)+(0,1)$ pair. Quantify the binding energy per unit length.
3. Restore general τ and verify from (7.9) that $T_{(dp',dq')} = d\,T_{(p',q')}$ for any integer d, so coprimality is the exact condition for a below-threshold bound state.

(Difficulty: routine, doable now; parts 1–2 are one-line checks and part 3 is a short line of algebra.)

Remark 7.1 (Beyond the lectures: the bound state as a worldsheet CFT). The (p,q)-strings are more than an algebraic prediction. Witten constructed the bound state of p fundamental strings and q D-strings directly, as the ground state of the $U(q)$ gauge theory living on q coincident D-strings with p units of electric flux switched on along them; the flux is the dissolved fundamental-string charge, and the energy of the flux tube reproduces (7.9) for coprime (p,q). Schwarz then read off the tension from the supergravity soliton. The two computations agree with the BPS formula, which is the first sharp test of $\mathrm{SL}(2,\mathbb{Z})$ beyond the single S that exchanges F1 and D1.

7.1.4 The rest of the multiplet: five-branes, the self-dual three-brane, seven-branes

$\mathrm{SL}(2,\mathbb{Z})$ acts on every object charged under the fields it rotates, and the doublet structure of $\mathcal{B} = (C_2, B_2)^{\mathsf{T}}$ organises the whole IIB brane spectrum.

The magnetic duals of the strings are the five-branes. The NS5-brane is the magnetic source of B_2 and the D5-brane the magnetic source of C_2, so they form a doublet in exactly the same way the F1 and D1 do, and S exchanges them:

$$S: \quad \text{F1} \longleftrightarrow \text{D1}, \qquad \text{NS5} \longleftrightarrow \text{D5}. \tag{7.11}$$

There is a (p,q) five-brane multiplet mirroring the string multiplet, with tension scaling g_s^{-1} for the D5 and g_s^{-2} for the NS5 in string frame (Section 5.6), the two again exchanged by $g_s \to 1/g_s$.

The D3-brane is different. It is the electric source of the four-form C_4, and C_4 is an $\mathrm{SL}(2,\mathbb{Z})$ singlet: (7.2) leaves it inert. The D3 is therefore *self-dual*, a fixed point of S-duality. This single fact has a large payoff in field theory.

Remark 7.2 (Beyond the lectures: Montonen-Olive duality on the D3-brane). The low-energy theory on N coincident D3-branes is four-dimensional $\mathcal{N} = 4$ super-Yang-Mills with gauge group $U(N)$ (Chapter 6). Its complexified gauge coupling is inherited directly from the bulk axio-dilaton,

$$\tau_{\mathrm{YM}} = \frac{\theta_{\mathrm{YM}}}{2\pi} + \frac{4\pi i}{g_{\mathrm{YM}}^2} = \tau,$$

so that $g_{\mathrm{YM}}^2 = 4\pi g_s$ fixes the coupling and $\theta_{\mathrm{YM}} = 2\pi C_0$ the theta angle (this is the standard $\mathcal{N} = 4$ normalisation, the one for which the S below acts as $g_{\mathrm{YM}} \to 4\pi/g_{\mathrm{YM}}$). Because the D3 is fixed by $\mathrm{SL}(2,\mathbb{Z})$, the bulk duality descends to an exact duality of the worldvolume theory: $S : \tau_{\mathrm{YM}} \to -1/\tau_{\mathrm{YM}}$ sends $g_{\mathrm{YM}} \to 4\pi/g_{\mathrm{YM}}$ and exchanges electric charges with magnetic ('t Hooft-Polyakov) monopoles, while T shifts the theta angle by 2π and dresses monopoles into dyons by the Witten effect, precisely the (p,q) story one dimension down. This is Montonen-Olive duality, conjectured for $\mathcal{N} = 4$ in the 1970s on the strength of its BPS spectrum and here delivered for free by string theory. The W-bosons and monopoles of the gauge theory are the endpoints of $(1,0)$ and $(0,1)$ strings stretched between the branes, so the field-theory duality is the same $\mathrm{SL}(2,\mathbb{Z})$ acting on the same (p,q)-strings, which makes $\mathcal{N} = 4$, under far tighter control than the full string theory, the sharpest available quantitative check of IIB S-duality.

Finally, the real part of τ is sourced magnetically by seven-branes. A D7-brane is a codimension-two object, and encircling it C_0 shifts by one, $\tau \to \tau + 1$, which is the monodromy T. General (p,q) seven-branes carry general $\mathrm{SL}(2,\mathbb{Z})$ monodromies, and once one allows several of them the field $\tau(x)$ becomes a multivalued function on the transverse plane whose monodromies live in $\mathrm{SL}(2,\mathbb{Z})$. That is precisely the data of an elliptic curve, of modular parameter τ, fibred over spacetime. Promoting τ to the modulus of a genuine two-torus is the starting point of F-theory.

Remark 7.3 (Beyond the lectures: seven-branes and F-theory). A single D7-brane cannot be the whole story: its monodromy T has infinite order, but a compact transverse space allows only monodromies whose product is trivial, and the deficit-angle bookkeeping of codimension-two objects caps the total charge. The resolution is that the fundamental seven-branes are the various (p,q) images of the D7 under $\mathrm{SL}(2,\mathbb{Z})$, and consistent configurations are read off from the singular fibres of an elliptically fibred geometry. This turns non-perturbative IIB backgrounds with order-one g_s into ordinary algebraic geometry, and it reaches gauge groups (the exceptional $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$) that no perturbative type II or open-string (D-brane and orientifold) construction can produce; those constructions give only unitary, orthogonal and symplectic groups. We develop it properly in Chapter 11; here it is enough that the seven-branes and their monodromies are one more consequence of the same $\mathrm{SL}(2,\mathbb{Z})$ that acts on τ.

7.1.5 Why we believe it

S-duality is a conjecture about a theory we cannot solve at strong coupling, so the case for it is circumstantial by construction. It is nonetheless strong, and every strand of it runs through quantities that supersymmetry protects.

First, the D-string exists and has the right tension. It is not put in to make the duality work: it is forced on us by T-duality and the existence of D-branes (Chapter 6), with tension $1/(2\pi\alpha' g_s)$ computed from first principles by Polchinski. That this is exactly the tension (7.9) of the S-image of the fundamental string is a nontrivial match: two independent calculations, the string-frame D-brane tension and the BPS image of F1, agree.

Second, the full (p, q) tower predicted by $\mathrm{SL}(2,\mathbb{Z})$ is present as genuine BPS bound states, with the tensions and coprimality structure of Section 7.1.3, verified both from the D-string worldvolume gauge theory and from the supergravity solitons. A whole spectrum is checked, not a single mass.

Third, the D3-brane hands the duality down to $\mathcal{N} = 4$ super-Yang-Mills, where it becomes Montonen-Olive duality and can be tested far more sharply, because the field theory is under control: the BPS masses of W-bosons and monopoles, the $\mathrm{SL}(2,\mathbb{Z})$ action on line operators, and the modular properties of partition functions on T^4 all come out as required.

None of these is a proof. What they share is the reason the duality is testable at all: BPS states have masses locked to their charges by the supersymmetry algebra, so a BPS mass computed at weak coupling continues, unrenormalised, to strong coupling, where it can be compared with the dual prediction. The fundamental string spectrum and the D-string spectrum are the same $\frac{1}{2}$-BPS multiplet seen from two sides. This is the mechanism that makes the whole web of this chapter trustworthy, and we set it out in general in Section 7.4. IIB S-duality is its first and cleanest instance, and it is not alone: the same logic identifies type I string theory with the SO(32) heterotic string at strong coupling, a second strong-weak duality introduced among the links of Section 7.3 and then derived on the type I D-string in Section 7.4.3.

Exercise 7.2 (The orbit of the fundamental string)**.** Work at $C_0 = 0$, $g_s = 1$, so $\tau = i$ (the fixed point of S).

1. Verify $S^2 = -\mathbf{1}$ and $(ST)^3 = -\mathbf{1}$ as matrices, and check that $\tau = i$ is fixed by S while $\tau = e^{2\pi i/3}$ is fixed by ST.
2. Starting from the fundamental string $(1, 0)$, apply words in S and T from (7.7) to generate the charges $(0, 1), (1, 1), (1, -1), (2, 1), (1, 2)$, and confirm each is coprime.
3. Show that the set of charges reachable from $(1, 0)$ is exactly the coprime pairs, so the F1 and every (p, q)-string lie in a single $\mathrm{SL}(2,\mathbb{Z})$ orbit. *(Hint: an integer matrix with a primitive first column and determinant one can be completed to an element of* $\mathrm{SL}(2,\mathbb{Z})$*.)*

(Difficulty: parts 1–2 routine and doable now; part 3 uses the standard number-theory fact quoted in the hint.)

What you should be able to do. By the end of this section:

- explain why charge quantisation cuts the classical $\mathrm{SL}(2,\mathbb{R})$ of Section 5.4.4 down to the modular group $\mathrm{SL}(2,\mathbb{Z})$;
- state what S and T do to τ, to the coupling g_s, and to the F1/D1 charge vector (p, q);
- write down the (p, q)-string tension (7.9) and check its $\mathrm{SL}(2,\mathbb{Z})$ covariance and the coprimality condition for a stable bound state;

- name the protected datum, the $\frac{1}{2}$-BPS tension, and say why its non-renormalisation is what makes the conjecture testable at strong coupling.

We have turned the classical scalar-coset symmetry of IIB supergravity into a conjectured exact symmetry of the string, the first duality that mixes weak and strong coupling and the first that trades an elementary string for a soliton. It leaves IIB looking less like a fixed weakly coupled theory and more like a single object viewed through a choice of duality frame. The next move is to ask the same question of the other type II theory: what does IIA look like at strong coupling? The answer is not another ten-dimensional string but an eleventh dimension, and it puts M-theory at the centre of the web (Section 7.2).

7.2 M-theory, the M2 and M5 branes

In Section 5.3 we built eleven-dimensional supergravity from the top down, as the unique maximal supergravity, and reduced it on a circle to recover type IIA. That reduction handed us an algebraic fact: a IIA background is an M-theory background with one small circle, and the string coupling is the size of that circle,

$$R_{11} = g_s\,\ell_s, \qquad \ell_p = g_s^{1/3}\,\ell_s, \qquad \text{equivalently} \quad R_{11} = g_s^{2/3}\,\ell_p \tag{7.12}$$

(Eq. (5.38)). Here we give that fact its physics and its name. The claim of this section is that the algebra is the shadow of a genuine strong-coupling statement about a full quantum theory. Strongly coupled type IIA is not some ten-dimensional theory we have yet to identify; it is an eleven-dimensional theory whose low-energy limit is the supergravity of Section 5.3. That theory is *M-theory*, and it sits at the centre of the duality web: type IIA is M-theory on a circle, and, as later sections show, everything else in the web descends from it by further duality.

M-theory as the hub

Eleven-dimensional supergravity is the low-energy limit of the theory whose weakly coupled corner is type IIA, not a curiosity bolted onto ten-dimensional string theory. As g_s grows the eleventh dimension Eq. (7.12) opens up, and the tower of D0-branes that becomes light is the Kaluza-Klein spectrum of that dimension. M-theory has no dimensionless coupling and no fundamental string: its dynamical objects are two branes, the M2 (electric source of C_3) and the M5 (its magnetic dual). Reduce them on the M-theory circle and the entire IIA brane spectrum falls out, including the fundamental string itself as a wrapped membrane. This is where the F1 and NS5 of Chapter 6 acquire an eleven-dimensional origin.

7.2.1 The eleventh dimension is a strong-coupling limit

The reduction of Section 5.3 is a statement about supergravity, valid when curvatures and field strengths are small. On its own it does not tell us that a full quantum type IIA at large g_s really grows a dimension; it only tells us that if it does, the low-energy physics is eleven-dimensional. The physical evidence for the "if" is a tower of states, and it repays a careful look, because it is the model for every strong-coupling argument in the chapter.

Momentum around the M-theory circle is quantised, $p_{11} = n/R_{11}$, so an eleven-dimensional observer's momentum modes look to a ten-dimensional observer like a tower of particles of mass

$$M_n = \frac{|n|}{R_{11}} = \frac{|n|}{g_s\,\ell_s}, \tag{7.13}$$

each carrying n units of charge under the Kaluza-Klein photon. That photon is the Ramond-Ramond one-form C_1 of IIA (Eq. (5.36)), and the objects charged under C_1 are the D0-branes,

with mass $1/(g_s\ell_s)$ (Chapter 6). The $n = 1$ state is a single D0-brane. For the interpretation Eq. (7.13) to hold, every $|n| > 1$ must also be present: there must exist, for each n, a threshold bound state of n D0-branes with mass exactly $|n|/(g_s\ell_s)$, a normalisable ground state of their supersymmetric quantum mechanics sitting precisely at the BPS mass so that the binding energy vanishes. The existence of exactly one such marginal bound state for every n is a sharp, nontrivial prediction of the eleven-dimensional picture. It has been checked in low cases and is believed in general; we take it as asserted and return to the logic of why such BPS counts survive from weak to strong coupling in Section 7.4.

Now watch the tower as a function of the coupling. At weak coupling, $g_s \to 0$, the D0-branes are heavy, the circle Eq. (7.12) is far smaller than ℓ_s, and the eleventh dimension is invisible: IIA looks honestly ten-dimensional and its perturbation theory is the genus expansion in g_s. As g_s grows the whole tower Eq. (7.13) comes down, the spacing $1/R_{11}$ shrinks, and at strong coupling the tower collapses to a continuum. A continuum of evenly spaced charged states of ever-smaller mass is the unmistakable signature of a decompactifying dimension. Strongly coupled type IIA is M-theory on a large circle, and eleven-dimensional supergravity governs it at low energies.

Careful

M-theory is not "string theory in eleven dimensions". There is no fundamental string in eleven dimensions and no worldsheet expansion, because there is no dimensionless coupling to expand in: the only scale in Section 5.3 is the Planck length ℓ_p. The eleventh dimension and the string coupling are the same datum viewed two ways, Eq. (7.12), so "weak coupling" and "eleven-dimensional" are opposite limits. We do not have a first-quantised definition of M-theory of the kind the perturbative string enjoys. What we have is a low-energy limit (11D supergravity), a set of branes, and a web of dualities that pins the theory down; that web, held rigid by BPS matching, is the working definition.

Remark 7.4 (Beyond the lectures: what "M" might mean operationally). The D0-brane tower is more than evidence. Its dynamics is a supersymmetric matrix quantum mechanics, the dimensional reduction of ten-dimensional SU(N) super-Yang-Mills to $0 + 1$ dimensions. Banks, Fischler, Shenker and Susskind (1996) conjectured that this quantum mechanics, in the large-N limit and in the infinite-momentum frame, *is* M-theory. In that proposal the eleventh dimension and the graviton scattering of eleven-dimensional supergravity are supposed to emerge from the collective dynamics of D0-branes. Matrix theory remains the most concrete candidate for a non-perturbative definition, and it is a direct descendant of the argument above: the same D0-branes whose bound states fill out Eq. (7.13) are proposed as the fundamental degrees of freedom. Witten's "M" was left deliberately open (membrane, matrix, mystery); the branes are the concrete part.

7.2.2 The two branes

M-theory has no D-branes, because it has no Ramond-Ramond fields and no string endpoints; the D-branes are a IIA notion. It has two branes, and they are forced on us by the single gauge field of the theory, the three-form potential C_3 with field strength $F_4 = \mathrm{d}C_3$ (Eq. (5.28)). A three-form couples electrically to a three-dimensional worldvolume and magnetically to a six-dimensional one, exactly as a Maxwell one-form couples electrically to a worldline. The electric source is the *M2-brane* (the membrane), the magnetic source is the *M5-brane*, and we met both in Section 5.3.4. Their tensions, in Planck units, are

$$T_{\mathrm{M2}} = \frac{1}{(2\pi)^2\,\ell_p^3}, \qquad T_{\mathrm{M5}} = \frac{1}{(2\pi)^5\,\ell_p^6}, \tag{7.14}$$

tied together by Dirac quantisation, $2\kappa_{11}^2\,T_{\mathrm{M2}}\,T_{\mathrm{M5}} = 2\pi$ (Eq. (5.33)). Both are half-BPS: each preserves sixteen of the thirty-two supercharges, so its tension equals its charge and two parallel

M2-branes, or two parallel M5-branes, feel no net force. That is the property, established in Section 5.6 and Appendix C, that will let us follow these objects from weak to strong coupling without their masses drifting.

There is nothing else. Every other object in IIA, and by the further dualities of this chapter every brane in the web, is one of these two branes wrapped or unwrapped on cycles, or a purely geometric excitation (Kaluza-Klein momentum, or a Kaluza-Klein monopole).

7.2.3 Reducing the branes on the M-theory circle

Put M-theory on the circle of radius R_{11} and ask what each brane looks like to the ten-dimensional IIA observer. The organising question is a single binary: does the brane's worldvolume wrap the circle or not? A wrapped p-brane loses one worldvolume dimension and becomes a $(p-1)$-brane in ten dimensions; an unwrapped one keeps all its dimensions and stays a p-brane. Combined with the tensions Eq. (7.14) and the dictionary Eq. (7.12), this fixes the type of the IIA object and its tension, coupling scaling and all.

Take the membrane first. Wrap it on the circle. Its two-dimensional worldvolume, with one leg around S^1, becomes a one-dimensional worldline in ten dimensions: a string. Its tension is the membrane tension times the length of the wrapped circle,

$$T_{\mathrm{F1}} = (2\pi R_{11})\, T_{\mathrm{M2}} = 2\pi R_{11} \cdot \frac{1}{(2\pi)^2\, \ell_p^3} = \frac{R_{11}}{2\pi\, \ell_p^3} = \frac{g_s\, \ell_s}{2\pi\, (g_s\, \ell_s^3)} = \frac{1}{2\pi\, \ell_s^2} = \frac{1}{2\pi\alpha'}, \tag{7.15}$$

where we used $\ell_p^3 = g_s\, \ell_s^3$ and $R_{11} = g_s \ell_s$ from Eq. (7.12), together with $\alpha' = \ell_s^2$. The coupling cancels completely. What comes out is exactly the fundamental-string tension $T = 1/(2\pi\alpha')$ of Appendix A, independent of g_s, which is precisely the property a fundamental string must have: its tension defines the perturbative expansion and cannot itself depend on the coupling. The wrapped membrane *is* the type IIA fundamental string. When $g_s \to 0$ the wrapping circle shrinks below ℓ_s and the tube of membrane becomes a thin worldsheet: the string we quantised in the first half of the course is a cylinder of M-theory membrane seen edge-on.

If instead the membrane sits at a point on the circle and extends in the other directions, it stays a two-brane. Its tension is unchanged at $T_{\mathrm{M2}} = 1/((2\pi)^2 \ell_p^3) = 1/((2\pi)^2 g_s \ell_s^3)$, which scales as $1/g_s$: this is the D2-brane. The same fork applied to the fivebrane gives the two remaining heavy objects. Wrapping the M5 on the circle turns a five-brane into a four-brane whose tension is $(2\pi R_{11}) T_{\mathrm{M5}} = 1/((2\pi)^4 g_s \ell_s^5)$, again scaling as $1/g_s$: the D4-brane. Leaving the M5 transverse to the circle keeps it a five-brane, with tension $T_{\mathrm{M5}} = 1/((2\pi)^5 g_s^2 \ell_s^6)$, scaling as $1/g_s^2$: the NS5-brane. Finally, Kaluza-Klein momentum on the circle is the D0-brane, Eq. (7.13). The whole harvest fits in one table.

IIA object	tension	g_s-scaling	M-theory origin
D0	$\lvert n\rvert/(g_s \ell_s)$	g_s^{-1}	Kaluza-Klein momentum n on S^1
F1	$1/(2\pi\alpha')$	g_s^{0}	M2 wrapping S^1
D2	$1/((2\pi)^2 g_s \ell_s^3)$	g_s^{-1}	M2 transverse to S^1
D4	$1/((2\pi)^4 g_s \ell_s^5)$	g_s^{-1}	M5 wrapping S^1
NS5	$1/((2\pi)^5 g_s^2 \ell_s^6)$	g_s^{-2}	M5 transverse to S^1
D6	$1/((2\pi)^6 g_s \ell_s^7)$	g_s^{-1}	Kaluza-Klein monopole (Taub-NUT)

Read the table as a story about the coupling. The fundamental string is the only object whose tension is g_s-independent, because it is the only brane whose wrapping cancels the coupling exactly; that is what makes it fundamental in perturbation theory. The D-branes all carry a single inverse power of g_s, the hallmark of a Ramond-Ramond soliton, whether they come from a wrapped or an unwrapped M-brane. The NS5, the magnetic dual of the fundamental string,

is the heaviest, going as $1/g_s^2$, which is why it is the last object to appear and the hardest to treat: it is a genuinely non-perturbative lump of the NS-NS sector. The D6 is the geometric completion, a Kaluza-Klein monopole for the M-theory circle (a Taub-NUT geometry whose asymptotic circle is the eleventh dimension); we quote it from Section 5.3.6 and do not need its details. The one IIA brane with no clean eleven-dimensional origin, the D8, is tied to a cosmological-constant deformation of the theory and is left to Appendix I.

The line to remember: the fundamental string and the NS5-brane of Chapter 6, which there looked like primitive ten-dimensional objects, are the wrapped membrane and the transverse fivebrane of M-theory.

Example 7.5 (The fundamental string is a wrapped membrane)**.** Set the tension of a membrane wrapped once on the M-theory circle equal to $(2\pi R_{11})\,T_{\mathrm{M2}}$ and substitute Eq. (7.12). The chain Eq. (7.15) gives $1/(2\pi\alpha')$ with every factor of g_s gone. Turn the logic around: demand that the wrapped membrane have exactly the fundamental-string tension, and you recover $R_{11} = g_s\ell_s$ without ever mentioning D0-branes. The two derivations of the radius, one from the D0 tower Eq. (7.13) and one from the wrapped membrane, agree, which is a first small instance of the BPS cross-checks of Section 7.4.

Exercise 7.3 (Every IIA brane from an M-brane)**.** Using $\ell_p = g_s^{1/3}\ell_s$ and $R_{11} = g_s\ell_s$, verify the four remaining entries of the table. (a) The transverse M2 gives $T_{\mathrm{D2}} = 1/((2\pi)^2 g_s\ell_s^3)$. (b) The wrapped M5 gives $T_{\mathrm{D4}} = (2\pi R_{11})T_{\mathrm{M5}} = 1/((2\pi)^4 g_s\ell_s^5)$. (c) The transverse M5 gives $T_{\mathrm{NS5}} = 1/((2\pi)^5 g_s^2\ell_s^6)$. (d) Kaluza-Klein momentum gives $M_{\mathrm{D0}} = 1/(g_s\ell_s)$. Confirm that each matches the standard Dp-brane tension $T_{\mathrm{D}p} = 1/((2\pi)^p g_s\ell_s^{p+1})$ for $p = 2, 4$, and that the NS5 scales as g_s^{-2} as a magnetic NS-NS object must. (Difficulty: routine bookkeeping, doable now, worth doing once.)

Remark 7.6 (Beyond the lectures: bounding the eleventh dimension by an interval)*.* The circle was one choice; an interval is another. Replace the M-theory circle by the orbifold $S^1/\mathbb{Z}_2$ that identifies x^{11} with $-x^{11}$. The quotient has two ten-dimensional fixed walls, one at each endpoint. Invariance of the eleven-dimensional theory forces C_3 to be odd under the reflection, and that projection removes half of the gravitino: sixteen of the thirty-two supercharges survive, exactly the $\mathcal{N} = 1$ supersymmetry of a ten-dimensional heterotic string. The projection also strands a gravitational anomaly on each wall, and cancelling it by inflow from the bulk forces a ten-dimensional E_8 vector multiplet to live on each of the two walls. The result is the $\mathrm{E}_8 \times \mathrm{E}_8$ heterotic string realised in eleven dimensions: gauge fields confined to the two boundaries, gravity propagating in the bulk between them, and the length of the interval playing the role the circle radius played for IIA. A short interval is weakly coupled and looks ten-dimensional; a long one is the strong-coupling regime where the eleventh dimension opens up, the heterotic analogue of the D0 tower that opened this section. This is the Hořava-Witten construction; we quote the anomaly-inflow result rather than derive it, and take up its place in the wider duality web in Section 7.3.

7.2.4 The membrane worldvolume theory

Reducing the branes tells us how they fit into the web. To use them as physical probes we also want the theory that lives on them, and here the two branes part ways sharply. Consider N coincident M2-branes. At low energies the fields on their common three-dimensional worldvolume decouple from the bulk, and what remains is a three-dimensional superconformal field theory. It has the maximal supersymmetry a three-dimensional theory can carry, $\mathcal{N} = 8$ (sixteen supercharges), enhanced by conformal invariance to the superconformal group $\mathrm{OSp}(8|4)$; the eight scalars are the transverse positions of the branes in eleven dimensions, and there is no dimensionless coupling, consistent with M-theory having none.

The number we care about is the count of low-energy degrees of freedom. For N coincident D-branes it is the N^2 of a U(N) gauge theory, the entries of an $N \times N$ matrix, one open string for each ordered pair of branes. For M2-branes the count is different and, at first sight, strange. Measured for instance by the thermal free energy, or the entropy of a stack of near-extremal M2-branes, it scales as

$$\#\,\text{degrees of freedom on } N \text{ M2-branes} \;\sim\; N^{3/2}. \tag{7.16}$$

This is read off from the near-horizon geometry of the M2 stack, $\mathrm{AdS}_4 \times S^7$, whose supergravity free energy carries the $N^{3/2}$; the logic of reading a field-theory quantity off a dual geometry is the subject of Chapter 13, and we quote the result here rather than derive it. The scaling Eq. (7.16) says the membrane theory is not a matrix model of N^2 gluons. It grows more slowly, and for a long time no one knew what the degrees of freedom were.

The Lagrangian answer arrived late. Aharony, Bergman, Jafferis and Maldacena (2008) wrote down a three-dimensional $\mathrm{U}(N)_k \times \mathrm{U}(N)_{-k}$ Chern-Simons-matter theory, the ABJM theory, which at levels $k = 1, 2$ has the full $\mathcal{N} = 8$ supersymmetry and describes N M2-branes (at a $\mathbb{C}^4/\mathbb{Z}_k$ orbifold point for general k). Localisation in that theory reproduces the $N^{3/2}$ of Eq. (7.16) from the field theory side, closing the loop. We name ABJM and go no further; the construction belongs to a course on Chern-Simons-matter theories.

7.2.5 The fivebrane worldvolume theory, and an honest open problem

The M5-brane is harder than the membrane, and it leaves us with an open problem. The low-energy theory on N coincident M5-branes is a six-dimensional superconformal field theory with $\mathcal{N} = (2, 0)$ supersymmetry, chiral (both supercharges have the same six-dimensional chirality). Its field content, on a single brane, is a *tensor multiplet*: five scalars (the transverse positions), fermions, and a two-form potential b_2 whose three-form field strength is self-dual,

$$h_3 = \mathrm{d}b_2, \qquad h_3 = \star_6\, h_3. \tag{7.17}$$

The self-duality is the source of all the trouble. A self-dual field strength has no simple Lorentz-covariant action: the naive kinetic term $h_3 \wedge \star_6 h_3$ vanishes identically once Eq. (7.17) is imposed, and writing a Lagrangian that yields the self-duality as an equation of motion forces you to break manifest Lorentz invariance or introduce auxiliary machinery. This is the same obstruction that afflicts the self-dual five-form of type IIB (Section 5.4). For a single fivebrane one can still work with the free tensor multiplet and its equations of motion. For $N > 1$ the theory is interacting and there is no known conventional Lagrangian at all, not even a non-covariant one: the $\mathcal{N} = (2, 0)$ theory is an isolated interacting fixed point with no marginal coupling to turn down, so there is no weakly coupled description to write an action for.

The degree-of-freedom count makes the mystery quantitative. From the near-horizon geometry $\mathrm{AdS}_7 \times S^4$, or equivalently from the coefficient of its conformal (Weyl) anomaly, the number of degrees of freedom on N coincident M5-branes scales as

$$\#\,\text{degrees of freedom on } N \text{ M5-branes} \;\sim\; N^3. \tag{7.18}$$

Again this is read from the dual supergravity and quoted rather than derived (Chapter 13). Compare the three counts: N^2 for N D-branes, $N^{3/2}$ for M2-branes, N^3 for M5-branes. The N^3 is the deepest puzzle of the three. A matrix model gives N^2; nobody has a simple picture of what carries the extra factor of N. That the fundamental degrees of freedom of the M5-brane theory are not known, and that the N^3 scaling has no elementary microscopic derivation, is one of the genuine open problems the whole subject has left us.

Remark 7.7 (Beyond the lectures: why the $(2,0)$ theory matters anyway). A theory with no Lagrangian can still be enormously useful, because you can compactify it. Put the $\mathcal{N}=(2,0)$ theory of type A_{N-1} (the label for N M5-branes, in the ADE classification of these theories) on a two-torus. Six dimensions minus two leaves four, and the low-energy theory is four-dimensional $\mathcal{N}=4$ super-Yang-Mills with gauge group $\mathrm{SU}(N)$. The complex structure τ of the torus becomes the complexified gauge coupling, and the geometric $\mathrm{SL}(2,\mathbb{Z})$ that acts on τ becomes the Montonen-Olive S-duality of the gauge theory: the same $\mathrm{SL}(2,\mathbb{Z})$ that runs the IIB S-duality of Section 7.1. Seen this way, four-dimensional S-duality is the modular group of a torus, promoted to field theory by an object that has no Lagrangian of its own. Compactifying the $(2,0)$ theory on more general Riemann surfaces generates the vast web of four-dimensional $\mathcal{N}=2$ dualities (the theories of class S). The fivebrane is hard, and it is also the parent of a great deal of what we understand about strongly coupled gauge theory.

Exercise 7.4 (Counting the M-brane degrees of freedom)**.** (a) For N coincident Dp-branes, argue from open strings stretched between branes that the massless worldvolume theory is $\mathrm{U}(N)$ super-Yang-Mills, with $O(N^2)$ fields. (b) Take the near-horizon geometries $\mathrm{AdS}_4\times S^7$ and $\mathrm{AdS}_7\times S^4$ as given, together with the fact (to be established in Chapter 13) that a free energy scales as the inverse of the effective Newton constant in the anti-de Sitter factor. Using the eleven-dimensional relation $2\kappa_{11}^2=(2\pi)^8\ell_p^9$ and the flux quantisation that fixes the anti-de Sitter radius in terms of N and ℓ_p, reproduce the powers $N^{3/2}$ for the M2 and N^3 for the M5. (Difficulty: part (a) is a warm-up, doable now; part (b) is a genuine AdS/CFT computation, optional here and best returned to after Chapter 13. Hint for (b): the AdS_{d+1} radius L obeys $L^9/\ell_p^9\sim N$ for the M2 and $L^9/\ell_p^9\sim N^3$ for the M5 up to numbers, and the free energy goes as $L^{\dim(\mathrm{AdS})-2}/\kappa_{11}^2$.)

7.2.6 The hub

Assemble what this section establishes. There is an eleven-dimensional theory, M-theory, with eleven-dimensional supergravity as its low-energy limit and no free coupling. It has two branes, the M2 and the M5, and two purely geometric excitations, Kaluza-Klein momentum and the Kaluza-Klein monopole. Put it on a circle and you get type IIA, with the size of the circle serving as the string coupling; the fundamental string is a wrapped membrane, the NS5 is a transverse fivebrane, and the full IIA brane spectrum is accounted for. This is the sense in which M-theory is the hub.

By the end of this section you should be able to:

- state that type IIA is M-theory on a circle, with $R_{11}=g_s\ell_s$ and $\ell_p=g_s^{1/3}\ell_s$, and explain why the D0-brane tower coming down at large g_s signals a decompactifying eleventh dimension;
- read the reduction table: name the IIA object produced by each M2 or M5 wrapped on or transverse to the circle, together with the g_s-scaling of its tension;
- explain why the wrapped membrane is the only object with g_s-independent tension, and hence why it is the type IIA fundamental string;
- quote the low-energy degree-of-freedom counts, $N^{3/2}$ for M2-branes and N^3 for M5-branes, and say which of them remains an open problem.

The rest of the chapter connects the remaining spokes to this centre. Bound the eleventh dimension by an interval rather than a circle and the $\mathrm{E}_8\times\mathrm{E}_8$ heterotic string appears, with a gauge E_8 on each of the two ten-dimensional walls; that is the Hořava-Witten construction of Theorem 7.6, whose role in the web is developed in Section 7.3. Reduce M-theory further and compare the resulting supergravities and their BPS spectra, and the S-duality of IIB (Section 7.1),

the type-I/heterotic duality, and the string-string duality that turns heterotic gauge symmetry into IIA geometry on K3 (Section 7.3) all fall into a single connected web, with the U-duality groups $\mathrm{E}_{n(n)}(\mathbb{Z})$ organising its toroidal reductions. Every one of those statements is a claim that two effective theories describe the same physics, and every one is trusted for the same reason: the BPS objects introduced here, the M2 and M5 and their descendants, carry masses fixed by their charges, so their spectra can be matched across the web from weak coupling to strong. That argument is the business of Section 7.4. The same branes reappear as the microscopic constituents of black holes in Chapter 12 and as the objects whose near-horizon throats define the holographic dualities of Chapter 13. The membrane and the fivebrane, and the circle that turns them into strings, are the through-line.

7.3 The web and gauge-symmetry-becomes-geometry

The big picture

The individual links are in hand: T-duality on circles, the S-dualities of Section 7.1, and the M-theory lift of Section 7.2. Snap them together and the five ten-dimensional strings plus eleven-dimensional supergravity become one connected theory, with M-theory at the hub. Compactify it and the discrete dualities fuse into the U-duality groups $\mathrm{E}_{n(n)}(\mathbb{Z})$. The single sharpest lesson of the web is a change of language: a gauge symmetry that is fundamental in one description can be pure geometry in another. The clean example is heterotic on T^4 against IIA on K3, where the rank-16 heterotic gauge symmetry reappears as the ADE singularities of a four-manifold.

By now we have three kinds of statement relating string theories, and it pays to keep them straight before we wire them together. They differ in what they do to the string coupling, and therefore in how hard they are to trust.

7.3.1 Three kinds of link

T-duality: a perturbative equivalence. Compactify a theory on a circle of radius R and the spectrum of momentum and winding modes is invariant under $R \leftrightarrow \alpha'/R$ with momentum and winding exchanged. This is an order-by-order statement: it holds at each order in the genus expansion and does not move the string coupling out of the weakly coupled regime (the dilaton shifts only by the finite amount needed to keep the lower-dimensional Newton constant fixed). Type IIA on a circle of radius R is type IIB on a circle of radius α'/R, which is why the two theories carry Ramond-Ramond potentials of opposite form-degree parity and why a wrapped Dp-brane maps to an unwrapped D$(p \mp 1)$-brane. The two heterotic strings are likewise a single theory on a circle: $\mathrm{E}_8 \times \mathrm{E}_8$ heterotic on a circle, with a Wilson line breaking each E_8, is SO(32) heterotic on the dual circle. We built T-duality in Chapter 6; here we only use the conclusion. For a d-torus the momentum and winding charges together fill the even self-dual Narain lattice $\Gamma_{d,d}$ (heterotic: $\Gamma_{d,d+16}$), and the perturbative duality group is the lattice automorphism group $O(d,d;\mathbb{Z})$ (heterotic: $O(d,d+16;\mathbb{Z})$). The lattices and these groups are assembled once in Appendix G; we cite them rather than rebuild them.

The right way to think of T-duality is as a redundancy of description, not a symmetry of a fixed background. Two geometric-looking data, radius R and radius α'/R, label the same physics. Once $R \sim \ell_s$ the notion of "the radius" stops being operationally meaningful, and this is the first hint that strings do not see spacetime the way point particles do.

S-duality: a strong-weak equivalence. S-dualities invert the coupling, $g_s \to 1/g_s$, and so exchange the perturbative expansion for something with no perturbative description at all. Two

such links live in ten dimensions. Type IIB is self-dual under $\mathrm{SL}(2,\mathbb{Z})$ acting on $\tau = C_0 + i/g_s$ of (5.53), exchanging the fundamental string with the D-string and, more generally, permuting the (p,q)-strings; this is Section 7.1. Type I and the SO(32) heterotic string are the weak- and strong-coupling faces of one theory: the heterotic string at coupling g_s is the Type I string at coupling $1/g_s$, with the heterotic fundamental string appearing at strong coupling as the Type I D-string. The worldsheet derivation of that identification, D-string fields equal heterotic fields, is Section 7.4.3. Unlike T-duality, an S-duality cannot be checked order by order. Its evidence is the matching of BPS spectra, which is what Section 7.4 is for.

The M-theory lift: growing a dimension. The remaining two theories are not linked to the others by a duality within ten dimensions; they grow an eleventh. Type IIA at strong coupling opens up a circle of radius $R_{11} \sim g_s^{2/3}\ell_p$, with the D0-branes becoming the Kaluza-Klein tower of eleven-dimensional supergravity, and the $\mathrm{E}_8 \times \mathrm{E}_8$ heterotic string opens up an interval $S^1/\mathbb{Z}_2$ with a gauge E_8 on each of the two ten-dimensional walls (Hořava-Witten). The circle lift is developed in Section 7.2; the interval lift is taken up in the remark below. The parent is eleven-dimensional supergravity, the unique maximal supergravity, whose reduction on a circle produced IIA in the first place (Section 5.3). M-theory is the name for its ultraviolet completion, defined operationally as "whatever has eleven-dimensional supergravity as its low-energy limit and IIA string theory as its circle reduction".

Remark 7.8 (Beyond the lectures: the Hořava-Witten wall). The interval lift is worth a closer look, because it is where the $\mathrm{E}_8 \times \mathrm{E}_8$ gauge group comes from. Put M-theory on $S^1/\mathbb{Z}_2$, the circle quotiented by the reflection $x^{11} \to -x^{11}$; the quotient is an orbifold with two ten-dimensional fixed planes, the walls, at the ends of the interval. The reflection acts on the supergravity three-form by a sign and projects out half of the 32 supercharges, leaving the 16 of a chiral ten-dimensional theory, precisely the supersymmetry of the heterotic string. That projection leaves a gravitational anomaly uncancelled on each wall, so consistency forces extra ten-dimensional degrees of freedom to live there; anomaly inflow fixes them uniquely to a single E_8 vector multiplet on each wall, and the product $\mathrm{E}_8 \times \mathrm{E}_8$ is the ten-dimensional gauge group. As the interval shrinks, the two walls collide and the theory becomes weakly coupled $\mathrm{E}_8 \times \mathrm{E}_8$ heterotic string theory, with the interval length L setting the coupling, $g_s^{\mathrm{het}} \sim (L/\ell_p)^{3/2}$, exactly as the circle radius set g_s^{IIA} in (7.12). An M2-brane stretched across the interval, with one boundary on each wall, is the heterotic string, and the wall degrees of freedom become its current algebra.

7.3.2 The web as one theory

Put the links on a page (Fig. 7.1). The five perturbative strings sit on a ring, connected around the perimeter by the T- and S-dualities and by the orientifold projection $\mathrm{I} = \mathrm{IIB}/\Omega$ that defines Type I as the worldsheet-parity quotient of IIB. Two spokes run to the centre: IIA and $\mathrm{E}_8 \times \mathrm{E}_8$ heterotic are M-theory on a circle and on an interval. IIB carries its own $\mathrm{SL}(2,\mathbb{Z})$ as a loop.

This is one theory with five weakly coupled corners, not five theories with bridges between them. Each "string theory" is the perturbative expansion of the single object around a different cusp of its moduli space, the region where some coupling or some radius goes to a limit and a weakly interacting string becomes visible. Away from the cusps there is no small parameter and no privileged string; there is just the moduli space, with eleven-dimensional supergravity governing the large-volume, large-radius interior. That the corners are genuinely connected, rather than five look-alike theories that happen to share a low-energy limit, is the content of the duality conjectures, and it is exactly what BPS matching establishes.

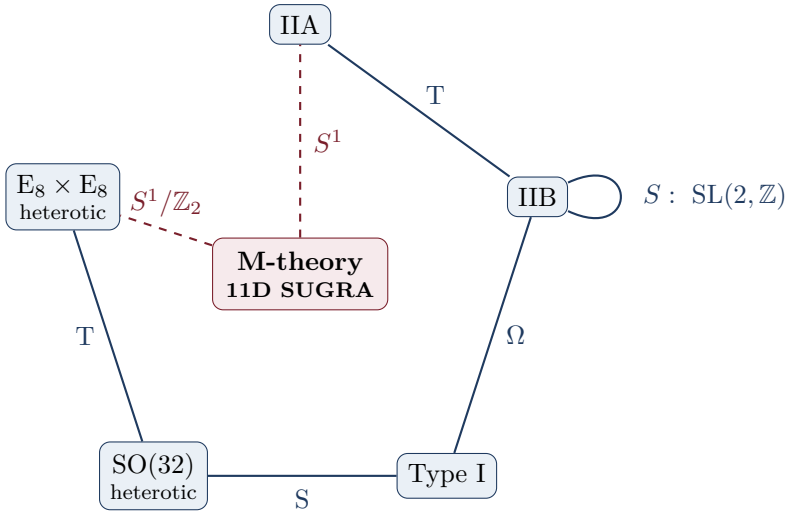


Figure 7.1: The web of dualities. Solid links are exact equivalences (T-duality, S-duality, the orientifold Ω); dashed spokes decompactify to eleven dimensions. The five weakly coupled string theories are cusps of one connected moduli space whose interior is M-theory.

These are conjectures backed by evidence

None of the non-perturbative links in Fig. 7.1 is proven in the sense of a derivation from a complete definition of the theory on both sides, because no such definition exists off the cusps. They are established the way good physics often is: two descriptions are claimed to be the same theory, one computes every quantity that is protected against the strong-coupling extrapolation, and one checks that the answers agree. The protected quantities are the BPS masses, charges, and degeneracies. Section 7.4 is the audit.

7.3.3 U-duality: S and T fuse under compactification

The dualities interact once you compactify. Reduce eleven-dimensional supergravity on T^n and the two-derivative action acquires a continuous global symmetry far larger than the obvious $\mathrm{GL}(n,\mathbb{R})$ of the torus: the exceptional group $\mathrm{E}_{n(n)}(\mathbb{R})$, the split real form, acting on the scalars through the coset $\mathrm{E}_{n(n)}/K$ with K its maximal compact subgroup. This is the Cremmer-Julia hidden symmetry, and it is a fact about the classical maximal supergravity in $D = 11 - n$ dimensions; the scalars from the metric, the three-form, and their duals conspire to fill exactly one $\mathrm{E}_{n(n)}$ coset. We take this as read from the reduction of Section 5.3; the derivation lives with the supergravity formulary in Appendix I.

Charge quantisation cuts $\mathrm{E}_{n(n)}(\mathbb{R})$ down to an arithmetic subgroup, exactly as the classical $\mathrm{SL}(2,\mathbb{R})$ of IIB was cut to $\mathrm{SL}(2,\mathbb{Z})$ in Section 7.1. The wrapped-brane charges live in a lattice, and only the subgroup $\mathrm{E}_{n(n)}(\mathbb{Z})$ preserving that lattice can be an exact symmetry. This discrete remnant is the *U-duality group*. It contains two subgroups we already know: the perturbative T-duality group $O(n-1,n-1;\mathbb{Z})$ of the type II string on T^{n-1}, and the non-perturbative S-duality $\mathrm{SL}(2,\mathbb{Z})$. Neither is normal in $\mathrm{E}_{n(n)}(\mathbb{Z})$, and they do not commute: conjugating a winding-number shift by an S-duality produces a transformation in neither factor, and iterating generates the whole exceptional group. U-duality is the statement that all of these, perturbative and non-perturbative alike, are exact equivalences of the compactified theory.

Table 7.1 lists the maximal case, where all 32 supercharges are kept. The exceptional groups grow with the number of compact dimensions and reach $\mathrm{E}_{8(8)}$ in three dimensions, where every propagating field can be dualised to a scalar and the entire bosonic sector becomes an $\mathrm{E}_{8(8)}/\mathrm{SO}(16)$

D	n	$\mathrm{E}_{n(n)}$	T-duality	U-duality $\mathrm{E}_{n(n)}(\mathbb{Z})$
9	2	$\mathrm{SL}(2)\times\mathbb{R}^+$	$\mathbb{Z}_2$	$\mathrm{SL}(2,\mathbb{Z})$
8	3	$\mathrm{SL}(3)\times\mathrm{SL}(2)$	$O(2,2;\mathbb{Z})$	$\mathrm{SL}(3,\mathbb{Z})\times\mathrm{SL}(2,\mathbb{Z})$
7	4	$\mathrm{SL}(5)$	$O(3,3;\mathbb{Z})$	$\mathrm{SL}(5,\mathbb{Z})$
6	5	$\mathrm{SO}(5,5)$	$O(4,4;\mathbb{Z})$	$\mathrm{SO}(5,5;\mathbb{Z})$
5	6	$\mathrm{E}_{6(6)}$	$O(5,5;\mathbb{Z})$	$\mathrm{E}_{6(6)}(\mathbb{Z})$
4	7	$\mathrm{E}_{7(7)}$	$O(6,6;\mathbb{Z})$	$\mathrm{E}_{7(7)}(\mathbb{Z})$
3	8	$\mathrm{E}_{8(8)}$	$O(7,7;\mathbb{Z})$	$\mathrm{E}_{8(8)}(\mathbb{Z})$

Table 7.1: U-duality of M-theory on T^n, equivalently the maximal (type II) string on T^{n-1}, in $D = 11-n$ dimensions. The T-duality column is the perturbative subgroup $O(n-1,n-1;\mathbb{Z})$; the full U-duality group is the arithmetic exceptional group $\mathrm{E}_{n(n)}(\mathbb{Z})$, quoted here from the Cremmer-Julia reduction and derived in Appendix I. The low-rank names $\mathrm{SL}(5)=\mathrm{E}_{4(4)}$ and $\mathrm{SO}(5,5)=\mathrm{E}_{5(5)}$ identify members of the exceptional series. The same groups appear in Section 7.4.6, indexed there by the M-theory torus dimension $d=n$.

sigma model. The heterotic and Type I strings preserve only half the supersymmetry, and their U-duality is correspondingly smaller: heterotic on T^d has the T-duality group $O(d,d+16;\mathbb{Z})$ combined with an S-duality that, in six dimensions, is not a self-duality at all but the string-string duality we turn to next. The full dictionary of which brane maps to which under each of these groups is collected in Appendix I.

7.3.4 K3 in one page

To state the string-string duality we need one manifold. K3 is the unique compact Calabi-Yau twofold: a complex surface with vanishing first Chern class ($c_1=0$, hence a Ricci-flat Kähler metric by Yau's theorem) and no continuous one-cycles ($b_1=0$). Every K3 is diffeomorphic to every other, so there is really one smooth four-manifold here, with a family of Ricci-flat metrics on it. Equivalently, K3 is the compact hyperkähler four-manifold: its holonomy is $\mathrm{SU}(2)\cong\mathrm{Sp}(1)$, the smallest special holonomy, and this is the whole reason it enters string duality. A metric with SU(2) holonomy admits covariantly constant spinors that preserve exactly half of whatever supersymmetry the higher-dimensional theory started with. K3 geometry is developed in Appendix E and Chapter 8; here is the minimum.

The cohomology carries the whole story. The Betti numbers are $b_0=b_4=1$, $b_1=b_3=0$, $b_2=22$, so the Euler characteristic is $\chi=24$. The Hodge numbers are $h^{2,0}=h^{0,2}=1$ and $h^{1,1}=20$. The middle cohomology $H^2(\mathrm{K3},\mathbb{Z})$, equipped with the intersection form, is an even self-dual lattice of signature $(3,19)$, and by the uniqueness of indefinite even self-dual lattices this is forced to be

$$H^2(\mathrm{K3},\mathbb{Z}) \;\cong\; U^{\oplus 3}\oplus(-\mathrm{E}_8)^{\oplus 2} \;:=\; \Gamma_{3,19}, \tag{7.19}$$

where $U=\Gamma_{1,1}$ is the hyperbolic plane $\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$ and $-\mathrm{E}_8$ is the E_8 root lattice with reversed sign. The two E_8's of the heterotic string are already visible, sitting inside the second cohomology of a four-manifold. That is not a coincidence, and Section 7.3.6 is where it pays off.

A concrete singular model helps. Take $T^4/\mathbb{Z}_2$, the four-torus quotiented by $x^m\to -x^m$. The quotient has 16 fixed points, and near each one it looks like $\mathbb{C}^2/\mathbb{Z}_2$, an A_1 orbifold singularity. Blowing up each fixed point replaces it with a two-sphere ($\mathbb{CP}^1$) of self-intersection -2, and the result is a smooth K3. The sixteen exceptional spheres, together with the six $\mathbb{Z}_2$-invariant two-cycles of the torus, span the 22-dimensional H^2. More general degenerations of K3 collapse configurations of such (-2)-spheres, and the possible collapses are classified by the ADE Dynkin diagrams. This is the K3 side of the ADE story of Appendix E, and it is the geometry that will

carry the gauge fields.

7.3.5 Heterotic on T^4 equals IIA on K3

Here is the cleanest non-trivial duality in the web, and the one to understand in full.

Supersymmetry. Heterotic string theory in ten dimensions has 16 supercharges. Compactify on T^4, which preserves all of them, and the result is six-dimensional supergravity with 16 supercharges: $\mathcal{N} = (1,1)$ in six dimensions. Type IIA has 32 supercharges. Compactify on K3, whose SU(2) holonomy preserves half, and the result is again six-dimensional $\mathcal{N} = (1,1)$ supergravity. The supersymmetry matches. This is necessary, not sufficient, but it is the first thing to check and it is the reason K3 rather than T^4 appears on the IIA side: reducing IIA on T^4 would keep all 32 supercharges and land on a maximal theory, the wrong amount of supersymmetry to meet the heterotic string.

Moduli space. Count the massless scalars. Heterotic on T^4 has the six-dimensional dilaton, and its remaining moduli are the Narain data of the internal conformal field theory: the metric and B-field on T^4 (16 real parameters) and the Wilson lines of the rank-16 gauge group around the four circles ($4 \times 16 = 64$ parameters), for 80 in all. These fill the Narain moduli space

$$\mathcal{M}_{4,20} \;=\; O(4,20;\mathbb{Z})\backslash O(4,20)/\big(O(4)\times O(20)\big), \qquad \dim_{\mathbb{R}} \mathcal{M}_{4,20} = 80, \tag{7.20}$$

with the discrete $O(4,20;\mathbb{Z})$ the T-duality group acting on the Narain lattice $\Gamma_{4,20} = \Gamma_{4,4} \oplus (\mathrm{E}_8 \oplus \mathrm{E}_8)$ (Appendix G). Now count on the IIA side. The moduli of the K3 sigma model are the Ricci-flat metrics on K3 and the B-field. The moduli space of unit-volume Einstein metrics on K3 is 57-dimensional; adding the overall volume gives 58, and the B-field, a class in $H^2(\mathrm{K3},\mathbb{R})/H^2(\mathrm{K3},\mathbb{Z})$, adds 22, for 80. These 80 real moduli fill precisely the same space (7.20). The reason is (7.19): the metric-and-B-field data of a K3 sigma model are governed by the even self-dual lattice

$$H^{\mathrm{even}}(\mathrm{K3},\mathbb{Z}) \;=\; H^0 \oplus H^2 \oplus H^4 \;\cong\; U \oplus \Gamma_{3,19} \;=\; \Gamma_{4,20}, \tag{7.21}$$

the same lattice, up to the unique isometry, as the heterotic Narain lattice. Two moduli spaces built on the same even self-dual lattice of signature $(4,20)$ are the same space. What is a torus-plus-Wilson-line datum on one side is a K3-metric-plus-B-field datum on the other.

The two dilatons are what makes this a duality of the interesting kind. Reduce both descriptions to the six-dimensional Einstein frame and the dilatons are related by

$$\phi_{\mathrm{IIA}} \;=\; -\,\phi_{\mathrm{het}}, \tag{7.22}$$

a strict coupling inversion, while the 80 Narain moduli are identified directly. Weakly coupled heterotic string theory is strongly coupled IIA on K3, and the other way round. This is an S-type duality in six dimensions, invisible to perturbation theory on either side, and its cornerstone piece of evidence is that the heterotic fundamental string is realised on the IIA side as the solitonic NS5-brane wrapped on the whole of K3; the tension and worldsheet bookkeeping behind that identification are in the remark at the end of Section 7.3.6.

The twenty-four gauge fields. Both theories carry an abelian gauge group of rank 24 at a generic point of (7.20). On the heterotic side these are the 16 Cartan gauge fields of the ten-dimensional group together with 4 Kaluza-Klein photons $g_{\mu m}$ and 4 from $B_{\mu m}$: $16 + 4 + 4 = 24$. On the IIA side the 24 are organised by (7.21). The Ramond-Ramond one-form C_1 gives one

vector; the three-form C_3 reduced on the 22 harmonic two-forms gives 22; and C_3 with all three legs in six dimensions is, in six dimensions, dual to a vector, supplying the last one. The count is

$$\underbrace{1}_{C_1} + \underbrace{22}_{C_3 \text{ on } H^2} + \underbrace{1}_{\star C_3 \text{ in 6D}} = 24, \tag{7.23}$$

one vector for each generator of $H^{\text{even}}(\text{K3})$, exactly matching $\Gamma_{4,20}$. The charges are carried by even-dimensional D-branes wrapping even cycles: a D0-brane (a point, the generator of H^4 by Poincaré duality), a D2-brane on each of the 22 two-cycles, and a D4-brane on all of K3 (the generator of H^0). The perturbative heterotic charge lattice and the D-brane charge lattice of IIA on K3 are the same $\Gamma_{4,20}$.

7.3.6 Gauge symmetry becomes geometry

The rank of the gauge group is 24 everywhere in (7.20), but on special loci extra vectors become massless and the group enhances to something non-abelian. What that enhancement *is*, physically, is completely different on the two sides, and comparing them is the payoff of the whole chapter.

The heterotic side: a lattice. Perturbative heterotic gauge bosons come from the left-moving sector. A state with right-moving momentum $p_R = 0$ (so that it is massless from the right) and left-moving momentum p_L in the Narain lattice with $p_L^2 = 2$ is a massless vector, and its charge is a root. At a generic point of (7.20) no lattice vector meets $p_R = 0$ with $p_L^2 = 2$, so the only massless vectors are the 24 Cartan generators and the gauge group is $\text{U}(1)^{24}$. On special subloci (where a radius hits its self-dual value, or a Wilson line switches off) extra lattice vectors of length-squared two appear, and U(1) factors enhance to non-abelian groups. Because every root has $p_L^2 = 2$, all roots have the same length, so the enhanced groups are always simply-laced: A_n, D_n, E_6, E_7, E_8. This is the level-one current algebra fact of Appendix G: a heterotic gauge symmetry is an ADE symmetry, read off from the roots sitting in a lattice.

The IIA side: a singularity. On the IIA side, a massless vector needs a massless charged state, and the charged states are wrapped D-branes. A D2-brane wrapping a holomorphic two-sphere Σ of area A is a particle in six dimensions with BPS mass set by the complexified area,

$$M_{\text{D2}} \propto \left| \int_\Sigma (B + iJ) \right| = |\, b_\Sigma + iA \,|, \tag{7.24}$$

where J is the Kähler form and $b_\Sigma = \int_\Sigma B$ the B-field period (D-brane BPS masses are derived in Chapter 6). Shrink the sphere, $A \to 0$, with the B-flux also tuned to zero, and this state becomes massless. A (-2)-sphere and its orientation reversal give a $W^\pm$ pair, and the U(1) from C_3 on the two-form dual to Σ enhances to SU(2). Collapse a configuration of (-2)-spheres whose intersection matrix is minus the Cartan matrix of an ADE algebra and the wrapped D2-branes fill out the roots of that algebra, enhancing the gauge group to the corresponding ADE group. A collection of (-2)-spheres contracting to a point is exactly a du Val (Kleinian) singularity $\mathbb{C}^2/\Gamma_{\text{ADE}}$ of the K3 surface, and the ADE classification of those singularities *is* the ADE classification of the gauge group. The $T^4/\mathbb{Z}_2$ model makes the smallest case concrete: each of the 16 blow-up spheres, collapsed with zero B-flux, is an A_1 singularity supporting an SU(2).

Example 7.9 (The A_1 singularity, end to end)**.** Take the simplest enhancement, a single collapsing sphere, using the $T^4/\mathbb{Z}_2$ model of Section 7.3.4. Resolve one fixed point into a (-2)-sphere Σ, so $\Sigma \cdot \Sigma = -2$, and set the B-period $b_\Sigma = \int_\Sigma B$ to zero. At finite area $A = \int_\Sigma J$ the IIA spectrum contains two BPS particles from (7.24): a D2-brane wrapping Σ and a D2-brane wrapping Σ with reversed orientation. They have equal mass $M_{\text{D2}} \propto A$ and opposite charge under the U(1) that comes from C_3 reduced on the two-form Poincaré-dual to Σ, so they are

a W^+ and a W^- waiting to happen. Now shrink the sphere, $A \to 0$. Both states become massless, and together with the original photon they fill out the three gauge bosons of SU(2): the $W^\pm$ from the two wrapping orientations and the W^0 from the C_3 photon itself. The U(1) has enhanced to SU(2), with Σ acting as the single simple root and $\Sigma \cdot \Sigma = -2$ reproducing the length-squared-two normalisation of an A_1 root. The collapsed geometry is the $\mathbb{C}^2/\mathbb{Z}_2$ singularity of Section 7.3.4, and SU(2) is its ADE label. Turn the B-flux back on, $b_\Sigma \neq 0$: the $W^\pm$ keep a mass $\propto |b_\Sigma|$ even at $A = 0$, and SU(2) breaks to U(1) without loss of rank, so the flux is a Coulomb-branch modulus. The A_{n-1} exercise below chains $n-1$ of these spheres together.

Gauge symmetry becomes geometry

The same six-dimensional physics, a point of enhanced ADE gauge symmetry, is on the heterotic side a set of length-squared-two vectors in the Narain lattice, a purely algebraic and perturbative datum, and on the IIA side a set of two-spheres collapsing to an ADE singularity of K3, a purely geometric and non-perturbative datum. The dictionary $\Gamma_{4,20} \leftrightarrow H^{\text{even}}(\text{K3})$ of (7.21) carries the roots of the gauge algebra onto the vanishing cycles of the singularity. A quantum number becomes a shape.

Two numbers need to be kept apart here. The rank-16 gauge group of the ten-dimensional heterotic string sits inside K3 as the two $(-\text{E}_8)$ summands of (7.19): there is room to collapse cycles into an $\text{E}_8 \times \text{E}_8$ singularity and recover the full heterotic group, and the same H^2 accommodates the D_{16}-type configuration matching SO(32). Independently, the negative-definite part of $\Gamma_{3,19}$ has rank 19, and this caps the largest ADE singularity a single K3 surface can carry. That geometric cap is rank 19, not rank 16; the two are statements about different things. In the six-dimensional duality the perturbative heterotic gauge symmetry can in fact reach rank 20, enhanced along the $\Gamma_{4,4}$ momentum and winding directions of (7.20) that have no counterpart in the collapse of K3 cycles. So the ten-dimensional bound (rank 16), the geometric cap on a K3 singularity (rank 19), and the maximal six-dimensional heterotic enhancement (rank 20) are three separate facts, and conflating them is a common slip.

This mechanism, a non-abelian gauge theory produced by branes probing a singularity, is not special to K3. Replacing K3 by a local ADE singularity fibred over a base, and D-branes by the appropriate wrapped branes, turns the recipe into a machine for engineering prescribed gauge theories and matter from geometry. That is geometric engineering, and Chapter 10 is built on the seed planted here.

Remark 7.10 (Beyond the lectures: the heterotic string from a wrapped fivebrane). The claim that $\phi_{\text{IIA}} = -\phi_{\text{het}}$ can be made almost tactile. The heterotic string tension in the IIA description comes out proportional to $1/g_s^{\text{IIA}}$, the hallmark of a soliton rather than a fundamental string, and the object with the right tension and charge is the IIA NS5-brane wrapped on K3: a five-brane has a six-dimensional worldvolume, four of whose directions wrap K3, leaving a string in six dimensions. Its worldsheet degrees of freedom, counted from the zero modes of the wrapped fivebrane, reproduce the left- and right-moving field content of the heterotic string on T^4, including the current algebra that carries the gauge group. The strong-coupling heterotic string is a brane that has eaten a K3.

Remark 7.11 (Beyond the lectures: self-duality of IIB on K3, and F-theory). The same K3 read through a different string gives a different lesson. IIB on K3 is a chiral six-dimensional $\mathcal{N} = (2,0)$ theory, and it is essentially self-dual under a duality group again controlled by the K3 lattice. Fibre the K3 elliptically and follow only the complex-structure data of the fibre, and one is varying the IIB axio-dilaton τ over a base: this is the seven-brane and (p,q)-string geometry of Section 7.1, promoted to F-theory (Chapter 11). K3, the smallest manifold on which gauge symmetry becomes geometry, is also the smallest arena in which the IIB $\text{SL}(2,\mathbb{Z})$ becomes the monodromy of an elliptic fibration.

A short checklist for the section. You should now be able to:

- name the link (T-duality, S-duality, the orientifold Ω, or an M-theory lift) joining any two nodes of Fig. 7.1, and say what each does to the string coupling;
- read off from Table 7.1 the U-duality group $\mathrm{E}_{n(n)}(\mathbb{Z})$ in a given dimension and locate its perturbative T-duality and non-perturbative S-duality subgroups;
- quote the K3 cohomology and the lattice (7.19), and use the Mukai identification (7.21) to match heterotic Narain data to K3 sigma-model data;
- describe an enhanced ADE gauge symmetry from both sides, as length-squared-two vectors in $\Gamma_{4,20}$ on the heterotic side and as a collapsing ADE configuration of (-2)-spheres on the IIA side.

Exercise 7.5 (Matching the twenty-four charges)**.** Work out both charge lattices and check that they agree.

1. On the heterotic side, list the 24 abelian gauge fields of the T^4 reduction (16 Cartan, 4 from $g_{\mu m}$, 4 from $B_{\mu m}$) and identify the lattice of electric charges as $\Gamma_{4,20} = \Gamma_{4,4} \oplus (\mathrm{E}_8 \oplus \mathrm{E}_8)$.
2. On the IIA side, reproduce the count (7.23) and identify each vector with a generator of $H^{\mathrm{even}}(\mathrm{K3})$. Which wrapped D-brane carries the charge dual to C_1?
3. Using (7.21), argue that no additional discrete data are needed to make the two lattices agree: an even self-dual lattice of signature $(4, 20)$ is unique up to isometry (Appendix G).

Exercise 7.6 (An A_{n-1} singularity gives $\mathrm{SU}(n)$)**.** The resolution of $\mathbb{C}^2/\mathbb{Z}_n$ contains a chain of $n-1$ rational curves $\Sigma_1, \dots, \Sigma_{n-1}$, each a (-2)-sphere, with $\Sigma_i \cdot \Sigma_{i+1} = 1$ and $\Sigma_i \cdot \Sigma_j = 0$ for $|i-j| > 1$. The A_1 case is worked in Theorem 7.9.

1. Write down the $(n-1) \times (n-1)$ intersection matrix and check that it equals $-C$, with C the Cartan matrix of A_{n-1}.
2. As all Σ_i collapse (areas and B-fluxes to zero), count the wrapped-D2 states of (7.24) that become massless, including bound wrappings on $\Sigma_i \cup \Sigma_{i+1} \cup \cdots$, and match them to the positive roots of A_{n-1}. Confirm the total, $n(n-1)$ non-zero roots plus $n-1$ Cartan generators, is $\dim \mathrm{SU}(n) = n^2 - 1$.
3. Explain why keeping a non-zero B-flux through some Σ_i leaves the corresponding W-bosons massive, and hence breaks $\mathrm{SU}(n)$ to a subgroup of the same rank. What is the maximal A-type singularity K3 can support, given (7.19)?

Exercise 7.7 (U-duality contains S and T)**.** Take type II on T^2, with U-duality group $\mathrm{SL}(3,\mathbb{Z}) \times \mathrm{SL}(2,\mathbb{Z})$ from Table 7.1.

1. Identify the $\mathrm{SL}(2,\mathbb{Z})$ factor with the IIB S-duality of Section 7.1, and locate the perturbative T-duality group $O(2,2;\mathbb{Z})$ inside $\mathrm{SL}(3,\mathbb{Z}) \times \mathrm{SL}(2,\mathbb{Z})$. (Recall $O(2,2) \cong \mathrm{SL}(2) \times \mathrm{SL}(2)$ up to discrete factors.)
2. The $\mathrm{SL}(3,\mathbb{Z})$ acts geometrically on M-theory on T^3. Which of its generators is invisible in the perturbative type IIB description, and why does its existence force the type II string coupling to be a geometric modulus of a hidden torus?

Where this is going: the web is only as trustworthy as the quantities that survive the strong-coupling extrapolations it rests on. Section 7.4 makes that precise by matching BPS states across every link, turning the conjectural equalities of this section into a tightly overdetermined and consistent system.

7.4 Why the web is trustworthy

The big picture

Every duality in this chapter is the same kind of claim: two Lagrangians that look nothing alike describe one physics, with the dictionary sending the weak-coupling end of one to the strong-coupling end of the other. That asymmetry is the difficulty. Perturbation theory reaches only small coupling, so no calculation ever sees both sides at once. The quantities that survive the trip from weak to strong are the masses and multiplicities of BPS states, frozen by supersymmetry (Appendix C.3). Computing them at weak coupling on each side and demanding agreement is the evidence for the web, and where a worldsheet or an index computation is available it hardens into proof. This section spends the protection theorem that Chapter 6 set up.

The three preceding sections asserted dualities. Type IIB is invariant under $\mathrm{SL}(2,\mathbb{Z})$ (Section 7.1); type IIA at strong coupling grows an eleventh dimension whose low-energy theory is eleven-dimensional supergravity (Section 7.2); heterotic strings on T^4 describe the same physics as type IIA on K3 (Section 7.3). Each is a statement about effective theories, and each equates a regime we can calculate with one we cannot. This section audits all three, together with two links they spawn: type I as the strong-coupling $\mathrm{SO}(32)$ heterotic string, and the U-duality multiplets that appear once we compactify. The question is why we believe any of it.

7.4.1 The problem, and the one quantity that survives it

Take a duality that sends coupling g_s to $1/g_s$. At $g_s \ll 1$ the left description is under perturbative control and the right is strongly coupled; at $g_s \gg 1$ the roles swap. There is no window in which both descriptions are weakly coupled, so a direct comparison of, say, a scattering amplitude is impossible: on either side one of the two calculations is a strong-coupling calculation, and strong-coupling calculations are exactly what we cannot do. A duality of this type can never be checked by comparing generic observables. It can only be checked on quantities that are the same at weak and at strong coupling.

Supersymmetry supplies such quantities. A BPS state saturates the mass bound of Appendix C.3,

$$M = |Z|, \tag{7.25}$$

with Z the central charge fixed, on an irreducible representation, by the quantised charges the state carries. Three facts, established in Appendix C.3.3, make Eq. (7.25) rigid under changes of coupling. The dimension of a short multiplet is an integer that cannot jump continuously, so a state that is BPS stays BPS unless it collides with the exact quantum numbers needed to pair up into a long multiplet. The central charge is a fixed function of the charges, which live on a lattice and cannot drift. And the only place a BPS state can appear or disappear is a wall of marginal stability, a real-codimension-one locus in moduli space. Away from such walls the mass Eq. (7.25) and the degeneracy at fixed charge are constants of the coupling, which is exactly the invariance a strong/weak duality can exploit.

The recipe is then forced on us. Identify a BPS state on side A, where it is an elementary or perturbative object, and its image on side B, where the dictionary may turn it into a soliton or a wrapped brane. Compute its mass at the *weak*-coupling end of each side, where each computation is honest, and demand

$$\underbrace{M_A(\text{charges}; g_A)}_{\text{elementary, } g_A \ll 1} \;=\; |Z| \;=\; \underbrace{M_B(\text{charges}; g_B)}_{\text{brane/soliton, } g_B \ll 1}, \tag{7.26}$$

after translating charges and couplings through the duality map. Both masses are protected, so each equals its own continuation to the regime where the state is heavy and inaccessible. The equality in Eq. (7.26) therefore compares two legitimate weak-coupling numbers while secretly connecting weak coupling to strong.

Careful

The protected quantity is an index, not a naive degeneracy. Only a signed count, a helicity supertrace $\mathrm{Tr}(-1)^{2J_3}(2J_3)^{2n}$ tuned to annihilate long multiplets, is guaranteed constant across moduli space (Appendix C.3.3). Two theories can have different numbers of BPS states at fixed charge and still be dual, provided the difference sits in multiplets that could recombine into long ones and so cancel in the index. Matching raw state counts is the right thing to do only for the shortest, $\frac{1}{2}$-BPS multiplets, where lengthening is impossible; below that, match the index.

What follows works this recipe on five checks, in rising order of difficulty, and then takes stock of how much has really been proved.

7.4.2 The cleanest check: (p, q)-string tensions in type IIB

The first check is one we have already done, now read as a BPS matching. Section 7.1.3 derived the tension of the (p, q)-string, the bound state of p fundamental strings and q D1-branes whose charge sits in the $\mathrm{SL}(2, \mathbb{Z})$ doublet (B_2, C_2),

$$T_{(p,q)} = \frac{1}{2\pi\alpha'}\sqrt{(p - C_0\, q)^2 + \frac{q^2}{g_s^2}} \qquad \text{(string frame)}, \tag{7.27}$$

with C_0 the RR axion (Schwarz, 1995). What matters here is what this formula does across the coupling. The $(1, 0)$ string has $T = 1/2\pi\alpha'$, the fundamental-string tension; the $(0, 1)$ string has $T = 1/2\pi\alpha' g_s$, the D1 tension of Chapter 6. Read at $g_s \ll 1$ the D-string is heavy and the fundamental string light; read at $g_s \gg 1$ the ordering flips. Neither reading is perturbatively available on both sides at once, yet both are exact, because Eq. (7.27) is a BPS tension: it is $|Z|$ for the doublet charge (the central charge of a doublet-charged string, Section 7.1.3), unrenormalised between the two regimes.

The $\mathrm{SL}(2, \mathbb{Z})$ covariance is cleanest in Einstein frame, where S-duality acts geometrically. Writing $\tau = C_0 + i/g_s$, the tension is the length of the charge doublet in the coset metric of Eq. (7.8),

$$T^2_{(p,q)} = \Big(\frac{1}{2\pi\alpha'}\Big)^2 \begin{pmatrix} p & q \end{pmatrix} \mathcal{M} \begin{pmatrix} p \\ q \end{pmatrix}, \qquad \mathcal{M} = \frac{1}{\mathrm{Im}\,\tau}\begin{pmatrix} 1 & -\mathrm{Re}\,\tau \\ -\mathrm{Re}\,\tau & |\tau|^2 \end{pmatrix}, \tag{7.28}$$

the same matrix as Eq. (7.8), with the $g_s^{1/2}$ Weyl factor between this and the string-frame Eq. (7.27) being the frame bookkeeping of Section 7.1.3. Under $S : \tau \to -1/\tau$ together with $\binom{p}{q} \to \Lambda\binom{p}{q}$ for $\Lambda \in \mathrm{SL}(2, \mathbb{Z})$ (Eq. (7.7)), the matrix transforms as $\mathcal{M} \to (\Lambda^{-1})^T \mathcal{M}\, \Lambda^{-1}$ and the tension is invariant. The transformation S carries the elementary $(1, 0)$ string to the $(0, 1)$ D-string, which is why the same object is a fundamental string in one duality frame and a soliton in another. The BPS spectrum maps to itself: the lattice of (p, q)-strings is a single $\mathrm{SL}(2, \mathbb{Z})$ orbit, and the duality only relabels its points.

The same $\mathrm{SL}(2, \mathbb{Z})$ organises the five-branes into a doublet (NS5, D5) and leaves the self-dual D3 a singlet, matching the central-charge structure catalogued in Appendix C.1.6. Stability sorts the string lattice as well: the coprimality and threshold arithmetic worked in Exercise 7.1 carries over unchanged, and the one new question, whether a coprime (p, q)-string can ever go marginally stable as (g_s, C_0) vary, is the subject of the exercise.

Exercise 7.8. Coprimality and the exact threshold $T_{(dp',dq')} = d\,T_{(p',q')}$ were established in Exercise 7.1; here pin down what stays stable. Work with the positive-definite metric $\mathcal{M}(\tau)$ of Eq. (7.28).

1. Show that $T_{(p,q)} < p\,T_{(1,0)} + q\,T_{(0,1)}$ whenever $p \neq 0$ and $q \neq 0$, so a coprime (p,q)-string cannot decay into separated fundamental strings and D-strings.
2. Show more generally that for any splitting $(p,q) = (p_1,q_1) + (p_2,q_2)$ into non-parallel pieces the tension is strictly subadditive, using only positive-definiteness of $\mathcal{M}$. Conclude that a coprime (p,q)-string is stable throughout the (g_s, C_0) upper half-plane, and that the sole marginal locus, the non-coprime (parallel) threshold, is independent of τ. Contrast this with the genuine walls of marginal stability that BPS particles cross when their central-charge phases align.

(Difficulty: short; each part is a one-line inequality.)

7.4.3 Type I is the strongly coupled heterotic string

The heterotic SO(32) string and type I share a low-energy theory, ten-dimensional $\mathcal{N} = 1$ supergravity coupled to SO(32) super-Yang-Mills. The duality (Section 7.3.1) identifies them with the coupling inverted, $g_s^{\text{het}} = 1/g_s^{\text{I}}$, so a strongly coupled heterotic string should reappear as a light object in weakly coupled type I. The heterotic string is fundamental, with a tension $T_{\text{het}} = 1/2\pi\alpha'_{\text{het}}$ that does not depend on its coupling. Its candidate image is the type I D1-brane. Under the string-frame dictionary $g_s^{\text{I}} = 1/g_s^{\text{het}}$ and $\alpha'_{\text{I}} = \alpha'_{\text{het}}\, g_s^{\text{het}}$ (fixed by matching the two ten-dimensional effective actions, Section 7.3.1; frame details in Appendix I), the D-string tension is

$$T_{\text{D1}}^{\text{I}} = \frac{1}{2\pi\alpha'_{\text{I}}\, g_s^{\text{I}}} = \frac{1}{2\pi\,(\alpha'_{\text{het}} g_s^{\text{het}})\,(1/g_s^{\text{het}})} = \frac{1}{2\pi\alpha'_{\text{het}}} = T_{\text{het}}. \tag{7.29}$$

A tension match is necessary but, on its own, weak evidence. Many objects could accidentally share a mass. The decisive check is that the type I D-string is not merely as heavy as the heterotic string but *is* the heterotic string, worldsheet and all.

Quantise the type I D1-brane (Polchinski and Witten, 1995). Its worldvolume carries the usual D-brane fields, but the open strings stretching between the D1 and the thirty-two D9-branes of type I, which carry the SO(32) gauge group fixed by the tadpole condition of Chapter 6, contribute thirty-two left-moving Majorana-Weyl fermions living on the D-string. Those fermions realise a level-one SO(32) current algebra, exactly the internal sector that builds the heterotic string. The D-string worldsheet reproduces the heterotic worldsheet: the same transverse coordinates, the same chiral split, the same gauge current algebra, and the same GSO and level-matching structure. Its oscillator excitations are the perturbative heterotic states, so the duality is derived on this BPS object rather than inferred from a coincidence of masses. This is the strongest form of evidence in the section, short of a nonperturbative definition of either theory.

7.4.4 The heterotic string as a fivebrane wrapped on K3

The six-dimensional duality of Section 7.3.5, heterotic on T^4 against type IIA on K3, is again a coupling inversion (Eq. (7.22)), so the same recipe applies. The heterotic side offers a fundamental string of tension $1/2\pi\alpha'_{\text{het}}$, flat in its coupling. That string cannot map to a IIA fundamental string, whose tension is likewise coupling-independent and so cannot invert. Its image must be a soliton, and the object with the right charge is the IIA NS5-brane wrapped on the whole of K3: a fivebrane has a six-dimensional worldvolume, four legs of which wrap K3, leaving a string in the six non-compact dimensions.

Its tension is the NS5 tension times the wrapped volume,

$$T^{\mathrm{IIA}}_{\mathrm{het}} \;=\; V_{\mathrm{K3}}\, T^{\mathrm{IIA}}_{\mathrm{NS5}} \;=\; \frac{V_{\mathrm{K3}}}{(2\pi)^5\,(g_s^{\mathrm{IIA}})^2\,\ell_s^6}, \tag{7.30}$$

with the NS5 tension read from the reduction table of Section 7.2.3. This carries inverse powers of g_s^{IIA}, the signature of a solitonic rather than a fundamental string. Passing to the six-dimensional Einstein frame and using the coupling inversion Eq. (7.22) turns it into the coupling-independent tension a heterotic fundamental string must have; the frame algebra is the Weyl bookkeeping of Appendix I.

The tension is once more only the necessary part. The sharper statement is that the wrapped fivebrane carries the heterotic worldsheet. Quantise its collective coordinates: the transverse oscillations give the embedding into the six non-compact directions, while the zero modes of the self-dual two-form b_2 expanded on the 22 two-cycles of K3, split by the $(3,19)$ signature of $H^2(\mathrm{K3})$ into right- and left-movers, supply the bulk of the internal chiral content, including a current algebra of the right rank to carry the gauge group. Together with the wrapping and momentum modes, these assemble the Narain data $\Gamma_{4,20}$ of the heterotic string on T^4 through the dictionary $H^{\mathrm{even}}(\mathrm{K3}) \cong \Gamma_{4,20}$ of Eq. (7.21).

This check is real but softer than the type I derivation of Section 7.4.3. The $\mathcal{N} = (2,0)$ fivebrane theory has no Lagrangian (Section 7.2.5), so the zero-mode content is read from the cohomology of K3 rather than quantised from an action. What it delivers is the match of charges, tension scaling, and worldsheet field content that the BPS argument needs; the remark in Section 7.3.6 sketches the same picture from the fivebrane side.

7.4.5 D0-branes and the eleventh dimension

The claim of Section 7.2 is that type IIA is eleven-dimensional M-theory on a circle, with the D0-brane the first Kaluza-Klein mode. The reduction of the eleven-dimensional superalgebra Eq. (C.15) makes half of this algebraic: the momentum $P_\natural$ around the circle becomes the IIA zero-form central charge Z, so "D0 charge" and "momentum in the eleventh direction" are the same generator. The D0-brane is the $\frac{1}{2}$-BPS carrier of that charge, and its mass saturates Eq. (C.41),

$$m_{\mathrm{D0}} = \frac{1}{g_s \ell_s} = \frac{1}{R_{11}}, \qquad R_{11} = g_s \ell_s = g_s^{2/3} \ell_p, \tag{7.31}$$

the mass of the first momentum mode on a circle of radius R_{11} (the two forms of R_{11} are the string-frame and eleven-dimensional Planck-frame readings of Section 7.2).

The identification is only consistent if the whole Kaluza-Klein tower is present. Level n must be a state of mass

$$m_n = \frac{|n|}{R_{11}} = \frac{|n|}{g_s \ell_s}, \tag{7.32}$$

and, being the reduction of the unique eleven-dimensional supergraviton, it must be a single $\frac{1}{2}$-BPS multiplet of $2^8 = 256$ states (the 44 of the graviton, the 84 of the three-form, and the 128 of the gravitino). For $n = 1$ this is one D0-brane, whose ground state is exactly a 256-state $\frac{1}{2}$-BPS multiplet. For $|n| \geq 2$ it must be a bound state of $|n|$ D0-branes forming one, and only one, such multiplet.

This is a hard requirement. D0-branes are mutually BPS, so they exert no static force and their moduli space is flat (the marginal-stability edge of Appendix C.3.3). The bound state therefore has zero binding energy and sits at the very bottom of the continuum, a threshold bound state, the most delicate kind to establish. Its existence is a normalisable zero-energy ground state of the $\mathrm{U}(N)$ supersymmetric quantum mechanics on N D0-branes (the dimensional reduction of

ten-dimensional super-Yang-Mills). The U(1) centre of mass carries the free supergraviton's 256 states; the relative SU(N) sector must contribute exactly one normalisable ground state, so that the total is a single multiplet. The Witten index of the relative problem must be

$$d(N) = 1 \qquad \text{for every } N. \tag{7.33}$$

Nothing about Eq. (7.33) is forced by a symmetry; it is a genuine computation, and it came out right (Yi, 1997; Sethi and Stern, 1998; supported by the instanton analysis of Green and Gutperle, 1997). Had $d(N)$ been 0 for some N the tower would have a gap, and had it been larger than 1 the tower would be overpopulated; either way eleven-dimensional Lorentz invariance, which needs a single clean momentum tower, would fail. The eleventh dimension of M-theory stands or falls on this index, and the index confirms it.

Remark 7.12 (Beyond the lectures). The same U(N) quantum mechanics is promoted from a consistency check to a definition in the BFSS proposal (Banks, Fischler, Shenker and Susskind, 1996): M-theory in the infinite-momentum frame *is* the large-N limit of the D0-brane matrix quantum mechanics, with the n units of light-cone momentum carried by n D0-branes. The threshold bound-state count Eq. (7.33) is then not an afterthought but a boundary datum the model must reproduce to contain the free graviton at all. The eleven-dimensional supergravity scattering of gravitons has been recovered from this matrix model at leading orders, a nonperturbative window that the BPS matching first pried open.

7.4.6 U-duality multiplets of branes

Toroidal compactification turns the pairwise checks above into a single organising structure. Reduce the type II string on T^d to $D = 10 - d$ dimensions (equivalently M-theory on T^{d+1}; this is the convention of Table 7.1). The two-derivative supergravity has a continuous global symmetry, the Cremmer-Julia group $\mathrm{E}_{d+1(d+1)}(\mathbb{R})$, acting on its scalars; charge quantisation breaks it to the arithmetic subgroup $\mathrm{E}_{d+1(d+1)}(\mathbb{Z})$, the U-duality group (Hull and Townsend, 1995), derived in Section 7.3. It contains the perturbative T-duality group $O(d,d;\mathbb{Z})$ acting on the Narain lattice (Appendix G) and the $\mathrm{SL}(2,\mathbb{Z})$ S-duality, and fuses them.

Perturbatively, the point-like BPS states in D dimensions arrive in families that look unrelated: Kaluza-Klein momenta on the torus, winding fundamental strings, wrapped D-branes, wrapped M-branes and NS5-branes, and Kaluza-Klein monopoles. U-duality assembles their charges into the weights of a single irreducible representation R_1 of $\mathrm{E}_{d+1(d+1)}$.

D	$\mathrm{E}_{d+1(d+1)}(\mathbb{Z})$	point-particle rep R_1	$\dim R_1$
8	$\mathrm{SL}(3,\mathbb{Z}) \times \mathrm{SL}(2,\mathbb{Z})$	$(\mathbf{3},\mathbf{2})$	6
7	$\mathrm{SL}(5,\mathbb{Z})$	$\mathbf{10}$	10
6	$\mathrm{SO}(5,5;\mathbb{Z})$	$\mathbf{16}_s$	16
5	$\mathrm{E}_{6(6)}(\mathbb{Z})$	$\mathbf{27}$	27
4	$\mathrm{E}_{7(7)}(\mathbb{Z})$	$\mathbf{56}$	56

The BPS mass is a U-invariant. Writing q for the integer charge vector in R_1 and $V(\phi) \in \mathrm{E}_{d+1(d+1)}/K$ for the coset representative built from the scalar moduli (K the maximal compact subgroup), the central charge is $Z = V(\phi)\, q$ and the $\tfrac{1}{2}$-BPS mass is

$$M^2_{\frac{1}{2}\text{-BPS}} = q^T\, \mathcal{M}(\phi)\, q, \qquad \mathcal{M} = V^T V, \tag{7.34}$$

the direct descendant of the SL(2) formula Eq. (7.28) with the doublet metric replaced by the $\mathrm{E}_{d+1(d+1)}/K$ coset metric (the full statement, including the $\tfrac{1}{4}$- and $\tfrac{1}{8}$-BPS cases governed by higher central-charge eigenvalues, is collected in Appendix I; see Obers and Pioline, 1999). Under $q \to Uq$ and $\mathcal{M} \to (U^{-1})^T \mathcal{M}\, U^{-1}$ the mass is invariant.

Read Eq. (7.34) as a prediction. A single formula covers momentum modes, winding modes and every wrapped brane, because they are U-images of one another and their masses coincide the moment they lie on a common orbit. From the existence of one BPS state, U-duality demands the whole orbit, with masses fixed. Every state so demanded is found among the branes of Chapter 6, and the tension scalings needed to complete the multiplet, an F1 at g_s^0, a Dp at g_s^{-1}, an NS5 at g_s^{-2} (Eq. (C.41)), are exactly those computed independently, one brane at a time, in Chapters 5 and 6. That the separately computed tensions assemble into $\mathrm{E}_{d+1(d+1)}$ representations is an over-determined check: the group has far more entries than free parameters, and they all agree.

Exercise 7.9. Compactify type IIA on T^6 to $D = 4$, where the U-duality group is $\mathrm{E}_{7(7)}(\mathbb{Z})$ with point-particle representation **56**. List the point-like $\frac{1}{2}$-BPS charges: 6 Kaluza-Klein momenta, 6 winding fundamental strings, one D0, D2-branes on the $\binom{6}{2} = 15$ two-cycles, D4-branes on the 15 four-cycles, one D6 on T^6, NS5-branes on the $\binom{6}{5} = 6$ five-cycles, and 6 Kaluza-Klein monopoles. Show that they total $6 + 6 + 1 + 15 + 15 + 1 + 6 + 6 = 56$, the dimension of the **56**. Which subgroup of $\mathrm{E}_{7(7)}(\mathbb{Z})$ is visible to the perturbative type IIA string, and which generators require branes? (Difficulty: routine counting, but it draws on the full IIA brane spectrum of Chapter 6.)

The representation R_1 splits into U-orbits labelled by the preserved fraction, $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, distinguished by U-invariant polynomials in q. For $\mathrm{E}_{6(6)}$ in $D = 5$ this is a cubic $I_3(q)$; for $\mathrm{E}_{7(7)}$ in $D = 4$ it is the quartic Cartan invariant $I_4(q)$. The $\frac{1}{8}$-BPS black holes built from these charges carry entropies

$$S_{D=5} \propto \sqrt{|I_3(q)|}, \qquad S_{D=4} \propto \sqrt{|I_4(q)|}, \tag{7.35}$$

manifestly U-invariant, and a microscopic count of BPS states at fixed charge reproduces them exactly (Chapter 12). A weak-coupling degeneracy, organised by nothing but the representation theory of $\mathrm{E}_{7(7)}$, matching a strong-coupling horizon area is the most quantitative single entry in the web.

7.4.7 What is proved, and what is checked

The evidence comes in tiers, and the distinction matters.

At the top sits derivation. The type I D-string is the heterotic string because its worldsheet fields are the heterotic worldsheet fields (Section 7.4.3); the D0 tower is the Kaluza-Klein tower because the index Eq. (7.33) computes the required bound states (Section 7.4.5). The six-dimensional heterotic/K3 duality sits just below, derived on the wrapped fivebrane but resting on the less controlled $\mathcal{N} = (2, 0)$ theory (Section 7.4.4). These are constructions, not analogies, and they turn the corresponding dualities into near-theorems about the BPS sector.

Next is non-renormalisation. In the maximally supersymmetric compactifications the $\frac{1}{2}$-BPS masses Eq. (7.34) receive no quantum corrections, so once the dictionary is fixed at a single point the entire spectrum matches automatically. The check is then not a run of coincidences but a single algebraic identity, U-duality invariance, holding for every charge at once.

At the base is the sheer weight of consistent numbers. Hundreds of BPS states across dozens of duality frames, tensions computed independently for each brane, all fall into $\mathrm{E}_{d+1(d+1)}$ multiplets with the masses Eq. (7.34) demands. Any one match could be an accident. The closure of the entire system under $\mathrm{E}_{d+1(d+1)}(\mathbb{Z})$ is not something a wrong web could fake.

Remark 7.13 (Beyond the lectures). The matching extends past the BPS spectrum to protected pieces of the effective action. The R^4 coupling in type IIB, for instance, has a coefficient that is a non-holomorphic $\mathrm{SL}(2, \mathbb{Z})$ Eisenstein series $E_{3/2}(\tau, \bar{\tau})$, whose weak-coupling expansion resolves into a tree-level term, a one-loop term, and an infinite series of D-instanton contributions, each of which is separately computable and each of which comes out as the Eisenstein series predicts

(Green, Gutperle and Vanhove, 1997). This is not the mass of a single state, so the BPS argument of Section 7.4.1 does not directly cover it, yet it is protected by the same supersymmetry and is exactly U-invariant. Such threshold couplings are the next layer of evidence out from the BPS spectrum, and they agree.

None of this constructs M-theory. What we have is eleven-dimensional supergravity as a low-energy limit, a catalogue of BPS branes with exact tensions, and a web of duality frames pinned together at a dense set of protected points. The BPS matching does not prove that a single consistent quantum theory interpolates all of these limits. It proves that *if* such a theory exists, its BPS sector is uniquely determined and internally consistent, over-constrained to the point where inconsistency would have shown up long ago and did not. That is the actual status of the web, and it is a great deal more than a conjecture resting on nothing.

This closes the BPS thread opened in Chapter 6. D-brane tensions were shown there to be protected; here that protection is spent, buying the trustworthiness of every arrow in the web. The same protected data return in Chapter 12, where the count reproduces a black-hole entropy, and underwrite the holographic duality of Chapter 13. The full T, S and U dictionary, with brane-charge tables and BPS spectra tabulated frame by frame, is collected as a Rosetta stone in Appendix I. With the web secured, Chapter 8 turns to the geometry, K3 and Calabi-Yau, on which the second half of the course is built.

Chapter 8

Compactification: Kaluza-Klein, Orbifolds, K3, and Calabi-Yau

8.1 Kaluza-Klein reduction

The web of Chapter 7 left us with a single theory. Its supergravity limits, the five ten-dimensional superstrings and eleven-dimensional M-theory, are corners of one structure, and we now have to make one of them look like the world: four large dimensions, minimal supersymmetry, chiral matter. The tool is compactification. Take the extra dimensions to be a small compact manifold K, so spacetime factorises as $M_4 \times K$, and ask what a four-dimensional observer, who cannot resolve K, actually sees.

This section builds the machinery on the simplest K there is, a circle, and then a torus, and then a general curved K. The physics is contained in one idea: a field on $M_4 \times K$ is a tower of four-dimensional fields, one for each mode of the internal Laplacian, and the light ones, the zero modes, are the low-energy content. Counting zero modes is counting harmonic objects on K, which is topology. By the end we will have reduced the question "which compactification gives minimal supersymmetry in four dimensions?" to "which K carries a covariantly constant spinor?", and handed that question to Section 8.2.

Everything here is field theory on a product manifold. It applies to the two-derivative supergravities of Chapter 5 as they stand. The stringy corrections, winding modes and the enhanced symmetries they bring, belong to Chapter 6 and are cited, not re-derived.

8.1.1 The mechanism: modes on a circle

Start with a single massless scalar Φ in D dimensions, on the product $M_{D-1} \times S^1$. Write the circle coordinate as y with period $2\pi R$, so $y \sim y + 2\pi R$, and keep the external coordinates x^μ. Because y is periodic, Φ is expanded in a Fourier series rather than a Fourier integral:

$$\Phi(x, y) \;=\; \sum_{n \in \mathbb{Z}} \phi_n(x)\, e^{iny/R}. \tag{8.1}$$

The D-dimensional wave equation splits. With $\Box_D = \Box_{D-1} + \partial_y^2$ and $\partial_y^2 e^{iny/R} = -(n/R)^2 e^{iny/R}$, the equation $\Box_D \Phi = 0$ becomes an infinite set of $(D-1)$-dimensional equations,

$$\Big(\Box_{D-1} - \frac{n^2}{R^2}\Big)\phi_n(x) = 0, \qquad \text{so} \qquad m_n^2 = \frac{n^2}{R^2}. \tag{8.2}$$

One higher-dimensional field has become a *Kaluza-Klein tower*: a massless mode ϕ_0, the zero mode, and above it an evenly spaced ladder of massive scalars with masses $|n|/R$. The tower is

the imprint of the extra dimension: its rungs are the internal momenta, quantised because the circle is compact.

The single lesson of this section is already visible. At energies well below the compactification scale $1/R$, only the zero mode is accessible; every massive rung is frozen out. The low-energy theory in $D-1$ dimensions keeps the y-independent part of each field. Compactification is, at leading order, the projection onto the kernel of the internal Laplacian, and the light spectrum is fixed by the harmonic data of K. Everything that follows is this statement, applied to fields with more indices and to internal manifolds with more structure.

Example 8.1 (Why we have not seen the extra dimensions)**.** The rung spacing $1/R$ is the mass of the lightest Kaluza-Klein excitation of every field. Collider searches for such a tower, universal because gravity itself furnishes one (see Section 8.1.2), currently place $1/R$ above roughly a TeV for the most weakly constrained scenarios, and far higher for fields with Standard-Model charges. In string compactifications R is typically near the string length $\ell_s = \sqrt{\alpha'}$ (with α', ℓ_s and the string scale $M_{\rm s} \sim 1/\ell_s$ as fixed in Chapter 6), putting $1/R$ close to $M_{\rm s}$, so the tower is invisible and the zero-mode truncation is the whole of accessible physics. Keeping only the zero modes is not an approximation; it defines the low-energy effective theory.

8.1.2 Gravity on a circle: graviton, graviphoton, radion

Now reduce the field we cannot switch off: the metric. In D dimensions the graviton is a symmetric tensor g_{MN}. Split the index $M = (\mu, y)$ into external and internal. Under the $(D-1)$-dimensional Lorentz group the pieces $g_{\mu\nu}$, $g_{\mu y}$ and g_{yy} transform as a spin-2 tensor, a vector and a scalar. Reduction turns one metric into a graviton, a gauge field and a modulus before we write a single equation. The content of the calculation is to make that split covariant and read off how the fields interact.

Keep only the zero modes, meaning fields independent of y, and write the reduction ansatz in the geometrically transparent form

$$\mathrm{d}s_D^2 \;=\; e^{2\alpha\varphi(x)}\, g_{\mu\nu}(x)\,\mathrm{d}x^\mu \mathrm{d}x^\nu \;+\; e^{2\beta\varphi(x)} \big(\mathrm{d}y + A_\mu(x)\,\mathrm{d}x^\mu\big)^2 . \tag{8.3}$$

The one-form $A = A_\mu\,\mathrm{d}x^\mu$ is the off-diagonal metric packaged as a connection; the scalar φ measures the size of the fibre and is the *radion*. It is a metric modulus, distinct from the ten-dimensional dilaton, which is a modulus of a different origin; both are massless scalars, and both will haunt the moduli problem. The constants α, β are pure bookkeeping, fixed below so that the external metric $g_{\mu\nu}$ is the Einstein-frame metric and φ is canonically normalised.

The gauge symmetry is a fossil of higher-dimensional diffeomorphism invariance. Consider a reparametrisation of the fibre coordinate that depends on the external point,

$$y \;\longrightarrow\; y + \lambda(x). \tag{8.4}$$

Then $\mathrm{d}y \to \mathrm{d}y + \partial_\mu\lambda\,\mathrm{d}x^\mu$, and the invariant combination $\mathrm{d}y + A$ is preserved provided

$$A_\mu \;\longrightarrow\; A_\mu - \partial_\mu\lambda. \tag{8.5}$$

This is an abelian gauge transformation. The U(1) gauge symmetry of A_μ is the isometry of the circle, promoted to a local symmetry by allowing the shift to vary over spacetime. The massive rungs ϕ_n carry charge n under it: under (8.4) the mode $e^{iny/R}$ picks up a phase $e^{in\lambda/R}$, so the Kaluza-Klein momentum *is* the electric charge. A tower of states with charge n and mass $|n|/R$ is a robust prediction of any circle compactification, the graviphoton coupled to its own momentum modes.

Substituting (8.3) into the Einstein-Hilbert action, integrating over the circle ($\int \mathrm{d}y = 2\pi R$), and choosing

$$\alpha^2 = \frac{1}{2(D-2)(D-3)}, \qquad \beta = -(D-3)\alpha \tag{8.6}$$

to land in Einstein frame with a canonical scalar, gives the reduced action

$$S_{D-1} \;=\; \frac{1}{2\kappa_{D-1}^2}\int \mathrm{d}^{D-1}x\,\sqrt{-g}\,\Big(R - \tfrac{1}{2}\,\partial_\mu\varphi\,\partial^\mu\varphi - \tfrac{1}{4}\,e^{-2(D-2)\alpha\varphi}\,F_{\mu\nu}F^{\mu\nu}\Big), \qquad F = \mathrm{d}A. \tag{8.7}$$

The three towers of (8.2), applied to $g_{\mu\nu}$, A_μ and φ, have collapsed at zero mode to a massless graviton, a Maxwell field and a massless scalar. The exponents (8.6) answer two demands, Einstein frame and canonical normalisation; Theorem 8.2 works them out, and Exercise 8.1 runs the whole reduction at $D = 5$.

Example 8.2 (Where the exponents come from)**.** Insert the ansatz (8.3) and factorise the determinant. The external block is $(D-1)$-dimensional and carries $e^{2\alpha\varphi}$, the fibre carries $e^{2\beta\varphi}$, so

$$\sqrt{-g_D} \;=\; e^{((D-1)\alpha+\beta)\varphi}\,\sqrt{-g_{D-1}}. \tag{8.8}$$

The external curvature enters the D-dimensional Ricci scalar through the inverse metric $g^{\mu\nu} = e^{-2\alpha\varphi}g^{\mu\nu}$, contributing a further $e^{-2\alpha\varphi}$, so the Einstein-Hilbert term reduces as $e^{((D-3)\alpha+\beta)\varphi}\sqrt{-g_{D-1}}\,R_{D-1}$. Einstein frame is the demand that this prefactor be absent,

$$(D-3)\alpha + \beta = 0 \qquad\Longrightarrow\qquad \beta = -(D-3)\alpha, \tag{8.9}$$

which fixes β with no reference to normalisation. The gradients of the two warp factors, together with the fibre-metric kinetic term, then assemble into $-\frac{1}{2}(\partial\varphi)^2$ exactly when $\alpha^2 = 1/[2(D-2)(D-3)]$; any other α leaves φ non-canonical. The gauge kinetic term collects $e^{2\beta\varphi}$ from the fibre norm, $e^{-4\alpha\varphi}$ from the two external inverse metrics, and the volume factor $e^{((D-1)\alpha+\beta)\varphi}$ of (8.8); with $\beta = -(D-3)\alpha$ these collapse to the $e^{-2(D-2)\alpha\varphi}$ of (8.7).

Three features of (8.7) carry the physics. First, the Newton constants. The prefactor bookkeeping gives

$$\frac{1}{2\kappa_{D-1}^2} = \frac{2\pi R}{2\kappa_D^2}, \qquad \text{equivalently} \qquad M_{\mathrm{pl},\,D-1}^{D-3} = M_{*,D}^{D-2}\;\mathrm{vol}(K), \tag{8.10}$$

with $M_{*,D}$ the higher-dimensional gravitational scale and $\mathrm{vol}(K) = 2\pi R$ here. The lower-dimensional Planck mass is the higher-dimensional one dressed by the internal volume. This one relation is why a large internal space can separate the four-dimensional Planck scale from the string scale: for compactification of ten-dimensional gravity on a six-manifold, $M_{\mathrm{pl}}^2 = M_{*,10}^8\,\mathrm{vol}(K)$, so a big $\mathrm{vol}(K)$ makes gravity weak in four dimensions without any small higher-dimensional coupling.

Second, the gauge coupling runs on the modulus. The factor $e^{-2(D-2)\alpha\varphi}$ in front of F^2 means the effective coupling of the graviphoton is set by the radion vacuum value, $g^{-2} \propto e^{-2(D-2)\alpha\langle\varphi\rangle}$. Physically the coupling tracks the circumference of the fibre. Since $\beta < 0$, the direction $\varphi \to +\infty$ is the one along which the circle shrinks, and the negative exponent then drives $g^{-2} \to 0$, so a shrinking internal circle is a strengthening four-dimensional coupling. This is the first instance of a general fact: in a compactification the low-energy couplings are not constants of nature but expectation values of moduli.

Third, and most important for the rest of the chapter, φ has no potential. The action (8.7) is flat in φ: its equation of motion is solved by any constant $\langle\varphi\rangle$, and the radius of the circle is undetermined by the dynamics. The radion is a *modulus*, a massless scalar parametrising a continuous family of vacua. This is the prototype of the moduli problem. A massless scalar with gravitational-strength couplings is a fifth force and a cosmological embarrassment, and

a flat direction means the compactification scale is not fixed. The whole apparatus of flux compactification in Chapter 9 exists to lift flat directions like this one.

Remark 8.3 (Beyond the lectures: is the truncation consistent?). We kept only zero modes and threw away every massive rung. Is that legal? A truncation is *consistent* if every solution of the truncated theory lifts to a solution of the full one, that is, if the discarded fields can be set to zero without their equations of motion forcing them back on. On a circle, and more generally on a torus or a group manifold reduced on its left-invariant forms, the truncation to the singlet (invariant) sector is consistent, because a product of two invariant modes is again invariant and cannot source a non-invariant one. On a general curved K this fails: two zero modes can source a massive mode through the nonlinear couplings, and the "consistent truncation" keeping only the massless fields is not guaranteed to solve the higher-dimensional equations exactly. What is always legitimate is the effective-field-theory statement: the light fields, with couplings read off from the reduction, correctly capture physics below $1/R$. That is all we need, and it is what (8.7) delivers.

8.1.3 Gauge fields from isometry, and Wilson lines

The circle taught the general rule. An isometry of the internal manifold, a continuous symmetry of its metric generated by a Killing vector, becomes a gauge symmetry of the reduced theory, with a gauge field sitting in the mixed components $g_{\mu m}$. If K has isometry group G, reduction produces a Yang-Mills theory with gauge group G in the lower dimension. This was Kaluza and Klein's original hope: geometry as the source of gauge interactions. A torus T^d has isometry group $\mathrm{U}(1)^d$ and yields d graviphotons; a round sphere S^n has isometry $\mathrm{SO}(n+1)$ and yields non-abelian gauge fields. The dream in its pure form does not reach the Standard Model, since obtaining chiral fermions this way is obstructed, but the mechanism is right, and string theory supplies the gauge groups by other means.

There is a second, independent way to get four-dimensional gauge and scalar fields, available whenever the higher-dimensional theory already carries a gauge field A_M, as the heterotic and type I strings do. Split $A_M = (A_\mu, A_m)$. The external components A_μ reduce to a lower-dimensional gauge field. The internal components A_m reduce to scalars. On the circle, a constant internal component $A_y = a$ looks locally like pure gauge, but on a topologically nontrivial cycle it is physical: the gauge-invariant datum is the holonomy, the *Wilson line*

$$W = \exp\Big(i \oint_{S^1} A_y \, \mathrm{d}y\Big) = \exp(2\pi i \, R\, a). \tag{8.11}$$

A Wilson line is a modulus: for an abelian gauge group it is a continuous angle $\vartheta := Ra \in [0, 1)$, valued in the flat connections on K. It does three things at once.

It shifts the Kaluza-Klein spectrum of charged fields. A field of charge q under the internal $\mathrm{U}(1)$ has internal covariant derivative $D_y = \partial_y - iqa$, so its modes carry $D_y\, e^{iny/R} = i(n/R - qa)\, e^{iny/R}$ and its tower masses become

$$m_n^2 = \frac{(n - q\vartheta)^2}{R^2}, \qquad \vartheta = Ra. \tag{8.12}$$

The Wilson line interpolates continuously between integer momentum sectors, and at generic ϑ no charged state is massless. It can break the gauge group. For non-abelian G, a Wilson line is a holonomy matrix, and the surviving gauge symmetry is its commutant, the subgroup that commutes with W. This is how continuous Wilson-line moduli Higgs a gauge group without a charged scalar getting a vacuum value: the breaking is stored in the geometry of the internal connection. And it contributes flat directions, more massless scalars with no potential, adding to the moduli count.

Remark 8.4 (Beyond the lectures: Scherk-Schwarz and twisted reductions). A Wilson line that is not flat but wound, so that fields come back to themselves only up to a global symmetry transformation as $y \to y + 2\pi R$, is a *Scherk-Schwarz* reduction. If the symmetry used is an R-symmetry, one obtains masses for the gravitini and breaks supersymmetry at the scale $1/R$; if it is an ordinary global symmetry, one gets a controlled tower of masses. This is the same shift (8.12) with ϑ tied to a charge under the twisting symmetry, generalised from an internal gauge field to any global symmetry of the higher-dimensional theory. Scherk-Schwarz breaking is the field-theory ancestor of the freely acting orbifolds of Section 8.3 and one of the few calculable ways to break supersymmetry by geometry.

8.1.4 The torus and its moduli

Stack circles. A torus $T^d = \mathbb{R}^d/\Lambda$, the quotient of $\mathbb{R}^d$ by a lattice Λ, carries a flat metric G_{mn} and, in the string theories, a constant internal B-field B_{mn}. Both are constant, both solve the internal equations trivially (a flat torus is Ricci-flat), and both reduce to towers whose zero modes are the moduli. Count them.

The internal metric is a constant symmetric $d \times d$ matrix, so it contributes

$$\#\, G\text{-moduli} = \tfrac{1}{2}d(d+1) \tag{8.13}$$

massless scalars: the size and shape of the torus. The internal B-field is a constant antisymmetric matrix,

$$\#\, B\text{-moduli} = \tfrac{1}{2}d(d-1). \tag{8.14}$$

The isometry group $\mathrm{U}(1)^d$ gives d graviphotons from $g_{\mu m}$, and the B-field gives a further d vectors from $B_{\mu m}$, so a toroidal reduction of ten-dimensional supergravity generically has gauge group $\mathrm{U}(1)^{2d}$ from the closed-string sector alone, enlarged by any Wilson lines of the gauge fields already present.

For the case that matters, $d = 6$ on the way to four dimensions,

$$\tfrac{1}{2}\cdot 6\cdot 7 = 21 \text{ metric moduli}, \qquad \tfrac{1}{2}\cdot 6\cdot 5 = 15 \ B\text{-field moduli}, \tag{8.15}$$

for 36 Neveu-Schwarz moduli before the gauge and Ramond-Ramond sectors are counted. A torus preserves all supersymmetry, so this is a maximally supersymmetric compactification, too symmetric to be realistic. Its value is as the exactly solvable base of the ladder. The next rungs, orbifolds and Calabi-Yau spaces, are deformations and quotients of this that break most of the supersymmetry while keeping some.

The string sees more than the field theory

Everything in this subsection is the reduction of the two-derivative supergravity. A string wrapping a cycle of T^d adds *winding* states, absent from field theory, with the momentum-winding spectrum and the $R \leftrightarrow \alpha'/R$ T-duality of Chapter 6. The metric and B-field moduli then combine into a single Narain moduli space $\mathrm{O}(d,d)/(\mathrm{O}(d)\times\mathrm{O}(d))$, acted on by the T-duality group $\mathrm{O}(d,d;\mathbb{Z})$, and at special points the $\mathrm{U}(1)^{2d}$ enhances to a non-abelian group. This structure, the Narain lattice $\Gamma_{d,d}$ and its duality group, is developed once in Appendix G and used in Chapter 7; we cite it rather than rebuild it. The counting (8.15) is the correct supergravity moduli count and the right starting point.

8.1.5 Curved internal spaces and the zero-mode dictionary

Flat tori keep all the supersymmetry because they are flat. To break some, curve the internal space. Take the product ansatz

$$\mathrm{d}s_{10}^2 = g_{\mu\nu}(x)\,\mathrm{d}x^\mu \mathrm{d}x^\nu + g_{mn}(y)\,\mathrm{d}y^m \mathrm{d}y^n, \qquad \mu = 0,\dots,3, \quad m = 1,\dots,6, \tag{8.16}$$

with M_4 maximally symmetric. Demanding a four-dimensional Minkowski vacuum from pure gravity forces the two factors to be separately Ricci-flat, $R_{\mu\nu} = 0$ externally and

$$R_{mn}(g) = 0 \tag{8.17}$$

on the internal manifold K. Ricci-flatness of K is the leading-order vacuum condition, and it is the geometric door to the rest of the chapter: the Calabi-Yau condition is Ricci-flatness plus a complex structure, and Section 8.5 arrives at it from exactly here.

The light spectrum is again zero modes, now zero modes of Laplacians on a curved K, and the count becomes topological. The rule generalises (8.2): a p-form field C_p in ten dimensions reduces to a set of four-dimensional fields, one for each harmonic form on K, because the massless internal profiles are the harmonic representatives. Expanding

$$C_p(x, y) = \sum_k \sum_{\omega_k} c^{(k)}_{p-k}(x) \wedge \omega_k(y), \qquad \omega_k \in H^k(K), \tag{8.18}$$

a k-form harmonic on K carries a $(p-k)$-form in four dimensions, and the number of these is the Betti number $b_k(K) = \dim H^k(K)$. A p-form spills out four-dimensional forms of every lower degree, weighted by the Betti numbers of K. The four-dimensional spectrum is read off the cohomology of K: scalars from b_0 and top-degree pieces, vectors from intermediate cohomology, and so on. This is the dictionary the rest of the chapter fills in, and it is why Hodge numbers become particle counts.

The metric brings its own moduli. Deformations δg_{mn} that preserve the Ricci-flat condition (8.17) are massless scalars, the geometric moduli of K: the sizes and shapes of its cycles. Counting them is a cohomology problem on K, and on a Calabi-Yau threefold it splits cleanly into the $h^{1,1}$ Kähler (size) moduli and the $h^{2,1}$ complex-structure (shape) moduli of Section 8.5. The apparatus, harmonic forms, Betti and Hodge numbers, the Hodge diamond, is the differential geometry of Appendices D and E, assumed as pre-reading. Here we have only established the mechanism: massless four-dimensional fields are harmonic modes on K, so their number is fixed by the topology of K, not by any coupling.

Ricci-flat is the leading condition, not the last word

The clean split (8.16) with a Ricci-flat K is the vacuum of pure gravity at the two-derivative level. Turning on background fluxes for the p-form fields, or including the α' corrections of Chapter 5, back-reacts on the geometry: the metric acquires a warp factor $\mathrm{d}s^2 = e^{2\mathcal{A}(y)}\eta_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu + \dots$ and K is deformed away from Ricci-flat. Those corrections are the tools that lift the moduli of this section, and they are the subject of Chapter 9. The unwarped Ricci-flat product is the correct starting point, the locus around which the flux expansion is organised.

8.1.6 Fermions and the supersymmetry question

Fermions reduce the same way, and they carry the decisive information. A ten-dimensional spinor decomposes under $\mathrm{SO}(1,3) \times \mathrm{SO}(6)$, the Lorentz group of M_4 times the tangent-space rotations of K, into a sum of four-dimensional spinors labelled by internal spinor components. The massless four-dimensional fermions are the zero modes of the internal Dirac operator $\not{D}_K$: a solution $\eta(y)$ of $\not{D}_K \eta = 0$ pairs with a four-dimensional spinor to give a massless field. The count of massless fermions is the count of harmonic spinors, the fermionic analogue of the Betti-number counting above.

Supersymmetry sharpens this. A supercharge survives compactification when the supersymmetry variation of every fermion vanishes in the vacuum. For the gravitino ψ_M, whose variation in a

bosonic background is, at leading order with fluxes off,

$$\delta\psi_M = \nabla_M \epsilon = 0, \tag{8.19}$$

this is the statement that the supersymmetry parameter ϵ is covariantly constant. The variation descends from the ten-dimensional supergravity of Chapter 5, and its full flux-dependent form is (8.23). Split $\epsilon(x, y) = \zeta(x) \otimes \eta(y)$ into a four-dimensional spinor and an internal one. The external part ζ is unconstrained (flat space); the internal part must satisfy

$$\nabla_m \eta = 0. \tag{8.20}$$

An unbroken supercharge is a covariantly constant, or *parallel*, spinor on K. The number of independent solutions of (8.20) is the number of preserved supercharges, a fixed property of the geometry of K.

That reframes the entire problem. "How much supersymmetry survives?" becomes "how many parallel spinors does K carry?", and parallel spinors are controlled by the holonomy group of K, the group of rotations a spinor undergoes on parallel transport around loops. A generic six-manifold has holonomy SO(6) and no parallel spinor, so it breaks all supersymmetry; a torus has trivial holonomy and keeps all of it. The interesting middle, where a single internal spinor is parallel and four-dimensional $\mathcal{N} = 1$ survives, requires the holonomy to sit inside a special subgroup of SO(6). Which subgroup, and why a parallel spinor forces the holonomy to shrink at all, is Berger's question. Section 8.2 takes it up, working from the condition (8.20) we have just derived and reading off the manifolds that satisfy it. The explicit spaces on which it is realised, the orbifolds of Section 8.3, K3 in Section 8.4 and the Calabi-Yau threefold of Section 8.5, then feed their harmonic data through the zero-mode dictionary (8.18) to become the four-dimensional spectrum.

Exercise 8.1 (Five dimensions to four: graviton, graviphoton, radion)**.** Carry out the reduction of Section 8.1.2 explicitly for $D = 5$. Take pure Einstein gravity in five dimensions, insert the ansatz (8.3) with $\alpha^2 = 1/12$ and $\beta = -2\alpha$ from (8.6), and reduce on the circle. Verify that the residual y-diffeomorphism (8.4) acts on A_μ as the gauge transformation (8.5). Show that the result is four-dimensional Einstein gravity coupled to a Maxwell field with kinetic term $-\frac{1}{4}e^{-\sqrt{3}\varphi}F^2$ and a single canonically normalised scalar with no potential, matching (8.7). Confirm the Newton-constant relation (8.10) and identify which sign of $\langle\varphi\rangle$ is the large-radius, weak-graviphoton-coupling limit.

Exercise 8.2 (The charged Kaluza-Klein tower and its Wilson line)**.** A complex scalar of charge q under an internal U(1) gauge field with constant component $A_y = a$ lives on $M_4 \times S^1$. Starting from its ten- (or five-) dimensional equation of motion, derive the shifted tower (8.12). Show that the spectrum is invariant under $\vartheta \to \vartheta + 1$ accompanied by a relabelling $n \to n + q$ of the momentum, so the Wilson line is genuinely an angle. For which values of ϑ does a charged massless mode exist, and how does the continuous modulus ϑ interpolate between them? Comment on how this realises gauge-symmetry breaking with no charged scalar acquiring a vacuum value.

Exercise 8.3 (Moduli counting on the torus)**.** Reduce ten-dimensional Neveu-Schwarz supergravity, the metric, B-field and dilaton, on T^6 keeping only constant internal profiles. Reproduce the counts (8.15): 21 metric moduli, 15 B-field moduli, one dilaton, and the $\mathrm{U}(1)^{12}$ of graviphotons and B-field vectors. Then, using (8.18), explain why passing from T^6 to a curved K replaces these fixed numbers by the Betti numbers $b_k(K)$, and identify which Betti numbers of the torus, $b_k(T^6) = \binom{6}{k}$, are responsible for each of the tallies above.

8.2 Reduced holonomy and preserved supersymmetry

The Kaluza-Klein reduction of Section 8.1 left one sharp condition. A supercharge survives compactification when the internal manifold carries a covariantly constant spinor, $\nabla_m \eta = 0$ in (8.20), and the number of surviving supercharges is the number of independent such spinors. On the flat torus every spinor is parallel, nothing is broken, and the four-dimensional theory keeps all of it: the maximal $\mathcal{N} = 4$ for the heterotic string and $\mathcal{N} = 8$ for type II, as we confirm in Section 8.2.4. That is the overabundance we set out to cure. Minimal $\mathcal{N} = 1$ in four dimensions calls for an internal geometry that admits exactly one parallel spinor and no more.

How many parallel spinors a manifold carries is controlled by a single geometric invariant, its holonomy group. This section supplies the middle links of the spine begun in Section 8.1,

$$\text{a preserved supercharge} \iff \text{a covariantly constant spinor} \iff \text{reduced holonomy}, \tag{8.21}$$

and specialises them to the case that yields $\mathcal{N} = 1$ in four dimensions: a six-manifold of SU(3) holonomy, a Calabi-Yau threefold. The equivalence is kinematical. It follows from the structure of the supersymmetry variations and a piece of Riemannian geometry, and it holds whichever string theory sits on top.

8.2.1 The supersymmetry condition in the vacuum

Take the ten-dimensional spacetime to be a direct product $M_{10} = M_{3,1} \times X_6$ with metric

$$\mathrm{d}s^2 = \eta_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu + g_{mn}(y)\,\mathrm{d}y^m\,\mathrm{d}y^n, \qquad \mu,\nu = 0,\dots,3, \quad m,n = 1,\dots,6. \tag{8.22}$$

Maximal four-dimensional Poincaré symmetry forces the external metric to be flat and the internal metric to depend only on the compact coordinates y. A supersymmetry generator that survives is a spinor parameter ϵ whose action leaves this background invariant. The background is purely bosonic, so the variations of the bosonic fields vanish automatically (each is proportional to a fermion, and the fermions are zero in the vacuum). The surviving conditions come from the fermion variations, which must vanish for ϵ to generate an unbroken symmetry.

For the heterotic string the fermions are the gravitino ψ_M, the dilatino λ and the gaugino χ. In the conventions of Appendix A their variations in a bosonic background read, schematically,

$$\delta\psi_M = \Big(\nabla_M - \tfrac{1}{8} H_{MNP}\,\Gamma^{NP}\Big)\epsilon, \tag{8.23}$$

$$\delta\lambda = \Big(\Gamma^M \nabla_M \phi - \tfrac{1}{12} H_{MNP}\,\Gamma^{MNP}\Big)\epsilon, \tag{8.24}$$

$$\delta\chi = -\tfrac{1}{4} F_{MN}\,\Gamma^{MN}\,\epsilon, \tag{8.25}$$

with ∇_M the Levi-Civita covariant derivative acting on spinors, $H = \mathrm{d}B$ the NS three-form, ϕ the dilaton and F the gauge field strength. Type II has the same structure, with two independent parameters ϵ_1, ϵ_2 in place of one.

Take the simplest vacuum: vanishing flux $H = 0$, constant dilaton $\nabla_M \phi = 0$ and vanishing gauge field $F = 0$. The dilatino variation (8.24) then vanishes identically, the gaugino variation (8.25) is trivially satisfied, and the gravitino variation (8.23) collapses to $\nabla_M \epsilon = 0$, the Killing-spinor condition (8.19) already met in the toroidal reduction. Split $\epsilon(x,y) = \zeta(x) \otimes \eta(y)$ (plus its conjugate) into a constant external spinor ζ, the would-be $\mathcal{N} = 1$ parameter, and an internal spinor η on X_6. The external part is annihilated by $\nabla_\mu = \partial_\mu$ on flat space, so the condition reduces to the internal parallel-spinor equation $\nabla_m \eta = 0$ of (8.20). The number of unbroken four-dimensional supersymmetries is the number of linearly independent parallel spinors η on X_6. That is the first arrow of (8.21), imported from Section 8.1 rather than re-derived here; the rest of the section is about the second.

Remark 8.5 (Beyond the lectures). Switching on flux ($H \neq 0$) or a varying dilaton replaces the Levi-Civita connection in (8.20) by a connection with torsion, and covariant constancy is imposed with respect to that modified connection, a *generalised holonomy*. The internal manifold is then no longer Ricci-flat, and generically no honest Calabi-Yau metric solves the equations. This is the entry point to flux compactification, taken up in Chapter 9. Throughout the present chapter we stay in the fluxless limit, where the geometry is Ricci-flat and the holonomy argument is exact.

8.2.2 Holonomy: parallel transport and invariant spinors

Parallel transport around a closed loop rotates a vector by an amount that measures the curvature the loop encloses. Fix a point $p \in X$ and a loop γ based at p. Transporting a tangent vector once around γ with the Levi-Civita connection returns it rotated, defining a linear map P_γ on T_pX. As γ ranges over all loops based at p, the maps P_γ compose into a group, the *holonomy group*

$$\mathrm{Hol}_p(X) \;\subseteq\; \mathrm{SO}(n), \tag{8.26}$$

a subgroup of the orthogonal group because parallel transport preserves lengths and, on an oriented manifold, orientation. On a connected manifold the groups at different base points are conjugate, so we drop the subscript and write $\mathrm{Hol}(X)$. A generic Riemannian six-manifold has the full $\mathrm{Hol}(X) = \mathrm{SO}(6)$. *Reduced holonomy* means $\mathrm{Hol}(X)$ is a proper subgroup, and that reduction is what supersymmetry needs.

The link to spinors is a dictionary between parallel fields and holonomy-invariant objects. A tensor or spinor field Ψ that is covariantly constant, $\nabla\Psi = 0$, is unchanged by parallel transport around any loop, hence invariant under every element of $\mathrm{Hol}(X)$. Conversely, a spinor η_p at p that is fixed by all of $\mathrm{Hol}_p(X)$ can be parallel transported to any other point along any path, and holonomy-invariance makes the result path-independent (on a simply connected X), defining a global covariantly constant spinor η. The two notions coincide:

$$\exists \text{ parallel spinor } \eta \text{ on } X \iff \mathrm{Hol}(X) \text{ fixes a spinor} \iff \mathrm{Hol}(X) \subseteq \mathrm{Stab}(\eta). \tag{8.27}$$

The existence of a preserved supercharge has become a purely group-theoretic question: for which subgroups of $\mathrm{SO}(n)$ does the spinor representation contain a singlet? Background on connections, parallel transport and the Ambrose-Singer theorem relating $\mathrm{Hol}(X)$ to the curvature is collected in Appendix D.

8.2.3 Berger's list

The reduced holonomy groups are not arbitrary. Berger classified the holonomy groups of simply connected Riemannian manifolds that are irreducible (not a metric product) and non-symmetric (symmetric spaces form a separate, fully classified family). The result is a short list.

$\mathrm{Hol}(X)$	$\dim X$	Geometry	Parallel spinor
$\mathrm{SO}(n)$	n	generic oriented Riemannian	no
$\mathrm{U}(n)$	$2n$	Kähler	no
$\mathrm{SU}(n)$	$2n$	Calabi-Yau (Ricci-flat Kähler)	**yes**
$\mathrm{Sp}(n)$	$4n$	hyperkähler	**yes**
$\mathrm{Sp}(n)\cdot\mathrm{Sp}(1)$	$4n$	quaternionic-Kähler (Einstein)	no
G_2	7	exceptional	**yes**
$\mathrm{Spin}(7)$	8	exceptional	**yes**

The last column is what we care about. The generic case $\mathrm{SO}(n)$ fixes no spinor, so a generic manifold breaks all supersymmetry. Holonomy $\mathrm{U}(n)$, Kähler but not Ricci-flat, also fixes no spinor, because the extra U(1) factor acts on it by a phase. The entries that do admit a parallel spinor, and so preserve some supersymmetry, are

$$\mathrm{SU}(n), \qquad \mathrm{Sp}(n) \subset \mathrm{SU}(2n), \qquad G_2, \qquad \mathrm{Spin}(7). \tag{8.28}$$

Since $\mathrm{Sp}(n) \subset \mathrm{SU}(2n)$, hyperkähler manifolds are a special case of Calabi-Yau. Each of these groups is the stabiliser in $\mathrm{Spin}(n)$ of one or more spinors, and each labels a class of internal geometries used somewhere in string or M-theory: SU(2) gives K3 (Section 8.4), SU(3) gives the Calabi-Yau threefold (Section 8.5), and G_2 and Spin(7) enter when one compactifies M-theory or F-theory, to which we return at the end of the section.

8.2.4 SU(3) holonomy gives four-dimensional $\mathcal{N} = 1$

Specialise the dictionary (8.27) to six real internal dimensions. The relevant spin group is $\mathrm{Spin}(6) \cong \mathrm{SU}(4)$, whose two irreducible spinor representations are the fundamental $\mathbf{4}$ and its conjugate $\bar{\mathbf{4}}$. Pick a nonzero spinor in the $\mathbf{4}$. The subgroup of SU(4) fixing it is

$$\mathrm{Stab}(\eta) = \mathrm{SU}(3) \subset \mathrm{SU}(4), \tag{8.29}$$

acting on the orthogonal complement. Under $\mathrm{SU}(3) \subset \mathrm{SU}(4)$ the spinors branch as

$$\mathbf{4} \;\to\; \mathbf{3} \oplus \mathbf{1}, \qquad \bar{\mathbf{4}} \;\to\; \bar{\mathbf{3}} \oplus \mathbf{1}, \tag{8.30}$$

so each Weyl spinor of Spin(6) contains exactly one SU(3) singlet, and the two singlets are complex conjugates. A six-manifold with $\mathrm{Hol}(X) = \mathrm{SU}(3)$ therefore carries one covariantly constant spinor η, together with its conjugate η^c, the minimal nonzero amount.

Now count four-dimensional supercharges. The heterotic and type I strings have a single Majorana-Weyl supercharge in ten dimensions, 16 real components sitting in the $\mathbf{16}$ of $\mathrm{Spin}(9,1)$. Under $\mathrm{Spin}(3,1) \times \mathrm{Spin}(6)$,

$$\mathbf{16} \;\to\; (\mathbf{2}, \mathbf{4}) \;\oplus\; (\bar{\mathbf{2}}, \bar{\mathbf{4}}), \tag{8.31}$$

with $\mathbf{2}, \bar{\mathbf{2}}$ the two four-dimensional Weyl spinors. The unbroken supercharges are those tensored with a *parallel* internal spinor, an SU(3) singlet in (8.30). There is one such singlet in the $\mathbf{4}$ and its conjugate in the $\bar{\mathbf{4}}$, so (8.31) leaves a single four-dimensional Weyl supercharge ($\mathbf{2}$) and its conjugate. That is one Majorana supercharge:

$$\boxed{\text{heterotic on an SU(3)-holonomy } X_6 \implies \mathcal{N} = 1 \text{ in four dimensions.}} \tag{8.32}$$

The general rule is $\mathcal{N}_{4\mathrm{d}} = \mathcal{N}_{10\mathrm{d}} \times$ (number of parallel internal spinors). Type IIA and IIB carry two ten-dimensional supercharges (32 components), so the same SU(3)-holonomy manifold gives $\mathcal{N} = 2$ in four dimensions. At the other extreme the flat torus T^6 has trivial holonomy, every spinor is parallel, and all four internal singlets survive: the heterotic string recovers $\mathcal{N} = 4$ and type II recovers $\mathcal{N} = 8$, the overabundance of Section 8.1. Reducing the holonomy from the trivial group to SU(3) is what dials the supersymmetry down from maximal to minimal.

8.2.5 Ricci-flatness from the integrability condition

A parallel spinor constrains the geometry, not merely the topology. Its integrability condition forces the manifold to be Ricci-flat, which is the internal Einstein equation of a fluxless vacuum, so the construction is consistent with the equations of motion.

Act on the parallel-spinor condition (8.20) with a second covariant derivative and antisymmetrise. The commutator of covariant derivatives on a spinor is the curvature two-form contracted into the Lorentz generators $\frac{1}{4}\Gamma^{pq}$,

$$0 = [\nabla_m, \nabla_n]\,\eta = \tfrac{1}{4}\,R_{mnpq}\,\Gamma^{pq}\,\eta. \tag{8.33}$$

Contract with Γ^n and use the Clifford identity $\Gamma^n\Gamma^{pq} = \Gamma^{npq} + \delta^{np}\Gamma^q - \delta^{nq}\Gamma^p$. The totally antisymmetric term drops out: $R_{mnpq}\,\Gamma^{npq}$ selects the antisymmetrisation $R_{m[npq]}$, which vanishes by the first Bianchi identity. The two remaining contractions combine, using the pair-exchange symmetry $R_{mnpq} = R_{pqmn}$, into the Ricci tensor,

$$R_{mn}\,\Gamma^n\,\eta = 0, \tag{8.34}$$

one equation for each m (the numerical factors follow the curvature conventions of Appendix A). A short Clifford argument finishes it. For any vector v_n and any nonzero spinor η, the slashed vector squares to the norm, $\not{v}^{\,2} = v_n v^n\,\mathbf{1}$, so $\not{v}\,\eta = 0$ forces $|v|^2 = 0$ and hence $v = 0$ in Euclidean signature. The map $v \mapsto \not{v}\,\eta$ is therefore injective, and the six vectors $\Gamma^n\eta$ are linearly independent. Equation (8.34) then forces every coefficient to vanish,

$$\boxed{R_{mn} = 0.} \tag{8.35}$$

A manifold with a parallel spinor is Ricci-flat. Together with the covariantly constant complex structure that the same spinor defines, this identifies X_6 as a Ricci-flat Kähler manifold, a Calabi-Yau threefold.

The spinor and the geometry carry the same content in two languages. The bilinears of η build covariantly constant tensors. The two-form

$$J_{mn} \;\sim\; \eta^\dagger\,\Gamma_{mn}\,\eta \tag{8.36}$$

is the Kähler form, and the three-form

$$\Omega_{mnp} \;\sim\; \eta^{cT}\,\Gamma_{mnp}\,\eta \tag{8.37}$$

is the holomorphic volume form. Both are covariantly constant because η is, so $\nabla J = 0$ (the manifold is Kähler) and $\nabla\Omega = 0$ (there is a nowhere-vanishing holomorphic top form, equivalently a trivial canonical bundle, $c_1 = 0$). That SU(3) holonomy is equivalent to the existence of J and Ω with these properties, and that a Kähler manifold with $c_1 = 0$ admits a unique Ricci-flat metric in each Kähler class (Yau's theorem), is developed in Appendix E. We invoke these results here and read the four-dimensional spectrum off J and Ω in Section 8.5.

The big picture

The keystone chain. For a fluxless product compactification $M_{3,1} \times X_6$, all four statements are equivalent:

$$\underbrace{\mathcal{N}=1 \text{ supercharge preserved}}_{\delta(\text{fermions})=0} \iff \underbrace{\nabla_m\eta = 0}_{\text{parallel spinor}} \iff \underbrace{\mathrm{Hol}(X_6) \subseteq \mathrm{SU}(3)}_{\text{reduced holonomy}} \iff \underbrace{\text{Ricci-flat Kähler}}_{\text{Calabi-Yau threefold}}.$$

The first arrow is the vanishing of the gravitino variation (imported from Section 8.1), the second the parallel-transport dictionary (8.27), the third the integrability condition forcing $R_{mn} = 0$ together with the covariantly constant J and Ω. Read in reverse, choosing a Calabi-Yau threefold guarantees exactly one preserved supercharge per ten-dimensional supercharge, hence $\mathcal{N} = 1$ heterotic and $\mathcal{N} = 2$ type II.

Containment, not equality

The condition is $\mathrm{Hol}(X) \subseteq \mathrm{SU}(3)$, containment and not equality. A manifold whose holonomy sits in a proper subgroup of SU(3) preserves *more* supersymmetry, not less. If $\mathrm{Hol}(X) = \mathrm{SU}(2)$ the six-manifold is (locally) K3 $\times\, T^2$ and preserves twice as many supercharges; if the holonomy is trivial it is a torus and preserves all of them. Getting exactly $\mathcal{N} = 1$ from the heterotic string requires strict SU(3) holonomy, a six-dimensional Calabi-Yau that does not factorise.

Remark 8.6 (Beyond the lectures). The same logic classifies the minimally supersymmetric compactifications of the other members of the duality web assembled in Chapter 7. M-theory has 32 supercharges in eleven dimensions; reducing on a seven-manifold of G_2 holonomy, whose stabiliser in Spin(7) leaves a single spinor invariant ($\mathbf{8} \to \mathbf{7} \oplus \mathbf{1}$), gives $\mathcal{N} = 1$ in four dimensions directly from the eleven-dimensional hub, with no heterotic detour. Compactifying to three dimensions on a Spin(7) eight-manifold ($\mathbf{8}_s \to \mathbf{7} \oplus \mathbf{1}$) is the analogous minimal choice. These exceptional-holonomy spaces have no complex structure and no worldsheet CFT description, one reason the Calabi-Yau case, solvable at special points through the orbifolds of Section 8.3, remains the workhorse.

Exercise 8.4 (Counting invariant spinors)**.** Take the fundamental $\mathbf{4}$ of $\mathrm{SU}(4) \cong \mathrm{Spin}(6)$ and a nonzero spinor η in it.

1. Show that the subgroup of SU(4) fixing η is SU(3), and reproduce the branching $\mathbf{4} \to \mathbf{3} \oplus \mathbf{1}$ of (8.30). Deduce that a six-manifold of SU(3) holonomy has exactly one parallel spinor up to conjugation.
2. Repeat for $\mathrm{SU}(2) \cong \mathrm{Spin}(3) \subset \mathrm{Spin}(4)$ acting on the spinors of a four-manifold. Show that SU(2) holonomy leaves *two* spinors invariant, so K3 preserves twice the supersymmetry of a generic Calabi-Yau. (This is the hyperkähler structure that Section 8.4 builds on.)
3. Using (8.31), translate the spinor counts of parts (1) and (2) into the number of unbroken four-dimensional (respectively six-dimensional) supercharges for the heterotic string.

Exercise 8.5 (From the parallel spinor to $R_{mn} = 0$)**.** Fill in the Clifford algebra behind Section 8.2.5.

1. Starting from $\nabla_m \eta = 0$, take a second covariant derivative and antisymmetrise to obtain $[\nabla_m, \nabla_n]\eta = \frac{1}{4} R_{mnpq} \Gamma^{pq} \eta = 0$, and say where the spin-connection curvature enters.
2. Contract with Γ^n and expand with $\Gamma^n \Gamma^{pq} = \Gamma^{npq} + \delta^{np}\Gamma^q - \delta^{nq}\Gamma^p$. Use the first Bianchi identity $R_{m[npq]} = 0$ to remove the three-index term and the pair symmetry $R_{mnpq} = R_{pqmn}$ to combine the remaining two into $R_{mn}\Gamma^n \eta = 0$.
3. From $\{\Gamma^m, \Gamma^n\} = 2\delta^{mn}$ show $\not{v}^{\,2} = v_n v^n \, \mathbf{1}$, and conclude that $v \mapsto \not{v}\,\eta$ is injective for $\eta \neq 0$ in Euclidean signature. Deduce $R_{mn} = 0$.

Exercise 8.6 (The other two variations)**.** Return to the dilatino and gaugino variations (8.24) and (8.25).

1. Check that in the fluxless vacuum ($H = 0$, constant dilaton, $F = 0$) both vanish identically, so the parallel-spinor condition from the gravitino is the only constraint.
2. Now keep the gauge field on ($F \neq 0$) with $H = 0$ and constant dilaton. The gaugino variation imposes $F_{mn}\Gamma^{mn}\eta = 0$. Using the branching $\mathbf{4} \to \mathbf{3} \oplus \mathbf{1}$ under SU(3), argue that this kills the SU(3)-invariant part of F and is solved when F is a $(1,1)$-form with $F_{mn}J^{mn} = 0$: the internal field strength is of type $(1,1)$ and primitive. These are the Hermitian-Yang-Mills conditions on the heterotic bundle discussed in Section 8.5.

8.3 Orbifolds: exactly solvable compactifications

The holonomy argument of Section 8.2 tells us where to look: a compactification preserving minimal four-dimensional supersymmetry needs an internal manifold with SU(3) holonomy, a Calabi-Yau threefold. That is a demanding request. No explicit Ricci-flat Kähler metric on a compact Calabi-Yau is known in closed form, and quantising a string in such a background means solving an interacting two-dimensional sigma model that we cannot solve. The torus of Section 8.1 is the opposite extreme: flat, exactly solvable, and useless for phenomenology because too much supersymmetry survives and the spectrum is non-chiral.

Orbifolds sit between these two extremes. Take the flat torus and quotient by a finite group of isometries. The quotient is singular at the fixed points of the group, and near each singular point it looks like a cone; away from them it is still flat. Because the sigma model is free everywhere except at measure-zero loci, the string is still exactly solvable: the worldsheet theory is the free torus CFT supplemented by a projection and by twisted boundary conditions. Yet the quotient can preserve as little supersymmetry as a smooth Calabi-Yau, and its massless spectrum is chiral. An orbifold is a Calabi-Yau seen at a special, singular point of its moduli space, the point where the geometry degenerates enough to become computable. Deforming away from that point (*blowing up* the singularities) lands on a smooth Calabi-Yau, and, as we will see, the deformation parameters are themselves string states living at the singular points.

This section develops the construction, extracts the Hodge numbers of the resolved space by hand, and then localises to a single $\mathbb{C}^3/\Gamma$ singularity, where the same orbifold projection acting on Chan-Paton labels turns a stack of D-branes into a *quiver gauge theory*. That last mechanism is the engine of Chapter 10; we build it here.

8.3.1 Building a toroidal orbifold

Start with the six-torus written as a complex three-torus,

$$T^6 = \mathbb{C}^3/\Lambda, \tag{8.38}$$

where Λ is a rank-six lattice and we have chosen a complex structure so that the flat coordinates group into $z^a = x^{2a-1} + i x^{2a}$, $a = 1, 2, 3$. Now pick a finite group Γ acting on $\mathbb{C}^3$ by isometries and quotient:

$$X = T^6/\Gamma. \tag{8.39}$$

For the quotient to make sense the action must respect the lattice, that is, each $g \in \Gamma$ must map Λ to itself. This is a crystallographic condition and it sharply restricts the allowed groups: only the point groups compatible with some six-dimensional lattice appear. The simplest and most important case is $\Gamma = \mathbb{Z}_N$ generated by a single rotation θ. Because θ commutes with the complex structure we can diagonalise it,

$$\theta : \quad z^a \longmapsto e^{2\pi i\, v_a}\, z^a, \qquad v_a \in \tfrac{1}{N}\mathbb{Z}, \tag{8.40}$$

and the data of the orbifold is packaged in the *twist vector* $v = (v_1, v_2, v_3)$.

The supersymmetry condition. In Section 8.2 preserved supersymmetry was tied to a covariantly constant spinor, and on the flat torus that spinor is simply constant. Quotienting keeps only what is invariant under Γ, so the surviving supercharges are the constant spinors left fixed by θ. Under $\mathrm{Spin}(6) \simeq \mathrm{SU}(4)$ the spinor decomposes as $\mathbf{4} \to \mathbf{3} \oplus \mathbf{1}$ with respect to the SU(3) generated by (8.40), and a fixed component exists when θ lies in that SU(3). The clean way to phrase this is through the holomorphic three-form, which under (8.40) transforms as

$$\Omega = \mathrm{d}z^1 \wedge \mathrm{d}z^2 \wedge \mathrm{d}z^3 \longmapsto e^{2\pi i(v_1+v_2+v_3)}\, \Omega, \tag{8.41}$$

so Ω is invariant, equivalently $\theta \in \mathrm{SU}(3)$ rather than merely U(3), when

$$v_1 + v_2 + v_3 = 0 \pmod 1. \tag{8.42}$$

An invariant Ω and a fixed spinor (the $\mathbf{1}$ in the branching) are the same condition; that equivalence is the holonomy chain of Section 8.2, and the Spin(6) spinor decomposition itself is in Appendix B. A $\mathbb{Z}_N$ twist obeying (8.42) with no $v_a = 0$ sits inside SU(3) but not inside any SU(2), and the branching $\mathbf{4} \to \mathbf{3} \oplus \mathbf{1}$ keeps one component in four, one quarter of the torus supercharges. For the heterotic string on T^6 that means $\mathcal{N} = 4 \to \mathcal{N} = 1$; for type II it means $\mathcal{N} = 8 \to \mathcal{N} = 2$, matching the smooth Calabi-Yau counting of Section 8.5. If instead one $v_a = 0$, the twist lies in an SU(2) acting on two of the planes with the third untouched: this is (locally) K3 times a two-torus, and it preserves twice as much, the subject of Section 8.4.

Definition 8.7 (Supersymmetric orbifold)**.** A toroidal orbifold T^6/Γ preserves minimal ($\mathcal{N} = 1$ heterotic, $\mathcal{N} = 2$ type II) four-dimensional supersymmetry iff the point group Γ acts as a subgroup of SU(3), equivalently iff it fixes the holomorphic three-form Ω. For a cyclic twist this is condition (8.42) together with $v_a \neq 0$ for every a.

The crystallographic and SU(3) conditions together leave a short list of cyclic orbifolds. A few of them:

N	twist $N\,v = (Nv_1, Nv_2, Nv_3)$	lattice	fixed points of θ
3	$(1,1,1)$	A_2^3	27
4	$(1,1,2)$	$A_1^2 \times A_3$ (e.g.)	16
6	$(1,1,4)$	$A_2 \times G_2$ (e.g.)	12
7	$(1,2,4)$	A_6	7

The entries are read modulo N, so that $\sum_a Nv_a \equiv 0$. We will run the whole machine on the first row, the $\mathbb{Z}_3$ orbifold, which is the cleanest example in the subject.

Where the curvature hides. The quotient (8.39) is flat wherever Γ acts freely. It fails to act freely at points fixed by some non-trivial g, and there the metric develops a conical singularity: all of the curvature of X is concentrated, as a delta function, at the fixed points. Ricci-flatness away from the fixed points is automatic (the space is locally flat), so an orbifold satisfies the Calabi-Yau condition in the distributional sense. The singularities are the price, and their local physics is what the rest of this section is about.

Singular geometry, smooth string

The supergravity metric on T^6/Γ is singular at the fixed points, and a naive Kaluza-Klein reduction would choke there. String theory does not: the extra, *twisted* states localised at each fixed point smooth out every observable. This is the recurring lesson of the duality web (Chapter 7): a singularity of the classical geometry is usually a signal that new light degrees of freedom must be included, not that the theory breaks down.

8.3.2 Untwisted and twisted sectors

Quantising a closed string on (8.39) is quantising it on T^6 and then keeping only Γ-invariant physics. Two things happen.

Untwisted sector. These are ordinary torus strings, $X(\sigma + 2\pi) = X(\sigma)$, kept only if invariant under Γ. Projection onto invariants is done by the group average

$$P = \frac{1}{|\Gamma|} \sum_{g \in \Gamma} g, \tag{8.43}$$

inserted into the trace. A torus state survives iff it is fixed by every θ-action on its oscillators and momenta.

Twisted sectors. On the quotient a string need only close up to a group element,

$$X(\sigma + 2\pi) = g \cdot X(\sigma), \qquad g \in \Gamma. \tag{8.44}$$

Such a string is not a closed string on the covering torus; it is a new object that exists only because we have quotiented. Its centre of mass is pinned to a fixed point of g (that is the only way (8.44) can be satisfied with finite energy), so twisted states are *localised* at the orbifold singularities. In the θ^k-twisted sector (8.44) uses $g = \theta^k$, whose eigenvalue on plane a is $e^{2\pi i\, k v_a}$, so the oscillators of that complex boson pick up fractional moding,

$$\alpha^a_n \;\longrightarrow\; \alpha^a_{n+kv_a}, \qquad \tilde{\alpha}^a_n \;\longrightarrow\; \tilde{\alpha}^a_{n-kv_a}, \qquad n \in \mathbb{Z}, \tag{8.45}$$

with the shift kv_a read modulo 1, and the zero-point energy of the sector shifts accordingly. The precise ground-state energies (and the fermionic shifts that go with them in the superstring) are computed in Appendix F; the outcome we need is that for an SU(3) twist the twisted ground state is massless and supplies the resolution modes discussed below.

Twisted sectors are not optional. Modular invariance of the one-loop amplitude forces them. Writing the projected partition function as a sum over boundary conditions around the two cycles of the torus,

$$Z(\tau) = \frac{1}{|\Gamma|} \sum_{g,h\,\in\,\Gamma} Z\begin{bmatrix} g \\ h \end{bmatrix}, \tag{8.46}$$

the label h is the projection insertion (8.43) and g is the twist (8.44). The modular transformation $\tau \to -1/\tau$ rotates the two cycles into each other, sending $Z[{}^{g}_{h}] \to Z[{}^{h}_{g}]$: a projection in the untwisted sector ($g = 1$) maps to a twisted sector ($h = 1,\ g \neq 1$). You cannot keep one without the other. This is the sense in which the orbifold string is exactly solvable: every box in (8.46) is a free-field trace with shifted moding, and the whole spectrum, including the twisted states, is fixed by the free theory plus modular invariance. (For $\mathbb{Z}_N$ this same consistency requires a compatible action on the gauge lattice of the heterotic string, the level-matching or "modular invariance of the twist" condition; see Appendix G.)

8.3.3 Hodge numbers by hand: the $T^6/\mathbb{Z}_3$ orbifold

The massless closed-string spectrum reorganises into the cohomology of the resolved Calabi-Yau, and we can read it straight off the two sectors. The untwisted sector supplies the Γ-invariant part of the torus cohomology; the twisted sectors supply one class per fixed point, the mode that resolves that singularity. Take $\Gamma = \mathbb{Z}_3$ with $v = \frac{1}{3}(1,1,1)$, so $\theta = \omega\,\mathbf{1}$ with $\omega = e^{2\pi i/3}$, acting on $T^6 = (T^2)^3$ built on the hexagonal (A_2) lattice.

Untwisted $(1,1)$-forms. The torus has nine harmonic $(1,1)$-forms $\mathrm{d}z^a \wedge \mathrm{d}\bar{z}^b$. Under θ each transforms as $\omega\,\bar{\omega} = 1$, so all nine are invariant:

$$h^{1,1}_{\text{unt}} = 9. \tag{8.47}$$

Untwisted $(2,1)$-forms. The torus has nine harmonic $(2,1)$-forms $\mathrm{d}z^a \wedge \mathrm{d}z^b \wedge \mathrm{d}\bar{z}^c$. Each transforms as $\omega^2\bar{\omega} = \omega \neq 1$, so none survive:

$$h^{2,1}_{\text{unt}} = 0. \tag{8.48}$$

The holomorphic three-form $\Omega = \mathrm{d}z^1 \wedge \mathrm{d}z^2 \wedge \mathrm{d}z^3$ transforms as $\omega^3 = 1$ and is kept, as it must be for a Calabi-Yau.

Twisted sectors. On each T^2 the rotation by ω has three fixed points (the order-three fixed points of the hexagonal lattice), so θ has $3^3 = 27$ fixed points on T^6. Near each one the geometry is the local model $\mathbb{C}^3/\mathbb{Z}_3$ with the $\omega(1,1,1)$ action. Its supersymmetric (crepant) resolution replaces the singular point with a single compact divisor, a $\mathbb{CP}^2$, contributing one $(1,1)$-class and no complex-structure deformation. String theory sees this resolution mode as the massless θ-twisted ground state sitting at that fixed point (the θ^2-twisted sector delivers its conjugate). Hence

$$h^{1,1}_{\text{tw}} = 27, \qquad h^{2,1}_{\text{tw}} = 0. \tag{8.49}$$

Adding the two sectors,

$$h^{1,1} = 9 + 27 = 36, \qquad h^{2,1} = 0 + 0 = 0, \tag{8.50}$$

and therefore

$$\chi = 2(h^{1,1} - h^{2,1}) = 72. \tag{8.51}$$

Through the heterotic standard embedding of Section 8.5, which breaks $\mathrm{E}_8 \to \mathrm{E}_6$, this orbifold gives $|\chi|/2 = 36$ net chiral generations. That rule, $|\chi|/2$ counting the net **27**'s of E_6 and so the number of chiral families, is derived in Section 8.5; here it turns the Euler number (8.51) into a physical count. It is not the Standard Model, but it is a fully computable string vacuum with $\mathcal{N} = 1$ supersymmetry and chiral matter, obtained without ever writing down a curved metric.

Remark 8.8 (Beyond the lectures: the stringy Euler number). The count (8.51) has a slick closed form due to Dixon, Harvey, Vafa and Witten. The orbifold Euler characteristic is

$$\chi_{\text{orb}} = \frac{1}{|\Gamma|} \sum_{gh=hg} \chi\big(\mathrm{Fix}(g) \cap \mathrm{Fix}(h)\big), \tag{8.52}$$

summed over commuting pairs, and it equals the ordinary Euler number of any crepant resolution. For $T^6/\mathbb{Z}_3$ the group is abelian, so all nine pairs (g,h) commute. The identity pair contributes $\chi(T^6) = 0$; each of the remaining eight pairs has $\mathrm{Fix}(g) \cap \mathrm{Fix}(h)$ equal to the same 27 points, because θ and θ^2 share their fixed locus, so each contributes 27. Hence $\chi_{\text{orb}} = \frac{1}{3}(8 \times 27) = \frac{1}{3} \cdot 216 = 72$, in agreement with (8.51). Formula (8.52) is the fast way to get the Euler number of an orbifold before doing any sector-by-sector bookkeeping.

Exercise 8.7. Consider $T^6/\mathbb{Z}_7$ with twist $v = \frac{1}{7}(1,2,4)$. Check that (8.42) holds and that no v_a vanishes, so the orbifold is a genuine SU(3) (not SU(2)) example. Since 7 is prime, show that every non-trivial power θ^k acts freely except at the same set of 7 fixed points, and feed this into the Dixon-Harvey-Vafa-Witten formula (8.52) to obtain $\chi = 48$. (Matching this to the resolved Hodge numbers $h^{1,1} = 24$, $h^{2,1} = 0$ needs the crepant resolution of the $\mathbb{C}^3/\mathbb{Z}_7$ singularity, whose exceptional set carries several divisors; see Appendix E.)

8.3.4 Local singularities and their resolution

The twisted-sector modes are not abstract: each is the string state that, when given a vacuum expectation value, blows up the singularity into smooth geometry. Turning on a twisted modulus is turning on the size of the exceptional cycle that replaces the fixed point. This is why the same orbifold appears both as a singular point (all twisted moduli set to zero) and, after deformation, as an interior point of Calabi-Yau moduli space. The local geometry near a fixed point is

$$\mathbb{C}^3/\Gamma, \qquad \Gamma \subset \mathrm{SU}(3), \tag{8.53}$$

and its resolution theory (crepant resolutions, exceptional divisors, the identification of blow-up moduli with $H^{1,1}$) is developed in Appendix E. What matters here is the dictionary:

twisted modulus at a fixed point $\longleftrightarrow$ exceptional class of the resolution	(8.54)

The two-complex-dimensional version of this is the one that recurs across the course. A finite subgroup $\Gamma \subset \mathrm{SU}(2)$ produces the quotient singularity $\mathbb{C}^2/\Gamma$, and these are classified by the ADE Dynkin diagrams:

$\Gamma \subset \mathrm{SU}(2)$	singularity	resolution graph
cyclic $\mathbb{Z}_{n+1}$	A_n	chain of n $\mathbb{CP}^1$'s
binary dihedral BD_{n-2}	D_n	D_n tree
binary tetrahedral	E_6	E_6 tree
binary octahedral	E_7	E_7 tree
binary icosahedral	E_8	E_8 tree

Resolving $\mathbb{C}^2/\Gamma$ replaces the singular point with a set of $\mathbb{CP}^1$'s whose intersection matrix is minus the Cartan matrix of the corresponding ADE algebra. These are the du Val or ALE singularities, and gluing them into a compact space is how K3 acquires its ADE structure in Section 8.4. The chain also runs the other way, into physics: wrapping branes on the shrinking $\mathbb{CP}^1$'s gives massless gauge bosons filling out the ADE algebra, which is the geometric-engineering statement of Chapter 10 and the origin of the enhanced gauge symmetry in the heterotic/IIA-on-K3 duality of Chapter 7.

8.3.5 Branes at singularities: quiver gauge theories

So far the orbifold acted on closed strings. Its most reused consequence appears when we add open strings, that is, when we place a stack of D-branes at a $\mathbb{C}^n/\Gamma$ singularity. The construction is due to Douglas and Moore, and it is mechanical.

A D-brane stack carries two labels the orbifold can act on: the geometric position/orientation in the transverse $\mathbb{C}^n$, and the Chan-Paton index that says which brane in the stack an open-string endpoint sits on. The orbifold acts on both. Let $R(g)$ be the geometric action on the transverse complex coordinates (the twist (8.40)) and let $\gamma(g)$ be a unitary representation of Γ on the Chan-Paton indices. A state survives the projection only if it is invariant under the *combined* action. For the gauge field A_μ (a geometric scalar) and for the transverse scalars Φ^a (in the "$\mathbf{n}$" of the geometric rotation) the conditions are

$$A_\mu = \gamma(g)\, A_\mu\, \gamma(g)^{-1}, \tag{8.55}$$

$$\Phi^a = e^{2\pi i v_a}\, \gamma(g)\, \Phi^a\, \gamma(g)^{-1}, \tag{8.56}$$

for every $g \in \Gamma$, with the fermions projected by the corresponding spinor phases. Equation (8.55) keeps only the gauge fields commuting with γ, breaking the original $\mathrm{U}(M)$ into a product of factors; (8.56) keeps only the off-diagonal scalars whose Chan-Paton phase cancels the geometric phase, and those become bifundamental matter connecting the factors.

Concretely, take $\Gamma = \mathbb{Z}_N$ and choose the Chan-Paton representation to be n copies of the regular representation, $M = nN$, so

$$\gamma(\theta) = \mathrm{diag}\left(\mathbf{1}_n,\, \omega\, \mathbf{1}_n,\, \omega^2\, \mathbf{1}_n,\, \ldots,\, \omega^{N-1}\, \mathbf{1}_n\right), \qquad \omega = e^{2\pi i/N}. \tag{8.57}$$

Then (8.55) leaves the gauge group

$$\mathrm{U}(M) \;\longrightarrow\; \prod_{i=0}^{N-1} \mathrm{U}(n), \tag{8.58}$$

one factor per irreducible representation (per "node"), and (8.56) keeps a scalar Φ^a_{ij} only when the phases match,

$$\omega^{\,N v_a}\, \omega^{\,i} = \omega^{\,j} \quad \Longleftrightarrow \quad j = i + N v_a \ (\mathrm{mod}\ N). \tag{8.59}$$

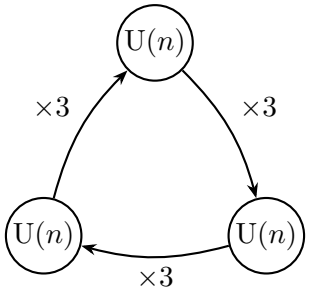


Figure 8.1: The $\mathbb{C}^3/\mathbb{Z}_3$ quiver. Three U(n) nodes, and three chiral bifundamentals along each cyclic arrow (×3), with the superpotential (8.61). This is $\mathcal{N}=4$ super Yang-Mills projected to $\mathcal{N}=1$ by the geometric $\mathbb{Z}_3 \subset \mathrm{SU}(3)$.

Each surviving Φ^a_{ij} is a chiral multiplet in the bifundamental $(\mathbf{n}_i, \overline{\mathbf{n}}_j)$. Draw a node for each gauge factor and an arrow $i \to j$ for each such field, and you have drawn the *quiver*. The rule (8.59) is the representation-theoretic McKay statement: the number of arrows $i \to j$ is the multiplicity a_{ij} in the decomposition of the transverse representation tensored with the i-th irrep,

$$\mathbf{Q} \otimes R_i = \bigoplus_j a_{ij}\, R_j, \tag{8.60}$$

with $\mathbf{Q}$ the representation of Γ on $\mathbb{C}^n$.

Example 8.9 (The $\mathbb{C}^3/\mathbb{Z}_3$ quiver)**.** Put n D-branes (say D3-branes, filling the four non-compact directions) at the tip of $\mathbb{C}^3/\mathbb{Z}_3$ with $v = \frac{1}{3}(1,1,1)$. The gauge group is $\mathrm{U}(n)^3$. Since every $v_a = \frac{1}{3}$, the rule (8.59) gives $j = i+1$ for all three transverse scalars, so there are three arrows $i \to i+1$ at each node, the three chiral bifundamentals $\Phi^a_{i,i+1}$, $a = 1,2,3$. The resulting quiver is a triangle with a tripled arrow on each edge (Fig. 8.1), and the surviving Yukawa couplings assemble into the superpotential

$$W = \epsilon_{abc}\ \mathrm{Tr}\left(\Phi^a_{12}\,\Phi^b_{23}\,\Phi^c_{31}\right). \tag{8.61}$$

This is $\mathcal{N}=4$ super Yang-Mills with gauge group U($3n$) projected to $\mathcal{N}=1$: the $\mathbb{Z}_3 \subset \mathrm{SU}(3)$ breaks three quarters of the supercharges, and the single adjoint of $\mathcal{N}=4$ splits into the three bifundamentals above. It is the workhorse quiver of Chapter 10 and reappears as a holographic example in the capstone.

The two-dimensional case $\mathbb{C}^2/\Gamma$ with $\Gamma \subset \mathrm{SU}(2)$ is where the McKay content is sharpest. There $\mathbf{Q}$ is the two-dimensional defining representation, it is self-conjugate, and the numbers a_{ij} in (8.60) are the adjacency matrix of the *affine* (extended) ADE Dynkin diagram whose finite part appeared in the resolution table above. The extra node is the trivial representation. Branes at $\mathbb{C}^2/\Gamma$ therefore realise the affine-ADE quiver gauge theory, with $\mathcal{N}=2$ supersymmetry, and this is McKay's correspondence made physical: the same Dynkin diagram governs the resolution of the singularity, the gauge algebra of wrapped branes, and the quiver of branes sitting at the point.

Remark 8.10 (Beyond the lectures: Fayet-Iliopoulos terms and the Kronheimer construction)*.* The two descriptions of a $\mathbb{C}^2/\Gamma$ singularity, the closed-string resolution and the open-string quiver, are the same statement about one gauge theory. The Higgs branch of the affine-ADE quiver gauge theory, obtained by solving its D- and F-term equations and dividing by the gauge group, is the ALE space resolving $\mathbb{C}^2/\Gamma$ (Kronheimer's hyperkähler quotient). The blow-up parameters that resolve the singularity enter the quiver as Fayet-Iliopoulos terms for the U(1) factors. Setting them to zero puts the branes at the singular point; turning them on moves onto the resolved space. The twisted-sector moduli of the closed string and the FI parameters of the

open string are two readings of the same geometry, which is one concrete face of open/closed string duality.

Exercise 8.8. For $\mathbb{C}^2/\mathbb{Z}_N$ with the SU(2) twist $v = \frac{1}{N}(1,-1)$, place n D-branes in the regular representation and use (8.59) to derive the quiver. Show the gauge group is $\mathrm{U}(n)^N$, that each node is joined to its two neighbours (from $j = i \pm 1$), and that the nodes close up into a ring, the affine A_{N-1} Dynkin diagram. Count the resulting bifundamental hypermultiplets and confirm the $\mathcal{N} = 2$ matter content of the A_{N-1} quiver.

8.3.6 Where this sits on the ladder

Orbifolds are the computable corner of Calabi-Yau space. They gave us a fully explicit $\mathcal{N} = 1$ vacuum with chiral matter and a Hodge-number count (8.50) done entirely by counting invariant forms and fixed points, no metric required. The twisted sectors are the localised modes that resolve the singularities and connect the orbifold to a smooth Calabi-Yau; the $\mathbb{C}^2/\Gamma$ singularities and their ADE structure are the seeds of K3 in Section 8.4; and the orbifold projection on Chan-Paton factors, (8.55)–(8.60), is the quiver machine that Chapter 10 runs in earnest. When we return to smooth Calabi-Yau threefolds the same accounting survives in a covariant guise: Hodge numbers count massless fields, and the twisted-sector modes of the orbifold become the exceptional $H^{1,1}$ classes of its resolution.

8.4 K3: the CY two-fold

We have climbed from the torus, where the holonomy is trivial, all of the supersymmetry survives and the internal space is flat, through orbifolds, which are flat away from a set of fixed points and quantum-mechanically exact but singular. The next rung is the first genuinely curved special-holonomy manifold that is still under complete control: K3, the unique Calabi-Yau two-fold. Real dimension four, complex dimension two. It is the smallest arena in which reduced holonomy does real work, and every feature we will need for the threefold appears here in a form we can compute by hand: a covariantly constant spinor, a Hodge diamond, moduli that split into shapes and sizes, and singularities that grow non-abelian gauge symmetry. K3 is also the geometry in which the Chapter 7 statement heterotic-on-T^4 = IIA-on-K3 becomes a statement about a manifold.

The holonomy chain, a preserved supercharge $\Leftrightarrow$ a covariantly constant spinor $\Leftrightarrow$ reduced holonomy $\Leftrightarrow$ Ricci-flat Kähler, is derived once and for all in Section 8.2. Here we specialise it to complex dimension two and read off the geometry. The complex, Kähler and Hodge-theoretic background is collected in Appendices D and E; the results I quote below have their proofs there.

8.4.1 Definition, uniqueness and SU(2) holonomy

A Calabi-Yau n-fold is a compact Kähler manifold of complex dimension n with holonomy contained in $\mathrm{SU}(n)$, equivalently a vanishing first Chern class $c_1 = 0$, equivalently (by Yau) a Ricci-flat Kähler metric. At $n = 2$ the definition is sharp enough to fix the manifold almost completely.

Definition 8.11 (K3 surface)**.** A *K3 surface* is a compact complex surface (complex dimension two, real dimension four) with trivial canonical bundle, $c_1 = 0$, and vanishing first Betti number, $b_1 = 0$.

The condition $b_1 = 0$ is what separates K3 from the other compact Ricci-flat Kähler surface, the four-torus T^4. The torus has $c_1 = 0$ as well, but $b_1(T^4) = 4$ and its holonomy is trivial, so it keeps all the supersymmetry and is really a fancy circle compactification. The extra condition $b_1 = 0$ rules that flat case out, leaving the curved SU(2)-holonomy geometry.

Proposition 8.12 (Uniqueness and holonomy)**.** *Every K3 surface is simply connected. All K3 surfaces are diffeomorphic to one another as smooth real four-manifolds: there is a single underlying smooth K3, and the various complex structures and metrics live on top of it. On any K3, Yau's theorem provides a Ricci-flat Kähler metric, unique in each Kähler class, whose Riemannian holonomy is exactly* SU(2)*.*

This is asserted, not proved here; the argument (Siu, Todorov, Yau) is in Appendix E. The uniqueness statement is stronger than anything available for threefolds, where distinct topological types proliferate. For K3 the topology is rigid and only the geometry moves. That rigidity is why K3 is the controllable rung: we can enumerate its cohomology once and be done.

The holonomy $\mathrm{SU}(2) \subset \mathrm{SO}(4)$ deserves a moment. An oriented Riemannian four-manifold has holonomy in SO(4), and the double cover splits as

$$\mathrm{Spin}(4) \;\cong\; \mathrm{SU}(2)_+ \times \mathrm{SU}(2)_-. \tag{8.62}$$

Reduced holonomy SU(2) means the connection sits inside one factor, say $\mathrm{SU}(2)_+$, and acts trivially on the other. A Dirac spinor of Spin(4) decomposes into two Weyl pieces, $\mathbf{4} = (\mathbf{2},\mathbf{1}) \oplus (\mathbf{1},\mathbf{2})$. The piece $(\mathbf{1},\mathbf{2})$ is a singlet under the holonomy $\mathrm{SU}(2)_+$, so it is covariantly constant: K3 carries two independent parallel spinors, forming a doublet of the surviving $\mathrm{SU}(2)_-$. By the chain of Section 8.2, two parallel internal spinors preserve half of the supercharges of whatever ten-dimensional theory we place on K3. For type IIA this leaves sixteen supercharges, the six-dimensional $\mathcal{N} = (1,1)$ algebra. The pair (n_+, n_-) counts the supercharges of each six-dimensional chirality, so $(1,1)$ is non-chiral, one of each handedness; that non-chiral count is the reason the string-string duality pairs IIA-on-K3 with heterotic-on-T^4 rather than with IIB, as we will see below.

8.4.2 Hyperkähler structure

In real dimension four the coincidence $\mathrm{SU}(2) \cong \mathrm{Sp}(1)$ has a consequence with no analogue in higher dimension: a Calabi-Yau two-fold is automatically *hyperkähler*. Holonomy Sp(1) means the metric is compatible with three integrable complex structures obeying the quaternion algebra,

$$I^2 = J^2 = K^2 = IJK = -\mathbf{1}, \qquad IJ = -JI = K, \tag{8.63}$$

each Kähler with respect to the same Ricci-flat metric g. In dimension $4n$ with $n \geq 2$ the inclusion $\mathrm{Sp}(n) \subset \mathrm{SU}(2n)$ is strict, and a generic Calabi-Yau is not hyperkähler; at $n = 1$ the two groups coincide and every K3 metric carries the full quaternionic structure.

Three complex structures give three Kähler forms $\omega_A(\,\cdot\,,\,\cdot\,) = g(A\,\cdot\,,\,\cdot\,)$ for $A = I, J, K$. Fix I as "the" complex structure. Then

$$J \equiv \omega_I, \qquad \Omega \equiv \omega_J + i\,\omega_K \tag{8.64}$$

are, respectively, the $(1,1)$ Kähler form and a nowhere-vanishing holomorphic $(2,0)$-form. The existence of a nowhere-zero Ω is the geometric face of $c_1 = 0$: it trivialises the canonical bundle. The three forms $\omega_I, \omega_J, \omega_K$ are the whole hyperkähler content, and rotating them among one another,

$$\omega \;=\; a\,\omega_I + b\,\omega_J + c\,\omega_K, \qquad a^2 + b^2 + c^2 = 1, \tag{8.65}$$

sweeps out a two-sphere of complex structures, all sharing the one metric. What is a Kähler class in one complex structure is (part of) the holomorphic two-form in another. This freedom is why K3 moduli refuse to split cleanly into "Kähler" and "complex structure" the way threefold moduli do; the hyperkähler rotation mixes them.

8.4.3 Cohomology, the K3 lattice, and moduli

Because the smooth manifold is unique, its Betti numbers are fixed numbers. They are

$$b_0 = b_4 = 1, \qquad b_1 = b_3 = 0, \qquad b_2 = 22, \tag{8.66}$$

so the Euler characteristic and signature are

$$\chi(\mathrm{K3}) = \sum_p (-1)^p b_p = 2 + b_2 = 24, \qquad \tau(\mathrm{K3}) = b_2^+ - b_2^- = 3 - 19 = -16. \tag{8.67}$$

The Hodge decomposition of H^2 refines $b_2 = 22$ into $h^{2,0} + h^{1,1} + h^{0,2} = 1 + 20 + 1$, giving the K3 Hodge diamond

$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 1 & & 20 & & 1 \\ & 0 & & 0 & \\ & & 1 & & \end{array} \tag{8.68}$$

The single entry $h^{2,0} = 1$ is the class of the holomorphic form Ω; its presence is equivalent to $c_1 = 0$. The twenty $(1,1)$-classes are the Kähler deformations. The absence of any odd cohomology ($h^{1,0} = h^{2,1} = 0$) is $b_1 = 0$ restated, and it kills the vector multiplets one might otherwise expect from reducing p-form fields on one-cycles.

The middle cohomology carries more than its dimension. With the intersection pairing (cup product), $H^2(\mathrm{K3}, \mathbb{Z})$ is an even, unimodular lattice of signature $(3, 19)$, the *K3 lattice*

$$H^2(\mathrm{K3}, \mathbb{Z}) \;\cong\; \Lambda_{\mathrm{K3}} \;=\; U^{\oplus 3} \;\oplus\; \mathrm{E}_8(-1)^{\oplus 2}, \tag{8.69}$$

where U is the hyperbolic plane $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $\mathrm{E}_8(-1)$ is the E_8 root lattice with its sign reversed to negative-definite (lattice conventions and E_8 live in Appendix G). The two copies of E_8 are the source of both the signature -16 and, on the physics side, the appearance of $\mathrm{E}_8 \times \mathrm{E}_8$ structure in the heterotic dual.

The signature $(3, 19)$ is the hyperkähler triple in disguise. The three self-dual harmonic two-forms are $\omega_I, \omega_J, \omega_K$; they span the positive-definite three-plane, and the remaining nineteen anti-self-dual forms span the negative part. A Ricci-flat metric is fixed (up to scale) by the choice of that oriented positive three-plane inside $H^2(\mathrm{K3}, \mathbb{R}) = \mathbb{R}^{3,19}$. Hence the moduli space of unit-volume Einstein metrics on K3 is the real Grassmannian

$$\frac{\mathrm{O}(3, 19)}{\mathrm{O}(3) \times \mathrm{O}(19)}, \qquad \dim = 3 \times 19 = 57, \tag{8.70}$$

and with the overall volume restored this is 58 real geometric moduli. (This is a statement of the Torelli theorem for K3: the metric is recovered from its periods, the position of the three-plane. See Appendix E.)

8.4.4 K3 as a resolved orbifold: the Kummer construction

The cleanest way to see all twenty-four of the Euler number, and to connect K3 to the machinery of Section 8.3, is to build one as an orbifold and then repair its singularities. Take

$$Y \;=\; T^4/\mathbb{Z}_2, \qquad \mathbb{Z}_2 :\; x^i \mapsto -x^i. \tag{8.71}$$

The reflection has $2^4 = 16$ fixed points, the half-lattice points, and near each one the space looks like $\mathbb{C}^2/\mathbb{Z}_2$. Away from them Y is smooth and flat.

Count the invariant cohomology first. On T^4 the one-forms $\mathrm{d}x^i$ are odd under $x \mapsto -x$, so all of H^1 and H^3 is projected out, consistent with $b_1(\mathrm{K3}) = 0$. The two-forms $\mathrm{d}x^i \wedge \mathrm{d}x^j$ are even, so all six of $H^2(T^4)$ survive: these are the untwisted classes, one $(2,0)$, four $(1,1)$ and one $(0,2)$. That already delivers $h^{2,0} = h^{0,2} = 1$ and four of the $(1,1)$-classes.

Now resolve. Each $\mathbb{C}^2/\mathbb{Z}_2$ point is the simplest ADE singularity, an A_1 point, cured by blowing up: the point is replaced by a single $\mathbb{CP}^1$ (a two-sphere) of self-intersection -2. The sixteen exceptional spheres are new $(1,1)$-classes, so

$$h^{1,1}(\mathrm{K3}) \;=\; \underbrace{4}_{\text{untwisted}} \;+\; \underbrace{16}_{\text{blow-ups}} \;=\; 20, \tag{8.72}$$

and the Hodge diamond (8.68) is reproduced in full. The Euler number tracks the surgery directly: the singular quotient has $\chi(Y) = \frac{1}{2}(\chi(T^4) - 16) + 16 = 8$, and each blow-up trades a point ($\chi = 1$) for a sphere ($\chi = 2$), adding 16, for $\chi = 24$.

The sixteen new classes are the *twisted sectors* of the $T^4/\mathbb{Z}_2$ conformal field theory made geometric. In the exact orbifold CFT they are states localised at the fixed points; blowing up gives their vacuum expectation values a geometric meaning as the areas of the exceptional spheres. The blow-up moduli are the twisted-sector marginal operators. This is the same twisted-sector-to-geometry dictionary that Section 8.3 sets up for $T^6/\mathbb{Z}_N$, here in its two-dimensional and fully controllable form.

Remark 8.13 (Beyond the lectures)*.* The Kummer construction gives the topology and complex structure honestly, but the metric it inherits is the singular flat orbifold metric. The smooth hyperkähler metric on the resolution is not known in closed form for compact K3; existence is Yau's theorem, not a formula. What *is* known explicitly is the local model near each singular point: the resolved $\mathbb{C}^2/\mathbb{Z}_2$ carries the Eguchi-Hanson metric, the simplest gravitational instanton, and the A_{n-1} resolutions carry the Gibbons-Hawking multi-centre metrics. These asymptotically locally Euclidean (ALE) spaces are the exact non-compact hyperkähler geometries that a K3 looks like in the neighbourhood of a shrinking cycle.

8.4.5 ADE singularities from $\mathbb{C}^2/\Gamma$

The A_1 point of the Kummer construction is the first entry in a classification that will run through the rest of the course. When a two-cycle of K3 shrinks to zero size, the manifold degenerates to a canonical singularity modelled on

$$\mathbb{C}^2/\Gamma, \qquad \Gamma \subset \mathrm{SU}(2) \text{ finite}. \tag{8.73}$$

The requirement $\Gamma \subset \mathrm{SU}(2)$, rather than the larger $\mathrm{U}(2)$, is forced by asking the quotient to remain a local piece of a Calabi-Yau. The holomorphic form $\Omega|_{\mathbb{C}^2} = \mathrm{d}z_1 \wedge \mathrm{d}z_2$ must survive the quotient, and $\gamma \in \mathrm{GL}(2,\mathbb{C})$ scales it by $\det\gamma$; invariance needs $\det\gamma = 1$, i.e. $\Gamma \subset \mathrm{SL}(2,\mathbb{C})$. Preserving the flat Kähler metric as well needs $\Gamma \subset \mathrm{U}(2)$. Together these give $\Gamma \subset \mathrm{SU}(2)$, the local statement of $\mathrm{SU}(2)$ holonomy. (Contrast the threefold case of Section 8.3, where the analogous local models are $\mathbb{C}^3/\Gamma$ with $\Gamma \subset \mathrm{SU}(3)$.)

The finite subgroups of $\mathrm{SU}(2)$ are classified, and the classification is the ADE list, one of the recurring miracles of the subject.

$\Gamma \subset \mathrm{SU}(2)$	order	singularity type
cyclic $\mathbb{Z}_n$	n	A_{n-1}
binary dihedral	$4(n-2)$	D_n
binary tetrahedral	24	E_6
binary octahedral	48	E_7
binary icosahedral	120	E_8

Resolving $\mathbb{C}^2/\Gamma$ replaces the singular point by a tree of $\mathbb{CP}^1$'s. Their number equals the rank of the corresponding simple Lie algebra, and their intersection pattern is the ADE Dynkin diagram: if C_a are the exceptional curves and A_{ab} is the Cartan matrix, then

$$C_a \cdot C_b \;=\; -A_{ab}, \qquad \text{in particular } C_a \cdot C_a = -2. \tag{8.74}$$

Every exceptional curve is a (-2)-sphere, and two of them meet transversally (intersection $+1$) exactly when the corresponding Dynkin nodes are joined. For the A_{n-1} case, $\mathbb{C}^2/\mathbb{Z}_n$ resolves to a chain of $n-1$ spheres,

$$\bullet\!-\!\bullet\!-\cdots-\!\bullet \qquad (n-1 \text{ nodes}),$$

which is the A_{n-1} diagram; at $n=2$ this is the single Eguchi-Hanson sphere of the Kummer points. That the resolution data of a quotient singularity should organise itself into a Lie-algebra Dynkin diagram is the McKay correspondence: the nodes are in bijection with the non-trivial irreducible representations of Γ, of which there are exactly rank many. The dictionary is developed in Appendix G.

8.4.6 Gauge symmetry from geometry: heterotic on T^4 equals IIA on K3

We can now make the Chapter 7 duality geometric. Compactify the $\mathrm{E}_8 \times \mathrm{E}_8$ or $\mathrm{SO}(32)$ heterotic string on T^4, and IIA on K3. Both give six-dimensional theories with sixteen supercharges, the non-chiral $\mathcal{N} = (1,1)$ algebra, and the claim is that they are the same theory.

Start with the moduli. The heterotic string on T^4 carries the metric G_{mn} (10 components), the B-field B_{mn} (6) and the Wilson lines A^I_m of the sixteen-dimensional gauge lattice ($4 \times 16 = 64$), a total of $10 + 6 + 64 = 80 = 4 \times (4 + 16)$ real moduli, organised by the Narain construction into

$$\mathcal{M}_{\text{het}/T^4} \;\cong\; \mathrm{O}(4,20;\mathbb{Z}) \backslash \frac{\mathrm{O}(4,20)}{\mathrm{O}(4) \times \mathrm{O}(20)}, \qquad \dim = 80, \tag{8.75}$$

the quotient of the Grassmannian of positive four-planes in $\mathbb{R}^{4,20}$ by the T-duality group $\mathrm{O}(4,20;\mathbb{Z})$, the automorphisms of the Narain lattice $\Gamma^{4,20}$ (see Appendix G). On the IIA-on-K3 side, the geometric moduli of (8.70) (fifty-seven for the unit-volume Einstein metric, plus one for the volume) combine with the twenty-two periods of the NS-NS B-field over $H^2(\mathrm{K3})$ to give $58 + 22 = 80$, and the enlarged structure that pairs metric and B-field is again (8.75). The lattice matches too:

$$\Gamma^{4,20} \;=\; \Lambda_{\mathrm{K3}} \oplus U \;=\; \left(U^{\oplus 3} \oplus \mathrm{E}_8(-1)^{\oplus 2}\right) \oplus U, \tag{8.76}$$

the K3 lattice enlarged by one hyperbolic plane, which on the IIA side accounts for the B-field-and-volume pair and on the heterotic side for the extra even self-dual directions. Two independent constructions give the same eighty-dimensional moduli space over the same arithmetic lattice, the strongest kinematic evidence for the duality.

The dynamics is where geometry earns its keep. Reducing IIA on K3 gives, generically, an abelian gauge group of rank twenty-four. The counting is

$$\underbrace{22}_{C_3 \text{ on } H^2(\mathrm{K3})} \;+\; \underbrace{2}_{\text{remaining RR fields}} \;=\; 24, \tag{8.77}$$

the twenty-two vectors coming from the RR three-form C_3 expanded on the twenty-two harmonic two-forms, and two more from the other RR fields (the one-form C_1 and the six-dimensional dual of C_3). On the heterotic side the same rank is $16 + 4 + 4$: the sixteen Cartan generators of $\mathrm{E}_8 \times \mathrm{E}_8$ or $\mathrm{SO}(32)$, plus four Kaluza-Klein photons from $G_{\mu m}$ and four from $B_{\mu m}$ on T^4. The generic gauge group is $\mathrm{U}(1)^{24}$ on both sides.

At special loci the two descriptions of enhancement line up point for point. On the heterotic side, at special radii and Wilson lines the Narain momenta hit lattice points where extra states

become massless and U(1)'s enhance to ADE gauge groups (the Frenkel-Kac-Segal mechanism of Appendix G). On the IIA side, those same points are where the K3 develops an ADE singularity of the matching type: a collection of two-cycles shrinks, and D2-branes wrapping the vanishing $\mathbb{CP}^1$'s become massless BPS particles. A wrapped D2-brane has mass equal to its tension $T_{\mathrm{D2}} \sim 1/(g_s\,\alpha'^{3/2})$, with g_s the string coupling, times the area it wraps,

$$M \;\sim\; \frac{\mathrm{Area}(C_a)}{g_s\,\alpha'^{3/2}} \;\longrightarrow\; 0. \tag{8.78}$$

These are the W-bosons: the simple roots correspond to the exceptional curves C_a of (8.74), and the full root system is reproduced by wrapped branes on effective curves. The gauge group of the six-dimensional theory is dictated by the singularity type of K3, and the Cartan matrix is read off from the intersection form. Blowing the singularity up (turning on the Kähler moduli, the areas) gives the W-bosons mass and moves onto the Coulomb branch. This is gauge symmetry from geometry in its cleanest form, the two-complex-dimensional seed of the geometric engineering programme that Chapter 10 builds on $\mathbb{C}^3/\Gamma$ and Calabi-Yau threefolds.

Shrinking a cycle is not enough

Enhanced gauge symmetry needs the geometric cycle to shrink *and* the B-field period through it to vanish. If $\int_{C_a} B \neq 0$ while $\mathrm{Area}(C_a) \to 0$, the wrapped D2-brane keeps a non-zero central charge and the K3 sigma model stays perfectly regular. The genuine singular point in the eighty-dimensional moduli space (8.75) is the locus where a whole positive-norm cycle degenerates, metric and B-field together. This is why the correct arena for the duality is the combined lattice $\Gamma^{4,20}$ rather than metric geometry alone.

Remark 8.14 (Beyond the lectures: IIB and F-theory on K3). The choice of IIA is what makes the duality work. Type IIB on K3 keeps the same sixteen supercharges but assembles them chirally into the six-dimensional $\mathcal{N} = (2,0)$ algebra, with a gravity multiplet and 21 tensor multiplets and moduli space $\mathrm{O}(5,21)/(\mathrm{O}(5)\times\mathrm{O}(21))$. The vectors are gone; the low-energy fields are self-dual tensors, and at an ADE point one finds not massless gauge bosons but tensionless self-dual strings, from D3-branes wrapping the vanishing cycles. There is no six-dimensional Lagrangian for that physics. Fibring K3 rather than using it as the whole internal space gives yet another face: F-theory on an elliptically fibred K3 is dual to heterotic on T^2 in eight dimensions, the base case of the heterotic/F-theory ladder taken up in Chapter 11, and it is there that the exceptional groups $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$ appear as gauge symmetries directly, from Kodaira fibres that a perturbative string cannot produce.

Where this leaves us

K3 delivers the whole Calabi-Yau template at a dimension where nothing is hidden: a unique smooth manifold, an explicit Hodge diamond, a hyperkähler metric guaranteed by Yau, and singularities that are classified by ADE and turn into gauge symmetry on the nose. The one feature it lacks is chirality: six-dimensional $\mathcal{N} = (1,1)$ is non-chiral, and $h^{2,0} = 1$ leaves no room for the complex-structure/Kähler split that will carry the four-dimensional spectrum. For that we go up one complex dimension, to the Calabi-Yau threefold of Section 8.5, where the holonomy is SU(3), the two-form Ω becomes a three-form, and the Hodge numbers $h^{1,1}$ and $h^{2,1}$ become the counting of the four-dimensional theory. K3 remains in play there as a building block: many phenomenologically interesting threefolds are K3-fibred, and the ADE dictionary we have just built is reused verbatim.

Exercise 8.9 (Kummer bookkeeping)**.** Work through the resolved orbifold $T^4/\mathbb{Z}_2$ in detail. (a) Verify that the $\mathbb{Z}_2$ action $x^i \mapsto -x^i$ has 16 fixed points and that $H^1(T^4)$ and $H^3(T^4)$ are entirely projected out, so $b_1 = b_3 = 0$. (b) Show that all six classes of $H^2(T^4)$ are invariant and identify

which are $(2,0)$, $(1,1)$ and $(0,2)$. (c) Adding one exceptional $\mathbb{CP}^1$ per fixed point, reproduce the Hodge diamond (8.68) and confirm $\chi = 24$ by the surgery count $\chi = \frac{1}{2}(\chi(T^4) - 16) + 2 \cdot 16$.

Exercise 8.10 (A_{n-1} and its Cartan matrix)**.** Consider $\mathbb{C}^2/\mathbb{Z}_n$ with the action $(z_1, z_2) \mapsto (\zeta z_1, \zeta^{-1} z_2)$, $\zeta = e^{2\pi i/n}$. (a) Check that $\zeta \in \mathrm{SU}(2)$ (not merely $\mathrm{U}(2)$) and hence that $\mathrm{d}z_1 \wedge \mathrm{d}z_2$ is invariant. (b) The minimal resolution replaces the origin by a chain of $n-1$ spheres. Assuming each is a (-2)-curve meeting only its neighbours, write down the $(n-1) \times (n-1)$ intersection matrix and confirm it equals minus the Cartan matrix of $\mathrm{SU}(n)$. (c) Explain, in one or two sentences, why wrapping IIA D2-branes on these curves gives the $\mathrm{SU}(n)$ W-bosons and how blowing up the curves corresponds to moving onto the Coulomb branch.

Exercise 8.11 (Two covariantly constant spinors)**.** Using the split $\mathrm{Spin}(4) \cong \mathrm{SU}(2)_+ \times \mathrm{SU}(2)_-$ of (8.62) and the decomposition of the Dirac spinor $\mathbf{4} = (\mathbf{2}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{2})$, show that holonomy $\mathrm{SU}(2)_+$ leaves exactly two covariantly constant spinors, transforming as a doublet of $\mathrm{SU}(2)_-$. Contrast with T^4 (trivial holonomy: four parallel spinors, all supersymmetry kept) and state, without repeating the derivation of Section 8.2, why IIA on K3 gives six-dimensional $\mathcal{N} = (1,1)$ rather than $(2,0)$.

8.5 Calabi-Yau threefolds and the 4D spectrum

The holonomy chain of Section 8.2 ended with a demand. To keep exactly one four-dimensional supercharge out of a ten-dimensional theory, compactify the six extra dimensions on a space whose Riemannian holonomy sits inside $\mathrm{SU}(3)$. That single group-theoretic condition, worked backwards through Berger's list, picks out a very particular kind of geometry: a compact, Ricci-flat, Kähler manifold of complex dimension three. This section is about that geometry and, more to the point, about reading four-dimensional physics off it. The metric we can rarely write down; the physics we can, because it is controlled by topology.

The big picture

A Calabi-Yau threefold X carries exactly two integers of physical weight: $h^{1,1}$, the number of independent $(1,1)$-forms, and $h^{2,1}$, the number of $(2,1)$-forms. Everything the four-dimensional effective theory sees at the massless level is counted by these. The $h^{1,1}$ deformations resize X (Kähler moduli); the $h^{2,1}$ deformations reshape it (complex-structure moduli). Heterotic strings on X give 4D $\mathcal{N} = 1$ with $|\chi|/2 = |h^{1,1} - h^{2,1}|$ net chiral generations; type II gives $\mathcal{N} = 2$ with vector and hypermultiplet counts set by the same two numbers. Mirror symmetry swaps $h^{1,1} \leftrightarrow h^{2,1}$, and behind that swap is an equality of physical theories that turns an impossible instanton sum on one side into a classical integral on the other.

8.5.1 What a Calabi-Yau threefold is

Four conditions, on a compact Kähler manifold, turn out to be the same condition. Learning why they coincide is most of the geometry; here is the statement, with the logic sketched and the proofs sent to Appendix E.

Definition 8.15 (Calabi-Yau n-fold)**.** A *Calabi-Yau n-fold* is a compact Kähler manifold X of complex dimension n satisfying any one (hence all) of the following equivalent conditions:

1. the restricted holonomy group of the Levi-Civita connection is contained in $\mathrm{SU}(n)$;
2. the first Chern class vanishes, $c_1(X) = 0$;
3. the canonical bundle $K_X = \Lambda^{n,0} T^*X$ is trivial, equivalently X admits a nowhere-vanishing holomorphic $(n,0)$-form Ω;

4. X admits a Ricci-flat Kähler metric.

The equivalences (1)⇔(3)⇔(4) are essentially linear algebra plus the fact that on a Kähler manifold the holonomy already sits in $\mathrm{U}(n)$; reducing it to $\mathrm{SU}(n)$ is exactly killing the U(1) part measured by the Ricci form. Write the Ricci form of a Kähler metric as

$$\rho = -\,i\,\partial\bar{\partial}\log\det(g_{i\bar{j}}), \qquad \left[\tfrac{\rho}{2\pi}\right] = c_1(X) \in H^2(X,\mathbb{Z}). \tag{8.79}$$

A Ricci-flat metric has $\rho = 0$, so $c_1(X) = 0$ follows immediately: (4)⇒(2) is a one-line consequence of (8.79). The content, and the reason the definition is useful at all, is the converse. Vanishing of the *topological* class $c_1(X)$ does not obviously produce a Ricci-flat *metric*; that it does is a theorem.

Theorem 8.16 (Calabi conjecture; Yau 1978)**.** *Let X be a compact Kähler manifold with $c_1(X) = 0$. Then each Kähler class contains a unique Ricci-flat Kähler metric.*

This is an existence-and-uniqueness statement, and it is all we get: Yau's proof is by a continuity/estimate argument for a complex Monge-Ampère equation and produces no closed form for the metric (Appendix E states the equation and the strategy). To this day no explicit Ricci-flat metric is known on any compact Calabi-Yau of dimension ≥ 2 except at special degenerate limits; numerical metrics are obtained by solving the Monge-Ampère equation on a computer. For string phenomenology this is less crippling than it sounds. The four-dimensional *spectrum* and its symmetries are topological and cohomological data, insensitive to the precise metric. The metric would be needed for the Kähler-potential normalisations of the low-energy Lagrangian, and there the topological skeleton plus a controlled expansion carries you a long way.

$c_1 = 0$ is a cohomology statement

Condition (2) is that $c_1(X)$ vanishes as a real (equivalently rational) cohomology class. For a simply-connected X this is clean and equivalent to strict $\mathrm{SU}(n)$ holonomy. When $\pi_1(X) \neq 0$ one must be careful: a finite fundamental group can make the canonical bundle a nontrivial torsion class even though $c_1 = 0$ in $H^2(X,\mathbb{Q})$, and the holonomy is then $\mathrm{SU}(n) \rtimes (\text{finite})$. We take Calabi-Yau to mean strict holonomy $\mathrm{SU}(n)$ and, unless stated otherwise, $\pi_1(X) = 0$. The finite-quotient case is how one lowers the generation count in model building.

For us $n = 3$: a Calabi-Yau threefold has real dimension six, the right size to reduce a ten-dimensional theory to four dimensions. Its holonomy is SU(3), and the parallel holomorphic three-form Ω is the geometric avatar of the covariantly constant spinor that Section 8.2 used to preserve one supercharge: schematically $\Omega_{ijk} \sim \eta^\dagger \Gamma_{ijk} \eta$ with η the singlet in $\mathbf{4} \to \mathbf{3} \oplus \mathbf{1}$ of $\mathrm{SU}(4) = \mathrm{Spin}(6)$ under SU(3). The threefold is the top rung of a ladder indexed by complex dimension and by holonomy. Complex dimension one is T^2, holonomy trivial, breaking no supersymmetry: a type II string keeps all 32 of its supercharges (the same trivial holonomy on T^6 is what yields 4D $\mathcal{N} = 8$). Complex dimension two is K3, the Calabi-Yau two-fold of Section 8.4, holonomy SU(2), keeping half. Complex dimension three is the threefold, holonomy SU(3), keeping a quarter.

The canonical worked example, which we return to repeatedly, is the *quintic*.

Example 8.17 (The quintic threefold)**.** Let $p_5(z_0,\dots,z_4)$ be a generic homogeneous polynomial of degree 5 in the projective coordinates of $\mathbb{CP}^4$, and set

$$X = \{\, p_5(z_0,\dots,z_4) = 0 \,\} \subset \mathbb{CP}^4. \tag{8.80}$$

For a smooth degree-d hypersurface in $\mathbb{CP}^n$, the adjunction formula gives $c_1(X) = (n+1-d)\,H|_X$, with H the hyperplane class. In $\mathbb{CP}^4$ ($n = 4$) a quintic ($d = 5$) has $n + 1 - d = 0$, so $c_1(X) = 0$:

the quintic is a Calabi-Yau threefold. The same arithmetic is the fastest CY-detector there is. A degree-$(n+1)$ hypersurface in $\mathbb{CP}^n$ is a Calabi-Yau $(n-1)$-fold; the cubic in $\mathbb{CP}^2$ is the torus T^2, the quartic in $\mathbb{CP}^3$ is a K3, the quintic in $\mathbb{CP}^4$ is the threefold.

8.5.2 The Hodge diamond

Massless four-dimensional fields come from harmonic forms on X: solving the ten-dimensional wave equation with no internal excitation means being annihilated by the internal Laplacian, and by Hodge's theorem the harmonic representatives are in one-to-one correspondence with cohomology classes (Appendix D). So before any physics we need to count cohomology. On a complex manifold the natural refinement is Dolbeault cohomology, and its dimensions are the Hodge numbers

$$h^{p,q}(X) = \dim_{\mathbb{C}} H^{p,q}_{\bar\partial}(X), \tag{8.81}$$

the number of independent harmonic (p,q)-forms. Three facts constrain them. Complex conjugation gives $h^{p,q} = h^{q,p}$. Serre (Poincaré) duality gives $h^{p,q} = h^{n-p,n-q}$. And the Calabi-Yau structure fixes the corners: X connected forces $h^{0,0} = 1$, the unique holomorphic three-form forces $h^{3,0} = 1$, and strict SU(3) holonomy forces $h^{1,0} = h^{2,0} = 0$ (a nonzero holomorphic one- or two-form would reduce the holonomy further, splitting off a torus or a K3). What survives is remarkably little. For a Calabi-Yau threefold the entire Hodge diamond is fixed by two integers,

$$\begin{array}{ccccccc}
 & & h^{0,0} & & & & \\
 & h^{1,0} & & h^{0,1} & & & \\
h^{2,0} & & h^{1,1} & & h^{0,2} & & \\
 & h^{2,1} & & h^{1,2} & & h^{0,3} & \\
h^{3,0} & & h^{2,2} & & h^{1,3} & & h^{3,3} \\
 & h^{3,1} & & h^{2,3} & & & \\
 & & h^{3,2} & & & &
\end{array}
\quad=\quad
\begin{array}{ccccccc}
 & & 1 & & & & \\
 & 0 & & 0 & & & \\
0 & & h^{1,1} & & 0 & & \\
 & h^{2,1} & & h^{2,1} & & 1 & \\
1 & & h^{1,1} & & 0 & & 1 \\
 & 0 & & 0 & & & \\
 & & 1 & & & &
\end{array}
\tag{8.82}$$

The Betti numbers follow by summing rows: $b_0 = b_6 = 1$, $b_1 = b_5 = 0$, $b_2 = b_4 = h^{1,1}$, and $b_3 = 2 + 2h^{2,1}$. The vanishing $b_1 = 0$ says X has no one-cycles for a gauge field to wrap; it is why toroidal compactification's extra U(1)'s (Section 8.1) do not appear here. The alternating sum is the Euler number,

$$\chi(X) = \sum_{k=0}^{6} (-1)^k b_k = 2(h^{1,1} - h^{2,1}). \tag{8.83}$$

This single combination will turn out to be the net number of chiral generations, up to a factor of two, in the heterotic string.

8.5.3 Moduli: the size and shape of X

Theorem 8.16 gives a Ricci-flat metric in each Kähler class, but not a single metric: it gives a whole family, and the parameters of that family are massless scalar fields in four dimensions. Ask which deformations $g \to g + \delta g$ keep the metric Ricci-flat and Kähler. Linearising the Einstein equation about a Ricci-flat background and splitting δg by holomorphic type gives two decoupled families.

Mixed deformations $\delta g_{i\bar\jmath}$ keep the complex structure fixed and change the Kähler form

$$J = i\, g_{i\bar\jmath}\, \mathrm{d}z^i \wedge \mathrm{d}\bar z^{\bar\jmath}. \tag{8.84}$$

The Ricci-flat ones are in bijection with harmonic $(1,1)$-forms, so there are $h^{1,1}$ real *Kähler moduli*. Geometrically they are the volumes of the $h^{1,1}$ independent two-cycles: they set the

size and shape of the box, and they are constrained to the Kähler cone, the region where every holomorphic curve and divisor has positive volume. In string theory each real Kähler modulus is complexified by the internal B-field, whose components on the same two-cycles are periodic scalars. Expanding both in a basis $\{\omega_a\}_{a=1}^{h^{1,1}}$ of $H^{1,1}(X)$,

$$B + i\,J = \sum_{a=1}^{h^{1,1}} \left(b^a + i\,t^a\right)\omega_a \;\coloneqq\; \sum_a T^a\,\omega_a, \tag{8.85}$$

gives $h^{1,1}$ *complexified* Kähler moduli T^a. The two-torus per modulus, $b^a \sim b^a + 1$, is a shift symmetry that will protect T^a from perturbative corrections.

Pure deformations δg_{ij} (and their conjugates $\delta g_{\bar{i}\bar{j}}$) cannot be absorbed by a coordinate change; they change the complex structure, deforming which combinations of real coordinates count as holomorphic. Contracting with the holomorphic three-form converts such a deformation into a harmonic $(2,1)$-form,

$$\delta g_{\bar{i}\bar{j}} \;\longmapsto\; \Omega_{ij}{}^{\bar{j}}\,\delta g_{\bar{k}\bar{j}}\,\mathrm{d}z^i \wedge \mathrm{d}z^j \wedge \mathrm{d}\bar{z}^{\bar{k}} \in H^{2,1}(X), \tag{8.86}$$

so the *complex-structure moduli* are counted by $h^{2,1}$ complex parameters. Equivalently, the Kodaira-Spencer map identifies them with $H^1(X, T_X) \cong H^{2,1}(X)$: the tangent-bundle-valued $(0,1)$-forms that deform X as a complex manifold. These moduli reshape X at fixed volume.

A clean way to package the complex-structure moduli is through the periods of Ω. Choose a symplectic basis $\{\Gamma_K\}$ of $H_3(X,\mathbb{Z})$, $K = 0, \ldots, h^{2,1}$ together with duals, and define

$$\Pi^K(z) = \int_{\Gamma_K} \Omega. \tag{8.87}$$

As the complex structure z varies, Ω (which lives in $H^{3,0}$, a line inside the fixed lattice $H^3(X,\mathbb{Z})$) rotates, and the periods Π^K trace out the complex-structure moduli space. They are projective coordinates on it and satisfy a system of Picard-Fuchs differential equations. This is not idle bookkeeping: these same periods are what the flux superpotential $W = \int G_3 \wedge \Omega$ of Chapter 9 evaluates, so the machinery here is the input to moduli stabilisation there.

The two moduli spaces carry different geometries. The complex-structure moduli space is a special-Kähler manifold with the Weil-Petersson metric; the complexified Kähler moduli space is special-Kähler as well but, on the worldsheet, receives instanton corrections. We meet both again below as the scalar geometries of the 4D theory, where the name special-Kähler takes on its $\mathcal{N} = 2$ meaning.

8.5.4 From cohomology to the four-dimensional spectrum

Now the dictionary. The rule is uniform: expand each ten-dimensional field in harmonic forms on X; each harmonic form of the right degree contributes one four-dimensional field, and the number of them is a Hodge number. The mechanism is the Kaluza-Klein reduction of Section 8.1, applied to a curved internal space where "constant mode" is replaced by "harmonic form". How the resulting fields organise into supermultiplets depends on how much supersymmetry survives, which we settled by counting parallel spinors.

Heterotic on X: 4D $\mathcal{N} = 1$

The heterotic string has 16 supercharges in ten dimensions. On X the internal Spin(6) spinor $\mathbf{4}$ decomposes as $\mathbf{3} \oplus \mathbf{1}$ under SU(3); the single singlet preserves one quarter of the supercharges, leaving four, that is 4D $\mathcal{N} = 1$. This is the phenomenologically interesting corner, because $\mathcal{N} = 1$ in four dimensions permits chiral fermions.

Cohomology / source	Multiplicity	4D $\mathcal{N}=1$ content
graviton, gravitino	1	supergravity multiplet
$\mathrm{E}_6 \times \mathrm{E}_8$	1	vector multiplets (gauge sector)
$H^1(X, TX) \cong H^{2,1}$	$h^{2,1}$	chiral matter in **27** of E_6
$H^1(X, \Omega^1) \cong H^{1,1}$	$h^{1,1}$	chiral matter in $\overline{\mathbf{27}}$
$H^{1,1}(X)$	$h^{1,1}$	Kähler moduli T^a (singlets)
$H^{2,1}(X)$	$h^{2,1}$	complex-structure moduli (singlets)
dilaton, $B_{\mu\nu}$	1	dilaton-axion S (singlet)
$H^1(X, \mathrm{End}\, TX)$	$h^1(\mathrm{End}\, TX)$	bundle moduli (singlets)

Table 8.1: Heterotic $\mathrm{E}_8 \times \mathrm{E}_8$ on a Calabi-Yau threefold with the standard embedding. Net generations $= |\chi|/2$.

Heterotic compactification needs one extra ingredient the metric alone does not supply: a gauge bundle. The ten-dimensional gauge field must be given a background on X, and consistency ties it to the geometry through the Bianchi identity for the three-form,

$$\mathrm{d}H = \frac{\alpha'}{4}\Big(\operatorname{tr} R \wedge R - \operatorname{tr} F \wedge F\Big), \tag{8.88}$$

whose cohomological integral forces the second Chern classes to match, $c_2(V) = c_2(TX)$, for the gauge bundle V (the coefficient and the Green-Schwarz story live in Appendix H). The cheapest solution is the *standard embedding*: set the gauge connection equal to the spin connection, so $V = TX$ and $F = R$. Then (8.88) holds identically, the SU(3) holonomy is embedded into one E_8, and supersymmetry is preserved because the tangent bundle of a Calabi-Yau is a stable holomorphic bundle satisfying the Hermitian-Yang-Mills equations.

Embedding SU(3) into E_8 breaks it to the commutant. The commutant of SU(3) in E_8 is E_6, and the adjoint decomposes as

$$\mathbf{248} \;\longrightarrow\; (\mathbf{1}, \mathbf{78}) \oplus (\mathbf{8}, \mathbf{1}) \oplus (\mathbf{3}, \mathbf{27}) \oplus (\bar{\mathbf{3}}, \overline{\mathbf{27}}). \tag{8.89}$$

The four-dimensional gauge group is $\mathrm{E}_6 \times \mathrm{E}_8$ (the second E_8 is untouched, a hidden sector). Matter multiplicities are again cohomology counts, now of the internal Dirac operator coupled to V (an index theorem, Appendix D). A field in the **3** of SU(3) is valued in $V = TX$, and its chiral zero modes are counted by $H^1(X, TX) \cong H^{2,1}(X)$; a field in the $\bar{\mathbf{3}}$ is valued in $V^* = \Omega^1_X$, counted by $H^1(X, \Omega^1_X) \cong H^{1,1}(X)$. Therefore

$$n_{\mathbf{27}} = h^{2,1}, \qquad n_{\overline{\mathbf{27}}} = h^{1,1}, \qquad n_{\mathrm{gen}} = |n_{\mathbf{27}} - n_{\overline{\mathbf{27}}}| = |h^{2,1} - h^{1,1}| = \frac{|\chi|}{2}. \tag{8.90}$$

The individual numbers 101 versus 1 depend on orientation conventions; the net chirality $|\chi|/2$ does not, because it is an index. Each net **27** is one chiral family: under $\mathrm{E}_6 \supset \mathrm{SO}(10) \supset \mathrm{SU}(5)$ the **27** contains a **16** of SO(10), which is one Standard Model generation with a right-handed neutrino. So the coarse phenomenology of a heterotic vacuum is read straight off the Euler number, and "three generations" becomes the geometric request $|h^{1,1} - h^{2,1}| = 3$.

The rest of the $\mathcal{N} = 1$ spectrum is moduli. The complexified Kähler moduli T^a and the complex-structure moduli both become gauge-singlet chiral multiplets, $h^{1,1}$ and $h^{2,1}$ of them; the four-dimensional dilaton-axion sits in one more chiral multiplet S; and the $(\mathbf{8}, \mathbf{1})$ piece of (8.89) gives further E_6-singlet bundle moduli, counted by $H^1(X, \mathrm{End}\, TX)$. Table 8.1 collects the dictionary.

Remark 8.18 (Beyond the lectures: general bundles and $(0,2)$ models). The standard embedding is a convenience, not a necessity, and it never gives realistic gauge groups on its own. The general

construction takes any stable holomorphic vector bundle V with $c_1(V) = 0$ and $c_2(V) = c_2(TX)$ (or $c_2(V) = c_2(TX) - [W]$ if one adds heterotic five-branes wrapping a curve $[W]$). By the Donaldson-Uhlenbeck-Yau theorem, slope-stability of V is equivalent to the existence of a Hermitian-Yang-Mills connection, which is the condition for the background to preserve $\mathcal{N} = 1$. A structure group SU(4) or SU(5) $\subset \mathrm{E}_8$ leaves SO(10) or SU(5) unbroken, and the generation count becomes $h^1(X, V)$ with the antigenerations $h^1(X, V^*)$; these are $(0, 2)$ worldsheet theories, not the $(2, 2)$ locus of the standard embedding. This is the engine of heterotic model building, and the source of the finite-quotient trick: a freely-acting $\mathbb{Z}_n$ on X divides χ, and Wilson lines of V around $\pi_1(X/\mathbb{Z}_n)$ break E_6 down towards the Standard Model.

Type II on X: 4D $\mathcal{N} = 2$

Type II strings have 32 supercharges in ten dimensions, and X preserves one quarter of them: 4D $\mathcal{N} = 2$, eight supercharges, no chiral matter, but the cleanest possible statement of the cohomology-to-spectrum map. Take type IIA. The bosonic fields are the metric, the B-field and dilaton in the NS sector, and the Ramond-Ramond forms C_1, C_3. Reduce them on the harmonic forms of X:

- the metric gives the 4D graviton, the $h^{1,1}$ Kähler moduli, and the $h^{2,1}$ complex-structure moduli;
- C_1 gives the single vector of the $\mathcal{N} = 2$ gravity multiplet, the graviphoton, which is therefore not counted among the n_V below;
- $C_3 = \sum_a A^a \wedge \omega_a + (\text{scalars on } H^3)$ gives $h^{1,1}$ vectors and $2h^{2,1} + 2$ real scalars;
- B and the dilaton give the $h^{1,1}$ axions b^a and the universal scalars.

Organised into $\mathcal{N} = 2$ multiplets this is

$$\text{IIA on } X: \quad n_V = h^{1,1} \ \text{(vector multiplets)}, \qquad n_H = h^{2,1} + 1 \ \text{(hypermultiplets)}. \tag{8.91}$$

Each vector multiplet pairs one of the C_3-vectors with a complexified Kähler modulus T^a; each hypermultiplet holds a complex-structure modulus dressed with Ramond-Ramond scalars, and the extra "+1" is the universal hypermultiplet containing the dilaton, the axion dual to $B_{\mu\nu}$, and the two scalars from C_3 on $\Omega, \bar{\Omega}$. The vector-multiplet moduli space is special-Kähler, the constrained geometry $\mathcal{N} = 2$ forces on vector-multiplet scalars, its metric fixed by a single holomorphic prepotential; the hypermultiplet moduli space is quaternionic-Kähler, the corresponding hypermultiplet geometry, of real dimension $4n_H$ and carrying three compatible complex structures. That the Kähler moduli sit in the special-Kähler (vector) sector is the fact mirror symmetry will exploit.

Type IIB is the mirror reduction. Now the self-dual C_4 on the three-cycles supplies the vectors, and the roles of the two Hodge numbers swap:

$$\text{IIB on } X: \quad n_V = h^{2,1}, \qquad n_H = h^{1,1} + 1. \tag{8.92}$$

In IIB the complex-structure moduli are the special-Kähler (vector) sector and the Kähler moduli are hypers, the reverse of IIA. This asymmetry between the two type II strings on the *same* X is the first hint of mirror symmetry: if IIA and IIB are to describe the same four-dimensional physics, it cannot be on the same X.

8.5.5 Mirror symmetry

Calabi-Yau threefolds come in pairs. For (almost) every X there is a *mirror* $\tilde{X}$ whose Hodge diamond is that of X reflected across a diagonal,

$$h^{1,1}(\tilde{X}) = h^{2,1}(X), \qquad h^{2,1}(\tilde{X}) = h^{1,1}(X), \qquad \chi(\tilde{X}) = -\chi(X). \tag{8.93}$$

Empirically this is visible in the plot of known Calabi-Yau threefolds in the $(h^{1,1}, h^{2,1})$ plane: it is symmetric about the diagonal. As a statement about numbers this is a curiosity. Its force is physical.

Type IIA on X is the same four-dimensional theory as type IIB on $\tilde{X}$, and vice versa. Compare the multiplet counts: (8.91) for IIA on X has $n_V = h^{1,1}(X)$, while (8.92) for IIB on $\tilde{X}$ has $n_V = h^{2,1}(\tilde{X}) = h^{1,1}(X)$. They match, and so do the hypermultiplets. The theories are identical. What makes this a theorem worth its name, rather than a coincidence of counting, is that the two sides compute the vector-multiplet metric by completely different means. On the IIA side the vector moduli are Kähler moduli, and their special-Kähler metric is corrected by worldsheet instantons: a Euclidean string wrapping a holomorphic curve k times is weighted by $e^{2\pi i\, k\, T^a}$, exponentially small in the area of the curve, so the metric is an infinite series in $e^{2\pi i\, T^a}$. On the IIB side the very same numbers are complex-structure moduli, whose special-Kähler metric is *classical*: complex structure does not couple to the worldsheet area, so there are no α' corrections at all, only the period integrals (8.87). Mirror symmetry equates a hard, all-orders instanton sum with a finite classical computation.

A- and B-model

The precise home of this statement is the topological string. The $\mathcal{N} = (2,2)$ worldsheet sigma model on a Calabi-Yau admits two inequivalent topological twists. The *A-model* depends only on the Kähler moduli and computes symplectic invariants: its correlators are Gromov-Witten invariants, counts of holomorphic maps (worldsheet instantons) from the sphere into X. The *B-model* depends only on the complex-structure moduli and computes periods and variations of Hodge structure, pure complex geometry with no instanton sum. Mirror symmetry is the statement

$$\text{A-model on } X \;\cong\; \text{B-model on } \tilde{X}, \tag{8.94}$$

which is why it exchanges "quantum" Kähler data for "classical" complex-structure data. The exchange of the two moduli spaces in (8.93) is the shadow at the level of dimensions of an isomorphism at the level of every correlator.

SYZ: mirror symmetry is T-duality

Why should such a partner $\tilde{X}$ exist at all? The physical mechanism is Strominger-Yau-Zaslow. Near a large-volume, large-complex-structure limit a Calabi-Yau threefold is fibred by special Lagrangian three-tori T^3 over a three-dimensional base,

$$T^3 \hookrightarrow X \longrightarrow B, \qquad \dim_{\mathbb{R}} B = 3, \tag{8.95}$$

and mirror symmetry is T-duality performed simultaneously on all three legs of the fibre (Chapter 6). Three T-dualities invert the volume of the T^3 and exchange the momentum and winding of strings on it; done fibrewise, they turn the symplectic (Kähler) data of X into the complex-structure data of the dual fibration $\tilde{X}$ and back. This is the sense in which mirror symmetry is not a new duality but a corner of the same T-duality that runs through the whole course: it is fibrewise T-duality, and D-branes wrapping the T^3 (which SYZ uses to reconstruct $\tilde{X}$ as a moduli space of branes) are what makes the dual geometry visible. SYZ is proven only in limits and remains a conjecture in general, but as a picture it is the reason the mirror exists.

Example 8.19 (The quintic, worked)**.** Return to the quintic of (8.80). Its Kähler class is inherited from the single hyperplane class of $\mathbb{CP}^4$ restricted to X, so

$$h^{1,1}(X) = 1, \tag{8.96}$$

a lone Kähler modulus, the overall size. Its complex-structure moduli are the coefficients of the defining polynomial, counted modulo the redundancies that do not change the hypersurface.

Degree-5 monomials in five variables number $\binom{5+4}{4} = 126$; subtract 1 for the overall rescaling $p_5 \to \lambda p_5$, and $\dim \mathrm{PGL}(5,\mathbb{C}) = 24$ for linear coordinate changes of $\mathbb{CP}^4$:

$$h^{2,1}(X) = 126 - 1 - 24 = 101. \tag{8.97}$$

Hence $\chi(X) = 2(1 - 101) = -200$. Heterotic $\mathrm{E}_8 \times \mathrm{E}_8$ with the standard embedding on the quintic gives, by (8.90), net $|\chi|/2 = 100$ chiral generations. That is gloriously wrong for the real world, and it is exactly why one needs Calabi-Yaus with small $|\chi|$: the quintic is the "hydrogen atom" of the subject, not a model of Nature.

The mirror quintic $\tilde{X}$ has $h^{1,1}(\tilde{X}) = 101$, $h^{2,1}(\tilde{X}) = 1$, and $\chi = +200$. Greene and Plesser built it as a resolution of the orbifold $X/(\mathbb{Z}_5)^3$, the first explicit mirror pair. Its punchline is computational. The single Kähler modulus of the original quintic receives worldsheet-instanton corrections whose coefficients are the numbers n_d of rational curves of degree d inside X. Extracting them directly is a fearsome enumerative problem; through mirror symmetry Candelas, de la Ossa, Green and Parkes computed them from a one-parameter period integral on $\tilde{X}$, obtaining

$$n_1 = 2875, \qquad n_2 = 609250, \qquad n_3 = 317206375, \quad \ldots \tag{8.98}$$

The first number, the count of lines on the quintic, was classical nineteenth-century geometry; the second, the conics, was known; the higher ones were not, and the mirror produced them all at once. This calculation, more than any formal argument, is why mirror symmetry was taken seriously, and it opened the field of Gromov-Witten theory in mathematics.

8.5.6 Where this is going: the moduli problem

We have built vacua and read off their spectra, and in doing so we have inherited a problem. Every Calabi-Yau vacuum comes with $h^{1,1} + h^{2,1}$ massless complex scalars, the moduli, plus the dilaton, and nothing in the construction gives them a potential. Massless scalars are not a decoration: their vacuum expectation values set the size and shape of X and, through them, every coupling of the four-dimensional theory, the gauge couplings and Yukawas included. Undetermined moduli mean undetermined physics, long-range fifth forces they would mediate, and, for the dilaton, an outright runaway to zero coupling where the potential slides to zero. A vacuum with flat directions is not a prediction.

Fixing this, giving the moduli a potential and a mass, is the subject of Chapter 9: turn on quantised fluxes of the p-form field strengths through the cycles of X, generate the superpotential $W = \int G_3 \wedge \Omega$ built from exactly the periods (8.87), and lift the flat directions. That the input to flux stabilisation is the cohomology of X is the reason this chapter spent its effort on Hodge numbers and periods rather than on a metric we cannot write.

Remark 8.20 (Beyond the lectures: the hub theory on G_2). The same logic compactifies eleven-dimensional M-theory, the hub of the duality web (Chapter 7). Reducing $11 \to 4$ preserves minimal $\mathcal{N} = 1$ when the seven-dimensional internal space has holonomy G_2, the exceptional entry of Berger's list that a single real spinor picks out, exactly as SU(3) was picked out here. A G_2-holonomy seven-manifold has its own two relevant Betti numbers, b_2 and b_3, which set the numbers of U(1) vector multiplets and neutral chiral multiplets; getting chiral charged matter and nonabelian gauge symmetry requires singularities of the G_2 space, the eleven-dimensional analogue of the ADE loci of Sections 8.3 and 8.4. The Calabi-Yau story is the string-frame slice of this bigger M-theoretic picture, and it shares the same unsolved moduli problem.

Exercise 8.12 (The CY-detector)**.** Using $c(T\mathbb{CP}^n) = (1 + H)^{n+1}$ and the adjunction sequence for a smooth hypersurface X of degree d, show that $c_1(X) = (n + 1 - d)\, H|_X$. Deduce that a hypersurface in $\mathbb{CP}^n$ is Calabi-Yau precisely when $d = n + 1$, and identify the resulting complex

dimension. Check that the cubic in $\mathbb{CP}^2$, the quartic in $\mathbb{CP}^3$ and the quintic in $\mathbb{CP}^4$ are the torus, K3 and the quintic threefold.

Exercise 8.13 (Euler number from the diamond)**.** From the Calabi-Yau three-fold Hodge diamond (8.82), write down all seven Betti numbers and verify $\chi = 2(h^{1,1} - h^{2,1})$. Then confirm that under the mirror map (8.93) the Euler number changes sign, and explain in one sentence why type IIA on X and type IIB on $\tilde{X}$ nonetheless have identical numbers of vector and hypermultiplets.

Exercise 8.14 (Counting the quintic's shapes)**.** Reproduce $h^{2,1} = 101$ for the quintic by counting degree-5 monomials in five variables and subtracting the coordinate redundancies, as in Theorem 8.19. How many net heterotic generations does the standard embedding give, and what value of $|h^{1,1} - h^{2,1}|$ would you need instead for three? (You need not find such a manifold; freely-acting quotients, as in the warning above, are one route.)

Exercise 8.15 (Which sector are the Kähler moduli in?)**.** In type IIA on X the complexified Kähler moduli sit in vector multiplets and the complex-structure moduli in hypermultiplets; in type IIB the assignment is reversed. Using this, argue that the worldsheet-instanton corrections to the special-Kähler metric of the vector multiplets appear on the IIA side but not the IIB side, and hence that mirror symmetry lets one compute IIA instanton sums by a classical period integral in IIB. Which of $h^{1,1}$ and $h^{2,1}$ controls the number of independent instanton-corrected couplings in each theory?

Chapter 9

Flux, Moduli Stabilisation, and the Landscape

9.1 The moduli problem

Chapter 8 handed us a way to make four-dimensional physics out of ten: take a superstring, wrap six of its dimensions on a Calabi-Yau (or one of its cousins), and read off the light spectrum from the topology of the internal space. The result is a genuine four-dimensional effective field theory with chiral matter, gauge groups and gravity. It also carries a structural defect that would sink any of these models as a description of nature: the compactification does not fix the size and shape of the internal manifold, nor the strength of the string coupling. Those data survive into four dimensions as *massless scalar fields*, the *moduli*. A theory with unfixed moduli predicts a continuum of vacua rather than a world, and each massless scalar is, on its own, in conflict with experiment. This section states the problem sharply; the rest of the chapter is the machinery that removes it.

The big picture

A Calabi-Yau compactification delivers a whole family of vacua, labelled by the continuous parameters (sizes, shapes, the dilaton) that leave the internal space Ricci-flat. In four dimensions those parameters are massless scalars with gravitational-strength couplings. Massless scalars are excluded three times over: they mediate long-range fifth forces that violate the equivalence principle, they leave every coupling constant of the low-energy theory undetermined and time-dependent, and their potential, when one is generated, generically runs away to zero coupling and infinite volume (the Dine-Seiberg problem). Supersymmetry protects the flat directions, and the same supersymmetry makes lifting them delicate: a perturbatively generated superpotential is forbidden by holomorphy, so the potential must come from quantised flux and from non-perturbative effects. Producing it, at values of the moduli where the calculation can be trusted, is the moduli stabilisation problem.

9.1.1 Moduli: the flat directions a compactification cannot avoid

Recall the mechanism. To preserve $\mathcal{N} = 1$ supersymmetry in four dimensions from the heterotic string (or $\mathcal{N} = 2$ from type II), the internal six-manifold must admit a covariantly constant spinor, which forces $SU(3)$ holonomy and hence a Ricci-flat Kähler metric: a Calabi-Yau threefold (Chapter 8). Yau's theorem guarantees that such a metric exists and is unique once you fix two sets of data: the cohomology class of the Kähler form J and the complex structure of the manifold. Neither is unique. You can rescale and reshape the internal space through a continuous

family of Ricci-flat metrics, and every member of that family is an equally good solution of the ten-dimensional equations of motion.

Deform the internal metric $g_{mn} \to g_{mn} + \delta g_{mn}$ while staying Ricci-flat. The allowed deformations split, by type, into two classes (the details are in Chapter 8 and Appendix E):

- mixed deformations $\delta g_{i\bar{j}}$, which change the Kähler form and are counted by $H^{1,1}(X)$: these are the *Kähler moduli*, and they control the volumes of the manifold and of its holomorphic cycles ($h^{1,1}$ of them);
- pure deformations δg_{ij}, which change the complex structure and are counted by $H^{2,1}(X)$: these are the *complex-structure moduli*, controlling the shape ($h^{2,1}$ of them).

On top of the metric moduli sit scalars from expanding the p-form gauge fields on internal cycles (axions from B_2 and, in type II, from the RR potentials), open-string and brane-position moduli when D-branes are present, and Wilson lines. And every string compactification carries one universal modulus that has nothing to do with the choice of X: the *dilaton* ϕ, whose vacuum value sets the string coupling,

$$g_s = e^{\langle \phi \rangle}. \tag{9.1}$$

In type IIB the dilaton pairs with the RR axion C_0 into the axio-dilaton $\tau = C_0 + ie^{-\phi}$ of Section 5.4.4, Eq. (5.53); in the heterotic string it pairs with the model-independent axion. In both descriptions the dilaton is present, and it counts among the moduli.

Why are these fields massless? Because they are exactly *flat directions* of the classical potential. Moving along a modulus takes one Ricci-flat solution to a neighbouring one at the same energy: the ten-dimensional action is stationary and its value does not change, so the four-dimensional potential is independent of the modulus. A scalar whose potential is identically flat has no mass term. Collecting the moduli into real fields φ^I, the two-derivative effective action is a massless sigma model coupled to gravity,

$$S_{\text{mod}} = \frac{1}{2\kappa_4^2} \int \mathrm{d}^4x \, \sqrt{-g} \, \Big[R - 2\, \mathcal{G}_{IJ}(\varphi)\, \partial_\mu \varphi^I \partial^\mu \varphi^J - 2\, V(\varphi) \Big], \qquad V \equiv 0, \tag{9.2}$$

where the target-space metric $\mathcal{G}_{IJ}$ is the (special-Kähler times quaternionic, for $\mathcal{N} = 2$) geometry of the moduli space computed in Chapter 8. The crucial and damning entry is the last one: $V \equiv 0$, identically, to all orders in perturbation theory along these directions. In $\mathcal{N} = 1$ language the moduli are the lowest components of chiral multiplets, $\mathcal{G}_{IJ}$ is the Kähler metric $K_{i\bar{j}}$ of Appendix C, and the vanishing of V is the statement that the superpotential does not depend on them.

Example 9.1 (Counting the moduli of the quintic)**.** The quintic threefold, the running example of Chapter 8, has $h^{1,1} = 1$ and $h^{2,1} = 101$. That is one Kähler modulus (the overall volume), 101 complex-structure moduli, and the universal dilaton: 103 complex scalar fields, 206 real degrees of freedom, none of which the compactification fixes and each of which the effective theory would like to be massless. A generic Calabi-Yau has hundreds of moduli. The problem is generic, the rule rather than the exception.

Remark 9.2 (Beyond the lectures: why flatness is protected, not accidental)*.* One might hope that quantum corrections quietly generate a mass. Supersymmetry blocks the easy routes. The moduli superpotential W is holomorphic in the chiral fields, and the perturbative non-renormalisation theorems forbid its generation order by order in g_s: a term such as $m\,\Phi^2$ cannot appear in W if it is absent classically. The Kähler potential K does receive perturbative (α' and g_s) corrections, but along a genuine flat direction these correct the kinetic term, not the (absent) potential. Lifting a modulus therefore requires an ingredient that evades holomorphy plus perturbation theory: topologically quantised background flux, which contributes to W at "tree level" but

only in discrete units (Section 9.2), or non-perturbative effects of order e^{-1/g_s} or $e^{-\mathrm{Vol}}$ such as instantons and gaugino condensation (Section 9.3). This is the deep reason the solution looks the way it does.

9.1.2 Why massless moduli are excluded by experiment

A massless scalar with gravitational-strength couplings is ruled out directly, by tabletop experiments and by astrophysics. There are two independent reasons.

Long-range fifth forces. A modulus φ couples to the Standard Model because the Standard Model parameters depend on it (see below). Expanding those couplings to linear order gives a trilinear vertex between φ and ordinary matter, so exchanging a φ quantum between two bodies produces a Yukawa force alongside Newtonian gravity,

$$V(r) = -\frac{G_N m_1 m_2}{r}\Big(1 + 2\alpha_\varphi\, e^{-m_\varphi r}\Big), \qquad \alpha_\varphi = \mathcal{O}(1)\,, \tag{9.3}$$

of range set by the Compton wavelength $\lambda = 1/m_\varphi$. For a truly massless modulus the range is infinite and the "fifth force" is a permanent correction to gravity of order-one relative strength. Worse, because φ couples to the gauge and Yukawa couplings, and hence to nuclear and electromagnetic binding energies, its charge is not proportional to mass: different materials fall at different rates. This violates the equivalence principle, which is tested to extraordinary precision. Lunar laser ranging and the torsion-balance experiments of the Eöt-Wash group, and the satellite test MICROSCOPE, bound the differential (composition-dependent) acceleration to $\eta \lesssim 10^{-13}$–10^{-15}; the same experiments verify the Newtonian inverse-square law down to distances of order $50\,\mu$m. A gravitational-strength scalar of longer range is excluded. Turning the inverse-square-law reach into a mass, a modulus must satisfy

$$m_\varphi \;\gtrsim\; \frac{1}{50\,\mu\mathrm{m}} \;\approx\; 4\times 10^{-3}\ \mathrm{eV}, \tag{9.4}$$

just to escape the fifth-force searches, and the equivalence-principle tests push the bound higher for composition-dependent couplings. A massless modulus misses this by every order of magnitude there is.

The couplings of nature are the moduli. This is the deeper problem, and it is there even if you could somehow switch off the fifth force. In a string compactification the parameters of the low-energy theory are outputs, fixed by the moduli vacuum expectation values rather than put in by hand. The string coupling is $g_s = e^{\langle\phi\rangle}$. A gauge coupling comes from the real part of the holomorphic gauge kinetic function of Appendix C,

$$\frac{1}{g_a^2} = \mathrm{Re}\, f_a(\text{moduli}), \tag{9.5}$$

which for a brane wrapping a cycle Σ_a is set by the cycle volume, $1/g_a^2 \sim \mathrm{Vol}(\Sigma_a)/g_s$, a Kähler modulus; the heterotic gauge coupling is set by the dilaton, $1/g^2 \sim \mathrm{Re}\, S$. Yukawa couplings are overlap integrals of internal wavefunctions and depend on the complex-structure moduli and on the geometry. Newton's constant in four dimensions is $M_{\mathrm{pl}}^2 \sim \mathrm{Vol}(X)/(g_s^2\,\alpha'^4)$, again a modulus. If the moduli are not fixed, none of these numbers is predicted: the "theory" is a family with a continuum of different fine-structure constants, different mass ratios, different gauge groups' strengths, one for each point of moduli space. There is nothing to compare with experiment.

And if the moduli are not fixed they are dynamical, so the couplings vary in time as the fields roll on the cosmological background. A time-varying φ makes the fine-structure constant time-dependent, $\dot\alpha/\alpha \neq 0$. Observations forbid this at a stringent level: quasar absorption spectra

bound $|\Delta\alpha/\alpha| \lesssim 10^{-5}$ over the last ten billion years, the Oklo natural fission reactor bounds $|\Delta\alpha/\alpha| \lesssim 10^{-7}$ over the past 1.8 Gyr, and atomic-clock comparisons give $|\dot\alpha/\alpha| \lesssim 10^{-17}\,\mathrm{yr}^{-1}$ today. Massless rolling moduli are inconsistent with a universe whose constants are, to very high accuracy, constant.

Exercise 9.1 (The mass a fifth force allows)**.** Using $\hbar c \approx 197$ eV nm, convert the $50\,\mu$m reach of the inverse-square-law tests into the mass bound (9.4). Then estimate the range of a modulus of mass $m_\varphi = 1$ TeV and confirm that it is far too short to be seen in any fifth-force experiment. What does this tell you about the mass scale a successful stabilisation should aim for?

9.1.3 The cosmological moduli problem

Giving the moduli a mass is necessary but, on its own, still not sufficient: a light modulus is a cosmological hazard. This is the *cosmological moduli problem* (the same disease as the older Polonyi problem of hidden-sector supersymmetry breaking), and it puts a surprisingly high lower bound on the moduli masses.

The argument is short. A modulus is displaced from its eventual minimum in the early universe by an amount of order $M_{\rm pl}$ (there is no reason for the pre-inflationary value to sit at the minimum of a potential generated later). While the Hubble rate exceeds the modulus mass, $H > m_\varphi$, Hubble friction pins the field; once H drops below m_φ the field begins to oscillate about the minimum. Coherent oscillations of a scalar about a quadratic minimum behave as pressureless matter, with energy density $\rho_\varphi \propto a^{-3}$, redshifting more slowly than the ambient radiation ($\rho_{\rm rad} \propto a^{-4}$). Starting from a Planckian displacement the modulus quickly comes to dominate the energy budget. It only decays much later, because its couplings are gravitational: the width takes the standard gravitational-strength form

$$\Gamma_\varphi \;\simeq\; \frac{1}{4\pi}\,\frac{m_\varphi^3}{M_{\rm pl}^2}, \tag{9.6}$$

and the decay reheats the universe to a temperature

$$T_{\rm reh} \;\simeq\; \left(\frac{90}{\pi^2 g_*}\right)^{1/4}\sqrt{\Gamma_\varphi\, M_{\rm pl}} \;\sim\; \left(\frac{m_\varphi}{M_{\rm pl}}\right)^{3/2} M_{\rm pl}. \tag{9.7}$$

Both are textbook estimates (see e.g. Kolb and Turner); the mass bound they force is the cosmological moduli problem of Coughlan, Fischler, Kolb, Raby and Ross, and of Banks, Kaplan and Nelson. If $T_{\rm reh}$ falls below the temperature of Big-Bang nucleosynthesis, $T_{\rm BBN} \sim$ a few MeV, the modulus decays *after* the light elements have formed, dumping entropy and high-energy particles into a delicate plasma and destroying the successful predictions of nucleosynthesis. Demanding $T_{\rm reh} \gtrsim T_{\rm BBN}$ turns (9.7) into a lower bound on the mass,

$$m_\varphi \;\gtrsim\; 50\text{–}100\ \text{TeV}. \tag{9.8}$$

A modulus lighter than this decays too late to be cosmologically safe. The target is therefore sharper than "give the moduli a mass": the lightest modulus must weigh at least tens of TeV, many orders of magnitude above the supersymmetry-breaking scale one might have hoped to build a model around. Stabilisation schemes are judged, in part, on whether their lightest modulus clears (9.8).

Exercise 9.2 (Deriving the moduli mass bound)**.** Take the reduced Planck mass $M_{\rm pl} = 2.4 \times 10^{18}$ GeV, $g_* \sim 10$ and $T_{\rm BBN} = 4$ MeV. Using (9.6) and (9.7), solve $T_{\rm reh} = T_{\rm BBN}$ for m_φ and reproduce the order of magnitude in (9.8). Why does a *weaker*-than-gravitational coupling of φ to the Standard Model make the problem worse, not better?

9.1.4 Why a potential is hard to make: the Dine-Seiberg problem

Suppose we accept that some dynamics must generate a potential $V(\varphi)$ for the moduli. There is a structural obstruction to doing this in the regime where we can compute. It is best seen for the fields that control the expansion parameters: the dilaton, whose value is the string coupling g_s, and the overall volume $\mathcal{V}$ (in string units), whose inverse controls the α' expansion. Perturbative string theory, and supergravity, are trustworthy only at

$$g_s \ll 1, \qquad \mathcal{V} \gg 1, \tag{9.9}$$

weak coupling and large volume. Every correction we know how to compute, string loops, α' corrections, brane and instanton effects, is a series whose terms carry positive powers of g_s or negative powers of $\mathcal{V}$, and whose non-perturbative pieces go as e^{-a/g_s} or $e^{-b\,\mathcal{V}^{1/3}}$. Every one of these terms vanishes as $g_s \to 0$ or $\mathcal{V} \to \infty$. Schematically the potential for the coupling-controlling modulus u (think $u = 1/g_s$ or $u = \mathcal{V}$) is a sum

$$V(u) = \sum_n c_n\, u^{-p_n}, \qquad p_n > 0, \quad V(u) \xrightarrow{u\to\infty} 0^- \text{ or } 0^+. \tag{9.10}$$

If a single term dominates, V is monotonic and its only extremum is at the boundary $u = \infty$: the field runs away to zero coupling and infinite volume, decompactifying the theory back to ten dimensions. This is the *Dine-Seiberg problem*: the generic behaviour of a modulus potential is a runaway to the free, decompactified limit. To get a minimum at finite u you need at least two terms of comparable magnitude and opposite tendency to balance. But two terms of a series in $1/u$ are comparable only when $u = \mathcal{O}(1)$, that is, at $g_s \sim 1$ or $\mathcal{V} \sim 1$, precisely where (9.9) fails and the calculation cannot be trusted.

This is the tension the whole subject lives with. The place where we can compute is the place where the potential wants to run away; the place where a minimum can form is the place where we lose control. A successful stabilisation must therefore engineer a *parametric* separation, arranging that the competing terms balance at $g_s \ll 1$ and $\mathcal{V} \gg 1$ through some small parameter, rather than at $u = \mathcal{O}(1)$. The two schemes of Section 9.3, KKLT and the Large Volume Scenario, are exactly two ways of manufacturing such a balance: KKLT tunes the flux superpotential W_0 small so that a single non-perturbative term can fight it at large volume, while the Large Volume Scenario plays an α' correction against a non-perturbative term to stabilise at exponentially large $\mathcal{V}$. Whether either balance is genuinely under control is one of the open questions the chapter ends on.

Control is where the trouble is

It is tempting to read "weak coupling, large volume" as the safe corner of the theory. For moduli stabilisation it is the dangerous corner: it is where every term in the potential is small and the field is pulled towards decompactification. A stabilised vacuum at large volume has to be built against the Dine-Seiberg tendency; it survives only when the small parameter holding it together is small enough to keep the competing terms balanced far out in the controlled regime.

Exercise 9.3 (A single term cannot stabilise)**.** Consider $V(u) = A\,u^{-p}$ with $A \neq 0$ and $p > 0$, modelling one contribution to a modulus potential in the variable $u = 1/g_s$ or $u = \mathcal{V}$. Show that V has no stationary point at finite $u > 0$ and determine the direction of the force on u. Now take $V(u) = A\,u^{-p} - B\,u^{-q}$ with $A, B > 0$ and $0 < q < p$; find the critical point

$$u_* = \left(\frac{pA}{qB}\right)^{1/(p-q)},$$

show it is a minimum, and show that $u_* = \mathcal{O}(1)$ unless the coefficients are hierarchically tuned, $A/B \gg 1$. Interpret the required tuning in the language of (9.9).

9.1.5 The statement of the problem

Assemble the requirements. A compactification worth calling a model of nature must come with a mechanism that does all of the following at once.

- **Lift every flat direction.** Generate a potential $V(\varphi)$ whose critical point fixes *all* of the moduli, geometric, dilaton, and axionic, at definite values. A single leftover massless modulus reinstates the fifth force and the undetermined coupling.
- **Stay in the controlled regime.** Locate that minimum at $g_s \ll 1$ and $\mathcal{V} \gg 1$, where the two-derivative supergravity plus its leading corrections is a reliable description, and do so against the Dine-Seiberg runaway of Section 9.1.4.
- **Give the moduli acceptable masses.** Make the lightest modulus heavy enough to evade both the fifth-force bound (9.4) and, more stringently, the cosmological bound (9.8) of order tens of TeV, while keeping the compactification scale below the Kaluza-Klein and string scales so the four-dimensional description holds.
- **Get the vacuum energy right.** Match the observed, tiny, *positive* cosmological constant. As Appendix C shows, a supersymmetric $\mathcal{N} = 1$ vacuum has $V_* = -3\, e^K |W|^2 \leq 0$, anti-de Sitter or Minkowski; reaching $V_* > 0$ forces supersymmetry breaking and a positive "uplift". This is the hardest requirement, and it is where the reservations of Section 9.4 concentrate.

The ingredients that lift the flat directions and hold the minimum in the controlled regime, the first two requirements, are the subject of the next two sections. Quantised three-form flux threading the internal cycles generates a superpotential for the complex-structure moduli and the axio-dilaton, the Gukov-Vafa-Witten superpotential $W = \int G_3 \wedge \Omega$ of Section 9.2, and stabilises them at tree level. The Kähler moduli survive that step, protected by a no-scale structure (Section 9.2, and Appendix C), and are fixed by the non-perturbative superpotential of Section 9.3. The mass requirement is less automatic. Flux makes the complex-structure moduli and the axio-dilaton heavy, but the lightest Kähler modulus can come out parametrically light, and in the Large Volume Scenario the volume modulus and its axion are light by construction; whether such a spectrum clears the cosmological bound (9.8) is a question we carry into Section 9.3. The positive vacuum energy is arranged, if it can be, by an uplift whose control is contested. The count of how many such vacua there are, and what the landscape/swampland debate makes of them, is Section 9.4.

We can now restate the slogan the chapter opened with, and mean it precisely. A compactification with massless moduli is a continuum of models with unfixed, time-varying couplings and long-range forces that experiment has already excluded. Turning that continuum into a discrete set of candidate vacua, each with fixed couplings and heavy moduli, is the job of the flux, the superpotential, and the F-term scalar potential

$$V = e^K \left(K^{i\bar{\jmath}}\, D_i W\, \overline{D_j W} - 3|W|^2 \right), \tag{9.11}$$

whose first term is the positive F-term energy of broken supersymmetry and whose second is the negative, gravitational contribution that drives the vacuum towards anti-de Sitter. The sign of the vacuum energy is set by the competition between the two, and the no-scale cancellation of Section 9.2 is one special way that competition can play out.

9.2 Flux and the $\mathcal{N} = 1$ scalar potential

The big picture

Compactify type IIB on a Calabi-Yau orientifold and the low-energy theory is a 4D $\mathcal{N} = 1$ supergravity whose scalar potential is the object $V = e^K(K^{i\bar{\jmath}}D_iW\,\overline{D_{\jmath}W} - 3|W|^2)$ of Appendix C.4, a contest between the positive F-term energy of broken supersymmetry and the negative $-3|W|^2$ of gravity. With no background flux, W vanishes and every modulus is a flat direction: the moduli problem of Section 9.1. Switch on quantised three-form flux and two things happen. The flux is a fixed integer cohomology class, but its energy depends on the complex structure and the dilaton through the Hodge star, so it lifts those flat directions. And it feeds a holomorphic superpotential $W = \int_X G_3 \wedge \Omega$, the Gukov-Vafa-Witten superpotential, whose F-flatness conditions fix all $h^{2,1}$ complex-structure moduli and the axio-dilaton in one stroke. The Kähler moduli survive at this order, protected by a no-scale cancellation, and are the business of Section 9.3. The price of the flux is a tadpole: its D3-charge must be cancelled by branes and orientifold planes, and that budget is finite.

By the end of this section you should be able to:

- write down the Gukov-Vafa-Witten superpotential and compute its F-terms for the complex-structure moduli and the axio-dilaton;
- derive the imaginary self-dual condition on G_3 and explain why it fixes those fields while leaving the Kähler moduli flat;
- state the no-scale cancellation that makes the GKP vacuum Minkowski, and identify the constant W_0 passed to the next section;
- state the tadpole constraint and explain why it makes the set of flux vacua finite.

We work in type IIB, the setting where flux stabilisation is best controlled, and throughout in the reduced-Planck units $M_{\rm pl} = 1$ of Appendix C.4. The ten-dimensional theory is the one of Section 5.4: the metric, the axio-dilaton $\tau = C_0 + ie^{-\phi}$ of Eq. (5.53), the NS-NS three-form $H_3 = \mathrm{d}B_2$, the RR three-form $F_3 = \mathrm{d}C_2$, and the self-dual five-form $\tilde{F}_5$. We compactify on a Calabi-Yau *orientifold* X: the quotient of the theory by worldsheet parity combined with a holomorphic involution of X, whose fixed loci are the orientifold planes introduced in Section 6.4. Those O-planes carry negative D3-charge, and that turns out to be what lets us switch on flux at all. Keeping the four-dimensional slice SO(3, 1)-invariant leaves a 4D $\mathcal{N} = 1$ theory whose moduli are the $h^{2,1}$ complex-structure deformations z^a, the axio-dilaton τ, and the $h^{1,1}$ Kähler moduli T_α. This section builds the potential that fixes the first two sets.

9.2.1 Background flux and Dirac quantisation

A p-form field strength can take a nonzero value in the vacuum, provided the value is compatible with the field being a legitimate gauge field. For an ordinary Maxwell field on a compact space, this is the statement that the magnetic flux through a two-cycle is quantised: $\frac{1}{2\pi}\int_{\Sigma_2} F_2 \in \mathbb{Z}$. The same Dirac argument applies to H_3 and F_3. Each is the field strength of a p-form potential (B_2 and C_2) that couples to a charged brane (the fundamental string and the D1-brane, Chapter 6), and consistency of the brane wavefunction in the presence of the flux forces the periods over integral three-cycles to be quantised. In the conventions of Appendix A, with $\ell_s = \sqrt{\alpha'}$,

$$\frac{1}{(2\pi)^2\alpha'}\int_\Sigma F_3 \;\in\; \mathbb{Z}, \qquad \frac{1}{(2\pi)^2\alpha'}\int_\Sigma H_3 \;\in\; \mathbb{Z}, \qquad \forall\,\Sigma \in H_3(X,\mathbb{Z}). \tag{9.12}$$

Pick a symplectic basis $\{A^I, B_I\}$, $I = 0, \ldots, h^{2,1}$, of $H_3(X, \mathbb{Z})$ with intersection pairing $A^I \cap B_J = \delta^I{}_J$. The two fluxes are then labelled by two integer vectors,

$$\frac{1}{(2\pi)^2\alpha'}\int_{A^I} F_3 = M^I, \quad \frac{1}{(2\pi)^2\alpha'}\int_{B_I} F_3 = M_I, \quad \frac{1}{(2\pi)^2\alpha'}\int_{A^I} H_3 = K^I, \quad \frac{1}{(2\pi)^2\alpha'}\int_{B_I} H_3 = K_I, \tag{9.13}$$

a total of $2(h^{2,1}+1)$ integers for each three-form. (The H_3 labels K^I, K_I are flux integers, not the Kähler potential K that appears below; the context always makes clear which is meant.) These integers are frozen: they are topological data, unchanged by any continuous deformation of the metric or the dilaton. What is *not* frozen is the energy the flux carries, and it is that energy that does the stabilising.

The energy density of a p-form flux is $\int F \wedge \star F$, and the Hodge star depends on the metric of X, hence on the moduli. A cohomology class is a fixed integral, but its representative of minimal norm, and the norm itself, move as the moduli move. Flux therefore assigns different energies to different points of moduli space, and the vacuum sits where that energy is least. This is the mechanism by which flux lifts flat directions, and it holds quite generally. The task of the rest of the section is to make this precise in the $\mathcal{N} = 1$ language, where the flux energy just described is repackaged as a holomorphic superpotential.

Remark 9.3 (Beyond the lectures: the Freed-Witten shift). Equation (9.12) is the leading statement. On a curved space the F_3 flux through a cycle wrapped by a D-brane can be shifted by a half-integer set by the third Stiefel-Whitney class of the cycle (Freed and Witten, 1999): the correct quantisation is $\frac{1}{(2\pi)^2\alpha'}\int_\Sigma F_3 + \frac{1}{2}w_3(\Sigma) \in \mathbb{Z}$, and analogously for H_3 on cycles wrapped by NS-branes. On the spin cycles relevant to the models below the shift is absent, and we use (9.12) as stated; the half-integer refinements matter for the exact tadpole bookkeeping and are tracked in the model-building literature.

9.2.2 The Gukov-Vafa-Witten superpotential

Assemble the two three-forms into the single complex flux that the $\mathrm{SL}(2,\mathbb{Z})$ of Section 5.4.4 rotates by a phase,

$$G_3 \;\coloneqq\; F_3 - \tau\, H_3. \tag{9.14}$$

This is the unnormalised combination; the field strength written in Eq. (5.56) is this one divided by $\sqrt{\mathrm{Im}\,\tau}$, a normalisation convenient for the kinetic term but not for the holomorphic pairing we now want. Because τ multiplies H_3, G_3 is a holomorphic function of the axio-dilaton, and its periods are fixed integers dressed by τ: from Eq. (9.13), $\frac{1}{(2\pi)^2\alpha'}\int_{A^I} G_3 = M^I - \tau K^I$, and likewise on the B-cycles.

The flux enters the four-dimensional effective theory through the Gukov-Vafa-Witten superpotential

$$\boxed{W \;=\; \frac{1}{(2\pi)^2\alpha'}\int_X G_3 \wedge \Omega \;=\; \frac{1}{(2\pi)^2\alpha'}\int_X (F_3 - \tau H_3)\wedge\Omega} \tag{9.15}$$

with Ω the holomorphic $(3,0)$-form of X (Chapter 8). We absorb the overall numerical constant (a convention-dependent factor times powers of α' and M_{pl}) into the normalisation of W; only the ratios of critical points and the location of extrema carry physics, and both are insensitive to it. The object (9.15) is holomorphic in the moduli: it depends on τ explicitly and on the complex-structure moduli z^a through $\Omega(z)$, and on nothing else. In particular it does not depend on the Kähler moduli T_α, which we will use repeatedly below.

I am quoting (9.15), not deriving it. It was written down for M-theory on an eightfold by Gukov, Vafa and Witten (1999) and transcribed to the IIB three-form flux by Taylor and Vafa (1999);

the clean way to see it is to reduce the ten-dimensional gravitino variation and read off the term proportional to a four-dimensional gravitino mass, which by Eq. (C.58) is $e^{K/2}|W|$. The reduction is carried out in Giddings, Kachru and Polchinski (2001) and in DeWolfe and Giddings (2002). Here we take (9.15) as the input and study its consequences.

The other input is the Kähler potential. At tree level in α' and g_s it splits into three decoupled blocks,

$$K \;=\; \underbrace{-\ln\Big(-i\int_X \Omega\wedge\bar\Omega\Big)}_{K_{\rm cs}(z,\bar z)} \;\underbrace{-\ln\big(-i(\tau-\bar\tau)\big)}_{K_{\rm dil}(\tau,\bar\tau)} \;\underbrace{-2\ln\mathcal{V}}_{K_{\rm K}(T,\bar T)}, \tag{9.16}$$

where $\mathcal{V}$ is the Einstein-frame volume of X in string units, a function of the Kähler moduli alone. The three arguments are each positive in the conventions of Appendix E: $-i\int_X \Omega\wedge\bar\Omega > 0$ for that orientation (equivalently $i\int_X \Omega\wedge\bar\Omega > 0$ under the opposite orientation), and $-i(\tau-\bar\tau) = 2\,\mathrm{Im}\,\tau = 2/g_s > 0$. Because (9.16) is a sum of a z-piece, a τ-piece and a T-piece, the Kähler metric $K_{i\bar\jmath}$ is block diagonal and the three sectors do not mix at this order. The complex-structure block $K_{\rm cs}$ is the special-Kähler metric on the moduli space of Ω; the axio-dilaton block $K_{\rm dil}$ is the hyperbolic metric of Section 5.4.4; the Kähler block $K_{\rm K}$ is no-scale, as we now use.

9.2.3 Special geometry and the F-flatness conditions

A word first on the complex geometry we are about to use; the facts are collected in Section 8.5 and Appendix E. The internal X is a complex threefold, so its differential forms split by holomorphic type into (p,q)-pieces, and the Hodge numbers $h^{p,q} = \dim H^{p,q}(X)$ count the independent harmonic forms of each type. Integrating over X keeps only the top $(3,3)$ piece of a form and discards the rest, so a wedge product survives the integral only when its holomorphic and antiholomorphic degrees each add up to three. The internal Hodge star $\star_6$ maps each (p,q) block of a middle-degree form to itself. Those are the only ingredients the derivation needs.

To differentiate (9.15) we need to know how Ω moves with the complex structure. The holomorphic three-form is not quite holomorphic in the moduli: rescaling $\Omega \to e^{-h(z)}\Omega$ is a Kähler transformation of Appendix C.4.2, so Ω is a section of a line bundle. Its derivative decomposes into a piece along itself and a genuine $(2,1)$-form,

$$\partial_a\Omega \;=\; k_a(z)\,\Omega \;+\; \chi_a, \qquad \chi_a \in H^{2,1}(X), \tag{9.17}$$

the Kodaira formula. Read $k_a(z)$ simply as the component of $\partial_a\Omega$ along Ω, and χ_a as the leftover $(2,1)$ part. The χ_a span $H^{2,1}$ as a runs over the $h^{2,1}$ moduli, which is why complex-structure deformations and $(2,1)$-forms are counted by the same Hodge number. Two orthogonality facts follow from counting holomorphic degree on a threefold, where the only nonzero top form is $(3,3)$:

$$\int_X \Omega\wedge\partial_a\Omega = 0 \quad \text{(Griffiths transversality)}, \qquad \int_X \chi_a\wedge\bar\Omega = 0. \tag{9.18}$$

The first holds because $\Omega\wedge\Omega$ and $\Omega\wedge\chi_a$ both vanish by degree; the second because $(2,1)\wedge(0,3)$ has holomorphic degree $2 \neq 3$. Now feed (9.17) into $K_{\rm cs} = -\ln(-i\int_X \Omega\wedge\bar\Omega)$. Differentiating and using the second identity $\int_X \chi_a\wedge\bar\Omega = 0$,

$$\partial_a K_{\rm cs} \;=\; -\,\frac{\int_X \partial_a\Omega\wedge\bar\Omega}{\int_X \Omega\wedge\bar\Omega} \;=\; -\,\frac{k_a\int_X \Omega\wedge\bar\Omega}{\int_X \Omega\wedge\bar\Omega} \;=\; -\,k_a, \tag{9.19}$$

so the Kähler connection of the complex-structure sector is exactly $-k_a$. That identity is what makes the connection cancel the $k_a\Omega$ term in the covariant derivative, as we now see.

Compute the F-terms $D_iW = \partial_i W + (\partial_i K)W$ of Eq. (C.48). For a complex-structure modulus, using (9.17) and $\partial_a K = \partial_a K_{\rm cs} = -k_a$ from (9.19),

$$D_aW \;=\; \int_X G_3 \wedge \partial_a\Omega + (\partial_a K)\int_X G_3 \wedge \Omega \;=\; k_a W + \int_X G_3 \wedge \chi_a - k_a W \;=\; \int_X G_3 \wedge \chi_a. \tag{9.20}$$

Since χ_a is a $(2,1)$-form, the wedge $G_3 \wedge \chi_a$ is a top form only through the $(1,2)$ part of G_3. So $D_aW = 0$ for every a is precisely the statement that G_3 has no $(1,2)$ piece. For the axio-dilaton, $\partial_\tau G_3 = -H_3$ and $\partial_\tau K_{\rm dil} = -1/(\tau-\bar\tau)$; writing $H_3 = (\bar G_3 - G_3)/(\tau-\bar\tau)$ and simplifying,

$$D_\tau W \;=\; -\int_X H_3 \wedge \Omega \;-\; \frac{1}{\tau-\bar\tau}\int_X G_3 \wedge \Omega \;=\; -\frac{1}{\tau-\bar\tau}\int_X \bar G_3 \wedge \Omega. \tag{9.21}$$

Here $\bar G_3 \wedge \Omega$ is a top form only through the $(0,3)$ part of $\bar G_3$, that is through the conjugate of the $(3,0)$ part of G_3. So $D_\tau W = 0$ says G_3 has no $(3,0)$ piece. Collecting the two conditions,

$$D_aW = 0\ (\forall a) \iff G^{(1,2)} = 0, \qquad D_\tau W = 0 \iff G^{(3,0)} = 0. \tag{9.22}$$

A generic three-form on a Calabi-Yau splits as $(3,0)\oplus(2,1)\oplus(1,2)\oplus(0,3)$. F-flatness for the complex-structure moduli and the dilaton kills the first and third, leaving

$$G_3 \;\in\; H^{2,1}\oplus H^{0,3}, \qquad \text{equivalently} \qquad \star_6\, G_3 = i\, G_3, \tag{9.23}$$

the *imaginary self-dual* (ISD) condition, since on a threefold the internal Hodge star acts as $+i$ on $(2,1)$ and $(0,3)$ forms and as $-i$ on $(3,0)$ and $(1,2)$. Imposing that the fixed cohomology class $[G_3]$ be ISD is $h^{2,1}+1$ complex conditions on the $h^{2,1}+1$ complex fields (z^a,τ). Generically the solutions are isolated points: every complex-structure modulus and the axio-dilaton are fixed by the flux.

Example 9.4 (A rigid Calabi-Yau fixes the dilaton)**.** Take X rigid, $h^{2,1}=0$, so there are no complex-structure moduli and Ω is fixed once and for all. Only the dilaton is dynamical. With a single A/B pair of three-cycles, define the flux periods $f = \int_X F_3\wedge\Omega$ and $h = \int_X H_3\wedge\Omega$, both fixed complex numbers determined by the integers (9.13). Then $W(\tau) = f - \tau h$, a linear function, and by (9.21) the F-flatness condition $D_\tau W = 0$ is $\int \bar G_3\wedge\Omega = f - \bar\tau\, h = 0$, so

$$\tau_* \;=\; \overline{\Big(\frac{f}{h}\Big)}. \tag{9.24}$$

The axio-dilaton is pinned at the ratio of the F_3 and H_3 periods. The vacuum is physical, $\operatorname{Im}\tau_* > 0$, only when those integers give the ratio the right phase, and it is weakly coupled, $\operatorname{Im}\tau_* = 1/g_s \gg 1$, only for a suitable near-degeneracy of f and h. Not every flux choice lands in the controlled region: this is the first sign that the counting of Section 9.4 runs over candidate vacua, most of which fail one control criterion or another.

Exercise 9.4 (The dilaton F-term and the $(3,0)$ piece)**.** Starting from $W = \int_X G_3\wedge\Omega$ with $G_3 = F_3 - \tau H_3$ and $K_{\rm dil} = -\ln(-i(\tau-\bar\tau))$, verify the intermediate steps of (9.21): compute $\partial_\tau W$, use $H_3 = (\bar G_3 - G_3)/(\tau-\bar\tau)$, and show $D_\tau W = -(\tau-\bar\tau)^{-1}\int_X \bar G_3\wedge\Omega$. Conclude that $D_\tau W = 0$ removes the $(3,0)$ part of G_3. Then combine with (9.20) to establish the ISD condition (9.23), and identify which Hodge type of G_3 leaves $W = \int_X G_3\wedge\Omega$ nonzero at the F-flat point.

9.2.4 The GKP vacuum and the no-scale structure

We have fixed the complex-structure moduli and the dilaton without ever mentioning the Kähler moduli. They are absent from W, and the reason the resulting potential is nonetheless well behaved is the no-scale structure of Appendix C.4.6. Write the full F-term potential with the index I running over all moduli (z^a,τ,T_α):

$$V = e^K\Big(K^{I\bar J} D_I W\, \overline{D_J W} - 3|W|^2\Big). \tag{9.25}$$

The two terms pull against each other: $K^{I\bar{J}} D_I W \, \overline{D_J W} \geq 0$ is the F-term energy stored in broken supersymmetry, while $-3|W|^2 \leq 0$ is the gravitational piece that on its own would drive the vacuum to anti-de Sitter, and which of them wins sets the sign of V. Because W is independent of T_α, the Kähler F-terms are pure connection, $D_\alpha W = K_\alpha W$. The internal volume $\mathcal{V}$ is homogeneous of degree 3/2 in the real parts of the Kähler moduli, so $e^{K_{\rm K}} = \mathcal{V}^{-2}$ scales homogeneously with them, and this homogeneity is what produces the no-scale identity $K^{\alpha\bar{\beta}} K_\alpha K_{\bar{\beta}} = 3$ of Eq. (C.60). With the block-diagonal metric of (9.16), the Kähler-moduli contribution to (9.25) is $K^{\alpha\bar{\beta}} K_\alpha K_{\bar{\beta}} |W|^2 = 3|W|^2$, which cancels the $-3|W|^2$ term exactly:

$$V \;=\; e^K \sum_{i \in \{z^a, \tau\}} K^{i\bar{\jmath}} \, D_i W \, \overline{D_j W} \;\geq\; 0. \tag{9.26}$$

The potential is positive semi-definite, it depends on the Kähler moduli only through the overall $e^K \sim \mathcal{V}^{-2}$ prefactor, and it is minimised at $V = 0$ precisely on the F-flat locus $D_a W = D_\tau W = 0$ of Section 9.2.3. This is the Giddings-Kachru-Polchinski (GKP) vacuum: a no-scale Minkowski solution in which the complex-structure moduli and the axio-dilaton are stabilised at the ISD point, the Kähler moduli remain exactly flat, and the vacuum energy vanishes at this order.

Two features of the leftover deserve emphasis. First, the value of W at the minimum is generically nonzero: an ISD flux with a $(0,3)$ component gives

$$W_0 \;:=\; \langle W \rangle \;=\; \int_X G^{(0,3)} \wedge \Omega \;\neq\; 0, \tag{9.27}$$

a fixed complex constant, set by the flux and by the stabilised values of (z, τ). This constant W_0 is what Section 9.3 takes as its input: it is the number that seeds Kähler-moduli stabilisation in KKLT and the Large Volume Scenario. A purely $(2,1)$ flux gives $W_0 = 0$ and a supersymmetric Minkowski vacuum; a flux with a $(0,3)$ part gives $W_0 \neq 0$, and then $D_\alpha W = K_\alpha W_0 \neq 0$, so $F^{T_\alpha} \neq 0$ and supersymmetry is broken along the Kähler directions even though $V = 0$. Second, the Kähler moduli stay flat because flux simply does not act on them; closing these directions needs the non-perturbative and higher-order effects of the next section.

Exercise 9.5 (The no-scale cancellation)**.** Take a single Kähler modulus with $K = -2\ln \mathcal{V}$ and $\mathcal{V} = (T + \bar{T})^{3/2}$. Show $K = -3\ln(T + \bar{T})$, compute $K_{T\bar{T}}$ and its inverse $K^{T\bar{T}}$, and verify the no-scale identity $K^{T\bar{T}} K_T K_{\bar{T}} = 3$ of Eq. (C.60). Then, with $W = W_0$ a T-independent constant, show $D_T W = K_T W_0$ and hence, from (9.26), that $V = 0$, even though $D_T W \neq 0$ breaks supersymmetry along this direction.

Remark 9.5 (Beyond the lectures: the ten-dimensional echo)*.* The ISD condition also drops out of the ten-dimensional equations, which is a useful check that the four-dimensional superpotential is capturing the right physics. Reducing the IIB action of Eq. (5.55) on X, the flux energy is, up to positive constants,

$$V_{\rm flux} \;\propto\; \frac{1}{\operatorname{Im}\tau} \int_X G_3 \wedge \star_6 \bar{G}_3 \;=\; \frac{2}{\operatorname{Im}\tau} \left\| G_3^{\rm IASD} \right\|^2 \;+\; \frac{\text{topological}}{\operatorname{Im}\tau} \int_X G_3 \wedge \bar{G}_3, \tag{9.28}$$

where the split follows from the projectors onto ISD and imaginary-anti-self-dual (IASD) parts and the fact that $\int G_3 \wedge \bar{G}_3$ is a topological pairing of fixed integers. The first term is a sum of squares, minimised to zero when $G_3^{\rm IASD} = 0$, that is when G_3 is ISD; the topological term is a constant that the five-form Bianchi identity and the warp factor conspire to cancel against, leaving $V = 0$. This is the same no-scale Minkowski vacuum seen in (9.26), now from the ten-dimensional side.

Warping. The GKP solution is genuinely warped. Ten-dimensional Poincaré invariance allows a metric

$$\mathrm{d}s_{10}^2 \;=\; e^{2A(y)} \, \eta_{\mu\nu} \, \mathrm{d}x^\mu \mathrm{d}x^\nu \;+\; e^{-2A(y)} \, \tilde{g}_{mn}(y) \, \mathrm{d}y^m \mathrm{d}y^n, \tag{9.29}$$

with $\tilde{g}$ conformal to the Calabi-Yau metric, and a self-dual five-form $\tilde{F}_5 = (1 + \star_{10})\,\mathrm{d}\Phi \wedge \mathrm{d}x^0 \wedge \cdots \wedge \mathrm{d}x^3$ built from a single scalar potential $\Phi(y)$. The Einstein equations then tie the warp factor to the flux and the local sources through a Poisson equation, schematically

$$-\tilde{\nabla}^2 e^{-4A} \;=\; \frac{G_{mnp}\,\bar{G}^{mnp}}{12\ \mathrm{Im}\,\tau} \;+\; (\text{local D3/O3 sources}), \tag{9.30}$$

and for ISD flux together with sources that saturate a BPS-like bound one finds the simple relation $\Phi = e^{4A}$. Strong warping, $e^{A} \ll 1$, occurs in regions where flux piles up, and it turns order-one flux integers into exponentially small energy scales. The redshift at the bottom of such a region is set by the flux integers (M, K) threading it: this is the Klebanov-Strassler throat (Klebanov and Strassler, 2000; Giddings, Kachru and Polchinski, 2001), which we quote rather than derive. It is the string realisation of a Randall-Sundrum hierarchy, and the region where the anti-brane uplift of Section 9.3 lives. For the leading stabilisation story the warping is a controlled correction: at large volume it acts back on the complex-structure sector only weakly, so computing W from (9.15) on the unwarped Calabi-Yau is self-consistent.

9.2.5 The tadpole constraint

Flux is not free. The five-form Bianchi identity, the C_4 equation of motion of Eq. (5.50), reads

$$\mathrm{d}\tilde{F}_5 \;=\; H_3 \wedge F_3 \;+\; 2\kappa_{10}^2\, T_3\, \rho_3^{\mathrm{loc}}, \tag{9.31}$$

where ρ_3^{loc} is the D3-charge density of the localised sources: D3-branes (charge $+1$ each), anti-D3-branes, O3-planes (charge $-\frac{1}{4}$ each), and the curvature-induced D3-charge carried by wrapped D7-branes and O7-planes. Integrate (9.31) over the compact X. The left-hand side is the integral of an exact form over a manifold without boundary, so it vanishes, and what remains is a global neutrality condition, a Gauss law: the total D3-charge on X must be zero. Write the flux contribution as an integer,

$$N_{\mathrm{flux}} \;\coloneqq\; \frac{1}{(2\pi)^4 \alpha'^2} \int_X H_3 \wedge F_3 \;=\; K^I M_I - K_I M^I \;\in\; \mathbb{Z}, \tag{9.32}$$

the antisymmetric symplectic pairing of the two integer flux vectors of (9.13). Collecting the positive charges (flux and D3-branes) on one side and the magnitudes of the negative charges on the other, the neutrality condition becomes

$$N_{\mathrm{flux}} + N_{\mathrm{D3}} \;=\; \tfrac{1}{4} N_{\mathrm{O3}} \;+\; (\text{curvature-induced D3-charge on wrapped D7/O7}), \tag{9.33}$$

where the right-hand side is the total available negative charge. Its two pieces are the O3-planes, each contributing $-\frac{1}{4}$, and the curvature term carried by the wrapped seven-branes; the latter is not a clean count of branes but is fixed by the topology of the divisors they wrap. In the F-theory description of Chapter 11, where the IIB orientifold with its D7/O7 content is packaged as an elliptically fibred Calabi-Yau fourfold Y_4, that entire negative budget collapses to a single number, and the constraint takes its cleanest form,

$$\boxed{N_{\mathrm{flux}} + N_{\mathrm{D3}} \;=\; \frac{\chi(Y_4)}{24}} \tag{9.34}$$

with $\chi(Y_4)$ the Euler number of the fourfold. Equations (9.33) and (9.34) are the same statement that the net D3-charge vanishes, rewritten with the negatively charged sources moved to the right.

Three consequences matter here. For ISD flux, N_{flux} has the same sign as a D3-brane charge, so it must be balanced by objects of negative charge: orientifold planes, or the curvature term.

Flux cannot be switched on in a theory that lacks them, which is one reason orientifolds and F-theory, not ordinary Calabi-Yau compactifications, are the arena. Second, the constraint is a genuine ceiling: since $N_{\rm D3} \geq 0$ for a supersymmetric configuration, $N_{\rm flux} \leq \chi(Y_4)/24$, so only finitely many flux vectors are allowed on a given geometry. The bound can be large, $\chi(Y_4)/24$ reaching into the tens of thousands for known fourfolds, but it is finite, and this finiteness is what makes the flux landscape a large-but-countable set rather than a continuum. Third, the same $H_3 \wedge F_3$ that fixes the moduli is the object that costs tadpole, so stabilising more moduli, or reaching a given hierarchy, draws on a fixed charge budget. Whether that budget suffices to fix everything one wants, in a controlled regime, is a live question, sharpened by the tadpole conjecture of Section 9.4.

Exercise 9.6 (Counting what flux fixes)**.** On a Calabi-Yau orientifold with $h^{2,1}$ complex-structure moduli, count the real field-space dimension of the moduli fixed by (9.23): $h^{2,1}$ complex z^a plus one complex τ, against $h^{2,1}+1$ complex F-flatness conditions. Explain why the Kähler moduli T_α are untouched at this order by inspecting the dependence of W in (9.15) on T_α, and why (9.26) is therefore flat in those directions. Finally, using (9.32), argue that adding flux to fix moduli more rigidly draws on the finite tadpole budget (9.34), so the two cannot be increased independently.

Where this leaves us: quantised three-form flux, through the Gukov-Vafa-Witten superpotential and the no-scale potential, stabilises all $h^{2,1}$ complex-structure moduli and the axio-dilaton at a warped, no-scale Minkowski vacuum, subject to a finite tadpole set by the orientifold and brane charges. It leaves the $h^{1,1}$ Kähler moduli flat and hands forward a single constant W_0. Closing the flat directions, and lifting the vacuum to positive energy, is the work of the non-perturbative effects, KKLT and the Large Volume Scenario, in Section 9.3; counting how many such vacua the flux integers can produce, and whether de Sitter survives the swampland critiques, is Section 9.4.

9.3 Moduli stabilisation: instantons, KKLT and LVS

The big picture

The flux of Section 9.2 fixes the complex-structure moduli and the axio-dilaton and freezes the flux superpotential to a constant W_0. It leaves the Kähler moduli untouched: the tree-level potential is no-scale, so the sizes of the internal cycles are exactly flat directions once the heavier moduli are integrated out. Giving them a mass needs a piece of the superpotential that depends on the Kähler moduli, and holomorphy together with an axionic shift symmetry forbids any such piece to all orders in perturbation theory. It has to be non-perturbative. Two microscopic sources supply it: a Euclidean D3-brane wrapping a four-cycle, and gaugino condensation on a stack of D7-branes, each contributing $W_{\rm np} = A(z)\, e^{-aT}$ with a prefactor $A(z)$ that is a Pfaffian. Two ways of using this term define the two standard schemes. KKLT balances $W_{\rm np}$ against a tuned, small W_0; the Large Volume Scenario balances it against the leading α' correction. Both produce a controlled anti-de Sitter minimum with every modulus fixed, and both then face the harder problem of lifting that minimum to de Sitter. That last step is where this section runs out of solid ground and Section 9.4 takes over.

9.3.1 What the flux leaves behind

The GKP construction of Section 9.2 stabilises the complex-structure moduli z^a and the axio-dilaton by imposing $D_a W = D_S W = 0$ on the Gukov-Vafa-Witten superpotential $W = \int_X G_3 \wedge \Omega$. A word on notation before we start: we now write the axio-dilaton as $S \equiv \tau$, the field of Section 9.2, with $S = C_0 + i\, e^{-\phi}$ so that $\mathrm{Im}\, S = e^{-\phi} = 1/g_s$ carries the coupling. Renaming it this way frees the letter τ for the four-cycle volumes introduced just below. These fields sit at the flux

scale, parametrically heavier than everything below, so at low energy we integrate them out and replace W by its stabilised value, a z- and S-independent constant

$$W_0 \coloneqq \langle \textstyle\int_X G_3 \wedge \Omega \rangle. \tag{9.35}$$

What survives are the Kähler moduli T_α, $\alpha = 1, \dots, h^{1,1}_+(X)$, with $h^{1,1}_+$ the orientifold-even part of $h^{1,1}$, one for each independent four-cycle (divisor) D_α. Each is complex,

$$T_\alpha = \tau_\alpha + i\,\rho_\alpha, \qquad \tau_\alpha = \operatorname{Re} T_\alpha = \frac{\operatorname{Vol}(D_\alpha)}{\ell_s^4}, \qquad \rho_\alpha = \int_{D_\alpha} C_4, \tag{9.36}$$

its real part the Einstein-frame volume of the divisor in string units, its imaginary part the axion from the RR four-form. The tree-level Kähler potential is $K = -2\ln\mathcal{V}$, with $\mathcal{V}$ the internal volume, a homogeneous function of degree $3/2$ in the τ_α. As set out in Appendix C.4.6, this K is *no-scale*: it satisfies $K^{\alpha\bar\beta}K_\alpha K_{\bar\beta} = 3$ identically (Eq. (C.60)). With $W = W_0$ independent of the T_α, the two terms of the potential Eq. (C.50) cancel and

$$V_F = e^K |W_0|^2 \big(K^{\alpha\bar\beta}K_\alpha K_{\bar\beta} - 3\big) = 0 \tag{9.37}$$

along every Kähler direction (Eq. (C.61)). The T_α are massless flat directions, supersymmetry is broken along them by $F^{T_\alpha} = e^{K/2}K^{\alpha\bar\beta}\overline{D_\beta W} \neq 0$, and the vacuum energy is exactly zero. The gravitino is nonetheless massive, $m_{3/2} = e^{K/2}|W_0| = |W_0|/\mathcal{V} \neq 0$ (Eq. (C.58)). This is the volume-modulus runaway of the appendix example transcribed to the flux vacuum: the classical IIB flux potential fixes the complex structure and the dilaton and says nothing at all about the size of the manifold.

Two qualifications sharpen the logic. First, the split into "heavy" flux-stabilised moduli and "light" Kähler moduli is only a controlled approximation, valid when the flux-induced masses of z^a, S sit well above the tiny Kähler-modulus masses generated below. That hierarchy is comfortable at large volume and small $|W_0|$, and it is precisely where both schemes below operate; it can fail, and when it does the two-step integrating-out is not reliable. Second, once the Kähler moduli are fixed and W_0 is generic the constant in Eq. (9.35) feeds directly into the depth of the resulting AdS vacuum through $V_* = -3m_{3/2}^2$, so the size of $|W_0|$ matters and we will have to control it.

Exercise 9.7 (The no-scale identity for one modulus)**.** Take a single Kähler modulus with volume $\mathcal{V} = \big(\frac{1}{2}(T+\bar T)\big)^{3/2}$, so that $K = -2\ln\mathcal{V} = -3\ln(T+\bar T)$. Compute K_T, the metric $K_{T\bar T}$ and its inverse $K^{T\bar T}$, and verify the no-scale identity $K^{T\bar T}K_T K_{\bar T} = 3$ of Eq. (C.60). With $W = W_0$ constant, substitute into Eq. (C.50) and confirm $V_F = 0$ as in Eq. (9.37), while $F^T = e^{K/2}K^{T\bar T}\overline{D_T W} \neq 0$, so supersymmetry is broken along the flat direction.

9.3.2 Why the Kähler moduli need non-perturbative physics

The axion $\rho_\alpha = \int_{D_\alpha} C_4$ inherits a continuous shift symmetry from the ten-dimensional gauge invariance $C_4 \to C_4 + \mathrm{d}\lambda_3$ of the RR four-form,

$$\rho_\alpha \longrightarrow \rho_\alpha + c_\alpha, \qquad c_\alpha \in \mathbb{R}. \tag{9.38}$$

To all orders in the g_s and α' expansions this is an exact symmetry: no local counterterm in the effective action can depend on ρ_α, because a shift of a massless axion is a gauge transformation of the ten-dimensional field. The superpotential is holomorphic in $T_\alpha = \tau_\alpha + i\rho_\alpha$, so any dependence of W on T_α is simultaneously a dependence on ρ_α, and Eq. (9.38) forbids it in perturbation theory. A single wrapped brane sees the full period $\rho_\alpha \to \rho_\alpha + 1$ of the axion, and the only

holomorphic functions compatible with that residual discrete symmetry are the exponentials $e^{-2\pi k T_\alpha}$, $k \in \mathbb{Z}_{>0}$. Hence

$$W_{\rm np} = \sum_{k\geq 1} A_k(z)\, e^{-2\pi k\, T_\alpha} = A(z)\, e^{-2\pi T_\alpha} + \mathcal{O}\Big(e^{-4\pi T_\alpha}\Big)\,, \tag{9.39}$$

dominated by the single-instanton term $k = 1$ at large volume. A confining SU(N) gauge sector on the same divisor is the one refinement of this statement: it realises the residual shift symmetry on an N-fold cover of the axion circle and so produces the fractional exponent $2\pi/N$ of Eq. (9.43), a fractional-instanton effect we come to in Section 9.3.3. The real part of the exponent is $2\pi\tau_\alpha = 2\pi\,\mathrm{Vol}(D_\alpha)/\ell_s^4$, the action of a Euclidean object wrapping D_α; the imaginary part $e^{-2\pi i\rho_\alpha}$ is the phase that turns the continuous shift into the residual discrete one. The T_α-dependence of W is weighted by an exponentially small instanton action, so it is invisible order by order in perturbation theory and can only be generated by an effect of order e^{-1/g_s} or $e^{-\mathrm{Vol}}$. Perturbation theory leaves the flat directions untouched; only instantons can lift them.

Exercise 9.8 (Axionic shift symmetry and the instanton exponent)**.** Show that a superpotential term $W \supset f(T)$ invariant under Eq. (9.38) up to the discrete identification $\rho \to \rho + 1$, and holomorphic in T, must be a sum of the exponentials in Eq. (9.39). Explain why the real, positive coefficient 2π in the exponent (rather than any smaller number) is fixed by the wrapped-brane action, and why a term $e^{+2\pi T}$ is excluded.

9.3.3 Two sources of $W_{\rm np}$: E3 instantons and gaugino condensation

Equation Eq. (9.39) is a symmetry statement; the microscopics supply the prefactor and the exact exponent. There are two mechanisms, and both give the same functional form $A(z)\, e^{-aT_\alpha}$.

Euclidean D3 (E3) instantons. A Euclidean D3-brane wrapping the divisor D_α is a genuine instanton of the four-dimensional theory: a localised, finite-action Euclidean configuration. Its Dirac-Born-Infeld action, complexified by the coupling to C_4, is exactly the complexified volume,

$$S_{\rm E3} = 2\pi\, T_\alpha, \tag{9.40}$$

which we take as the normalisation of T_α. The instanton has two universal fermionic zero modes, the Goldstini θ_α of the two supercharges it breaks; the Grassmann integral $\int \mathrm{d}^2\theta$ over them produces an F-term, that is, a contribution to the superpotential,

$$W_{\rm np} \supset A(z)\, e^{-2\pi T_\alpha}, \qquad a = 2\pi. \tag{9.41}$$

Whether the E3 actually contributes, and what $A(z)$ is, are the content of Section 9.3.4.

Gaugino condensation on D7-branes. Put a stack of N D7-branes on the *same* divisor D_α (wrapping D_α and filling the four non-compact directions). Their worldvolume theory is a four-dimensional $\mathcal{N} = 1$ gauge theory with group SU(N), and its holomorphic gauge kinetic function is the Kähler modulus itself,

$$f_\alpha = T_\alpha, \qquad \frac{1}{g^2} = \mathrm{Re}\, f_\alpha = \tau_\alpha. \tag{9.42}$$

The one-instanton factor of this gauge theory, $e^{-2\pi f_\alpha} = e^{-2\pi T_\alpha}$, is exactly the action of a single E3 on D_α; a Euclidean D3 is a gauge instanton of the D7 theory. If the geometry leaves the theory as pure super-Yang-Mills (no light charged matter), it confines, and the gaugino bilinear condenses. Dimensional transmutation sets the holomorphic scale by $\Lambda^{3N} \propto e^{-2\pi T_\alpha}$, so the condensate and the effective superpotential it generates are

$$\langle\lambda\lambda\rangle \sim \Lambda^3 \propto e^{-2\pi T_\alpha/N}, \qquad W_{\rm np} = A\, e^{-2\pi T_\alpha/N}, \qquad a = \frac{2\pi}{N}. \tag{9.43}$$

This is the fractional-instanton effect promised above: the condensate is the N-th root of the one-instanton factor, so the exponent is shallower by $1/N$. A large stack gives a small a, hence a longer-range potential and a modulus stabilised at smaller volume, which is often useful for keeping the exponential under control. The two mechanisms are two limits of the same wrapped-brane physics rather than independent alternatives, and both are captured by writing $W_{\rm np} = A(z)\, e^{-aT_\alpha}$ with $a = 2\pi$ for a single E3 and $a = 2\pi/N$ for SU(N) gaugino condensation.

9.3.4 The prefactor is a Pfaffian, and it can vanish

The prefactor $A(z)$ is not a free constant. Holomorphy of W forces it to depend holomorphically on the complex-structure moduli z and on the positions of any branes the instanton sees. Witten computed it, in the F-theory / M-theory lift where the E3 becomes a Euclidean M5-brane wrapping a divisor of a fourfold (Witten 1996). The answer is a *Pfaffian* of the Dirac operator $\mathcal{D}$ governing the instanton's charged fermionic fluctuations,

$$A(z) = \mathrm{Pf}\,(\mathcal{D}), \tag{9.44}$$

twisted by the relevant bundle data. It is a Pfaffian rather than a determinant because $\mathcal{D}$ acts on a chiral (Majorana) set of zero modes, and $\mathrm{Pf}(\mathcal{D})^2 = \det \mathcal{D}$.

Not every divisor contributes. The $\int \mathrm{d}^2\theta$ that produces an F-term needs *exactly* the two universal Goldstino zero modes and no others; any extra fermionic zero mode would demand extra field insertions to saturate the Grassmann integral, and the leftover contributes to higher-derivative terms or to the Kähler potential, not to W. The extra zero modes are counted by the Dolbeault cohomology of the wrapped surface D. A holomorphic $(1,0)$-form on D is a Wilson-line mode of the instanton, counted by $h^{1,0}(D)$; a deformation of D inside X is a section of its normal bundle, and by the adjunction formula on a Calabi-Yau threefold $N_D = K_D$, so deformations are holomorphic $(2,0)$-forms, counted by $h^{2,0}(D)$. The clean invariant that packages both is the holomorphic Euler characteristic, the *arithmetic genus*,

$$\chi(\mathcal{O}_D) \;=\; \sum_{i=0}^{2} (-1)^i\, h^{0,i}(D) \;=\; h^{0,0}(D) - h^{1,0}(D) + h^{2,0}(D). \tag{9.45}$$

The instanton contributes to W (generically) precisely when the divisor is *rigid* with the minimal zero-mode content,

$$\boxed{\chi(\mathcal{O}_D) = 1, \qquad h^{0,0}(D) = 1, \;\; h^{1,0}(D) = h^{2,0}(D) = 0.} \tag{9.46}$$

The conditions read off physically: $h^{0,0} = 1$ says D is connected and supplies the two universal modes; $h^{1,0} = 0$ removes the Wilson lines; $h^{2,0} = 0$ says D cannot be deformed, which is what "rigid" means. The workhorse examples are the del Pezzo surfaces, which have $h^{1,0} = h^{2,0} = 0$ and $\chi(\mathcal{O}_{dP}) = 1$ automatically. A K3 divisor, by contrast, has $h^{2,0} = 1$ and $\chi(\mathcal{O}_{K3}) = 2$: it carries two extra zero modes and does not contribute to W unless they are lifted by worldvolume or background flux.

Even for a rigid divisor the story is not finished, because Eq. (9.44) is a nontrivial holomorphic function that can *vanish*. On codimension-one subloci of the complex-structure and brane-position moduli space, $\mathrm{Pf}(\mathcal{D}) = 0$, an extra pair of zero modes becomes normalisable, and the contribution switches off exactly there (Witten 1996; Ganor 1997). Physically these are the loci where charged matter localised on the instanton, for example an open string stretched to a nearby brane, becomes massless. The lesson for model building is sharp: rigidity of D is necessary but not sufficient. One must check that $A(z) \neq 0$ at the intended vacuum, and a naive count of rigid divisors overstates how many actually generate a superpotential.

Exercise 9.9 (Arithmetic genus and instanton zero modes)**.** Let D be a smooth complex surface with $h^{1,0}(D) = h^{2,0}(D) = 0$. Show $\chi(\mathcal{O}_D) = 1$ from Eq. (9.45). Now take D a K3 surface, with $h^{1,0} = 0$, $h^{2,0} = 1$; compute $\chi(\mathcal{O}_D)$ and identify the geometric object (a deformation of D inside X) responsible for the extra zero mode. Using the adjunction formula $N_D = K_D$ on a Calabi-Yau threefold, explain why $h^{2,0}(D)$ counts exactly the deformations of D.

Remark 9.6 (Beyond the lectures: zero modes in a flux background)*.* The condition Eq. (9.46) is the statement in the absence of flux. Worldvolume flux $\mathcal{F}$ on the E3 and background three-form flux G_3 both modify the Dirac operator and can lift would-be extra zero modes, so a divisor with $h^{2,0}(D) > 0$ can be made to contribute after all, and conversely flux can lift a rigid divisor out of contention through the Freed-Witten anomaly, which requires a half-integral worldvolume flux on a non-spin cycle. The general criterion is cohomological and background-dependent; see Blumenhagen, Cvetič, Kachru and Weigand (2009) for the systematic treatment. The clean $\chi(\mathcal{O}_D) = 1$ rule is the right first cut, not the last word.

9.3.5 KKLT

Kachru, Kallosh, Linde and Trivedi (2003) assembled the first explicit scheme in which every modulus is fixed. It uses the non-perturbative term at face value and tunes the flux to make it work. Take the simplest case, a Calabi-Yau orientifold with a single Kähler modulus, $h^{1,1}_{+} = 1$. For one modulus the volume is $\mathcal{V} = \left(\frac{1}{2}(T+\bar{T})\right)^{3/2}$, so the no-scale potential $K = -2\ln\mathcal{V}$ becomes $K = -3\ln(T+\bar{T})$ up to an additive constant, and

$$K = -3\ln(T+\bar{T}), \qquad W = W_0 + A\,e^{-aT}, \tag{9.47}$$

with W_0 the tuned flux constant and $A\,e^{-aT}$ from Section 9.3.3. Look for a supersymmetric critical point, $D_T W = 0$. Writing $T = \tau + i\,\rho$ and choosing the axion ρ so that $A\,e^{-aT}$ is real and opposes W_0, set $\rho = 0$ and use $K_T = -3/(T+\bar{T}) = -3/(2\tau)$:

$$D_T W = -aA\,e^{-a\tau} - \frac{3}{2\tau}(W_0 + A\,e^{-a\tau}) = 0 \implies W_0 = -A\,e^{-a\tau}\Big(1 + \tfrac{2}{3}\,a\tau\Big). \tag{9.48}$$

This is one real equation fixing $\tau = \mathrm{Re}\,T$. For a small flux constant, $|W_0| \ll 1$, the balance forces $a\tau$ large,

$$a\,\tau_* \approx \ln\frac{|A|}{|W_0|}, \tag{9.49}$$

up to a logarithm, so the divisor is stabilised at a volume that grows as $|W_0|$ shrinks. This is why control (large volume, tiny $e^{-a\tau}$, a trustworthy α' expansion) demands an exponentially small $|W_0|$: the scheme rests on tuning the flux to deliver one. At the critical point $D_T W = 0$, so by Eq. (C.59) the vacuum energy is

$$V_* = -3\,e^{K}|W|^2 = -3\,m_{3/2}^2 < 0, \tag{9.50}$$

a supersymmetric anti-de Sitter minimum with all moduli fixed. Given Eq. (9.47) and a small enough $|W_0|$, the modulus is stabilised and the vacuum is AdS, with no further approximation.

Example 9.7 (KKLT numerics for a single modulus)**.** Fix $A = 1$, $a = 2\pi$ (a single E3, $N = 1$) and $W_0 = -10^{-4}$, all real. The supersymmetric condition Eq. (9.48) reads $10^{-4} = e^{-2\pi\tau}\big(1 + \frac{4\pi}{3}\tau\big)$, a transcendental equation for $\tau = \mathrm{Re}\,T$. Because the balance depends on τ mainly through $u \coloneqq a\tau$, it is cleanest to solve $e^{-u}(1 + \frac{2}{3}u) = 10^{-4}$ once: a few Newton steps give $u_* \simeq 11.4$, hence $\tau_* = u_*/a \simeq 1.8$. The volume is only marginally in the supergravity regime, $\tau_* \gtrsim 1$; Exercise 9.10 shows how a gaugino condensate with $a = 2\pi/N$ raises $\tau_* = u_*/a$ by a factor N and restores control. At the minimum $W = W_0 + Ae^{-a\tau_*} \simeq -10^{-4} + 1.2\times10^{-5} \simeq -8.8\times10^{-5}$, and with $K = -3\ln(2\tau_*)$,

$$m_{3/2} = e^{K/2}|W| = (2\tau_*)^{-3/2}|W| \simeq 1.3\times10^{-5} \quad \text{(Planck units)},$$

so $V_* = -3m_{3/2}^2 \simeq -5\times10^{-10} < 0$, an AdS vacuum as Eq. (9.50) requires.

De Sitter needs one more ingredient, an *uplift* that adds a small positive energy without spoiling the stabilisation. KKLT use an anti-D3-brane placed at the tip of a warped Klebanov-Strassler throat (Klebanov and Strassler 2000; Giddings, Kachru and Polchinski 2002), a region of large redshift that the flux itself produces (Section 9.2). The anti-brane breaks supersymmetry and contributes a positive term to the potential, redshifted by the warp factor e^{4A_0} at the tip,

$$V_{\rm up} = \frac{D}{(T+\bar T)^2}, \qquad D = 2\,T_3\,e^{4A_0}, \qquad e^{4A_0} \sim e^{-8\pi K_{\rm flux}/(3g_sM)}, \tag{9.51}$$

with M and $K_{\rm flux}$ the F_3 and H_3 flux integers of the throat. The exponent in Eq. (9.51) is the standard Klebanov-Strassler redshift set by those integers, quoted from Section 9.2 rather than rederived here; note that the flux quantum $K_{\rm flux}$ is an integer three-form charge and has nothing to do with the Kähler potential K. Because e^{4A_0} is exponentially small and controlled by discrete flux, D is tunable, and one chooses it to lift the AdS depth of Eq. (9.50) up to a small positive residue: a metastable de Sitter vacuum. The word "metastable" carries weight here, and every step from Eq. (9.51) onward is where the reservations of Section 9.3.8 collect.

Remark 9.8 (Beyond the lectures: the anti-brane as a constrained superfield). Treating the anti-D3 as an explicit supersymmetry-breaking source, as in Eq. (9.51), is a shortcut. Its low-energy content, a single Goldstino with no scalar superpartner, is captured supersymmetrically by a *nilpotent* chiral superfield $\mathbf{X}$ with $\mathbf{X}^2 = 0$, coupled through a linear superpotential $W \supset \mu^2\,\mathbf{X}$ (Ferrara, Kallosh and Linde 2014; Kallosh and Linde). This recasts the uplift as a spontaneous, rather than explicit, breaking and makes the anti-brane part of a bona fide four-dimensional $\mathcal{N}=1$ theory, which is what one needs to trust the effective potential. It does not by itself settle whether the ten-dimensional configuration is stable; see the ledger.

Exercise 9.10 (KKLT numerics: single E3 vs gaugino condensate)**.** Building on Theorem 9.7, take the KKLT model Eq. (9.47) with $A=1$, $a=2\pi/N$ for a stack of N D7-branes, and $W_0 = -10^{-4}$ real. Solve Eq. (9.48) numerically for τ_* with $N=1$ and with $N=10$, and comment on which choice better warrants the supergravity approximation $\tau_* \gg 1$. Then evaluate $m_{3/2} = e^{K/2}|W|$ at the minimum and confirm $V_* = -3m_{3/2}^2 < 0$ from Eq. (9.50).

9.3.6 The Large Volume Scenario

The Large Volume Scenario, due to Balasubramanian, Berglund, Conlon and Quevedo (2005), takes a different route: instead of tuning W_0 small, it balances the non-perturbative term against the leading α' correction to the Kähler potential and lets the volume stabilise itself, exponentially large, for generic W_0. The price is a slightly richer geometry.

Three ingredients. First, a "Swiss-cheese" Calabi-Yau with at least two Kähler moduli, one large cycle τ_b controlling the overall size and one small rigid (del Pezzo) blow-up cycle τ_s supporting the instanton,

$$\mathcal{V} = \tau_b^{3/2} - \tau_s^{3/2} + \dots \tag{9.52}$$

Second, the α'^3 correction to the Kähler potential (Becker, Becker, Haack and Louis 2002), which breaks the no-scale structure,

$$K = -2\ln\!\Big(\mathcal{V} + \tfrac{1}{2}\,\hat\xi\Big), \qquad \hat\xi = -\frac{\zeta(3)\,\chi(X)}{2\,(2\pi)^3\,g_s^{3/2}}, \qquad \chi(X) = 2\big(h^{1,1} - h^{2,1}\big). \tag{9.53}$$

The correction has a definite sign only if $\chi(X) < 0$, that is $h^{2,1} > h^{1,1}$, which the scenario requires. Third, the non-perturbative term on the small cycle, $W = W_0 + A_s\,e^{-a_sT_s}$; the large cycle's own instanton is negligible because $e^{-a_b\tau_b}$ is doubly exponentially small at large volume.

Expanding the potential Eq. (C.50) at large $\mathcal{V}$ leaves three competing terms,

$$V_{\rm LVS} \simeq \underbrace{\frac{c_1\,(a_s A_s)^2\sqrt{\tau_s}}{\mathcal{V}}\,e^{-2a_s\tau_s}}_{\text{E3 self-energy}} - \underbrace{\frac{c_2\,a_s A_s\,|W_0|\,\tau_s}{\mathcal{V}^2}\,e^{-a_s\tau_s}}_{\text{cross term}} + \underbrace{\frac{c_3\,\hat{\xi}\,|W_0|^2}{\mathcal{V}^3}}_{\alpha'^3}, \tag{9.54}$$

with positive constants c_i set by the intersection numbers (Conlon, Quevedo and Suruliz 2005). The first two terms fix the small cycle and relate the instanton to the volume, $e^{-a_s\tau_s} \sim \sqrt{\tau_s}\,|W_0|/(a_s A_s \mathcal{V})$; feeding that into the balance against the α'^3 term stabilises

$$a_s\tau_s \sim \hat{\xi}^{2/3} \quad \text{(moderate)}, \qquad \mathcal{V}_* \sim \frac{|W_0|\sqrt{\tau_s}}{a_s A_s}\,e^{\,a_s\tau_s} \quad \text{(exponentially large)}. \tag{9.55}$$

The volume is stabilised exponentially large in $a_s\tau_s \sim \hat{\xi}^{2/3}$, with no tuning of W_0; values $W_0 = \mathcal{O}(1)$ or larger are fine. The minimum is anti-de Sitter, $V_* \sim -|W_0|^2/\mathcal{V}^3 < 0$, and crucially it is *non*-supersymmetric: $D_{T_b}W \neq 0$ for the large modulus, so it fails the F-flatness conditions Eq. (C.57) and is instead a minimum of the full potential.

The payoff of an exponentially large $\mathcal{V}$ is that the α' expansion is under excellent control and the mass scales split into a natural hierarchy: the gravitino mass $m_{3/2} = |W_0|/\mathcal{V}\cdot M_{\rm pl}$ is suppressed, the string scale is $M_{\rm s} \sim M_{\rm pl}/\sqrt{\mathcal{V}}$, and the small cycle τ_s is heavy, $m_{\tau_s} \sim m_{3/2}\ln\mathcal{V}$, while the volume modulus and its axion are light. The lightest field is the volume modulus τ_b, with

$$m_{\tau_b} \sim \frac{m_{3/2}}{\sqrt{\mathcal{V}}} \sim \frac{|W_0|\,M_{\rm pl}}{\mathcal{V}^{3/2}}, \tag{9.56}$$

a factor $\mathcal{V}^{-1/2}$ below the gravitino; its axionic partner is lighter still, gaining a potential only from the doubly suppressed large-cycle instanton. This is where LVS meets the cosmological moduli bound (9.8). Requiring $m_{\tau_b} \gtrsim$ tens of TeV forces $\mathcal{V} \lesssim 10^9$ for $|W_0| = \mathcal{O}(1)$; the larger volumes that make the string and supersymmetry-breaking scales phenomenologically low drive m_{τ_b} below the bound, so LVS with low-scale supersymmetry runs straight into the cosmological moduli problem of Section 9.1.3. That is a real constraint on the scenario, revisited in the ledger. What LVS shares with KKLT is the ending: the vacuum is AdS, and reaching de Sitter still needs an uplift, with all the same reservations. What it trades is fine-tuning of W_0 for the geometric requirements $\chi(X) < 0$ and a rigid del Pezzo cycle, together with reliance on the leading α' correction dominating the ones that are not yet known.

Exercise 9.11 (The Large Volume minimum)**.** Work with the two-modulus data of Eqs. (9.52) and (9.53) and $W = W_0 + A_s\,e^{-a_s T_s}$. Expanding V_F of Eq. (C.50) to leading order at large $\mathcal{V}$ and fixed τ_s, reproduce the three terms of Eq. (9.54) and identify which power of $1/\mathcal{V}$ each carries. Treating $\mathcal{V}$ and τ_s as independent variables, extremise in two steps: first solve $\partial V/\partial\tau_s = 0$ to get $e^{-a_s\tau_s} \sim \sqrt{\tau_s}\,|W_0|/(a_s A_s\mathcal{V})$, then substitute into $\partial V/\partial\mathcal{V} = 0$ and show the balance against the α'^3 term fixes $a_s\tau_s \sim \hat{\xi}^{2/3}$ and $\mathcal{V}_* \sim (|W_0|\sqrt{\tau_s}/a_s A_s)\,e^{a_s\tau_s}$, exponentially large. Evaluate V at the minimum, confirm $V_* \sim -|W_0|^2/\mathcal{V}^3 < 0$, and explain why the minimum needs $\hat{\xi} > 0$, that is $\chi(X) < 0$. Finally, using Eq. (9.56), estimate the volume modulus mass at $\mathcal{V} = 10^7$ and compare it with the bound (9.8).

9.3.7 The heterotic counterpart

The IIB story has an exact analogue on the heterotic side, and the parallel is close enough to spell out: the structure, a Pfaffian prefactor multiplying an instanton exponential, is identical, while the microscopic object is different. This subsection goes a little beyond lecture pace; the results, not the derivations, are what to carry away.

Remark 9.9 (Beyond the lectures: heterotic non-perturbative superpotentials). Compactify the $E_8 \times E_8$ heterotic string on a Calabi-Yau threefold X with a holomorphic gauge bundle V breaking the first E_8 to the visible gauge group. Two non-perturbative effects generate a superpotential.

(i) Worldsheet instantons. The fundamental heterotic string can wrap a holomorphic curve $C \subset X$. For an isolated rational curve, $C \cong \mathbb{CP}^1$ and rigid, each such curve contributes

$$W_{\rm np} \supset A_C(z)\, e^{-2\pi t_C}, \qquad t_C = \frac{1}{\alpha'} \int_C (J + i\, B), \tag{9.57}$$

with t_C the complexified Kähler parameter of the curve (its area plus the B-field through it) and the prefactor a Pfaffian

$$A_C(z) = \mathrm{Pf}(\mathcal{D}_{V|_C}), \tag{9.58}$$

the Pfaffian of the worldsheet Dirac operator for the right-moving fermions twisted by the gauge bundle V restricted to C, holomorphic in the bundle and complex-structure moduli (Dine, Seiberg, Wen and Witten 1986, 1987; Distler and Greene; Silverstein and Witten; Buchbinder, Donagi and Ovrut 2002). This is the precise mirror of the E3 Pfaffian Eq. (9.44): same object, Dirac operator twisted by the relevant bundle, same holomorphy, and the same capacity to vanish.

(ii) Gaugino condensation in the hidden E_8. The second E_8 typically remains as a strongly coupled hidden sector. Its gauge coupling is the heterotic dilaton (the field we call S in the heterotic frame), so condensation gives $W_{\rm np} \propto e^{-aS}$, which stabilises that dilaton, the field whose runaway is the heterotic version of the moduli problem (Dine, Rohm, Seiberg and Witten 1985; Derendinger, Ibáñez and Nilles 1985). Racetrack and Kähler-stabilisation refinements exist because a single condensate gives a runaway rather than a minimum.

The sharp result is due to Beasley and Witten (2003). Summing the worldsheet-instanton contributions Eq. (9.57) over all curves in a fixed homology class, the total can *cancel*. For the standard embedding, $V = TX$, where the worldsheet theory has $(2,2)$ supersymmetry, the contributions cancel identically and no superpotential is generated, consistent with $(2,2)$ non-renormalisation. Beasley and Witten proved this as a residue theorem: the sum over the moduli space of curves is a total residue that integrates to zero. For a generic $(0,2)$ bundle, away from the standard embedding, the cancellation fails and $W_{\rm np} \neq 0$. This is the heterotic analogue of the statement that the E3 Pfaffian can vanish. In both frames, whether a given wrapped cycle actually generates a superpotential is a delicate cohomological question, settled by a Pfaffian on the IIB side and by a residue sum on the heterotic side, and not by the mere existence of the cycle.

9.3.8 An honest ledger

Separate what these schemes establish from what they assume. The following is a summary, not a verdict.

Schemes, not theorems

KKLT and LVS are *constructions*: each assembles ingredients that are individually understood, flux stabilisation of z and S, a non-perturbative $W(T)$, in LVS an α'^3 correction, and an uplift, into a candidate vacuum. Neither is a single, fully explicit, fully back-reacted ten-dimensional solution with every α' and g_s correction demonstrably under control. Treat the boxed results above as the leading terms of an expansion whose convergence is argued, not proved.

What is solid. Flux fixes the complex-structure moduli and the axio-dilaton (Section 9.2). Non-perturbative effects fix the Kähler moduli, by an argument (Section 9.3.2) that is symmetry-based and robust. The resulting AdS minima, Eq. (9.50) for KKLT and Eq. (9.55) for LVS, are as reliable as the effective supergravity that produces them, and at large volume or small W_0 that supergravity is weakly coupled.

What is open, roughly in order of severity.

- *The de Sitter uplift.* This is the weakest link. The anti-D3 backreacts on the throat, and whether the uplifted state is a controlled metastable vacuum or an artefact of an over-simplified effective description is not settled. Brane-flux annihilation gives it a finite lifetime (Kachru, Pearson and Verlinde 2002); singular flux densities near the anti-brane raise questions about the ten-dimensional solution (Bena, Graña and collaborators); and a direct ten-dimensional analysis has been argued to obstruct the KKLT de Sitter vacuum (Moritz, Retolaza and Westphal 2018), with rebuttals maintaining that the four-dimensional derivation stands (Kachru, Kim, McAllister and collaborators). The debate is live. It is also exactly the arena of the de Sitter swampland conjecture (Section 9.4).
- *Control of the approximations.* Both schemes need the neglected corrections to be truly subleading. In KKLT the small-W_0 tuning can push the Kähler-modulus mass down towards the complex-structure masses, straining the two-step integrating-out of Section 9.3.1; in LVS the whole construction rests on the single α'^3 term Eq. (9.53) dominating the higher, incompletely known, corrections.
- *Light moduli and cosmology.* A stabilisation is only viable if its lightest modulus clears the cosmological bound (9.8) of tens of TeV (Section 9.1.3). A KKLT vacuum with tuned W_0 can sit above it, but LVS stabilises a volume modulus of mass $m_{\tau_b} \sim |W_0| M_{\rm pl}/\mathcal{V}^{3/2}$, Eq. (9.56), that drops below the bound at the large volumes wanted for low-scale supersymmetry. The cosmological moduli problem is therefore a live constraint on any light-volume scheme, and the nearly massless volume axion adds a dark-radiation worry on top.
- *Tadpole tension.* The flux that stabilises moduli, and in KKLT the special flux that yields a small W_0, must fit inside the O-plane and D-brane tadpole budget of Section 9.2. Whether a small enough W_0 is compatible with the available tadpole in genuine models has been questioned (the tadpole conjecture; Bena, Blåbäck, Graña and Lüst 2020).
- *Microscopic inputs are not automatic.* The scheme needs rigid divisors with $\chi(\mathcal{O}_D) = 1$ and a non-vanishing Pfaffian $A(z)$ at the vacuum (Section 9.3.4), and LVS additionally needs $\chi(X) < 0$ and a del Pezzo blow-up. Arranging all of these at once, in a compact model with the tadpole and flux constraints satisfied, is nontrivial.

Set against these, there has been real progress worth stating plainly. Explicit Calabi-Yau flux vacua with exponentially small $|W_0|$ have been constructed (Demirtas, Kim, McAllister and Moritz 2019), addressing the KKLT tuning at least in principle; explicit LVS models and large systematic scans exist; and the nilpotent-Goldstino description has put the anti-brane uplift on firmer four-dimensional footing. The state of play is a set of well-motivated candidate constructions with an unresolved question at their heart, whether four-dimensional de Sitter space survives in a controlled string compactification. That question is the hinge on which Section 9.4 turns.

9.4 The landscape and the swampland

The previous three sections built a working machine. The moduli problem (Section 9.1) said a Calabi-Yau compactification hands us hundreds of massless scalars and no dynamics for them.

Flux and the Gukov-Vafa-Witten superpotential (Section 9.2) freeze the complex-structure moduli and the axio-dilaton; instantons and gaugino condensation, dressed up in KKLT and the Large Volume Scenario (Section 9.3), freeze the Kähler moduli and, with an uplift, aim at a positive vacuum energy. A model with massless moduli is not a model, and this machinery is how you get rid of them.

Two questions survive the construction, and they pull in opposite directions. First: given that a single choice of quantised flux fixes a vacuum, and that there are astronomically many allowed flux choices, how many vacua are there? The answer, an estimate rather than a theorem, is a number so large it has become the emblem of the subject: the *landscape*. Second: does every effective theory that looks internally consistent, and in particular every metastable de Sitter vacuum the machine appears to produce, actually descend from a consistent theory of quantum gravity? The conjecture that most of them do not, and that the good ones satisfy sharp and surprising constraints, is the *swampland* programme. The landscape says string theory predicts almost anything; the swampland says quantum gravity forbids almost everything. They cannot both be as strong as their loudest advocates claim, and where the truth sits between them is not known. This section teaches that tension without resolving it, because resolving it is current research, not settled physics.

By the end of this section you should be able to: estimate how many flux vacua the tadpole bound allows and state the caveats on that estimate; explain the discretuum of cosmological constants and the anthropic argument built on it; and state the main swampland conjectures with a grade of the evidence for each.

The big picture

Count the flux choices consistent with the tadpole bound of Section 9.2 and you get an enormous *discretuum* of vacua, with a correspondingly fine spacing of possible cosmological constants: the landscape, and with it the anthropic temptation to explain the smallness of Λ by selection. Against this, the swampland programme argues that consistency with quantum gravity, not just with effective field theory, is a strong extra filter: no exact global symmetries, a Weak Gravity state in every gauge theory, an infinite tower of states becoming light at every infinite-distance limit of moduli space, and (most contentiously) no long-lived de Sitter vacua at all. The de Sitter conjecture sits squarely on top of KKLT. The defensible position is that moduli stabilisation is real, the counting is a genuine multiplicity, some swampland conjectures are close to theorems and others are guesses, and whether controlled string de Sitter exists is open.

9.4.1 Counting flux vacua

Fix a Calabi-Yau orientifold. A flux vacuum in the sense of Section 9.2 is a choice of integer three-form fluxes F_3 and H_3, each an element of $H^3(X,\mathbb{Z})$, subject to the F-flatness conditions $D_i W = 0$ and the single tadpole (Gauss) constraint set by the orientifold and brane charges,

$$N_{\text{flux}} \;=\; \frac{1}{(2\pi)^4 \alpha'^2}\int_X H_3 \wedge F_3 \;\le\; L, \tag{9.59}$$

with L the positive number fixed by the O-plane and D3-brane budget (in the F-theory description $L = \chi(Y_4)/24$; see Section 9.2). Different fluxes stabilise the moduli at different points and leave different vacuum energies. So each lattice point in flux space that solves the F-flatness conditions and respects Eq. (9.59) is, to leading approximation, a separate four-dimensional vacuum with its own physics. The question is how many there are.

Two features make the count both large and useful. The flux lattice has high rank: the number of independent three-cycles is $b_3(X) = 2\,h^{2,1}(X) + 2$, typically a few hundred for a generic

Calabi-Yau, and you get one integer for each cycle in each of F_3 and H_3. The constraint Eq. (9.59) is a single inequality, so it carves a large region, not a point, out of that lattice. Counting the lattice points inside becomes a problem in the geometry of numbers.

The discretuum of cosmological constants. Before the full count, take the toy model that makes the mechanism transparent, due to Bousso and Polchinski (2000). Suppose the vacuum energy is a sum over J independent flux quanta,

$$\Lambda(\vec{n}) \;=\; \Lambda_0 + \frac{1}{2}\sum_{i=1}^{J} q_i^2\, n_i^2, \qquad n_i \in \mathbb{Z}, \tag{9.60}$$

with $\Lambda_0 < 0$ a bare (negative) contribution and q_i the "charge" each unit of flux i adds. For a single flux, the allowed values of Λ are spaced by $\sim q_i^2$, far too coarse to land near the observed value. With many fluxes the spacing collapses. The values $\{\Lambda(\vec{n})\}$ that fall in a thin shell $\Lambda \le \Lambda(\vec{n}) \le \Lambda + \mathrm{d}\Lambda$ are the lattice points between two ellipsoids in $\mathbb{R}^J$, and their number is the shell volume. For J of order a few hundred and generic incommensurate q_i, the achievable values of Λ near zero are spaced more finely than 10^{-120} in Planck units, dense enough that some vacuum sits within observational distance of the measured $\Lambda \approx 10^{-120} M_{\mathrm{pl}}^4$ purely by the pigeonhole principle. This is the discretuum: a grid too fine to resolve by eye, yet genuinely discrete, dense enough for the purpose of tuning a single number.

Nothing in Eq. (9.60) explains *why* we sit in a vacuum with small Λ; it only shows that such vacua exist in the ensemble. The explanatory step, if you want one, is anthropic and is taken up in Section 9.4.2.

How many vacua. Counting the actual supersymmetric flux vacua, not just their energies, is the subject of the statistical programme of Douglas (2003) and Ashok and Douglas (2004). The idea is to stop treating the fluxes as integers and treat them as continuous, replacing the sum over the flux lattice by an integral weighted by the density of solutions to $D_i W = 0$. The F-flatness equations are, for each flux, a set of holomorphic conditions on the complex-structure moduli and the dilaton, and the number of their solutions is captured by an index (a signed count of critical points, via a Gauss-Bonnet-type formula). Integrating that index density over the region Eq. (9.59) gives the leading estimate

$$N_{\mathrm{vac}}(N_{\mathrm{flux}} \le L) \;\approx\; \frac{(2\pi L)^{\,b_3}}{b_3!} \int_{\mathcal{M}_{\mathrm{cs}}} \rho_{\mathrm{geom}}, \tag{9.61}$$

where $\mathcal{M}_{\mathrm{cs}}$ is the complex-structure moduli space and ρ_{geom} is an order-one geometric density (an integral of a curvature form of $\mathcal{M}_{\mathrm{cs}}$) that does not affect the scaling. The content is the combinatorial factor: with L of order a few hundred and b_3 of order several hundred, $(2\pi L)^{b_3}/b_3!$ is of order 10^{500}. That figure, quoted so often it has lost its shock value, is the origin of "the landscape has 10^{500} vacua".

The 10^{500} is an estimate, not a theorem

Read Eq. (9.61) as a heuristic. Three approximations sit underneath it. (i) The fluxes are integers, and treating them as continuous overcounts or undercounts near the boundary of the allowed region; the formula is a smooth density, not an integer count. (ii) The index counts vacua with signs, so it is a lower bound on the number of vacua and can differ from it. (iii) The formula counts solutions to the classical F-flatness conditions and says nothing about whether each survives the α' and g_s corrections, the Kähler-moduli stabilisation, and the uplift that a full construction needs. What is robust is the *scaling*: the number of flux choices consistent with Eq. (9.59) grows factorially fast in b_3, so the multiplicity is real and

enormous even if no one has enumerated a single complete example. Treat 10^{500} as "a lot," not as a measured quantity.

Remark 9.10 (Beyond the lectures: how big can it get, and does it survive). The 10^{500} figure is conservative. Taylor and Wang (2015) exhibited a single F-theory base geometry whose flux count, estimated from Eq. (9.61), is around 10^{272000}, driven by an exceptionally large b_3. Pulling the other way, the *tadpole conjecture* of Bena, Blåbäck, Graña and Lüst (2020) proposes that stabilising a large number of moduli generically requires a flux tadpole growing *faster* than the available L, roughly $N_{\text{flux}} \gtrsim \frac{1}{3}$ (number of stabilised moduli). If true, the fully-stabilised vacua at very large b_3, the ones responsible for the biggest numbers, may not fit inside Eq. (9.59) at all, and the effective count is far smaller than Eq. (9.61) suggests. The conjecture is supported by explicit computations in asymptotic regions of moduli space and is itself an active question. The lesson: the size of the landscape is not a settled number, and the defensible range is "vast, with the upper reaches in dispute."

Exercise 9.12 (The fineness of the discretuum)**.** Take the Bousso-Polchinski energy Eq. (9.60) with all $q_i = q$ equal. Approximating the n_i as continuous, show that the number of flux choices with $\Lambda(\vec{n}) - \Lambda_0$ below a cutoff Λ_c is the volume of a J-ball of radius $\sqrt{2\Lambda_c}/q$,

$$\mathcal{N}(<\Lambda_c) \;=\; \frac{\pi^{J/2}}{\Gamma(J/2+1)} \left(\frac{\sqrt{2\Lambda_c}}{q}\right)^{J}.$$

Differentiate to get the density $\mathrm{d}\mathcal{N}/\mathrm{d}\Lambda$ near $\Lambda_c \sim \mathcal{O}\left(M_{\text{pl}}^4\right)$, and hence the typical spacing $\delta\Lambda \sim (\mathrm{d}\mathcal{N}/\mathrm{d}\Lambda)^{-1}$ of achievable values. With $q \sim M_{\text{pl}}^2$ and J a few hundred, estimate J such that some vacuum lies within $10^{-120} M_{\text{pl}}^4$ of $\Lambda = 0$ without any tuning of the q_i. (You should find that a few hundred fluxes suffice, which is the entire point.)

Exercise 9.13 (Reading the scaling)**.** From Eq. (9.61), take $L = 500$ and $b_3 = 370$ and confirm that $(2\pi L)^{b_3}/b_3!$ is of order 10^{500} (use Stirling's approximation $\ln b_3! \approx b_3 \ln b_3 - b_3$). Then show that the count responds far more strongly to b_3 than to L: doubling b_3 moves $\ln N_{\text{vac}}$ by more than doubling L does, and by about as much as raising L tenfold. This is why the search for the largest landscapes is a search for Calabi-Yau (or F-theory) geometries with the largest b_3, and why the tadpole conjecture, which limits how much of that b_3 can actually be stabilised, matters so much.

9.4.2 The anthropic temptation

A discretuum of cosmological constants invites an old idea. Weinberg (1987) pointed out, before any of this string machinery existed, that Λ cannot be much larger than the density of matter at the epoch of structure formation without preventing galaxies (hence observers) from forming: too much positive vacuum energy blows the universe apart before anything can clump. The bound is a window, not a point, and the observed Λ sits comfortably inside it, a couple of orders of magnitude below the upper edge. Weinberg read this as a successful *prediction*: if Λ is selected anthropically from a range of possibilities, its most likely value is of order the largest value compatible with observers, which is roughly what we see.

The landscape supplies the range of possibilities that Weinberg's argument needs. Combine Eq. (9.60) with eternal inflation, which populates different vacua in different causally-disconnected regions, and you have a physical ensemble: the discretuum is realised somewhere, and we necessarily find ourselves in a member where observers can exist. On this reading the smallness of Λ is a selection effect rather than a dynamical fact to be derived, no more mysterious than the fact that we live on a planet at a habitable temperature.

I will not pretend this is uncontroversial, and you should not either. The argument has real problems, and they are worth naming because they carry the actual content of the position.

- **The measure problem.** Eternal inflation produces infinitely many pocket universes of every allowed type. To extract a probability distribution over Λ you must regulate a ratio of two infinities, and the answer depends on the regulator (the "measure"). No preferred measure is known, and different natural-looking choices give wildly different predictions, including paradoxes such as the youngness and Boltzmann-brain problems. Until the measure is fixed, "the anthropic prediction" is not a single number.
- **Testability.** A theory that can accommodate almost any low-energy physics somewhere in its landscape makes few sharp predictions about *our* vacuum. The anthropic explanation of Λ is at best a statistical statement about an ensemble we cannot survey. Whether this counts as physics or as an abdication of it is genuinely debated, and reasonable physicists land on both sides.
- **It assumes the vacua exist.** The whole argument is downstream of the landscape being real, which is exactly what the swampland programme questions.

The defensible stance presents the anthropic reading as a coherent, historically successful idea that remains methodologically contested, neither an established explanation nor a non-scientific evasion. Susskind's (2003) phrase "the anthropic landscape" gave the picture a memorable name, which is not the same as an argument for it.

9.4.3 The swampland

Now the counter-current. Effective field theory is permissive: write down any field content, any potential, any gauge group with any spectrum, add gravity as a weakly-coupled graviton, and you have a theory that is consistent as an EFT below its cutoff. The swampland programme, initiated by Vafa (2005) and Ooguri and Vafa (2006), asserts that the vast majority of such EFTs cannot be completed into a consistent theory of quantum gravity in the ultraviolet. The ones that can are the *landscape*; the ones that look fine as EFTs but have no quantum-gravity completion are the *swampland*. The programme is an attempt to draw the boundary between them by sharp conjectures, each motivated by black-hole physics, by holography, and by the recurring patterns of actual string compactifications.

The conjectures come with very different levels of support. I will grade them as I go, because pretending they are equally secure would mislead you.

No global symmetries (near-theorem). Quantum gravity has no exact global symmetries, continuous or discrete: every symmetry is either gauged or explicitly broken. The oldest argument is that a black hole can swallow global charge and then evaporate, so global charge is not conserved once gravity is dynamical. In AdS/CFT this has been sharpened by Harlow and Ooguri (2018) into something close to a theorem, using the structure of the boundary algebra. This is the most secure entry on the list, and it already has teeth: it forbids, for example, an exactly stable proton protected by a global baryon number.

The Weak Gravity Conjecture (well-supported). In any U(1) gauge theory coupled to gravity there exists a charged state whose charge-to-mass ratio is at least that of an extremal black hole,

$$m \;\leq\; \sqrt{2}\, g\, q\, M_{\mathrm{pl}}, \tag{9.62}$$

with g the gauge coupling and the numerical factor fixed by the extremality bound in four dimensions. The slogan is "gravity is the weakest force": there is a state on which the gauge

repulsion beats the gravitational attraction. The motivation, from Arkani-Hamed, Motl, Nicolis and Vafa (2007), is that without such a state an extremal charged black hole cannot discharge, and one is left with an infinite tower of stable extremal remnants, which is widely believed to be pathological. The conjecture has passed a large number of independent string-theory checks and has spawned an entire web of refinements (lattice, tower, and magnetic versions). Take it as well-supported though not proved.

The Swampland Distance Conjecture (well-supported). This one speaks directly to the moduli of Chapter 8 and to the runaway of Section 9.1. Moduli space is never compact, and it has boundaries at infinite geodesic distance (large volume, weak coupling, large complex structure). The conjecture of Ooguri and Vafa (2006) states that as a modulus ϕ traverses a geodesic distance Δ towards such a boundary, an infinite tower of states becomes exponentially light,

$$m(\Delta) \;\simeq\; m(0)\, e^{-\alpha\, \Delta/M_{\rm pl}}, \qquad \alpha = \mathcal{O}(1) > 0. \tag{9.63}$$

Once the tower drops below the EFT cutoff, the four-dimensional theory with a fixed finite spectrum stops making sense: you are forced to include the tower, which usually means the theory is decompactifying or a new weakly-coupled string is emerging. The physical upshot is a bound on how far any scalar can roll before the EFT you started with breaks down, generically a distance of order $M_{\rm pl}$. The evidence in string theory is overwhelming: every infinite-distance limit examined has produced exactly such a tower.

Example 9.11 (The distance conjecture on the volume modulus)**.** Take the no-scale volume modulus of Appendix C.4.6, $K = -3\ln(T+\bar T)$, and write $T = \tau + ia$ with $\tau = \mathrm{Re}\, T$ the (four-cycle) volume and a its axion. The Kähler metric $K_{T\bar T} = 3/(T+\bar T)^2 = 3/(4\tau^2)$ gives the field-space line element (Eq. (C.44))

$$\mathrm{d}s^2 = 2K_{T\bar T}\,\mathrm{d}T\,\mathrm{d}\bar T = \frac{3}{2\tau^2}\left(\mathrm{d}\tau^2 + \mathrm{d}a^2\right).$$

Along the axion-fixed direction the geodesic distance from τ_0 out to volume τ is

$$\Delta = \sqrt{\tfrac{3}{2}}\int_{\tau_0}^{\tau}\frac{\mathrm{d}\tau'}{\tau'} = \sqrt{\tfrac{3}{2}}\,\ln\frac{\tau}{\tau_0}, \tag{9.64}$$

which diverges as $\tau\to\infty$: large volume is an infinite-distance point. Now track the Kaluza-Klein tower. With overall volume $\mathcal{V}\sim\tau^{3/2}$ in string units, the KK mass in four-dimensional Planck units scales as $m_{\rm KK}/M_{\rm pl}\sim\mathcal{V}^{-2/3}\sim\tau^{-1}$, so inverting Eq. (9.64),

$$\frac{m_{\rm KK}}{M_{\rm pl}} \;\sim\; \tau^{-1} \;=\; e^{-\sqrt{2/3}\,\Delta/M_{\rm pl}}.$$

This is exactly Eq. (9.63) with $\alpha=\sqrt{2/3}\approx 0.82$, an order-one number, and the tower is the KK modes of the decompactifying internal space. The precise coefficient depends on which tower you follow and on the g_s-dependence, but the exponential lightening with geodesic distance is robust and is the content of the conjecture. The same computation reframes the moduli problem of Section 9.1: the runaway to large volume is more than a flat direction, it is an infinite-distance boundary where the four-dimensional description dissolves into a higher-dimensional one.

The de Sitter conjecture (contested). The sharpest and least secure claim. In its original form, from Obied, Ooguri, Spodyneiko and Vafa (2018), a scalar potential in a consistent quantum-gravity EFT obeys

$$|\nabla V| \;\geq\; \frac{c}{M_{\rm pl}}\,V, \qquad c=\mathcal{O}(1)>0, \tag{9.65}$$

wherever $V > 0$. Here $|\nabla V| \equiv \sqrt{K^{i\bar{\jmath}}\,\nabla_i V\,\nabla_{\bar{\jmath}} V}$ is the norm of the gradient measured in the field-space (moduli-space) metric. A metastable de Sitter vacuum has $V > 0$ and $\nabla V = 0$ at its minimum, so Eq. (9.65) forbids it outright: the conjecture says string theory has no long-lived de Sitter vacua, only quintessence-like slopes. Because Eq. (9.65) is too strong (it also forbids the perfectly ordinary maximum of a Higgs-like potential), a refined version by Ooguri, Palti, Shiu and Vafa (2018) replaces it with a disjunction: either $|\nabla V| \geq c\,V/M_{\rm pl}$, or the potential has a sufficiently tachyonic direction, $\min(\nabla_i \nabla_j V) \leq -c'\,V/M_{\rm pl}^2$, where $\min(\nabla_i \nabla_j V)$ denotes the least eigenvalue of the Hessian. Both versions kill a metastable de Sitter minimum. This conjecture stands on much weaker footing than the WGC or the distance conjecture: it has fewer clean string derivations, several proposed counterexamples and rebuttals, and it sits in direct contradiction with the KKLT and LVS constructions of Section 9.3. That contradiction is the crux of the whole subject, and I take it up in Section 9.4.4.

Remark 9.12 (Beyond the lectures: the emergent string conjecture and asymptotic dark energy). Two threads deepen the picture. The *emergent string conjecture* of Lee, Lerche and Weigand (2019) classifies every infinite-distance limit: the light tower of Eq. (9.63) is always either a Kaluza-Klein tower (the theory decompactifies) or the excitations of a weakly-coupled, asymptotically tensionless critical string (a dual string emerges). There is no third option and no "fully four-dimensional" infinite-distance limit; this is a strong structural statement about how EFTs die at the boundary of moduli space. Separately, if the de Sitter conjecture holds, our observed accelerated expansion cannot be a true cosmological constant and must be *quintessence*, a slowly-rolling scalar, which would have to satisfy Eq. (9.65) with c small enough to fit the data. Current observations put pressure on that: fitting cosmic acceleration to a rolling field with $c = \mathcal{O}(1)$ is tight, which is itself used as an argument against the strong form of the conjecture. The review by Palti (2019) surveys the conjectures; Agrawal, Obied, Steinhardt and Vafa (2018) work out the observational pressure the de Sitter conjecture puts on dark energy read as quintessence. Expect the numbers to move.

9.4.4 The tension

Put the two halves of this section side by side. The construction of Section 9.3 produces, or appears to produce, metastable de Sitter vacua: GKP fixes the complex structure and dilaton, a non-perturbative superpotential fixes the Kähler modulus at a supersymmetric anti-de Sitter minimum, and an anti-brane in a warped throat uplifts it to $V > 0$. The de Sitter conjecture Eq. (9.65) says such vacua do not exist. One of these statements is wrong, and which one is not known.

The tension is sharp because both sides have soft spots in exactly the same place: the control of approximations. KKLT is not a solution of ten-dimensional supergravity that anyone has written down in closed form; it is an effective four-dimensional argument assembled from ingredients each computed in its own regime, and stitching them together requires that several small parameters stay small simultaneously. The most scrutinised joint is the anti-brane uplift. A decade of work has asked whether the anti-D3-brane at the tip of the Klebanov-Strassler throat is the innocuous, calculable, metastable object KKLT assumes, or whether its backreaction destabilises the throat. Bena, Graña and collaborators found singular flux densities near the anti-brane and argued for trouble; Moritz, Retolaza and Westphal (2018) argued that a careful ten-dimensional accounting of the gaugino-condensate superpotential removes the de Sitter minimum; Kachru, Kim, McAllister, Zimet and others rebutted, arguing the effect is a scheme artefact and the vacuum survives. The debate is technical, it is not fully resolved, and its status is "open," not "settled in favour of my side." The Large Volume Scenario, which controls its expansion by a large volume rather than a warped throat, faces its own version of the same questions about the reliability of the uplift and the neglected corrections.

Here is where each claim now stands, which is what I most want you to take away. The ledger at the close of Section 9.3 already sorted the stabilisation machinery; this section adds two rows.

- **Carried over from Section 9.3.** Flux stabilisation of the complex-structure moduli and the axio-dilaton is solid; Kähler-moduli stabilisation by instantons and gaugino condensation (KKLT, LVS) is plausible but less tightly controlled; and a fully controlled, metastable de Sitter uplift is contested. That last row is where the anti-brane debate lives and where the de Sitter conjecture bites.
- **A genuine multiplicity, imprecisely known.** The landscape count is a real phenomenon rather than an artefact: the flux multiplicity grows factorially in b_3. Its size is estimated rather than enumerated, and its upper reaches are in dispute (the tadpole conjecture).
- **Graded conjectures.** No global symmetries is a near-theorem. The Weak Gravity and distance conjectures are well-supported by black-hole arguments and pervasive string evidence. The de Sitter conjecture is a live guess, with counterexamples claimed on both sides.

Why insist on the ledger rather than a verdict? Because both of the easy verdicts are wrong. The triumphalist reading, "string theory has 10^{500} vacua including ours, and the cosmological constant is explained," oversells an estimate as a count and an interpretation as a mechanism, and it ignores that the de Sitter vacua it needs are the least controlled part of the whole edifice. The dismissive reading, "string theory predicts anything and therefore nothing, and the swampland is a set of vague guesses," ignores that some swampland statements are close to theorems, that they are falsifiable in a way the bare landscape is not (they forbid things), and that they have already ruled out otherwise-consistent effective theories. The productive position treats the landscape and the swampland as two sides of one research programme: the landscape is the set of EFTs with a quantum-gravity completion, the swampland conjectures are proposed characterisations of that set, and the job is to make both precise enough to test. That job is not finished.

What "unresolved" does and does not license

Unresolved does not mean "anything goes." The no-global-symmetry and Weak Gravity constraints are strong enough to kill real model-building attempts, and you should apply them. It does mean that a claimed de Sitter construction deserves scrutiny of its approximations rather than either automatic acceptance or automatic rejection, and that the existence of controlled string de Sitter is a question you may live to see decided, in either direction. Teaching it as decided now, either way, would be teaching you something that is not known.

Exercise 9.14 (Weak gravity from black-hole discharge)**.** Model an extremal Reissner-Nordström black hole in four dimensions with mass M and charge Q satisfying the extremality relation $M = Q$ in units where the extremal bound is saturated at equality (absorb the $M_{\rm pl}$ and g factors into Q). Suppose it decays by emitting a single quantum of mass m and charge q, leaving a black hole of mass $M - m$ and charge $Q - q$. Requiring the remnant to be non-superextremal, $M - m \geq Q - q$, show that the decay is kinematically possible only if $q \geq m$ in the same units, i.e. only if a state exists with charge-to-mass ratio at least that of the extremal black hole. Restoring units, this is Eq. (9.62). Explain why the *absence* of such a state would leave a tower of stable extremal remnants of every charge, and why that is regarded as pathological.

Exercise 9.15 (Infinite distance at weak coupling)**.** Repeat Theorem 9.11 for the axio-dilaton. Write it as $S = C_0 + ie^{-\phi}$, the field τ of Section 9.2 under the name it carries in Section 9.3, so that $\operatorname{Im} S = e^{-\phi} = 1/g_s$ and $\operatorname{Re} S = C_0$, with Kähler potential $K = -\ln(-i(S - \bar{S}))$. Show that the field-space metric (the factor-of-two convention of Eq. (C.44)) is

$$\mathrm{d}s^2 = -\frac{2}{(S-\bar{S})^2}\,\mathrm{d}S\,\mathrm{d}\bar{S} = \frac{\mathrm{d}C_0^2 + \mathrm{d}(\operatorname{Im} S)^2}{2\,(\operatorname{Im} S)^2},$$

that the geodesic distance to weak coupling $g_s = e^{\phi} \to 0$, i.e. $\mathrm{Im}\, S \to \infty$, is again logarithmic and hence infinite, and identify a tower of states whose mass falls exponentially in that distance. (Consider wrapped branes, or the tower of the S-dual description; a weakly-coupled corner always comes with something becoming light.) Conclude that the large-volume and weak-coupling limits, the two regimes where we have computational control, are exactly the infinite-distance boundaries where the distance conjecture predicts the four-dimensional EFT must fail. Comment on why this makes controlled de Sitter constructions structurally difficult: control lives where the towers threaten.

Where this is going. This chapter turned a compactification with hundreds of massless moduli into a discrete set of stabilised vacua, and then asked how many there are and whether we can trust them. The techniques (fluxes, an $\mathcal{N} = 1$ potential, instanton superpotentials, warped throats) are the working vocabulary of string phenomenology, and the open questions (controlled de Sitter, the reach of the swampland) are among the most active in the field. From here the course turns to explicit constructions of gauge theories from geometry, geometric engineering and F-theory (Chapters 10 and 11), where the same flux and brane ingredients are put to work building the Standard Model rather than counting vacua.

Chapter 10

Geometric Engineering

10.1 Gauge symmetry from ADE singularities

A gauge theory is normally something you write down. Choose a group, choose the matter representations, fix the couplings, and quantise. String theory offers a second route to the same object. Put the theory on a background carrying a controlled singularity, and a gauge sector appears on its own, with the group, the charged spectrum and the couplings all read off from the geometry rather than posted by hand. The programme of turning geometric data into a field theory this way is *geometric engineering*, and this chapter builds its dictionary. The cleanest entry, and the one we treat first, is a sharp statement: a two-complex-dimensional ADE (du Val) singularity in an M-theory or type IIA background carries an ADE gauge symmetry. The Cartan generators come from reducing the eleven-dimensional three-form on the vanishing cycles of the singularity, and the charged W-bosons are branes wrapping those same cycles.

We have already met the phenomenon once, dressed as a duality. In Chapter 7 the equivalence of the heterotic string on T^4 with type IIA on K3 turned the perturbative gauge enhancement of the heterotic side into pure IIA geometry: the loci in K3 moduli space where the heterotic gauge symmetry jumps to a non-abelian group are exactly the loci where K3 develops an ADE singularity. Here we localise that statement. Strip away the compact K3, keep only a neighbourhood of the singular point, and study the singularity by itself. What survives is a self-contained gauge theory engineered by the local geometry, with gravity switched off.

The big picture

Take M-theory on a four-real-dimensional space with an ADE singularity, $\mathbb{C}^2/\Gamma$ for a finite $\Gamma \subset \mathrm{SU}(2)$. Resolving the singularity replaces the singular point by a set of two-spheres whose intersection matrix is minus the Cartan matrix of an ADE Lie algebra $\mathfrak{g}$. Reduce the eleven-dimensional three-form on these spheres and you get $\operatorname{rank}\mathfrak{g}$ abelian gauge fields: the Cartan. Wrap M2-branes on the spheres and you get the charged W-bosons, one BPS particle per root. When the spheres shrink to zero size the W-bosons become massless and the symmetry is enhanced from $\mathrm{U}(1)^{\mathrm{rank}}$ to the full ADE group. The sizes of the spheres are the Coulomb-branch moduli of the gauge theory.

By the end of this section you should be able to:

- read the ADE gauge algebra directly off a du Val singularity $\mathbb{C}^2/\Gamma$ (Section 10.1.2);
- obtain the Cartan gauge fields by reducing the eleven-dimensional three-form on the resolution two-cycles, and the charged W-bosons from wrapped M2-branes (Sections 10.1.3 and 10.1.4);

- identify the resolution moduli of the ALE space with the Coulomb branch of the seven-dimensional gauge theory (Eq. (10.6));
- say when gravity decouples, so that what is left is a rigid quantum field theory (Section 10.1.1).

10.1.1 Localising gauge enhancement: from K3 to $\mathbb{C}^2/\Gamma$

The simplest gauge enhancement in the whole subject needs no geometry at all. A single boson on a circle of radius R carries momentum and winding modes; at the self-dual radius $R = \ell_s$ a tower of these becomes massless, and the generic $\mathrm{U}(1)_L \times \mathrm{U}(1)_R$ is enhanced to $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ (Chapter 7). The extra gauge bosons are winding strings that happen to go light at one special radius. Everything in this section is the same mechanism with a four-dimensional internal space: a modulus is tuned, some wrapped objects go massless, and they fill out the adjoint of a non-abelian group. What changes is that the special locus is now a genuine geometric singularity, and the light states are wrapped branes rather than winding strings.

Gauge dynamics of this kind is local in the internal space. Near a point of K3 where a two-cycle collapses, the K3 metric does not care about the global geometry; it approaches the metric of a non-compact hyper-Kähler four-manifold asymptotic to $\mathbb{C}^2/\Gamma$. Such a manifold is called *asymptotically locally Euclidean* (ALE): it looks like flat $\mathbb{R}^4/\Gamma$ near infinity, with corrections falling off as r^{-4}, and it is smooth in the interior where a compact geometry has resolved the orbifold point.

Definition 10.1 (ALE space)**.** An ALE space of type Γ is a complete hyper-Kähler four-manifold $\widetilde{\mathbb{C}^2/\Gamma}$ whose metric approaches the flat orbifold metric on $\mathbb{C}^2/\Gamma = \mathbb{R}^4/\Gamma$ at infinity, with the leading correction of order r^{-4}. It is the minimal resolution (or a smooth deformation) of the singular quotient $\mathbb{C}^2/\Gamma$.

Two properties of ALE spaces do the physical work. First, they are hyper-Kähler, so they preserve supersymmetry: a hyper-Kähler four-manifold has holonomy in $\mathrm{SU}(2) \cong \mathrm{Sp}(1)$, which leaves half of the supercharges of the ambient theory unbroken. Concretely $\Gamma \subset \mathrm{SU}(2)$ has unit determinant, so it fixes the holomorphic two-form $\Omega = \mathrm{d}z_1 \wedge \mathrm{d}z_2$; the quotient inherits a covariantly constant Ω and stays Ricci-flat. This is why the relevant orbifold groups are subgroups of SU(2) and not of the full U(2): only the SU(2) subgroups keep the space Calabi-Yau, hence supersymmetric. Second, all of the interesting topology sits in the interior, near where the orbifold point used to be, so the gauge physics decouples from whatever the geometry does far away.

That decoupling is the geometric-engineering philosophy stated precisely. Embed the ALE space inside a large compact geometry, keep the light degrees of freedom the singularity supports, and send the overall volume (equivalently M_{pl}) to infinity so that gravity switches off. What remains is a rigid quantum field theory whose data are geometric. The rest of this section works out that field theory for the ADE singularities.

10.1.2 The du Val singularities and the ADE classification

Which quotients $\mathbb{C}^2/\Gamma$ can occur is a classical piece of mathematics, and the answer is the reason the letters A, D, E appear at all.

Definition 10.2 (du Val singularity)**.** A *du Val* (equivalently ADE, Kleinian, or rational double point) singularity is a quotient $\mathbb{C}^2/\Gamma$ by a finite subgroup $\Gamma \subset \mathrm{SU}(2)$ acting freely on $\mathbb{C}^2 \setminus \{0\}$. Every such quotient is a hypersurface $\{f(x, y, z) = 0\} \subset \mathbb{C}^3$ with an isolated singular point at the origin.

Type	$\Gamma \subset \mathrm{SU}(2)$	$\lvert\Gamma\rvert$	hypersurface $f=0$ in $\mathbb{C}^3$	gauge algebra
A_{n-1}	cyclic $\mathbb{Z}_n$	n	$xy = z^n$	$\mathfrak{su}(n)$
D_n	binary dihedral	$4(n-2)$	$x^2 + y^2 z + z^{n-1} = 0$	$\mathfrak{so}(2n)$
E_6	binary tetrahedral	24	$x^2 + y^3 + z^4 = 0$	$\mathfrak{e}_6$
E_7	binary octahedral	48	$x^2 + y^3 + yz^3 = 0$	$\mathfrak{e}_7$
E_8	binary icosahedral	120	$x^2 + y^3 + z^5 = 0$	$\mathfrak{e}_8$

Table 10.1: The du Val singularities, their orbifold groups, defining equations and the ADE gauge algebra each engineers. The rank of the algebra equals the number of exceptional two-spheres in the resolution: $n-1$ for A_{n-1}, n for D_n, and $6, 7, 8$ for the exceptional cases.



Figure 10.1: The dual graph of the resolved A_4 singularity $\mathbb{C}^2/\mathbb{Z}_5$: four exceptional spheres Σ_i, each of self-intersection -2, meeting in a chain. The graph is the A_4 Dynkin diagram, and the intersection matrix (10.2) is minus the SU(5) Cartan matrix. The D and E singularities give the branched Dynkin graphs.

The finite subgroups of SU(2) were classified in the nineteenth century, and they fall into exactly three families: the cyclic groups $\mathbb{Z}_n$, the binary dihedral groups, and three exceptional cases (the binary tetrahedral, octahedral and icosahedral groups, the double covers of the symmetry groups of the platonic solids). These are in one-to-one correspondence with the simply-laced Lie algebras, and Table 10.1 collects the dictionary. The invariant ring of Γ acting on $\mathbb{C}^2$ is generated by three polynomials x, y, z subject to a single relation $f = 0$, which is the hypersurface equation.

Take the A series as the concrete model. The cyclic group $\mathbb{Z}_n$ acts by

$$(z_1, z_2) \;\longmapsto\; (\omega\, z_1,\ \omega^{-1} z_2), \qquad \omega = e^{2\pi i/n}, \tag{10.1}$$

which is in SU(2) because the two eigenvalues multiply to one. The invariants are $x = z_1^n$, $y = z_2^n$ and $z = z_1 z_2$, and they satisfy $xy = z^n$, the A_{n-1} equation. For $n = 2$ this is $xy = z^2$, the orbifold group is $\Gamma = \mathbb{Z}_2$, the engineered gauge algebra is $\mathfrak{su}(2)$, and the resolved space is the Eguchi-Hanson metric on $T^*\mathbb{CP}^1$.

Resolution and the intersection matrix. Blow up the singular point and it is replaced by a union of complex projective lines $\Sigma_1, \dots, \Sigma_r$, the *exceptional divisors*, with r equal to the rank in Table 10.1. Each Σ_i is a two-sphere $\mathbb{CP}^1$ of self-intersection -2, and two of them meet transversally in a single point precisely when they are adjacent, otherwise they do not meet. The intersection numbers therefore assemble into

$$\Sigma_i \cdot \Sigma_j \;=\; -\,C_{ij}, \tag{10.2}$$

where C_{ij} is the Cartan matrix of the corresponding ADE algebra: $C_{ii} = 2$, and $C_{ij} = -1$ if nodes i and j are linked in the Dynkin diagram, zero otherwise (Appendix G). The dual graph of the resolution, one node per sphere and one edge per intersection, is the ADE Dynkin diagram itself (Fig. 10.1). The spheres are the simple roots; their intersection form is minus the Cartan matrix.

The correspondence between the representations of Γ and the nodes of the diagram is the *McKay correspondence*: the non-trivial irreducible representations of Γ are in bijection with the simple roots, and tensoring each with the defining two-dimensional representation of SU(2) reproduces the adjacency of the affine Dynkin diagram (the trivial representation supplies the extra affine

node). We state this rather than prove it; the proof and the marked diagrams are in Appendix G. The one feature we do use repeatedly is (10.2).

There is a structural reason only the simply-laced groups appear. The intersection form of curves on a complex surface is symmetric, $\Sigma_i \cdot \Sigma_j = \Sigma_j \cdot \Sigma_i$, so (10.2) can only ever produce a *symmetric* Cartan matrix. Symmetric Cartan matrices are exactly those of the simply-laced algebras A, D, E. The non-simply-laced algebras have asymmetric Cartan matrices, with roots of two different lengths, and no plain intersection form can deliver them. Two extra ingredients reach them. An orientifold projection produces $B_n = \mathrm{SO}(2n+1)$ and $C_n = \mathrm{Sp}(n)$ perturbatively (Section 10.3); the genuinely exceptional F_4 and G_2 instead need a monodromy acting on the collapsing cycles as they are transported around a base, which is the province of F-theory in Chapter 11.

10.1.3 M-theory on an ALE space: seven-dimensional super-Yang-Mills

Now put M-theory on the ALE space. Eleven dimensions minus the four of the ALE space leaves a seven-dimensional spacetime, and the claim is that the low-energy physics there is seven-dimensional super-Yang-Mills with gauge algebra $\mathfrak{g}$ of the ADE type set by Γ. Let us assemble the evidence piece by piece, because each piece is a line of the geometry-to-field-theory dictionary.

Supersymmetry. Eleven-dimensional supergravity has 32 supercharges. The ALE space is hyper-Kähler, holonomy SU(2), so it preserves exactly half, leaving 16 supercharges in seven dimensions. That is the minimal amount of supersymmetry in 7D, and its vector multiplet is rigid: once the gauge group is fixed, the two- and three-point couplings are determined. So it is enough to pin down the gauge group and the massless spectrum; supersymmetry fixes the interactions.

The Cartan gauge fields. Resolve the singularity to a smooth ALE space. Its second homology is spanned by the r exceptional spheres, and dually it carries r normalisable harmonic two-forms $\omega_1, \ldots, \omega_r$, which are square-integrable and anti-self-dual (both facts follow from the hyper-Kähler structure). Reduce the eleven-dimensional three-form C_3 on this basis,

$$C_3 \;=\; \sum_{i=1}^{r} A^i \wedge \omega_i \;+\; \cdots\,, \tag{10.3}$$

and each coefficient $A^i(x)$ is a seven-dimensional abelian gauge field. These r fields are the Cartan generators $\mathrm{U}(1)^r$. Their kinetic term comes from reducing the C_3 kinetic term of eleven-dimensional supergravity,

$$-\frac{1}{4\kappa_{11}^2}\int_{11} \mathrm{d}C_3 \wedge \star\,\mathrm{d}C_3 \;\longrightarrow\; -\sum_{i,j}\frac{\tau_{ij}}{2}\int_7 F^i \wedge \star F^j, \qquad \tau_{ij} \;\propto\; \int_{\mathrm{ALE}} \omega_i \wedge \star\,\omega_j, \tag{10.4}$$

with $F^i = \mathrm{d}A^i$. The overlap integral is finite because the ω_i are normalisable, so the gauge coupling matrix τ_{ij} is finite even though the ALE space is non-compact. Gravity, by contrast, comes with the non-normalisable volume of the whole space and decouples: the seven-dimensional Planck mass runs off to infinity in the local limit.

The Coulomb branch. A seven-dimensional vector multiplet with 16 supercharges contains, besides the gauge field, three real scalars, transforming as a triplet of the $\mathrm{SU}(2)_R$ R-symmetry. Geometrically these are the resolution parameters. A hyper-Kähler four-manifold has not one Kähler form but a triplet $\vec{J} = (J_1, J_2, J_3)$, rotated into each other by the SU(2) that acts on its

complex structures. That same SU(2) is the R-symmetry. Each exceptional sphere then carries a triplet of periods,

$$\vec{\zeta}_i \;=\; \int_{\Sigma_i} \vec{J} \;\in\; \mathbb{R}^3, \tag{10.5}$$

one real modulus for the Kähler size and two more for the hyper-Kähler deformation. Turning on $\vec{\zeta}_i$ blows the sphere up to finite size and resolves the singularity; the $3r$ real parameters $\{\vec{\zeta}_i\}$ are precisely the $3r$ scalars of r vector multiplets. Switching them on Higgses the gauge group from $\mathfrak{g}$ down to $\mathrm{U}(1)^r$, so

$$\boxed{\text{resolution moduli of the ALE space} \;=\; \text{Coulomb branch of the 7D gauge theory.}} \tag{10.6}$$

The origin of the Coulomb branch, $\vec{\zeta}_i = 0$, is the unresolved singularity, and there the full non-abelian symmetry is restored. What restores it is the subject of the next part.

10.1.4 W-bosons from wrapped M2-branes

The Cartan fields (10.3) are only the rank of the group. The roots, the charged W-bosons that complete $\mathrm{U}(1)^r$ to the non-abelian $\mathfrak{g}$, come from the one ingredient we have not yet used: the M2-brane.

M-theory contains a supersymmetric membrane, the M2-brane, with a three-dimensional worldvolume (Chapter 7). Wrap two of those dimensions on a holomorphic two-cycle Σ inside the ALE space, and the one remaining worldvolume dimension is a worldline in the seven-dimensional spacetime. The wrapped M2 is a particle. Because a holomorphic curve is calibrated by the Kähler form, its area is $\int_\Sigma J$ and the wrapped brane is BPS, with mass equal to tension times area:

$$M \;=\; T_{\mathrm{M2}} \int_\Sigma J \;=\; T_{\mathrm{M2}} \sum_i n_i\, a_i, \qquad a_i = \int_{\Sigma_i} J, \tag{10.7}$$

for a curve in the homology class $[\Sigma] = \sum_i n_i [\Sigma_i]$ with areas a_i. Its electric charges under the Cartan fields follow from the M2 coupling $\int_{\mathrm{M2}} C_3$: inserting (10.3) and integrating over Σ,

$$\int_{\mathrm{M2}} C_3 \;=\; \int_{\mathbb{R}_t} A^i \left(\int_\Sigma \omega_i \right) \;=\; q_i \int_{\mathbb{R}_t} A^i, \qquad q_i = \int_\Sigma \omega_i, \tag{10.8}$$

so the charge vector $(q_1, \ldots, q_r)$ lives in the root lattice. The homology classes that actually contain a holomorphic curve, hence a BPS particle, are the positive integer combinations $\sum_i n_i [\Sigma_i]$ that assemble the exceptional spheres into a connected chain. These are exactly the positive roots $\alpha = \sum_i n_i \alpha_i$ of $\mathfrak{g}$. An M2 wrapping such a curve is a W-boson of charge $+\alpha$; the anti-M2, the same curve with opposite orientation, carries $-\alpha$.

Promote the areas in (10.7) to the full triplet of Coulomb-branch scalars, $a_i \to \vec{\zeta}_i$, and the mass becomes

$$M_\alpha \;=\; T_{\mathrm{M2}} \left| \sum_i n_i\, \vec{\zeta}_i \right| \;=\; T_{\mathrm{M2}}\, |\vec{\alpha} \cdot \vec{\phi}|. \tag{10.9}$$

Read the two vectors in this contraction carefully. The adjoint scalar $\vec{\phi}$ collects the r triplets $\vec{\zeta}_i$, one for each Cartan direction, so its value along the i-th simple coroot is $\vec{\zeta}_i$. The root $\vec{\alpha} = (n_1, \ldots, n_r)$ is the expansion of α in the simple roots, the same integers n_i that fixed the homology class $[\Sigma] = \sum_i n_i [\Sigma_i]$. The dot in $\vec{\alpha} \cdot \vec{\phi} = \sum_i n_i\, \vec{\zeta}_i$ contracts the rank-r Cartan index, and $|\cdot|$ is the ordinary norm in $\mathbb{R}^3$ of the resulting $\mathrm{SU}(2)_R$ triplet. What comes out is the textbook mass of a W-boson on the Coulomb branch of a gauge theory, $M = |\vec{\alpha} \cdot \vec{\phi}|$: the geometric area of the wrapped cycle is literally the adjoint Higgs field, so (10.6) is an equality of physical quantities, not a loose picture.

Now take the singular limit $\vec{\zeta}_i \to 0$. Every W-boson mass (10.9) goes to zero, and the massless vectors are

$$\underbrace{r}_{\text{Cartan, from } C_3} + \underbrace{\#\{\text{roots}\}}_{\text{wrapped M2/anti-M2}} = r + (\dim \mathfrak{g} - r) = \dim \mathfrak{g}, \tag{10.10}$$

exactly the adjoint of $\mathfrak{g}$. When the cycles collapse the abelian $\mathrm{U}(1)^r$ of the resolved geometry is enhanced to the full ADE group, with the Cartan coming from a supergravity form field and the roots from wrapped membranes that go light together.

Example 10.3 (A_1: SU(2) from one collapsing sphere)**.** Take $\mathbb{C}^2/\mathbb{Z}_2$, resolved to Eguchi-Hanson space with a single exceptional sphere Σ_1 of self-intersection -2. There is one Cartan field A^1 from C_3, and SU(2) has one pair of roots $\pm\alpha$. The two states of charge $\pm\alpha$ are the M2 and the anti-M2 wrapping Σ_1; their mass on the Coulomb branch is $M_W = T_{\mathrm{M2}}\,|\vec{\zeta}_1|$, and it vanishes linearly as the sphere shrinks. This $r = 1$ case is the cleanest check on (10.5): the single seven-dimensional vector multiplet carries exactly three real scalars, and they are the three periods $\vec{\zeta}_1 = \int_{\Sigma_1} \vec{J}$, one Kähler size of the sphere and two hyper-Kähler deformations. Counting massless vectors at $\vec{\zeta}_1 = 0$: two W-bosons plus one Cartan equals three, the dimension of SU(2), filling out the adjoint. This is the K3 analogue of the circle at self-dual radius: there a winding string became a massless vector and enhanced U(1) to SU(2); here a wrapped M2 does the same job, and the special radius has become a collapsing two-sphere.

This gauge symmetry is not carried by open strings

There are no D-branes anywhere in this construction. The gauge group is a closed-string, purely geometric effect: the Cartan comes from the supergravity three-form C_3, and the W-bosons are wrapped M2-branes, which are solitonic BPS particles rather than perturbative open-string oscillator states. Like the heterotic gauge fields of Chapter 3, this is a closed-string effect rather than an open-string one, with one difference: there the whole gauge symmetry is perturbative current-algebra data, whereas here only the Cartan is a perturbative closed-string mode and the roots arise non-perturbatively as wrapped M2-branes. It contrasts sharply with the next section, where D-branes sit *at* a singularity and the gauge theory lives on their worldvolume as open-string modes. The two mechanisms are physically different and are related to each other only through the web of dualities.

10.1.5 The type IIA picture, and where this is going

Reducing on the M-theory circle turns every statement above into a type IIA one. M-theory on $\mathbb{C}^2/\Gamma \times S^1$ is IIA on $\mathbb{C}^2/\Gamma$, now a six-dimensional theory. The Cartan gauge fields come from the IIA Ramond-Ramond three-form reduced on the same harmonic forms ω_i, and the W-bosons are D2-branes wrapping the collapsing cycles: an M2 that does not wrap the circle descends to a D2, and wrapping its remaining two dimensions on Σ_i again leaves a particle. Assembling the ALE singularities into a compact K3 recovers the statement we started from: type IIA on K3 has ADE gauge enhancement at every du Val point of the surface, and this is the geometric face of the heterotic-on-T^4 gauge symmetry established in Chapter 7. The heterotic gauge bosons, which were momentum and winding modes of a perturbative string on T^4, are the wrapped D2-branes of the dual IIA description.

Two directions lead out of here, and both are picked up later.

The first is matter and chirality. A pure ADE singularity gives the gauge fields and nothing charged under them beyond the adjoint. To get charged matter you let the singularity vary over a base so that its type jumps at special points, and cycles that collapse only over those points give localised hypermultiplets. Sweeping the whole construction down to four dimensions, and reading matter and Yukawa couplings off the geometry, is the geometry-to-field-theory dictionary

of Section 10.3. The alternative engineering of charged and chiral matter from D-branes at a singularity is Section 10.2, and from intersecting or magnetised branes, Section 10.3.

The second is the non-simply-laced and exceptional groups. We saw that a plain intersection form only ever yields symmetric Cartan matrices, hence ADE. Folding a Dynkin diagram by an outer automorphism, realised as a monodromy acting on the vanishing cycles as they are transported around a base, produces the non-simply-laced algebras, and organising all of this over an elliptically fibred base is F-theory. The present section is the local seed; Chapter 11 makes it global, four-dimensional, and rich enough for model building. F-theory is, in a precise sense, the statement that the gauge group is read off from the ADE (and monodromy-folded) degeneration of an elliptic fibration, which is why the exceptional groups, so hard to reach with perturbative branes, are natural there.

Remark 10.4 (Beyond the lectures: ALE spaces as hyper-Kähler quotients). The ALE metrics are not just asserted to exist; Kronheimer constructed them explicitly as hyper-Kähler quotients built from the representation theory of Γ. The data of that quotient are exactly a quiver, the McKay quiver of Γ, whose nodes are the irreducible representations and whose arrows come from tensoring with the defining SU(2) doublet, and the resolution parameters $\vec{\zeta}_i$ enter as the Fayet-Iliopoulos terms of the quotient. This is the same quiver, with the same FI terms, that governs a D-brane probing $\mathbb{C}^2/\Gamma$ in Section 10.2: the Higgs branch of the probe's worldvolume gauge theory reproduces the ALE space it moves in. The geometry a brane sees is engineered by its own vacuum moduli space, and the two halves of this chapter meet at that point.

Remark 10.5 (Beyond the lectures: the same ADE data in four-dimensional $\mathcal{N}=2$). Fibring the ALE singularity over a complex line inside a Calabi-Yau threefold reduces the seven-dimensional gauge theory to four dimensions with $\mathcal{N}=2$ supersymmetry, and the geometry then computes the low-energy solution. The sizes of the wrapped cycles become the Coulomb-branch coordinates u, the wrapped M2 and its bound states become the BPS spectrum, and the pattern of loci where cycles degenerate reproduces the singularities of the Seiberg-Witten solution of the gauge theory. Geometric engineering in this stronger sense, building a specific asymptotically free gauge theory by a specific local Calabi-Yau and recovering its exact quantum dynamics, is due to Katz-Klemm-Vafa; the ADE singularity is the classical limit of that story.

Exercise 10.1 (Why SU(2) and not U(2); Warm-up, ∼10 min)**.** Let $\Gamma \subset \mathrm{U}(2)$ act on $\mathbb{C}^2$ with a generator of determinant $e^{i\theta} \neq 1$. (a) Show that this generator rescales the holomorphic two-form $\Omega = \mathrm{d}z_1 \wedge \mathrm{d}z_2$ by $e^{i\theta}$, so the quotient does not inherit a covariantly constant Ω. (b) Argue that the quotient is then not Ricci-flat and preserves no supersymmetry, so it does not engineer a supersymmetric gauge theory. (c) Conclude that the du Val list is exactly the set of *supersymmetry-preserving* isolated quotient singularities in complex dimension two, and that this is why the classification is by finite subgroups of SU(2) rather than of U(2).

Exercise 10.2 (The wrapped-M2 charge; Warm-up, ∼15 min)**.** Starting from the M2 worldvolume coupling $\int_{\mathrm{M2}} C_3$ and the reduction (10.3), derive (10.8) and show that the electric charge of an M2 wrapping $[\Sigma] = \sum_i n_i[\Sigma_i]$ under the Cartan field A^j is $q_j = \int_\Sigma \omega_j$. Choosing the basis $\int_{\Sigma_i} \omega_j = \delta_{ij}$, show that the charge vector is $(n_1, \ldots, n_r)$ and explain why the anti-M2 carries the opposite charge, so that the two orientations of a single collapsing sphere supply the pair of roots $\pm\alpha$ needed to enhance U(1) to SU(2).

Exercise 10.3 (A_2 and the eight of SU(3); Standard, ∼25 min)**.** Consider $\mathbb{C}^2/\mathbb{Z}_3$, the A_2 singularity, with two exceptional spheres Σ_1, Σ_2. (a) Write the $\mathbb{Z}_3$ action in the form (10.1) and find the three invariants and the relation between them; check it matches the A_2 entry of Table 10.1. (b) Using $\Sigma_i \cdot \Sigma_j = -C_{ij}$, write down the 2×2 intersection matrix and confirm it is minus the SU(3) Cartan matrix. (c) The positive roots of SU(3) are $\alpha_1, \alpha_2, \alpha_1 + \alpha_2$. Identify the three holomorphic curves that carry them, and hence list the six W-bosons. (d) Count the

massless vectors at the singular point and check you get $\dim \mathrm{SU}(3) = 8$. Then, matching the moduli, verify that the six real periods $\vec{\zeta}_1, \vec{\zeta}_2$ of (10.5) are exactly the 3×2 scalars of the two Cartan vector multiplets.

10.2 Branes at singularities and quiver gauge theories

In Section 10.1 the gauge symmetry came out of geometry alone. An ADE singularity sat in the compact space, branes wrapping the collapsing cycles supplied the W-bosons, and the low-energy theory inherited an ADE gauge group without a single D-brane being placed by hand. That is the top-down story. This section turns it around and makes the branes the active ingredient. Put a stack of D-branes at the tip of a singular space, let open strings end on them, and the massless open-string modes assemble into a gauge theory whose entire field content is fixed by the representation theory of the discrete group that defines the singularity. Gauge group, matter, and superpotential are all read off one labelled diagram, the *quiver*.

The construction only ever refers to the neighbourhood of the singularity. It never asks how that neighbourhood is glued into a compact Calabi-Yau. That locality is the point: the gauge sector is engineered on its own, decoupled from the bulk, and gravity is reattached afterwards as a controlled correction. This is the engine of bottom-up model building, and it is where chiral matter and Yukawa interactions first appear in this course as outputs of a geometric recipe rather than inputs to a Lagrangian.

The big picture

Take N D3-branes filling four-dimensional spacetime and sitting at a point of the transverse $\mathbb{C}^3$. In flat $\mathbb{C}^3$ their worldvolume theory is $\mathcal{N} = 4$ super Yang-Mills with gauge group $\mathrm{U}(N)$. Now quotient the transverse space by a finite group $\Gamma \subset \mathrm{SU}(3)$. The quotient acts both on the geometry and on the Chan-Paton labels, and keeping only the invariant open-string states projects the parent theory down to a *quiver gauge theory*:

- one gauge factor $\mathrm{U}(N_i)$ for each irreducible representation R_i of Γ;
- chiral bifundamentals counted by how the transverse **3** mixes the irreps, $R_{(3)} \otimes R_i = \bigoplus_j a_{ij} R_j$;
- a cubic superpotential inherited from the $\mathcal{N} = 4$ parent, summed over the closed loops of the diagram.

Because $\Gamma \subset \mathrm{SU}(3)$ the quotient is a local Calabi-Yau, four supercharges survive, and the matter is generically chiral. That is the whole engine; the rest of the section is derivation and worked examples.

10.2.1 The parent theory on the covering space

Start with N coincident D3-branes in flat ten-dimensional type IIB string theory, filling the four spacetime directions x^μ and localised at the origin of the transverse $\mathbb{R}^6$. The low-energy worldvolume dynamics is the dimensional reduction of ten-dimensional $\mathrm{U}(N)$ super Yang-Mills to four dimensions, established from the disc amplitude and the Chan-Paton construction in Chapter 6. Every field is an $N \times N$ matrix in the adjoint of $\mathrm{U}(N)$.

It is cleanest to organise the fields in $\mathcal{N} = 1$ language, because that is what survives the projection. Group the six transverse coordinates into three complex scalars

$$\Phi^a \;=\; \frac{1}{\sqrt{2}}\bigl(X^{2a+2} + iX^{2a+3}\bigr), \qquad a = 1, 2, 3, \tag{10.11}$$

so that a point of the transverse space is a triple $(\Phi^1, \Phi^2, \Phi^3) \in \mathbb{C}^3$ and the rotations that preserve the complex structure form $\mathrm{U}(3) \subset \mathrm{SO}(6)$. The worldvolume content is then a vector

multiplet (A_μ, λ) together with three adjoint chiral multiplets Φ^a, all valued in $N \times N$ matrices. The sixteen supercharges make this $\mathcal{N} = 4$, and the interactions are fixed by that: besides the gauge couplings there is the superpotential

$$W_{\mathcal{N}=4} \;=\; \frac{3g}{\sqrt{2}}\,\mathrm{Tr}\big(\Phi^1[\Phi^2,\Phi^3]\big) \;=\; \frac{g}{\sqrt{2}}\,\epsilon_{abc}\,\mathrm{Tr}\big(\Phi^a\Phi^b\Phi^c\big), \tag{10.12}$$

the two forms agreeing because $\epsilon_{abc}\,\mathrm{Tr}(\Phi^a\Phi^b\Phi^c) = 3\,\mathrm{Tr}(\Phi^1[\Phi^2,\Phi^3])$, by cyclicity of the trace and the antisymmetry of ϵ_{abc}. The tensor ϵ_{abc} is the invariant of $\mathrm{SU}(3) \subset \mathrm{U}(3)$: it is the holomorphic three-form Ω on $\mathbb{C}^3$ written in components. Hold onto that fact, because it is the reason the transverse rotation group we are allowed to orbifold by is SU(3) and not all of SO(6).

The R-symmetry of the parent is $\mathrm{SU}(4) \cong \mathrm{Spin}(6)$, acting on the four supercharges Q and rotating the transverse space. Under the subgroup $\mathrm{SU}(3) \subset \mathrm{SU}(4)$ that preserves Φ's complex structure, the fundamental branches as

$$\mathbf{4} \;\longrightarrow\; \mathbf{3} \oplus \mathbf{1}. \tag{10.13}$$

The singlet is the $\mathcal{N} = 1$ gaugino λ, partner of A_μ; the triplet holds the three matter fermions, partners of Φ^a. Equation (10.13) is the whole reason the orbifold below preserves supersymmetry when $\Gamma \subset \mathrm{SU}(3)$: the surviving supercharges are exactly those left invariant, and a Γ that sits inside SU(3) fixes the singlet in (10.13), hence one four-dimensional $\mathcal{N} = 1$ worth of supersymmetry.

10.2.2 The orbifold projection on Chan-Paton factors

A finite group $\Gamma \subset \mathrm{SU}(3)$ acts on the transverse $\mathbb{C}^3$ by the three-dimensional representation $R_{(3)}$, taking $\Phi^a \mapsto (R_{(3)}(g))^a{}_b\,\Phi^b$. To define string theory on the quotient we mod out by this action, exactly the orbifold procedure of Chapter 8: keep only the Γ-invariant closed-string states, and add twisted sectors localised at the fixed point. The new ingredient with branes present is that Γ must also act on the open strings, and therefore on the Chan-Paton labels they carry at their endpoints.

The physical picture makes the rule unavoidable. A single D-brane sitting away from the fixed point has $|\Gamma|$ images, one for each group element, and a Γ-invariant configuration must include the whole orbit. Open strings stretch between the images, so their Chan-Paton labels are permuted by Γ. The permutation is implemented by a set of unitary matrices $\gamma(g) \in \mathrm{U}(N)$, one per group element. Consistency of the string amplitudes in general lets the $\gamma(g)$ represent the group law only up to a phase, a projective representation whose obstruction is the discrete torsion in $H^2(\Gamma, \mathrm{U}(1))$ (Chapter 8). For the groups used here that cohomology is trivial, the phase can be removed, and the $\gamma(g)$ obey

$$\gamma(g_1)\,\gamma(g_2) \;=\; \gamma(g_1 g_2) \qquad \text{for all } g_1, g_2 \in \Gamma, \tag{10.14}$$

that is, $\gamma : \Gamma \to \mathrm{U}(N)$ is an honest N-dimensional representation. This is the Douglas-Moore prescription. A worldsheet derivation, tracking the orbifold action through the boundary state and the Chan-Paton trace, is collected in Chapter 8; here we take (10.14) as the defining data and work out its consequences.

An open-string field Ψ is an $N \times N$ matrix, so Γ acts on it by conjugation, $\Psi \mapsto \gamma(g)\,\Psi\,\gamma(g)^{-1}$, combined with whatever geometric action the field carries from its transverse-space index. The orbifold projection keeps the invariants. Concretely, the projector onto surviving modes is the group average

$$P \;=\; \frac{1}{|\Gamma|}\sum_{g\in\Gamma} (\text{geometric action of } g) \otimes (\text{conjugation by } \gamma(g)), \tag{10.15}$$

and the light spectrum of the quiver theory is its image. Apply it field by field.

Gauge field. The vector A_μ carries no transverse index, so its geometric action is trivial and the projection reads

$$A_\mu \;=\; \gamma(g)\, A_\mu\, \gamma(g)^{-1} \qquad \text{for all } g \in \Gamma. \tag{10.16}$$

The surviving A_μ are the matrices that commute with the entire representation $\gamma(\Gamma)$: its commutant. Decompose γ into irreducibles, writing R_i for the irreducible representations of Γ with dimensions d_i and letting N_i be the multiplicity with which R_i appears,

$$\gamma \;=\; \bigoplus_i N_i\, R_i, \qquad N \;=\; \sum_i N_i\, d_i. \tag{10.17}$$

By Schur's lemma the matrices commuting with γ act as an arbitrary $N_i \times N_i$ block on the multiplicity space of each irrep and as the identity on the irrep itself. The commutant is therefore $\bigoplus_i \mathfrak{u}(N_i)$, and the unbroken gauge group is a product of unitary factors,

$$\boxed{\,G \;=\; \prod_i \mathrm{U}(N_i)\,} \tag{10.18}$$

one factor for each irreducible representation of Γ, with rank equal to the multiplicity of that irrep in the Chan-Paton representation. Each factor is a *node* of the quiver.

Matter. The scalars Φ^a carry the transverse index a, which transforms in $R_{(3)}$, so their projection couples the geometric and Chan-Paton actions:

$$\Phi^a \;=\; (R_{(3)}(g))^a{}_b\, \gamma(g)\, \Phi^b\, \gamma(g)^{-1} \qquad \text{for all } g \in \Gamma. \tag{10.19}$$

A matrix Φ^a living in the $\gamma \otimes \bar{\gamma}$ representation of Γ (fundamental at one endpoint, antifundamental at the other) survives only in the piece where the transverse $R_{(3)}$ is undone by the Chan-Paton conjugation. The counting is pure representation theory. Decompose the tensor product of the transverse triplet with each irrep,

$$R_{(3)} \otimes R_i \;=\; \bigoplus_j a_{ij}\, R_j, \qquad a_{ij} \;=\; \dim \mathrm{Hom}_\Gamma\big(R_j,\, R_{(3)} \otimes R_i\big), \tag{10.20}$$

where Hom_Γ is defined in Appendix G. Then (10.19) leaves exactly a_{ij} independent scalars connecting the i block to the j block, each an $N_i \times N_j$ matrix. These are a_{ij} chiral multiplets in the bifundamental $(N_i, \bar{N}_j)$ of $\mathrm{U}(N_i) \times \mathrm{U}(N_j)$. In quiver language they are a_{ij} *arrows* pointing from node i to node j. The fermionic partners come along automatically: the matter fermions live in the **3** of (10.13), which is the same $R_{(3)}$, so they land in the same bifundamentals and fill out $\mathcal{N} = 1$ chiral multiplets.

Chirality is now a statement about Γ. If $R_{(3)}$ is a complex representation, meaning $R_{(3)} \neq \overline{R_{(3)}}$, then generically $a_{ij} \neq a_{ji}$ and the arrow from i to j is not matched by an equal number back. The net number of chiral multiplets in $(N_i, \bar{N}_j)$, namely $a_{ij} - a_{ji}$, is nonzero, and the matter is chiral. For $\Gamma \subset \mathrm{SU}(2)$ the defining representation is pseudoreal, $\overline{R_{(2)}} \cong R_{(2)}$, forcing $a_{ij} = a_{ji}$ and non-chiral matter, the $\mathcal{N} = 2$ case of Section 10.2.6.

10.2.3 Which representation? Regular branes and fractional branes

The projection determines the theory once the Chan-Paton representation γ is chosen, and (10.17) shows that a choice of γ is a choice of the ranks N_i. Not every choice describes the same physical object.

The distinguished choice is the *regular representation*, in which every irrep appears with multiplicity equal to its own dimension,

$$R_{\mathrm{reg}} \;=\; \bigoplus_i d_i\, R_i, \qquad \dim R_{\mathrm{reg}} \;=\; \sum_i d_i^2 \;=\; |\Gamma|. \tag{10.21}$$

Representation theory of Γ	Quiver gauge theory
irreducible rep R_i, dimension d_i	node i, gauge factor $\mathrm{U}(N_i)$
Chan-Paton rep $\gamma = \bigoplus_i N_i R_i$	multiplicities N_i (ranks)
$R_{(3)} \otimes R_i = \bigoplus_j a_{ij} R_j$	a_{ij} arrows $i \to j$: chiral $(N_i, \bar{N}_j)$
invariant ϵ_{abc} ($\Gamma \subset \mathrm{SU}(3)$)	cubic superpotential over closed loops
regular representation, $N_i = k\, d_i$	k mobile (bulk) D3-branes; $\sum_i d_i^2 = \lvert\Gamma\rvert$

Table 10.2: The read-off dictionary for D3-branes at a $\mathbb{C}^3/\Gamma$ singularity. Every entry on the right is fixed by the entry on the left; nothing depends on how the singularity is embedded in a compact space.

This is the representation carried by a single D-brane living in the bulk together with its $|\Gamma|$ images: on the covering space the images fill out Γ acting on itself, which is the regular representation by definition. A brane assigned R_{reg} can be pulled off the fixed point and moved freely into $\mathbb{C}^3/\Gamma$, because up on the cover it is just one ordinary brane and its orbit. For k such mobile branes the ranks are $N_i = k\, d_i$, and the gauge group is $\prod_i \mathrm{U}(k\, d_i)$.

At the fixed point the mobile brane can do something the covering-space picture forbids: it can fall apart into constituents that are separately stuck. These are the *fractional branes*, one species for each node i, carrying the Chan-Paton representation R_i alone. A single fractional brane has ranks $N_i = 1$ and $N_{j\neq i} = 0$; it cannot leave the singularity, because a lone irrep R_i is not the orbit of any bulk brane. The regular brane is the bound state of one fractional brane of each type weighted by dimension, $R_{\mathrm{reg}} = \bigoplus_i d_i R_i$, which is why it is exactly the combination that reassembles into something mobile. Fractional branes are the true elementary objects at the singularity, and the freedom to choose the ranks N_i independently is the freedom to stack fractional branes in any numbers, subject only to the consistency conditions of Section 10.2.7.

Remark 10.6 (Beyond the lectures: fractional branes as wrapped branes). A fractional D3-brane is a D3-brane dressed with extra, geometrically induced charges. When the singularity is resolved to a smooth space with exceptional cycles, the fractional brane is realised as a bound state of D5-, D3- and D1-branes wrapping the collapsed even cycles of the resolution, with the wrapping numbers fixed by the Chern character of the corresponding fractional-brane sheaf. In the $\mathbb{C}^2/\Gamma$ case of Section 10.2.6 this is transparent: the fractional branes are D5-branes wrapping the same shrinking two-cycles that carry the W-bosons of Section 10.1, where those W-bosons are wrapped M2/D2-branes in the dual M-theory/IIA frame. What the two constructions share is the collapsing two-cycle, carrying a fractional brane on one side and a W-boson on the other, not one and the same wrapped brane.

10.2.4 The superpotential

The interactions descend from the parent (10.12) by the same projection. Only the Γ-invariant part of $\epsilon_{abc}\,\mathrm{Tr}(\Phi^a\Phi^b\Phi^c)$ survives, and because $\Gamma \subset \mathrm{SU}(3)$ preserves ϵ_{abc}, the invariance falls entirely on the Chan-Paton and quiver structure. After projection each Φ^a is a bifundamental Φ^a_{ij} attached to an arrow, so a nonzero contribution to the trace requires the three fields to close into a loop $i \to j \to k \to i$ of the quiver, matching indices head to tail. The surviving superpotential is

$$\boxed{W \;=\; \frac{g}{\sqrt{2}}\,\epsilon_{abc} \sum_{\text{closed loops } i\to j\to k\to i} \mathrm{Tr}\big(\Phi^a_{ij}\,\Phi^b_{jk}\,\Phi^c_{ki}\big),} \tag{10.22}$$

where the sum runs over the oriented triangles of the quiver and the ϵ_{abc} enforces the antisymmetrisation inherited from Ω. A sum of gauge-invariant closed traces is automatically gauge invariant, so (10.22) is consistent by construction. The couplings are not free parameters

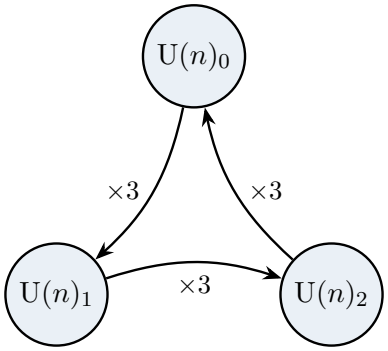


Figure 10.2: The $\mathbb{C}^3/\mathbb{Z}_3$ quiver for n regular D3-branes. Each node is a U(n) gauge factor, one per irrep of $\mathbb{Z}_3$. Each arrow carries three chiral multiplets in the bifundamental $(n_i, \bar{n}_{i+1})$, from $R_{(3)} = 3R_1$. The three arrows close a single oriented triangle, which is the loop summed in the superpotential.

here; they are all equal to the parent gauge coupling g, a rigidity that is a hallmark of the orbifold point and that relaxes once the singularity is deformed.

The requirement $\Gamma \subset \mathrm{SU}(3)$ now has a clean field-theory reading. Were Γ to sit in SU(4) but not SU(3), the tensor ϵ_{abc} would fail to be invariant, the cubic (10.12) would be projected out entirely, and with it the last supercharge: the theory would be non-supersymmetric. Preserving Ω, that is, keeping the local geometry Calabi-Yau, is the same condition as keeping a holomorphic superpotential and $\mathcal{N} = 1$ supersymmetry. This is the brane-probe echo of the holonomy argument of Chapter 8.

10.2.5 Worked example: the $\mathbb{C}^3/\mathbb{Z}_3$ quiver

The smallest chiral example is $\Gamma = \mathbb{Z}_3$. Let $\omega = e^{2\pi i/3}$ and act on the transverse coordinates by

$$g : \ (\Phi^1, \Phi^2, \Phi^3) \ \longmapsto \ (\omega\, \Phi^1, \ \omega\, \Phi^2, \ \omega\, \Phi^3), \tag{10.23}$$

so that $\det = \omega^3 = 1$ and $g \in \mathrm{SU}(3)$: the quotient is a local Calabi-Yau. The group $\mathbb{Z}_3$ has three one-dimensional irreps R_0, R_1, R_2 on which g acts as $1, \omega, \omega^2$, all with $d_i = 1$, and $\sum_i d_i^2 = 3 = |\Gamma|$ as (10.21) demands. The regular representation is $R_0 \oplus R_1 \oplus R_2$, so a single mobile D3-brane gives ranks $N_0 = N_1 = N_2 = 1$; a stack of n regular branes gives

$$G \ = \ \mathrm{U}(n)_0 \times \mathrm{U}(n)_1 \times \mathrm{U}(n)_2. \tag{10.24}$$

The transverse triplet carries charge $+1$ under $\mathbb{Z}_3$ on each of its three components, so $R_{(3)} = R_1 \oplus R_1 \oplus R_1 = 3R_1$. Tensoring,

$$R_{(3)} \otimes R_i \ = \ 3\, R_{i+1} \quad (\text{indices mod } 3) \implies a_{i,i+1} = 3, \ \text{all other } a_{ij} = 0. \tag{10.25}$$

The quiver is a triangle: three nodes, three arrows running around it in one direction, each arrow tripled (Fig. 10.2). The matter is three chiral multiplets in each of $(n_0, \bar{n}_1)$, $(n_1, \bar{n}_2)$, $(n_2, \bar{n}_0)$. Since $a_{i,i+1} = 3$ while $a_{i+1,i} = 0$, the matter is maximally chiral, as promised for a complex $R_{(3)}$.

There is one oriented triangle, $0 \to 1 \to 2 \to 0$, so the superpotential (10.22) is a single term,

$$W \ = \ \frac{g}{\sqrt{2}}\, \epsilon_{abc}\ \mathrm{Tr}\big(\Phi^a_{01}\, \Phi^b_{12}\, \Phi^c_{20}\big), \tag{10.26}$$

with a, b, c labelling the three arrows on each edge. The interaction is a genuine cubic Yukawa coupling between three distinct bifundamentals, engineered purely by the geometry. For $n = 3$

the gauge group is $\mathrm{U}(3)^3$, and stripping the abelian factors (Section 10.2.7) leaves $\mathrm{SU}(3)^3$ with matter

$$3 \times \left[(3, \bar{3}, 1) + (1, 3, \bar{3}) + (\bar{3}, 1, 3)\right],$$

the trinification group carrying three full copies of its **27**. That factor of three is the tripled arrow of (10.25), one copy of each bifundamental for each arrow on an edge, so $n = 3$ delivers three generations, the phenomenological feature that first drew attention to the model. This chiral, three-generation trinification, gauge group and cubic couplings and generation number together, is the original observation behind bottom-up model building of Aldazabal, Ibáñez, Quevedo and Uranga.

10.2.6 The $\mathbb{C}^2/\Gamma$ case and the McKay correspondence

Take $\Gamma \subset \mathrm{SU}(2)$ instead, acting on two of the three complex coordinates and leaving the third, Φ^3, untouched. The transverse space is $\mathbb{C}^2/\Gamma \times \mathbb{C}$. The extra flat $\mathbb{C}$ means one adjoint chiral multiplet at each node survives unprojected and pairs with the gauge field into an $\mathcal{N} = 2$ vector multiplet, so the whole theory has $\mathcal{N} = 2$ supersymmetry: eight supercharges, in line with $\Gamma \subset \mathrm{SU}(2)$ preserving twice as much as $\Gamma \subset \mathrm{SU}(3)$. The two-dimensional defining representation $R_{(2)}$ is pseudoreal, so $a_{ij} = a_{ji}$: the arrows come in balanced pairs and the bifundamentals assemble into $\mathcal{N} = 2$ hypermultiplets. No chirality, as expected of an eight-supercharge theory.

The data $\{d_i\}$ and $\{a_{ij}\}$ now have a name. The *McKay correspondence* states that for a finite $\Gamma \subset \mathrm{SU}(2)$ the tensoring rule

$$R_{(2)} \otimes R_i \;=\; \bigoplus_j A_{ij}\, R_j \tag{10.27}$$

has A_{ij} equal to the adjacency matrix of an *affine* ADE Dynkin diagram: the extended diagram with one extra node adjoined. The irreps R_i are the nodes, the tensoring (10.27) draws the links, the trivial representation R_0 is the affine node, and the dimensions d_i are the Dynkin marks (Coxeter labels) of that node. The dictionary between finite subgroups and diagrams is

$\Gamma \subset \mathrm{SU}(2)$	affine diagram	$\lvert\Gamma\rvert$
cyclic $\mathbb{Z}_N$	$\widehat{A}_{N-1}$	N
binary dihedral (dicyclic)	$\widehat{D}_N$	$4(N-2)$
binary tetrahedral	$\widehat{E}_6$	24
binary octahedral	$\widehat{E}_7$	48
binary icosahedral	$\widehat{E}_8$	120

The equality $\sum_i d_i^2 = |\Gamma|$ of (10.21) is, node by node, the statement that the squared Dynkin marks of an affine ADE diagram sum to the order of the group, one of McKay's original identities. Written out, (10.27) reads $\sum_j A_{ij}\, d_j = 2\, d_i$, which is precisely the null-eigenvector equation $(2\delta_{ij} - A_{ij})d_j = 0$ for the affine Cartan matrix: the marks are the imaginary-root multiplicities, collected in Appendix G.

This is the bridge back to Section 10.1. There the ADE gauge symmetry came from an ADE singularity in the geometry; here the same ADE diagram appears as the quiver of D-branes probing that singularity, with the affine node the mark of the brane charge. The two constructions are the same ADE data read from opposite ends, one from the collapsing cycles and one from the fractional branes wrapping them.

Example 10.7 (The $\mathbb{C}^2/\mathbb{Z}_N$ necklace)**.** Let $\Gamma = \mathbb{Z}_N$ act on $\mathbb{C}^2$ by $(\Phi^1, \Phi^2) \mapsto (\omega\, \Phi^1, \omega^{-1}\Phi^2)$ with $\omega = e^{2\pi i/N}$; the opposite charges keep $\det = 1$, so $\Gamma \subset \mathrm{SU}(2)$. There are N one-dimensional irreps R_i on which g acts as ω^i, and $R_{(2)} = R_1 \oplus R_{-1}$. Hence $R_{(2)} \otimes R_i = R_{i+1} \oplus R_{i-1}$: each node connects to its two neighbours around a ring of N nodes, the affine $\widehat{A}_{N-1}$ diagram. For

one regular brane every rank is $d_i = 1$, so $G = \mathrm{U}(1)^N$; for k regular branes $G = \mathrm{U}(k)^N$ with an $\mathcal{N} = 2$ hypermultiplet on each of the N links. This is the standard $\mathcal{N} = 2$ necklace quiver, the worldvolume theory of D3-branes on the A_{N-1} ALE space of Chapter 7.

Remark 10.8 (Beyond the lectures: the quiver moduli space is the resolved geometry). The quiver gauge theory does more than reproduce a spectrum; its space of supersymmetric vacua reconstructs the transverse geometry. For a single regular brane the D-term and F-term equations of the quiver, solved modulo the gauge group, cut out a moduli space that is exactly $\mathbb{C}^3/\Gamma$, the position of the brane in the covering space. Turning on Fayet-Iliopoulos parameters ζ_i for the abelian factors, which arise physically from twisted-sector B-field moduli at the fixed point, resolves the vacuum manifold into a smooth crepant resolution of the singularity, and the ζ_i map to its Kähler (blow-up) parameters. This is the physicist's form of the McKay correspondence promoted to three dimensions: the derived category of coherent sheaves on the resolution is equivalent to the representations of the quiver with its superpotential relations (Bridgeland, King and Reid).

10.2.7 Consistency: anomalies and tadpoles

A chiral gauge theory is only consistent if its non-abelian anomalies cancel. For node i the $\mathrm{SU}(N_i)^3$ anomaly counts fundamentals against antifundamentals. Arrows out of i contribute fundamentals of $\mathrm{U}(N_i)$ with multiplicity $\sum_j a_{ij} N_j$; arrows into i contribute antifundamentals with multiplicity $\sum_j a_{ji} N_j$. Cancellation requires

$$\sum_j (a_{ij} - a_{ji})\, N_j \;=\; 0 \qquad \text{for every node } i. \tag{10.28}$$

The orbifold ranks satisfy this identically. For the regular representation, $N_j = k\, d_j$, and the row sums of (10.20) weighted by dimension give $\sum_j a_{ij} d_j = \dim(R_{(3)} \otimes R_i) = 3\, d_i$. The transpose sum is equal: using $a_{ji} = \dim \mathrm{Hom}_\Gamma(R_i, R_{(3)} \otimes R_j) = \dim \mathrm{Hom}_\Gamma(R_j, \overline{R_{(3)}} \otimes R_i)$ one finds $\sum_j a_{ji} d_j = \dim(\overline{R_{(3)}} \otimes R_i) = 3\, d_i$ as well. The two are equal, so (10.28) holds for any multiple of the regular representation.

In the string theory this field-theory statement is a *tadpole* condition. The twisted-sector closed strings localised at the fixed point couple to the branes, and a would-be gauge anomaly shows up as an uncancelled tadpole for a twisted Ramond-Ramond field. Requiring those localised tadpoles to vanish reproduces (10.28) exactly. These are the *local* tadpoles, sensitive only to the physics at the singularity, and they are automatically satisfied by regular-brane configurations. The separate, *global* tadpole, the total Ramond-Ramond charge integrated over the compact space, is what forces orientifold planes and constrains the full model; that is taken up in Section 10.3.

The abelian factors are subtle

The gauge group (10.18) is a product of $\mathrm{U}(N_i)$, not $\mathrm{SU}(N_i)$, and the abelian pieces need care. The diagonal U(1), under which every bifundamental is neutral, decouples completely and can be dropped. The remaining U(1)'s are generically anomalous through mixed anomalies, and their gauge bosons acquire a mass by a Green-Schwarz mechanism: a twisted-sector axion shifts under the U(1) and the $B \wedge F$ coupling eats the photon, exactly as in Appendix H. The upshot is that the light non-abelian gauge group is $\prod_i \mathrm{SU}(N_i)$ together with whatever anomaly-free U(1)'s remain massless. When we quoted $\mathrm{SU}(3)^3$ trinification from the $\mathbb{C}^3/\mathbb{Z}_3$ quiver in Section 10.2.5, this is the mechanism that removed the extra abelian factors.

10.2.8 Why this is a local model

Nothing in the construction used a compact space. The projection (10.15), the gauge group (10.18), the matter (10.20) and the superpotential (10.22) are all fixed by Γ and the choice of

ranks alone. The singularity $\mathbb{C}^3/\Gamma$ is a non-compact cone; the gauge theory lives at its tip and knows only the geometry in an arbitrarily small neighbourhood.

That is what makes the approach powerful. To build a would-be Standard Model one first engineers the gauge sector locally, choosing Γ and the fractional-brane ranks to hit the desired group and chiral spectrum, without ever specifying the compact Calabi-Yau. Gravity decouples in this step because the four-dimensional Planck mass is set by the volume of the whole bulk,

$$M_{\rm pl}^2 \;\sim\; \frac{\mathrm{vol}_6}{g_s^2\,\ell_s^8}, \tag{10.29}$$

so taking $\mathrm{vol}_6 \to \infty$ at fixed gauge coupling sends $M_{\rm pl} \to \infty$ and leaves the theory at the singularity untouched. The decoupling limit, together with the scaling of $g_{\rm YM}$ that holds the gauge sector finite, is worked out in Section 10.3.3; here it is enough that a large bulk switches gravity off. Only afterwards does one embed the singularity in a compact space, where the global Ramond-Ramond tadpole must cancel, orientifolds enter, and the moduli must be stabilised by the fluxes of Chapter 9. Local matter here, global consistency there: that division of labour organises the rest of the chapter, and it frames the wrapped- and intersecting-brane constructions of Section 10.3.

Exercise 10.4 (Gauge group from Schur's lemma; Standard, ∼20 min)**.** Let Γ have irreps R_i of dimension d_i and take a Chan-Paton representation $\gamma = \bigoplus_i N_i R_i$. Show directly, using Schur's lemma, that the space of matrices commuting with all $\gamma(g)$ is $\bigoplus_i \mathfrak{gl}(N_i,\mathbb{C})$, and hence that the unitary gauge symmetry preserved by the projection (10.16) is $\prod_i \mathrm{U}(N_i)$. Confirm that the adjoint of this group has dimension $\sum_i N_i^2$, and check this against a direct count of the invariant components of an $N \times N$ matrix for $\Gamma = \mathbb{Z}_3$ with $N = 3$.

Exercise 10.5 (The A_1 quiver from $\mathbb{C}^2/\mathbb{Z}_2$; Warm-up, ∼15 min)**.** Let $\mathbb{Z}_2$ act on $\mathbb{C}^2$ by $(\Phi^1,\Phi^2) \mapsto (-\Phi^1,-\Phi^2)$, so $\Gamma \subset \mathrm{SU}(2)$. Write down the two irreps, verify $\sum_i d_i^2 = 2 = |\Gamma|$, and use (10.27) to show that the quiver is the affine $\widehat{A}_1$ diagram: two nodes joined by a doubled link in each direction. For k regular branes deduce the gauge group $\mathrm{U}(k) \times \mathrm{U}(k)$ and the two bifundamental hypermultiplets. Explain why the matter is non-chiral, and identify which $\mathcal{N} = 2$ theory this is.

Exercise 10.6 (Anomaly cancellation from characters; Standard, ∼25 min)**.** Prove the identity used below (10.28), namely $a_{ji} = \dim \mathrm{Hom}_\Gamma(R_j, \overline{R_{(3)}} \otimes R_i)$, by expressing both sides as character inner products $\langle \chi_j, \overline{\chi}_{(3)}\chi_i \rangle$ over Γ and using that multiplicities are real. Conclude that $\sum_j a_{ij} d_j = \sum_j a_{ji} d_j = 3 d_i$, so that the anomaly condition (10.28) holds for every multiple of the regular representation. Then check (10.28) by hand for the $\mathbb{C}^3/\mathbb{Z}_3$ quiver of Section 10.2.5 with arbitrary ranks (N_0, N_1, N_2), and show that away from $N_0 = N_1 = N_2$ the theory is anomalous.

With the local gauge theory in hand, read off a diagram and free of the bulk, the remaining question is how the same chiral matter arises when branes are wrapped and intersected in a compact space, and what global condition keeps that consistent. Both are taken up in the geometry-to-field-theory dictionary of Section 10.3, which reattaches the bulk and brings in the orientifold planes and the global tadpole condition.

10.3 The geometry-to-field-theory dictionary

The last two sections each built a machine. In Section 10.1 an ADE singularity in the internal geometry produced an ADE gauge group, with the W-bosons of the broken Cartan supplied by M2-branes wrapping the collapsing cycles of the resolution. In Section 10.2 a stack of branes probing an orbifold produced a quiver gauge theory, with the orbifold projection on the Chan-Paton factors producing chiral matter and anomaly and tadpole cancellation guaranteed, for regular branes, by the regular representation. Both machines run the same programme:

write down a singular internal space, place branes (or read off the collapsed cycles), and the four-dimensional gauge sector falls out. This section collects that programme into a single dictionary, states how far it reaches, and marks where it stops.

The organising idea is cheap to state and worth taking seriously: the field theory is read off the geometry *by codimension*. The gauge group is fixed by the singularity along the whole locus that supports it. Charged matter lives one complex codimension deeper, on curves where the singularity gets worse. Yukawa couplings live one deeper still, at the points where those curves cross. Everything else, the couplings, the number of generations, the presence or absence of an offending anomaly, is bookkeeping on top of these three strata. Once you see the pattern you can run it forwards (build a spectrum from a geometry) or backwards (demand a spectrum and solve for the geometry). Chapter 11 runs it backwards, with one extra dial turned on: the axio-dilaton is allowed to vary, and the same three strata reappear as the degenerations of an elliptic fibre.

The big picture

Engineer a gauge theory on a complex surface S carrying an ADE singularity. Then *gauge group* $\leftrightarrow$ singularity on S (codim 0), *matter* $\leftrightarrow$ enhancement on a curve $\Sigma \subset S$ (codim 1), *Yukawa* $\leftrightarrow$ enhancement at a point $p \in \Sigma$ (codim 2).
Gravity decouples when the compact volume is sent to infinity while the volume of S is held fixed: $M_{\rm pl} \to \infty$ at fixed gauge coupling, so the gauge dynamics can be studied on S alone. This "local model" is the object F-theory globalises in Chapter 11, where the singularity is the degeneration of an elliptic fibre and the enhancement chain reaches $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$.

10.3.1 The dictionary

Fix the arena. Take a stack of seven-branes wrapping a complex surface S, a divisor of the internal space, and filling the four non-compact directions. In the geometric (brane-free) language of Section 10.1 the same data is an ADE singularity fibred over S; the two descriptions are the open- and closed-string faces of one object, related by the transitions of Chapter 7. Either way the gauge dynamics is eight-dimensional to begin with, four non-compact directions plus the two complex directions of S, and the four-dimensional theory is what survives after reducing on S.

Over a generic point of S the transverse ADE singularity is of a fixed type, say $\mathfrak{g}$. That is the gauge algebra: $A_{N-1} \to \mathrm{SU}(N)$, $D_N \to \mathrm{SO}(2N)$, $E_n \to \mathrm{E}_n$. The vector multiplet fills the adjoint of $\mathfrak{g}$ and lives over all of S, codimension zero in the surface. This is the content of Section 10.1, now read as the first line of a table rather than as an endpoint.

Charged matter appears where the singularity is worse than generic. Along a curve $\Sigma \subset S$ the transverse singularity can degenerate further, enhancing $\mathfrak{g} \to \mathfrak{g}'$ with $\mathfrak{g}' \supset \mathfrak{g}$ of one higher rank. The rule for what matter this produces is the Katz-Vafa mechanism, which we state and use rather than derive: decompose the adjoint of the larger algebra under the smaller one,

$$\mathrm{adj}\,\mathfrak{g}' \;=\; \mathrm{adj}\,\mathfrak{g} \;\oplus\; \mathrm{adj}\,\mathfrak{h} \;\oplus\; \bigoplus_i (R_i,\, r_i), \tag{10.30}$$

where $\mathfrak{h}$ is the commutant of $\mathfrak{g}$ in $\mathfrak{g}'$. The mechanism is due to Katz and Vafa; the branching rules and commutant data it needs are collected in Appendix G. The gauge bosons of $\mathfrak{g}$ stay in the first summand; the off-diagonal pieces (R_i, r_i) are the new states that become light along Σ, and they are the localised matter in the representation R_i of the gauge group. A curve of enhancement is therefore a curve of hypermultiplets, and because Σ has complex dimension one the matter is genuinely four-dimensional after reduction. Its multiplicity (the number of generations) is counted by the topology of Σ together with any flux threading it, a point we return to in Chapter 11.

field theory	geometry	where on S	codim	example
gauge group $\mathfrak{g}$	ADE singularity on S	everywhere	0	$A_4 \to \mathrm{SU}(5)$
matter R (hyper/chiral)	$\mathfrak{g} \to \mathfrak{g}'$ on a curve Σ	curve	1	$\mathrm{SU}(5) \to \mathrm{SO}(10)$ gives $\mathbf{10}$
Yukawa $R_1R_2R_3$	$\mathfrak{g} \to \mathfrak{g}''$ at a point p	point	2	$\mathrm{SU}(5) \to \mathrm{E}_6$ gives $\mathbf{10}\cdot\mathbf{10}\cdot\mathbf{5}$

Table 10.3: The local dictionary for a gauge theory engineered on a surface S with an ADE singularity. Each field-theory datum sits on a stratum of S of definite codimension, and the representation content is read from the enhancement of the singularity there via (10.30). The examples are the SU(5) grand-unified chain worked in the text.

Yukawa couplings appear where matter curves meet. At an isolated point $p \in S$ two enhancement curves can cross, and the singularity there enhances by rank two, $\mathfrak{g} \to \mathfrak{g}''$. Three matter representations become simultaneously light at p, and if their tensor product contains a singlet,

$$R_1 \otimes R_2 \otimes R_3 \;\supset\; \mathbf{1}, \tag{10.31}$$

gauge invariance permits a cubic coupling $\lambda\,\Phi_1\Phi_2\Phi_3$ in the superpotential. The coupling is localised at p because that is the only place where all three wavefunctions have support at once. The three strata, surface, curve and point, are complex codimension $0, 1, 2$ in S, and they carry the vector multiplet, the matter and the Yukawa in turn. Table 10.3 collects the dictionary.

Remark 10.9 (Beyond the lectures: why matter is a difference of adjoints). Equation (10.30) looks like a group-theory accident but has a clean geometric origin. Near the enhancement curve the singular geometry is an unfolding of the $\mathfrak{g}'$ singularity, and the collapsing two-cycles of the resolution are in one-to-one correspondence with the roots of $\mathfrak{g}'$. Wrapping an M2-brane on a cycle whose class is a root of $\mathfrak{g}'$ but *not* of $\mathfrak{g}$ gives a state charged under the Cartan of $\mathfrak{g}$: that is exactly an off-diagonal entry of (10.30). The W-boson story of Section 10.1 and the matter story here are the same wrapped-brane counting, applied to the roots that survive and the roots that do not.

10.3.2 Matter on curves, Yukawa couplings at points

The dictionary tells you which couplings are allowed. It also, with more work, tells you their values, because a localised wavefunction is a concrete object and a Yukawa is an overlap of three of them.

Reduce the eight-dimensional gauge theory on S. The eight-dimensional fields include a gauge connection and an adjoint-valued Higgs field Φ that is a $(2,0)$-form on S, the field whose profile records how the seven-brane is embedded. A four-dimensional chiral field is a zero mode of the internal Dirac operator, and (10.30) says these zero modes are supported near the matter curves. Each mode Φ_i has two pieces of internal wavefunction that enter here: a $(0,1)$-form fluctuation a_i of the connection, counted by a Dolbeault cohomology group on S, and a $(2,0)$-form fluctuation φ_i of the Higgs field. The cubic term of the eight-dimensional superpotential, $\int_S \mathrm{Tr}(\Phi \wedge F)$, pairs two connection modes with one Higgs mode, so the holomorphic Yukawa coupling is the triple overlap

$$\lambda_{ijk} \;=\; \int_S \mathrm{Tr}(a_i \wedge a_j \wedge \varphi_k). \tag{10.32}$$

The degrees add up to a top form: two $(0,1)$-forms and one $(2,0)$-form make a $(2,2)$-form, which is exactly what a surface integrates. (Writing all three factors as $(0,1)$-forms would give a $(0,3)$-form, identically zero on a complex surface; the Higgs profile is what makes the integrand a genuine top form.) Each wavefunction is peaked on its own matter curve, so the integrand has support only where all three curves come close, that is at the Yukawa point p, and the integral

collapses to a local quantity there. Following Cecotti, Cheng, Heckman and Vafa and, in the intersecting-brane frame, Font and Ibanez, the collapse is exact: (10.32) equals a residue,

$$\lambda_{ijk} \;=\; \mathrm{Res}_p\!\left[\frac{h_i\, h_j\, h_k}{\partial_1 f\;\partial_2 f}\right], \tag{10.33}$$

where $f = 0$ cuts out the enhancement and the h's are holomorphic data of the three modes. We quote (10.33) rather than derive it; the derivation is a local computation in the Higgs-bundle description of the seven-brane, due to Cecotti, Cheng, Heckman and Vafa and to Font and Ibanez, and it is taken up in Chapter 11. Two features survive without any computation. The coupling is holomorphic, so it is protected and computable in the topological subsector. And at leading order the Yukawa matrix built from a single point has rank one: only one linear combination of generations couples, a geometric explanation for why one family (the top quark, in a grand-unified reading) is so much heavier than the rest, with the lighter families lifted by subleading corrections to the point-like picture.

Example 10.10 (The SU(5) chain)**.** Take $\mathfrak{g} = \mathfrak{su}(5)$, an A_4 singularity on S. This is the prototype of a geometrically engineered grand-unified theory. Run the enhancements through (10.30).

Along a curve where $\mathrm{SU}(5) \to \mathrm{SU}(6)$, decompose $\mathbf{35} = \mathbf{24} \oplus \mathbf{1} \oplus \mathbf{5} \oplus \bar{\mathbf{5}}$. The off-diagonal piece is a $\mathbf{5}$ (with its conjugate), so this is a $\bar{\mathbf{5}}$ matter curve: the place where the Higgs and the lepton doublets live.

Along a curve where $\mathrm{SU}(5) \to \mathrm{SO}(10)$, decompose $\mathbf{45} = \mathbf{24} \oplus \mathbf{1} \oplus \mathbf{10} \oplus \overline{\mathbf{10}}$. The off-diagonal piece is a $\mathbf{10}$, the curve carrying the quark doublets and the up-type singlets.

At a point where three curves meet and $\mathrm{SU}(5) \to \mathrm{E}_6$, decompose the adjoint $\mathbf{78}$ under SU(5). The extra states organise into $\mathbf{10}$'s and $\mathbf{5}$'s whose triple product contains a singlet, and the surviving cubic invariant is $\mathbf{10}\cdot\mathbf{10}\cdot\mathbf{5}$. This is the top Yukawa. The down-type coupling $\mathbf{10}\cdot\bar{\mathbf{5}}\cdot\bar{\mathbf{5}}$ comes instead from an SO(12) point. The largest coupling in the Standard Model thus sits at the most symmetric point of the geometry, a statement Chapter 11 makes precise.

Remark 10.11 (Beyond the lectures: the same Yukawa from a triangle)*.* In the T-dual, intersecting-brane frame of Section 10.3.4, chiral matter sits at brane intersections and a Yukawa is a disc amplitude with three boundary-changing insertions, one at each intersection. The result is a sum over worldsheet triangles with those three intersections as vertices,

$$\lambda_{ijk} \;\sim\; \sum_{\triangle} \exp\!\Big(-\tfrac{1}{2\pi\alpha'}\,\mathrm{Area}(\triangle)\Big), \tag{10.34}$$

computed for toroidal intersections by Cremades, Ibanez and Marchesano as a ratio of theta functions. The exponential of a holomorphic area in (10.34) and the residue in (10.33) are the same coupling in mirror frames: matter at points, Yukawa from a localised overlap, holomorphy protecting the answer. The hierarchy is small triangles against large ones rather than leading against subleading residues.

10.3.3 Decoupling gravity: the local model

The dictionary of Section 10.3.1 refers only to a neighbourhood of the singular locus. Nothing in (10.30) or (10.33) knows about the rest of the compact space. That is a licence to work locally, and the licence has to be earned: you may forget the bulk only if gravity, whose strength is set by the bulk, has decoupled from the gauge sector, whose strength is set by S.

The two scales come from the same dimensional reduction, evaluated on two different cycles.

The four-dimensional Planck mass is set by the whole internal volume,

$$M_{\rm pl}^2 \;\sim\; \frac{\mathrm{vol}_6}{g_s^2\,\ell_s^8}, \tag{10.35}$$

while the gauge coupling of a brane wrapping S is set by the volume of S alone,

$$\frac{1}{g_{\rm YM}^2} \;\sim\; \frac{\mathrm{vol}(S)}{g_s\,\ell_s^4}. \tag{10.36}$$

These are the scalings of Chapter 6, quoted here up to numerical factors that do not affect the limit. Now hold the gauge sector fixed, $\mathrm{vol}(S)$ and g_s constant so that $g_{\rm YM}$ is constant, and send the transverse directions large, $\mathrm{vol}_6 \to \infty$. Then

$$\frac{M_{\rm pl}^2}{M_{\rm s}^2} \;\sim\; \frac{\mathrm{vol}_6}{g_s^2\,\ell_s^6} \;\longrightarrow\; \infty, \qquad g_{\rm YM}\ \text{fixed}, \tag{10.37}$$

with $M_{\rm s} = \ell_s^{-1}$. Gravity switches off, $M_{\rm pl} \to \infty$, while the gauge theory keeps its coupling and its spectrum. In the strict limit the gauge sector is a rigid four-dimensional field theory, the bulk supergravity a decoupled spectator, and the local dictionary is exact. This is the "local model" philosophy: engineer the physics you want on the singularity, take the decoupling limit, and study the field theory on its own before worrying about how to glue S into a compact space.

The gluing is not free, and the constraint is geometric. For the limit (10.37) to exist, S must be a surface that can shrink to zero size at finite distance in moduli space, the four-real-dimensional analogue of the collapsing two-cycle that made the ALE space of Section 10.1 singular in the first place. A divisor that shrinks to a point has a normal bundle equal to its own canonical bundle: adjunction in a Calabi-Yau X (where K_X is trivial) gives

$$N_{S/X} \;\cong\; \big(K_X \otimes \mathcal{O}_X(S)\big)\big|_S \;=\; \mathcal{O}_X(S)\big|_S \;=\; K_S, \tag{10.38}$$

and shrinkability to a point requires this normal bundle to be negative, that is $-K_S$ ample. Surfaces with $-K_S$ ample are the del Pezzo surfaces dP_n ($n = 0, \dots, 8$), together with $\mathbb{CP}^1 \times \mathbb{CP}^1$. These are the contractible divisors, and they are exactly the ones that support a gauge theory decoupled from gravity. A surface with trivial or positive canonical class cannot be shrunk, its gauge dynamics cannot be separated from the bulk, and the local dictionary does not apply to it. We assert (10.38) and the del Pezzo criterion here; the surface theory behind them is collected in Appendix E.

What the local model throws away

Decoupling gravity is a physical statement about a limit, not a promise that the model exists. The local dictionary is silent about three things the compact theory cannot be. First, tadpoles: the flux and brane charges sourced on S must be absorbed somewhere, and in a compact space there is no "somewhere" unless the total charge cancels, which is the RR tadpole condition made explicit for intersecting branes in Section 10.3.4 and in general in Section 6.4. Second, moduli: vol_6, g_s and the shape of the bulk are dynamical fields whose stabilisation is a separate problem (Chapter 9), and the decoupling limit treats them as frozen by hand. Third, global symmetries: a symmetry that looks exact on S is generically gauged or broken once S sits in a compact space, because its would-be gauge field propagates in the bulk. A local model is a necessary ingredient, not a complete vacuum: its global completion is a separate construction, and not every local model admits one.

10.3.4 Chiral matter from intersecting branes, and the global tadpole

The local model of Section 10.3.3 declared the global tadpole out of scope. It is time to pay that bill, and the cleanest place to see it is a perturbative construction that also delivers chirality without any singularity at all: branes wrapping cycles that intersect. This is the T-dual face of the D-branes-at-a-point story of Section 10.2, and it makes the orientifold planes and the RR-charge budget concrete.

Work in type IIA. A stack of N_a D6-branes fills the four non-compact directions and wraps a three-cycle Π_a in the internal six-manifold; a second stack of N_b D6-branes wraps a three-cycle Π_b. Two three-cycles in a six-manifold generically meet in isolated points, and at each such point the open string stretched across the angle between the branes has a Ramond ground state that is chiral, a single four-dimensional Weyl fermion in the bifundamental $(N_a, \bar{N}_b)$. This is the mechanism of Berkooz, Douglas and Leigh: an angle between two branes traps a chiral fermion at their intersection.

The number that survives deformation is topological. Moving the cycles shifts the intersection points, but points of opposite orientation annihilate in pairs, so the net chirality is the signed count, the intersection number

$$I_{ab} \;=\; [\Pi_a]\cdot[\Pi_b] \;\in\; \mathbb{Z}. \tag{10.39}$$

The net number of chiral generations in $(N_a, \bar{N}_b)$ is I_{ab}, read off from homology alone. This is the intersecting-brane analogue of $a_{ij} - a_{ji}$ in the quiver of Section 10.2 and of the enhancement curve in the dictionary. In the T-dual type IIB frame the D6-branes at angles become D9- (or lower D7-) branes carrying worldvolume magnetic flux F_a, and I_{ab} becomes an integral of a product of first Chern classes; the chiral fermion is a Landau level of the flux, and the physics is identical.

Now compactify, and the charge has to balance. A D6-brane wrapping Π_a sources the Ramond-Ramond seven-form, and on a compact internal space the total D6 charge must vanish, Gauss's law with nowhere for the flux to escape (Section 6.4). The sink is the orientifold: an O6-plane, the fixed locus of an antiholomorphic involution combined with worldsheet parity, wraps a three-cycle Π_{O6} and carries -4 units of D6 charge in covering-space units, the 2^{p-4} of Section 6.4.3 at $p=6$. Cancellation is a single equation in homology,

$$\sum_a N_a\big([\Pi_a] + [\Pi_a']\big) \;=\; 4\,[\Pi_{\mathrm{O6}}], \tag{10.40}$$

where Π_a' is the orientifold image of Π_a. This is the RR tadpole condition (6.57) specialised to intersecting D6-branes and O6-planes. It constrains the homology classes of the whole brane configuration at once, not any single brane, and it is exactly the global tadpole the local model set aside.

The two constraints are one. Contracting (10.40) with a cycle $[\Pi_b]$ turns the RR-charge budget into a sum of intersection numbers, and the result is that the cubic non-abelian anomaly of each $\mathrm{SU}(N_b)$ cancels once the orientifold images are included. Tadpole cancellation and gauge-anomaly cancellation are the same homology identity, the intersecting-brane counterpart of the quiver statement (10.28). A consistent intersecting-brane model is therefore a choice of wrapping numbers solving (10.40) whose intersection numbers (10.39) reproduce a desired chiral spectrum. This is the framework in which the first globally consistent, three-generation Standard-Model-like spectra were built, by Blumenhagen, Cvetic, Langacker, Shiu and others.

10.3.5 The bridge to F-theory

The dictionary as stated has a perturbative ceiling. In type IIB with D-branes and orientifolds, the gauge groups reachable on a brane stack are the A-series from $\mathrm{SU}(N)$ stacks and, with an

O-plane, the B, C, D series from SO and Sp. The exceptional algebras $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$ do not occur: there is no perturbative brane configuration whose open strings furnish the exceptional adjoint. Yet the enhancement chain of Theorem 10.10 ran straight into E_6 at the top-Yukawa point, and the ADE classification of Section 10.1 put the E-series on exactly the same footing as the A and D series. The dictionary demands the exceptional groups; perturbative IIB cannot supply them.

The missing ingredient is the axio-dilaton. A D7-brane is a source for $\tau = C_0 + ie^{-\phi}$, and encircling it shifts C_0 by one, so τ is not a constant but a multivalued function on the space transverse to the branes, with $\mathrm{SL}(2,\mathbb{Z})$ monodromy around each. Once τ varies there is no frame in which every seven-brane is a weakly coupled D7-brane at once. Mutually non-local (p,q) seven-branes, the ones already glimpsed in the (p,q)-string discussion of Chapter 5, can collide to produce precisely the exceptional singularities. Encoding the varying τ as the complex-structure modulus of an auxiliary two-torus turns the transverse geometry into an elliptic fibration, and the type of degeneration of the fibre over each brane, the Kodaira classification, reproduces the ADE list and completes it with the exceptional types $\mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$ for $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$. That is F-theory.

The point for this chapter is that F-theory changes nothing in the dictionary. It runs Table 10.3 verbatim, with the ADE singularity on S now a Kodaira degeneration of the elliptic fibre over S. The gauge group is still the singularity type over the surface. Matter still lives on the curves where the fibre degenerates further, read off by the same Katz-Vafa decomposition (10.30). Yukawa couplings still sit at the points where matter curves cross and the fibre degenerates most, computed by the same residue (10.33). Gravity still decouples in the same large-volume limit (10.37), and the surface still has to be a contractible del Pezzo for the limit to exist. The new thing is that the enhancement chain now reaches the exceptional groups, and with them the grand-unified structures, the $\mathbf{10}\cdot\mathbf{10}\cdot\mathbf{5}$ top Yukawa among them, that perturbative strings leave out of reach. Chapter 11 builds the elliptic fibration, states the Kodaira-to-ADE dictionary (collected in Appendix E), and turns the local model of this section into a candidate for a Standard-Model-like spectrum.

Exercise 10.7 (Reading matter off the adjoint; Standard, ∼25 min)**.** Use (10.30) to confirm the entries of Table 10.3. (a) For $\mathrm{SU}(N) \to \mathrm{SU}(N+1)$, branch the adjoint $\mathbf{N^2} + \mathbf{2N}$ of $\mathrm{SU}(N+1)$ under $\mathrm{SU}(N) \times \mathrm{U}(1)$ and identify the off-diagonal piece as a fundamental $\mathbf{N}$ plus its conjugate. (b) For $\mathrm{SU}(5) \to \mathrm{SO}(10)$, branch $\mathbf{45}$ under $\mathrm{SU}(5) \times \mathrm{U}(1)$ and identify the $\mathbf{10}$. (c) For the Yukawa point, branch the E_6 adjoint $\mathbf{78}$ under $\mathrm{SU}(5) \times \mathrm{U}(1) \times \mathrm{U}(1)$ and show that the surviving cubic gauge invariant among the localised states is $\mathbf{10}\cdot\mathbf{10}\cdot\mathbf{5}$. Which enhancement, $\mathrm{SO}(12)$ or E_6, gives $\mathbf{10}\cdot\bar{\mathbf{5}}\cdot\bar{\mathbf{5}}$ instead?

Exercise 10.8 (Holomorphy and the rank-one Yukawa; Warm-up, ∼15 min)**.** Take the residue (10.33) at a single Yukawa point, with the holomorphic data of a mode labelled by a generation index and written $h_i^{(m)} = c_i^{(m)}\, h_i$ for a common local factor h_i and generation-dependent constants $c_i^{(m)}$. (a) Show that the Yukawa matrix $\lambda^{(mn\ell)} = \mathrm{Res}_p[\, c_1^{(m)} c_2^{(n)} c_3^{(\ell)}\, h_1 h_2 h_3/(\partial_1 f\, \partial_2 f)]$ factorises as $c_1^{(m)} c_2^{(n)} c_3^{(\ell)}$ times a single number, and hence that as a matrix in generation space it has rank one. (b) Explain why holomorphy of the integrand forbids any dependence on the Kähler moduli, so the leading Yukawa is protected, and state in one sentence what physical effect must lift the two massless generations left by the rank-one texture.

Exercise 10.9 (Decoupling gravity; Standard, ∼20 min)**.** Starting from (10.35) and (10.36), hold $\mathrm{vol}(S)$ and g_s fixed and write $\mathrm{vol}_6 = \mathrm{vol}(S)\cdot L^2$ for a transverse size L^2. (a) Show $M_{\mathrm{pl}}^2/M_{\mathrm{s}}^2 \to \infty$ as $L \to \infty$ while g_{YM} stays finite, so gravity decouples and the gauge theory does not. (b) Show that the opposite limit, sending $\mathrm{vol}(S) \to \infty$ at fixed vol_6, sends $g_{\mathrm{YM}} \to 0$: the gauge theory becomes free and the local dictionary trivialises. Explain why the decoupled field theory of interest lives at fixed g_{YM} and large L. (c) A grand-unified coupling $\alpha_{\mathrm{GUT}} = g_{\mathrm{YM}}^2/4\pi \approx 1/25$

fixes $\mathrm{vol}(S)$ in string units. Estimate how large L/ℓ_s must be for $M_{\rm pl}/M_{\rm s} \gtrsim 10^2$, and comment on whether the separation of scales is comfortable or marginal.

Exercise 10.10 (Intersection numbers and the tadpole; Standard, ∼25 min)**.** Take the factorised six-torus $T^6 = T^2 \times T^2 \times T^2$, and let a D6-brane wrap a straight line of winding numbers (n_a^I, m_a^I) on the I-th T^2, $I = 1, 2, 3$. (a) On a single T^2 two lines (n_a, m_a) and (n_b, m_b) meet in $|n_a m_b - m_a n_b|$ points. Deduce that on T^6 the intersection number (10.39) is $I_{ab} = \prod_{I=1}^{3}(n_a^I m_b^I - m_a^I n_b^I)$, and explain why the sign of I_{ab} is the net chirality rather than a count of intersections. (b) With an O6-plane wrapping $(1,0)(1,0)(1,0)$ and its images, write the tadpole condition (10.40) as a linear relation on the wrapping numbers $\sum_a N_a n_a^1 n_a^2 n_a^3 = \text{const}$, and interpret the constant. (c) Explain in one sentence why a would-be model with a single stack and no O-plane cannot satisfy (10.40) on a compact T^6.

Exercise 10.11 (The contractible surface; Advanced, optional, ∼30 min)**.** Let S be a smooth divisor in a Calabi-Yau threefold X. (a) Derive (10.38), $N_{S/X} \cong K_S$, from the adjunction formula $K_S = (K_X \otimes \mathcal{O}_X(S))|_S$ and triviality of K_X. (b) The self-intersection of a shrinkable exceptional curve inside a surface is negative (the analogue of a (-1)-curve). State the two-complex-dimensional counterpart: a surface contracts to a point when $-K_S$ is ample. List the del Pezzo surfaces and check that $\mathbb{CP}^2$ and $\mathbb{CP}^1 \times \mathbb{CP}^1$ qualify while an abelian surface (with K_S trivial) does not. (c) Explain in one paragraph why a surface with K_S trivial cannot host a gauge theory decoupled from gravity, connecting your answer to which term in (10.37) fails to diverge.

Chapter 11

F-Theory: Elliptic Fibrations, 7-Branes, and Exceptional GUTs

11.1 IIB with varying axio-dilaton

Every compactification so far has held the string coupling fixed. We chose a weakly coupled vacuum, expanded in g_s, and never asked what happens where the coupling is large. That was a safe habit because the coupling is a modulus: the expectation value of a field, free to sit wherever the potential lets it. In a genuine compactification with localised sources the coupling has no reason to stay constant, and once it varies over the internal space it will generically pass through the region where g_s is of order one and perturbation theory says nothing. F-theory is the statement that in type IIB one can follow the coupling into that region and keep control, not by solving the strongly coupled dynamics but by trading it for geometry. The key is the exact $\mathrm{SL}(2,\mathbb{Z})$ duality of Section 7.1: the axio-dilaton is not an ordinary complex field but one identified under a modular group, and a field valued in $\mathbb{H}/\mathrm{SL}(2,\mathbb{Z})$ is exactly the complex-structure modulus of a two-torus. Let the torus vary over the internal space, and its degenerations are the seven-branes: the non-perturbative physics becomes the geometry of a fibred elliptic curve.

Varying τ as a fibred torus

Type IIB has a complex scalar $\tau = C_0 + ie^{-\phi}$, with $\mathrm{Im}\,\tau = 1/g_s$, subject to the gauge redundancy $\tau \sim \frac{a\tau+b}{c\tau+d}$ for $\mathrm{SL}(2,\mathbb{Z})$. Compactify on a base B and let τ vary holomorphically over it. Wherever a seven-brane sits, τ has non-trivial $\mathrm{SL}(2,\mathbb{Z})$ monodromy: encircle a D7-brane and $C_0 \to C_0 + 1$. Because τ is defined only up to $\mathrm{SL}(2,\mathbb{Z})$, that data is precisely the data of a torus E_b whose complex structure is $\tau(b)$, fibred over B. F-theory is the resulting elliptic fibration $Y \to B$: type IIB on B, with the seven-branes located on the discriminant where the fibre pinches, and with g_s allowed to be arbitrarily large. The fibre is auxiliary: only its shape, the value of τ, carries physics, and its size never enters.

By the end of this section you should be able to:

- explain why a varying axio-dilaton forces you out of perturbation theory, and why the $\mathrm{SL}(2,\mathbb{Z})$ redundancy is what lets you carry on anyway (Section 11.1.1);
- solve for the profile of τ around a seven-brane, read off the monodromy, and see why a single brane in a non-compact plane is not a consistent background (Section 11.1.2);
- run the deficit-angle bookkeeping that caps the number of seven-branes and closes the transverse space (Section 11.1.3);

- state the dictionary that turns type IIB with varying τ into an elliptic fibration, and say precisely what F-theory is and is not (Sections 11.1.4 and 11.1.5).

11.1.1 Letting the coupling vary

Recall the two type IIB scalars. The Ramond-Ramond zero-form C_0 and the dilaton ϕ combine into the axio-dilaton

$$\tau \;=\; C_0 + i\, e^{-\phi}, \qquad \operatorname{Im}\tau = e^{-\phi} = \frac{1}{g_s}, \tag{11.1}$$

a point of the upper half-plane $\mathbb{H} = \{\operatorname{Im}\tau > 0\}$, exactly as in Eqs. (5.53) and (7.1). In the Einstein frame the two-derivative action for τ is the $\mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ coset sigma model, with the Poincaré metric $\mathrm{d}\tau\,\mathrm{d}\bar\tau/(\operatorname{Im}\tau)^2$ as target, and charge quantisation cuts the classical $\mathrm{SL}(2,\mathbb{R})$ down to the modular group $\mathrm{SL}(2,\mathbb{Z})$ of Eq. (7.3). Two facts from Section 7.1 do all the work here. First, $\mathrm{SL}(2,\mathbb{Z})$ is a gauge identification on the space of vacua: τ and $\frac{a\tau+b}{c\tau+d}$ label the same physics, so τ is really valued in the quotient $\mathbb{H}/\mathrm{SL}(2,\mathbb{Z})$. Second, the generator $S : \tau \to -1/\tau$ sends $g_s \to 1/g_s$, so weak and strong coupling are the same theory in different frames.

Now let τ depend on position in the internal space. Nothing forbids it: τ is a field, and a solution of the equations of motion in which it varies is as legitimate as one in which it is constant. The moment it varies, though, $g_s = 1/\operatorname{Im}\tau$ varies too, and on any compact internal space $\operatorname{Im}\tau$ cannot stay large everywhere. Somewhere the coupling reaches order one. A perturbative string expansion, an asymptotic series about $g_s = 0$, has nothing to say at such a point. This is not a technical nuisance to be pushed to higher loops: it is the ordinary situation in a compactification with localised charges, and it is where the interesting gauge dynamics lives.

The way through is to change what we track. Instead of the pair (C_0, ϕ), follow the $\mathrm{SL}(2,\mathbb{Z})$-invariant content of τ. The modular group has a single holomorphic invariant, Klein's j-function. On the open quotient it is a biholomorphism

$$j :\; \mathbb{H}/\mathrm{SL}(2,\mathbb{Z}) \;\xrightarrow{\;\sim\;}\; \mathbb{C}, \qquad j(\tau) = j\Big(\tfrac{a\tau+b}{c\tau+d}\Big)\,, \tag{11.2}$$

and adding the weak-coupling cusp $\operatorname{Im}\tau \to \infty$ compactifies both sides: j extends to an isomorphism of $\overline{\mathbb{H}/\mathrm{SL}(2,\mathbb{Z})}$ with $\mathbb{CP}^1$, with its single pole at that cusp, where $j \sim e^{-2\pi i\tau} \to \infty$. It is normalised so that $j(i) = 1728$ and $j(e^{2\pi i/3}) = 0$ at the two orbifold points of the quotient. The composite $j(\tau(b))$ is then a single-valued meromorphic function on the internal space, even though $\tau(b)$ itself jumps by $\mathrm{SL}(2,\mathbb{Z})$ across branch cuts, and its poles mark the weakly coupled regions where $\operatorname{Im}\tau \to \infty$. Reconstructing τ from j leaves the modular ambiguity, and packaging that ambiguity is what the geometry will do for us. The map $\tau \mapsto j(\tau)$ is also the bridge to algebraic geometry, since j is the classical invariant that labels a torus up to isomorphism; the Weierstrass presentation that makes this precise is the subject of Section 11.2 and Appendix E.

11.1.2 Seven-branes and the holomorphic profile of τ

The objects that make τ vary are seven-branes. A D7-brane fills eight of the ten spacetime dimensions and has two transverse ones, so it is a point in the transverse plane, which we coordinatise by

$$z \;=\; x^8 + i\, x^9 \;\in\; \mathbb{C}. \tag{11.3}$$

A D7 is a magnetic source for C_0, in the same way a vortex sources a periodic scalar. Its field strength is $F_1 = \mathrm{d}C_0$, closed away from the brane; but the brane punctures the transverse plane and modifies the Bianchi identity to $\mathrm{d}F_1 = \delta^{(2)}(z - z_1)$, a unit point source. Integrating over a small disc, $\oint_{S^1} F_1 = \oint \mathrm{d}C_0 = 1$ on any circuit enclosing the brane, so C_0 is not single-valued: it steps up by one each time round,

$$z \text{ around a D7}: \qquad C_0 \to C_0 + 1, \qquad \text{i.e.} \qquad \tau \to \tau + 1, \tag{11.4}$$

the T monodromy of Eq. (7.4). This is the statement met at the end of Section 7.1, now the starting point rather than an aside.

The profile of τ between the branes is fixed by supersymmetry. With the seven-branes filling the eight non-transverse directions, a background preserving half the IIB supersymmetry has τ depending on z alone, and the vanishing of the dilatino and gravitino variations forces τ to be a holomorphic function of z,

$$\partial_{\bar{z}}\,\tau \;=\; 0 \tag{11.5}$$

(the spinor calculation is the standard BPS analysis of Appendix C). One can see the same conclusion without spinors. The equation of motion for the coset scalar in the flat transverse plane is the harmonic-map equation into $\mathbb{H}$,

$$\partial_z\partial_{\bar{z}}\,\tau \;=\; \frac{2\,\partial_z\tau\,\partial_{\bar{z}}\tau}{\tau-\bar{\tau}}, \tag{11.6}$$

whose only non-linearity is the Christoffel term of the Poincaré metric. A holomorphic $\tau(z)$ kills both sides at once, and among the solutions of Eq. (11.6) it is the one whose stress tensor saturates the BPS bound. Antiholomorphic τ solves it too, and preserves the opposite half of the supersymmetry; a background with both would break all of it.

Holomorphy plus the monodromy Eq. (11.4) fixes the local form of τ near a brane. A holomorphic function whose imaginary part is single-valued but whose real part shifts by one around a point must have a logarithmic branch there:

$$\tau(z) \;=\; \tau_0 + \frac{1}{2\pi i}\,\ln\frac{z-z_1}{\lambda} + \mathcal{O}(z-z_1)\,, \tag{11.7}$$

with z_1 the brane location and λ a scale. Sending $z-z_1 \to e^{2\pi i}(z-z_1)$ shifts $\ln(z-z_1)$ by $2\pi i$ and so shifts τ by one, reproducing T. The imaginary part is

$$\operatorname{Im}\tau(z) \;=\; \operatorname{Im}\tau_0 - \frac{1}{2\pi}\,\ln\Big|\frac{z-z_1}{\lambda}\Big|, \tag{11.8}$$

which grows without bound as $z \to z_1$: the coupling goes to zero at the brane, and the neighbourhood of a D7 is weakly coupled. That is the good news. The bad news is the sign. Move away from the brane and $\operatorname{Im}\tau$ decreases, reaches zero at $|z-z_1| = \lambda\, e^{2\pi \operatorname{Im}\tau_0}$, and then goes negative. A negative $\operatorname{Im}\tau$ is a negative g_s, which is unphysical: τ has left the upper half-plane.

Example 11.1 (A single D7-brane is not a vacuum)**.** Put one D7 at the origin, so $\tau(z) = \frac{1}{2\pi i}\ln(z/\lambda)$. The monodromy round $z=0$ is $\tau \to \tau+1$, as required. The coupling is $g_s = 1/\operatorname{Im}\tau = 2\pi/\ln|\lambda/z|$, weak near the origin and diverging on the circle $|z|=\lambda$, beyond which $\operatorname{Im}\tau < 0$. There is no way to complete $\tau(z) = \frac{1}{2\pi i}\ln(z/\lambda)$ to a globally sensible field on the whole plane: a single logarithm cannot stay in $\mathbb{H}$. A single seven-brane in an infinite transverse plane is therefore not a consistent supersymmetric background. What saves it is putting enough branes together that their combined monodromy closes up and the transverse space curls into something compact, which is the content of Section 11.1.3.

The T monodromy is special to the D7. A general seven-brane is an $\mathrm{SL}(2,\mathbb{Z})$ image of it, on which a (p,q)-string can end (the bound state of p fundamental strings and q D-strings of Section 7.1), and its monodromy is the corresponding conjugate of T. If $g \in \mathrm{SL}(2,\mathbb{Z})$ maps $(1,0)$ to the coprime charge (p,q), the monodromy of the $[p,q]$ seven-brane is $g\,T\,g^{-1}$, which works out to

$$M_{[p,q]} \;=\; \begin{pmatrix} 1-pq & p^2 \\ -q^2 & 1+pq \end{pmatrix}, \tag{11.9}$$

an $\mathrm{SL}(2,\mathbb{Z})$ element of unit determinant and trace two, fixing the charge vector (p,q) as its invariant eigenvector. The invariance of (p,q) is the statement that the (p,q)-string ending

on the brane keeps its identity as it is dragged around it. Two seven-branes with charges (p, q) and (p', q') have commuting monodromies, and can be rotated to two ordinary D7-branes simultaneously, only when the antisymmetric pairing $pq' - qp'$ vanishes. When it does not, the branes are *mutually non-local*: no duality frame turns them both into D7-branes at once. Perturbative type IIB, which sees only D7-branes and their orientifold planes, cannot describe such a configuration, and this is the first hint of where the exceptional gauge groups will come from (Section 11.3).

11.1.3 Backreaction: deficit angles and a compact base

A seven-brane is a codimension-two source, and codimension-two sources curve the transverse plane into a cone. That back-reaction is what makes many seven-branes globally consistent when one is not. Run the bookkeeping carefully, because done naively it puts the deficit in the wrong place. Solving the Einstein equations with the holomorphic $\tau(z)$ as source gives a transverse metric

$$\mathrm{d}s_\perp^2 \;=\; e^{2B(z,\bar z)}\,\mathrm{d}z\,\mathrm{d}\bar z, \qquad e^{2B} \;=\; \mathrm{Im}\,\tau(z)\;|\eta(\tau)|^4 \prod_{i=1}^{N} |z - z_i|^{-1/6}, \tag{11.10}$$

with η the Dedekind eta function and the product over the N brane locations. This is asserted, not derived: the calculation is due to Greene, Shapere, Vafa and Yau, and is reproduced in Appendix E. The combination $\mathrm{Im}\,\tau\,|\eta(\tau)|^4$ is invariant under $\mathrm{SL}(2,\mathbb{Z})$ acting on τ, because η is a modular form of weight $1/2$: under $\tau \to (a\tau + b)/(c\tau + d)$ the factor $|\eta|^4$ scales by $|c\tau + d|^2$ and cancels the $|c\tau + d|^{-2}$ picked up by $\mathrm{Im}\,\tau$ (Appendix G). The metric is therefore single-valued even though τ is not, and the physical geometry never sees the branch cuts.

The last factor $\prod |z - z_i|^{-1/6}$ looks singular at $z = z_i$, as though there were a cone at each brane. There is not, and seeing why fixes where the deficit actually sits. Package the metric as $e^{2B}|\mathrm{d}z|^2 = |\omega|^2$ with the single-valued one-form

$$\omega \;=\; (\mathrm{Im}\,\tau)^{1/2}\,\eta(\tau)^2 \prod_{i=1}^{N} (z - z_i)^{-1/12}\,\mathrm{d}z. \tag{11.11}$$

Near the i-th brane the local profile Eq. (11.7) makes the nome $q = e^{2\pi i\tau} \sim (z - z_i)/\lambda$, so $\eta(\tau)^2 \sim q^{1/12} \sim (z - z_i)^{1/12}$, which cancels the $(z - z_i)^{-1/12}$ in the product exactly. The holomorphic part of ω is regular and non-zero at z_i; the only thing that happens there is the slow logarithmic growth of $\mathrm{Im}\,\tau$ from Eq. (11.8). So $e^{2B} \sim \mathrm{Im}\,\tau \sim \ln|\lambda/(z - z_i)|$: the neighbourhood of a brane is weakly coupled and only logarithmically curved, not a $\pi/6$ cone. The $|\eta|^4$ that made the metric single-valued is the same factor that dissolves the would-be cone.

The curvature sits at infinity instead. Far from all the branes $\tau \to \tau_\infty$ is constant, $\eta(\tau)^2$ tends to a constant, and $\omega \sim z^{-N/12}\,\mathrm{d}z$, so

$$e^{2B} \;\sim\; |z|^{-N/6} \qquad (|z| \to \infty). \tag{11.12}$$

Set $z = re^{i\theta}$ and pass to the proper radial coordinate $\rho = \int^r e^B\,\mathrm{d}r' \propto r^{\,1-N/12}$; the metric becomes $\mathrm{d}\rho^2 + (1 - \frac{N}{12})^2\rho^2\,\mathrm{d}\theta^2$. The angular circle is short by a total deficit

$$\delta_{\mathrm{tot}} \;=\; 2\pi \cdot \frac{N}{12} \;=\; N \cdot \frac{\pi}{6}, \tag{11.13}$$

one factor of $\pi/6$ for each brane. That per-brane $\pi/6$ is 2π times the modular weight of η^2: under $T : \tau \to \tau + 1$ one has $\eta^2 \to e^{i\pi/6}\eta^2$, and it is this phase, carried around the branch cut rather than localised at the brane, that curves the plane. To close the transverse plane into a

two-sphere the deficit accumulated out to infinity must equal 4π, the total curvature of $\mathbb{CP}^1$ by Gauss-Bonnet, which forces

$$N \;=\; \frac{4\pi}{\pi/6} \;=\; 24. \tag{11.14}$$

Equivalently, in the coordinate $w = 1/z$ near infinity the metric Eq. (11.12) reads $|w|^{\,N/6-4}|\mathrm{d}w|^2$, smooth and cone-free at $w = 0$ precisely when $N = 24$. With twenty-four seven-branes the plane wraps up into a $\mathbb{CP}^1$, the axio-dilaton varies over that sphere as a well-defined section of the $\mathrm{SL}(2,\mathbb{Z})$ bundle, and the coupling can be strong wherever it likes with no inconsistency. The same 24 returns from pure geometry once the fibration is algebraic: the Calabi-Yau condition puts the degenerations in the class $-12K_B$, of degree 24 over $\mathbb{CP}^1$ (Section 11.2), so physics and geometry count the branes the same way.

11.1.4 τ as the complex structure of a fibre torus

The pieces now assemble. On the internal space B we have a field $\tau(b)$ valued in $\mathbb{H}$, defined only up to $\mathrm{SL}(2,\mathbb{Z})$, holomorphic away from the branes, and with prescribed $\mathrm{SL}(2,\mathbb{Z})$ monodromy around each brane. Compare this with the oldest example of a modular parameter. A two-torus $\mathbb{C}/(\mathbb{Z}+\tau\mathbb{Z})$ is built by quotienting the plane by a lattice with generators 1 and τ; its shape, the complex-structure modulus, is the ratio $\tau \in \mathbb{H}$, and two values of τ related by $\mathrm{SL}(2,\mathbb{Z})$ give the same torus because they are two bases for the same lattice. The moduli space of complex structures on T^2 is therefore

$$\mathcal{M}_{T^2} \;=\; \mathbb{H}/\mathrm{SL}(2,\mathbb{Z}), \tag{11.15}$$

the very space in which the axio-dilaton lives. The group $\mathrm{SL}(2,\mathbb{Z})$ is not a coincidence on the two sides: it is the mapping class group of the torus, the ways of cutting along a one-cycle and regluing with a twist, acting on the homology basis (α,β) of $H_1(T^2,\mathbb{Z})$, so the type IIB S-duality group and the geometric symmetry group of a torus are one and the same.

That coincidence of groups is what the whole construction rests on. Attach to each point $b \in B$ a torus E_b whose complex structure is the local value of the axio-dilaton, $\tau(E_b) = \tau(b)$. The result is a family of tori over B, an elliptic fibration

$$\pi :\; Y \;\longrightarrow\; B, \qquad \pi^{-1}(b) = E_b, \tag{11.16}$$

whose total space Y has two complex dimensions more than the base. Everything about the varying axio-dilaton is now geometry. Where $\mathrm{Im}\,\tau \to \infty$, the torus E_b degenerates into a long thin cylinder as one of its one-cycles shrinks; those are the weakly coupled regions. Where a seven-brane sits, the fibre pinches off completely and E_b becomes singular; the locus of such points in B is the discriminant. The $\mathrm{SL}(2,\mathbb{Z})$ monodromy of τ around a brane is the monodromy of the homology $H_1(E_b,\mathbb{Z})$ as the fibre is carried around the discriminant, and which one-cycle pinches records which $[p,q]$ seven-brane is present: the cycle $p\,\alpha + q\,\beta$ collapses at a $[p,q]$ brane.

Definition 11.2 (F-theory on an elliptic fibration)**.** Let $\pi : Y \to B$ be an elliptic fibration with a chosen zero-section (a holomorphic choice of one point on each fibre, supplying the identity element of the torus; made precise in Section 11.2). *F-theory on Y* is type IIB string theory compactified on the base B, with axio-dilaton $\tau(b)$ equal to the complex-structure modulus of the fibre E_b, with seven-branes wrapping the discriminant $\Delta \subset B$ where the fibre degenerates, and with the S-duality bundle of τ identified with the monodromy representation of $\mathrm{SL}(2,\mathbb{Z})$ on $H_1(E_b,\mathbb{Z})$. The complex structure of the fibre is physical; its area is not part of the data.

Table 11.1 collects the dictionary. Each line is a physical statement about type IIB with a varying coupling, rewritten as a geometric statement about the fibration. The last two lines point forward: the map from the type of degeneration to the gauge algebra it carries is Kodaira's classification (Section 11.2), and the requirement that the whole background be supersymmetric is the requirement that Y be Calabi-Yau.

Type IIB with varying τ	Elliptic fibration $\pi : Y \to B$
axio-dilaton $\tau(b) = C_0 + ie^{-\phi}$	complex-structure modulus of the fibre E_b
$\mathrm{SL}(2,\mathbb{Z})$ S-duality	mapping class group acting on $H_1(E_b,\mathbb{Z})$
weak coupling $\mathrm{Im}\,\tau \to \infty$	a one-cycle of E_b shrinks (long thin fibre)
seven-brane	component of the discriminant $\Delta \subset B$
$[p,q]$ seven-brane	the cycle $p\,\alpha + q\,\beta$ pinches there
monodromy of τ around a brane	monodromy of $H_1(E_b,\mathbb{Z})$ around Δ
mutually non-local branes	degenerations that cannot be undone in one frame
gauge algebra on a brane stack	Kodaira type of the fibre over that component (Section 11.2)
supersymmetric background	Y Calabi-Yau

Table 11.1: The F-theory dictionary in its most basic form: type IIB with a position-dependent axio-dilaton, read as the geometry of an elliptic fibration. The final two rows are developed in Section 11.2.

11.1.5 What F-theory is, and what it is not

The construction adds two real dimensions to spacetime, the fibre E_b, but F-theory is not a twelve-dimensional theory. The fibre carries no physical size. Only its complex structure, the shape τ, is a field of the type IIB background; its area is not a modulus of any ten-dimensional theory and does not appear in Theorem 11.2. There is no twelve-dimensional metric to vary, no Kaluza-Klein tower from the fibre, no twelve-dimensional Lorentz symmetry. F-theory is ten-dimensional type IIB, written in a language where the strongly coupled data of the axio-dilaton is stored in an auxiliary torus.

The extra dimensions are bookkeeping

F-theory is not quantum gravity in twelve dimensions. The two extra dimensions of the fibre have no independent dynamics: the fibre's complex structure is the type IIB coupling, and its size is frozen (formally sent to zero). A Calabi-Yau fourfold Y therefore gives an F-theory compactification to $10 - 2 \times 3 = 4$ dimensions, not to $12 - 8 = 4$ by counting real dimensions of a would-be twelve-dimensional theory. The base B is the physical internal space; it is a threefold, six real dimensions, and $4 + 6 = 10$, with the fibre extra bookkeeping on top.

For the background to be a supersymmetric vacuum the total space Y must be Calabi-Yau, for the same reason a heterotic or type II compactification demands it (Section 8.5): a covariantly constant spinor on Y is what preserves supersymmetry, and it forces $c_1(Y) = 0$. The Calabi-Yau condition ties the twisting of the fibre to the geometry of the base and pins down the discriminant, which is how the count of twenty-four branes in Eq. (11.14) is reproduced without ever solving an Einstein equation. Working out that constraint needs the Weierstrass presentation of the fibration, and it is the business of Section 11.2.

Remark 11.3 (Beyond the lectures: where the name comes from, and the M-theory definition). The auxiliary torus was introduced by Vafa in 1996, and the letter F was chosen as a sequel to M with no committed meaning. There is a microscopic definition that removes any mystery about the extra dimensions. Recall from Chapter 7 that M-theory on a two-torus of complex structure τ and area A is dual to type IIB on a circle, with the IIB axio-dilaton equal to τ and the IIB circle radius set by A; shrinking $A \to 0$ decompactifies the IIB circle and yields ten-dimensional type IIB with coupling τ. Applied fibrewise: M-theory on the elliptically fibred Calabi-Yau Y, in the limit where every fibre shrinks to zero area, is F-theory on Y. This is why F-theory is a well-defined corner of a theory we already have rather than a new twelve-dimensional postulate, and it is the definition one actually uses to derive the spectrum, since the M-theory side is

under Lagrangian control at finite fibre size. The shrinking limit is singular from the M-theory viewpoint, which is the price of decompactifying a dimension, and reading physics off it is where the elliptic fibration's singular fibres do their work.

The payoff of all this is stated most sharply by contrast with the perturbative string. An ordinary type IIB or type I construction builds gauge groups from stacks of D-branes and orientifold planes, and those produce only the classical series: unitary, orthogonal and symplectic groups, $\mathrm{SU}(n)$, $\mathrm{SO}(n)$, $\mathrm{Sp}(n)$. The exceptional groups $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$ never appear, because there is no stack of mutually local branes whose open strings fill out an exceptional adjoint. In F-theory those groups are ordinary. They are the gauge algebras carried by particular collisions of mutually non-local seven-branes, encoded in particular singular fibres, and they cost no more than the classical groups to write down. That is the reason the chapter exists: the geometry of a degenerating torus reaches gauge symmetry that a weakly coupled string cannot, and it does so with the top-quark Yukawa coupling and grand-unified structure sitting naturally at the singular fibres. The mechanism is Kodaira's classification of degenerate fibres, which Section 11.2 sets up and Section 11.3 turns into gauge theory, with the exceptional cases explained by the heterotic dual of Section 11.4.

We have turned type IIB with an arbitrary, possibly strong, position-dependent coupling into the geometry of an elliptic fibration: the axio-dilaton is the shape of a fibre torus, the seven-branes are the points where the torus degenerates, and the non-perturbative $\mathrm{SL}(2,\mathbb{Z})$ monodromy is the ordinary monodromy of a homology basis. What remains is to make the fibration algebraic, so that the singular fibres and their gauge content can be read off by hand. That is a presentation in Weierstrass form and the classification of its degenerations, which is where Section 11.2 begins.

Exercise 11.1 (Monodromy of a $[p,q]$ seven-brane; Standard, ∼20 min)**.** Let $g = \left(\begin{smallmatrix} p & r \\ q & s \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{Z})$ carry $(1,0)$ to (p,q), so $ps - qr = 1$.

1. Compute $g\,T\,g^{-1}$ with $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and verify that it equals $M_{[p,q]}$ of Eq. (11.9), independent of r, s.
2. Check that $\det M_{[p,q]} = 1$ and $\operatorname{tr} M_{[p,q]} = 2$, and that $(p,q)^{\mathsf{T}}$ is the eigenvector with eigenvalue one. Interpret the invariance of (p,q) physically.
3. Show that $M_{[p,q]}$ and $M_{[p',q']}$ commute if and only if $pq' - qp' = 0$, and conclude that two seven-branes can be mapped to a pair of D7-branes in one duality frame precisely when their charges are parallel.

Exercise 11.2 (The single-brane profile and the deficit angle; Standard, ∼25 min)**.** Take one seven-brane at the origin with the naive profile $\tau(z) = \frac{1}{2\pi i}\ln(z/\lambda)$.

1. Verify the monodromy $\tau \to \tau + 1$ on a circuit of the origin, and find the radius $|z|$ at which $\operatorname{Im}\tau$ reaches zero, confirming that this profile leaves the upper half-plane at finite distance.
2. The $\mathrm{SL}(2,\mathbb{Z})$-invariant prefactor $\operatorname{Im}\tau\,|\eta|^4$ smooths the brane itself (Section 11.1.3). Strip it and keep only the discriminant factor, the *reduced* metric $e^{2B} = |z|^{-1/6}$ standing for one brane's contribution. With $z = re^{i\theta}$ and $\rho = \int_0^r e^B\,\mathrm{d}r'$, show $\rho \propto r^{11/12}$ and that the metric becomes $\mathrm{d}\rho^2 + (\frac{11}{12})^2\rho^2\,\mathrm{d}\theta^2$, a deficit angle $\pi/6$.
3. N branes contribute N such factors, so at large $|z|$, where τ is constant, $e^{2B} \sim |z|^{-N/6}$. Show that this closes smoothly into a two-sphere, either by Gauss-Bonnet ($N\cdot\pi/6 = 4\pi$) or by checking that $w = 1/z$ is a regular point, only for $N = 24$. This is the F-theory-on-K3 background whose heterotic dual is Section 11.4.

Exercise 11.3 (Is F-theory twelve-dimensional?; Warm-up, ∼10 min)**.** Decide whether each statement is true or false, with one sentence of justification.

1. A varying axio-dilaton on a compact base must reach $g_s \sim 1$ somewhere, so no single perturbative expansion covers the whole internal space.
2. The fibre torus E_b contributes a Kaluza-Klein tower to the lower-dimensional spectrum.
3. An elliptically fibred Calabi-Yau fourfold gives an F-theory vacuum in four spacetime dimensions.
4. The area of the fibre is a modulus of the ten-dimensional type IIB theory.

(Use Theorem 11.2 and the warning box: the fibre's complex structure is physical, its area is not.)

11.2 Elliptic fibrations and the Kodaira dictionary

Section 11.1 left us with a type IIB background in which the axio-dilaton $\tau = C_0 + ie^{-\phi}$ of (7.1) is not a constant but a multivalued function $\tau(z)$ on the space transverse to the seven-branes. Each time we encircle a brane, τ jumps by an element of $\mathrm{SL}(2,\mathbb{Z})$: a D7-brane produces the monodromy $T : \tau \to \tau + 1$, a general (p, q) seven-brane a general element of the modular group, and once the branes are mutually non-local there is no duality frame in which all of them are weakly coupled at once. As a field configuration this is an awkward object: a multivalued section of a bundle whose fibre is the fundamental domain of $\mathrm{SL}(2,\mathbb{Z})$. The move that makes it tractable, and that defines F-theory, is to stop treating τ as a field and start treating it as the shape of a geometry.

Geometrising the axio-dilaton

A value of τ in the upper half-plane, read modulo $\mathrm{SL}(2,\mathbb{Z})$, is the same data as the complex structure of a two-torus. So a IIB background with varying τ is a torus fibred over the physical space, and the fibration is packaged algebraically by a *Weierstrass model*

$$y^2 = x^3 + f\,x + g, \qquad \Delta = 4f^3 + 27g^2.$$

The seven-branes sit on the discriminant locus $\{\Delta = 0\}$, where the torus fibre pinches. How badly the fibre degenerates there is classified by Kodaira into a finite list of types, and each type carries an ADE (or exceptional) Dynkin diagram: that is the gauge algebra of the seven-branes wrapping the locus. Perturbative IIB reaches only the A, B, C, D series; the exceptional fibres $\mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$ for $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$ are what F-theory adds, and they have no weakly coupled description.

By the end of this section you should be able to:

- recast a IIB background of varying τ as an elliptically fibred space, and say what is physical about the fibre and what is auxiliary (Section 11.2.1);
- write a Weierstrass model $y^2 = x^3 + fx + g$, identify (f, g) as sections of $\bar{K}_B^4, \bar{K}_B^6$, and recover τ from (f, g) through the j-function (Section 11.2.2);
- locate the seven-branes on the discriminant $\Delta = 4f^3 + 27g^2 = 0$ and read its class $12\,c_1(B)$ as a charge constraint (Section 11.2.3);
- read a Kodaira type off the vanishing orders $(\mathrm{ord}_S f, \mathrm{ord}_S g, \mathrm{ord}_S \Delta)$ of a Weierstrass model (Section 11.2.4);
- turn that fibre type into an ADE or exceptional gauge algebra, and explain why $\mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8$ have no perturbative description (Section 11.2.5).

What you need from algebraic geometry. A handful of objects recur below; the fuller treatment is in Appendix E. A holomorphic *line bundle* L over B is the natural home for a holomorphic field that is only locally a function; its *sections* are those fields, and $H^0(B, L)$ is the finite-dimensional space of global holomorphic sections (for $B = \mathbb{CP}^1$ and $L = \mathcal{O}(d)$ these are the polynomials of degree d). The *canonical bundle* K_B is the bundle of holomorphic top-forms; its inverse $\bar{K}_B$, the *anticanonical bundle*, has first Chern class $c_1(B) = -c_1(K_B)$ and controls the Calabi-Yau condition. A *divisor* is a codimension-one locus, the zero set of a section, and its *class* records only its topology. The one computation we borrow is *adjunction*, which relates the canonical bundle of a hypersurface to that of the space it sits in.

11.2.1 From a varying coupling to a fibred torus

Recall from Section 11.1 that a point τ of the upper half-plane, read modulo $\mathrm{SL}(2,\mathbb{Z})$, is exactly the complex-structure modulus of a two-torus

$$E_\tau \;=\; \mathbb{C}/(\mathbb{Z} + \tau\,\mathbb{Z}), \qquad \tau \in \mathbb{H}, \tag{11.17}$$

a parallelogram with sides 1 and τ and opposite edges identified. The $\mathrm{SL}(2,\mathbb{Z})$ that made τ multivalued is the freedom to relabel the torus's one-cycles, its mapping class group; it acts by $\tau \mapsto (a\tau + b)/(c\tau + d)$ and leaves the lattice $\mathbb{Z} + \tau\mathbb{Z}$ untouched. A IIB background with a varying $\tau(z)$ is therefore a torus varying over the transverse space, and the seven-brane monodromy of τ becomes the monodromy of the fibre's one-cycles as they are carried around a brane. A (p, q) seven-brane is the locus where the one-cycle $p\,a + q\,b$ pinches to zero length. This turns a multivalued field into single-valued geometry.

Definition 11.4 (Elliptic fibration)**.** An *elliptic fibration* is a holomorphic map $\pi : Y \to B$ whose generic fibre $\pi^{-1}(z)$ is a smooth elliptic curve (a torus with a marked point), together with a *section* $\sigma : B \to Y$ meeting each fibre once, which supplies the marked point fibrewise. The base B is the physical space; the fibre is the auxiliary torus $E_{\tau(z)}$.

F-theory is the statement that IIB with this varying-τ profile is captured by the total space Y, and that a supersymmetric vacuum forces Y to be Calabi-Yau. If B has complex dimension n, then Y has complex dimension $n + 1$, the IIB spacetime is the base B times $\mathbb{R}^{1,9-2n}$, and the effective theory is $(10 - 2n)$-dimensional. The cases that matter here are

$$\begin{aligned} &Y \text{ an elliptic K3 (a CY twofold), } B = \mathbb{CP}^1 : && \text{eight-dimensional,} \\ &Y \text{ an elliptic CY threefold, } \dim_{\mathbb{C}} B = 2 : && \text{six-dimensional,} \\ &Y \text{ an elliptic CY fourfold, } \dim_{\mathbb{C}} B = 3 : && \text{four-dimensional.} \end{aligned} \tag{11.18}$$

The fourfold is the one that yields a four-dimensional GUT, and Section 11.5 builds on it; the K3 case is the arena for the heterotic duality of Section 11.4. We assert the Calabi-Yau condition now and justify it geometrically in Section 11.2.4 and Appendix E.

The fibre is auxiliary

Only the fibre's *complex structure* τ is physical (it is the IIB axio-dilaton); its area, the Kähler modulus, is not a IIB field at all and is sent to zero in the F-theory limit. So $Y \to B$ is not a compactification with two extra dimensions, and F-theory is not twelve-dimensional: there is no supergravity and no momentum tower on the fibre. The one honest circle is in the M-theory dual of Theorem 11.6, where the fibre area is the radius of an M-theory circle.

11.2.2 The Weierstrass model

To do algebraic geometry with the fibration we need equations, not a parallelogram. Every elliptic curve with a marked point can be embedded in $\mathbb{CP}^2$ as a cubic, and by shifting and

rescaling the coordinates the cubic can always be brought to *Weierstrass form*

$$y^2 \;=\; x^3 + f\,x + g, \tag{11.19}$$

with the marked point at the point at infinity $[x : y : z] = [0 : 1 : 0]$ of the projective completion. The pair (f, g) are two complex numbers, and (11.19) is a torus for generic (f, g).

Fibring over B means letting f and g vary. They cannot be arbitrary functions: for Y to be a well-defined compact space they are holomorphic sections of line bundles over B, and for Y to be Calabi-Yau those bundles are pinned down (the surface-geometry derivation is in Appendix E). Writing $L = \bar{K}_B$ for the anticanonical bundle of the base, the Calabi-Yau condition is

$$f \in H^0(B,\, L^4) = H^0(B,\, \bar{K}_B^{\,4}), \qquad g \in H^0(B,\, L^6) = H^0(B,\, \bar{K}_B^{\,6}). \tag{11.20}$$

The counting behind (11.20) is short: promoting x and y to sections of L^2 and L^3 makes the three terms of (11.19) sections of L^6, and adjunction on the resulting hypersurface gives $K_Y = \pi^*(K_B \otimes L)$, which is trivial exactly when $L = \bar{K}_B$.

The dictionary back to the physical axio-dilaton runs through the modular j-function. For the curve (11.19) the value of τ is fixed implicitly by

$$j(\tau) \;=\; 1728\,\frac{4f^3}{4f^3 + 27g^2}, \tag{11.21}$$

where j is the modular-invariant function with the q-expansion $j = q^{-1} + 744 + \mathcal{O}(q)$, $q = e^{2\pi i\tau}$. Because j is $\mathrm{SL}(2,\mathbb{Z})$-invariant, the right-hand side of (11.21) is single-valued in z while its inverse $\tau(z)$ is multivalued, and the multivaluedness is precisely the seven-brane monodromy of Section 11.1. The Weierstrass data (f, g) is therefore a single-valued, algebraic repackaging of the multivalued coupling. Three special values anchor the map, and we return to them in Exercise 11.5: $g = 0$ gives $j = 1728$ and the square torus $\tau = i$; $f = 0$ gives $j = 0$ and the hexagonal torus $\tau = e^{2\pi i/3}$; and $4f^3 + 27g^2 \to 0$ with $f \neq 0$ sends $j \to \infty$, that is $\tau \to i\infty$ and $g_s \to 0$.

11.2.3 The discriminant locates the seven-branes

The torus (11.19) degenerates when the cubic $x^3 + fx + g$ develops a repeated root, because then the branch structure of $y = \sqrt{x^3 + fx + g}$ collapses and a one-cycle of the fibre pinches. A double root x_0 requires the cubic and its derivative to vanish together,

$$x_0^3 + f x_0 + g = 0, \qquad 3x_0^2 + f = 0. \tag{11.22}$$

The second equation gives $x_0^2 = -f/3$; using it in the first gives $x_0 = -3g/(2f)$, and squaring this and setting it equal to $-f/3$ eliminates x_0 to leave $27g^2 = -4f^3$. The locus where the fibre is singular is therefore the vanishing of the *discriminant*

$$\boxed{\;\Delta \;=\; 4f^3 + 27g^2 \;=\; 0\;} \tag{11.23}$$

(the discriminant of the cubic is $-\Delta$; the sign is convention). At a point of $\{\Delta = 0\}$ with $f \neq 0$ the cubic factorises as $(x - x_0)^2(x - x_1)$ and the fibre is the *nodal cubic* $y^2 = (x - x_0)^2(x - x_1)$, a torus with one one-cycle shrunk to a point, topologically a sphere with two points identified (Fig. 11.1). This is a single seven-brane.

Globally, Δ is a section of $L^{12} = \bar{K}_B^{\,12}$, so the discriminant is a divisor in the class

$$[\Delta] \;=\; -12\,K_B \;=\; 12\,c_1(B). \tag{11.24}$$

Its topology, and hence the total seven-brane content, is fixed by the base alone. This is the geometric form of the codimension-two tadpole that Section 11.1 phrased in IIB language: the seven-brane charges cannot be chosen freely, their sum is pinned by $c_1(B)$.

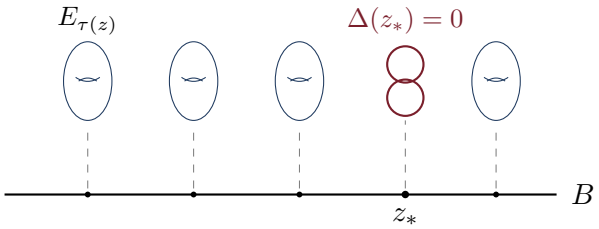


Figure 11.1: An elliptic fibration $\pi : Y \to B$. Over a generic point the fibre is a smooth torus $E_{\tau(z)}$; over a point z_* of the discriminant $\{\Delta = 0\}$ a one-cycle pinches and the fibre becomes nodal (drawn as a figure-eight). The seven-branes of the IIB picture sit on $\{\Delta = 0\}$, and the vanishing one-cycle labels the (p, q) type of the brane.

Example 11.5 (Twenty-four seven-branes on $\mathbb{CP}^1$)**.** Take $B = \mathbb{CP}^1$, the base of an elliptic K3. Then $\bar{K}_{\mathbb{CP}^1} = \mathcal{O}(2)$, so (11.20) makes f a degree-8 polynomial (nine coefficients) and g a degree-12 polynomial (thirteen coefficients), while

$$\Delta = 4f^3 + 27g^2 \in H^0\big(\mathbb{CP}^1, \mathcal{O}(24)\big) \tag{11.25}$$

has degree 24 and therefore 24 zeros. Generically these are 24 distinct I_1 points, that is 24 mutually non-local (p, q) seven-branes scattered over the sphere. The number is not an accident: 24 is the Euler characteristic of K3, and it is the same 24 that appears when one asks how many (p, q) seven-branes can consistently be placed on a compact transverse $\mathbb{CP}^1$ before their deficit angles close it up. The product of the 24 monodromies is trivial in $\mathrm{SL}(2, \mathbb{Z})$, which is what makes the configuration compact. This elliptic K3 is exactly the geometry that Section 11.4 matches to the heterotic string on T^2.

Near a generic I_1 point the profile of τ is controlled by inverting (11.21). With the brane at $z = z_*$ one finds the logarithmic behaviour

$$\tau(z) \;\simeq\; \tau_0 + \frac{1}{2\pi i}\,\ln(z - z_*), \tag{11.26}$$

so $\operatorname{Im}\tau \to +\infty$ and $g_s \to 0$ as $z \to z_*$, and encircling the brane, $\ln(z - z_*) \to \ln(z - z_*) + 2\pi i$, gives $\tau \to \tau + 1$, the monodromy T of a D7-brane (a $(1, 0)$ seven-brane). The vanishing cycle in this frame is the a-cycle. Everything in (11.26) is the Greene-Shapere-Vafa-Yau solution of Section 11.1, now read off the geometry rather than solved for as a field.

11.2.4 Kodaira's classification of singular fibres

The nodal fibre is the mildest degeneration. Where f and g vanish to higher order along a component of the discriminant, the fibre can collapse more drastically: several one-cycles pinch, the singular fibre breaks into several rational curves, and the monodromy grows. Kodaira classified every way a fibre of an elliptic surface can degenerate over a point of the base, and the classification is a short, closed list. In a higher-dimensional base the same list applies fibrewise along each irreducible component of $\{\Delta = 0\}$, which is a divisor rather than a point, so the fibre type is now attached to a codimension-one locus in B.

Each Kodaira type is fixed by three integers, the orders of vanishing $(\mathrm{ord}_S f,\ \mathrm{ord}_S g,\ \mathrm{ord}_S \Delta)$ of the sections along the component S, and it carries a definite monodromy conjugacy class in $\mathrm{SL}(2, \mathbb{Z})$. The organising fact, which we assert and refer to Appendix E for, is geometric: when the singular fibre is resolved it becomes a collection of $\mathbb{CP}^1$'s (rational curves), and the dual graph of that collection, one node per curve and one edge per intersection, is an *affine* ADE Dynkin diagram. Table 11.2 collects the list.

Kodaira type	$(\mathrm{ord}_S f,\ \mathrm{ord}_S g,\ \mathrm{ord}_S \Delta)$	monodromy	fibre graph	algebra $\mathfrak{g}$
I_0	$(\geq 0,\ \geq 0,\ 0)$	$\mathbf{1}$	smooth	–
I_1	$(0,\ 0,\ 1)$	$\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$	nodal	–
I_n $(n \geq 2)$	$(0,\ 0,\ n)$	$\left(\begin{smallmatrix}1&n\\0&1\end{smallmatrix}\right)$	$\widehat{A}_{n-1}$	$\mathfrak{su}(n)$
II	$(\geq 1,\ 1,\ 2)$	$\left(\begin{smallmatrix}1&1\\-1&0\end{smallmatrix}\right)$	cusp	–
III	$(1,\ \geq 2,\ 3)$	$\left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)$	$\widehat{A}_1$	$\mathfrak{su}(2)$
IV	$(\geq 2,\ 2,\ 4)$	$\left(\begin{smallmatrix}0&1\\-1&-1\end{smallmatrix}\right)$	$\widehat{A}_2$	$\mathfrak{su}(3)$
I_0^*	$(\geq 2,\ \geq 3,\ 6)$	$\left(\begin{smallmatrix}-1&0\\0&-1\end{smallmatrix}\right)$	$\widehat{D}_4$	$\mathfrak{so}(8)$
I_n^* $(n \geq 1)$	$(2,\ 3,\ 6+n)$	$\left(\begin{smallmatrix}-1&-n\\0&-1\end{smallmatrix}\right)$	$\widehat{D}_{n+4}$	$\mathfrak{so}(2n{+}8)$
IV^*	$(\geq 3,\ 4,\ 8)$	$\left(\begin{smallmatrix}-1&-1\\1&0\end{smallmatrix}\right)$	$\widehat{E}_6$	$\mathfrak{e}_6$
III^*	$(3,\ \geq 5,\ 9)$	$\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$	$\widehat{E}_7$	$\mathfrak{e}_7$
II^*	$(\geq 4,\ 5,\ 10)$	$\left(\begin{smallmatrix}0&-1\\1&1\end{smallmatrix}\right)$	$\widehat{E}_8$	$\mathfrak{e}_8$
non-minimal	$(\geq 4,\ \geq 6,\ \geq 12)$	–	–	(blow up B)

Table 11.2: Kodaira's classification of singular elliptic fibres. Along a component S of the discriminant the fibre degenerates according to the vanishing orders of (f, g, Δ); the monodromy is a representative of the $\mathrm{SL}(2,\mathbb{Z})$ conjugacy class; the resolved fibre is a set of $\mathbb{CP}^1$'s whose dual graph is the affine Dynkin diagram shown; and the gauge algebra is the corresponding finite algebra (the "split" case, before the folding of Theorem 11.7). A dash (–) marks fibres carrying no non-abelian gauge symmetry. The final row is not a gauge enhancement but a signal that the base must be blown up.

11.2.5 From fibre type to gauge algebra

Now the physics. Along a component S of $\{\Delta = 0\}$ carrying a Kodaira fibre of a given type, a stack of seven-branes wraps S and supports a non-abelian gauge theory. The gauge algebra is read from the fibre in one step: the resolved singular fibre is an *affine* Dynkin diagram, and deleting the affine node, the unique component that meets the zero section σ of Theorem 11.4, leaves an ordinary ADE Dynkin diagram. That finite diagram is the gauge algebra. The last column of Table 11.2 is the result: $\mathrm{I}_n \to \mathfrak{su}(n)$, $\mathrm{I}_n^* \to \mathfrak{so}(2n{+}8)$, and the three exceptional fibres $\mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^* \to \mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8$.

This is not a new mechanism. It is the ADE story of Section 10.1 sitting inside the elliptic fibre. Transverse to S, the total space Y develops a du Val singularity $\mathbb{C}^2/\Gamma$ of Theorem 10.2, fibred over S: the pinching of the elliptic fibre is a two-complex-dimensional ADE singularity in Y, of exactly the type in Table 10.1. Resolving it replaces the singular point by the exceptional $\mathbb{CP}^1$'s, which are the finite Kodaira fibre components, and their intersection matrix is minus the ADE Cartan matrix, the relation (10.2) now realised inside the elliptic fibre. The Kodaira-to-ADE map and the singularity-to-ADE map of Chapter 10 are one map, and Section 11.3 runs the rest of the local dictionary, the matter on curves and the Yukawas at points, in this language.

Remark 11.6 (Beyond the lectures: the M-theory derivation). The shrinking-T^2 picture of Section 11.1 also settles the two things we have only asserted. Compactify M-theory on the smooth elliptic Calabi-Yau Y: the background is supersymmetric *because* Y is Calabi-Yau, which is where that condition comes from. Sending the fibre area to zero returns F-theory on Y, one dimension larger, with the fibre's complex structure τ playing the IIB axio-dilaton. Where Y carries an ADE singularity, M-theory on the local $\mathbb{C}^2/\Gamma$ produces the ADE gauge symmetry by the wrapped-M2 mechanism of Section 10.1.3, and the F-theory limit lifts that gauge theory

onto the seven-brane. This is the derivation behind Table 11.2; the full M-theory analysis is in Chapters 7 and 10.

Two features of the last column need care. First, the intersection form of curves on a surface is symmetric, so the naive resolution can only ever produce a simply-laced algebra, exactly the obstruction noted at the end of Section 10.1.2. The non-simply-laced algebras appear through the extra ingredient F-theory has and Section 10.1 did not: monodromy in the base.

Proposition 11.7 (Non-simply-laced algebras by monodromy)**.** *When the exceptional $\mathbb{CP}^1$'s of the resolved fibre are permuted by the monodromy of the fibration as one moves along the divisor S, the affine Dynkin diagram is folded by the corresponding outer automorphism, and the gauge algebra is the non-simply-laced algebra invariant under the folding. The relevant cases are*

$$\mathrm{I}_0^*:\ \mathfrak{so}(8)\to\mathfrak{so}(7)\to\mathfrak{g}_2,\qquad \mathrm{IV}^*:\ \mathfrak{e}_6\to\mathfrak{f}_4,\qquad \mathrm{IV}:\ \mathfrak{su}(3)\to\mathfrak{su}(2),\qquad \mathrm{I}_n:\ \mathfrak{su}(n)\to\mathfrak{sp}(\lfloor n/2\rfloor). \tag{11.27}$$

Whether the folding occurs is fixed by whether certain polynomials built from (f,g) factorise over the function field of S, the "split / semi-split / non-split" distinction of Tate's algorithm.

We state Theorem 11.7 rather than prove it; the folding data and the Tate table are collected in Appendix G, and the algorithm that reads the refined algebra off (f,g) is worked in Appendix E. This delivers the promise made in Section 10.1.2: the genuinely exceptional $\mathfrak{f}_4$ and $\mathfrak{g}_2$, which no intersection form on a surface can produce, are reached in F-theory by monodromy acting on the collapsing cycles as they are carried around the base.

Second, and this is the point of the whole chapter, the exceptional *simply-laced* algebras $\mathfrak{e}_6,\mathfrak{e}_7,\mathfrak{e}_8$ have no perturbative description. The A series comes from mutually local D7-branes (I_n is n coincident $(1,0)$-branes), and an O7-plane adds the SO and Sp groups, so a perturbative IIB orientifold reaches SU, SO and Sp and stops (Section 10.3, Section 10.3.5). The exceptional fibres require mutually non-local (p,q) seven-branes to collide at a point where g_s is of order one, a configuration with no weakly coupled frame; the exceptional adjoint simply cannot be assembled from open strings stretched between D-branes. F-theory produces it because it works directly with the geometry of τ, where a collision of (p,q)-branes is just a fibre that pinches to Kodaira type IV^*, III^* or II^*. That is what makes the exceptional GUTs of Section 11.3 available, and why the top-quark Yukawa of Section 11.3.4 sits where it does.

Remark 11.8 (Beyond the lectures: Sen's weak-coupling limit)*.* There is a controlled limit in which an F-theory background degenerates to a perturbative IIB orientifold, and it makes the perturbative ceiling precise. Following Sen, tune the sections to

$$f=-3h^2+\epsilon\,\eta,\qquad g=-2h^3+\epsilon\,h\,\eta-\tfrac{1}{12}\epsilon^2\chi,$$

with h,η,χ sections of appropriate bundles on B and ϵ a small parameter. The leading terms are arranged to cancel in the discriminant, $4(-3h^2)^3+27(-2h^3)^2=0$, so Δ is pushed to order ϵ^2 and factorises,

$$\Delta\ =\ -9\,\epsilon^2\,h^2\,(\eta^2-h\chi)+\mathcal{O}\big(\epsilon^3\big)\,.$$

Away from $h=0$ the string coupling runs to zero, $\mathrm{Im}\,\tau\to\infty$, and the physics is a weakly coupled IIB orientifold on the double cover of B branched over $h=0$. The O7-plane sits at $h=0$ and the D7-branes on $\eta^2=h\chi$; where four D7-branes meet the O7 the fibre enhances to I_0^*, the perturbative SO(8). The exceptional groups live at $\epsilon\sim 1$, far from this limit, which is another way to see that they are invisible to perturbation theory. The O-plane and D7 charges here are the tadpole bookkeeping of Section 6.4, and this limit is the bridge Section 11.5 uses to import perturbative intuition into the F-theory ledger.

Remark 11.9 (Beyond the lectures: non-minimal points and the base)*.* The last row of Table 11.2, where $(\mathrm{ord}_S f,\mathrm{ord}_S g,\mathrm{ord}_S\Delta)\geq(4,6,12)$, is not a gauge group. There the Weierstrass model

is too singular to admit a crepant resolution, so Y fails to be a good Calabi-Yau F-theory background as written. The remedy is to blow up the base B at that point, which lowers the vanishing orders below the threshold and restores a Calabi-Yau elliptic fibration over the new base. Physically these loci are where the seven-brane physics is genuinely strongly coupled: they support tensionless strings and superconformal sectors, and in six dimensions they are the origin of the strongly coupled fixed points that need no Lagrangian. We flag them as a research-frontier feature and return to their role only in passing in Section 11.5.

Where this leaves us

The varying axio-dilaton of Section 11.1 is now a piece of algebraic geometry: an elliptically fibred Calabi-Yau $Y \to B$, given concretely by a Weierstrass model $y^2 = x^3 + fx + g$ whose discriminant $\Delta = 4f^3 + 27g^2$ locates the seven-branes, and whose fibre degenerations are classified by Kodaira into the ADE and exceptional types of Table 11.2. The gauge algebra over any component of the discriminant is read off in one line, and F-theory reaches the exceptional algebras that perturbative IIB cannot. Section 11.3 takes the exceptional types seriously as GUT gauge groups, promotes the local dictionary of matter curves and Yukawa points into this setting, and Section 11.4 explains, through the duality with the heterotic string, why the exceptional structure and the top-quark Yukawa are natural rather than accidental.

Exercise 11.4 (The discriminant of the cubic; Standard, 25 min)**.** Fill in the elimination behind (11.23).

1. From the double-root conditions (11.22), show $x_0 = -3g/(2f)$ (assume $f \neq 0$) and hence, using $x_0^2 = -f/3$, derive $4f^3 + 27g^2 = 0$.
2. Show directly that the cubic $x^3 + fx + g$ has three distinct roots iff $4f^3 + 27g^2 \neq 0$, by writing the discriminant as $\prod_{i<j}(x_i - x_j)^2$ up to sign and expressing it in the elementary symmetric polynomials $e_1 = 0,\ e_2 = f,\ e_3 = -g$.
3. Explain in one sentence why a double root of the cubic makes the Riemann surface $y^2 = x^3 + fx + g$ singular, and which cycle of the torus pinches.

Exercise 11.5 (Special values of τ from the j-function; Warm-up, 15 min)**.** Use (11.21) and the standard values $j(i) = 1728$, $j(e^{2\pi i/3}) = 0$.

1. Show that $g = 0$ forces $\tau = i$ at that fibre, and $f = 0$ forces $\tau = e^{2\pi i/3}$. Interpret these as the square and hexagonal tori of (11.17).
2. Show that as $\Delta = 4f^3 + 27g^2 \to 0$ with $f \neq 0$ one has $j \to \infty$, hence $\operatorname{Im}\tau \to \infty$ and $g_s \to 0$. Reconcile this with the local profile (11.26).
3. A Kodaira II fibre has $\operatorname{ord}_S f \geq 1,\ \operatorname{ord}_S g = 1$ and a IV* fibre has $\operatorname{ord}_S f \geq 3,\ \operatorname{ord}_S g = 4$. Show that along both loci f^3 vanishes faster than g^2, so $j \to 0$ and the fibre sits at $\tau = e^{2\pi i/3}$. (The IV* and II* fibres, E_6 and E_8, thus live at the most symmetric point of the torus moduli space; Section 11.4 makes the E_6 point the geometric home of the top Yukawa.)

Exercise 11.6 (Counting seven-branes on a base; Warm-up, 15 min)**.** 1. For $B = \mathbb{CP}^1$, verify from (11.20) that f and g have degrees 8 and 12 and that Δ has degree 24, reproducing Theorem 11.5.

2. For a base surface B (the sixfold case, complex dimension two), the discriminant is in the class $12\,c_1(B)$ by (11.24). Take $B = \mathbb{CP}^2$, where $c_1 = 3H$ with H the hyperplane class, and compute the degree of the discriminant curve and its self-intersection $[\Delta] \cdot [\Delta]$.

3. Explain why (11.24) is a constraint, not a choice: the total seven-brane charge is fixed by the topology of B, so engineering a large gauge group on one divisor is paid for elsewhere in $\{\Delta = 0\}$.

Exercise 11.7 (Reading Kodaira types off the vanishing orders; Standard, 20 min)**.** Use Table 11.2.

1. Along a divisor $S = \{s = 0\}$ suppose f and g are finite and non-zero on S while Δ vanishes to order 5. Identify the Kodaira type and the gauge algebra, and draw the affine Dynkin diagram of the resolved fibre.
2. Suppose instead $\mathrm{ord}_S f = 2$, $\mathrm{ord}_S g = 3$, $\mathrm{ord}_S \Delta = 6$. Identify the type and its simply-laced algebra, then say which non-simply-laced algebras the folding of Theorem 11.7 can produce from it, and what feature of (f, g) decides between $\mathfrak{so}(8)$, $\mathfrak{so}(7)$ and $\mathfrak{g}_2$.
3. Suppose $\mathrm{ord}_S f = 4$, $\mathrm{ord}_S g = 6$, $\mathrm{ord}_S \Delta = 12$. Explain why this is not $\mathfrak{e}_8$ enhancement but a signal that B must be blown up, and state what Theorem 11.9 says the locus supports.

11.3 Exceptional gauge groups and GUTs

The Kodaira dictionary of Section 11.2 assigns a gauge algebra to every way an elliptic fibre can degenerate over the base, and its list runs across the full ADE series and past it. Most of that list is familiar from geometric engineering: the A and D entries are the $\mathrm{SU}(N)$ and $\mathrm{SO}(2N)$ groups that D-branes and orientifolds already produce. Three entries are not familiar. The fibres IV*, III* and II* carry the exceptional algebras $\mathfrak{e}_6$, $\mathfrak{e}_7$ and $\mathfrak{e}_8$, and no stack of D-branes, with or without orientifold planes, can produce these. They are the reason F-theory earns a chapter of its own. Perturbative type IIB gives you the classical Lie algebras and stops; letting the axio-dilaton vary opens the door to the exceptional ones, and with them to a class of grand-unified structures, most sharply the top-quark Yukawa coupling, that the perturbative constructions of Chapter 10 leave out of reach.

This section is a taste, not a survey. It does three things. It says precisely why the exceptional groups are inaccessible perturbatively and how F-theory reaches them. It sets up the local grand-unified model: the gauge group on a divisor, matter on the curves where seven-branes intersect, Yukawa couplings at the points where three matter curves meet. And it isolates the one coupling, the up-type Yukawa, whose very existence signals an exceptional point in the geometry. The deep reason the exceptional structures are natural at all is heterotic/F-theory duality, which the next section makes precise; here we take the Kodaira dictionary as given and read off its physics.

The big picture

Perturbative open strings furnish only the classical algebras: a D-brane stack gives $\mathfrak{u}(N)$, an orientifold projection gives $\mathfrak{so}(N)$ or $\mathfrak{sp}(N)$. The exceptional $\mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8$ have no such realisation. F-theory reaches them because a varying axio-dilaton allows mutually *non-local* (p, q) seven-branes, and when several of them collide the elliptic fibre degenerates to an exceptional Kodaira type IV*, III*, II*. Wrap such a seven-brane on a complex surface S and you engineer a grand-unified theory: the gauge group is the fibre type over S, charged matter lives on the curves in S where the fibre degenerates further, and Yukawa couplings sit at the points where those curves cross. The prize coupling is the up-type Yukawa $\mathbf{10} \cdot \mathbf{10} \cdot \mathbf{5}$, which requires an E_6 point and is forbidden in perturbation theory.

By the end of this section you should be able to:

- explain why perturbative branes deliver only $\mathrm{SU}, \mathrm{SO}, \mathrm{Sp}$ and never $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$ (Section 11.3.1);
- read the exceptional gauge algebra off the vanishing orders of a Weierstrass model, and picture the fibre as a collision of mutually non-local seven-branes (Section 11.3.2);
- lay out a local SU(5) grand-unified model on a divisor, with matter on curves and Yukawa couplings at points (Section 11.3.3);
- state why the top Yukawa needs an E_6 point, and why that coupling is absent in perturbative type IIB (Section 11.3.4).

11.3.1 Where perturbative branes stop

Everything about a perturbative gauge group traces back to a Chan-Paton index. Open strings ending on a stack of N coincident D-branes carry a label $i = 1, \ldots, N$ at each endpoint, the massless vector is an $N \times N$ matrix, and the gauge algebra is $\mathfrak{u}(N)$ (Chapter 6). Gauging worldsheet parity to make an orientifold identifies the two endpoints up to a symmetric or antisymmetric bilinear form on the Chan-Paton space, and the surviving vectors are the matrices preserved by that form: $\mathfrak{so}(N)$ or $\mathfrak{sp}(N)$ (Section 6.4). That is the entire perturbative menu. Written by their Cartan-Killing type it is the classical series A_{N-1}, B_N, C_N, D_N, and nothing about placing branes on singular geometry changes it: the ADE-singularity construction of Section 10.1 and the intersecting-brane models of Section 10.3 realise the same A and D groups from different angles.

The exceptional algebras $\mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8$ (together with $\mathfrak{f}_4, \mathfrak{g}_2$) are exactly the ones that are *not* algebras of matrices preserving a bilinear form. There is no Chan-Paton space of any dimension whose parity-invariant endomorphisms close on $\mathfrak{e}_6$. So there is no perturbative open-string stack that produces an exceptional gauge boson, and this is not a failure of ingenuity but a structural fact about which algebras the open-string construction can reach. We assert it here; the group theory is collected in Appendix G.

One perturbative string does carry an exceptional group: the heterotic string has gauge algebra $\mathfrak{e}_8 \oplus \mathfrak{e}_8$ built from a chiral current algebra, not from Chan-Paton factors (Chapter 3). That is a genuine exceptional symmetry, but it sits at the string scale in ten dimensions, spread over the whole spacetime rather than localised on a brane you can wrap on a small cycle and break to a Standard-Model group. Model building wants the opposite: an exceptional group confined to a compact piece of the internal space, with the flavour structure controlled by how that piece sits inside the geometry. F-theory delivers precisely that, and, as Section 11.4 shows, the reason it can is that F-theory *is* the heterotic string in disguise, fibrewise. The exceptional current algebra of one description becomes the exceptional Kodaira fibre of the other.

The ingredient that breaks the perturbative ceiling is the one this chapter is built on: a varying axio-dilaton. A single D7-brane is a source for $\tau = C_0 + ie^{-\phi}$, and circling it sends $\tau \to \tau + 1$ (Section 11.1). Once several seven-branes are present, no single duality frame can keep them all weakly coupled at once, and the objects that fill out an exceptional gauge multiplet are not ordinary open strings on a single stack. The next subsection makes that concrete.

11.3.2 Exceptional fibres from colliding seven-branes

Recall the Weierstrass presentation from Section 11.2. The elliptic fibration over a base coordinate u is

$$y^2 \;=\; x^3 + f(u)\,x + g(u), \qquad \Delta \;=\; 4f^3 + 27g^2, \tag{11.28}$$

with f, g sections of powers of the anticanonical bundle of the base and Δ the discriminant (up to an overall constant). Seven-branes sit on the zero locus $\Delta = 0$, and the gauge algebra along a

component of Δ is fixed by how fast f, g and Δ vanish there. The Kodaira/Tate classification turns those three vanishing orders into a fibre type and an algebra. The exceptional entries are the last rows of that table:

fibre	$\mathrm{ord}(f)$	$\mathrm{ord}(g)$	$\mathrm{ord}(\Delta)$	algebra
I_0^*	≥ 2	≥ 3	6	$\mathfrak{so}(8)$
IV^*	≥ 3	4	8	$\mathfrak{e}_6$
III^*	3	≥ 5	9	$\mathfrak{e}_7$
II^*	≥ 4	5	10	$\mathfrak{e}_8$

(11.29)

The full table, including the split/non-split refinements that distinguish $\mathfrak{so}$ from $\mathfrak{sp}$ and $\mathfrak{e}_6$ from $\mathfrak{f}_4$, is in Section 11.2 and Appendix E. What matters here is that the three exceptional algebras are on exactly the same footing as A_{N-1} and D_N: each is a Kodaira degeneration of one elliptic fibre, read off from a pair of integers, and the classification specifies $\mathfrak{e}_6$ no differently from A_{N-1}.

The physics behind the last three rows is a collision of seven-branes. A single D7-brane is an I_1 fibre, Δ vanishing to first order, with $\mathrm{SL}(2,\mathbb{Z})$ monodromy $T : \tau \to \tau + 1$. Bring N mutually local D7-branes together and the fibre becomes I_N, algebra $\mathfrak{su}(N)$: this is still perturbative, because all N branes are D7-branes in one common frame. An exceptional fibre is different. It is a coincidence of several seven-branes of *different* (p,q) types that are mutually non-local. Two seven-branes of charges (p,q) and (p',q') are mutually non-local when

$$\det \begin{pmatrix} p & p' \\ q & q' \end{pmatrix} \;=\; p\,q' - q\,p' \;\neq\; 0, \tag{11.30}$$

and for such a pair there is no $\mathrm{SL}(2,\mathbb{Z})$ transformation carrying both to $(1,0)$ at once. An E_6, E_7 or E_8 point involves seven-branes no common frame can turn into D7-branes, so at such a point the string coupling is genuinely of order one and there is no weakly coupled open-string description to be had. This is the precise sense in which the exceptional groups are non-perturbative. The Kodaira classification packages the whole non-perturbative configuration into a single piece of algebraic geometry, the vanishing orders (11.29), which is why they are as easy to write down as the perturbative cases.

The charged gauge bosons follow the same logic as the wrapped-brane W-bosons of Section 10.1, generalised. There, an A or D enhancement came from M2-branes (or, in IIA, D2-branes) wrapping the two-cycles of the resolved singularity, one BPS vector per root. Here the object that stretches between mutually non-local seven-branes is a *string junction*: a network of (p,q)-string segments meeting at a trivalent vertex, with a prong ending on each seven-brane and charge conserved at the vertex. The junctions carrying the right asymptotic charges fill out the adjoint of the exceptional algebra, one junction per root, and the resolved exceptional fibre is a configuration of $\mathrm{rank}(\mathfrak{g}) + 1$ two-spheres whose dual graph is the affine Dynkin diagram $\widehat{E}_6, \widehat{E}_7, \widehat{E}_8$, exactly the ADE-from-geometry pattern of Section 10.1 carried up to the exceptional cases. We state this rather than derive it; the junction calculus is a subject in itself.

Remark 11.10 (Beyond the lectures: string junctions and the E_n lattice). The E_n seven-brane configurations were constructed explicitly by DeWolfe, Hauer, Iqbal and Zwiebach as collections of a few elementary branes, conventionally $\mathbf{A} = (1,0)$, $\mathbf{B} = (1,-1)$, $\mathbf{C} = (1,1)$, with an E_n point realised by a specific string $\mathbf{A}^{\,k}\mathbf{B}\mathbf{C}\cdots$. The lattice of string junctions ending on such a configuration, equipped with its natural intersection pairing, is isomorphic to the root lattice of E_n, and the junctions of self-intersection -2 are the roots. This makes the exceptional gauge symmetry completely explicit in the seven-brane picture, and it reproduces, junction by junction, the same E_n that heterotic/F-theory duality delivers wholesale in Section 11.4. The two derivations are independent and they agree, which is a large part of why one trusts the F-theory dictionary at strong coupling.

11.3.3 The local GUT: gauge group, matter curves, Yukawa points

Now build a four-dimensional model. The internal space is an elliptically fibred Calabi-Yau fourfold X_4 over a three-complex-dimensional base B_3. The seven-branes wrap divisors of B_3, complex surfaces filling the four non-compact dimensions. Pick one such surface S and arrange the fibre to degenerate to a fixed Kodaira type all along it. That type is the gauge group of the grand-unified theory. For definiteness take the SU(5) model: an I_5 fibre, algebra $\mathfrak{su}(5)$, over the whole of S. This is the codimension-zero datum, the vector multiplet living over S, and it is the F-theory realisation of the local model of Section 10.3. The base B_3 is now genuinely three-dimensional, so S is a surface and its curves and points are honest submanifolds where the extra structure lives.

The dictionary that turns the geometry into a spectrum is the one established in Section 10.3, read here in the elliptic-fibration language. The three strata are:

Gauge group (codimension 0, the surface S). The fibre type over generic points of S gives the algebra $\mathfrak{g}$. For the SU(5) model the vector multiplet fills the adjoint **24** over all of S.

Matter (codimension 1, curves in S). Where S meets another component of the discriminant, the fibre degenerates further and the singularity enhances by one rank, $\mathfrak{g} \to \mathfrak{g}'$. The extra states that become light along that curve are read off by the Katz-Vafa branching (10.30), decomposing the adjoint of $\mathfrak{g}'$ under $\mathfrak{g}$. For SU(5) the two relevant enhancements are

$$\mathrm{SU}(5) \to \mathrm{SO}(10) \text{ on a curve } \Sigma_{\mathbf{10}}, \qquad \mathrm{SU}(5) \to \mathrm{SU}(6) \text{ on a curve } \Sigma_{\mathbf{5}}, \tag{11.31}$$

giving a **10** on $\Sigma_{\mathbf{10}}$ (the quark doublets and up-type singlets) and a **5** on $\Sigma_{\mathbf{5}}$ (the Higgs and, in the conjugate, the lepton doublets and down-type singlets). Each matter curve is a Riemann surface inside S, and the number of generations it carries is not fixed by the enhancement alone: it is a zero-mode count, set by the topology of the curve together with the gauge flux threading it. That counting, and the flux data that makes it come out three, belongs to the ledger of Section 11.5.

Yukawa couplings (codimension 2, points in S). Where two matter curves cross, the singularity enhances by rank two, $\mathfrak{g} \to \mathfrak{g}''$, and three matter representations become simultaneously light. If their tensor product contains a gauge singlet the superpotential gets a cubic term, localised at that point because it is the only place where all three wavefunctions have support at once. For SU(5) the two couplings the Standard Model needs come from two different rank-two points,

$$\mathrm{SU}(5) \to \mathrm{E}_6: \qquad \mathbf{10} \cdot \mathbf{10} \cdot \mathbf{5} \quad \text{(up-type)}, \tag{11.32}$$

$$\mathrm{SU}(5) \to \mathrm{SO}(12): \qquad \mathbf{10} \cdot \bar{\mathbf{5}} \cdot \bar{\mathbf{5}} \quad \text{(down-type and charged-lepton)}. \tag{11.33}$$

The value of each coupling is the overlap integral (10.32) of the three localised wavefunctions, which collapses to the residue (10.33) at the point. Because a single point produces a rank-one Yukawa matrix, only one generation gets an order-one mass from each point, with the lighter families lifted by subleading corrections. Figure 11.2 shows the two points and the curves that meet at them.

Example 11.11 (An $\mathrm{I}_5 \to \mathrm{IV}^*$ collision and the top coupling)**.** Run the enhancement of the SU(5) model up to the up-type Yukawa point one fibre at a time. Over generic points of S the fibre is I_5, with vanishing orders $(\mathrm{ord}(f), \mathrm{ord}(g), \mathrm{ord}(\Delta)) = (0, 0, 5)$ and algebra $\mathfrak{su}(5)$. Two matter curves pass through the point p of interest.

- $\Sigma_{\mathbf{10}}$: the fibre worsens to I_1^* ($\mathfrak{su}(5) \to \mathfrak{so}(10)$, $\mathrm{ord}(\Delta) = 7$), and Katz-Vafa (10.30) branches $\mathbf{45} = \mathbf{24} \oplus \mathbf{1} \oplus \mathbf{10} \oplus \overline{\mathbf{10}}$, depositing a **10**.

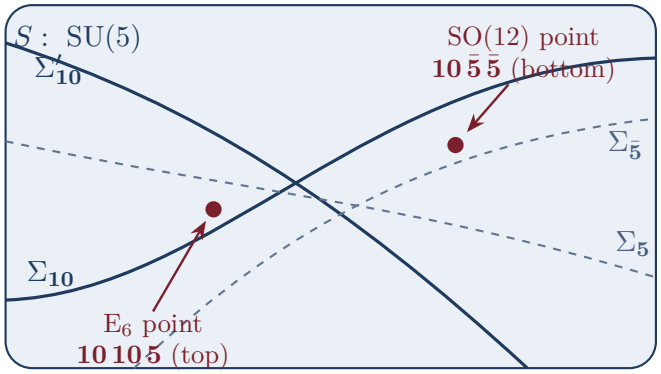


Figure 11.2: Schematic of the local SU(5) model on a surface S. The gauge group lives over all of S; matter **10** and **5** (solid and dashed curves) is trapped on the curves where the fibre enhances to SO(10) and SU(6). Two matter curves crossing at a point enhance the fibre by rank two: the up-type Yukawa $\mathbf{10}\cdot\mathbf{10}\cdot\mathbf{5}$ sits at an E_6 point, the down-type $\mathbf{10}\cdot\bar{\mathbf{5}}\cdot\bar{\mathbf{5}}$ at an SO(12) point. The exceptional enhancement of the up-type point is the feature no perturbative brane construction reproduces.

- $\Sigma_{\mathbf{5}}$: the fibre worsens to I_6 ($\mathfrak{su}(5)\to\mathfrak{su}(6)$, $\mathrm{ord}(\Delta)=6$), and $\mathbf{35}=\mathbf{24}\oplus\mathbf{1}\oplus\mathbf{5}\oplus\bar{\mathbf{5}}$ deposits a **5**.

At p the two enhancements superpose and the rank climbs by two. The smallest simple algebra containing both $\mathfrak{so}(10)$ and $\mathfrak{su}(6)$ with the common $\mathfrak{su}(5)$ is $\mathfrak{e}_6$; in Kodaira terms the fibre is IV*, with orders $(\geq 3,4,8)$ from (11.29). The discriminant order tracks the collision: 5 over generic S, then 7 and 6 along the two curves, then 8 at the point.

Now read off the coupling. The states made light at p are the two **10**'s brought in by the pair of **10**-curves through it (Fig. 11.2) and the **5** from $\Sigma_{\mathbf{5}}$. Branching **78** under SU(5) (Theorem 10.10), the only SU(5) singlet in $\mathbf{10}\otimes\mathbf{10}\otimes\mathbf{5}$ is $\epsilon_{abcde}\,\mathbf{10}^{ab}\mathbf{10}^{cd}\mathbf{5}^{e}$. That singlet sits inside the $\mathfrak{e}_6$ commutators, so the $\mathfrak{e}_6$ gauge theory carries it with an order-one coefficient, evaluated by the residue (10.33). Replacing $\Sigma_{\mathbf{5}}$ by a $\Sigma_{\bar{\mathbf{5}}}$ curve enhances the point to $\mathfrak{so}(12)$ instead (fibre I_2^*, an ordinary D-type degeneration), and gives the neutral down-type coupling $\mathbf{10}\cdot\bar{\mathbf{5}}\cdot\bar{\mathbf{5}}$.

Nothing in this construction is new relative to the codimension dictionary of Section 10.3; what is new is that the singularity is now a Kodaira degeneration of an elliptic fibre rather than an ADE singularity put in by hand, so the enhancement chain is free to run up into the exceptional groups. That is what lets (11.32) exist at all, and it is the subject of the next subsection.

11.3.4 The top Yukawa, and what perturbation theory misses

Compare the two Yukawa points (11.32) and (11.33). The down-type coupling sits at an SO(12) point, and SO(12) is an ordinary D-type group, reachable perturbatively. The up-type coupling sits at an E_6 point, and E_6 is exceptional. That asymmetry is not cosmetic: in a grand-unified theory the single largest coupling in the Standard Model, the top-quark Yukawa, is tied to an exceptional structure in the geometry.

To see why the up-type coupling is genuinely absent perturbatively, put an SU(5) on a stack of seven-branes in the perturbative regime, where the branes are honest D7-branes. A stack of D-branes never gives a bare SU(5); it gives $\mathrm{U}(5)=\mathrm{SU}(5)\times\mathrm{U}(1)$, with a diagonal U(1) under which every open string is charged. Normalise this U(1) so a fundamental **5** has charge $+1$. Then

$$q(\mathbf{5})=+1,\qquad q(\bar{\mathbf{5}})=-1,\qquad q(\mathbf{10})=+2, \tag{11.34}$$

the $\mathbf{10}=\Lambda^2\mathbf{5}$ carrying twice the charge of a fundamental because it has two upper indices. The

down-type coupling $\mathbf{10}\cdot\bar{\mathbf{5}}\cdot\bar{\mathbf{5}}$ has total charge $2-1-1=0$: it is U(1)-neutral and perturbatively allowed. The up-type coupling $\mathbf{10}\cdot\mathbf{10}\cdot\mathbf{5}$ needs the antisymmetric invariant ϵ_{abcde} to contract five SU(5) indices, and the three matter fields carry total charge $2+2+1=+5$. A perturbative open-string amplitude conserves this diagonal U(1), so a coupling that violates it by five units cannot be generated at any order in g_s. The U(1) gauge boson itself is massive, eaten by a Green-Schwarz axion, so the symmetry is not visible in the low-energy gauge group, but it survives as a perturbative selection rule and it forbids the top Yukawa. Only non-perturbative effects, Euclidean D-brane instantons of Chapter 9, can produce the coupling, suppressed by e^{-1/g_s} and generically far too small to be the top quark.

The E_6 point removes the obstruction at a stroke. There the SU(5) is not the SU(5) factor of a U(5) stack; it is a genuine subgroup of E_6, and E_6 has no such diagonal U(1) commuting with it. The decomposition of the E_6 adjoint under SU(5) contains the states whose triple product is the cubic invariant $\mathbf{10}\cdot\mathbf{10}\cdot\mathbf{5}$ (worked in Theorem 10.10), and the coupling is there at order one, computed by the residue (10.33) with no exponential suppression. The exceptional enhancement is exactly what buys the perturbatively forbidden coupling. This is the observation of Beasley, Heckman and Vafa that launched F-theory grand unification: the up-type Yukawa is a diagnostic for an exceptional point, and a construction that cannot reach the exceptional groups cannot produce a realistic top mass.

This is local data, not a finished vacuum

Everything in this section is a statement about a neighbourhood of the surface S: the gauge group, the matter curves, the Yukawa points and their couplings are all fixed by the local geometry, and gravity decouples in the large-volume limit of Section 10.3. Whether a compact Calabi-Yau fourfold exists with this S embedded in it, whether the flux that counts generations can be switched on consistently, whether the tadpoles cancel and the moduli are stabilised, and whether exactly three chiral families with the right hypercharge survive, are separate and much harder questions. The exceptional gauge structure and the Yukawa pattern are robust and geometric; the global completion into a Standard-Model-like vacuum is where the honest caveats live, and that is the business of Section 11.5.

Remark 11.12 (Beyond the lectures: everything from a single E_8). The rank-two points E_6 and SO(12) are themselves contained in a single rank-four enhancement to E_8, and one can build an entire F-theory grand-unified model by unfolding the geometry around one point of E_8 symmetry. The chain $\mathrm{E}_8 \supset \mathrm{SU}(5)_{\mathrm{GUT}} \times \mathrm{SU}(5)_{\perp}$ splits the adjoint **248** into the gauge fields of the visible SU(5) and a set of representations whose $\mathrm{SU}(5)_{\perp}$ labels are the flavour indices; the Higgs field that resolves the E_8 point away from the origin then determines the matter curves, the Yukawa points and, through its profile, the fermion mass hierarchies. This "local E_8" organising principle, due to Heckman and Vafa, is one reason the exceptional groups are viewed as central rather than exotic in F-theory model building. It is also as far as the local approach reaches on its own: turning the unfolding into a compact model with three generations and no exotics is exactly the open problem flagged above.

The exceptional groups, the matter curves and the Yukawa points assemble into a local grand-unified theory that no perturbative string construction can match, and the top Yukawa is the sharpest marker of that. What we have not explained is *why* the exceptional structures should be there in the first place, rather than being a coincidence of the Kodaira list. The answer is that F-theory on an elliptic K3 is the heterotic string on T^2, and fibrewise F-theory on an elliptic Calabi-Yau is the heterotic string on one dimension fewer. The $\mathfrak{e}_8 \oplus \mathfrak{e}_8$ of the perturbative heterotic string is then the same object as the exceptional Kodaira fibres, and the top Yukawa a heterotic worldsheet coupling seen from the F-theory side. That duality, cheap now that K3 is in hand from Chapter 7, is the next section. The concrete Standard-Model-like ledger, with the

flux that counts three generations and the tadpoles that must cancel, is the section after.

Exercise 11.8 (Vanishing orders of the exceptional fibres; Warm-up, ∼15 min)**.** Use the Weierstrass discriminant $\Delta = 4f^3 + 27g^2$ of (11.28). (a) For the orders $(\mathrm{ord}(f), \mathrm{ord}(g))$ listed in (11.29) for IV*, III* and II*, compute $\mathrm{ord}(4f^3)$ and $\mathrm{ord}(27g^2)$ and verify that the smaller of the two gives $\mathrm{ord}(\Delta) = 8, 9, 10$ respectively. Which term dominates in each case? (b) The resolved singular fibre of type IV*, III*, II* is a configuration of $\mathrm{rank}(\mathfrak{g}) + 1$ two-spheres whose dual graph is the affine Dynkin diagram. Deduce that $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$ give $7, 8, 9$ components, and match this to the ADE-from-geometry count of Section 10.1, where an A_{N-1} fibre had N components. (c) A fibre with $\mathrm{ord}(f) \geq 4$, $\mathrm{ord}(g) \geq 6$, $\mathrm{ord}(\Delta) \geq 12$ is not a valid Kodaira degeneration: it signals a base point that must be blown up. Check that II* sits just below this bound, and comment on what it means that E_8 is the largest gauge group F-theory produces on a divisor.

Exercise 11.9 (The local SU(5) layout; Warm-up, ∼10 min)**.** Without consulting Fig. 11.2, sketch the surface S carrying $\mathfrak{su}(5)$ and mark the local grand-unified data on it. (a) Draw the two kinds of matter curve, labelling each by its rank-one enhancement (SO(10) for the **10**, SU(6) for the **5**) and by the Standard-Model fields it traps. (b) Mark the two Yukawa points where three curves cross, labelling the rank-two enhancement at each (E_6 for the up-type $\mathbf{10}\cdot\mathbf{10}\cdot\mathbf{5}$, SO(12) for the down-type $\mathbf{10}\cdot\bar{\mathbf{5}}\cdot\bar{\mathbf{5}}$). (c) State in one sentence which of the two points has no perturbative-brane counterpart, and why. Check your sketch against Theorem 11.11.

Exercise 11.10 (The perturbative selection rule; Standard, ∼20 min)**.** Take a perturbative $\mathrm{U}(5) = \mathrm{SU}(5) \times \mathrm{U}(1)$ seven-brane stack with the charge assignment (11.34). (a) Confirm that $q(\mathbf{10}) = +2$ by writing $\mathbf{10} = \Lambda^2\mathbf{5}$ and adding the charges of its two upper indices, and that $q(\bar{\mathbf{5}}) = -1$. (b) Show that the down-type coupling $\mathbf{10}\cdot\bar{\mathbf{5}}\cdot\bar{\mathbf{5}}$ is U(1)-neutral, so it is allowed perturbatively, while the up-type coupling $\mathbf{10}\cdot\mathbf{10}\cdot\mathbf{5}$ carries net charge $+5$ and is forbidden. Identify the SU(5) invariant tensor that would have to supply the compensating -5 and explain why a perturbative open-string amplitude cannot furnish it. (c) A Euclidean D-brane instanton can violate the U(1) by absorbing charged fermion zero modes. Estimate the resulting top Yukawa as $\lambda_t \sim e^{-S_{\mathrm{inst}}}$ with $S_{\mathrm{inst}} \sim 1/g_s$, and comment on why this generically fails to reproduce $\lambda_t \approx 1$. Contrast with the E_6 point, where no U(1) selection rule is present.

11.4 Heterotic/F-theory duality

The previous section left a promissory note. The exceptional Kodaira fibres IV*, III*, II* deliver E_6, E_7, E_8, and the top Yukawa follows, but nothing so far explains why those degenerations should be present and physical rather than an accident of the Weierstrass discriminant. Perturbative type II branes reach only the classical groups, SU from a stack and SO or Sp after an orientifold (Section 11.3). The one perturbative string that carries an exceptional symmetry is the $\mathrm{E}_8 \times \mathrm{E}_8$ heterotic string, and it carries it as a ten-dimensional current algebra spread over all of spacetime, not localised on any brane you can wrap and break. The claim of this section is that these are the same fact seen from two sides. F-theory on a suitable Calabi-Yau is the heterotic string in disguise, and the exceptional Kodaira fibres are its two E_8's turned into geometry.

The big picture

F-theory on a Calabi-Yau X_{n+1} that is *both* elliptically fibred and K3-fibred is the same theory as the $\mathrm{E}_8 \times \mathrm{E}_8$ heterotic string on an elliptically fibred Calabi-Yau X_n over the common base. In the cleanest case, eight dimensions, this reads: heterotic on T^2 equals F-theory on an elliptic K3. The heterotic string already has two E_8's, sitting in an even self-dual lattice (Appendix G); the duality turns each of them into geometry, an E_8 hidden in the second cohomology of half of the K3. That is why F-theory sees $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$ where

perturbative branes cannot: the groups are on loan from the heterotic string.

By the end of this section you should be able to:

- state the heterotic/F-theory duality and check that it matches supersymmetry, moduli space and BPS spectrum dimension by dimension, with heterotic on T^2 equal to F-theory on an elliptic K3 as the base case (Sections 11.4.1 and 11.4.2);
- match the eight-dimensional moduli, identifying the Narain lattice $\Gamma_{2,18}$ with the complex structure of an elliptic K3 with section and placing the heterotic coupling as the size of the base $\mathbb{CP}^1$ (Section 11.4.2);
- use stable degeneration to see the two E_8's as two rational elliptic surfaces, and reassemble $\Gamma_{3,19}$ from two copies of $U \oplus (-\mathrm{E}_8)$ plus the neck (Section 11.4.3);
- read off, through the spectral cover and the commutant chain, why F-theory reaches the exceptional groups and where the top Yukawa comes from, geometric on the F-theory side and current-algebraic on the heterotic side (Sections 11.4.4 and 11.4.5).

11.4.1 The statement

Two compactifications land in the same lower-dimensional theory when they share supersymmetry, moduli space and BPS spectrum. The heterotic/F-theory pair is built so that all three match dimension by dimension, and the mechanism is a fibrewise version of a single eight-dimensional identity.

Proposition 11.13 (Heterotic/F-theory duality)**.** *Let B_{n-1} be a base of complex dimension $n-1$. The $\mathrm{E}_8 \times \mathrm{E}_8$ heterotic string compactified on an elliptically fibred Calabi-Yau $X_n \to B_{n-1}$, with a choice of gauge bundle, is dual to F-theory on an elliptically fibred Calabi-Yau $X_{n+1} \to B_n$, where B_n is a $\mathbb{CP}^1$-bundle over B_{n-1}. Equivalently X_{n+1} is fibred by elliptic K3 surfaces over B_{n-1}, and the duality is applied fibrewise over B_{n-1}, reducing everywhere to the eight-dimensional base case*

$$\text{het } \textit{on } T^2 \quad = \quad \text{F } \textit{on an elliptic K3}. \tag{11.35}$$

The dimension count is the first check. F-theory on X_{n+1}, whose base B_n has complex dimension n, occupies $10-2n$ non-compact spacetime dimensions; heterotic on X_n occupies the same $10-2n$. The compact spaces are rigidly parallel too: both are elliptic over their own base, and the F-theory base B_n is exactly one $\mathbb{CP}^1$ taller than the heterotic base B_{n-1}. That extra $\mathbb{CP}^1$ carries the whole story, because F-theory over it, fibre by fibre, is the eight-dimensional duality (11.35). The three cases that matter are

n	spacetime	heterotic on X_n	F-theory on X_{n+1}
1	8D	T^2	elliptic K3
2	6D	elliptic K3	elliptic CY_3, K3-fibred over $\mathbb{CP}^1$
3	4D	elliptic CY_3	elliptic CY_4, K3-fibred over a surface

The $n=3$ row is the one the model-building section (Section 11.5) lives on. We develop the base case in full, because everything else is this case fibred over B_{n-1}, and then say precisely how the fibration works.

11.4.2 Eight dimensions: heterotic on T^2 equals F-theory on K3

The heterotic side. Compactify the $\mathrm{E}_8 \times \mathrm{E}_8$ heterotic string on a two-torus. All sixteen supercharges survive, giving half-maximal eight-dimensional supergravity. The massless data

beyond the gravity multiplet is a set of moduli, and they organise into two decoupled pieces. First, the internal conformal field theory is a Narain compactification (Chapter 7) on the even self-dual lattice

$$\Gamma_{2,18} \;=\; \Gamma_{2,2} \,\oplus\, (\mathrm{E}_8 \oplus \mathrm{E}_8), \tag{11.36}$$

of signature $(2,18)$: two right-movers and eighteen left-movers, the eighteen being the two left-moving directions of the T^2 together with the sixteen Cartan directions of $\mathrm{E}_8 \times \mathrm{E}_8$. Its moduli fill

$$\mathcal{M}_{2,18} \;=\; O(2,18;\mathbb{Z})\backslash O(2,18)/\big(O(2)\times O(18)\big), \qquad \dim_{\mathbb{C}} \mathcal{M}_{2,18} = 18. \tag{11.37}$$

These eighteen complex moduli are the complex structure τ and complexified Kähler modulus $\rho = B_{T^2} + i\,\mathrm{vol}(T^2)$ of the torus, together with 16 complex Wilson lines, one for each Cartan generator of the gauge group wrapped on the two cycles. Second, and decoupled from (11.37), there is the eight-dimensional axio-dilaton

$$S \;=\; a + \frac{i}{g_8^2} \;\in\; \frac{\mathrm{SL}(2,\mathbb{Z})\backslash\mathrm{SL}(2,\mathbb{R})}{\mathrm{U}(1)}, \tag{11.38}$$

with a the axion dual in eight dimensions to $B_{\mu\nu}$ and g_8 the eight-dimensional string coupling. The generic gauge group is $\mathrm{U}(1)^{18}$ from the eighteen left-movers, plus two graviphotons from the right-movers that sit in the gravity multiplet and never enhance. On the special loci of (11.37) where left-moving lattice vectors of length-squared two appear, the group enhances to a rank-18 non-abelian group, and because the roots all have the same length these are the simply-laced ADE groups, exactly as in the heterotic-on-T^4 story of Section 7.3.6, now one torus smaller.

The F-theory side. An elliptic K3 is a K3 surface presented as an elliptic fibration over a $\mathbb{CP}^1$ base. In Weierstrass form (Section 11.2),

$$y^2 \;=\; x^3 + f(u)\,x + g(u), \qquad f \in H^0\big(\mathbb{CP}^1, \mathcal{O}(8)\big), \quad g \in H^0\big(\mathbb{CP}^1, \mathcal{O}(12)\big), \tag{11.39}$$

where u is a coordinate on the base and the degrees 8 and 12 are forced by the Calabi-Yau (canonical-class) condition on the total space. The discriminant

$$\Delta \;=\; 4f^3 + 27g^2 \;\in\; H^0\big(\mathbb{CP}^1, \mathcal{O}(24)\big) \tag{11.40}$$

has degree 24, so the fibre degenerates over 24 points of the base. Those 24 points are the 24 seven-branes whose total charge and deficit angle close up the sphere, the tadpole statement of Section 11.1, and $24 = \chi(\mathrm{K3})$ is the same number seen from the Euler characteristic. In F-theory the fibre area is sent to zero, so the Kähler class of the elliptic K3 is not physical; the only geometric moduli are its complex structure, and for an elliptic K3 *with section* these fill

$$\mathcal{M}_{\mathrm{cs}}(\mathrm{K3}_{\mathrm{ell}}) \;=\; O(2,18;\mathbb{Z})\backslash O(2,18)/\big(O(2)\times O(18)\big). \tag{11.41}$$

This is (11.37) on the nose. The two $O(2)$ directions are the holomorphic two-form of the K3; the 18 negative directions are the part of $H^2(\mathrm{K3},\mathbb{Z}) = \Gamma_{3,19}$ orthogonal to the fibre and section classes, a $\Gamma_{2,18}$ that is literally the heterotic Narain lattice (11.36).

The dictionary. Matching (11.37), (11.38) and (11.41) fixes the map. The eighteen Narain moduli become the eighteen complex-structure moduli of the elliptic K3, and the heterotic coupling, absent from (11.41), has to reappear as the one geometric datum F-theory has not yet used: the size of the base $\mathbb{CP}^1$. The geometric entries read most cleanly in the stable-degeneration limit of Section 11.4.3, where the K3 splits into two halves glued along a torus E; the table below is phrased in that language, which the next subsection sets up.

heterotic on T^2	F-theory on elliptic K3
complex structure τ of T^2	complex structure of the gluing fibre E
Kähler modulus $\rho = B + i$ vol of T^2	relative modulus of the two halves (gluing datum)
16 Wilson lines	16 complex-structure deformations of the branes
8D axio-dilaton $S = a + i/g_8^2$	complexified size of the base $\mathbb{CP}^1$
enhanced ADE point	K3 acquires an ADE Kodaira singularity

The last row carries the physics. A heterotic gauge enhancement, a purely lattice-theoretic and perturbative event on the left, becomes a Kodaira degeneration of the elliptic K3 on the right, the same statement made in Section 7.3.6 for heterotic/IIA, now with the fibre of an F-theory geometry doing the collapsing. The bottom entry of the previous table, the heterotic coupling mapping to the base size, is the seed of the whole higher-dimensional construction: fibre this identity over a base and the heterotic coupling becomes a local modulus, free to vary, which is why F-theory can describe strongly coupled heterotic vacua that no perturbative expansion reaches.

Example 11.14 (The eighteen moduli as a coset dimension)**.** The Narain moduli space (11.37) is a coset, so its dimension is pure group theory. Using $\dim O(p,q) = \frac{1}{2}(p+q)(p+q-1)$,

$$\dim_{\mathbb{R}} \frac{O(2,18)}{O(2)\times O(18)} \;=\; \dim O(20) - \dim O(2) - \dim O(18) \;=\; 190 - 1 - 153 \;=\; 36,$$

and the space is Hermitian symmetric, so this is 18 complex moduli. They split as the two torus moduli τ and ρ plus one complex Wilson line for each of the 16 Cartan generators of $\mathrm{E}_8 \times \mathrm{E}_8$, giving $2 + 16 = 18$. The elliptic K3 with section must therefore carry exactly 18 complex-structure moduli. A direct Weierstrass count of (11.39) confirms this in Exercise 11.11; the two answers agree because (11.37) and (11.41) are the same coset written twice.

11.4.3 Stable degeneration

The dictionary above is an equality of moduli spaces. To see the two E_8's appear as geometry, and to make the map local rather than a global lattice isomorphism, one goes to a corner of moduli space where the K3 falls apart in a controlled way. This is the *stable degeneration* limit, and it is the heterotic weak-coupling limit in disguise: large base $\mathbb{CP}^1$, which by the dictionary is small g_8.

Take the base $\mathbb{CP}^1$ and stretch it until it pinches, splitting into two spheres $\mathbb{CP}^1_+$ and $\mathbb{CP}^1_-$ joined at a node. The elliptic K3 fibred over it splits with it,

$$\mathrm{K3} \;\longrightarrow\; X_+ \cup_E X_-, \tag{11.42}$$

into two surfaces glued along a single common elliptic fibre E sitting over the node. Each half $X_\pm$ is a *rational elliptic surface*, the projective plane blown up at nine points, also written dP_9 and known as half of a K3. As an elliptic fibration over its $\mathbb{CP}^1_\pm$ it is

$$y^2 = x^3 + f_\pm(z)\,x + g_\pm(z), \qquad (\deg f_\pm, \deg g_\pm, \deg \Delta_\pm) = (4,6,12), \tag{11.43}$$

so each half carries 12 of the original 24 seven-branes, and $12 + 12 = 24$ as it must. The gluing curve E is the heterotic T^2: its complex structure is τ, and the parameter controlling how the two halves are glued is ρ.

The two E_8's are now visible. A rational elliptic surface has

$$H^2(\mathrm{dP}_9, \mathbb{Z}) \;\cong\; U \oplus (-\mathrm{E}_8), \tag{11.44}$$

of signature $(1,9)$, with an E_8 root lattice sitting inside it. Gluing two of them along E reassembles the K3 lattice: the two copies of $U \oplus (-\mathrm{E}_8)$ supply $U^{\oplus 2} \oplus (-\mathrm{E}_8)^{\oplus 2}$, and the vanishing cycle of the neck contributes the third hyperbolic plane, recovering

$$H^2(\mathrm{K3},\mathbb{Z}) \;\cong\; U^{\oplus 3} \oplus (-\mathrm{E}_8)^{\oplus 2} \;=\; \Gamma_{3,19} \tag{11.45}$$

of Section 7.3.4, one $(-\mathrm{E}_8)$ landing in each half. The two E_8's that (11.45) always displayed in the cohomology of K3 are the two E_8's of the heterotic string, one per rational elliptic surface. If you set all Wilson lines in one E_8 to zero, that half degenerates maximally: $f_\pm$ acquires a zero of order four and $g_\pm$ a zero of order five at the far fibre $z \to \infty$, an II^* Kodaira fibre, and the unbroken gauge group is the full E_8. Switching Wilson lines on deforms the surface away from II^* and Higgses the E_8 down, matching the heterotic statement that a Wilson line breaks the gauge group to the commutant of its holonomy, so the whole E_8 enhancement is trapped at one point of one half of the degenerate K3.

Remark 11.15 (Beyond the lectures: the fibrewise version and the $(12,12)$ split). For general n the stable degeneration is applied over every point of B_{n-1} at once. The F-theory Calabi-Yau splits as $X_{n+1} \to W_+ \cup_{X_n} W_-$, where $W_\pm$ are families of rational elliptic surfaces over B_{n-1} and they meet along the heterotic Calabi-Yau X_n, the common elliptic fibration over B_{n-1}. Each $W_\pm$ encodes one E_8 bundle on X_n. The $\mathbb{CP}^1$-bundle $B_n \to B_{n-1}$ of Theorem 11.13 is the family of pinching base curves, and its two sections are where the two E_8's sit. The six-dimensional case ($n=2$, heterotic on K3) is the sharpest illustration: anomaly cancellation forces the two E_8 bundles to carry instanton numbers summing to $c_2(\mathrm{K3}) = 24$, a split (n_1, n_2) with $n_1 + n_2 = 24$. On the F-theory side the base B_2 is the Hirzebruch surface $\mathbb{F}_{|n_1-n_2|/2}$, so the twisting is set by the asymmetry of the split; the symmetric split $(12,12)$ gives $\mathbb{F}_0 = \mathbb{CP}^1 \times \mathbb{CP}^1$. The two dozen instantons are the two dozen seven-branes, twelve to each half, one rational elliptic surface apiece.

11.4.4 How the two E_8 bundles appear

On the heterotic side the data hidden in each half $W_\pm$ is a gauge bundle $V_\pm$ with structure group inside E_8. For X_n elliptically fibred, these bundles have an efficient description, the spectral cover, and it is the technology that makes the duality computable in four dimensions. It is genuinely beyond lecture scope; the sketch below is for orientation, with the construction in the references.

Remark 11.16 (Beyond the lectures: spectral covers). A holomorphic bundle on an elliptic curve, of structure group $\mathrm{SU}(N)$ and vanishing first Chern class, is a sum of N line bundles of degree zero, and degree-zero line bundles on an elliptic curve are labelled by points of the dual curve. So the bundle on a single fibre is fixed by N points on that fibre summing to zero. Letting the fibre vary over the base B_{n-1}, these N points sweep out a divisor

$$\mathcal{C} \;\subset\; X_n, \qquad \mathcal{C} \;\text{ an } N\text{-fold cover of } B_{n-1}, \tag{11.46}$$

the *spectral cover*, an N-section of the elliptic fibration. Together with a line bundle on $\mathcal{C}$ it reconstructs V, the Friedman-Morgan-Witten correspondence. The unbroken gauge group in the heterotic vacuum, and equally the F-theory gauge group over B_{n-1}, is the commutant of the structure group of V inside E_8:

$$\mathrm{SU}(5) \subset \mathrm{E}_8 \;\Rightarrow\; \mathrm{SU}(5)_{\mathrm{GUT}}, \qquad \mathrm{SU}(4) \subset \mathrm{E}_8 \;\Rightarrow\; \mathrm{SO}(10), \qquad \mathrm{SU}(3) \subset \mathrm{E}_8 \;\Rightarrow\; \mathrm{E}_6. \tag{11.47}$$

On the F-theory side the same spectral cover reappears as the detailed shape of the Weierstrass model near the degenerate fibre, and the matter curves and Yukawa points of Section 11.5 are read off its ramification and intersections. Smaller structure group means larger commutant means more of E_8 left unbroken as gauge symmetry.

11.4.5 Why F-theory sees the exceptional groups

Now the question from the top of the section answers itself. The commutant chain (11.47) runs the other way too. Shrink the structure group of the bundle towards nothing and the commutant grows through

$$\mathrm{SU}(5)_{\mathrm{GUT}} \subset \mathrm{SO}(10) \subset \mathrm{E}_6 \subset \mathrm{E}_7 \subset \mathrm{E}_8, \tag{11.48}$$

so the exceptional groups are exactly what the heterotic string produces as it un-Higgses one of its E_8's. Through the duality each rung of (11.48) is a Kodaira fibre: I_5, I_1^*, IV^*, III^*, II^* for SU(5), SO(10), E_6, E_7, E_8, the dictionary of Section 11.2. The exceptional degenerations are not accidents of Weierstrass algebra. They are the geometric shadow of the heterotic E_8, which was there from the start as a level-one current algebra built on an even self-dual lattice (Appendix G), a structure the type II string does not have.

This is the precise sense in which F-theory reaches past the perturbative type II string. The perturbative ceiling, that a Chan-Paton index builds only the classical algebras, was established in Section 11.3; within F-theory an exceptional gauge symmetry instead requires mutually non-local (p, q) seven-branes to collide, a configuration with no weakly coupled description in any duality frame (Section 10.3.5). F-theory handles the collision because it never chooses a frame; it tracks the $\mathrm{SL}(2,\mathbb{Z})$ monodromy of τ directly, as the monodromy of an elliptic fibre. What the heterotic dual adds is a guarantee. It certifies that these exceptional points are honest string vacua and not merely singular geometries someone drew: whatever F-theory computes at an E_6 point must agree, in the stable degeneration limit, with a heterotic vacuum whose E_8 has been Higgsed to E_6, and heterotic vacua are ones we can construct and count independently. The exceptional groups are geometric on the F-theory side and current-algebraic on the heterotic side, and the duality says these are the same physics.

The concrete prize is the top-quark Yukawa $\mathbf{10}\cdot\mathbf{10}\cdot\mathbf{5}$. That it needs an E_6 point, that it is killed by the perturbative U(1) selection rule of a U(5) brane stack, and that it descends from the single E_6-invariant $\mathbf{27}\cdot\mathbf{27}\cdot\mathbf{27}$ was the argument of Section 11.3.4; Section 11.5 places it on a Standard-Model-like spectrum. What the duality adds is the reason the E_6 point is there to be used: it is the geometric image of a heterotic E_8 Higgsed down to E_6, and heterotic vacua with that unbroken group are ones we already know how to build.

Remark 11.17 (Beyond the lectures: the same K3, three readings). The elliptic K3 of (11.39) has now appeared three times, once for each string that can be compactified on a K3. IIA on K3 is dual to heterotic on T^4 and turns gauge symmetry into the ADE singularities of the surface (Section 7.3.5). IIB on K3 is a chiral six-dimensional theory whose enhanced points carry tensionless strings rather than gauge bosons (Theorem 8.14). F-theory reads the *same* K3 yet again, keeping only the complex structure of the elliptic fibre and promoting the IIB $\mathrm{SL}(2,\mathbb{Z})$ to fibre monodromy, and it is this third reading that is dual to heterotic on T^2 and that reaches the exceptional groups: one manifold read three ways, with the exceptional groups visible only in the F-theory reading.

Exercise 11.11 (Counting the moduli match in eight dimensions; Standard, ∼20 min)**.** Verify the moduli count behind (11.41).

1. A Weierstrass model (11.39) has f with 9 coefficients and g with 13. Subtract the redundancies: the PGL(2) acting on the base $\mathbb{CP}^1$ (3 parameters) and the $\mathbb{C}^*$ rescaling $(x, y, f, g) \to (\lambda^2 x, \lambda^3 y, \lambda^4 f, \lambda^6 g)$ (1 parameter). Show the number of complex-structure moduli is $9 + 13 - 3 - 1 = 18$, matching $\dim_{\mathbb{C}} \mathcal{M}_{2,18}$ and the coset count of Theorem 11.14.
2. On the heterotic side, list the 18: the complex structure τ and Kähler modulus ρ of T^2, and 16 Wilson lines. Which two of the 18 Weierstrass moduli are τ and ρ, in the stable degeneration picture of Section 11.4.3?

3. The heterotic coupling S of (11.38) is a nineteenth modulus not counted above. Explain why it is invisible in the F-theory complex structure and where it reappears.

Exercise 11.12 (Two halves make a K3; Standard, ∼25 min)**.** Using (11.44), carry the stable-degeneration lattice gluing through to $\Gamma_{3,19}$.

1. Show that gluing two rational elliptic surfaces along a common fibre produces a lattice containing $U^{\oplus 2} \oplus (-\mathrm{E}_8)^{\oplus 2}$, and check the signatures: $(1,9) + (1,9)$ from the two halves against $(3,19)$ for K3.
2. Argue that the class of the neck fibre together with a dual cycle supplies the missing third hyperbolic plane U, recovering $\Gamma_{3,19}$ of (11.45).
3. Locate the two heterotic E_8 gauge factors in the reassembled lattice, and show how setting one half to an II^* fibre recovers the full E_8 there while the other half stays generic.

11.5 A Standard-Model-like ledger

Everything in this chapter has been machinery. The elliptic fibration of Section 11.2 turned varying τ into geometry, the Kodaira dictionary attached a gauge algebra to each degeneration, Section 11.3 put an exceptional group at the singular point where the top Yukawa lives, and Section 11.4 explained through the heterotic dual why those exceptional structures are there at all. Now we spend it. The question is blunt: can this reach of geometry write down something that looks like the Standard Model, gauge group and three chiral families and a Higgs and the right Yukawa pattern, all read off a compactification rather than typed in by hand?

The honest answer is a qualified yes, and the qualifications are the point. What follows is one construction, an SU(5) grand-unified theory on a seven-brane, broken to the Standard Model by a flux rather than by a Higgs field. It reproduces the correct gauge group and chiral spectrum, it forbids the couplings a good grand-unified theory should forbid, and it makes the top quark heavy for a geometric reason. It also leans on assumptions that a complete vacuum would have to discharge, and those are laid out explicitly at the end. Treat this as a proof of principle: the ingredients exist and fit together, while a fully realistic and moduli-stabilised example remains a research problem. The references throughout are the founding F-theory model-building papers of Beasley, Heckman and Vafa and of Donagi and Wijnholt.

The big picture

Wrap a seven-brane on a contractible surface S (a del Pezzo) inside the base B_3 of the elliptic fourfold, with an I_5 (that is A_4) Kodaira degeneration along S so the gauge group is SU(5). Matter lives on curves in S where the singularity enhances: the **10** on an SO(10) curve, the $\overline{\mathbf{5}}$ on an SU(6) curve. Chirality comes from flux through those curves. Break SU(5) to $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y$ with a *hypercharge flux*: a line bundle along the hypercharge generator whose class is non-trivial on S but trivial in B_3, so the hypercharge photon stays massless. The same flux splits the Higgs doublet from its coloured partner, solving doublet-triplet splitting for free. The Yukawas sit at points where curves cross and the singularity jumps to E_6 or SO(12).

By the end of this section you should be able to:

- lay out the local SU(5) seven-brane on a del Pezzo surface and place its matter and Higgs curves (Section 11.5.1);
- count chiral families as flux degrees on the matter curves (Section 11.5.2);

- break SU(5) to the Standard Model with hypercharge flux, keep the hypercharge photon massless, and split the Higgs doublet from its colour triplet (Section 11.5.3);
- assemble the Standard-Model ledger and locate its two Yukawa couplings (Section 11.5.4);
- state the global assumptions a complete vacuum must still discharge (Section 11.5.5).

11.5.1 The gauge brane and its matter

Put F-theory on an elliptically fibred Calabi-Yau fourfold $Y \to B_3$ over a complex three-dimensional base. The local SU(5) model is already in hand from Section 11.3.3: a seven-brane fills the four non-compact directions and wraps a complex surface $S \subset B_3$, the fibre degenerates to an I_5 (A_4) Kodaira type all along S so the gauge algebra is $\mathfrak{su}(5)$, and matter sits on the curves where the singularity enhances, the $\mathbf{10}$ on the SO(10) curve $\Sigma_{\mathbf{10}}$ and the $\overline{\mathbf{5}}$ on the SU(6) curve $\Sigma_{\overline{\mathbf{5}}}$ of (11.31). Two things quoted there we now need in detail: the constraint that fixes S, and the flux that has been missing from the story so far.

The choice of S is not free. For the gauge dynamics to decouple from four-dimensional gravity, S must shrink to zero size at finite distance in moduli space, which forces $-K_S$ ample, exactly the contractibility criterion (10.38) of the previous chapter. In plain terms, $-K_S$ ample means the surface curves positively on average and can be collapsed to a point; the surfaces that qualify are the del Pezzos dP_n, the projective plane $\mathbb{CP}^2$ blown up at $n \le 8$ generic points, together with $\mathbb{CP}^1 \times \mathbb{CP}^1$. That single requirement pays two dividends we use at once. A del Pezzo has $h^{2,0}(S) = 0$ (no holomorphic two-forms) and $\pi_1(S) = 0$ (simply connected). The first kills adjoint matter: the number of SU(5) adjoints is $h^{2,0}(S)$, the deformations of the brane inside the fourfold, and it vanishes. The second kills Wilson lines. So the two textbook ways to break a grand-unified group, an adjoint Higgs vacuum expectation value and a discrete Wilson line, are both unavailable here by construction. Whatever breaks SU(5) has to be the flux, and we come to it in Section 11.5.3.

One distinction the local model of Section 11.3.3 did not yet need. The Higgs fields are also $\overline{\mathbf{5}}$ and $\mathbf{5}$ multiplets, but they live on their own curves rather than the matter curves: $\mathbf{5}_H$ on a curve $\Sigma_{\mathbf{5}_H}$ and $\overline{\mathbf{5}}_H$ on $\Sigma_{\overline{\mathbf{5}}_H}$. A $\overline{\mathbf{5}}$ curve carrying matter and one carrying a Higgs are the same type of geometry; what distinguishes them is the flux they see, which we now turn on.

11.5.2 Chirality from flux

An unbroken I_5 singularity along S with no flux gives a non-chiral spectrum: each matter curve carries a hypermultiplet, a vector-like pair. The Standard Model is chiral, so something must lift half of each pair. In the intersecting-brane picture of Section 10.3.4 chirality was the signed intersection number I_{ab} of (10.39). Here it is the same idea in cohomological dress: a background gauge flux on the seven-brane, valued in the Cartan directions of SU(5) and of the U(1)s that sit inside the ambient exceptional structure, restricts to a line bundle L_R on each matter curve, and the net chirality is its degree,

$$n_R - n_{\bar{R}} \;=\; \int_{\Sigma_R} c_1(L_R) \;=\; \deg\!\left(L_R|_{\Sigma_R}\right). \tag{11.49}$$

This counts holomorphic zero modes minus antiholomorphic ones on the curve. The precise statement involves the Freed-Witten twist by $K_\Sigma^{1/2}$, so the zero modes live in $H^0(\Sigma_R, L_R \otimes K_\Sigma^{1/2})$ and their conjugates in H^1; the index that survives Riemann-Roch is the degree in (11.49). We quote this; the derivation is the Dirac-operator index on the matter curve and is standard.

To land three generations, choose the fluxes so that

$$\deg\!\left(L_{\mathbf{10}}|_{\Sigma_{\mathbf{10}}}\right) \;=\; 3, \qquad \deg\!\left(L_{\overline{\mathbf{5}}}|_{\Sigma_{\overline{\mathbf{5}}}}\right) \;=\; 3. \tag{11.50}$$

Three copies of the **10** and three of the $\overline{\mathbf{5}}$: the chiral content of one grand-unified generation, repeated three times. The generation number is a topological integer, not a tuned parameter, which is the whole appeal of getting chirality from geometry. What the local model cannot do is guarantee that a global fourfold exists with matter curves of the right genus and homology class to support these bundles; that is the first entry on the assumptions list.

11.5.3 Breaking to the Standard Model with hypercharge flux

We have an SU(5) theory with three chiral families and a Higgs sector, and no adjoint and no Wilson line to break it. The remaining tool is a second kind of flux, switched on along the hypercharge generator inside SU(5),

$$Y \;=\; \tfrac{1}{\sqrt{60}}\,\mathrm{diag}(-2,-2,-2,+3,+3), \tag{11.51}$$

acting on the fundamental **5**. Turning on a background field strength $\langle F_Y\rangle$ proportional to Y, threading a two-cycle of S, breaks the gauge group to the commutant of Y,

$$\mathrm{SU}(5) \;\xrightarrow{\langle F_Y\rangle}\; \mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)_Y. \tag{11.52}$$

The breaking is done by a flux in the internal space rather than by a four-dimensional Higgs field: the gauge symmetry is broken because the internal wavefunctions of the would-be off-diagonal gauge bosons, the $\mathrm{SU}(5)/(\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1))$ generators, are lifted by the flux and have no massless zero mode. It is the higher-dimensional analogue of a monopole background breaking a gauge symmetry by the Hosotani mechanism, encoded here in the topology of a line bundle L_Y with $c_1(L_Y) = [\langle F_Y\rangle]/2\pi$.

The mechanism has a catch that is also its cleverest feature. A generic internal flux gives the corresponding four-dimensional gauge boson a mass. Reducing the Ramond-Ramond four-form C_4 on the flux produces a Stückelberg coupling of the schematic form $\int C_4\wedge F_Y \supset (\text{number})\, B_2\wedge \mathrm{d}A_Y$, where B_2 is a two-form descending from C_4 and the number is the pairing of $c_1(L_Y)$ with a transverse cycle. If that number is non-zero, A_Y eats B_2 and the hypercharge photon is massive. That is fatal: we want $\mathrm{U}(1)_Y$ in the massless spectrum. The photon stays massless precisely when the hypercharge flux is homologically non-trivial inside S but trivial when pushed into the base,

$$c_1(L_Y)\neq 0 \text{ in } H^2(S,\mathbb{Z}), \qquad \iota_*\, c_1(L_Y) = 0 \text{ in } H_2(B_3,\mathbb{Q}), \tag{11.53}$$

where $\iota : S \hookrightarrow B_3$ is the inclusion. The class must live in the kernel of the pushforward: a two-cycle that is real inside the brane but bounds in the ambient base. A del Pezzo surface has $b_2(\mathrm{dP}_n) = n+1$ independent two-cycles to play with, and for a suitable embedding some of them are trivial in B_3. This is a genuinely global statement: whether such a divisor sits inside an honest compact base is invisible to the local model and is the second entry on the assumptions list. When it holds, $\mathrm{U}(1)_Y$ survives and the flux (11.52) breaks SU(5) without a Green-Schwarz mass for hypercharge. F-theory can do this because the relevant cohomology is available; a purely four-dimensional grand-unified theory has no analogue.

Doublet-triplet splitting. The same flux does a second job. A grand-unified Higgs sits in a **5**, which contains the weak doublet $H_u = (\mathbf{1},\mathbf{2})_{1/2}$ we want and a colour triplet $T = (\mathbf{3},\mathbf{1})_{-1/3}$ we do not. The triplet mediates dimension-six and, paired up, dimension-five proton decay, and it spoils gauge coupling unification if light. Splitting the light doublet from the heavy triplet is the doublet-triplet problem, and four-dimensional grand-unified theories solve it only by fine-tuning or elaborate model-building. Here it is automatic. The hypercharge flux restricts to the Higgs curve and, because the doublet and the triplet carry different hypercharge, it shifts their zero-mode counts differently. Within a multiplet R, a component ρ picks up a multiplicity

that departs from the bulk flux $M_R \equiv \deg(L_R|_{\Sigma_R})$ by an amount set by the hypercharge-flux degree $N_R \equiv \deg(L_Y|_{\Sigma_R}) = c_1(L_Y)\cdot\Sigma_R$. For the $\mathbf{5}_H$ curve the two components split as

$$n_T = M_{\mathbf{5}_H}, \qquad n_{H_u} = M_{\mathbf{5}_H} + N_{\mathbf{5}_H}. \tag{11.54}$$

Choose $M_{\mathbf{5}_H} = 0$ and $N_{\mathbf{5}_H} = 1$: no triplet, one doublet. Topology removes the coloured Higgs from the massless spectrum outright, without giving it a large mass. The down-type Higgs $\overline{\mathbf{5}}_H$ works the same way with the opposite sign, giving one H_d and no anti-triplet.

Keeping the families complete. The flux that helps the Higgs would hurt the matter. If the hypercharge flux restricted non-trivially to the $\mathbf{10}$ and $\overline{\mathbf{5}}$ matter curves, it would split each family into pieces with unequal multiplicities and, worse, would ruin the automatic anomaly cancellation that comes from complete SU(5) multiplets. The fix is a condition: demand that the hypercharge flux be trivial on the matter curves,

$$N_{\mathbf{10}} \equiv c_1(L_Y)\cdot\Sigma_{\mathbf{10}} = 0, \qquad N_{\overline{\mathbf{5}}} \equiv c_1(L_Y)\cdot\Sigma_{\overline{\mathbf{5}}} = 0. \tag{11.55}$$

Then each $\mathbf{10}$ splits into (Q, u^c, e^c) with a common multiplicity $M_{\mathbf{10}} = 3$, and each $\overline{\mathbf{5}}$ into (d^c, L) with $M_{\overline{\mathbf{5}}} = 3$: three complete families. Because they are complete SU(5) representations, hypercharge is automatically non-anomalous, and the notorious $\mathrm{U}(1)_Y$-gravitational and cubic anomalies cancel with no extra work. The hypercharge flux is thus asked to vanish on some curves and not on others, and (11.53), (11.54) and (11.55) together are a system of topological constraints on one line bundle L_Y. That such a bundle exists on a globally consistent fourfold is a strong requirement. It can be met in examples; whether it can be met alongside everything else is what the assumptions list below accounts for.

Example 11.18 (A hypercharge flux on dP_n)**.** Take $S = \mathrm{dP}_n$ with $n \geq 2$. Its second cohomology is spanned by the hyperplane class H and the exceptional classes $E_1, \ldots, E_n$ of the n blown-up points, with intersection form $H^2 = 1$, $E_i \cdot E_j = -\delta_{ij}$, $H \cdot E_i = 0$, and canonical class $K_S = -3H + \sum_i E_i$. Put the hypercharge flux in the class

$$c_1(L_Y) \;=\; E_1 - E_2.$$

It has self-intersection $(E_1 - E_2)^2 = -2 \neq 0$, so it is non-trivial in $H^2(S, \mathbb{Z})$, the first half of (11.53). It is also orthogonal to the canonical class, $(E_1 - E_2)\cdot K_S = 0$, so it is a root of the E_n lattice sitting inside S; for a suitable embedding $\iota : S \hookrightarrow B_3$ a class of this kind bounds in the base, giving $\iota_* c_1(L_Y) = 0$ and a massless hypercharge photon.

Now place the curves against this flux. Since $(E_1 - E_2)\cdot(aH - \sum_i b_i E_i) = b_1 - b_2$, any curve whose class is symmetric in E_1 and E_2 pairs to zero with the flux. Put the matter curves $\Sigma_{\mathbf{10}}$ and $\Sigma_{\overline{\mathbf{5}}}$ in such symmetric classes; then $N_{\mathbf{10}} = N_{\overline{\mathbf{5}}} = 0$ and (11.55) holds, so the families stay complete. Put the up-Higgs curve in a class with $b_1 - b_2 = +1$, so $N_{\mathbf{5}_H} = +1$, and take $M_{\mathbf{5}_H} = \deg(L_{\mathbf{5}_H}|_{\Sigma_{\mathbf{5}_H}}) = 0$; then (11.54) gives $n_T = 0$ and $n_{H_u} = 1$, one doublet and no triplet. The down-Higgs curve with $b_1 - b_2 = -1$ and $M = 0$ gives one H_d and no anti-triplet. Finally switch on the chirality fluxes with $\deg(L_{\mathbf{10}}|_{\Sigma_{\mathbf{10}}}) = \deg(L_{\overline{\mathbf{5}}}|_{\Sigma_{\overline{\mathbf{5}}}}) = 3$. The output is exactly the ledger of Table 11.3: three complete families, one Higgs doublet of each type, no coloured triplet, all from one del Pezzo, one line bundle L_Y in the class $E_1 - E_2$, and two chirality fluxes. What the example does not settle is whether $E_1 - E_2$ really pushes to zero in a compact B_3; that is the global input the local data cannot supply.

11.5.4 The ledger

Collecting the output of (11.50), (11.54) and (11.55) gives the massless charged spectrum in Table 11.3. It is the Standard Model: three families of quarks and leptons in the correct

Field	SU(5) origin	$(\mathrm{SU}(3),\mathrm{SU}(2))_Y$	Multiplicity	Curve
Q	$\mathbf{10}$	$(\mathbf{3},\mathbf{2})_{+1/6}$	3	$\Sigma_{\mathbf{10}}$
u^c	$\mathbf{10}$	$(\overline{\mathbf{3}},\mathbf{1})_{-2/3}$	3	$\Sigma_{\mathbf{10}}$
e^c	$\mathbf{10}$	$(\mathbf{1},\mathbf{1})_{+1}$	3	$\Sigma_{\mathbf{10}}$
d^c	$\overline{\mathbf{5}}$	$(\overline{\mathbf{3}},\mathbf{1})_{+1/3}$	3	$\Sigma_{\overline{\mathbf{5}}}$
L	$\overline{\mathbf{5}}$	$(\mathbf{1},\mathbf{2})_{-1/2}$	3	$\Sigma_{\overline{\mathbf{5}}}$
H_u	$\mathbf{5}_H$	$(\mathbf{1},\mathbf{2})_{+1/2}$	1	$\Sigma_{\mathbf{5}_H}$
H_d	$\overline{\mathbf{5}}_H$	$(\mathbf{1},\mathbf{2})_{-1/2}$	1	$\Sigma_{\overline{\mathbf{5}}_H}$
$T,\bar{T}$	$\mathbf{5}_H,\overline{\mathbf{5}}_H$	$(\mathbf{3},\mathbf{1})_{\mp 1/3}$	0	removed by L_Y

Table 11.3: The Standard-Model-like ledger from an SU(5) seven-brane broken by hypercharge flux. Matter multiplicities are the flux degrees (11.50) with the completeness condition (11.55); the Higgs multiplicities are the split (11.54). The coloured Higgs triplets are projected out.

representations with the correct hypercharges, one Higgs doublet of each type, and no coloured Higgs triplet. The gauge group is $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)_Y$, engineered from an I_5 fibre and a single flux.

The interactions are set by where the curves meet, by the same residue overlap (10.33) used in Section 11.3.3. The two couplings the Standard Model needs sit at the two enhancement points already identified there, written here in Standard-Model components:

$$\text{E}_6\text{ point}: \quad \mathbf{10}\cdot\mathbf{10}\cdot\mathbf{5}_H \;\supset\; Q\,u^c H_u \qquad \text{(up-type / top Yukawa)}, \tag{11.56}$$

$$\text{SO(12) point}: \quad \mathbf{10}\cdot\overline{\mathbf{5}}\cdot\overline{\mathbf{5}}_H \;\supset\; Q\,d^c H_d + L\,e^c H_d \qquad \text{(down / charged-lepton Yukawa)}. \tag{11.57}$$

The up-type coupling (11.56) is the perturbatively forbidden one: it requires the fibre to degenerate to E_6 where the two $\mathbf{10}$ curves and the up-Higgs curve meet, ruled out in perturbative type IIB by the selection rule established in Section 11.3.4. At the E_6 point it is present at order one, which is why the top quark is heavy. The down and charged-lepton couplings come from the milder SO(12) point that perturbative constructions can also build. What the ledger adds is placement: it fixes which curves cross where, so the up coupling lands on the two $\mathbf{10}$ curves and the up-Higgs curve, and the down coupling on the $\mathbf{10}$, $\overline{\mathbf{5}}$ and down-Higgs curves, and it records the Standard-Model fields each cubic invariant becomes.

There is a texture built into (10.33) that we met in Exercise 10.8: a single enhancement point gives a rank-one Yukawa matrix. Only one linear combination of the three generations couples, so at leading order one family is massive and two are massless. Read in the grand-unified language this is a virtue: the top quark is heavy, the up and charm are light, and the observed hierarchy is the statement that the leading geometry is rank one and the lighter families are lifted by subleading effects, non-perturbative corrections and the finite spread of the wavefunctions away from the idealised point. Getting the full mass matrix and the mixing angles right is a quantitative programme, not a one-line consequence, and it is an active research direction rather than a solved problem.

11.5.5 What we are assuming

The ledger is real, and so are its debts. The construction above is local: it lives on the surface S and its curves, in the limit where gravity has decoupled. A complete four-dimensional vacuum has to embed this local data in a compact fourfold and pay several bills the local model deferred.

The assumptions, on the table

1. **A global completion exists.** The matter curves, their genera, and the line bundles L_R and L_Y satisfying (11.53)–(11.55) must be realised on an actual compact elliptic fourfold $Y \to B_3$ with S a contractible divisor. Local consistency does not imply a global model exists, and constructing explicit bases that carry a hypercharge flux trivial in B_3 is non-trivial.
2. **The D3-brane and flux tadpole balances.** Tadpole cancellation is Gauss's law on a compact space: charge sourced by branes and flux has nowhere to escape to, so it must sum to zero. Gauge flux carries D3-brane charge, and on a compact fourfold the total must cancel,
$$N_{\mathrm{D3}} + \tfrac{1}{2}\int_Y G \wedge G \;=\; \frac{\chi(Y)}{24}, \tag{11.58}$$
with $N_{\mathrm{D3}} \geq 0$ mobile D3-branes and G the four-form flux (Chapter 9). The fluxes generating three families and breaking the gauge group contribute to $\int G \wedge G$ and must fit inside the geometric budget $\chi(Y)/24$.
3. **Moduli are stabilised.** The construction treats the volume of S, the axio-dilaton profile and the shape of B_3 as fixed. Stabilising them, with the flux and non-perturbative superpotentials of Chapter 9, is a separate problem, and a wrong answer there can undo everything, for instance by driving S to infinite volume where the gauge coupling runs to zero.
4. **Unification survives the flux.** Hypercharge flux threads the whole surface, so it induces different threshold corrections for the three Standard Model couplings and generically splits their unification at the high scale. Whether precise gauge coupling unification survives is a quantitative constraint on the flux, not a guarantee.
5. **Proton decay and the μ-term are controlled.** Removing the coloured triplet by flux kills the worst dimension-five operators, but dimension-four baryon- and lepton-number violation and the size of the Higgs μ-term depend on the extra U(1) selection rules from the fluxes and on the global geometry. These are model-dependent and among the sharpest phenomenological tests.
6. **No exotics and the right hierarchy.** Extra vector-like states from other curves must be absent or heavy, and the rank-one Yukawa texture must be corrected into three realistic generations with the observed masses and mixings. Both are worked on; neither is automatic.

None of these is a fatal objection, and each has been met in examples in the literature. What no single construction has done, as of writing, is discharge all of them at once in a fully explicit, moduli-stabilised, globally consistent model with exactly the Standard Model and nothing else. That is the research frontier, and an honest account of F-theory model-building names it as such.

Remark 11.19 (Beyond the lectures: spectral covers and the U(1)s). The fluxes L_R that give chirality are not arbitrary. In the semi-local description the seven-brane data near the SU(5) locus is packaged by a *spectral cover*, a five-sheeted branched cover of S that records how the SU(5) sits inside an ambient E_8, together with a line bundle on the cover. The matter curves are where sheets collide, the fluxes are the line bundle restricted to those curves, and the extra U(1) symmetries that police proton decay and the μ-term are the traceless combinations of sheet exchanges. This Higgs-bundle picture, due to Donagi-Wijnholt and to Hayashi, Tatar, Toda, Watari and Yamazaki, is where the residue Yukawa (10.33) and the constraints (11.53)–(11.55) are actually computed, and it is the natural next stop for anyone who wants to build these models rather than admire them.

Where this is going

That closes the model-building arc. Geometry has taken us from a Kaluza-Klein circle to Calabi-Yau compactification to branes at singularities to, here, a seven-brane wrapping a del Pezzo with a flux that writes down the Standard Model gauge group and its three families. The lever throughout was a duality or a piece of geometry that reorganised strongly coupled physics into something classical: T-duality, the ADE singularity, the elliptic fibre and its exceptional degenerations.

The remaining chapters turn the same lever on gravity itself. The BPS branes we have counted all along carry mass and charge, and a bound state of them is a black hole. Chapter 12 counts the microstates of the D1-D5-P system and reproduces the Bekenstein-Hawking entropy $S = A/4G$ from a brane partition function, the Strominger-Vafa result. Chapter 13 takes the near-horizon limit of the same branes and finds a spacetime emerging holographically from a gauge theory, with the genus expansion of Chapter 4 becoming the $1/N$ expansion. The geometry that engineered gauge theories in this chapter becomes, in those, a gauge theory that engineers geometry.

Exercise 11.13 (Doublet-triplet splitting from flux; Warm-up, ∼15 min)**.** Use the split (11.54) and its $\overline{\mathbf{5}}_H$ counterpart. (a) Show that the choice $M_{\mathbf{5}_H} = 0$, $N_{\mathbf{5}_H} = +1$ gives exactly one weak doublet H_u and no colour triplet, and find the analogous choice on the $\overline{\mathbf{5}}_H$ curve giving one H_d and no anti-triplet. (b) Explain why the same flux is required to satisfy (11.55) on the matter curves, and what would go wrong with the chiral spectrum if $N_{\mathbf{10}} \neq 0$: split the **10** into (Q, u^c, e^c) multiplicities $(M, M-N, M+N)$ and say which Standard Model anomaly is threatened.

Exercise 11.14 (Why hypercharge stays massless; Standard, ∼20 min)**.** The hypercharge photon A_Y gets a Stückelberg mass from the reduction of C_4 on the internal flux. (a) Argue on dimensional grounds that the coupling $\int_{\mathbb{R}^{1,3}\times S} C_4 \wedge F_Y$ reduces to a four-dimensional term $\propto (\iota_* c_1(L_Y))\, B_2 \wedge \mathrm{d}A_Y$, and hence that A_Y is massless iff (11.53) holds. (b) Contrast the two unavailable alternatives. Explain why $h^{2,0}(\mathrm{dP}_n) = 0$ forbids an adjoint Higgs and $\pi_1(\mathrm{dP}_n) = 0$ forbids a Wilson line, so that on a del Pezzo GUT brane flux breaking is the only option. (c) Explain in one sentence why a compact four-dimensional grand-unified theory has no analogue of (11.53): what geometric ingredient is it missing?

Exercise 11.15 (Complete multiplets and anomalies; Warm-up, ∼15 min)**.** (a) Verify that one **10** plus one $\overline{\mathbf{5}}$ of SU(5) contains exactly the Standard Model fermions of one generation, and check the hypercharges in Table 11.3 against Y in (11.51). (b) Show that the cubic $\mathrm{U}(1)_Y^3$ and mixed $\mathrm{U}(1)_Y$-gravitational anomalies cancel within a single complete generation, and explain why this makes the completeness condition (11.55) the natural one to impose: hypercharge flux that respects it can never generate a gauge anomaly.

Exercise 11.16 (The tadpole budget; Standard, ∼20 min)**.** Take the D3 tadpole (11.58) with supersymmetric self-dual flux, so $\frac{1}{2}\int_Y G \wedge G \geq 0$ and $N_{\mathrm{D3}} \geq 0$. (a) Deduce the bound $\frac{1}{2}\int_Y G \wedge G \leq \chi(Y)/24$ and explain why a large Euler characteristic is helpful for accommodating flux. (b) Take, illustratively, a fourfold with $\chi(Y) = 288$, so the geometric budget is $\chi(Y)/24 = 12$. If the chirality-generating flux and the hypercharge flux together contribute $\frac{1}{2}\int_Y G \wedge G = 5$, how many mobile D3-branes N_{D3} must sit in the vacuum? Show that a fourfold with $\chi(Y) = 96$ (budget 4) has no room for the same flux with $N_{\mathrm{D3}} \geq 0$. (c) The chirality-generating flux and the hypercharge flux both contribute to $\int G \wedge G$. Argue qualitatively that demanding three generations sets a lower bound on the flux, and hence on $\chi(Y)$, so that not every fourfold with an SU(5) brane has room for the model; relate this to why $\chi(Y)/24$ being an integer is itself a non-trivial geometric fact.

Chapter 12

Black-Hole Microstates and BPS Entropy

12.1 BPS black holes in supergravity

The big picture

A black hole carries an entropy $S = A/4G$, and for anything of astrophysical size that number is colossal: a solar-mass hole has $S \sim 10^{77}$. Thermodynamics reads an entropy that large as a count of microstates, $S = \log \Omega$, so the smooth horizon of general relativity ought to be the coarse-grained face of the $e^{A/4G}$ underlying microstates. Relativity gives no hint what they are. This chapter identifies them for one family, and the present section builds the target. We take the most tractable black holes, the extremal supersymmetric ones, write them as supergravity solutions governed by harmonic functions, and read off the horizon area. The number to be reproduced is

$$S_{\mathrm{BH}} \;=\; \frac{A}{4G}.$$

Two properties make these holes tractable. They saturate the BPS bound $M = |Z|$, so their quantum numbers are protected as the coupling is dialled from weak to strong. And their horizon area depends only on the quantised charges, not on any modulus, by the attractor mechanism. The first fact lets us count the states in a weakly coupled brane description; the second guarantees that the count and the area are functions of the very same integers. The count itself, and the exact match, are the business of Sections 12.2 to 12.4.

12.1.1 Black holes are thermodynamic systems

The starting point is older than string theory. A stationary black hole obeys laws that look exactly like the laws of thermodynamics, with the horizon area playing the role of entropy and the surface gravity the role of temperature. Bekenstein (1972–1973) argued on information-theoretic grounds that the entropy must be proportional to the area, and Hawking (1975) fixed the coefficient by showing that a black hole radiates as a body at temperature

$$T_{\mathrm{H}} \;=\; \frac{\kappa}{2\pi}, \tag{12.1}$$

with κ the surface gravity of the horizon. Feeding Eq. (12.1) into the first law $\mathrm{d}M = T_{\mathrm{H}}\,\mathrm{d}S + \Phi\,\mathrm{d}Q + \Omega\,\mathrm{d}J$ fixes the entropy to be one quarter of the horizon area in Planck units,

$$S_{\mathrm{BH}} \;=\; \frac{A}{4G} \;=\; \frac{A\,c^3}{4G\hbar}\Big|_{\hbar=c=1}, \tag{12.2}$$

where A is the area of the spatial cross-section of the horizon and G is Newton's constant in whatever spacetime dimension we work in. The formula is universal: it holds for a Schwarzschild hole, a charged hole, a rotating hole, and for black branes, with the same numerical $\frac{1}{4}$.

Taken as a statement about a quantum theory of gravity, Eq. (12.2) is a demand. If S_{BH} is a genuine entropy, there must be a Hilbert space of $e^{S_{\mathrm{BH}}}$ states carrying the same mass and charges as the black hole, indistinguishable to an outside observer once the horizon has formed. Reproducing that number from an independent, microscopic count is the goal of the chapter. Everything hinges on finding a black hole for which the counting can actually be done, and that is what supersymmetry provides.

The clue is the temperature. A generic black hole is hot, and a hot system is hard to enumerate state by state: one is counting a thermal ensemble at finite energy above the ground state. Extremal black holes are different. They sit at $T_{\mathrm{H}} = 0$ yet keep a finite, macroscopic area, hence a finite entropy Eq. (12.2). A zero-temperature system with a large entropy is a system with an enormous ground-state degeneracy. That degeneracy is a pure number, a count of quantum ground states at fixed charge, and it is exactly the kind of object a protected, supersymmetric argument can compute. The rest of the section makes "extremal" precise and connects it to "BPS".

12.1.2 Extremal Reissner-Nordström and the throat

The cleanest example lives in four-dimensional Einstein-Maxwell theory, the bosonic sector of $\mathcal{N} = 2$ supergravity. A static, spherically symmetric, electrically charged solution is the Reissner-Nordström (RN) black hole. In units $G = 1$, and with the charge normalised so that the gauge potential is $A_t = Q/r$,

$$\mathrm{d}s^2 = -f(r)\,\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{f(r)} + r^2\,\mathrm{d}\Omega_2^2, \qquad f(r) = 1 - \frac{2M}{r} + \frac{Q^2}{r^2}. \tag{12.3}$$

There is no dilaton in this theory, so the string and Einstein frames coincide and no frame label is needed here. The blackening factor factorises, $f = (r - r_+)(r - r_-)/r^2$, with two horizons at

$$r_\pm = M \pm \sqrt{M^2 - Q^2}. \tag{12.4}$$

A regular horizon needs $M \geq |Q|$. The surface gravity is $\kappa = f'(r_+)/2 = (r_+ - r_-)/(2r_+^2)$, so the Hawking temperature is

$$T_{\mathrm{H}} = \frac{r_+ - r_-}{4\pi r_+^2}. \tag{12.5}$$

As $|Q|$ climbs towards M the two horizons approach one another and the temperature drops. At

$$M = |Q| \qquad \text{(extremality)} \tag{12.6}$$

they coincide, $r_+ = r_- = M$, the temperature reaches zero, and yet the horizon has not shrunk: its area is

$$A = 4\pi r_+^2 = 4\pi M^2 = 4\pi Q^2, \qquad S_{\mathrm{BH}} = \frac{A}{4} = \pi Q^2. \tag{12.7}$$

Restoring G this is $S = A/4G = \pi Q^2/G$, and Q is the geometrised charge with units of length. Here is the promised object: a zero-temperature black hole with a macroscopic entropy, a ground-state degeneracy of order $e^{\pi Q^2/G}$ to be accounted for.

The geometry near an extremal horizon has a characteristic shape that will recur throughout the chapter. Shift to an isotropic radial coordinate $\rho = r - M$, so the horizon sits at $\rho = 0$. Extremality collapses Eq. (12.3) into a form built from a single function,

$$\mathrm{d}s^2 = -H(\rho)^{-2}\,\mathrm{d}t^2 + H(\rho)^2(\mathrm{d}\rho^2 + \rho^2\,\mathrm{d}\Omega_2^2), \qquad H(\rho) = 1 + \frac{Q}{\rho}, \tag{12.8}$$

with the gauge field $A_t = 1 - H^{-1}$ read off the same H. The function H is harmonic on the flat transverse $\mathbb{R}^3$, $\nabla^2 H = 0$ away from the source at $\rho = 0$, which is the whole content of the field equations for this solution. This harmonic structure, taken up in Section 12.1.3, underlies every BPS black hole and black brane.

Zoom in on the horizon, $\rho \to 0$, where $H \simeq Q/\rho$. The metric Eq. (12.8) becomes

$$\mathrm{d}s^2 \;\longrightarrow\; -\frac{\rho^2}{Q^2}\,\mathrm{d}t^2 + \frac{Q^2}{\rho^2}\,\mathrm{d}\rho^2 + Q^2\,\mathrm{d}\Omega_2^2, \tag{12.9}$$

the product $\mathrm{AdS}_2 \times S^2$ with both factors of radius Q, the Bertotti-Robinson geometry (1959). The S^2 keeps a fixed radius Q all the way down: the horizon does not pinch off, and its area Eq. (12.7) is finite and smooth. The proper distance to the horizon, $\int H\,\mathrm{d}\rho \sim \int Q\,\mathrm{d}\rho/\rho$, diverges logarithmically as $\rho \to 0$, so the throat is infinitely long. An infalling observer travels down an AdS_2 tube of fixed transverse size, and the near-horizon region decouples from the asymptotically flat exterior. This decoupling makes the throat a self-contained holographic system, foreshadowing the AdS near-horizon geometries of Chapter 13.

12.1.3 The harmonic-function ansatz

The rewriting Eq. (12.8) is not a trick special to four dimensions. Every extremal BPS solution in supergravity is assembled from harmonic functions on the flat space transverse to the source, and this is why such solutions are so easy to write down and to superpose.

The mechanism is the no-force property of BPS objects (Section 5.6). A pair of identical extremal RN holes feels no net force: the gravitational and electric potentials cancel exactly, term by term, because the extremality condition Eq. (12.6) ties the mass to the charge. Static equilibrium at arbitrary separation is therefore possible, and the corresponding solutions are the Majumdar-Papapetrou geometries. They take the identical form Eq. (12.8), now with a multi-centre harmonic function

$$H(\vec{x}) = 1 + \sum_a \frac{Q_a}{|\vec{x} - \vec{x}_a|}, \qquad \nabla^2 H = 0 \;\text{ away from the centres}, \tag{12.10}$$

one pole per black hole. The centres $\vec{x}_a$ are free: nothing holds the holes apart and nothing pulls them together. Linearity of Laplace's equation, and the fact that the sources can be added freely, is the field-theory face of the algebraic no-force statement that one set of supercharges annihilates any number of like-charged BPS objects (Section 5.6).

The same ansatz builds the extremal black branes of ten-dimensional supergravity, the objects that will carry the charges of the microscopic system. A single extremal Dp-brane is sourced by one harmonic function H_p on its $(9-p)$-dimensional transverse space. These brane metrics are written in string frame, the metric the fundamental string actually sees; it is related to the Einstein frame, in which the gravitational action is canonically normalised, by a dilaton-dependent Weyl rescaling, $g^{\text{string}}_{MN} = e^{\phi/2}\, g^{\text{Einstein}}_{MN}$ in ten dimensions. That change of frame is why the dilaton powers in Eq. (12.11) differ from those in the Einstein-frame metric Eq. (12.16) below. In string frame,

$$\mathrm{d}s^2 = H_p^{-1/2}\,\eta_{ab}\,\mathrm{d}x^a\mathrm{d}x^b + H_p^{1/2}\,\delta_{ij}\,\mathrm{d}x^i\mathrm{d}x^j, \tag{12.11}$$

$$e^{\phi} = H_p^{(3-p)/4}, \qquad C_{01\cdots p} = H_p^{-1} - 1, \qquad H_p(r) = 1 + \frac{d_p\, g_s N\, \ell_s^{7-p}}{r^{\,7-p}}, \tag{12.12}$$

where x^a ($a = 0, \ldots, p$) run along the worldvolume, x^i are the $9-p$ transverse coordinates with $r^2 = \delta_{ij} x^i x^j$, and $\nabla_\perp^2 H_p = 0$ makes H_p harmonic on $\mathbb{R}^{9-p}$; the constant d_p and the charge N (the number of branes) are fixed by Dirac quantisation and collected in Appendix I. The solution interpolates between flat space at large r, where $H_p \to 1$, and a near-horizon throat at small r, where $H_p \simeq (\text{const})/r^{7-p}$. These are the BPS branes of Section 5.6.3 and Chapter 6, now seen as gravitational solutions rather than as loci where open strings end. The three-brane is the special case: $p = 3$ makes e^{ϕ} constant and the throat is the regular $\mathrm{AdS}_5 \times S^5$ that anchors Chapter 13.

The point that carries into the black-hole construction is superposition. When several mutually BPS branes are placed on top of one another, each still no-forces against the others, and the supergravity solution is obtained by giving each brane its own harmonic function and combining them by simple rules: metric factors multiply, dilaton contributions add. A configuration of k intersecting branes is governed by k harmonic functions $H_1, \ldots, H_k$. The horizon area of the resulting black object, and therefore its entropy, is a product of the charges through those functions. Getting a *finite* area out of this product turns out to require a specific number of charges, which we come to in Section 12.1.5.

12.1.4 The BPS bound saturated

Extremality Eq. (12.6) is a statement about a metric. Its supersymmetric meaning is that the extremal RN black hole saturates the BPS bound and preserves part of the supersymmetry. Embed Einstein-Maxwell in $\mathcal{N} = 2$ supergravity, where the Maxwell field is the graviphoton sitting in the same multiplet as the metric. The central charge of the $\mathcal{N} = 2$ algebra is then the graviphoton charge of the state, and for a purely electric configuration $|Z| = |Q|$ in the normalisation of Eq. (12.3). The general bound of Section 5.6 and Appendix C,

$$M \;\geq\; |Z|, \tag{12.13}$$

reads $M \geq |Q|$: the same inequality as the regularity condition $M \geq |Q|$ below Eq. (12.4). For these black holes the extremal limit and the BPS limit are the same limit,

$$\text{extremal } (M = |Q|) \iff \text{BPS } (M = |Z|), \tag{12.14}$$

and the extremal RN hole is a genuine BPS state of the supergravity theory.

Concretely, "BPS" means the background admits Killing spinors. Just as the flat vacuum solves $\partial_M \epsilon = 0$ with 32 constant spinors (Section 5.3), the extremal RN geometry solves the gravitino Killing-spinor equation $\delta\psi_M = 0$ for a nonzero set of ϵ: the metric Eq. (12.8) and its gauge field preserve half of the $\mathcal{N} = 2$ supersymmetry. The unbroken supercharges are those obeying a projection $\Gamma\,\epsilon = \epsilon$ built from the direction of the charge, exactly the brane projectors Eq. (C.42) of Appendix C. The non-extremal hole $M > |Q|$ admits no Killing spinor and breaks all of it.

Careful

Extremal and BPS are logically distinct, and coincide here only because the central charge is the electric charge. Extremality is the statement $T_{\mathrm{H}} = 0$, a degenerate horizon. BPS is the statement $M = |Z|$, a preserved supersymmetry. In theories with less symmetry, or for black holes whose charges do not fill out the central charge, one can have extremal non-BPS solutions and (in unstable situations) BPS non-extremal ones. What we use in this chapter is the supersymmetric case, where the two conditions lock together and the machinery of BPS protection applies.

The payoff of saturation is protection, and this is the hinge of the whole argument. A BPS black hole sits in a short representation of the supersymmetry algebra. Short multiplets cannot lengthen under a continuous deformation without discontinuously gaining 2^{2k} states, with $2k$ the number of supercharges that would have to be broken (Appendix C.3.3), so the relation $M = |Z|$ and the quantum numbers of the state survive as the string coupling g_s, or any other modulus, is varied. This licenses the central manoeuvre of the chapter. At weak coupling $g_s N \ll 1$ the object is a flat, weakly interacting bound state of branes on which we can count states; at strong coupling $g_s N \gg 1$ the same charges back-react into the black hole Eq. (12.11) with a large smooth horizon. Because the count is a protected index, it does not change between the two regimes. Whatever we enumerate at weak coupling is what the horizon must hide at strong coupling.

12.1.5 Three charges and a finite horizon

We now assemble the target number. The black hole to be matched lives in five dimensions, obtained by wrapping ten-dimensional branes on an internal $T^4 \times S^1$ and looking at the non-compact $4+1$ directions; the specific D1-D5-P realisation is set up in Section 12.2. Here we need only its supergravity form and its area. Each of the three charges reduces to a point charge in the five-dimensional spacetime, sharing the same transverse $\mathbb{R}^4$, and so contributes one harmonic function

$$H_i(r) = 1 + \frac{Q_i}{r^2}, \qquad \nabla^2_{\mathbb{R}^4} H_i = 0, \qquad i = 1, 2, 3, \tag{12.15}$$

harmonic on $\mathbb{R}^4$ because $1/r^2$ is the Green's function there. Superposing them by the multiplication rule of Section 12.1.3, the five-dimensional Einstein-frame metric of the three-charge black hole is

$$\mathrm{d}s_5^2 = -(H_1 H_2 H_3)^{-2/3}\,\mathrm{d}t^2 + (H_1 H_2 H_3)^{1/3}(\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega_3^2). \tag{12.16}$$

Send $r \to 0$, where $H_i \simeq Q_i/r^2$ and $H_1 H_2 H_3 \simeq Q_1 Q_2 Q_3 / r^6$. The warp factors conspire:

$$\mathrm{d}s_5^2 \;\longrightarrow\; -\frac{r^4}{(Q_1 Q_2 Q_3)^{2/3}}\,\mathrm{d}t^2 + (Q_1 Q_2 Q_3)^{1/3}\,\frac{\mathrm{d}r^2}{r^2} + (Q_1 Q_2 Q_3)^{1/3}\,\mathrm{d}\Omega_3^2. \tag{12.17}$$

The first two terms are an AdS_2; the last is a round S^3 of fixed radius-squared $R_H^2 = (Q_1 Q_2 Q_3)^{1/3}$. The near-horizon geometry is $\mathrm{AdS}_2 \times S^3$, the five-dimensional analogue of the Bertotti-Robinson throat Eq. (12.9), and the S^3 does not collapse. Its area is finite,

$$A_3 = 2\pi^2 R_H^3 = 2\pi^2 \sqrt{Q_1 Q_2 Q_3}, \tag{12.18}$$

using $\mathrm{vol}(S^3) = 2\pi^2$ for a unit three-sphere. The Bekenstein-Hawking entropy is

$$\boxed{S_{\mathrm{BH}} = \frac{A_3}{4G_5} = \frac{\pi^2}{2G_5}\sqrt{Q_1 Q_2 Q_3}\,.} \tag{12.19}$$

When the charges Q_i and the five-dimensional Newton constant G_5 are expressed in terms of the integer brane numbers N_1, N_5, N_p and the string data, the numerical dust settles into

$$S_{\mathrm{BH}} = 2\pi\sqrt{N_1\, N_5\, N_p}, \tag{12.20}$$

the target of the microscopic count. The translation from Eq. (12.19) to Eq. (12.20), which needs the D1-D5-P charge dictionary and the volume of $T^4 \times S^1$, is carried out in Section 12.4 on the brane system set up in Section 12.2; the count that reproduces the right-hand side is Section 12.3.

Why three charges? The area Eq. (12.18) came out finite because the transverse warp factor $(H_1H_2H_3)^{1/3}$ scaled as r^{-2} near the horizon, exactly cancelling the r^2 in $r^2\mathrm{d}\Omega_3^2$. Count the powers for a general number of charges. With n harmonic functions each $\sim r^{-2}$, the S^3 radius-squared behaves as $(\prod Q_i)^{1/3}\, r^{\,2-2n/3}$ as $r \to 0$. Only $n = 3$ gives a finite, nonzero limit:

$$R_H^2 \sim r^{\,2-2n/3} \xrightarrow{r\to 0} \begin{cases} 0 & n = 1, 2 \ \ (\text{area} \to 0), \\ \text{finite} & n = 3, \\ \infty & n \geq 4. \end{cases} \tag{12.21}$$

A one- or two-charge black hole in five dimensions has a horizon of zero area at this level of the supergravity description: the naive solution degenerates to a null singularity, a "small" black hole whose entropy is not visible in two-derivative gravity. Three charges are the minimum for a regular, macroscopic horizon. This is the structural reason the microscopic system must be D1-D5-P: two charges, D1 and D5, source a geometry but no macroscopic entropy; the third charge, momentum P along the common circle, opens up the horizon. The four-dimensional analogue needs four charges. In both cases the combination under the square root is the U-duality-invariant of the charges, so the entropy comes out duality-covariant: the cubic invariant I_3 in five dimensions, giving the $\sqrt{Q_1Q_2Q_3}$ here, and the quartic invariant $I_4(q)$ in four dimensions, giving $S \propto \sqrt{I_4(q)}$ (Eq. (7.35)).

Remark 12.1 (Beyond the lectures: an AdS_3 hides in the throat). In the six-dimensional duality frame where the momentum P is a wave along the common circle, the two-charge D1-D5 near-horizon is $\mathrm{AdS}_3 \times S^3$, and adding the third charge P turns the AdS_3 factor into a rotating BTZ black hole. Dimensionally reducing on that circle is what produces the $\mathrm{AdS}_2 \times S^3$ of Eq. (12.17). The virtue of keeping the AdS_3 manifest is that its boundary carries a two-dimensional conformal field theory, and the entropy Eq. (12.20) is the Cardy growth of states in that CFT. The macroscopic area computed here and the microscopic Cardy count of Section 12.3 are the two sides of $\mathrm{AdS}_3/\mathrm{CFT}_2$, which is why the match is exact rather than merely parametric.

12.1.6 The attractor mechanism

One feature of the entropy Eq. (12.19) is easy to miss and central to everything. It depends only on the charges Q_i. It does not depend on the values of the scalar moduli, the size of T^4, the string coupling, or the asymptotic values of any field. Yet the moduli are physical, and they certainly appear in the full solution Eq. (12.16) away from the horizon. Why do they drop out of the area?

The answer is the attractor mechanism (Ferrara, Kallosh and Strominger, 1995; Ferrara and Kallosh, 1996). In an $\mathcal{N} = 2$ (or more) supergravity the vector-multiplet scalars are not constant across the solution: they run as functions of the radius, driven by the flow down the throat. The BPS equations force this radial flow to be a gradient flow of the central charge $|Z|$, and its endpoint at the horizon is a fixed point. Whatever the scalars are asymptotically, at the horizon they are dragged to values ϕ_* determined entirely by the charges, through the attractor equations

$$\partial_\phi |Z(\phi;\, q, p)|\Big|_{\phi=\phi_*} = 0, \tag{12.22}$$

which extremise the central charge over the moduli at fixed charges. The horizon is an attractor: nearby flows are funnelled to the same ϕ_*. Because the horizon area is set by the geometry there, and that geometry knows only ϕ_*, the area is a function of the charges alone. In four dimensions the clean statement is

$$S_{\mathrm{BH}} = \frac{A}{4G} = \pi\, |Z(\phi_*)|^2 = \pi\, |Z|^2_{\text{extremised}}, \tag{12.23}$$

the entropy is π times the extremised central charge squared, a function of the quantised charges and nothing else. For the single-charge extremal RN hole Eq. (12.7) there are no moduli to run and Eq. (12.23) reduces to $S = \pi|Z|^2 = \pi Q^2$. For the multi-charge holes it reproduces the invariants $\sqrt{I_3}$ and $\sqrt{I_4}$ quoted above.

The attractor mechanism is the second pillar under the microscopic count, standing beside BPS protection. Protection (Section 12.1.4) says the number of states does not change as the coupling is dialled from the weakly coupled brane regime to the black-hole regime. Attractor moduli-independence says the entropy we are matching is a function of the charges only, so it does not matter that the weak-coupling count and the strong-coupling geometry sit at different points in moduli space. The count is done wherever it is convenient, and the attractor mechanism guarantees it lands on the same integers that fix the area.

Remark 12.2 (Beyond the lectures). The attractor story generalises well past the two-derivative, large-charge regime used here. Sen's entropy function packages the near-horizon extremisation Eq. (12.22) as a Legendre transform of the on-shell Lagrangian evaluated on the AdS_2 throat, and it accommodates higher-derivative corrections through Wald's generalisation of $A/4G$. Those corrections produce subleading terms in S beyond Eq. (12.20), and for certain $\mathcal{N} = 4$ and $\mathcal{N} = 8$ black holes the entire series is captured by a single modular object, the count of $\frac{1}{4}$-BPS dyons being reproduced to all orders by a Siegel modular form (the Dijkgraaf-Verlinde-Verlinde partition function). A separate and more radical line of thought, the fuzzball programme (Mathur; Bena and Warner), asks whether the individual microstates counted here are themselves smooth, horizonless supergravity geometries, so that the black hole is a coarse-grained average over an ensemble of such geometries rather than a single solution with an interior. Both directions take the finite $S = A/4G$ of this section as their point of departure.

Where this is going

We have a target. The three-charge extremal black hole of Section 12.1.5 is a regular, zero-temperature, half-BPS supergravity solution with a smooth horizon of area Eq. (12.18), and its Bekenstein-Hawking entropy is the number Eq. (12.20). Two supersymmetric facts make that number computable from the outside: BPS saturation protects the state count against changes in coupling, and the attractor mechanism makes the entropy a function of the quantised charges alone. What remains is to build the microscopic system carrying those charges, the D1-D5-P bound state (Section 12.2), count its ground states with the Cardy formula (Section 12.3), and check that the count is Eq. (12.20) on the nose (Section 12.4).

Exercise 12.1 (The extremal throat)**.** Start from the Reissner-Nordström metric Eq. (12.3). (a) Verify that at extremality $M = |Q|$ the substitution $\rho = r - M$ brings the metric to the harmonic form Eq. (12.8) with $H = 1 + Q/\rho$. (b) Take $\rho \to 0$ and confirm the near-horizon geometry Eq. (12.9), an $AdS_2 \times S^2$ with both radii equal to Q. (c) Compute the proper radial distance $\int H \, d\rho$ from a finite ρ_0 down to the horizon and show that it diverges, so the throat is infinitely long. (d) Evaluate the Hawking temperature Eq. (12.5) as $M \to |Q|$ and confirm that it vanishes while the area Eq. (12.7) stays finite.

Exercise 12.2 (Why exactly three charges)**.** Consider the five-dimensional metric Eq. (12.16) with the same warp exponents, $-2/3$ on dt^2 and $+1/3$ on the transverse space, but with a product of n harmonic functions $H_i = 1 + Q_i/r^2$ in place of three. Treat this as a power-counting heuristic: an actual regular supergravity solution of this form exists only at $n = 3$. (a) Show that near the horizon $(H_1 \cdots H_n)^{1/3} \sim r^{-2n/3}$, so the S^3 radius-squared scales as $r^{\,2-2n/3}$, reproducing the trichotomy Eq. (12.21). (b) Conclude that the horizon area vanishes for one or two charges and diverges for four or more, so $n = 3$ is the unique choice with a finite macroscopic horizon in five dimensions. (c) Repeat the power-counting for a four-dimensional black hole (transverse $\mathbb{R}^3$, harmonic functions $1 + Q_i/r$, warp factors $(\prod_i H_i)^{-1/2}$ on dt^2 and $(\prod_i H_i)^{1/2}$ on the transverse

space, sphere S^2) and show that the S^2 radius-squared scales as $r^{\,2-n/2}$, so four charges are required there.

Exercise 12.3 (Multi-centre no-force)**.** For the Majumdar-Papapetrou solution Eq. (12.10) with two centres, argue that the absence of any constraint relating the positions $\vec{x}_1, \vec{x}_2$ is the gravitational statement of the BPS no-force condition of Section 5.6. Estimate the Newtonian potential energy of the two-centre configuration by expanding the metric at large separation, and check that the gravitational attraction $-Gm_1m_2/|\vec{x}_1-\vec{x}_2|$ is cancelled by the electrostatic repulsion $+Q_1Q_2/|\vec{x}_1-\vec{x}_2|$ once $m_a = Q_a$. Where does the calculation break down if the charges have opposite sign?

Exercise 12.4 (Attractor for one modulus)**.** (a) The single-charge extremal Reissner-Nordström hole has no vector-multiplet moduli. Show that the attractor formula Eq. (12.23), $S_{\rm BH} = \pi|Z|^2$, reduces to the entropy Eq. (12.7), $S = \pi Q^2$. (b) Take a toy central charge with one real modulus ϕ and two charges,

$$|Z(\phi; q_1, q_2)|^2 = e^{-\phi}\, q_1^2 + e^{\phi}\, q_2^2.$$

Solve the attractor equation Eq. (12.22), $\partial_\phi |Z| = 0$, for the horizon value ϕ_*, and show that $\phi_* = \frac{1}{2}\log(q_1^2/q_2^2)$ depends only on the charges. (c) Evaluate $|Z(\phi_*)|^2 = 2|q_1 q_2|$ and hence $S_{\rm BH} = 2\pi|q_1 q_2|$, a function of the quantised charges alone and independent of the asymptotic value of ϕ. This is the moduli-independence used in Section 12.1.6.

12.2 The D1-D5 system

The previous section built a genuine black hole out of supergravity: an extremal, BPS solution with a regular horizon of area A and a Bekenstein-Hawking entropy $A/4G$. That entropy is a statement about a huge number of microstates, $\exp(A/4G)$ of them, yet the supergravity description gives no hint of what those states are. To find them we change our vantage point. At weak string coupling the same charges that source the black hole are carried by a configuration of D-branes, and the branes are objects we can quantise. The plan for the rest of the chapter is to build that configuration here, count its bound states in the next section, and match the count to $A/4G$ in the last. The bridge that makes the weak-coupling count relevant to the strong-coupling black hole is BPS protection (Appendix C); we return to it at the end.

The big picture

Compactify type IIB on $\mathcal{M}_4 \times S^1$, with $\mathcal{M}_4 = T^4$ or K3. Wrap N_5 D5-branes on $\mathcal{M}_4 \times S^1$ and N_1 D1-branes on the S^1. The two brane systems share only the circle, so their bound-state dynamics is a field theory in the 1+1 dimensions (t, y) of that circle: an effective string. Its low-energy limit is a two-dimensional superconformal field theory whose target is the moduli space of the bound state, a hyper-Kähler manifold of real dimension $4N_1N_5$, and whose central charge is therefore

$$c \;=\; 6\,N_1N_5.$$

Two charges alone make a black hole with a singular, zero-area horizon. Adding a third charge, momentum N_p running along the circle, carried by left-moving waves on the effective string, opens the horizon to finite size. Counting the ways to share that momentum among the $c = 6N_1N_5$ modes is the microstate count of Section 12.3.

12.2.1 The brane configuration

Work in type IIB string theory compactified on $\mathcal{M}_4 \times S^1$, where $\mathcal{M}_4$ is a four-manifold, either the four-torus T^4 or the K3 surface of Theorem 8.11, and S^1 is a circle of radius R in the direction $y := x^5$. Five dimensions stay non-compact: time x^0 and the four spatial directions

$x^1, \ldots, x^4$. Any black hole we build lives in this $\mathbb{R}^{1,4}$, with transverse space $\mathbb{R}^4$ and horizons of topology S^3. The internal manifold occupies $x^6, \ldots, x^9$; write its volume as $V_4 = \mathrm{vol}(\mathcal{M}_4)$.

Place the branes as in Table 12.1. The N_5 D5-branes wrap all five compact directions, $\mathcal{M}_4 \times S^1$; the N_1 D1-branes wrap only the circle and sit at a point of $\mathcal{M}_4$ and at the origin of $\mathbb{R}^4$. Both brane types are pointlike in the non-compact space, so from the five-dimensional point of view they sit on top of each other and look like a single charged object.

	x^0	x^1	x^2	x^3	x^4	$y{=}x^5$	x^6	x^7	x^8	x^9
	t	$\mathbb{R}^4$ transverse				S^1	$\mathcal{M}_4$			
N_1 D1	•	–	–	–	–	•	–	–	–	–
N_5 D5	•	–	–	–	–	•	•	•	•	•
P (momentum)						→				

Table 12.1: The D1-D5-P array in type IIB on $\mathcal{M}_4 \times S^1$. A bullet marks a worldvolume direction, a dash a transverse one. The momentum P is a wave running along the common circle y. The four directions $x^{6,7,8,9}$ are Neumann for the D5 and Dirichlet for the D1: these are the four ND directions that control the open-string spectrum.

The reason for this particular pair of branes is supersymmetry. A single Dp-brane preserves half of the thirty-two supercharges of IIB, through a projector $\Gamma_{\text{brane}}\epsilon = \epsilon$ built from the gamma matrices along its worldvolume (Eq. (C.42)). Two branes preserve the spinors annihilated by both projectors (Eq. (C.43)). For the D1 along $\{0, 5\}$ and the D5 along $\{0, 5, 6, 7, 8, 9\}$ the two projectors commute, so half of the D1's supersymmetry survives: the D1-D5 pair is $\frac{1}{4}$-BPS, eight supercharges. What makes the projectors compatible is the count of directions in which the two branes differ: exactly four, the $x^{6,7,8,9}$ that are along the D5 and transverse to the D1. A configuration with four such mixed (ND) directions is supersymmetric; with a different number it would not be. Adding the momentum wave, which is itself $\frac{1}{2}$-BPS, imposes a third commuting projector and halves the count once more, to four supercharges. The D1-D5-P system is $\frac{1}{8}$-BPS, exactly the intersection anticipated in Appendix C.

Because the state is BPS, its mass saturates the bound $M = |Z|$ and the three central charges add in phase,

$$M \;=\; M_{\mathrm{D1}} + M_{\mathrm{D5}} + M_P, \qquad M_{\mathrm{D1}} \sim \frac{N_1 R}{g_s \ell_s^2}, \quad M_{\mathrm{D5}} \sim \frac{N_5 R\, V_4}{g_s \ell_s^6}, \quad M_P = \frac{N_p}{R}, \tag{12.24}$$

each term a brane tension (Eq. (6.35)) times the volume it wraps, and M_P the energy of N_p units of momentum on a circle of radius R. The numerical factors of 2π are collected in Appendix I and play no role in what follows. The masses adding with no binding energy is the hallmark of a threshold bound state: the three constituents can be assembled at no cost in energy, so all of the interesting physics lives in the *degeneracy* of ways to assemble them at fixed charge.

12.2.2 Which states are bound: the Higgs branch

Charges alone do not make a bound state. A pile of D1-branes sitting next to a stack of D5-branes carries the same charges as a bound state but is not one; it can fall apart into its pieces at no cost. The counting problem is to identify the genuine bound states, the ones that cannot be separated in the non-compact $\mathbb{R}^4$, and to count them at fixed (N_1, N_5, N_p). This is a question about the open strings stretched between the branes, and it is cleanest to read it off the low-energy gauge theory living on the common (t, y).

Four sectors of open string contribute. D1-D1 strings give the gauge field and transverse scalars of a U(N_1) theory; D5-D5 strings give a U(N_5) theory; and the D1-D5 and D5-D1 strings, stretched

across the four ND directions, give matter in the bifundamental $(N_1, \bar{N}_5)$. The four ND directions are what make this matter well-behaved: the ground-state energies of the Neveu-Schwarz and Ramond sectors line up so that the lightest D1-D5 states are massless and there is no tachyon, and they assemble into hypermultiplets of the $(4,4)$ supersymmetry preserved on the string (Exercise 12.6). From the D5 point of view the whole system is a six-dimensional $\mathrm{U}(N_5)$ gauge theory, and the D1-branes are solitons in it: the Ramond-Ramond charge of a D1 is a unit of instanton number of the D5 gauge field, spread over $\mathcal{M}_4$. That a D$(p{-}4)$-brane inside a Dp-brane is an instanton of the Dp gauge theory is a general fact (Douglas; Witten's small-instanton analysis), and binding the D1-branes means dissolving them into the D5 worldvolume as a smooth instanton configuration.

The two options, D1-branes dissolved into the D5 or D1-branes pulled off into $\mathbb{R}^4$, are the Higgs and Coulomb branches of the 1+1-dimensional gauge theory. On the Coulomb branch the D1-D5 hypermultiplets are massive and the branes are free to separate; these are the unbound states. On the Higgs branch the hypermultiplets condense, the branes bind, and the low-energy theory is a sigma model whose target is the space of inequivalent instanton configurations,

$$\mathcal{M}_{N_1,N_5} \;=\; \Big\{\; N_1 \text{ instantons of } \mathrm{U}(N_5) \text{ on } \mathcal{M}_4 \;\Big\}\big/\text{ gauge}, \tag{12.25}$$

the moduli space of the bound state. Its dimension is fixed by an index theorem: N_1 instantons of $\mathrm{U}(N_5)$ on a four-manifold have, to leading order in the charges,

$$\dim_{\mathbb{R}} \mathcal{M}_{N_1,N_5} \;=\; 4\,N_1 N_5. \tag{12.26}$$

We quote this rather than derive it; the ADHM construction and the index computation behind (12.26) are standard instanton technology, and the factor of four is the quaternionic dimension that makes $\mathcal{M}_{N_1,N_5}$ hyper-Kähler, as $(4,4)$ supersymmetry demands. The number $4N_1N_5$ is the one piece of geometry that the entropy will depend on.

Remark 12.3 (Beyond the lectures). As a complex manifold $\mathcal{M}_{N_1,N_5}$ is deformation-equivalent to the symmetric product $\mathrm{Sym}^{N_1N_5}(\mathcal{M}_4) = (\mathcal{M}_4)^{N_1N_5}/S_{N_1N_5}$, more precisely to its resolution, the Hilbert scheme of N_1N_5 points. The symmetric-product picture is the one that makes the microstate count transparent, and it is where the "long effective string" of Section 12.2.3 comes from. The exact dimension for K3 carries a small correction to (12.26) from the extra cohomology of the Mukai lattice; for T^4 the four translational zero modes of the centre of mass decouple. Both shifts are subleading in the charges and are dealt with in Section 12.3; the leading count uses (12.26) as it stands.

12.2.3 The effective conformal field theory and its central charge

Flow the Higgs-branch gauge theory to low energies. The gauge fields on the string are non-dynamical in 1+1 dimensions and freeze out, leaving a sigma model whose fields are coordinates on $\mathcal{M}_{N_1,N_5}$ and their $(4,4)$ superpartners. At its infrared fixed point this is a two-dimensional superconformal field theory, and everything we need about it is packaged in one number, its central charge.

Count degrees of freedom. A sigma model on a target of real dimension d has d non-chiral bosons, and $(4,4)$ supersymmetry adds d Majorana fermions in each of the left- and right-moving sectors. A free boson contributes 1 to the central charge and a Majorana-Weyl fermion contributes $\tfrac{1}{2}$, so

$$c \;=\; \underbrace{\dim_{\mathbb{R}} \mathcal{M}_{N_1,N_5}}_{\text{bosons}} \;+\; \underbrace{\tfrac{1}{2}\dim_{\mathbb{R}} \mathcal{M}_{N_1,N_5}}_{\text{fermions}} \;=\; \tfrac{3}{2}\dim_{\mathbb{R}} \mathcal{M}_{N_1,N_5} \;=\; \tfrac{3}{2}\,(4N_1N_5) \;=\; 6\,N_1N_5, \tag{12.27}$$

and by the same counting in the right-moving sector $c_L = c_R = 6N_1N_5$. This is the central result of the section: the microscopic data of the black hole, two large integers N_1 and N_5, enter the effective description only through the single combination $c = 6N_1N_5$.

The symmetric-product description gives a picture worth carrying to the count. The N_1N_5 copies of $\mathcal{M}_4$ in $\mathrm{Sym}^{N_1N_5}(\mathcal{M}_4)$ are permuted by $S_{N_1N_5}$, and the twisted sectors of that orbifold join the copies into cycles. The dominant configuration is a single cycle of maximal length: one long effective string that winds N_1N_5 times around y before closing. On this long string the momentum modes are fractionated, carrying momentum in units of $1/(N_1N_5)$ rather than $1/R$, and it is precisely this fractionation that produces enough low-lying states to account for the entropy. We set up the fractionated spectrum here and count it in Section 12.3.

Example 12.4 (A single free T^4)**.** Take $N_1 = N_5 = 1$ with $\mathcal{M}_4 = T^4$. The bound state is one D1 inside one D5, the moduli space is a single copy of T^4 (where the D1 sits inside the internal manifold), $\dim_{\mathbb{R}} = 4$, and (12.27) gives $c = 6$: four bosons for the four coordinates of T^4, plus four left-moving and four right-moving fermions, $4 + \frac{1}{2}(4) = 6$. The general result is N_1N_5 copies of this building block.

12.2.4 Adding momentum: the third charge and a finite horizon

With only the D1 and D5 charges the effective string sits in its ground state and the corresponding black hole is degenerate: solving the supergravity equations for the two-charge solution gives a horizon of *zero* area. The geometry pinches to a singular point, and there is nothing macroscopic to assign an entropy to. Something must excite the string.

That something is momentum. Give the bound state N_p units of momentum along the circle,

$$P \;=\; \frac{N_p}{R}, \qquad N_p \in \mathbb{Z}, \tag{12.28}$$

carried not by moving the branes rigidly but by waves travelling along the effective string, the pp-wave that the third projector describes. Momentum on a two-dimensional CFT is the difference of left- and right-moving levels, $PR = L_0 - \tilde{L}_0$. Here is where the BPS condition earns its keep. Of the two ways to carry a given momentum, only one is supersymmetric: the third projector is compatible with the first two only for a wave moving in a single direction, say left-moving, which forces the right-movers into their ground state,

$$\tilde{L}_0 = 0, \qquad L_0 = N_p. \tag{12.29}$$

A right-moving excitation would preserve no supersymmetry and would let the state decay. So the whole of the momentum sits in the left-moving sector of a CFT with $c_L = 6N_1N_5$, and the microstate count is the number of ways to build a left-moving state at level N_p.

For large level that number is controlled by the central charge alone, through the Cardy formula. Anticipating the derivation of Section 12.3,

$$S_{\mathrm{micro}} \;=\; \log d(N_p) \;\simeq\; 2\pi\sqrt{\frac{c_L\,N_p}{6}} \;=\; 2\pi\sqrt{N_1N_5N_p}, \tag{12.30}$$

with $c_L = 6N_1N_5$ from (12.27). The three charges enter symmetrically, which reflects the U-duality relating them.

Turning the momentum on also cures the geometry. The three-charge solution has a horizon whose area no longer vanishes: the S^3 in $\mathbb{R}^4$ acquires a finite radius set by all three charges, and carries a Bekenstein-Hawking entropy

$$S_{\mathrm{BH}} \;=\; \frac{A}{4G_5}, \tag{12.31}$$

in the five-dimensional Newton constant G_5 obtained by reducing on $\mathcal{M}_4 \times S^1$. The horizon geometry and the extraction of A belong to Section 12.1; comparing the resulting S_{BH} with the

microscopic count (12.30), coefficient and all, is the task of Section 12.4. What matters now is the structure: each charge plays a distinct role, two to build the string and set its central charge, one to excite it, and all three are needed for a horizon of finite size.

Remark 12.5 (Beyond the lectures). The near-horizon limit of the three-charge geometry factorises into $\mathrm{AdS}_3 \times S^3 \times \mathcal{M}_4$, and the AdS_3 throat is holographically dual to the very two-dimensional CFT constructed here. The relation between the Cardy growth of states and the black-hole area is then a special case of the AdS/CFT dictionary, the subject of Chapter 13. Read in that light, the effective string is the holographic boundary theory whose thermodynamics is the black hole.

12.2.5 Why the weak-coupling count survives

The whole construction lives at weak string coupling, $g_s N_1, g_s N_5 \ll 1$, where the branes are heavy and well described by the flat-space gauge theory, while the black hole with its horizon lives at $g_s N_1, g_s N_5 \gg 1$. The object we have set up to count is the number of $\frac{1}{8}$-BPS states at fixed charge (N_1, N_5, N_p), and the reason a flat-space count can speak for a strong-coupling black hole is that this number is protected. BPS multiplets are short (Appendix C), so the properly protected quantity is an index in which states that could recombine into long multiplets cancel, and an index does not vary continuously with g_s. We only flag the mechanism here. Its technical home, where the relevant index is written down explicitly as an elliptic genus, is Section 12.3, and that is what licenses carrying the count (12.30) from the gauge theory to the horizon.

Exercise 12.5. Using the projector rule (C.43), write the three commuting projectors for the D1 (along 05), the D5 (along 056789) and a left-moving wave along 5. Show that each independent projector halves the number of preserved spinors, so that starting from the thirty-two supercharges of IIB the counts run $32 \to 16 \to 8 \to 4$ (the D1 projector alone gives the $\frac{1}{2}$-BPS count 16, the D5 halves it to 8, and the wave to 4), and the D1-D5-P system preserves $\frac{1}{8}$ of the supercharges. Explain why a *right*-moving wave gives an incompatible projector and breaks all supersymmetry.

Exercise 12.6. The masslessness of the D1-D5 strings, quoted in Section 12.2.2, is fixed by the four ND directions; here you compute it. In light-cone gauge the D1-D5 open string has eight transverse worldsheet bosons X^i and their fermionic partners ψ^i. With ν ND directions, ν of the X^i are half-integer moded and $8-\nu$ integer moded (the same in both sectors), while the fermions are half-integer moded along NN/DD and integer moded along ND in the Neveu-Schwarz sector, and the reverse in the Ramond sector. Using the zeta-regularised zero-point energy per real field,

$$\begin{aligned} &\text{boson, integer moded: } -\tfrac{1}{24}, \quad &&\text{boson, half-integer moded: } +\tfrac{1}{48}, \\ &\text{fermion, integer moded: } +\tfrac{1}{24}, \quad &&\text{fermion, half-integer moded: } -\tfrac{1}{48}, \end{aligned}$$

sum the contributions to show that the ground-state energies are

$$a_{\mathrm{NS}} = \frac{\nu - 4}{8}, \qquad a_{\mathrm{R}} = 0, \qquad \text{with} \qquad \alpha' M^2 = N + a.$$

Deduce that for $\nu = 4$ the Neveu-Schwarz ground state is massless and the usual tachyon (recovered at $\nu = 0$, where $a_{\mathrm{NS}} = -\frac{1}{2}$) is gone, that the Ramond ground state is massless for every ν, and hence that at $\nu = 4$ the two sectors supply the bosons and fermions of a single $(4,4)$ hypermultiplet.

Exercise 12.7. Verify (12.27) by counting fields. A $(4,4)$ sigma model on a hyper-Kähler target of real dimension d has d real scalars and d Majorana fermions in each chirality. Using $c_{\mathrm{boson}} = 1$ and $c_{\mathrm{MW\ fermion}} = \frac{1}{2}$, obtain $c = \frac{3}{2}d$, and hence $c = 6N_1N_5$ for $d = 4N_1N_5$. Confirm that a single copy of T^4 gives $c = 6$, and that this matches the central charge of a free superconformal sigma model on T^4.

Exercise 12.8. Assume the Cardy formula $S = 2\pi\sqrt{c_L N_p/6}$ for the number of left-moving states at level N_p in a CFT of central charge c_L. With $c_L = 6N_1N_5$, reproduce $S_{\text{micro}} = 2\pi\sqrt{N_1N_5N_p}$. Comment on the symmetry of the result under permutations of (N_1, N_5, N_p) and relate it to the freedom to reach the same black hole through a U-duality that exchanges the three charges.

12.3 Counting the microstates

We now have all the pieces of the microscopic problem. From Section 12.2 the low-energy dynamics of the D1-D5 bound state, once the branes have merged into a single object, is a two-dimensional superconformal field theory living on the common circle S^1 of radius R. It has $\mathcal{N} = (4, 4)$ supersymmetry and central charge

$$c \;=\; 6\,N_1N_5, \tag{12.32}$$

the sigma model whose target is (a resolution of) the symmetric product $\mathrm{Sym}^{N_1N_5}(T^4) = (T^4)^{N_1N_5}/S_{N_1N_5}$. The third charge, momentum, is the quantised momentum $P = N_p/R$ carried by the excitations of this CFT along the circle. The BPS condition worked out in Section 12.1, three commuting brane projectors leaving four of the thirty-two supercharges unbroken (Appendix C.3.4), forces the right-movers into their ground state and puts *all* of the momentum into the left-moving sector,

$$L_0 \;=\; N_p, \qquad \bar{L}_0 \;=\; 0. \tag{12.33}$$

The physics question is now a counting question, and a clean one. Fix the three integers N_1, N_5, N_p. How many distinct states does the CFT carry with left-moving level N_p, right-movers in their ground state, in a theory of central charge Eq. (12.32)? Call that number $d(N_1, N_5, N_p)$. Its logarithm is the microscopic entropy,

$$S_{\text{micro}} \;=\; \log d(N_1, N_5, N_p), \tag{12.34}$$

and the claim to be established in this section is

$$S_{\text{micro}} \;=\; 2\pi\sqrt{N_1N_5N_p} \qquad (N_1, N_5, N_p \gg 1). \tag{12.35}$$

Nothing about a black hole enters the calculation. It is a partition count in a free-field theory, controlled entirely by the central charge and the level. The comparison with the Bekenstein-Hawking area is the job of Section 12.4; here we do the count.

12.3.1 The counting problem is a density of states

Strip the CFT down to what the count needs. In the left-moving sector the $(4, 4)$ sigma model on a target of real dimension $4N_1N_5$ provides $4N_1N_5$ real bosonic oscillators and, by worldsheet supersymmetry, $4N_1N_5$ real fermionic oscillators. Each field carries a tower of creation operators labelled by a positive mode number n, and acting with a mode of number n raises L_0 by n. A state at level N_p is any way of writing N_p as a sum of such mode numbers, one term per excited oscillator, with the fermionic modes used at most once each. The degeneracy $d(N_1, N_5, N_p)$ is the number of such partitions.

Package the count in a generating function. With q a formal variable, the partition function of the left-movers is

$$Z(q) \;=\; \mathrm{Tr}\; q^{L_0} \;=\; \sum_{N=0}^{\infty} d(N)\, q^N, \tag{12.36}$$

where $d(N)$ is the number of states at level N and we suppress the fixed labels N_1, N_5. A single bosonic tower contributes $\prod_{n\geq 1}(1-q^n)^{-1}$, because each mode n can be excited any number of times, and a single fermionic tower contributes $\prod_n(1+q^n)$, because Fermi statistics allow each mode zero or one excitation. For the D1-D5 content,

$$Z(q) \;=\; \prod_{n=1}^{\infty} \frac{(1+q^n)^{4N_1N_5}}{(1-q^n)^{4N_1N_5}} \times (\text{zero-mode factor}), \tag{12.37}$$

the zero-mode factor being a finite multiplicity from the fermionic ground state that does not affect the leading asymptotics. Recovering $d(N)$ is contour extraction,

$$d(N) \;=\; \frac{1}{2\pi i}\oint \frac{\mathrm{d}q}{q^{N+1}}\, Z(q), \tag{12.38}$$

the integral running around $q=0$. Everything now is asymptotics of Eq. (12.38) at large N, and the answer is a universal one that depends on Z only through the central charge. That universality is the Cardy formula. We build it up from the simplest case.

12.3.2 Warm-up: partitions of a single boson

Take one real boson, $c=1$. Then $d(N)=p(N)$, the number of partitions of the integer N into positive parts, and

$$Z_B(q) \;=\; \prod_{n=1}^{\infty}\frac{1}{1-q^n} \;=\; \sum_{N=0}^{\infty} p(N)\, q^N. \tag{12.39}$$

The generating function is Euler's, and the large-N behaviour of $p(N)$ is the Hardy-Ramanujan result. We derive it by the saddle-point method, because that method is exactly what generalises to any central charge.

Write $q=e^{-\beta}$ and think of β as an inverse temperature. The number of states at level N is fixed by how $\log Z_B$ behaves as $\beta\to 0$, the high-temperature limit where all the modes contribute. Take the logarithm and expand:

$$\log Z_B(\beta) \;=\; -\sum_{n=1}^{\infty}\log(1-e^{-\beta n}) \;=\; \sum_{n=1}^{\infty}\sum_{k=1}^{\infty}\frac{e^{-\beta n k}}{k}. \tag{12.40}$$

As $\beta\to 0$ the sum over n becomes an integral, $\sum_n \to \int_0^\infty \mathrm{d}x/\beta$, and

$$\log Z_B(\beta) \;\xrightarrow{\beta\to 0}\; \frac{1}{\beta}\int_0^\infty \mathrm{d}x\,\big[-\log(1-e^{-x})\big] \;=\; \frac{1}{\beta}\sum_{k=1}^{\infty}\frac{1}{k^2} \;=\; \frac{\pi^2}{6\beta}. \tag{12.41}$$

The $\pi^2/6=\zeta(2)$ is the whole content: a free boson has free energy $\log Z\sim \pi^2/(6\beta)$. Now feed this into the inverse transform Eq. (12.38), written as

$$p(N) \;=\; \frac{1}{2\pi i}\int \mathrm{d}\beta\, e^{\beta N}\, Z_B(\beta) \;\approx\; \frac{1}{2\pi i}\int \mathrm{d}\beta\, \exp\!\Big(\beta N+\frac{\pi^2}{6\beta}\Big). \tag{12.42}$$

The exponent $f(\beta)=\beta N+\pi^2/(6\beta)$ has a saddle where $f'(\beta)=0$, that is $N=\pi^2/(6\beta^2)$, so

$$\beta_* \;=\; \pi\sqrt{\frac{1}{6N}}, \qquad f(\beta_*) \;=\; \pi\sqrt{\tfrac{N}{6}}+\pi\sqrt{\tfrac{N}{6}} \;=\; 2\pi\sqrt{\tfrac{N}{6}}. \tag{12.43}$$

The two terms in f contribute equally at the saddle, a feature that will recur. Hence

$$p(N) \;\sim\; \exp\!\Big(2\pi\sqrt{\tfrac{N}{6}}\Big), \tag{12.44}$$

the Hardy-Ramanujan asymptotic. The entropy of a single free boson at level N is $\log p(N)\sim 2\pi\sqrt{N/6}$, which is the Cardy formula with $c=1$.

12.3.3 The Cardy formula

The saddle-point calculation used only one fact about Z: its small-β free energy. So the generalisation is immediate. Suppose the chiral CFT has central charge c. The free energy at high temperature is

$$\log Z(\beta) \xrightarrow{\beta\to 0} \frac{\pi^2 c}{6\beta}, \tag{12.45}$$

which for $c = 1$ is Eq. (12.41). The coefficient is additive in the fields: each real boson contributes $\pi^2/(6\beta)$ as computed, and each real fermion contributes half as much, because the fermionic sum

$$\log Z_F(\beta) \;=\; \sum_n \log(1+e^{-\beta n}) \;\xrightarrow{\beta\to 0}\; \frac{1}{\beta}\int_0^\infty \mathrm{d}x\, \log(1+e^{-x}) \;=\; \frac{1}{\beta}\sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k^2} \;=\; \frac{\pi^2}{12\beta} \tag{12.46}$$

carries the alternating $\zeta(2)$, half of the bosonic value. So the effective central charge that controls the growth is

$$c \;=\; (\#\text{ real bosons}) + \tfrac{1}{2}\,(\#\text{ real fermions}), \tag{12.47}$$

which for the D1-D5 fields, $4N_1N_5$ bosons and $4N_1N_5$ fermions, reproduces $c = 4N_1N_5+2N_1N_5 = 6N_1N_5$, consistent with Eq. (12.32). Running the saddle point with Eq. (12.45) in place of the $c = 1$ free energy,

$$\beta_* = \pi\sqrt{\frac{c}{6N}}, \qquad f(\beta_*) = \beta_* N + \frac{\pi^2 c}{6\beta_*} = 2\pi\sqrt{\frac{cN}{6}}, \tag{12.48}$$

gives the density of states in the form we need.

Proposition 12.6 (Cardy formula)**.** *A unitary chiral conformal field theory of central charge c, whose spectrum is bounded below by a unique ground state, has asymptotic degeneracy at level N*

$$d(N) \;\sim\; \exp\left(2\pi\sqrt{\frac{c\,N}{6}}\right), \qquad N\to\infty, \tag{12.49}$$

so that the entropy is $S(N) = \log d(N) = 2\pi\sqrt{cN/6}$.

The free-energy law Eq. (12.45) is not a lucky feature of free fields. It is a consequence of modular invariance of the torus partition function (Section 2.3.4). The S transformation $\tau \to -1/\tau$ exchanges the high- and low-temperature ends of the trace, and the low-temperature end is dominated by the ground state of energy $-c/24$; mapping that back gives $\log Z(\beta) \sim (c/24)(4\pi^2/\beta) = \pi^2 c/(6\beta)$ for *any* unitary CFT, interacting or free, with the same c. Cardy's original derivation is exactly this modular argument, and the free-field partition count above makes it concrete.

One refinement matters later. The object that appears in Eq. (12.45) is strictly the *effective* central charge $c_{\text{eff}} = c - 24h_{\min}$, with $h_{\min}$ the lowest weight in the sector being counted. For the D1-D5 CFT the momentum-carrying sector has a unique normalisable ground state at $h_{\min} = 0$, so $c_{\text{eff}} = c$ and there is no distinction. We use c throughout.

Careful

The Cardy asymptotics Eq. (12.49) is a large-N statement. At small N the exact $d(N)$ from Eq. (12.37) differs, and the crossover is set by $N \gtrsim c$. For the black hole this is harmless: the interesting regime is precisely macroscopic charges, $N_1, N_5, N_p \to \infty$, where the horizon is large and semiclassical gravity applies. The match to $A/4G$ is an asymptotic match, exact in the leading exponent.

12.3.4 The D1-D5-P degeneracy

Now specialise. The counting sector is the left-movers of the $(4,4)$ CFT, central charge $c = 6N_1N_5$ from Eq. (12.32), and the level is the momentum $N = N_p$ from Eq. (12.33). Substituting into Eq. (12.49),

$$S_{\text{micro}} \;=\; 2\pi\sqrt{\frac{(6N_1N_5)\,N_p}{6}} \;=\; 2\pi\sqrt{N_1N_5N_p}. \tag{12.50}$$

$$\boxed{S_{\text{micro}} \;=\; \log d(N_1,N_5,N_p) \;=\; 2\pi\sqrt{N_1N_5N_p}} \tag{12.51}$$

That is the entire microscopic derivation. The degeneracy of D1-D5-P bound states at fixed charges, computed in the free CFT of the branes, grows as $\exp\left(2\pi\sqrt{N_1N_5N_p}\right)$. The result is symmetric in the three charges, as it must be: the U-duality group of the compactification permutes D1, D5, and momentum, so no honest count could prefer one over another (Section 7.4 sets up this permutation symmetry). The symmetry is a strong internal check: a count that gave, say, $N_1^2N_5N_p$ would be wrong on its face.

The long-string picture. There is a second, more physical way to see the same number, and it explains where the large factor N_1N_5 inside the square root comes from. In the bound-state regime the N_1 D1-branes and N_5 D5-branes do not behave as N_1N_5 separate objects but merge into a single *effective string* that winds the circle N_1N_5 times, of total length $2\pi R\,N_1N_5$. Its transverse oscillations are the four bosons and four fermions of the T^4 directions, so it is a $c = 6$ system, not $c = 6N_1N_5$. The catch is that the momentum modes on a string of N_1N_5 times the length come in fractional units $1/(N_1N_5)$ of the ordinary momentum quantum. Carrying total momentum N_p therefore means filling the fractionated modes up to an effective oscillator level

$$N_{\text{eff}} \;=\; N_1N_5\,N_p, \tag{12.52}$$

and Cardy with $c = 6$ at level N_{eff} gives $2\pi\sqrt{6\cdot N_1N_5N_p/6} = 2\pi\sqrt{N_1N_5N_p}$, the same answer. The Cardy entropy depends on c and N only through the product cN, and both descriptions have

$$cN \;=\; (6N_1N_5)(N_p) \;=\; (6)(N_1N_5N_p), \tag{12.53}$$

so the symmetric-orbifold and long-string pictures are the same computation seen from two sides. Fractionation is the mechanism, first stressed by Das and Mathur and by Maldacena and Susskind, that makes the entropy large: the momentum is spread over N_1N_5 times as many available modes as a single string would offer, and the number of partitions grows accordingly.

Example 12.7 (Why $\sqrt{N_1N_5N_p}$ and not $N_1N_5N_p$)**.** The square root is the signature of a one-dimensional gas. A gas of massless quanta in one spatial dimension at energy E on a segment of size L has, by the standard thermodynamic estimate, energy $E \sim c\,L\,T^2$ and entropy $S \sim c\,L\,T$, so $S \sim \sqrt{c\,L\,E}$. Here the left-movers are exactly such a gas: $c = 6N_1N_5$ species, $L = 2\pi R$, and total energy $E = N_p/R$ set by the momentum. Then $S \sim \sqrt{(6N_1N_5)(2\pi R)(N_p/R)} \sim \sqrt{N_1N_5N_p}$, up to the numerical factor that the exact Cardy calculation fixes to 2π. The area law of a black hole hides a one-dimensional gas of brane excitations.

12.3.5 What is really protected: the elliptic genus

There is a gap in the argument so far, and closing it is what licenses the whole construction. The count Eq. (12.51) is done in the free CFT of the branes, which is the D1-D5-P system at *weak* string coupling, $g_sN \ll 1$, where the branes are light and the backreaction is negligible. The black hole with a horizon lives at *strong* coupling, $g_sN \gg 1$, where the same charges gravitate.

In between, the branes bind, interact, and the free-field description Eq. (12.37) is no longer literally valid. Why should a degeneracy computed at weak coupling survive to strong coupling and equal the horizon count?

The naive degeneracy $d(N)$ need not survive; a genuine number of states can change as states pair up and lift when the coupling is turned on. What survives is a supersymmetric index, a count of BPS states weighted by $(-1)^F$, from which the lifting pairs cancel. For the D1-D5 CFT the relevant index is the *elliptic genus*

$$\chi(q,y) \;=\; \mathrm{Tr}_{\mathrm{RR}}\left[(-1)^F\, q^{L_0-c/24}\,\bar{q}^{\,\bar{L}_0-c/24}\,y^{J_0}\right], \tag{12.54}$$

where F is the fermion number, J_0 an R-symmetry charge, and the trace is over the Ramond sector. Because $(-1)^F$ and $\bar{q}^{\,\bar{L}_0-c/24}$ combine so that only right-moving ground states survive the trace, χ is holomorphic in q, independent of $\bar{q}$: it counts exactly the states of Eq. (12.33), weighted by signs. This is the object protected by the argument of Appendix C.3.3: a topological invariant of the CFT, constant along the moduli that connect weak and strong coupling, so it may be computed in the free description and read off in the black-hole regime.

One subtlety attaches to our running target. The elliptic genus of a single T^4 vanishes identically, $\chi(T^4)=0$: four fermion zero modes tied to the one-cycles of T^4 (its H^1) are never lifted, and $(-1)^F$ pairs them off to zero. The plain genus protects nothing here because it counts nothing, and the DMVV product below degenerates to 1. The remedy, due to Maldacena, Moore and Strominger, is a modified index that inserts extra fermion-number to saturate those zero modes,

$$\Omega \;=\; \mathrm{Tr}_{\mathrm{RR}}\left[(-1)^F\, F^2\, q^{L_0-c/24}\,\bar{q}^{\,\bar{L}_0-c/24}\right], \tag{12.55}$$

after which one is again counting 1/8-BPS states with signs. A $K3$ target carries no one-cycles and hence no such zero modes, so its elliptic genus Eq. (12.54) is already a good index and needs no modification. In either case the protected quantity is a moduli-independent count of the 1/8-BPS states of Eq. (12.33), and its leading large-charge growth is governed by the same Cardy formula, since the modular argument of Theorem 12.6 applies to the index as well; both reproduce Eq. (12.51). The T^4 modification changes the prefactor and the subleading terms, not the leading $2\pi\sqrt{N_1N_5N_p}$. The interacting theory cannot spoil the count because it cannot spoil the index.

Remark 12.8 (Beyond the lectures: Jacobi forms and modularity). For a $K3$ target the elliptic genus is not merely modular invariant; it is a weak Jacobi form of weight 0 and index $m=N_1N_5$ under $\mathrm{SL}(2,\mathbb{Z})$ acting on τ (with $q=e^{2\pi i\tau}$) together with elliptic transformations of the fugacity y. That rigidity is powerful. For the symmetric product $\mathrm{Sym}^N(K3)$ the full set of elliptic genera is packaged, by the DMVV formula (Dijkgraaf, Moore, Verlinde and Verlinde), into a single generating function

$$\sum_{N\geq 0} p^N\,\chi(\mathrm{Sym}^N(K3);q,y) \;=\; \prod_{n\geq 1,\,m\geq 0,\,\ell}\frac{1}{(1-p^nq^my^\ell)^{c(nm,\ell)}},$$

with $c(k,\ell)$ the Fourier coefficients of the single-copy genus. The Cardy limit of this expression is Eq. (12.51), and its subleading structure encodes the exact black-hole degeneracy, not just its logarithm. For a T^4 target the same construction runs with the modified index Eq. (12.55) in place of χ, since the plain genus vanishes. For the four-charge $\frac{1}{4}$-BPS dyons of $\mathcal{N}=4$ compactifications the exact index is a Fourier coefficient of a Siegel modular form, the reciprocal of the Igusa cusp form Φ_{10} (Dijkgraaf, Verlinde and Verlinde; sharpened by Sen and by David, Jatkar and Sen), and its asymptotics reproduces the area entropy together with an infinite series of computable corrections; the $\frac{1}{8}$-BPS count in $\mathcal{N}=8$ is governed by a different modular object (Maldacena, Moore and Strominger; Sen). This is one of the sharpest quantitative successes in the subject.

Remark 12.9 (Beyond the lectures: the subleading term). Keeping the Gaussian fluctuation around the saddle Eq. (12.48) gives a bare prefactor for the density of states,

$$d(N) \;\approx\; \left(\frac{c}{96\,N^3}\right)^{1/4} \exp\left(2\pi\sqrt{\frac{cN}{6}}\right).$$

This $N^{-3/4}$ is not yet the honest coefficient. The same S transformation that produced the free energy $\pi^2 c/(6\beta)$ also carries a modular (Dedekind-η) Jacobian $\sim\sqrt{\beta_*}\sim N^{-1/4}$, which multiplies the bare $N^{-3/4}$ up to N^{-1}. For a single boson this reproduces the exact Hardy-Ramanujan prefactor, $p(N)\sim(4\sqrt{3}\,N)^{-1}\exp(2\pi\sqrt{N/6})$, whose logarithm therefore carries $-\log N$, not $-\frac{3}{4}\log N$. The correction to the entropy is logarithmic either way; summed over both chiral sectors and dressed by the full modular measure it combines into a coefficient times $\log(A/G)$ that a one-loop computation in the near-horizon supergravity reproduces exactly (Sen and collaborators). The leading Bekenstein-Hawking term is the classical saddle; the logarithm is its first quantum correction.

We have produced a number. The degeneracy of the weakly coupled D1-D5-P system grows as $\exp\left(2\pi\sqrt{N_1N_5N_p}\right)$, the count is protected by a supersymmetric index and so carries to strong coupling, and the whole computation used nothing but partitions and a saddle point. What remains is to build the black hole these charges make, read off its horizon area, and check that $A/4G$ equals Eq. (12.51). That match is Section 12.4.

Exercise 12.9 (The Cardy saddle in full)**.** Starting from $d(N)=\frac{1}{2\pi i}\int \mathrm{d}\beta\,\exp[\beta N+\pi^2 c/(6\beta)]$, (a) locate the saddle β_* and confirm $f(\beta_*)=2\pi\sqrt{cN/6}$, checking that the two terms of f are equal there. (b) Compute $f''(\beta_*)$ and show the Gaussian integral produces the bare prefactor $(c/96N^3)^{1/4}$ quoted in the remark above. (c) Explain why the saddle sits at real $\beta_*>0$ and why $\beta_*\to 0$ as $N\to\infty$, so that the high-temperature free energy Eq. (12.45) is the only input needed.

Exercise 12.10 (Effective central charge by hand)**.** Consider the left-movers of a single copy of the T^4 sigma model: four real bosons and four real fermions, $c=6$. (a) Write the level generating function $Z(q)=\prod_{n\geq 1}(1+q^n)^4/(1-q^n)^4$ and expand to q^4, listing $d(0),\ldots,d(4)$. (b) Using Eqs. (12.41) and (12.46), confirm $\log Z(\beta)\to\pi^2/\beta$ as $\beta\to 0$, i.e. $c=6$. (c) Compare $\log d(N)$ from part (a) with $2\pi\sqrt{N}$ for $N=10,100$ (you may compute $d(N)$ numerically), and comment on how the crossover to Cardy behaviour sets in around $N\sim c$.

Exercise 12.11 (Fractionation and U-duality)**.** (a) Redo the count in the long-string description: a $c=6$ system at effective level $N_{\text{eff}}=N_1N_5N_p$, and verify it reproduces Eq. (12.51). (b) Show that the entropy Eq. (12.51) is symmetric under any permutation of (N_1,N_5,N_p), and explain in one sentence why U-duality (Section 7.4) demands this. (c) A student proposes that a two-charge system, a fundamental string carrying momentum N_p and winding N_1 with no D-branes, should have an entropy of the same $\sqrt{N_1N_p}$ form by the same logic. Identify the central charge and level for the string's transverse oscillators, obtain the parametric form of the entropy, and state what is missing for this two-charge system to develop a macroscopic, two-derivative horizon (this is why the black hole with a finite classical area needs the *third* charge; see Section 12.4).

12.4 Strominger-Vafa: the match

By now we have done the same sum twice with completely different tools. On the gravity side (Sections 12.1 and 12.2) we built a supersymmetric black hole out of D1-branes, D5-branes and momentum, solved the supergravity equations, and read off the area of its horizon. On the brane side (Section 12.3) we forgot about gravity entirely, treated the same charges as a weakly coupled gas of open strings, organised its low-energy dynamics into a two-dimensional conformal

field theory, and counted how many quantum states sit at the given energy and charge. One calculation is geometry, the other is combinatorics. This section puts the two answers next to each other. They are equal, digit for digit, with no adjustable constant left over. That equality is the result of Strominger and Vafa, and it is the first time anyone computed the entropy of a black hole by counting its microstates.

The big picture

Bekenstein and Hawking told us a black hole has an entropy $S = A/4G$, a number attached to a smooth classical geometry with no obvious states to count. Boltzmann told us entropy is log of a degeneracy. For over twenty years those two statements sat side by side with nothing connecting them. The D1-D5-P system connects them: the degeneracy of brane bound states at fixed charge is $\exp\!\big(2\pi\sqrt{N_1N_5N_p}\big)$, and the horizon area of the black hole those same charges make is exactly $4G$ times $2\pi\sqrt{N_1N_5N_p}$. The bridge that lets a weak-coupling count speak for a strong-coupling black hole is supersymmetry: the states are BPS, so their number is protected as the coupling is dialled from the brane regime to the black-hole regime.

12.4.1 Two computations of one number

Start with the gravity answer, in the compact form we will need. The three-charge extremal solution of Section 12.1, dimensionally reduced on $T^4 \times S^1$ to five non-compact dimensions and written in five-dimensional Einstein frame (the frame in which gravity is the ordinary Einstein-Hilbert action $R/16\pi G_5$), is

$$\mathrm{d}s_5^2 = -\lambda^{-2/3}\,\mathrm{d}t^2 + \lambda^{1/3}(\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega_3^2), \qquad \lambda = f_1 f_5 f_p, \qquad f_i = 1 + \frac{r_i^2}{r^2}, \tag{12.56}$$

with $\mathrm{d}\Omega_3^2$ the round metric on the unit S^3 and r the radial coordinate of the transverse $\mathbb{R}^4$. This is (12.16); the three lengths r_i^2 are the charges Q_i of the harmonic functions (12.15), one apiece for the D1, the D5 and the momentum. The $r \to 0$ analysis of Section 12.1 showed that the throat does not pinch off. Each $f_i \to r_i^2/r^2$, so $\lambda \to (r_1r_5r_p)^2/r^6$ and the angular part $\lambda^{1/3}r^2\,\mathrm{d}\Omega_3^2 \to (r_1r_5r_p)^{2/3}\,\mathrm{d}\Omega_3^2$: the throat opens onto a three-sphere of constant radius $(r_1r_5r_p)^{1/3}$. Using $\mathrm{vol}(S^3_{\mathrm{unit}}) = 2\pi^2$, the horizon area is

$$A = 2\pi^2\,(r_1r_5r_p). \tag{12.57}$$

The new work of this section is to turn those three lengths into charges and to watch the moduli cancel.

The lengths r_i are fixed by matching (12.56) to the D-brane sources; the DBI tensions and charge quantisation that produce them are collected in Appendix I. With $V \coloneqq \mathrm{vol}(T^4)/(2\pi)^4$ the internal four-torus volume and R the radius of the S^1 that carries the momentum, the dictionary is

$$r_1^2 = \frac{g_s N_1\,\alpha'^3}{V}, \qquad r_5^2 = g_s N_5\,\alpha', \qquad r_p^2 = \frac{g_s^2 N_p\,\alpha'^4}{VR^2}, \tag{12.58}$$

and the five-dimensional Newton constant follows from $2\kappa_{10}^2 = (2\pi)^7 g_s^2\alpha'^4$ (Appendix A) by dividing out the compact volume $(2\pi)^4 V \cdot 2\pi R$,

$$G_5 = \frac{G_{10}}{(2\pi)^4 V\,(2\pi R)} = \frac{\pi\, g_s^2\,\alpha'^4}{4\,VR}. \tag{12.59}$$

Now assemble the Bekenstein-Hawking entropy $S_{\mathrm{BH}} = A/4G_5$ and watch what survives. From (12.58),

$$r_1r_5r_p = \sqrt{r_1^2r_5^2r_p^2} = \frac{g_s^2\,\alpha'^4}{VR}\,\sqrt{N_1N_5N_p}\,, \tag{12.60}$$

so that

$$S_{\rm BH} = \frac{A}{4G_5} = \frac{2\pi^2}{4}\,\frac{r_1 r_5 r_p}{G_5} = \frac{2\pi^2}{4}\cdot\frac{g_s^2\alpha'^4\sqrt{N_1N_5N_p}}{VR}\cdot\frac{4VR}{\pi\, g_s^2\alpha'^4} = 2\pi\sqrt{N_1N_5N_p}\,. \tag{12.61}$$

Every dimensionful modulus has cancelled. The string coupling g_s, the torus volume V and the circle radius R all disappear, leaving a pure number times the square root of a product of three integers. Section 12.4.3 explains why the moduli had to drop out.

Now the brane answer. In Section 12.3 the low-energy bound-state dynamics reduced to a $(4,4)$ superconformal field theory on the effective string, with target the symmetric product $\mathrm{Sym}^{N_1N_5}(T^4)$ and central charge

$$c = 6\,N_1N_5. \tag{12.62}$$

The momentum N_p along the S^1 is the level L_0 carried by one set of movers, the other set held in its ground state so that four supercharges survive. Cardy's formula for the asymptotic degeneracy of a single sector,

$$S_{\rm micro} = 2\pi\sqrt{\frac{c\,L_0}{6}} = 2\pi\sqrt{\frac{6\,N_1N_5\cdot N_p}{6}} = 2\pi\sqrt{N_1N_5N_p}\,, \tag{12.63}$$

returns the same number as (12.61). Setting the two side by side,

$$\boxed{S_{\rm micro} \;=\; \log d(N_1,N_5,N_p) \;=\; 2\pi\sqrt{N_1N_5N_p} \;=\; \frac{A}{4G} \;=\; S_{\rm BH}} \tag{12.64}$$

in the regime where both computations are valid. Boltzmann's $S = \log\Omega$ and the area law $S = A/4G$ are the same statement about the same object.

12.4.2 Why the weak-coupling count is the black hole's count

The equality (12.64) is assembled from two sides computed in regimes that do not overlap. Whether the D1-D5-P charges make a flat-space gas of branes or a black hole is controlled by the dimensionless combination g_sN: gravity back-reacts on a length scale $r_i^2 \sim g_sN\,\alpha'$, so the branes dig out a horizon only once $g_sN \gg 1$. The microscopic count (12.63) is done at $g_sN \ll 1$, where the gravitational radii are sub-stringy and spacetime is essentially flat; the area (12.61) is read off at $g_sN \gg 1$, where the throat is deep and curvatures at the horizon are small. In a generic quantum system nothing lets a weak-coupling count survive that extrapolation: as the interaction is turned on, levels cross and states mix, and the degeneracy at fixed energy is not a smooth function of the coupling.

What carries the count across is that the D1-D5-P bound state is $\frac{1}{8}$-BPS: three commuting brane projectors (the kappa-symmetry conditions of Appendix C, the Killing-spinor projectors used to build the solution in Section 12.1) leave four of the thirty-two supercharges unbroken, (C.39). The quantity that is rigorously constant along the path from branes to black hole is the protected index built from these states. Its machinery, short multiplets, walls of marginal stability, and the gap between the signed index and the unsigned degeneracy d, was set up once in Section 12.3.5 and Appendix C.3.3, and this section simply borrows the conclusion. At the large charges of a macroscopic hole the signed-versus-unsigned distinction is invisible at the order we work: $\log(\text{index}) = \log d = 2\pi\sqrt{N_1N_5N_p}$ to Cardy accuracy. The same largeness that makes Cardy's asymptotics reliable makes the horizon big in string and Planck units, so both approximations are controlled by one limit. The match is a statement about macroscopic black holes, not about a single quantum of charge.

What licenses the extrapolation, and what does not

The chain of logic is: (i) count BPS states at $g_sN \ll 1$; (ii) invoke the index of Appendix C.3.3 to carry that number, unchanged, to $g_sN \gg 1$; (iii) compare with $A/4G$ computed there. Step (ii) is the load-bearing one, and it needs unbroken supersymmetry along the whole path. Nothing here derives the entropy of a Schwarzschild black hole, which is not BPS and whose microstate count is not protected. Strominger-Vafa is an exact result about a measure-zero, supersymmetric corner of the space of black holes. That it works at all is the evidence that a horizon is made of states; extending the lesson off the BPS locus is the open problem, not the theorem.

12.4.3 Moduli independence and the attractor mechanism

The cancellation in (12.61) is forced, and the same fact shows up on both sides of the match. On the brane side, $d(N_1, N_5, N_p)$ is the dimension of a Hilbert space labelled by three integers, so it cannot depend on the continuous parameters g_s, V, R or the shape of the T^4. On the gravity side that statement is the attractor mechanism of Section 12.1.6: the extremal BPS flow drags the scalars, g_s, V and the internal moduli among them, down the throat to horizon values fixed by the charges alone, whatever their asymptotic values, so the near-horizon geometry and hence the area depend only on the N_i. That is exactly what (12.61) shows: the asymptotic moduli enter the intermediate expressions and then cancel out of $A/4G_5$. Both sides are functions of the integer charges and of nothing else, and a mismatch in the moduli dependence would have killed the correspondence before any coefficient was checked.

Remark 12.10 (Beyond the lectures: reading the attractor value off the entropy). For extremal black holes the horizon values of the scalars extremise the central charge $|Z|$ over moduli space at fixed charge, and the entropy is the extremal value, $S = \pi|Z_\star|^2$ in a normalisation suited to four dimensions. This is Ferrara-Kallosh-Strominger attractor flow, later packaged as Sen's entropy function: extremise a Legendre transform of the near-horizon Lagrangian over the electric fields and geometric parameters, and its value is the entropy, higher-derivative corrections included. The moduli independence we used is the statement that this extremum does not know about the asymptotic scalars.

12.4.4 What has been shown

The horizon of a black hole carries an entropy that scales with its area, not its volume, and for the D1-D5-P family we can now say what that entropy counts: the number of ways to distribute the momentum N_p among the oscillations of a bound string of D1- and D5-branes. The geometry $A/4G$ and the microscopics $\log d$ are two descriptions of one degeneracy, related by turning up the gravitational coupling. That is the content of (12.64), and it is the first derivation of $S = A/4G$ from a genuine state count.

It shows that a black hole has microstates and that their number is the exponential of the Bekenstein-Hawking entropy, which fixes where the information in a black hole lives. It does not, by itself, tell you how those microstates are arranged in the interior, nor how information gets out of an evaporating hole; the states are counted, not resolved into distinct geometries. And it is restricted to the BPS locus, where supersymmetry protects the count. The reach of the result is set entirely by how far protection reaches, which is why the near-extremal extension in the next remark matters: it is the first evidence that the same states control the black hole slightly away from BPS, where nothing forces the answer.

Remark 12.11 (Beyond the lectures: near-extremal dynamics and greybody factors). Add a little energy above extremality by exciting the previously dormant left-moving sector as well as the right. The effective string now radiates: a left-mover and a right-mover collide, form a

closed string, and leave the brane. Computing this emission in the CFT reproduces, at weak coupling, a thermal spectrum at temperature $T_{\rm H}$ with left and right temperatures $T_{\rm L}, T_{\rm R}$ set by the two oscillator energies, and even the frequency-dependent *greybody factors*, the deviation of the spectrum from a pure blackbody due to the potential barrier around the horizon. The gravity side gives the same $T_{\rm H}$ and the same greybody factors from wave scattering off the near-extremal geometry (Callan-Maldacena; Das-Mathur; Maldacena-Strominger). Matching a whole function of frequency, not just one number, is much stronger evidence than the entropy match alone that the CFT states are the black-hole states. It is the first hint that a black hole is dual to an ordinary thermal quantum system, which is the theme of Chapter 13.

Remark 12.12 (Beyond the lectures: the near-horizon CFT and exact degeneracies). The extremal throat of (12.56) has near-horizon geometry $\mathrm{AdS}_3 \times S^3 \times T^4$. The isometries of AdS_3 enhance to two copies of the Virasoro algebra with a Brown-Henneaux central charge $c = 6N_1N_5$, the very central charge (12.62) we fed into Cardy. Strominger's later observation is that one can then read the entropy off the gravitational asymptotic symmetry algebra directly, without a microscopic Lagrangian: Cardy applied to the boundary Virasoro reproduces $A/4G$. This is the microscopic count reorganised as $\mathrm{AdS}_3/\mathrm{CFT}_2$, and it is the sharpest version of the match. Pushing to subleading orders, the exact BPS degeneracies are Fourier coefficients of modular forms: the elliptic genus of $\mathrm{Sym}^N(K3)$ for the five-dimensional case, and for the four-dimensional four-charge black hole of $\mathcal{N} = 4$ string theory the Igusa cusp form $1/\Phi_{10}$ of Dijkgraaf-Verlinde-Verlinde, counting $\frac{1}{4}$-BPS dyons. Their asymptotics reproduce the Wald entropy, higher-derivative R^2 corrections to $A/4G$ included, and go well beyond the leading area; the all-orders relation is conjecturally the Ooguri-Strominger-Vafa formula $Z_{\rm BH} = |Z_{\rm top}|^2$ tying the black hole to the topological string. The original Strominger-Vafa count used Type IIB on $K3$ rather than T^4 because this modularity is cleaner for $K3$, whose elliptic genus is non-zero; the T^4 case needs a modified index because of extra fermion zero modes, though the leading $2\pi\sqrt{N_1N_5N_p}$ is untouched.

Exercise 12.12 (The cancellation, in full)**.** Take the dictionary (12.58) and the Newton constant (12.59) as given.

1. From (12.56), verify that the near-horizon S^3 has constant radius $(r_1r_5r_p)^{1/3}$ and hence recover the area (12.57). (Show that the $\mathrm{d}t^2$ and $\mathrm{d}r^2$ pieces stay finite in appropriate coordinates, so that the geometry is a genuine horizon and not a curvature singularity.)
2. Assemble $A/4G_5$ and confirm (12.61), keeping every factor. Check the dimensions: each r_i^2 has units of length^2 and G_5 has units of length^3, so A/G_5 is dimensionless, as an entropy must be.
3. Identify precisely which combination of g_s, V and R cancels between A and G_5, and state the physical reason (Section 12.4.3) that it had to.

Exercise 12.13 (Why four charges in four dimensions)**.** The five-dimensional black hole needed three charges to reach a finite horizon; two charges give $A = 0$. In four non-compact dimensions the harmonic functions behave as $f_i = 1 + r_i/r$ (transverse $\mathbb{R}^3$), and the Einstein-frame metric is $\mathrm{d}s_4^2 = -\lambda^{-1/2}\mathrm{d}t^2 + \lambda^{1/2}(\mathrm{d}r^2 + r^2\mathrm{d}\Omega_2^2)$ with $\lambda = \prod_i f_i$.

1. With three charges, $\lambda = f_1f_2f_3 \to (r_1r_2r_3)/r^3$ near $r = 0$. Show that the horizon two-sphere then has area scaling as $\lambda^{1/2}r^2 \to (r_1r_2r_3)^{1/2}\, r^{1/2} \to 0$, so three charges do not suffice in four dimensions.
2. Show that a fourth charge, $\lambda = f_1f_2f_3f_4 \to (r_1r_2r_3r_4)/r^4$, gives a finite area and hence
$$S_{\rm BH} = 2\pi\sqrt{N_1N_2N_3N_4}\,.$$
3. The microscopic realisation adds a fourth stack (for instance a Kaluza-Klein monopole) to

D1-D5-P. Explain in one line why the extremal four-charge black hole still preserves four supercharges, so the count is protected exactly as before. (In its $\mathcal{N}=4$ realisation, IIB on $K3\times T^2$, this is the $\frac{1}{4}$-BPS dyon whose *exact* degeneracy is the $1/\Phi_{10}$ of the remark above.)

Chapter 13

The AdS/CFT Correspondence and Emergent Spacetime

13.1 The near-horizon limit of D3-branes

The big picture

A stack of N coincident D3-branes admits two descriptions. Seen from the worldvolume it is a hyperplane where open strings end, carrying a four-dimensional U(N) gauge theory. Seen from far away it is a heavy, charged source that curves the ten-dimensional geometry, digging a throat whose interior is $\mathrm{AdS}_5 \times S^5$ threaded by N units of five-form flux. Both are descriptions of the same object; which one is under perturbative control is decided by the single dimensionless combination $g_s N$. This section builds the geometry and reads off the two length scales, the string length ℓ_s and the curvature radius L, that organise everything after it.

We built the D3-brane twice in Chapter 6: once as the surface on which open strings end, and once as a $\frac{1}{2}$-BPS carrier of Ramond-Ramond charge. The first construction hands us a field theory. At low energies the open strings on N coincident branes reduce to a U(N) gauge theory in $3+1$ dimensions, and for the D3-brane the worldvolume coupling

$$g_{\mathrm{YM}}^2 = 2\pi g_s \tag{13.1}$$

is dimensionless (Eq. (6.37)), so this is an honest, marginal four-dimensional gauge theory rather than an effective description with a built-in cut-off. That theory is $\mathcal{N} = 4$ super Yang-Mills, and we return to it in Sections 13.2 and 13.3.

The second construction is the subject of this section. A single D3-brane is light in the sense that its backreaction on spacetime is of order g_s; a stack of N of them curves spacetime by an amount controlled by $g_s N$, and once $g_s N$ is large the flat-space picture of a thin sheet in Minkowski space is no longer the right one. The branes have dug a well in the metric, and we want its shape, in particular what it looks like far down the throat.

13.1.1 The extremal D3-brane solution

The metric, dilaton and Ramond-Ramond field sourced by N coincident static D3-branes solve the type IIB equations of motion of Section 5.4. Write the ten coordinates as four worldvolume directions x^μ ($\mu = 0, 1, 2, 3$) along the branes and six transverse directions, and let r be the radial distance in the transverse $\mathbb{R}^6$ with $\mathrm{d}\Omega_5^2$ the metric on the unit five-sphere at fixed r. In

the string frame the solution is

$$\mathrm{d}s^2 \;=\; H(r)^{-1/2}\,\eta_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu \;+\; H(r)^{1/2}\big(\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega_5^2\big), \tag{13.2}$$

with a constant dilaton and a self-dual five-form,

$$e^{\phi} = g_s, \qquad F_5 = (1+\star)\,\mathrm{d}C_4, \qquad C_4 = \big(H(r)^{-1} - 1\big)\,\mathrm{d}x^0 \wedge \mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3. \tag{13.3}$$

Because the dilaton is constant, $e^{\phi} = g_s$, the string frame and the Einstein frame here differ only by an overall constant Weyl rescaling, so nothing below depends on which one we work in. The single function that carries all the remaining information is $H(r)$. It is harmonic in the transverse space,

$$H(r) = 1 + \frac{L^4}{r^4}, \qquad \nabla^2_{(6)}\,H = 0 \quad (r \neq 0), \tag{13.4}$$

because L^4/r^4 is exactly the Green's function of the Laplacian in the six transverse dimensions (the potential of a point charge falls as $1/r^{d-2} = 1/r^4$ in $d = 6$). The constant term is the boundary condition that space is asymptotically flat: as $r \to \infty$, $H \to 1$ and Eq. (13.2) becomes ten-dimensional Minkowski space. The delta-function source at $r = 0$ is the stack of branes.

Three features of Eqs. (13.2) to (13.3) deserve comment before we take any limit. The warp factor $H^{-1/2}$ multiplying the worldvolume is the gravitational redshift: a clock at fixed r runs slow relative to one at infinity by $H^{-1/4}$, and the redshift grows without bound as $r \to 0$. The five-form is electric along the worldvolume through C_4 and, by the self-duality Eq. (5.50) of type IIB, magnetic through the transverse sphere; in ten-dimensional Lorentzian signature $\star^2 = +1$ on a five-form, so a real self-dual F_5 is consistent (see Section 5.4), and the D3-brane carries equal electric and magnetic five-form charge. The dilaton does not run: $e^{\phi} = g_s$ everywhere. That last fact is special to $p = 3$, and it is the reason the D3-brane, alone among the Dp-branes, produces a clean anti-de Sitter geometry rather than a geometry with a position-dependent string coupling.

The coefficient L^4 is fixed by how much charge sits at $r = 0$. Gauss's law for the five-form reads

$$\frac{1}{2\kappa_{10}^2}\int_{S^5} \star F_5 \;=\; N\,T_3, \tag{13.5}$$

where S^5 is any sphere enclosing the branes, $T_3 = \big((2\pi)^3 g_s \alpha'^2\big)^{-1}$ is the D3-brane tension of Eq. (6.35) and, by the BPS equality $\mu_3 = T_3$, also its Ramond-Ramond charge, and the integer N is forced by Dirac quantisation: the flux through S^5 can only come in whole units of a single brane's charge. Balancing the two sides gives the master relation of the whole subject,

$$\boxed{L^4 = 4\pi\, g_s N\, \alpha'^2}\,. \tag{13.6}$$

The size of the geometry the branes create is set by $g_s N$, not by g_s or N separately, and it is measured in string units $\alpha' = \ell_s^2$. The worked example below fills in the flux integral; for most of what follows the parametric statement $L^4 \sim g_s N\,\alpha'^2$ is enough.

Example 13.1 (Flux quantisation fixes the radius)**.** The five-form Eq. (13.3) has an electric part along the worldvolume and, by self-duality, a magnetic part through the transverse sphere. Writing the electric piece as

$$\mathrm{d}C_4 = \partial_r\big(H^{-1}\big)\,\mathrm{d}r \wedge \mathrm{d}x^0 \wedge \mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3, \qquad \partial_r\big(H^{-1}\big) = \frac{4L^4}{r^5 H^2}, \tag{13.7}$$

the magnetic part is its Hodge dual $\star\,\mathrm{d}C_4$, a five-form on the angular directions. Taking the star with the warp factors of Eq. (13.2), the powers of H and the explicit r-dependence cancel, leaving

$$\star F_5\big|_{S^5} = 4L^4\,\omega_{S^5}, \tag{13.8}$$

with ω_{S^5} the volume form of the *unit* five-sphere. The cancellation is not an accident: $\mathrm{d}F_5 = 0$ away from the source makes $\int_{S^5} \star F_5$ independent of the radius of the sphere we integrate over. Using $\mathrm{vol}(S^5) = \pi^3$,

$$\int_{S^5} \star F_5 = 4L^4\ \mathrm{vol}(S^5) = 4\pi^3 L^4. \tag{13.9}$$

Insert this into Gauss's law Eq. (13.5) with $2\kappa_{10}^2 = (2\pi)^7 \alpha'^4 g_s^2$ of Eq. (A.6) and $T_3 = ((2\pi)^3 g_s \alpha'^2)^{-1}$:

$$\frac{4\pi^3 L^4}{(2\pi)^7 \alpha'^4 g_s^2} = \frac{N}{(2\pi)^3 g_s \alpha'^2} \quad \Longrightarrow \quad L^4 = \frac{(2\pi)^4 \alpha'^2 g_s}{4\pi^3}\, N = 4\pi\, g_s N\, \alpha'^2, \tag{13.10}$$

reproducing Eq. (13.6).

Remark 13.2 (Beyond the lectures: why $p = 3$ is the clean case). For a general Dp-brane the harmonic function is $H = 1 + (L/r)^{7-p}$ in the $9-p$ transverse dimensions, and the dilaton profile is $e^{\phi} = g_s\, H^{(3-p)/4}$. The exponent vanishes precisely at $p = 3$, which is why the D3 dilaton is constant. For $p \neq 3$ the string coupling varies across the throat, the near-horizon geometry is only conformally $\mathrm{AdS}_{p+2} \times S^{8-p}$, and the dual field theory is not conformal: its coupling $g_{\mathrm{YM}}^2 \sim g_s\, \alpha'^{(p-3)/2}$ is dimensionful, so it runs. Those cases (the Dp/(p+1)-dimensional super Yang-Mills dualities) are perfectly real and useful, but they lack the exact scale invariance that makes $\mathrm{AdS}_5 \times S^5$ and $\mathcal{N} = 4$ super Yang-Mills the sharpest example. The constant dilaton, the exact anti-de Sitter factor and the marginal gauge coupling are three faces of the same accident at $p = 3$; Exercise 13.5 works the general case out.

13.1.2 The near-horizon throat

The solution Eq. (13.2) has two regimes separated by the crossover scale $r \sim L$. For $r \gg L$ the harmonic function is essentially 1 and the geometry is flat ten-dimensional space with a barely perceptible dent. For $r \ll L$ the 1 in H is negligible against L^4/r^4, and the metric changes character completely. Dropping the 1,

$$\mathrm{d}s^2 \ \longrightarrow\ \frac{r^2}{L^2}\,\eta_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu \ +\ \frac{L^2}{r^2}\,\mathrm{d}r^2 \ +\ L^2\,\mathrm{d}\Omega_5^2. \tag{13.11}$$

The last term is a round five-sphere of fixed radius L: it does not shrink as we go down the throat. The first two terms are anti-de Sitter space. Introducing the coordinate $z = L^2/r$, so that the brane region $r \to 0$ becomes $z \to \infty$ and the crossover $r \to \infty$ becomes the boundary $z \to 0$, they become

$$\mathrm{d}s^2_{\mathrm{AdS}_5} \ =\ \frac{L^2}{z^2}\Big(\eta_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu + \mathrm{d}z^2\Big), \tag{13.12}$$

the Poincaré patch of AdS_5 with curvature radius L. The near-horizon geometry of the D3-brane stack is therefore

$$\mathrm{AdS}_5 \times S^5, \qquad R_{\mathrm{AdS}_5} = R_{S^5} = L, \tag{13.13}$$

threaded by the same N units of five-form flux Eq. (13.5) that fixed L: two factors of equal radius L, with the flux holding up the sphere. The two radii are equal for a physical reason, not by a choice of notation: the same H warps both factors, and the five-form flux that curves the sphere positively is exactly what curves the anti-de Sitter factor negatively by the same amount. This is the Freund-Rubin mechanism, and Exercise 13.4 checks that the ten-dimensional Einstein equations force $R_{\mathrm{AdS}_5} = R_{S^5}$.

Two properties of this throat matter for the story. First, it is completely regular, and infinitely deep. The proper radial distance from a point at r up to a fixed reference radius r_0 in the throat,

$$\int_r^{r_0} \sqrt{g_{rr}}\,\mathrm{d}r' = \int_r^{r_0} \frac{L}{r'}\,\mathrm{d}r' = L \ln \frac{r_0}{r} \ \longrightarrow\ +\infty \quad \text{as } r \to 0, \tag{13.14}$$

diverges: the region $r = 0$ sits at infinite proper distance down a throat of constant cross-sectional radius L. There is no curvature singularity, unlike the interiors of the other extremal branes. Second, the redshift noted below Eq. (13.3) becomes infinite here: an excitation of fixed proper energy sitting at radius r deep in the throat is seen from the asymptotic region to have energy suppressed by $H^{-1/4} \to r/L \to 0$. Modes far down the throat are therefore arbitrarily soft to an outside observer, and that observation is the physical seed of Maldacena's decoupling argument, which we develop in Section 13.2.

The near-horizon limit also enhances the supersymmetry. The sixteen supercharges preserved by the full solution Eq. (13.2) are the standard $\frac{1}{2}$-BPS count of Chapter 6, half of the thirty-two of type IIB. That they double back to thirty-two in the limit Eq. (13.11) is a Killing-spinor computation with the type IIB supersymmetry variations; we state it rather than derive it here (see the BPS discussion of Chapter 6). With the full complement restored, $\mathrm{AdS}_5 \times S^5$ is, together with flat space and a plane-wave limit of it, one of the maximally supersymmetric backgrounds of type IIB. The extra sixteen supercharges are conformal supersymmetries, and their appearance is the geometric shadow of the fact that the dual gauge theory is superconformal. The isometry group of the throat is likewise larger than that of the asymptotically flat brane: AdS_5 has isometry $\mathrm{SO}(4,2)$ and S^5 has isometry $\mathrm{SO}(6)$, and we will match these onto the conformal group and the R-symmetry of $\mathcal{N} = 4$ super Yang-Mills in Section 13.3. For now they are simply the symmetries of the geometry we have just derived.

Remark 13.3 (Beyond the lectures: the ten-dimensional cousin of the extremal throat). The infinitely deep, regular throat of Eq. (13.14) is the higher-dimensional analogue of the near-horizon region of an extremal Reissner-Nordström black hole, whose geometry factorises into $\mathrm{AdS}_2 \times S^2$ (the Bertotti-Robinson universe) at fixed sphere radius. In both cases extremality is what keeps the sphere from collapsing and turns the horizon into an infinite throat rather than a point. The extremal Reissner-Nordström throat is where the microstate counting of Chapter 12 lives; the D3 throat of this section is where holography lives. They are two instances of the same statement that a BPS horizon grows an AdS factor.

13.1.3 Two length scales, two descriptions

The geometry carries two lengths. One is the string length $\ell_s = \sqrt{\alpha'}$, below which the point-particle description of the supergravity fields fails and the full tower of string states must be kept. The other is the curvature radius L of the throat. Their ratio is fixed by Eq. (13.6),

$$\left(\frac{L}{\ell_s}\right)^4 = 4\pi\, g_s N = 2\lambda, \qquad \lambda \coloneqq g_{\mathrm{YM}}^2 N = 2\pi g_s N, \tag{13.15}$$

where λ is the 't Hooft coupling built from Eq. (13.1). Classical supergravity on the throat is trustworthy when its curvature is gentle in string units, $L \gg \ell_s$, which by Eq. (13.15) means $g_s N \gg 1$, equivalently $\lambda \gg 1$. That is precisely the regime in which the α' corrections to supergravity (higher-derivative terms sourced by massive string states) are small. String loop corrections are separately controlled by g_s: for the classical geometry to make sense on top of $L \gg \ell_s$ we also need $g_s \ll 1$. The two conditions coexist only if

$$g_s \ll 1 \ll g_s N, \qquad \text{i.e.} \qquad N \to \infty \text{ with } \lambda = 2\pi g_s N \text{ large and fixed.} \tag{13.16}$$

Classical, weakly curved supergravity on $\mathrm{AdS}_5 \times S^5$ is the description of N D3-branes in the large-N, strong-'t Hooft-coupling corner.

A factor of two in the conventions

We follow the D-brane normalisation of Chapter 6, in which $g_{\mathrm{YM}}^2 = 2\pi g_s$, so that $L^4/\alpha'^2 = 4\pi g_s N = 2\lambda$. Many references (including Maldacena's) absorb a factor of two into the gauge coupling, writing $g_{\mathrm{YM}}^2 = 4\pi g_s$ and hence the tidier $L^4/\alpha'^2 = g_{\mathrm{YM}}^2 N = \lambda$. The geometry is the same; only the dictionary entry $g_{\mathrm{YM}}^2(g_s)$ differs. When you compare Eq. (13.15) with an outside source, check which convention for g_{YM}^2 it uses before trusting a bare numerical factor. The flux relation Eq. (13.6), written in g_s and N alone, is convention-independent.

Put the two constructions of the D3-brane side by side. The open-string picture, a U(N) gauge theory on a flat sheet, is weakly coupled and reliable when the effective planar coupling $g_s N = \lambda/2\pi$ is small. The closed-string picture, the curved throat Eq. (13.13), is weakly curved and reliable when $g_s N$ is large. The same object is under perturbative control in complementary regimes of the one parameter $g_s N$, and there is no window in which both descriptions are simultaneously weakly coupled. This is what gives the correspondence its power: a hard, strongly coupled question on one side becomes an easy, classical question on the other. The decoupling argument is what turns "two descriptions of one object" into "one theory equals another theory", and we take it up next in Section 13.2. The dictionary of Section 13.3 then matches single-trace operators to single supergravity states and pins down the leading large-N counting; the full reconnection of the gauge theory's $1/N$ expansion with the closed-string genus expansion of Chapter 4 is the business of Section 13.4, which closes the chapter.

Exercise 13.1 (The throat is anti-de Sitter)**.** *(Short.)*

1. Verify that L^4/r^4 is harmonic in $\mathbb{R}^6$, i.e. that $\nabla^2_{(6)}(r^{-4}) = 0$ for $r \neq 0$, using the radial Laplacian $\nabla^2_{(6)} = r^{-5}\partial_r(r^5\partial_r)$. Why does the power r^{-4} appear rather than r^{-2}?
2. Drop the 1 in Eq. (13.4) and reduce Eq. (13.2) to Eq. (13.11). Then substitute $z = L^2/r$ and confirm Eq. (13.12), reading off that the AdS_5 and S^5 radii are both L.
3. Compute the Ricci tensor of Eq. (13.12) and show $R_{\mu\nu} = -\frac{4}{L^2} g_{\mu\nu}$ on the AdS_5 factor, the defining relation of a five-dimensional space of curvature radius L.

Exercise 13.2 (The magnetic flux through the sphere)**.** *(Moderate.)* Theorem 13.1 quoted $\star F_5|_{S^5} = 4L^4\,\omega_{S^5}$; here you supply the Hodge star and check the counting.

1. Starting from $\mathrm{d}C_4 = \partial_r(H^{-1})\,\mathrm{d}r \wedge \mathrm{d}x^0 \wedge \mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3$, compute its Hodge dual on the metric Eq. (13.2) and confirm that the magnetic part of F_5 is $4L^4$ times the unit-sphere volume form, with the explicit r-dependence cancelling.
2. Repeat the flux integral on the full solution, not just the near-horizon region, and show that $\int_{S^5} \star F_5$ takes the same value for every sphere enclosing the branes, so that the integer N is well defined.
3. Rewrite $L^4 = 4\pi g_s N \alpha'^2$ as $(L/\ell_s)^4 = 4\pi g_s N$ and explain why the supergravity description improves, rather than degrades, as the number of branes grows.

Exercise 13.3 (Redshift down the throat)**.** *(Short.)* A static excitation deep in the geometry Eq. (13.2) has proper energy E_{proper} measured by a local observer at radius r. Using the timelike warp factor $g_{tt} = -H^{-1/2}$, show that the energy E_∞ assigned to it by an observer at $r \to \infty$ is $E_\infty = H(r)^{-1/4} E_{\mathrm{proper}}$. Evaluate $E_\infty/E_{\mathrm{proper}}$ in the near-horizon region and show it tends to $r/L \to 0$ as $r \to 0$. Explain in one sentence why this makes the throat modes look like a decoupled sector of arbitrarily low energy to the asymptotically flat region, anticipating Section 13.2.

Exercise 13.4 (Freund-Rubin and the equal radii)**.** *(Moderate.)* Take the ten-dimensional

Einstein equations sourced only by a self-dual five-form, $R_{MN} = \frac{1}{4\cdot 4!} F_{MPQRS} F_N{}^{PQRS}$ (the trace piece cancels for a self-dual five-form in ten dimensions). Adopt the product ansatz $\mathrm{AdS}_5 \times S^5$ with radii a and b and a five-form proportional to the S^5 volume form with N units of flux. Show that the Einstein equations force $R_{\mu\nu} = -\frac{4}{a^2} g_{\mu\nu}$ on AdS_5 and $R_{mn} = +\frac{4}{b^2} g_{mn}$ on S^5 with $a = b$, so that the two curvature radii are equal, and relate the common radius to N. This is the Freund-Rubin statement that flux supports the geometry with the anti-de Sitter and sphere curvatures balanced.

Exercise 13.5 (The general Dp-brane throat)**.** *(Long.)* A stack of N Dp-branes has $9-p$ transverse dimensions and harmonic function $H = 1 + (L/r)^{7-p}$, string-frame dilaton $e^{\phi} = g_s\, H^{(3-p)/4}$, and the metric Eq. (13.2) with $H^{-1/2}$ over the $p+1$ worldvolume directions and $H^{1/2}$ over the transverse $\mathbb{R}^{9-p}$.

1. Show that H is harmonic away from the branes, i.e. that $r^{-(7-p)}$ solves the flat Laplacian $\nabla^2_{(9-p)} = r^{-(8-p)} \partial_r \big(r^{8-p} \partial_r \big)$ for $r \neq 0$. Recover the D3 case r^{-4}.
2. Take the near-horizon limit $r \ll L$ and show that the metric is conformal to $\mathrm{AdS}_{p+2} \times S^{8-p}$, becoming exactly $\mathrm{AdS}_{p+2} \times S^{8-p}$ (with constant dilaton) only at $p = 3$. Identify the power of H that spoils the anti-de Sitter form for $p \neq 3$.
3. The dual $(p+1)$-dimensional super Yang-Mills coupling is $g^2_{\mathrm{YM}} \sim g_s\, \alpha'^{(p-3)/2}$. Show it is dimensionful for $p \neq 3$ and dimensionless only at $p = 3$, and connect this to whether the dual coupling runs or is exactly scale invariant. Which single feature at $p = 3$ ties together the constant dilaton, the exact AdS factor and the marginal coupling?

13.2 Maldacena's decoupling argument

The big picture

A stack of N D3-branes is one physical object with two honest descriptions. Seen through the open strings that end on it, it is a four-dimensional gauge theory: $\mathcal{N} = 4$ super-Yang-Mills with gauge group $\mathrm{U}(N)$. Seen through the closed strings it sources, it is a lump of curved spacetime whose throat is $\mathrm{AdS}_5 \times S^5$. Take the low-energy limit of each description. In both, the physics splits into a piece that lives far from the brane, free supergravity in flat space, and a piece that carries all the interesting dynamics. The far piece is literally the same in the two descriptions. Cancel it, and what is left over on each side must agree:

$$\mathcal{N} = 4 \ \mathrm{U}(N) \text{ super-Yang-Mills} \quad \Longleftrightarrow \quad \text{type IIB string theory on } \mathrm{AdS}_5 \times S^5.$$

This section runs that argument and records the map between the two sets of parameters.

Section 13.1 did the geometry: N coincident D3-branes in type IIB backreact into the extremal three-brane solution, and the region close to the branes, the throat, is $\mathrm{AdS}_5 \times S^5$ with the two factors sharing one radius L set by

$$L^4 = 4\pi\, g_s\, N\, \alpha'^2, \tag{13.17}$$

and N units of five-form flux threading the S^5. That is a statement about a classical background. Maldacena's argument turns it into a statement about a *duality*: the near-horizon string theory and the gauge theory on the branes are two names for one quantum theory. The logic is a piece of physical reasoning, not a theorem, and I will flag where it asserts rather than proves.

13.2.1 A D-brane has two faces

The argument exploits a tension built up in Sections 6.2 and 6.3 and already made quantitative in Section 13.1. A stack of N D3-branes wears two faces at once. Through the open strings that end on it (Section 6.2) it is a U(N) gauge theory, the massless open-string modes assembling into ten-dimensional $\mathcal{N}=1$ super-Yang-Mills reduced to the $3+1$ worldvolume, that is $\mathcal{N}=4$ super-Yang-Mills, with the action (6.32) and dimensionless coupling

$$g_{\rm YM}^2 = 2\pi g_s \tag{13.18}$$

read off from the Dirac-Born-Infeld expansion (6.37). Through the closed strings it sources (Section 6.3) the same stack is the electric source of the Ramond-Ramond potential C_4, a BPS object of tension $T_3 \sim 1/(g_s \ell_s^4)$, and its backreaction curves spacetime into the extremal three-brane solution of Section 13.1. Both faces are real at once.

Which of them is the *useful* description is fixed by a single dimensionless number, $g_s N$, the strength with which the stack gravitates. Section 13.1 settled this: the throat radius (13.17), in string units $L^4/\ell_s^4 = 4\pi g_s N$, is small when $g_s N \ll 1$, so the geometry outside the branes is barely curved and the honest description is the open-string one, a gauge theory on a brane in nearly flat space; it is large when $g_s N \gg 1$, so the throat is deep and weakly curved and the honest description is the closed-string one, strings in a smooth $\mathrm{AdS}_5 \times S^5$. The two faces are accurate in opposite corners of $g_s N$, with no obvious window where both hold.

What the decoupling argument adds is a way to compare them anyway. Rather than sit at fixed $g_s N$, take the *low-energy* limit of each description. In that limit each one splits into a piece far from the branes and a piece carrying the near-brane dynamics, the far piece turns out to be identical on the two sides, and removing it leaves the two near-brane pieces facing each other. Running that manoeuvre carefully is the business of the rest of the section.

13.2.2 The system, and the limit we take

Fix the setup once. We have N coincident D3-branes in type IIB string theory in flat ten-dimensional Minkowski space. This is a fully-fledged (if formidable) quantum system: closed strings roaming the bulk, open strings ending on the branes, and every interaction string theory allows between them. We are going to ask for its *low-energy* physics, meaning we keep only excitations whose energy E stays fixed while we send the string scale to infinity,

$$\alpha' \to 0 \qquad \text{with the gauge-theory data } g_{\rm YM},\ N,\ \text{and the energies } E \text{ held fixed.} \tag{13.19}$$

Holding $g_{\rm YM}^2 = 2\pi g_s$ fixed keeps g_s fixed by (13.18). The combination we send to zero is α', equivalently ℓ_s, equivalently the energy $1/\ell_s$ of the first massive string state. This is the same low-energy limit that reduces any string theory to its massless field theory; the novelty is watching what it does to the brane system in each of the two descriptions.

13.2.3 Description A: open strings, and the gauge theory

Start from the branes-in-flat-space picture. The full Lagrangian has three kinds of term,

$$\mathcal{L} = \mathcal{L}_{\rm bulk} + \mathcal{L}_{\rm brane} + \mathcal{L}_{\rm int}, \tag{13.20}$$

the closed strings in the bulk, the open strings on the branes, and the couplings between them. Track each through the limit (13.19).

The bulk term $\mathcal{L}_{\rm bulk}$ is the ten-dimensional closed-string theory. At low energy its massive strings are frozen out and it collapses to type IIB supergravity. Expanding the supergravity action around flat space, every interaction vertex carries a power of the gravitational coupling

$\kappa_{10} \sim g_s \alpha'^2$, so as $\alpha' \to 0$ the gravitons stop interacting: the bulk becomes *free* type IIB supergravity in flat ten-dimensional space.

The brane term $\mathcal{L}_{\text{brane}}$ is the Dirac-Born-Infeld plus Wess-Zumino action of Section 6.2. Its massless content is $\mathcal{N} = 4$ super-Yang-Mills with coupling (13.18); its higher terms are the infinite tower of α' corrections resummed by the Born-Infeld square root, each suppressed by a power of α' times a worldvolume field strength. In the limit those corrections switch off and the brane theory becomes *exactly* $\mathcal{N} = 4$ super-Yang-Mills, with no stringy tail. Because g_{YM} is dimensionless the gauge dynamics is untouched by $\alpha' \to 0$, so what survives is an interacting four-dimensional theory that comes through the limit intact.

The interaction term $\mathcal{L}_{\text{int}}$ couples brane fields to bulk gravitons. It too is controlled by $\kappa_{10} \sim g_s \alpha'^2$. Restore dimensions: the leading coupling of a canonically normalised bulk graviton h to the brane stress tensor is $\kappa_{10} \int h\, T_{\text{brane}}$, and $\kappa_{10} \to 0$ as $\alpha' \to 0$. The brane and the bulk stop talking. This is the decoupling. Newton's constant vanishes with κ_{10} as $\alpha' \to 0$, while the dimensionless gauge coupling is untouched, so gravity switches itself off and the gauge theory does not.

What remains are two systems that no longer interact:

$$\boxed{\text{(A)} \qquad \underbrace{\mathcal{N} = 4 \text{ U}(N) \text{ super-Yang-Mills}}_{\text{interacting}} \quad \oplus \quad \underbrace{\text{free type IIB supergravity on } \mathbb{R}^{1,9}}_{\text{decoupled bulk}}} \tag{13.21}$$

The gauge group is $\text{U}(N)$. Its diagonal $\text{U}(1)$ is the centre-of-mass position of the whole stack, a free abelian sector that decouples from the rest; the interacting dynamics lives in the $\text{SU}(N)$ part, and it is that part the dictionary of Section 13.3 maps to the string theory. We keep the full $\text{U}(N)$ here, following the brane count, and return to the free $\text{U}(1)$ in the remark at the end of the section.

13.2.4 Description B: closed strings, and the geometry

Now describe the very same N D3-branes as the supergravity solution they source. We write the solution in string frame, the metric the string worldsheet couples to directly, related to the Einstein-frame metric by a dilaton-dependent Weyl rescaling (Chapter 5). For the D3-brane the dilaton is constant, so the two frames differ only by an overall constant factor, and every conclusion below, the $\text{AdS}_5 \times S^5$ throat and the redshift that keeps it, is frame-independent. The extremal three-brane metric of Section 13.1 is

$$\mathrm{d}s^2 = H(r)^{-1/2}\, \eta_{\mu\nu}\, \mathrm{d}x^\mu \mathrm{d}x^\nu + H(r)^{1/2} (\mathrm{d}r^2 + r^2\, \mathrm{d}\Omega_5^2), \qquad H(r) = 1 + \frac{L^4}{r^4}, \tag{13.22}$$

with $\mu, \nu = 0, \ldots, 3$ along the brane and r the transverse radial coordinate. There is no brane theory to add by hand here: in this description the branes *are* the geometry, and everything is closed strings moving in (13.22). Ask again for the low-energy excitations, energy measured by an observer at infinity where the space is asymptotically flat and $H \to 1$.

Two kinds of excitation look soft to that asymptotic observer. The first is obvious: long-wavelength closed strings out in the flat asymptotic region, whose energy is small because their wavelength is large. The second is the interesting one and turns on the gravitational redshift. A static clock at radius r runs at proper rate

$$\mathrm{d}\tau = \sqrt{-g_{tt}}\, \mathrm{d}t = H(r)^{-1/4}\, \mathrm{d}t, \tag{13.23}$$

so an excitation of proper (locally measured) energy E_{loc} registers at infinity, where energy is conjugate to the Killing time t, as

$$E_\infty = H(r)^{-1/4}\, E_{\text{loc}}. \tag{13.24}$$

Deep in the throat, $r \ll L$, we have $H \simeq L^4/r^4$ and $H^{-1/4} \simeq r/L$, so

$$E_\infty \simeq \frac{r}{L} E_{\rm loc} \xrightarrow{r\to 0} 0. \tag{13.25}$$

An object of *any* proper energy, including a heavily excited string of energy $\sim 1/\ell_s$, looks arbitrarily soft from infinity once it sits deep enough in the throat. So the low-energy spectrum seen from infinity contains the entire tower of throat excitations, no matter how energetic they are locally, provided they live at small enough r. We are obliged to keep them: they are a genuine part of the low-energy physics, not an approximation we are free to drop.

These two families decouple from each other in the limit. The long-wavelength bulk modes have wavelengths far larger than L; to them the throat is a point, and the low-energy cross-section for a bulk field of frequency ω to be absorbed into it scales as $\sigma \propto \omega^3 L^8 \to 0$ (the greybody computation of Klebanov (1997) and of Gubser, Klebanov and Tseytlin; see Exercise 13.7). From the other side, a throat excitation must climb an ever-steepening gravitational potential to escape, and cannot. The mouth of the throat pinches off, and what survives is again two non-interacting systems:

$$\boxed{\text{(B)} \qquad \underbrace{\text{full type IIB string theory on } \mathrm{AdS}_5 \times S^5}_{\text{throat}} \;\oplus\; \underbrace{\text{free type IIB supergravity on } \mathbb{R}^{1,9}}_{\text{decoupled bulk}}} \tag{13.26}$$

The throat factor is the whole string theory in the near-horizon region, including its massive modes, precisely because the softening (13.25) let us keep arbitrarily energetic local excitations. The bulk factor is, once more, free supergravity in asymptotically flat space.

13.2.5 Equating the two descriptions

Line up (13.21) and (13.26). Each is the low-energy limit of one and the same system, N D3-branes in type IIB, so they must describe the same physics. Each has split into two decoupled factors, and one factor is identical on the two sides: free type IIB supergravity in flat ten-dimensional space, the modes that live far from the branes and never knew about them. Remove that common spectator. The remaining factors have nowhere else to go but to each other:

$$\boxed{\begin{aligned} &\mathcal{N}=4 \text{ U}(N) \text{ super-Yang-Mills in } 3+1 \text{ dimensions} \\ &\qquad = \quad \text{type IIB string theory on } \mathrm{AdS}_5 \times S^5. \end{aligned}} \tag{13.27}$$

This is Maldacena's correspondence. The gauge theory has no gravity in it and lives in four flat dimensions; the string theory is a full ten-dimensional theory of quantum gravity. The equality (13.27) says they are the same quantum theory written in two languages. The four dimensions of the gauge theory are the four flat directions x^μ shared by the branes and by the boundary of AdS_5; the extra dimensions of the string theory, the radial direction of AdS_5 and the S^5, are reconstructed by the gauge theory rather than put in by hand. That reconstruction is the subject of Sections 13.3 and 13.4.

This is an argument, not a proof

Two steps in the reasoning are physical rather than rigorous. First, that the low-energy limit cleanly decouples the throat from the asymptotic bulk on *both* sides is checked to leading order (the vanishing absorption cross-section, the redshift) and believed to survive, but it is not proven to all orders. Second, and more basic, descriptions A and B are each reliable only in their own corner of g_sN, and (13.27) extrapolates the identification across the whole range. The correspondence is a conjecture. What makes it credible is that the D3

system is half-BPS, so a great deal of its structure is protected against the extrapolation (Section 5.6.4), and that every quantity anyone has managed to compute on both sides has agreed. It has not been derived from more basic axioms.

13.2.6 The parameter map

Each side of (13.27) carries two dimensionless parameters, and the correspondence identifies them. On the gauge side they are the coupling $g_{\rm YM}$ and the rank N, most usefully packaged as the 't Hooft coupling and the rank,

$$\lambda \coloneqq g_{\rm YM}^2 N, \qquad N. \tag{13.28}$$

On the string side they are the string coupling g_s and the size of the geometry in string units, L/ℓ_s, equivalently L^4/α'^2. The two identifications we already have, the D-brane coupling (13.18) and the throat radius (13.17), are exactly the map:

$$g_s = \frac{g_{\rm YM}^2}{2\pi}, \qquad \frac{L^4}{\alpha'^2} = 4\pi\, g_s N = 2\, g_{\rm YM}^2 N = 2\,\lambda. \tag{13.29}$$

Parametrically this is the pair of relations the correspondence is usually quoted with,

$$g_s \sim g_{\rm YM}^2, \qquad \frac{L^4}{\alpha'^2} \sim g_{\rm YM}^2 N = \lambda. \tag{13.30}$$

The order-one coefficients in (13.29) are convention-dependent: some references absorb factors of 2π into the definition of $g_{\rm YM}$ so that $L^4/\alpha'^2 = \lambda$ exactly. We track only the parametric content, which is coefficient-independent and is what carries the physics. The consequence that matters most is what (13.29) says about the string coupling at fixed λ: eliminating g_s in favour of λ and N,

$$g_s = \frac{g_{\rm YM}^2}{2\pi} = \frac{\lambda}{2\pi N} \quad\Longrightarrow\quad g_s \sim \frac{\lambda}{N} \xrightarrow{\ \lambda \text{ fixed}\ } \frac{1}{N}. \tag{13.31}$$

Holding λ fixed and sending $N \to \infty$ drives $g_s \to 0$: the 't Hooft limit of the gauge theory is the classical (tree-level, no handles) limit of the string. This is the hinge on which Section 13.4 hangs the identification of the $1/N$ expansion with the genus expansion of Section 4.1.2; we set it up here and develop it there.

When is classical gravity a good description? The point of (13.27) that gets used in practice is that classical type IIB supergravity on $\mathrm{AdS}_5 \times S^5$ computes the strongly-coupled, large-N gauge theory. Two conditions must hold for that classical description to be reliable. The geometry must be weakly curved, so that α' corrections (higher-derivative terms, massive strings) are small, which by (13.29) means

$$\frac{L}{\ell_s} = (2\lambda)^{1/4} \gg 1 \quad\Longleftrightarrow\quad \lambda \gg 1. \tag{13.32}$$

And the string coupling must be small, so that string loops (handles) are suppressed, which by (13.31) means $g_s = \lambda/2\pi N \ll 1$, that is $\lambda \ll N$. Together,

$$1 \ll \lambda \ll N. \tag{13.33}$$

In this window the gauge theory is strongly coupled ($\lambda \gg 1$) and effectively planar ($N \to \infty$), exactly the regime where its Feynman diagrams are useless, while the dual is weakly curved, weakly coupled classical gravity, exactly where *its* calculations are easy. This inversion, hard on the gauge side and easy on the gravity side, is what makes the correspondence a working tool.

Regime	Gauge theory	String theory
$\lambda \ll 1$	perturbative $\mathcal{N}=4$ SYM	$L \ll \ell_s$: strongly curved, no classical geometry
$1 \ll \lambda \ll N$	strong coupling, planar	$L \gg \ell_s$, $g_s \ll 1$: classical IIB supergravity
$\lambda \lesssim N$, finite N	$1/N$ corrections	$g_s \sim \lambda/N$: string loops (genus expansion)

Remark 13.4 (Beyond the lectures: how strong a claim is being made). Equation (13.27) comes in graded strengths, and the literature is careful to say which is meant. The *strongest* form asserts exact equality of the two theories for all $g_{\rm YM}$ and all N, including finite N and all α' and g_s corrections. A *milder* form keeps $N \to \infty$ but all λ, matching the full classical string ($g_s = 0$) on $\mathrm{AdS}_5 \times S^5$ to the planar gauge theory. The *weakest* and most-tested form is the supergravity limit (13.33), matching two-derivative IIB supergravity to strongly-coupled planar SYM. Essentially all sharp checks live in the last regime or exploit BPS/supersymmetric non-renormalisation to reach the first. The decoupling argument as given supports the strong form; the evidence is strongest for the weak one. Two technical points a careful treatment does not sweep away: the free centre-of-mass U(1) noted under (13.21) is a genuine subtlety in matching U(N) against SU(N) across the duality, and the clean decoupling of the throat can be spoilt for non-conformal branes (Dp with $p \neq 3$), where the dual geometry is not AdS and the field theory is not a fixed-point CFT. The D3 case is special because $g_{\rm YM}$ is dimensionless and the theory is exactly conformal, its β function vanishing to all orders (Sohnius and West, 1981; see Section 13.3), which is what makes the near-horizon geometry pure AdS.

Exercise 13.6 (How deep the throat must be). *(Short.)* The redshift relation (13.24) was set up from the metric in Section 13.1; here we push it into the decoupling limit. A string oscillator of proper energy $E_{\rm loc} \sim 1/\ell_s$ sitting at radius $r \ll L$ registers at infinity as $E_\infty \sim r/(L\ell_s)$ by (13.25). Holding E_∞ fixed as $\alpha' \to 0$, and using $L \propto \alpha'^{1/2}(g_s N)^{1/4}$ from (13.17), find how r must scale with α'. Explain in one sentence why the ability to keep such excitations, at ever smaller r, is what lets the throat description (13.26) retain the full string theory rather than only its supergravity modes.

Exercise 13.7 (Gravity switches itself off). *(Moderate.)* This runs the mechanism that (13.21) and (13.26) rely on.

(a) The leading coupling of a canonically normalised bulk graviton $h_{\mu\nu}$ to the branes is $\kappa_{10}\int \mathrm{d}^4x\, h_{\mu\nu}\, T^{\mu\nu}_{\rm brane}$, with $2\kappa_{10}^2 = (2\pi)^7 \alpha'^4 g_s^2$. Show $\kappa_{10} \propto g_s\, \alpha'^2$, and hence that at fixed $g_{\rm YM}$ (equivalently fixed g_s) it vanishes as $\alpha' \to 0$. Conclude that $\mathcal{L}_{\rm int}$ in (13.20) switches off, so the brane decouples from the bulk.

(b) From the closed-string side the same statement is the vanishing of the throat's absorption. The low-energy cross-section for a bulk field of frequency ω to be captured by the branes is $\sigma \propto \omega^3 L^8$ (Klebanov, 1997). Using $L^4 = 4\pi g_s N\, \alpha'^2$, show that at fixed $g_s N$ and fixed ω one has $\sigma \propto \alpha'^4 \to 0$, so the throat and the asymptotic bulk stop exchanging energy.

(c) Both (13.21) and (13.26) are a direct sum of two non-interacting theories. Identify the factor common to the two, and explain why removing it forces the remaining factor of (A) to equal the remaining factor of (B).

Exercise 13.8 (The supergravity window). *(Short.)* Using the map (13.29) and (13.31), express both the curvature radius in string units, L/ℓ_s, and the string coupling g_s in terms of λ and N. Show that demanding $L \gg \ell_s$ and $g_s \ll 1$ gives the double inequality (13.33). Which inequality fails first as you turn λ down at fixed large N, and what stringy effect takes over there? Which fails as you lower N at fixed large λ, and what does that correspond to on the string side?

Where this is going

We now have the central claim, (13.27), and the dictionary of couplings, (13.29). What we have not done is say how a specific string field maps to a specific gauge-invariant operator, nor why the symmetries match: the isometries of $\mathrm{AdS}_5 \times S^5$ had better be the conformal and R-symmetries of $\mathcal{N} = 4$ super-Yang-Mills, and they are. That is Section 13.3. After it, Section 13.4 takes the parametric fact (13.31), $g_s \sim 1/N$ at fixed λ, and turns it into the statement that the closed-string genus expansion of Section 4.1.2 *is* the $1/N$ expansion of the gauge theory, so that the extra dimensions and their gravity emerge, order by order, from a theory with neither.

13.3 The dictionary

The big picture

The near-horizon limit (Section 13.1) and the decoupling argument (Section 13.2) delivered a statement of equality: type IIB string theory on $\mathrm{AdS}_5 \times S^5$, threaded by N units of five-form flux, is the same theory as four-dimensional $\mathcal{N} = 4$ super-Yang-Mills with gauge group $\mathrm{SU}(N)$. A statement of equality is not yet a way to compute. This section builds the translation table. Three things have to be pinned down before the equality can be used. The symmetries on the two sides must be literally the same group, so that a state on one side and its image on the other carry the same quantum numbers. Every field propagating in the bulk must be matched to a gauge-invariant operator on the boundary. And there must be a rule that turns a bulk calculation into a boundary correlation function. The first is a counting exercise, the second is representation theory, and the third is the Gubser-Klebanov-Polyakov-Witten relation, which we state and use but do not derive.

Throughout this section it helps to keep one picture of the geometry in mind. Written in the Poincaré coordinates of Section 13.1, the AdS_5 factor is

$$\mathrm{d}s^2_{\mathrm{AdS}_5} = \frac{L^2}{z^2}\Big(\mathrm{d}z^2 + \eta_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu\Big), \qquad z > 0, \quad \mu,\nu = 0,1,2,3, \tag{13.34}$$

with the four-dimensional Minkowski space on which the gauge theory lives sitting at the conformal boundary $z \to 0$. The radius L is the one fixed by the flux in Section 13.1,

$$L^4 = 4\pi g_s N\, \alpha'^2, \tag{13.35}$$

and the same combination controls the gauge theory through $g^2_{\mathrm{YM}} = 2\pi g_s$ (Eq. (6.37)) and the 't Hooft coupling $\lambda := g^2_{\mathrm{YM}} N = 2\pi g_s N$, so that

$$\frac{L^4}{\alpha'^2} = 4\pi g_s N = 2\lambda. \tag{13.36}$$

The gauge group is $\mathrm{SU}(N)$ throughout: the overall centre-of-mass U(1) of the D3-brane stack decoupled in Section 13.2, leaving the interacting $\mathrm{SU}(N)$ theory that the dictionary maps. The bulk is weakly curved, $L \gg \ell_s$, exactly when $\lambda \gg 1$: classical geometry on the string side is strong coupling on the gauge side, which is what makes the correspondence useful, and hard to prove. We will not need the radial direction z to mean anything beyond "distance into the bulk" here; its reading as a renormalisation-group scale is the business of Section 13.4.

13.3.1 The symmetries are the same group

Start with the bosonic symmetries, because they are the ones you can count on your fingers. A geometry has an isometry group, the transformations that leave its metric invariant, and $\mathrm{AdS}_5 \times S^5$ is a product, so its isometry group is a product.

The five-sphere is the easy factor. Its isometries are the rotations of the $\mathbb{R}^6$ it sits inside, the group SO(6). There are $\binom{6}{2} = 15$ of them.

The anti-de Sitter factor is the one worth pausing on. AdS_5 is the hyperboloid $-X_{-1}^2 - X_0^2 + X_1^2 + \cdots + X_4^2 = -L^2$ inside $\mathbb{R}^{2,4}$, and the linear transformations of $\mathbb{R}^{2,4}$ that preserve the signature-$(2,4)$ form preserve the hyperboloid. That group is $\mathrm{SO}(4,2)$, again with 15 generators. From the point of view of the boundary theory these fifteen are not an accident: $\mathrm{SO}(4,2)$ is exactly the conformal group of four-dimensional Minkowski space. It contains the six Lorentz rotations $M_{\mu\nu}$, the four translations P_μ, the four special conformal transformations K_μ, and the single dilatation D, and $6 + 4 + 4 + 1 = 15$. The isometries of the bulk geometry are the conformal transformations of the boundary. An isometry of AdS_5 moves points in the interior and, in the limit $z \to 0$, acts on the boundary Minkowski space as a conformal (angle-preserving, not length-preserving) map. The dilatation D is the one to watch: it is the isometry $z \to \Lambda z$, $x^\mu \to \Lambda x^\mu$ that leaves Eq. (13.34) invariant, and on the boundary it is the scale transformation $x^\mu \to \Lambda x^\mu$.

Now the boundary theory has to supply both factors. That $\mathcal{N} = 4$ super-Yang-Mills is exactly conformal is a theorem: its beta function vanishes identically, to all orders in perturbation theory and beyond (the all-order statement was established by Mandelstam and by Brink, Lindgren and Nilsson), so g_{YM} is an exactly marginal coupling and the theory is scale invariant for every value of it. A scale-invariant, Poincaré-invariant, local and unitary quantum field theory is conformally invariant, so the symmetry group is the full $\mathrm{SO}(4,2)$, strictly larger than Poincaré times dilatations. That accounts for the AdS_5 isometries. The S^5 isometries are accounted for by the R-symmetry. The theory has six real scalars X^i (Appendix C.2.5), the transverse position fields of the D3-brane stack, and they rotate into one another under an SO(6) that acts on the index i. This is the R-symmetry of $\mathcal{N} = 4$, and it is an $\mathrm{SO}(6) \cong \mathrm{SU}(4)$ that rotates the four supercharges as a **4** and the six scalars as a **6** (Appendix C.1.5). The rotations of the sphere the branes span are, on the gauge side, the rotations of the scalars that record where the branes sit.

$$\underbrace{\mathrm{SO}(4,2)}_{\text{isometries of } \mathrm{AdS}_5} \times \underbrace{\mathrm{SO}(6)}_{\text{isometries of } S^5} = \underbrace{\mathrm{SO}(4,2)}_{\text{conformal group}} \times \underbrace{\mathrm{SU}(4)_R}_{\text{R-symmetry}} . \tag{13.37}$$

The supersymmetry closes the match and makes it rigid. $\mathrm{AdS}_5 \times S^5$ with N units of five-form flux is a maximally supersymmetric background of type IIB supergravity: it preserves all 32 supercharges, the same number as flat ten-dimensional space. The flat D3-brane stack preserves only 16; the doubling to the full 32 is a feature of the near-horizon limit (Section 13.1). On the gauge side, $\mathcal{N} = 4$ super-Yang-Mills has 16 Poincaré supercharges, and a superconformal theory has more. The commutator of a Poincaré supercharge Q with a special conformal transformation K produces a new fermionic generator S, the superconformal charge, and there are 16 of these too. Bosonic and fermionic generators together close into a single simple superalgebra,

$$\mathfrak{psu}(2,2|4), \tag{13.38}$$

whose bosonic part is $\mathrm{SO}(4,2) \times \mathrm{SU}(4)_R$ (the 30 generators above) and whose fermionic part is the 32 supercharges Q and S. The isometry supergroup of the $\mathrm{AdS}_5 \times S^5$ background and the superconformal group of $\mathcal{N} = 4$ super-Yang-Mills are the same supergroup $\mathrm{PSU}(2,2|4)$. Every statement in the rest of the dictionary is organised by its representations: a bulk field and its dual operator sit in the same representation of Eq. (13.38), and matching those representations is most of what "checking AdS/CFT" means in practice.

Remark 13.5 (Beyond the lectures: why $\mathfrak{psu}$ and not $\mathfrak{su}$). The superconformal algebra of $\mathcal{N} = 4$ is the projective special unitary superalgebra $\mathfrak{psu}(2,2|4)$, not $\mathfrak{su}(2,2|4)$. The supergroup $\mathrm{SU}(2,2|4)$ contains a U(1) generated by the identity in the fermionic block; this is the anomalous axial

R-symmetry that is broken in the quantum theory. Removing it (the "P") leaves the algebra actually realised on operators. The same U(1) is the one flagged as anomalous for $\mathcal{N}=4$ in Appendix C.1.5, which is why the honest R-symmetry is SU(4) and not U(4). That the algebra is simple, with no ordinary conformal or flavour symmetry factored off, is special to $\mathcal{N}=4$ and is part of why it is the most symmetric interacting theory available in four dimensions.

13.3.2 Fields in the bulk, operators on the boundary

A local operator in the gauge theory carries a scaling dimension Δ, a Lorentz spin, and an $\mathrm{SU}(4)_R$ representation. A field propagating in $\mathrm{AdS}_5 \times S^5$ carries a mass, a spin, and an $\mathrm{SU}(4)_R$ representation coming from how it sits on the S^5. The correspondence pairs them: to each bulk field there is one boundary operator with matching quantum numbers, and the pairing is a bijection on the physical (gauge-invariant, colour-singlet) sector.

Gauge invariance is the crucial filter on the boundary side. The elementary fields of $\mathcal{N}=4$, the gauge field A_μ, the scalars X^i, the fermions, are all $N \times N$ matrices in the adjoint of $\mathrm{SU}(N)$ and are not themselves observables; only colour singlets are. The singlets one builds are traces of products of fields, the single-trace operators

$$\mathcal{O}(x) = \mathrm{Tr}\big[\,\Phi_{a_1}(x)\,\Phi_{a_2}(x)\cdots\Phi_{a_n}(x)\,\big], \tag{13.39}$$

where each Φ_a stands for one of the elementary fields or a covariant derivative of one, together with products of such traces. The single-trace operators are the primaries dual to single-particle states in the bulk; a product of k traces is dual to a k-particle state. This split is exact only at $N=\infty$ and is corrected in $1/N$, the corrections being the interactions that let one bulk particle decay into several; the systematic reading of $1/N$ as the closed-string genus expansion is developed in Section 13.4. Here we need only the leading dictionary, single trace to single particle.

The way a bulk field encodes its operator is through its boundary behaviour. A field $\phi(z,x)$ in AdS_5 does not have a single value at the boundary; it has a fall-off. Solving its wave equation near $z=0$ produces two independent powers of z, and the general solution behaves as

$$\phi(z,x) \;\xrightarrow{z\to 0}\; z^{\,d-\Delta}\big(\phi_0(x) + \mathcal{O}\big(z^2\big)\big) \;+\; z^{\,\Delta}\big(A(x) + \mathcal{O}\big(z^2\big)\big), \qquad d=4. \tag{13.40}$$

The two coefficients play completely different roles. The leading (slower-decaying) coefficient $\phi_0(x)$ is a boundary condition we are free to impose; it is the *source* for the dual operator $\mathcal{O}$. The subleading coefficient $A(x)$ is not free once regularity in the interior is imposed; it is proportional to the *expectation value* $\langle\mathcal{O}(x)\rangle$ in the state prepared by that source. The exponent Δ is the scaling dimension of $\mathcal{O}$, and the next subsection fixes it in terms of the mass of ϕ. For now, read Eq. (13.40) as the statement that a single bulk field carries exactly the data of one boundary operator: the freedom to turn on a source, and the response of the operator to it.

A handful of entries are forced by symmetry alone and are worth knowing by heart. In the table below the first three rows, the stress tensor, the R-current, and the marginal coupling, are fixed by the conserved currents and the modulus of the theory; the last two, the chiral-primary tower and the unprotected operators typified by Konishi, are the ones defined in the paragraphs that follow.

bulk field in AdS_5	boundary operator	Δ	$\mathrm{SU}(4)_R$	origin
graviton $g_{\mu\nu}$	stress tensor $T_{\mu\nu}$	4	**1**	$\mathrm{SO}(4,2)$
$\mathrm{SO}(6)$ gauge field $A_\mu^{[ij]}$	R-current $J^\mu_{[ij]}$	3	**15**	$\mathrm{SU}(4)_R$
axio-dilaton τ_{IIB}	on-shell Lagrangian $\mathcal{O}_\tau$	4	**1**	marginal coupling
S^5 scalar harmonics	chiral primaries $\mathcal{O}_k$	k	$[0,k,0]$	KK tower
massive string modes	Konishi, long operators	$\sim \lambda^{1/4}$	various	stringy

The first two rows are pure kinematics. Every local quantum field theory has a conserved stress tensor $T_{\mu\nu}$, a symmetric conserved current of spin two and dimension $\Delta = d = 4$; it generates the conformal transformations, so it must be dual to the field that generates the bulk isometries, the graviton. Likewise the conserved R-symmetry current $J^\mu_{[ij]}$, of spin one and dimension $\Delta = d - 1 = 3$, sits in the adjoint **15** of $\mathrm{SU}(4)_R$ and is dual to the $\mathrm{SO}(6)$ gauge fields that arise in the bulk from the isometries of the S^5: a conserved global current on the boundary is dual to a gauge field in the bulk.

The third row records the exactly marginal coupling. Because g_{YM} runs to nothing, the operator $\mathcal{O}_\tau \sim \mathrm{Tr}(F_{\mu\nu}F^{\mu\nu} + \dots)$ that deforms the coupling is exactly marginal, dimension $\Delta = 4$, and it is dual to the type IIB axio-dilaton $\tau_{\mathrm{IIB}} = C_0 + ie^{-\phi}$, whose boundary value is the modulus of the theory. The complexified gauge coupling of $\mathcal{N} = 4$ is identified with τ_{IIB} only up to the same factor of two that separates g^2_{YM} from g_s in this chapter's convention $g^2_{\mathrm{YM}} = 2\pi g_s$ (flagged in the factor-of-two warning of Section 13.1), so the identification is proportional rather than literal. Under it, the $\mathrm{SL}(2,\mathbb{Z})$ S-duality of IIB acting on τ_{IIB} (Chapter 7) is the Montonen-Olive S-duality of $\mathcal{N} = 4$ acting on g_{YM}.

The fourth row is the workhorse. Reducing type IIB supergravity on the S^5 produces an infinite Kaluza-Klein tower: each ten-dimensional field becomes a tower of five-dimensional fields labelled by the spherical harmonic k it carries on the S^5, and a harmonic of degree k transforms in the rank-k symmetric traceless representation of $\mathrm{SO}(6)$, written $[0,k,0]$ in the Dynkin labels of $\mathrm{SO}(6) \cong \mathrm{SU}(4)$ (so $k = 1$ is the vector **6** and $k = 2$ is the **20′**). The spectrum, including the mass formula used below, was computed by Kim, Romans and van Nieuwenhuizen. The bottom of the graviton's own tower reproduces the stress tensor, and the higher rungs are new. The scalar rungs are dual to the *chiral primary operators*

$$\mathcal{O}_k(x) = \mathrm{Tr}\big[X^{\{i_1}X^{i_2}\cdots X^{i_k\}}\big](x), \qquad k \geq 2, \tag{13.41}$$

the symmetrised traceless products of the six scalars, with dimension $\Delta = k$. These are $\frac{1}{2}$-BPS: they sit in short representations of $\mathfrak{psu}(2,2|4)$, their dimensions are protected against quantum corrections, and $\Delta = k$ holds at any coupling. That protection is what lets a weak-coupling gauge-theory quantity be trusted at the strong coupling where the geometry is classical, and it is why the supergravity fields match precisely the protected operators. The lowest, $\mathcal{O}_2$ in the **20′** (the rank-2 symmetric traceless of $\mathrm{SO}(6)$, distinct from the **20** of $\mathrm{SU}(4)$), is the superconformal primary of the whole stress-tensor multiplet: acting with the 32 supercharges builds the R-current and the stress tensor out of it, and on the bulk side the same multiplet is the graviton supermultiplet of five-dimensional gauged supergravity.

The last row is where the strings live. An unprotected operator, the simplest being the Konishi operator $\mathcal{K} = \mathrm{Tr}(X^iX^i)$, the $\mathrm{SO}(6)$ singlet trace part left out of Eq. (13.41), has a dimension that runs with the coupling. At weak coupling $\Delta_{\mathcal{K}} = 2 + \frac{3\lambda}{4\pi^2} + \dots$, an honest anomalous dimension (the one-loop coefficient is a standard gauge-theory computation, quoted here from the $\mathcal{N} = 4$ literature). At strong coupling it grows without bound, $\Delta_{\mathcal{K}} \simeq 2\lambda^{1/4}$; the coefficient is fixed by integrability, while the mass estimate in Theorem 13.6 fixes only the $\lambda^{1/4}$ scaling. On the bulk

side such operators are dual to massive excited string states, with masses of order the string scale, $m \sim 1/\ell_s$. The next subsection shows that a bulk mass $m \sim 1/\ell_s$ gives $\Delta \sim mL \sim L/\ell_s \sim \lambda^{1/4}$, matching the gauge-theory growth: long operators are heavy strings. When $\lambda \gg 1$ these states are parametrically heavier than the supergravity fields and decouple from the low-energy bulk theory, which is the operator-language version of the statement that classical supergravity captures only the protected, chiral-primary sector of the gauge theory.

13.3.3 Operator dimensions from bulk masses

The exponent Δ in the fall-off Eq. (13.40) is fixed by the bulk mass. Take a scalar ϕ of mass m in AdS_{d+1} and write its wave equation $(\Box - m^2)\phi = 0$ in the metric Eq. (13.34). Using $\sqrt{-g} = (L/z)^{d+1}$ and $g^{zz} = z^2/L^2$, the Laplacian acting on a mode that is slowly varying along the boundary is

$$\Box\phi = \frac{z^{d+1}}{L^2}\,\partial_z\!\left(z^{1-d}\,\partial_z\phi\right) + \mathcal{O}\!\left(z^2\,\partial_x^2\phi\right). \tag{13.42}$$

Trying a power $\phi \sim z^{\Delta}$ near the boundary, the derivative terms give $\Delta(\Delta - d)\,z^{\Delta}/L^2$, and dropping the subleading ∂_x^2 piece the wave equation collapses to an algebraic condition,

$$\boxed{\Delta(\Delta - d) = m^2L^2} \qquad \Longrightarrow \qquad \Delta_{\pm} = \frac{d}{2} \pm \sqrt{\frac{d^2}{4} + m^2L^2}\,. \tag{13.43}$$

The two roots are the two exponents in Eq. (13.40): $\Delta_- = d - \Delta_+$ is the source fall-off, and Δ_+, which we write simply as Δ, is the dimension of the operator. For $d = 4$, $\Delta_{\pm} = 2 \pm \sqrt{4 + m^2L^2}$. A massless bulk scalar ($m = 0$) gives $\Delta = 4$; heavier fields give larger Δ; the general pattern that "heavier in the bulk is more irrelevant on the boundary" is written into Eq. (13.43).

Two features deserve a comment. First, m^2L^2 is allowed to be negative. The square root in Eq. (13.43) stays real, and Δ stays real, as long as

$$m^2L^2 \geq -\frac{d^2}{4}, \tag{13.44}$$

the Breitenlohner-Freedman bound. A tachyonic mass in anti-de Sitter space is not an instability, provided it is not too tachyonic: the negative curvature acts like a box that lifts the would-be runaway. Below the bound Δ goes complex and the mode oscillates unboundedly, the genuine instability. The chiral primary $\mathcal{O}_k$ sits at $m^2L^2 = k(k-4)$, and the bottom rung $k = 2$ has $m^2L^2 = -4 = -d^2/4$: it saturates the bound, the two roots collide at $\Delta_{\pm} = 2$, and this is the mode dual to the relevant $\mathbf{20'}$ scalar $\mathcal{O}_2$ (relevant because $\Delta = 2 < 4$). Negative mass-squared fields are common here, and make up much of the supergravity spectrum.

Second, spin shifts the formula. The scalar relation Eq. (13.43) generalises: a bulk field of spin s and mass m is dual to an operator of dimension fixed by the same algebra with s-dependent shifts, and for the conserved cases the answer is forced. A massless spin-two field gives $\Delta = d = 4$, the stress tensor; a massless spin-one field gives $\Delta = d - 1 = 3$, a conserved current; a massless spin-$\frac{3}{2}$ gives $\Delta = d - \frac{1}{2} = \frac{7}{2}$, the supersymmetry current. Masslessness in the bulk is conservation on the boundary. The precise spin-s and p-form versions are quoted in the references; the scalar case carries the idea.

Example 13.6 (Reading a dimension off a mass, and back)**.** Take the AdS_5 graviton, a massless spin-two field, $m = 0$. The spin-two version of Eq. (13.43) for a conserved current gives $\Delta = d = 4$, and indeed the four-dimensional stress tensor has dimension 4: its components are energy densities, so $[\,T_{00}\,] = \text{energy/volume} = (\text{mass})^4$ in four dimensions. Now run it the other way for the Konishi operator. At strong coupling its bulk dual is an excited string state with

mass of order the string scale, $m^2 \sim 1/\alpha'$, so $m^2 L^2 \sim L^2/\alpha' = \sqrt{2\lambda}$ by Eq. (13.36). Feeding this into Eq. (13.43) with $m^2 L^2 \gg 1$,

$$\Delta \simeq \sqrt{m^2 L^2} \sim \lambda^{1/4}. \tag{13.45}$$

The estimate reproduces the $\lambda^{1/4}$ growth and nothing sharper: the numerical coefficient depends on the exact string level (the first massive closed-string mode has $m^2 = 4/\alpha'$, not $1/\alpha'$) and on the normalisation of λ, and the true value $\Delta_{\mathcal{K}} \simeq 2\lambda^{1/4}$ is an integrability result the mass estimate cannot deliver. The single relation Eq. (13.43) turns the supergravity mass gap into the protected sector and the string mass tower into the growing anomalous dimensions.

13.3.4 The generating-functional relation

The fall-off Eq. (13.40) already told us that the boundary value ϕ_0 of a bulk field is the source for the dual operator $\mathcal{O}$. The Gubser-Klebanov-Polyakov-Witten relation makes this an equation between two generating functionals. In its cleanest (Euclidean) form,

$$Z_{\text{string}}\big[\,\phi \to \phi_0\,\big] \;=\; \Big\langle \exp\Big(\int \mathrm{d}^4x\; \phi_0(x)\, \mathcal{O}(x) \Big) \Big\rangle_{\text{CFT}}. \tag{13.46}$$

The left side is the full type IIB string partition function on the space that is asymptotically $\mathrm{AdS}_5 \times S^5$, computed with the boundary condition that each bulk field ϕ approaches its prescribed boundary profile ϕ_0 (with the $z^{d-\Delta}$ dressing of Eq. (13.40) stripped off). The right side is the generating functional of the gauge theory, with ϕ_0 turned on as a source multiplying the dual operator $\mathcal{O}$ in the action. Boundary values of bulk fields are sources for boundary operators, and the two partition functions, computed by completely different means, are declared equal.

We state Eq. (13.46) rather than derive it. It is the defining equation of the correspondence in its calculational form, proposed by Gubser, Klebanov and Polyakov and, independently, by Witten in 1998. (The acronym clashes with the Giddings-Kachru-Polchinski flux vacuum of Section 9.2.4; the two "GKP"s are unrelated.) A derivation from first principles would be a proof of AdS/CFT, which nobody has; what one has is a large body of consistency checks, of which the protected quantities above are the cleanest.

What Eq. (13.46) buys is correlation functions. Differentiating both sides with respect to the source and setting it to zero pulls down insertions of $\mathcal{O}$,

$$\langle \mathcal{O}(x_1) \cdots \mathcal{O}(x_n) \rangle = \frac{\delta^n}{\delta\phi_0(x_1) \cdots \delta\phi_0(x_n)} \log Z_{\text{string}}[\phi_0] \Big|_{\phi_0 = 0}. \tag{13.47}$$

In the regime where the geometry is weakly curved and weakly coupled, $\lambda \gg 1$ and $N \gg 1$, the string partition function is dominated by the classical supergravity action evaluated on the solution with boundary data ϕ_0,

$$\log Z_{\text{string}}[\phi_0] \;\simeq\; -\, S^{\text{on-shell}}_{\text{sugra}}[\phi_0]. \tag{13.48}$$

The strongly coupled gauge-theory correlator is then computed by solving classical field equations in the bulk, imposing ϕ_0 at the boundary and regularity in the interior, evaluating the action, and differentiating. This is the concrete sense in which classical gravity computes strong-coupling field theory. The two-point function, worked out this way, comes out proportional to $|x_1 - x_2|^{-2\Delta}$ with exactly the Δ of Eq. (13.43), the conformally required form for a primary of dimension Δ; the dimensions read off from masses in the previous subsection are the linearised shadow of the full relation Eq. (13.46).

Careful

The on-shell action in Eq. (13.48) is divergent. The volume of anti-de Sitter space is infinite, the divergence comes entirely from the boundary region $z \to 0$, and it must be removed by boundary counterterms before Eq. (13.47) yields finite correlators. The divergences are local in the source ϕ_0, so the counterterms are local boundary terms, and subtracting them is the exact bulk avatar of ultraviolet renormalisation in the boundary theory: the near-boundary region is the ultraviolet, and cutting the bulk off at small z is a short-distance cut-off on the field theory. This is holographic renormalisation, and it is why correlators computed from Eq. (13.46) carry the same scheme ambiguities, contact terms and conformal anomalies, as any renormalised field theory. We flag the machinery and do not run it.

Remark 13.7 (Beyond the lectures: the two roots and alternative quantisation). When both roots $\Delta_\pm$ of Eq. (13.43) exceed the unitarity bound, that is in the window $-d^2/4 \le m^2L^2 < -d^2/4+1$, either can be the operator dimension. The standard choice reads Δ_+ as the dimension and ϕ_0 as the source; the alternative quantisation swaps their roles, treating the z^{Δ_-} coefficient as the vev of an operator of dimension Δ_-. The two are related by a Legendre transform on the boundary, physically by gauging or ungauging a global symmetry, and the flow between them is a standard example of a boundary renormalisation-group flow driven by a double-trace deformation. This is the mechanism that lets the lowest chiral primaries sit where they do, and it is the cleanest place to see that the "source or vev" split in Eq. (13.40) is a choice of boundary condition, not a fact about the field.

13.3.5 What the dictionary says

Three matchings carry the section. The symmetries agree on the nose: the isometry supergroup $\mathrm{PSU}(2,2|4)$ of $\mathrm{AdS}_5 \times S^5$ is the superconformal group of $\mathcal{N}=4$ super-Yang-Mills, with $\mathrm{SO}(4,2)$ the conformal group, $\mathrm{SU}(4)_R$ the R-symmetry, and 32 supercharges on each side. Fields and operators are paired representation by representation: single-particle bulk states to single-trace gauge-invariant operators, supergravity fields to protected chiral primaries with $\Delta = k$, massive string modes to unprotected operators with $\Delta \sim \lambda^{1/4}$, and the fall-off of a bulk field carries the source and the vev of its operator. Dimensions come from masses through $\Delta(\Delta - d) = m^2L^2$, and the whole map is packaged in the generating-functional relation Eq. (13.46), with correlators computed from a classical bulk action at strong coupling. What we have not yet used is the meaning of the radial coordinate z, nor why the $1/N$ counting of the gauge theory reproduces the genus counting of Chapter 4. Those are the two threads that turn a dictionary into an emergence, and Section 13.4 picks them up.

Exercise 13.9 (Counting the superconformal generators). *(Short.)* Add up the generators of $\mathfrak{psu}(2,2|4)$ and match them across the duality. (a) On the conformal side, list $M_{\mu\nu}, P_\mu, K_\mu, D$ and confirm $\dim \mathrm{SO}(4,2) = 15$; do the same for $\mathrm{SO}(6)$. (b) Identify the $15+15$ bosonic generators with the isometries of the AdS_5 and S^5 factors. (c) The fermionic generators are the 16 Poincaré supercharges Q and the 16 superconformal charges S. Explain, using the schematic commutator $[Q,K] \sim S$, why a theory with Poincaré supersymmetry and conformal symmetry must also carry the S's, so that superconformal symmetry always doubles the count of supercharges. (d) Check that 30 bosonic plus 32 fermionic generators is consistent with the maximal supersymmetry 32 on the gravity side, and say in one line why $\mathrm{AdS}_5 \times S^5$ is called maximally supersymmetric.

Exercise 13.10 (Dimensions across the spectrum). *(Moderate.)* Use Eq. (13.43) throughout, with $d = 4$. (a) For the chiral primary tower, take $m^2L^2 = k(k-4)$ and show $\Delta_+ = k$ for all $k \ge 2$, and that $k = 2$ saturates the Breitenlohner-Freedman bound Eq. (13.44). (b) A relevant deformation of the gauge theory is an operator with $\Delta < 4$; an irrelevant one has $\Delta > 4$. Translate these into signs of m^2 for the dual bulk scalar, and confirm that a marginal operator

($\Delta = 4$) is dual to a massless bulk field. (c) The unitarity bound for a scalar primary in four dimensions is $\Delta \geq 1$. Show that this corresponds to $\Delta_- \leq 1$, and identify the value of m^2L^2 at which $\Delta_+ = \Delta_-$.

Exercise 13.11 (The two-point function)**.** *(Moderate; uses Eqs. (13.46) and (13.48).)* For a free massive scalar in AdS_5, the classical solution with boundary source ϕ_0 regular in the interior is fixed by Eqs. (13.34) and (13.40). Sketch, without evaluating every integral, why the on-shell action Eq. (13.48) is quadratic in ϕ_0 and how its second functional derivative produces a two-point function of the form $\langle \mathcal{O}(x)\mathcal{O}(0)\rangle \propto |x|^{-2\Delta}$. Explain which term in the near-boundary expansion Eq. (13.40) carries the physical, non-contact part of the answer, and why the leading $z^{d-\Delta}$ piece contributes only a divergent, source-local contact term that holographic renormalisation subtracts. State clearly the one input you are taking from the references rather than computing.

Exercise 13.12 (The holographically renormalised two-point function)**.** *(Hard; carries out Exercise 13.11 in full.)* Do the computation explicitly for a massless bulk scalar in AdS_5, so $m = 0$, $\Delta = 4$, $d = 4$. (a) Fourier transform along the boundary, $\phi(z, x) = \int \mathrm{d}^4k\, e^{ik\cdot x}\, \phi_k(z)$, and use Eq. (13.42) to show that the radial equation is solved, for the branch regular in the interior $z \to \infty$, by $\phi_k(z) \propto z^2 K_2(kz)$ with $k = |k|$ (Euclidean momentum) and K_2 a modified Bessel function. (b) Expand $z^2K_2(kz)$ about $z = 0$, read off the two coefficients of Eq. (13.40), and show that the ratio of the $z^\Delta = z^4$ coefficient to the $z^{d-\Delta} = z^0$ coefficient contains a term $\propto k^4 \log k$ alongside terms analytic in k^2. (c) Reduce the on-shell action Eq. (13.48) to a boundary term at small z, insert the regular solution, and argue that it is quadratic in ϕ_0 with a coefficient set by that ratio. (d) Identify the analytic-in-k^2 pieces as source-local contact terms removed by holographic renormalisation, and show that the $k^4 \log k$ term Fourier transforms back to $\langle \mathcal{O}(x)\mathcal{O}(0)\rangle \propto |x|^{-2\Delta} = |x|^{-8}$. State clearly the one overall normalisation you take from the references rather than fixing yourself.

13.4 Emergent spacetime

The big picture

The two descriptions of a stack of D3-branes are the same physics written twice: a four-dimensional gauge theory with no gravity, and type IIB string theory on $AdS_5 \times S^5$ with all of ten-dimensional gravity in it. This section explains *why* they can be the same. The $1/N$ expansion of the gauge theory is organised by exactly the topological invariant that organises the closed-string genus expansion of Chapter 4. Resumming gauge-theory Feynman diagrams builds a worldsheet; the worldsheet builds the graviton; the graviton builds the geometry. Gravity and the five extra dimensions are what a gauge theory with no gravity looks like when you sum its planar diagrams and read the answer as a spacetime. The radial direction of that spacetime is the renormalisation-group scale of the gauge theory. This is the sense in which spacetime is emergent, and it is where the course ends.

By now we have the correspondence as a statement. The near-horizon throat of N coincident D3-branes is $AdS_5 \times S^5$ (Section 13.1); Maldacena's decoupling argument gives one system two descriptions (Section 13.2); and the dictionary matches the $SO(4,2) \times SU(4)_R$ isometries to the conformal and R-symmetry of $\mathcal{N} = 4$ super-Yang-Mills, together with the field-operator map and the GKP-Witten generating functional (Section 13.3). What we have not yet explained is the mechanism. Two theories can share a symmetry group by coincidence. These two share far more: an entire perturbative expansion, term by term, with the same combinatorial skeleton, and seeing that is what turns the correspondence from a surprising equality into an account of where spacetime comes from.

The engine is an observation 't Hooft made in 1974, two decades before Maldacena, and for an

entirely different reason. He was trying to find a solvable limit of quantum chromodynamics. What he found was that a gauge theory with gauge group of rank N, expanded in $1/N$, reorganises its Feynman diagrams by their *topology*, and that this topological expansion looks exactly like the perturbation series of a string theory. He could not say which string. AdS/CFT answers that: for $\mathcal{N}=4$ SYM the string is the type IIB string moving in $\mathrm{AdS}_5\times S^5$, and $1/N$ is its coupling. We build the argument in stages.

13.4.1 The 't Hooft limit and the double-line expansion

The fields of $\mathcal{N}=4$ SYM are all in the adjoint of $\mathrm{SU}(N)$: the gauge field A_μ, six real scalars ϕ^I, and four Weyl fermions, each an $N\times N$ traceless Hermitian matrix. Write a generic adjoint field as $\Phi^a{}_b$ with a fundamental index a up and an antifundamental index b down. The Lagrangian is a single trace,

$$\mathcal{L} \;=\; \frac{1}{g_{\mathrm{YM}}^2}\,\mathrm{Tr}\Big(\tfrac{1}{2}(D\Phi)^2+\Phi^3+\Phi^4+\cdots\Big), \tag{13.49}$$

where the schematic Φ^n stand for the gauge, Yukawa and quartic vertices with their contractions, and we have scaled the coupling out front so that all fields are order one. The overall $1/g_{\mathrm{YM}}^2$ in front means every propagator carries a factor g_{YM}^2 and every interaction vertex a factor $1/g_{\mathrm{YM}}^2$.

The trick that makes large N tractable is bookkeeping. Because a propagator $\langle\Phi^a{}_b\,\Phi^c{}_d\rangle$ contracts an upper index with a lower index, $\propto\delta^a_d\,\delta^c_b$, it carries *two* colour lines, one for the fundamental index and one for the antifundamental, running antiparallel. Draw the propagator not as a single line but as a double line, a thin ribbon, one edge for each index. At a vertex the ribbons join so that colour indices flow continuously: the trace structure of (13.49) guarantees that an outgoing a on one field meets an incoming a on the next. A Feynman diagram becomes a network of ribbons, and the colour indices run along the edges, close into loops, and each closed index loop is a free sum over N colours, contributing a factor of N.

Now count the N-dependence of a connected vacuum diagram (no external legs; the same counting with a fixed correction governs correlators). Introduce the 't Hooft coupling

$$\lambda \;\coloneqq\; g_{\mathrm{YM}}^2\,N, \tag{13.50}$$

and hold it fixed as $N\to\infty$. Since $g_{\mathrm{YM}}^2=\lambda/N$, a diagram with V vertices, E propagators (edges of the ribbon graph) and F closed index loops (faces) scales as

$$\Big(\tfrac{1}{g_{\mathrm{YM}}^2}\Big)^{V}\Big(g_{\mathrm{YM}}^2\Big)^{E}N^{F} \;=\; \Big(\tfrac{N}{\lambda}\Big)^{V}\Big(\tfrac{\lambda}{N}\Big)^{E}N^{F} \;=\; N^{\,V-E+F}\,\lambda^{\,E-V}. \tag{13.51}$$

The exponent of N is $V-E+F$. That combination is the Euler characteristic of the surface you get by gluing a face into every closed index loop of the ribbon graph, filling the diagram in until it becomes a closed orientable surface with V vertices, E edges and F two-dimensional faces. For such a triangulation,

$$V-E+F \;=\; \chi \;=\; 2-2h, \tag{13.52}$$

with h the number of handles of the surface. Compare (4.1): the same Euler characteristic, now built out of a Feynman diagram rather than out of a worldsheet. Every ribbon graph draws itself on a definite two-dimensional surface, the one of least genus on which its lines do not cross, and the power of N it carries is fixed by that genus alone.

Example 13.8 (The planar sunset)**.** Take the simplest interacting vacuum graph: two cubic vertices joined by three propagators, the "sunset". Read the powers of N straight off the double-line rules. Each vertex carries $1/g_{\mathrm{YM}}^2=N/\lambda$ and each propagator $g_{\mathrm{YM}}^2=\lambda/N$, so the couplings alone contribute $(N/\lambda)^2(\lambda/N)^3=\lambda/N$. The colour factor is what the topology decides.

Drawn on the plane with none of the three ribbons twisted (the left panel of Fig. 13.1), the graph has $V = 2$, $E = 3$ and $F = 3$ closed index loops, each a free sum worth N, so the colour trace gives N^3. Putting the two together,

$$\frac{\lambda}{N} \times N^3 \;=\; N^2\,\lambda \;=\; N^{\,V-E+F}\,\lambda^{\,E-V}, \tag{13.53}$$

and $V - E + F = 2 - 3 + 3 = 2 = \chi(S^2)$: the planar sunset triangulates the sphere and sits at the leading order N^2. Twist one ribbon before gluing, as in the right panel of the figure, and two index loops merge into one, F drops from 3 to 1, χ falls to 0, and the same graph is demoted to $N^0\lambda$, down by $1/N^2$. The twist is a handle. Exercise 13.13 runs the same count in the other common normalisation; the bookkeeping looks different, the topology and the answer do not.

Summing all diagrams at fixed genus and organising by handles, the free energy of the gauge theory is

$$\boxed{\ln Z_{\text{gauge}} \;=\; \sum_{h=0}^{\infty} N^{\,2-2h}\, f_h(\lambda), \qquad f_h(\lambda) = \sum_{\text{genus-}h\ \text{graphs}} \lambda^{\,E-V}.} \tag{13.54}$$

The leading term, $h = 0$, is order N^2: these are the *planar* diagrams, the ones you can draw on the plane (or sphere) without any line crossing another. Each handle you add costs a factor N^{-2}. Within a fixed genus the remaining sum over λ is still an infinite set of ordinary Feynman diagrams, and $f_h(\lambda)$ is generically a complicated function of the 't Hooft coupling; even the planar function $f_0(\lambda)$ is not known in closed form for a generic theory. What the large-N limit has bought is not the evaluation of f_h but the *sorting*: the double infinity of Feynman diagrams has been split into a controlled expansion in $1/N^2$, one topology per order.

Remark 13.9 (The large-N limit is a classical limit). Holding λ fixed while $N \to \infty$ suppresses handles, and (13.54) is dominated by the single planar term $N^2 f_0(\lambda)$. A quantity whose fluctuations vanish as some parameter goes to zero is behaving classically, and here that parameter is $1/N$. Single-trace operators become commuting classical variables at $N = \infty$ (their connected correlators are suppressed by powers of $1/N$), and the planar theory is a classical theory in a precise sense, a "master field". This is the field-theory shadow of the fact that the dual string becomes classical, its loops switched off, in the same limit. We make the identification of parameters below.

13.4.2 Ribbon graphs are worldsheets

Set the two expansions side by side. The gauge theory gives (13.54). The closed string gives the genus expansion of Section 4.1.2, which for a vacuum amplitude (free energy) reads

$$\ln Z_{\text{string}} \;=\; \sum_{g=0}^{\infty} g_s^{\,2g-2}\, F_g, \tag{13.55}$$

with g_s^{2g-2} the weight of a genus-g worldsheet from (4.2) and F_g the integral over the moduli space of that surface. The two series have the same shape. Term by term they match under the identification

$$\frac{1}{N} \;\longleftrightarrow\; g_s, \qquad h \;\longleftrightarrow\; g, \qquad f_h(\lambda) \;\longleftrightarrow\; F_g. \tag{13.56}$$

A gauge-theory ribbon graph of genus h sits at the same order as a string worldsheet of genus $g = h$. Adding a handle costs $1/N^2$ on one side and g_s^2 on the other, and the planar diagrams that dominate the gauge theory are the sphere that dominates the string. This is more than an analogy dressed up in matching symbols. The dictionary of Section 13.3 supplies the coupling

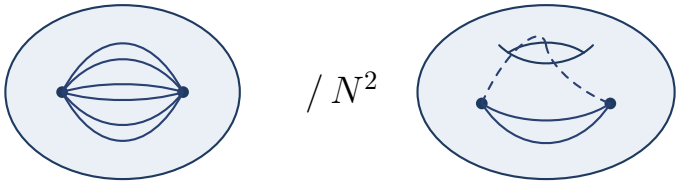


Figure 13.1: The same two-vertex vacuum graph drawn twice. On the left the three ribbon propagators lie flat: the graph fits on the sphere, its index loops are the faces of a genus-zero triangulation, and it carries the maximal power N^2. On the right one propagator is routed so that it must cross the others; it can only be drawn without a crossing by threading it over a handle, so the graph lives on a torus and is suppressed by N^{-2}. Topology of the ribbon graph, not the number of loops, controls the power of N. This is the gauge-theory image of the genus expansion of Fig. 4.1.

map $g_{\rm YM}^2 = 2\pi g_s$ (Eq. (6.37)) that turns the first of these identifications into a number; the term-by-term genus reconnection is the work of the present section. With $\lambda = g_{\rm YM}^2 N$,

$$g_s \;=\; \frac{g_{\rm YM}^2}{2\pi} \;=\; \frac{\lambda}{2\pi N}, \tag{13.57}$$

so at fixed 't Hooft coupling the string coupling g_s is set by the inverse rank $1/N$, exactly as (13.56) requires.

Here is the picture to hold onto. A ribbon graph is a two-dimensional surface with a very coarse triangulation on it, one face per index loop (Fig. 13.1). As you go to higher orders in the gauge theory you include graphs with more and more vertices and propagators, which is to say finer and finer triangulations of the same surfaces. In the strict sense of a lattice becoming a continuum, the sum over all ribbon graphs of a fixed genus *is* a sum over all shapes of a continuous worldsheet of that genus: the discrete triangulation refines into the smooth surface, and the sum over graphs turns into an integral over the moduli space of Riemann surfaces, the very object $\mathcal{M}_{g,n}$ that carries the string amplitude (4.3). The 't Hooft coupling λ controls how finely the surface is triangulated, and it resurfaces on the string side as the size of the geometry in string units. From the dictionary, $L^4/\alpha'^2 = 4\pi g_s N = 2\lambda$, so

$$\left(\frac{L}{\ell_s}\right)^4 \;=\; 2\lambda. \tag{13.58}$$

Strong 't Hooft coupling is a large, smooth, weakly curved worldsheet target; weak 't Hooft coupling is a target smaller than the string, where the geometric picture dissolves. The function $f_h(\lambda)$, an infinite sum of Feynman diagrams, is repackaged as F_g, an integral over one worldsheet, and that repackaging is where the string emerges from the gauge theory.

The emergent string is not made of the gauge fields

It is tempting to say the closed string "is" a colour-electric flux tube of the gauge theory, and at strong coupling in a confining theory there is truth to that. But the worldsheet that emerges here lives in ten dimensions, five of them (AdS_5 radial plus S^5) with no counterpart among the four dimensions the gauge theory occupies. The ribbon graph is a two-dimensional surface abstractly, in the space of colour indices; the identification of that abstract surface with a worldsheet embedded in $\mathrm{AdS}_5 \times S^5$ is the content of the duality, not something visible in the Feynman rules. 't Hooft's counting tells you a string is hiding in the large-N expansion, but not that it moves in five extra dimensions; that is what the D-brane construction of

Section 13.1 supplies.

Exercise 13.13 (Counting powers of N)**.** *(Moderate.)* This exercise uses the alternative "N out front" normalisation, with the coupling in the action rather than scaled out of the fields as in (13.49). The propagator and vertex factors come out differently from the main text, but the topological counting N^{V-E+F} is the same. Consider a single Hermitian matrix field $\Phi^a{}_b$ with action $S = N\ \mathrm{Tr}\left(\frac{1}{2}\Phi^2 + \frac{g}{3}\Phi^3\right)$, so $\lambda := g^2$ is held fixed as $N \to \infty$.

(a) Show the propagator carries a factor $1/N$ and each cubic vertex a factor N, in double-line notation.

(b) The "sunset" vacuum graph has two cubic vertices joined by three propagators. Draw its ribbon version with no crossings and count the closed index loops F; verify $V - E + F = 2$ and that the graph scales as $N^2\lambda$.

(c) Now twist one of the three ribbons before joining, so a line must cross. Show the number of index loops drops to $F = 1$, that $V - E + F = 0$, and that the graph is suppressed to $N^0\lambda$, a relative factor $1/N^2$. Interpret the twist as routing a propagator over a handle.

(d) Conclude that at $N = \infty$ only planar graphs survive, and that $1/N^2$ is the handle-counting parameter, the gauge-theory analogue of g_s^2.

13.4.3 The radial direction is a renormalisation-group scale

The gauge theory lives in four dimensions; the geometry has five (plus the sphere). Where does the extra dimension come from, and what does it mean? There is a one-line answer, and it organises much of what follows: the radial direction of AdS_5 is the energy scale of the gauge theory, and moving into the bulk is running the renormalisation group.

Write AdS_5 in Poincaré coordinates,

$$\mathrm{d}s^2 \;=\; \frac{L^2}{z^2}\Big(\mathrm{d}z^2 + \eta_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu\Big), \tag{13.59}$$

with $z > 0$ the radial coordinate and x^μ the four coordinates the gauge theory shares. The boundary where the field theory lives is at $z \to 0$. This metric has an isometry that will do all the work: the simultaneous rescaling

$$x^\mu \;\to\; a\,x^\mu, \qquad z \;\to\; a\,z, \tag{13.60}$$

leaves (13.59) invariant, because the $1/z^2$ prefactor compensates the rescaling of $\mathrm{d}x$ and $\mathrm{d}z$ together. On the boundary this is precisely a scale transformation $x \to ax$, an element of the conformal group, matched to the bulk isometry in Section 13.3. But the bulk realises it geometrically, and it acts on the radial coordinate as well: a dilatation of the field theory is a rigid rescaling of z. The radial position therefore carries dimensions of length and transforms like a distance scale in the gauge theory. Small z is short distances, the ultraviolet; large z is long distances, the infrared.

Make it quantitative through the redshift. A local process in the bulk at radial slice z, with some fixed proper energy set by the local physics (say a string oscillator, proper energy of order $1/\ell_s$), is seen by the boundary observer with the conserved energy

$$E \;=\; \sqrt{-g_{tt}}\,E_{\text{proper}} \;=\; \frac{L}{z}\,E_{\text{proper}} \;\propto\; \frac{1}{z}. \tag{13.61}$$

A bulk excitation deep in the interior (large z) redshifts down to low boundary energy; one near the boundary (small z) is blue-shifted to high energy. High-energy gauge-theory physics happens

near the boundary, low-energy physics deep inside. In the alternative coordinate $r = L^2/z$, which appears when the near-horizon limit is taken in Section 13.1, the relation reads $E \propto r$: energy grows as you climb out of the throat, and the field theory sits at its top.

This turns the extra dimension into a geometrisation of the renormalisation group. The boundary value of a bulk field is the source (equivalently the coupling) for the dual operator, by the GKP-Witten relation of Section 13.3. Following that field inward from $z \to 0$ to larger z is following the coupling as you integrate out high-energy modes: the bulk equations of motion in the radial direction are, read the right way, the renormalisation-group flow equations of the coupling. A field theory with a genuine mass gap or confinement scale is dual to a geometry that ends, or caps off, at some z_{IR}; a conformal theory like $\mathcal{N} = 4$ SYM, whose coupling is exactly marginal so that no scale is generated (Section 13.3), has a dual geometry that is scale-invariant all the way down, which is exactly the statement that (13.60) is an exact isometry with no preferred z. The absence of a scale in the gauge theory is the homogeneity of AdS_5 along the radial direction.

Remark 13.10 (Holographic renormalisation, in one line). The near-boundary expansion of a bulk field in powers of z has two independent pieces: a leading term z^{Δ_-} fixed by the source, and a subleading term z^{Δ_+} whose coefficient is the operator expectation value $\langle \mathcal{O} \rangle$, with $\Delta_\pm$ the two roots of the mass-dimension relation quoted in Section 13.3. Reading off $\langle \mathcal{O} \rangle$ requires subtracting the divergent source-dependent terms, and the counterterms that do so are local functionals on the boundary: they are the field-theory counterterms of ordinary renormalisation, now derived from geometry. This machinery, holographic renormalisation, is what makes the GKP-Witten prescription finite and computable. We only need its existence.

The radial direction also makes the holographic principle concrete, and this is worth a short calculation. 't Hooft and Susskind argued on general grounds that a theory of quantum gravity in a region should be describable by degrees of freedom living on the region's boundary, with a density of at most one per Planck area: the maximal entropy is the area in Planck units, not the volume, the lesson of the Bekenstein-Hawking formula $S = A/4G$ of Chapter 12. AdS/CFT realises this literally. The bulk gravity of $\mathrm{AdS}_5 \times S^5$ is captured by a field theory on the four-dimensional boundary, one dimension fewer. Count the degrees of freedom on each side. The gauge theory has of order N^2 fields, the entries of the adjoint matrices. The gravity side measures degrees of freedom by area over Newton's constant. Reduce on the S^5 of radius L to get a five-dimensional Newton constant G_5, then estimate the number of bulk degrees of freedom per boundary cell:

$$\frac{1}{G_5} \;\sim\; \frac{\mathrm{vol}(S^5)\, L^5}{G_{10}} \;\sim\; \frac{L^5}{G_{10}}, \qquad G_{10} \;\sim\; g_s^2\, \alpha'^4 \;\; (\text{from } 2\kappa_{10}^2 = (2\pi)^7 \alpha'^4 g_s^2), \tag{13.62}$$

with $\mathrm{vol}(S^5) = \pi^3$ the volume of the unit five-sphere, so that the number of degrees of freedom associated with a boundary region of size L is

$$\frac{L^3}{G_5} \;\sim\; \frac{L^8}{G_{10}} \;\sim\; \frac{(g_s N\, \alpha'^2)^2}{g_s^2\, \alpha'^4} \;=\; N^2. \tag{13.63}$$

The area law of the bulk in Planck units returns the N^2 degrees of freedom of the boundary matrices. The parameter dictionary delivers the holographic principle on its own, with no extra assumption: a gauge theory holds, on its four-dimensional support, exactly the number of degrees of freedom that five-dimensional gravity assigns to the region it bounds.

Exercise 13.14 (The central charge is N^2)**.** *(Short.)* Using $L^4 = 4\pi g_s N\, \alpha'^2$ and $2\kappa_{10}^2 = (2\pi)^7 \alpha'^4 g_s^2 = 16\pi G_{10}$, and $G_5 = G_{10}/(\mathrm{vol}(S^5)\, L^5)$ with $\mathrm{vol}(S^5) = \pi^3$ the unit five-sphere volume, show that L^3/G_5 is proportional to N^2 with a pure-number coefficient, and find that coefficient.

	Gauge theory	**Gravity dual**
handles / loops	$1/N^2$ per handle	$g_s^2 = (\lambda/2\pi N)^2$ per handle
$N \to \infty$, λ fixed	planar limit	classical strings, $g_s \to 0$
$\lambda \gg 1$	strong 't Hooft coupling	large radius $L \gg \ell_s$, drop α'
$\lambda \ll 1$	weak coupling, Feynman diagrams	sub-string-scale, geometry dissolves

Table 13.1: The regimes of the correspondence. Classical supergravity is reliable in the bottom-left-to-top-right corner: large N and large λ. That is precisely where the gauge theory cannot be computed perturbatively. The duality trades the hard problem on one side for an easy one on the other, and never lets both be easy together.

(You should get $L^3/G_5 = 2N^2/\pi$ up to your conventions for G_5.) Comment on why an N-independent numerical prefactor is exactly what a comparison with the gauge-theory conformal anomaly coefficient, which scales as N^2 at large N, would require.

13.4.4 What the correspondence buys

Two payoffs above all, and they are why the correspondence is a working tool rather than a curiosity.

Strong coupling from classical gravity. Classical supergravity is a good description of the bulk when two conditions hold: the curvature radius is large in string units, $L \gg \ell_s$, so that α' corrections are negligible and the string sees a smooth geometry; and the string coupling is small, $g_s \ll 1$, so that worldsheet handles, the quantum-gravity loops, are suppressed. Translate both through the dictionary. From (13.58), $L \gg \ell_s$ means $\lambda \gg 1$: the gauge theory is at *strong* 't Hooft coupling. From (13.57), $g_s \ll 1$ at fixed λ means $N \gg \lambda$, i.e. $N \to \infty$: the gauge theory is *planar*. Putting the two together,

$$\text{classical gravity on } \mathrm{AdS}_5 \times S^5 \quad \Longleftrightarrow \quad N \to \infty, \quad \lambda \to \infty. \tag{13.64}$$

The regime where the geometry is smooth and classical is the same regime where the gauge theory is strongly coupled and hard to reach by Feynman diagrams, and that is what makes the correspondence useful. A non-perturbative gauge-theory question, the spectrum of anomalous dimensions, a transport coefficient of the plasma, the free energy at strong coupling, becomes a classical boundary-value problem in Einstein gravity. The correspondence is a strong-weak duality (Table 13.1), and like every strong-weak duality it cuts both ways: the two sides are never both weakly coupled at once, which is what lets it compute quantities no perturbative method reaches, and also why a direct proof stays out of reach and the evidence, though extensive, is circumstantial.

Black-hole thermodynamics is thermal field theory. Heat the gauge theory to temperature T. On the gravity side, a field theory at finite temperature is dual to a bulk geometry with a black hole in it, the planar AdS-Schwarzschild black brane, whose Hawking temperature is T and whose horizon fills the infrared end of the radial direction. Everything thermodynamic now translates. The black hole's Bekenstein-Hawking entropy $S = A/4G_5$ is the thermal entropy of the gauge plasma. Its free energy is the gauge theory's free energy. Dimensional analysis in the conformal theory fixes the form: with only T and the volume V_3 available, and the N^2 degrees of freedom counted above,

$$F \;\sim\; -N^2\, T^4\, V_3, \tag{13.65}$$

and the black-brane free energy reproduces exactly this scaling, with a definite coefficient. Comparing that coefficient against the free plasma at $\lambda = 0$ gives the celebrated result of Gubser, Klebanov and Tseytlin: the strongly coupled entropy is $\frac{3}{4}$ of the free-field entropy,

$$\frac{S_{\lambda=\infty}}{S_{\lambda=0}} \;=\; \frac{3}{4}, \tag{13.66}$$

a prediction for strongly coupled $\mathcal{N} = 4$ SYM extracted from the area of a black hole. We state it rather than derive it; the computation is a textbook exercise in the AdS-Schwarzschild geometry, and the factor $\frac{3}{4}$ has stood up to every independent check.

This closes a circle opened in Chapter 12. There, the Bekenstein-Hawking entropy of a supersymmetric black hole was matched to a microscopic count of BPS brane bound states, a count protected by the BPS bound so that it could be extrapolated from weak to strong coupling. Here the black hole is not BPS and not protected: it is a hot, dynamical object, and its entropy is the entropy of an ordinary thermal state in a gauge theory. AdS/CFT gives the entropy $S = A/4G$ a completely general microscopic meaning as a count of field-theory states, and it identifies the Hawking-Page transition between thermal AdS and the big black hole with the confinement-deconfinement transition of the gauge theory. The two halves of the capstone, microstate counting and holography, express the same statement: a black hole is a thermal density matrix of an ordinary quantum system, and its gravity is that system's coarse-grained hydrodynamics.

Exercise 13.15 (Free energy of the black brane)**.** *(Hard.)* The planar AdS_5-Schwarzschild black brane, the geometry dual to $\mathcal{N} = 4$ SYM at temperature T, is

$$\mathrm{d}s^2 \;=\; \frac{L^2}{z^2}\Big(-f(z)\,\mathrm{d}t^2 + \mathrm{d}\vec{x}^{\,2} + \frac{\mathrm{d}z^2}{f(z)}\Big), \qquad f(z) \;=\; 1 - \Big(\frac{z}{z_h}\Big)^4,$$

with a horizon at $z = z_h$ and $\vec{x} \in \mathbb{R}^3$ the three field-theory space directions.

(a) Continue to Euclidean time $t \to -i\tau$ and demand that the (τ, z) geometry close smoothly at $z = z_h$ with no conical defect. Expanding f near the horizon, show that this fixes the period of τ and gives the temperature $T = 1/(\pi z_h)$.

(b) The horizon is the constant-t, $z = z_h$ slice. Compute its induced area per unit coordinate volume V_3 of the $\vec{x}$ directions, and hence the Bekenstein-Hawking entropy density $s = S/V_3 = A/(4G_5V_3)$. Show $s = (L/z_h)^3/(4G_5) = (\pi^3/4G_5)\,L^3T^3$.

(c) Insert $L^3/G_5 \sim N^2$ from Exercise 13.14 to get $s \sim N^2T^3$. Integrate the thermodynamic identity $s = -V_3^{-1}\,\partial F/\partial T$ (equivalently use $F = E - TS$ with E, S of a conformal gas) to recover $F \sim -N^2T^4V_3$, the scaling asserted in (13.65), and read off that the coefficient is a pure number.

(d) This coefficient is the strong-coupling ($\lambda = \infty$) value. Explain why comparing it against the free-field ($\lambda = 0$) theory, where the N^2 adjoint fields are a gas of free particles, is a clean test at fixed N, and identify which quantity the ratio (13.66) reports. You need not compute the free-field side; state what it counts.

Remark 13.11 (Beyond the lectures: geometry from entanglement)*.* The modern refinement of "spacetime is emergent" sharpens the slogan from *gravity emerges from a gauge theory* to *geometry emerges from entanglement.* Ryu and Takayanagi (2006) showed that the entanglement entropy of a spatial region A of the boundary theory equals the area of the minimal bulk surface anchored on ∂A, divided by $4G$:

$$S_{\mathrm{ent}}(A) \;=\; \frac{\mathrm{Area}(\gamma_A)}{4G}, \qquad \partial\gamma_A = \partial A.$$

This is the Bekenstein-Hawking formula applied not to a horizon but to any minimal surface, and it ties a purely quantum-information quantity in the gauge theory to a purely geometric one in the bulk. Van Raamsdonk pushed the logic further: dial down the entanglement between two halves of the boundary and the bulk that connects them pinches off and disconnects, suggesting that the smooth interior geometry, the very connectivity of space, is *built* out of boundary entanglement. The radial direction, in this language, is a scale of coarse-graining, and tensor-network models (MERA) reproduce its structure as a circuit that disentangles the state layer by layer as it moves inward. The ER=EPR proposal of Maldacena and Susskind, that entangled particles are joined by microscopic wormholes, is the same idea at its most provocative. None of this is settled, and it is where a large part of the field now lives.

13.4.5 What "emergent" means, and what it does not

Be precise about the claim, because it is easy to overstate and easy to understate. What emerges is the *bulk*: the closed string and hence the graviton, the ten-dimensional geometry $\mathrm{AdS}_5 \times S^5$, its dynamics, and its five extra dimensions, all reconstructed from a four-dimensional gauge theory that contains no metric, no graviton and no gravity in its definition. The gauge-theory path integral, a perfectly ordinary quantum field theory computed by summing diagrams, *is* the quantum-gravity partition function of the bulk, through GKP-Witten. In that exact sense gravity here is derived rather than fundamental: a reorganisation, valid at large N and large λ, of a non-gravitational theory.

What does *not* emerge is equally important. The four-dimensional Minkowski space on which the gauge theory lives is put in by hand: $\mathcal{N} = 4$ SYM is defined on a fixed $\mathbb{R}^{1,3}$, and that spacetime is background, not output. So this is emergence of the bulk from the boundary, of the extra radial and sphere directions from the field-theory data, and not the emergence of everything from nothing. The correspondence is also stated for fixed boundary conditions, asymptotically AdS. It is a complete, non-perturbative *definition* of quantum gravity with those asymptotics, arguably the only one we possess, but it is not a background-independent formulation, and it does not by itself tell us how to do quantum gravity in a cosmology like ours.

The status of the claim

AdS/CFT is a conjecture. It is unproven in the mathematical sense, and no derivation reduces one side to the other from first principles. What it has is evidence of a quality that few conjectures in physics enjoy: exact matches of protected quantities, agreement of entire symmetry representations, successful strong-coupling predictions later confirmed by the field-theory integrability of planar $\mathcal{N} = 4$, and the internal consistency of ten years of computations that could have failed and did not. Treat it as extremely well founded, not as a theorem. And remember that the emergent geometry is sharp only in the classical limit (13.64); away from large N and large λ, "spacetime" becomes stringy, then quantum, and finally, at small λ, ceases to be a useful notion at all. Geometry is what the gauge theory looks like in one particular corner of its parameter space.

Even hedged this way, the statement is remarkable. A property we take to be the stage on which physics happens, the existence and shape of spacetime, is here a derived, collective, approximate description of a system that does not fundamentally contain it. Gravity is emergent in the way that hydrodynamics is emergent from molecules, or thermodynamics from statistical mechanics: real, predictive, and not the bottom layer, which here is a gauge theory.

13.4.6 Where the course ends

Stand back and look at the arc. We began on the worldsheet, quantising a single relativistic string and finding that consistency alone, the vanishing of the Weyl anomaly, forced twenty-six

dimensions and then, with the superstring, ten, and forced the spectrum to contain a massless spin-two particle whose low-energy dynamics is Einstein gravity (Chapters 1 to 4). Gravity appeared first as a *law*, the requirement that the background solve Einstein's equations for the worldsheet theory to be conformal. We then built the ten- and eleven-dimensional supergravities, wove them into a single web by the dualities T, S and U, and found the theory's objects to be a whole democracy of branes, with strings only the lightest members (Chapters 5 to 7). We compactified on Calabi-Yau spaces, stabilised moduli with flux, and used branes and singularities to engineer gauge theories and something like the Standard Model (Chapters 8 to 11). And in the capstone we cashed two of the theory's oldest promissory notes: black-hole entropy from microstate counting (Chapter 12), and now spacetime itself as an emergent, holographic description of a gauge theory.

Gravity has appeared three times in these notes, and the progression is worth naming. As a *law*, in the Weyl anomaly. As a *state*, a coherent condensate of gravitons in the string spectrum. And here as an *emergent phenomenon*, the large-N shadow of a theory with no gravity in it. The three descriptions agree; they are the same physics seen at increasing depth.

We have deliberately stopped short of a great deal. Precision holography, where the factor $\frac{3}{4}$ becomes an interpolating function checked against integrability, is a subject of its own. The conformal bootstrap constrains $\mathcal{N}=4$ and its cousins non-perturbatively without ever mentioning a bulk. The applications reach into the physics of the quark-gluon plasma, of strange metals and quantum critical points, wherever a strongly coupled system needs a computable dual. And the open problems are the deep ones. We have a theory of quantum gravity in a box with negative curvature; the universe we live in is expanding and, as far as we can tell, has positive vacuum energy, and no controlled holographic description of de Sitter space exists. Whether string theory contains such vacua at all is the swampland question, unresolved. Whether spacetime can be reconstructed from entanglement in general, whether the interior of a black hole is smooth, whether holography survives in asymptotically flat space where we actually do our experiments: these are live, and they are yours to take up.

What the string has delivered, over the length of this course, is not a final theory. It is a working, mathematically rich, internally consistent theory of quantum gravity that reproduces general relativity and gauge theory at low energy, resolves the black-hole information paradox in at least one clean example, and, in AdS/CFT, teaches that spacetime and gravity may not be fundamental ingredients of the world but emergent ones, arising at large N from a quantum system that gets along without them. Where that lesson finally points, we do not yet know. That is a good place for a course to end.

Exercise 13.16 (The whole dictionary on one line)**.** *(Moderate.)* Assemble the parameter dictionary and check its internal consistency. Starting from $g_{\rm YM}^2 = 2\pi g_s$, $\lambda = g_{\rm YM}^2 N$, $L^4 = 4\pi g_s N\,\alpha'^2$:

(a) Express g_s and $(L/\ell_s)^4$ in terms of λ and N, and confirm $g_s = \lambda/(2\pi N)$ and $(L/\ell_s)^4 = 2\lambda$.

(b) Show that the two conditions for classical supergravity, $g_s \ll 1$ and $L \gg \ell_s$, are $N \gg \lambda \gg 1$, and sketch the corner of the (N,λ) plane they occupy.

(c) Show that a single worldsheet handle (g_s^2) and a single gauge-theory handle ($1/N^2$) agree at fixed λ, and identify the proportionality constant.

(d) Using the ten-dimensional Planck length $\ell_p^8 \sim G_{10} \sim g_s^2 \alpha'^4$, show that $L/\ell_p \sim N^{1/4}$, so the geometry is large in Planck units precisely when N is large. Where does the semiclassical bulk description break down as N is lowered?

Appendix A

Conventions (the single source of truth)

Every chapter points here. When a formula elsewhere in the notes looks off by a sign or a factor of 2π, this appendix is what it is meant to agree with. The conventions follow the mostly-plus, string-theory mainstream (close to Polchinski and to Becker, Becker and Schwarz), and are chosen so that the spacetime effective actions of Chapter 5 come out clean.

A.1 Units, signature, indices

We set $\hbar = c = 1$ throughout. The spacetime metric has *mostly-plus* signature,

$$\eta_{\mu\nu} = \mathrm{diag}(-, +, +, \ldots, +), \tag{A.1}$$

in D spacetime dimensions ($D = 26$ for the bosonic string, $D = 10$ for the superstring, $D = 11$ for M-theory). A timelike vector has $V^\mu V_\mu < 0$. This is the signature in which a Nambu-Goto area and a supergravity Einstein-Hilbert term carry their natural signs; the price is the occasional extra minus sign relative to the particle-physics mostly-minus convention, and we pay it consistently.

Index conventions are collected once here and used everywhere:

Symbol	Range / meaning
$\mu, \nu, \rho, \ldots$	curved spacetime, $0, 1, \ldots, D-1$
$a, b, c, \ldots$ (early Latin)	flat/frame (tangent-space) spacetime, raised with η_{ab}
α, β (on the worldsheet)	worldsheet coordinates $\sigma^0 = \tau$, $\sigma^1 = \sigma$
$i, j, k, \ldots$	transverse or spatial directions, context permitting
$m, n, \ldots$	real internal (compact) directions
$I, J, \ldots$	gauge or flavour indices
$A, B, \ldots$	spinor indices (see Appendix B)
$M, N, \ldots$	the full set in $D = 11$, or an $O(d,d)$ multiplet, per context

On the complex internal manifolds of Chapter 8 we use holomorphic indices $i, \bar{\jmath} = 1, \ldots, n$ (with $n = 3$ for a Calabi-Yau threefold); the bar denotes antiholomorphic. Symmetrisation and antisymmetrisation carry unit weight,

$$T_{(\mu\nu)} = \tfrac{1}{2}(T_{\mu\nu} + T_{\nu\mu})\,, \qquad T_{[\mu\nu]} = \tfrac{1}{2}(T_{\mu\nu} - T_{\nu\mu})\,, \tag{A.2}$$

and generally $T_{[\mu_1 \cdots \mu_p]}$ carries a prefactor $1/p!$.

A.2 The string scale

The one dimensionful parameter of a string is its tension. We write the tension as

$$T \;=\; \frac{1}{2\pi\alpha'}, \tag{A.3}$$

with α' ("alpha-prime", the Regge slope) carrying units of length squared. The *string length* and *string mass* are

$$\ell_s \coloneqq \sqrt{\alpha'}, \qquad M_{\rm s} \coloneqq \frac{1}{\ell_s} = \frac{1}{\sqrt{\alpha'}}. \tag{A.4}$$

Excited string states sit on Regge trajectories with masses $M^2 \sim n/\alpha'$, so ℓ_s is the length at which stringy physics turns on. Some sources use $\ell_s = 2\pi\sqrt{\alpha'}$ or $\sqrt{2\alpha'}$; we do not, and any imported formula is converted to $\ell_s = \sqrt{\alpha'}$ before it appears in these notes.

The dimensionless *string coupling* is set by the vacuum value of the dilaton ϕ,

$$g_s \coloneqq e^{\langle\phi\rangle}. \tag{A.5}$$

It is not a parameter put in by hand: it is the expectation value of a field, which is why strong and weak coupling can be related by dualities that move $\langle\phi\rangle$ (Chapter 7). The genus-g term in a closed-string amplitude comes weighted by g_s^{2g-2} (Chapter 4).

Newton's constant in D dimensions is written $2\kappa_D^2 = 16\pi G_D$. In ten dimensions the type II value in terms of string data is

$$2\kappa_{10}^2 \;=\; (2\pi)^7 \alpha'^4 g_s^2, \tag{A.6}$$

which fixes the ten-dimensional Planck length against ℓ_s and g_s; the eleven-dimensional version and the reduction relations are collected in Appendix I.

A.3 Levi-Civita symbol and tensor

The totally antisymmetric *symbol* $\epsilon^{\mu_1\cdots\mu_D}$ is defined by $\epsilon^{01\cdots(D-1)} = +1$ and takes values $0, \pm 1$ in every frame. The Levi-Civita *tensor* carries the metric determinant,

$$\varepsilon_{\mu_1\cdots\mu_D} = \sqrt{-g}\,\epsilon_{\mu_1\cdots\mu_D}, \qquad \varepsilon^{\mu_1\cdots\mu_D} = \frac{\operatorname{sgn}(g)}{\sqrt{-g}}\,\epsilon^{\mu_1\cdots\mu_D}, \tag{A.7}$$

so that ε is a genuine tensor under general coordinate transformations. In Lorentzian signature $\operatorname{sgn}(g) = -1$.

A.4 Differential forms and the Hodge star

A p-form is expanded with a $1/p!$,

$$\omega = \frac{1}{p!}\,\omega_{\mu_1\cdots\mu_p}\,\mathrm{d}x^{\mu_1}\wedge\cdots\wedge\mathrm{d}x^{\mu_p}. \tag{A.8}$$

The exterior derivative d raises the degree by one and is nilpotent, $\mathrm{d}^2 = 0$; on components,

$$(\mathrm{d}\omega)_{\mu_0\cdots\mu_p} = (p+1)\,\partial_{[\mu_0}\omega_{\mu_1\cdots\mu_p]}. \tag{A.9}$$

The wedge product of a p-form and a q-form satisfies $\omega_p\wedge\omega_q = (-1)^{pq}\,\omega_q\wedge\omega_p$. The Hodge star $\star$ maps a p-form to a $(D-p)$-form,

$$(\star\omega)_{\mu_1\cdots\mu_{D-p}} = \frac{1}{p!}\,\varepsilon_{\mu_1\cdots\mu_{D-p}}{}^{\nu_1\cdots\nu_p}\,\omega_{\nu_1\cdots\nu_p}, \tag{A.10}$$

and on a p-form in Lorentzian signature obeys $\star\star\,\omega = -(-1)^{p(D-p)}\omega$ (the overall minus is the Lorentzian one; it becomes a plus in Euclidean signature). The invariant kinetic term of a p-form gauge field is

$$\omega \wedge \star\omega = \frac{1}{p!}\,\omega_{\mu_1\cdots\mu_p}\omega^{\mu_1\cdots\mu_p}\sqrt{-g}\,\mathrm{d}^D x. \tag{A.11}$$

Integration by parts uses $\int \mathrm{d}(\cdots) = 0$ on a closed manifold, and Stokes' theorem in the form $\int_M \mathrm{d}\omega = \int_{\partial M}\omega$.

A.5 Gauge fields and field strengths

A p-form gauge potential C_p has field strength

$$F_{p+1} = \mathrm{d}C_p, \qquad \mathrm{d}F_{p+1} = 0 \ \ \text{(Bianchi)}, \tag{A.12}$$

invariant under $C_p \to C_p + \mathrm{d}\lambda_{p-1}$. For the ordinary gauge field, $F = \mathrm{d}A$ with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$; for a non-abelian field, $F = \mathrm{d}A + A\wedge A$ with $A = A^I T_I$, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]$, and the trace normalisation $\mathrm{Tr}(T_I T_J) = \frac{1}{2}\delta_{IJ}$ for the defining representation of $\mathrm{SU}(N)$.

The Ramond-Ramond sector of type II is written democratically where useful: the RR potentials C_p (with p even in IIA, odd in IIB) assemble into a polyform $C = \sum_p C_p$, with improved field strength

$$F = \mathrm{d}C - H\wedge C, \qquad H = \mathrm{d}B, \tag{A.13}$$

where B is the NS-NS two-form. The self-duality $F_p = (-1)^{\lfloor p/2\rfloor}\star F_{D-p}$ is imposed at the level of the equations of motion. The Chern-Simons and Wess-Zumino couplings that go with these fields, including the $\sqrt{\hat{A}}$ curvature corrections to brane charge, are stated where they are used (Chapter 6) and collected in Appendix I.

A.6 Curvature

We use the conventions in which the Riemann tensor is

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}, \tag{A.14}$$

with the Levi-Civita connection $\Gamma^\rho_{\mu\nu} = \frac{1}{2}g^{\rho\lambda}(\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu})$. The Ricci tensor is $R_{\mu\nu} = R^\rho{}_{\mu\rho\nu}$ and the Ricci scalar $R = g^{\mu\nu}R_{\mu\nu}$. With these signs a round sphere has $R > 0$, and the Einstein equations from $S = \frac{1}{2\kappa^2}\int\sqrt{-g}\,R + S_{\text{matter}}$ read $R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \kappa^2 T_{\mu\nu}$.

A.7 Spinors and Clifford algebras

The Clifford algebra in a flat frame is

$$\{\Gamma^a, \Gamma^b\} = 2\,\eta^{ab}, \tag{A.15}$$

with curved gammas $\Gamma^\mu = e^\mu{}_a\Gamma^a$ built from the vielbein $e^\mu{}_a$ ($g_{\mu\nu} = e_\mu{}^a e_\nu{}^b\eta_{ab}$). Antisymmetrised products are written $\Gamma^{a_1\cdots a_n} = \Gamma^{[a_1}\cdots\Gamma^{a_n]}$. In even D the chirality operator is $\Gamma_* \propto \Gamma^0\Gamma^1\cdots\Gamma^{D-1}$, normalised to $\Gamma_*^2 = 1$. Which of the reality (Majorana), chirality (Weyl) and combined Majorana-Weyl conditions can be imposed depends on D mod 8; the dimension-by-dimension table, the reason $D = 10$ admits Majorana-Weyl spinors, and the Fierz identities live in Appendix B. The Dirac slash is $\not{p} = \Gamma^\mu p_\mu$.

A.8 Worldsheet and mode conventions

On the closed-string worldsheet we use $\sigma \sim \sigma + 2\pi$ and complex coordinates in the Euclidean (radial) picture

$$z = e^{\tau_E - i\sigma}, \qquad \bar{z} = e^{\tau_E + i\sigma}, \tag{A.16}$$

so that time ordering becomes radial ordering. Holomorphic (left-moving) and antiholomorphic (right-moving) sectors are labelled $X_{\mathrm{L}}, X_{\mathrm{R}}$. A free boson has the holomorphic current and the standard propagator

$$\partial X^\mu(z) = -i\sqrt{\frac{\alpha'}{2}} \sum_{n\in\mathbb{Z}} \frac{\alpha_n^\mu}{z^{n+1}}, \qquad \langle X^\mu(z,\bar{z})\, X^\nu(0)\rangle = -\frac{\alpha'}{2}\,\eta^{\mu\nu} \ln|z|^2, \tag{A.17}$$

with $\alpha_0^\mu = \sqrt{\alpha'/2}\, p^\mu$ for the closed string and oscillator algebra $[\alpha_m^\mu, \alpha_n^\nu] = m\,\delta_{m+n,0}\,\eta^{\mu\nu}$. Normal ordering (annihilators to the right) is denoted $:\cdot:$. A holomorphic operator of weight h has the Laurent expansion $\mathcal{O}(z) = \sum_n \mathcal{O}_n\, z^{-n-h}$. The Fourier convention for spacetime fields is $f(x) = \int \frac{\mathrm{d}^D k}{(2\pi)^D}\, e^{ik\cdot x} \tilde{f}(k)$.

A.9 Frequently used symbols

Symbol	Meaning
$\alpha',\ \ell_s = \sqrt{\alpha'},\ M_{\mathrm{s}} = 1/\ell_s$	Regge slope, string length, string mass
$T = 1/(2\pi\alpha')$	fundamental string tension
$g_s = e^{\langle\phi\rangle}$	string coupling
$g_{\mu\nu},\ B_{\mu\nu},\ \phi$	NS-NS graviton, Kalb-Ramond field, dilaton
$C_p,\ F_{p+1} = \mathrm{d}C_p$	RR p-form potential and field strength
$H = \mathrm{d}B$	NS-NS three-form field strength
$2\kappa_D^2 = 16\pi G_D$	gravitational coupling in D dimensions
M_{pl}	(reduced) Planck mass in the relevant dimension
$\Gamma^a,\ \Gamma_*$	Clifford generators, chirality operator
$L_n,\ \tilde{L}_n$	Virasoro generators (left, right)
c	central charge / Weyl anomaly coefficient
Q	BRST charge; also a brane/particle charge, per context
$J,\ \Omega$	Kähler form, holomorphic top form on a Calabi-Yau
$h^{p,q}$	Hodge numbers of the internal manifold

Remark A.1. Two habits save grief later. First, α' appears only through $\ell_s = \sqrt{\alpha'}$ and the tension $1/(2\pi\alpha')$, never as a bare "α'" with a hidden factor. Second, all RR fields are treated in the democratic formulation of Eq. (A.13), with self-duality imposed at the end, so a C_p and its dual C_{D-2-p} are never double-counted in an action.

Appendix B

Spinors and Clifford Algebras in Diverse Dimensions

B.1 Clifford algebras and gamma matrices in D dimensions

A spinor is the object on which the square root of the Klein-Gordon operator acts. Dirac's trick, $(\not{\partial})^2 = \Box$, needs matrices Γ^a whose symmetrised products reproduce the metric. The same construction, carried out in the relevant D, underlies everything spinorial in these notes: the Ramond ground state of Chapter 3, the gravitino and dilatino of ten- and eleven-dimensional supergravity, the supercharges of Appendix C, and the κ-symmetric brane actions of Chapter 6. This section builds those matrices dimension by dimension, records the size $2^{\lfloor D/2 \rfloor}$ of a Dirac spinor, isolates the chirality operator that exists only in even D, and defines the charge conjugation matrix together with the properties that the reality analysis of Appendix B.2 and the bilinear catalogue of Appendix B.3 take as input. Conventions follow Appendix A: mostly-plus signature $\eta_{ab} = \mathrm{diag}(-, +, \ldots, +)$ with a single time direction, and the Clifford relation

$$\{\Gamma^a, \Gamma^b\} \;=\; 2\,\eta^{ab}\,\mathbf{1}. \tag{B.1}$$

The big picture

Everything here is governed by how D enters two invariants. The smallest matrices obeying Eq. (B.1) have size $2^{\lfloor D/2 \rfloor}$, the number of complex components of a Dirac spinor. Whether that representation is irreducible or splits is set by D mod 2: odd D gives a single irreducible spinor with no chirality, even D splits it into two Weyl halves under a chirality operator Γ_*. A third object, the charge conjugation matrix C, sits on top, and the symmetry of its transpose runs with period eight in D. That period-eight pattern decides which spinors can be made real: it is what lets $D = 10$ carry Majorana-Weyl spinors and makes $D = 11$ the ceiling for supergravity. Here we assemble Γ^a, Γ_* and C; the reality conditions they allow are read off in Appendix B.2 and the supercharge count in Appendix B.4.

B.1.1 The Clifford algebra and the size of a Dirac spinor

The associative algebra generated by symbols Γ^a subject only to Eq. (B.1) is the Clifford algebra. For counting its representations the signature is irrelevant: complexifying, any choice of η_{ab} with the same D gives isomorphic algebras, because rescaling a generator by i flips the sign of its square. Signature returns only when we ask which representations are *real*, the subject of Appendix B.2. So the representation theory is a statement about D alone.

A vector-space basis is furnished by the antisymmetrised products

$$\Gamma^{a_1\cdots a_n} \;:=\; \Gamma^{[a_1}\Gamma^{a_2}\cdots\Gamma^{a_n]}, \qquad n=0,1,\ldots,D, \tag{B.2}$$

with $\Gamma^{()}=\mathbf{1}$ at $n=0$. The relation Eq. (B.1) lets any word in the Γ^a be reordered and its repeated indices contracted, reducing it to a combination of the Eq. (B.2); these are linearly independent because they carry distinct antisymmetric index structures. Counting them,

$$\dim_{\mathbb{C}}\,\mathrm{Cl}(D) \;=\; \sum_{n=0}^{D}\binom{D}{n} \;=\; 2^D. \tag{B.3}$$

An irreducible representation by $N\times N$ complex matrices spans, by Burnside's theorem, the full matrix algebra of dimension N^2. Matching against Eq. (B.3) fixes N.

Proposition B.1 (Dirac representation)**.** *Over* $\mathbb{C}$ *the Clifford algebra of Eq. (B.1) is*

$$\mathrm{Cl}(2k)\otimes\mathbb{C}\cong \mathrm{Mat}_{2^k}(\mathbb{C}), \qquad \mathrm{Cl}(2k{+}1)\otimes\mathbb{C}\cong \mathrm{Mat}_{2^k}(\mathbb{C})\oplus\mathrm{Mat}_{2^k}(\mathbb{C}). \tag{B.4}$$

For $D=2k$ *even there is a unique irreducible representation up to equivalence, the* Dirac representation*, of complex dimension*

$$N \;=\; 2^{\lfloor D/2\rfloor}. \tag{B.5}$$

For $D=2k{+}1$ *odd there are two inequivalent irreducible representations, each of the same dimension Eq. (B.5), related by* $\Gamma^a\mapsto-\Gamma^a$*.*

The even case is the counting above: $2^D=(2^k)^2$ forces $N=2^k$, and Burnside plus Schur give uniqueness. The odd case is quickest to see from the top product $\Gamma^0\cdots\Gamma^{D-1}$, which for odd D commutes with every Γ^a and so, by Schur, is a scalar in any irreducible representation; the two possible scalars are the two inequivalent representations. The general classification Eq. (B.4), including its real forms, is the Brauer-Weyl (1935) and Atiyah-Bott-Shapiro (1964) theory; we quote it and use it. The values that matter are small:

D	2	3	4	5	6	7	8	9	10	11
$\lfloor D/2\rfloor$	1	1	2	2	3	3	4	4	5	5
$N=2^{\lfloor D/2\rfloor}$	2	2	4	4	8	8	16	16	32	32

A $D=11$ Dirac spinor carries 32 complex components; imposing the Majorana condition of Appendix B.2 leaves 32 real ones, the supercharges of eleven-dimensional supergravity. That this is the largest N compatible with a graviton of spin ≤ 2 is the counting of Appendix B.4.

B.1.2 Explicit gamma matrices, dimension by dimension

The representation is built by tensoring Pauli matrices, adding two dimensions at a time. Write $\sigma^1,\sigma^2,\sigma^3$ for the Pauli matrices and start in $D=2$ with the single time direction already in place,

$$\Gamma^0_{(2)}=i\sigma^2=\begin{pmatrix}0&1\\-1&0\end{pmatrix}, \qquad \Gamma^1_{(2)}=\sigma^1=\begin{pmatrix}0&1\\1&0\end{pmatrix}, \tag{B.6}$$

which obey $(\Gamma^0)^2=-\mathbf{1}$, $(\Gamma^1)^2=+\mathbf{1}$ and $\{\Gamma^0,\Gamma^1\}=0$, as required by mostly-plus. The recursion raises D by two while keeping the one time direction.

Proposition B.2 (Doubling)**.** *Let* $\{\Gamma^a_{(D)}\}_{a=0}^{D-1}$ *be an* $N\times N$ *representation of Eq. (B.1) in* D *dimensions. Then*

$$\Gamma^a_{(D+2)}=\Gamma^a_{(D)}\otimes\sigma^3\ (a<D), \qquad \Gamma^D_{(D+2)}=\mathbf{1}_N\otimes\sigma^1, \qquad \Gamma^{D+1}_{(D+2)}=\mathbf{1}_N\otimes\sigma^2, \tag{B.7}$$

is a $2N\times 2N$ *representation in* $D+2$ *dimensions, the two new directions being spacelike.*

The checks are one line each: $\{\sigma^3,\sigma^1\}=\{\sigma^3,\sigma^2\}=\{\sigma^1,\sigma^2\}=0$ decouples the old block from the new, $(\sigma^3)^2=\mathbf{1}$ preserves the old anticommutators, and $(\sigma^1)^2=(\sigma^2)^2=\mathbf{1}$ makes both new generators square to $+\mathbf{1}$. Iterating Eq. (B.7) from Eq. (B.6) produces the Dirac matrices in every even D, of size $2^{D/2}$ as promised. For $D=4$ one instance is the chiral (Weyl) basis

$$\Gamma^a_{(4)}=\begin{pmatrix}0 & \sigma^a\\ \bar\sigma^a & 0\end{pmatrix},\qquad \sigma^a=(\,\mathbf{1},\sigma^i\,),\quad \bar\sigma^a=(-\mathbf{1},\sigma^i\,), \tag{B.8}$$

block off-diagonal, the form used in Appendix C to split a four-dimensional Dirac spinor into two-component Weyl pieces.

Proposition B.3 (Unitary gauge and the Dirac adjoint)**.** *The representation built from Eqs. (B.6) and (B.7) has Γ^0 anti-Hermitian and each spatial Γ^i Hermitian, so covariantly*

$$(\Gamma^a)^\dagger \;=\; \Gamma^0\,\Gamma^a\,\Gamma^0,\qquad (\Gamma^0)^2=-\mathbf{1}. \tag{B.9}$$

The Dirac adjoint is built from the Hermitian intertwiner

$$A\;\coloneqq\; i\,\Gamma^0,\qquad A^\dagger=A,\quad A^2=\mathbf{1},\quad A\,\Gamma^a A^{-1}=-(\Gamma^a)^\dagger,\qquad \bar\psi\;\coloneqq\;\psi^\dagger A, \tag{B.10}$$

and this A is the one used throughout Appendix B.3.

The two forms are compatible: $A^2=\mathbf{1}$ follows from $(\Gamma^0)^2=-\mathbf{1}$, and $A\Gamma^aA^{-1}=i\Gamma^0\Gamma^a(i\Gamma^0)^{-1}=-\Gamma^0\Gamma^a\Gamma^0=-(\Gamma^a)^\dagger$ by Eq. (B.9). The Hermitian choice $A=i\Gamma^0$ is fixed once and for all here; Appendices B.2 and B.3 use no other.

B.1.3 The chirality operator in even dimensions

In even $D=2k$ the ordered product of all generators anticommutes with each of them (moving a single Γ^a through the other $D-1$ factors costs $D-1$ odd sign flips). Normalised to square to the identity it is the chirality operator

$$\Gamma_*\;\coloneqq\;(-i)^{\,k-1}\,\Gamma^0\Gamma^1\cdots\Gamma^{D-1},\qquad D=2k. \tag{B.11}$$

The phase is fixed by the square. Using Eq. (B.1) to reverse the product, $(\Gamma^0\cdots\Gamma^{D-1})^2=(-1)^{D(D-1)/2}\prod_a\eta^{aa}=(-1)^{k+1}$ for one time direction, so the factor $(-i)^{k-1}$ delivers

$$\Gamma_*^2=\mathbf{1},\qquad \{\Gamma_*,\Gamma^a\}=0,\qquad \Gamma_*^\dagger=\Gamma_*. \tag{B.12}$$

Because Γ_* commutes with the Lorentz generators $\frac14[\Gamma^a,\Gamma^b]=\frac12\Gamma^{ab}$ (it anticommutes with each factor twice), it is a Lorentz invariant, and its eigenvalues ±1 label the two *Weyl* (chiral) representations. The projectors

$$P_\pm=\tfrac12\,(\mathbf{1}\pm\Gamma_*)\,,\qquad P_\pm^2=P_\pm,\quad P_++P_-=\mathbf{1}, \tag{B.13}$$

split the Dirac spinor into two Weyl spinors of $2^{D/2-1}$ complex components each. This is the split that GSO acts on (Chapter 3) and that distinguishes the non-chiral IIA spectrum (opposite chiralities on the two sides) from the chiral IIB spectrum (equal chiralities); the physics of that choice is developed there and in Appendix B.2.

Because Γ_* anticommutes with each generator, it commutes with even-rank products and anticommutes with odd-rank ones,

$$\Gamma_*\,\Gamma^{a_1\cdots a_n}\;=\;(-1)^n\,\Gamma^{a_1\cdots a_n}\,\Gamma_*, \tag{B.14}$$

and it implements Hodge duality on the antisymmetric basis,

$$\Gamma^{a_1\cdots a_n}\,\Gamma_*\;\propto\;\varepsilon^{a_1\cdots a_n}{}_{b_1\cdots b_{D-n}}\,\Gamma^{b_1\cdots b_{D-n}}, \tag{B.15}$$

so in even D the rank-n and rank-$(D-n)$ products are not independent once Γ_* is included; the proportionality constant, a signature- and dimension-dependent phase, is fixed in Appendix A. The chirality selection rule Eq. (B.14) is what the bilinear catalogue of Appendix B.3 uses to sort $\bar{\psi}\,\Gamma^{(n)}\chi$ by chirality.

For $D = 10$ the phase in Eq. (B.11) is $(-i)^4 = 1$, so $\Gamma_* = \Gamma^0\Gamma^1\cdots\Gamma^9$, the object often written Γ^{11} or $\bar{\Gamma}$ in the string literature. For $D = 4$ it is $\Gamma_* = -i\,\Gamma^0\Gamma^1\Gamma^2\Gamma^3$, the mostly-plus counterpart of the familiar γ^5.

Odd dimensions are reached not by doubling but by adjoining Γ_* as an extra generator. Since $\Gamma_*^2 = \mathbf{1}$, $\{\Gamma_*, \Gamma^a\} = 0$ and Γ_* squares like a spacelike direction, the set

$$\{\Gamma^0, \ldots, \Gamma^{2k-1},\ \Gamma^{2k} \coloneqq \Gamma_*\} \tag{B.16}$$

represents Eq. (B.1) in $D = 2k{+}1$ at the same size $2^k = 2^{\lfloor D/2 \rfloor}$. The opposite sign $\Gamma^{2k} = -\Gamma_*$ is the second, inequivalent representation of Theorem B.1.

Careful

There is no chirality operator inside odd D. The natural candidate $\Gamma^0\cdots\Gamma^{D-1}$ is central, a multiple of $\mathbf{1}$ by Theorem B.1, so it cannot project, and a would-be Weyl condition is empty: the Dirac spinor is already irreducible. Chiral fermions, and with them GSO-type projections and net gauge anomalies, therefore exist only in even spacetime dimensions.

B.1.4 The charge conjugation matrix

Charge conjugation supplies the second piece of index gymnastics a spinor needs. To make a Lorentz scalar or tensor out of two spinors one lowers a spinor index, forming $\psi^T C\,\Gamma^{(n)}\chi$, and the same matrix is what lets a reality condition be imposed. What this requires is a matrix intertwining each Γ^a with its transpose. Transposing Eq. (B.1) shows that $\{(\Gamma^a)^T\}$ and equally $\{-(\Gamma^a)^T\}$ satisfy the same Clifford algebra. In even D the representation is unique (Theorem B.1), so each transposed set must be conjugate to the original by an invertible matrix. These intertwiners are the two charge conjugation matrices.

Definition B.4 (Charge conjugation matrices)**.** $C_\pm$ are the invertible matrices, unique up to scale, obeying

$$C_\pm\,\Gamma^a\,C_\pm^{-1} \;=\; \pm\,(\Gamma^a)^T. \tag{B.17}$$

In even D both C_+ and C_- exist, and they are related by the chirality operator,

$$C_+ \;\propto\; C_-\,\Gamma_*, \tag{B.18}$$

since conjugating Γ^a by Γ_* supplies the extra minus sign. In odd D the sign in Eq. (B.17) is not free: applying it to the central element $\Gamma^0\cdots\Gamma^{D-1}$ (a scalar, so equal to its own transpose) is consistent only for

$$\pm \;=\; (-1)^{(D-1)/2}, \tag{B.19}$$

so exactly one of $C_\pm$ survives when D is odd. The other transposed representation is the inequivalent one.

The property that does the work in Appendix B.3 is the transpose symmetry of C itself. Reversing the n factors in Eq. (B.2) costs $(-1)^{n(n-1)/2}$, and each transpose supplies a sign $\pm$ through Eq. (B.17), giving the master relation

$$\left(C_\pm\,\Gamma^{a_1\cdots a_n}\right)^T \;=\; \xi_\pm\,(\pm1)^n\,(-1)^{n(n-1)/2}\,C_\pm\,\Gamma^{a_1\cdots a_n}, \qquad C_\pm^T = \xi_\pm\,C_\pm. \tag{B.20}$$

Here $\xi_\pm = \pm 1$ is the symmetry of $C_\pm$, and Eq. (B.20) is what the bilinear catalogue reads off to decide which $\bar\psi\,\Gamma^{(n)}\chi$ are symmetric or antisymmetric under exchange. In the notation of Appendix B.3 the symmetry $\xi_\pm$ is written ξ and the intertwiner sign $\pm$ is written t, so the same relation appears there as $\sigma_n = \xi\, t^{\,n}(-1)^{n(n-1)/2}$; the letter η is reserved throughout for the metric.

The symmetry $\xi_\pm$ is not free either. It is fixed rank by rank by evaluating Eq. (B.20) at each n, subject to the constraint that among the 2^D basis matrices Eq. (B.2) the symmetric ones number $N(N+1)/2$ (equivalently, exceed the antisymmetric by $N = 2^{\lfloor D/2\rfloor}$). For even D the solution is the closed form

$$\xi_\pm \;=\; \operatorname{sgn}\Big[\cos\tfrac{\pi(D\mp 1)}{4}\Big], \qquad D \text{ even}, \tag{B.21}$$

periodic with period eight in D; for odd D only one C survives, and its symmetry follows from Eq. (B.20) together with the existence rule Eq. (B.19). The rank-by-rank count behind Eq. (B.21) is carried out in Freedman and Van Proeyen (2012); we quote the result. Its values give the following table (S denotes $\xi = +1$, symmetric; A denotes $\xi = -1$, antisymmetric; $\varnothing$ marks the C that does not exist):

D	1	2	3	4	5	6	7	8	9	10	11
C_+	S	S	$\varnothing$	A	A	A	$\varnothing$	S	S	S	$\varnothing$
C_-	$\varnothing$	A	A	A	$\varnothing$	S	S	S	$\varnothing$	A	A

The columns repeat with period eight, the shadow of Bott periodicity. The two dimensions where C is antisymmetric and $C_-\Gamma^a$ is symmetric, namely $D = 2$ and $D = 10$, are precisely the ones admitting Majorana-Weyl spinors and hence a Lorentz-invariant kinetic term for a real chiral fermion; Appendix B.2 works this out, and it is what makes a ten-dimensional super-Yang-Mills theory possible. For $D = 11$ the surviving $C = C_-$ is antisymmetric with $C\Gamma^a C^{-1} = -(\Gamma^a)^T$, the convention in which the eleven-dimensional gravitino Lagrangian is written.

Two further properties are used repeatedly. First, in the unitary gauge of Theorem B.3 C can be chosen unitary, $C^\dagger = C^{-1}$. Second, combining C with the Dirac-adjoint matrix $A = i\Gamma^0$ of Eq. (B.10) produces the matrix $B \propto C\,A^T$ that intertwines the complex conjugate representation, $(\Gamma^a)^* = \pm B\,\Gamma^a B^{-1}$. Whether B can be chosen to satisfy $B^*B = +\mathbf{1}$, the condition for a Majorana (real) spinor to exist, is exactly the reality question of Appendix B.2; Eqs. (B.17), (B.20) and (B.21) are the ingredients it consumes.

Remark B.5 (Beyond the lectures). The period-eight pattern in Eq. (B.21) and the table above is the real Bott periodicity of the Clifford algebras $\mathrm{Cl}(s,t)$, whose Morita classes depend only on $s - t \bmod 8$. The eight-fold ladder of commutant division algebras,

$$\mathbb{R},\ \mathbb{C},\ \mathbb{H},\ \mathbb{H}\oplus\mathbb{H},\ \mathbb{H},\ \mathbb{C},\ \mathbb{R},\ \mathbb{R}\oplus\mathbb{R},$$

is the data Atiyah, Bott and Shapiro (1964) organised into KO-theory. The same ladder underlies the division-algebra description of minimal supersymmetry in $D = 3, 4, 6, 10$ (Kugo and Townsend 1983), developed in Appendix B.3 where the Fierz identity closes the super-Yang-Mills algebra. Appendix B.2 uses only the conclusions modulo eight; the algebraic source is this periodicity.

B.2 Majorana, Weyl, and Majorana-Weyl by dimension

The Dirac spinor of $\mathrm{Spin}(1, D-1)$, the $2^{\lfloor D/2\rfloor}$-dimensional module built in Appendix B.1, is almost never the representation a physical theory actually carries. Two independent operations trim it. A *reality* (Majorana) condition ties a spinor to its complex conjugate and halves the real

component count; a *chirality* (Weyl) projection, available whenever D is even, halves it again. Which of the two can be imposed, and whether they can be imposed together, is fixed entirely by D mod 8. Almost everything special about ten dimensions, from the GSO projection to the split between the two type II theories, follows from the single fact that $D = 10$ is where both conditions hold at once.

Throughout this section spacetime is Lorentzian with a single time direction and signature $(-,+,\dots,+)$, as in Appendix A. The reality type of a spinor depends on the signature and not on D alone; the general rule in signature (t,s) is quoted at the end of the section.

B.2.1 The conjugation intertwiner and the Majorana condition

Complex conjugation turns the Clifford generators into Γ^{a*}. Both $+\Gamma^{a*}$ and $-\Gamma^{a*}$ satisfy the same algebra $\{\Gamma^a,\Gamma^b\} = 2\eta^{ab}$ as the Γ^a themselves, so by the uniqueness of the irreducible Clifford module (Theorem B.1) there exist invertible matrices $B_\pm$, the *conjugation intertwiners*, with

$$B_\pm\,\Gamma^a\,B_\pm^{-1} \;=\; \pm\,\Gamma^{a*}. \tag{B.22}$$

In even D both signs are realised on the irreducible module (they are related by $B_- \propto B_+\Gamma_*$); in odd D only one of the two is available, fixed by D mod 4 through the sign of the central element $\Gamma^0\cdots\Gamma^{D-1}$. This B is the complex-conjugation intertwiner assembled in Appendix B.1.4 from the charge conjugation matrix C and the Hermitian Dirac-adjoint matrix $A = i\Gamma^0$ of Eq. (B.10) ($A^\dagger = A$, $A^2 = \mathbf{1}$), as $B \propto C\,A^T$; we work with B directly here because reality is naturally a statement about complex conjugation.

The Lorentz generators $\Sigma^{ab} = \frac{1}{4}[\Gamma^a,\Gamma^b]$ inherit from Eq. (B.22) the property

$$B\,\Sigma^{ab}\,B^{-1} \;=\; \Sigma^{ab*}, \tag{B.23}$$

for either choice $B = B_\pm$, since the sign in Eq. (B.22) appears squared. This is what makes the following condition Lorentz covariant. A *Majorana spinor* is one that equals its own conjugate,

$$\psi^* \;=\; B\,\psi, \tag{B.24}$$

which by Eq. (B.23) is preserved by Lorentz transformations. Whether Eq. (B.24) has nonzero solutions is a matter of consistency: conjugating it once more gives $\psi = B^*\psi^* = B^*B\,\psi$, so the condition is admissible only if

$$B^*B \;=\; \mathbf{1}. \tag{B.25}$$

When instead $B^*B = -\mathbf{1}$ the reality condition Eq. (B.24) has only the trivial solution and no genuine Majorana spinor exists; the fix is the symplectic construction of Appendix B.2.4.

Everything hinges on the sign of B^*B, which is periodic in D. That sign is fixed by the transpose-symmetry sign $\xi_\pm$ of the charge conjugation matrix (Eqs. (B.20) and (B.21)), which recurs with period eight; we state the outcome and refer to the standard treatments (Kugo and Townsend; van Nieuwenhuizen; Freedman and Van Proeyen, Chapter 3; Green, Schwarz and Witten, Appendix 5.A; Polchinski, Appendix B) for the book-keeping.

Proposition B.6 (Reality by dimension)**.** *In Lorentzian signature $(-,+,\dots,+)$ there is a choice of B in Eq. (B.22) with $B^*B = +\mathbf{1}$, so that a Majorana spinor Eq. (B.24) exists, precisely for*

$$D \bmod 8 \in \{0,1,2,3,4\}.$$

For D mod $8 \in \{5,6,7\}$ every available B has $B^*B = -\mathbf{1}$, no Majorana spinor exists, and the weakest reality condition is the symplectic one of Appendix B.2.4.

A Majorana spinor has half the real components of a Dirac spinor, namely $2^{\lfloor D/2\rfloor}$. Nothing here has used chirality, so the statement holds in odd D as well; $D = 11$, with $11 \equiv 3$, is Majorana.

B.2.2 Weyl spinors

Chirality is the second truncation, and it exists only in even D. There the chirality operator Γ_* of Eq. (B.11) anticommutes with every Γ^a, squares to $\mathbf{1}$, and through the projectors $P_\pm = \frac{1}{2}(\mathbf{1} \pm \Gamma_*)$ splits the Dirac spinor into two *Weyl* spinors of $2^{D/2-1}$ complex components each. In odd D the same product is central (Appendix B.1.3), so there is no chirality and the Dirac spinor is already irreducible. A Weyl condition is purely complex and signature-blind: it can be imposed in every even dimension and in no odd one. Whether that projection is compatible with the reality condition of Appendix B.2.1 is the subject of the next subsection.

B.2.3 Majorana-Weyl: when reality respects chirality

The two truncations coexist only if the reality condition maps a Weyl spinor to a Weyl spinor of the *same* chirality. The charge conjugate of ψ is $\psi^c = B^{-1}\psi^*$, and its chirality is read off from

$$\Gamma_*\,\psi^c \;=\; B^{-1}(B\,\Gamma_*\,B^{-1})\psi^* \;=\; B^{-1}(\pm\,\Gamma_*)\psi^*, \tag{B.26}$$

where the sign is that of $B\Gamma_*B^{-1} = \pm\Gamma_*$, which follows from Eqs. (B.11) and (B.22) once the reality of Γ_* is fixed by D mod 8. A Majorana-Weyl spinor exists exactly when all three requirements meet: D even (so that Γ_* projects), $B^*B = +\mathbf{1}$ (so that Majorana is consistent), and $B\Gamma_*B^{-1} = +\Gamma_*$ (so that conjugation preserves chirality). Evaluating the last two signs across the even dimensions gives a clean verdict.

Proposition B.7 (Majorana-Weyl)**.** *In Lorentzian signature* $(-,+,\dots,+)$ *a Majorana-Weyl spinor exists if and only if*

$$D \equiv 2 \pmod 8.$$

Its real dimension is $2^{D/2-1}$*, one quarter of the real Dirac spinor. At* $D \equiv 0$ *and* $D \equiv 4$ *the Majorana and Weyl conditions are each available but incompatible: conjugation flips chirality, so no Majorana-Weyl spinor exists. At* $D \equiv 6$ *conjugation preserves chirality but Majorana fails* ($B^*B = -\mathbf{1}$)*, leaving only the symplectic-Majorana-Weyl spinor.*

In the range that matters for supergravity and strings, $D \equiv 2 \pmod 8$ picks out $D = 2$ and $D = 10$. The $D = 2$ case is why worldsheet fermions can be chiral, and underlies the 32 left-moving Majorana-Weyl fermions of the heterotic construction (Chapter 3); the $D = 10$ case is the subject of Appendix B.2.5.

B.2.4 The master table

Collecting Theorems B.6 and B.7 with the Weyl count gives the table every calculation in these notes eventually consults. In it, a bullet marks a condition that can be imposed on a single spinor, "s" marks a condition available only in the symplectic (paired) form of Eq. (B.27) below, and a blank marks a condition that is unavailable. "$\dim_{\mathbb{C}}$ Dirac" is the complex dimension $2^{\lfloor D/2 \rfloor}$ of the Dirac spinor, and $\dim_{\mathbb{R}}$ is the real dimension of the *minimal* spinor in that dimension.

D	D mod 8	$\dim_{\mathbb{C}}$ Dirac	Weyl	Maj.	Maj.-Weyl	minimal spinor	$\dim_{\mathbb{R}}$
2	2	2	•	•	•	Majorana-Weyl	1
3	3	2		•		Majorana	2
4	4	4	•	•		Majorana ($\cong$ Weyl)	4
5	5	4		s		Dirac (sympl.-Majorana)	8
6	6	8	•	s	s	Weyl (sympl.-MW)	8
7	7	8		s		Dirac (sympl.-Majorana)	16
8	0	16	•	•		Majorana or Weyl	16
9	1	16		•		Majorana	16
10	**2**	32	•	•	•	**Majorana-Weyl**	**16**
11	**3**	32		•		**Majorana**	**32**
12	4	64	•	•		Majorana or Weyl	64

The pattern is periodic with period eight; the two rows in bold are the superstring and M-theory dimensions. Reading off $\dim_{\mathbb{R}}$: a Majorana or a Weyl condition individually gives $2^{\lfloor D/2 \rfloor}$ real components, a Majorana-Weyl condition gives $2^{\lfloor D/2 \rfloor - 1}$, and where neither Majorana nor Weyl is available (as at $D = 5, 7$) the minimal spinor is the full Dirac spinor with $2^{\lfloor D/2 \rfloor + 1}$ real components. At $D = 6$ the minimal spinor is a complex Weyl spinor, already 8 real; the symplectic-Majorana-Weyl condition marked "s" is its reality refinement, not a further truncation.

Symplectic-Majorana. When $B^*B = -\mathbf{1}$ the single-spinor condition Eq. (B.24) is empty, but a reality condition can still be imposed on an *even* number of spinors ψ^i, $i = 1, \ldots, 2n$, by pairing them with the invariant antisymmetric form Ω^{ij}_{sp} of $\mathrm{USp}(2n)$ ($\Omega^2_{\text{sp}} = -\mathbf{1}$),

$$\psi^{i*} \;=\; \Omega^{ij}_{\text{sp}}\, B\, \psi^j. \tag{B.27}$$

Conjugating twice and using $\Omega^2_{\text{sp}} = -\mathbf{1}$ turns the obstruction around: consistency now demands

$$B^*B \;=\; -\mathbf{1}, \tag{B.28}$$

which is exactly the case Majorana fails. The condition halves the real content of the multiplet, so a symplectic-Majorana pair carries the same real dimension as a single Dirac spinor. This is the reality structure of the minimal spinors in $D = 5, 6, 7 \bmod 8$, and it underlies the $\mathrm{USp}(2\mathcal{N})$ R-symmetry that rotates the supercharges of five- and six-dimensional supergravities (Appendix B.4). In $D = 6$, where chirality is also available, the plain complex Weyl spinor is already the 8-real minimal object; adding reality refines it to the symplectic-Majorana-Weyl spinor of the $(2, 0)$ and $(1, 1)$ theories.

B.2.5 Why ten dimensions

The Ramond ground state of the superstring is forced to be a spacetime spinor by the zero-mode algebra $\{\psi_0^\mu, \psi_0^\nu\} = \eta^{\mu\nu}$, which is the Clifford algebra of $\mathrm{SO}(1, 9)$ up to normalisation (Chapter 3). The irreducible module is the $2^5 = 32$ complex-component (64 real) Dirac spinor. Left on its own it is far too big: a 64-real-component spinor cannot pair with anything in a supersymmetric spectrum, and its states do not match the NS bosons. Both defects are cured by the same two truncations, and $D = 10$ is the one dimension where both are legal.

Majorana-Weyl in $D = 10$ and the type II split

The 32-complex-component Dirac spinor of $\mathrm{SO}(1, 9)$ collapses in three steps, each halving a count.

- **Majorana** ($10 \equiv 2$, Theorem B.6): impose $\psi^* = B\psi$. The 32 complex components

become 32 real.
- **Weyl** (D even): the GSO projection is the spacetime chirality projection $P_{\pm}$. It is compatible with Majorana because $10 \equiv 2 \pmod 8$ (Theorem B.7), and survives as a genuine *Majorana-Weyl* spinor of 16 real components.
- **On-shell**: the massless Dirac equation $\not{p}\psi = 0$, which in the R sector is the physical-state condition $G_0 |\psi\rangle = 0$ (G_0 the R-sector super-Virasoro zero mode, Chapter 3), removes half again, leaving 8 physical fermions in the $\mathbf{8}_s$ or $\mathbf{8}_c$ of the transverse Spin(8), its two inequivalent spinor representations swapped by triality.

That final 8 matches the 8 transverse polarisations of the NS vector, the $\mathbf{8}_v$ of Spin(8): the boson-fermion balance of ten-dimensional supersymmetry. Every step is legal at $D = 10$ because reality survives the chirality projection there, the content of $10 \equiv 2 \pmod 8$.

The same arithmetic has two downstream consequences.

Two chiralities, freely chosen. Because conjugation preserves chirality at $D \equiv 2 \pmod 8$, the Majorana-Weyl spinors come in two genuinely inequivalent kinds, the $\mathbf{8}_s$ and $\mathbf{8}_c$ of Spin(8), exchanged by a spacetime reflection. The GSO projection can keep either one in each of the two Ramond sectors of the closed string. Aligning the two choices gives the chiral theory IIB with $\mathcal{N} = (2, 0)$ supersymmetry; opposing them gives the non-chiral IIA with $\mathcal{N} = (1, 1)$. That these are physically distinct, not related by any field redefinition, is the statement $\mathbf{8}_s \neq \mathbf{8}_c$ (Chapter 3), and it is what makes the IIA/IIB dichotomy possible.

Ten-dimensional super-Yang-Mills. A gauge field A_μ in $D = 10$ has $D - 2 = 8$ on-shell real bosonic degrees of freedom. A minimal fermion with which it can be paired must also carry 8 on-shell, hence 16 real components off-shell. Reading the table, that is exactly one Majorana-Weyl spinor,

$$\underbrace{8}_{A_\mu,\ \text{on-shell}} \quad = \quad \underbrace{8}_{\lambda,\ \text{Majorana-Weyl on-shell}}, \qquad 16 = 2^{D/2-1}\big|_{D=10}. \tag{B.29}$$

A Dirac, a plain Majorana, or a plain Weyl gaugino would each carry too many components for the match. Minimal ($\mathcal{N} = 1$) super-Yang-Mills therefore exists as a classical field theory in $D = 10$, and this is the field content the heterotic and type I strings deliver. Closure of the supersymmetry algebra on these fields needs, in addition to the counting Eq. (B.29), the cubic gamma-matrix (Fierz) identity that holds only in $D = 3, 4, 6, 10$; that identity is stated in Appendix B.3.4 (Eq. (B.41)), with the proof deferred to the references cited there.

B.2.6 Eleven dimensions, and the ceiling

Eleven dimensions has no chirality (D odd) but does admit a Majorana spinor, since $11 \equiv 3$. It carries $2^5 = 32$ real components, the same 32 as a ten-dimensional Majorana-Weyl spinor doubled. Reducing $\mathrm{SO}(1, 10) \to \mathrm{SO}(1, 9)$ splits it into the two ten-dimensional chiralities,

$$\mathbf{32}_{\mathrm{SO}(1,10)} \longrightarrow \mathbf{16} \oplus \mathbf{16}', \tag{B.30}$$

one Majorana-Weyl spinor of each handedness. This is the spinorial reason eleven-dimensional supergravity reduces to the non-chiral IIA theory: its single 32-component supercharge becomes the $16 + 16'$ of $\mathcal{N} = (1, 1)$ (Chapter 7, Appendix B.4).

Eleven is also the ceiling. A single real supercharge with more than 32 components cannot act on a massless multiplet without generating fields of spin greater than two, so a Lorentzian supergravity is bounded by 32 real supercharges. The minimal spinor grows past 32 real components for the first time at $D = 12$ (where it is 64), so $D = 11$, whose minimal Majorana spinor is exactly 32 real, is the largest dimension in which supergravity exists. The supercharge bound and this counting are the content of Appendix B.4 and Appendix C; the classification of the underlying superalgebras is due to Nahm.

Exercise B.1. Use the table to check the following boson-fermion balances at the level of on-shell (physical, little-group) degrees of freedom.

1. $D = 10$ *super-Yang-Mills.* A Majorana-Weyl gaugino (16 real off-shell, 8 on-shell) balances a ten-dimensional gauge field (8 on-shell).
2. $D = 11$ *supergravity.* On-shell the gravitino ψ_μ is a gamma-traceless vector-spinor of the little group SO(9). The on-shell spinor is 16 real (the 32-component Majorana halved by its field equation) and the transverse vector index runs over 9 values, so removing the single gamma-trace leaves $9 \times 16 - 16 = (9-1) \times 16 = 128$ physical states. Check that this matches the bosons: the graviton $g_{\mu\nu}$ (traceless symmetric transverse tensor of SO(9), 44) plus the three-form C_3 (transverse antisymmetric, $\binom{9}{3} = 84$), a total of 128.
3. $D = 6$ *tensor multiplet.* In the $(2,0)$ theory a symplectic-Majorana-Weyl fermion balances a self-dual two-form and five scalars. On-shell the self-dual two-form contributes 3 and the five scalars 5, a bosonic total of 8; confirm that the fermion supplies the matching 8.

Remark B.8 (Beyond the lectures: signature and division algebras)*.* The classification extends in two directions. First, it depends on the signature (t, s) only through $s - t$ mod 8, the real Bott periodicity that Appendix B.1 traces to the Clifford algebras $\mathrm{Cl}(s, t)$. Majorana-Weyl spinors exist iff $s - t \equiv 0 \pmod 8$: for one time direction this is $D \equiv 2 \pmod 8$, but it also holds in Euclidean $D = 8$ (where the triality reps $\mathbf{8}_s, \mathbf{8}_c$ are real) and in the split signature $(2, 10)$, which is why F-theory can be phrased in twelve dimensions with two times (Chapter 11). The signature-general table is in Freedman and Van Proeyen and in Kugo and Townsend.

Second, minimal super-Yang-Mills exists precisely in $D = 3, 4, 6, 10$, the dimensions $2 + \dim_{\mathbb{R}} \mathbb{K}$ of the four division algebras $\mathbb{K} = \mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ of real dimension $1, 2, 4, 8$; the transverse little groups $\mathrm{Spin}(1), \mathrm{Spin}(2), \mathrm{Spin}(4), \mathrm{Spin}(8)$ act on the physical spinor as multiplication in $\mathbb{K}$. The Fierz identity that closes the algebra in these dimensions, and its reading as the statement that $\mathbb{K}$ is a division algebra, are in Appendix B.3.4; the top case $\mathbb{O}$, whose non-associativity singles out the octonionic dimension, is $D = 10$.

B.3 Bilinears and Fierz identities

A spinor bilinear is a Lorentz-covariant number built from two spinors and a string of gamma matrices. Every fermion mass term, every supersymmetry variation, every central charge in the algebra of Appendix C, and every calibration form on a supersymmetric background is one of these objects. Two facts control all of their manipulations: the symmetry of a bilinear under exchange of its two spinors, and the Fierz identity that re-couples a product of two bilinears. This section assembles both, in the conventions of Appendix A. The gamma-matrix construction, the chirality operator Γ_* and the charge-conjugation matrix C live in Appendix B.1; the reality (Majorana) conditions are worked out in Appendix B.2; the supercharge count that these identities feed is Appendix B.4.

Throughout, lower Greek indices $\alpha, \beta, \gamma, \delta$ label spinor components (the abstract spinor indices of Appendix A); the letter C is reserved for the charge-conjugation matrix. We write $\Gamma^{(n)} := \Gamma^{a_1 \cdots a_n} = \Gamma^{[a_1} \cdots \Gamma^{a_n]}$ for the antisymmetrised product of n gammas.

B.3.1 Conjugate spinors and the bilinear basis

To contract a spinor into a scalar one needs a conjugate. The *Dirac conjugate* uses the Hermitian intertwiner $A = i\,\Gamma^0$ built in Eq. (B.10),

$$\bar{\psi} = \psi^\dagger A, \qquad A\,\Gamma^a A^{-1} = -\,(\Gamma^a)^\dagger, \qquad A^\dagger = A, \quad A^2 = \mathbf{1}, \tag{B.31}$$

so that A intertwines the gammas with their Hermitian conjugates. In the mostly-plus signature of Appendix A one has $\Gamma^{0\dagger} = -\Gamma^0$ and $\Gamma^{i\dagger} = +\Gamma^i$, so $A = i\,\Gamma^0$ solves Eq. (B.31) (Hermitian, $A^2 = \mathbf{1}$); this is the choice fixed once and for all in Appendix B.1.2, and no other is used here. The point of A is that $\bar\psi\Gamma^{(n)}\chi$ then transforms as a genuine rank-n Lorentz tensor.

The *charge-conjugation matrix* C, constructed in Appendix B.1.4, turns a spinor into its transpose. It obeys

$$C\,\Gamma^a C^{-1} = t\,(\Gamma^a)^T, \qquad C^T = \xi\,C, \qquad t, \xi \in \{+1, -1\}, \tag{B.32}$$

where t is the intertwiner sign and ξ the transpose symmetry; both are fixed by D mod 8 (and, in even D, by which of the two available $C_\pm$ one picks). These are the $\pm$ and $\xi_\pm$ of the master relation Eq. (B.20), tabulated in Appendix B.1.4. For a Majorana spinor the Dirac and Majorana conjugates coincide, so we may write $\bar\psi = \psi^T C$ and let C carry all of the symmetry data. That is the case of interest for the supercharges and Killing spinors below.

In D dimensions the antisymmetrised products $\{\mathbf{1}, \Gamma^a, \Gamma^{ab}, \dots\}$ span the Clifford algebra, and the bilinears

$$(\bar\psi_1\,\Gamma^{a_1\cdots a_n}\psi_2) \quad \text{transform as an antisymmetric rank-}n\text{ tensor } (n\text{-form}). \tag{B.33}$$

In even $D = 2m$ there are $\sum_{n=0}^{D}\binom{D}{n} = 2^D = (2^m)^2$ independent matrices $\Gamma^{(n)}$, exactly enough to span the full $2^m \times 2^m$ matrix algebra. In odd D the top gamma is proportional to the identity, $\Gamma^{(n)} \propto \Gamma^{(D-n)}$, and only $n = 0, \dots, (D-1)/2$ are independent. The basis is orthogonal under the trace,

$$\mathrm{Tr}\big(\Gamma^{a_1\cdots a_p}\,\Gamma_{b_q\cdots b_1}\big) = 2^{\lfloor D/2\rfloor}\,\delta_{pq}\,\delta^{a_1\cdots a_p}_{b_1\cdots b_p}, \tag{B.34}$$

with $\delta^{a_1\cdots a_p}_{b_1\cdots b_p} = \delta^{[a_1}_{b_1}\cdots\delta^{a_p]}_{b_p}$ the unit-weight antisymmetrised Kronecker delta and $\mathrm{Tr}\,\mathbf{1} = 2^{\lfloor D/2\rfloor}$. Orthogonality (Eq. (B.34)) and completeness together are the Fierz machinery of Appendix B.3.3.

B.3.2 Symmetry and reality of bilinears

Transpose the matrix $C\Gamma^{(n)}$. Combining Eq. (B.32), the reversal of n distinct anticommuting gammas ($(-1)^{n(n-1)/2}$) and $C^T = \xi C$ gives the master relation Eq. (B.20) in the present notation,

$$\big(C\,\Gamma^{a_1\cdots a_n}\big)^T = \sigma_n\,C\,\Gamma^{a_1\cdots a_n}, \qquad \boxed{\sigma_n = \xi\,t^{\,n}\,(-1)^{n(n-1)/2}} \tag{B.35}$$

So $C\Gamma^{(n)}$ is symmetric when $\sigma_n = +1$ and antisymmetric when $\sigma_n = -1$. The sign is periodic in n with period 4, and the matrices with $\sigma_n = +1$ are exactly those that may stand on the right of an anticommutator $\{Q, Q\}$, so Eq. (B.35) fixes the momentum and brane-charge structure of the superalgebra in Appendix C.

The same sign controls the exchange of the two spinors in a bilinear. For anticommuting (Grassmann) Majorana spinors λ, χ, moving one past the other costs a minus sign, and Eq. (B.35) supplies the rest:

$$\bar\lambda\,\Gamma^{a_1\cdots a_n}\chi = -\,\sigma_n\,\bar\chi\,\Gamma^{a_1\cdots a_n}\lambda \qquad (\lambda, \chi \text{ anticommuting}). \tag{B.36}$$

For commuting (c-number) spinors the Grassmann sign is absent and the flip sign is $+\sigma_n$. This is the *Majorana flip* used on nearly every line of a supergravity variation. A single anticommuting Majorana spinor therefore has $\bar\lambda\Gamma^{(n)}\lambda \neq 0$ only when $-\sigma_n = +1$, that is $\sigma_n = -1$; for commuting spinors the nonzero bilinears are the complementary set $\sigma_n = +1$.

In even $D = 2m$ a second selection rule comes from chirality. Since Γ_* anticommutes with each Γ^a, one has $\Gamma_*\Gamma^{(n)} = (-1)^n\Gamma^{(n)}\Gamma_*$ (Eq. (B.14)). Take the bilinears in the Majorana form

D	minimal spinor	(ξ, t)	symmetric $C\Gamma^{(n)}$, i.e. $\sigma_n = +1$
4	Majorana	$(-,-)$	$n \equiv 1, 2 \pmod 4$
10	Majorana-Weyl	$(-,-)$	$n = 1, 5, 9$ (odd, self-dual)
11	Majorana	$(-,-)$	$n \equiv 1, 2 \pmod 4$: $n = 1, 2, 5, 6, 9, 10$

Table B.1: Symmetry of the bilinear-forming matrices $C\Gamma^{(n)}$ from Eq. (B.35), in the dimensions used in the course. In $D = 4$ the leading symmetric matrix $C\Gamma^{\mu}$ carries P_μ; in $D = 11$ the symmetric set $n = 1, 2, 5$ carries P_μ, the M2 two-form and the M5 five-form charges. In $D = 10$ we take $C = C_-$, giving $(\xi, t) = (-,-)$: the chirality selection rule Eq. (B.37) restricts to odd n, and self-duality then leaves $C\Gamma^{\mu}$ (P_μ) and $C\Gamma^{(5)}$ (self-dual five-form) as the independent symmetric bilinears of a single Majorana-Weyl spinor. Full D mod 8 data are in Appendix B.1.4; the pattern follows Van Proeyen (1999) and Freedman and Van Proeyen (2012).

$\bar\psi = \psi^T C$ used above; the relation $C^{-1}\Gamma_*^T C = (-1)^m \Gamma_*$ moves the projector across C, and inserting $P_\pm = \frac{1}{2}(\mathbf{1} \pm \Gamma_*)$ on both spinors gives

$$\bar\psi_\pm \, \Gamma^{(n)} \, \chi_\pm \neq 0 \;\Rightarrow\; n \equiv m \pmod 2, \qquad \bar\psi_\pm \, \Gamma^{(n)} \, \chi_\mp \neq 0 \;\Rightarrow\; n \equiv m + 1 \pmod 2, \tag{B.37}$$

with $\psi_\pm = P_\pm \psi$. The rule as written uses the Majorana bar $\bar\psi = \psi^T C$. With the Dirac bar $\bar\psi = \psi^\dagger A$ the parities invert: since $A = i\Gamma^0$ anticommutes with Γ_*, one has $\overline{P_\pm \psi} = \bar\psi \, P_\mp$, so same-chirality Dirac bilinears are nonzero for the opposite n.

The two bars therefore read oppositely, and one must be named. In $D = 4$ ($m = 2$), under the Majorana bar the scalar $\bar\psi_\pm \chi_\pm$ and the tensor $\bar\psi_\pm \Gamma^{\mu\nu} \chi_\pm$ survive (both even rank) while the current $\bar\psi_\pm \Gamma^{\mu} \chi_\pm$ vanishes and needs opposite chiralities. Under the Dirac bar the assignment is the one familiar from four-dimensional QFT, and it is the opposite: $\bar\psi_\pm \chi_\pm = \bar\psi_\pm \Gamma^{\mu\nu} \chi_\pm = 0$ while the chiral current $\bar\psi_\pm \Gamma^{\mu} \chi_\pm \neq 0$. In $D = 10$ ($m = 5$), under the Majorana bar, same-chirality bilinears carry only odd rank, the origin of the $n = 1, 5, 9$ entry in Table B.1 and of the fact that IIB pairs two spinors of equal chirality while IIA pairs opposite ones (Appendix B.2.5).

Careful

The flip sign in Eq. (B.36) differs by an overall minus between anticommuting and commuting spinors. Killing spinors and rigid supersymmetry parameters are usually treated as commuting c-numbers, dynamical gauginos and gravitini as anticommuting. Using the wrong version silently swaps "symmetric" for "antisymmetric" and is the most common single-sign error in these manipulations.

B.3.3 Completeness and the Fierz rearrangement

The Fierz identity is completeness of the basis Eq. (B.33) written as a resolution of the identity on the tensor product of two spinor spaces. In even $D = 2m$,

$$\delta_\alpha{}^\delta \, \delta_\gamma{}^\beta = \frac{1}{2^m} \sum_{n=0}^{2m} \frac{1}{n!} \left(\Gamma_{a_1 \cdots a_n}\right)_\alpha{}^\beta \left(\Gamma^{a_n \cdots a_1}\right)_\gamma{}^\delta . \tag{B.38}$$

The reversed index order $a_n \cdots a_1$ on the second factor is what makes the coefficient a clean $1/2^m$ with no rank-dependent sign; the normalisation is fixed by tracing both sides against $\mathbf{1}$. Sandwiching Eq. (B.38) between four anticommuting spinors and paying one transposition sign for the swap $\psi_2 \leftrightarrow \psi_4$ gives the rearrangement

$$(\bar\psi_1 \psi_2)(\bar\psi_3 \psi_4) = -\frac{1}{2^m} \sum_{n=0}^{2m} \frac{1}{n!} \left(\bar\psi_1 \, \Gamma_{a_1 \cdots a_n} \, \psi_4\right) \left(\bar\psi_3 \, \Gamma^{a_n \cdots a_1} \, \psi_2\right). \tag{B.39}$$

Inserting arbitrary matrices M, N in the two bilinears and re-expanding the same way produces the general Fierz identity; Eq. (B.39) is the case $M = N = \mathbf{1}$ and is the one needed most often. In odd D the sum runs only to $n = (D-1)/2$ and $2^m \to 2^{\lfloor D/2 \rfloor}$.

Specialising Eq. (B.38) to $D = 4$ ($m = 2$), the rank-dependent signs $(-1)^{n(n-1)/2}$ from undoing the index reversal give the explicit kernel

$$\delta_\alpha{}^\delta\,\delta_\gamma{}^\beta = \frac{1}{4}\Big[\,\delta_\alpha{}^\beta\delta_\gamma{}^\delta + (\Gamma_a)_\alpha{}^\beta(\Gamma^a)_\gamma{}^\delta - \tfrac{1}{2}(\Gamma_{ab})_\alpha{}^\beta(\Gamma^{ab})_\gamma{}^\delta - \tfrac{1}{6}(\Gamma_{abc})_\alpha{}^\beta(\Gamma^{abc})_\gamma{}^\delta + \tfrac{1}{24}(\Gamma_{abcd})_\alpha{}^\beta(\Gamma^{abcd})_\gamma{}^\delta\,\Big]. \tag{B.40}$$

Dualising the two top terms through $\Gamma_{abc} \propto \varepsilon_{abcd}\Gamma^d\Gamma_*$ and $\Gamma_{abcd} \propto \varepsilon_{abcd}\Gamma_*$ (signs per Eq. (B.15)) repackages Eq. (B.40) into the classic scalar/vector/tensor/axial/pseudoscalar Fierz table of Itzykson and Zuber; we keep the antisymmetrised-rank form because it is convention-proof.

B.3.4 The cyclic identity and ten-dimensional super-Yang-Mills

A single cubic Fierz identity decides whether minimal super-Yang-Mills exists. For the minimal spinor, define the symmetric matrix $\gamma^\mu_{\alpha\beta} \coloneqq (C\Gamma^\mu)_{\alpha\beta}$. Then

$$\gamma^\mu_{(\alpha\beta}\,\gamma_{\mu\,\gamma)\delta} = 0 \qquad \text{in} \qquad D = 3,\ 4,\ 6,\ 10, \tag{B.41}$$

with the symmetrisation over the three indices $\alpha\beta\gamma$ (the fourth index δ is a spectator), and the minimal spinor being Majorana in $D = 3, 4$, Weyl in $D = 6$, and Majorana-Weyl in $D = 10$. Equivalently, for a single anticommuting minimal spinor λ,

$$\Gamma_\mu\,\lambda\;(\bar\lambda\,\Gamma^\mu\lambda) = 0. \tag{B.42}$$

The identity fails in every other dimension. It is the reason $\mathcal{N} = 1$ super-Yang-Mills exists classically only in $D = 3, 4, 6, 10$: with the on-shell transformations $\delta A_\mu = \bar\epsilon\,\Gamma_\mu\lambda$ and $\delta\lambda = \frac{1}{2}F_{\mu\nu}\Gamma^{\mu\nu}\epsilon$, the variation of the action closes up to a single term cubic in λ, of the form $(\bar\epsilon\,\Gamma^\mu\lambda)(\bar\lambda\,\Gamma_\mu\lambda)$, and Eq. (B.42) is exactly what makes it vanish. The top case $D = 10$ is the maximal such gauge theory; its dimensional reductions give super-Yang-Mills in every $D \le 10$, and it is the theory living on the D-branes of Chapter 6. The four special dimensions are $D - 2 = 1, 2, 4, 8$, the real dimensions of the division algebras $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$, and Eq. (B.41) is the algebraic statement that $\mathbb{K}$ is a division algebra (Kugo and Townsend 1983). We record the identity here; the dimension-by-dimension proof is in Brink, Schwarz and Scherk (1977) and Green, Schwarz and Witten (1987).

Remark B.9 (Beyond the lectures). For a *commuting* Killing spinor ϵ (the chirality selection rule and the $+\sigma_n$ flip both apply) the nonzero bilinears $\omega^{(n)}_{\mu_1\cdots\mu_n} \coloneqq \bar\epsilon\,\Gamma_{\mu_1\cdots\mu_n}\epsilon$ form a preferred set of p-forms on the background, and they carry its geometry directly. The one-form $K_\mu = \bar\epsilon\,\Gamma_\mu\epsilon$ is a Killing vector, timelike or null; the higher $\omega^{(n)}$ are calibrations fixing the G-structure of the geometry, in $D = 11$ the two-form and five-form of Table B.1. Their mutual algebra is a web of Fierz identities of the type in Eq. (B.38), and solving that web is the modern classification of supersymmetric supergravity backgrounds (Gauntlett and collaborators).

Exercise B.2. (a) Take the $D = 4$ values $(\xi, t) = (-, -)$ in Eq. (B.35) and verify the anticommuting Majorana flips $\bar\lambda\chi = +\bar\chi\lambda$, $\bar\lambda\Gamma_*\chi = +\bar\chi\Gamma_*\lambda$, $\bar\lambda\Gamma^\mu\chi = -\bar\chi\Gamma^\mu\lambda$ and $\bar\lambda\Gamma^{\mu\nu}\chi = -\bar\chi\Gamma^{\mu\nu}\lambda$. Which of $\bar\lambda\lambda$, $\bar\lambda\Gamma^\mu\lambda$, $\bar\lambda\Gamma^{\mu\nu}\lambda$ vanish for a single such spinor? (b) Contract Eq. (B.38) on $(\alpha = \beta,\ \gamma = \delta)$ and separately on $(\alpha = \delta,\ \gamma = \beta)$ and confirm both sides give 2^m and 2^{2m} respectively, fixing the coefficient $1/2^m$.

B.4 Supercharge counting under reduction

The reality tables of Appendix B.2 fix, dimension by dimension, the number of real components in the minimal spinor. Turned into supercharges, those numbers fix how much supersymmetry

D	11	10	9	8	7	6	5	4	3
$\delta(D)$	32	16	16	16	16	8	8	4	2
$\mathcal{N}_{\max}$ (gravity)	1	2	2	2	2	4	4	8	16

Table B.2: Real dimension $\delta(D)$ of the minimal spinor and the maximal number of them compatible with a graviton, $\mathcal{N}_{\max} = 32/\delta(D)$. Reality types and the derivation are in Appendix B.2; the same data organise Table C.1. In $D = 6, 10$ the minimal spinor is chiral, so $\mathcal{N}$ splits by chirality, $\mathcal{N} = (\mathcal{N}_+, \mathcal{N}_-)$.

a theory carries and how much survives a reduction of spacetime. This section restates the count as a single invariant, follows the 32 supercharges of eleven dimensions down to ten and to four, and collects the reference tables that Chapters 5 and 8 draw on. The mechanism behind the tables, the vanishing of a fermion variation and the reduction of holonomy, is derived in Section 8.2; here we assume it and count.

The big picture

The controlling quantity is $\mathcal{Q}$, the total number of real supercharges: a spacetime scalar, invariant under dualities, never raised by a geometric reduction, only kept or lowered. Toroidal reduction keeps all of $\mathcal{Q}$ and repackages it into a different $\mathcal{N}$; compactification on a manifold of reduced holonomy multiplies $\mathcal{Q}$ by a fraction $\nu \leq 1$ set by how many spinors the internal holonomy leaves invariant. Eleven-dimensional supergravity sits at the ceiling, $\mathcal{Q} = 32$, and every maximally supersymmetric theory in lower dimensions is related by dualities to 11D on a torus. Everything with less follows by compactification on a manifold of reduced holonomy.

B.4.1 The invariant count

Write $\delta(D)$ for the number of real components of the minimal spinor of $\mathrm{Spin}(1, D-1)$, as fixed in Appendix B.2. A theory with $\mathcal{N}$ such minimal spinors worth of supersymmetry has

$$\mathcal{Q} \;=\; \mathcal{N} \cdot \delta(D) \tag{B.43}$$

real supercharges (the same $\mathcal{Q}$ as in Eq. (C.11)). The values of $\delta(D)$ that we need are collected in Table B.2; they run with $D \bmod 8$ through the Bott periodicity of Appendix B.2, and the corresponding minimal spinor is Majorana, Weyl, Majorana-Weyl or symplectic-Majorana according to the same table.

Two ceilings bound Eq. (B.43), both proved from the massless representations in Theorem C.2: a theory with a graviton has $\mathcal{Q} \leq 32$, and a theory with top spin one (super-Yang-Mills, no gravity) has $\mathcal{Q} \leq 16$. A single Majorana spinor saturates the first cap in $D = 11$; because $\delta(12) = 64$ already exceeds 32, no interacting supergravity lives above eleven dimensions, which pins it as the top. A single Majorana-Weyl spinor saturates the second in $D = 10$, where ten-dimensional super-Yang-Mills therefore exists.

The distinction between $\mathcal{Q}$ and $\mathcal{N}$ matters under reduction. The physical, duality-invariant statement is always about $\mathcal{Q}$. The label $\mathcal{N}$ is $\mathcal{Q}$ divided by the minimal spinor *of the dimension you have landed in*, so the same $\mathcal{Q}$ wears a different $\mathcal{N}$ in each dimension. Sixteen supercharges are $\mathcal{N} = 1$ in ten dimensions, $\mathcal{N} = 2$ in six, and $\mathcal{N} = 4$ in four; they are the same sixteen supercharges throughout.

B.4.2 How a spinor splits under reduction

Reduce from D to d dimensions on an internal manifold X of real dimension $n = D - d$. The Lorentz group breaks as

$$\mathrm{Spin}(1, D-1) \;\supset\; \mathrm{Spin}(1, d-1) \times \mathrm{Spin}(n), \tag{B.44}$$

with $\mathrm{Spin}(n)$ the Euclidean rotation group of the internal tangent space. A supercharge is a spinor of the left-hand group; under Eq. (B.44) it decomposes into external spinors tensored with internal ones,

$$\mathbf{S}_D \;\longrightarrow\; \bigoplus_i \,(\text{spinor of } \mathrm{Spin}(1, d-1)) \otimes (\text{spinor of } \mathrm{Spin}(n)), \tag{B.45}$$

with the number of internal spinor slots set by

$$\Delta(n) \;=\; 2^{\lfloor n/2 \rfloor}, \tag{B.46}$$

the dimension of the $\mathrm{Spin}(n)$ Dirac spinor, equal to the number of linearly independent constant spinors on the flat torus T^n. Component counting is multiplicative up to a reality correction. The naive product $\delta(d)\,\Delta(n)$ equals $\delta(D)$ when the higher-dimensional minimal spinor is non-chiral, as at $D = 11$ where $\delta(4)\,\Delta(7) = 4 \cdot 8 = 32 = \delta(3)\,\Delta(8) = 2 \cdot 16$, but it overshoots by a factor of two when that spinor is chiral, as at $D = 10$ where $\delta(4)\,\Delta(6) = 32 \neq 16 = \delta(10)$; the excess is removed by the external/internal chirality correlation. The physical statement is the ratio in Eq. (B.48), which is blind to this factor. The correlation of chiralities and the reality book-keeping that decides which components are independent over $\mathbb{R}$ follow the rules of Appendix B.2; the ten-to-four example $\mathbf{16} \to (\mathbf{2}, \mathbf{4}) \oplus (\bar{\mathbf{2}}, \bar{\mathbf{4}})$ is worked out explicitly in Eq. (8.31).

On flat space every internal component is a genuine supercharge. On a curved X only the covariantly constant internal spinors, $\nabla_m \eta = 0$ with ∇ the Levi-Civita connection, survive; the rest are projected out because the corresponding gravitino variation cannot be made to vanish. That is the content of the chain

$$\text{preserved supercharge} \iff \text{parallel spinor on } X \iff \text{reduced holonomy of } X, \tag{B.47}$$

imported from Eq. (8.21) and not re-derived here. Let $n_\parallel(X)$ be the number of parallel spinors on X, equivalently the number of $\mathrm{Hol}(X)$-singlets inside the $\mathrm{Spin}(n)$ spinor. The preserved supercharge count is external components times parallel internal spinors,

$$\boxed{\mathcal{Q}_d \;=\; \mathcal{Q}_D \,\frac{n_\parallel(X)}{\Delta(n)} \;=\; \mathcal{Q}_D \,\nu(X), \qquad \nu(X) \coloneqq \frac{n_\parallel(X)}{\Delta(n)}.} \tag{B.48}$$

The *preserved fraction* $\nu(X)$ runs from 1 (every internal spinor parallel) down to 0 (none), and it depends on X only through its holonomy group. Since $n_\parallel \leq \Delta(n)$ always, Eq. (B.48) can never raise $\mathcal{Q}$: geometric reduction is monotone downward in supersymmetry.

B.4.3 Toroidal reduction keeps everything

The flat torus T^n has trivial holonomy, so every constant spinor is parallel: $n_\parallel = \Delta(n)$ and $\nu = 1$. All of $\mathcal{Q}$ survives. The supercharges are simply reorganised, $\Delta(n)$ internal slots turning one higher-dimensional spinor into $\Delta(n)$ lower-dimensional ones, so $\mathcal{N}$ climbs while $\mathcal{Q}$ holds fixed. Reduction and truncation to the zero modes commute with this statement; keeping the whole Kaluza-Klein tower would only add massive multiplets built on the same supercharges.

This single fact organises maximal supergravity. Start from $\mathcal{Q} = 32$ in $D = 11$ and reduce on T^{11-d}:

$$32 \xrightarrow{S^1} \mathrm{IIA}_{10}\ (32) \xrightarrow{T^6} 4\mathrm{D}\ \mathcal{N} = 8\ (32), \tag{B.49}$$

$\mathrm{Hol}(X)$	geometry	$n = \dim_{\mathbb{R}} X$	$\Delta(n)$	$n_{\parallel}$	$\nu = n_{\parallel}/\Delta(n)$
$\{1\}$	torus T^n	n	$2^{\lfloor n/2 \rfloor}$	$2^{\lfloor n/2 \rfloor}$	1
SU(2)	K3 (CY_2), hyperkähler	4	4	2	$\frac{1}{2}$
SU(3)	Calabi-Yau CY_3	6	8	2	$\frac{1}{4}$
SU(4)	Calabi-Yau CY_4	8	16	2	$\frac{1}{8}$
Sp(k)	hyperkähler $4k$-fold	$4k$	2^{2k}	$k+1$	$(k+1)/2^{2k}$
G_2	G_2 manifold	7	8	1	$\frac{1}{8}$
Spin(7)	Spin(7) manifold	8	16	1	$\frac{1}{16}$
SO(n)	generic	n	$2^{\lfloor n/2 \rfloor}$	0	0

Table B.3: Preserved fraction ν by internal holonomy. $\Delta(n) = 2^{\lfloor n/2 \rfloor}$ is the number of constant spinors on T^n; $n_{\parallel}$ is the number left invariant by $\mathrm{Hol}(X)$, equivalently the number of parallel spinors on X. Calabi-Yau n-folds (SU(n) holonomy) carry exactly two parallel spinors, η and its conjugate. The hyperkähler row includes K3 at $k = 1$. Applying $\mathcal{Q}_d = \mathcal{Q}_D\,\nu$ (Eq. (B.48)) to these fractions produces Table B.4.

and every intermediate dimension inherits a maximal theory with the same 32 supercharges, wearing $\mathcal{N} = (1,1)$ in ten dimensions, $\mathcal{N} = 2$ in nine, up to $\mathcal{N} = 8$ in four. Type IIB, also 32, enters the same web without being a circle reduction of 11D: it is the T^2 reduction of M-theory in the limit where the torus shrinks, T-dual to IIA on the other circle (Chapter 7). The heterotic and type I strings enter this chain one rung lower, with $\mathcal{Q} = 16$: reduced on T^6 they give four-dimensional $\mathcal{N} = 4$.

B.4.4 Reduced holonomy: the preserved fraction

To get less than maximal supersymmetry, choose X so that its holonomy fixes only some of the internal spinors. The count $n_{\parallel}(X)$ is a group-theory question, how many singlets sit in the Spin(n) spinor under $\mathrm{Hol}(X)$, and Berger's list (Appendix E) enumerates the possibilities. Table B.3 collects the ones used in string and M-theory; the singlet counts are standard (Wang 1989 classifies parallel spinors; Joyce 2000 for the exceptional cases).

Read three features off the table. The generic six-manifold has Hol = SO(6) and no parallel spinor, so a random compactification breaks all supersymmetry; special holonomy is a fine-tuning, and generic holonomy is the default. The Calabi-Yau n-folds all carry precisely two parallel spinors regardless of n, so their fraction $\nu = 2^{1-n}$ halves with each complex dimension added. The exceptional holonomies G_2 and Spin(7) have no complex structure and no worldsheet sigma-model description, yet they fit the same counting and supply the minimal M-theory and F-theory reductions.

Careful

The holonomy condition is containment, $\mathrm{Hol}(X) \subseteq H$, not equality. A manifold whose holonomy sits in a proper subgroup of H preserves *more* supersymmetry, because a smaller group fixes more spinors. A six-manifold with Hol = SU(2) is locally $\mathrm{K3} \times T^2$ and gives $\nu = \frac{1}{2}$, not $\frac{1}{4}$; strict SU(3) holonomy, a Calabi-Yau that does not factorise, is what pins the fraction at $\frac{1}{4}$. The same caveat is spelled out at Section 8.2.

B.4.5 The fate of the thirty-two

Feeding Table B.3 into Eq. (B.48) and reading off $\mathcal{N} = \mathcal{Q}_d/\delta(d)$ from Table B.2 gives the master list, Table B.4. It is the reference that Chapters 5 and 8 quote when they name the supersymmetry of a compactification.

from	$\mathcal{Q}_D$	on X	$\to d$	ν	$\mathcal{Q}_d$	susy in d dimensions
M (11D)	32	S^1	10	1	32	IIA, $\mathcal{N}=(1,1)$
M	32	T^7	4	1	32	$\mathcal{N}=8$
M	32	K3	7	$\frac{1}{2}$	16	$\mathcal{N}=1$
M	32	CY_3	5	$\frac{1}{4}$	8	minimal (8 supercharges)
M	32	$\mathrm{CY}_3\times S^1$	4	$\frac{1}{4}$	8	$\mathcal{N}=2$
M	32	G_2	4	$\frac{1}{8}$	4	$\mathcal{N}=1$
M	32	Spin(7)	3	$\frac{1}{16}$	2	minimal (2 supercharges)
II (10D)	32	T^6	4	1	32	$\mathcal{N}=8$
II	32	K3	6	$\frac{1}{2}$	16	$\mathcal{N}=(1,1)$ (IIA), $(2,0)$ (IIB)
II	32	K3$\times T^2$	4	$\frac{1}{2}$	16	$\mathcal{N}=4$
II	32	CY_3	4	$\frac{1}{4}$	8	$\mathcal{N}=2$
het/I	16	T^6	4	1	16	$\mathcal{N}=4$
het/I	16	K3	6	$\frac{1}{2}$	8	$\mathcal{N}=(1,0)$
het/I	16	CY_3	4	$\frac{1}{4}$	4	$\mathcal{N}=1$
F	32	ell. CY_4	4	$\frac{1}{8}$	4	$\mathcal{N}=1$

Table B.4: The 32 supercharges of eleven and ten dimensions and their fate under reduction. $\mathcal{Q}_d=\mathcal{Q}_D\,\nu$ from Eq. (B.48); the target $\mathcal{N}$ is $\mathcal{Q}_d/\delta(d)$ with $\delta(d)$ from Table B.2. In $d=5$ the minimal spinor is symplectic-Majorana with $\delta=8$, so eight supercharges is minimal ("$\mathcal{N}=1$" here, called $\mathcal{N}=2$ by some authors); the F-theory entry is IIB with varying axio-dilaton, whose count is discussed in Chapter 11.

A few rows carry most of the weight downstream. M-theory on a G_2 manifold reaches four-dimensional $\mathcal{N}=1$ directly from the eleven-dimensional hub, $32\times\frac{1}{8}=4$, with no string and no Calabi-Yau in sight. The heterotic string on a Calabi-Yau threefold reaches the same $\mathcal{N}=1$ by a different route, $16\times\frac{1}{4}=4$, so the two are dual descriptions of overlapping four-dimensional physics (Chapter 7). Type II on the same threefold keeps twice as much, $\mathcal{N}=2$, because it starts with twice the supercharges. Type II on K3 lands on the six-dimensional theories, chiral $(2,0)$ for IIB and non-chiral $(1,1)$ for IIA, whose distinction is exactly the chirality book-keeping of Appendix B.2.

Example B.10 (Heterotic on a Calabi-Yau threefold)**.** Start with $\mathcal{Q}_{10}=16$, one Majorana-Weyl supercharge in the **16** of Spin$(1,9)$. Under Spin$(1,3)\times$Spin(6) it branches as $\mathbf{16}\to(\mathbf{2},\mathbf{4})\oplus(\bar{\mathbf{2}},\bar{\mathbf{4}})$, so each of the $\Delta(6)=8$ internal spinor slots would give a four-dimensional Weyl supercharge on a torus, recovering $\mathcal{N}=4$. On a threefold of SU(3) holonomy the **4** decomposes as $\mathbf{3}\oplus\mathbf{1}$: exactly one internal singlet in **4** and its conjugate in $\bar{\mathbf{4}}$, so $n_{\|}=2$ and $\nu=\frac{1}{4}$. The surviving supercharge is $(\mathbf{2},\mathbf{1})\oplus(\bar{\mathbf{2}},\bar{\mathbf{1}})$, a single four-dimensional Majorana spinor: $\mathcal{Q}_4=16\times\frac{1}{4}=4$, that is $\mathcal{N}=1$. The full branching is in Eq. (8.31).

Exercise B.3. Use Eq. (B.48) and Tables B.2 and B.3.

1. M-theory on a Spin(7) eight-manifold reduces to three dimensions. Compute $\mathcal{Q}$ and identify $\mathcal{N}$. Confirm it is the minimal three-dimensional supersymmetry.

2. Type IIA on K3 gives $\mathcal{Q}=16$ in six dimensions. Explain why the result is $(1,1)$ rather than $(2,0)$, while IIB on the same K3 gives $(2,0)$. The two K3 parallel spinors have the same Spin(4) chirality, so the surviving six-dimensional supercharges inherit the relative chirality of the two ten-dimensional ones (Appendix B.2): opposite for IIA, giving non-chiral $(1,1)$; equal for IIB, giving chiral $(2,0)$. As a check, count the multiplets: the $(2,0)$ theory needs a self-dual two-form, which sits in the IIB spectrum on K3 but not the IIA one.

3. Show that no fluxless geometric compactification of the heterotic string can give four-dimensional $\mathcal{N} = 2$ from a manifold of strict SU(3) holonomy, but that K3 $\times T^2$ (holonomy SU(2)) does. Which fraction ν is responsible?

Remark B.11 (Beyond the lectures). Two escapes from Eq. (B.48) matter in practice. First, switching on flux or a running dilaton replaces the Levi-Civita connection in $\nabla_m \eta = 0$ by one with torsion, so the relevant invariant is a *generalised holonomy* and the internal manifold need not be Ricci-flat; the parallel-spinor count is then read from a G-structure rather than from Table B.3, and the fraction can differ from the metric-holonomy value. This is the entry point to flux compactification (Chapter 9). Second, Scherk-Schwarz reduction twists the fields by a symmetry as one goes around a cycle; if the symmetry is an R-symmetry it gives the gravitini a mass and breaks the supersymmetry that Eq. (B.48) would otherwise preserve, at a scale set by the inverse cycle length (Section 8.1). Both are ways the naive holonomy count is corrected once the background is not a fluxless metric product; the counting above is exact only in that limit.

Appendix C

Supersymmetry, Supergravity, BPS Multiplets, and the 4D $\mathcal{N} = 1$ Potential

C.1 The super-Poincaré algebra

Supersymmetry is the statement that the symmetry algebra of a relativistic quantum theory can carry *fermionic* generators: operators that turn a boson into a fermion and back, and whose anticommutators close on the ordinary spacetime symmetries. This section fixes the algebra, in four dimensions and in general D, and reads off what its structure demands. Everything downstream in this appendix, and much of the spacetime story in Chapters 5, 6 and 12, is representation theory of the algebra written here.

The big picture

Three facts do most of the work later. First, the anticommutator of two supercharges is a translation, $\{Q, \bar{Q}\} \sim P$, so a supersymmetric theory has $E \geq 0$ built into its algebra. Second, the algebra admits *central charges* Z sitting on the right-hand side alongside P; in a theory of extended objects these Z are literally the charges of branes. Third, whenever a state saturates the bound $M \geq |Z|$ that the algebra forces, some supercharges annihilate it, the representation shortens, and the resulting count is rigid against continuous deformations. The explicit multiplets are built in Appendix C.2 and the bound and shortening are derived in Appendix C.3; here we assemble the algebra that makes both possible.

C.1.1 Why the algebra has the shape it does

Start from the Poincaré algebra: translations P_μ and Lorentz generators $M_{\mu\nu}$,

$$[P_\mu, P_\nu] = 0, \qquad [M_{\mu\nu}, P_\rho] = i\,(\eta_{\nu\rho} P_\mu - \eta_{\mu\rho} P_\nu), \tag{C.1}$$

$$[M_{\mu\nu}, M_{\rho\sigma}] = i\,(\eta_{\nu\rho} M_{\mu\sigma} - \eta_{\mu\rho} M_{\nu\sigma} - \eta_{\nu\sigma} M_{\mu\rho} + \eta_{\mu\sigma} M_{\nu\rho}). \tag{C.2}$$

A natural question in the 1960s was how much this can be enlarged by generators that commute into it, that is, by combining spacetime and internal symmetries into a single simple group. The answer is a sharp no. The Coleman-Mandula theorem (Coleman and Mandula, 1967) states that in a theory with a mass gap, a nontrivial analytic S-matrix, and finitely many particle types below any mass, the most general *bosonic* symmetry algebra is a direct product of the Poincaré

algebra with an internal (compact Lie) algebra whose generators are Lorentz scalars. Spacetime and internal symmetries do not mix: there is no conserved tensor charge $Q_{\mu\nu}$ beyond P_μ and $M_{\mu\nu}$ that acts nontrivially on scattering.

The loophole is that Coleman-Mandula assumes the generators are bosonic, closing under commutators into a Lie algebra. Allow generators that close under *anti*commutators and the theorem no longer applies. Haag, Łopuszański and Sohnius (1975) classified the possibilities and found essentially one: the new generators are spinorial charges Q in the spin-$\frac{1}{2}$ representation of the Lorentz group, and the enveloping structure is a *graded* (or super) Lie algebra. Such an algebra splits into an even (bosonic) part $\mathfrak{g}_{\bar{0}}$ and an odd (fermionic) part $\mathfrak{g}_{\bar{1}}$, with brackets

$$[\mathfrak{g}_{\bar{0}},\mathfrak{g}_{\bar{0}}]\subseteq\mathfrak{g}_{\bar{0}},\qquad [\mathfrak{g}_{\bar{0}},\mathfrak{g}_{\bar{1}}]\subseteq\mathfrak{g}_{\bar{1}},\qquad \{\mathfrak{g}_{\bar{1}},\mathfrak{g}_{\bar{1}}\}\subseteq\mathfrak{g}_{\bar{0}}, \tag{C.3}$$

the last an anticommutator. Consistency is the graded Jacobi identity, in which the sign $(-1)^{|X||Y|}$ appears on interchanging two odd elements X,Y. Supersymmetry is the unique such extension of Poincaré compatible with an interacting S-matrix, which is why the algebra below is forced on us rather than chosen.

C.1.2 Minimal supersymmetry in four dimensions

The smallest four-dimensional spinor is a Majorana (equivalently Weyl) spinor with four real components. In two-component (Weyl) notation the single supercharge is Q_α with $\alpha=1,2$, and its Hermitian conjugate $\bar{Q}_{\dot\alpha}=(Q_\alpha)^\dagger$. The $\mathcal{N}=1$ super-Poincaré algebra adds to Eqs. (C.1) and (C.2)

$$\{Q_\alpha,\bar{Q}_{\dot\beta}\}=-2\,\sigma^\mu_{\alpha\dot\beta}\,P_\mu,\qquad \{Q_\alpha,Q_\beta\}=\{\bar{Q}_{\dot\alpha},\bar{Q}_{\dot\beta}\}=0, \tag{C.4}$$

$$[P_\mu,Q_\alpha]=0,\qquad [M_{\mu\nu},Q_\alpha]=-(\sigma_{\mu\nu})_\alpha{}^\beta\,Q_\beta, \tag{C.5}$$

with $\sigma^\mu=(\mathbf{1},\sigma^i)$, $\bar\sigma^\mu=(\mathbf{1},-\sigma^i)$, $\sigma_{\mu\nu}=\frac{1}{4}(\sigma_\mu\bar\sigma_\nu-\sigma_\nu\bar\sigma_\mu)$, and the index and signature conventions fixed in Appendix B. The minus sign on the first anticommutator is the mostly-plus choice that makes the positivity below come out right; it is fixed once and used consistently here and in Appendix C.3. The content is read off line by line. The last commutator says Q_α is a Lorentz spinor. The relation $[P,Q]=0$ says supersymmetry commutes with translations, so all members of a supermultiplet share a mass: P^2 is still a Casimir. The first anticommutator is the heart of the matter, since two supersymmetry transformations compose into a spacetime translation.

That single line carries the positivity that makes supersymmetry powerful. Set $\dot\beta=\dot\alpha$ and sum over $\alpha=\dot\alpha$, placing the dotted index of $Q^\dagger_\alpha=\bar{Q}_{\dot\alpha}$ at the same numerical value as the undotted α; then

$$\sum_\alpha\{Q_\alpha,Q_\alpha^\dagger\}\;=\;-2\,\mathrm{tr}(\sigma^\mu)\,P_\mu\;=\;-4\,P_0\;=\;4E\;\geq\;0, \tag{C.6}$$

using $\mathrm{tr}\,\sigma^0=2$, $\mathrm{tr}\,\sigma^i=0$, and $P_0=-P^0=-E$ in the mostly-plus signature (consistent with the $P^2=-M^2$ used throughout). Each operator $\{Q_\alpha,Q_\alpha^\dagger\}$ on the left is positive semi-definite, so $E\geq 0$ in any supersymmetric theory, with $E=0$ if and only if $Q_\alpha|\psi\rangle=0$ for all α: a supersymmetric vacuum has exactly zero energy, and any zero-energy state is supersymmetric. This is the seed of the BPS bound of Appendix C.3 and of the vacuum-energy analysis behind the scalar potential of Appendix C.4.

The algebra Eqs. (C.4) and (C.5) has an outer automorphism that rephases the supercharge, $Q_\alpha\to e^{i\varphi}Q_\alpha$, $\bar{Q}_{\dot\alpha}\to e^{-i\varphi}\bar{Q}_{\dot\alpha}$, leaving P and M fixed. This is the $\mathrm{U}(1)_R$ *R-symmetry*. It is a genuine symmetry of the algebra, and where a theory realises it, different members of a multiplet carry different R-charge.

C.1.3 Extended supersymmetry and central charges

Nothing stops us adding more supercharges. Take $\mathcal{N}$ copies Q^I_α, $I=1,\dots,\mathcal{N}$, with conjugates $\bar{Q}_{\dot\alpha\, I}$. The anticommutators generalise to

$$\{Q^I_\alpha, \bar{Q}_{\dot\beta\, J}\} = -2\,\sigma^\mu_{\alpha\dot\beta}\,P_\mu\,\delta^I{}_J, \tag{C.7}$$

$$\{Q^I_\alpha, Q^J_\beta\} = 2\,\epsilon_{\alpha\beta}\,Z^{IJ}, \qquad \{\bar{Q}_{\dot\alpha\, I}, \bar{Q}_{\dot\beta\, J}\} = 2\,\epsilon_{\dot\alpha\dot\beta}\,\bar{Z}_{IJ}, \tag{C.8}$$

where the Z^{IJ} are the *central charges*. They are Lorentz scalars: the $\epsilon_{\alpha\beta}$ carries all the spinor structure, and since it is antisymmetric the charge must be antisymmetric too, $Z^{IJ}=-Z^{JI}$. Being central means they commute with everything, with P_μ, $M_{\mu\nu}$, the supercharges, and among themselves,

$$[Z^{IJ},\,\cdot\,] = 0. \tag{C.9}$$

For $\mathcal{N}=1$ there is no antisymmetric Z^{IJ} to write down, which is why minimal four-dimensional supersymmetry has no central charge; central charges are a feature of extended supersymmetry, $\mathcal{N}\geq 2$.

Because they commute among themselves the Z^{IJ} are simultaneously diagonalisable, and an R-symmetry rotation brings them to a standard skew-diagonal form with real, non-negative eigenvalues $q_1,\dots,q_{\lfloor\mathcal{N}/2\rfloor}$. Each is a physical charge, and the algebra pins the mass of any state to $M\geq \max_a q_a$, saturated exactly when a supercharge annihilates the state. That is the BPS bound; its derivation and the fraction of supersymmetry preserved at saturation are the business of Appendix C.3. What matters for the algebra is that the possibility exists at all, and it exists only because the right-hand side of $\{Q,Q\}$ can carry more than P_μ.

Careful

The phrase "central charge" is overloaded. Here Z is a central element of a *spacetime supersymmetry* algebra, a spacetime scalar (or, below, a spacetime tensor) that commutes with the supercharges. It has nothing to do with the Virasoro central charge c of Appendix A, which measures a two-dimensional worldsheet Weyl anomaly. Different objects, same two words.

Nonzero central charges shrink the R-symmetry. Without them the $\mathcal{N}$ supercharges of Eq. (C.7) can be rotated into one another by the full $\mathrm{U}(\mathcal{N})$; a given Z^{IJ} is invariant only under the subgroup preserving its skew-normal form, typically a product of USp and U(1) factors. The unbroken R-symmetry is therefore charge-dependent.

C.1.4 The algebra in general dimension

In D spacetime dimensions the supercharge is a spinor of $\mathrm{Spin}(D-1,1)$, and it is cleanest to package it as a single (possibly reducible) real spinor Q_A and use the charge-conjugation matrix C of Appendix B. The super-translation part of the algebra reads

$$\{Q_A, Q_B\} = (C\,\Gamma^M)_{AB}\,P_M \;+\; (\text{tensor central charges}), \tag{C.10}$$

with $[P_M, Q]=0$ and Q a spinor under M_{MN} as before. The matrix $C\Gamma^M$ is symmetric in AB precisely in the dimensions where a Majorana (or symplectic-Majorana) supercharge exists, so that the anticommutator on the left matches. The number of independent *real* supercharges is

$$\mathcal{Q} = \mathcal{N}\cdot\delta(D), \tag{C.11}$$

with $\delta(D)$ the number of real components of the minimal spinor in D dimensions and $\mathcal{N}$ the number of minimal spinors. The minimal spinor and its reality type run with D mod 8; the pattern is derived in Appendix B and the data used throughout are collected in Table C.1.

D	minimal spinor	$\delta(D)$	$\mathcal{N}_{\max}^{\text{grav}}$	R-symmetry H_R
11	Majorana	32	1	trivial
10	Majorana-Weyl	16	2	$\mathrm{SO}(\mathcal{N}_+)\times\mathrm{SO}(\mathcal{N}_-)$
9	Majorana	16	2	$\mathrm{SO}(\mathcal{N})$
8	Majorana	16	2	$\mathrm{U}(\mathcal{N})$
7	symplectic-Majorana	16	2	$\mathrm{USp}(2\mathcal{N})$
6	symplectic-Majorana-Weyl	8	4	$\mathrm{USp}(2\mathcal{N}_+)\times\mathrm{USp}(2\mathcal{N}_-)$
5	symplectic-Majorana	8	4	$\mathrm{USp}(2\mathcal{N})$
4	Majorana (= Weyl)	4	8	$\mathrm{U}(\mathcal{N})$
3	Majorana	2	16	$\mathrm{SO}(\mathcal{N})$

Table C.1: Minimal spinors, their real dimension $\delta(D)$, the maximal number of minimal spinors $\mathcal{N}_{\max}^{\text{grav}} = 32/\delta(D)$ compatible with a graviton, and the compact R-symmetry (automorphism) group of the $\mathcal{N}$-extended super-Poincaré algebra. In $D = 6, 10$ the supercharge is chiral, so $\mathcal{N} = (\mathcal{N}_+, \mathcal{N}_-)$ counts each chirality separately. Reality types are derived in Appendix B; the H_R column follows Strathdee (1987) and Freedman and Van Proeyen (2012). In $D = 11$ the algebra admits only $\mathcal{N} = 1$ and has no R-symmetry.

Two ceilings cap Eq. (C.11), both read off from the massless representations of Appendix C.2. A multiplet whose highest state is a graviton (helicity ≤ 2) carries at most $\mathcal{Q} = 32$ real supercharges; one capped at a gauge field (helicity ≤ 1, no gravity) at most 16. The count is the same in both: a massless multiplet has $\mathcal{Q}/4$ fermionic creation operators, each lowering the helicity by $\frac{1}{2}$, so it spans a helicity range $\mathcal{Q}/8$, and fitting this inside $|h| \leq h_{\max}$ (width $2h_{\max}$) forces $\mathcal{Q} \leq 16\, h_{\max}$, giving 32 for $h_{\max} = 2$ and 16 for $h_{\max} = 1$ (Appendix C.2). Through $\mathcal{N}_{\max} = 32/\delta(D)$ the first ceiling caps every row of Table C.1.

Reading Table C.1 against $\mathcal{Q} \leq 32$ fixes the top of each row. Eleven dimensions with a single Majorana spinor is the maximal supergravity (supergravity being local supersymmetry, defined in Appendix C.2.2), 32 supercharges, and nothing sits above it. Ten dimensions allows $\mathcal{N} = (1,1)$ (type IIA) and $(2,0)$ (type IIB), both 32, and the minimal $(1,0)$ with 16 is the type I and heterotic case. The cap $\mathcal{Q} \leq 16$ picks out ten-dimensional super-Yang-Mills as the maximal gauge theory without gravity, the seed of the D-brane worldvolume theories of Chapter 6.

C.1.5 R-symmetry as an automorphism group

R-symmetry is defined intrinsically: it is the group of automorphisms of the super-Poincaré algebra that act trivially on P_μ and $M_{\mu\nu}$ (so they commute with the Lorentz group) while rotating the supercharges among themselves. It is an outer symmetry of the algebra, and whether a given theory realises it as a symmetry of the dynamics is a separate question. In four dimensions the automorphism group is $\mathrm{U}(\mathcal{N})$, as we saw. The R-symmetry actually realised on a massless multiplet can be smaller: the U(1) factor is anomalous for $\mathcal{N} = 4$ super-Yang-Mills and absent for $\mathcal{N} = 8$ supergravity, leaving $\mathrm{SU}(4) \cong \mathrm{SO}(6)$ and $\mathrm{SU}(8)$ respectively (Appendix C.2). In general dimension the automorphism group is $\mathrm{SO}(\mathcal{N})$, $\mathrm{U}(\mathcal{N})$ or $\mathrm{USp}(2\mathcal{N})$ according to whether the supercharges organise into real, complex or quaternionic multiplets; the choice runs with $D \bmod 8$, the same Bott periodicity that governs spinor reality, and is tabulated in the last column of Table C.1.

Three cases recur. Type IIB, $\mathcal{N} = (2,0)$, has compact R-symmetry $\mathrm{SO}(2) \cong \mathrm{U}(1)$; this is the compact part of the on-shell $\mathrm{SL}(2,\mathbb{R})$ that acts on the axio-dilaton (Chapter 5). Type IIA, $\mathcal{N} = (1,1)$, has no compact R-symmetry. Eleven-dimensional supergravity has none at all: its single supercharge cannot be rotated into anything, so M-theory has no R-symmetry, a fact that

makes its algebra especially rigid.

C.1.6 Brane charges as central charges

Return to the "tensor central charges" hidden in Eq. (C.10). Whether $C\Gamma^{(p)} \coloneqq C\Gamma^{M_1\cdots M_p}$ is symmetric or antisymmetric under transpose of the spinor indices is a fact about the Clifford algebra, and in a given dimension only certain ranks p are symmetric. Each symmetric rank supplies a Lorentz tensor that may appear on the right of $\{Q,Q\}$. The four-dimensional lesson, that $\{Q,Q\}$ carries more than P, becomes in higher dimensions the statement that it carries a tower of antisymmetric-tensor charges $Z_{M_1\cdots M_p}$,

$$\{Q_A, Q_B\} = (C\Gamma^M)_{AB}\, P_M + \sum_{p\geq 2} \frac{1}{p!}\,(C\Gamma^{M_1\cdots M_p})_{AB}\, Z_{M_1\cdots M_p}. \tag{C.12}$$

For $p \geq 1$ these carry Lorentz indices, so they are not central in the strict sense of commuting with M_{MN}; they are central only within the super-translation subalgebra, that is, they commute with the Q's and with P. Calling them "central charges" is a standard abuse. They were written down abstractly by van Holten and Van Proeyen (1982), and their physical meaning was pinned down by de Azcárraga, Gauntlett, Izquierdo and Townsend (1989): $Z_{M_1\cdots M_p}$ is the charge of a p-dimensional extended object, measured as a surface integral of a $(p{+}1)$-form flux at spatial infinity, exactly the way electric charge is measured. The tensor central charges are *brane charges*.

The eleven-dimensional case is the cleanest, because $\mathcal{N}=1$ leaves no R-symmetry indices and the Clifford symmetry is unambiguous. With a single 32-component Majorana Q_A, the symmetric gamma-matrix structures are $C\Gamma^M$, $C\Gamma^{MN}$ and $C\Gamma^{M_1\cdots M_5}$, and the full anticommutator is

$$\{Q_A, Q_B\} = (C\Gamma^M)_{AB}\, P_M + \tfrac{1}{2}(C\Gamma^{MN})_{AB}\, Z_{MN} + \tfrac{1}{5!}(C\Gamma^{M_1\cdots M_5})_{AB}\, Y_{M_1\cdots M_5}. \tag{C.13}$$

The counting is exact. The left-hand side is a symmetric 32×32 matrix, with $32\cdot 33/2 = 528$ independent entries, and the right-hand side matches them as

$$528 \;=\; \underbrace{11}_{P_M} \;+\; \underbrace{55}_{Z_{MN}} \;+\; \underbrace{462}_{Y_{M_1\cdots M_5}}\,. \tag{C.14}$$

The rank-0 structure C is antisymmetric in $D=11$, so no Lorentz-scalar central charge $C\,M$ appears: the only "mass" available is P_M itself. The two-form charge Z_{MN} is carried by the M2-brane and the five-form charge $Y_{M_1\cdots M_5}$ by the M5-brane (Chapters 5 and 6). This is often called the M-theory superalgebra, and a surprising amount of the brane spectrum can be extracted from it before any dynamics is specified (Townsend, 1995).

Reducing Eq. (C.13) on a circle gives the type IIA algebra directly: the component of each eleven-dimensional charge along the circle drops in rank. Splitting $M=(m,\natural)$ with $\natural$ the circle direction,

$$P_\natural \to Z, \qquad Z_{m\natural} \to Z_m, \qquad Y_{m_1\cdots m_4\natural} \to Y_{m_1\cdots m_4}, \tag{C.15}$$

so type IIA carries the ten-dimensional momentum P_m, a 0-form charge Z (the D0-brane, that is Kaluza-Klein momentum on the M-theory circle), a 1-form (the fundamental string, F1), the surviving 2-form (D2), a 4-form (D4) and a 5-form (NS5), matching Eq. (C.14) term by term,

$$\underbrace{10}_{P_m} + \underbrace{1}_{Z\,(\mathrm{D0})} + \underbrace{10}_{\mathrm{F1}} + \underbrace{45}_{\mathrm{D2}} + \underbrace{210}_{\mathrm{D4}} + \underbrace{252}_{\mathrm{NS5}} \;=\; 528. \tag{C.16}$$

Type IIB is chiral, $\mathcal{N}=(2,0)$, and its central charges organise into $\mathrm{SL}(2,\mathbb{R})$ multiplets: the 1-form charges form a doublet (F1 and D1), the 5-form charges another doublet (NS5 and D5),

and the self-dual 3-form charge is an $\mathrm{SL}(2,\mathbb{R})$ singlet (D3). The brane-by-brane dictionary is set out in Chapter 6; the point here is that every brane in the theory appears as a term on the right of $\{Q,Q\}$, and its tension is bounded below by that charge through the BPS inequality of Appendix C.3.

Remark C.1 (Beyond the lectures). The brane charges enter the algebra as boundary terms, which is why they can be nonzero in a topologically nontrivial sector while $\{Q,Q\}\sim P$ still holds locally. Concretely, the supercurrent S^μ_α obeys, schematically, $\{Q_\beta, S^0_\alpha\}\sim (C\Gamma^\mu)_{\alpha\beta}\,\mathcal{T}_{\mu 0}+\partial_i\, k^i_{\alpha\beta}$, with $\mathcal{T}$ the stress tensor and k^i a total-derivative (superpotential) term. Integrating over space, the divergence piece becomes a flux at infinity and reproduces the tensor charges of Eq. (C.12). This is the sense in which they are "topological": insensitive to local deformations, they count the branes threading the boundary. See de Azcárraga et al. (1989) for the original argument and Townsend (1997) for the M-theory version.

C.1.7 Representations

Because $[P_\mu, Q]=0$, momentum is conserved within a supermultiplet and $P^2=-M^2$ still labels irreducible representations. Ordinary spin does not: the second Poincaré Casimir W^2 (the square of the Pauli-Lubanski vector) fails to commute with the supercharges, because Q shifts the spin by half a unit. A supermultiplet therefore collects several spins, and the invariant that labels it is a *superspin* Casimir built from a supersymmetric modification of the Pauli-Lubanski vector (Sohnius, 1985); its eigenvalue replaces spin as the second quantum number. An irreducible representation is thus not a single spin but a short tower of them.

The tower is constructed by fixing a reference momentum and its little group, the subgroup of Lorentz transformations that fix it, then treating the surviving supercharges as fermionic raising and lowering operators. The outcome depends on three cases.

- *Massless.* The little group is $\mathrm{SO}(D-2)$. For a null reference momentum the contraction $\not{p}$ is nilpotent, $\not{p}^{\,2}=p^2=0$, so it has half-maximal rank; hence half of the $\mathcal{Q}$ real supercharges anticommute to zero and act trivially, leaving $\mathcal{Q}/4$ creation operators and a Fock space of $2^{\mathcal{Q}/4}$ states. These are the short multiplets that build supergravity and super-Yang-Mills; the explicit content in 11, 10 and 4 dimensions is worked out in Appendix C.2.
- *Massive, no central charge.* The little group is $\mathrm{SO}(D-1)$ and all $\mathcal{Q}$ supercharges are active, giving $\mathcal{Q}/2$ creation operators and a long multiplet of $2^{\mathcal{Q}/2}$ states, exponentially larger than the massless case.
- *Massive BPS.* With central charge saturating $M=|Z|$, some of the otherwise-active supercharges become null and drop out, and the multiplet shortens to the size of a massless one. The exact counting and the preserved fraction $(\frac{1}{2},\frac{1}{4},\frac{1}{8},\dots)$ are the content of Appendix C.3.

In all three cases the states split evenly into bosons and fermions, since each fermionic creation operator flips statistics. The stability of the massive-BPS count against continuous deformation is what makes these representations the workhorses of Chapters 6 and 12.

Exercise C.1. In eleven dimensions ($D=11$, one time direction) with an antisymmetric charge-conjugation matrix C, the products $C\Gamma^{(p)}$ are symmetric for $p=1,2,5$ and antisymmetric for $p=0,3,4$. Using $\binom{11}{1}=11$, $\binom{11}{2}=55$, $\binom{11}{5}=462$, check that the symmetric ranks fill the 528 independent entries of the symmetric 32×32 matrix on the left of Eq. (C.13), and that the antisymmetric rank C leaves no room for a Lorentz-scalar central charge in $D=11$.

The algebra is now in hand: its shape forced by Coleman-Mandula and Haag-Łopuszański-Sohnius, its right-hand side dressed with central and brane charges, its automorphisms giving

the R-symmetry, and its representations sorted into long and short by how many supercharges act. Appendix C.2 fills in the explicit massless multiplets that build supergravity in 4, 10 and 11 dimensions; Appendix C.3 then turns the positivity of $\{Q, Q^\dagger\}$ into the BPS bound and quantifies the shortening.

C.2 Massless multiplets and supergravity

The big picture

Fix the super-Poincaré algebra of Appendix C.1 and ask what one-particle states it can act on. For a massless particle, half the supercharges are represented trivially and the other half become fermionic oscillators. Acting on a ground state they build a small tower of states, $2^{\mathcal{Q}/4}$ of them for $\mathcal{Q}$ real supercharges, organised into representations of the massless little group and of the R-symmetry. These towers are the massless multiplets. When the algebra is gauged, that is, when the constant spinor parameter is allowed to depend on position, one of the multiplets is forced to contain the graviton and its partner the gravitino, and the theory becomes a theory of gravity. That is supergravity (local supersymmetry; Appendix C.2.2). This section lists the multiplets that Chapter 5 builds and quotes: the $D=11$ multiplet, the four $D=10$ theories (IIA, IIB, type I and heterotic), and the $D=4$ multiplets for $\mathcal{N}=1,2,4,8$.

C.2.1 Massless one-particle multiplets

A massless momentum can be brought to the frame $p^\mu = E(1,0,\dots,0,1)$. Its little group, the subgroup of the Lorentz group fixing p^μ, is $\mathrm{ISO}(D-2)$; finite multiplets carry trivial charge under the translation part, so the states are labelled by representations of the compact factor

$$G_{\text{little}} = \mathrm{SO}(D-2). \tag{C.17}$$

In $D=4$ this is $\mathrm{SO}(2)=\mathrm{U}(1)$ and states are labelled by a single helicity h; in $D=10$ it is $\mathrm{SO}(8)$, whose triality between the vector $\mathbf{8}_v$ and the two spinors $\mathbf{8}_s, \mathbf{8}_c$ controls the whole spectrum; in $D=11$ it is $\mathrm{SO}(9)$.

Let $\mathcal{Q}$ be the number of real supercharges (Appendix B fixes the minimal spinor and hence $\mathcal{Q}$ dimension by dimension: 32 in $D=11$, 16 per Majorana-Weyl spinor in $D=10$, 4 per Majorana or Weyl spinor in $D=4$). Evaluate the anticommutator $\{Q, \bar{Q}\} \sim \not{p}$ of Appendix C.1 on a massless state. For a null momentum $\not{p}^2 = p^2 = 0$, so $\not{p}$ is nilpotent and its rank is half-maximal, $\mathcal{Q}/2$; half of the supercharges therefore anticommute to zero and are represented trivially. The surviving $\mathcal{Q}/2$ real supercharges pair into

$$n = \tfrac{1}{4}\mathcal{Q} \tag{C.18}$$

complex fermionic creation and annihilation operators, $\{a_i, a_j^\dagger\} = \delta_{ij}$, $i = 1,\dots,n$, transforming in a spinor of $\mathrm{SO}(D-2)$ and in a fundamental of the R-symmetry. Starting from a "Clifford vacuum" $|\Omega\rangle$ annihilated by every a_i and sitting in some little-group representation R_Ω, repeated application of the $a_i^\dagger$ generates

$$\dim(\text{multiplet}) = 2^n \cdot \dim R_\Omega = 2^{\mathcal{Q}/4}\, \dim R_\Omega. \tag{C.19}$$

Each raising operator carries a little-group spinor weight, so successive states step through the little-group representations. A theory is CPT invariant only if the set of multiplets is closed under helicity reversal, so a multiplet that is not self-conjugate must be accompanied by its CPT mirror.

For maximal supersymmetry, $\mathcal{Q}=32$, the vacuum can be a singlet and Eq. (C.19) already gives $2^8=256$ states, a set that necessarily reaches spin two: the graviton sits in the basic multiplet. For less supersymmetry one builds several distinct multiplets by choosing $|\Omega\rangle$ in different little-group representations. Spinning up the vacuum is how the 10D $\mathcal{N}=1$ supergravity multiplet, with 128 states, is obtained from the 16-state Clifford module tensored with a vector $\mathbf{8}_v$ (see below).

The $D=4$ case is cleanest in helicity language. The $n=\mathcal{N}$ complex oscillators each lower the helicity by $\frac{1}{2}$; on a vacuum of helicity $h_{\max}$ one finds helicities $h_{\max}, h_{\max}-\frac{1}{2}, \dots, h_{\max}-\frac{\mathcal{N}}{2}$ with multiplicity $\binom{\mathcal{N}}{k}$ at the k-th step. Choosing $h_{\max}$ and then adding the CPT mirror gives every massless multiplet in four dimensions.

Proposition C.2 (Spin-two bound and the ceilings on supersymmetry)**.** *An interacting massless theory contains no particle of helicity greater than* 2 *(no consistent long-range interactions for higher spin in flat space, barring the higher-spin loophole of Theorem C.4). Since each of the* $n=\mathcal{Q}/4$ *oscillators shifts the helicity by* $\frac{1}{2}$*, a single multiplet spans a helicity range* $\frac{n}{2}=\mathcal{Q}/8$*. A theory with a graviton reaches helicity* 2*, so the range runs from* $+2$ *to* -2*, a width of* 4*; requiring* $\mathcal{Q}/8\le 4$ *gives* $\mathcal{Q}\le 32$*. Requiring instead a top helicity of* 1 *(a gauge theory with no graviton) caps the width at* 2*, so* $\mathcal{Q}/8\le 2$*, that is* $n\le 4$ *and* $\mathcal{Q}\le 16$*. Hence the maximal number of real supercharges is* 32 *with gravity and* 16 *without, realised in* $D=11$ *at the top, in* $D=4$ *as* $\mathcal{N}=8$*, and nowhere above eleven dimensions (Nahm 1978).*

C.2.2 Local supersymmetry is supergravity

Global supersymmetry has a conserved spin-$\frac{3}{2}$ current, the supercurrent S^μ_α, with $\partial_\mu S^\mu_\alpha=0$ and a constant spinor parameter ϵ^α. Promote the parameter to a local one, $\epsilon(x)$, exactly as one gauges an internal symmetry. The action is no longer invariant: its variation is proportional to $\int(\partial_\mu\bar\epsilon)\,S^\mu$. To cancel this one introduces a gauge field for supersymmetry, a vector-spinor ψ_μ, coupled through $\int\bar\psi_\mu S^\mu$ and transforming inhomogeneously,

$$\delta\psi_\mu=\tfrac{1}{\kappa}\,D_\mu\epsilon+\dots,\qquad D_\mu\epsilon=\Big(\partial_\mu+\tfrac{1}{4}\,\omega_\mu{}^{ab}\Gamma_{ab}\Big)\epsilon, \tag{C.20}$$

with $\kappa^2=8\pi G_D$. The field ψ_μ is the gravitino, and the $\partial_\mu\epsilon$ piece is the tell-tale of a gauge connection. Here is the twist that internal gauging does not have: the supersymmetry algebra closes on translations, $\{Q,\bar Q\}\sim\Gamma^\mu P_\mu$, so making supersymmetry local forces local translations, that is, diffeomorphisms, and with them a dynamical metric. The vielbein $e_\mu{}^a$ enters as the gauge field of translations and its transformation is the superpartner of Eq. (C.20),

$$\delta e_\mu{}^a=\kappa\,\bar\epsilon\,\Gamma^a\psi_\mu. \tag{C.21}$$

Iterating the Noether procedure to close the algebra generates the Einstein-Hilbert term together with the Rarita-Schwinger kinetic term for the gravitino,

$$e^{-1}\mathcal{L}=\frac{1}{2\kappa^2}\,R(e,\omega)\;-\;\tfrac{1}{2}\,\bar\psi_\mu\,\Gamma^{\mu\nu\rho}\,D_\nu\psi_\rho\;+\;\dots, \tag{C.22}$$

with $\Gamma^{\mu\nu\rho}$ the antisymmetrised product of three gammas. The graviton and gravitino carry matching on-shell counts (in $D=4$, two states each), as any pair of superpartners must. Coefficients and the precise higher-order terms are convention dependent and are collected, dimension by dimension, in Freedman and Van Proeyen's *Supergravity* and in Appendix I; the statement that closure forces Eq. (C.22) is the content of the original $\mathcal{N}=1$ constructions (Freedman, van Nieuwenhuizen and Ferrara 1976; Deser and Zumino 1976).

The gravitino is thus the gauge field of local supersymmetry, one for each independent supercharge: a theory with $\mathcal{N}$ supersymmetries has $\mathcal{N}$ gravitini. Off shell the local algebra closes only after

auxiliary fields are added; for $D=4$, $\mathcal{N}=1$ the minimal set is a complex scalar and a real vector (the old-minimal auxiliary fields), deferred to Appendix C.4, where they organise the Kähler and superpotential data.

Remark C.3 (Beyond the lectures). Gauging a subgroup of the R-symmetry, or of an isometry acting on the scalars, promotes some of the vectors already present in an extended multiplet to non-abelian gauge fields and generates a scalar potential without spoiling supersymmetry. This is *gauged* supergravity; it is the mechanism behind the flux-induced potentials of Chapter 9 and the anti-de Sitter vacua of holography. The ungauged multiplets below are the starting point.

C.2.3 Eleven dimensions

There is a single spinor, Majorana, with $\mathcal{Q}=32$, so by Theorem C.2 eleven is the top dimension and the multiplet is unique. The little group is SO(9) and Eq. (C.19) gives $2^8=256$ states, split $128+128$. The field content is

$$(\, g_{MN},\ C_{MNP},\ \Psi_M \,), \tag{C.23}$$

a metric, a three-form potential and a Majorana gravitino. The on-shell counting under SO(9) is

field	SO(9) irrep	states	type
graviton g_{MN}	**44** (symmetric traceless)	44	boson
three-form C_{MNP}	$\mathbf{84}=\Lambda^3\mathbf{9}$	84	boson
gravitino Ψ_M	**128** (gamma-traceless vector-spinor)	128	fermion
total		256	

The three-form is the potential sourced by the M2-brane, with its magnetic dual the M5, and its Chern-Simons self-coupling $C_3\wedge F_4\wedge F_4$ is fixed by supersymmetry (Cremmer, Julia and Scherk 1978). This is the multiplet that Chapter 5 turns into the two-derivative action, and its reduction on a circle produces the IIA fields below.

C.2.4 Ten dimensions

Now $\mathcal{Q}=16$ per Majorana-Weyl spinor and the little group is SO(8), with $\mathbf{8}_v,\mathbf{8}_s,\mathbf{8}_c$ its three eight-dimensional representations. Triality, the outer automorphism of SO(8) that permutes these three (Appendix B), relates the tensor products below and is why the same SO(8) representations can play different physical roles in different theories. The relevant products are

$$\mathbf{8}_v\otimes\mathbf{8}_v=\mathbf{1}\oplus\mathbf{28}\oplus\mathbf{35}_v, \qquad \mathbf{8}_v\otimes\mathbf{8}_s=\mathbf{8}_c\oplus\mathbf{56}_c, \tag{C.24}$$

$$\mathbf{8}_s\otimes\mathbf{8}_c=\mathbf{8}_v\oplus\mathbf{56}_v, \qquad \mathbf{8}_s\otimes\mathbf{8}_s=\mathbf{1}\oplus\mathbf{28}\oplus\mathbf{35}_s, \tag{C.25}$$

where $\mathbf{35}_v$ is the graviton, $\mathbf{28}$ a two-form, $\mathbf{56}_v$ a three-form, $\mathbf{56}_{s,c}$ the gravitino and $\mathbf{35}_s$ a self-dual four-form.

Type II ($\mathcal{Q}=32$, maximal). The type II multiplets are the product of a left-moving and a right-moving factor, each $\mathbf{8}_v\oplus\mathbf{8}_{s\text{ or }c}$, giving $16\times16=256$ states. IIA takes opposite chiralities (non-chiral, $\mathcal{N}=(1,1)$), IIB the same chirality (chiral, $\mathcal{N}=(2,0)$). The NS-NS sector, common to both, is $\mathbf{8}_v\otimes\mathbf{8}_v=\mathbf{1}\oplus\mathbf{28}\oplus\mathbf{35}_v$, that is the dilaton ϕ, the Kalb-Ramond field B_2 and the graviton g. The sectors differ in the Ramond-Ramond forms and in the chirality of the two gravitini and two dilatini:

sector	IIA	IIB
NS-NS	$\phi,\ B_2,\ g$	$\phi,\ B_2,\ g$
R-R	$C_1,\ C_3\ (\mathbf{8}_v \oplus \mathbf{56}_v)$	$C_0,\ C_2,\ C_4^+\ (\mathbf{1} \oplus \mathbf{28} \oplus \mathbf{35}_s)$
NS-R $\oplus$ R-NS	$\psi_M^s, \psi_M^c;\ \lambda^c, \lambda^s$	$\psi_M^c, \psi_M^c;\ \lambda^s, \lambda^s$
R-symmetry (compact)	trivial	$\mathrm{SO}(2) \simeq \mathrm{U}(1)$

The IIA gravitini have opposite chirality, the IIB gravitini the same chirality. Only IIB carries a compact R-symmetry, $\mathrm{SO}(2) \simeq \mathrm{U}(1)$; IIA has none (the non-compact $\mathrm{SO}(1,1)$ rotating its two supercharges does not act on the spectrum), in agreement with the algebra of Appendix C.1 and its spinor table. The self-dual field strength of C_4^+ is why IIB has no covariant action and is written with a self-duality constraint imposed by hand (Chapter 5, Appendix I). Both give 128 bosonic and 128 fermionic states.

Type I and heterotic ($\mathcal{Q}=16$, $\mathcal{N}=1$). With half the supersymmetry there are two multiplets, and the physical spectra of type I, heterotic SO(32) and heterotic $\mathrm{E}_8 \times \mathrm{E}_8$ share the same two-derivative content: one $\mathcal{N}=1$ supergravity multiplet and one super-Yang-Mills multiplet in the adjoint of the gauge group.

multiplet	bosons	fermions	states
supergravity	$g_{MN}(\mathbf{35}_v),\ B_{MN}(\mathbf{28}),\ \phi(\mathbf{1})$	gravitino $\psi_M(\mathbf{56})$, dilatino $\lambda(\mathbf{8})$	$64+64$
super-Yang-Mills	gauge field $A_M(\mathbf{8}_v)$	gaugino $\chi(\mathbf{8}_s)$	$8+8$

The supergravity multiplet is the 16-state Clifford module of Eq. (C.19) built on an $\mathbf{8}_v$ vacuum, hence $16 \times 8 = 128$ states; the gauge multiplet is built on a singlet, hence 16. The gauge group is SO(32) for type I and heterotic SO(32), and $\mathrm{E}_8 \times \mathrm{E}_8$ for the other heterotic string; both have $\dim = 496$, the number selected by anomaly cancellation (Appendix H). The B-field here has the modified Bianchi identity $\mathrm{d}H = \frac{\alpha'}{4}(\operatorname{tr} R \wedge R - \operatorname{tr} F \wedge F)$, with α' the Regge slope of Appendix A, and this identity ties the two multiplets together (Chapter 5).

C.2.5 Four dimensions

The little group is U(1) and states are labelled by helicity. Each $\mathcal{N}$ counts four real supercharges, so $\mathcal{N}=1,2,4,8$ correspond to $\mathcal{Q}=4,8,16,32$, the last saturating Theorem C.2. The automorphism group of the superalgebra is $\mathrm{U}(\mathcal{N})$ (Appendix C.1); the R-symmetry realised on the massless states is $\mathrm{U}(1)_R$ for $\mathcal{N}=1$, U(2) for $\mathcal{N}=2$, $\mathrm{SU}(4) \simeq \mathrm{SO}(6)$ for $\mathcal{N}=4$ and SU(8) for $\mathcal{N}=8$, the overall U(1) factor being anomalous in $\mathcal{N}=4$ super-Yang-Mills and absent in $\mathcal{N}=8$ supergravity. The physical multiplets, with helicities before CPT completion, are listed below. A multiplet whose helicities are not symmetric about zero is understood to be accompanied by its CPT mirror.

$\mathcal{N}=1$.

multiplet	helicities	content
chiral	$(0, \frac{1}{2})$	complex scalar ϕ, Weyl fermion ψ
vector	$(\frac{1}{2}, 1)$	gauge field A_μ, gaugino λ
gravity	$(\frac{3}{2}, 2)$	graviton $g_{\mu\nu}$, gravitino ψ_μ

The chiral and vector multiplets carry the matter and gauge data of Chapter 9; their couplings are the Kähler potential, superpotential and gauge kinetic function of Appendix C.4.

$\mathcal{N}=2$.

multiplet	content
hypermultiplet	4 real scalars, 1 Dirac (two Weyl) fermion
vector	gauge field A_μ, 2 gauginos, 1 complex scalar
gravity	graviton, 2 gravitini, 1 graviphoton A_μ

The graviphoton is the vector that lives in the gravity multiplet rather than in a vector multiplet. The scalars of the vector multiplets and of the hypermultiplets live on separate factors of the moduli space, a special-Kähler manifold times a quaternionic-Kähler one; this is the structure behind Calabi-Yau vector- and hyper-moduli in Chapter 8.

$\mathcal{N}=4$.

multiplet	content
vector	1 gauge field, 4 Weyl fermions, 6 real scalars
gravity	graviton, 4 gravitini, 6 vectors, 4 spin-$\frac{1}{2}$, 1 complex scalar

The vector multiplet is the field content of $\mathcal{N}=4$ super-Yang-Mills, the maximal rigid supersymmetric gauge theory in four dimensions and the conformal theory on the far side of the duality of Chapter 13. The six scalars sit in the **6** of the SO(6) R-symmetry.

$\mathcal{N}=8$. The unique multiplet, with $2^8=256$ states, is the gravity multiplet, and it is CPT self-conjugate:

helicity $\lvert h\rvert$	2	$\frac{3}{2}$	1	$\frac{1}{2}$	0
$\binom{8}{k}$	1	8	28	56	70
field	graviton	gravitini	vectors	spin-$\frac{1}{2}$	scalars

So $\mathcal{N}=8$ supergravity has 1 graviton, 8 gravitini, 28 vectors, 56 spin-$\frac{1}{2}$ fermions and 70 real scalars, giving 128 bosonic and 128 fermionic states, the same 256 as in eleven dimensions. That is no accident: reducing $D=11$ supergravity on T^7 (or IIA/IIB on T^6) gives exactly this content, with the 70 scalars parametrising the coset $\mathrm{E}_{7(7)}/\mathrm{SU}(8)$, whose subscript (7) labels the split, maximally non-compact real form of E_7 (Cremmer and Julia 1979).

Exercise C.2. Using Eq. (C.19) with a scalar vacuum and $n=\mathcal{N}=8$ oscillators, each lowering the helicity by $\frac{1}{2}$ from $h_{\max}=2$, reproduce the multiplicities $\binom{8}{k}$ in the table and check that the multiplet is symmetric about $h=0$, hence CPT self-conjugate. Confirm the $128+128$ split. Repeat for $\mathcal{N}=4$ with $h_{\max}=1$ to recover the super-Yang-Mills vector multiplet, and note that its CPT mirror is itself.

Exercise C.3. In $D=11$ the gravitino is a vector-spinor of SO(9). The SO(9) spinor has dimension 16, so the reducible vector-spinor has $9\times 16=144$ components. Show that removing the gamma-trace $\Gamma^M\Psi_M$, itself a spinor, leaves $144-16=128$ physical states, matching the $44+84$ bosonic states of Eq. (C.23).

Remark C.4 (Beyond the lectures). Theorem C.2 assumes flat-space, finitely many fields and standard minimal coupling. Vasiliev's higher-spin theories evade it with an infinite tower of massless fields on an anti-de Sitter background, and are conjectured to be dual to free or critical vector models. They do not describe the massive higher-spin excitations of the string, which are not protected and lift at generic α'. The bound $\mathcal{Q}\le 32$ for theories with a finite number of fields and a well-defined flat-space S-matrix stands.

Remark C.5 (Beyond the lectures). Six dimensions ($\mathrm{SO}(4)\simeq\mathrm{SU}(2)\times\mathrm{SU}(2)$ little group) host chiral multiplets built on self-dual tensors. The $(2,0)$ theory living on coincident M5-branes has

a tensor multiplet with a self-dual two-form, five scalars and fermions, and no known Lagrangian description; the $(1,1)$ theory on D5/NS5 systems is an ordinary vector multiplet. These control the strong-coupling limits discussed in Chapter 7.

C.3 The BPS bound and short multiplets

Recall from Appendix C.1.3 that the right-hand side of the super-Poincaré algebra can carry *central charges*, operators that commute with every generator and so reduce to fixed numbers on any irreducible representation. Here we turn their presence into a mass bound. The same positivity that guarantees a sensible Hilbert space forces a lower bound on the mass of every state, the *BPS bound*

$$M \;\geq\; |Z|, \tag{C.26}$$

with Z built from the central charges. States that saturate it, $M = |Z|$, sit in *short* multiplets: some supercharges act trivially on them, so the representation has fewer states than a generic massive one. Shortness is rigid. A short multiplet cannot deform into a long one without discontinuously gaining states, so the relation $M = |Z|$ and the quantum numbers of a BPS state survive continuation from weak to strong coupling. This is the single most useful fact in the second half of these notes: it is why D-brane tensions are exact (Chapter 6), why the duality web can be tested by matching spectra (Chapter 7), and why a weak-coupling state count reproduces a strong-coupling black-hole entropy (Chapter 12).

The big picture

The logic runs in four steps, and each chapter that uses BPS states relies on all four.

1. **Positivity.** $\{Q, Q^\dagger\}$ is a sum of squares, so its eigenvalues are non-negative. Feeding in the central charge turns this into $M \geq |Z|$.
2. **Saturation kills supercharges.** At $M = |Z|$ a set of supercharges anticommutes to zero and therefore annihilates every state in the multiplet.
3. **Shortening.** Those supercharges drop out of the creation algebra, so the multiplet is smaller, by a factor 2^2 per saturated central charge.
4. **Protection.** Multiplet dimension is an integer; it cannot change under continuous deformation. Hence $M = |Z|$ is exact, and BPS states can leave the spectrum only by pairing up into long multiplets at walls of marginal stability.

The name records the origin. Bogomol'nyi (1976) and Prasad and Sommerfield (1975) found that solitons in certain field theories saturate an energy bound, $E \geq |q_{\text{top}}|$, set by a topological charge, with saturation equivalent to a first-order "self-duality" equation. Witten and Olive (1978) showed that in a supersymmetric completion that topological charge *is* the central charge of the algebra, so the Bogomol'nyi bound and the algebraic bound Eq. (C.26) are the same statement. We derive the algebraic version, which needs nothing but the anticommutators.

C.3.1 The bound from positivity of the anticommutator

We work in four dimensions with $\mathcal{N}$ supercharges; the argument is identical in any dimension once the algebra is written in the appropriate spinor language (Appendix B). From Appendix C.1, in two-component notation and mostly-plus signature,

$$\{Q^I_\alpha,\; \bar{Q}_{\dot\beta\, J}\} = -2\,\sigma^\mu_{\alpha\dot\beta}\,P_\mu\,\delta^I_J, \tag{C.27}$$

$$\{Q^I_\alpha,\; Q^J_\beta\} = 2\,\epsilon_{\alpha\beta}\,Z^{IJ}, \qquad Z^{IJ} = -Z^{JI}, \tag{C.28}$$

where $I, J = 1, \ldots, \mathcal{N}$ label the extended supercharges, $\bar{Q}_{\dot\beta J} = (Q^J_\beta)^\dagger$, and Z^{IJ} is the complex antisymmetric matrix of central charges, commuting with every generator and hence numerical

on an irreducible representation. The explicit minus sign in Eq. (C.27) is the seed of positivity, and in mostly-plus signature it is forced: with $\sigma^0 = \mathbb{1}$ and $P_0 = -P^0 = -E$, setting $I = J$, contracting $\dot\beta$ with α and summing gives

$$\sum_{\alpha} \{Q^I_\alpha,\ (Q^I_\alpha)^\dagger\} = -2\ \mathrm{Tr}(\sigma^\mu)\, P_\mu = -4\, P_0 = 4E\ \geq\ 0, \tag{C.29}$$

the left-hand side being a sum of $\|\cdot\|^2$. So $E \geq 0$; feeding in the central charge sharpens this to Eq. (C.26). Fixing the sign here rather than deferring to Appendix B keeps σ^0 the same in the algebra and in the rest frame below.

Skew-diagonalising Z. The R-symmetry group of Appendix C.1 acts on the index I. A $\mathrm{U}(\mathcal{N})$ rotation brings the antisymmetric matrix Z^{IJ} to standard skew-diagonal form,

$$Z\ \longrightarrow\ \begin{pmatrix} 0 & Z_1 \\ -Z_1 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & Z_2 \\ -Z_2 & 0 \end{pmatrix} \oplus \cdots, \qquad Z_1 \geq Z_2 \geq \cdots \geq 0, \tag{C.30}$$

with $\lfloor \mathcal{N}/2 \rfloor$ real, non-negative *skew-eigenvalues* Z_n (an extra zero appears for odd $\mathcal{N}$). The phases removed in this step are the reason we may take each $Z_n \geq 0$. It is enough to analyse one 2×2 block, so from here take $\mathcal{N} = 2$ with a single central charge $Z_1 \coloneqq Z \geq 0$ and supercharges Q^1, Q^2; the general case is a direct sum of such blocks.

Rest frame. For a massive state go to the rest frame $P^\mu = (M, \vec{0})$, so $P_0 = -M$ in mostly-plus signature. With $\sigma^0 = \mathbb{1}$, Eq. (C.27) collapses to

$$\{Q^I_\alpha,\ (Q^J_\beta)^\dagger\} = 2M\, \delta_{\alpha\beta}\, \delta^I_J, \tag{C.31}$$

the little group SU(2) identifying dotted and undotted indices; the explicit minus sign of Eq. (C.27) and $P_0 = -M$ cancel to give the positive right-hand side $+2M$ (spinor conventions of Appendix B). Now form two sets of fermionic oscillators out of the four supercharges,

$$A_\alpha \coloneqq \tfrac{1}{\sqrt{2}}\Big(Q^1_\alpha + \epsilon_{\alpha\beta}\, (Q^2_\beta)^\dagger\Big), \qquad B_\alpha \coloneqq \tfrac{1}{\sqrt{2}}\Big(Q^1_\alpha - \epsilon_{\alpha\beta}\, (Q^2_\beta)^\dagger\Big), \tag{C.32}$$

with $\epsilon_{12} = +1$. Using Eqs. (C.28) and (C.31) and the identities $\epsilon_{\alpha\gamma}\epsilon_{\beta\gamma} = \delta_{\alpha\beta}$, $\epsilon_{\alpha\gamma}\epsilon_{\gamma\beta} = -\delta_{\alpha\beta}$, the four cross terms in each anticommutator combine cleanly:

$$\{A_\alpha,\ A^\dagger_\beta\} = 2\,(M + Z)\,\delta_{\alpha\beta}, \qquad \{B_\alpha,\ B^\dagger_\beta\} = 2\,(M - Z)\,\delta_{\alpha\beta}, \tag{C.33}$$

while $\{A, A\} = \{A, B^\dagger\} = \{B, B\} = 0$. The right-hand sides are numbers times the identity, so Eq. (C.33) is a pair of decoupled fermionic oscillator algebras. Take the diagonal element $\alpha = \beta$ and sandwich it in any physical state $|\psi\rangle$:

$$2(M - Z)\, \|\psi\|^2 = \langle\psi|\, \{B_\alpha, B^\dagger_\alpha\}\, |\psi\rangle = \|B_\alpha\, |\psi\rangle\|^2 + \|B^\dagger_\alpha\, |\psi\rangle\|^2\ \geq\ 0. \tag{C.34}$$

A positive-definite Hilbert space therefore forces $M \geq Z$. Restoring the phase and the general block structure of Eq. (C.30), the bound reads $M \geq Z_n$ for every skew-eigenvalue, hence

$$\boxed{M\ \geq\ \max_n |Z_n|\ \coloneqq\ |Z|\,.} \tag{C.35}$$

Nothing in the derivation used dynamics: the bound is kinematical, a property of the representation, true state by state.

Careful

The Z in Eq. (C.35) is the exact quantum central charge, not its classical value. In theories where the central charge depends on couplings or moduli (for example $Z = n_e\, a(u) + n_m\, a_D(u)$ in Seiberg-Witten theory, or $Z = \int_\Sigma \Omega$ for a wrapped brane), the equality $M = |Z|$ holds with the fully corrected Z. Saturating the bound is also not the same as being stable: a BPS state can still sit above a multi-particle threshold and decay. Stability is the separate marginal-stability question of Appendix C.3.3.

C.3.2 Shortening and preserved fractions

The size of a massive multiplet is fixed by counting fermionic creation operators. In the rest frame the $2\mathcal{N}$ supercharges organise, block by block, into the oscillators Eq. (C.32): there are $\mathcal{N}$ operators of A-type and $\mathcal{N}$ of B-type (two spinor components for each of the $\mathcal{N}/2$ blocks). For a generic state, $M > |Z|$, all $2\mathcal{N}$ of them have non-zero norm by Eq. (C.33) and act as independent raising operators on a lowest state. The Fock space they generate is the *long* multiplet,

$$\dim(\text{long massive}) = 2^{\,2\mathcal{N}} \times (\text{dim of the lowest-spin state}). \tag{C.36}$$

Now saturate one skew-eigenvalue, $M = Z_n$. Then $\{B_\alpha, B_\alpha^\dagger\} = 0$ in that block, and Eq. (C.34) forces

$$B_\alpha\,|\psi\rangle = 0 = B_\alpha^\dagger\,|\psi\rangle \qquad (\alpha = 1, 2) \tag{C.37}$$

on *every* state of the multiplet: the two B-oscillators of that block are represented trivially and cannot raise anything. They are the *unbroken* supersymmetries, the supercharges under which the BPS state is invariant. Each saturated block removes two creation operators, shrinking the multiplet by 2^2. If k of the $\mathcal{N}/2$ skew-eigenvalues are saturated,

$$\dim(k\text{-fold BPS}) = 2^{\,2\mathcal{N}-2k} \times (\text{dim of the lowest-spin state}). \tag{C.38}$$

Counting the supercharges annihilating the state, each saturated block contributes B_1, B_2 and their conjugates, four real supercharges out of the $4\mathcal{N}$ real supercharges in four dimensions, so the *preserved fraction* is

$$\frac{\#\,\text{unbroken}}{\#\,\text{total}} = \frac{4k}{4\mathcal{N}} = \frac{k}{\mathcal{N}}. \tag{C.39}$$

The maximally short case is $k = \mathcal{N}/2$, all skew-eigenvalues equal to M: half the supercharges are unbroken, the multiplet is $\frac{1}{2}$-BPS, and its dimension $2^{\mathcal{N}}$ coincides with that of a massless multiplet (Appendix C.2). That coincidence is not an accident. A massless representation is already short, because $P^2 = 0$ makes half the supercharges nilpotent in the light-cone frame regardless of any central charge; the shortest massive BPS multiplet is what a massless one becomes when it acquires a mass without losing states.

$\mathcal{N}$	saturated k	preserved fraction	states
2	0 (long)	0	$2^4 = 16$
2	1	1/2	$2^2 = 4$
4	0 (long)	0	$2^8 = 256$
4	1	1/4	$2^6 = 64$
4	2	1/2	$2^4 = 16$
8	1, 2, 3, 4	1/8, 1/4, 3/8, 1/2	2^{14}, 2^{12}, 2^{10}, 2^8

The gap in size between a long and a short multiplet, a factor 2^{2k}, is the whole engine of protection: it is an integer jump that no continuous deformation can produce.

Exercise C.4. Take $\mathcal{N}=4$ with skew-eigenvalues $Z_1 \geq Z_2 \geq 0$. Show that $M = Z_1 > Z_2$ preserves one quarter of the supersymmetry and gives 2^6 states, while $M = Z_1 = Z_2$ preserves one half and gives 2^4. Verify that the $\frac{1}{2}$-BPS count matches the dimension of the massless $\mathcal{N}=4$ multiplet from Appendix C.2, and explain why $\frac{3}{4}$-BPS ($k=3$ of $\mathcal{N}/2=2$) is impossible.

C.3.3 Why BPS data are protected

Three independent facts conspire to make BPS quantities computable exactly, each doing a distinct job.

Representation theory. The dimension of a supermultiplet is an integer that can only change discontinuously. Suppose a short multiplet is carried around a loop in the space of couplings or moduli. As long as it stays short, $M = |Z|$ holds identically, because a short representation with $M \neq |Z|$ does not exist: the two are tied together by Eq. (C.37). For M to drift above $|Z|$ the multiplet would have to grow by a factor 2^{2k} in a single step, which is impossible along a continuous path. The only escape is for several short multiplets to have exactly the right quantum numbers to recombine into one long multiplet. This "pairing up" is the supersymmetric analogue of levels colliding, and it is rare: it needs a coincidence of charges and masses. Away from such loci the BPS mass formula receives no quantum corrections. This is the argument Chapter 12 invokes to carry a weak-coupling degeneracy to strong coupling.

The central charge is fixed by charges. Being central, Z commutes with every generator, so on an irreducible representation it is a single complex number set by the superselection charges, the electric and magnetic charges, momenta, windings and brane charges of the state. Those charges are quantised, so Z lives on a lattice and cannot vary continuously. The couplings enter only through the fixed function that assembles Z from the charges (a period integral, a gauge coupling matrix), and once that function is known the mass $M = |Z|$ of every BPS state follows arithmetically. Appendix C.1 gives the general form of these charges, including the p-form central charges that carry brane charge.

Marginal stability. A BPS state of charge γ can decay into $\gamma_1 + \gamma_2$ only if energy allows it, $M(\gamma) \geq M(\gamma_1) + M(\gamma_2)$. For BPS masses this reads $|Z(\gamma)| \geq |Z(\gamma_1)| + |Z(\gamma_2)|$, and because Z is additive in the charge, $Z(\gamma) = Z(\gamma_1) + Z(\gamma_2)$, the triangle inequality is saturated precisely when the two central charges are aligned in phase. The locus where $\arg Z(\gamma_1) = \arg Z(\gamma_2)$ is a real-codimension-one *wall of marginal stability*; a BPS state can appear or disappear only on crossing it (Seiberg-Witten; Denef; Denef-Moore for the wall-crossing formula). On either side the index counting BPS states is locally constant, which is exactly what makes it a reliable invariant to compare across dualities (Chapter 7) and to match against a black-hole entropy (Chapter 12).

Remark C.6 (Beyond the lectures). Protection is sharpest for the shortest, $\frac{1}{2}$-BPS multiplets and weakens as the multiplet lengthens. Intermediate ($\frac{1}{4}$-BPS and lower) states can undergo further shortening or lengthening transitions among themselves, and their protected quantities are indices, signed counts weighted by $(-1)^F$, rather than naive degeneracies, so that the pairs which recombine into long multiplets cancel. The helicity supertrace $\Omega_n = \mathrm{Tr}(-1)^{2J_3}(2J_3)^{2n}$ is engineered to annihilate long multiplets while remaining sensitive to short ones, and it is the correct protected quantity to track; the ordinary degeneracy is not.

C.3.4 Central charges as brane charges, and Killing-spinor projectors

In ten and eleven dimensions the maximal algebras carry no scalar central charge but instead antisymmetric *p-form charges* (Appendix C.1),

$$\{Q_\alpha,\ Q_\beta\} = (C\,\Gamma^M)_{\alpha\beta}\,P_M + \sum_p \frac{1}{p!}(C\,\Gamma^{M_1\cdots M_p})_{\alpha\beta}\,Z_{M_1\cdots M_p}, \tag{C.40}$$

with C the charge conjugation matrix. The tensor $Z_{M_1\cdots M_p}$ is sourced by a p-dimensional extended object: the fundamental string and NS5-brane sit in the charges of the NS sector, the Dp-branes in the RR charges, and the M2 and M5 in the two eleven-dimensional charges Z_{MN} and Z_{MNPQR}. Repeating the positivity argument of Appendix C.3.1 on Eq. (C.40) for a static, infinitely extended brane turns Eq. (C.35) into a bound on the *tension* (energy per unit worldvolume),

$$T \;\geq\; |z|, \qquad z \coloneqq (\text{charge density carried by the brane}), \tag{C.41}$$

and a brane is BPS when it saturates it. The tension scalings that make this concrete, $T_{\mathrm{F1}} \sim g_s^0$, $T_{\mathrm{D}p} \sim 1/g_s$, $T_{\mathrm{NS5}} \sim 1/g_s^2$ (the string coupling g_s of Appendix A), are established in Chapters 5 and 6; here we only note that each is a case of Eq. (C.41).

The unbroken supersymmetries Eq. (C.37) now become an algebraic condition on constant spinors. For a flat brane the surviving supercharges act through a *projector* built from the gamma matrices along the worldvolume,

$$\mathcal{P} = \tfrac{1}{2}(1+\Gamma_{\text{brane}}), \qquad \Gamma_{\text{brane}}^2 = 1, \tag{C.42}$$

where for a p-brane stretched along directions $0,1,\ldots,p$ one has, schematically, $\Gamma_{\text{brane}} \propto \Gamma^0\Gamma^1\cdots\Gamma^p$ (with the phase fixed by the brane's charge and, for D-branes, a factor from the RR sector). Because Γ_{brane} squares to one, $\mathcal{P}$ projects onto half the spinor components: the preserved supersymmetry parameters are those with $\Gamma_{\text{brane}}\,\epsilon = \epsilon$, so a single brane is $\tfrac{1}{2}$-BPS. This is the spacetime shadow of worldvolume κ-symmetry (Bergshoeff-Sezgin-Townsend; Becker-Becker-Strominger), and it is the tool used in Chapters 6 and 12.

For a configuration of several branes the surviving spinors must satisfy one projection per brane,

$$\Gamma^{(1)}\epsilon = \epsilon, \quad \Gamma^{(2)}\epsilon = \epsilon, \quad \ldots, \tag{C.43}$$

and each independent, commuting projector that is not implied by the others halves the number of solutions again: two give $\tfrac{1}{4}$-BPS, three give $\tfrac{1}{8}$-BPS, matching Eq. (C.39). The D1-D5-P system that reproduces the black-hole entropy of Chapter 12 is exactly a $\tfrac{1}{8}$-BPS intersection of this kind, three commuting projectors leaving four of the thirty-two supercharges unbroken. If two projectors fail to commute the count is different, and if they anticommute the configuration preserves no supersymmetry at all; the compatibility of the projectors is what distinguishes a genuine BPS bound state from a non-supersymmetric pile of branes.

Remark C.7 (Beyond the lectures). The condition $\Gamma_{\text{brane}}\,\epsilon = \epsilon$ is the linearised, flat-space version of a *Killing spinor equation*. In a curved supergravity background a bosonic solution preserves the supersymmetries whose parameters $\epsilon(x)$ make the fermion variations vanish, $\delta\psi_M = \nabla_M\epsilon + \cdots = 0$ and $\delta\lambda = \cdots = 0$. These first-order equations are the direct descendants of the Bogomol'nyi self-duality equations mentioned at the start of the section, and solving them is both easier than the second-order field equations and, for a supersymmetric solution, sufficient to imply them (given the Bianchi identities). The number of independent Killing spinors a background admits is its preserved fraction, computed geometrically rather than algebraically. Appendix I collects the fermion variations for the ten- and eleven-dimensional theories.

C.4 4D $\mathcal{N}=1$ supergravity and the scalar potential

The big picture

Compactifying a ten- or eleven-dimensional supergravity on a Calabi-Yau (or one of its cousins) produces, at low energy, a four-dimensional $\mathcal{N}=1$ supergravity. The whole two-derivative bosonic theory of its scalars is fixed by three functions of those scalars: a real

Kähler potential K, a holomorphic superpotential W, and a holomorphic gauge kinetic function f_{ab}. From K and W one builds the scalar potential

$$V_F = e^K \left(K^{i\bar{\jmath}}\, D_i W\, \overline{D_j W} - 3|W|^2 \right), \qquad D_i W = \partial_i W + (\partial_i K)\, W,$$

whose critical points are the vacua. This is the object Chapter 9 uses to stabilise moduli. The rest of this section says what each symbol means, why the $-3|W|^2$ is there, and when it cancels.

We work in units $M_{\rm pl} = 1$ throughout, with $M_{\rm pl}$ the *reduced* four-dimensional Planck mass, $M_{\rm pl}^{-2} = \kappa_4^2 = 8\pi G_4$. Fields, K and W are then dimensionless; the potential is measured in $M_{\rm pl}^4$ and the gravitino mass in $M_{\rm pl}$, and factors of $M_{\rm pl}$ are restored at the end of the section. The multiplet content (chiral, vector and gravity multiplets) is the one assembled in Appendix C.2; here we only need its bosons.

C.4.1 The data and the bosonic action

A 4D $\mathcal{N}=1$ supergravity coupled to matter is specified by:

Datum	Type	Role
ϕ^i $(i = 1, \dots, n)$	complex scalars	lowest components of n chiral multiplets
$K(\phi, \bar{\phi})$	real, dimensionless	Kähler potential; metric on scalar space
$W(\phi)$	holomorphic	superpotential; encodes masses and couplings
A_μ^a	real vectors	gauge fields of a group G with generators T_a
$f_{ab}(\phi)$	holomorphic, symmetric	gauge kinetic function

The scalars ϕ^i are coordinates on a target manifold $\mathcal{M}$, and $\mathcal{N}=1$ supersymmetry forces $\mathcal{M}$ to be Kähler: its metric is the second derivative of a potential,

$$K_{i\bar{\jmath}} \coloneqq \partial_i \partial_{\bar{\jmath}} K, \qquad \mathrm{d}s^2 = 2\, K_{i\bar{\jmath}}\, \mathrm{d}\phi^i\, \mathrm{d}\bar{\phi}^{\bar{\jmath}}, \tag{C.44}$$

positive definite for a physical (ghost-free) theory. We write $K^{i\bar{\jmath}}$ for the inverse, $K^{i\bar{\jmath}} K_{k\bar{\jmath}} = \delta^i{}_k$, and abbreviate $\partial_i K = K_i$, $\partial_i W = W_i$.

Collecting the bosons, the two-derivative Lagrangian in the Einstein frame is

$$e^{-1}\mathcal{L} = \tfrac{1}{2} R - K_{i\bar{\jmath}}\, \nabla_\mu \phi^i\, \nabla^\mu \bar{\phi}^{\bar{\jmath}} - \tfrac{1}{4}\, \mathrm{Re}\, f_{ab}\, F^a_{\mu\nu} F^{b\,\mu\nu} + \tfrac{1}{4}\, \mathrm{Im}\, f_{ab}\, F^a_{\mu\nu} \widetilde{F}^{b\,\mu\nu} - V, \tag{C.45}$$

with $e = \sqrt{-g}$, $\widetilde{F}^{\mu\nu} = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}$, and $\nabla_\mu \phi^i$ the gauge-covariant derivative when the isometries are gauged (Appendix C.4.4). The real part $\mathrm{Re}\, f_{ab}$ is the inverse gauge coupling matrix and must be positive definite; the imaginary part supplies the theta angles. The potential $V = V_F + V_D$ is what the remaining subsections build. The full derivation, including the fermions, is long; it is carried out in Cremmer, Ferrara, Girardello and Van Proeyen (1983) and set out textbook-style in Wess and Bagger (1992) and in Freedman and Van Proeyen (2012). We state the result and use it.

C.4.2 Kähler invariance and the covariant derivative

Neither K nor W is separately physical. The metric of Eq. (C.44) is unchanged by a *Kähler transformation*

$$K \longrightarrow K + h(\phi) + \bar{h}(\bar{\phi}), \qquad W \longrightarrow e^{-h(\phi)}\, W, \tag{C.46}$$

with h holomorphic. The action of Eq. (C.45) is invariant because the two combine into a single object,

$$G \coloneqq K + \ln|W|^2, \tag{C.47}$$

which Eq. (C.46) leaves fixed. The superpotential is thus not a function on $\mathcal{M}$ but a section of a line bundle, of *Kähler weight* 1; a plain derivative $\partial_i W$ is not covariant under Eq. (C.46). The correct Kähler-covariant derivative adds the Kähler connection K_i,

$$D_i W \coloneqq \partial_i W + (\partial_i K)\, W = W_i + K_i\, W, \qquad \overline{D_j W} \coloneqq D_{\bar\jmath}\bar W = \bar W_{\bar\jmath} + K_{\bar\jmath}\,\bar W, \tag{C.48}$$

and one checks directly from Eq. (C.46) that it carries the same weight as W itself:

$$D_i W \to \partial_i(e^{-h}W) + (K_i + \partial_i h)e^{-h}W = e^{-h}(W_i - h_i W + K_i W + h_i W) = e^{-h}\, D_i W. \tag{C.49}$$

Consequently $e^K K^{i\bar\jmath} D_i W\, \overline{D_j W}$ and $e^K |W|^2$ are each invariant, which is why the potential below is well defined. Equivalently $D_i W = W\, \partial_i G$ wherever $W \neq 0$, so the whole potential can be written in terms of G alone.

C.4.3 The F-term scalar potential

The superpotential contribution to V is the central formula of this appendix:

$$\boxed{V_F = e^K \left(K^{i\bar\jmath}\, D_i W\, \overline{D_j W} - 3|W|^2 \right)} \tag{C.50}$$

Two pieces of physics are packaged here. The first term is positive and generalises the rigid-supersymmetry $|F|^2$; it comes from eliminating the auxiliary fields F^i of the chiral multiplets, whose on-shell values are

$$F^i = e^{K/2}\, K^{i\bar\jmath}\, \overline{D_j W}. \tag{C.51}$$

The second term, $-3e^K|W|^2$, is genuinely gravitational: it descends from the auxiliary complex scalar of the supergravity multiplet in the old-minimal formulation (auxiliary fields S, P, A_μ), and has no analogue in global supersymmetry. It is the reason V_F can be negative. In terms of the invariant G of Eq. (C.47) the same object reads

$$V_F = e^G (G^{i\bar\jmath}\, G_i\, G_{\bar\jmath} - 3), \qquad G^{i\bar\jmath} \coloneqq K^{i\bar\jmath}, \tag{C.52}$$

which makes the Kähler invariance manifest and shows that a vacuum is controlled by the single dimensionless combination $G^{i\bar\jmath} G_i G_{\bar\jmath}$ against the number 3.

The exponential prefactor e^K is not optional. Under a Kähler transformation it compensates the weight of $|W|^2$; physically it also means that a constant shift of K rescales W, so only the product $e^{K/2}W$ carries physical content. That product is the gravitino mass, computed in Appendix C.4.5.

C.4.4 D-terms and gauged isometries

If a subgroup of the isometries of $\mathcal{M}$ is gauged by the vectors A^a_μ, the scalars pick up a second, non-negative contribution. Gauging means promoting a set of holomorphic Killing vectors $X_a = X^i_a(\phi)\,\partial_i$ of the Kähler metric to local symmetries, $\nabla_\mu \phi^i = \partial_\mu \phi^i - A^a_\mu X^i_a$. To each Killing vector belongs a real *moment map* (Killing potential) $\mathcal{P}_a$, fixed up to a constant by

$$K_{i\bar\jmath}\, X^i_a = i\, \partial_{\bar\jmath} \mathcal{P}_a. \tag{C.53}$$

Just as V_F came from eliminating the chiral auxiliaries F^i, the D-term comes from eliminating the real auxiliary field D^a of the vector multiplets, whose on-shell value is

$$D^a = (\operatorname{Re} f)^{-1\,ab}\, \mathcal{P}_b. \tag{C.54}$$

The moment map $\mathcal{P}_a$ is thus the vector-multiplet auxiliary. Substituting it back gives

$$V_D = \tfrac{1}{2}\, (\operatorname{Re} f)^{-1\,ab}\, \mathcal{P}_a\, \mathcal{P}_b, \qquad V = V_F + V_D, \tag{C.55}$$

the exact D-term analogue of the $|F|^2$ piece of V_F. For fields in a linear representation, $\delta\phi^i = -\epsilon^a (T_a)^i{}_j \phi^j$, the moment map is $\mathcal{P}_a = K_i\,(T_a)^i{}_j \phi^j + \xi_a$ (up to a convention-dependent overall sign), and the constant ξ_a, a Fayet-Iliopoulos term, is allowed only for abelian factors. Gauge invariance of the superpotential requires

$$X_a^i\,\partial_i W = 0 \tag{C.56}$$

when $\xi_a = 0$; the constant-FI case is subtler and is flagged in the remark below.

Remark C.8 (Beyond the lectures: FI terms are R-symmetric). When $\xi_a \neq 0$ the invariance condition of Eq. (C.56) is replaced by the equivariance condition $X_a^i \partial_i W = i\,\xi_a\,W$: the superpotential is no longer invariant but rotates by a phase, so the gauged U(1) must act as an *R-symmetry*. Unlike in rigid supersymmetry, a field-independent Fayet-Iliopoulos parameter in supergravity is not a free coupling but a gauged R-symmetry with a fixed charge tied to the gravitino. See Komargodski and Seiberg (2009) and Freedman and Van Proeyen (2012).

C.4.5 Vacua, the gravitino mass, and supersymmetry breaking

Vacua are critical points of V, $\partial_i V = 0$. The order parameters for supersymmetry breaking are the auxiliary fields of Eq. (C.51) and the D-terms $\mathcal{P}_a$: a vacuum preserves $\mathcal{N}=1$ if and only if all of them vanish,

$$D_i W = 0 \quad (\forall i), \qquad \mathcal{P}_a = 0 \quad (\forall a). \tag{C.57}$$

The conditions $D_i W = 0$ are the *F-flatness* equations and $\mathcal{P}_a = 0$ the *D-flatness* equations; together they are the workhorse of moduli stabilisation. Any solution of Eq. (C.57) is automatically a stationary point of V, so one solves first-order equations rather than the full $\partial_i V = 0$.

Whether or not supersymmetry is broken, the gravitino acquires a mass set by the invariant G,

$$m_{3/2} = e^{K/2}\,|W| = e^{G/2}, \tag{C.58}$$

so with Eq. (C.51) the F-part of the potential is a difference of squares,

$$V_F = K_{i\bar{\jmath}}\,F^i \bar{F}^{\bar{\jmath}} - 3\,m_{3/2}^2. \tag{C.59}$$

At a supersymmetric vacuum $F^i = 0$ and $V_* = -3\,m_{3/2}^2 \leq 0$: the vacuum is anti-de Sitter unless W also vanishes there, in which case it is Minkowski. Negative V_* is not a pathology. Unbroken supersymmetry in AdS is perfectly unitary, and $m_{3/2}$ then sets the AdS curvature scale; this is the AdS avatar of the BPS structure of Appendix C.3. To reach $V_* > 0$, de Sitter, one *must* break supersymmetry, $F^i \neq 0$, and pay the $-3m_{3/2}^2$. Doing this in a controlled way is hard, and that difficulty is the subject of Chapter 9.

When supersymmetry is spontaneously broken the goldstino is eaten by the gravitino (the super-Higgs mechanism) and the scale of breaking is $\sqrt{K_{i\bar{\jmath}}F^i\bar{F}^{\bar{\jmath}}} = \sqrt{3}\,m_{3/2}$ at $V = 0$. The resulting soft terms in the observable sector are read off by expanding V around the vacuum; we do not need them here.

C.4.6 The no-scale structure

A Kähler potential is called *no-scale* when its metric satisfies

$$K^{i\bar{\jmath}}\,K_i\,K_{\bar{\jmath}} = 3 \tag{C.60}$$

identically on some subset of the fields. Split the scalars into those no-scale moduli T^α obeying Eq. (C.60) and the rest ϕ^a. If W is independent of the T^α, then $D_\alpha W = K_\alpha W$ and the two terms of Eq. (C.50) cancel along T^α:

$$V_F = e^K |W|^2 (K^{\alpha\bar\beta} K_\alpha K_{\bar\beta} - 3) + e^K K^{a\bar b} D_a W \, \overline{D_b W} = e^K K^{a\bar b} D_a W \, \overline{D_b W} \;\geq\; 0. \tag{C.61}$$

The dangerous negative term is gone: V_F is positive semi-definite, the T^α are flat directions at this order, and supersymmetry is generically broken along them by $F^\alpha \neq 0$ with vanishing vacuum energy. This is exactly the accident that makes the classical type IIB flux potential of Chapter 9 positive semi-definite and leaves the Kähler moduli unfixed at leading order.

Example C.9 (The volume modulus)**.** Take one modulus with $K = -3\ln(T+\bar T)$ and $W = W_0$ constant. Then

$$K_T = -\frac{3}{T+\bar T}, \qquad K_{T\bar T} = \frac{3}{(T+\bar T)^2}, \qquad K^{T\bar T} = \frac{(T+\bar T)^2}{3},$$

so $K^{T\bar T} K_T K_{\bar T} = \dfrac{(T+\bar T)^2}{3} \cdot \dfrac{9}{(T+\bar T)^2} = 3$, and Eq. (C.61) gives $V_F \equiv 0$. Yet $D_T W = K_T W_0 \neq 0$, so $F^T \neq 0$ and $m_{3/2} = e^{K/2}|W_0| = |W_0|/(T+\bar T)^{3/2} \neq 0$: supersymmetry is broken with a massless modulus and exactly zero energy. The runaway is lifted only once W depends on T (non-perturbative effects) or the no-scale condition is corrected (Chapter 9).

Remark C.10 (Beyond the lectures: where no-scale comes from)*.* The relation in Eq. (C.60) is not an accident of one example. For the tree-level Kähler potential of a Calabi-Yau orientifold, $K = -2\ln\mathcal{V}$ with $\mathcal{V}$ the Einstein-frame volume, a homogeneous degree-3/2 function of the Kähler moduli $\mathrm{Re}\, T^\alpha$, Euler's theorem on homogeneous functions gives Eq. (C.60) automatically. The structure was isolated by Cremmer, Ferrara, Kounnas and Nanopoulos (1983); see Lahanas and Nanopoulos (1987) for the phenomenology. It is broken by α' and string-loop corrections to K, which is how the Large Volume Scenario stabilises $\mathcal{V}$.

C.4.7 The rigid limit

Restoring $M_{\rm pl}$ by dimensional analysis, Eqs. (C.48) and (C.50) become

$$D_i W = \partial_i W + \frac{K_i}{M_{\rm pl}^2} W, \qquad V_F = e^{K/M_{\rm pl}^2}\left(K^{i\bar\jmath}\, D_i W \, \overline{D_j W} - \frac{3}{M_{\rm pl}^2}|W|^2 \right), \tag{C.62}$$

and the gravitino mass is $m_{3/2} = e^{K/2M_{\rm pl}^2}|W|/M_{\rm pl}^2$. Sending $M_{\rm pl} \to \infty$ with the fields held fixed switches off gravity: $e^{K/M_{\rm pl}^2} \to 1$, the Kähler connection in $D_i W$ drops, and the $-3|W|^2/M_{\rm pl}^2$ term vanishes, leaving the familiar rigid $\mathcal{N}=1$ potential

$$V \xrightarrow{M_{\rm pl}\to\infty} K^{i\bar\jmath}\, \partial_i W \, \overline{\partial_j W} + \tfrac{1}{2}\,(\mathrm{Re}\, f)^{-1\,ab} \mathcal{P}_a \mathcal{P}_b = |F|^2 + \tfrac{1}{2}\, g^2 D^2. \tag{C.63}$$

The three specifically supergravitational features, the e^K dressing, the Kähler-connection term in $D_i W$, and above all the negative $-3|W|^2$, are precisely the pieces that survive at finite $M_{\rm pl}$ and that make gravitational moduli stabilisation both possible and delicate.

Three things that trip people up

(i) V_F uses the Kähler-*covariant* $D_i W$, not $\partial_i W$; the difference $K_i W$ matters exactly when $W \neq 0$, which is the interesting case. (ii) V_F is not positive: the $-3|W|^2$ can and does make supersymmetric vacua anti-de Sitter. (iii) K and W are individually meaningless; only $G = K + \ln|W|^2$, equivalently the pair up to Eq. (C.46), is physical, so always check that a

claimed result is Kähler-invariant.

Exercise C.5. For the STU model $K = -\sum_{\alpha=1}^{3} \ln(T^\alpha + \bar{T}^\alpha)$, show that each factor contributes 1 to the left-hand side of Eq. (C.60), so that $K^{\alpha\bar{\beta}} K_\alpha K_{\bar{\beta}} = 3$ and the model is no-scale. Deduce that a superpotential $W = W(\phi^a)$ independent of the T^α gives $V_F \geq 0$. Then take $K = -\ln(S+\bar{S}) - 3\ln(T+\bar{T})$ and show instead that $K^{i\bar{j}} K_i K_{\bar{j}} = 4$, so a T- and S-independent W produces the runaway $V_F = e^K |W|^2 > 0$ rather than a flat direction.

Appendix D

Differential Geometry, Cohomology, and Characteristic Classes

D.1 Manifolds, forms, and cohomology

The internal spaces of string compactification are smooth manifolds, and most of what the low-energy physicist reads off them is a topological invariant assembled from differential forms: the massless spectrum, the number of moduli (the massless scalars parametrising the compactification's continuous vacuum choices), the fluxes that can thread the geometry, the tadpole conditions that must cancel. This section is a self-contained development of the machinery that produces those invariants, assuming only the p-form expansion, the exterior derivative on components and the Hodge star fixed once and for all in Appendix A. On that base it builds the cohomological apparatus in full: the exterior algebra and de Rham cohomology, the Hodge star, codifferential and Laplacian, the Hodge decomposition (stated as a theorem), the Betti numbers that count everything, and the Dolbeault refinement used for complex internal spaces. It is meant to be consulted on its own, so the central results are given complete, with proofs cited to the literature rather than reproduced. Connections, curvature and characteristic classes belong to Appendix D.2, and the index theorems to Appendix D.3; the forms and Hodge theory here underpin the moduli and flux counting of Chapter 8, and the Dolbeault material feeds Appendix E. Standard references at this level are Nakahara, *Geometry, Topology and Physics*, Bott and Tu, *Differential Forms in Algebraic Topology*, and the review of Eguchi, Gilkey and Hanson (1980); for Hodge theory and the complex case, Griffiths and Harris, *Principles of Algebraic Geometry*, and Warner, *Foundations of Differentiable Manifolds*.

The concrete skills this section supports:

- compute the de Rham groups H^p_{dR} and Betti numbers of S^n, T^n and a genus-g surface, and read off the Euler characteristic;
- find the harmonic representative of a de Rham class, and say why it is unique;
- explain why $b_p(X)$ counts the massless modes obtained by reducing a p-form on the internal space X;
- split Betti numbers into Hodge numbers on a compact Kähler manifold.

The notes use two metric signatures. Hodge theory (the inner product on forms, the Laplacian, the split into harmonic pieces) is a Riemannian statement: it needs a positive-definite metric. That is exactly the signature of the compact internal factor X in a compactification $M_D = M_d \times X$, so every Hodge-theoretic statement below is made for a *compact, oriented, Riemannian* manifold.

The Lorentzian formulae of Appendix A govern the full spacetime.

The big picture

The one idea to carry away: on a compact manifold, taking a p-form field and expanding it in the harmonic p-forms of the internal space is how a higher-dimensional field turns into a finite list of massless lower-dimensional fields. The number of such fields is the Betti number b_p, a purely topological integer. de Rham cohomology is the bookkeeping that makes this precise and metric-independent; the Hodge theorem is what guarantees a unique harmonic representative to expand in. Everything in Chapter 8 about counting moduli, and everything in Appendix D.2 about which fluxes are allowed, runs on this.

D.1.1 Manifolds, tangent spaces, tensor fields

A smooth n-dimensional manifold M is a space that looks locally like $\mathbb{R}^n$: it is covered by coordinate charts $x^\mu : U \to \mathbb{R}^n$ whose overlaps are smooth. At each point $p \in M$ the tangent space T_pM is the n-dimensional real vector space of directional derivatives, with coordinate basis $\partial_\mu \coloneqq \partial/\partial x^\mu$. A vector field $V = V^\mu \partial_\mu$ assigns a tangent vector smoothly across M. The dual space T_p^*M, the cotangent space, has the basis $\mathrm{d}x^\mu$ dual to ∂_μ, meaning $\mathrm{d}x^\mu(\partial_\nu) = \delta^\mu{}_\nu$. A one-form is a section $\omega = \omega_\mu \, \mathrm{d}x^\mu$ of the cotangent bundle. Tensor fields of type (r, s) are built by tensoring r copies of TM with s copies of T^*M; their components transform in the familiar way under a change of chart $x \to x'$, with one factor of $\partial x'/\partial x$ per upper index and one of $\partial x/\partial x'$ per lower index.

Two structures on M are optional and independent. A *Riemannian metric* $g_{\mu\nu}$ is a smooth, symmetric, positive-definite $(0, 2)$ tensor; it lets one measure lengths, raise and lower indices, and (as below) define the Hodge star. An *orientation* is a consistent global choice of "handedness", equivalently a nowhere-vanishing top-degree form; not every manifold admits one (the Möbius band and $\mathbb{RP}^2$ do not). Integration of top forms and Poincaré duality both require orientability.

The Lie bracket of two vector fields, $[X, Y]^\mu = X^\nu \partial_\nu Y^\mu - Y^\nu \partial_\nu X^\mu$, is again a vector field and measures the failure of their flows to commute. It needs no metric and enters the coordinate-free definition of the exterior derivative below.

D.1.2 The exterior algebra and the exterior derivative

A differential p-form is a totally antisymmetric $(0, p)$ tensor field, written in the convention of Appendix A as

$$\omega = \frac{1}{p!} \, \omega_{\mu_1 \cdots \mu_p} \, \mathrm{d}x^{\mu_1} \wedge \cdots \wedge \mathrm{d}x^{\mu_p}, \tag{D.1}$$

with $\omega_{\mu_1 \cdots \mu_p}$ totally antisymmetric. The space of p-forms is denoted $\Omega^p(M)$; it is nonzero only for $0 \le p \le n$, with $\dim \Omega^p_{(\text{pointwise})} = \binom{n}{p}$. Zero-forms are smooth functions, n-forms are top forms. The wedge product turns the graded vector space $\Omega^\bullet(M) = \bigoplus_{p=0}^{n} \Omega^p(M)$ into an associative, graded-commutative algebra,

$$\alpha_p \wedge \beta_q = (-1)^{pq} \, \beta_q \wedge \alpha_p, \tag{D.2}$$

so odd-degree forms anticommute and an odd form wedged with itself vanishes.

The *exterior derivative* $\mathrm{d} : \Omega^p \to \Omega^{p+1}$ is the one first-order differential operator that exists on any manifold without extra structure. On components,

$$(\mathrm{d}\omega)_{\mu_0 \mu_1 \cdots \mu_p} = (p + 1) \, \partial_{[\mu_0} \omega_{\mu_1 \cdots \mu_p]}, \tag{D.3}$$

and it is characterised, independently of any coordinates, by three properties: it is $\mathbb{R}$-linear, it acts on a function as the ordinary differential $\mathrm{d}f = \partial_\mu f \, \mathrm{d}x^\mu$, and it is a graded (or "anti-")

derivation that squares to zero,

$$\mathrm{d}(\alpha_p \wedge \beta) = \mathrm{d}\alpha_p \wedge \beta + (-1)^p\, \alpha_p \wedge \mathrm{d}\beta, \tag{D.4}$$

$$\mathrm{d}^2 = 0. \tag{D.5}$$

Nilpotency (D.5) is the statement $\partial_{[\mu}\partial_{\nu]} = 0$ dressed up, and it is the algebraic engine of everything that follows. There is also a fully invariant formula, useful because it shows d never refers to a metric or a connection:

$$\begin{aligned}\mathrm{d}\omega(X_0,\dots,X_p) = &\sum_{i=0}^{p}(-1)^i\, X_i\, \omega(X_0,\dots,\widehat{X_i},\dots,X_p)\\ &+\sum_{i<j}(-1)^{i+j}\,\omega([X_i,X_j],X_0,\dots,\widehat{X_i},\dots,\widehat{X_j},\dots,X_p),\end{aligned} \tag{D.6}$$

where a hat denotes an omitted argument. Under a smooth map $f : M \to N$ the pullback f^* on forms commutes with d,

$$f^*(\mathrm{d}\omega) = \mathrm{d}(f^*\omega), \tag{D.7}$$

which is why de Rham cohomology will turn out to be a homotopy invariant.

A p-form is *closed* if $\mathrm{d}\omega = 0$ and *exact* if $\omega = \mathrm{d}\eta$ for some $(p-1)$-form η. By (D.5) every exact form is closed. The converse fails globally, and that failure is what de Rham cohomology measures.

D.1.3 Interior product, Lie derivative, Cartan's formula

The calculus of forms is rounded out by two operations that recur throughout the notes. The *interior product* (contraction) with a vector field X lowers the degree by one,

$$(\iota_X\omega)_{\mu_1\cdots\mu_{p-1}} = X^\nu\, \omega_{\nu\mu_1\cdots\mu_{p-1}}, \tag{D.8}$$

and is itself a graded derivation, $\iota_X(\alpha_p\wedge\beta) = \iota_X\alpha_p\wedge\beta + (-1)^p\alpha_p\wedge\iota_X\beta$, with $\iota_X^2 = 0$. The *Lie derivative* $\mathcal{L}_X$ measures the change of a form along the flow of X and is tied to the other two operators by Cartan's magic formula,

$$\mathcal{L}_X = \mathrm{d}\,\iota_X + \iota_X\,\mathrm{d}. \tag{D.9}$$

This one line does a lot of work: it gives $\mathcal{L}_X\mathrm{d} = \mathrm{d}\mathcal{L}_X$ for free, it turns the proof of the Poincaré lemma (stated in Appendix D.1.5) into a two-line homotopy, and it is the quickest route to the statement that a form is invariant under a symmetry generated by X precisely when $\mathcal{L}_X\omega = 0$.

D.1.4 Integration and Stokes' theorem

Forms are the objects one integrates without a metric. On an oriented n-manifold a top form $\omega_n = \frac{1}{n!}\omega_{\mu_1\cdots\mu_n}\mathrm{d}x^{\mu_1}\wedge\cdots\wedge\mathrm{d}x^{\mu_n}$ integrates to a number $\int_M \omega_n$, defined chart by chart and glued with a partition of unity; the antisymmetry of the wedge supplies exactly the Jacobian sign that makes the answer coordinate-independent (up to the overall orientation). More generally a p-form is integrated over an oriented p-dimensional submanifold or cycle. The central identity is *Stokes' theorem*: for an $(n-1)$-form ω on an oriented manifold with boundary,

$$\int_M \mathrm{d}\omega = \int_{\partial M}\omega, \tag{D.10}$$

with ∂M carrying the induced orientation. Two corollaries are used constantly. On a *closed* manifold (compact, no boundary) $\int_M \mathrm{d}\omega = 0$, which is integration by parts for forms. And the integral of a closed form over a cycle depends only on the homology class of the cycle and the cohomology class of the form, because shifting either by a boundary or by an exact piece changes the integral by $\int \mathrm{d}(\cdots) = 0$. These metric-free numbers are the *periods*; Appendix D.1.5 makes them the bridge from forms to topology.

D.1.5 de Rham cohomology

Because $\mathrm{d}^2 = 0$, the sequence

$$0 \to \Omega^0(M) \xrightarrow{\mathrm{d}} \Omega^1(M) \xrightarrow{\mathrm{d}} \cdots \xrightarrow{\mathrm{d}} \Omega^n(M) \to 0 \tag{D.11}$$

is a cochain complex: the image of one arrow lies in the kernel of the next. The p-th *de Rham cohomology* is the quotient

$$H^p_{\mathrm{dR}}(M) \coloneqq \frac{\ker\left(\mathrm{d} : \Omega^p \to \Omega^{p+1}\right)}{\mathrm{Im}\left(\mathrm{d} : \Omega^{p-1} \to \Omega^p\right)} = \frac{\{\text{closed } p\text{-forms}\}}{\{\text{exact } p\text{-forms}\}}. \tag{D.12}$$

Its elements are equivalence classes $[\omega]$ of closed forms, two being identified when they differ by an exact form. Cohomology measures the global obstruction to a closed form being exact. Locally there is no obstruction: the *Poincaré lemma* states that on a contractible (for instance star-shaped) open set every closed p-form with $p \geq 1$ is exact, so $H^p_{\mathrm{dR}} = 0$ there for $p \geq 1$. A nonzero class is therefore a genuinely global fact about M. In physics language, a source-free field strength always has a local potential; whether it has a global one is a cohomology question, and when it does not one has flux.

A few structural facts, stated rather than proved (see Bott and Tu, or Nakahara ch. 6):

- $H^0_{\mathrm{dR}}(M) \cong \mathbb{R}^c$, where c is the number of connected components: a closed zero-form is a locally constant function.
- The groups are *homotopy invariants*. If M and N are homotopy equivalent then $H^p_{\mathrm{dR}}(M) \cong H^p_{\mathrm{dR}}(N)$; this follows from (D.7) and a homotopy operator built from (D.9). In particular cohomology cannot see a smooth deformation.
- The wedge product descends to cohomology, $[\alpha] \smile [\beta] \coloneqq [\alpha \wedge \beta]$, making $H^\bullet_{\mathrm{dR}}(M) = \bigoplus_p H^p_{\mathrm{dR}}(M)$ a graded-commutative ring, the *cohomology ring*. Products of generators encode intersection data.
- *de Rham's theorem*: integration of forms over cycles, $([\omega], \gamma) \mapsto \int_\gamma \omega$, is a nondegenerate pairing of $H^p_{\mathrm{dR}}(M)$ with the singular homology $H_p(M;\mathbb{R})$. Hence $H^p_{\mathrm{dR}}(M) \cong H^p(M;\mathbb{R})$, the smooth de Rham theory computes the same real cohomology as the topologist's singular theory.
- *Künneth*: for a product, $H^k_{\mathrm{dR}}(M \times N) \cong \bigoplus_{i+j=k} H^i_{\mathrm{dR}}(M) \otimes H^j_{\mathrm{dR}}(N)$, which is how the cohomology of a torus is built from that of a circle.

The pairing in de Rham's theorem is where quantisation enters. Inside the real cohomology $H^p(M;\mathbb{R})$ sits the image of the *integral* cohomology $H^p(M;\mathbb{Z})$: a class is integral when all of its periods over integer cycles are integers. A field strength whose flux is quantised represents such an integral class. The precise statement, that $\frac{1}{2\pi}\int_\Sigma F \in \mathbb{Z}$ because $F/2\pi$ is the first Chern class of a line bundle, belongs to Appendix D.2; de Rham cohomology is the arena in which it is made.

Example D.1 (Cohomology of the sphere)**.** For the n-sphere with $n \geq 1$,

$$H^p_{\mathrm{dR}}(S^n) = \begin{cases} \mathbb{R} & p = 0,\ p = n, \\ 0 & \text{otherwise.} \end{cases} \tag{D.13}$$

The top class is represented by the normalised volume form: on S^2 the two-form $\frac{1}{4\pi}\sin\theta\,\mathrm{d}\theta \wedge \mathrm{d}\varphi$ is closed (it is a top form) but not exact, since its integral over S^2 is $1 \neq 0$, and an exact form would integrate to zero by Stokes. This same class, pulled back to the worldvolume, is the unit of magnetic flux whose integrality is Dirac quantisation.

D.1.6 Betti numbers, Poincaré duality, the Euler characteristic

On a compact manifold each $H^p_{\mathrm{dR}}(M)$ is finite-dimensional. Its dimension is the *p-th Betti number*,

$$b_p(M) \coloneqq \dim_{\mathbb{R}} H^p_{\mathrm{dR}}(M), \tag{D.14}$$

the number of independent nontrivial p-cycles, and the basic count of the low-energy spectrum in any compactification. The generating function $P_M(t) = \sum_p b_p\, t^p$ is the Poincaré polynomial; Künneth reads $P_{M\times N} = P_M\, P_N$.

For a *closed, oriented* n-manifold, wedging complementary-degree classes and integrating,

$$H^p_{\mathrm{dR}}(M) \times H^{n-p}_{\mathrm{dR}}(M) \to \mathbb{R}, \qquad ([\alpha],[\beta]) \mapsto \int_M \alpha \wedge \beta, \tag{D.15}$$

is a nondegenerate pairing. This is *Poincaré duality*, and it forces

$$b_p(M) = b_{n-p}(M). \tag{D.16}$$

On the middle cohomology of an even-dimensional manifold the pairing (D.15) becomes the *intersection form*, symmetric for $n \equiv 0 \bmod 4$ and antisymmetric for $n \equiv 2 \bmod 4$; it controls, for instance, the lattice of two-cycles of K3 and the Dirac pairing of wrapped-brane charges. On a four-manifold this form is symmetric, and diagonalising it splits the middle Betti number as

$$b_2 = b_2^+ + b_2^-, \qquad \tau(M) = b_2^+ - b_2^-, \tag{D.17}$$

where $b_2^\pm$ count the self-dual and anti-self-dual harmonic two-forms (the ± 1 eigenspaces of $\star$, which squares to $+1$ in this degree) and τ is the signature. For K3 one has $b_2^+ = 3$, $b_2^- = 19$ and $\tau = -16$.

The alternating sum of the Betti numbers is the *Euler characteristic*,

$$\chi(M) = \sum_{p=0}^{n} (-1)^p\, b_p(M). \tag{D.18}$$

By (D.16) an odd-dimensional closed manifold has $\chi = 0$. The Euler characteristic is also computed by the integral of the Euler class of TM (Gauss-Bonnet), and it counts induced brane charge in the tadpole conditions; those readings are developed in Appendix D.2 and Appendix D.3.

Manifold	Betti numbers $(b_0, b_1, \dots)$	Poincaré poly. $P_M(t)$	χ
point	1	1	1
S^n $(n \geq 1)$	$1, 0, \dots, 0, 1$	$1 + t^n$	$1 + (-1)^n$
T^n	$b_p = \binom{n}{p}$	$(1+t)^n$	0
Σ_g (genus g)	$1,\ 2g,\ 1$	$1 + 2g\,t + t^2$	$2 - 2g$
$\mathbb{CP}^n$	$b_{2k} = 1,\ b_{\mathrm{odd}} = 0$	$1 + t^2 + \cdots + t^{2n}$	$n + 1$
K3	$1, 0, 22, 0, 1$	$1 + 22t^2 + t^4$	24

The K3 numbers are quoted here as topology; their refinement into Hodge numbers, and the special role of K3 as the unique nontrivial Calabi-Yau twofold, are developed in Appendix E.

D.1.7 The Hodge star, the codifferential, and the Laplacian

Introduce now a Riemannian metric g and an orientation. The *Hodge star* of Appendix A maps $\Omega^p \to \Omega^{n-p}$,

$$(\star\omega)_{\mu_1\cdots\mu_{n-p}} = \frac{1}{p!}\, \varepsilon_{\mu_1\cdots\mu_{n-p}}{}^{\nu_1\cdots\nu_p}\, \omega_{\nu_1\cdots\nu_p}, \tag{D.19}$$

with $\varepsilon = \sqrt{g}\,\epsilon$ the Levi-Civita tensor. In Euclidean signature it squares to

$$\star\star\,\omega = (-1)^{p(n-p)}\,\omega \qquad \text{(Riemannian)}, \tag{D.20}$$

the sign being the Euclidean counterpart of the Lorentzian result in Appendix A. The star packages the metric into an L^2 inner product on forms of a given degree,

$$(\alpha,\beta) \coloneqq \int_M \alpha\wedge\star\beta = \int_M \frac{1}{p!}\,\alpha_{\mu_1\cdots\mu_p}\beta^{\mu_1\cdots\mu_p}\,\sqrt{g}\,\mathrm{d}^n x, \tag{D.21}$$

which is symmetric and positive-definite because g is Riemannian. Positivity is lost in Lorentzian signature, where the construction breaks down.

The formal adjoint of d under (D.21) is the *codifferential* $\mathrm{d}^\dagger : \Omega^p \to \Omega^{p-1}$, defined by $(\mathrm{d}\alpha,\beta) = (\alpha,\mathrm{d}^\dagger\beta)$ on a closed manifold and given explicitly by

$$\mathrm{d}^\dagger = (-1)^{\,n(p+1)+1}\,\star\,\mathrm{d}\,\star \qquad \text{on } \Omega^p, \tag{D.22}$$

which for even n (the relevant case, real dimension 4 or 6) collapses to the tidy $\mathrm{d}^\dagger = -\star\,\mathrm{d}\star$ in every degree. Since $(\mathrm{d})^2 = 0$ one has $(\mathrm{d}^\dagger)^2 = 0$ as well. As a concrete instance, on flat $\mathbb{R}^n$ the codifferential of a one-form $\omega = \omega_\mu\,\mathrm{d}x^\mu$ is minus its divergence, $\mathrm{d}^\dagger\omega = -\partial^\mu\omega_\mu$. The *Hodge-de Rham Laplacian* is the anticommutator,

$$\Delta \coloneqq \mathrm{d}\,\mathrm{d}^\dagger + \mathrm{d}^\dagger\,\mathrm{d} = (\mathrm{d}+\mathrm{d}^\dagger)^2, \tag{D.23}$$

a second-order elliptic operator preserving degree that reduces on functions to the ordinary (positive) Laplace-Beltrami operator $-\frac{1}{\sqrt{g}}\partial_\mu(\sqrt{g}\,g^{\mu\nu}\partial_\nu)$. A form is *harmonic* if $\Delta\omega = 0$. On a compact manifold the positivity of (D.21) gives the key equivalence,

$$\Delta\omega = 0 \iff \mathrm{d}\omega = 0 \ \text{ and } \ \mathrm{d}^\dagger\omega = 0, \tag{D.24}$$

since $(\Delta\omega,\omega) = \|\mathrm{d}\omega\|^2 + \|\mathrm{d}^\dagger\omega\|^2 \geq 0$ vanishes only when both terms do. A harmonic form is closed and co-closed at once. Write $\mathcal{H}^p(M) = \ker\Delta|_{\Omega^p}$ for the space of harmonic p-forms.

D.1.8 Hodge decomposition and harmonic representatives

Theorem D.2 (Hodge decomposition)**.** *On a compact, oriented, Riemannian manifold M the space of p-forms splits as an L^2-orthogonal direct sum*

$$\Omega^p(M) = \mathcal{H}^p(M) \ \oplus\ \mathrm{d}\,\Omega^{p-1}(M) \ \oplus\ \mathrm{d}^\dagger\Omega^{p+1}(M), \tag{D.25}$$

the space of harmonic forms $\mathcal{H}^p(M)$ is finite-dimensional, and every de Rham class contains exactly one harmonic representative. Consequently

$$H^p_{\mathrm{dR}}(M) \ \cong\ \mathcal{H}^p(M), \qquad b_p(M) = \dim\mathcal{H}^p(M). \tag{D.26}$$

A proof rests on the elliptic theory of Δ and is given in Warner or Griffiths and Harris; we take it as known. The content is that abstract cohomology classes have concrete representatives, the solutions of a linear PDE $\Delta\omega = 0$. A closed form ω decomposes as $\omega = \omega_{\mathrm{harm}} + \mathrm{d}\eta$, its harmonic part being the unique element of least norm in the class. The metric enters the representative but not the dimension: the class and its Betti number are topological, the harmonic form that stands for it is not.

Example D.3 (Harmonic forms on the circle)**.** Take $S^1 = \mathbb{R}/\mathbb{Z}$ with the flat metric $g = \mathrm{d}\theta^2$, $\theta \sim \theta + 1$. The Laplacian acts on a function as $\Delta f = -f''$ and on a one-form as $\Delta(a(\theta)\,\mathrm{d}\theta) = -a''\,\mathrm{d}\theta$. Expand in Fourier modes $e^{2\pi \mathrm{i} k\theta}$, $k \in \mathbb{Z}$: the mode of wavenumber k has Δ-eigenvalue $(2\pi k)^2$, which vanishes only for $k = 0$. Harmonic forms are therefore the constants, so $\mathcal{H}^0(S^1)$ is spanned by the constant function and $\mathcal{H}^1(S^1)$ by $\mathrm{d}\theta$, giving $b_0 = b_1 = 1$. Every nonzero mode sits in the exact or co-exact summand of (D.25), since $\mathrm{d}\, e^{2\pi \mathrm{i} k\theta} = 2\pi \mathrm{i} k\, e^{2\pi \mathrm{i} k\theta}\mathrm{d}\theta$. The torus exercise below runs this same computation in n coordinates.

Two consequences follow at once. Poincaré duality (D.16) becomes transparent: (D.22) gives $\star\,\mathrm{d} = \pm\,\mathrm{d}^\dagger\star$ and $\star\,\mathrm{d}^\dagger = \pm\,\mathrm{d}\star$, so $\star$ commutes with Δ; being invertible by (D.20), it maps $\mathcal{H}^p(M)$ isomorphically onto $\mathcal{H}^{n-p}(M)$ and forces $b_p = b_{n-p}$. And harmonic forms are the zero modes of the internal Laplacian, hence the massless fields of a Kaluza-Klein reduction, spelled out next.

The big picture

Reduce a D-dimensional p-form field A_p on $M_d \times X$ by expanding it in a basis of eigenforms of Δ_X. Each internal eigenvalue λ becomes a mass^2 in the d-dimensional theory, so the massless sector is spanned by the harmonic forms, $\ker \Delta_X$. The number of massless d-dimensional fields that descend from A_p this way is $b_p(X)$, and the moduli of the compactification are counted the same way. This is why the low-energy spectrum of a string compactification is read off from the Betti (and, in the complex case, Hodge) numbers of the internal space, worked out in detail in Chapter 8.

Careful

The decomposition (D.25) and the isomorphism (D.26) are Riemannian and compact statements. In Lorentzian signature the inner product (D.21) is indefinite, Δ is the hyperbolic wave operator rather than an elliptic one, and "harmonic" no longer means "closed and co-closed". Apply Hodge theory to the compact internal factor X, never to the spacetime M_d.

Exercise D.1 (The flat torus)**.** Take $T^n = \mathbb{R}^n/\mathbb{Z}^n$ with the flat metric and coordinates $\theta^i \sim \theta^i + 1$. Show that the constant-coefficient forms $\mathrm{d}\theta^{i_1} \wedge \cdots \wedge \mathrm{d}\theta^{i_p}$ are harmonic, that no nonconstant Fourier mode is, and hence that $\mathcal{H}^p(T^n)$ is spanned by these $\binom{n}{p}$ forms. Conclude $b_p(T^n) = \binom{n}{p}$, in agreement with Künneth applied to $T^n = (S^1)^n$, and check $\chi(T^n) = \sum_p (-1)^p \binom{n}{p} = 0$. Finally, reducing a higher-dimensional graviton on T^n, the mixed metric components $g_{\mu i}$ pair with the $b_1 = n$ harmonic one-forms $\mathrm{d}\theta^i$, so all n of them survive as $U(1)$ Kaluza-Klein gauge fields.

D.1.9 Complex manifolds and Dolbeault cohomology

The internal spaces that preserve supersymmetry are complex, and on a complex manifold the de Rham complex refines. This subsection states the refinement so the appendix is self-contained; the geometry of Kähler and Calabi-Yau manifolds, the Hodge diamond and the counting of moduli are developed in Appendix E.

A complex manifold of complex dimension n (real dimension $2n$) has holomorphic coordinates z^i and antiholomorphic $\bar{z}^{\bar{\jmath}}$, $i, \bar{\jmath} = 1, \ldots, n$. Complex-valued forms split by holomorphic and antiholomorphic degree: a (p, q)-form has p holomorphic and q antiholomorphic legs,

$$\omega = \frac{1}{p!\,q!}\,\omega_{i_1 \cdots i_p \bar{\jmath}_1 \cdots \bar{\jmath}_q}\,\mathrm{d}z^{i_1} \wedge \cdots \wedge \mathrm{d}z^{i_p} \wedge \mathrm{d}\bar{z}^{\bar{\jmath}_1} \wedge \cdots \wedge \mathrm{d}\bar{z}^{\bar{\jmath}_q}, \tag{D.27}$$

and $\Omega^k \otimes \mathbb{C} = \bigoplus_{p+q=k} \Omega^{p,q}$. The exterior derivative splits as $\mathrm{d} = \partial + \bar{\partial}$ with $\partial : \Omega^{p,q} \to \Omega^{p+1,q}$ and $\bar{\partial} : \Omega^{p,q} \to \Omega^{p,q+1}$. Nilpotency of d and the bidegree bookkeeping give

$$\partial^2 = 0, \qquad \bar{\partial}^{\,2} = 0, \qquad \partial\bar{\partial} + \bar{\partial}\partial = 0. \tag{D.28}$$

The $\bar\partial$-complex defines *Dolbeault cohomology*,

$$H^{p,q}_{\bar\partial}(M) \coloneqq \frac{\ker\left(\bar\partial : \Omega^{p,q} \to \Omega^{p,q+1}\right)}{\operatorname{Im}\left(\bar\partial : \Omega^{p,q-1} \to \Omega^{p,q}\right)}, \qquad h^{p,q} \coloneqq \dim_{\mathbb{C}} H^{p,q}_{\bar\partial}(M), \tag{D.29}$$

whose dimensions $h^{p,q}$ are the *Hodge numbers*. On a general complex manifold Dolbeault and de Rham cohomology are not simply related. On a compact *Kähler* manifold, however (the setting for Calabi-Yau compactification), the Hodge theorem refines: there is a $\bar\partial$-Laplacian whose harmonic forms carry a definite bidegree, and the de Rham groups split accordingly,

$$H^k_{\mathrm{dR}}(M;\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}_{\bar\partial}(M), \qquad H^{p,q}_{\bar\partial} = \overline{H^{q,p}_{\bar\partial}}, \tag{D.30}$$

so that the Betti numbers refine into Hodge numbers,

$$b_k = \sum_{p+q=k} h^{p,q}, \qquad h^{p,q} = h^{q,p}, \qquad h^{p,q} = h^{n-p,n-q}. \tag{D.31}$$

The first symmetry is complex conjugation, the second Serre duality (the complex refinement of Poincaré duality). Arranged in the Hodge diamond these numbers organise the massless spectrum of a Calabi-Yau compactification, which is where Appendix E takes up the story.

Example D.4 (Hodge diamond of $\mathbb{CP}^1$)**.** $\mathbb{CP}^1 \cong S^2$ has complex dimension 1. Its only nonzero Dolbeault groups are $h^{0,0} = h^{1,1} = 1$, with $h^{1,0} = h^{0,1} = 0$: a holomorphic function on a compact connected complex manifold is constant, and $H^{1,1}$ is generated by the Fubini-Study Kähler form. Then (D.31) gives $b_0 = h^{0,0} = 1$, $b_1 = h^{1,0} + h^{0,1} = 0$ and $b_2 = h^{1,1} = 1$, reproducing the S^2 row of the table. The complex torus T^2 carries the same $b_0 = b_2 = 1$ but has $h^{1,0} = h^{0,1} = 1$, so $b_1 = 2$: the same Betti numbers, split differently across bidegree.

Exercise D.2 (Hodge numbers against Betti numbers)**.** Use (D.31) to recover the Betti table from Hodge data. For $\mathbb{CP}^n$, $h^{p,p} = 1$ and $h^{p,q} = 0$ for $p \neq q$, giving $b_{2k} = 1$ and $b_{\mathrm{odd}} = 0$. For K3, $h^{0,0} = h^{2,2} = 1$, $h^{2,0} = h^{0,2} = 1$ and $h^{1,1} = 20$; confirm $b_2 = h^{2,0} + h^{1,1} + h^{0,2} = 22$ and $\chi = \sum_{p,q}(-1)^{p+q} h^{p,q} = 24$, matching the table.

Remark D.5 (Beyond the lectures). The equalities (D.30) rest on the Kähler identities and can fail on complex manifolds that admit no Kähler metric; the Hopf surface, with $b_1 = 1$, is the standard counterexample. More generally the difference between $\sum h^{p,q}$ and b_k is measured by the Frölicher spectral sequence, which degenerates at its first page precisely in the Kähler case. A parallel refinement of de Rham theory in the presence of a closed three-form H is *twisted cohomology*, with differential $\mathrm{d}_H = \mathrm{d} - H\wedge$ obeying $\mathrm{d}_H^2 = 0$ when $\mathrm{d}H = 0$; its classes are the natural home for Ramond-Ramond fluxes in the democratic formulation of Appendix A, and the corresponding K-theory sharpening of charge quantisation is flagged in Appendix D.2.

D.2 Bundles, connections, and characteristic classes

The forms of Appendix D.1 live on a single manifold. Physics puts extra data at each point: a gauge field is a rule for comparing charged fields at neighbouring points, a spinor is a section of a bundle twisted by the tangent frame, a Ramond-Ramond flux is a curvature whose integral is quantised. All of this is the geometry of *bundles*. This section fixes the language, defines connections and their curvature, and builds the characteristic classes (Chern, Euler, Pontryagin, the Chern character and the $\hat{A}$ genus) that turn a smooth geometric object into an integer. Those integers are what quantise Ramond-Ramond charge, count the induced lower-brane charges that enter tadpole cancellation, and organise anomalies.

The big picture

A characteristic class is a recipe that takes a bundle with connection, feeds its curvature F into an invariant polynomial, and returns a closed form whose cohomology class does not depend on the connection. Two facts make this the backbone of the rest of the notes. First, these classes are *integral*: their periods over cycles are integers, which is charge quantisation before any string enters. Second, they are *multiplicative and additive* under the natural bundle operations, so a wrapped-brane charge, an anomaly polynomial, or a Dirac index all reduce to reading coefficients off a product of power series in F. Get the normalisation of that power series right (the factors of i and 2π below) and the rest is bookkeeping.

D.2.1 Fibre bundles and vector bundles

A *fibre bundle* is a space E that looks locally like a product of a base M with a fixed fibre F, glued together in a possibly twisted way.

Definition D.6 (Fibre bundle)**.** A fibre bundle is a smooth surjection $\pi : E \to M$ (total space, projection, base) with typical fibre F and structure group G acting on F, together with an open cover $\{U_\alpha\}$ of M and diffeomorphisms (*local trivialisations*) $\phi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times F$ commuting with the projection to U_α. On overlaps the trivialisations differ by *transition functions* $g_{\alpha\beta} : U_\alpha \cap U_\beta \to G$,

$$\phi_\alpha \circ \phi_\beta^{-1}(x, f) = (x,\, g_{\alpha\beta}(x) \cdot f), \tag{D.32}$$

which satisfy the *cocycle condition* $g_{\alpha\beta} g_{\beta\gamma} g_{\gamma\alpha} = 1$ on triple overlaps and $g_{\alpha\alpha} = 1$.

The cocycle data determines the bundle completely: give a cover, a group, and functions $g_{\alpha\beta}$ obeying Eq. (D.32), and E is reconstructed by gluing $U_\alpha \times F$. A bundle is *trivial* if it is globally a product $M \times F$, equivalently if the $g_{\alpha\beta}$ can be gauged to the identity. Twisting is measured by the failure of that, and characteristic classes are the topological invariants of the twist.

Three special cases carry the physics that follows.

Vector bundles. The fibre is a vector space $\mathbb{R}^r$ or $\mathbb{C}^r$ and $G \subseteq \mathrm{GL}(r, \mathbb{R})$ or $\mathrm{GL}(r, \mathbb{C})$ acts linearly; r is the *rank*. A rank-1 complex bundle is a *line bundle*. A *section* is a smooth map $s : M \to E$ with $\pi \circ s = \mathrm{id}$, that is, a choice of one vector in each fibre; the sections form a module $\Gamma(E)$ over the functions on M. A local *frame* is r sections that are pointwise a basis. The tangent bundle TM, cotangent bundle T^*M, and their tensor and exterior powers $\Lambda^p T^*M$ (whose sections are the p-forms of Appendix D.1) are the archetypes; their structure group is $\mathrm{GL}(D, \mathbb{R})$, reduced to $\mathrm{SO}(D)$ by a metric and orientation and to $\mathrm{Spin}(D)$ when a spin structure exists (Appendix B).

Principal bundles. The fibre is the group G itself, acting on itself by right multiplication. A principal G-bundle $P \to M$ is the natural home of a gauge field: the transition functions are G-valued, and there is no extra data beyond the group. A gauge transformation is a change of local trivialisation, i.e. a bundle automorphism covering the identity on M.

Associated bundles. Given a principal G-bundle P and a representation $\rho : G \to \mathrm{GL}(V)$, the *associated* vector bundle $P \times_\rho V$ has the same transition functions $g_{\alpha\beta}$ acting through ρ. Matter in representation ρ is a section of $P \times_\rho V$; the gauge field lives on P; the two share their cocycle. This is the precise sense in which "a field in the **3** of SU(3)" is a geometric object.

Bundles inherit the linear-algebra operations fibrewise, and their transition functions transform accordingly (Table D.1). The determinant and the pullback deserve separate mention. The *determinant line bundle* $\det E = \Lambda^r E$ collects the rank into a single line bundle whose twist is the "overall" twist of E; its first Chern class (the c_1 of Appendix D.2.4 below) equals that of E. The *pullback* f^*E along $f : N \to M$ transports a bundle to another base by composing

Operation	Fibre	Transition function
dual E^*	V^*	$(g_{\alpha\beta}^{-1})^{\mathsf{T}}$
direct (Whitney) sum $E \oplus E'$	$V \oplus V'$	$g_{\alpha\beta} \oplus g'_{\alpha\beta}$
tensor $E \otimes E'$	$V \otimes V'$	$g_{\alpha\beta} \otimes g'_{\alpha\beta}$
homomorphisms $\mathrm{Hom}(E, E')$	$V^* \otimes V'$	$g'_{\alpha\beta}(\cdot)g_{\alpha\beta}^{-1}$
determinant $\det E = \Lambda^r E$	$\Lambda^r V$	$\det g_{\alpha\beta}$
pullback f^*E, $f : N \to M$	V	$g_{\alpha\beta} \circ f$

Table D.1: Operations on vector bundles and the induced transition functions. Each is defined fibrewise, so any construction that is natural in vector spaces (kernels aside) is natural in bundles.

transition functions with f, and it commutes with every characteristic class, $c(f^*E) = f^*c(E)$, which is why wrapping and embedding computations pull back cleanly.

Example D.7 (Line bundles on projective space)**.** On $\mathbb{CP}^n$ the *tautological* line bundle $\mathcal{O}(-1)$ assigns to each point (a line through the origin in $\mathbb{C}^{n+1}$) that very line as its fibre. Its dual is the *hyperplane* bundle $\mathcal{O}(1)$, and tensor powers give $\mathcal{O}(k)$ for all $k \in \mathbb{Z}$. Every holomorphic line bundle on $\mathbb{CP}^n$ is one of these, so the group of line bundles under $\otimes$ (the *Picard group*) is $\mathbb{Z}$, labelled by the integer $k = \int_{\mathbb{CP}^1} c_1(\mathcal{O}(k))$ with c_1 the first Chern class of Appendix D.2.4. The homogeneous coordinates $z^0, \dots, z^n$ are the holomorphic sections of $\mathcal{O}(1)$, and degree-d polynomials are sections of $\mathcal{O}(d)$: the quintic of Appendix E is the zero locus of a section of $\mathcal{O}(5)$ over $\mathbb{CP}^4$.

D.2.2 Connections and curvature

Sections of a bundle cannot be differentiated in a coordinate-invariant way: the difference of fibres at two points is not defined until we say how to identify them. A *connection* supplies that identification infinitesimally.

Definition D.8 (Connection)**.** A connection on a vector bundle E is a linear map $\nabla : \Gamma(E) \to \Gamma(T^*M \otimes E)$ obeying the Leibniz rule $\nabla(fs) = \mathrm{d}f \otimes s + f\,\nabla s$ for functions f and sections s. In a local frame $\{e_a\}$ it is encoded by a matrix of one-forms A, the *connection* (or gauge) form,

$$\nabla e_b = A^a{}_b\, e_a, \qquad \nabla(s^a e_a) = (\mathrm{d}s^a + A^a{}_b\, s^b)e_a. \tag{D.33}$$

Under a change of frame $e_a \to e_b\,(g^{-1})^b{}_a$ (a gauge transformation $g : U \to G$) the connection form transforms inhomogeneously,

$$A \;\longmapsto\; gAg^{-1} + g\,\mathrm{d}g^{-1} \;=\; gAg^{-1} - (\mathrm{d}g)\,g^{-1}, \tag{D.34}$$

the familiar gauge-field transformation. On a principal G-bundle the same object is a $\mathfrak{g}$-valued one-form A on the total space (a splitting of each tangent space into horizontal and vertical parts); Eq. (D.34) is how its pullback to M by different local sections is related.

The *curvature* measures the failure of covariant derivatives to commute, $[\nabla_\mu, \nabla_\nu]s = F_{\mu\nu}s$, and is the gauge-covariant two-form

$$F \;=\; \mathrm{d}A + A \wedge A, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \tag{D.35}$$

matching the field-strength convention of Appendix A (for a matrix-valued one-form $A \wedge A = \frac{1}{2}[A_\mu, A_\nu]\mathrm{d}x^\mu \wedge \mathrm{d}x^\nu$ is not zero). Under gauge transformations $F \mapsto gFg^{-1}$: the curvature is covariant, not invariant, so it is its *invariant* polynomials, and not F itself, that carry topological data. Differentiating Eq. (D.35) gives the *Bianchi identity*

$$\nabla F \;=\; \mathrm{d}F + [A, F] \;=\; 0, \tag{D.36}$$

the geometric content of "$\mathrm{d}F = 0$" for an abelian field and of $\nabla_{[\mu}F_{\nu\rho]} = 0$ in general.

The tangent bundle. For $E = TM$ with the metric-compatible, torsion-free (Levi-Civita) connection, Eq. (D.33) is $\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu{}_{\mu\rho}V^\rho$ with the Christoffel symbols of Appendix A. In an orthonormal frame $e^a = e^a{}_\mu \mathrm{d}x^\mu$ the connection form is the *spin connection* $\omega^a{}_b$, an $\mathrm{SO}(D-1,1)$-valued one-form fixed by $\mathrm{d}e^a + \omega^a{}_b \wedge e^b = 0$, and the curvature two-form is

$$\mathcal{R}^a{}_b \;=\; \mathrm{d}\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b \;=\; \tfrac{1}{2}\, R^a{}_{b\mu\nu}\, \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu, \tag{D.37}$$

with $R^a{}_{b\mu\nu}$ the Riemann tensor of Appendix A in frame indices. Gauge theory and gravity are the same construction with structure group G and $\mathrm{SO}(D-1,1)$ (or Spin) respectively; this is the bridge that lets the same $\hat{A}$ genus appear for a gauge bundle and for the tangent bundle.

Remark D.9 (Beyond the lectures: variation of the curvature). This identity is needed only for the effective-action manipulations of Chapter 5, and is collected here as the one curvature-variation fact the notes rely on. Varying an Einstein-Hilbert-type action needs the response of the curvature to a metric variation. The Palatini identity states that the variation of the Ricci tensor is a total covariant derivative of the connection variation,

$$\delta R_{\mu\nu} \;=\; \nabla_\rho\, \delta\Gamma^\rho{}_{\mu\nu} - \nabla_\nu\, \delta\Gamma^\rho{}_{\rho\mu}, \tag{D.38}$$

so that

$$\delta(\sqrt{-g}\, R) = \sqrt{-g}\left(R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R\right)\delta g^{\mu\nu} + \sqrt{-g}\,\nabla_\mu v^\mu, \qquad v^\mu = g^{\alpha\beta}\delta\Gamma^\mu{}_{\alpha\beta} - g^{\mu\beta}\delta\Gamma^\alpha{}_{\alpha\beta}. \tag{D.39}$$

On a closed manifold the last term drops. When R is multiplied by a non-constant factor $f(\phi)$, as in the string-frame action of Chapter 5, that term is kept and integrated by parts against $\mathrm{d}f$, producing the $\nabla_\mu\nabla_\nu f - g_{\mu\nu}\Box f$ contributions to the graviton equation.

D.2.3 Holonomy

Parallel transport of a section around a closed loop γ returns it rotated by an element of G: the *holonomy* $\mathrm{Hol}_\gamma(\nabla) \in G$. Ranging over all loops based at a point p gives a subgroup $\mathrm{Hol}_p(\nabla) \subseteq G$, the *holonomy group*; restricting to contractible loops gives the *restricted holonomy group* Hol_p^0, a connected Lie subgroup independent of the base point up to conjugation. Holonomy is curvature integrated: for an infinitesimal loop spanning $\mathrm{d}x^\mu \wedge \mathrm{d}x^\nu$ the transport is $1 - \frac{1}{2}F_{\mu\nu}\,\mathrm{d}x^\mu \wedge \mathrm{d}x^\nu + \cdots$, and the global statement is the Ambrose-Singer theorem.

Theorem D.10 (Ambrose-Singer). *The Lie algebra of the restricted holonomy group* $\mathrm{Hol}_p^0(\nabla)$ *is spanned by the curvature endomorphisms* $F_{\mu\nu}(q)$ *at all points* q *reachable from* p*, parallel-transported back to* p*. In particular a flat connection* ($F \equiv 0$) *has discrete holonomy, determined by* $\pi_1(M)$*.*

For the tangent bundle of a Riemannian manifold, a reduction of $\mathrm{Hol}(\nabla)$ below the generic $\mathrm{SO}(D)$ signals extra parallel tensors and, when a covariantly constant spinor survives, preserved supersymmetry. Berger's classification of the possible reduced holonomies, the chain "reduced holonomy $\Leftrightarrow$ parallel spinor $\Leftrightarrow$ preserved SUSY", and the entries $\mathrm{SU}(n)$, G_2, $\mathrm{Spin}(7)$ relevant to compactification are the subject of Appendix E and Chapter 8; the connection-and-curvature machinery those arguments rest on is exactly this subsection.

D.2.4 Chern-Weil theory and the Chern classes

Feed the curvature into a polynomial invariant under $F \mapsto gFg^{-1}$ and the result is closed, with a de Rham class independent of the connection. This is the Chern-Weil construction, the source of every characteristic class below.

Theorem D.11 (Chern-Weil)**.** *Let P be an* Ad*-invariant symmetric polynomial on $\mathfrak{g}$ of degree k. For any connection with curvature F, the $2k$-form $P(F)$ is closed, $\mathrm{d}\,P(F) = 0$, and its cohomology class $[P(F)] \in H^{2k}(M, \mathbb{R})$ is independent of the connection. The map $P \mapsto [P(F)]$ is a ring homomorphism from invariant polynomials to de Rham cohomology.*

Closure follows from the Bianchi identity Eq. (D.36) and invariance; connection-independence follows from the transgression formula. If F_0, F_1 are the curvatures of two connections A_0, A_1 and $A_t = A_0 + t(A_1 - A_0)$, then

$$P(F_1) - P(F_0) = \mathrm{d}\,\mathcal{T}P, \qquad \mathcal{T}P = k \int_0^1 \mathrm{d}t\; P(A_1 - A_0,\, F_t, \ldots, F_t), \tag{D.40}$$

an exact form, so the class does not move. Here $P(\cdot, \ldots, \cdot)$ denotes the symmetric multilinear form (the polarisation) of the degree-k invariant polynomial P, evaluated on k arguments. The transgression form $\mathcal{T}P$ is the Chern-Simons form when $P = \operatorname{Tr} F^k$; it is the object the anomaly descent of Appendix H manipulates, and the local counterterm that trades a bulk characteristic class for a boundary coupling.

For a complex vector bundle the invariant polynomials are generated by the coefficients of the characteristic polynomial of F. The determinant below packages all of them into one object: its lowest piece is $c_1 = [\operatorname{Tr}(iF/2\pi)]$, the overall flux already met for line bundles and the twist of the determinant line bundle $\det E$ of Table D.1, while its top piece c_r is the Euler class, which counts the zeros of a generic section. Define the *total Chern class* and the *Chern character* by

$$c(E) = \det\!\left(1 + \frac{iF}{2\pi}\right) = \sum_{k\geq 0} c_k(E), \qquad \mathrm{ch}(E) = \operatorname{Tr}\exp\!\left(\frac{iF}{2\pi}\right) = \sum_{k\geq 0} \mathrm{ch}_k(E), \tag{D.41}$$

with c_k a $2k$-form and $\mathrm{ch}_k = \frac{1}{k!}\operatorname{Tr}(iF/2\pi)^k$. The factor $i/2\pi$ is chosen so that the classes are *real* and *integral*: $c_k(E) \in H^{2k}(M, \mathbb{Z})$ after passing to cohomology (integrality follows from comparison with integral Čech cohomology; see Bott and Tu, or Nakahara ch. 11). Splitting F into formal eigenvalue two-forms, the *Chern roots* $x_1, \ldots, x_r$, turns every class into a symmetric function,

$$c(E) = \prod_{j=1}^{r} (1 + x_j), \qquad \mathrm{ch}(E) = \sum_{j=1}^{r} e^{x_j}, \tag{D.42}$$

so $c_1 = \sum_j x_j$, $c_2 = \sum_{i<j} x_i x_j$, and so on. The low-order expansions collected in Table D.2 are just Eq. (D.42) multiplied out.

Factors of i and 2π

Conventions differ by where the i, the 2π, and the sign live. We fix once: $c(E) = \det(1 + iF/2\pi)$, so that for a unitary (anti-Hermitian F) connection $iF/2\pi$ is a real, integrally quantised two-form. In terms of the ordinary field strength of an abelian gauge field, $c_1 = [F/2\pi]$ with $\int_\Sigma F/2\pi \in \mathbb{Z}$ over any closed surface. In that abelian formula the symbol F is the ordinary (Hermitian, real) field strength, the i having been absorbed; the two uses of F, anti-Hermitian bundle curvature in Eq. (D.41) and real physical field strength here, differ by exactly this factor of i. A source that writes F Hermitian throughout, or absorbs the 2π into F, will differ from us by exactly these factors; convert before comparing.

The Chern character is a *ring homomorphism*,

$$\mathrm{ch}(E \oplus E') = \mathrm{ch}(E) + \mathrm{ch}(E'), \qquad \mathrm{ch}(E \otimes E') = \mathrm{ch}(E)\,\mathrm{ch}(E'), \tag{D.43}$$

which is why it, rather than the total Chern class, is the natural object for K-theory and for the brane charge of Appendix D.2.6. The total Chern class instead multiplies on Whitney sums,

$c(E \oplus E') = c(E)\,c(E')$ (the *Whitney sum formula*), and the *splitting principle* guarantees that any identity provable for a sum of line bundles holds in general, which is what licenses the Chern-root manipulations.

Example D.12 (First Chern class and line bundles)**.** For a line bundle L the only class is $c_1(L) = [F/2\pi] \in H^2(M,\mathbb{Z})$, and it is a complete invariant: two line bundles are isomorphic iff they share c_1, so the Picard group is $H^2(M,\mathbb{Z})$ (as in Theorem D.7, where it is $\mathbb{Z}$). Integrality of c_1 is Dirac's quantisation condition: the flux of a monopole, or of a Ramond-Ramond field strength, through a closed two-cycle is an integer multiple of 2π. This turns the smooth field strength F of Appendix A into a quantised charge, and it is the entry point to Appendix D.2.6.

D.2.5 Euler and Pontryagin classes; the Hirzebruch genera

Real bundles carry their own invariants. Complexifying E (tensoring with $\mathbb{C}$) and taking Chern classes gives the *Pontryagin classes*

$$p_k(E) = (-1)^k c_{2k}(E \otimes \mathbb{C}) \in H^{4k}(M,\mathbb{Z}), \tag{D.44}$$

the odd Chern classes of $E\otimes\mathbb{C}$ (which satisfy $2c_{2k+1}(E\otimes\mathbb{C}) = 0$ and so vanish in real cohomology) being dropped. In curvature terms the first is

$$p_1(E) = -\frac{1}{8\pi^2}\,\mathrm{Tr}(F \wedge F), \tag{D.45}$$

with the sign chosen to match the $\hat{A}$ expansion below; for the tangent bundle $F = \mathcal{R}$ of Eq. (D.37), so $p_1(TM)$ is built from $\mathrm{Tr}(\mathcal{R}\wedge\mathcal{R})$, the object that appears in the Green-Schwarz mechanism and the brane curvature couplings.

For an oriented real bundle of *even* rank $2n$ there is a finer invariant than p_n, the *Euler class*, the Pfaffian of the curvature,

$$e(E) = \frac{1}{(2\pi)^n}\,\mathrm{Pf}(F), \qquad e(E)^2 = p_n(E). \tag{D.46}$$

For the tangent bundle of a closed oriented manifold its integral is the Euler characteristic, the Gauss-Bonnet theorem,

$$\int_M e(TM) = \chi(M) = \sum_p (-1)^p b_p, \tag{D.47}$$

tying the curvature integral to the Betti numbers of Appendix D.1. The Euler class is multiplicative on Whitney sums, $e(E\oplus E') = e(E)\,e(E')$, and vanishes for a bundle with a nowhere-zero section (its zeros are what it counts).

Genera. A *multiplicative genus* is a characteristic class of the form $\prod_j Q(x_j)$ for a power series Q with $Q(0) = 1$, built on the Chern roots x_j (for complex bundles) or on the Pontryagin roots y_j defined by $p(E) = \prod_j (1 + y_j^2)$ (for real bundles). Multiplicativity $Q(E\oplus E') = Q(E)\,Q(E')$ is automatic. Three choices of Q generate everything used in these notes, listed with their low-order expansions in Table D.2:

$$\mathrm{Td}(E) = \prod_j \frac{x_j}{1 - e^{-x_j}} \qquad \text{(Todd, complex bundles)}, \tag{D.48}$$

$$\hat{A}(E) = \prod_j \frac{y_j/2}{\sinh(y_j/2)} \qquad (\hat{A}\text{ genus, real bundles}), \tag{D.49}$$

$$L(E) = \prod_j \frac{y_j}{\tanh y_j} \qquad \text{(Hirzebruch } L\text{, real bundles)}. \tag{D.50}$$

Class / genus	Expansion in the underlying curvature classes
$c(E)$	$1 + c_1 + c_2 + c_3 + \cdots$
$\mathrm{ch}(E)$	$r + c_1 + \frac{1}{2}(c_1^2 - 2c_2) + \frac{1}{6}(c_1^3 - 3c_1c_2 + 3c_3) + \cdots$
$\mathrm{Td}(E)$	$1 + \frac{1}{2}c_1 + \frac{1}{12}(c_1^2 + c_2) + \frac{1}{24}c_1c_2 + \cdots$
$\hat{A}(E)$	$1 - \frac{1}{24}p_1 + \frac{1}{5760}(7p_1^2 - 4p_2) + \cdots$
$\sqrt{\hat{A}(E)}$	$1 - \frac{1}{48}p_1 + \cdots$
$L(E)$	$1 + \frac{1}{3}p_1 + \frac{1}{45}(7p_2 - p_1^2) + \cdots$
$e(E)$	$\mathrm{Pf}(F/2\pi), \quad \int_M e(TM) = \chi(M)$

Table D.2: Low-order expansions of the characteristic classes and genera, with $r = \operatorname{rank} E$. The c_k are $2k$-forms and the p_k are $4k$-forms; each row is the corresponding power series of Eqs. (D.41) and (D.48) to (D.50) multiplied out on the Chern (or Pontryagin) roots. Read the degree-$2n$ piece off the n-th column when integrating over a $2n$-cycle.

The Todd class is the multiplier in Hirzebruch-Riemann-Roch, the $\hat{A}$ genus is the density in the Dirac index, and the L-genus is the density in the signature theorem; all three are stated as index theorems in Appendix D.3. The reason $\hat{A}$ rather than any other series controls brane charge is that it is the genus of the Dirac operator, and D-branes are sources built by a Dirac-type index (anomaly inflow), a point made precise in Appendix D.3 and Appendix H.

Exercise D.3. Using $\mathrm{ch} = \sum_j e^{x_j}$ and $c_1 = \sum_j x_j$, $c_2 = \sum_{i<j} x_i x_j$, verify $\mathrm{ch}_2 = \frac{1}{2}(c_1^2 - 2c_2)$. Then, for a rank-2 bundle, confirm the multiplicative property Eq. (D.43) at degree two by expanding $\mathrm{ch}(E \otimes E')$ and comparing with $\mathrm{ch}(E)\,\mathrm{ch}(E')$. (The splitting principle lets you treat each factor as a sum of line bundles.)

D.2.6 Integrality, charge quantisation, and the brane Wess-Zumino coupling

The single physical fact that makes this section load-bearing is *integrality*: the periods of c_k, p_k, e over integer cycles are integers (Bott and Tu; Nakahara ch. 11). Charge quantisation is this statement applied to a field strength. A U(1) flux obeys $\int_\Sigma F/2\pi \in \mathbb{Z}$ (Theorem D.12); a Ramond-Ramond field strength $F_{p+2} = \mathrm{d}C_{p+1}$ of Appendix A obeys the analogous condition, so D-brane charge is quantised without any dynamical input. The refinement that Ramond-Ramond charge is classified by K-theory rather than plain cohomology, which is why the Chern character Eq. (D.43) is the right variable, is flagged in Theorem D.13.

The place these classes enter the string dynamics is the Wess-Zumino coupling of a Dp-brane, written in Chapter 6 as

$$S_{\mathrm{WZ}} = \mu_p^{(\mathrm{WZ})} \int_{\mathcal{W}} \Big(\bigoplus_q C_q \Big) \wedge e^{\,B + 2\pi\alpha' F} \wedge \sqrt{\frac{\hat{A}(\mathcal{R}_N)}{\hat{A}(\mathcal{R}_T)}}\,, \tag{D.51}$$

the completion of Eq. (6.38). Every factor is now defined. The polyform $C = \bigoplus_q C_q$ is pulled back to the worldvolume $\mathcal{W}$; the integral keeps the $(p+1)$-form part. The exponent contains the NS-NS two-form B and the Regge slope α' of Appendix A, and for a non-abelian worldvolume bundle the exponential is read as the trace $\mathrm{Tr}\, e^{\,B+2\pi\alpha' F}$, that is, the Chern character ch of Eq. (D.41) with the $i/2\pi$ absorbed into the brane-literature (Hermitian) normalisation of F. Expanding it reproduces the $C_{p-1} \wedge (B + 2\pi\alpha' F)$ and $\frac{1}{2}C_{p-3} \wedge (B + 2\pi\alpha' F)^2$ terms of Eq. (6.41). Each such term is a lower Ramond-Ramond potential, so a Dp-brane with $k = \int F/2\pi$ units of worldvolume flux on a two-cycle carries k units of D$(p-2)$ charge: the Chern character is the induced-charge generating function.

The last factor is the geometric analogue. It is the ratio of $\hat{A}$ genera of the normal bundle $\mathcal{R}_N$ and the tangent bundle $\mathcal{R}_T$ of the brane (conventions that order the ratio the other way differ by the overall sign of the induced curvature charge, exactly the kind of factor to convert before comparing sources), and by Eq. (D.49) it is a sum of $4k$-forms starting at 1,

$$\sqrt{\frac{\hat{A}(\mathcal{R}_N)}{\hat{A}(\mathcal{R}_T)}} = 1 - \frac{1}{48}(p_1(N) - p_1(T)) + \cdots , \tag{D.52}$$

so a brane wrapping a curved cycle picks up induced lower-brane charges set by the Pontryagin numbers of that cycle, with no worldvolume flux needed. For a four-cycle W the leading term is controlled by the adjunction relation $p_1(N) - p_1(T) = 2\,c_2(TW)$. The tangent Pontryagin number alone, $\int_W p_1(T) = 3\,\sigma(W)$, gives the signature, not the Euler number; $\chi(W)$ appears only once the normal-bundle piece is included, through $\int_W c_2(TW) = \chi(W)$. The overall sign is the one fixed by the anomaly-inflow matching of Appendix H, and carried out with all conventions the induced charge equals $-\chi(W)/24$ (Bershadsky, Sadov and Vafa 1996; Minasian and Moore 1997). A D-brane wrapping a K3 surface therefore carries

$$-\frac{\chi(\mathrm{K3})}{24} = -1 \tag{D.53}$$

unit of the charge of the D-brane four dimensions lower. That single unit, summed over the geometry, is the origin of the $24 = \chi(\mathrm{K3})$ units of lower-brane charge (equivalently gauge instanton number) that a consistent K3 compactification must supply. This is the honest version of "induced charge". It is not a correction to be waved away; the tadpole conditions of Chapter 9 sum over exactly these terms.

Tadpoles are curvature bookkeeping

On a compact internal space the total Ramond-Ramond charge must vanish: the flux sourced by a C_{p+1} has nowhere to escape, so its integrated source, branes plus orientifold planes plus these induced curvature charges plus background flux, is forced to zero. A miscounted $\sqrt{\hat{A}}$ term Eq. (D.52) is a miscounted tadpole. The explicit conditions are assembled in Chapters 9 to 11 and the brane and orientifold charge tables in Appendix I.

That the power series in Eq. (D.51) is exactly $\sqrt{\hat{A}}$, and not some other genus, is not put in by hand. It is fixed by demanding that the worldvolume anomaly of the chiral fermions on intersecting or wrapped branes be cancelled by inflow from the bulk Ramond-Ramond fields. The index density that controls this is developed in Appendix D.3; the anomaly-inflow derivation that pins the factor to $\sqrt{\hat{A}}$, together with the anomaly-polynomial and descent technology it uses, is carried out in Appendix H, where the same $\hat{A}$ appears as the density of the Dirac index.

Remark D.13 (Beyond the lectures: K-theory and the Chern character). Ramond-Ramond charge is more finely classified than by cohomology. A brane and antibrane of the same type can annihilate, so what is conserved is a bundle up to the addition and subtraction of trivial bundles, which is exactly the K-theory group $K(M)$ (Minasian and Moore 1997; Witten 1998). The Chern character is the natural map $\mathrm{ch} : K(M) \to H^{\mathrm{even}}(M, \mathbb{Q})$, an isomorphism after tensoring with $\mathbb{Q}$, which is why Eq. (D.51) is written with ch: it converts the K-theoretic brane charge into the cohomological charge that couples to the Ramond-Ramond forms. The $\sqrt{\hat{A}}$ factor is the correction that makes the map an *isometry* for the natural pairings, so integrality of the charge is preserved. Torsion classes in $K(M)$, invisible to ch, are the charges of the unstable and torsion branes that pure cohomology misses.

D.3 Index theorems

An elliptic differential operator on a compact manifold has a finite-dimensional kernel, and the integer that measures the imbalance between its zero modes and those of its adjoint does not move when the operator is deformed smoothly. That rigid integer is an integral of the characteristic classes of Appendix D.2. This is the mechanism that turns a physics question, "how many chiral fermions does this background produce", into a topological count that no continuous choice of metric, connection or complex structure can spoil. Every generation-counting statement in Chapters 8 and 11, and the anomaly bookkeeping of Appendix H, is one specialisation of the same theorem.

One idea, three uses

The Dirac operator on a compact space K coupled to a background gauge bundle V has an index

$$\operatorname{ind} D_V \;=\; \dim\ker D_V^{+} - \dim\ker D_V^{-} \;=\; \int_K \hat{A}(TK)\,\mathrm{ch}(V),$$

a signed count of chiral zero modes. Read on the internal space of a compactification it gives the *net number of chiral families*. Read as the integrated divergence of an axial current it gives the *anomaly*. Read on a brane worldvolume it gives the *induced lower brane charge* (Appendix D.2).

D.3.1 Elliptic operators and the analytic index

Let E, F be vector bundles over a compact oriented manifold M (no boundary) and $D : \Gamma(E) \to \Gamma(F)$ a linear differential operator. Its *symbol* $\sigma(D)$ replaces each derivative ∂_μ by a cotangent vector ik_μ and keeps the top-derivative part; D is *elliptic* when $\sigma(D)(k)$ is invertible for every $k \neq 0$. Ellipticity makes D Fredholm: $\ker D$ and $\operatorname{coker} D \cong \ker D^\dagger$ are finite dimensional. The *analytic index* is

$$\operatorname{ind} D \;:=\; \dim\ker D - \dim\ker D^\dagger. \tag{D.54}$$

The point is stability. Deforming D through elliptic operators, or the metric and connections it is built from, can create or destroy zero modes only in $\pm$ pairs that cross between $\ker D$ and $\ker D^\dagger$, so (D.54) is a deformation invariant. It depends on D only through the homotopy class of its symbol, that is, through topology.

D.3.2 The Atiyah-Singer theorem and its four classical cases

Theorem D.14 (Atiyah and Singer 1968)**.** *For an elliptic operator D on a compact manifold M of dimension n without boundary, the analytic index* (D.54) *equals the* topological index*, a characteristic number built from the symbol,*

$$\operatorname{ind} D \;=\; (-1)^n \int_M \mathrm{ch}\big(\sigma(D)\big)\,\frac{\mathrm{Td}(TM\otimes\mathbb{C})}{e(TM)}. \tag{D.55}$$

The right-hand side uses the Chern character, Todd class and Euler class of Appendix D.2, with $\mathrm{ch}(\sigma(D))$ the Chern character of the symbol class, a compactly supported class on the cotangent bundle T^*M; the orientation sign is fixed case by case and we never apply (D.55) in this raw form. Its value is that four choices of elliptic complex reproduce the four classical characteristic numbers at once. These are the cases we actually use, collected in Table D.3. Each row is a theorem in its own right, older than the general statement that unifies them.

Elliptic complex	Analytic index	$= \int_M$ of	Name
de Rham, $\mathrm{d}+\mathrm{d}^\dagger$	$\chi(M)=\sum_p(-1)^p b_p$	$e(TM)$	Gauss-Bonnet-Chern
signature, self-dual part	$\sigma(M)=b_2^+-b_2^-$	$L(TM)$	Hirzebruch signature
Dolbeault, $\bar\partial+\bar\partial^\dagger$	$\chi(X,\mathcal{O})=\sum_q(-1)^q h^{0,q}$	$\mathrm{Td}(TX)$	Riemann-Roch
Dirac, D	n_+-n_-	$\hat A(TM)$	spin index

Table D.3: The four classical specialisations of Theorem D.14. The first is the topological Euler characteristic; the third holds on a compact complex manifold X and the fourth on a spin manifold. Betti numbers b_p and Hodge numbers $h^{p,q}$ are defined in Appendix D.1; the self-dual and anti-self-dual harmonic two-form counts $b_2^\pm$, whose difference is the signature $\sigma=b_2^+-b_2^-$, are defined there too. L is the Hirzebruch L-genus and $\hat A$ the A-roof genus of Appendix D.2.

D.3.3 The Dirac index

Take M spin of even dimension $n=2k$, with the chirality operator Γ_* of Appendix A splitting the spinor bundle $S=S^+\oplus S^-$. The Dirac operator $D=\not\nabla$ anticommutes with Γ_*, so it maps $D^\pm:\Gamma(S^\pm)\to\Gamma(S^\mp)$, and $\ker D^\pm$ are the positive and negative chirality zero modes. Coupling to a gauge bundle V with connection (a fermion in the representation carried by V) gives the twisted operator D_V, and the index is

$$\operatorname{ind} D_V \;=\; \dim\ker D_V^+ - \dim\ker D_V^- \;=\; \int_M \hat A(TM)\,\mathrm{ch}(V). \tag{D.56}$$

This is the master formula for chiral zero-mode counting. To low order the two factors are (see Appendix D.2)

$$\hat A(TM) = 1 - \frac{p_1}{24} + \frac{7p_1^2-4p_2}{5760} + \cdots, \tag{D.57}$$

$$\mathrm{ch}(V) = \operatorname{rank} V + c_1 + \tfrac12(c_1^2-2c_2) + \tfrac16(c_1^3-3c_1c_2+3c_3) + \cdots, \tag{D.58}$$

with $p_i=p_i(TM)$ the Pontryagin classes and $c_i=c_i(V)$ the Chern classes. On a four-manifold only the degree-four part survives,

$$\operatorname{ind} D_V\big|_{n=4} = \int_M \Big[\mathrm{ch}_2(V) - \operatorname{rank} V\,\frac{p_1(TM)}{24}\Big] = \int_M \Big[\tfrac12(c_1^2-2c_2) - \operatorname{rank} V\,\frac{p_1}{24}\Big]. \tag{D.59}$$

Because the left side is an integer, (D.56) is also a source of integrality constraints on characteristic numbers. With V trivial it says $\int_M \hat A\in\mathbb{Z}$ on a spin manifold; in four dimensions $\int_M\hat A = -\sigma(M)/8$, so the signature of a spin four-manifold is divisible by 8, and in fact by 16 (Rokhlin's theorem, a refinement the index alone does not see).

Exercise D.4 (The Dirac index on K3)**.** K3 is the compact spin four-manifold of Chapter 8 with Betti numbers $b_0=b_4=1$, $b_1=b_3=0$, $b_2=22$ split as $b_2^+=3$, $b_2^-=19$. Use the Hirzebruch signature theorem $\sigma=\frac13\int p_1$ to show $\int_{\mathrm{K3}}p_1=-48$, and hence that the untwisted Dirac index is $\int_{\mathrm{K3}}\hat A=-\frac{1}{24}\int p_1=2$. K3 is Calabi-Yau, so $c_1(T)=0$ and the Dolbeault row of Table D.3 gives the same number as the arithmetic genus $\chi(\mathcal{O})=h^{0,0}-h^{0,1}+h^{0,2}=1-0+1=2$. The two theorems agree because on a Kähler manifold they are the same operator, as Appendix D.3.4 explains.

D.3.4 Riemann-Roch and the Dolbeault index

On a compact complex manifold X of complex dimension n the natural elliptic complex is Dolbeault, $\bar\partial:\Omega^{0,q}\to\Omega^{0,q+1}$, twisted by a holomorphic vector bundle E. Its index is the

holomorphic Euler characteristic,

$$\chi(X,E) \;=\; \sum_{q=0}^{n} (-1)^q \dim H^q(X,E), \tag{D.60}$$

where $H^q(X,E)$ is Dolbeault cohomology valued in E, the group $H^{0,q}_{\bar\partial}(E)$ of Appendix D.1, and the structure sheaf $\mathcal{O}$ (the case $E=\mathcal{O}$) is the sheaf of holomorphic functions, so $\chi(X,\mathcal{O}) = \sum_q (-1)^q h^{0,q}$. The index theorem is the Hirzebruch-Riemann-Roch formula.

Theorem D.15 (Hirzebruch-Riemann-Roch)**.** *For a holomorphic bundle E over a compact complex manifold X,*

$$\chi(X,E) \;=\; \int_X \mathrm{ch}(E)\,\mathrm{Td}(TX), \qquad \mathrm{Td}(TX) = 1 + \frac{c_1}{2} + \frac{c_1^2+c_2}{12} + \frac{c_1 c_2}{24} + \cdots, \tag{D.61}$$

with $c_i = c_i(TX)$ the Chern classes of the holomorphic tangent bundle.

On a curve (complex dimension one, real genus g) the classical Riemann-Roch theorem is the degree-one part: for a line bundle L,

$$h^0(\Sigma,L) - h^1(\Sigma,L) = \int_\Sigma \left(c_1(L) + \tfrac{1}{2}c_1(T\Sigma)\right) = \deg L + 1 - g, \tag{D.62}$$

using $\int_\Sigma c_1(T\Sigma) = \chi(\Sigma) = 2-2g$. This is the index that sets the chirality on the F-theory matter curves of Section 11.5.2.

The Dolbeault and Dirac rows of Table D.3 are not independent. On a Kähler manifold the operator $\bar\partial + \bar\partial^\dagger$ is the Dirac operator for the canonical spin^c structure (a spin structure twisted by a U(1) line bundle, which exists whenever $w_2(TX)$ is the mod-two reduction of an integral class; Appendix B), and the two characteristic classes are related by

$$\mathrm{Td}(TX) \;=\; e^{\,c_1(TX)/2}\,\hat{A}(TX) \;=\; \mathrm{ch}\big(K_X^{-1/2}\big)\,\hat{A}(TX), \tag{D.63}$$

where $K_X = \det T^*X$ is the canonical bundle, $c_1(K_X) = -c_1(TX)$. Feeding (D.63) into (D.61) rewrites the Dolbeault index of E as a genuine Dirac index twisted by $E\otimes K_X^{-1/2}$:

$$\chi(X,E) \;=\; \int_X \mathrm{ch}\big(E\otimes K_X^{-1/2}\big)\,\hat{A}(TX) \;=\; \mathrm{ind}\, D_{E\otimes K_X^{-1/2}}. \tag{D.64}$$

The half-canonical twist $K_X^{1/2}$ is the reason zero modes on a brane cycle carry the Freed-Witten factor: a D-brane wrapping a cycle whose canonical bundle has no square root is anomalous, and its worldvolume fermions couple to $K^{1/2}$ (Chapters 9 and 11). Concretely, a fermion in a bundle L on a curve Σ has honest Dirac zero modes counted by $\chi(\Sigma, L\otimes K_\Sigma^{1/2})$, whose H^0 and H^1 are the modes and their conjugates. By (D.62) and $\deg K_\Sigma^{1/2} = g-1$,

$$\mathrm{ind}\, D_L\big|_\Sigma = \chi(\Sigma,\, L\otimes K_\Sigma^{1/2}) = \deg L + (g-1) + 1 - g = \deg L, \tag{D.65}$$

exactly the net chirality $n_R - n_{\bar R} = \int_{\Sigma_R} c_1(L_R)$ quoted at (11.49): the genus dependence cancels and only the degree survives.

D.3.5 Chiral zero modes and fermion generations

Compactify a higher-dimensional chiral fermion on a compact internal space K with a background gauge bundle V. The Dirac operator factorises into a four-dimensional piece and an internal piece, and a massless four-dimensional mode requires an internal zero mode. A four-dimensional

fermion is chiral precisely when the internal zero mode has definite internal chirality, so the *net* number of four-dimensional left-handed minus right-handed fermions is the internal Dirac index

$$n_L - n_R \;=\; \operatorname{ind} D_V\big|_K \;=\; \int_K \hat{A}(TK)\,\mathrm{ch}(V). \tag{D.66}$$

The separate numbers n_L and n_R depend on the detailed spectrum and are not protected; their difference is topological. This is what the phrase "generations come from geometry" amounts to.

Example D.16 (Heterotic generations on a Calabi-Yau threefold)**.** Take the internal space to be a Calabi-Yau threefold X (real dimension six, $c_1(TX) = 0$) and a gauge bundle V with $c_1(V) = 0$, as in Chapter 8. In the degree-six part of $\hat{A}(TX)\,\mathrm{ch}(V)$ the term $-\frac{p_1}{24}c_1(V)$ dies because $c_1(V) = 0$, and $\hat{A}$ has no degree-two piece, so only $\mathrm{ch}_3(V)$ survives:

$$\operatorname{ind} D_V = \int_X \mathrm{ch}_3(V) = \int_X \tfrac{1}{6}(c_1^3 - 3c_1c_2 + 3c_3)(V) = \tfrac{1}{2}\int_X c_3(V). \tag{D.67}$$

For the standard embedding $V = TX$ the top Chern class integrates to the Euler characteristic, $\int_X c_3(TX) = \chi(X)$, so the net number of chiral families is $\mathrm{ind} = \frac{1}{2}\chi(X)$, and $n_{\mathrm{gen}} = \frac{1}{2}|\chi|$. This is (8.90), now derived. The same number is the holomorphic Euler characteristic $\chi(X, TX) = h^{1,1} - h^{2,1} = \frac{1}{2}\chi$ by Theorem D.15, the Dolbeault route to the count of **27** minus $\overline{\mathbf{27}}$ matter multiplets.

D.3.6 The link to anomalies

The chiral anomaly is the index theorem read as a statement about a current. Under a constant axial rotation $\psi \to e^{i\alpha\Gamma_*}\psi$ the fermion path-integral measure is not invariant; its Jacobian is the integrated index density (Fujikawa 1979), and the axial current divergence in a four-dimensional gauge background is

$$\partial_\mu j_5^\mu = \frac{1}{16\pi^2}\,\epsilon^{\mu\nu\rho\sigma}\,\mathrm{Tr}(F_{\mu\nu}F_{\rho\sigma}) = 2\,\big[\hat{A}(R)\,\mathrm{ch}(F)\big]_4 \text{ (density)}, \tag{D.68}$$

whose integral is $2(n_+ - n_-)$. In (D.68), and in the normalisation $c_1 = [F/2\pi]$ of Appendix D.2, the symbol F is the Hermitian (real) field strength, with the factor of i of the Chern-character convention absorbed; kept anti-Hermitian, $2[\hat{A}(R)\,\mathrm{ch}(F)]_4$ carries the opposite sign, and either bookkeeping returns the same integer $2(n_+ - n_-)$. The number of fermion zero modes in an instanton background is therefore fixed by the instanton number, $n_+ - n_- = \int_M \mathrm{ch}_2(F)$, the four-dimensional case of (D.59) with a trivial gravitational part. This is the counting behind the 't Hooft vertex and every anomaly-driven selection rule.

Gauge and gravitational anomalies of a chiral theory in $D = 2n$ dimensions are packaged the same way. The anomaly is not itself gauge invariant, but its variation is governed by a gauge-invariant $(2n+2)$-form, the *anomaly polynomial*, and that form is the index density of a Dirac operator two dimensions up. For a positive-chirality spin-$\frac{1}{2}$ fermion (Alvarez-Gaumé and Witten 1984),

$$\hat{I}_{2n+2} \;=\; \big[\hat{A}(R)\,\mathrm{ch}(F)\big]_{2n+2}, \tag{D.69}$$

with modified densities for the spin-$\frac{3}{2}$ gravitino and the self-dual tensor (the $\hat{A}\,(\mathrm{Tr}\,e^{iR/2\pi} - 1)$ and $-\frac{1}{8}L$ contributions respectively, the -1 removing the spurious spin-$\frac{1}{2}$ piece). The physical anomaly is recovered from (D.69) by the Stora-Zumino descent,

$$\hat{I}_{2n+2} = \mathrm{d}\,\omega_{2n+1}, \qquad \delta\,\omega_{2n+1} = \mathrm{d}\,\omega^1_{2n}, \tag{D.70}$$

where ω^1_{2n} is the anomalous variation of the effective action. The Green-Schwarz cancellation of Appendix H is the statement that in ten dimensions $\hat{I}_{12}$ factorises so that a $B \wedge X_8$ counterterm

removes it. The same $\hat{A}$ and ch densities appear in the D-brane Wess-Zumino coupling of Chapter 6; the induced brane charges computed from that coupling in Appendix D.2 and the worldvolume anomaly they inflow are two readings of the one index density. The inflow that fixes the Wess-Zumino genus to $\sqrt{\hat{A}}$ is an index computation of exactly this kind, and it is carried out in Appendix H.

Remark D.17 (Beyond the lectures: refinements used elsewhere). The following refinements are used elsewhere in the notes. (i) On a manifold *with boundary* the index picks up a boundary correction, the Atiyah-Patodi-Singer η-invariant (Atiyah, Patodi and Singer 1975), a spectral asymmetry that is not an integral of local densities. It controls the parity anomaly in odd dimensions and the phase of the M-theory partition function. (ii) When the Dirac operator is real or quaternionic, an extra $\mathbb{Z}_2$-valued *mod-two* index survives even when the integer index vanishes; it detects the discrete brane charges that make D-brane charge live in K-theory rather than cohomology (Chapter 6), and the global anomalies that the polynomial (D.69) cannot see. (iii) On an orbifold or in the presence of a symmetry, the equivariant (Atiyah-Bott-Lefschetz) fixed-point form of the theorem localises the index onto the fixed loci, which is how twisted-sector spectra and twisted-sector zero-mode counts of Chapter 8 are obtained. The family version (Bismut and Freed 1986), where the index becomes a determinant line bundle over a parameter space, is the geometry underneath (D.70).

Appendix E

Calabi-Yau Geometry, Special Holonomy, and Elliptic Fibrations

E.1 Complex and Kähler geometry

The internal spaces that carry the compactifications of Chapter 8 are complex, and the ones that preserve supersymmetry are Kähler, indeed Calabi-Yau. This section fixes that language: complex manifolds, the almost complex structure, Hermitian and Kähler metrics, the Kähler form, Dolbeault cohomology, and the Hodge decomposition that turns the topology of a Kähler manifold into a diamond of integers. The four Calabi-Yau definitions, Yau's theorem and the holonomy chain build on this and are developed in Appendices E.2 and E.3. De Rham cohomology, connections on bundles and the Chern classes are the subject of Appendix D; we quote them where needed. Standard references at this level are Griffiths and Harris, Huybrechts, and Nakahara; for the physics-facing account, the Trieste lectures of Candelas and volume two of Green, Schwarz and Witten. By the end you can pass between the complex structure, the metric and the Kähler form in components, read the Hodge diamond of a compact Kähler manifold off its two symmetries, and place c_1 and the Ricci form in this language.

One letter, two objects

The symbol J does double duty, following Appendix A. As an endomorphism $J^m{}_n$ of the tangent space it is the complex structure, with $J^2 = -\mathbb{1}$. As a two-form J_{mn} it is the Kähler form. They are the same tensor with an index lowered, $J_{mn} = g_{mp}J^p{}_n$, and the index structure tells you which is meant. In holomorphic-coordinate and index-free equations we write this two-form as J, which is the same letter J. We never use J for anything else.

E.1.1 Complex manifolds and the almost complex structure

A *complex manifold* of complex dimension n is a real $2n$-dimensional manifold with an atlas whose charts map to open sets of $\mathbb{C}^n$ and whose transition functions are holomorphic. On overlaps the coordinates $z^i = x^{2i-1} + \mathrm{i}\, x^{2i}$, $i = 1, \ldots, n$, are related by holomorphic maps, so "holomorphic function" is a well-defined notion globally. We write $\bar{z}^{\bar{\imath}}$ for the conjugates and abbreviate $\partial_i = \partial/\partial z^i$, $\partial_{\bar{\imath}} = \partial/\partial \bar{z}^{\bar{\imath}}$. Throughout, n is the *complex* dimension unless we say otherwise.

The infinitesimal content of a complex structure is a tensor. An *almost complex structure* on a real manifold is a smooth bundle endomorphism $J\colon TM \to TM$ with

$$J^2 = -\mathbb{1}, \qquad J^m{}_p\, J^p{}_n = -\delta^m{}_n. \tag{E.1}$$

Its existence already forces $\dim_{\mathbb{R}} M = 2n$ even. Complexifying the tangent bundle, J has eigenvalues $\pm\mathrm{i}$, and the eigenspaces split

$$T_{\mathbb{C}}M \;=\; T^{1,0}M \oplus T^{0,1}M, \qquad T^{1,0}M = \ker(J-\mathrm{i}), \quad T^{0,1}M = \ker(J+\mathrm{i}), \tag{E.2}$$

into the *holomorphic* and *antiholomorphic* tangent bundles, exchanged by complex conjugation. The projectors are $P^{\pm} = \tfrac{1}{2}(\mathbb{1} \mp \mathrm{i}J)$. On a genuine complex manifold $T^{1,0}M$ is spanned by the ∂_i and

$$J = \mathrm{i}\,\mathrm{d}z^i \otimes \partial_i - \mathrm{i}\,\mathrm{d}\bar{z}^{\bar{\imath}} \otimes \partial_{\bar{\imath}}, \qquad J^i{}_j = \mathrm{i}\,\delta^i{}_j, \quad J^{\bar{\imath}}{}_{\bar{\jmath}} = -\mathrm{i}\,\delta^{\bar{\imath}}{}_{\bar{\jmath}}, \tag{E.3}$$

with the mixed components vanishing.

Not every almost complex structure comes from a complex atlas. It does exactly when the holomorphic distribution $T^{1,0}M$ is closed under the Lie bracket, equivalently when the *Nijenhuis tensor*

$$N(X,Y) = [X,Y] + J[JX,Y] + J[X,JY] - [JX,JY] \tag{E.4}$$

vanishes identically. An almost complex structure with $N \equiv 0$ is called *integrable*.

Theorem E.1 (Newlander-Nirenberg)**.** *An almost complex structure is induced by a complex-manifold atlas if and only if it is integrable, $N \equiv 0$. Vanishing of N is then equivalent to $T^{1,0}M$ being involutive and to $\bar{\partial}^{\,2} = 0$ (Eq. (E.9) in the next subsection, once $\bar{\partial}$ has been defined).*

The theorem is asserted here; proofs are in Newlander and Nirenberg (1957), or in Nakahara and Huybrechts. Vanishing of N is a real condition and not automatic, and we take our manifolds to be complex from the outset, so $N = 0$ throughout.

E.1.2 Forms of type (p,q) and the Dolbeault operators

The splitting (E.2) dualises to the cotangent bundle and propagates to all exterior powers. A complex-valued k-form decomposes into pieces carrying p holomorphic and q antiholomorphic legs, $p+q=k$,

$$\Lambda^k T^*_{\mathbb{C}}M \;=\; \bigoplus_{p+q=k} \Lambda^{p,q}M, \tag{E.5}$$

and a form in $\Lambda^{p,q}M$, a *(p,q)-form*, is written in the $1/p!$ convention of Appendix A as

$$\alpha = \frac{1}{p!\,q!}\,\alpha_{i_1\cdots i_p\,\bar{\jmath}_1\cdots\bar{\jmath}_q}\,\mathrm{d}z^{i_1}\wedge\cdots\wedge\mathrm{d}z^{i_p}\wedge\mathrm{d}\bar{z}^{\bar{\jmath}_1}\wedge\cdots\wedge\mathrm{d}\bar{z}^{\bar{\jmath}_q}. \tag{E.6}$$

Applying the exterior derivative and sorting by type, d splits into two pieces,

$$\mathrm{d} = \partial + \bar{\partial}, \qquad \partial\colon \Lambda^{p,q} \to \Lambda^{p+1,q}, \qquad \bar{\partial}\colon \Lambda^{p,q} \to \Lambda^{p,q+1}, \tag{E.7}$$

the *Dolbeault operators*. On components ∂ differentiates and antisymmetrises the holomorphic index, $\bar{\partial}$ the antiholomorphic one,

$$(\bar{\partial}\alpha)_{i_1\cdots i_p\,\bar{k}\,\bar{\jmath}_1\cdots\bar{\jmath}_q} = (q+1)\,\partial_{[\bar{k}}\,\alpha_{i_1\cdots i_p\,\bar{\jmath}_1\cdots\bar{\jmath}_q]}, \tag{E.8}$$

where the bracket antisymmetrises $\bar{k}$ together with the q indices $\bar{\jmath}_1\cdots\bar{\jmath}_q$, that is over the $q+1$ antiholomorphic indices only, leaving the holomorphic $i_1\cdots i_p$ untouched. That d has only these two components, with no part raising q by two or lowering it, is precisely integrability. Squaring $\mathrm{d} = \partial + \bar{\partial}$ and matching types gives

$$\partial^2 = 0, \qquad \bar{\partial}^{\,2} = 0, \qquad \partial\bar{\partial} + \bar{\partial}\partial = 0. \tag{E.9}$$

Remark E.2 (Beyond the lectures: the non-integrable case). On a merely almost complex manifold d acquires four components, mapping $\Lambda^{p,q}$ to $\Lambda^{p+2,q-1}$, $\Lambda^{p+1,q}$, $\Lambda^{p,q+1}$ and $\Lambda^{p-1,q+2}$; the two extreme pieces are measured by the Nijenhuis tensor. Integrability is the statement that they vanish, leaving (E.7). The heterotic and flux backgrounds of Chapter 9 sometimes leave Kähler geometry but keep a complex structure, so $\mathrm{d} = \partial + \bar{\partial}$ survives even when $\mathrm{d}J \neq 0$.

E.1.3 Dolbeault cohomology

The nilpotency $\bar\partial^{\,2}=0$ defines a cohomology on each column of the type decomposition. The *Dolbeault cohomology groups* are

$$H^{p,q}_{\bar\partial}(M)=\frac{\ker\bigl(\bar\partial\colon \Lambda^{p,q}\to\Lambda^{p,q+1}\bigr)}{\operatorname{im}\bigl(\bar\partial\colon \Lambda^{p,q-1}\to\Lambda^{p,q}\bigr)},\qquad h^{p,q}(M)\coloneqq\dim_{\mathbb{C}}H^{p,q}_{\bar\partial}(M),\tag{E.10}$$

and their dimensions $h^{p,q}$ are the *Hodge numbers*. For M compact they are finite. Two of them read off structure directly: $H^{p,0}_{\bar\partial}(M)$ is the space of global holomorphic p-forms (those α with $\bar\partial\alpha=0$ and no antiholomorphic legs), and $H^{n,0}_{\bar\partial}(M)$ is the space of holomorphic top forms, whose dimension is 1 exactly when the canonical bundle $K_M=\Lambda^{n,0}T^*M$ is trivial. That last condition is the algebraic face of the Calabi-Yau property, developed in Appendix E.2; the trivialising section is the holomorphic volume form Ω.

Dolbeault's theorem identifies these analytic groups with sheaf cohomology,

$$H^{p,q}_{\bar\partial}(M)\;\cong\;H^{q}(M,\Omega^{p}_{M}),\tag{E.11}$$

the q-th cohomology of the sheaf Ω^p_M of holomorphic p-forms. Sheaf cohomology $H^q(M,-)$ is standard algebraic-geometry machinery (Griffiths and Harris, Huybrechts); we invoke it only as an alternative name for the same Dolbeault groups, and this is the bridge to the algebraic-geometry computations of Appendix E.2. Relating $h^{p,q}$ to the topological Betti numbers $b_k=\dim H^k_{\mathrm{dR}}(M,\mathbb{R})$ needs a metric, and needs that metric to be Kähler. We turn to that.

E.1.4 Hermitian metrics and the Kähler form

A Riemannian metric g on a complex manifold is *Hermitian* if it is compatible with the complex structure,

$$g(JX,JY)=g(X,Y),\qquad\text{equivalently}\qquad J^{p}{}_{m}J^{q}{}_{n}\,g_{pq}=g_{mn}.\tag{E.12}$$

In holomorphic coordinates (E.12) kills the pure components $g_{ij}=g_{\bar\imath\bar\jmath}=0$, leaving only the mixed block, and the line element is

$$\mathrm{d}s^2=2\,g_{i\bar\jmath}\,\mathrm{d}z^i\,\mathrm{d}\bar z^{\bar\jmath},\qquad g_{i\bar\jmath}=\overline{g_{j\bar\imath}}\ \text{ a positive Hermitian matrix.}\tag{E.13}$$

Every complex manifold admits Hermitian metrics; averaging any metric over $\mathbb{K}$ and J produces one. Contracting the metric with the complex structure produces a two-form, the *fundamental* or *Kähler form*,

$$J(X,Y)=g(JX,Y),\qquad J_{mn}=g_{mp}\,J^{p}{}_{n},\qquad J=\mathrm{i}\,g_{i\bar\jmath}\,\mathrm{d}z^i\wedge\mathrm{d}\bar z^{\bar\jmath},\tag{E.14}$$

a real form of type $(1,1)$; its components $J_{mn}=g_{mp}J^p{}_n$ are the complex structure with one index lowered, so this is exactly the two-form J_{mn} of the opening warning. Its top power is the Riemannian volume, up to the standard combinatorial factor,

$$\mathrm{vol}_g=\frac{J^{\,n}}{n!}=\det(g_{i\bar\jmath})\prod_{k=1}^{n}\bigl(\mathrm{i}\,\mathrm{d}z^k\wedge\mathrm{d}\bar z^{\bar k}\bigr),\tag{E.15}$$

each factor $\mathrm{i}\,\mathrm{d}z^k\wedge\mathrm{d}\bar z^{\bar k}=2\,\mathrm{d}x^{2k-1}\wedge\mathrm{d}x^{2k}$ being real and positively oriented. So far nothing constrains $\mathrm{d}J$; a general Hermitian manifold has $\mathrm{d}J\neq0$.

E.1.5 Kähler manifolds

A Hermitian manifold is *Kähler* when its Kähler form is closed,

$$\mathrm{d}J = 0. \tag{E.16}$$

This innocuous-looking equation is equivalent to a bundle of statements that individually look much stronger. Writing $\mathrm{d}J = (\partial + \bar\partial)J$ and separating types, closure is the local symmetry

$$\partial_i\, g_{j\bar k} = \partial_j\, g_{i\bar k} \qquad \text{(and its conjugate)}, \tag{E.17}$$

which by the Poincaré lemma is solved locally by a single real function, the *Kähler potential* K,

$$g_{i\bar\jmath} = \partial_i \partial_{\bar\jmath} K, \qquad J = \mathrm{i}\,\partial\bar\partial K. \tag{E.18}$$

The potential is defined up to a *Kähler transformation* $K \to K + f(z) + \overline{f(z)}$ with f holomorphic, which leaves $g_{i\bar\jmath}$ and J unchanged. Because the metric is a second derivative of a scalar, its Levi-Civita connection is unusually simple: the only non-vanishing Christoffel symbols are the purely holomorphic ones and their conjugates,

$$\Gamma^{i}_{jk} = g^{i\bar l}\,\partial_j g_{k\bar l}, \qquad \Gamma^{\bar\imath}_{\bar\jmath\bar k} = \overline{\Gamma^{i}_{jk}}, \tag{E.19}$$

with every mixed symbol zero. Parallel transport therefore never rotates a holomorphic vector into an antiholomorphic one: the complex structure is covariantly constant.

Proposition E.3 (Equivalent characterisations of the Kähler condition)**.** *For a Hermitian manifold (M, J, g) the following are equivalent: (i) $\mathrm{d}J = 0$; (ii) the complex structure is parallel, $\nabla J = 0$, for the Levi-Civita connection ∇; (iii) a local Kähler potential* (E.18) *exists; (iv) the Levi-Civita and Chern connections coincide, the Chern connection being the unique connection on $T^{1,0}M$ compatible with both g and J whose $(0,1)$-part equals $\bar\partial$; (v) the holonomy group, generated by parallel transport around closed loops and defined in Appendix E.3, is contained in $\mathrm{U}(n) \subset \mathrm{SO}(2n)$.*

The equivalences are standard; see Kobayashi and Nomizu, Moroianu, or Nakahara. Item (v) is the entry point to Appendix E.3: reducing the holonomy further, from $\mathrm{U}(n)$ to $\mathrm{SU}(n)$, is the Calabi-Yau condition and the origin of the parallel spinor. On a Kähler manifold the curvature inherits the pure-type structure of (E.19). The Ricci tensor collapses to a derivative of the metric determinant,

$$R_{i\bar\jmath} = -\partial_i \partial_{\bar\jmath} \log\det\left(g_{k\bar l}\right), \tag{E.20}$$

and packages into the *Ricci form* ρ,

$$\rho = \mathrm{i}\, R_{i\bar\jmath}\, \mathrm{d}z^i \wedge \mathrm{d}\bar z^{\bar\jmath} = -\mathrm{i}\,\partial\bar\partial \log\det\left(g_{k\bar l}\right), \tag{E.21}$$

a closed real $(1,1)$-form (we use ρ throughout, including in Appendices E.2 and E.3; do not confuse it with the spacetime index ρ of Appendix A). Its de Rham class is metric-independent and equals, up to normalisation, the first Chern class of the manifold,

$$\left[\tfrac{1}{2\pi}\rho\right] = c_1(M) \coloneqq c_1(T^{1,0}M) \in H^2(M, \mathbb{Z}), \tag{E.22}$$

with c_1 the Chern class of Appendix D. This is an integral class; the Calabi-Yau condition of Appendix E.2 imposes its vanishing in $H^2(M, \mathbb{R})$, a real vanishing that still permits torsion in the integral class. Ricci-flatness $R_{i\bar\jmath} = 0$ thus forces $c_1(M) = 0$, and Yau's theorem supplies the converse on a compact Kähler manifold; this is the whole content of Appendix E.2, foreshadowed here only to place ρ and c_1 in the geometry. Finally, the Kähler form itself defines a class $[J] \in H^{1,1}(M, \mathbb{R})$, the *Kähler class*, and the classes represented by genuine (positive) Kähler metrics form an open cone, the Kähler cone, whose geometry is a moduli space discussed in Appendix E.2.

Example E.4 (Projective space is Kähler)**.** On the patch $z^a = Z^a/Z^0$ of $\mathbb{CP}^n$ the *Fubini-Study* potential

$$K_{\mathrm{FS}} = \log\Bigl(1 + \textstyle\sum_{a=1}^{n} |z^a|^2\Bigr), \qquad J_{\mathrm{FS}} = \mathrm{i}\,\partial\bar\partial K_{\mathrm{FS}}, \tag{E.23}$$

gives a globally defined closed $(1,1)$-form (the ambiguity between patches is a Kähler transformation), so $\mathbb{CP}^n$ is Kähler. Any submanifold of a Kähler manifold is Kähler in the induced metric, so every smooth projective variety, in particular the complete intersections of Appendix E.2, carries a Kähler metric for free. For $n = 1$, (E.23) is the round metric on $S^2 \cong \mathbb{CP}^1$ up to scale.

E.1.6 Hodge theory on Kähler manifolds

The metric turns each exterior derivative into a Laplacian. On a compact manifold the L^2-adjoints (formal codifferentials; in even real dimension $\mathrm{d}^* = -\star\,\mathrm{d}\star$) build three second-order operators,

$$\Delta_{\mathrm{d}} = \mathrm{d}\,\mathrm{d}^* + \mathrm{d}^*\mathrm{d}, \qquad \Delta_{\partial} = \partial\,\partial^* + \partial^*\partial, \qquad \Delta_{\bar\partial} = \bar\partial\,\bar\partial^* + \bar\partial^*\bar\partial, \tag{E.24}$$

whose kernels are the harmonic forms. Hodge's theorem (Riemannian, Appendix D) makes each cohomology class the span of a unique harmonic representative, so $H^k_{\mathrm{dR}}(M,\mathbb{C}) \cong \mathcal{H}^k_{\mathrm{d}}$ and $H^{p,q}_{\bar\partial}(M) \cong \mathcal{H}^{p,q}_{\bar\partial}$. On a general Hermitian manifold the three Laplacians are unrelated. The Kähler condition ties them together, and everything downstream follows.

Theorem E.5 (Kähler identities)**.** *On a compact Kähler manifold the three Laplacians agree up to a factor of two,*

$$\Delta_{\mathrm{d}} = 2\,\Delta_{\partial} = 2\,\Delta_{\bar\partial}. \tag{E.25}$$

In particular Δ_{d} commutes with the projection onto type (p,q), so the real and imaginary parts of a harmonic form are harmonic and every harmonic k-form splits into harmonic (p,q)-pieces.

The identities follow from the commutation relations between ∂, $\bar\partial$ and the Lefschetz operator $L = J \wedge (\,\cdot\,)$ (see the remark below, where L is used again); proofs in Griffiths and Harris, Huybrechts, or Nakahara. Their immediate corollary is the central structural result of the subject.

Theorem E.6 (Hodge decomposition)**.** *On a compact Kähler manifold the de Rham cohomology decomposes into Dolbeault pieces,*

$$H^k(M,\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(M), \qquad \overline{H^{p,q}(M)} = H^{q,p}(M), \tag{E.26}$$

where $H^{p,q}(M) = \mathcal{H}^{p,q}_{\bar\partial} \cong H^{p,q}_{\bar\partial}(M)$. Consequently

$$b_k = \sum_{p+q=k} h^{p,q}. \tag{E.27}$$

Two symmetries act on the $h^{p,q}$. Complex conjugation of (E.26) reflects the type,

$$h^{p,q} = h^{q,p}. \tag{E.28}$$

The Hodge star maps $\Lambda^{p,q} \to \Lambda^{n-q,n-p}$ and commutes with Δ_{d}, so on harmonic representatives $h^{p,q} = h^{n-q,n-p}$; composing this with the conjugation (E.28) gives the Serre-duality relation

$$h^{p,q} = h^{n-p,n-q}. \tag{E.29}$$

Together they fix a great deal. Odd Betti numbers are even,

$$b_1 = 2\,h^{1,0}, \qquad b_{2k+1} \in 2\mathbb{Z}, \tag{E.30}$$

which is a genuine obstruction: a compact complex manifold with odd b_1 carries no Kähler metric at all. The number $h^{n,0} = \dim H^0(M, K_M)$ is the geometric genus, the count of holomorphic top forms, and $h^{p,0}$ counts holomorphic p-forms.

Remark E.7 (Beyond the lectures: hard Lefschetz and $\partial\bar\partial$). Wedging with the Kähler class, $L = J \wedge (\,\cdot\,)$, is an isomorphism $L^k\colon H^{n-k}(M) \to H^{n+k}(M)$ (the hard Lefschetz theorem), which forces $h^{p,q} \geq h^{p-1,q-1}$ for $p + q \leq n$ and organises cohomology into primitive pieces. A second consequence of the Kähler identities is the $\partial\bar\partial$-lemma: a form that is d-exact and of pure type is $\partial\bar\partial$-exact, so d-, ∂- and $\bar\partial$-cohomology all coincide and compact Kähler manifolds are *formal*. Both statements fail on non-Kähler complex manifolds and are the technical engine behind period integrals and Griffiths transversality in Chapter 9.

E.1.7 The Hodge diamond

Arranging the Hodge numbers with $k = p + q$ increasing downward, p to the lower left and q to the lower right, gives the *Hodge diamond*. For a complex threefold, the case Chapter 8 needs, the unconstrained diamond is

$$\begin{array}{ccccccc} & & & h^{0,0} & & & \\ & & h^{1,0} & & h^{0,1} & & \\ & h^{2,0} & & h^{1,1} & & h^{0,2} & \\ h^{3,0} & & h^{2,1} & & h^{1,2} & & h^{0,3} \\ & h^{3,1} & & h^{2,2} & & h^{1,3} & \\ & & h^{3,2} & & h^{2,3} & & \\ & & & h^{3,3} & & & \end{array} \tag{E.31}$$

The conjugation symmetry (E.28) reflects the diamond in its vertical axis; Serre duality (E.29) reflects it through its centre. A connected manifold has $h^{0,0} = 1$, and duality then fixes $h^{n,n} = 1$. Summing the k-th row reproduces b_k by (E.27), and the alternating sum is the Euler characteristic,

$$\chi(M) = \sum_{k=0}^{2n} (-1)^k b_k = \sum_{p,q} (-1)^{p+q} h^{p,q}. \tag{E.32}$$

The physical dictionary that reads massless four-dimensional fields off harmonic forms is set up in Chapter 8: on a Calabi-Yau threefold $h^{1,1}$ counts Kähler (size) moduli and $h^{2,1}$ counts complex-structure (shape) moduli. The Calabi-Yau specialisation of (E.31), where SU(3) holonomy collapses the diamond to the two integers $h^{1,1}$ and $h^{2,1}$, is derived in Appendix E.2. Here we record the diamonds of the simplest Kähler manifolds, which need no such input.

Manifold	Hodge diamond	Betti / Euler
Genus-g curve Σ_g $(n = 1)$	$\begin{array}{ccc} & 1 & \\ g & & g \\ & 1 & \end{array}$	$b = (1, 2g, 1),\ \chi = 2 - 2g$
$\mathbb{CP}^2$ $(n = 2)$	$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 0 & & 1 & & 0 \\ & 0 & & 0 & \\ & & 1 & & \end{array}$	$b = (1, 0, 1, 0, 1),\ \chi = 3$
Complex torus $\mathbb{C}^n/\Lambda$	$h^{p,q} = \binom{n}{p}\binom{n}{q}$	$b_k = \binom{2n}{k},\ \chi = 0$

For $\mathbb{CP}^n$ the pattern continues: $h^{p,p} = 1$ for $0 \le p \le n$ and all off-diagonal entries vanish, so $b_{2k} = 1$, $b_{2k+1} = 0$ and $\chi(\mathbb{CP}^n) = n + 1$. The complex torus is the opposite extreme, with a full binomial diamond and $\chi = 0$; its $n = 1$ case is the elliptic curve, the genus-1 entry above, which reappears as the fibre of the elliptic fibrations in Appendix E.4.

Remark E.8 (Beyond the lectures: a complex manifold that is not Kähler). The Hopf surface $S^1 \times S^3$ carries a complex structure (it is $(\mathbb{C}^2 \setminus \{0\})/\mathbb{Z}$, the quotient by $z \mapsto 2z$), yet it is not Kähler: it has $b_1 = 1$, odd, in violation of (E.30). Its Dolbeault and de Rham numbers do not assemble into a Hodge decomposition. This is the sharp sense in which "Kähler" is a real constraint and not a formality, and it is why the internal spaces that admit a worldsheet CFT description and a clean spectrum are Kähler.

Exercise E.1 (The Kähler condition in components)**.** Starting from $J = \mathrm{i}\, g_{i\bar{\jmath}}\, \mathrm{d}z^i \wedge \mathrm{d}\bar{z}^{\bar{\jmath}}$, compute $\mathrm{d}J = (\partial + \bar{\partial})J$ and show that closure is equivalent to the two conditions $\partial_{[i} g_{j]\bar{k}} = 0$ and $\partial_{[\bar{\jmath}} g_{|i|\bar{k}]} = 0$. Deduce the local existence of a Kähler potential with $g_{i\bar{\jmath}} = \partial_i \partial_{\bar{\jmath}} K$.

Exercise E.2 (Reconstructing a Calabi-Yau diamond)**.** Take a compact connected Kähler threefold with $h^{1,0} = h^{2,0} = 0$ and $h^{3,0} = 1$ (the Calabi-Yau inputs of Appendix E.2). Using only $h^{0,0} = 1$ and the two symmetries (E.28) and (E.29), fill (E.31) in terms of the two integers $h^{1,1}$ and $h^{2,1}$. Confirm $b_0 = b_6 = 1$, $b_1 = b_5 = 0$, $b_2 = b_4 = h^{1,1}$, $b_3 = 2 + 2h^{2,1}$, and hence $\chi = 2\,(h^{1,1} - h^{2,1})$.

Exercise E.3 (Fubini-Study on $\mathbb{CP}^1$)**.** For $n = 1$ evaluate (E.23) to find $J_{\mathrm{FS}} = \mathrm{i}\,\mathrm{d}z \wedge \mathrm{d}\bar{z}/(1+|z|^2)^2$, and by integrating in polar coordinates show $\int_{\mathbb{CP}^1} J_{\mathrm{FS}} = 2\pi$, so $[J_{\mathrm{FS}}/2\pi]$ is the positive generator of $H^2(\mathbb{CP}^1, \mathbb{Z})$. This normalisation of the Kähler class is the one used for volume and central-charge counting in Chapter 8.

E.2 Calabi-Yau manifolds

The complex and Kähler geometry of Appendix E.1 and the holonomy story of Appendix E.3 converge on a single class of spaces. A Calabi-Yau manifold is the geometry a supersymmetric string vacuum wants: compact, Ricci-flat, and complex, with just enough parallel structure to keep a fraction of the supercharges alive. This section states what such a manifold is, in four interchangeable languages; records Yau's theorem, which is what makes the definition non-vacuous; describes the moduli spaces that become the scalar fields of the four-dimensional theory; collects K3 and the du Val (ADE) singularities that reappear fibrewise in Appendix E.4; lists the constructions that actually produce examples; and sets out mirror symmetry. The holonomy and spinor facts invoked below are established in Appendix E.3, and the complex- and Kähler-geometry background in Appendix E.1. The physics that reads a four-dimensional spectrum off these data lives in Chapters 8 and 9; here we assemble the geometry those chapters invoke. By the end a reader can state the defining conditions, read Hodge data off a construction, and locate the moduli, singularities and mirror map that a compactification will use.

The big picture

Four conditions on a compact Kähler manifold X of complex dimension n turn out to be the same condition: holonomy in $\mathrm{SU}(n)$; vanishing first Chern class $c_1(X) = 0$; a covariantly constant spinor; and a nowhere-vanishing holomorphic $(n, 0)$-form Ω. The bridge between the topological statement ($c_1 = 0$) and the metric statement (Ricci-flat) is Yau's theorem, an existence-and-uniqueness result with no closed-form output. The Ricci-flat metrics come in a family, and its parameters are the moduli: $h^{1,1}$ *Kähler (size)* deformations and $h^{n-1,1}$ *complex-structure (shape)* deformations, which for a threefold means $h^{1,1}$ and $h^{2,1}$. In complex dimension two the same package specialises to K3 and its du Val (ADE) degenerations,

the local singularities whose resolutions carry the only Calabi-Yau metrics anyone can write down explicitly. Explicit Calabi-Yaus come from cutting hypersurfaces or complete intersections out of projective and toric ambient spaces. Mirror symmetry pairs X with $\tilde{X}$ so that $h^{1,1}(X) = h^{2,1}(\tilde{X})$, trading a quantum computation on one side for a classical one on the other.

E.2.1 The four characterisations

Definition E.9 (Calabi-Yau n-fold)**.** A *Calabi-Yau n-fold* is a compact Kähler manifold X of complex dimension n satisfying any one, hence all, of the equivalent conditions:

(C1) the restricted holonomy group of the Levi-Civita connection lies in $\mathrm{SU}(n)$;

(C2) the first Chern class vanishes, $c_1(X) = 0 \in H^2(X, \mathbb{R})$;

(C3) X admits a nowhere-vanishing holomorphic $(n, 0)$-form Ω, equivalently the canonical bundle $K_X = \Lambda^{n,0} T^*X$ is trivial;

(C4) X admits a covariantly constant (parallel) spinor.

Any such X carries a unique Ricci-flat Kähler metric in each Kähler class; this analytic fifth face of the definition is the content of Yau's theorem (Theorem E.11).

Conditions (C2) and (C3) are set up in Appendix E.1; the holonomy statement (C1) and the parallel-spinor statement (C4) rest on Berger's classification and the spinor-stabiliser fact, both established in Appendix E.3. The equivalences are almost all linear algebra on top of one hard theorem. Start from the fact that on a Kähler manifold the Levi-Civita connection preserves the complex structure, so the holonomy already sits inside $\mathrm{U}(n)$. The residual $\mathrm{U}(1) = \mathrm{U}(n)/\mathrm{SU}(n)$ is exactly the piece that rotates the canonical bundle, and it is measured by the Ricci form, the closed real $(1,1)$-form of Eq. (E.21), which we denote ρ,

$$\rho \;=\; -\,i\,\partial\bar{\partial}\log\det(g_{i\bar{j}}), \qquad \rho_{i\bar{j}} = i\,R_{i\bar{j}}, \qquad \left[\tfrac{\rho}{2\pi}\right] = c_1(X) \in H^2(X, \mathbb{Z}), \tag{E.33}$$

the curvature of the canonical bundle K_X, representing c_1. This ρ is the same form met in Appendix E.1; we keep the one symbol for it here. Its class $c_1(X)$ is integral, whereas condition (C2) asks only that it vanish over $\mathbb{R}$, a distinction that matters when $\pi_1(X) \neq 0$ (see the warning below). Reading Eq. (E.33) in both directions organises the whole web:

$$\begin{array}{ccc} \text{(C4) parallel spinor } \eta & \xLeftrightarrow{\ \text{Berger}\ } & \text{(C1) Hol} \subseteq \mathrm{SU}(n) \\ \text{bilinears}\ \updownarrow & & \updownarrow \\ \text{(C3) parallel } \Omega,\ K_X \text{ trivial} & \Longrightarrow & \text{(C2) } c_1 = 0 \\ & \overset{\text{Yau}}{\longleftarrow} & \end{array}$$

Ricci-flatness gives $\rho = 0$, hence $c_1(X) = 0$ in one line: this is the cheap implication. Reducing $\mathrm{U}(n)$ to $\mathrm{SU}(n)$ is the same as making ρ vanish, which by the Bochner argument on a compact Kähler manifold produces a covariantly constant section of K_X, that is a parallel Ω; the mechanism is the Bochner-Weitzenböck identity, by which a harmonic section of a bundle of vanishing curvature is parallel (Appendix D). Conversely a parallel Ω trivialises K_X and reduces the holonomy. Berger's classification (Appendix E.3) supplies (C1)⇔(C4): $\mathrm{SU}(n) \subset \mathrm{Spin}(2n)$ is precisely the stabiliser of a spinor, so reduced holonomy and a parallel spinor are the same datum. The single implication that is not linear algebra is the converse of the cheap one, (C2)⇒Ricci-flat: vanishing of the *topological* class c_1 does not visibly produce a Ricci-flat *metric*. That it does is Yau's theorem, and it closes the loop.

Remark E.10 (The parallel spinor builds Ω and J). The equivalence (C3)⇔(C4) is constructive. Pick the covariantly constant spinor η (of definite chirality, normalised $\eta^\dagger\eta = 1$). Its bilinears are parallel because η is, and they assemble the Kähler form and the holomorphic top-form,

$$J_{i\bar{\jmath}} \;\propto\; -i\,\eta^\dagger \Gamma_{i\bar{\jmath}}\,\eta, \qquad \Omega_{i_1\cdots i_n} \;\propto\; \eta^T C\,\Gamma_{i_1\cdots i_n}\,\eta, \tag{E.34}$$

with C the charge-conjugation matrix. The chirality projector forces Ω to have purely holomorphic indices, which is why it is an $(n,0)$-form; the precise Clifford identities are in Appendix B. This bilinear is the geometric origin of the covariantly constant spinor that Section 8.2 uses to preserve one four-dimensional supercharge: Ω is the tensor avatar of η.

$c_1 = 0$ is a statement about a cohomology class

Condition (C2) asks that $c_1(X)$ vanish in $H^2(X,\mathbb{R})$. For simply connected X this is equivalent to strict SU(n) holonomy and to a genuinely trivial K_X. When $\pi_1(X) \neq 0$ a subtlety intrudes: a finite fundamental group can leave K_X a non-trivial *torsion* class even though $c_1 = 0$ in $H^2(X,\mathbb{Q})$, and the holonomy is then SU(n) extended by a finite group. Throughout we take Calabi-Yau to mean strict holonomy SU(n) and, unless stated, $\pi_1(X) = 0$. The free-quotient case X/Γ with Γ finite and freely acting is the standard device for lowering the generation count in heterotic model building (Chapter 8); it keeps $c_1 = 0$ but introduces Wilson lines and torsion.

For string compactification $n = 3$: a Calabi-Yau threefold has real dimension six, holonomy SU(3), and one covariantly constant spinor, hence keeps a quarter of the supercharges. The Hodge diamond of such an X collapses to the two integers $h^{1,1}$ and $h^{2,1}$, with $h^{p,0} = 0$ for $0 < p < n$ forced by strict holonomy; the diamond and the Dolbeault machinery behind it are set up in Appendix E.1 and Section 8.5.

E.2.2 Yau's theorem

Theorem E.11 (Calabi conjecture; Yau 1978)**.** *Let X be a compact Kähler manifold of complex dimension n with $c_1(X) = 0$. Then each Kähler class $[J_0] \in H^{1,1}(X,\mathbb{R})$ contains a unique Ricci-flat Kähler metric.*

More generally Calabi asked, and Yau proved, that any real $(1,1)$-form representing $c_1(X)$ is the Ricci form of a unique metric in a given Kähler class; Theorem E.11 is the case $c_1 = 0$. The proof is not an explicit construction. Fix a reference Kähler metric g_0 with Kähler form J_0 in the chosen class; since $c_1 = 0$, the Ricci form of g_0 is exact, $\rho_0 = i\,\partial\bar\partial f$ for a smooth real function f fixed up to a constant by $\int_X (e^f - 1)\,J_0^n = 0$. Every other metric in the class has Kähler form $J = J_0 + i\,\partial\bar\partial\phi$ for a real potential ϕ (the $\partial\bar\partial$-lemma), and demanding $\rho = 0$ becomes a single scalar equation for ϕ, the *complex Monge-Ampère equation*

$$(J_0 + i\,\partial\bar\partial\phi)^n \;=\; e^f\,J_0^n, \qquad \text{equivalently} \qquad \det\!\Big(g_{0\,i\bar{\jmath}} + \partial_i\partial_{\bar{\jmath}}\phi\Big) = e^f\,\det\!\big(g_{0\,i\bar{\jmath}}\big), \tag{E.35}$$

with the positivity constraint $J_0 + i\,\partial\bar\partial\phi > 0$. Yau solves Eq. (E.35) by the continuity method: deform the right-hand side from the trivial case to the target through a one-parameter family, and show the set of parameters for which a solution exists is both open (implicit function theorem) and closed (a chain of a priori C^0, C^2 and $C^{2,\alpha}$ estimates on ϕ). Uniqueness follows from the maximum principle. The proof is in Yau (1978); a parallel real Monge-Ampère result is Aubin (1978), and textbook accounts are in Joyce (2000) and Gross, Huybrechts and Joyce (2003).

Remark E.12 (Beyond the lectures: no metric in closed form). Theorem E.11 guarantees a metric but hands you no formula for it. To date no explicit Ricci-flat metric is known on

any compact Calabi-Yau of dimension ≥ 2 away from degenerate limits (an orbifold point, a large-complex-structure or large-volume boundary). This is not a defect of the proof: the metric is transcendental, not algebraic. What saves phenomenology is that the four-dimensional *spectrum* and its symmetries are cohomological, insensitive to the metric; the metric enters only the Kähler-potential normalisations of the low-energy Lagrangian. When those are needed, Eq. (E.35) is solved numerically, historically by Donaldson's balanced-metric algorithm and by direct energy-functional minimisation, more recently with neural-network parametrisations of ϕ (the `cymetric` line of work). The metrics on non-compact Calabi-Yaus (the conifold, ALE and ALF spaces of Appendix E.2.4) *are* explicit, because there the extra symmetry reduces Eq. (E.35) to an ordinary differential equation.

E.2.3 Moduli space

Yau's theorem gives one Ricci-flat metric per Kähler class and complex structure, so the space of Ricci-flat metrics is a family, and its coordinates are massless scalars in four dimensions. Linearising the Ricci-flatness condition about a background and splitting the metric perturbation δg by holomorphic type gives two decoupled sets of zero-modes, and the moduli space is, locally, their product

$$\mathcal{M}(X) \;\cong\; \mathcal{M}_{\mathrm{K}}(X) \times \mathcal{M}_{\mathrm{cs}}(X), \qquad \dim_{\mathbb{R}} \mathcal{M}_{\mathrm{K}} = h^{1,1}, \quad \dim_{\mathbb{C}} \mathcal{M}_{\mathrm{cs}} = h^{n-1,1}. \tag{E.36}$$

The mixed perturbations $\delta g_{i\bar{j}}$ keep the complex structure and deform the Kähler form; they are harmonic $(1,1)$-forms, giving the *Kähler moduli*. The pure perturbations δg_{ij} cannot be undone by a coordinate change and deform the complex structure; contracting with Ω turns each into a harmonic $(n-1,1)$-form, giving the *complex-structure moduli*. For a threefold that is $h^{1,1}$ real Kähler moduli and $h^{2,1}$ complex complex-structure moduli. The two factors carry sharply different geometries.

Complex-structure moduli. These deform X as a complex manifold. Infinitesimally they are elements of $H^1(X, T_X)$, the tangent-bundle-valued $(0,1)$-forms of the Kodaira-Spencer deformation theory (the sheaf cohomology group $H^1(X,T_X)$ is the Dolbeault group $H^{0,1}(X,T_X)$ under another name, Appendix E.1), and on a Calabi-Yau the contraction with Ω gives the canonical isomorphism

$$H^1(X, T_X) \;\xrightarrow{\;\lrcorner\,\Omega\;}\; H^{n-1,1}(X), \tag{E.37}$$

which for $n = 3$ identifies the $h^{2,1}$ deformations with $H^{2,1}(X)$. A structural fact makes the moduli space well behaved:

Theorem E.13 (Unobstructedness; Bogomolov 1978, Tian 1987, Todorov 1989)**.** *The complex-structure moduli space of a compact Calabi-Yau manifold is smooth of dimension $h^{n-1,1}$: every first-order deformation in $H^1(X,T_X)$ integrates to a genuine deformation, with no obstruction.*

The clean way to package $\mathcal{M}_{\mathrm{cs}}$ is through the periods of Ω. Choose a symplectic basis $\{A^I, B_I\}$ of $H_3(X,\mathbb{Z})$, $I = 0, \ldots, h^{2,1}$, and set

$$X^I = \int_{A^I} \Omega, \qquad F_I = \int_{B_I} \Omega. \tag{E.38}$$

As the complex structure varies, the line $H^{3,0} \subset H^3(X,\mathbb{C})$ rotates inside the fixed lattice $H^3(X,\mathbb{Z})$, and the periods trace out $\mathcal{M}_{\mathrm{cs}}$. The X^I are projective coordinates and the $F_I = \partial F/\partial X^I$ derive from a single holomorphic *prepotential* $F(X)$, homogeneous of degree two; the periods satisfy the Picard-Fuchs system, a set of linear ODEs in the moduli. The metric on $\mathcal{M}_{\mathrm{cs}}$ is the Weil-Petersson metric, which is Kähler with potential

$$K_{\mathrm{cs}} \;=\; -\log\Big(-\,i\int_X \Omega\wedge\bar{\Omega}\Big) \;=\; -\log\Big[\,i(\bar{X}^I F_I - X^I \bar{F}_I)\Big], \tag{E.39}$$

where the orientation is fixed so the argument is positive (matching Chapter 9). Writing $\partial_a\Omega = k_a\Omega + \chi_a$ with $\chi_a \in H^{2,1}$ the Kodaira derivative (Section 9.2.3), the Weil-Petersson metric is

$$G_{a\bar{b}} = \partial_a \partial_{\bar{b}} K_{\rm cs} = -\frac{\int_X \chi_a \wedge \bar{\chi}_{\bar{b}}}{\int_X \Omega \wedge \bar{\Omega}}\,. \tag{E.40}$$

The structure defined by Eqs. (E.38) and (E.39) is *special Kähler* geometry, the same rigid geometry $\mathcal{N}=2$ supersymmetry imposes on vector-multiplet scalars (Appendix C); the periods are the input to the flux superpotential $W = \int_X G_3 \wedge \Omega$ of Chapter 9.

Kähler moduli and the Kähler cone. A Ricci-flat metric requires a Kähler class in which volumes are positive. The allowed classes form the open *Kähler cone*

$$\mathcal{K}(X) = \Big\{\, [J] \in H^{1,1}(X,\mathbb{R}) \;:\; \int_C J > 0,\ \int_S J\wedge J > 0,\ \int_X J^n > 0 \,\Big\}, \tag{E.41}$$

the conditions running over all holomorphic curves C, divisors S and X itself (this positivity-over-all-subvarieties characterisation of the transcendental Kähler cone is the Demailly-Paun theorem; the Nakai-Moishezon criterion is its algebraic analogue, testing ampleness of integral divisor classes). Its walls are where some subvariety collapses: a curve shrinking to a point signals a flop, a divisor collapsing signals a more drastic degeneration, and a node forming is the conifold locus of Appendix E.2.4. Expand $J = \sum_a t^a \omega_a$ in an integral basis $\{\omega_a\}$ of $H^{1,1}(X,\mathbb{Z})$; the volume is the cubic

$$\mathcal{V} = \frac{1}{n!}\int_X J^n = \frac{1}{3!}\,\kappa_{abc}\,t^a t^b t^c \quad (n=3), \qquad \kappa_{abc} = \int_X \omega_a\wedge\omega_b\wedge\omega_c, \tag{E.42}$$

in terms of the triple intersection numbers κ_{abc}. In string theory each real Kähler modulus t^a is complexified by the internal B-field, $T^a = b^a + i\,t^a$ with $B + iJ = \sum_a T^a\omega_a$, and the periodicity $b^a \sim b^a + 1$ is a shift symmetry protecting T^a perturbatively. The Kähler moduli space metric follows from the potential

$$K_{\rm K} = -2\log\mathcal{V}, \tag{E.43}$$

the $-2\log(\text{volume})$ that produces the no-scale structure of Section 9.2.4.

Remark E.14 (Beyond the lectures: the stringy Kähler moduli space). The geometric cone Eq. (E.41) is not the whole story for a string. Worldsheet instantons make $\mathcal{M}_{\rm K}$ a globally non-trivial special-Kähler manifold, and mirror symmetry identifies it with the *complex-structure* moduli space of the mirror $\tilde{X}$, which is a single connected space with distinguished points: the large-volume point, the conifold locus where a curve degenerates (Appendix E.2.4), and Landau-Ginzburg or orbifold points where the geometric description fails altogether. Crossing a wall of $\mathcal{K}(X)$ by a flop passes to a birationally distinct Calabi-Yau X' with the same Hodge numbers but different κ_{abc}; the union of the Kähler cones of all birational models is the *movable cone*, and string theory glues them into one smooth moduli space. The upshot is that "the" Kähler moduli space is larger and better behaved quantum mechanically than the classical cone Eq. (E.41) of any single geometry.

E.2.4 K3, du Val singularities and ALE spaces

One complex dimension down, the whole apparatus collapses to a single manifold and a clean list of singularities. This is the reference material that the fibrewise degenerations of Appendix E.4 and the enhanced gauge symmetry of Section 10.1 rest on.

type	$\Gamma \subset \mathrm{SU}(2)$	hypersurface $f = 0$	rank r
A_{n-1}	cyclic $\mathbb{Z}_n$	$x^2 + y^2 + z^n = 0$	$n-1$
D_n	binary dihedral	$x^2 + y^2 z + z^{n-1} = 0$	n
E_6	binary tetrahedral	$x^2 + y^3 + z^4 = 0$	6
E_7	binary octahedral	$x^2 + y^3 + yz^3 = 0$	7
E_8	binary icosahedral	$x^2 + y^3 + z^5 = 0$	8

Table E.1: The du Val singularities $\mathbb{C}^2/\Gamma$, their orbifold groups and defining equations. The rank r is the number of exceptional spheres in the minimal resolution and the rank of the corresponding ADE algebra; the same data drive the elliptic-fibre gauge algebras of Appendix E.4.

K3. Up to diffeomorphism there is a unique compact Calabi-Yau twofold: the K3 surface, simply connected, with holonomy exactly $\mathrm{SU}(2) = \mathrm{Sp}(1)$, hence hyperkähler. Triviality of K_X forces $h^{2,0} = h^{0,2} = 1$, and simple connectivity forces $h^{1,0} = 0$, so the Hodge diamond and Betti numbers are

$$h^{0,0} = h^{2,2} = 1, \quad h^{2,0} = h^{0,2} = 1, \quad h^{1,1} = 20, \qquad b_2 = 22, \qquad \chi(\mathrm{K3}) = 24. \tag{E.44}$$

With the cup-product pairing, $H^2(\mathrm{K3}, \mathbb{Z})$ is the even, unimodular lattice of signature $(3, 19)$,

$$\Lambda_{\mathrm{K3}} \;\cong\; U^{\oplus 3} \;\oplus\; \mathrm{E}_8(-1)^{\oplus 2}, \tag{E.45}$$

with U the hyperbolic plane and $\mathrm{E}_8(-1)$ the negative-definite E_8 root lattice (lattice conventions in Appendix G). Because $h^{2,0} = 1$ there is no clean $(1,1)/(2,1)$ split of the moduli and the six-dimensional theory it feeds is non-chiral; chirality is the reason string phenomenology climbs to the threefold. The physics of K3 compactification, its lattice and its moduli is developed in Section 8.4.

Du Val (ADE) singularities. A K3, or any Calabi-Yau surface, degenerates through isolated *du Val* points, the local model

$$\mathbb{C}^2/\Gamma, \qquad \Gamma \subset \mathrm{SU}(2) \text{ finite}, \tag{E.46}$$

also called Kleinian, ADE or rational double points (Theorem 10.2). The finite subgroups of $\mathrm{SU}(2)$ were classified in the nineteenth century and stand in one-to-one correspondence with the simply-laced Lie algebras, Table E.1. Each quotient is a hypersurface $\{f(x, y, z) = 0\} \subset \mathbb{C}^3$ with an isolated singular point at the origin. The minimal (crepant) resolution $\widetilde{\mathbb{C}^2/\Gamma}$ replaces the singular point by a tree of r rational curves $\Theta_1, \ldots, \Theta_r$, each a $\mathbb{CP}^1$ of self-intersection -2, meeting according to the ADE Dynkin diagram. Their intersection form is minus the Cartan matrix,

$$\Theta_i \cdot \Theta_j \;=\; -C_{ij}, \qquad C_{ii} = 2, \quad C_{ij} = -1 \text{ iff nodes } i, j \text{ adjacent}, \tag{E.47}$$

so the spheres are the simple roots (Appendix G). Because the resolution is crepant it leaves K trivial, so a du Val point on a Calabi-Yau resolves to a Calabi-Yau. This single relation Eq. (E.47) is the geometry behind gauge symmetry from singularities: wrapping branes on the shrinking Θ_i builds the ADE algebra (Section 10.1 and Chapter 11).

ALE and ALF spaces. The resolved du Val space carries an explicit complete metric, unlike its compact cousins. An *ALE* (asymptotically locally Euclidean) space is a complete hyperkähler four-manifold asymptotic to $\mathbb{C}^2/\Gamma$ at infinity, with curvature falling off as for flat space; these are exactly the minimal resolutions of Eq. (E.46) equipped with a Ricci-flat metric. The A-series metrics are the Gibbons-Hawking multi-centre family (the A_1 case is Eguchi-Hanson on

$T^*\mathbb{CP}^1$), and Kronheimer constructed the full ADE list by a hyperkähler quotient. Their *ALF* (asymptotically locally flat) analogues approach a circle fibration (Taub-NUT) at infinity rather than a cone, with volume growth r^3 instead of r^4, and describe Kaluza-Klein monopoles in M-theory. In every case the extra symmetry reduces Eq. (E.35) to an ODE, which is why the metric is known in closed form.

The conifold. The three-dimensional analogue of a du Val point is the *conifold*, the non-compact Calabi-Yau threefold

$$\left\{ z_1^2 + z_2^2 + z_3^2 + z_4^2 = 0 \right\} \subset \mathbb{C}^4, \tag{E.48}$$

a real cone over $T^{1,1} = S^2 \times S^3$ with an isolated node (ordinary double point) at the origin, carrying an explicit Ricci-flat Kähler metric (Candelas and de la Ossa 1990). A compact threefold develops a node exactly where a three-cycle or a rational curve collapses. The node has two distinct smoothings: the *deformation* $\sum_i z_i^2 = \epsilon$, which grows a finite S^3 (a complex-structure modulus), and the *small resolution*, which grows a finite $\mathbb{CP}^1 = S^2$ (a Kähler modulus). Passing from one to the other through the singular point is a *conifold transition*; it changes the Hodge numbers and connects topologically distinct Calabi-Yaus inside a single moduli space, the geometric face of the conifold locus met in Appendix E.2.3.

E.2.5 Constructions

There is no classification of Calabi-Yau threefolds and no proof that the number of topological types is finite. What we have instead are construction machines that manufacture examples in bulk. All of them work the same way: realise X as a zero locus inside an ambient space A whose geometry is understood, and use *adjunction* to reduce the Calabi-Yau condition to arithmetic on degrees. For a smooth hypersurface $X \subset A$ cut out by a section of a line bundle $\mathcal{O}_A(X)$,

$$K_X = \left(K_A \otimes \mathcal{O}_A(X)\right)\big|_X, \tag{E.49}$$

so X is Calabi-Yau precisely when this restriction is trivial. Here $\mathcal{O}_A(D)$ is the holomorphic line bundle of a divisor D and K_A the canonical bundle (Appendix D). On $\mathbb{CP}^N$ every line bundle is a power $\mathcal{O}(d) = \mathcal{O}(1)^{\otimes d}$ of the hyperplane bundle $\mathcal{O}(1)$: its holomorphic sections are the degree-d homogeneous polynomials, its first Chern class is $d\,H$ with H the hyperplane class, and the canonical bundle is $K_{\mathbb{CP}^N} = \mathcal{O}(-N-1)$. Those few facts drive every degree count below.

Hypersurfaces in projective space. Take $A = \mathbb{CP}^{n+1}$, whose canonical bundle is $K_A = \mathcal{O}(-n-2)$, and let X be a smooth degree-d hypersurface, $\mathcal{O}_A(X) = \mathcal{O}(d)$. Then Eq. (E.49) gives $K_X = \mathcal{O}(d-n-2)|_X$ and $c_1(X) = (n+2-d)\,H|_X$, so

$$X \text{ Calabi-Yau} \iff d = n+2. \tag{E.50}$$

A degree-$(m+1)$ hypersurface in $\mathbb{CP}^m$ is a Calabi-Yau $(m-1)$-fold: the cubic in $\mathbb{CP}^2$ is T^2, the quartic in $\mathbb{CP}^3$ is K3, and the quintic in $\mathbb{CP}^4$ is the canonical threefold. Allowing weighted projective spaces $\mathbb{CP}^{n+1}_{[w_0,\dots,w_{n+1}]}$ enlarges the list; the Calabi-Yau condition becomes $d = \sum_i w_i$ (after a crepant resolution of the quotient singularities the weights introduce).

Example E.15 (The quintic and its Hodge numbers)**.** The quintic $X = \{p_5 = 0\} \subset \mathbb{CP}^4$ has $d = 5 = n+2$, hence $c_1 = 0$. Its Kähler class is the restriction of the single hyperplane class of $\mathbb{CP}^4$, so $h^{1,1}(X) = 1$ by the Lefschetz hyperplane theorem (for a smooth hypersurface of complex dimension ≥ 3 the ambient $H^{1,1}$ restricts isomorphically). The complex-structure moduli are the coefficients of the degree-five polynomial modulo linear redefinitions of the coordinates. A degree-five polynomial in five variables has $\binom{5+4}{4} = 126$ monomials; the reparametrisation group

$\mathrm{GL}(5,\mathbb{C})$ has dimension $5^2 = 25$, and its scalar centre $z_i \mapsto \lambda z_i$ already realises the overall rescaling $p_5 \mapsto \lambda^5 p_5$, so nothing further is subtracted. Hence

$$h^{2,1}(X) = 126 - 25 = 101. \tag{E.51}$$

For the quintic the polynomial deformations are complete: every complex-structure modulus is realised by a monomial deformation of p_5, so this naive count is exact. Thus $(h^{1,1}, h^{2,1}) = (1, 101)$ and $\chi = 2(h^{1,1} - h^{2,1}) = -200$.

Complete intersections (CICYs). Cut X out of a product of projective spaces $A = \mathbb{CP}^{n_1} \times \cdots \times \mathbb{CP}^{n_m}$ by k homogeneous equations, the a-th of multidegree $(q_1^a, \ldots, q_m^a)$. The data are recorded in a *configuration matrix*

$$X \in \left[\begin{array}{c|ccc} n_1 & q_1^1 & \cdots & q_1^k \\ \vdots & \vdots & & \vdots \\ n_m & q_m^1 & \cdots & q_m^k \end{array}\right], \qquad \dim_{\mathbb{C}} X = \sum_{r=1}^{m} n_r - k, \tag{E.52}$$

and adjunction, applied once per equation, makes X Calabi-Yau exactly when the degrees balance the dimension in each projective factor,

$$\sum_{a=1}^{k} q_r^a \;=\; n_r + 1 \qquad \text{for every } r = 1, \ldots, m. \tag{E.53}$$

Imposing $\dim_{\mathbb{C}} X = 3$ and Eq. (E.53) leaves a finite search. Candelas, Dale, Lütken and Schimmrigk (1988) enumerated all 7890 configuration matrices producing smooth Calabi-Yau threefolds, realising 266 distinct Hodge-number pairs. The quintic is the shortest CICY, $[\,4 \,\|\, 5\,]$; the "bicubic" $[\,2\ 2 \,\|\, 3\ 3\,]$ in $\mathbb{CP}^2 \times \mathbb{CP}^2$ has $(h^{1,1}, h^{2,1}) = (2, 83)$; two cubics in $\mathbb{CP}^5$, $[\,5 \,\|\, 3\ 3\,]$, give $(1, 73)$.

Toric hypersurfaces (Batyrev). The largest known dataset comes from toric geometry. Let $\Delta \subset M_{\mathbb{R}} \cong \mathbb{R}^{n+1}$ be a *reflexive* lattice polytope: it contains the origin as its unique interior lattice point, and its polar dual

$$\Delta^\circ = \{\, y \in N_{\mathbb{R}} \;:\; \langle y, x\rangle \geq -1 \;\; \forall\, x \in \Delta \,\} \tag{E.54}$$

is again a lattice polytope. The toric variety $\mathbb{P}_\Delta$ built from the normal fan of Δ is Gorenstein Fano, and Batyrev (1994) showed that a generic anticanonical hypersurface in $\mathbb{P}_\Delta$, defined by $\sum_{m \in \Delta \cap M} c_m\, x^m = 0$, is a Calabi-Yau n-fold after a crepant resolution of the ambient singularities. Reflexivity of Δ is the toric avatar of $c_1 = 0$. In four dimensions Kreuzer and Skarke (2000) classified all reflexive polytopes: there are 473,800,776 of them, and the Calabi-Yau threefolds they produce cover 30,108 distinct $(h^{1,1}, h^{2,1})$ pairs, with $h^{1,1} + h^{2,1}$ reaching 502. This is the ambient space in which most explicit string compactifications live.

Exercise E.4 (Reading a configuration matrix)**.** For the bicubic $[\,2\ 2 \,\|\, 3\ 3\,]$: (a) verify the dimension formula Eq. (E.52) and the Calabi-Yau condition Eq. (E.53) row by row. (b) The two Kähler classes are the hyperplane classes H_1, H_2 of the two $\mathbb{CP}^2$ factors, so $h^{1,1} = 2$; using the fact that on the ambient product $H_r^3 = 0$ and $\int_{\mathbb{CP}^2 \times \mathbb{CP}^2} H_1^2 H_2^2 = 1$, together with the class $[X] = 3H_1 + 3H_2$ of the hypersurface, compute the triple intersection numbers $\kappa_{abc} = \int_X H_a H_b H_c$ of Eq. (E.42). (c) Given $(h^{1,1}, h^{2,1}) = (2, 83)$, compute $\chi = 2(h^{1,1} - h^{2,1})$ and confirm the entry in Table E.2.

X	ambient / configuration	type	$h^{1,1}$	$h^{2,1}$	χ
quintic	$[\,4 \,\|\| \, 5\,] \subset \mathbb{CP}^4$	hypersurface	1	101	-200
mirror quintic	$X/(\mathbb{Z}_5)^3$ resolved	Greene-Plesser	101	1	$+200$
two cubics	$[\,5 \,\|\| \, 3\ 3\,] \subset \mathbb{CP}^5$	complete intersection	1	73	-144
bicubic	$[\,2\ 2 \,\|\| \, 3\ 3\,] \subset \mathbb{CP}^2 \times \mathbb{CP}^2$	complete intersection	2	83	-162
sextic	$[\,4_{[1,1,1,1,2]} \,\|\| \, 6\,]$	weighted hypersurface	1	103	-204

Table E.2: A handful of Calabi-Yau threefolds and their Hodge data. Mirror pairs exchange $h^{1,1} \leftrightarrow h^{2,1}$ and flip the sign of χ.

E.2.6 Mirror symmetry

Calabi-Yau threefolds come in pairs. A *mirror pair* $(X, \tilde{X})$ has Hodge diamonds reflected across the diagonal,

$$h^{1,1}(X) = h^{2,1}(\tilde{X}), \qquad h^{2,1}(X) = h^{1,1}(\tilde{X}), \qquad \chi(\tilde{X}) = -\chi(X), \tag{E.55}$$

and the physical statement behind the numerology is an equivalence of string theories:

$$\text{type IIA on } X \;\cong\; \text{type IIB on } \tilde{X} \tag{E.56}$$

as four-dimensional theories. The symmetry of Table E.2 under $h^{1,1} \leftrightarrow h^{2,1}$ is the visible face; the content is that a single physics is computed two ways, one hard and one easy.

A-model and B-model. The precise home of mirror symmetry is the topological string. The $\mathcal{N} = (2,2)$ worldsheet sigma model on a Calabi-Yau admits two inequivalent topological twists.

- The *A-model* depends only on the complexified Kähler moduli. Its correlators are *symplectic* invariants: genus-zero Gromov-Witten invariants n_β counting holomorphic maps (worldsheet instantons) of the sphere into X in the curve class $\beta \in H_2(X, \mathbb{Z})$.
- The *B-model* depends only on the complex-structure moduli. Its correlators are *period integrals* and variations of Hodge structure, pure complex geometry with no instanton sum.

Mirror symmetry is the statement that the A-model on X equals the B-model on $\tilde{X}$. Because the B-model is classical, this trades an infinite instanton series for a finite computation. Concretely, the normalised three-point functions (the $\mathcal{N} = 2$ Yukawa couplings) match. On the A-side, on the Kähler moduli,

$$\kappa^{\mathrm{A}}_{abc}(t) = \kappa_{abc} + \sum_{\beta \neq 0} n_\beta \, d_a d_b d_c \, \frac{q^\beta}{1 - q^\beta}, \qquad q^\beta = e^{2\pi i \, t \cdot \beta}, \quad d_a = \textstyle\int_\beta \omega_a, \tag{E.57}$$

a classical intersection number Eq. (E.42) corrected by instantons; on the B-side, on the complex-structure moduli of $\tilde{X}$,

$$\kappa^{\mathrm{B}}_{abc}(z) = \int_{\tilde{X}} \Omega \wedge \partial_a \partial_b \partial_c \Omega, \tag{E.58}$$

a pure period integral with no corrections. The two are equated by the *mirror map* $t = t(z)$, read off from the ratios of periods of Ω on $\tilde{X}$. Equivalently, with the convention $\kappa^{\mathrm{A}}_{abc} = \partial_a \partial_b \partial_c F_{\mathrm{A}}$, the instanton-corrected A-model prepotential is

$$F_{\mathrm{A}}(t) = \frac{1}{6} \kappa_{abc} \, t^a t^b t^c + (\text{polynomial}) + \frac{1}{(2\pi i)^3} \sum_\beta n_\beta \, \mathrm{Li}_3(q^\beta), \tag{E.59}$$

whose third derivative reproduces Eq. (E.57) term by term, and mirror symmetry computes it from the classical prepotential of $\tilde{X}$.

The quintic mirror. The first explicit mirror was built by Greene and Plesser as a crepant resolution of the orbifold $X/(\mathbb{Z}_5)^3$, giving $\tilde{X}$ with $(h^{1,1}, h^{2,1}) = (101, 1)$. The single complex-structure modulus ψ of $\tilde{X}$ (the deformation $\sum z_i^5 - 5\psi \prod z_i$) obeys a one-parameter Picard-Fuchs equation, and Candelas, de la Ossa, Green and Parkes (1991) solved it and matched to Eq. (E.57), extracting the numbers n_d of rational curves of degree d on the quintic:

$$n_1 = 2875, \qquad n_2 = 609250, \qquad n_3 = 317206375, \quad \dots \tag{E.60}$$

The count of lines $n_1 = 2875$ was classical nineteenth-century enumerative geometry, the conics n_2 were known, and the higher n_d were new; the mirror produced them all from one integral. This calculation, more than any formal argument, established mirror symmetry and launched Gromov-Witten theory; the physics narrative is in Section 8.5.

SYZ: mirror symmetry as T-duality. The reason a mirror exists at all is geometric. Strominger, Yau and Zaslow (1996) argued that near a large-complex-structure limit a Calabi-Yau threefold is fibred by *special Lagrangian* three-tori over a base $B \cong S^3$,

$$T^3 \hookrightarrow X \xrightarrow{\pi} B, \qquad J\big|_L = 0, \quad \operatorname{Im}\Omega\big|_L = 0, \tag{E.61}$$

the two conditions defining a special Lagrangian submanifold L of real dimension n: Lagrangian for the symplectic form J, and calibrated by $\operatorname{Re}\Omega$ (so that $\operatorname{Re}\Omega$ restricts to the volume form on L). Mirror symmetry is T-duality (Chapter 6) performed simultaneously on all three legs of the fibre. Three T-dualities invert the volume of the T^3 and exchange momentum with winding; done fibrewise, they turn the symplectic (Kähler) data of X into the complex-structure data of the dual fibration $\tilde{X}$, which is why Eq. (E.55) holds. The dual $\tilde{X}$ is reconstructed as the moduli space of a D-brane wrapping a fibre T^3. SYZ is proven only in limits and remains a conjecture in general, but it is the picture that explains the mirror.

Exercise E.5 (Mirror numerology)**.** (a) The mirror quintic has $(h^{1,1}, h^{2,1}) = (101, 1)$; verify $\chi = +200$, the flip of the quintic's $\chi = -200$ predicted by Eq. (E.55). (b) Using $\chi = 2(h^{1,1} - h^{2,1})$, show that a self-mirror threefold $\tilde{X} \cong X$ must have $\chi = 0$, hence $h^{1,1} = h^{2,1}$.

Remark E.16 (Beyond the lectures: homological mirror symmetry)*.* Kontsevich (1994) proposed the deepest form of the correspondence, a statement about the categories of D-branes on the two sides. The B-branes on X, holomorphic objects, are the derived category of coherent sheaves $D^b(\operatorname{Coh} X)$; the A-branes on $\tilde{X}$, special Lagrangians decorated with flat connections, are the (derived) Fukaya category $\operatorname{Fuk}(\tilde{X})$. Homological mirror symmetry is the equivalence of triangulated categories

$$D^b(\operatorname{Coh} X) \;\cong\; D^\pi \operatorname{Fuk}(\tilde{X}), \tag{E.62}$$

which contains the Hodge-number statement Eq. (E.55) and the enumerative statement Eq. (E.60) as shadows on cohomology. It has been proven for elliptic curves, abelian varieties, and (following Sheridan) the quintic. For the course, the operational content is the one line Eq. (E.56): a IIA computation of Kähler-moduli couplings, corrected by worldsheet instantons, is a IIB computation of complex-structure periods on the mirror, with no corrections at all.

E.3 Special holonomy

The holonomy group of a Riemannian manifold records what parallel transport does to a vector, or to a spinor, carried once around a closed loop. It is the invariant that decides which tensor and spinor fields can be covariantly constant, and through the gravitino variation of Section 8.2 it fixes how much supersymmetry a fluxless compactification preserves. This section is the geometric reference behind that statement: Berger's classification of the possible holonomy groups, the parallel forms and spinors each carries, and the two exceptional cases G_2 and Spin(7)

that have no complex structure. The supercharge bookkeeping that turns a parallel-spinor count into a value of $\mathcal{N}$ lives in Appendix B.4; here we supply the geometry it quotes. The Calabi-Yau case in particular is developed from four equivalent viewpoints in Appendix E.2, and the Hodge theory used to count harmonic forms in Appendix E.1. We work throughout with the Levi-Civita connection ∇ of a Riemannian metric g on a manifold X of real dimension $n = \dim_{\mathbb{R}} X$. In this section n always means the *real* dimension; in Appendices E.1 and E.2 the same letter denotes the *complex* dimension of a Calabi-Yau, so a Calabi-Yau k-fold has $n = 2k$ here.

E.3.1 The holonomy group

Fix a point $p \in X$. Parallel transport along a piecewise-smooth loop γ based at p is a linear isometry $P_\gamma \colon T_pX \to T_pX$, and composing loops composes the maps. The set of all such maps is a group,

$$\mathrm{Hol}_p(X) \;=\; \{\, P_\gamma : \gamma \text{ a loop based at } p \,\} \;\subseteq\; \mathrm{O}(T_pX) \cong \mathrm{O}(n), \tag{E.63}$$

a subgroup of the orthogonal group because $\nabla g = 0$ means transport preserves the inner product, and of $\mathrm{SO}(n)$ when X is oriented. Restricting to contractible loops gives the *restricted holonomy group* $\mathrm{Hol}^0_p(X)$, the identity component; it is a closed connected Lie subgroup of $\mathrm{SO}(n)$, and the quotient $\mathrm{Hol}_p/\mathrm{Hol}^0_p$ is a countable group carrying the image of $\pi_1(X)$. On a connected X the groups at different base points are conjugate inside $\mathrm{O}(n)$, so we suppress the base point and write $\mathrm{Hol}(X)$, understood as a conjugacy class of subgroups. On a simply connected X the two coincide, $\mathrm{Hol} = \mathrm{Hol}^0$.

The curvature is the infinitesimal generator of holonomy. The Ambrose-Singer theorem states that the Lie algebra $\mathfrak{hol}_p \subseteq \mathfrak{so}(n)$ of Hol^0_p is spanned by the curvature endomorphisms

$$\mathfrak{hol}_p \;=\; \mathrm{span}\Big\{\, P_\gamma^{-1}\, R(v,w)\, P_\gamma \;:\; \gamma \text{ a path from } p,\; v,w \in T_{\gamma(1)}X \,\Big\}, \tag{E.64}$$

the Riemann tensor $R(v,w) \in \mathfrak{so}(n)$ transported back to p over all paths (Ambrose and Singer 1953). Flat metrics have trivial restricted holonomy; curvature is exactly the obstruction. Connections, parallel transport and a fuller account of Eq. (E.64) are in Appendix D.

What makes holonomy the right invariant for our purposes is that it controls parallel fields.

Proposition E.17 (The holonomy principle)**.** *Let X be connected. Evaluation at p gives a bijection between parallel tensor (or spinor) fields on X and tensors (or spinors) in the fibre at p that are invariant under $\mathrm{Hol}_p(X)$. On a simply connected X, invariance under $\mathrm{Hol}^0_p(X)$ suffices.*

A parallel field is unchanged by transport around every loop, so its value at p is holonomy-invariant; conversely an invariant tensor at p transports to a well-defined global parallel field, path-independence being guaranteed by invariance (Besse 1987; Joyce 2000). Reduced holonomy and extra parallel fields are two names for the same thing, and the search for parallel spinors becomes the search for Hol-invariant spinors.

Two reductions of the general classification problem come first. If the holonomy representation on T_pX is reducible, the de Rham decomposition theorem says X is locally a Riemannian product and Hol factorises accordingly; a complete simply connected X is globally such a product. It is therefore enough to classify the *irreducible* holonomy groups. Second, the Riemannian symmetric spaces, those with $\nabla R = 0$, have holonomy equal to the isotropy group of the symmetric pair and were classified completely by Cartan. Setting these two families aside leaves Berger's problem.

E.3.2 Berger's classification

Theorem E.18 (Berger 1955)**.** *Let X be a simply connected Riemannian manifold of dimension n whose holonomy representation is irreducible, and suppose X is not locally symmetric. Then $\mathrm{Hol}(X)$ is one of the groups in Table E.3.*

$\mathrm{Hol}(X)$	$n = \dim_{\mathbb{R}} X$	geometry	curvature	parallel forms
$\mathrm{SO}(n)$	n	generic oriented	–	none
$\mathrm{U}(k)$	$2k$	Kähler	–	J
$\mathrm{SU}(k)$	$2k$	Calabi-Yau	Ricci-flat	J, Ω
$\mathrm{Sp}(k)$	$4k$	hyperkähler	Ricci-flat	$\omega_I, \omega_J, \omega_K$
$\mathrm{Sp}(k)\cdot\mathrm{Sp}(1)$	$4k$	quaternionic-Kähler	Einstein, $\Lambda \neq 0$	Θ (4-form)
G_2	7	exceptional	Ricci-flat	φ, $\star\varphi$
$\mathrm{Spin}(7)$	8	exceptional	Ricci-flat	Ψ (4-form)

Table E.3: Berger's list of irreducible, non-symmetric Riemannian holonomy groups, with the geometry each defines, its Einstein type, and the parallel forms it fixes: the Kähler form J, the holomorphic volume form Ω, the hyperkähler triple $\omega_I, \omega_J, \omega_K$, the quaternionic 4-form Θ, the associative 3-form φ with its coassociative dual $\star\varphi$, and the Cayley 4-form Ψ. Ranges: $k \geq 1$ for $\mathrm{U}(k)$; $k \geq 2$ for $\mathrm{SU}(k)$ and $\mathrm{Sp}(k)$ (the cases $\mathrm{SU}(1), \mathrm{Sp}(1)$ degenerate or coincide with entries above); $k \geq 2$ for $\mathrm{Sp}(k)\cdot\mathrm{Sp}(1)$. A dash in the curvature column marks the absence of any Einstein condition. The Ricci-flat entries $\mathrm{SU}(k), \mathrm{Sp}(k), G_2, \mathrm{Spin}(7)$ are exactly those admitting a parallel spinor (Table E.4).

Three comments fix the reading of Table E.3. Berger's original list carried an eighth entry, $\mathrm{Spin}(9) \subset \mathrm{SO}(16)$; Alekseevskii (1968), and independently Brown and Gray, showed that $\mathrm{Spin}(9)$ holonomy forces the metric to be symmetric (the Cayley projective plane $\mathbb{O}P^2$ and its noncompact dual), so it drops out of the non-symmetric list. The inclusions among the entries are geometric: $\mathrm{SU}(k) \subset \mathrm{U}(k)$ removes the trace part of the Kähler holonomy, and $\mathrm{Sp}(k) \subset \mathrm{SU}(2k)$ makes every hyperkähler manifold a Calabi-Yau. Curvature splits the list in two. The entries $\mathrm{U}(k)$ and $\mathrm{Sp}(k)\cdot\mathrm{Sp}(1)$ carry nonzero Ricci curvature (a Kähler manifold need not be Einstein; a quaternionic-Kähler one is Einstein with $\Lambda \neq 0$), while

$$\mathrm{SU}(k), \qquad \mathrm{Sp}(k), \qquad G_2, \qquad \mathrm{Spin}(7) \tag{E.65}$$

are Ricci-flat. These four are what physics means by "special holonomy": they solve the internal vacuum Einstein equation $R_{mn} = 0$ of a fluxless compactification, and, as Appendix E.3.4 shows, they are precisely the groups that fix a spinor.

E.3.3 Parallel forms and G-structures

By Theorem E.17 each reduced holonomy group is the stabiliser of one or more parallel differential forms, listed in the last column of Table E.3. These forms are the calibrating tensors of the geometry.

- $\mathrm{U}(k)$: the Kähler form $J = g(J\,\cdot, \cdot)$ built from the parallel complex structure J, with $\nabla J = 0$. Its powers $J^p/p!$ are parallel too.
- $\mathrm{SU}(k)$: in addition to J, a nowhere-vanishing parallel holomorphic $(k,0)$-form Ω, the holomorphic volume form. Vanishing of the $\mathrm{U}(1)$ trace part of the connection is exactly $\nabla\Omega = 0$, equivalently triviality of the canonical bundle, $c_1(X) = 0$.
- $\mathrm{Sp}(k)$: a triple of parallel complex structures I, J, K with $IJ = K$ obeying the quaternion relations, and their three Kähler forms $\omega_I, \omega_J, \omega_K$ (the single Kähler form J of the entries above is now just one member ω_J of the triple). Any $aI + bJ + cK$ with $a^2 + b^2 + c^2 = 1$ is again a complex structure: a whole S^2 of them.

$\mathrm{Hol}(X)$	geometry	n	$n_\parallel$	chirality of the parallel spinors
$\mathrm{SU}(k)$	Calabi-Yau k-fold	$2k$	2	η, η^c; same chirality for k even, opposite for k odd
$\mathrm{Sp}(k)$	hyperkähler $4k$-fold	$4k$	$k+1$	all of one chirality
G_2	G_2 manifold	7	1	(no chirality in odd dimension)
Spin(7)	Spin(7) manifold	8	1	one chirality

Table E.4: Parallel spinors by holonomy (Wang 1989). $n_\parallel$ is the number of singlets in the $\mathrm{Spin}(n)$ spinor under $\mathrm{Hol}(X)$, equivalently the dimension of the space of covariantly constant spinors. The special isomorphisms $\mathrm{SU}(2) = \mathrm{Sp}(1)$ (K3) and $\mathrm{Sp}(1) = \mathrm{Spin}(3)$ make the K3 row appear as both $\mathrm{SU}(2)$ and $\mathrm{Sp}(1)$, with $n_\parallel = 2$. The generic $\mathrm{SO}(n)$ and the non-Ricci-flat $\mathrm{U}(k), \mathrm{Sp}(k)\cdot\mathrm{Sp}(1)$ fix no spinor.

- $\mathrm{Sp}(k)\cdot\mathrm{Sp}(1)$: here the individual $\omega_I, \omega_J, \omega_K$ are not globally defined, but their combination $\Theta = \omega_I \wedge \omega_I + \omega_J \wedge \omega_J + \omega_K \wedge \omega_K$ is a parallel 4-form.
- G_2 and Spin(7): the associative 3-form φ and the Cayley 4-form Ψ of Appendix E.3.6.

The distinction between having such a form and having reduced holonomy is the distinction between a G-structure and its integrability. A *G-structure* is a reduction of the frame bundle to a subgroup $G \subseteq \mathrm{SO}(n)$, equivalently a choice of the defining tensor at every point with no parallel condition imposed; almost complex, almost Kähler and G_2-structures are examples. The holonomy sits in G if and only if that tensor is parallel, if and only if the *intrinsic torsion* of the G-structure vanishes. For the Kähler case this is the statement $\nabla J = 0$ developed in Appendices E.1 and E.2; for G_2 it becomes a pair of closed-form conditions below.

E.3.4 Parallel spinors and the supersymmetry count

A covariantly constant spinor η, $\nabla_m \eta = 0$, is by Theorem E.17 a spinor in the fibre invariant under $\mathrm{Hol}(X)$. So the number of parallel spinors is the number of singlets in the branching of the $\mathrm{Spin}(n)$ spinor representation under $\mathrm{Hol}(X)$:

$$n_\parallel(X) \;=\; \#\{\text{singlets of } \mathrm{Hol}(X) \text{ in the spinor of } \mathrm{Spin}(n)\} \;=\; \dim\left(\ker \nabla\big|_{\text{spinors}}\right). \tag{E.66}$$

Which subgroups of $\mathrm{SO}(n)$ fix a spinor is a closed question, answered by Wang (1989): on an irreducible simply connected Riemannian spin manifold the possibilities are exactly the Ricci-flat entries Eq. (E.65), with the counts of Table E.4. The parallel forms of Appendix E.3.3 are then bilinears in these parallel spinors, so the "form" and "spinor" descriptions of a special holonomy are two readings of the same reduction (the SU(3) bilinears $J \sim \eta^\dagger \Gamma_{mn} \eta$ and $\Omega \sim \eta^{cT} \Gamma_{mnp} \eta$ are written out at Eqs. (8.36) and (8.37)).

The Calabi-Yau count is the workhorse. For $\mathrm{SU}(k) \subset \mathrm{Spin}(2k)$ a nonzero spinor in one chiral half is fixed together with its charge conjugate, giving $n_\parallel = 2$ for every k: the pair η, η^c, of the same chirality when k is even and of opposite chirality when k is odd. The exceptional groups fix a single spinor, which is what makes them the minimally supersymmetric choices. The hyperkähler count $k+1$ exceeds the naive 2 because $\mathrm{Sp}(k)$ is smaller than $\mathrm{SU}(2k)$ and leaves more invariant; at $k = 1$ it reproduces the K3 value 2.

A parallel spinor pins down the metric as well as the topology.

Proposition E.19 (Parallel spinor implies Ricci-flat)**.** *A Riemannian manifold admitting a nonzero parallel spinor is Ricci-flat.*

The integrability condition $0 = [\nabla_m, \nabla_n]\eta = \frac{1}{4} R_{mnpq}\Gamma^{pq}\eta$, contracted with Γ^n and reduced with the first Bianchi identity, gives $R_{mn}\Gamma^n\eta = 0$; since $v \mapsto \not{v}\,\eta$ is injective for $\eta \neq 0$ in Euclidean signature, $R_{mn} = 0$. The Clifford algebra is spelled out at Eqs. (8.33) and (8.34). This closes the geometry: the same reduction that supplies a preserved supercharge also guarantees the internal Einstein equation of the fluxless vacuum.

Now the physics count, which Appendix B.4 carries out in full. A ten- or eleven-dimensional supercharge is a spinor of the higher Lorentz group; it splits under $\mathrm{Spin}(1, d-1) \times \mathrm{Spin}(n)$ into external spinors tensored with the $2^{\lfloor n/2 \rfloor}$ internal slots, and only the $n_\parallel$ parallel ones survive in the vacuum. The preserved fraction is therefore purely geometric,

$$\nu(X) \;=\; \frac{n_\parallel(X)}{2^{\lfloor n/2 \rfloor}}, \qquad \mathcal{Q}_d \;=\; \nu(X)\,\mathcal{Q}_D, \tag{E.67}$$

so a Calabi-Yau k-fold keeps $\nu = 2^{1-k}$ of the supersymmetry ($\frac{1}{2}$ for K3, $\frac{1}{4}$ for the threefold, $\frac{1}{8}$ for the fourfold), G_2 keeps $\frac{1}{8}$ and Spin(7) keeps $\frac{1}{16}$. The table of which theory on which manifold lands on which $\mathcal{N}$, running 32 supercharges down, is Tables B.3 and B.4; the derivation of the first arrow, the vanishing gravitino variation, is Section 8.2.

The keystone chain

For a fluxless product compactification the following are equivalent, holonomy group by holonomy group:

$$\underbrace{\text{preserved supercharge}}_{\delta\psi_M = 0} \iff \underbrace{\nabla_m \eta = 0}_{\text{parallel spinor}} \iff \underbrace{\mathrm{Hol}(X) \subseteq G}_{\text{reduced holonomy}}$$
$$\iff \underbrace{\text{parallel calibrating form}}_{J,\ \Omega,\ \varphi,\ \Psi,\ldots} \implies \underbrace{R_{mn} = 0}_{\text{fluxless vacuum}} \ .$$

The first equivalence is the gravitino variation of Section 8.2; the second and third are the holonomy principle of Theorem E.17; the implication is Theorem E.19. The number of preserved supercharges per higher-dimensional supercharge is $n_\parallel(X)$ from Table E.4. Special holonomy is the geometric content of a supersymmetric vacuum.

Containment, not equality

The condition is $\mathrm{Hol}(X) \subseteq G$. A smaller holonomy fixes *more* spinors and preserves *more* supersymmetry. A six-manifold with $\mathrm{Hol} = \mathrm{SU}(2)$ is locally K3 $\times\, T^2$ and keeps $\nu = \frac{1}{2}$, not the $\frac{1}{4}$ of a genuine SU(3) threefold; trivial holonomy is a torus and keeps everything. Getting exactly $\mathcal{N} = 1$ requires strict, non-factorising holonomy.

E.3.5 $\mathrm{SU}(k)$: Calabi-Yau

Holonomy $\mathrm{SU}(k)$ on a $2k$-manifold is one of the four equivalent definitions of a Calabi-Yau manifold, alongside vanishing first Chern class, existence of a Ricci-flat Kähler metric, and existence of a nowhere-vanishing holomorphic volume form. Their equivalence, and Yau's theorem that a compact Kähler manifold with $c_1 = 0$ carries a unique Ricci-flat metric in each Kähler class, are the subject of Appendix E.2; we use the holonomy characterisation here and defer the rest. The low-dimensional cases are governed by the exceptional isomorphisms of the spin groups.

- $k = 1$: SU(1) is trivial, so a Calabi-Yau onefold is a flat torus T^2, the elliptic curve. Holonomy is trivial and all supersymmetry survives.

- $k = 2$: $\mathrm{SU}(2) = \mathrm{Sp}(1)$, so a Calabi-Yau twofold is hyperkähler. Up to the torus T^4 the only compact example is K3, whose SU(2) holonomy, du Val (ADE) singularities and ALE resolutions are treated in Appendix E.2. Two parallel spinors of the same chirality make K3 preserve half the supersymmetry.
- $k = 3$: the Calabi-Yau threefold, holonomy $\mathrm{SU}(3) \subset \mathrm{Spin}(6) = \mathrm{SU}(4)$, the case that gives four-dimensional $\mathcal{N} = 1$ from the heterotic string and $\mathcal{N} = 2$ from type II. The branching $\mathbf{4} \to \mathbf{3} \oplus \mathbf{1}$ and its consequences are worked out at Section 8.2.4 and in Theorem E.21 below.
- $k = 4$: the Calabi-Yau fourfold, holonomy $\mathrm{SU}(4) \subset \mathrm{Spin}(8)$, the geometry of F-theory (Appendix E.4) and of M-theory to three dimensions.

Since $\mathrm{Sp}(k) \subset \mathrm{SU}(2k)$, every hyperkähler manifold is a Calabi-Yau of even complex dimension carrying extra parallel spinors; K3 at $k = 1$ is where the two families overlap.

E.3.6 G_2 and Spin(7)

The two exceptional holonomies live in seven and eight dimensions and have no complex structure, no Kähler form and no worldsheet sigma-model of the kind Calabi-Yau spaces enjoy. Both are built on the octonions $\mathbb{O}$, the real normed division algebra of dimension 8.

The compact exceptional group G_2 is the automorphism group of $\mathbb{O}$, of dimension 14 and rank 2. Acting on the imaginary octonions $\mathrm{Im}\,\mathbb{O} = \mathbb{R}^7$ it preserves the cross product, encoded in a 3-form. On $\mathbb{R}^7$ with orthonormal coframe $e^1, \dots, e^7$ and $e^{ijk} = e^i \wedge e^j \wedge e^k$ the standard *associative 3-form* is

$$\varphi \;=\; e^{123} + e^{145} + e^{167} + e^{246} - e^{257} - e^{347} - e^{356}, \tag{E.68}$$

and G_2 is exactly its stabiliser in $\mathrm{GL}(7, \mathbb{R})$, which lands inside SO(7) (Bryant 1987). Its Hodge dual $\star\varphi$ is the *coassociative* 4-form. A G_2-structure, a choice of φ pointwise, exists on any oriented spin 7-manifold. Its holonomy sits in G_2 if and only if φ is parallel, and Fernández and Gray (1982) reduced this to two closed-form conditions,

$$\nabla\varphi = 0 \quad\Longleftrightarrow\quad \mathrm{d}\varphi = 0 \;\text{ and }\; \mathrm{d}\star\varphi = 0. \tag{E.69}$$

Neither condition alone suffices: $\mathrm{d}\varphi = 0$ defines a calibrated ("closed") G_2-structure and $\mathrm{d}\star\varphi = 0$ a cocalibrated one, but torsion-free needs both. A parallel φ forces $R_{mn} = 0$ by Theorem E.19, and the holonomy is the full G_2 (not a proper subgroup) precisely when $\pi_1(X)$ is finite (Joyce 2000).

The group Spin(7) enters as a subgroup of SO(8) through its real 8-dimensional spinor representation, an embedding distinct from the vector one; it has dimension 21. It is the stabiliser in $\mathrm{GL}(8, \mathbb{R})$ of the self-dual *Cayley 4-form* Ψ. Writing $\mathbb{R}^8 = \mathbb{R} \oplus \mathbb{R}^7$ with the extra coframe direction e^8, the Cayley form is assembled from the G_2 data on $\mathbb{R}^7$,

$$\Psi \;=\; e^8 \wedge \varphi \;+\; \star_7\varphi, \qquad \star_8\Psi = \Psi, \tag{E.70}$$

a 14-term self-dual form. A Spin(7)-structure exists on an 8-manifold under a spin and characteristic-class condition, and it is torsion-free, holonomy in Spin(7), if and only if

$$\mathrm{d}\Psi = 0, \tag{E.71}$$

the single condition being enough because self-duality turns $\mathrm{d}\star\Psi$ into $\mathrm{d}\Psi$ (Fernández 1986). Again $\nabla\Psi = 0$ gives $R_{mn} = 0$, and full Spin(7) holonomy, rather than a subgroup $\mathrm{SU}(4), G_2, \dots$, requires the manifold to be simply connected with the right characteristic numbers.

Explicit metrics of exceptional holonomy are hard to write down, because there is no analogue of Yau's theorem reducing existence to a class calculation. The first complete noncompact examples were the cohomogeneity-one metrics of Bryant and Salamon (1989) on total spaces of bundles over $S^3, S^4, \mathbb{CP}^2$. The first compact examples came from resolving flat orbifolds T^7/Γ and T^8/Γ (Joyce 1996), later greatly extended by the twisted connected sum construction for G_2 (Kovalev 2003); Joyce (2000) is the standard reference. In physics these are the internal spaces of M-theory: on a G_2 manifold it gives four-dimensional $\mathcal{N}=1$ directly from eleven dimensions ($\mathbf{8} \to \mathbf{7} \oplus \mathbf{1}$ under $G_2 \subset \mathrm{Spin}(7)$, one singlet), and on a Spin(7) manifold three-dimensional $\mathcal{N}=1$ ($\mathbf{8}_s \to \mathbf{7} \oplus \mathbf{1}$ under $\mathrm{Spin}(7) \subset \mathrm{Spin}(8)$). Chiral matter and non-abelian gauge symmetry then come from codimension-seven and codimension-four singularities of the G_2 space, the M-theory analogue of the ADE story on K3.

Remark E.20 (Beyond the lectures: calibrations and wrapped branes). Each parallel form of Table E.3 is a *calibration* in the sense of Harvey and Lawson (1982): a closed p-form ϕ with $\phi|_S \le \mathrm{vol}_S$ on every oriented tangent p-plane, with equality on the *calibrated* submanifolds, which are then homologically volume-minimising. The calibrated cycles are where supersymmetric branes wrap. On a Calabi-Yau, $J^p/p!$ calibrates holomorphic (even-dimensional) cycles and $\mathrm{Re}(e^{i\theta}\Omega)$ calibrates the special Lagrangian ones central to mirror symmetry; on a G_2 manifold, φ calibrates *associative* 3-cycles and $\star\varphi$ *coassociative* 4-cycles; on a Spin(7) manifold, Ψ calibrates *Cayley* 4-cycles. A brane wrapping a calibrated cycle preserves the fraction of supersymmetry set by the same parallel spinor, which is why these cycles organise the BPS spectrum of Chapters 10 and 11.

Example E.21 (A singlet count by branching: the Calabi-Yau threefold)**.** The counts of Table E.4 come from branching the $\mathrm{Spin}(n)$ spinor under $\mathrm{Hol}(X)$ and counting singlets; the exceptional cases below are this same count with a different subgroup. Take the threefold, $\mathrm{Hol} = \mathrm{SU}(3) \subset \mathrm{Spin}(6)$. The two chiral spinors of Spin(6) are the $\mathbf{4}$ and $\bar{\mathbf{4}}$ of the isomorphic SU(4), and under $\mathrm{SU}(3) \subset \mathrm{SU}(4)$ they branch as

$$\mathbf{4} \;\to\; \mathbf{3} \oplus \mathbf{1}, \qquad \bar{\mathbf{4}} \;\to\; \bar{\mathbf{3}} \oplus \mathbf{1}.$$

The singlet in the $\mathbf{4}$ is a covariantly constant spinor η of one chirality; the singlet in the $\bar{\mathbf{4}}$ is its charge conjugate η^c, of the *opposite* chirality because $6 \equiv 2 \pmod 4$ ($k=3$ odd). Two singlets, so $n_\parallel = 2$, the SU(k) entry, and $\nu = 2/2^3 = \frac{1}{4}$. The bilinears $J \sim \eta^\dagger \Gamma_{mn} \eta$ and $\Omega \sim \eta^{cT} \Gamma_{mnp} \eta$ recover the parallel forms of the same reduction.

Exercise E.6 (Counting parallel spinors by branching)**.** Reproduce the remaining entries of Table E.4 by the method of Theorem E.21. All required branchings are given, so this is rep theory, not geometry.

(a) *Warm-up, the fourfold.* For $\mathrm{SU}(4) \subset \mathrm{Spin}(8)$, branch the chiral $\mathbf{8}_s$ using $\mathbf{8}_s \to \mathbf{1} \oplus \mathbf{6} \oplus \mathbf{1}$ under SU(4). Read off $n_\parallel = 2$, now of the *same* chirality ($k=4$ even, $8 \equiv 0 \pmod 4$), and confirm $\nu = \frac{1}{8}$.

(b) G_2. Branch the real $\mathbf{8}$ (the spinor of Spin(7)) under $G_2 \subset \mathrm{Spin}(7)$ using $\mathbf{8} \to \mathbf{7} \oplus \mathbf{1}$. Deduce $n_\parallel = 1$ and $\nu = 1/2^{\lfloor 7/2 \rfloor} = \frac{1}{8}$.

(c) Spin(7). The three eight-dimensional representations $\mathbf{8}_v, \mathbf{8}_s, \mathbf{8}_c$ of Spin(8) are permuted by triality. Under the spinor embedding $\mathrm{Spin}(7) \subset \mathrm{Spin}(8)$ one of them stays irreducible (it is the spinor $\mathbf{8}$ of Spin(7)) while the other two branch as $\mathbf{7} \oplus \mathbf{1}$. Deduce $n_\parallel = 1$ and $\nu = \frac{1}{16}$.

Confirm that the fractions $\frac{1}{8}$ and $\frac{1}{16}$ agree with Eq. (E.67).

E.4 Elliptic fibrations and Kodaira's classification

An elliptically fibred Calabi-Yau is the geometry F-theory runs on (Chapter 11): a family of two-tori varying holomorphically over a base B, whose degenerations record seven-branes and the gauge symmetry they carry. This section is the geometric reference for that construction. It fixes the Weierstrass presentation, the Calabi-Yau condition through adjunction, the discriminant locus, Kodaira's list of singular fibres, and Tate's refinement that reads a gauge algebra, including the non-simply-laced ones, directly off the Weierstrass coefficients. The physics that quotes all of this, the varying axio-dilaton, the $\mathrm{SL}(2,\mathbb{Z})$ monodromy, the matter on curves and the Yukawa couplings at points, is developed in Chapter 11; here we supply the algebraic geometry. The complex-structure and Kähler data of Appendices E.1 and E.2 are assumed throughout, and the du Val (ADE) singularities of Appendix E.2 return fibrewise inside the elliptic fibre. After this section you can read a Kodaira fibre type off the vanishing orders of (f, g, Δ), name the gauge algebra it carries (simply-laced from the resolved fibre, non-simply-laced from monodromy), and place charged matter and Yukawa couplings by codimension.

E.4.1 Elliptic curves and the modular parameter

An *elliptic curve* is a complex torus $E = \mathbb{C}/\Lambda$ carrying a marked point, the image of 0. The lattice can be normalised to $\Lambda = \mathbb{Z} \oplus \tau\mathbb{Z}$ with $\tau \in \mathbb{H}$ the upper half-plane, and two values τ, τ' give isomorphic curves precisely when they are related by a modular transformation

$$\tau' = \frac{a\tau + b}{c\tau + d}, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathbb{Z}). \tag{E.72}$$

The moduli space of elliptic curves is therefore the quotient $\mathbb{H}/\mathrm{SL}(2,\mathbb{Z})$, a single complex dimension. The Weierstrass function $\wp(z;\Lambda)$ and its derivative embed E into $\mathbb{CP}^2$ as a smooth plane cubic, which after shifting and rescaling coordinates is brought to the short *Weierstrass form*

$$y^2 = x^3 + f\,x + g, \tag{E.73}$$

with the marked point sent to the flex $[x : y : z] = [0 : 1 : 0]$ at infinity of the projective completion. The pair (f, g) is two complex numbers and (E.73) is a smooth torus for generic values.

The curve is singular exactly when the cubic on the right develops a repeated root, that is when the *discriminant* vanishes,

$$\Delta \;=\; 4f^3 + 27g^2 \;=\; 0. \tag{E.74}$$

Away from that locus the complex-structure parameter τ is recovered from (f, g) through the $\mathrm{SL}(2,\mathbb{Z})$-invariant j-function,

$$j(\tau) \;=\; 1728\,\frac{4f^3}{4f^3 + 27g^2} \;=\; 1728\,\frac{4f^3}{\Delta}, \qquad j = q^{-1} + 744 + \mathcal{O}(q)\,, \quad q = e^{2\pi i\tau}. \tag{E.75}$$

Three points anchor the map and recur below: $g = 0$ gives $j = 1728$, the square torus $\tau = i$ (extra $\mathbb{Z}_4$ automorphism); $f = 0$ gives $j = 0$, the hexagonal torus $\tau = e^{2\pi i/3}$ (extra $\mathbb{Z}_6$); and $\Delta \to 0$ with $f \neq 0$ sends $j \to \infty$, that is $\tau \to i\infty$, the degenerate long-thin torus.

E.4.2 Elliptic fibrations and Weierstrass models

Definition E.22 (Elliptic fibration with section)**.** An *elliptic fibration* is a proper holomorphic surjection $\pi : Y \to B$ between complex manifolds whose generic fibre $\pi^{-1}(b)$ is a smooth elliptic curve, together with a holomorphic *section* $\sigma : B \to Y$ meeting each fibre in exactly one point, supplying the marked point (and hence the group identity) fibrewise. If $\dim_{\mathbb{C}} B = n$ then $\dim_{\mathbb{C}} Y = n + 1$. The section $\sigma(B) \subset Y$ is the *zero section*.

The existence of a section is what lets the fibration be written algebraically. Every fibre with its marked point embeds in a $\mathbb{CP}^2$, and globalising this embeds Y as a hypersurface in a $\mathbb{CP}^2$-bundle over B. Concretely, let $L \to B$ be a line bundle, promote the fibre coordinates to sections

$$x \in \Gamma(L^2), \qquad y \in \Gamma(L^3), \qquad z \in \Gamma(\mathcal{O}_B), \tag{E.76}$$

and take $Y = \{y^2 z = x^3 + f\, x\, z^2 + g\, z^3\}$ inside the projective bundle $\mathbb{P}(\mathcal{O} \oplus L^2 \oplus L^3) \to B$. This is the *Weierstrass model* of the fibration. Balancing the weights in (E.76) forces

$$f \in \Gamma(L^4), \qquad g \in \Gamma(L^6), \qquad \Delta = 4f^3 + 27g^2 \in \Gamma(L^{12}), \tag{E.77}$$

so that all three terms of Eq. (E.73) are sections of L^6 and Δ of L^{12}.

The Calabi-Yau condition. The one datum still free is L. Fixing it is a one-line adjunction computation, which we state and refer to Morrison and Vafa (1996) for the details. On the hypersurface $Y \subset \mathbb{P}(\mathcal{O} \oplus L^2 \oplus L^3)$ adjunction gives

$$K_Y \;=\; \pi^*\big(K_B \otimes L\big), \tag{E.78}$$

because the canonical class of the $\mathbb{CP}^2$-bundle and the class of the Weierstrass hypersurface conspire to cancel the fibre direction and leave only a pullback from the base. The Calabi-Yau condition $K_Y = \mathcal{O}_Y$ (Appendix E.2) is then simply

$$\boxed{L = K_B^{-1} = \bar{K}_B} \tag{E.79}$$

the anticanonical bundle of the base. From here on $f \in \Gamma(\bar{K}_B^{\,4})$, $g \in \Gamma(\bar{K}_B^{\,6})$ and $\Delta \in \Gamma(\bar{K}_B^{\,12})$. The Calabi-Yau condition is thus not an extra tuning but a bundle assignment: once the base is chosen, the degrees of f, g and Δ are fixed, and with them the seven-brane content. The three cases used in F-theory are

$$\dim_{\mathbb{C}} B = 1 \;\; \text{(elliptic K3)}, \qquad \dim_{\mathbb{C}} B = 2 \;\; \text{(CY threefold)}, \qquad \dim_{\mathbb{C}} B = 3 \;\; \text{(CY fourfold)}, \tag{E.80}$$

giving eight-, six- and four-dimensional F-theory vacua respectively.

Only the *complex structure* of the fibre, the value of τ, is physical in F-theory; the fibre area, a Kähler modulus of Y, is sent to zero and is not a field of the ten-dimensional type IIB background. This is why $Y \to B$ with two extra real dimensions is nonetheless a compactification on the base B alone, and why F-theory is not a twelve-dimensional theory (Chapter 11).

Remark E.23 (Beyond the lectures: the Mordell-Weil group and abelian factors). The sections of an elliptic fibration form an abelian group under fibrewise addition on the elliptic curve, the *Mordell-Weil group* $\mathrm{MW}(Y)$, with the zero section as identity. Its free rank counts the abelian gauge factors of the F-theory vacuum: each independent rational section beyond the zero section contributes a U(1), and its torsion part constrains the global form (as opposed to the algebra) of the gauge group. A Weierstrass model with generic (f, g) has $\mathrm{MW} = 0$ and no U(1); abelian symmetry is engineered by tuning (f, g) so that extra sections appear, for instance the Morrison-Park U(1) model. The interplay of Mordell-Weil rank, the resolved fibre components and the Hodge numbers is packaged in the Shioda-Tate-Wazir relation of Appendix E.4.7.

E.4.3 The discriminant locus and the seven-branes

The fibre degenerates over the vanishing of the discriminant, and by Eqs. (E.77) and (E.79) the *discriminant divisor* sits in a class fixed by the base alone,

$$\{\Delta = 0\} \subset B, \qquad [\Delta] \;=\; 12\, c_1(\bar{K}_B) \;=\; 12\, c_1(B) \;=\; -12\, c_1(K_B). \tag{E.81}$$

Over a generic point of $\{\Delta = 0\}$ with $f \neq 0$ the cubic factorises as $(x - x_0)^2(x - x_1)$: one one-cycle of the torus pinches and the fibre is a *nodal cubic*, a sphere with two points glued. That is the mildest degeneration and corresponds to a single seven-brane. Higher vanishing orders of f and g along components of $\{\Delta = 0\}$ produce the worse degenerations classified in Appendix E.4.4. The class (E.81) is the geometric form of the seven-brane charge constraint: the total degeneration is pinned by $c_1(B)$ and cannot be chosen freely, so gauge symmetry engineered on one divisor is paid for elsewhere in $\{\Delta = 0\}$.

Example E.24 (The elliptic K3 and its twenty-four branes)**.** Take $B = \mathbb{CP}^1$, so $\bar{K}_{\mathbb{CP}^1} = \mathcal{O}(2)$ and Eq. (E.77) makes f a degree-8 and g a degree-12 polynomial, while $\Delta \in H^0(\mathbb{CP}^1, \mathcal{O}(24))$ has degree 24. The total space Y is an elliptic K3, and generically Δ has 24 distinct simple zeros: 24 mutually non-local seven-branes on the sphere. The count is topological, $24 = \chi(\text{K3})$, and it is the same 24 that closes up the transverse plane in the Greene-Shapere-Vafa-Yau solution (Chapter 11). The product of the 24 local monodromies is trivial in $\mathrm{SL}(2,\mathbb{Z})$, which is what makes the configuration compact.

E.4.4 Kodaira's classification of singular fibres

Kodaira classified every way the fibre of a relatively minimal elliptic surface can degenerate over a point of the base, and the list is short and closed. For a higher-dimensional base the same list applies along each irreducible component S of the discriminant, which is now a divisor rather than a point, so a fibre type is attached to a codimension-one locus in B. The type over S is fixed by the three vanishing orders

$$(\mathrm{ord}_S f,\ \mathrm{ord}_S g,\ \mathrm{ord}_S \Delta), \tag{E.82}$$

and carries a definite local monodromy conjugacy class in $\mathrm{SL}(2,\mathbb{Z})$, the image of a small loop around S acting on H_1 of the fibre. Table E.5 is the classification.

Theorem E.25 (Kodaira 1963)**.** *The singular fibres of a relatively minimal elliptic fibration with section are exactly those of Table E.5. Each singular fibre is a divisor $F = \sum_i m_i \Theta_i$ whose components Θ_i are smooth rational curves with $\Theta_i \cdot \Theta_i = -2$; the intersection matrix of the components is minus the Cartan matrix of the corresponding affine ADE algebra, the multiplicities m_i are the marks (dual Kac labels) of that affine diagram, and the class of the fibre equals the imaginary root $\delta = \sum_i m_i \alpha_i$.*

Two structural facts do all the work of turning geometry into gauge theory. First, resolving the total space Y along a component S replaces the pinched fibre by the tree of $\mathbb{CP}^1$'s of Theorem E.25: transverse to S, Y develops precisely a du Val singularity $\mathbb{C}^2/\Gamma$ of Appendix E.2 fibred over S, and its resolution is the ADE story of Chapter 10 sitting inside the elliptic fibre. The relation $\Theta_i \cdot \Theta_j = -C_{ij}$, minus the ADE Cartan matrix, is the same equation that appeared for the exceptional spheres of a resolved du Val point. Second, one component of the resolved fibre, the one meeting the zero section σ, has multiplicity $m_0 = 1$ and is the *affine node*. Deleting it turns the affine Dynkin diagram $\widehat{\mathrm{ADE}}$ into the ordinary finite diagram, and that finite diagram is the gauge algebra: $\mathrm{I}_n \to \mathfrak{su}(n)$, $\mathrm{I}_n^* \to \mathfrak{so}(2n{+}8)$, and $\mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^* \to \mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8$ (Fig. E.1). The fibres I_1, II (nodal and cuspidal) are irreducible and carry no non-abelian symmetry.

The correspondence between fibre type and monodromy has a clean reading through the j-function. The I_n fibres have $j \to \infty$ with an order-n pole (multiplicative degeneration, $\tau \to i\infty$), and their monodromy is the parabolic T^n. The remaining types have finite j (additive degeneration): $\mathrm{II}, \mathrm{IV}, \mathrm{IV}^*, \mathrm{II}^*$ sit at $j = 0$, the hexagonal point $\tau = e^{2\pi i/3}$, and $\mathrm{III}, \mathrm{III}^*$ at $j = 1728$, the square point $\tau = i$; their monodromies are elliptic or the central $-\mathbf{1}$. This is why the exceptional $\mathfrak{e}_6, \mathfrak{e}_8$ fibres always live at the most symmetric torus.

Type	$(\mathrm{ord}_S f,\ \mathrm{ord}_S g,\ \mathrm{ord}_S \Delta)$	# comp.	monodromy	fibre graph	$\mathfrak{g}$ (split)
I_0	$(\geq 0,\ \geq 0,\ 0)$	1	$\mathbf{1}$	smooth	–
I_1	$(0,\ 0,\ 1)$	1	$\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$	nodal	–
I_n $(n\geq 2)$	$(0,\ 0,\ n)$	n	$\left(\begin{smallmatrix}1&n\\0&1\end{smallmatrix}\right)$	$\widehat{A}_{n-1}$	$\mathfrak{su}(n)$
II	$(\geq 1,\ 1,\ 2)$	1	$\left(\begin{smallmatrix}1&1\\-1&0\end{smallmatrix}\right)$	cuspidal	–
III	$(1,\ \geq 2,\ 3)$	2	$\left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)$	$\widehat{A}_1$	$\mathfrak{su}(2)$
IV	$(\geq 2,\ 2,\ 4)$	3	$\left(\begin{smallmatrix}0&1\\-1&-1\end{smallmatrix}\right)$	$\widehat{A}_2$	$\mathfrak{su}(3)$
I_0^*	$(\geq 2,\ \geq 3,\ 6)$	5	$\left(\begin{smallmatrix}-1&0\\0&-1\end{smallmatrix}\right)$	$\widehat{D}_4$	$\mathfrak{so}(8)$
I_n^* $(n\geq 1)$	$(2,\ 3,\ 6+n)$	$5+n$	$\left(\begin{smallmatrix}-1&-n\\0&-1\end{smallmatrix}\right)$	$\widehat{D}_{n+4}$	$\mathfrak{so}(2n{+}8)$
IV^*	$(\geq 3,\ 4,\ 8)$	7	$\left(\begin{smallmatrix}-1&-1\\1&0\end{smallmatrix}\right)$	$\widehat{E}_6$	$\mathfrak{e}_6$
III^*	$(3,\ \geq 5,\ 9)$	8	$\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$	$\widehat{E}_7$	$\mathfrak{e}_7$
II^*	$(\geq 4,\ 5,\ 10)$	9	$\left(\begin{smallmatrix}0&-1\\1&1\end{smallmatrix}\right)$	$\widehat{E}_8$	$\mathfrak{e}_8$
non-minimal	$(\geq 4,\ \geq 6,\ \geq 12)$	–	–	–	(blow up B)

Table E.5: Kodaira's classification of singular elliptic fibres. Along a discriminant component S the fibre type is fixed by the vanishing orders of (f, g, Δ); "# comp." is the number of irreducible $\mathbb{CP}^1$ components of the resolved fibre; the monodromy is a representative of the $\mathrm{SL}(2,\mathbb{Z})$ conjugacy class; the resolved fibre is a configuration of rational curves whose dual graph is the *affine* Dynkin diagram shown; and the last column is the gauge algebra in the split case (Appendix E.4.5). For every algebra-carrying type # comp. $=$ rank $\mathfrak{g}+1$ and $\mathrm{ord}_S\,\Delta$ equals the Euler number of the singular fibre. The final row is not a gauge enhancement but a signal that B must be blown up.

E.4.5 Tate's algorithm and non-simply-laced algebras

The intersection form of curves on a surface is symmetric, so the resolution of Theorem E.25 produces only simply-laced algebras. The non-simply-laced algebras $\mathfrak{so}(2k{+}1)$, $\mathfrak{sp}(k)$, $\mathfrak{f}_4$, $\mathfrak{g}_2$ appear when the exceptional $\mathbb{CP}^1$'s of the fibre are permuted by monodromy as one moves along S: the affine diagram is folded by the corresponding outer automorphism, and the algebra is the invariant non-simply-laced one. Whether this happens is read off the Weierstrass coefficients by *Tate's algorithm*, which is the practical tool for building a model with a prescribed gauge algebra.

The algorithm works from the long Weierstrass (Tate) form

$$y^2 + a_1\,xy + a_3\,y \;=\; x^3 + a_2\,x^2 + a_4\,x + a_6, \qquad a_i \in \Gamma(\bar{K}_B^{\,i}), \tag{E.83}$$

related to Eq. (E.73) by completing the square and cube. The intermediate quantities b_2, b_4, b_6 are the standard Weierstrass invariants assembled from the a_i; the passage to the short form (f, g) reads

$$b_2 = a_1^2{+}4a_2, \quad b_4 = a_1a_3{+}2a_4, \quad b_6 = a_3^2{+}4a_6, \quad f = -\tfrac{1}{48}(b_2^2{-}24b_4), \quad g = -\tfrac{1}{864}(-b_2^3{+}36b_2b_4{-}216b_6). \tag{E.84}$$

It is the vanishing orders of the a_i along S, funnelled through (E.84) into orders of (f, g, Δ), that Tate's algorithm converts into a Kodaira type. Prescribing those orders along a divisor $S = \{\zeta = 0\}$ fixes the type, and a discrete refinement, "split / semi-split / non-split", fixes the folding. Table E.6 lists the standard assignments (Bershadsky, Intriligator, Kachru, Morrison, Sadov and Vafa 1996; the complete table with all I_n^* refinements is in Katz, Morrison, Schafer-Nameki and Sethi 2011).

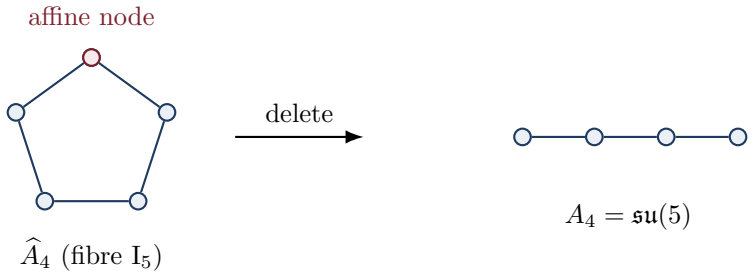


Figure E.1: An I_5 fibre resolves into five $\mathbb{CP}^1$'s in a necklace, the affine diagram $\widehat{A}_4$. The unique component meeting the zero section (the affine node, red) is deleted, leaving the finite A_4 diagram: the gauge algebra $\mathfrak{su}(5)$. The general rule is the same for every entry of Table E.5.

Kodaira type	$(\operatorname{ord} a_1, \operatorname{ord} a_2, \operatorname{ord} a_3, \operatorname{ord} a_4, \operatorname{ord} a_6)$	split $\mathfrak{g}$	non-/semi-split $\mathfrak{g}$
I_2	$(0,0,1,1,2)$	$\mathfrak{su}(2)$	–
I_{2k} $(k \geq 2)$	$(0,1,k,k,2k)$	$\mathfrak{su}(2k)$	$\mathfrak{sp}(k)$
I_{2k+1}	$(0,1,k,k{+}1,2k{+}1)$	$\mathfrak{su}(2k{+}1)$	$\mathfrak{sp}(k)$
III	$(1,1,1,1,2)$	$\mathfrak{su}(2)$	–
IV	$(1,1,1,2,3)$	$\mathfrak{su}(3)$	$\mathfrak{sp}(1)$
I_0^*	$(1,1,2,2,4)$	$\mathfrak{so}(8)$	$\mathfrak{so}(7)$ / $\mathfrak{g}_2$
I_n^* $(n \geq 1)$	$(1,1,2,3,4{+}\lfloor n/2 \rfloor)$	$\mathfrak{so}(2n{+}8)$	$\mathfrak{so}(2n{+}7)$
IV^*	$(1,2,2,3,5)$	$\mathfrak{e}_6$	$\mathfrak{f}_4$
III^*	$(1,2,3,3,5)$	$\mathfrak{e}_7$	–
II^*	$(1,2,3,4,5)$	$\mathfrak{e}_8$	–

Table E.6: Tate's algorithm: vanishing orders of the coefficients a_i of Eq. (E.83) along S, and the resulting split and non-split (or semi-split) gauge algebras. In the non-split rows the order of a_6 (or a_4) is lowered by one so that the relevant monodromy cover fails to factorise. The $\mathrm{III}^*, \mathrm{II}^*$ rows have no non-split variant because $\mathfrak{e}_7, \mathfrak{e}_8$ have no outer automorphism.

Proposition E.26 (Non-simply-laced algebras by monodromy)**.** *The split / semi-split / non-split distinction is whether a* monodromy cover*, a branched cover of S built from the leading Weierstrass coefficients, is reducible over the function field of S. Concretely a polynomial (a perfect-square test for the* I*,* I**,* IV*,* IV* *types, a cubic-factorisation test for* I_0^**) either factorises, leaving the simply-laced split algebra, or is irreducible, and its Galois monodromy folds the affine diagram to the non-simply-laced invariant algebra. The relevant foldings are*

$$\mathrm{I}_{2k}: \ \mathfrak{su}(2k) \to \mathfrak{sp}(k), \quad \mathrm{IV}: \ \mathfrak{su}(3) \to \mathfrak{su}(2), \quad \mathrm{I}_0^*: \ \mathfrak{so}(8) \to \mathfrak{so}(7) \to \mathfrak{g}_2, \quad \mathrm{IV}^*: \ \mathfrak{e}_6 \to \mathfrak{f}_4. \tag{E.85}$$

This is the delivery of the promise made in Chapter 10: the genuinely non-simply-laced $\mathfrak{f}_4$ and $\mathfrak{g}_2$, which no intersection form on a surface can produce, are reached because monodromy in the base permutes the collapsing cycles. The precise cover polynomials, the Grassi-Morrison presentation, and the extension to Mordell-Weil torsion are collected in Appendix G. The exceptional simply-laced algebras $\mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8$ need no folding; they are ordinary entries of Table E.5 and are exactly the ones no perturbative type IIB brane stack can produce, which is the point of Chapter 11.

Example E.27 (Reading a type off a model)**.** Let $S = \{\zeta = 0\}$ and suppose the Weierstrass

sections behave as $f = f_0 + \mathcal{O}(\zeta)$ with $f_0 \neq 0$, $g = g_0\zeta^4 + \mathcal{O}(\zeta^5)$, so $\mathrm{ord}_S f = 0$ and $\mathrm{ord}_S g = 4$. Then $\Delta = 4f^3 + 27g^2 = 4f_0^3 + \mathcal{O}(\zeta^8)$ is generically non-zero: no gauge symmetry, an I_0 fibre. Now tune $f_0 \to 0$ so that instead $\mathrm{ord}_S f = 2$, $\mathrm{ord}_S g = 3$. The leading $4f^3 \sim \zeta^6$ and $27g^2 \sim \zeta^6$ can combine to give $\mathrm{ord}_S \Delta = 6$: this is I_0^*, algebra $\mathfrak{so}(8)$ in the split case. Whether it is instead $\mathfrak{so}(7)$ or $\mathfrak{g}_2$ is decided by how the cubic $\psi^3 + f_2\psi + g_3$ in the leading coefficients $f_2 = (f/\zeta^2)|_S$, $g_3 = (g/\zeta^3)|_S$ factorises over S: three roots gives $\mathfrak{so}(8)$, one root gives $\mathfrak{so}(7)$, none gives $\mathfrak{g}_2$.

E.4.6 Matter and Yukawa loci

Gauge symmetry lives on the codimension-one components of $\{\Delta = 0\}$. Charged matter and its couplings live where those components meet, and the geometry of the enhancement fixes the representations. We record the geometric statements; the physics, the zero-mode counting, the fluxes, and the family number, is the business of Chapter 11.

Along a component S carrying algebra $\mathfrak{g}$, the fibre degenerates further where S meets another discriminant component, a curve $\Sigma \subset S$ of one higher codimension in B. There the singularity enhances by (at least) one rank, $\mathfrak{g} \to \mathfrak{g}'$, and localised matter appears in the representation dictated by the *Katz-Vafa* rule (Katz and Vafa 1996): decompose the adjoint of $\mathfrak{g}'$ under $\mathfrak{g} \oplus \mathfrak{u}(1)$ and the off-diagonal pieces are the matter multiplets trapped on Σ. For an $\mathfrak{su}(5)$ model the two enhancements $\mathfrak{su}(5) \to \mathfrak{so}(10)$ and $\mathfrak{su}(5) \to \mathfrak{su}(6)$ deposit a $\mathbf{10}$ and a $\mathbf{5}$ on the respective matter curves. At a codimension-three point where two such curves cross, the rank enhances by two, $\mathfrak{g} \to \mathfrak{g}''$, three matter multiplets become simultaneously light, and if their tensor product contains a singlet a cubic Yukawa coupling is localised there. The up-type coupling $\mathbf{10} \cdot \mathbf{10} \cdot \mathbf{5}$ requires an $\mathfrak{e}_6$ point and is the diagnostic for an exceptional enhancement (Chapter 11). This nested structure, gauge on divisors, matter on curves, Yukawas at points, is the same codimension dictionary as the geometric engineering of Chapter 10, now carried by the singular fibres of the elliptic fibration.

E.4.7 Global constraints: Hodge numbers, Euler characteristics, tadpoles

The resolved fibre components are new algebraic cycles, so the gauge data feeds back into the topology of Y. For a smooth elliptic Calabi-Yau $\hat{Y} \to B$ with gauge algebras $\mathfrak{g}_i$ on the discriminant components, the second Betti number is the Shioda-Tate-Wazir relation

$$h^{1,1}(\hat{Y}) \;=\; h^{1,1}(B) + 1 + \mathrm{rank}\ \mathrm{MW}(\hat{Y}) + \sum_i \mathrm{rank}\ \mathfrak{g}_i, \tag{E.86}$$

where the $+1$ is the zero section (equivalently the fibre class), the Mordell-Weil rank counts abelian factors (Theorem E.23), and each simple factor contributes its rank in resolved fibre components. This is the geometric bookkeeping behind the gauge group of the vacuum. Strictly, the right-hand side of (E.86) is the Néron-Severi (algebraic) rank $\rho(\hat{Y})$; it agrees with $h^{1,1}$ precisely when $h^{2,0}(\hat{Y}) = 0$, which holds for the elliptic threefold and fourfold but not for an elliptic K3, where the same count returns the Picard number $\rho = 2$ rather than $h^{1,1} = 20$ (Exercise E.7).

For the fourfold case $\dim_{\mathbb{C}} B = 3$, the Euler characteristic of a smooth Weierstrass model over a base B_3 is (Sethi, Vafa and Witten 1996; Klemm, Lian, Roan and Yau 1996)

$$\chi(\hat{Y}_4) \;=\; 12\int_{B_3} c_1(B_3)\, c_2(B_3) \;+\; 360\int_{B_3} c_1(B_3)^3, \tag{E.87}$$

for instance $\chi = 23328$ for $B_3 = \mathbb{CP}^3$. This number sets the fourform-flux tadpole. In F-theory on $\hat{Y}_4$ the G_4-flux obeys the Witten shifted quantisation $[G_4] + \frac{1}{2}c_2(\hat{Y}_4) \in H^4(\hat{Y}_4, \mathbb{Z})$, and consistency of the dual M-theory three-form (Chapter 11) requires

$$N_{\mathrm{D3}} + \frac{1}{2}\int_{\hat{Y}_4} G_4 \wedge G_4 \;=\; \frac{\chi(\hat{Y}_4)}{24}, \tag{E.88}$$

the number of spacetime-filling D3-branes plus the flux contribution being fixed by the geometry. The flux physics and its role in moduli stabilisation belong to Chapter 9; here (E.88) is the geometric constraint an elliptic fourfold must satisfy.

Remark E.28 (Beyond the lectures: non-minimal points and resolutions). The Weierstrass model of Appendix E.4.2 is singular over $\{\Delta = 0\}$, and F-theory requires a *crepant* (Calabi-Yau-preserving) resolution $\hat{Y} \to Y$. Such a resolution exists over codimension-one and codimension-two loci for every Kodaira type, but it can be obstructed at codimension three, where the fibre may be non-flat (a full surface component appears over a point). The last row of Table E.5, $(\mathrm{ord}_S f, \mathrm{ord}_S g, \mathrm{ord}_S \Delta) \geq (4, 6, 12)$, is a locus where no crepant resolution keeps Y Calabi-Yau: the remedy is to blow up the base B, which lowers the orders below the threshold. Physically these are not gauge enhancements but the sites of tensionless strings and, in six dimensions, of the strongly coupled superconformal fixed points that have no Lagrangian. They are a research frontier and we only flag them here.

Exercise E.7 (Fibre type from vanishing orders; ~20 min). Use Tables E.5 and E.6.

1. Along $S = \{\zeta = 0\}$ suppose f, g are finite and non-zero on S while Δ vanishes to order 6. Identify the Kodaira type and the split gauge algebra, and draw the resolved fibre as an affine diagram.
2. For $\mathrm{ord}_S f = 2$, $\mathrm{ord}_S g = 3$, $\mathrm{ord}_S \Delta = 6$, name the type, its split algebra, and the two non-simply-laced algebras the folding of Theorem E.26 can produce. State the cubic whose factorisation decides among $\mathfrak{so}(8), \mathfrak{so}(7), \mathfrak{g}_2$.
3. For $\mathrm{ord}_S f = 4$, $\mathrm{ord}_S g = 6$, $\mathrm{ord}_S \Delta = 12$, explain why this is not $\mathfrak{e}_8$ enhancement but a signal that B must be blown up (Theorem E.28).
4. Verify Theorem E.24: for $B = \mathbb{CP}^1$ the discriminant has degree 24. Then feed a generic (all-I_1) elliptic K3, which has MW $= 0$ and no gauge enhancement, into (E.86) and read off $h^{1,1}(\mathbb{CP}^1) + 1 = 2$. This is the *Picard number* (Néron-Severi rank) $\rho = 2$, generated by the zero section and the fibre class, *not* the Hodge number: every K3 has $h^{1,1} = 20$ (and $h^{2,0} = 1$). Confirm that (E.86) returns the algebraic rank, which coincides with $h^{1,1}$ only when $h^{2,0} = 0$, so it computes $h^{1,1}$ directly for the elliptic threefold and fourfold but only the Picard number for K3.

Appendix F

Worldsheet CFT Reference Card

F.1 Conformal field theory in two dimensions

This section is the toolbox that Chapters 1 and 2 draw on and that the rest of this appendix specialises. It collects, in reference form, the machinery of a two-dimensional conformal field theory (CFT): the conformal algebra, radial quantisation, the operator product expansion (OPE), primaries and descendants, the stress tensor and central charge, and the Ward identities. The free-field computations that produce the specific central charges live in Appendix F.2; the torus and modular invariance in Appendix F.3; the BRST cohomology and the no-ghost theorem in Appendix F.4. Conventions are those of Appendix A: Euclidean worldsheet, $z = e^{\tau_E - i\sigma}$, propagator $\langle X^\mu X^\nu \rangle = -\frac{\alpha'}{2}\eta^{\mu\nu}\ln|z|^2$, with α' the Regge slope. Following Appendix A, the antiholomorphic (right-moving) copy of any generator or field carries a tilde, $\tilde{\ell}_n, \tilde{L}_n, \tilde{c}, \tilde{T}$; a bar denotes complex conjugation, $\bar{z}, \bar{q}, \bar{\tau}$, and the antiholomorphic weight $\bar{h}$. Standard treatments at the same level are Belavin, Polyakov and Zamolodchikov (1984), abbreviated BPZ; Polchinski (1998, Vol. 1, Ch. 2 and 6); Ginsparg's Les Houches lectures (1988); and Di Francesco, Mathieu and Sénéchal (1997).

The big picture

In two dimensions the local conformal algebra is infinite dimensional: every holomorphic reparametrisation is a symmetry, so the theory factorises into a holomorphic and an antiholomorphic sector, each governed by a Virasoro algebra. Radial quantisation turns the apparatus into contour calculus, with operators and states the same data, symmetries acting by contour integrals of currents, and short-distance products captured by the OPE. A single number, the central charge c (defined in Appendix F.1.5), controls the quantum failure of scale invariance and appears identically in the TT OPE, in the anomalous transformation of T, and in the trace (Weyl) anomaly.

F.1.1 The conformal group in two dimensions

A conformal transformation preserves the metric up to a local scale, $g_{\alpha\beta} \to \Omega^2 g_{\alpha\beta}$. In $d > 2$ dimensions the solutions form the finite-dimensional group $\mathrm{SO}(d,2)$ (Lorentzian) or $\mathrm{SO}(d+1,1)$ (Euclidean), of dimension $\frac{1}{2}(d+1)(d+2)$: translations, rotations, one dilatation, and d special conformal transformations. In $d = 2$ the conformal Killing equation degenerates into the Cauchy-Riemann equations, so *every* holomorphic change of coordinate is conformal, and the local algebra is infinite dimensional.

Work in complex coordinates $z, \bar{z}$ and treat them as independent. A local field is a function of both; the equations of motion split it into a holomorphic (left-moving) part depending on z and

an antiholomorphic (right-moving) part depending on $\bar{z}$, and the two sectors decouple. For the free boson this factorisation is explicit: $\partial\bar{\partial}X^\mu = 0$ integrates to $X^\mu = X^\mu_{\mathrm{L}}(z) + X^\mu_{\mathrm{R}}(\bar{z})$, solved in Chapter 1. Infinitesimal holomorphic maps $z \to z + \epsilon(z)$ with $\epsilon(z) = -\sum_n \epsilon_n z^{n+1}$ are generated by

$$\ell_n = -z^{n+1}\partial_z, \qquad \tilde{\ell}_n = -\bar{z}^{n+1}\partial_{\bar{z}} \qquad (n \in \mathbb{Z}), \tag{F.1}$$

which obey two commuting copies of the *Witt algebra*,

$$[\ell_m, \ell_n] = (m-n)\,\ell_{m+n}, \qquad [\tilde{\ell}_m, \tilde{\ell}_n] = (m-n)\,\tilde{\ell}_{m+n}, \qquad [\ell_m, \tilde{\ell}_n] = 0. \tag{F.2}$$

The vector field ℓ_n is regular at $z = 0$ for $n \geq -1$ and, after the inversion $z \to 1/z$, regular at $z = \infty$ for $n \leq 1$. Only $\ell_{-1}, \ell_0, \ell_1$ (with their tilde counterparts) extend to globally defined holomorphic vector fields on the Riemann sphere $\mathbb{CP}^1 = \mathbb{C} \cup \{\infty\}$. They span $\mathfrak{sl}(2,\mathbb{C}) \oplus \overline{\mathfrak{sl}(2,\mathbb{C})}$ and exponentiate to the *global conformal group*, the Möbius transformations

$$z \;\longmapsto\; \frac{az+b}{cz+d}, \qquad ad - bc = 1, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{PSL}(2,\mathbb{C}). \tag{F.3}$$

Here $\ell_{-1} = -\partial_z$ is translation, $\ell_0 = -z\partial_z$ is (complex) scaling, and $\ell_1 = -z^2\partial_z$ is a special conformal transformation. The full Witt/Virasoro algebra is the local symmetry that constrains the operator algebra; its global $\mathrm{PSL}(2,\mathbb{C})$ subgroup is the anomaly-free symmetry of the sphere, and it fixes the low-point functions and the vacuum $|0\rangle$, the state named $\mathrm{SL}(2,\mathbb{C})$-invariant after that global group.

On radial quantisation (Appendix F.1.2) each ℓ_n is represented by an operator L_n, the mode of the stress tensor T defined in Eq. (F.4), with $\tilde{L}_n$ the antiholomorphic copy. Anticipating that passage: on the plane of Eq. (1.48), $L_0 + \tilde{L}_0$ generates dilatations and $i(L_0 - \tilde{L}_0)$ generates rotations, so on a state (Theorem 2.2) the eigenvalue of $L_0 + \tilde{L}_0$ is the scaling dimension $\Delta = h + \bar{h}$ and that of $L_0 - \tilde{L}_0$ is the spin $s = h - \bar{h}$.

F.1.2 Radial quantisation, modes, and conjugation

Radial quantisation was set up in Section 1.3.1: Euclidean time slices are circles about the origin, dilatation is the Hamiltonian, and time ordering becomes *radial ordering*, the operator at larger $|z|$ standing to the left. A holomorphic field $\mathcal{O}(z)$ of weight h has the Laurent expansion and inverse

$$\mathcal{O}(z) = \sum_{n\in\mathbb{Z}} \mathcal{O}_n\, z^{-n-h}, \qquad \mathcal{O}_n = \oint_0 \frac{\mathrm{d}z}{2\pi i}\, z^{\,n+h-1}\,\mathcal{O}(z), \tag{F.4}$$

with an identical antiholomorphic expansion in $\bar{z}$. The mode $\mathcal{O}_n$ lowers the L_0 eigenvalue by n. For the stress tensor ($h = 2$) this reproduces $L_n = \oint \frac{\mathrm{d}z}{2\pi i} z^{n+1} T(z)$ of Eq. (1.58); for the current ∂X^μ ($h = 1$) it reproduces the oscillator modes α^μ_n of Eq. (1.53).

Cylinder frame and Casimir energy. The plane is not the cylinder. Under the map $z = e^{-iw}$ to the cylinder coordinate $w \sim w + 2\pi$, the stress tensor picks up the anomalous shift derived in Appendix F.1.5,

$$T_{\mathrm{cyl}}(w) = -z^2\, T(z) + \frac{c}{24}, \tag{F.5}$$

so the generator of cylinder time translations, the physical worldsheet Hamiltonian, is

$$H = L_0 + \tilde{L}_0 - \frac{c + \tilde{c}}{24}. \tag{F.6}$$

The constant $-c/24$, with c the central charge of Appendix F.1.5, is the Casimir energy of the CFT on a circle; it fixes the intercept, the normal-ordering constant a, of Section 1.4.5, and seeds the η-function in the partition function of Appendix F.3.

Conjugation and the inner product. Radial quantisation places past and future at 0 and ∞, so Hermitian conjugation must map $z \to 1/\bar{z}$ (i.e. $\tau_E \to -\tau_E$). For a Hermitian holomorphic field of weight h,

$$[\mathcal{O}(z)]^\dagger = \bar{z}^{-2h}\,\mathcal{O}(1/\bar{z}) \qquad \Longrightarrow \qquad \mathcal{O}_n^\dagger = \mathcal{O}_{-n}, \tag{F.7}$$

the second form obtained by substituting the mode expansion Eq. (F.4). In particular $L_n^\dagger = L_{-n}$ and $\alpha_n^{\mu\dagger} = \alpha_{-n}^\mu$; unitarity of the CFT is the statement that this adjoint has no negative-norm states in its physical subspace, the content of the no-ghost theorem of Appendix F.4. The out-state is prepared at infinity by the inversion $z \to 1/z$: for a primary $\mathcal{O}$ of weight $(h, \bar{h})$,

$$\langle \mathcal{O}| = \lim_{z,\bar{z}\to\infty} z^{2h}\,\bar{z}^{2\bar{h}}\ \langle 0|\ \mathcal{O}(z,\bar{z}), \tag{F.8}$$

the BPZ conjugate of Eq. (2.12). The powers $z^{2h}\bar{z}^{2\bar{h}}$ are the Jacobian of the inversion acting on a weight-$(h,\bar{h})$ tensor; they are exactly what makes $\langle\mathcal{O}|\mathcal{O}\rangle$ finite and coordinate independent. The resulting bilinear pairing is the two-point function of Appendix F.1.6, normalised so that $\langle\mathcal{O}|\mathcal{O}\rangle = C$ with C the two-point coefficient.

F.1.3 The operator product expansion

The OPE writes a radially ordered product of two local operators, as their separation shrinks, as a Laurent series in the separation with local-operator coefficients, Eq. (1.52). Two facts promote it from a short-distance approximation to an exact operator identity inside correlators.

Proposition F.1 (Convergence and associativity)**.** *In a unitary CFT with a discrete, bounded-below spectrum, the OPE of two operators inserted at w_1, w_2 converges, as an expansion of the state $\mathcal{O}_1(w_1)\mathcal{O}_2(w_2)|0\rangle$ in Virasoro descendants (defined in Appendix F.1.4), on any disc about w_2 that excludes every other insertion; its radius is the distance to the nearest other operator. The expansion is associative: $(\mathcal{O}_1\mathcal{O}_2)\mathcal{O}_3 = \mathcal{O}_1(\mathcal{O}_2\mathcal{O}_3)$, which as a constraint on the coefficients is crossing symmetry.*

This is the convergence statement deferred from Section 2.1. It follows from the completeness of the highest-weight decomposition (Appendix F.1.4) and reflection positivity of the radially quantised inner product; see Polchinski §2.1–2.2 and, for the axiomatic version, the vertex-operator-algebra literature. We use it whenever an operator product is replaced by a sum of local terms inside a correlator.

Normal ordering. The regular part of the self-OPE defines the normal-ordered product $:AB:(w)$: subtract the singular terms of $A(z)B(w)$ before taking $z \to w$. Equivalently, in modes, push annihilators to the right (Appendix A). For free fields the two prescriptions agree, and Wick's theorem gives the entire product as the normal-ordered product plus all contractions,

$$A(z)\,B(w) = :A(z)B(w): + \ \text{(sum over contractions)}, \qquad \text{contraction} = \langle A(z)B(w)\rangle, \tag{F.9}$$

each contraction replacing a pair of fields by their two-point function; this drives every free-field OPE of Appendix F.2.

OPE as an algebra. The singular terms of an OPE are equivalent to the (anti)commutators of the modes. For holomorphic fields A, B of weights h_A, h_B,

$$[A_m, B_n\} = \oint_0 \frac{\mathrm{d}w}{2\pi i}\, w^{\,n+h_B-1} \oint_w \frac{\mathrm{d}z}{2\pi i}\, z^{\,m+h_A-1}\, A(z)\,B(w), \tag{F.10}$$

with the anticommutator taken when both fields are fermionic. The inner contour picks up the residues of the singular OPE terms; the outer contour reassembles them into modes. Feeding the $\partial X\,\partial X$ OPE of Eq. (F.22) into Eq. (F.10) gives $[\alpha_m^\mu, \alpha_n^\nu] = m\,\delta_{m+n,0}\eta^{\mu\nu}$, and feeding the TT OPE of Eq. (F.12) gives the Virasoro algebra of Eq. (F.13), worked out in Theorem F.3.

F.1.4 Primaries, quasi-primaries, and descendants

A *primary* of weight $(h,\bar{h})$ transforms as a tensor density under every local conformal map $z \to z'(z)$, $\bar{z} \to \bar{z}'(\bar{z})$,

$$\mathcal{O}(z,\bar{z}) = \left(\frac{\partial z'}{\partial z}\right)^{h} \left(\frac{\partial \bar{z}'}{\partial \bar{z}}\right)^{\bar{h}} \mathcal{O}'(z',\bar{z}'), \tag{F.11}$$

the finite form of the defining OPE $T(z)\mathcal{O}(w) \sim h\,\mathcal{O}(w)/(z-w)^2 + \partial\mathcal{O}(w)/(z-w)$, Eq. (2.7). Under the state-operator map (Theorem 2.1) a primary corresponds to a highest-weight state, $L_0\,|\mathcal{O}\rangle = h\,|\mathcal{O}\rangle$ with $L_{n\geq 1}\,|\mathcal{O}\rangle = 0$, and likewise for the antiholomorphic (tilde) sector. A *quasi-primary* transforms as Eq. (F.11) only under the global $\mathrm{PSL}(2,\mathbb{C})$, equivalently is an L_0 eigenstate annihilated by L_1 (but not necessarily by $L_{n\geq 2}$). Every primary is quasi-primary; the stress tensor is quasi-primary but not primary once $c \neq 0$, because it acquires the Schwarzian anomaly of Appendix F.1.5.

Acting with the raising modes L_{-n} $(n \geq 1)$ on a primary builds its *descendants*. The primary and its descendants form a *conformal family* (Verma module), graded by level $N = \sum_i n_i$; $L_{-1}\mathcal{O} = \partial\mathcal{O}$ is the first descendant, Eq. (2.10). The full Hilbert space is a sum of such modules, one per primary. Characters $\mathrm{Tr}\; q^{L_0 - c/24}$ of these modules assemble the partition function of Appendix F.3.

Remark F.2 (Beyond the lectures: unitarity and null states). For a Verma module to have no negative-norm states, unitarity requires $c \geq 0$ and $h \geq 0$. When $0 \leq c < 1$ the allowed weights are quantised: $c = 1 - \frac{6}{m(m+1)}$ with $m = 3, 4, \ldots$ and h on the Kac grid, the minimal models. A Verma module contains a null descendant (a state orthogonal to everything, which generates a submodule to quotient out) exactly when the Kac determinant vanishes; the resulting decoupling equation is a differential equation for correlators. None of the worldsheet CFTs of these notes is a minimal model, but the null-state machinery is the same one that makes the physical-state (BRST) analysis of Appendix F.4 work: spurious states are null descendants.

F.1.5 The stress tensor, central charge, and anomalies

The holomorphic stress tensor $T(z)$ generates conformal transformations, and its self-OPE defines the central charge,

$$T(z)\,T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2\,T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}, \tag{F.12}$$

recalled from Eq. (1.57) with c now generic. Through Eq. (F.10) this is the Virasoro algebra

$$[L_m, L_n] = (m-n)\,L_{m+n} + \frac{c}{12}\,m(m^2-1)\,\delta_{m+n,0}, \tag{F.13}$$

Eq. (1.61), and an identical copy for $\tilde{L}_n$ with $\tilde{c}$.

Example F.3 (Virasoro central term from the TT OPE)**.** Put $A = B = T$, so $h_A = h_B = 2$, in the double contour Eq. (F.10). The inner integral of $z^{m+1}T(z)T(w)$ around w collects the residues of the four terms in Eq. (F.12),

$$\oint_w \frac{\mathrm{d}z}{2\pi i}\, z^{m+1}\,T(z)\,T(w) = \frac{c}{12}\,(m^3-m)\,w^{m-2} + 2(m+1)\,w^m\,T(w) + w^{m+1}\,\partial T(w).$$

The outer integral $\oint_0 \frac{\mathrm{d}w}{2\pi i}\, w^{n+1}(\cdots)$ then gives $\frac{c}{12}(m^3-m)\,\delta_{m+n,0}$ from the first term and, after one integration by parts on the third, $[\,2(m+1) - (m+n+2)\,]L_{m+n} = (m-n)L_{m+n}$ from the other two. This is the Virasoro algebra Eq. (F.13); the $(z-w)^{-4}$ term of the OPE is the sole source of the central charge.

Central charges of decoupled sectors add. The free boson has $c = 1$ per scalar (so $c = D$ for X^μ); the values for the bc, $\beta\gamma$ and free-fermion systems are assembled in Appendix F.2, and their sum against the ghost sector is the critical-dimension condition of Appendix F.4.

Finite transformation and the Schwarzian. The $(z-w)^{-4}$ term spoils the tensor law: under a finite map $z \mapsto w(z)$, with $w' = \mathrm{d}w/\mathrm{d}z$,

$$(w')^2\, T'(w) = T(z) - \frac{c}{12}\,\{w;z\}, \qquad \{w;z\} := \frac{w'''}{w'} - \frac{3}{2}\left(\frac{w''}{w'}\right)^2, \tag{F.14}$$

where $\{w;z\}$ is the Schwarzian derivative. It vanishes precisely on the Möbius maps Eq. (F.3), so T is a good quasi-primary tensor under the global group and anomalous only under the rest. The plane-to-cylinder map $w = i\ln z$ has $\{w;z\} = 1/(2z^2)$, which inserted into Eq. (F.14) gives the shift Eq. (F.5) and hence the Casimir energy Eq. (F.6).

Example F.4 (The cylinder Casimir energy)**.** Take $w = i\ln z$, so $w' = i/z$, $w'' = -i/z^2$, $w''' = 2i/z^3$. Then $w'''/w' = 2/z^2$ and $(w''/w')^2 = 1/z^2$, giving $\{w;z\} = 2/z^2 - \frac{3}{2}\cdot 1/z^2 = 1/(2z^2)$. Substituting into Eq. (F.14), $T'(w) = (w')^{-2}\big[T - \frac{c}{12}\{w;z\}\big] = -z^2\,T(z) + c/24$, which is Eq. (F.5). Reading off the zero mode, $L_0^{\mathrm{cyl}} = L_0 - c/24$.

The trace (Weyl) anomaly. The same c measures the failure of Weyl invariance on a curved worldsheet. With the curvature convention of Appendix A (a round sphere has $R^{(2)} > 0$),

$$\langle T^a{}_a \rangle = -\frac{c}{12}\,R^{(2)}, \tag{F.15}$$

the anomaly asserted at Eq. (1.62). Classically $T^a{}_a = 0$; the anomaly is a one-loop effect, and integrating it against a Weyl factor $g_{\alpha\beta} = e^{2\omega}\hat{g}_{\alpha\beta}$ produces the Liouville action

$$S_{\mathrm{L}}[\omega] = \frac{c}{24\pi}\int \mathrm{d}^2\sigma\,\sqrt{\hat{g}}\,\big(\hat{g}^{\alpha\beta}\partial_\alpha\omega\,\partial_\beta\omega + \hat{R}^{(2)}\omega\big), \tag{F.16}$$

whose coefficient is fixed by the same c (the Weyl integration is carried out in Polchinski §3.4). Two derivations give Eq. (F.15): the heat-kernel/Seeley-DeWitt computation of the regularised determinant (Polchinski §3.4), and the short-distance $\langle T(z)T(0)\rangle = c/(2z^4)$ continued to a curved background. The three appearances of c, in Eqs. (F.12), (F.14) and (F.15), are one anomaly seen at short distance, under finite maps, and on curved backgrounds. Its cancellation between matter and ghosts is what makes the gauge-fixed string independent of the fiducial metric.

Careful

Sign and normalisation conventions for the Schwarzian term, the Casimir shift, and the trace anomaly differ across the literature by the orientation of the plane-cylinder map and the curvature convention. The set used here, Eqs. (F.5), (F.14) and (F.15), is internally consistent: it yields $L_0^{\mathrm{cyl}} = L_0 - c/24$ and the sign of Eq. (1.62). Convert any imported formula to this set before combining it with these notes.

F.1.6 Ward identities and low-point functions

A continuous symmetry with holomorphic current $j(z)$ acts on a field by the contour charge $\delta_\epsilon\mathcal{O}(w) = \oint_w \frac{\mathrm{d}z}{2\pi i}\,\epsilon(z)\,j(z)\,\mathcal{O}(w)$, the residue of the current-field OPE. Applied to $j = T$ and a set of primaries, contour deformation gives the master *conformal Ward identity*: as a function of z, the correlator of T with primaries is meromorphic with poles only at the insertions,

$$\Big\langle T(z)\prod_i \mathcal{O}_i(w_i,\bar{w}_i)\Big\rangle = \sum_i\left[\frac{h_i}{(z-w_i)^2} + \frac{1}{z-w_i}\,\partial_{w_i}\right]\Big\langle\prod_j \mathcal{O}_j(w_j,\bar{w}_j)\Big\rangle, \tag{F.17}$$

plus the antiholomorphic copy with $\tilde{T}$ and $\bar{h}_i$. This single identity generates every correlator involving descendants from the correlator of primaries; Chapter 4 uses it without re-deriving it.

Global constraints. Since T has weight 2, the left side of Eq. (F.17) falls off as z^{-4} at infinity. Demanding that the right side match, the coefficients of z^{-1}, z^{-2}, z^{-3} in its large-z expansion must vanish, which are the $\mathfrak{sl}(2)$ Ward identities

$$\sum_i \partial_{w_i}\langle\cdots\rangle = 0, \qquad \sum_i \left(w_i\partial_{w_i} + h_i\right)\langle\cdots\rangle = 0, \qquad \sum_i \left(w_i^2\partial_{w_i} + 2h_i w_i\right)\langle\cdots\rangle = 0, \tag{F.18}$$

from translations, dilatations and special conformal transformations respectively. They fix the low-point functions completely. A one-point function vanishes unless the operator is the identity; the two- and three-point functions of quasi-primaries are pinned to

$$\left\langle \mathcal{O}_1(z_1,\bar{z}_1)\,\mathcal{O}_2(z_2,\bar{z}_2)\right\rangle = \frac{C_{12}\,\delta_{h_1h_2}\,\delta_{\bar{h}_1\bar{h}_2}}{z_{12}^{2h_1}\,\bar{z}_{12}^{2\bar{h}_1}}, \tag{F.19}$$

$$\left\langle \mathcal{O}_1\,\mathcal{O}_2\,\mathcal{O}_3\right\rangle = C_{123}\prod_{i<j} z_{ij}^{-(h_i+h_j-h_k)}\,\bar{z}_{ij}^{-(\bar{h}_i+\bar{h}_j-\bar{h}_k)}, \tag{F.20}$$

with $z_{ij} = z_i - z_j$ and k the index distinct from i, j. Beyond three points the position dependence is fixed only up to a function of the Möbius-invariant cross-ratios; the four-point function carries the dynamical data, which for free fields is the Koba-Nielsen product of Eq. (2.30).

Internal symmetries. The same contour argument applied to an internal current gives ordinary charge conservation. The translation symmetry $X^\mu \to X^\mu + a^\mu$ of the free boson has current ∂X^μ, and its Ward identity is the statement that a correlator of exponentials vanishes unless total momentum is conserved: the $\delta^D(\sum_i k_i)$ in front of Eq. (2.30). That is the only Ward identity of the matter CFT the amplitudes of Chapter 4 need beyond the conformal one.

Exercise F.1 (Low-point functions from the global algebra)**.** *(Warm-up.)* Take holomorphic quasi-primaries for brevity. (a) From the first identity in Eq. (F.18) deduce that $\langle\mathcal{O}_1\mathcal{O}_2\rangle$ depends only on z_{12}; from the second that it scales as $z_{12}^{-(h_1+h_2)}$; from the third that it vanishes unless $h_1 = h_2$, recovering Eq. (F.19). (b) Repeat for three points: show Eq. (F.18) forces the exponents in Eq. (F.20) to be $-(h_i + h_j - h_k)$, so the three-point function is fixed up to the single constant C_{123}. (c) Explain why the four-point function is not fixed, by exhibiting the Möbius-invariant cross-ratio $x = z_{12}z_{34}/(z_{13}z_{24})$ that the identities cannot constrain.

Exercise F.2 (Central charge and oscillators of the free boson)**.** *(Challenging.)* For D free bosons the holomorphic stress tensor is $T = -\frac{1}{\alpha'}{:}\partial X^\mu \partial X_\mu{:}$ (Appendix F.2). (a) Wick-contract $\langle T(z)T(0)\rangle$ with the propagator of Appendix A; keeping the double contraction, read off $c = D$ from the z^{-4} coefficient against Eq. (F.12). (b) Feed the $\partial X\,\partial X$ OPE of Eq. (F.22) through the double contour Eq. (F.10), following Theorem F.3, to recover $[\alpha_m^\mu, \alpha_n^\nu] = m\,\delta_{m+n,0}\eta^{\mu\nu}$, and identify which OPE term supplies the central $(m\,\delta)$ piece.

Object	Reference formula
Witt/Virasoro algebra	Eq. (F.2), Eq. (F.13)
Global group	$\mathrm{PSL}(2,\mathbb{C})$, Möbius maps Eq. (F.3)
Mode expansion (weight h)	$\mathcal{O}(z)=\sum_n \mathcal{O}_n z^{-n-h}$, Eq. (F.4)
Mode (anti)commutator	double contour Eq. (F.10)
Primary transformation	tensor law Eq. (F.11)
T transformation	Schwarzian Eq. (F.14); cylinder $L_0 - c/24$
Central charge, three faces	TT OPE, Schwarzian, trace anomaly Eq. (F.15)
Conformal Ward identity	T-insertion Eq. (F.17)
Fixed correlators	two- and three-point Eqs. (F.19) and (F.20)

The specific systems that realise this toolbox, with their central charges and propagators, follow in Appendix F.2.

F.2 Free-field systems and their central charges

Every worldsheet theory quantised in these notes is a product of a handful of free conformal field theories, glued at the zero modes and constrained by a gauge symmetry. This section is the parts list. For each system we record the action, the propagator, the holomorphic stress tensor, the central charge, the mode algebra and the vacua, in a form that can be looked up without reading the derivation. The machinery that gives these words meaning, the operator product expansion, radial quantisation and the definition of c through the TT OPE, is Appendix F.1; the statement that the total c must cancel, and its equivalence to nilpotency of the BRST charge, is Appendix F.4. Here we take that statement as given and do the bookkeeping it demands.

Two facts organise everything. Central charges of decoupled sectors add, because a decoupled sector contributes its own $c/2$ to the quartic pole of the total TT OPE and nothing to the cross terms. And the holomorphic and antiholomorphic sectors carry independent copies of the Virasoro algebra, so c and $\tilde{c}$ are separate ledgers that must each balance. All conventions (the sign of the propagator, the complex frame $z = e^{\tau_E - i\sigma}$, the mode normalisations) are those of Appendix A. Throughout, the antiholomorphic (right-moving) copy of any mode or central charge carries a tilde ($\tilde{\alpha}^\mu_n$, $\tilde{L}_n$, $\tilde{c}$), as in Appendix A; a bar denotes complex conjugation ($\bar{z}$, $\bar{\tau}$) and antiholomorphic weights ($\bar{h}$).

Four numbers

The whole of the critical-dimension arithmetic is four central charges, one per building block:

$$\underbrace{X^\mu}_{c=1} \qquad \underbrace{\psi^\mu}_{c=\frac{1}{2}} \qquad \underbrace{bc}_{c=-26} \qquad \underbrace{\beta\gamma}_{c=+11} .$$

A free scalar contributes $+1$, a free Majorana fermion $+\frac{1}{2}$, the reparametrisation ghosts -26, the superconformal ghosts $+11$. The bosonic string balances $D - 26 = 0$ and the superstring $\frac{3}{2}D - 26 + 11 = 0$; nothing else enters the count.

F.2.1 The free boson X^μ

The matter of the bosonic string is D free scalars $X^\mu(z,\bar z)$, $\mu = 0,\dots,D-1$, with the propagator fixed in Appendix A,

$$\langle X^\mu(z,\bar z)\, X^\nu(w,\bar w)\rangle = -\frac{\alpha'}{2}\,\eta^{\mu\nu}\ln|z-w|^2. \tag{F.21}$$

The field itself is not a good conformal operator (the logarithm is not single-valued and carries no definite weight); the good local operators are its holomorphic current $\partial X^\mu(z)$, its antiholomorphic current $\bar\partial X^\mu(\bar z)$, and the normal-ordered exponentials $:e^{ik\cdot X}:$. Differentiating Eq. (F.21),

$$\partial X^\mu(z)\,\partial X^\nu(w) \sim -\frac{\alpha'}{2}\,\frac{\eta^{\mu\nu}}{(z-w)^2}, \qquad T(z) = -\frac{1}{\alpha'}\,:\partial X^\mu\,\partial X_\mu:(z), \tag{F.22}$$

so ∂X^μ is a primary of weight $(1,0)$ and $:e^{ik\cdot X}:$ a primary of weight $(\frac{\alpha'}{4}k^2, \frac{\alpha'}{4}k^2)$. Feeding T to itself gives the quartic pole $\frac{D/2}{(z-w)^4}$ (Theorem F.5), hence

$$c = \tilde c = D, \qquad \text{one unit of central charge per scalar.} \tag{F.23}$$

Example F.5 (Central charge from a TT double contraction)**.** The quartic pole in Eq. (F.23) is a single Wick contraction. With $T = -\frac{1}{\alpha'}:\partial X^\mu \partial X_\mu:$ and the current OPE Eq. (F.22), the only c-number term in $T(z)T(w)$ is the double contraction of both currents in $T(z)$ against both in $T(w)$. There are two independent pairings, and the two propagator signs cancel:

$$T(z)\,T(w)\Big|_{\text{double}} = \frac{1}{\alpha'^2}\cdot 2\,\langle\partial X^\mu(z)\,\partial X^\nu(w)\rangle\,\langle\partial X_\mu(z)\,\partial X_\nu(w)\rangle = \frac{1}{\alpha'^2}\cdot 2\left(\frac{\alpha'}{2}\right)^2\frac{\eta^{\mu\nu}\eta_{\mu\nu}}{(z-w)^4} = \frac{D/2}{(z-w)^4}, \tag{F.24}$$

using $\eta^{\mu\nu}\eta_{\mu\nu} = D$. Matching to the standard normalisation $T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \dots$ of Appendix F.1 gives $c = D$. The same contraction, with the appropriate propagator and statistics, delivers every entry of Table F.1; Exercise F.4 runs it for the fermion and the two first-order systems.

The current expands in modes as in Appendix A,

$$\partial X^\mu(z) = -i\sqrt{\tfrac{\alpha'}{2}}\sum_{n\in\mathbb{Z}}\frac{\alpha^\mu_n}{z^{n+1}}, \qquad [\alpha^\mu_m,\alpha^\nu_n] = m\,\delta_{m+n,0}\,\eta^{\mu\nu}, \tag{F.25}$$

with $\alpha^\mu_0 = \sqrt{\alpha'/2}\,p^\mu$ for the closed string and an independent antiholomorphic copy $\tilde\alpha^\mu_n$. The open string carries a single set, with $\alpha^\mu_0 = \sqrt{2\alpha'}\,p^\mu$ and the left and right movers identified at the endpoints. The $\mathrm{SL}(2,\mathbb{C})$-invariant vacuum $|0\rangle$, so named for the cover of the global conformal group $\mathrm{PSL}(2,\mathbb{C})$ that fixes it, is the zero-momentum state annihilated by α^μ_n for $n \geq 0$ (and $\tilde\alpha^\mu_n$, $n \geq 0$); the momentum ground states $|0;k\rangle = :e^{ik\cdot X}:(0)|0\rangle$ are its ∂X-descendant-free cousins, one for each k^μ.

The timelike direction is the source of all the trouble the ghosts exist to fix. Because $\eta^{00} = -1$, the timelike oscillators obey $[\alpha^0_m,\alpha^0_{-m}] = -m$, so $\alpha^0_{-m}|0\rangle$ has negative norm. The free boson theory on its own is not unitary; unitarity is restored only after the Virasoro (equivalently BRST) constraints remove the timelike and longitudinal excitations, the content of the no-ghost theorem in Appendix F.4.

Remark F.6 (Beyond the lectures: compactification and the Narain lattice)*.* When a coordinate is compactified on a circle of radius R, the zero mode splits into independent left and right momenta $p_{\mathrm{L},R} = \frac{n}{R} \pm \frac{wR}{\alpha'}$ with n (momentum) and w (winding) integers, and the allowed $(p_\mathrm{L}, p_\mathrm{R})$ fill an even self-dual lattice of signature $(1,1)$. On a d-torus this is the Narain lattice $\Gamma_{d,d}$, even and self-dual by construction, and the moduli space of inequivalent tori is $O(d,d)/[O(d)\times O(d)]$

quotiented by the duality group $O(d,d;\mathbb{Z})$. The central charge stays $c=\tilde{c}=d$: compactification changes the spectrum of momenta, not the number of bosons. Modular invariance of the torus partition function is exactly the statement that the momentum lattice be even and self-dual, and that is the subject of Appendix F.3, with the lattice theory in Appendix G.

F.2.2 First-order systems: the bc ghosts

The reparametrisation ghosts are the leading example of a first-order system: an anticommuting pair $b(z), c(z)$ of holomorphic weights $(\lambda, 1-\lambda)$, with action, propagator and stress tensor

$$S=\frac{1}{2\pi}\int \mathrm{d}^2z\, b\,\bar{\partial}c, \qquad b(z)\,c(w)\sim\frac{1}{z-w}\sim c(z)\,b(w), \qquad T=-\lambda:b\,\partial c:+(1-\lambda):(\partial b)\,c:, \tag{F.26}$$

with $b\,b$ and $c\,c$ regular. The self-OPE of T has a quartic pole whose coefficient is $c/2$; the anticommuting statistics fix its sign:

$$\boxed{c_{bc}(\lambda)=1-3(2\lambda-1)^2 \;=\; -2\,(6\lambda^2-6\lambda+1).} \tag{F.27}$$

This is the double contraction of Eq. (F.26) with itself, with the sign set by the fermionic reordering; it is worked in Exercise F.4, and in the literature in Polchinski (Vol. 1, §2.5). The reparametrisation ghosts have $\lambda=2$ (the antighost $b_{\alpha\beta}$ is the symmetric traceless conjugate to the metric fluctuation, a weight-2 tensor; the diffeomorphism ghost c^{α} is a weight-(-1) vector), so $c_{bc}(2)=-26$. The value is not tuned: it is the tensor type of the symmetry being gauged. Their modes and vacua are

$$b(z)=\sum_{n\in\mathbb{Z}}\frac{b_n}{z^{n+2}}, \qquad c(z)=\sum_{n\in\mathbb{Z}}\frac{c_n}{z^{n-1}}, \qquad \{b_m,c_n\}=\delta_{m+n,0}, \tag{F.28}$$

all other anticommutators vanishing. The $\mathrm{SL}(2,\mathbb{C})$-invariant vacuum $|0\rangle$ is annihilated by b_n for $n\geq -1$ and c_n for $n\geq 2$. The three modes c_{-1},c_0,c_1 neither create nor annihilate it cleanly; they are the operator shadow of the three conformal Killing vectors of the sphere. Acting on $|0\rangle$ they build two degenerate ground states of the mode algebra,

$$|{\downarrow}\rangle=c_1|0\rangle, \qquad |{\uparrow}\rangle=c_0\,c_1|0\rangle=c_0\,|{\downarrow}\rangle\,, \qquad b_0\,|{\downarrow}\rangle=0, \quad b_0\,|{\uparrow}\rangle=|{\downarrow}\rangle\,, \tag{F.29}$$

exchanged by the c_0 zero mode. The physical closed-string ground state pairs the two chiralities: it is $c_1\tilde{c}_1\,|0;k\rangle$, the $c\bar{c}$ dressing that makes a matter primary into a fixed vertex operator (Chapter 4). The ghost-number current $j_{\mathrm{gh}}=-:bc:$ assigns charge $+1$ to c and -1 to b; it is anomalous, with a background charge $Q=1-2\lambda$ appearing as a cubic pole $T(z)j_{\mathrm{gh}}(w)\sim\frac{1-2\lambda}{(z-w)^3}+\dots$, which for $\lambda=2$ reads $Q=-3$, whose magnitude reappears as the three c zero modes counted by Eq. (F.30).

Remark F.7 (Beyond the lectures: zero modes and the Riemann-Roch count). On a genus-g surface the ghost zero modes are counted by the index theorem

$$\#(c\text{ zero modes})-\#(b\text{ zero modes})=(2\lambda-1)(1-g), \tag{F.30}$$

which for $\lambda=2$ gives $3(1-g)$: three c zero modes and no b zero mode on the sphere (the three conformal Killing vectors that fix three vertex operators), one of each on the torus (one modulus, one Killing vector). This mismatch is what the $c\bar{c}$ dressings and the b-ghost insertions in the amplitude measure are there to soak up; the counting is used without comment in Chapter 4. The full statement, and the bosonisation $b=e^{-\phi_{bc}}$, $c=e^{+\phi_{bc}}$ (a boson ϕ_{bc} distinct from the worldsheet coordinate) that turns ghost number into a momentum, is standard (Polchinski, chapters 6 and 10).

F.2.3 First-order systems: the $\beta\gamma$ superghosts

Gauging worldsheet supersymmetry and super-Weyl invariance produces a second first-order system, the superconformal ghosts $\beta(z), \gamma(z)$. They are the exact superpartners of (b, c) with one decisive difference: they *commute*. A commuting first-order pair of weights $(\lambda, 1-\lambda)$ has the same stress tensor structure as Eq. (F.26) but the opposite-sign central term,

$$\gamma(z)\,\beta(w) \sim \frac{1}{z-w}, \quad \beta(z)\,\gamma(w) \sim -\frac{1}{z-w}, \qquad \boxed{c_{\beta\gamma}(\lambda) = -1 + 3(2\lambda-1)^2.} \tag{F.31}$$

The sign flip relative to Eq. (F.27) is the commuting statistics; the same contraction gives Eq. (F.31), again in Exercise F.4. The superconformal ghosts have weights $(\frac{3}{2}, -\frac{1}{2})$, the values forced by pairing them with the weight-$\frac{3}{2}$ supercurrent G exactly as (b, c) pair with the weight-2 stress tensor T. At $\lambda = \frac{3}{2}$,

$$c_{\beta\gamma} = -1 + 3(2\cdot\tfrac{3}{2} - 1)^2 = -1 + 12 = +11, \tag{F.32}$$

positive, which is what allows it to cancel part of the -26 from the bc system. The moding follows the sector of the fermion it partners: Neveu-Schwarz gives half-integer modes, Ramond gives integer modes,

$$\beta(z) = \sum_r \frac{\beta_r}{z^{r+3/2}}, \quad \gamma(z) = \sum_r \frac{\gamma_r}{z^{r-1/2}}, \qquad [\gamma_r, \beta_s] = \delta_{r+s,0}, \tag{F.33}$$

with $r \in \mathbb{Z} + \frac{1}{2}$ in NS and $r \in \mathbb{Z}$ in R (the commutator, not the anticommutator, because β, γ are bosonic).

Being commuting fields of half-integer weight, the $\beta\gamma$ system has no single Fock vacuum. Its ground states come in an infinite family labelled by *picture number*, and the choice of picture is part of the data of a superstring vertex operator (Chapter 4). The cleanest handle on this, and the quickest recomputation of $+11$, is bosonisation.

Remark F.8 (Beyond the lectures: bosonisation and the $+11$). Following Friedan, Martinec and Shenker, trade the first-order pair for a chiral boson ϕ and a weight-$(1,0)$ fermionic pair $\xi\eta$ (here η is a fermion, not the Dedekind η of Appendix F.3),

$$\beta = e^{-\phi}\,\partial\xi, \qquad \gamma = \eta\, e^{\phi}, \qquad \langle\phi(z)\phi(w)\rangle = -\ln(z-w). \tag{F.34}$$

The boson carries a background charge $Q_\phi = -2$, so its stress tensor is $T_\phi = -\frac{1}{2}{:}\partial\phi\,\partial\phi{:} - \frac{Q_\phi}{2}\partial^2\phi$ with central charge $c_\phi = 1 + 3Q_\phi^2 = 13$; the $\xi\eta$ system is a $\lambda = 1$ instance of Eq. (F.27), $c_{\xi\eta} = 1 - 3 = -2$. Their sum $13 - 2 = 11$ reproduces Eq. (F.32). Picture number is the ϕ-momentum, the charge of $e^{q\phi}$, and the Ramond ground state (a spacetime spinor) is built from the spin field $e^{-\phi/2}S_\alpha$. This is the machinery behind the superstring correlators of Chapter 4; it is used sparingly and not re-derived.

F.2.4 Free fermions ψ^μ

The RNS string adds to each coordinate a worldsheet Majorana fermion $\psi^\mu(z)$, a spacetime vector but a two-dimensional spinor, of weight $\frac{1}{2}$. Its free data are

$$S = \frac{1}{2\pi}\int \mathrm{d}^2z\; \psi^\mu\,\bar\partial\psi_\mu, \qquad \psi^\mu(z)\,\psi^\nu(w) \sim \frac{\eta^{\mu\nu}}{z-w}, \qquad T = -\tfrac{1}{2}{:}\psi^\mu\,\partial\psi_\mu{:}, \tag{F.35}$$

with the simple pole (rather than the double pole of $\partial X\,\partial X$) and the symmetric numerator marking a weight-$\frac{1}{2}$ anticommuting field. The TT OPE has quartic pole $\frac{D/4}{(z-w)^4}$ (Exercise F.4), so

$$c = \tilde c = \tfrac{1}{2} \quad \text{per Majorana fermion}, \qquad c = \tfrac{D}{2} \text{ for } D \text{ of them}. \tag{F.36}$$

The moding is set by the boundary condition around the string: antiperiodic (Neveu-Schwarz) gives half-integer modes, periodic (Ramond) gives integer modes,

$$\psi^\mu(z) = \sum_r \frac{\psi^\mu_r}{z^{r+1/2}}, \qquad \{\psi^\mu_r, \psi^\nu_s\} = \eta^{\mu\nu}\delta_{r+s,0}, \qquad (\psi^\mu_r)^\dagger = \psi^\mu_{-r}, \tag{F.37}$$

with $r \in \mathbb{Z} + \frac{1}{2}$ (NS) or $r \in \mathbb{Z}$ (R). The two sectors have qualitatively different vacua, and this is where spacetime fermions come from. The NS ground state is unique, annihilated by every ψ^μ_r with $r > 0$ and non-degenerate. The R sector has zero modes ψ^μ_0 obeying the Clifford algebra $\{\psi^\mu_0, \psi^\nu_0\} = \eta^{\mu\nu}$; these cannot annihilate the vacuum, so the R ground state carries a representation of the D-dimensional Clifford algebra, that is, it is a spacetime spinor. The full sector structure, the spin field and the GSO projection are the business of Chapter 3 and Appendix B.

Remark F.9 (Beyond the lectures: complex fermions and fermion-boson equivalence). Two Majorana fermions combine into a complex (Dirac) fermion $\Psi = \frac{1}{\sqrt{2}}(\psi^1 + i\psi^2)$ with $\Psi(z)\Psi^\dagger(w) \sim (z-w)^{-1}$, central charge $c = 1$ and a U(1) charge. In two dimensions such a fermion is equivalent to a compact boson at the self-dual radius, $\Psi \sim e^{iH}$ with H a chiral scalar, $c = 1$ either way. This bosonisation is what lets the heterotic string trade 32 left-moving Majorana fermions for 16 chiral bosons on a lattice (Appendix G), and it is the free-field root of the current algebras that carry the gauge symmetry.

F.2.5 Central-charge bookkeeping: $D = 26$ and $D = 10$

A gauge-fixed string is consistent only if the Weyl anomaly cancels, equivalently only if the total holomorphic central charge vanishes, $c_{\text{tot}} = 0$, and the same for $\tilde{c}$. That $c_{\text{tot}} = 0$ is precisely the condition for the BRST charge to be nilpotent, $Q^2 = 0$, is proven in Appendix F.4; here it is the balance sheet. Central charges add, so the sum runs over the free systems above.

For the bosonic string the matter is D free bosons and the ghosts are one bc pair,

$$c_{\text{tot}} = D \cdot 1 + c_{bc}(2) = D - 26 = 0 \quad \Longrightarrow \quad D = 26. \tag{F.38}$$

For the superstring, on each chirality, the matter is D bosons and D Majorana fermions, and the enlarged gauge symmetry brings a second ghost system,

$$c_{\text{tot}} = D\Big(1 + \tfrac{1}{2}\Big) + \underbrace{c_{bc}}_{-26} + \underbrace{c_{\beta\gamma}}_{+11} = \tfrac{3}{2}D - 15 = 0 \quad \Longrightarrow \quad D = 10. \tag{F.39}$$

The critical value factorises as $15 = 10 \times (1 + \frac{1}{2})$, ten dimensions each carrying a coordinate boson and its fermion, against ghosts summing to $-26 + 11 = -15$. This is the covariant version of the count. A light-cone route reaches the same $D = 10$ by demanding that the anomaly in the transverse Lorentz commutator $[J^{i-}, J^{j-}]$ cancel; it is carried out for the superstring in Chapter 3. For the bosonic string these notes stay covariant throughout (Chapter 1), so the 26 here is read off the ghost central charge rather than from a transverse count.

The heterotic string is the asymmetric case: the two sides carry different central charge. The right movers are a superstring, $c_{\text{R}} = 15$; the left movers are a bosonic string, $c_{\text{L}} = 26$. Ten spacetime coordinates are shared, contributing $(c_{\text{L}}, c_{\text{R}}) = (10, 10)$, and the right-moving fermions supply the remaining $\frac{1}{2} \cdot 10 = 5$ to make $c_{\text{R}} = 15$. That leaves

$$c_{\text{L}}^{\text{internal}} = 26 - 10 = 16 \tag{F.40}$$

of purely left-moving internal central charge. Modular invariance of the torus (Appendix F.3) forces the internal theory to be a chiral CFT with $c = 16$ whose momenta lie on an even self-dual

System	Statistics	Weights	c (each)	Moding	Role
X^μ	commuting	$(1,0)$	$+1$	$n \in \mathbb{Z}$	matter coordinate
ψ^μ	anticommuting	$(\frac{1}{2},0)$	$+\frac{1}{2}$	NS $r \in \mathbb{Z}+\frac{1}{2}$, R $r \in \mathbb{Z}$	RNS matter
bc	anticommuting	$(2,-1)$	-26	$n \in \mathbb{Z}$	diff. $\times$ Weyl ghosts
$\beta\gamma$	commuting	$(\frac{3}{2},-\frac{1}{2})$	$+11$	NS $r \in \mathbb{Z}+\frac{1}{2}$, R $r \in \mathbb{Z}$	susy $\times$ super-Weyl ghosts
general $bc(\lambda)$	anticommuting	$(\lambda, 1-\lambda)$	$1-3(2\lambda-1)^2$		
general $\beta\gamma(\lambda)$	commuting	$(\lambda, 1-\lambda)$	$-1+3(2\lambda-1)^2$		

Table F.1: The free-field systems of the notes and their holomorphic central charges. Each has an identical antiholomorphic copy with $c \to \tilde{c}$. The last two rows are the general first-order formulae Eqs. (F.27) and (F.31), from which the four physical values are the entries $\lambda = 2$ (bc) and $\lambda = \frac{3}{2}$ ($\beta\gamma$).

lattice, and in rank 16 there are exactly two: $\Gamma_8 \oplus \Gamma_8$ and Γ_{16}, giving the gauge groups $\mathrm{E}_8 \times \mathrm{E}_8$ and $\mathrm{Spin}(32)/\mathbb{Z}_2$. The arithmetic of those lattices is Appendix G, and the string construction is Section 3.5.

Exercise F.3 (The four numbers from two formulae)**.** *(Warm-up.)* (a) Evaluate the general first-order central charges Eqs. (F.27) and (F.31) at the physical weights and recover $c_{bc}(2) = -26$ and $c_{\beta\gamma}(\frac{3}{2}) = +11$. (b) Check that a *commuting* pair of weights $(2,-1)$ would give $c = +26$ and so fail to cancel the matter of the bosonic string, and that an *anticommuting* pair of weights $(\frac{3}{2},-\frac{1}{2})$ would give $c = -11$, so the sign of c turns on the statistics as much as on the weights. (c) Assemble Eqs. (F.38) and (F.39) and confirm that the ghost totals -26 and -15 are the negatives of the matter central charges at the critical dimension. (d) For the heterotic internal sector, verify that 16 left-moving Majorana fermions ($c = 8$) fall short of Eq. (F.40), and that 32 of them, or equivalently 16 chiral bosons, supply the required $c = 16$.

Exercise F.4 (Central charges from the TT OPE)**.** *(Challenging.)* Repeat the contraction of Theorem F.5 for the remaining free systems, reading the central charge off the quartic pole $c/2$. (a) For a single Majorana fermion, with T and the propagator of Eq. (F.35), obtain the pole $\frac{1/4}{(z-w)^4}$ and hence $c = \frac{1}{2}$. (b) For an anticommuting first-order pair of weights $(\lambda, 1-\lambda)$, contract T of Eq. (F.26) with itself and recover $c_{bc}(\lambda) = 1-3(2\lambda-1)^2$; track where the fermionic minus sign from reordering the contractions enters. (c) Repeat for the commuting $\beta\gamma$ pair and show that the same steps produce the opposite sign, $-1+3(2\lambda-1)^2$. This is the general first-order derivation that Chapter 1 defers to this appendix (Polchinski, Vol. 1, §2.5).

F.3 The torus partition function and modular invariance

One loop for a closed string is a torus: the worldsheet swept out by a string that appears from the vacuum, propagates once around a loop, and disappears. Its amplitude is the string version of the Coleman-Weinberg vacuum energy, a sum of ground-state energies over the whole spectrum. What has no analogue in point-particle field theory is the consistency condition that rides along with it. A torus is fixed by a single complex parameter, and different values of that parameter can describe the very same torus. Physics cannot depend on the redundant description. That invariance, modular invariance, is a constraint on the spectrum, and it is the common root of the GSO projection and the even self-dual heterotic lattice of Chapter 3. This section assembles the machinery those arguments quote. The trace formulae use the radial-quantisation and state-operator apparatus of Appendix F.1, and the central charges of the individual systems are collected in Appendix F.2; here we combine them.

F.3.1 The torus and the modular group

A torus is the plane quotiented by a rank-two lattice. Up to an overall scale and rotation, which the conformal symmetry of the worldsheet renders irrelevant, that lattice is generated by 1 and a single complex number,

$$T^2 \;=\; \mathbb{C}/(\mathbb{Z}+\tau\,\mathbb{Z}), \qquad \tau=\tau_1+i\tau_2, \quad \tau_2>0, \tag{F.41}$$

the *modular parameter* (or Teichmüller parameter) τ living in the upper half plane $\mathbb{H}$. The two independent one-cycles are the a-cycle (from 0 to 1) and the b-cycle (from 0 to τ). This one complex number is the single modulus left after diffeomorphisms and Weyl rescalings have been used up (Chapter 1); the torus is the simplest surface with a nontrivial moduli space.

Different τ can give the same torus, because the choice of basis for the lattice is not unique. Any $\mathrm{SL}(2,\mathbb{Z})$ change of basis leaves the lattice, hence the torus, untouched:

$$\tau \;\longmapsto\; \frac{a\tau+b}{c\tau+d}, \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathbb{Z}), \quad ad-bc=1. \tag{F.42}$$

This is the *modular group*, the mapping class group of T^2. Since the matrix and its negative act identically, the group that acts faithfully is $\mathrm{PSL}(2,\mathbb{Z})=\mathrm{SL}(2,\mathbb{Z})/\{\pm\mathbf{1}\}$. It is generated by two elements,

$$T:\ \tau\mapsto\tau+1, \qquad S:\ \tau\mapsto-\frac{1}{\tau}, \tag{F.43}$$

subject to $S^2=(ST)^3=\mathbf{1}$ in $\mathrm{PSL}(2,\mathbb{Z})$. Geometrically T is a Dehn twist, a relabelling of the b-cycle by $b\to b+a$; S exchanges the two cycles, $a\leftrightarrow b$, which is what makes it the powerful generator: it swaps the roles of worldsheet space and worldsheet time.

Quotienting $\mathbb{H}$ by Eq. (F.42) leaves a *fundamental domain*, the standard choice being

$$\mathcal{F} \;=\; \Big\{\,\tau\in\mathbb{H}\ :\ |\tau_1|\le\tfrac{1}{2},\ \ |\tau|\ge 1\,\Big\}. \tag{F.44}$$

Every torus is represented exactly once in $\mathcal{F}$ (up to boundary identifications). The unique $\mathrm{SL}(2,\mathbb{R})$-invariant measure on $\mathbb{H}$, which is therefore also modular invariant, is

$$\mathrm{d}\mu \;=\; \frac{\mathrm{d}^2\tau}{\tau_2^{\,2}}, \qquad \mathrm{d}^2\tau\coloneqq \mathrm{d}\tau_1\,\mathrm{d}\tau_2. \tag{F.45}$$

The region $\tau_2\to 0$, where the torus degenerates and a field theory would develop its ultraviolet divergence, sits outside $\mathcal{F}$: it is a modular image of the safe interior. String amplitudes integrate only over $\mathcal{F}$, so there is no short-distance region to diverge. The improved ultraviolet behaviour of string perturbation theory is built into the geometry of moduli space, provided the integrand is genuinely a function on that space, that is, modular invariant.

F.3.2 The partition function as a trace

Cut the torus along the a-cycle and it becomes a cylinder: a closed string of circumference 2π, propagated for Euclidean time $2\pi\tau_2$ and simultaneously translated around its length by $2\pi\tau_1$ before the ends are glued back. The two operators that generate these motions are the worldsheet Hamiltonian and momentum,

$$H \;=\; L_0+\tilde{L}_0-\frac{c+\tilde{c}}{24}, \qquad P \;=\; L_0-\tilde{L}_0, \tag{F.46}$$

where the shift $-c/24$ is the Casimir energy picked up in mapping the plane to the cylinder (Appendix F.1). Throughout, a tilde marks the antiholomorphic (right-moving) copy of an

operator or central charge; the bar is reserved for complex conjugation and for antiholomorphic weights $\bar{h}$. Gluing is a trace, so the one-loop vacuum amplitude is

$$Z(\tau,\bar{\tau}) \;=\; \mathrm{Tr}_{\mathcal{H}}\left[\, e^{2\pi i \tau_1 P}\, e^{-2\pi\tau_2 H}\,\right] \;=\; \mathrm{Tr}_{\mathcal{H}}\left[\, q^{\,L_0 - c/24}\;\bar{q}^{\,\tilde{L}_0 - \tilde{c}/24}\,\right], \qquad q \coloneqq e^{2\pi i\tau}. \tag{F.47}$$

The holomorphic and antiholomorphic sectors factorise inside the trace whenever the Hilbert space does. For a chiral sector built on a primary of weight h, the generating function of the descendant multiplicities is the *character*

$$\chi_h(\tau) \;=\; \mathrm{Tr}_{\mathcal{H}_h}\, q^{\,L_0 - c/24} \;=\; q^{\,h - c/24} \sum_{N \geq 0} d_N\, q^N, \tag{F.48}$$

with d_N the number of states at level N. A modular invariant partition function is a diagonal or block combination $\sum_{h,\bar{h}} \chi_h(\tau)\,\overline{\chi_{\bar{h}}(\tau)}$ of these characters, and the whole content of this section is which combinations are allowed.

The vacuum energy density (the one-loop cosmological constant) follows by integrating over inequivalent tori,

$$\Lambda_1 \;\propto\; -\int_{\mathcal{F}} \frac{\mathrm{d}^2\tau}{\tau_2^{\,2}}\; \tau_2\, Z(\tau,\bar{\tau}), \tag{F.49}$$

with Z the full trace Eq. (F.47), momentum zero modes included. The lone factor of τ_2 is not a second zero-mode integral: it is the Jacobian relating the invariant measure to the physical one-loop measure $\mathrm{d}^2\tau/\tau_2$, equivalently the factor that makes $\tau_2\, Z$ a modular scalar, as one checks for the bosonic string in Eq. (F.56). The precise normalisation is fixed in Polchinski (§7.3); we track the integrand, not the prefactor.

F.3.3 Modular invariance as a spectral constraint

For Eq. (F.49) to be a well-defined integral over $\mathcal{F}$, the integrand must be a function on moduli space: invariant under Eq. (F.42). Because T and S generate the group, it is enough to impose invariance under those two. Each carries a different piece of physics.

Proposition F.10 (The T constraint is level matching)**.** *Under $T : \tau \to \tau+1$ the trace Eq. (F.47) picks up the phase* $\exp\big[\,2\pi i\big((L_0 - \tilde{L}_0) - (c-\tilde{c})/24\big)\big]$ *on each state. Invariance requires*

$$L_0 - \tilde{L}_0 - \frac{c-\tilde{c}}{24} \;\in\; \mathbb{Z} \qquad \textit{on every state.} \tag{F.50}$$

For a non-chiral theory with $c = \tilde{c}$ this is level matching, the quantum promotion of the classical constraint of Chapter 1. The combination $c - \tilde{c}$ is the gravitational anomaly. For a purely bosonic, integer-moded spectrum containing the identity, T-invariance forces $c - \tilde{c} \equiv 0 \pmod{24}$; once worldsheet fermions and a spin-structure sum are present this is relaxed, and the heterotic string of Appendix F.3.7 realises transverse $(c,\tilde{c}) = (24,12)$, i.e. $c - \tilde{c} = 12$.

S-invariance is the deep condition. It exchanges $\tau_2 \to \tau_2/|\tau|^2$, trading the ultraviolet ($\tau_2 \to 0$) for the infrared ($\tau_2 \to \infty$), and so it relates the density of states at high energy to the low-lying spectrum. It is automatic for a single free field but becomes a genuine constraint the moment several sectors are assembled, which is exactly the situation for worldsheet fermions and for internal lattices. The two applications in Appendices F.3.6 and F.3.7 are S-invariance made to do work.

A second reading fixes the critical dimension: modular invariance is the global (large-diffeomorphism) counterpart of local Weyl invariance on the worldsheet. The obstruction to local Weyl invariance is the conformal anomaly, proportional to the total central charge $c_{\mathrm{matter}} + c_{\mathrm{ghost}}$. When that fails to vanish the torus amplitude depends on which representative metric one picks inside the conformal class, and in particular ceases to be modular invariant. Requiring modular invariance therefore forces $c_{\mathrm{tot}} = 0$, that is $D = 26$ for the bosonic string and $D = 10$ for the superstring, the same numbers the anomaly delivered locally in Appendix F.4.

	η	θ_2	θ_3	θ_4
$T:\ \tau\to\tau+1$	$e^{i\pi/12}\,\eta$	$e^{i\pi/4}\,\theta_2$	θ_4	θ_3
$S:\ \tau\to-1/\tau$	$\sqrt{-i\tau}\,\eta$	$\sqrt{-i\tau}\,\theta_4$	$\sqrt{-i\tau}\,\theta_3$	$\sqrt{-i\tau}\,\theta_2$

Table F.2: Modular transformations of the building blocks at zero argument. T swaps $\theta_3 \leftrightarrow \theta_4$ and S swaps $\theta_2 \leftrightarrow \theta_4$; together they generate the full permutation group S_3 of the three even spin structures. Each function has modular weight $\frac{1}{2}$, so a ratio θ_i^k/η^k is weight zero.

F.3.4 Building blocks: the free boson, η, and θ

Every string partition function is assembled from two special functions.

The free boson. A single noncompact boson X contributes a continuous momentum and an oscillator tower. The oscillators give the chiral character

$$q^{-1/24}\,\mathrm{Tr}\;q^{N} \;=\; q^{-1/24}\prod_{n\geq 1}\frac{1}{1-q^{n}} \;=\; \frac{1}{\eta(\tau)}, \qquad \eta(\tau)\coloneqq q^{1/24}\prod_{n\geq 1}(1-q^{n}), \tag{F.51}$$

η being the Dedekind eta function, of central charge $c=1$. The zero-mode momentum integral supplies a factor of τ_2:

$$\int\frac{V_1\,\mathrm{d}k}{2\pi}\;(q\bar{q})^{\alpha' k^2/4} \;=\; \frac{V_1}{\left(4\pi^2\alpha'\tau_2\right)^{1/2}}. \tag{F.52}$$

A full noncompact boson, both movers, therefore weighs $V_1\,(4\pi^2\alpha'\tau_2)^{-1/2}\,|\eta|^{-2}$. The transformations of η are the master formulae from which everything else descends:

$$\eta(\tau+1)=e^{i\pi/12}\,\eta(\tau), \qquad \eta(-1/\tau)=\sqrt{-i\tau}\;\eta(\tau). \tag{F.53}$$

Under S, η carries modular weight $\frac{1}{2}$, and the combination $\sqrt{\tau_2}\,|\eta|^2$ is invariant, which is why Eq. (F.52) pairs with the oscillators to make a modular scalar per boson.

The free fermion. A worldsheet fermion must be assigned a boundary condition, periodic (Ramond) or antiperiodic (Neveu-Schwarz), around *each* of the two cycles of the torus. Four choices result, the four *spin structures*. Their characters are the Jacobi theta functions, which at zero argument (the Nullwerte) read

$$\theta_2=\sum_{n\in\mathbb{Z}} q^{\frac{1}{2}(n+\frac{1}{2})^2}, \quad \theta_3=\sum_{n\in\mathbb{Z}} q^{\frac{1}{2}n^2}, \quad \theta_4=\sum_{n\in\mathbb{Z}}(-1)^n q^{\frac{1}{2}n^2}, \quad \theta_1=0. \tag{F.54}$$

Their product forms make the fermion content transparent,

$$\frac{\theta_3}{\eta}=q^{-1/24}\prod_{n\geq 1}\left(1+q^{n-1/2}\right)^2, \quad \frac{\theta_4}{\eta}=q^{-1/24}\prod_{n\geq 1}\left(1-q^{n-1/2}\right)^2, \quad \frac{\theta_2}{\eta}=2\,q^{1/12}\prod_{n\geq 1}\left(1+q^{n}\right)^2, \tag{F.55}$$

so that $(\theta_3/\eta)^{1/2}$ is the NS character of one real fermion, $(\theta_4/\eta)^{1/2}$ inserts $(-1)^F$ in the NS sector, and $(\theta_2/\eta)^{1/2}$ is the R character; the fourth, with $(-1)^F$ in the R sector, is $(\theta_1/\eta)^{1/2}\equiv 0$, the two Ramond ground-state chiralities cancelling. The modular transformations are collected in Table F.2; they say that S and T permute the three even spin structures $(\theta_2,\theta_3,\theta_4)$ among themselves, which is the arithmetic heart of GSO.

Example F.11 (One entry of Table F.2)**.** The T entries follow directly from the Nullwerte Eq. (F.54). Writing $\theta_3(\tau)=\sum_n q^{n^2/2}=\sum_n e^{i\pi n^2\tau}$, under $\tau\to\tau+1$ each term acquires $e^{i\pi n^2}=(-1)^{n^2}=(-1)^n$, so

$$\theta_3(\tau+1)=\sum_{n\in\mathbb{Z}}(-1)^n\,e^{i\pi n^2\tau}=\theta_4(\tau),$$

and the same argument on θ_4 returns θ_3. The S entries are not visible term by term: they come from Poisson resummation of the Gaussian sum, of which Eq. (F.61) is the general form.

F.3.5 The bosonic string: the reference amplitude

Assemble the pieces for the 26-dimensional bosonic string. The bc ghosts (central charge -26, Appendix F.2) remove two of the 26 oscillator sets and cancel the corresponding zero modes, leaving 24 transverse oscillators per chirality. With Eqs. (F.51) and (F.52) the integrand is

$$Z_{T^2} \;=\; V_{26}\int_{\mathcal{F}}\frac{\mathrm{d}^2\tau}{4\tau_2}\,\left(4\pi^2\alpha'\tau_2\right)^{-13}\,\left|\eta(\tau)\right|^{-48}, \tag{F.56}$$

where $|\eta|^{-48}=|\eta|^{-2\cdot 24}$ is the 24 transverse oscillators and $(4\pi^2\alpha'\tau_2)^{-13}=(\cdots)^{-D/2}$ the 26 momenta. Against the invariant measure Eq. (F.45) the integrand is $\tau_2^{-12}|\eta|^{-48}$, and using Eq. (F.53) one checks directly that this is modular invariant: under S, $\tau_2^{-12}\to\tau_2^{-12}|\tau|^{24}$ while $|\eta|^{-48}\to|\tau|^{-24}|\eta|^{-48}$, and the powers of $|\tau|$ cancel. The balance $24=24$ is $D-2=24$ transverse fields matched against the modular weight; it is the statement that the ghosts have done their job, and it works only because the same $D=26$ that cancels the conformal anomaly also fixes the exponent. The integral over $\mathcal{F}$ converges in the ultraviolet by construction. It diverges in the infrared, $\tau_2\to\infty$, because the q^{-1} from the tachyon in $|\eta|^{-48}=q^{-1}\bar{q}^{-1}(1+\cdots)$ grows without bound: the bosonic vacuum energy is genuinely ill-defined, a symptom of the tachyonic instability that the superstring removes.

F.3.6 Spin structures and the GSO projection

For the superstring the transverse fields are eight bosons X^i and eight fermions ψ^i per side. The fermionic factor is a sum over the four spin structures, one term per boundary condition around the time cycle for each sector. Writing $Z_i\coloneqq\theta_i^4/\eta^4$ for the eight-fermion character in spin structure i (weight zero, by Table F.2), the physically correct combination is

$$Z_\psi(\tau) \;=\; \frac{1}{2\,\eta(\tau)^4}\Big[\,\theta_3^4\;-\;\theta_4^4\;-\;\theta_2^4\;-\;\theta_1^4\,\Big] \;=\; \frac{1}{2}(Z_3-Z_4-Z_2-Z_1), \tag{F.57}$$

the object Chapter 3 quotes in the GSO discussion. The relative coefficients $(+,-,-,-)$ are the unique modular-invariant assignment, and reading off the surviving states reproduces the projector $P_{\mathrm{GSO}}=\frac{1}{2}(1+(-1)^F)$.

The argument is a short orbit calculation. Start with the NS sector without insertion, Z_3. Under T (Table F.2) it maps to Z_4: modular invariance forces the NS-with-$(-1)^F$ term into the sum, and $(-1)^F$ acting on the NS states is precisely the GSO operator in that sector. Apply S to Z_4 and it maps to Z_2, dragging in the Ramond sector. The three even structures thus form a single $\mathrm{SL}(2,\mathbb{Z})$ orbit; a modular-invariant sum must contain all three with coefficients equal up to the signs that the transformation phases dictate. Fixing those signs by demanding invariance yields exactly Eq. (F.57). Concretely, under T the bracket obeys $\theta_3^4-\theta_4^4-\theta_2^4\to-(\theta_3^4-\theta_4^4-\theta_2^4)$ while $\eta^4\to e^{i\pi/3}\eta^4$, so $Z_\psi\to-e^{-i\pi/3}Z_\psi$; the residual phase is cancelled by the eight transverse bosons, whose chiral factor η^{-8} transforms by $e^{-2\pi i/3}$, leaving the full transverse partition function

$$Z_{\mathrm{trans}}(\tau,\bar\tau) \;=\; \frac{1}{\tau_2^4}\,\frac{Z_\psi(\tau)}{\eta(\tau)^8}\,\frac{\overline{Z_\psi(\tau)}}{\bar\eta(\tau)^8} \tag{F.58}$$

modular invariant. The exponents record $D=10$: eight transverse fields give oscillator weight -4 per side, matched by the τ_2^{-4} from eight transverse momenta.

The physics of Eq. (F.57) sits in two properties, both consequences of the theta transformations above. First, $\theta_1\equiv 0$, so the $(-1)^F$-inserted Ramond term is absent: there is nothing to weigh

once the two Ramond chiralities have been paired off. Second, Jacobi's identity, verified directly from Eq. (F.54),

$$\theta_3(\tau)^4 \;=\; \theta_2(\tau)^4 + \theta_4(\tau)^4, \tag{F.59}$$

makes the entire bracket vanish, $Z_\psi \equiv 0$. Expanded in powers of q, coefficient by coefficient, this is the equality of NS and R state counts at every mass level: spacetime supersymmetry written as an identity between modular forms. The reading of Eq. (F.57) as a spectrum, and the alternative modular-invariant sign choices that give the non-supersymmetric type 0 strings, belong to Chapter 3; what the reference card supplies is the derivation that modular invariance fixes the coefficients and that Eq. (F.59) is why the amplitude vanishes.

F.3.7 Lattice sums and the even self-dual condition

The heterotic string carries sixteen purely left-moving bosons whose momenta lie on a Euclidean lattice Λ of rank 16 (Chapter 3). Which lattice is allowed is again decided by modular invariance, and the calculation is the cleanest illustration of the S and T split. The internal momenta enter the partition function only through the lattice theta function

$$\Theta_\Lambda(\tau) \;=\; \sum_{p\in\Lambda} q^{\,p^2/2}, \tag{F.60}$$

divided by η^{16} from the sixteen chiral oscillators. Impose the two generators.

Under $T:\tau\to\tau+1$ each term of Eq. (F.60) acquires $e^{i\pi p^2}$, so Θ_Λ is T-invariant iff $p^2\in 2\mathbb{Z}$ for every $p\in\Lambda$. The lattice must be **even**. By Theorem F.10 this is level matching for the internal sector: the chiral contribution to $L_0-\tilde L_0$ is an integer.

Under $S:\tau\to -1/\tau$, Poisson resummation converts a sum over Λ into a sum over the dual lattice $\Lambda^* = \{\, w\in\mathbb{R}^{16} : w\cdot p\in\mathbb{Z}\ \forall p\in\Lambda\,\}$,

$$\Theta_\Lambda(-1/\tau) \;=\; \frac{(-i\tau)^{\,r/2}}{\mathrm{vol}(\Lambda)}\,\Theta_{\Lambda^*}(\tau), \qquad r=\operatorname{rank}\Lambda, \tag{F.61}$$

with $\mathrm{vol}(\Lambda)$ the covolume of the unit cell. Matching this against the η^{-16} factor, whose S-weight is -8 from Eq. (F.53), forces $\Lambda^*=\Lambda$ and $\mathrm{vol}(\Lambda)=1$. The lattice must be **self-dual** (unimodular). Then $\Theta_\Lambda(-1/\tau)=(-i\tau)^8\Theta_\Lambda(\tau)$ of weight 8, exactly cancelling the oscillator weight, so Θ_Λ/η^{16} is a modular scalar.

Even self-dual Euclidean lattices are rigid objects. They exist only when the rank is a multiple of 8, and the residual T-phase $e^{-i\pi r/12}$ from η^{-r} combines with the rest of the heterotic partition function precisely for $r=16$. In rank 16 there are exactly two,

$$\Gamma_8\oplus\Gamma_8 \qquad\text{and}\qquad \Gamma_{16}=D_{16}^{+}, \tag{F.62}$$

giving the gauge groups $\mathrm{E}_8\times\mathrm{E}_8$ and $\mathrm{Spin}(32)/\mathbb{Z}_2$. The classification, and the arithmetic of the norm-two vectors that become massless gauge bosons, is stated in Appendix G and used in Chapter 3; here we have supplied only the modular reason there is a list at all.

Remark F.12 (Beyond the lectures: Narain lattices). Compactifying d coordinates of a closed string on a torus makes both movers carry internal momenta, $(p_{\mathrm L},p_{\mathrm R})\in\mathbb{R}^{d,d}$, with the level-matching combination $p_{\mathrm L}^2-p_{\mathrm R}^2$ appearing in T and the sum $p_{\mathrm L}^2+p_{\mathrm R}^2$ in the Hamiltonian. Modular invariance now demands that $(p_{\mathrm L},p_{\mathrm R})$ fill an even self-dual lattice with respect to the *Lorentzian* inner product of signature (d,d), the Narain lattice $\Gamma_{d,d}$. Unlike the Euclidean case these are not rigid: they form a continuous family parametrised by $O(d,d)/(O(d)\times O(d))$, the moduli of the torus and the B-field, with the discrete $O(d,d;\mathbb{Z})$ acting as T-duality. This is the lattice-theoretic home of T-duality and of the $O(d,d)$ structure used in Chapter 6 and Chapter 8; the classification lives in Appendix G.

Remark F.13 (Beyond the lectures: the partition function as a spectrum counter). Read as a power series, $Z(\tau,\bar\tau)=\sum_{h,\bar h}N_{h,\bar h}\,q^{h-c/24}\bar q^{\bar h-\tilde c/24}$ is the generating function of the spectrum, and S-invariance ties its two ends together. Sending $\tau_2\to 0$ and using S to map back to large τ_2 expresses the high-energy density of states in terms of the ground-state energy $-c/24$ alone. The result is Cardy's formula, an exponential Hagedorn growth $N(E)\sim\exp\!\big(2\pi\sqrt{cE/6}\,\big)$ in the level, which underlies both the Hagedorn temperature of the string and, in the AdS_3 context, the Bekenstein-Hawking entropy of the black holes of Chapter 12.

Exercise F.5 (The three even theta functions and Jacobi's identity). *(Standard.)* From the product forms Eq. (F.55) and η, and the transformations Eq. (F.53), verify each entry of Table F.2. Then show that $Z_i=\theta_i^4/\eta^4$ are weight-zero and that S fixes Z_3 while swapping $Z_2\leftrightarrow Z_4$, and T swaps $Z_3\leftrightarrow Z_4$ up to the phase $e^{-i\pi/3}$ while sending $Z_2\to e^{2\pi i/3}Z_2$. Deduce that $\frac12(Z_3-Z_4-Z_2)$ is S-invariant and T-covariant with phase $-e^{-i\pi/3}$, and that this phase is cancelled by η^{-8}, making Eq. (F.58) modular invariant. Finally expand Eq. (F.59) to order q^2 and read off the equal NS and R state counts at the first two mass levels.

Exercise F.6 (Even and self-dual from S and T). *(Standard.)* Prove Eq. (F.61) from the Poisson summation formula $\sum_{p\in\Lambda}f(p)=\mathrm{vol}(\Lambda)^{-1}\sum_{w\in\Lambda^*}\tilde f(w)$ applied to the Gaussian $f(p)=e^{i\pi\tau p^2}$. Conclude that T-invariance of Θ_Λ forces Λ even and S-invariance of Θ_Λ/η^{16} forces Λ self-dual. Check that Γ_8, the E_8 root lattice, is even (all roots have $p^2=2$) and unimodular, and that its theta function is the weight-four Eisenstein series $E_4=\Theta_{\Gamma_8}$, the unique such form up to scale.

Exercise F.7 (Narain lattices and the $O(d,d)$ S-transformation). *(Challenging.)* For d coordinates on a torus with constant metric G_{ij} and B-field B_{ij}, write the left- and right-moving momenta $(p_{\mathrm L},p_{\mathrm R})$ and show that the level-matching integer $\frac12(p_{\mathrm L}^2-p_{\mathrm R}^2)$ is T-invariance, i.e. evenness of $\Gamma_{d,d}$ under the signature-(d,d) inner product. Then Poisson-resum the Narain theta function and show that S-invariance of $\Theta_{\Gamma_{d,d}}/(\eta^d\bar\eta^d)$ forces self-duality; because the Lorentzian signature makes even self-dual lattices non-rigid, deduce that the partition function is modular invariant for *every* (G,B), with $O(d,d;\mathbb Z)$ acting as T-duality. [Reference: Polchinski §8.4; Appendix G.]

F.4 BRST cohomology and the no-ghost theorem

Covariant quantisation keeps every X^μ, including the timelike X^0 whose oscillators build negative-norm states. The claim that the string spectrum is nonetheless unitary is the no-ghost theorem, stated in Chapter 1 (Theorem 1.5) and proved here. The setup, the bc ghost system, the BRST current, and the fact that $Q^2=0$ forces $c_{\mathrm{tot}}=0$, was built in Section 1.4; this section takes it as given, records the cohomology machinery compactly, and then supplies the proof deferred from the main text. Everything is stated for one chirality (the holomorphic sector); the antiholomorphic copy is identical, with its Virasoro modes, central charge and oscillators carrying a tilde $(\tilde L_n,\tilde c,\tilde\alpha)$ and the bar reserved for complex conjugation. The closed-string modifications are collected at the end.

BRST in one box

Gauge fixing turns the constraints $L_n=0$ into a single nilpotent operator Q. Physical states are its cohomology: closed ($Q\,|\psi\rangle=0$) modulo exact ($|\psi\rangle\sim|\psi\rangle+Q\,|\chi\rangle$). Exact states have zero norm and decouple, so the inner product descends to cohomology. The content of the no-ghost theorem is that this cohomology, at ghost number one and on the mass shell, is isomorphic to the Fock space of the $D-2$ *transverse* oscillators, which is manifestly positive definite. The timelike and longitudinal oscillators do not disappear; they cancel in quartets against the ghosts. The isomorphism holds precisely when $c_{\mathrm{tot}}=0$, that is, at $D=26$ (bosonic) or $D=10$ (superstring).

F.4.1 BRST data, for reference

The holomorphic BRST current and charge are Eqs. (1.73) and (1.74),

$$j_B = c\,T^m + \tfrac{1}{2}\,{:}c\,T^{\mathrm{gh}}{:} + \tfrac{3}{2}\,\partial^2 c, \qquad Q = \oint \frac{\mathrm{d}z}{2\pi i}\, j_B(z) = \sum_n c_{-n} L_n^m + \sum_{m,n} \frac{m-n}{2} {:}c_{-m} c_{-n} b_{m+n}{:} - a\,c_0, \tag{F.63}$$

with the bc modes $\{b_m, c_n\} = \delta_{m+n,0}$ and weights $(h_b, h_c) = (2, -1)$ of Eq. (1.69). The single fact that drives everything is nilpotency, $Q^2 = \frac{1}{2}\{Q, Q\} = 0$, which holds iff the total central charge $c_{\mathrm{tot}} = c_m + c_{\mathrm{gh}} = c_m - 26$ vanishes (Eq. (1.77)). Two mode relations are used repeatedly. The current $j_B(z)\,b(w)$ has the total stress tensor as its single pole, so

$$\{Q,\, b_n\} = L_n^{\mathrm{tot}} = L_n^m + L_n^{\mathrm{gh}}, \qquad \text{in particular} \qquad \{Q,\, b_0\} = L_0^{\mathrm{tot}}. \tag{F.64}$$

The ghost-number current $j_{\mathrm{g}} = -{:}bc{:}$ grades the Fock space: each c_{-n} carries ghost number $+1$, each b_{-n} carries -1, and the grading operator N_{g} satisfies

$$[\,N_{\mathrm{g}},\, Q\,] = +Q, \tag{F.65}$$

so Q raises ghost number by one. We normalise N_{g} so that the b_0-annihilated ghost vacuum sits at ghost number 1 (the convention of Theorem 1.5).

The ghost zero modes b_0, c_0 anticommute with each other and square to zero, so they act on a two-dimensional ground-state doublet,

$$|{\downarrow}\rangle \coloneqq c_1|0\rangle = c(0)|0\rangle, \qquad |{\uparrow}\rangle \coloneqq c_0\,|{\downarrow}\rangle\,, \qquad b_0\,|{\downarrow}\rangle = 0, \quad c_0\,|{\downarrow}\rangle = |{\uparrow}\rangle\,, \quad b_0\,|{\uparrow}\rangle = |{\downarrow}\rangle\,. \tag{F.66}$$

Physical string states are built on $|{\downarrow}\rangle$ (dressed by matter), not on the $\mathrm{SL}(2,\mathbb{C})$-invariant vacuum $|0\rangle$: this is the standard name for the state annihilated by the global conformal (Möbius) group, which acts faithfully on the sphere as $\mathrm{PSL}(2,\mathbb{C})$ and lifts to $\mathrm{SL}(2,\mathbb{C})$ on the state space, with $|0\rangle$ its invariant representative. The BPZ inner product (the conformal, or radial-quantisation, adjoint) that makes Q Hermitian saturates the ghost zero modes, $\langle{\downarrow}|\,c_0\,|{\downarrow}\rangle = 1$; this is the origin of the c_0 insertion in every string amplitude and of the three fixed vertex positions on the sphere (see Appendix F.2 and Chapter 4).

F.4.2 Physical states as cohomology

A state is physical if it is Q-closed, and two closed states are identified if they differ by a Q-exact state,

$$\mathcal{H}_{\mathrm{phys}} = \frac{\ker Q}{\mathrm{im}\, Q} = H^{\bullet}(Q), \tag{F.67}$$

graded by ghost number. Hermiticity $Q^\dagger = Q$ together with $Q^2 = 0$ makes this well posed: an exact state $|s\rangle = Q\,|\chi\rangle$ is closed, has zero norm, and is orthogonal to every closed state, so the inner product descends to Eq. (F.67) (Exercise 1.10). Gauge artefacts are exactly the null directions of the naive Fock inner product.

Relative cohomology. Because $\{Q, b_0\} = L_0^{\mathrm{tot}}$, the operator b_0 is a contracting homotopy away from the mass shell: if $Q\,|\psi\rangle = 0$ and $L_0^{\mathrm{tot}}\,|\psi\rangle = \lambda\,|\psi\rangle$ with $\lambda \neq 0$, then $|\psi\rangle = \lambda^{-1}\{Q, b_0\}\,|\psi\rangle = Q(\lambda^{-1} b_0\,|\psi\rangle\,)$ is exact. All cohomology therefore lives at $L_0^{\mathrm{tot}} = 0$, the mass-shell condition, which on a matter state tensored with $|{\downarrow}\rangle$ reads $(L_0^m - a)\,|\psi\rangle = 0$ with the intercept $a = 1$ of Eq. (1.83). On this subspace one restricts to the *relative* complex,

$$\mathcal{H}_{\mathrm{rel}} = \{|\psi\rangle : b_0\,|\psi\rangle = 0,\ L_0^{\mathrm{tot}}\,|\psi\rangle = 0\}, \qquad \hat{Q} \coloneqq Q\big|_{\mathrm{rel}}, \tag{F.68}$$

on which $\hat{Q}$ is again nilpotent. The full (absolute) cohomology is recovered from the relative one by the ghost zero mode,

$$H^g_{\text{abs}}(Q) \;\cong\; H^g_{\text{rel}}(\hat{Q}) \;\oplus\; c_0\, H^{g-1}_{\text{rel}}(\hat{Q}), \tag{F.69}$$

a doubling that pairs each relative class with its c_0-partner. Physical string states are the relative classes at ghost number one; the c_0-image is not new physics. Working relative to b_0 is what makes the covariant physical-state conditions coincide with the old-covariant $L^m_n\,|\psi\rangle = 0$ $(n > 0)$, $(L^m_0 - 1)\,|\psi\rangle = 0$ dressed by the fixed ghost vacuum (Exercise 1.11).

Remark F.14 (Beyond the lectures: semi-infinite cohomology). $H^\bullet(Q)$ is a physicist's name for the semi-infinite (BRST) cohomology of the Virasoro algebra, made rigorous by Frenkel-Garland-Zuckerman (1986). Their vanishing theorem states that for a suitable $c = 26$ matter module the relative cohomology is concentrated at a single ghost number, so Eq. (F.69) exhausts the answer and there are no accidental classes at other degrees. The doubling Eq. (F.69) and the mass-shell localisation are the two structural inputs; the no-ghost theorem is the positivity statement layered on top.

F.4.3 The no-ghost theorem

Theorem F.15 (No-ghost theorem). *Let the matter CFT be $D = 26$ free bosons X^μ of signature $(-,+,\dots,+)$, and take $a = 1$. Pass to the null frame $X^\pm = (X^0 \pm X^1)/\sqrt{2}$ of Appendix F.4.4, with light-cone momentum components $k^\pm$. At any fixed on-shell momentum k^μ with $k^+ \neq 0$, the relative BRST cohomology Eq. (F.68) at ghost number one is isomorphic, as an inner-product space, to the Fock space $\mathcal{H}_\perp$ generated by the $D-2$ transverse oscillators α^i_{-n} ($i = 2,\dots,D-1$; $n \geq 1$) acting on the momentum ground state. That space carries a positive-definite inner product. The cohomology vanishes at every other ghost number.*

The isomorphism is an isomorphism of the physical inner product, so *positive definite* is the whole point: no negative-norm states survive, and the only null states are the Q-exact ones already quotiented out. The theorem is critical. For $D < 26$ the matter spectrum can still be made ghost-free (the original Goddard-Thorn statement holds for $D \leq 26$), but the conformal factor no longer decouples and Q is not nilpotent, so Eq. (F.67) is not even defined; for $D > 26$ negative-norm states enter the physical spectrum. Only at $c_{\text{tot}} = 0$ do nilpotency, anomaly freedom and unitarity hold together.

The result was first proved by Goddard and Thorn (1972) and independently by Brower (1972) in the old-covariant framework, using the DDF operators of Appendix F.4.5. The BRST-cohomology version, which is the one Chapter 1 promised, is due to Kato and Ogawa (1983); the clean textbook account is Polchinski (1998), §4.4. Two proofs follow: the homological one (a quartet mechanism) and the constructive one (an explicit positive-norm basis).

F.4.4 Proof by the quartet mechanism

Work at fixed momentum k^μ on the relative complex Eq. (F.68). Pick two null target directions and set $X^\pm = (X^0 \pm X^1)/\sqrt{2}$, so that $\eta_{+-} = \eta_{-+} = -1$, $\eta_{ij} = \delta_{ij}$ for the transverse $i = 2,\dots,D-1$, and

$$[\alpha^+_m, \alpha^-_n] = -\,m\,\delta_{m+n,0}, \qquad [\alpha^i_m, \alpha^j_n] = m\,\delta^{ij}\delta_{m+n,0}, \qquad \alpha^+_0 = \sqrt{\tfrac{\alpha'}{2}}\,k^+. \tag{F.70}$$

The hypothesis $k^+ \neq 0$ makes α^+_0 a nonzero c-number, and this is the lever the whole proof turns on. (Momenta with $k^+ = 0$ form a measure-zero set reached by continuity, or by choosing the null frame differently.)

Grade the relative complex by a degree that counts longitudinal ($\pm$) matter and ghost oscillator excitations, chosen so that the reduced charge splits as $\hat{Q} = Q_{(1)} + Q_{(0)} + \dots$ with $Q_{(1)}$ the

leading, α_0^+-proportional piece (Polchinski, §4.4; Kato and Ogawa, 1983),

$$Q_{(1)} = \sqrt{\tfrac{\alpha'}{2}}\, k^+ \sum_{n\neq 0} c_{-n}\, \alpha_n^- . \tag{F.71}$$

This operator squares to zero on its own: the c's anticommute among themselves and the α^-'s commute ($\eta^{--}=0$), so $Q_{(1)}^2 \propto \sum_{m,n} \alpha_m^- \alpha_n^- \{c_{-m}, c_{-n}\} = 0$. Its action on the longitudinal and ghost oscillators is read off from two (anti)commutators,

$$\{Q_{(1)},\, b_{-n}\} = \sqrt{\tfrac{\alpha'}{2}}\, k^+\, \alpha_{-n}^-, \qquad [\,Q_{(1)},\, \alpha_{-n}^+\,] = -\,n\,\sqrt{\tfrac{\alpha'}{2}}\, k^+\, c_{-n}, \qquad (n \geq 1). \tag{F.72}$$

Everything transverse commutes with $Q_{(1)}$, since α^i has no $\alpha^\pm$ or ghost content. Read Eq. (F.72) as a pairing of the four oscillators $(\alpha_{-n}^+, \alpha_{-n}^-, b_{-n}, c_{-n})$ at each level n into two $Q_{(1)}$-doublets:

- $b_{-n} \xrightarrow{Q_{(1)}} \alpha_{-n}^-$: every α_{-n}^- excitation is $Q_{(1)}$-exact, and every b_{-n} excitation fails to be closed.
- $\alpha_{-n}^+ \xrightarrow{Q_{(1)}} c_{-n}$: every c_{-n} excitation is $Q_{(1)}$-exact, and every α_{-n}^+ excitation fails to be closed.

A closed-modulo-exact count then removes all four: the longitudinal and ghost oscillators either drop out of $\ker Q_{(1)}$ or lie in $\operatorname{im} Q_{(1)}$. This is the Kugo-Ojima quartet mechanism, transplanted to the worldsheet: unphysical modes come in BRST quartets that are invisible in cohomology. What remains is

$$H^{\,1}(Q_{(1)}) \;=\; \mathcal{H}_\perp \otimes |{\downarrow}; k\rangle\,, \tag{F.73}$$

the transverse Fock space on the ghost ground state, with no $\alpha^\pm$, b or c excitations left.

The passage from $H(Q_{(1)})$ to the cohomology of the full $\hat{Q}$ is a spectral-sequence (filtration) argument: the lower-degree pieces $Q_{(0)}, \ldots$ act within Eq. (F.73), but since the leading cohomology is already concentrated at a single degree the sequence collapses at the first step and $H^1(\hat{Q}) \cong H^1(Q_{(1)})$. The inner product on $\mathcal{H}_\perp$ is built from $[\alpha_m^i, \alpha_{-m}^j] = m\,\delta^{ij} > 0$, so it is positive definite. The vanishing at other ghost numbers follows because Eq. (F.72) leaves nothing at those degrees. This proves Theorem F.15. ∎

Example F.16 (The level-one quartet)**.** Take the leading charge Eq. (F.71) and the four oscillators at level one on $|{\downarrow}; k\rangle$, writing $\kappa = \sqrt{\alpha'/2}\, k^+ \neq 0$. From Eq. (F.72),

$$Q_{(1)}\, b_{-1}\, |{\downarrow}; k\rangle = \kappa\, \alpha_{-1}^-\, |{\downarrow}; k\rangle\,, \qquad Q_{(1)}\, \alpha_{-1}^+\, |{\downarrow}; k\rangle = -\,\kappa\, c_{-1}\, |{\downarrow}; k\rangle\,,$$

while $Q_{(1)}$ annihilates $\alpha_{-1}^-\,|{\downarrow}; k\rangle$ and $c_{-1}\,|{\downarrow}; k\rangle$ (there is nothing at the next ghost number for them to map to). In the ghost-number grading of Eq. (F.65), with $|{\downarrow}\rangle$ at ghost number one, the state $c_{-1}\,|{\downarrow}; k\rangle$ sits at ghost number two and $b_{-1}\,|{\downarrow}; k\rangle$ at ghost number zero. So $\alpha_{-1}^-\,|{\downarrow}; k\rangle$ and $c_{-1}\,|{\downarrow}; k\rangle$ are exact, $\alpha_{-1}^+\,|{\downarrow}; k\rangle$ and $b_{-1}\,|{\downarrow}; k\rangle$ fail to close, and the four cancel in $H^1(Q_{(1)})$. The transverse states $\alpha_{-1}^i\,|{\downarrow}; k\rangle$ commute with $Q_{(1)}$ and are not exact, so they survive; their $D-2$ polarisations match the massless level of Section 1.5 (the graviton/B/dilaton row for the closed string, the photon for the open string).

Exercise F.8 (Quartet (anti)commutators)**.** (Standard.) Verify the two relations Eq. (F.72) directly from Eqs. (F.70) and (F.71), using $\{c_{-m}, b_{-n}\} = \delta_{m+n,0}$ and $[\alpha_m^-, \alpha_{-n}^+] = -\,n\,\delta_{m,n}$. Then extend Theorem F.16 to oscillator level two: list the longitudinal and ghost states, decide which are exact and which fail to close, and confirm that only the transverse combinations survive in $H^1(Q_{(1)})$.

Remark F.17 (Beyond the lectures: generality of the theorem). The proof used only that two of the matter bosons are noncompact free fields (to build $X^{\pm}$ and the nonzero α_0^+) and that the rest form a positive-definite $c = 24$ system. The Frenkel-Garland-Zuckerman theorem packages this: for any unitary matter CFT of total central charge 26 built on a positive-definite space and containing at least the two flat noncompact directions X^0, X^1, the relative cohomology at ghost number one is the positive-definite "transverse" subspace. This is what lets the compactifications of Chapter 8 replace 24 of the bosons by an internal CFT without reopening the ghost question, provided that internal CFT is unitary with $c = 24$ (heterotic: a $c = 16$ even self-dual lattice plus eight transverse bosons; see Appendix F.3 and Appendix G).

F.4.5 DDF operators: an explicit positive-norm basis

The quartet argument shows the physical space is *isomorphic* to $\mathcal{H}_\perp$; the DDF operators (Del Giudice, Di Vecchia and Fubini, 1972) exhibit the isomorphism by constructing the physical states directly. Fix a null reference momentum q^μ, $q^2 = 0$, normalised against the state momentum k by $2\alpha' q \cdot k = 1$, so that the shifted momenta $k + nq$ land on successive physical mass shells. The transverse DDF operators are the contour integrals of weight-one currents,

$$A_n^i = \sqrt{\tfrac{2}{\alpha'}} \oint \frac{\mathrm{d}z}{2\pi}\, \partial X^i(z)\, e^{\,i\,n\,q\cdot X(z)}, \qquad i = 2, \ldots, D-1. \tag{F.74}$$

The contour measure $\mathrm{d}z/2\pi$ is written without the usual i on purpose: the missing factor is supplied by the $-i$ in the mode expansion $\partial X^i(z) = -i\sqrt{\alpha'/2}\sum_m \alpha_m^i z^{-m-1}$, so that at $n = 0$ one recovers $A_0^i = \alpha_0^i$. The exponent is a null-momentum vertex, $(nq)^2 = 0$, so $e^{inq\cdot X}$ has weight zero and $\partial X^i\, e^{inq\cdot X}$ has weight one; the contour integral therefore commutes with the Virasoro generators,

$$[\,L_m,\, A_n^i\,] = 0 \quad \Longrightarrow \quad [\,Q,\, A_n^i\,] = 0, \tag{F.75}$$

so the A_n^i map physical states to physical states, shifting the momentum by nq. They realise a transverse oscillator algebra on the nose,

$$[\,A_m^i,\, A_n^j\,] = m\,\delta^{ij}\,\delta_{m+n,0}. \tag{F.76}$$

Acting with the raising operators A_{-n}^i on a physical ground state generates a copy of $\mathcal{H}_\perp$ built entirely from transverse (spacelike) indices, hence of manifestly positive norm. That this spans the whole physical spectrum is the Goddard-Thorn theorem. The DDF construction is the constructive face of Theorem F.15: it exhibits an explicit unitary basis, every physical state in manifestly positive-norm form, and it is the tool that makes covariant vertex operators and their factorisation tractable in Chapter 4.

Exercise F.9 (The DDF algebra)**.** (Challenging.) Derive Eq. (F.76) for two transverse DDF operators Eq. (F.74) from the free-field OPE $\partial X^i(z)\,\partial X^j(w) \sim -\frac{\alpha'}{2}\,\eta^{ij}(z-w)^{-2}$ together with the OPE of the exponentials $e^{imq\cdot X}$, $e^{inq\cdot X}$, using $q^2 = 0$ and $2\alpha' q\cdot k = 1$. Show that the cross-contraction of the two exponentials integrates to a total derivative, so the null momentum q drops out and the algebra closes on $m\,\delta^{ij}\delta_{m+n,0}$ with no central term.

F.4.6 The superstring: super-BRST and the superghost cancellation

The RNS string gauges a worldsheet supersymmetry as well as diffeomorphisms and Weyl, so the matter is D superfields with $c_m = \frac{3}{2}D$ and the constraints are the super-Virasoro generators, the stress tensor T^m and the supercurrent G^m of Eq. (3.10). Gauge fixing produces two ghost systems: the anticommuting (b, c) with $c_{bc} = -26$, and the commuting superconformal (β, γ) of weights $(\frac{3}{2}, -\frac{1}{2})$ with $c_{\beta\gamma} = +11$ (both derived as first-order systems in Appendix F.2; see also Exercise 1.9). The super-BRST charge is

$$Q = \oint \frac{\mathrm{d}z}{2\pi i}\Big[\,c\,T^m + \gamma\,G^m + \tfrac{1}{2}\,{:}c\,T^{\mathrm{gh}}{:} + (\text{ghost self-couplings})\,\Big], \qquad T^{\mathrm{gh}} = T^{bc} + T^{\beta\gamma}, \tag{F.77}$$

and, exactly as in Eq. (1.76), it is nilpotent iff the total superconformal anomaly vanishes,

$$Q^2 = 0 \iff c_m + c_{bc} + c_{\beta\gamma} = \tfrac{3}{2}D - 26 + 11 = 0 \iff D = 10. \tag{F.78}$$

The value -26 is the same weight-two reparametrisation ghost as before; the new $+11$ is the price of gauging supersymmetry, and it is what shifts the critical dimension from 26 to 10. The super no-ghost theorem holds in the same form as Theorem F.15: the relative cohomology is the Fock space of the $D-2=8$ transverse bosons α^i_{-n} and transverse fermions ψ^i_{-r}, positive definite, with the extra longitudinal fermions $\psi^{\pm}$ paired against $\beta\gamma$ excitations by a second, superconformal quartet. Constructively, the DDF construction Eq. (F.74) extends to superstring DDF operators built from the transverse $\partial X^i, \psi^i$ (Brower and Friedman, 1973).

Remark F.18 (Beyond the lectures: pictures). The $\beta\gamma$ system is commuting and half-integer weighted, so it has inequivalent vacua labelled by *picture* number, and the BRST cohomology is graded by it. A given physical state has representatives in every picture, related by the picture-changing operator $X = \{Q, \xi\}$ built from the bosonisation $\beta = e^{-\phi}\partial\xi$, $\gamma = e^{\phi}\eta$; the canonical choices are picture -1 (NS) and $-\frac{1}{2}$ (R) for vertex operators, with total picture -2 required on the sphere to soak up the $\beta\gamma$ zero modes. The bosonisation and the Ramond spin field belong to the free-field toolkit of Appendix F.2; here the point is only that "physical state = BRST class" acquires a picture label in the superstring, and that the no-ghost count is unchanged by it.

F.4.7 Closed strings and level matching

For the closed string one has two commuting copies, Q and $\tilde{Q}$, and physical states are classes of $Q + \tilde{Q}$ on the tensor product. The relative complex is defined with respect to both b_0 and $\tilde{b}_0$, and the natural combinations are

$$b_0^{\pm} := b_0 \pm \tilde{b}_0, \qquad \{Q + \tilde{Q},\, b_0^-\} = L_0^{\mathrm{tot}} - \tilde{L}_0^{\mathrm{tot}}, \tag{F.79}$$

so that imposing $b_0^- \,|\psi\rangle = 0$ enforces level matching $L_0^{\mathrm{tot}} = \tilde{L}_0^{\mathrm{tot}}$ in cohomology, while $b_0^+ \,|\psi\rangle = 0$ enforces the mass shell. The physical Hilbert space is the tensor product of two transverse Fock spaces subject to level matching, again positive definite. This is the covariant statement behind the transverse counting used throughout Chapters 1 and 3, and it is the spectrum that the torus partition function of Appendix F.3 sums, spin structure by spin structure.

Appendix G

Lie Algebras, ADE Classification, and Lattices

G.1 Lie algebras: roots, weights, Dynkin diagrams

Gauge symmetry in string theory is carried by a handful of compact Lie groups, SU(N), SO(N), and the exceptional $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$, and the physics of gauge enhancement, symmetry breaking and grand unification is the physics of how their algebras contain one another. This section is the reference recap: the Cartan subalgebra, root systems, simple roots and Dynkin diagrams, the Weyl group and the root and weight lattices, representations by highest weight, and the branching machinery used when a gauge group breaks. The exceptional algebras and the explicit GUT branchings are treated in the section on exceptional groups; the ADE meta-pattern and its appearance in singularities is treated in the section on the ADE classification; the even self-dual lattices $\Gamma_8 \oplus \Gamma_8$ and Γ_{16}, whose gauge content is $\mathrm{E}_8 \times \mathrm{E}_8$ and SO(32), live in the section on lattices. Everything here is standard; the compact treatments closest to this course are Georgi (1999), Cahn (1984), Fuchs and Schweigert (1997), and, for branching rules, Slansky's report (1981). The mathematical statements are proved in Humphreys (1972) and Fulton and Harris (1991); the definitive tables are Bourbaki (1968).

The big picture

A compact simple Lie algebra is fixed, up to isomorphism, by a small combinatorial gadget: a Dynkin diagram, a graph on r nodes with single, double or triple bonds. The number r is the rank, the dimension of the maximal torus, and it counts the simultaneously measurable charges (the Cartan generators). All the other generators are raising and lowering operators labelled by roots, points of a lattice in r-dimensional Euclidean space whose geometry, lengths and $60°/90°/120°$ angles (and, for multiple bonds, $135°/150°$), is what the diagram records. Representations are stacks of weights (charge vectors) built down from a highest weight; the highest weight is r non-negative integers, the Dynkin labels. Breaking the gauge group to a subgroup is deleting or adding nodes: delete a node of the ordinary diagram and you lose a simple root but keep the rank via a spectator U(1); delete a node of the extended diagram and you land on a maximal subalgebra of the same rank. This one picture runs the heterotic gauge groups of Chapter 3, the enhancement web of Chapters 6 and 7, and the geometric engineering of Chapters 10 and 11.

G.1.1 Gauge algebras and the Cartan-Weyl basis

A gauge theory assigns to its group G a Lie algebra $\mathfrak{g}$, spanned in the physicist's convention by Hermitian generators T^a with

$$[T^a, T^b] = i f^{abc}\, T^c, \qquad a = 1, \dots, \dim \mathfrak{g}, \tag{G.1}$$

and real structure constants f^{abc}, totally antisymmetric once indices are lowered with the Killing metric below. The gauge bosons live in the *adjoint* representation, $(\mathrm{ad}\, X)Y = [X, Y]$, of dimension $\dim \mathfrak{g}$. We take $\mathfrak{g}$ *simple* (no non-trivial ideals, so no independent gauge coupling factorises off); a general gauge algebra is a direct sum of simple pieces and abelian $\mathfrak{u}(1)$'s.

The Killing form

$$\kappa(X, Y) = \mathrm{Tr}(\mathrm{ad}_X\, \mathrm{ad}_Y) \tag{G.2}$$

is symmetric and ad-invariant, $\kappa([X, Z], Y) + \kappa(X, [Z, Y]) = 0$. Cartan's criterion states that κ is non-degenerate iff $\mathfrak{g}$ is semisimple; for the *compact real form* relevant to gauge theory, the real algebra spanned by the anti-Hermitian iT^a whose group is compact, it is negative definite on the iT^a, equivalently κ is a positive multiple of δ^{ab} on the T^a. This positive-definite metric is what lets us treat roots and weights as vectors in a Euclidean space, and it is the reason a compact gauge group has no ghosts in its group indices.

To expose the structure, complexify and pass to the *Cartan-Weyl basis*. Choose a *Cartan subalgebra* (CSA) $\mathfrak{h} \subset \mathfrak{g}$: a maximal abelian subalgebra of simultaneously diagonalisable elements. Its dimension,

$$r \coloneqq \mathrm{rank}\, \mathfrak{g} = \dim \mathfrak{h}, \tag{G.3}$$

is the *rank*. Physically $\mathfrak{h}$ is the maximal set of mutually commuting charges: the third component of isospin and the hypercharge of the Standard Model, the two independent colour charges of SU(3), the electric charge of U(1). Simultaneously diagonalising ad_H for all $H \in \mathfrak{h}$ splits the algebra into common eigenspaces,

$$[H, E_\alpha] = \alpha(H)\, E_\alpha, \qquad H \in \mathfrak{h}, \tag{G.4}$$

labelled by linear functionals $\alpha \in \mathfrak{h}^*$. The non-zero ones are the *roots*; the set of them is Δ. Each root space $\mathfrak{g}_\alpha = \mathrm{span}(E_\alpha)$ is one-dimensional, giving the *root space decomposition*

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_\alpha, \qquad \dim \mathfrak{g} = r + |\Delta|. \tag{G.5}$$

Using the Killing form to identify $\mathfrak{h}^* \cong \mathfrak{h}$, every root α becomes a vector, and (α, β) denotes the induced inner product on $\mathfrak{h}^*$. The remaining brackets are

$$[E_\alpha, E_\beta] = \begin{cases} N_{\alpha\beta}\, E_{\alpha+\beta}, & \alpha + \beta \in \Delta, \\ H_\alpha, & \beta = -\alpha, \\ 0, & \text{otherwise}, \end{cases} \tag{G.6}$$

with $H_\alpha = [E_\alpha, E_{-\alpha}] \in \mathfrak{h}$, whose normalisation as the coroot is fixed in Eq. (G.7), and $N_{\alpha\beta}$ fixed (up to sign conventions) by the root data. The $E_{\pm\alpha}$ are the raising and lowering operators: acting on a state of *weight* μ (the vector of Cartan charges of a state, Appendix G.1.5) they shift the charge to $\mu \pm \alpha$, the ladder logic of angular momentum now running on a lattice of ladders.

Example G.1 ($\mathfrak{su}(2)$ and $\mathfrak{su}(3)$)**.** For $\mathfrak{su}(2)$, $r = 1$, one positive root α, and $\{E_\alpha, E_{-\alpha}, H_\alpha\} = \{J_+, J_-, 2J_3\}$ is the familiar spin ladder. For $\mathfrak{su}(3)$, $r = 2$: the CSA is spanned by λ_3, λ_8 (the diagonal Gell-Mann matrices), there are six roots forming a regular hexagon, and the six E_α are the off-diagonal steps $T_\pm, U_\pm, V_\pm$ of the eightfold way.

G.1.2 Root systems and the $\mathfrak{sl}(2)$ strings

For each root α the three elements $\{E_\alpha, E_{-\alpha}, H_\alpha\}$ close into an $\mathfrak{su}(2)$ (an $\mathfrak{sl}(2,\mathbb{C})$ after complexifying). Its quantisation forces the roots to sit on a rigid lattice. Define the *coroot*

$$\alpha^\vee \coloneqq \frac{2\alpha}{(\alpha,\alpha)}, \tag{G.7}$$

normalised so that $H_\alpha = \alpha^\vee$ generates the $\mathfrak{su}(2)$ with $[H_\alpha, E_{\pm\alpha}] = \pm 2\, E_{\pm\alpha}$; the triplet $\{E_{-\alpha}, H_\alpha, E_\alpha\}$ is its spin-1 (adjoint) representation, on which $\mathrm{ad}_{H_\alpha} = 2J_3$ has eigenvalues $\{-2, 0, +2\}$. Applying the raising and lowering operators of the α string to another root β shows that the projection of β on α is half-integral: the *Cartan integers*

$$\langle \beta, \alpha^\vee \rangle \coloneqq \frac{2(\beta,\alpha)}{(\alpha,\alpha)} \in \mathbb{Z} \tag{G.8}$$

are integers for every pair of roots. Equivalently the α-string through β, the set $\beta + n\alpha \in \Delta$, is an unbroken run $n = -p, \ldots, q$ with $p - q = \langle \beta, \alpha^\vee \rangle$.

The integrality Eq. (G.8) is severe. Writing the angle between two roots as θ,

$$\langle \alpha, \beta^\vee \rangle \, \langle \beta, \alpha^\vee \rangle = \frac{4(\alpha,\beta)^2}{(\alpha,\alpha)(\beta,\beta)} = 4\cos^2\theta \in \{0, 1, 2, 3\} \tag{G.9}$$

for two non-parallel roots, so the only possible angles are $30°, 45°, 60°, 90°$ and their obtuse partners $120°, 135°, 150°$; between two *simple* roots only the right or obtuse ones occur. Whenever the angle is not $90°$ the squared length ratio of the two roots is fixed to 1, 2 or 3. At most two distinct root lengths occur in a simple algebra (three would need two independent ratios). An algebra with a single root length is called *simply-laced*; these are the A, D and E types, the ones that arise from lattices (Appendix G.4) and from surface singularities (Appendix G.3).

A *root system* is the abstract package: a finite set Δ spanning a Euclidean space, closed under the reflections

$$s_\alpha(\beta) = \beta - \langle \beta, \alpha^\vee \rangle \, \alpha, \tag{G.10}$$

with all Cartan integers integral and with $\pm\alpha$ the only multiples of α in Δ. Every simple Lie algebra gives one, and by Serre's theorem the root system reconstructs the algebra.

G.1.3 Simple roots, the Cartan matrix, and Dynkin diagrams

Pick any hyperplane through the origin missing all roots; it splits Δ into *positive* and *negative* roots, $\Delta = \Delta_+ \sqcup \Delta_-$. A positive root is *simple* if it is not the sum of two positive roots. The simple roots

$$\Pi = \{\alpha_1, \ldots, \alpha_r\} \tag{G.11}$$

are a basis of $\mathfrak{h}^*$, and every positive root is a *non-negative* integer combination $\sum_i n_i \alpha_i$; the number $\sum_i n_i$ is its height. All of the root data are then packed into the *Cartan matrix*

$$A_{ij} \coloneqq \langle \alpha_i, \alpha_j^\vee \rangle = \frac{2(\alpha_i,\alpha_j)}{(\alpha_j,\alpha_j)}. \tag{G.12}$$

Its diagonal is $A_{ii} = 2$; its off-diagonal entries are non-positive integers with $A_{ij} = 0 \iff A_{ji} = 0$; and for a finite-dimensional simple algebra it is positive definite (in particular $\det A > 0$, and $\det A$ turns out to be the order of the centre, Appendix G.1.4). The convention in Eq. (G.12) normalises on the *column* root; a transpose convention also circulates, and the two agree for simply-laced algebras where $A = A^{\mathsf{T}}$.

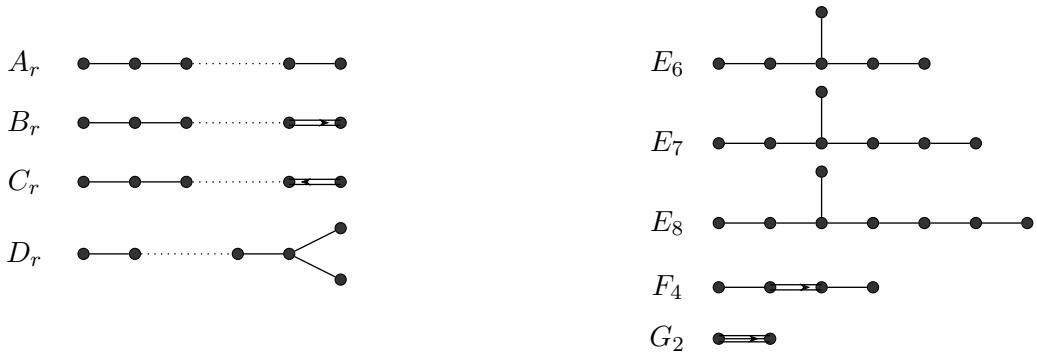


Figure G.1: The connected Dynkin diagrams. Left: the classical families $A_r = \mathfrak{su}(r+1)$, $B_r = \mathfrak{so}(2r+1)$, $C_r = \mathfrak{sp}(r)$, $D_r = \mathfrak{so}(2r)$. Right: the exceptionals. A double or triple bond carries an arrow from the long root to the short root. The simply-laced cases $A_r, D_r, E_{6,7,8}$ have all bonds single.

The Cartan matrix is drawn as a *Dynkin diagram*: one node per simple root, and between nodes $i \neq j$ a bond of

$$A_{ij}A_{ji} = 4\cos^2\theta_{ij} \in \{0, 1, 2, 3\} \tag{G.13}$$

lines, with an arrow on any multiple bond pointing from the long root to the short one. A single bond is a 120° pair of equal-length roots; a double bond a 135° pair with length ratio $\sqrt{2}$; a triple bond a 150° pair with ratio $\sqrt{3}$. A connected diagram is a simple algebra; a disconnected one is a direct sum of simple factors.

Theorem G.2 (Killing-Cartan classification)**.** *The connected Dynkin diagrams, equivalently the simple complex Lie algebras up to isomorphism, are the four infinite families $A_r\,(r \geq 1)$, $B_r\,(r \geq 2)$, $C_r\,(r \geq 3)$, $D_r\,(r \geq 4)$ and the five exceptional cases E_6, E_7, E_8, F_4, G_2, shown in Fig. G.1. The restrictions on r remove the coincidences $A_1 = B_1 = C_1$, $B_2 = C_2$, $A_3 = D_3$ and the reducible $D_2 = A_1 \times A_1$.*

The identification of each diagram with a matrix algebra, together with the invariants used throughout the notes, is collected in Table G.1. Two identities hold for every simple algebra: $\dim \mathfrak{g} = r(h+1)$ and $|\Delta| = rh$, where $h = |\Delta|/r$ is the *Coxeter number* (Appendix G.1.4). Thus E_8 has $\dim = 8 \cdot 31 = 248$ and 240 roots.

G.1.4 The Weyl group and the root and weight lattices

The reflections Eq. (G.10) in the simple roots generate a finite group,

$$W = \langle s_1, \ldots, s_r \rangle, \qquad s_i \coloneqq s_{\alpha_i}, \tag{G.14}$$

the *Weyl group*. It permutes the roots, preserves the inner product, and acts as the residual discrete gauge symmetry on the Cartan charges: two weights in the same W-orbit are physically the same set of charges seen in different Cartan frames. For A_r, $W = S_{r+1}$ permutes the $r+1$ eigenvalues, $|W| = (r+1)!$; at the other extreme $|W(E_8)| = 696\,729\,600$. The closure of a fundamental domain for W, the *fundamental Weyl chamber*, is $\{\lambda : (\lambda, \alpha_i) \geq 0\ \forall i\}$, and it is where the highest weights (Appendix G.1.5) live.

Two lattices sit in $\mathfrak{h}^*$. The *root lattice* is the integer span of the roots,

$$Q = \bigoplus_{i=1}^{r} \mathbb{Z}\,\alpha_i, \tag{G.15}$$

Cartan	algebra $\mathfrak{g}$	compact G	r	$\dim \mathfrak{g}$	centre P/Q	h	$h^\vee$
A_r	$\mathfrak{su}(r{+}1)$	$\mathrm{SU}(r{+}1)$	r	$r(r{+}2)$	$\mathbb{Z}_{r+1}$	$r{+}1$	$r{+}1$
B_r	$\mathfrak{so}(2r{+}1)$	$\mathrm{Spin}(2r{+}1)$	r	$r(2r{+}1)$	$\mathbb{Z}_2$	$2r$	$2r{-}1$
C_r	$\mathfrak{sp}(r)$	$\mathrm{USp}(2r)$	r	$r(2r{+}1)$	$\mathbb{Z}_2$	$2r$	$r{+}1$
D_r	$\mathfrak{so}(2r)$	$\mathrm{Spin}(2r)$	r	$r(2r{-}1)$	$\mathbb{Z}_4$ or $\mathbb{Z}_2^2$	$2r{-}2$	$2r{-}2$
E_6	$\mathfrak{e}_6$	E_6	6	78	$\mathbb{Z}_3$	12	12
E_7	$\mathfrak{e}_7$	E_7	7	133	$\mathbb{Z}_2$	18	18
E_8	$\mathfrak{e}_8$	E_8	8	248	1	30	30
F_4	$\mathfrak{f}_4$	F_4	4	52	1	12	9
G_2	$\mathfrak{g}_2$	G_2	2	14	1	6	4

Table G.1: Data for the simple Lie algebras: rank r, dimension, the centre $P/Q = Z(\tilde{G})$ of the simply connected group, and the Coxeter and dual Coxeter numbers $h, h^\vee$ (all three defined in Appendix G.1.4). For D_r the centre is $\mathbb{Z}_4$ for r odd and $\mathbb{Z}_2 \times \mathbb{Z}_2$ for r even. Simply-laced algebras have $h = h^\vee$.

and the *weight lattice* P is its dual with respect to the coroots,

$$P = \{\lambda \in \mathfrak{h}^* : \langle \lambda, \alpha_i^\vee \rangle \in \mathbb{Z} \ \ \forall i\}. \tag{G.16}$$

A basis of P is furnished by the *fundamental weights* ω_i, defined by

$$\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}, \qquad \text{so} \qquad \alpha_i = \sum_j A_{ij}\, \omega_j. \tag{G.17}$$

Because the change of basis from the ω_i to the α_i is the Cartan matrix, $Q \subseteq P$ with

$$|P/Q| = \det A = |Z(\tilde{G})|, \tag{G.18}$$

the order of the centre of the simply connected group $\tilde{G}$ (the column in Table G.1). The quotient P/Q labels the allowed global forms of the gauge group and the electric-magnetic superselection sectors; a weight in P is an admissible charge for the simply connected group, and Q is the sublattice realised by the adjoint (hence by any purely gluonic state). One combination appears constantly, the *Weyl vector*

$$\rho = \sum_{i=1}^{r} \omega_i = \tfrac{1}{2} \sum_{\alpha \in \Delta_+} \alpha. \tag{G.19}$$

Finally, normalise. String applications fix the long roots to

$$(\theta, \theta) = 2, \tag{G.20}$$

with θ the highest root, i.e. the highest weight of the adjoint (Appendix G.1.5). Then the simply-laced roots all have length squared 2 and the root lattice Q is even (Appendices G.3.1 and G.4). The *dual Coxeter number* is

$$h^\vee = 1 + \sum_{i=1}^{r} a_i^\vee, \qquad \theta = \sum_i a_i^\vee\, \alpha_i^\vee \ \text{(comarks)}, \tag{G.21}$$

and it sets the normalisation of the Killing form, $\kappa = 2h^\vee\,(\,\cdot\,,\cdot\,)$ in the basis where $(\theta, \theta) = 2$, controls the one-loop gauge beta function, and enters the Sugawara central charge $c = k \dim \mathfrak{g}/(k{+}h^\vee)$ of the level-k current algebra used in Chapter 3. A current algebra is the infinite-dimensional

worldsheet symmetry whose zero modes close on $\mathfrak{g}$; the *level* k is a non-negative integer labelling its central extension, and c is the Virasoro central charge of Appendix F. The heterotic gauge symmetry sits at $k = 1$; the level-one currents are built in Appendix G.4, and the general construction is in Di Francesco, Mathieu and Sénéchal (1997, Ch. 14).

G.1.5 Representations and weights

Matter sits in representations. A finite-dimensional representation R of $\mathfrak{g}$ decomposes under the CSA into weight spaces,

$$\mathfrak{h}\,|\mu\rangle = \mu\,|\mu\rangle\,, \qquad \mu \in \Pi(R) \subset P, \tag{G.22}$$

so a *weight* μ is just the vector of Cartan charges of a state, and every weight of every representation lies in the weight lattice P. The $E_{\pm\alpha}$ move between weight spaces, $E_\alpha\,|\mu\rangle \propto |\mu+\alpha\rangle$, so $\Pi(R)$ is a union of root strings, symmetric under W. There is a unique *highest weight* Λ, the one annihilated by all raising operators E_{α_i}; it is dominant, meaning

$$\Lambda = \sum_{i=1}^{r} \lambda_i\,\omega_i, \qquad \lambda_i = \langle \Lambda, \alpha_i^\vee \rangle \in \mathbb{Z}_{\geq 0}, \tag{G.23}$$

and the r non-negative integers $[\lambda_1, \ldots, \lambda_r]$ are the *Dynkin labels*. The theorem of the highest weight (Cartan; see Fulton and Harris 1991) states that

$$\{\text{irreducible representations of } \mathfrak{g}\} \;\longleftrightarrow\; \{\text{dominant weights } \Lambda \in P_+\}, \tag{G.24}$$

a bijection. The fundamental representations are $\Lambda = \omega_i$, one per node of the Dynkin diagram; the adjoint is $\Lambda = \theta$, the highest root; the singlet is $\Lambda = 0$. For $\mathfrak{su}(N)$ the defining representation is $\omega_1 = \mathbf{N}$ and its conjugate is $\omega_{N-1} = \overline{\mathbf{N}}$. A representation is real, complex or pseudoreal according to how the *outer automorphism* (a symmetry of the Dynkin diagram) acts on Λ; whether it is complex decides whether it can carry chiral fermions, the criterion that singles out SU(5), SO(10) and E_6 for unification (Appendix G.2).

Dimensions come from the *Weyl dimension formula*,

$$\dim R_\Lambda = \prod_{\alpha \in \Delta_+} \frac{(\Lambda + \rho, \alpha)}{(\rho, \alpha)}, \tag{G.25}$$

with ρ the Weyl vector. Two invariants govern the gauge bookkeeping. The *Dynkin index* $T(R)$ and the quadratic *Casimir* $C_2(R)$ are defined by

$$\mathrm{Tr}_R(T^a T^b) = T(R)\,\delta^{ab}, \qquad \sum_a T^a T^a = C_2(R)\,\mathbf{1}, \tag{G.26}$$

tied together by $T(R)\,\dim\mathfrak{g} = C_2(R)\,\dim R$. We fix the generators as in Appendix A, $\mathrm{Tr}(T_I T_J) = \tfrac{1}{2}\delta_{IJ}$ on the $\mathfrak{su}(N)$ fundamental, so that $T(\text{fund}) = \tfrac{1}{2}$, $C_2(\text{fund}) = (N^2-1)/2N$, and the adjoint has

$$T(\text{adj}) = C_2(\text{adj}) = h^\vee, \tag{G.27}$$

which is why $h^\vee$ controls the one-loop beta function. In the geometric normalisation $(\theta,\theta) = 2$ for weights the Casimir eigenvalue is read off as $C_2(R_\Lambda) = \tfrac{1}{2}(\Lambda, \Lambda + 2\rho)$; the bare combination $(\Lambda, \Lambda+2\rho)$ used in some texts is twice this, corresponding to the generator normalisation $\mathrm{Tr}(T_I T_J) = \delta_{IJ}$ (in which $T(\text{adj}) = 2h^\vee$). This Appendix-A normalisation is the one carried throughout the appendix, including the embedding indices of Appendix G.2. The index is additive over direct sums and multiplicative, $T(R_1 \otimes R_2) = T(R_1)\dim R_2 + \dim R_1\, T(R_2)$, which is the arithmetic behind the level counting of Chapter 3.

Remark G.3 (Beyond the lectures: reference traces and the E_8 quartic). The anomaly polynomials of Chapter 5 and Appendix H need a fixed *reference trace* tr for each group, distinct from the abstract index $T(R)$. For the classical groups tr is the trace in the defining representation: the vector **32** for $\mathfrak{so}(32)$, the fundamental **N** for $\mathfrak{su}(N)$. E_8 has no representation below the adjoint **248**, so one sets

$$\operatorname{tr} F^2 := \tfrac{1}{30}\,\operatorname{Tr}_{\text{adj}} F^2, \qquad \operatorname{Tr}_{\text{adj}} F^4 = \tfrac{1}{100}\big(\operatorname{Tr}_{\text{adj}} F^2\big)^2, \tag{G.28}$$

the factor $\frac{1}{30} = 1/h^\vee$ normalising against the adjoint. The second identity, that E_8 has no independent quartic Casimir, is what lets the $E_8 \times E_8$ and $\mathfrak{so}(32)$ anomaly twelve-forms factorise. These per-group conventions are collected in Appendix H and must not be mixed across groups.

Example G.4 (Weights of the **3** and **8** of $\mathfrak{su}(3)$)**.** Take A_2 with simple roots α_1, α_2 at 120°, each of length squared 2, so $A = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ and $\rho = \omega_1 + \omega_2 = \alpha_1 + \alpha_2$. The fundamental $\mathbf{3} = [1,0]$ has highest weight ω_1 and weights $\{\omega_1,\ \omega_1 - \alpha_1,\ \omega_1 - \alpha_1 - \alpha_2\}$, the three corners of a triangle. The adjoint $\mathbf{8} = [1,1]$ has highest weight $\theta = \alpha_1 + \alpha_2$; its weights are the six roots plus the origin with multiplicity two (the Cartan). For the adjoint $\Lambda = \theta = \rho$, so $\Lambda + \rho = 2\rho$ and Eq. (G.25) collapses to $\prod_{\alpha\in\Delta_+}(2\rho,\alpha)/(\rho,\alpha) = 2^{|\Delta_+|} = 2^3 = 8$; the same formula returns $\dim[1,0] = 3$.

G.1.6 Branching and gauge symmetry breaking

When a gauge group G breaks to a subgroup G', every representation of G becomes a (generally reducible) sum of representations of G'. This *branching* is the arithmetic of unification and of gauge enhancement, and it is worked entirely with weights. An embedding $\mathfrak{g}' \subset \mathfrak{g}$ is encoded by a *projection matrix* P that sends each weight $\mu \in \mathfrak{h}^*$ of $\mathfrak{g}$ to a weight of $\mathfrak{g}'$; applying P to the weight system $\Pi(R)$ and re-grouping into $\mathfrak{g}'$ highest weights gives the branching. The extensive tables are Slansky (1981).

Two mechanisms cover almost every case in the notes.

Adjoint breaking (regular, rank preserved). Giving a vacuum expectation value to a scalar in the adjoint along a Cartan direction $\langle\Phi\rangle \in \mathfrak{h}$ leaves unbroken exactly the generators commuting with it. Since $\langle\Phi\rangle$ is in the CSA, the full rank r survives: the unbroken algebra is $\mathfrak{g}' \oplus \mathfrak{u}(1)^k$ where the semisimple part $\mathfrak{g}'$ is generated by the roots orthogonal to $\langle\Phi\rangle$. Diagrammatically, deleting one node from the ordinary Dynkin diagram gives a maximal such $\mathfrak{g}'$ of rank $r-1$, with one leftover $\mathfrak{u}(1)$. This is the Standard-Model breaking $\mathfrak{su}(5) \to \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$: delete one of the two inner nodes of the A_4 chain (node 2 or node 3, each leaving an A_2 and an A_1).

Maximal regular subalgebras (same rank, no spare U(1)*).* Adjoin the lowest root $-\theta$ to the simple roots to form the *extended* (affine) Dynkin diagram, then delete one node. The result is a maximal regular subalgebra of the *same* rank r (Borel-de Siebenthal). This is the source of $\mathfrak{e}_8 \supset \mathfrak{e}_7 \oplus \mathfrak{su}(2)$, $\mathfrak{e}_8 \supset \mathfrak{e}_6 \oplus \mathfrak{su}(3)$, $\mathfrak{e}_8 \supset \mathfrak{so}(16)$, and $\mathfrak{so}(10) \supset \mathfrak{su}(5) \oplus \mathfrak{u}(1)$, the decompositions that organise the heterotic gauge sector and the F-theory GUT branchings; the exceptional cases are worked out in Appendix G.2. Not every subalgebra is regular: the *special* embeddings, such as $\mathfrak{su}(2) \subset \mathfrak{su}(3)$ via the spin-1 (adjoint) $\mathfrak{su}(2)$, do not come from deleting nodes and are read off from index and dimension data.

Example G.5 ($\mathfrak{su}(3) \to \mathfrak{su}(2) \oplus \mathfrak{u}(1)$)**.** Delete node α_2 of A_2. The surviving simple root α_1 generates the $\mathfrak{su}(2)$, and the orthogonal Cartan direction is the hypercharge Y; the projection P keeps the α_1-component of a weight and reads off Y from the component along $\langle\Phi\rangle$. Projecting the weights of the previous example, the fundamental and adjoint branch as

$$\mathbf{3} \;\to\; \mathbf{2}_{+1} \;\oplus\; \mathbf{1}_{-2}, \tag{G.29}$$

$$\mathbf{8} \;\to\; \mathbf{3}_{0} \;\oplus\; \mathbf{1}_{0} \;\oplus\; \mathbf{2}_{+3} \;\oplus\; \mathbf{2}_{-3}, \tag{G.30}$$

with Y normalised to integer charges. The $\mathbf{2}_{\pm 3}$ in the adjoint are the broken generators, the analogue of the X, Y bosons in a genuine GUT; the traceless-charge condition $\sum Y = 0$ over each multiplet is the fingerprint of a regular, rank-preserving embedding.

Remark G.6 (Beyond the lectures: anomalies and the cubic index). For complex representations a third invariant matters, the cubic *anomaly coefficient* $A(R)$ defined by $\mathrm{Tr}_R\{T^a, T^b\}T^c = A(R)\, d^{abc}$ with d^{abc} the symmetric invariant. Only $\mathfrak{su}(N \geq 3)$ has a non-zero d^{abc}; all other simple algebras are automatically free of the perturbative gauge anomaly, which is one reason $\mathfrak{so}$ and exceptional gauge groups are attractive in ten dimensions. The interplay of $A(R)$ with the Green-Schwarz mechanism is the subject of Appendix H.

Exercise G.1. Using Eq. (G.12), write the Cartan matrix of G_2 (two simple roots, a triple bond, one long and one short root with $(\alpha_\ell, \alpha_\ell) = 2$, $(\alpha_s, \alpha_s) = \frac{2}{3}$). Verify $\det A = 1$, consistent with the trivial centre in Table G.1, and use Eq. (G.17) to find the fundamental weights ω_1, ω_2 in the basis of simple roots.

Exercise G.2. For $\mathfrak{su}(N) = A_{N-1}$ show from Eq. (G.18) that $P/Q = \mathbb{Z}_N$, and identify the generator of P/Q with the N-ality (the number of boxes of a Young tableau modulo N). Confirm that the adjoint has N-ality 0, so gluons never change the centre charge.

G.2 Exceptional groups and GUT branchings

There are five exceptional simple Lie algebras, $\mathfrak{g}_2 \subset \mathfrak{f}_4 \subset \mathfrak{e}_6 \subset \mathfrak{e}_7 \subset \mathfrak{e}_8$, sitting outside the four infinite families A_n, B_n, C_n, D_n. Three of them, together with the large orthogonal algebra $\mathfrak{so}(32)$, carry the whole weight of string phenomenology: the heterotic string of Chapter 3 has gauge algebra $\mathfrak{e}_8 \oplus \mathfrak{e}_8$ or $\mathfrak{so}(32)$, and the exceptional Kodaira fibres of F-theory (Section 11.3) realise $\mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8$ as localised gauge symmetries. This section collects the data one actually needs about these algebras: their invariants, their maximal subalgebras, and the representation branchings that carry a chiral generation down the grand-unified chain $\mathrm{E}_8 \to \mathrm{E}_6 \to \mathrm{SO}(10) \to \mathrm{SU}(5) \to G_{\mathrm{SM}}$. The general machinery of roots, weights and Dynkin diagrams lives in Appendix G.1; the simply-laced ADE story and its McKay correspondence in Appendix G.3; the even self-dual and Narain lattices in Appendix G.4. Here the emphasis is representation theory in the service of model building.

Why grand unification likes the exceptional groups

A single Standard-Model generation is fifteen Weyl fermions with a lopsided set of hypercharges, plus a sterile neutrino makes sixteen. Grand unification is the observation that these sixteen states are one irreducible representation of a larger group: the **16** of SO(10), or the **27** of E_6. For that to describe chiral fermions the representation must be *complex* (inequivalent to its conjugate), and among the candidate unifying groups only SU(5), SO(10) and E_6 have complex representations of the right size. E_7 and E_8 do not, so a chiral theory built from them must first descend to E_6 or below. The exceptional groups earn their place because the descent $\mathrm{E}_8 \to \mathrm{E}_6 \to \mathrm{SO}(10) \to \mathrm{SU}(5)$ is a chain of rank-preserving or rank-lowering embeddings that keep a generation inside a single irreducible representation as far down as SO(10), where it is the **16**. Below that it splits only as much as group theory forces, into the $\mathbf{10} \oplus \overline{\mathbf{5}}$ of SU(5), and every branching is fixed once and for all.

G.2.1 Invariants of the exceptional algebras

Table G.2 lists the rank r, dimension, Coxeter number h, dual Coxeter number $h^\vee$, and the smallest non-trivial and adjoint representations of the exceptional algebras, with $\mathfrak{so}(32) = \mathfrak{d}_{16}$

$\mathfrak{g}$	r	$\dim \mathfrak{g}$	h	$h^\vee$	smallest rep.	adjoint
$\mathfrak{g}_2$	2	14	6	4	**7** (real)	**14**
$\mathfrak{f}_4$	4	52	12	9	**26** (real)	**52**
$\mathfrak{e}_6$	6	78	12	12	**27** (complex)	**78**
$\mathfrak{e}_7$	7	133	18	18	**56** (pseudoreal)	**133**
$\mathfrak{e}_8$	8	248	30	30	**248** (real)	**248**
$\mathfrak{so}(32)$	16	496	30	30	**32** (real)	**496**

Table G.2: Invariants of the exceptional algebras and of $\mathfrak{so}(32)$. The reality type of the smallest representation controls whether the algebra can support chiral fermions: only $\mathfrak{e}_6$ (and its subalgebras $\mathfrak{so}(10)$, $\mathfrak{su}(5)$) has a complex fundamental. The two spinors of $\mathfrak{so}(32)$ have dimension $2^{15} = 32768$.

included for reference. For a simply-laced algebra $h = h^\vee$ and all roots have the same length; $\mathfrak{f}_4$ and $\mathfrak{g}_2$ are not simply-laced and carry two root lengths, which is why $h \neq h^\vee$ there.

Three facts from the table do most of the work later. First, $\mathfrak{e}_8$ is the only simple algebra whose smallest representation is the adjoint: it has no representation smaller than **248**, which is why the E_8 gauge bosons of the heterotic string come with nothing lighter charged under them, and why E_8 must be broken before it can carry matter. Second, the smallest representations grow as $7, 26, 27, 56, 248$ across the exceptional series, and only the **27** of E_6 is complex. Third, the dual Coxeter numbers $h^\vee(E_6, E_7, E_8) = (12, 18, 30)$ set the central charges of the level-one current algebras that realise these groups on the heterotic worldsheet (Appendix G.2.5).

G.2.2 Maximal subalgebras: regular and special

A subalgebra $\mathfrak{g}' \subset \mathfrak{g}$ is *maximal* if no proper subalgebra sits strictly between them. Maximal subalgebras come in two kinds. A *regular* subalgebra is generated by a subset of the root system of $\mathfrak{g}$; its Cartan subalgebra is that of $\mathfrak{g}$, and it has the same or lower rank. A *special* (non-regular) subalgebra is embedded some other way, and its roots are not a subset of those of $\mathfrak{g}$. The regular ones are found by a diagrammatic recipe; the special ones have to be tabulated.

Maximal-rank regular subalgebras from the extended diagram. Recall from Appendix G.1.6 that adjoining the affine node associated with the lowest root $-\theta$ to the Dynkin diagram of $\mathfrak{g}$ and then deleting one node yields the Dynkin diagram of a maximal-rank semisimple subalgebra; this is the Borel-de Siebenthal construction. What is new for the exceptional algebras is the arithmetic that decides which deletions are useful. Label each node i with its *mark* a_i, the coefficient of the simple root α_i in $\theta = \sum_i a_i \alpha_i$, and give the affine node $a_0 = 1$. The marks sum to the Coxeter number, $\sum_{i \geq 0} a_i = h$. Deleting the affine node returns $\mathfrak{g}$ itself; deleting any node with mark $a_i = 1$ gives another copy of $\mathfrak{g}$; deleting a node with prime mark $a_i = p$ gives a genuine maximal subalgebra; a node of composite mark gives a non-maximal one.

Figure G.2 shows the extended diagram of E_8, whose marks are $(1, 2, 3, 4, 5, 6, 4, 2)$ along the chain with a 3 on the branch, summing to $30 = h$. Reading off the deletions gives the maximal-rank regular subalgebras of $\mathfrak{e}_8$,

$$\mathfrak{e}_8 \supset \mathfrak{e}_7 \oplus \mathfrak{su}(2), \quad \mathfrak{e}_6 \oplus \mathfrak{su}(3), \quad \mathfrak{su}(9), \quad \mathfrak{so}(16), \quad \mathfrak{su}(5) \oplus \mathfrak{su}(5), \tag{G.31}$$

each of rank 8. The same procedure applied to $\mathfrak{e}_7$ and $\mathfrak{e}_6$ gives

$$\mathfrak{e}_7 \supset \mathfrak{su}(8), \quad \mathfrak{so}(12) \oplus \mathfrak{su}(2), \quad \mathfrak{su}(6) \oplus \mathfrak{su}(3), \quad \mathfrak{e}_6 \oplus \mathfrak{u}(1), \tag{G.32}$$

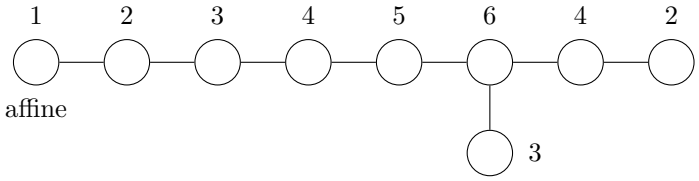


Figure G.2: Extended (affine) Dynkin diagram of E_8, with the mark a_i printed by each node. The leftmost node is the affine node ($a_0 = 1$), and the trivalent node carries the maximal mark 6. Deleting a node yields a maximal-rank subalgebra: removing the mark-2 node adjacent to the affine node gives $\mathfrak{e}_7 \oplus \mathfrak{su}(2)$; removing the mark-3 node on the chain gives $\mathfrak{e}_6 \oplus \mathfrak{su}(3)$; removing the mark-3 branch node gives $\mathfrak{su}(9)$; removing the mark-5 node gives $\mathfrak{su}(5) \oplus \mathfrak{su}(5)$; removing the mark-2 node at the far (right-hand) end gives $\mathfrak{so}(16)$. The trivalent mark-6 node has composite mark and its deletion gives a non-maximal subalgebra.

$$\mathfrak{e}_6 \supset \ \mathfrak{su}(6) \oplus \mathfrak{su}(2), \quad \mathfrak{su}(3) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(3), \quad \mathfrak{so}(10) \oplus \mathfrak{u}(1). \tag{G.33}$$

The subalgebras with a $\mathfrak{u}(1)$ factor, $\mathfrak{e}_6 \oplus \mathfrak{u}(1) \subset \mathfrak{e}_7$ and $\mathfrak{so}(10) \oplus \mathfrak{u}(1) \subset \mathfrak{e}_6$, are obtained by deleting a node from the *ordinary* (unextended) diagram rather than the affine one: removing one node leaves a rank-$(r-1)$ semisimple part, and the missing Cartan direction becomes the abelian factor. These are exactly the steps of the grand-unified chain, because the surviving $\mathfrak{u}(1)$ is a candidate for extra hypercharge structure.

Example G.7 (A branching from a node deletion)**.** Take the SO(16) of (G.31), obtained by deleting the far mark-2 node so that the extended E_8 diagram becomes the D_8 diagram. The 240 roots of E_8 split according to whether they lie in the $\mathfrak{so}(16)$ root system. In an orthonormal basis they are the 112 vectors $\pm e_i \pm e_j$ $(i < j)$, which are the roots of $\mathfrak{so}(16)$, together with the 128 vectors $\frac{1}{2}(\pm 1, \ldots, \pm 1)$ with an even number of minus signs. The first set plus the 8 Cartan generators is the adjoint **120** of SO(16); the second set is a chiral spinor **128**. Hence

$$\mathbf{248} = \mathbf{120} \oplus \mathbf{128},$$

which is (G.38). The same bookkeeping, roots of $\mathfrak{g}$ sorted into roots of the subalgebra plus a remainder that fills out a definite representation of it, produces every entry of Eqs. (G.36) to (G.39).

Special subalgebras and foldings. The exceptional non-simply-laced algebras $\mathfrak{f}_4$ and $\mathfrak{g}_2$ arise as fixed-point subalgebras of an outer automorphism, the operation known as *folding*. An order-n symmetry of the Dynkin diagram identifies the roots it permutes and produces a smaller diagram with roots of two lengths:

$$\mathfrak{a}_{2n-1} \to \mathfrak{c}_n, \quad \mathfrak{d}_{n+1} \to \mathfrak{b}_n, \quad \mathfrak{d}_4 \xrightarrow{\mathbb{Z}_3} \mathfrak{g}_2, \quad \mathfrak{e}_6 \xrightarrow{\mathbb{Z}_2} \mathfrak{f}_4. \tag{G.34}$$

These give the special maximal subalgebras $\mathfrak{f}_4 \subset \mathfrak{e}_6$ and $\mathfrak{g}_2 \subset \mathfrak{so}(8)$. The folding is the group-theoretic content of the F-theory statement (Section 10.1) that F_4 and G_2 need a monodromy: an ADE singularity transported around a base can come back rotated by the diagram automorphism, and the invariant part of the gauge algebra is the folded, non-simply-laced one. The Tate table that reads $\mathfrak{f}_4, \mathfrak{g}_2$ and the other refined algebras off the Weierstrass data (f, g) belongs with the F-theory construction (Chapter 11 and Appendix E); here we record only the folding. The remaining special maximal subalgebras of the exceptional algebras, the principal $\mathfrak{su}(2)$'s and a short list of products, are tabulated exhaustively by Dynkin; the standard reference tables, with branchings for every representation one meets in practice, are those of Slansky and, in modern completeness, Yamatsu.

Embedding index. Branchings carry one numerical invariant worth naming. For an embedding $\mathfrak{g}' \hookrightarrow \mathfrak{g}$ under which a representation R of $\mathfrak{g}$ decomposes as $\bigoplus_a R'_a$, the *embedding index* (Dynkin index of the embedding) is

$$j_{\mathfrak{g}'\hookrightarrow\mathfrak{g}} \;=\; \frac{\sum_a T(R'_a)}{T(R)}, \tag{G.35}$$

where $T(R)$ is the Dynkin index of a representation in the trace normalisation $\mathrm{Tr}(T_I T_J) = \frac{1}{2}\delta_{IJ}$ of Appendix A, so that $T(\text{fundamental of SU}(N)) = \frac{1}{2}$ and $T(\text{adjoint}) = h^\vee$. This is half the value in the $(\theta,\theta) = 2$ convention of Appendix G.1.5, where $T(\text{fund}) = 1$ and $T(\text{adjoint}) = 2h^\vee$; the ratio (G.35) is the same in either. The ratio is independent of R. Every regular maximal-rank embedding in Eqs. (G.31) to (G.33) has $j = 1$ on each simple factor, and $\mathfrak{f}_4 \subset \mathfrak{e}_6$ likewise has $j = 1$. An index-one embedding maps long roots to long roots and preserves the level of a current algebra, $k' = j\,k$; this is why the heterotic gauge symmetry, realised at level one (Appendix G.2.5), stays at level one all the way down the GUT chain, and it is the algebraic reason the Katz-Vafa enhancement of Chapter 11 works representation by representation.

G.2.3 The grand-unified chain $\mathrm{E}_8 \to \mathrm{E}_6 \to \mathrm{SO}(10) \to \mathrm{SU}(5)$

The physically decisive branchings are those that carry the descent from E_8 to the Standard Model. We list them step by step. Abelian charges are written as subscripts in the normalisation of Slansky's tables; an overall rescaling of each $\mathfrak{u}(1)$ is a convention, but the ratios of charges within a multiplet are physical.

E_8 down to E_6 and E_7. The adjoint **248** branches under the maximal-rank subalgebras of Eq. (G.31) as

$$\mathrm{E}_8 \supset \mathrm{E}_7 \times \mathrm{SU}(2): \quad \mathbf{248} = (\mathbf{133},\mathbf{1}) \oplus (\mathbf{56},\mathbf{2}) \oplus (\mathbf{1},\mathbf{3}), \tag{G.36}$$

$$\mathrm{E}_8 \supset \mathrm{E}_6 \times \mathrm{SU}(3): \quad \mathbf{248} = (\mathbf{78},\mathbf{1}) \oplus (\mathbf{1},\mathbf{8}) \oplus (\mathbf{27},\mathbf{3}) \oplus (\overline{\mathbf{27}},\overline{\mathbf{3}}), \tag{G.37}$$

$$\mathrm{E}_8 \supset \mathrm{SO}(16): \quad \mathbf{248} = \mathbf{120} \oplus \mathbf{128}, \tag{G.38}$$

$$\mathrm{E}_8 \supset \mathrm{SU}(5) \times \mathrm{SU}(5): \quad \mathbf{248} = (\mathbf{24},\mathbf{1}) \oplus (\mathbf{1},\mathbf{24}) \oplus (\mathbf{10},\mathbf{5}) \oplus (\overline{\mathbf{5}},\mathbf{10}) \oplus (\overline{\mathbf{10}},\overline{\mathbf{5}}) \oplus (\mathbf{5},\overline{\mathbf{10}}). \tag{G.39}$$

The commutant of SU(3) in E_8 is E_6, and the $(\mathbf{27},\mathbf{3})$ of (G.37) is the seed of every heterotic E_6 model: breaking one E_8 by an SU(3) bundle on a Calabi-Yau (the standard embedding of Chapter 8) leaves E_6 unbroken and turns the **3** index of the $(\mathbf{27},\mathbf{3})$ into a bundle label, so the number of **27**'s counts bundle cohomology. The decomposition (G.39) is the "local E_8" of F-theory model building: one SU(5) is the GUT group, the other is a spectral $\mathrm{SU}(5)_\perp$ whose labels are flavour indices (Section 11.3).

$\mathrm{E}_7 \to \mathrm{E}_6 \times \mathrm{U}(1)$. The step that peels a U(1) off E_7 is

$$\mathbf{133} = \mathbf{78}_0 \oplus \mathbf{1}_0 \oplus \mathbf{27}_{-2} \oplus \overline{\mathbf{27}}_{+2}, \tag{G.40}$$

$$\mathbf{56} = \mathbf{27}_{+1} \oplus \overline{\mathbf{27}}_{-1} \oplus \mathbf{1}_{+3} \oplus \mathbf{1}_{-3}. \tag{G.41}$$

$\mathrm{E}_6 \to \mathrm{SO}(10) \times \mathrm{U}(1)$. This is the first genuinely chiral step, since SO(10) has the complex spinor **16**:

$$\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}, \tag{G.42}$$

$$\mathbf{27} = \mathbf{16}_{+1} \oplus \mathbf{10}_{-2} \oplus \mathbf{1}_{+4}. \tag{G.43}$$

The U(1) here is the standard $\mathrm{U}(1)_\psi$. Equation (G.43) already contains the physics: a single **27** of E_6 is one **16** of SO(10), a full Standard-Model generation with a right-handed neutrino, plus a vectorlike **10** and a singlet that can be given large masses.

$\mathrm{SO}(10) \to \mathrm{SU}(5) \times \mathrm{U}(1)$**.** The Georgi-Glashow SU(5) sits inside SO(10) with a commuting $\mathrm{U}(1)_\chi$:

$$\mathbf{45} = \mathbf{24}_0 \oplus \mathbf{1}_0 \oplus \mathbf{10}_{+4} \oplus \overline{\mathbf{10}}_{-4}, \tag{G.44}$$

$$\mathbf{16} = \mathbf{10}_{-1} \oplus \overline{\mathbf{5}}_{+3} \oplus \mathbf{1}_{-5}, \tag{G.45}$$

$$\mathbf{10} = \mathbf{5}_{+2} \oplus \overline{\mathbf{5}}_{-2}. \tag{G.46}$$

The **16** splits into exactly the SU(5) representations of one family, $\mathbf{10} \oplus \overline{\mathbf{5}}$, plus the singlet **1** that is the right-handed neutrino.

$\mathrm{SU}(5) \to G_{\mathbf{SM}} = \mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$**.** The last step distributes colour and isospin, with hypercharge Y generated inside SU(5):

$$\mathbf{24} = (\mathbf{8},\mathbf{1})_0 \oplus (\mathbf{1},\mathbf{3})_0 \oplus (\mathbf{1},\mathbf{1})_0 \oplus (\mathbf{3},\mathbf{2})_{-\frac{5}{6}} \oplus (\overline{\mathbf{3}},\mathbf{2})_{+\frac{5}{6}}, \tag{G.47}$$

$$\mathbf{5} = (\mathbf{3},\mathbf{1})_{-\frac{1}{3}} \oplus (\mathbf{1},\mathbf{2})_{+\frac{1}{2}}, \tag{G.48}$$

$$\overline{\mathbf{5}} = (\overline{\mathbf{3}},\mathbf{1})_{+\frac{1}{3}} \oplus (\mathbf{1},\mathbf{2})_{-\frac{1}{2}}, \tag{G.49}$$

$$\mathbf{10} = (\mathbf{3},\mathbf{2})_{+\frac{1}{6}} \oplus (\overline{\mathbf{3}},\mathbf{1})_{-\frac{2}{3}} \oplus (\mathbf{1},\mathbf{1})_{+1}. \tag{G.50}$$

The last two lines are one generation: $\overline{\mathbf{5}} = \{d^c, L\}$ and $\mathbf{10} = \{Q, u^c, e^c\}$, with $Q = (\mathbf{3},\mathbf{2})_{1/6}$, $u^c = (\overline{\mathbf{3}},\mathbf{1})_{-2/3}$, $e^c = (\mathbf{1},\mathbf{1})_{+1}$, $d^c = (\overline{\mathbf{3}},\mathbf{1})_{+1/3}$, $L = (\mathbf{1},\mathbf{2})_{-1/2}$. The twelve extra gauge bosons in the **24**, the $(\mathbf{3},\mathbf{2})_{\mp 5/6}$ leptoquarks, are the X, Y bosons whose exchange mediates proton decay.

Example G.8 (One generation in a single line)**.** Compose the last three steps. Starting from the **27** of E_6 and using Eqs. (G.43) to (G.50),

$$\mathbf{27} \xrightarrow{\mathrm{SO}(10)} \mathbf{16} \oplus \mathbf{10} \oplus \mathbf{1} \xrightarrow{\mathrm{SU}(5)} \underbrace{\mathbf{10} \oplus \overline{\mathbf{5}} \oplus \mathbf{1}}_{\text{chiral family}} \oplus \underbrace{\mathbf{5} \oplus \overline{\mathbf{5}}}_{\text{vectorlike}} \oplus \mathbf{1}. \tag{G.51}$$

The chiral part $\mathbf{10} \oplus \overline{\mathbf{5}}$ is the quarks and leptons; the vectorlike $\mathbf{5} \oplus \overline{\mathbf{5}}$ and the extra singlets are the fields that must be lifted to the GUT scale for the low-energy theory to look like the Standard Model. Anomaly cancellation of the chiral part is automatic: within SU(5) the anomaly coefficients satisfy $A(\mathbf{10}) + A(\overline{\mathbf{5}}) = 1 + (-1) = 0$, which is the statement that the **16** of SO(10) is anomaly-free because SO(10) has no independent cubic Casimir.

G.2.4 Other embeddings used in model building

Not every unified model runs through Georgi-Glashow SU(5). Three alternative branches recur, and all three are read off the same subalgebra tables.

Pati-Salam. The maximal-rank subalgebra $\mathrm{SO}(10) \supset \mathrm{SU}(4) \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ (delete the central node of the extended D_5 diagram) gives lepton number as a fourth colour:

$$\mathbf{16} = (\mathbf{4},\mathbf{2},\mathbf{1}) \oplus (\overline{\mathbf{4}},\mathbf{1},\mathbf{2}), \qquad \mathbf{10} = (\mathbf{6},\mathbf{1},\mathbf{1}) \oplus (\mathbf{1},\mathbf{2},\mathbf{2}). \tag{G.52}$$

The SU(4) contains $\mathrm{SU}(3)_c \times \mathrm{U}(1)_{B-L}$, so a generation is $(\mathbf{4},\mathbf{2},\mathbf{1}) = \{Q, L\}$ and $(\overline{\mathbf{4}},\mathbf{1},\mathbf{2}) = \{u^c, d^c, e^c, \nu^c\}$.

Trinification. The maximal subalgebra $\mathrm{E}_6 \supset \mathrm{SU}(3)_c \times \mathrm{SU}(3)_L \times \mathrm{SU}(3)_R$ of Eq. (G.33) branches the fundamental cyclically,

$$\mathbf{27} = (\mathbf{3},\overline{\mathbf{3}},\mathbf{1}) \oplus (\mathbf{1},\mathbf{3},\overline{\mathbf{3}}) \oplus (\overline{\mathbf{3}},\mathbf{1},\mathbf{3}), \tag{G.53}$$

with the adjoint $\mathbf{78} = (\mathbf{8},\mathbf{1},\mathbf{1}) \oplus (\mathbf{1},\mathbf{8},\mathbf{1}) \oplus (\mathbf{1},\mathbf{1},\mathbf{8}) \oplus (\mathbf{3},\mathbf{3},\mathbf{3}) \oplus (\overline{\mathbf{3}},\overline{\mathbf{3}},\overline{\mathbf{3}})$. This is the group of the $\mathbb{Z}_3$ orbifold quiver of Chapter 10: the D-brane construction there delivers exactly trinification with three copies of the **27**, one per orbifold twisted sector.

Flipped SU(5)**.** The same $\mathrm{SU}(5)\times\mathrm{U}(1)$ subalgebra of SO(10) as in Eqs. (G.44) to (G.46), but with the physical hypercharge taken as a linear combination of the SU(5) hypercharge and $\mathrm{U}(1)_\chi$. The relabelling swaps $u^c \leftrightarrow d^c$ and $e^c \leftrightarrow \nu^c$ between the **10** and $\overline{\mathbf{5}}$, which lets a single SO(10) adjoint Higgs break the group and gives a different doublet-triplet splitting; the representation content is unchanged, only the $\mathrm{U}(1)_Y$ direction moves.

G.2.5 Realisation in string theory

Heterotic. The gauge algebra $\mathfrak{e}_8\oplus\mathfrak{e}_8$ or $\mathfrak{so}(32)$ of Chapter 3 sits on the left-moving worldsheet as a current algebra at level $k=1$. For a simply-laced $\mathfrak{g}$ at level one the Sugawara central charge is

$$c=\left.\frac{k\,\dim\mathfrak{g}}{k+h^\vee}\right|_{k=1}=\frac{\dim\mathfrak{g}}{1+h^\vee}, \tag{G.54}$$

which gives $c(\mathrm{E}_8)=248/31=8$ and $c(\mathfrak{so}(32))=496/31=16$, matching the sixteen internal left-moving bosons (Chapter 3 and Appendix G.4). Compactifying on a Calabi-Yau threefold and breaking E_8 by a vector bundle is the standard route to a four-dimensional GUT: an SU(3) bundle (the standard embedding) leaves E_6 by Eq. (G.37), with the net number of chiral **27**'s equal to $|\chi|/2$ for the Calabi-Yau (Chapter 8); an SU(4) or SU(5) bundle leaves SO(10) or SU(5) directly, using Eq. (G.31). The commutant of the structure group inside E_8 is the unbroken gauge group, and the branching of **248** under (commutant)×(structure group) is the list of matter representations, one factor per bundle-cohomology class.

The Frenkel-Kac-Segal construction. The level-one realisation is explicit, and this is where the current algebra actually comes from. Let $X^I_{\mathrm{L}}(z)$, $I=1,\dots,r$, be the chiral bosons of the internal lattice CFT, normalised so that $X^I_{\mathrm{L}}(z)X^J_{\mathrm{L}}(w)\sim-\delta^{IJ}\ln(z-w)$. The r derivatives $J^I(z)=i\partial X^I_{\mathrm{L}}(z)$ are abelian currents, $J^I(z)J^J(w)\sim\delta^{IJ}(z-w)^{-2}$, and they realise the Cartan subalgebra. For each norm-two vector α of the lattice ($\alpha^2=2$, a root of the simply-laced $\mathfrak{g}$) define the vertex operator

$$J^\alpha(z)=c_\alpha\,{:}e^{i\alpha\cdot X_{\mathrm{L}}(z)}{:},\qquad \alpha^2=2, \tag{G.55}$$

with c_α a cocycle fixing the relative signs. Together with the Cartan currents these close on the level-one operator products

$$J^I(z)\,J^\alpha(w)\sim\frac{\alpha^I}{z-w}\,J^\alpha(w),\qquad J^\alpha(z)\,J^\beta(w)\sim\begin{cases}\dfrac{1}{(z-w)^2}+\dfrac{i\,\alpha\cdot\partial X_{\mathrm{L}}(w)}{z-w}, & \beta=-\alpha,\\[2mm] \dfrac{\epsilon(\alpha,\beta)}{z-w}\,J^{\alpha+\beta}(w), & \alpha\cdot\beta=-1,\\[2mm] \text{regular}, & \alpha\cdot\beta\geq 0,\end{cases} \tag{G.56}$$

with $\epsilon(\alpha,\beta)=\pm1$ the structure constants supplied by the cocycle. These are the defining relations of the affine algebra $\hat{\mathfrak{g}}$ at level $k=1$. This is the Frenkel-Kac-Segal theorem: an even lattice whose norm-two vectors form the root system of a simply-laced $\mathfrak{g}$ carries a level-one current algebra for $\mathfrak{g}$, and the Sugawara central charge (G.54) equals the rank, $c=r$ (for simply-laced $\mathfrak{g}$ one has $\dim\mathfrak{g}=r(1+h^\vee)$). The $\Gamma_8\oplus\Gamma_8$ and D_{16}^+ lattices of the heterotic string are the rank-16 instances (Appendix G.4). The same mechanism runs at $r=1$: a single boson at the self-dual radius $R=\sqrt{\alpha'}$ has momentum-winding vectors reaching $p_{\mathrm{L}}^2=2$, $p_{\mathrm{R}}=0$, so two extra currents ${:}e^{\pm i\sqrt{2}\,X_{\mathrm{L}}}{:}$ join $i\partial X_{\mathrm{L}}$ and enhance $\mathrm{U}(1)\to\mathrm{SU}(2)$ at level one, the enhancement quoted in Chapter 6.

F-theory and geometric engineering. Here the exceptional groups are not spread over spacetime but localised on a divisor, and the same branchings run in reverse. A gauge algebra $\mathfrak{g}$

on a surface S enhances to $\mathfrak{g}'$ of one higher rank along a matter curve, and the Katz-Vafa rule reads the charged matter off the branching of the adjoint of $\mathfrak{g}'$ under $\mathfrak{g}$ (Chapters 10 and 11). For the SU(5) GUT the two enhancements $\mathrm{SU}(5) \to \mathrm{SO}(10)$ and $\mathrm{SU}(5) \to \mathrm{SU}(6)$ deposit a **10** and a **5**, exactly the off-diagonal blocks of Eqs. (G.45) to (G.46), and the top Yukawa $\mathbf{10} \cdot \mathbf{10} \cdot \mathbf{5}$ lives at the rank-two point where the fibre becomes E_6, the cubic invariant sitting inside the **78** of Eq. (G.42). No perturbative brane stack produces $\mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8$, because these are not algebras of matrices preserving a bilinear form, and that is what forces the coupling to be non-perturbative outside F-theory (Section 11.3).

SO(32) and Spin(32)/$\mathbb{Z}_2$. The other heterotic and the Type I string carry $\mathfrak{so}(32)$, whose useful maximal-rank subalgebras are $\mathfrak{su}(16) \oplus \mathfrak{u}(1)$, $\mathfrak{so}(2p) \oplus \mathfrak{so}(32-2p)$, and $\mathfrak{so}(16) \oplus \mathfrak{so}(16)$. Breaking $\mathrm{Spin}(32)/\mathbb{Z}_2$ by a bundle typically leaves an $\mathrm{SO}(2n)$ or $\mathrm{U}(n)$ gauge group rather than an exceptional one, so this string is less natural for exceptional-group unification; its phenomenology runs through the $\mathfrak{su}(16) \oplus \mathfrak{u}(1)$ branch and orthogonal subgroups. The global form of the group, $\mathrm{Spin}(32)/\mathbb{Z}_2$ rather than $\mathrm{SO}(32)$, is fixed by the charge lattice D_{16}^{+} carrying a spinor but no vector conjugacy class, a lattice statement developed in Appendix G.4.

Remark G.9 (Beyond the lectures: level-one integrable representations). Which representations a level-k current algebra admits is a sharper constraint than which are allowed by the finite Lie algebra. A dominant weight λ is integrable at level k only if $(\lambda, \theta^\vee) \leq k$, where $\theta^\vee$ is the highest coroot. At level one this forces the highest weight to be either zero or a fundamental weight ω_i whose comark $a_i^\vee$ equals one. For $\mathfrak{e}_8$ the only comark-one node is the affine node, so the unique level-one integrable representation is the vacuum module itself: the E_8 current algebra at level one has a single primary, and its character is the theta function of the Γ_8 lattice divided by η^8 (Appendix G.4). This is why the E_8 factor of the heterotic string has no charged states below the string scale beyond the adjoint currents, and why an E_8 gauge symmetry cannot be given light matter without first breaking it. For $\mathfrak{so}(32)$ the comark-one fundamental weights are the vector and the two spinors, so there are four level-one modules, vacuum, vector, spinor and conjugate spinor; the $\mathrm{Spin}(32)/\mathbb{Z}_2$ projection keeps the vacuum and one spinor, again matching the lattice D_{16}^{+}.

Exercise G.3 (The SO(10) spinor as a family)**.** Starting from Eqs. (G.45), (G.46), (G.48) and (G.49), decompose the **16** of SO(10) all the way to $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ and identify each of the sixteen states with a Standard-Model field. Verify that the sum of hypercharges over the multiplet vanishes and that the cubic sum $\sum Y^3$ also vanishes, so the family is free of both the mixed and the pure hypercharge anomaly. Then repeat for the Pati-Salam split (G.52) and check that the two descriptions of the generation agree state by state.

G.3 The ADE classification

Several classification problems that have nothing obvious to do with one another produce the same answer: two infinite families labelled A_n and D_n, and three exceptional cases E_6, E_7, E_8. The simply-laced Lie algebras are one such problem. The finite subgroups of SU(2) are another. The isolated surface singularities $\mathbb{C}^2/\Gamma$ that a Calabi-Yau twofold can degenerate to are a third. The McKay correspondence is the statement that these three lists share more than labels: an explicit dictionary ties them together, and it is that dictionary that makes gauge symmetry appear out of geometry in Chapters 7, 8 and 10. This section collects the dictionary. The general theory of roots, weights and Dynkin diagrams is in Appendix G.1, and the exceptional algebras themselves in Appendix G.2; here we use that machinery rather than rebuild it.

Type	rank r	$\mathfrak{g}$	$\dim \mathfrak{g}$	$h = h^\vee$	simple roots
A_n	n	$\mathfrak{su}(n{+}1)$	$n(n{+}2)$	$n{+}1$	chain
D_n	n	$\mathfrak{so}(2n)$	$n(2n{-}1)$	$2n{-}2$	chain with a two-pronged fork
E_6	6	$\mathfrak{e}_6$	78	12	three legs $2, 2, 1$
E_7	7	$\mathfrak{e}_7$	133	18	three legs $3, 2, 1$
E_8	8	$\mathfrak{e}_8$	248	30	three legs $4, 2, 1$

Table G.3: The simply-laced (ADE) simple Lie algebras. The last column records the shape of the Dynkin diagram as leg lengths measured from the branch node (a plain chain for A_n); the diagrams themselves are drawn in Appendix G.1. For all of these $h = h^\vee$.

The big picture

A symmetric integer matrix A with $A_{ii} = 2$ and off-diagonal entries 0 or -1 is positive definite exactly when it is the Cartan matrix of one of A_n, D_n, E_6, E_7, E_8, and positive semidefinite with a one-dimensional kernel exactly when it is the corresponding affine (extended) matrix. Equivalently, the graph with adjacency matrix $2\mathbf{1} - A$ has largest eigenvalue below 2 in the first case and equal to 2 in the second. Every problem whose data can be packaged as such a matrix, or as an even lattice generated by norm-two vectors, or as a graph of small spectral radius, therefore lands on the ADE list. The finite subgroups of SU(2), the du Val singularities and the ADE algebras are three instances of the same constraint, and the McKay correspondence is the map between them.

G.3.1 Simply-laced algebras and the ADE list

Recall from Appendix G.1 that a simple Lie algebra is simply-laced when all of its roots have the same length: its Dynkin diagram then has only single bonds, and its Cartan matrix $A_{ij} = 2(\alpha_i, \alpha_j)/(\alpha_j, \alpha_j)$ is symmetric with $A_{ii} = 2$ and $A_{ij} \in \{0, -1\}$ off the diagonal. The complete list of connected simply-laced Dynkin diagrams is

$$A_n \ (n \geq 1), \qquad D_n \ (n \geq 4), \qquad E_6, \quad E_7, \quad E_8, \tag{G.57}$$

where $A_n = \mathfrak{su}(n{+}1)$, $D_n = \mathfrak{so}(2n)$, and the three exceptional cases are as in Appendix G.2. The range restrictions remove the low-rank coincidences $D_2 \cong A_1 \oplus A_1$, $D_3 \cong A_3$, $E_3 \cong A_2 \oplus A_1$, $E_4 \cong A_4$, $E_5 \cong D_5$; kept within their stated ranges the diagrams in Eq. (G.57) are distinct and label distinct algebras. Because the algebra is simply-laced its Coxeter number and dual Coxeter number coincide, $h = h^\vee$, and the long/short distinction that plagues B_n, C_n, F_4, G_2 never arises. Table G.3 collects the numerical data.

The root lattice of an ADE algebra, the $\mathbb{Z}$-span of the roots with the Killing form normalised so that roots have length squared two, is a positive-definite even lattice generated by its norm-two vectors. The ADE root lattices are exactly the even lattices with that property, which is why E_8 reappears in Appendix G.4 as the unique even self-dual lattice of rank eight. The unifying reason behind the whole classification is a statement about the corresponding graphs.

Proposition G.10 (Spectral characterisation)**.** *Let G be a finite connected graph without loops or multiple edges, $\mathcal{A}$ its adjacency matrix, and $\lambda_{\max}(\mathcal{A})$ its largest eigenvalue. Then*

1. *$\lambda_{\max}(\mathcal{A}) < 2$ if and only if G is one of the Dynkin diagrams A_n, D_n, E_6, E_7, E_8;*
2. *$\lambda_{\max}(\mathcal{A}) = 2$ if and only if G is one of the affine diagrams $\widehat{A}_n, \widehat{D}_n, \widehat{E}_6, \widehat{E}_7, \widehat{E}_8$, in which case the Perron-Frobenius eigenvector has positive integer entries, the* marks *of the diagram.*

Type	$\Gamma \subset \mathrm{SU}(2)$	$\lvert\Gamma\rvert$	# irreps	$\mathbb{C}^2/\Gamma$ equation $f=0$	marks ($\dim \rho_i$)
A_{n-1}	cyclic $\mathbb{Z}_n$	n	n	$xy = z^n$	$1, 1, \dots, 1$
D_n	binary dihedral	$4(n-2)$	$n+1$	$x^2 + y^2 z + z^{n-1} = 0$	$1, 1, 2, \dots, 2, 1, 1$
E_6	binary tetrahedral	24	7	$x^2 + y^3 + z^4 = 0$	$1, 2, 3, 2, 1, 2, 1$
E_7	binary octahedral	48	8	$x^2 + y^3 + yz^3 = 0$	$1, 2, 3, 4, 3, 2, 1, 2$
E_8	binary icosahedral	120	9	$x^2 + y^3 + z^5 = 0$	$1, 2, 3, 4, 5, 6, 4, 2, 3$

Table G.4: The ADE dictionary: finite subgroups of SU(2), their orders, the number of irreducible representations, the Kleinian singularity $\mathbb{C}^2/\Gamma$ as a hypersurface in $\mathbb{C}^3$, and the McKay marks (the dimensions of the irreducibles). The marks are the components of the null vector of the affine Cartan matrix and satisfy Eq. (G.62). This is the reference version of the dictionary quoted in Table 10.1. The A series is indexed so that $\mathbb{Z}_n \leftrightarrow A_{n-1}$.

Equivalently, writing $A = 2\mathbf{1} - \mathcal{A}$, case (i) makes A positive definite and case (ii) makes it positive semidefinite with a one-dimensional kernel spanned by the marks. We state this rather than prove it; the argument is a finite check that any graph outside the list contains a subgraph with $\lambda_{\max} \geq 2$, and it is carried out in Kac, *Infinite-dimensional Lie algebras*, Ch. 4, and in Hazewinkel, Hesselink, Siersma and Veldkamp (1977), whose title, "the ubiquity of Coxeter-Dynkin diagrams", names the phenomenon this section is about.

G.3.2 Finite subgroups of SU(2)

The second appearance of the list is group-theoretic. Recall the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$ with kernel the centre $\{\pm\mathbf{1}\}$. A finite subgroup $\Gamma \subset \mathrm{SU}(2)$ projects to a finite subgroup of the rotation group, and the finite subgroups of SO(3) were classified by Klein: the cyclic groups $\mathbb{Z}_k$, the dihedral groups, and the three rotation groups of the platonic solids (tetrahedral, octahedral, icosahedral). Pulling back through the double cover gives the classification in SU(2).

Theorem G.11 (Klein, 1884)**.** *Every finite subgroup $\Gamma \subset \mathrm{SU}(2)$ is, up to conjugacy, one of*

- *the cyclic group $\mathbb{Z}_n$;*
- *the binary dihedral (dicyclic) group BD_{n-2} of order $4(n-2)$;*
- *the binary tetrahedral, octahedral or icosahedral group, of orders 24, 48, 120, the preimages in SU(2) of the rotation groups of the tetrahedron, octahedron and icosahedron.*

Each acts freely on $\mathbb{C}^2 \setminus \{0\}$ and contains the centre $-\mathbf{1}$ (except the odd cyclic groups, which meet it trivially).

These are the *binary polyhedral groups*. The number of conjugacy classes of Γ, equal to the number of its inequivalent irreducible complex representations, is $r+1$ where r is the rank in Table G.3: one more than the ADE rank, the extra one to be accounted for below. The dictionary between the two families is fixed in Table G.4.

G.3.3 Kleinian (du Val) singularities

The quotient $\mathbb{C}^2/\Gamma$ is a singular complex surface with one singular point at the image of the origin. It carries several equivalent names: *Kleinian* singularity, *du Val* singularity, *rational double point*, and *simple surface singularity*. Concretely it is a hypersurface. The ring of Γ-invariant polynomials on $\mathbb{C}^2$ is generated by three elements x, y, z subject to a single relation $f(x,y,z) = 0$, so

$$\mathbb{C}^2/\Gamma \;\cong\; \{\,(x,y,z) \in \mathbb{C}^3 :\; f(x,y,z) = 0\,\}, \tag{G.58}$$

with f read off from Table G.4. The A_{n-1} case is the cleanest: $\mathbb{Z}_n$ acts by $(z_1, z_2) \mapsto (\omega z_1, \omega^{-1} z_2)$ with $\omega = e^{2\pi i/n}$, the invariants are $x = z_1^n$, $y = z_2^n$, $z = z_1 z_2$, and the relation is $xy = z^n$.

These singularities are the two-complex-dimensional *canonical* (Gorenstein) singularities: the holomorphic two-form on the smooth part extends across the resolution without zeros or poles, so the quotient is Calabi-Yau away from a set of complex codimension two, and gravity does not obstruct placing it inside a compact $K3$ or Calabi-Yau (Chapter 8). Two invariants pin down the algebra. Resolving the singular point, replacing it by a collection of holomorphic curves, produces r exceptional $\mathbb{CP}^1$'s $\Sigma_1, \ldots, \Sigma_r$, each of self-intersection -2, whose intersection matrix is minus the Cartan matrix,

$$\Sigma_i \cdot \Sigma_j \;=\; -A_{ij}, \tag{G.59}$$

so the dual graph of the resolution is the ADE Dynkin diagram and r is the rank. This is derived on the geometry side in Section 10.1, around Eq. (10.2); the symmetry of an intersection form of curves on a surface is the structural reason only the simply-laced algebras occur, since Eq. (G.59) can never deliver an asymmetric Cartan matrix. The second invariant is deformation-theoretic: the singularity admits a versal deformation whose base has dimension equal to the Milnor number, and for a du Val singularity the Milnor number equals the rank r. Both the resolution parameters (Kähler, blowing the cycles up to finite size) and the deformation parameters (complex structure, smoothing the hypersurface) are r-dimensional, and they are the two branches of the moduli space of the engineered gauge theory in Chapter 10.

Remark G.12 (Beyond the lectures). The ADE surface singularities are the transverse slices of the nilpotent variety of the corresponding simple Lie algebra, and their simultaneous resolution is governed by the Weyl group. This is Brieskorn's theorem, conjectured by Grothendieck: the semiuniversal deformation of $\mathbb{C}^2/\Gamma$ is the restriction of the adjoint quotient $\mathfrak{g} \to \mathfrak{t}/W$ to a transverse slice at a subregular nilpotent, so the discriminant of the singularity is the reflection-arrangement discriminant of W. Slodowy's monograph is the reference. This is the sharpest form of the statement that the Lie algebra is hidden inside the singularity, and it is why the same E_8 controls both the algebra of Appendix G.2 and the $x^2 + y^3 + z^5$ singularity.

G.3.4 The McKay correspondence

So far the three lists share labels. McKay's observation is that a single elementary construction turns the representation theory of Γ into the affine Dynkin diagram, with no reference to the singularity or to Lie theory. Let $\rho_0 = \mathbf{1}, \rho_1, \ldots, \rho_r$ be the irreducible representations of $\Gamma \subset \mathrm{SU}(2)$, with ρ_0 the trivial one, and let R be the defining two-dimensional representation, the inclusion $\Gamma \hookrightarrow \mathrm{SU}(2)$ itself. Decompose each tensor product into irreducibles,

$$R \otimes \rho_i \;=\; \bigoplus_j \mathcal{A}_{ij}\, \rho_j, \tag{G.60}$$

so that $\mathcal{A}_{ij} = \dim \mathrm{Hom}_\Gamma(\rho_j, R \otimes \rho_i)$ are non-negative integers. Because R is two-dimensional and (being a genuine $\mathrm{SU}(2)$ representation) self-dual, the matrix $\mathcal{A}_{ij}$ is symmetric with zero diagonal and entries 0 or 1 (with the single exception of the $\mathbb{Z}_2$ case, that is $\widehat{A}_1$, where the single bond is doubled): it is the adjacency matrix of a graph on the $r+1$ nodes ρ_i.

Theorem G.13 (McKay, 1980)**.** *The graph with adjacency matrix $\mathcal{A}_{ij}$ defined by Eq. (G.60) is the affine Dynkin diagram $\widehat{\mathfrak{g}}$ of the ADE type assigned to Γ in Table G.4. The trivial representation ρ_0 is the affine node; deleting it leaves the finite ADE diagram of $\mathfrak{g}$. The dimensions $a_i := \dim \rho_i$ are the marks of $\widehat{\mathfrak{g}}$: they are the entries of the Perron-Frobenius eigenvector of $\mathcal{A}_{ij}$, equivalently the unique (up to scale) null vector of the affine Cartan matrix $\widehat{A} = 2\mathbf{1} - \mathcal{A}$,*

$$\sum_j \widehat{A}_{ij}\, a_j = 0, \qquad a_0 = 1, \tag{G.61}$$

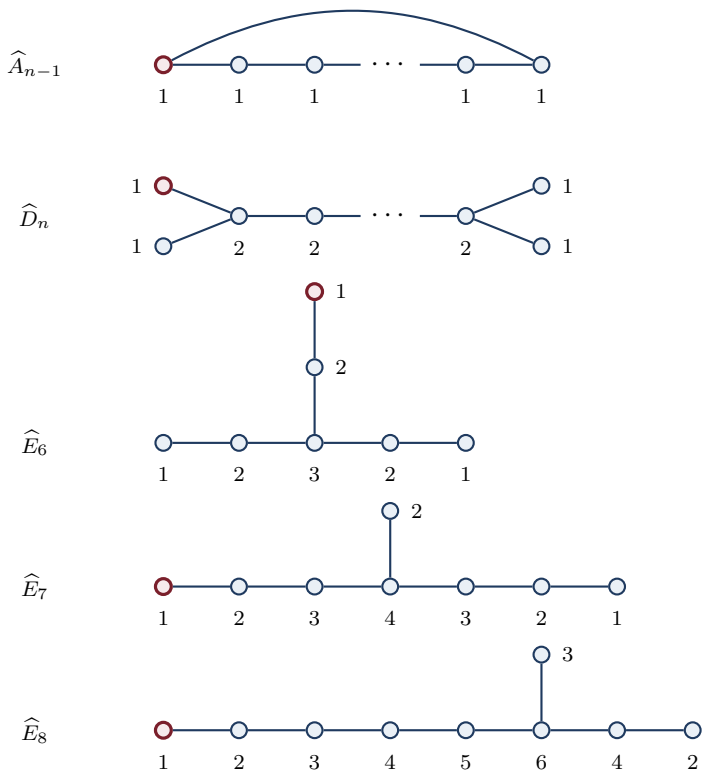


Figure G.3: The affine (extended) ADE Dynkin diagrams, which are the McKay quivers of the finite subgroups $\Gamma \subset \mathrm{SU}(2)$: one node per irreducible representation of Γ, one edge per copy of ρ_j in $R \otimes \rho_i$. The integer next to each node is its mark $a_i = \dim \rho_i$. The affine node $\rho_0 = \mathbf{1}$ (mark one) is drawn in red; deleting it leaves the finite ADE diagram of Table G.3. For $\widehat{A}_{n-1}$ the diagram is the n-cycle; the degenerate case $\widehat{A}_1$ (from $\mathbb{Z}_2$) instead has its two nodes joined by a doubled bond.

normalised so the affine node has mark one.

The finite ADE diagrams are drawn in Appendix G.1; the affine diagrams with their marks are drawn in Fig. G.3, and they are the "marked diagrams" promised in Section 10.1. Two identities make the correspondence quantitative. Summing the marks reproduces the Coxeter number, and summing their squares reproduces the order of the group:

$$\sum_{i=0}^{r} a_i \;=\; h, \qquad \sum_{i=0}^{r} a_i^2 \;=\; |\Gamma|. \tag{G.62}$$

The first is the standard expression of the Coxeter number as the sum of marks of the affine diagram (Appendix G.1); the second is Burnside's identity $\sum_i (\dim \rho_i)^2 = |\Gamma|$ read off the same list of dimensions. For the binary icosahedral group the marks are $1, 2, 3, 4, 5, 6, 4, 2, 3$, and indeed $1 + 4 + 9 + 16 + 25 + 36 + 16 + 4 + 9 = 120$ while $1 + 2 + 3 + 4 + 5 + 6 + 4 + 2 + 3 = 30 = h(E_8)$.

Example G.14 (The A series)**.** Take $\Gamma = \mathbb{Z}_n$, generated by $\mathrm{diag}(\omega, \omega^{-1})$ with $\omega = e^{2\pi i/n}$. Its irreducibles are the n one-dimensional characters $\rho_k : \mathrm{diag}(\omega, \omega^{-1}) \mapsto \omega^k$, for $k = 0, 1, \ldots, n-1$. The defining representation restricts to $R = \rho_1 \oplus \rho_{-1}$, so

$$R \otimes \rho_k = \rho_{k+1} \oplus \rho_{k-1} \quad (\text{indices mod } n), \tag{G.63}$$

and node k is joined to $k \pm 1$. The graph is the n-cycle, which is the affine diagram $\widehat{A}_{n-1}$; deleting the trivial node ρ_0 opens the cycle into the chain A_{n-1}. Every mark is $a_k = \dim \rho_k = 1$, so Eq. (G.62) reads $\sum_k a_k = n = h(A_{n-1})$ and $\sum_k a_k^2 = n = |\mathbb{Z}_n|$. Compare the resolution of $xy = z^n$: the $n-1$ exceptional spheres form a chain, exactly the deleted diagram.

Remark G.15 (Beyond the lectures). McKay's is a combinatorial correspondence. Its geometric lift, due to Gonzalez-Sprinberg and Verdier, realises the nodes as a basis of the (co)homology of the resolution $\widetilde{\mathbb{C}^2/\Gamma}$: the non-trivial irreducibles ρ_i index a basis of tautological bundles whose first Chern classes are dual to the exceptional curves Σ_i, so Eq. (G.60) and Eq. (G.59) are two faces of one statement. Kronheimer constructed the hyper-Kähler ALE metrics on these resolutions as hyper-Kähler quotients built from the representation theory of Γ, giving the physical moduli of Section 10.1 directly, and Ito and Nakamura recovered the resolution as the Γ-Hilbert scheme, one point of which is each Γ-orbit. The higher-dimensional analogue for $\Gamma \subset \mathrm{SU}(3)$ (the Bridgeland-King-Reid derived equivalence) is the mathematics behind the $\mathbb{C}^3/\Gamma$ quiver gauge theories of Chapter 8.

Exercise G.4. The binary tetrahedral group has order 24 and irreducible representations of dimensions $1, 1, 1, 2, 2, 2, 3$ (three one-dimensional characters from the abelianisation $\mathbb{Z}_3$, three two-dimensional, one three-dimensional). Using $\sum_i (\dim \rho_i)^2 = 24$ as a check, and the fact that tensoring with the two-dimensional defining representation gives the adjacency Eq. (G.60), show that the McKay quiver is $\widehat{E}_6$: a central node of mark 3 with three legs of length two, marks $3, 2, 1$ along each leg. Confirm $\sum_i a_i = 12 = h(E_6)$.

Exercise G.5. The binary dihedral group $\mathrm{BD}_{n-2} \subset \mathrm{SU}(2)$ has order $4(n-2)$ and $n+1$ irreducible representations: four of dimension one and $n-3$ of dimension two. Tensoring with the defining representation R and using Eq. (G.60), show that the McKay quiver is $\widehat{D}_n$, a spine of $n-3$ two-dimensional irreps of mark 2 capped at each end by a fork of two one-dimensional irreps of mark 1. Check $\sum_i (\dim \rho_i)^2 = 4 + 4(n-3) = 4(n-2) = |\Gamma|$ and $\sum_i a_i = 2n-2 = h(D_n)$. Deleting the trivial node leaves the finite D_n diagram, matching the n exceptional spheres in the resolution of $x^2 + y^2 z + z^{n-1} = 0$.

G.3.5 ADE as a universal pattern

The three lists this section ties together are the ones the rest of the notes need, but the same classification governs a longer catalogue: the finite-type quivers of Gabriel's theorem (the ones with finitely many indecomposable representations), the modular-invariant partition functions of the SU(2) WZW model classified by Cappelli, Itzykson and Zuber (1987), the subfactors of small index, and Arnold's simple (zero-modulus) singularities in catastrophe theory. In each case the underlying constraint is the one in Theorem G.10: a positive-definite even quadratic form generated by norm-two vectors, or a graph of spectral radius below two.

Inside these notes the thread runs through gauge enhancement. The heterotic string on T^4 develops non-abelian gauge symmetry on the loci where its Narain lattice acquires extra norm-two vectors (Appendix G.4), and under the duality to type IIA on $K3$ those loci are exactly where $K3$ develops an ADE singularity $\mathbb{C}^2/\Gamma$ (Sections 7.3.4 and 7.3.6). Blowing the singularity down localises the statement: M-theory or type IIA on the ALE space $\widetilde{\mathbb{C}^2/\Gamma}$ carries ADE gauge symmetry, with the Cartan from the supergravity three-form on the exceptional cycles and the charged W-bosons from membranes wrapping them, one root per cycle (Section 10.1). The intersection form Eq. (G.59) is what makes the collapsing geometry reproduce the algebra. Only the simply-laced groups arise geometrically because a surface intersection form is symmetric; the non-simply-laced and genuinely exceptional enhancements B_n, C_n, F_4, G_2 need an orientifold or an F-theory monodromy folding the ADE diagram, which is where Chapter 11 takes over.

G.4 Even self-dual and Narain lattices

Two constructions in these notes stand or fall on the arithmetic of integral lattices. The first is the internal sector of the heterotic string, where sixteen chiral bosons carry momenta on a

Euclidean even self-dual lattice of rank sixteen, and the two admissible lattices deliver the gauge groups $\mathrm{E}_8 \times \mathrm{E}_8$ and $\mathrm{Spin}(32)/\mathbb{Z}_2$ (Chapter 3). The second is toroidal compactification, where the momenta and windings of a string on T^d fill an even self-dual lattice of Lorentzian signature, the Narain lattice, whose automorphism group is the T-duality group $O(d,d;\mathbb{Z})$ (Chapters 6 to 8). These are one statement read in two signatures: modular invariance of the one-loop partition function forces the momentum lattice to be even and self-dual. This section collects the definitions, the two classification theorems, the Narain construction, and the level-one current algebra it supports. The modular machinery that produces the two conditions is in Appendix F, specifically Appendix F.3.7; the root systems of E_8 and of the ADE series live in Appendices G.1 to G.3.

G.4.1 Lattices, duals, and the two conditions

A *lattice* Λ of rank n is a free abelian group $\mathbb{Z}^n$ equipped with a symmetric bilinear form $\langle\cdot,\cdot\rangle$, which we take to be real-valued and non-degenerate. Fixing a basis $\{e_i\}$ records the form in the *Gram matrix* $\mathsf{G}_{ij} = \langle e_i, e_j\rangle$. We write $x^2 := \langle x, x\rangle$. If G is positive-definite the lattice is *Euclidean*; if it has p positive and q negative eigenvalues it has *signature* (p,q) and is called indefinite (Lorentzian when one sign is isolated).

Definition G.16 (Integral, even, self-dual)**.** The *dual lattice* is $\Lambda^* = \{y \in \Lambda \otimes \mathbb{R} : \langle y, x\rangle \in \mathbb{Z}\ \forall x \in \Lambda\}$. A lattice is

- *integral* if $\Lambda \subseteq \Lambda^*$, equivalently $\langle x, y\rangle \in \mathbb{Z}$ for all $x, y \in \Lambda$;
- *even* if $x^2 \in 2\mathbb{Z}$ for every $x \in \Lambda$ (an even lattice is automatically integral, by polarisation);
- *unimodular*, or *self-dual*, if $\Lambda = \Lambda^*$, equivalently $\det \mathsf{G} = \pm 1$.

The two conditions we care about, evenness and self-duality, are exactly the two consequences of modular invariance. Attach to a Euclidean lattice its *theta function*

$$\Theta_\Lambda(\tau) \;=\; \sum_{p\in\Lambda} q^{\frac{1}{2}p^2}, \qquad q = e^{2\pi i\tau}, \tag{G.64}$$

which counts lattice points weighted by norm and appears in the heterotic partition function as the internal contribution. Under the two generators of the modular group $\mathrm{SL}(2,\mathbb{Z})$ (Appendix F.3.1):

- $T : \tau \to \tau + 1$ multiplies each term by $e^{i\pi p^2}$, so Θ_Λ is T-invariant precisely when $p^2 \in 2\mathbb{Z}$. This is **evenness**, and physically it is level matching: with $L_0, \tilde{L}_0$ the left- and right-moving Virasoro zero modes (Appendix F), the internal contribution to $L_0 - \tilde{L}_0$ is an integer.
- $S : \tau \to -1/\tau$ acts by Poisson resummation,

$$\Theta_\Lambda(-1/\tau) \;=\; \frac{(-i\tau)^{n/2}}{\mathrm{vol}(\mathbb{R}^n/\Lambda)}\,\Theta_{\Lambda^*}(\tau), \tag{G.65}$$

  turning the sum over Λ into a sum over Λ^*. Invariance forces $\Lambda^* = \Lambda$ and unit covolume. This is **self-duality**.

For an even self-dual Λ, so that $\Lambda^* = \Lambda$ and $\mathrm{vol}(\mathbb{R}^n/\Lambda) = 1$, (G.65) reads $\Theta_\Lambda(-1/\tau) = (-i)^{n/2}\tau^{n/2}\,\Theta_\Lambda(\tau)$. Thus Θ_Λ transforms with weight $n/2$ under S, but carries the extra phase $(-i)^{n/2}$; under T it is invariant. A holomorphic modular form for the full group $\mathrm{SL}(2,\mathbb{Z})$ with trivial multiplier system exists only in even weight, and this residual S-phase is trivial precisely when $n/2 \equiv 0 \pmod 4$. Hence

$$n \equiv 0 \pmod 8, \tag{G.66}$$

and a Euclidean even self-dual lattice exists only in ranks $8, 16, 24, \ldots$; for such n the theta function is a genuine weight-$n/2$ form (for instance E_4 at $n = 8$). The number-theoretic proof is

in Serre, *A Course in Arithmetic* (Ch. V); the lattice-theory reference throughout this section is Conway and Sloane, *Sphere Packings, Lattices and Groups* (henceforth SPLG).

G.4.2 Euclidean even self-dual lattices

In the definite case the classification is rigid at low rank and explodes thereafter.

Theorem G.17 (Classification in low rank)**.** *Euclidean even self-dual lattices exist exactly when the rank is a multiple of* 8*. Up to isometry:*

- *rank* 8*: a unique lattice, the root lattice* Γ_8 *of* E_8*;*
- *rank* 16*: exactly two,* $\Gamma_8 \oplus \Gamma_8$ *and* $\Gamma_{16} \coloneqq D_{16}^+$*;*
- *rank* 24*: exactly* 24*, the* Niemeier lattices*, distinguished by their root systems, with the* Leech lattice *the unique one having no vectors of norm* 2*.*

The rank-8 and rank-16 cases are what Chapter 3 quotes. Two presentations make the rank-16 lattices concrete. The lattice $\Gamma_{16} = D_{16}^+$ is the D_{16} root lattice (integer vectors $x \in \mathbb{Z}^{16}$ with $\sum_i x_i$ even) enlarged by one spinor conjugacy class, the half-integer vectors $(\pm\frac{1}{2})^{16}$ with an even number of minus signs:

$$\Gamma_{16} \;=\; D_{16} \;\cup\; (D_{16} + s), \qquad s = (\tfrac{1}{2}, \ldots, \tfrac{1}{2}). \tag{G.67}$$

Its norm-2 vectors are the $2\binom{16}{2} \cdot 2 = 480$ roots of D_{16} (the spinor class starts at norm 4), giving the algebra $\mathfrak{so}(32)$; the added spinor class fixes the *global* form of the group as $\mathrm{Spin}(32)/\mathbb{Z}_2$ rather than $\mathrm{SO}(32)$ or $\mathrm{Spin}(32)$. The lattice Γ_8 has 240 norm-2 vectors, the roots of E_8, so $\Gamma_8 \oplus \Gamma_8$ carries $\mathrm{E}_8 \times \mathrm{E}_8$. Both root counts, and the way each pins the gauge algebra, are worked in the exercises of Chapter 3; the root systems themselves are in Appendix G.2.

Remark G.18 (Beyond the lectures: mass formulae and higher rank)*.* The count does not stay small. The Minkowski-Siegel *mass formula* computes the weighted total $\sum_\Lambda 1/|\mathrm{Aut}\,\Lambda|$ over isometry classes of a given rank, and in rank 32 this already forces more than 10^7 distinct even self-dual lattices (SPLG, Ch. 16). Only ranks 8 and 16 are small enough, and their lattices constrained enough, to give a manageable and physical answer, which is one reason the heterotic string lives at rank 16. The theta series of the low-rank cases are themselves distinguished modular forms: $\Theta_{\Gamma_8} = E_4$, the weight-4 Eisenstein series, whose q-expansion $1+240\sum_{k\geq 1}\sigma_3(k)q^k$ has leading coefficient 240, the number of E_8 roots. In rank 16 both lattices share the same theta series E_4^2, so they are *isospectral*: the partition functions of the two heterotic strings agree term by term in ten dimensions, and the theories are distinguished only by their charge lattices, not by any Euclidean invariant.

G.4.3 Indefinite even self-dual lattices: Milnor's theorem

The indefinite case is where the physics of moduli lives, and there the situation inverts: existence is easy and, crucially, the lattice is *unique*. The building block is the *hyperbolic plane* U, the rank-2 lattice with Gram matrix

$$U: \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{G.68}$$

which is even (diagonal entries vanish), self-dual ($\det = -1$), and of signature $(1,1)$. Write $\mathrm{E}_8(-1)$ for the E_8 root lattice with its form reversed to negative-definite.

Theorem G.19 (Milnor)**.** *An indefinite even self-dual lattice of signature* (p,q) *with* $p,q \geq 1$ *exists if and only if*

$$p \equiv q \pmod 8, \tag{G.69}$$

and when it exists it is unique up to isometry. *Explicitly, for* $p \le q$,

$$\Gamma_{p,q} \;\cong\; U^{\oplus p} \;\oplus\; \mathrm{E}_8(-1)^{\oplus (q-p)/8}, \tag{G.70}$$

and symmetrically with $\mathrm{E}_8(+1)$ *when* $p > q$.

The proof is in Milnor and Husemoller, *Symmetric Bilinear Forms* (Ch. II), and in Serre (Ch. V). Uniqueness is the load-bearing fact for everything that follows: two even self-dual lattices of the same indefinite signature are the *same* lattice, so any physical structure controlled by such a lattice is connected and has no discrete choices left. This is what makes toroidal moduli spaces connected, and it is the reason the two heterotic strings, distinct in ten dimensions, merge on a circle (Appendix G.4.7). The contrast with Theorem G.17 is sharp: switching on even a single direction of the opposite sign collapses a rich definite classification to a unique answer.

We use the shorthand $\Gamma_{p,q}$ for the (unique) indefinite even self-dual lattice of signature (p,q), and Γ_n for a Euclidean one of rank n. The orientation (p,q) versus (q,p) is a matter of which sign one calls positive; the chapters cite the same object in both orderings, for instance $\Gamma_{17,1}$ in Chapter 3 and $\Gamma_{4,20}$ in Chapter 8, and $\Gamma_{p,q} \cong \Gamma_{q,p}$ once the overall sign of the form is reversed.

G.4.4 The Narain lattice of a torus

Compactify d coordinates on a torus $T^d = \mathbb{R}^d/2\pi\mathbb{Z}^d$ carrying a constant metric G_{mn} and antisymmetric B-field B_{mn}, both measured in units of α'. A closed string now has two integer zero-mode quantum numbers per circle: momentum $n_m \in \mathbb{Z}$ and winding $w^m \in \mathbb{Z}$. The single-circle version of this splitting, and the physics of winding, is Chapter 6; here we give the T^d generalisation that chapter defers to this appendix. The left- and right-moving internal momenta are

$$p^a_{L,R} = \frac{1}{\sqrt{2}}\, e^{*ma}\Big[\, n_m \;\pm\; (G_{mn} \mp B_{mn})\, w^n \Big], \tag{G.71}$$

with $e_m{}^a$ a vielbein for the torus metric, $G_{mn} = e_m{}^a e_n{}^a$, and e^{*ma} its inverse; here the upper sign gives p_L and the lower p_R. The pair $(p_L, p_R) \in \mathbb{R}^d \oplus \mathbb{R}^d$ ranges, as (n,w) run over $\mathbb{Z}^{2d}$, over a lattice $\Gamma_{d,d}$ of signature (d,d), on which the natural pairing is $\langle (p_L,p_R),(p_L,p_R)\rangle = p_L^2 - p_R^2$. A short computation from (G.71) gives

$$p_L^2 - p_R^2 \;=\; 2\, n_m w^m \in 2\mathbb{Z}, \tag{G.72}$$

independent of the moduli (G,B): the lattice is *even*, and evenness is again level matching. One checks that it is self-dual with respect to this pairing. So $\Gamma_{d,d}$ is an even self-dual lattice of signature (d,d), and by Theorem G.19 it is the unique such lattice,

$$\Gamma_{d,d} \;\cong\; U^{\oplus d}. \tag{G.73}$$

The cleanest way to hold this is to separate the moduli-independent lattice from the moduli-dependent embedding. Collect the charges into $Z = (w^m, n_m)^{\mathsf{T}} \in \mathbb{Z}^{2d}$. The pairing (G.72) is

$$\langle Z, Z\rangle \;=\; Z^{\mathsf{T}} \eta\, Z \;=\; 2\, w^m n_m, \qquad \eta = \begin{pmatrix} 0 & \mathbf{1}_d \\ \mathbf{1}_d & 0 \end{pmatrix}, \tag{G.74}$$

a fixed $O(d,d)$ form, the same $U^{\oplus d}$ of (G.73) written in the momentum-winding basis. The metric and B-field enter only through the positive combination

$$p_L^2 + p_R^2 \;=\; Z^{\mathsf{T}} \mathcal{H}(G,B)\, Z, \qquad \mathcal{H} = \begin{pmatrix} G - BG^{-1}B & BG^{-1} \\ -G^{-1}B & G^{-1} \end{pmatrix}, \tag{G.75}$$

the *generalised metric*, an $O(d,d)$ element with $\mathcal{H}^{\mathsf{T}}\eta\mathcal{H}=\eta$ (quoted here in the standard form; see Giveon, Porrati and Rabinovici, *Phys. Rept.* 244 (1994) 77). The abstract lattice $(\mathbb{Z}^{2d},\eta)$ is frozen; the moduli are the choice of positive-definite d-plane inside $\mathbb{R}^{d,d}$ on which $\mathcal{H}$ measures p_L, equivalently the way the fixed even self-dual lattice is boosted relative to the left-right split. This is the Narain description of toroidal compactification (Narain, *Phys. Lett.* B169 (1986) 41; Narain, Sarmadi and Witten, *Nucl. Phys.* B279 (1987) 369).

The inequivalent choices of $\mathcal{H}$ form the *Narain moduli space*. The continuous data (G,B) parametrise the coset $O(d,d;\mathbb{R})/(O(d)\times O(d))$, and physically distinct compactifications are counted after quotienting by the lattice automorphisms:

$$\mathcal{M}_{d,d} \;=\; O(d,d;\mathbb{Z})\backslash O(d,d;\mathbb{R})\,/\,(O(d)\times O(d)), \qquad \dim_{\mathbb{R}}\mathcal{M}_{d,d}=d^2. \tag{G.76}$$

The dimension matches the moduli count directly: G_{mn} supplies $\frac{1}{2}d(d+1)$ parameters and B_{mn} supplies $\frac{1}{2}d(d-1)$, together d^2.

G.4.5 The T-duality group $O(d,d;\mathbb{Z})$

The discrete group in (G.76) is the group of automorphisms of $\Gamma_{d,d}$, the integer matrices preserving η:

$$O(d,d;\mathbb{Z}) \;=\; \{\, M\in \mathrm{GL}(2d,\mathbb{Z}) \;:\; M^{\mathsf{T}}\eta M=\eta \,\}. \tag{G.77}$$

Two backgrounds related by such an M have identical spectra, because M permutes the charge vectors while preserving both pairings (G.74) and (G.75): this is T-duality, exact order by order in g_s. The group is generated by three families of transformations, each with a transparent physical reading:

- **Basis changes** $A\in\mathrm{GL}(d,\mathbb{Z})$, acting as $G\to A^{\mathsf{T}}GA$, $B\to A^{\mathsf{T}}BA$: the large diffeomorphisms of T^d, relabelling momentum and winding by the same integral change of one-cycle basis.
- **B-shifts** $B\to B+N$ with N an integer antisymmetric matrix: the periodicity of the B-field, invisible to the closed string except through the theta-angle-like shift of momenta.
- **Factorised dualities**, the elementary inversions $R\to\alpha'/R$ acting on a single circle, exchanging $n\leftrightarrow w$ on that circle. The $d=1$ case is $O(1,1;\mathbb{Z})=\mathbb{Z}_2$, the single-circle T-duality of Chapter 6.

Example G.20 (T-duality as an $O(1,1;\mathbb{Z})$ rotation)**.** Take $d=1$ with $B=0$ and $G=R^2/\alpha'$, so the charge vector is $Z=(w,n)^{\mathsf{T}}$ and (G.75) reduces to $\mathcal{H}=\mathrm{diag}(R^2/\alpha',\ \alpha'/R^2)$. The single factorised duality is

$$M=\begin{pmatrix}0&1\\1&0\end{pmatrix}=\eta, \qquad M^{\mathsf{T}}\eta M=\eta, \tag{G.78}$$

so $M\in O(1,1;\mathbb{Z})=\mathbb{Z}_2$. It acts on charges by $Z\to MZ$, exchanging $w\leftrightarrow n$, and on the modulus by $\mathcal{H}\to M^{\mathsf{T}}\mathcal{H}M=\mathrm{diag}(\alpha'/R^2,\ R^2/\alpha')$, that is $R\to\alpha'/R$. The spectrum depends on the charges only through the two invariants $Z^{\mathsf{T}}\eta Z=2wn$ and $Z^{\mathsf{T}}\mathcal{H}Z=w^2R^2/\alpha'+n^2\alpha'/R^2$; relabelling $w\leftrightarrow n$ is a bijection of $\mathbb{Z}^2$ that carries the R spectrum onto the α'/R spectrum term by term, so the two backgrounds are physically identical. The self-dual radius $R=\sqrt{\alpha'}$ is the fixed point $\mathcal{H}=\mathbf{1}$, where the enhancement of Appendix G.4.6 occurs.

At a generic point of $\mathcal{M}_{d,d}$ the only massless vectors are the $2d$ Cartan gauge bosons of $\mathrm{U}(1)^d_L\times\mathrm{U}(1)^d_R$. On the sublocus where extra lattice vectors reach $(p_L^2,p_R^2)=(2,0)$ (or $(0,2)$), those states become massless charged vectors and the abelian symmetry enhances to a non-abelian group. Because every such root has $p_L^2=2$, all roots have equal length: the enhanced groups are always simply-laced, and they are exactly the ADE groups. The currents that realise

them are constructed in Appendix G.4.6, and this enhancement is the mechanism behind gauge symmetry at self-dual radii (Chapter 6) and at ADE points of the heterotic and K3 moduli spaces (Chapters 7, 8 and 10).

G.4.6 Level-one current algebras from the lattice

The enhanced symmetries of the previous subsection are realised on the worldsheet by an explicit set of holomorphic currents, the Frenkel-Kac-Segal construction. It turns any even lattice Λ into a chiral (left-moving) current algebra at level one, and specialises both to the heterotic gauge currents of Chapter 3 and to the enhancement at self-dual radius of Chapter 6.

Let $X_L^I(z)$, $I = 1, \dots, n$, be n chiral bosons in the self-dual normalisation

$$X_L^I(z)\, X_L^J(w) \;\sim\; -\delta^{IJ} \ln(z-w), \qquad \partial X_L^I(z)\, \partial X_L^J(w) \;\sim\; -\frac{\delta^{IJ}}{(z-w)^2}. \tag{G.79}$$

The n *Cartan currents* $H^I(z) = i\,\partial X_L^I(z)$ have weight one and OPE $H^I(z)H^J(w) \sim \delta^{IJ}/(z-w)^2$: an abelian (Heisenberg) current algebra generating $\mathrm{U}(1)_L^n$. For each lattice vector $\alpha \in \Lambda$ form the vertex operator

$$J_\alpha(z) \;=\; c_\alpha \,{:}e^{\,i\,\alpha\cdot X_L(z)}{:}, \tag{G.80}$$

of conformal weight $\frac{1}{2}\alpha^2$, where $c_\alpha = \pm 1$ is a cocycle inserted so that the algebra closes with consistent signs (SPLG Ch. 30; Goddard and Olive, *Int. J. Mod. Phys.* A1 (1986) 303). J_α has weight one exactly when $\alpha^2 = 2$, so the weight-one charged currents are labelled by the *roots* of Λ, its norm-two vectors. The free-field contraction $(z-w)^{\alpha\cdot\beta}$ fixes all OPEs:

$$H^I(z)\, J_\alpha(w) \;\sim\; \frac{\alpha^I}{z-w}\, J_\alpha(w), \tag{G.81}$$

$$J_\alpha(z)\, J_\beta(w) \;\sim\; \begin{cases} \dfrac{1}{(z-w)^2} + \dfrac{\alpha\cdot H(w)}{z-w}, & \beta = -\alpha, \\[2mm] \dfrac{\epsilon(\alpha,\beta)}{z-w}\, J_{\alpha+\beta}(w), & \alpha\cdot\beta = -1, \\[2mm] \text{regular}, & \alpha\cdot\beta \geq 0, \end{cases} \tag{G.82}$$

with $\epsilon(\alpha,\beta) = \pm 1$ from the cocycles. Since $\alpha^2 = \beta^2 = 2$, the integer $\alpha\cdot\beta$ lies in $\{-2,-1,0,1,2\}$, and $\alpha\cdot\beta = -2$ forces $\beta = -\alpha$; when $\alpha\cdot\beta = -1$ the sum $\alpha+\beta$ is again a root, so these three cases exhaust the possibilities. The double pole in the first line has coefficient 1: the level is $k = 1$. Together $\{H^I, J_\alpha\}$ close on the level-one current algebra of the simply-laced algebra whose root system is the set of norm-two vectors of Λ, which is why the enhanced groups are ADE.

Two specialisations recover the physics quoted elsewhere. Taking $n = 16$ chiral bosons on $\Lambda = \Gamma_{16}$ builds the level-one $\mathfrak{so}(32)$ current algebra from its 480 roots, while $\Lambda = \Gamma_8 \oplus \Gamma_8$ builds $\mathfrak{e}_8 \oplus \mathfrak{e}_8$ at level one from its $240 + 240$ roots: these are the current algebras carrying the ten-dimensional heterotic gauge symmetry (Chapter 3). Taking a single circle at the self-dual radius, the two states with $(p_L^2, p_R^2) = (2, 0)$ found in Exercise G.6(c) are $J_\pm = {:}e^{\pm i\sqrt{2}\,X_L}{:}$, and with $J^3 = \frac{1}{\sqrt{2}}H$ they close on the level-one $\mathfrak{su}(2)$ algebra, enhancing $\mathrm{U}(1)_L \to \mathrm{SU}(2)_L$ (Chapter 6).

G.4.7 The heterotic Narain lattice

The heterotic string carries sixteen extra left-moving bosons on the even self-dual lattice Γ_{16} of Appendix G.4.2. Compactifying it on T^d combines those with the momentum-winding lattice

and with the Wilson lines A^I_m of the gauge group around the d cycles. The charges fill an even self-dual lattice of signature $(d+16, d)$,

$$\Gamma_{d+16,d} \;\cong\; U^{\oplus d} \oplus \Gamma_{16} \;\cong\; U^{\oplus d} \oplus \mathrm{E}_8 \oplus \mathrm{E}_8, \tag{G.83}$$

unique up to isometry by Theorem G.19 since $(d+16) - d = 16 \equiv 0 \pmod 8$. The last isomorphism is the algebraic heart of heterotic T-duality: whether one starts from $\Gamma_8 \oplus \Gamma_8$ or from $\Gamma_{16} = D_{16}^+$, adjoining a single hyperbolic plane U (one compact circle, with its winding) lands on the same lattice. The moduli space is

$$\mathcal{M}_{d+16,d} \;=\; O(d, d+16;\mathbb{Z}) \backslash O(d, d+16;\mathbb{R}) \,/\, \big(O(d) \times O(d+16)\big), \qquad \dim_{\mathbb{R}} = d(d+16), \tag{G.84}$$

with the T-duality group now $O(d, d+16;\mathbb{Z})$. For $d = 1$ the two ten-dimensional heterotic strings, distinct at the boundary of the moduli space, are interior points of one nine-dimensional theory governed by $\Gamma_{17,1}$; $\mathrm{E}_8 \times \mathrm{E}_8$ heterotic on a circle with a suitable Wilson line is $\mathrm{Spin}(32)/\mathbb{Z}_2$ heterotic on the dual circle. The dimension count in (G.84) is $\frac{1}{2}d(d{+}1)$ from G, $\frac{1}{2}d(d{-}1)$ from B, and $16d$ from the Wilson lines, for $d^2 + 16d = d(d+16)$. This is the structure Chapters 7 and 8 use, where for instance the heterotic string on T^4 has $\Gamma_{20,4} \cong U^{\oplus 4} \oplus \mathrm{E}_8 \oplus \mathrm{E}_8$ (written $\Gamma_{4,20}$ in the opposite orientation) matching the K3 lattice of the dual IIA description.

Remark G.21 (Beyond the lectures: the Siegel-Narain theta and the K3 lattice)*.* Because the Narain form is indefinite, the sum (G.64) needs a Gaussian regulator to converge. The correct object is the *Siegel-Narain theta function*

$$\Theta_{\Gamma_{d,d}}(\tau, \bar\tau) = \sum_{Z \in \Gamma_{d,d}} q^{\frac{1}{2}p_L^2}\, \bar q^{\frac{1}{2}p_R^2}, \tag{G.85}$$

a non-holomorphic modular form of weight $(d/2, d/2)$, and even self-duality is exactly the statement that (G.85), divided by the appropriate power of η, is modular invariant. The same lattice arithmetic surfaces geometrically. The middle cohomology $H^2(\mathrm{K3},\mathbb{Z})$ with its intersection form is the even self-dual lattice $\Gamma_{3,19} \cong U^{\oplus 3} \oplus \mathrm{E}_8(-1)^{\oplus 2}$ (Chapter 8), and $H^{\mathrm{even}}(\mathrm{K3},\mathbb{Z}) \cong \Gamma_{4,20}$ is what heterotic/IIA duality on K3 identifies with the heterotic Narain lattice (G.83) at $d = 4$. Uniqueness (Theorem G.19) is what guarantees there is nothing to check beyond the signature: two even self-dual lattices of signature $(4, 20)$ are the same lattice, so the two eighty-dimensional moduli spaces coincide.

Exercise G.6 (Evenness and the circle lattice)**.** (a) Verify (G.72) directly from (G.71), using $e^{*ma} e^{*na} = G^{mn}$, and confirm that the (G, B)-dependent cross terms cancel so that $p_L^2 - p_R^2 = 2\, n_m w^m$ is a moduli-independent, even integer. (b) Specialise to $d = 1$ with $B = 0$ and $G_{11} = R^2/\alpha'$, so that $p_{L,R} = \frac{1}{\sqrt{2}}(n/R \pm wR/\alpha')$ in the conventions of Chapter 6. Show that (p_L, p_R) ranges over an even self-dual lattice of signature $(1, 1)$, hence $\Gamma_{1,1} \cong U$, for every R. (c) Show that the states with $(p_L^2, p_R^2) = (2, 0)$, which enhance $\mathrm{U}(1)_L$ to $\mathrm{SU}(2)_L$, occur only at the self-dual radius $R = \sqrt{\alpha'}$, and identify the two extra roots as the unit-winding, unit-momentum states $(n, w) = \pm(1, 1)$.

Appendix H

Anomalies and the Green-Schwarz Mechanism

H.1 Anomalies and anomaly polynomials

A symmetry of a classical action need not survive quantisation. When the offending symmetry is a global one, the breakdown is a physical prediction: the neutral pion decays to two photons because the axial current is anomalous. When the symmetry is gauged, or is a diffeomorphism, the same breakdown is fatal. Gauge invariance is what removes the unphysical polarisations of a gauge field and the longitudinal modes of the graviton; lose it at one loop and the theory is non-unitary and inconsistent. A chiral gauge or gravitational theory is therefore consistent only if its anomaly cancels.

This section sets up the machinery in the language a QFT student already owns, the chiral (triangle) anomaly, and then repackages it in the form that makes the ten-dimensional problem tractable: a single characteristic class, the *anomaly polynomial*, from which the actual anomaly is read off by a mechanical descent. The three chiral fields that occur in ten-dimensional supergravity (a spin-$\frac{1}{2}$ fermion, a gravitino, and a self-dual tensor) each get their own polynomial. Assembling them for the $\mathcal{N}=1$ theories, factorising the result, and asking which gauge groups let it be cancelled is the Green-Schwarz story of Appendix H.2; here we build and stock the toolbox.

The big picture

An anomaly in $D=2n$ spacetime dimensions is controlled by a formal, gauge-invariant, closed $(2n{+}2)$-form I_{2n+2} built from the gauge field strength F and the curvature two-form $\mathcal{R}$. Two facts do all the work. First, I_{2n+2} is an *index density*: it is fixed by the Atiyah-Singer index theorem applied to a Dirac operator two dimensions up, which is why one works with a form of degree $2n{+}2$ rather than $2n$. Second, the *descent procedure* turns I_{2n+2} into the anomalous variation of the effective action by two applications of d. Cancelling the anomaly means making I_{2n+2} vanish, or (Green-Schwarz) making it a reducible product that a local counterterm can kill.

H.1.1 Chiral anomalies, from the triangle to the hexagon

The prototype is the Adler-Bell-Jackiw anomaly. A single Dirac fermion coupled to an abelian gauge field has a classically conserved axial current $j^\mu_5 = \bar\psi\Gamma^\mu\Gamma_*\psi$, with Γ^μ the Dirac matrices and Γ_* the chirality operator (the higher-dimensional analogue of γ_5, Appendix A). Quantisation

spoils the conservation law by an amount that is one-loop exact (Adler and Bardeen),

$$\partial_\mu j_5^\mu \;=\; \frac{1}{16\pi^2}\,\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}, \tag{H.1}$$

where the fermion carries unit charge and the gauge coupling is absorbed into F. The right-hand side is the triangle graph with one axial and two vector vertices. Only chiral fermions feel it: a left-handed and a right-handed Weyl fermion of the same charge contribute with opposite signs, so a vector-like (Dirac) theory is safe and the anomaly is an obstruction specific to chiral matter.

Promote the axial symmetry to a chiral *gauge* symmetry and the same triangle makes the gauge current non-conserved, which is now a disease. For a non-abelian group the one-loop gauge variation of the effective action $W[A]$ is

$$\delta_\lambda W \;=\; \frac{i}{24\pi^2}\int_{M_4}\mathrm{Tr}\big[\lambda\,\mathrm{d}\big(A\,\mathrm{d}A+\tfrac{1}{2}A^3\big)\big], \tag{H.2}$$

the consistent non-abelian anomaly of Bardeen, with $\lambda=\lambda^I T_I$ the gauge parameter. Its group-theory content is the totally symmetric invariant $d_{IJK}=\mathrm{Tr}_{\mathrm{f}}\left(T_I\{T_J,T_K\}\right)$, evaluated in a fixed reference (defining) representation f. The anomaly of a Weyl fermion in a general representation r is measured by the cubic index $A(r)$, defined by $\mathrm{Tr}_r\left(T_I\{T_J,T_K\}\right)=A(r)\,d_{IJK}$ (so $A(\mathrm{f})=1$), and cancellation in four dimensions is the familiar condition $\sum_r A(r)=0$.

Two distinctions matter before we go up in dimension. A *gauge* or *gravitational* anomaly, one carried by a current coupled to a dynamical field, must cancel outright. A *global* anomaly, carried by a current with no dynamical gauge field, need not: it is a genuine feature of the theory and is the content of 't Hooft's anomaly-matching constraints. Everything below is about the first kind, the kind that has to be made to vanish.

The pattern generalises cleanly. In $D=2n$ dimensions the dangerous diagram is the one-loop polygon with $n{+}1$ external gauge or graviton legs: the triangle in $D=4$ ($n=2$), the box in $D=6$, the pentagon in $D=8$, and the *hexagon* in $D=10$ ($n=5$). Its gauge part is governed by the single trace $\mathrm{Tr}_r\,F^{\,n+1}$, the top piece of the Chern character of the fermion's representation (defined in Appendix H.1.4). That is the thread we will pull: the $D=4$ triangle lives on $\mathrm{Tr}\,F^3$, the $D=10$ hexagon on $\mathrm{Tr}\,F^6$.

Gravitational anomalies work the same way with external stress tensors in place of gauge currents, but they are more selective about dimension. A pure gravitational anomaly, or a mixed gauge-gravitational one, can arise *only* in $D=4k+2$, that is in $D=2,6,10,\ldots$ (Alvarez-Gaumé and Witten). The first case, $D=2$, is one the reader has already met on the worldsheet: the two-dimensional gravitational anomaly measures c_L-c_R, the left-right asymmetry of the central charge, and its cancellation is why a consistent conformal field theory balances its chiralities. That $D=10$ is a $4k{+}2$ dimension is exactly why ten-dimensional supergravity has a gravitational anomaly to worry about at all, and why the whole Green-Schwarz question exists.

H.1.2 The consistent anomaly and Wess-Zumino consistency

Write the anomaly abstractly as the gauge (or BRST) variation of the effective action, $\mathcal{G}[\lambda]\coloneqq\delta_\lambda W$. Because two successive symmetry transformations close into the algebra, $\mathcal{G}$ cannot be an arbitrary functional. It must satisfy the *Wess-Zumino consistency condition*

$$\delta_{\lambda_1}\mathcal{G}[\lambda_2]-\delta_{\lambda_2}\mathcal{G}[\lambda_1] \;=\; \mathcal{G}\big[[\lambda_1,\lambda_2]\big], \tag{H.3}$$

which is just the integrability statement $\delta(\delta W)=0$ rewritten. In BRST language, promoting λ to the ghost c and δ_λ to the nilpotent operator s, it reads $s\,\mathcal{G}=0$: a consistent anomaly is a BRST-closed local functional. Anomalies that are BRST-exact, $\mathcal{G}=s(\text{local})$, can be removed by

adding that local counterterm to W and are not real. *The genuine anomalies are the cohomology of s on local functionals*, and this cohomological problem is what the descent procedure of the next subsection solves.

One practical wrinkle. The current obtained by naively differentiating W has an anomaly, the *consistent* anomaly of Eq. (H.2), that satisfies Eq. (H.3) but is not gauge-covariant. One can instead define a gauge-*covariant* current; its divergence is the *covariant* anomaly, which is simpler in form but does not satisfy the consistency condition. The two differ by a definite local polynomial in the fields, the Bardeen-Zumino term, and either may be used as long as one is consistent. The descent construction below produces the consistent anomaly, which is the one whose vanishing is the true consistency requirement.

H.1.3 Anomaly polynomials and the descent procedure

The organising object is a formal differential form of degree $2n{+}2$ in $D = 2n$ dimensions, the *anomaly polynomial* $I_{2n+2}(F, \mathcal{R})$. It is a gauge-invariant, closed polynomial in the gauge field strength F and the curvature two-form $\mathcal{R}$, of total form-degree two higher than spacetime. It is not itself the anomaly. It is the generating object from which the anomaly descends.

Because I_{2n+2} is closed and gauge invariant it is locally exact, and its potential is a Chern-Simons form:

$$I_{2n+2} \;=\; \mathrm{d}\, I^{(0)}_{2n+1}. \tag{H.4}$$

The Chern-Simons form $I^{(0)}_{2n+1}$ is gauge invariant only up to a total derivative; under an infinitesimal gauge or local-Lorentz transformation with parameter λ it shifts by an exact piece,

$$\delta_\lambda I^{(0)}_{2n+1} \;=\; \mathrm{d}\, I^{(1)}_{2n}. \tag{H.5}$$

The $2n$-form $I^{(1)}_{2n}$ is linear in λ, lives in the physical spacetime dimension, and *is* the anomaly: the anomalous variation of the effective action is

$$\delta_\lambda W \;=\; 2\pi \int_{M_{2n}} I^{(1)}_{2n}. \tag{H.6}$$

Equations (H.4) and (H.5) are the Stora-Zumino *descent equations*. The overall normalisation in Eq. (H.6) is a convention, fixed once and for all by demanding that the characteristic numbers appearing in I_{2n+2} (instanton numbers, Euler and Pontryagin numbers) be integers; we quote the descent result and do not re-fix it here.

Two comments make the procedure usable. First, the Chern-Simons form and the descended anomaly are ambiguous up to the addition of exact and local pieces, and that ambiguity is exactly the freedom to add local counterterms to W. This is the same freedom that lets one trade the consistent anomaly for the covariant one, and it is the freedom the Green-Schwarz counterterm will exploit. Second, the whole construction reduces the physics to a single algebraic question: what is the polynomial I_{2n+2}? Cancelling the anomaly means either $I_{2n+2} = 0$, or, when I_{2n+2} factorises into a product a counterterm can absorb, the reducible situation handled in Appendix H.2.

Remark H.1 (Beyond the lectures: why two dimensions up). The appearance of a $(2n{+}2)$-form for a $2n$-dimensional anomaly is not bookkeeping. The anomaly polynomial is literally an *index density*. By the Atiyah-Singer family index theorem, the anomalous phase of the chiral determinant is computed by the index of a Dirac operator for a family of gauge and gravitational backgrounds fibred over a two-dimensional parameter space; the natural base of that family is a $(2n{+}2)$-dimensional total space (spacetime times a two-sphere, or a disc bundle over it), not the $2n$-dimensional spacetime. That index is the integral of a characteristic class, and I_{2n+2} is its

integrand. This is what fixes the anomaly polynomial to be precisely the $\hat{A}$-genus and Chern character combinations written below, with no free coefficients. We assert this rather than prove it; the index theorem and the characteristic classes it needs are collected in Appendix D, and the anomaly-inflow reading, in which the Green-Schwarz term supplies the anomaly from a brane or a bulk, is used in Appendix H.2 and Chapter 6.

H.1.4 The index density: $\hat{A}$-genus, Chern character, and the three fields

Everything reduces to two characteristic classes and their product. The gauge field enters through the *Chern character* of the fermion's representation r,

$$\mathrm{ch}_r(F) \;=\; \mathrm{Tr}_r \exp\!\Big(\frac{iF}{2\pi}\Big) \;=\; \dim r + \frac{i}{2\pi}\,\mathrm{Tr}_r\, F - \frac{1}{8\pi^2}\,\mathrm{Tr}_r\, F^2 + \cdots\,, \tag{H.7}$$

whose degree-$(2k)$ piece is $\mathrm{ch}_k = \frac{1}{k!}\,\mathrm{Tr}_r(iF/2\pi)^k$; the top piece $\mathrm{ch}_{n+1} \propto \mathrm{Tr}_r\, F^{n+1}$ is the hexagon of Appendix H.1.1. The tangent bundle enters through the *$\hat{A}$-genus* (the A-roof, or Dirac, genus)

$$\hat{A}(\mathcal{R}) \;=\; \prod_a \frac{x_a/2}{\sinh(x_a/2)} \;=\; 1 - \frac{p_1}{24} + \frac{7p_1^2 - 4p_2}{5760} + \frac{-31\,p_1^3 + 44\,p_1p_2 - 16\,p_3}{967680} + \cdots\,, \tag{H.8}$$

where the x_a are the skew eigenvalues of $\mathcal{R}/2\pi$ and p_k are the Pontryagin classes. A second combination, the Hirzebruch L-genus, controls the self-dual tensor:

$$L(\mathcal{R}) \;=\; \prod_a \frac{x_a}{\tanh x_a} \;=\; 1 + \frac{p_1}{3} + \frac{7p_2 - p_1^2}{45} + \frac{62\,p_3 - 13\,p_1p_2 + 2\,p_1^3}{945} + \cdots\,. \tag{H.9}$$

For explicit ten-dimensional work one needs the Pontryagin classes as traces of the curvature two-form $\mathcal{R}$ (an antisymmetric matrix of two-forms, traced over the tangent-space vector index). Direct expansion of $\prod_a(1 + x_a^2)$ gives

$$p_1 = -\frac{1}{8\pi^2}\,\mathrm{tr}\,\mathcal{R}^2, \tag{H.10}$$

$$p_2 = \frac{1}{(2\pi)^4}\Big[\tfrac{1}{8}(\mathrm{tr}\,\mathcal{R}^2)^2 - \tfrac{1}{4}\,\mathrm{tr}\,\mathcal{R}^4\Big], \tag{H.11}$$

$$p_3 = \frac{1}{(2\pi)^6}\Big[-\tfrac{1}{48}(\mathrm{tr}\,\mathcal{R}^2)^3 + \tfrac{1}{8}\,\mathrm{tr}\,\mathcal{R}^2\;\mathrm{tr}\,\mathcal{R}^4 - \tfrac{1}{6}\,\mathrm{tr}\,\mathcal{R}^6\Big], \tag{H.12}$$

with the shorthand $\mathrm{tr}\,\mathcal{R}^{2k} \coloneqq \mathrm{tr}(\mathcal{R}\wedge\cdots\wedge\mathcal{R})$ of Chapter 5. Definitions of the classes live in Appendix D; the reference normalisation of the gauge trace tr (and its relation to representation traces Tr_r) lives in Appendix G.

The index theorem assigns to each type of chiral field its anomaly polynomial, the degree-$(2n{+}2)$ part of a fixed characteristic class (Alvarez-Gaumé and Witten). For a positive-chirality Weyl fermion of spin $\frac{1}{2}$ in representation r,

$$\hat{I}_{1/2}(r) \;=\; \Big[\hat{A}(\mathcal{R})\,\mathrm{ch}_r(F)\Big]_{2n+2}. \tag{H.13}$$

For a gauge-singlet fermion $\mathrm{ch}_{\mathbf{1}} = 1$, and this collapses to the purely gravitational $\hat{I}_{1/2}(\mathbf{1}) = [\hat{A}(\mathcal{R})]_{2n+2}$. For a gravitino (spin $\frac{3}{2}$, gauge-neutral in the theories of interest), the Rarita-Schwinger field is a spin-$\frac{1}{2}$ field valued in the tangent bundle, corrected by the removal of one spurious spin-$\frac{1}{2}$ mode:

$$\hat{I}_{3/2} \;=\; \Big[\hat{A}(\mathcal{R})\,\big(\mathrm{tr}\;e^{\,i\mathcal{R}/2\pi} - 1\big)\Big]_{2n+2}, \tag{H.14}$$

where tr $e^{i\mathcal{R}/2\pi}$ is the trace over the D-dimensional tangent vector and the -1 subtracts the ghost. For a chiral (self-dual) antisymmetric tensor, the middle-rank $(2k)$-form potential with self-dual field strength that exists in $D = 4k + 2$,

$$\hat{I}_A \;=\; -\tfrac{1}{8}\Big[L(\mathcal{R})\Big]_{2n+2}. \tag{H.15}$$

The overall sign of each polynomial is the field's chirality; the opposite chirality flips it, and a Majorana-Weyl field contributes half of the Weyl value in Eqs. (H.13) to (H.15). These three formulae are the entire input. Everything specific to ten dimensions is now a matter of expanding them to twelve form-degree.

Trace normalisations do not mix

Two different traces appear and must be kept apart. The tangent-space trace in $\hat{A}$, L and tr $e^{i\mathcal{R}/2\pi}$ runs over the D-dimensional Lorentz vector. The gauge trace in ch_r runs over the representation r, and the reference symbol tr F^{2k} used in Appendix H.2 is a specific normalised trace (the vector of SO(32), and $\frac{1}{30}\,\mathrm{Tr}_{\mathrm{adj}}$ for E_8), fixed in Appendix G. Converting Tr_r to the reference tr for each group is where the group-dependent factors, and eventually the selection of the gauge group, enter.

H.1.5 Ten dimensions

In $D = 10$ the anomaly polynomial is a twelve-form. Expanding Eqs. (H.13) to (H.15) with Eqs. (H.8) to (H.12) (these degree-twelve expansions are standard; see Alvarez-Gaumé and Witten, and Appendix D), the purely gravitational parts of the three fields are

$$\hat{I}_{1/2}(\mathbf{1}) = \frac{1}{(2\pi)^6}\left[\frac{(\mathrm{tr}\,\mathcal{R}^2)^3}{663552} + \frac{\mathrm{tr}\,\mathcal{R}^2\;\mathrm{tr}\,\mathcal{R}^4}{276480} + \frac{\mathrm{tr}\,\mathcal{R}^6}{362880}\right], \tag{H.16}$$

$$\hat{I}_{3/2} = \frac{1}{(2\pi)^6}\left[-\frac{7\,(\mathrm{tr}\,\mathcal{R}^2)^3}{73728} + \frac{5\;\mathrm{tr}\,\mathcal{R}^2\;\mathrm{tr}\,\mathcal{R}^4}{6144} - \frac{11\;\mathrm{tr}\,\mathcal{R}^6}{8064}\right], \tag{H.17}$$

$$\hat{I}_A = \frac{1}{(2\pi)^6}\left[\frac{(\mathrm{tr}\,\mathcal{R}^2)^3}{10368} - \frac{7\;\mathrm{tr}\,\mathcal{R}^2\;\mathrm{tr}\,\mathcal{R}^4}{8640} + \frac{31\;\mathrm{tr}\,\mathcal{R}^6}{22680}\right]. \tag{H.18}$$

Three invariants can appear at this order: the irreducible tr $\mathcal{R}^6$, and the two reducible products tr $\mathcal{R}^2$ tr $\mathcal{R}^4$ and $(\mathrm{tr}\,\mathcal{R}^2)^3$. The irreducible one is the obstruction. No wedge of lower-degree forms equals tr $\mathcal{R}^6$, so no counterterm of the Green-Schwarz type can remove it, and any field content whose total tr $\mathcal{R}^6$ coefficient is non-zero is simply inconsistent.

The clean illustration is type IIB, the only anomaly-free chiral case among the maximal theories (type IIA is non-chiral, hence automatically safe). Its chiral content is two Majorana-Weyl gravitini of one chirality, two Majorana-Weyl dilatini of the opposite chirality, and a four-form potential with self-dual five-form field strength. The polynomials $\hat{I}$ are normalised to a single Weyl field, and two Majorana-Weyl fields make one Weyl field: the pair of gravitini enters as one $\hat{I}_{3/2}$, the pair of dilatini as one $\hat{I}_{1/2}(\mathbf{1})$, and the tensor (not a fermion) as $\hat{I}_A$. The net twelve-form is

$$\hat{I}_{\mathrm{IIB}} \;=\; \hat{I}_{3/2} - \hat{I}_{1/2}(\mathbf{1}) + \hat{I}_A \;=\; 0, \tag{H.19}$$

and it vanishes structure by structure, not just in total. The point is sharpest for the irreducible invariant: in units of $1/(362880\,(2\pi)^6)$ the coefficients of tr $\mathcal{R}^6$ in Eqs. (H.16) to (H.18) are

$$\big(\hat{I}_{1/2}(\mathbf{1}),\;\hat{I}_{3/2},\;\hat{I}_A\big)\big|_{\mathrm{tr}\,\mathcal{R}^6} \;=\; (\,1,\;-495,\;+496\,), \qquad -495 - 1 + 496 = 0. \tag{H.20}$$

The gravitino alone contributes -495, the self-dual tensor $+496$, the dilatino -1, and the books close. The two reducible structures cancel in Eq. (H.19) as well; the reader can check them from Eqs. (H.16) to (H.18). This is the Alvarez-Gaumé-Witten result that IIB needs no mechanism.

For the $\mathcal{N}=1$ theories, type I and the two heterotic strings, the content is one gravitino, one dilatino of the opposite chirality, and $\dim G$ gauginos in the adjoint of the gauge group, of the same chirality as the gravitino. The total twelve-form is

$$I_{12} \;=\; \tfrac{1}{2}\Big[\,\hat{I}_{3/2} - \hat{I}_{1/2}(\mathbf{1}) + \hat{I}_{1/2}(\mathrm{adj})\,\Big], \tag{H.21}$$

the overall $\frac{1}{2}$ being the Majorana-Weyl halving; in Appendix H.2 this factor is absorbed into the overall normalisation constant, so the two forms of I_{12} coincide. Here the gaugino term carries the full Chern character $\mathrm{ch}_{\mathrm{adj}}(F)$ of Eq. (H.7), so Eq. (H.21) contains gauge invariants $\mathrm{Tr}_{\mathrm{adj}}\,F^2,\dots,\mathrm{Tr}_{\mathrm{adj}}\,F^6$ and mixed terms alongside the gravitational ones. Unlike Eq. (H.19), this does *not* vanish on its own, and its irreducible pieces already carry a sharp consequence.

Look first at the pure-gravitational $\mathrm{tr}\,\mathcal{R}^6$. The gravitino and dilatino contribute -495 and -1 in the units of Eq. (H.20), and there is no self-dual tensor to supply the $+496$ that saved IIB. The gauginos do contribute here: the degree-zero piece of $\mathrm{ch}_{\mathrm{adj}}$ is $\dim G$, so the $\dim G$ gauginos add $\dim G$ copies of the spin-$\frac{1}{2}$ gravitational polynomial $\hat{I}_{1/2}(\mathbf{1})$, hence $+\dim G$ units of $\mathrm{tr}\,\mathcal{R}^6$. The irreducible coefficient is therefore

$$\big[I_{12}\big]_{\mathrm{tr}\,\mathcal{R}^6} \;\propto\; \dim G - 496, \tag{H.22}$$

and since no counterterm can remove an irreducible term, consistency forces $\dim G = 496$. The number is fixed here, by the gravitational anomaly alone, before any gauge group is chosen.

That is only half of the story, and the remaining half is where the group is actually selected. The gaugino also produces a pure-gauge irreducible term $\mathrm{Tr}_{\mathrm{adj}}\,F^6$, which no gravitational field can cancel; it can be disposed of only if the algebra satisfies a trace identity that rewrites $\mathrm{Tr}_{\mathrm{adj}}\,F^6$ in terms of lower Casimirs, and only then does the leftover I_{12} collapse to the reducible product $X_4 \wedge X_8$ that a Green-Schwarz counterterm removes. Which dimension-496 algebras pass that second test, why the answer is exactly $\mathrm{SO}(32)$ and $\mathrm{E}_8\times\mathrm{E}_8$, and how the counterterm works, is Appendix H.2. The polynomials Eqs. (H.13) to (H.15) and their ten-dimensional expansions Eqs. (H.16) to (H.18) are the input that section works from.

Remark H.2 (Beyond the lectures: what the polynomial does not see). The descent construction captures *local*, perturbative anomalies, the ones visible in a Feynman diagram and encoded in I_{2n+2}. It is blind to *global* anomalies, where the obstruction is a large gauge transformation not connected to the identity and the relevant invariant is a discrete phase rather than a polynomial (the classic example is Witten's $\mathrm{SU}(2)$ anomaly in four dimensions, valued in $\pi_4(\mathrm{SU}(2)) = \mathbb{Z}_2$, detected by an eta-invariant on the mapping torus). Ten dimensions has its own global anomalies, and consistency of the type I and heterotic strings requires them to cancel too; the modern statement is that all of these, local and global, are unified as a cobordism invariant. That refinement sits beyond these notes, but $I_{12} = X_4 \wedge X_8$ is only a necessary condition, not by itself the whole of consistency.

Exercise H.1 ($\dim G = 496$ from the gravitational anomaly)**.** *(Short.)* Using the $\mathrm{tr}\,\mathcal{R}^6$ coefficients Eq. (H.20), confirm that the IIB combination Eq. (H.19) has no irreducible piece. Then show that the $\mathcal{N}=1$ content of Eq. (H.21), one gravitino, one opposite-chirality dilatino, and $\dim G$ gauginos, gives a $\mathrm{tr}\,\mathcal{R}^6$ coefficient proportional to $\dim G - 496$, using that the degree-zero part of $\mathrm{ch}_{\mathrm{adj}}$ is $\dim G$, so the gauginos contribute $\dim G$ copies of $\hat{I}_{1/2}(\mathbf{1})$. Conclude that consistency requires $\dim G = 496$ regardless of which group it is. State in one sentence what further condition, handled in Appendix H.2, then singles out $\mathrm{SO}(32)$ and $\mathrm{E}_8\times\mathrm{E}_8$ among the algebras of that dimension.

Exercise H.2 (Descent in four dimensions)**.** *(Moderate.)* Take the six-form $I_6 = \frac{1}{3!}\operatorname{Tr}\left(\frac{iF}{2\pi}\right)^3 = -\frac{i}{3!\,(2\pi)^3}\operatorname{Tr} F^3$, the $D = 4$ gauge anomaly polynomial. Using $F = \mathrm{d}A + A^2$ and $\mathrm{d}\operatorname{Tr} F^k = 0$, verify $I_6 = \mathrm{d}I_5^{(0)}$ with the Chern-Simons five-form $I_5^{(0)} \propto \operatorname{Tr}\left(A(\mathrm{d}A)^2 + \frac{3}{2}A^3\mathrm{d}A + \frac{3}{5}A^5\right)$, and then that its gauge variation is exact, $\delta_\lambda I_5^{(0)} = \mathrm{d}I_4^{(1)}$, with $I_4^{(1)}$ reproducing the Bardeen anomaly Eq. (H.2) up to normalisation. This is the descent Eqs. (H.4) to (H.6) in its simplest non-trivial case.

H.2 The Green-Schwarz mechanism

A ten-dimensional $\mathcal{N} = 1$ supergravity coupled to super-Yang-Mills is chiral: its gravitino, dilatino and gauginos are Majorana-Weyl, and chiral fermions in $D = 10$ produce a hexagon anomaly, encoded in a twelve-form anomaly polynomial I_{12} by the descent formalism of Appendix H.1. Generically $I_{12} \neq 0$ and the theory is inconsistent. Green and Schwarz observed that for two special gauge groups the polynomial *factorises* into a wedge product of a four-form and an eight-form, and that a factorised anomaly can be removed by a local counterterm built from the antisymmetric tensor B. This section assembles I_{12} from the field-by-field densities of Appendix H.1, states the factorisation, gives the two conditions it imposes on the gauge algebra, and shows why exactly $\mathrm{SO}(32)$ and $\mathrm{E}_8 \times \mathrm{E}_8$ survive. The anomaly polynomials of a Weyl fermion, a gravitino and a self-dual tensor (in ten dimensions a four-form potential with self-dual five-form field strength), together with the $\hat{A}$-genus, the Hirzebruch L-genus and the descent equations, are established in Appendix H.1; here they are inputs. The result is quoted in Chapter 3 and reused in Chapter 5.

H.2.1 The chiral content and its anomaly

The massless $\mathcal{N} = 1$ multiplets contribute three sets of chiral fermions (Chapter 5):

- the *gravitino* ψ_M, a positive-chirality spin-$\frac{3}{2}$ field, gauge singlet;
- the *dilatino* λ, a spin-$\frac{1}{2}$ field of the *opposite* chirality, gauge singlet;
- $\dim G$ *gauginos* χ^I, positive-chirality spin-$\frac{1}{2}$ fields in the adjoint of G.

Write $\hat{I}_{3/2}$ and $\hat{I}_{1/2}(r)$ for the twelve-form anomaly densities of Appendix H.1: a spin-$\frac{3}{2}$ and a spin-$\frac{1}{2}$ field of positive chirality, the latter in representation r, with the singlet written $\mathbf{1}$. These densities are normalised per Weyl fermion, so a pair of Majorana-Weyl fields counts as one. The total polynomial is the chirality-weighted sum

$$I_{12} \;=\; \hat{I}_{3/2} \;-\; \hat{I}_{1/2}(\mathbf{1}) \;+\; \hat{I}_{1/2}(\mathrm{adj}), \tag{H.23}$$

the minus sign carrying the reversed chirality of the dilatino. The overall Majorana-Weyl factor $\frac{1}{2}$ of (H.21) is folded into the constant $\mathcal{N}$ below, so (H.23) and (H.21) denote the same object up to that constant. No self-dual tensor sits in the $\mathcal{N} = 1$ spectrum, so $\hat{I}_A$ does not appear; that density is what makes type IIB anomaly-free on its own (Appendix H.1.5; Alvarez-Gaumé and Witten), and it is absent here.

Two traces run through the computation, and they must not be mixed. Throughout this section $\operatorname{Tr} := \operatorname{Tr}_{\mathrm{adj}}$ denotes the trace over the adjoint, the representation the gauginos actually run, while $\operatorname{tr}\mathcal{R}^{2k} := \operatorname{tr}\left(\mathcal{R}^{\wedge k}\right)$ is the trace over the tangent-space vector of the curvature two-form $\mathcal{R} = \mathrm{d}\Omega + \Omega \wedge \Omega$ (Chapter 5, not the Ricci scalar). The four-form and eight-form below use a *reduced* gauge trace $\operatorname{tr} F$. For $\mathfrak{so}(N)$ this is the trace in the defining vector $\mathbf{N}$,

$$\operatorname{tr} F^2 \;:=\; \operatorname{tr}_{\mathbf{N}} F^2 \;=\; \frac{1}{N-2}\operatorname{Tr} F^2, \tag{H.24}$$

and for E_8, which has no representation below the adjoint **248**, the reduced trace is *defined* by $\mathrm{tr} := \frac{1}{30}\,\mathrm{Tr}$. Both SO(32) and E_8 have dual Coxeter number 30, so in each case $\mathrm{Tr}\,F^2 = 30\,\mathrm{tr}\,F^2$ and one tr serves both groups. The normalisations are collected in Appendix G, and the vector/adjoint distinction is set up in Appendix H.1.

Feeding the $\hat{A}$-genus expansion of Appendix H.1 into (H.23) and collecting by trace structure gives, up to an overall non-zero constant $\mathcal{N}$ that descent fixes but which cancels out of every statement below,

$$\begin{aligned}\mathcal{N}^{-1} I_{12} \;=\; &-\tfrac{1}{720}\,\mathrm{Tr}\,F^6 \;+\; \tfrac{1}{1152}\,\mathrm{Tr}\,F^4\,\mathrm{tr}\,\mathcal{R}^2 \;-\; \tfrac{1}{9216}\,\mathrm{Tr}\,F^2\,(\mathrm{tr}\,\mathcal{R}^2)^2 \;+\; \tfrac{\dim G-64}{663552}\,(\mathrm{tr}\,\mathcal{R}^2)^3 \\ &+\Big(\tfrac{\dim G+224}{276480}\,\mathrm{tr}\,\mathcal{R}^2 \;-\; \tfrac{1}{11520}\,\mathrm{Tr}\,F^2\Big)\,\mathrm{tr}\,\mathcal{R}^4 \;+\; \tfrac{\dim G-496}{362880}\,\mathrm{tr}\,\mathcal{R}^6 .\end{aligned} \tag{H.25}$$

The degree-twelve expansion behind (H.25) is carried out in Appendix H.1; the standard external references are Alvarez-Gaumé and Witten, Green, Schwarz and Witten (vol. 2), and Polchinski (vol. 2). The pure-gravitational $\mathrm{tr}\,\mathcal{R}^6$ and the pure-gauge $\mathrm{Tr}\,F^6$ in (H.25) are *irreducible*: neither can be written as a wedge product of lower-degree invariant forms. No local counterterm removes an irreducible anomaly, so both must vanish before factorisation is even attempted. Everything else in (H.25) is reducible, and is fair game for the counterterm of Appendix H.2.5.

H.2.2 Factorisation $I_{12} = X_4 \wedge X_8$

Definition H.3 (Factorised anomaly)**.** The polynomial I_{12} *factorises* if there exist a gauge-invariant closed four-form X_4 and eight-form X_8 with

$$I_{12} \;=\; X_4 \wedge X_8 . \tag{H.26}$$

For the $\mathcal{N} = 1$ system the only candidate four-form built from $\mathrm{tr}\,\mathcal{R}^2$ and $\mathrm{tr}\,F^2$ is, up to normalisation,

$$X_4 \;=\; \mathrm{tr}\,\mathcal{R}^2 - \mathrm{tr}\,F^2 , \tag{H.27}$$

with $\mathrm{tr}\,F^2$ the reduced trace (H.24).

Removing the two irreducible sextics makes factorisation possible, but does not compel it. The reducible remainder of (H.25) collapses onto a single product $X_4 \wedge X_8$ only because its coefficients conspire: after the reduction $\mathrm{Tr}\,F^2 = 30\,\mathrm{tr}\,F^2$ the mixed and gravitational coefficients have to line up, for instance the $\mathrm{tr}\,F^2\,\mathrm{tr}\,\mathcal{R}^4$ coefficient must equal the gravitational $\alpha = 1/384$. That coincidence is a non-trivial fact, verified group by group in Appendix H.2.4, not a consequence of dropping the sextics. When it holds, the eight-form is

$$X_8 \;=\; \alpha\,\mathrm{tr}\,\mathcal{R}^4 + \beta\,(\mathrm{tr}\,\mathcal{R}^2)^2 + \gamma\,\mathrm{tr}\,\mathcal{R}^2\,\mathrm{tr}\,F^2 + \delta\,\mathrm{tr}\,F^4 + \varepsilon\,(\mathrm{tr}\,F^2)^2 , \tag{H.28}$$

whose coefficients are read off group by group in Appendix H.2.4. The four-form (H.27) is exactly the combination that appears on the right-hand side of the modified Bianchi identity (H.35) below, and that identification, spelled out in Appendix H.2.5, is what turns factorisation into a cancellation.

H.2.3 The two conditions on the gauge algebra

Dimension. The irreducible $\mathrm{tr}\,\mathcal{R}^6$ in (H.25) carries coefficient $(\dim G - 496)/362880$, which vanishes at a single value.

Proposition H.4 (The 496 count)**.** *The gravitational irreducible anomaly* $\mathrm{tr}\,\mathcal{R}^6$ *cancels among gravitino, dilatino and gauginos if and only if*

$$\dim G \;=\; 496 . \tag{H.29}$$

This is the $\operatorname{tr}\mathcal{R}^6$ count already carried out in (H.22): the gravitino and dilatino leave a fixed residue, and the $\dim G$ gauge-singlet copies inside $\hat{I}_{1/2}(\mathrm{adj})$ cancel it only at $\dim G = 496$. The statement is blind to the group structure, and it is the first of the two conditions.

Trace reduction. The irreducible $\operatorname{Tr} F^6$ in (H.25) has a fixed, $\dim G$-independent coefficient, so (H.29) does nothing for it. It must be killed by the invariant theory of the algebra itself: the sixth-order adjoint trace has to be reducible, expressible through the quadratic and quartic Casimirs,

$$\operatorname{Tr} F^6 \;=\; a\,(\operatorname{Tr} F^2)(\operatorname{Tr} F^4) + b\,(\operatorname{Tr} F^2)^3 \tag{H.30}$$

with no independent sixth-order invariant left over; in representation-theoretic terms, the algebra must have no independent sixth-order Casimir. Most simple algebras carry one and fail this outright, which makes it the more selective of the two conditions.

Which trace goes where

The four-form (H.27) and eight-form (H.28) are written in the reduced trace $\operatorname{tr} F$; the polynomial (H.25) and the condition (H.30) are written in the adjoint trace $\operatorname{Tr} = \operatorname{Tr}_{\mathrm{adj}}$, because that is what the gauginos run. The bridge is (H.24), and it differs from group to group. Mixing the two normalisations is the commonest error in this computation.

H.2.4 Which algebras pass

The trace-reduction condition (H.30) is settled by the standard $\mathrm{SO}(N)$ and E_8 trace identities (Appendix G), collected in Table H.1.

Algebra	$\operatorname{Tr} F^2$	$\operatorname{Tr} F^4$	$\operatorname{Tr} F^6$
$\mathfrak{so}(N)$	$(N-2)\,\operatorname{tr} F^2$	$(N-8)\,\operatorname{tr} F^4 + 3(\operatorname{tr} F^2)^2$	$(N-32)\,\operatorname{tr} F^6 + 15\,\operatorname{tr} F^2 \operatorname{tr} F^4$
$\mathfrak{so}(32)$	$30\,\operatorname{tr} F^2$	$24\,\operatorname{tr} F^4 + 3(\operatorname{tr} F^2)^2$	$15\,\operatorname{tr} F^2 \operatorname{tr} F^4$
$\mathfrak{e}_8$	$30\,\operatorname{tr} F^2$	$9\,(\operatorname{tr} F^2)^2$	$\frac{15}{4}\,(\operatorname{tr} F^2)^3$

Table H.1: Adjoint traces $\operatorname{Tr} = \operatorname{Tr}_{\mathrm{adj}}$ in terms of the reduced trace (H.24). For $\mathfrak{so}(N)$, tr is the vector ($\mathbf{N}$) trace; for $\mathfrak{e}_8$ there is no smaller representation and $\operatorname{tr} := \frac{1}{30}\operatorname{Tr}$, so $\mathfrak{e}_8$ has no independent quartic or sextic Casimir.

The decisive entries sit in the sixth-order column. For $\mathfrak{so}(N)$ the irreducible $\operatorname{tr} F^6$ enters with coefficient $(N-32)$, which vanishes precisely at $N = 32$, so $\mathrm{SO}(32)$, and no other orthogonal group, loses its irreducible sextic. For $\mathfrak{e}_8$ there is no independent quartic Casimir to begin with, so $\operatorname{Tr} F^4$ and $\operatorname{Tr} F^6$ both collapse onto powers of $\operatorname{Tr} F^2$ and the sextic is reducible for free. These are the only two mechanisms available; together with (H.29) they give a short list, Table H.2.

Algebra	$\dim G$	Passes (H.30)?	Realised by a string?
$\mathfrak{so}(32)$	496	yes ($N = 32$)	type I; heterotic $\mathrm{SO}(32)$
$\mathfrak{e}_8 \oplus \mathfrak{e}_8$	496	yes (no sextic Casimir)	heterotic $\mathrm{E}_8 \times \mathrm{E}_8$
$\mathfrak{e}_8 \oplus \mathfrak{u}(1)^{248}$	496	yes	none
$\mathfrak{u}(1)^{496}$	496	yes	none

Table H.2: The four dimension-496 solutions of the anomaly conditions. Only the first two are non-abelian and arise from a consistent chiral string.

Substituting Table H.1 into (H.25) at $\dim G = 496$ and dividing out X_4 gives X_8 explicitly. The

two non-abelian solutions yield

$$X_8^{\mathrm{SO}(32)} = \frac{1}{384}\Big[\ \mathrm{tr}\,\mathcal{R}^4 + \tfrac{1}{4}(\mathrm{tr}\,\mathcal{R}^2)^2 - \mathrm{tr}\,F^2\ \mathrm{tr}\,\mathcal{R}^2 + 8\ \mathrm{tr}\,F^4\ \Big], \tag{H.31}$$

$$X_8^{\mathrm{E}_8\times\mathrm{E}_8} = \frac{1}{384}\Big[\ \mathrm{tr}\,\mathcal{R}^4 + \tfrac{1}{4}(\mathrm{tr}\,\mathcal{R}^2)^2 - \mathrm{tr}\,F^2\ \mathrm{tr}\,\mathcal{R}^2 + 2(\mathrm{tr}\,F^2)^2 - 6\ \mathrm{tr}\,F_1^2\ \mathrm{tr}\,F_2^2\ \Big], \tag{H.32}$$

where in (H.32) the total gauge trace splits over the two E_8 factors, $\mathrm{tr}\,F^2 = \mathrm{tr}\,F_1^2 + \mathrm{tr}\,F_2^2$. Reading (H.31) against the template (H.28) fixes $(\alpha,\beta,\gamma,\delta,\varepsilon) = \frac{1}{384}(1,\frac{1}{4},-1,8,0)$ for $\mathrm{SO}(32)$. The exceptional case has no independent $\mathrm{tr}\,F^4$, so its quartic sits in $(\mathrm{tr}\,F^2)^2$ together with the cross term $\mathrm{tr}\,F_1^2\ \mathrm{tr}\,F_2^2$; no simple group can produce that cross term, which is the fingerprint of the two commuting E_8's. Both eight-forms are checked to satisfy (H.26) identically. The gravitational part $\mathrm{tr}\,\mathcal{R}^4 + \frac{1}{4}(\mathrm{tr}\,\mathcal{R}^2)^2$ is common to both, and it is the same X_8 that reappears as the one-loop $B\wedge X_8$ coupling of the heterotic string and in the M5-brane anomaly (Chapter 6).

Example H.5 (Why SO(32) and not SO(31) or SO(33))**.** Dimension alone already isolates $N=32$ inside the orthogonal series, since $\frac{1}{2}N(N-1)=496$ forces $N=32$. Independently, the sextic condition selects $N=32$ through the $(N-32)$ factor in Table H.1; the two requirements agree, a first sign that the constraints are not generic. The other classical series do not help even where a dimension-496 member exists: $\mathrm{SU}(N)$ and $\mathrm{Sp}(N)$ retain an independent sixth-order Casimir and fail (H.30) whatever their dimension. (In fact no $\mathrm{SU}(N)$ or $\mathrm{Sp}(N)$ has dimension exactly 496.)

H.2.5 The counterterm and the modified B-field

Factorisation makes the anomaly cancellable, but a field performs the cancellation, not fiat. The four-form (H.27) is exact,

$$X_4 \;=\; \mathrm{d}(\omega_3^{\mathrm{L}} - \omega_3^{\mathrm{YM}}), \qquad \mathrm{d}\omega_3^{\mathrm{YM}} = \mathrm{tr}\,F^2, \quad \mathrm{d}\omega_3^{\mathrm{L}} = \mathrm{tr}\,\mathcal{R}^2, \tag{H.33}$$

with $\omega_3^{\mathrm{YM}}, \omega_3^{\mathrm{L}}$ the gauge and Lorentz Chern-Simons three-forms $\omega_{\mathrm{YM}}, \omega_{\mathrm{L}}$ of Chapter 5. The NS-NS two-form (the RR two-form in type I) carries a modified field strength

$$H \;=\; \mathrm{d}B - \frac{\alpha'}{4}(\omega_3^{\mathrm{YM}} - \omega_3^{\mathrm{L}}), \tag{H.34}$$

with α' the Regge slope, $\ell_s^2 = \alpha'$ (Appendix A). Its Bianchi identity is no longer $\mathrm{d}H = 0$ but

$$\boxed{\mathrm{d}H \;=\; \frac{\alpha'}{4}\big(\,\mathrm{tr}\,\mathcal{R}^2 - \mathrm{tr}\,F^2\big) \;=\; \frac{\alpha'}{4}\,X_4\ .} \tag{H.35}$$

One four-form governs both the anomaly, through (H.26), and the failure of H to close, through (H.35).

The cancellation runs through descent (Appendix H.1). Because X_8 is closed and gauge-invariant, (H.26) gives $I_{12} = \mathrm{d}\big[(\omega_3^{\mathrm{L}} - \omega_3^{\mathrm{YM}})\wedge X_8\big]$, so the descent partners are

$$I_{11}^{(0)} = (\omega_3^{\mathrm{L}} - \omega_3^{\mathrm{YM}})\wedge X_8, \qquad I_{10}^{(1)} = -\,\omega_2^{(1)}\wedge X_8, \tag{H.36}$$

where $\omega_2^{(1)}$ is the first-order descendant of $\omega_3^{\mathrm{YM}} - \omega_3^{\mathrm{L}}$ under a gauge or local Lorentz transformation, $\delta(\omega_3^{\mathrm{YM}} - \omega_3^{\mathrm{L}}) = \mathrm{d}\omega_2^{(1)}$; the sign of $I_{10}^{(1)}$ follows because $I_{11}^{(0)}$ carries the opposite ordering. The one-loop effective action shifts by $\delta\Gamma \propto \int I_{10}^{(1)}$. Now demand that H in (H.34) stay invariant. The Chern-Simons forms shift by exact pieces, $\delta(\omega_3^{\mathrm{YM}} - \omega_3^{\mathrm{L}}) = \mathrm{d}\omega_2^{(1)}$, so the two-form must absorb the shift,

$$\delta B \;=\; \frac{\alpha'}{4}\,\omega_2^{(1)}. \tag{H.37}$$

The *Green-Schwarz counterterm*

$$S_{\mathrm{GS}} \;\propto\; \int B \wedge X_8 \tag{H.38}$$

then varies, under (H.37), by $\delta S_{\mathrm{GS}} \propto \int \omega_2^{(1)} \wedge X_8 = -\int I_{10}^{(1)}$, and its coefficient, sign included, is fixed so that this exactly cancels $\delta\Gamma \propto \int I_{10}^{(1)}$. Diagrammatically, a tree-level exchange of B between the vertex (H.38) and the anomalous Chern-Simons coupling in (H.34) cancels the one-loop hexagon. The Lorentz Chern-Simons term ω_3^{L} in (H.34) is present for exactly this reason: it supplies the $\operatorname{tr}\mathcal{R}^2$ half of X_4, without which the gravitational part of the anomaly would not descend correctly.

Integrating (H.35) over a closed four-cycle gives a topological constraint that has nothing further to do with anomalies and everything to do with compactification. With the trace normalised so that periods of $\operatorname{tr} F^2$ and $\operatorname{tr}\mathcal{R}^2$ are integer instanton numbers,

$$\int_{C_4} \operatorname{tr} F^2 \;=\; \int_{C_4} \operatorname{tr}\mathcal{R}^2, \tag{H.39}$$

since $\int_{C_4} \mathrm{d}H = 0$. This is the tadpole condition behind "24 instantons on K3" and behind $c_2(V) = c_2(TX)$ for heterotic backgrounds (Chapter 5, Chapter 8).

The mechanism in four moves

(i) A chiral $D = 10$, $\mathcal{N} = 1$ theory has a hexagon anomaly I_{12}. (ii) It factorises, $I_{12} = X_4 \wedge X_8$ with $X_4 = \operatorname{tr}\mathcal{R}^2 - \operatorname{tr} F^2$, only if $\dim G = 496$ (kills $\operatorname{tr}\mathcal{R}^6$) and the algebra has no independent sixth-order Casimir (kills $\operatorname{Tr} F^6$). (iii) The counterterm $\int B \wedge X_8$, together with the anomalous shift $\delta B = \frac{\alpha'}{4}\omega_2^{(1)}$ forced by the modified field strength (H.34), cancels the descended anomaly. (iv) Exactly two compact non-abelian groups pass, $\mathrm{SO}(32)$ and $\mathrm{E}_8 \times \mathrm{E}_8$, and both are delivered independently by the heterotic worldsheet and the type I tadpole.

H.2.6 The 496 miracle

Both hurdles are cleared by two unrelated Lie algebras:

$$\dim \mathrm{SO}(32) = \frac{32 \cdot 31}{2} = 496, \qquad \dim\left(\mathrm{E}_8 \times \mathrm{E}_8\right) = 2 \cdot 248 = 496. \tag{H.40}$$

An orthogonal group and a product of the largest exceptional group with itself, unrelated as abstract groups, share a dimension and each satisfies the sextic condition by its own route: $\mathrm{SO}(32)$ because $(N-32)$ vanishes, $\mathrm{E}_8 \times \mathrm{E}_8$ because E_8 has no independent quartic Casimir. This is the 496 *miracle*. Both are realised: heterotic and type I string theory produce $\mathrm{SO}(32)$, and the heterotic string also produces $\mathrm{E}_8 \times \mathrm{E}_8$ (Chapter 3). The remaining dimension-496 solutions of Table H.2, $\mathfrak{e}_8 \oplus \mathfrak{u}(1)^{248}$ and $\mathfrak{u}(1)^{496}$, are consistent as field theories but arise from no chiral string.

Remark H.6 (Beyond the lectures: the worldsheet says the same thing). The heterotic string encodes its gauge degrees of freedom in sixteen internal left-moving bosons on an even self-dual Euclidean lattice, and modular invariance of the torus partition function requires precisely a rank-16 even self-dual lattice. There are exactly two, Γ_{16} and $\Gamma_8 \oplus \Gamma_8$, with automorphism data giving $\mathrm{Spin}(32)/\mathbb{Z}_2$ and $\mathrm{E}_8 \times \mathrm{E}_8$ (Chapter 3, Appendix G). This worldsheet count is logically independent of the spacetime anomaly count, and it returns the same two groups. The anomaly polynomial sees only the algebra $\mathfrak{so}(32)$, so (H.40) cannot distinguish $\mathrm{SO}(32)$ from the string's actual $\mathrm{Spin}(32)/\mathbb{Z}_2$; the spectrum can.

Remark H.7 (Beyond the lectures: the $\mathrm{E}_8 \times \mathrm{E}_8$ factorisation is even nicer). The cross term in (H.32) lets the $\mathrm{E}_8 \times \mathrm{E}_8$ anomaly factorise a second way. Writing $u = \operatorname{tr} F_1^2$, $v = \operatorname{tr} F_2^2$, one can

trade the single product $X_4 \wedge X_8$ for a manifestly symmetric form in which the difference $u - v$ appears, reflecting the two commuting simple factors and the freedom to place the counterterm on either. This is the structural reason $\mathrm{E}_8 \times \mathrm{E}_8$ tolerates a Wilson line splitting one E_8 without spoiling cancellation, which matters for model building in Chapter 8.

Remark H.8 (Beyond the lectures: inflow and the ruled-out solutions). The coupling (H.38) has a modern reading as anomaly inflow: the two-form transports the anomaly from the "bulk" of field space, and the same $B \wedge X_8$ structure assigns the correct anomalous couplings to the NS5-brane and to the D-branes that source these forms (Chapter 6). The two abelian-contaminated solutions of Table H.2 are anomaly-free field theories with no string origin; swampland arguments now give independent evidence that they admit no ultraviolet completion into quantum gravity at all, so string consistency is the sharper filter.

Exercise H.3 (The two conditions by hand)**.** (Short, computational.) (a) From Table H.1, verify that the irreducible $\operatorname{tr} F^6$ in the adjoint sextic $\operatorname{Tr} F^6$ of $\mathfrak{so}(N)$ carries coefficient $(N - 32)$, and hence vanishes only at $N = 32$. (b) Using $\operatorname{Tr} F^4 = \frac{1}{100}(\operatorname{Tr} F^2)^2$ and $\operatorname{Tr} F^6 = \frac{1}{7200}(\operatorname{Tr} F^2)^3$ for E_8, together with $\operatorname{Tr} F^2 = 30 \operatorname{tr} F^2$, show that the reduced trace gives $\operatorname{tr} F^4 = \frac{3}{10}(\operatorname{tr} F^2)^2$ and confirm the E_8 entries in Table H.1. (c) Check the four dimensions in Table H.2 and name, in one sentence, the physical requirement beyond the field-theory anomaly count that removes the two abelian cases. (d) (Longer.) Substitute Table H.1 at $N = 32$ into (H.25), set $\dim G = 496$, and divide by X_4 to reproduce (H.31), thereby verifying $I_{12} = X_4 \wedge X_8$ for $\mathrm{SO}(32)$.

Exercise H.4 (Descent of the counterterm)**.** (Moderate.) Starting from $I_{12} = X_4 \wedge X_8$ with $X_4 = \mathrm{d}(\omega_3^{\mathrm{L}} - \omega_3^{\mathrm{YM}})$ and $\mathrm{d}X_8 = 0$, derive the descendants (H.36). Then, using the invariance of H in (H.34) to fix δB, show that the variation of $\int B \wedge X_8$ reproduces $\int I_{10}^{(1)}$ up to sign, and identify the sign of the counterterm needed for cancellation.

Appendix I

Supergravity Formulary, Duality Dictionary, and p-Brane Catalogue

I.1 The supergravity actions in one convention

This section collects the bosonic two-derivative actions of the six supergravities used in the notes: eleven-dimensional supergravity and the four ten-dimensional theories IIA, IIB, type I and heterotic, together with the massive (Romans) deformation of IIA. Chapter 5 builds and motivates all of these; here they are lined up in a single convention set so that a sign or a factor of 2π can be looked up without re-deriving it. Everything follows Appendix A: mostly-plus signature, the p-form norm and Hodge star of Appendix A.4, the democratic Ramond-Ramond field strengths of Eq. (A.13), and $2\kappa_D^2 = 16\pi G_D$.

Unless stated otherwise every action is written in *string frame*, the frame in which the NS-NS fields multiply a common $e^{-2\phi}$ and the worldsheet metric appears directly. The one place we depart is the manifestly $\mathrm{SL}(2,\mathbb{R})$-covariant form of IIB, which is natural only in Einstein frame; the conversion rule is given once, in Appendix I.1.1, and applies to all of them. Field content is tabulated in Appendix I.1.7, and the four-dimensional reductions are noted in Appendix I.1.8. Brane tensions, orientifold data and the harmonic-function solutions are the business of Appendix I.2; the T/S/U maps are in Appendix I.3.

I.1.1 Conventions, constants, and frames

Everything in this block follows Appendix A. The recurring scales are the string length $\ell_s = \sqrt{\alpha'}$ (the square root of the Regge slope), the string coupling $g_s = e^{\langle\phi\rangle}$, and the eleven-dimensional Planck length ℓ_p; the two Newton constants are fixed just below. A reader after a single formula may jump straight to the action needed (Appendix I.1.2 onward) and consult this block only for a normalisation.

Form norm. A p-form field strength is contracted with the norm

$$|F_p|^2 \;\coloneqq\; \frac{1}{p!}\, F_{\mu_1\cdots\mu_p} F^{\mu_1\cdots\mu_p}, \qquad F_p \wedge \star F_p \;=\; |F_p|^2 \,\sqrt{-g}\,\mathrm{d}^D x, \tag{I.1}$$

so that the NS-NS three-form term reads $-\frac{1}{2}|H_3|^2 = -\frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho}$ with $H_3 = \mathrm{d}B_2$. This is the norm of Appendix A.4.

Democratic Ramond-Ramond strengths. Assemble the RR potentials into a polyform $C = \sum_p C_p$ (p odd in IIA, even in IIB) with improved strength

$$F \;=\; \mathrm{d}C - H_3 \wedge C, \qquad F_n \;=\; (-1)^{\lfloor n/2 \rfloor} \star F_{10-n}, \tag{I.2}$$

the self-duality imposed on the equations of motion. In an action one keeps only the lower half of the ladder and lets (I.2) supply the dual potentials. The explicit strengths below are the degree-by-degree components of F, with one caveat: the IIB $\tilde{F}_5$ coincides with the degree-five component only after the conventional C_4 redefinition that renders it self-dual, the form written in (I.15).

Gravitational couplings. The two Newton constants, and the relation that ties them across the M-theory circle, are

$$2\kappa_{11}^2 = (2\pi)^8\, \ell_p^9, \tag{I.3}$$

$$2\kappa_{10}^2 = (2\pi)^7\, \alpha'^4\, g_s^2 \;=\; (2\pi)^7\, \ell_s^8\, g_s^2, \tag{I.4}$$

$$2\kappa_{10}^2 = \frac{2\kappa_{11}^2}{2\pi R_{11}}, \qquad R_{11} = g_s\, \ell_s, \quad \ell_p = g_s^{1/3} \ell_s. \tag{I.5}$$

Equation (I.4) is the value quoted in Eq. (A.6); (I.5) is its consistency with (I.3) once the eleventh circle has radius $R_{11} = g_s \ell_s$, and reproducing that chain is Exercise I.1. Two conventions for the dilaton circulate; we use the *full* dilaton ϕ (vacuum value included) everywhere, so the axio-dilaton (I.17) has $\mathrm{Im}\,\tau = e^{-\phi}$, equal to $1/g_s$ at the vacuum. The vacuum weight $e^{2\langle\phi\rangle} = g_s^2$ is exactly the g_s^2 displayed in (I.4): it has been pulled out of the action into $1/2\kappa_{10}^2$, so that against this normalisation the NS-NS prefactor enters as $e^{-2\phi} \to e^{-2(\phi - \langle\phi\rangle)}$ and g_s is counted once, not twice.

String frame to Einstein frame. In D dimensions the Weyl rescaling that removes the $e^{-2\phi}$ from the Einstein-Hilbert term is

$$g_{\mu\nu} = e^{4\phi/(D-2)}\, \tilde{g}_{\mu\nu}, \qquad \text{i.e. at } D = 10: \quad g_{\mu\nu} = e^{\phi/2}\, \tilde{g}^{\mathrm{E}}_{\mu\nu}, \tag{I.6}$$

so that $4\phi/(D-2) = \phi/2$ at $D = 10$. Under (I.6) the universal NS-NS action becomes, at $D = 10$,

$$S^{\mathrm{E}}_{\mathrm{NS}} = \frac{1}{2\kappa_{10}^2} \int \mathrm{d}^{10}x\, \sqrt{-\tilde{g}}\, \Big(\tilde{R} - \tfrac{1}{2}\, \tilde{\partial}_\mu \phi\, \tilde{\partial}^\mu \phi - \tfrac{1}{2}\, e^{-\phi}\, |H_3|^2 \Big). \tag{I.7}$$

The string-frame $+4(\partial\phi)^2$ turns into the canonical $-\frac{1}{2}(\partial\phi)^2$ under the rescaling, as the example below records. A Ramond-Ramond n-form strength, which carries no dilaton prefactor in string frame, picks up a definite weight in Einstein frame,

$$-\tfrac{1}{2}\, |F_n|^2 \text{ (string frame)} \;\longmapsto\; -\tfrac{1}{2}\, e^{(5-n)\phi/2}\, |F_n|^2 \text{ (Einstein frame)}. \tag{I.8}$$

The NS three-form is *not* an instance of (I.8): it sits under the extra $e^{-2\phi}$ NS-NS prefactor, so its net Einstein-frame weight is $-2\phi + (5-3)\phi/2 = -\phi$, i.e. the $e^{-\phi}|H_3|^2$ of (I.7), and differs from the prefactor-free value $e^{+\phi}$ one would get by naively setting $n = 3$ in the rule. Throughout, $(\partial\phi)^2 \coloneqq \partial_\mu\phi\, \partial^\mu\phi$.

Example I.1 (The NS-NS sector in Einstein frame)**.** At $D = 10$ the map $g_{\mu\nu} = e^{\phi/2}\tilde{g}_{\mu\nu}$ gives $\sqrt{-g} = e^{5\phi/2}\sqrt{-\tilde{g}}$, $(\partial\phi)^2_g = e^{-\phi/2}(\partial\phi)^2_{\tilde{g}}$, and $R = e^{-\phi/2}(\tilde{R} - \frac{9}{2}\tilde{\nabla}^2\phi - \frac{9}{2}(\partial\phi)^2)$. Feeding these into $e^{-2\phi}\sqrt{-g}\,(R + 4(\partial\phi)^2)$ and dropping the total derivative $\tilde{\nabla}^2\phi$, the string-frame $+4$ and the $-\frac{9}{2}$ induced by the rescaling of R collapse to the canonical $-\frac{1}{2}$ of (I.7). The three-form dresses as $e^{-\phi}$ by the count above, and the same term-by-term step applied to the RR strengths gives (I.8); that RR calculation is Exercise I.1.

I.1.2 Eleven-dimensional supergravity

The Cremmer-Julia-Scherk theory is the unique supergravity in eleven dimensions. Its $\mathcal{N}=1$ multiplet (32 real supercharges) has no scalar, hence no dimensionless coupling; the only scale is ℓ_p. On-shell degrees of freedom are counted in the transverse little group, Spin(9) here and Spin(8) in ten dimensions; the consolidation in Appendix I.1.7 collects both.

Field	Spin(9) content	on-shell dof
g_{MN} (graviton)	symmetric traceless	44
C_{MNP} (three-form)	antisymmetric rank 3	84
ψ_M (gravitino)	Γ-traceless vector-spinor	128
bosons / fermions		$128+128$

The bosonic action, with $F_4=\mathrm{d}C_3$ (so $\mathrm{d}F_4=0$) and $|F_4|^2=\frac{1}{4!}F_{MNPQ}F^{MNPQ}$, is

$$\boxed{S_{11}=\frac{1}{2\kappa_{11}^2}\int \mathrm{d}^{11}x\,\sqrt{-g}\left(R-\tfrac{1}{2}\,|F_4|^2\right)\;-\;\frac{1}{12\kappa_{11}^2}\int C_3\wedge F_4\wedge F_4\,,} \tag{I.9}$$

with $2\kappa_{11}^2=(2\pi)^8\ell_p^9$. The metric-free Chern-Simons term and its coefficient are fixed by closure of the local supersymmetry algebra, not by bosonic reasoning; the fermion completion and the gravitino variation live in Appendix C. The C_3 equation of motion,

$$\mathrm{d}\star F_4+\tfrac{1}{2}\,F_4\wedge F_4=0, \tag{I.10}$$

sources the four-form by itself: this is the fingerprint of the Chern-Simons coupling and the seed of the IIA Chern-Simons term after reduction. The electric and magnetic sources of C_3 are the M2 and M5, catalogued in Appendix I.2. Reduction on a circle following the ansatz of Eq. (5.36) lands on IIA in string frame, with g_s traded for R_{11} through (I.5).

I.1.3 Type IIA supergravity

Type IIA is the non-chiral $\mathcal{N}=(1,1)$ theory (32 supercharges): two gravitini of opposite ten-dimensional chirality, RR potentials C_1, C_3. The bosonic content is the NS-NS triplet plus the odd RR forms.

Field	Spin(8) irrep	dof	sector
$g_{\mu\nu}$, $B_{\mu\nu}$, ϕ	$\mathbf{35}_v, \mathbf{28}, \mathbf{1}$	64	NS-NS
C_1	$\mathbf{8}_v$	8	RR
C_3	$\mathbf{56}_v$	56	RR

In string frame, to two derivatives,

$$\begin{aligned} S_{\mathrm{IIA}} \;=\; \frac{1}{2\kappa_{10}^2}\int \mathrm{d}^{10}x\,\sqrt{-g}\,\Big[\,e^{-2\phi}\Big(R+4(\partial\phi)^2-\tfrac{1}{2}|H_3|^2\Big)-\tfrac{1}{2}|F_2|^2-\tfrac{1}{2}|\tilde{F}_4|^2\,\Big] \\ -\;\frac{1}{4\kappa_{10}^2}\int B_2\wedge F_4\wedge F_4, \end{aligned} \tag{I.11}$$

with field strengths and Bianchi identities

$$F_2=\mathrm{d}C_1,\quad F_4=\mathrm{d}C_3,\quad \tilde{F}_4=\mathrm{d}C_3-H_3\wedge C_1,\quad H_3=\mathrm{d}B_2;\qquad \mathrm{d}\tilde{F}_4=H_3\wedge F_2. \tag{I.12}$$

The Chern-Simons term uses the bare $F_4=\mathrm{d}C_3$, not $\tilde{F}_4$; the RR kinetic terms carry no $e^{-2\phi}$, which is why the RR sector is cleanest in Einstein frame. Applying (I.8) with $n=2,4$ and (I.7),

$$S_{\mathrm{IIA}}^{\mathrm{E}}=\frac{1}{2\kappa_{10}^2}\int \mathrm{d}^{10}x\,\sqrt{-\tilde{g}}\left(\tilde{R}-\tfrac{1}{2}(\partial\phi)^2-\tfrac{1}{2}e^{-\phi}|H_3|^2-\tfrac{1}{2}e^{3\phi/2}|F_2|^2-\tfrac{1}{2}e^{\phi/2}|\tilde{F}_4|^2\right)-\frac{1}{4\kappa_{10}^2}\int B_2\wedge F_4\wedge F_4. \tag{I.13}$$

Remark I.2 (Massive (Romans) IIA). There is a one-parameter deformation carrying a zero-form strength $F_0 = m$, quantised as the RR flux a D8-brane sources. The RR sector of (I.11) is replaced by

$$-\tfrac{1}{2}\,m^2 - \tfrac{1}{2}|F_2|^2 - \tfrac{1}{2}|\tilde{F}_4|^2, \qquad F_2 = \mathrm{d}C_1 + m\,B_2, \quad \tilde{F}_4 = \mathrm{d}C_3 - H_3 \wedge C_1 + \tfrac{1}{2} m\, B_2 \wedge B_2,$$

with the Chern-Simons sector completed accordingly. The two-form eats C_1 by a Stückelberg mechanism and becomes massive. Massive IIA has no known lift to eleven dimensions at the level of a Lagrangian; the standard $m = 0$ theory (I.11) is the special point used elsewhere in these notes.

I.1.4 Type IIB supergravity

Type IIB is the chiral $\mathcal{N} = (2,0)$ theory (32 supercharges): two gravitini of the same chirality, RR potentials C_0, C_2, C_4 with C_4 carrying a self-dual five-form strength.

Field	Spin(8) irrep	dof	sector
$g_{\mu\nu}$, $B_{\mu\nu}$, ϕ	$\mathbf{35}_v, \mathbf{28}, \mathbf{1}$	64	NS-NS
C_0	$\mathbf{1}$	1	RR
C_2	$\mathbf{28}$	28	RR
C_4 (self-dual F_5)	$\mathbf{35}_+$	35	RR

Because a self-dual five-form has no covariant polynomial kinetic term, one writes a *pseudo-action* whose Euler-Lagrange equations are supplemented by the self-duality relation imposed by hand afterwards. In string frame,

$$\begin{aligned} S_{\mathrm{IIB}} \;=\; \frac{1}{2\kappa_{10}^2}\int \mathrm{d}^{10}x\,\sqrt{-g}\,\Big[\, e^{-2\phi}\Big(R + 4(\partial\phi)^2 - \tfrac{1}{2}|H_3|^2\Big) - \tfrac{1}{2}|F_1|^2 - \tfrac{1}{2}|\tilde{F}_3|^2 - \tfrac{1}{4}|\tilde{F}_5|^2 \,\Big] \\ - \frac{1}{4\kappa_{10}^2}\int C_4 \wedge H_3 \wedge F_3, \end{aligned} \tag{I.14}$$

with

$$F_1 = \mathrm{d}C_0, \quad F_3 = \mathrm{d}C_2, \quad \tilde{F}_3 = F_3 - C_0\,H_3, \quad \tilde{F}_5 = \mathrm{d}C_4 - \tfrac{1}{2}C_2 \wedge H_3 + \tfrac{1}{2}B_2 \wedge F_3, \tag{I.15}$$

and the constraint

$$\boxed{\tilde{F}_5 = \star\,\tilde{F}_5} \qquad \text{imposed after varying.} \tag{I.16}$$

The $-\frac{1}{4}$ in front of $|\tilde{F}_5|^2$ (rather than $-\frac{1}{2}$) compensates the double-count that self-duality would otherwise force. The Bianchi identity is $\mathrm{d}\tilde{F}_5 = H_3 \wedge F_3$, which coincides with the C_4 equation of motion once (I.16) is used; that coincidence is what makes imposing the constraint consistent.

The pseudo-action is not an ordinary action

Vary (I.14) with $\tilde{F}_5$ unconstrained, obtain the equations of motion, and only then set $\tilde{F}_5 = \star\tilde{F}_5$. Imposing self-duality inside the functional or before varying makes the five-form kinetic density vanish identically, since $\tilde{F}_5 \wedge \tilde{F}_5 = 0$ for any five-form in ten dimensions.

The IIB scalars combine into the axio-dilaton and the symmetry becomes manifest in Einstein frame. Set $g^{\mathrm{E}}_{\mu\nu} = e^{-\phi/2} g_{\mu\nu}$ and

$$\tau \;\coloneqq\; C_0 + i\,e^{-\phi}, \qquad \mathrm{Im}\,\tau = e^{-\phi}\ (= 1/g_s \text{ at } \langle\phi\rangle), \qquad G_3 = \frac{F_3 - \tau H_3}{\sqrt{\mathrm{Im}\,\tau}}. \tag{I.17}$$

Then

$$S^{\mathrm{E}}_{\mathrm{IIB}} = \frac{1}{2\kappa_{10}^2}\int \mathrm{d}^{10}x\,\sqrt{-g_{\mathrm{E}}}\Big(R_{\mathrm{E}} - \frac{\partial_\mu\bar{\tau}\,\partial^\mu\tau}{2(\mathrm{Im}\,\tau)^2} - \tfrac{1}{2}|G_3|^2 - \tfrac{1}{4}|\tilde{F}_5|^2\Big) - \frac{i}{8\kappa_{10}^2}\int C_4 \wedge G_3 \wedge \bar{G}_3, \tag{I.18}$$

invariant under $\mathrm{SL}(2,\mathbb{R})$ acting by $\tau \mapsto (a\tau+b)/(c\tau+d)$ and $(C_2, B_2)^{\mathsf{T}} \mapsto \Lambda (C_2, B_2)^{\mathsf{T}}$ with $g^{\mathrm{E}}_{\mu\nu}$ and C_4 inert; G_3 picks up a phase, so $|G_3|^2$ and $G_3 \wedge \bar{G}_3 = 2i\, F_3 \wedge H_3$ are invariant. The scalar term is the Poincaré metric on the $\mathrm{SL}(2,\mathbb{R})/\mathrm{SO}(2)$ upper half-plane. Charge quantisation breaks this to the arithmetic $\mathrm{SL}(2,\mathbb{Z})$, the exact S-duality developed in Chapter 7 and tabulated in Appendix I.3.

I.1.5 Type I and heterotic supergravity

Three of the five superstrings realise ten-dimensional $\mathcal{N}=1$ supergravity (16 supercharges, one Majorana-Weyl parameter) coupled to super-Yang-Mills: type I and heterotic with gauge algebra $\mathfrak{so}(32)$, and heterotic with $\mathfrak{e}_8 \oplus \mathfrak{e}_8$. The massless fields split into a gravity multiplet and $\dim G$ vector multiplets.

Multiplet	Bosons ; fermions	dof
gravity	$g_{MN}, B_{MN}, \phi\,;\ \psi_M, \lambda$	$64+64$
vector ($\times \dim G$)	$A^a_M\,;\ \chi^a$	$(8+8)\times \dim G$

Here ψ_M and χ^a are Majorana-Weyl of one chirality and λ of the other, the reality and chirality that $D=10$ uniquely permits (Appendix B). Anomaly cancellation forces $\dim G = 496$ and the trace identity that singles out exactly $\mathrm{SO}(32)$ and $\mathrm{E}_8 \times \mathrm{E}_8$; the Green-Schwarz mechanism and the anomaly polynomial are in Appendix H, the trace normalisations in Appendix G.

The heterotic bosonic action bolts a Yang-Mills sector onto the NS-NS action under the common $e^{-2\phi}$,

$$\boxed{S_{\mathrm{het}} = \frac{1}{2\kappa_{10}^2}\int \mathrm{d}^{10}x\,\sqrt{-g}\,e^{-2\phi}\Big[\,R + 4(\partial\phi)^2 - \tfrac{1}{2}|H_3|^2 - \tfrac{\alpha'}{4}\,\mathrm{tr}|F_2|^2\,\Big]\,,} \tag{I.19}$$

with $F = \mathrm{d}A + A\wedge A$, $A = A^a T_a$, and tr the reference-normalised gauge trace of Appendix G. Type I, the orientifold of IIB, carries the *RR* two-form C_2 in place of B_2 and gets its gauge fields from the disk, so the Yang-Mills term enters at order $e^{-\phi}$ and the two-form kinetic term at e^0,

$$S_{\mathrm{I}} = \frac{1}{2\kappa_{10}^2}\int \mathrm{d}^{10}x\,\sqrt{-g}\Big[\,e^{-2\phi}(R + 4(\partial\phi)^2) - \tfrac{1}{2}|\tilde{F}_3|^2 - \tfrac{\alpha'}{4}\,e^{-\phi}\,\mathrm{tr}|F_2|^2\,\Big], \tag{I.20}$$

where $\tilde{F}_3 = \mathrm{d}C_2 - \frac{\alpha'}{4}(\omega_3^{\mathrm{YM}} - \omega_3^{\mathrm{L}})$ is the Chern-Simons-modified RR three-form, the type I counterpart of (I.21). In both theories the three-form is not $\mathrm{d}B_2$ (respectively $\mathrm{d}C_2$) alone but carries these Chern-Simons corrections,

$$\boxed{H_3 = \mathrm{d}B_2 - \tfrac{\alpha'}{4}(\omega_3^{\mathrm{YM}} - \omega_3^{\mathrm{L}})\,,} \qquad \mathrm{d}H_3 = \tfrac{\alpha'}{4}\big(\,\mathrm{tr}\,\mathcal{R}\wedge\mathcal{R} - \mathrm{tr}\,F\wedge F\big), \tag{I.21}$$

with $\omega_3^{\mathrm{YM}} = \mathrm{tr}(A\wedge\mathrm{d}A + \frac{2}{3}A^3)$, $\omega_3^{\mathrm{L}} = \mathrm{tr}(\Omega\wedge\mathrm{d}\Omega + \frac{2}{3}\Omega^3)$, Ω the spin connection one-form, and $\mathcal{R} = \mathrm{d}\Omega + \Omega\wedge\Omega$ the curvature two-form (distinct from the Ricci scalar R). The modified Bianchi (I.21) is the low-energy shadow of Green-Schwarz: its right-hand side is the four-form $X_4 = \mathrm{tr}\,\mathcal{R}^2 - \mathrm{tr}\,F^2$ whose product with X_8 is the factorised anomaly. Integrated over a closed four-cycle it forces $\int \mathrm{tr}\,F^2 = \int \mathrm{tr}\,\mathcal{R}^2$, the tadpole condition of Chapter 8. Passing (I.19) for $\mathrm{SO}(32)$ to Einstein frame and sending $\phi \to -\phi$ maps it onto (I.20): this is type I / heterotic-$\mathrm{SO}(32)$ S-duality (Chapter 7, Appendix I.3).

I.1.6 The actions at a glance

Theory	SUSY (real Q)	Bosonic fields	Action / gauge term
11D	$\mathcal{N}=1$ (32)	g_{MN}, C_3	(I.9); $C_3 \wedge F_4 \wedge F_4$
IIA	$\mathcal{N}=(1,1)$ (32)	g, B, ϕ; C_1, C_3	(I.11); $B_2 \wedge F_4 \wedge F_4$
IIB	$\mathcal{N}=(2,0)$ (32)	g, B, ϕ; C_0, C_2, C_4^+	(I.14); $C_4 \wedge H_3 \wedge F_3$
het	$\mathcal{N}=1$ (16)	g, B, ϕ; A (SO(32) or $\mathrm{E}_8 \times \mathrm{E}_8$)	(I.19); $e^{-2\phi}\,\mathrm{tr}\,F^2$
type I	$\mathcal{N}=1$ (16)	g, ϕ; C_2; A (SO(32))	(I.20); $e^{-\phi}\,\mathrm{tr}\,F^2$

The three type II and heterotic actions Chapter 5 quotes are (I.11), (I.14), (I.19) and (I.20); the eleven-dimensional (I.9) is the fifth. All share the universal NS-NS core $e^{-2\phi}(R+4(\partial\phi)^2-\frac{1}{2}|H_3|^2)$, differing only in what is bolted onto it.

I.1.7 Ten- and eleven-dimensional multiplet content

Consolidating the counts above, and matching bosons to fermions on shell as any supersymmetric theory requires:

Theory	bosons	fermions	total
11D	$g_{MN}(44) \oplus C_3(84)$	$\psi_M(128)$	$128+128$
IIA	NS-NS$(64) \oplus C_1, C_3(64)$	$2\psi_M(2\times 56) \oplus 2\lambda(2\times 8)$	$128+128$
IIB	NS-NS$(64) \oplus C_0, C_2, C_4^+(64)$	$2\psi_M(2\times 56) \oplus 2\lambda(2\times 8)$	$128+128$
10D $\mathcal{N}=1$ grav.	$g, B, \phi(64)$	$\psi_M(56) \oplus \lambda(8)$	$64+64$
10D $\mathcal{N}=1$ vector	$A_M^a(8)$	$\chi^a(8)$	$8+8$

On-shell degrees of freedom are counted in the transverse little group, Spin(9) in eleven dimensions and Spin(8) in ten, as already used in the field tables above. The two IIA gravitini have opposite chirality (non-chiral), the two IIB gravitini the same chirality (chiral); the type II theories are the two inequivalent completions of the NS-NS sector at 32 supercharges. The representation theory behind these tallies is in Appendix C.

I.1.8 Four-dimensional reductions

Reducing any of these actions on a compact internal manifold $\mathcal{M}$ gives a four-dimensional supergravity whose amount of supersymmetry is fixed by the holonomy of $\mathcal{M}$. The number of preserved supercharges is

$$Q_4 \;=\; f(\mathcal{M}) \times Q_{10/11}, \qquad \mathcal{N}_4 \;=\; \frac{Q_4}{4}, \tag{I.22}$$

with f the fraction of covariantly constant spinors $\mathcal{M}$ carries (a minimal 4D Majorana supercharge has four real components). The fractions follow from Berger's list (Appendix E): $f=1$ for a torus (trivial holonomy), $\frac{1}{2}$ for $K3$ ($SU(2)$), $\frac{1}{4}$ for a Calabi-Yau threefold ($SU(3)$), $\frac{1}{8}$ for a G_2 sevenfold, and $\frac{1}{16}$ for a Spin(7) eightfold. The last reduces an eleven-dimensional parent to $D=3$ rather than $D=4$ and is listed only to complete Berger's list. An orientifold or a heterotic bundle that breaks a further half is applied on top. The standard cases:

Parent	Internal $\mathcal{M}$	$\mathcal{N}_4$	4D theory
11D	T^7	8	maximal supergravity
11D	G_2 sevenfold	1	$\mathcal{N}=1$ SUGRA + matter
IIA/IIB	T^6	8	maximal supergravity
IIA/IIB	CY_3	2	$\mathcal{N}=2$ (vector + hyper multiplets)
IIA/IIB	CY_3 orientifold	1	$\mathcal{N}=1$ (chiral + vector)
het / type I	T^6	4	$\mathcal{N}=4$ super-Yang-Mills-gravity
het / type I	$K3\times T^2$	2	$\mathcal{N}=2$
het / type I	CY_3	1	$\mathcal{N}=1$ GUT-like

These entries are noted, not derived: the reductions, their moduli spaces, and the resulting matter content are the subject of Chapter 8. The four-dimensional $\mathcal{N}=1$ scalar potential

$$V = e^{K}\left(K^{i\bar{\jmath}} D_i W\, \overline{D_j W} - 3|W|^2\right), \qquad D_i W = \partial_i W + (\partial_i K)\, W, \tag{I.23}$$

with K the Kähler potential, W the holomorphic superpotential and D_iW the Kähler-covariant derivative, governs moduli stabilisation and supersymmetry breaking; it is assembled in Appendix C and used in Chapter 9. The four-dimensional Planck mass is set by the internal volume through $M_{\mathrm{pl}}^2 = \mathrm{Vol}(\mathcal{M})/\kappa_{10}^2$ (respectively $/\kappa_{11}^2$), the reduction of the Einstein-Hilbert term of whichever parent action one starts from.

Exercise I.1 (The two Newton constants)**.** Insert $\ell_p = g_s^{1/3}\ell_s$ into (I.3) and reduce on the eleventh circle of radius $R_{11} = g_s\ell_s$ using $2\kappa_{10}^2 = 2\kappa_{11}^2/(2\pi R_{11})$. Verify that the result is exactly (I.4), $2\kappa_{10}^2 = (2\pi)^7\ell_s^8 g_s^2$. Then check the dilaton dressing rule (I.8) by carrying the string-frame RR term $-\frac{1}{2}|F_n|^2$ through the Weyl rescaling (I.6), using $\sqrt{-g} = e^{5\phi/2}\sqrt{-\tilde{g}}$ and $|F_n|_g^2 = e^{-n\phi/2}|F_n|_{\tilde{g}}^2$ at $D=10$.

I.2 The p-brane solution catalogue

Every antisymmetric tensor gauge field of the supergravities collected in Appendix I.1 is sourced by an extended object. This section catalogues the classical solutions those sources produce: the extremal, BPS p-brane backgrounds of the F1, NS5, Dp, M2 and M5 branes, together with the orientifold planes. The physics that puts these objects in the spectrum, and the BPS bound $M \geq |Z|$ that makes them rigid, is developed in Section 5.6 and Appendix C; the microscopic D-brane construction and the tension computation are in Chapter 6; the D3 near-horizon limit that launches holography is worked in Section 13.1. Here we assemble the solutions in one convention, with their metrics, charges, tensions and near-horizon geometries, as a reference the later chapters can quote.

Every entry is a single-charge, static, extremal (BPS) solution, controlled by one harmonic function H. The reasons this is the generic shape are three, and each is an instance of a fact proved elsewhere. Extremality is BPS saturation, so the mass equals the charge and the solution preserves half the supersymmetries (Section 5.6.2). BPS saturation is the no-force condition (Section 5.6), so identical parallel branes coexist at arbitrary separation, which is why the field equations linearise onto a harmonic superposition. And the transverse Poisson equation is what a harmonic function solves. We state the solutions and check the ansatz; the proof that each solves the full nonlinear equations of the actions in Appendix I.1 is in the references cited with each brane.

I.2.1 The harmonic-function ansatz

A p-brane in D spacetime dimensions has a $(p+1)$-dimensional worldvolume and breaks the Lorentz group to

$$\mathrm{ISO}(1,p)\times\mathrm{SO}(D-p-1), \tag{I.24}$$

the Poincaré symmetry along the brane times rotations of the transverse space. Split the coordinates into worldvolume x^a $(a=0,\dots,p)$ and transverse y^i $(i=1,\dots,D-p-1)$, with radial coordinate $r^2=y^iy^i$ and $\mathrm{d}\Omega^2_{D-p-2}$ the round metric on the transverse unit sphere. Two integers organise every formula:

$$d\coloneqq p{+}1\quad\text{(worldvolume dimension)},\qquad \tilde d\coloneqq D{-}p{-}3=D{-}d{-}2\quad\text{(dual worldvolume dimension)}. \tag{I.25}$$

The transverse space is $(\tilde d+2)$-dimensional, so the surrounding sphere is $S^{\tilde d+1}$ and a point source falls off as $1/r^{\tilde d}$. We write $\Omega_n=\mathrm{vol}(S^n)=2\pi^{(n+1)/2}/\Gamma(\frac{n+1}{2})$ for the volume of the unit n-sphere.

A $(p+1)$-form potential A_{p+1} with field strength $F_{p+2}=\mathrm{d}A_{p+1}$ couples *electrically* to a p-brane through $\int_{\mathcal{W}}A_{p+1}$; its *magnetic* dual, defined by $\star F_{p+2}=F_{D-p-2}=\mathrm{d}\widetilde{A}_{D-p-3}$, is a $(D-p-4)$-brane. In $D=10$ this pairs a p-brane with a $(6-p)$-brane, and in $D=11$ the M2 with the M5.

Take the Einstein-frame form of the relevant action to be a graviton, a canonically normalised dilaton ϕ, and one n-form field strength with dilaton coupling a,

$$S=\frac{1}{2\kappa_D^2}\int \mathrm{d}^Dx\,\sqrt{-g_{\mathrm{E}}}\left(R_{\mathrm{E}}-\tfrac12(\nabla\phi)^2-\frac{1}{2\,n!}\,e^{a\phi}F_n^2\right),\qquad n=p+2. \tag{I.26}$$

Everything about the solution is fixed by one dimensionless combination,

$$\Delta\coloneqq a^2+\frac{2\,d\,\tilde d}{D-2}. \tag{I.27}$$

For every brane in this course $\Delta=4$ (Exercise I.2); we quote the solution at that value and give the general-Δ dressing afterwards. The single-centre extremal electric p-brane solving (I.26) is, for $\Delta=4$,

$$\boxed{\begin{aligned}&\mathrm{d}s^2_{\mathrm{E}}=H^{-\tilde d/(D-2)}\,\eta_{ab}\,\mathrm{d}x^a\mathrm{d}x^b\;+\;H^{\,d/(D-2)}\,\delta_{ij}\,\mathrm{d}y^i\mathrm{d}y^j,\\ &e^{\phi}=H^{a/2},\qquad A_{01\cdots p}=H^{-1}-1,\qquad \nabla^2_{\perp}H=0,\end{aligned}} \tag{I.28}$$

where $\nabla^2_\perp$ is the flat Laplacian in the transverse $\mathbb{R}^{\tilde d+2}$. The general-Δ solution replaces the metric exponents by $-4\tilde d/[\Delta(D-2)]$ and $4d/[\Delta(D-2)]$, keeps the dilaton $e^\phi=H^{a/2}$ unchanged, and replaces the electric potential by $A_{01\cdots p}=\frac{2}{\sqrt{\Delta}}(H^{-1}-1)$. The magnetic solution is obtained by $a\to -a$ (so the electric-formula dilaton becomes $e^\phi=H^{-a/2}$, with a the n-form coupling of (I.26)) and by replacing the electric potential with a magnetic field strength

$$F_{i_1\cdots i_{\tilde d+1}}=\pm\,\epsilon_{i_1\cdots i_{\tilde d+1}}{}^{j}\,\partial_j H\qquad\text{(transverse indices)}, \tag{I.29}$$

so the flux threads the transverse sphere. The uniform framework of (I.26) to (I.27) is due to Horowitz and Strominger (1991) and Duff and Lu; the review by Stelle (*BPS branes in supergravity*, 1998) and the string-soliton review of Duff, Khuri and Lu (1995) are the standard references.

The harmonic function and superposition. The only content of the field equations beyond the ansatz is that H is harmonic away from the source. For a single brane at the origin,

$$H(r)=1+\frac{r_h^{\tilde d}}{r^{\tilde d}}, \tag{I.30}$$

where the constant 1 is the boundary condition of asymptotic flatness ($H \to 1$, $\phi \to \phi_\infty$, metric $\to$ Minkowski) and r_h is the charge radius fixed below. Because the equations are linear in H, several parallel branes superpose:

$$H(y) = 1 + \sum_A \frac{r_h^{\tilde d}}{|y - y_A|^{\tilde d}}, \tag{I.31}$$

with the centres y_A at arbitrary transverse positions. This is the no-force condition made geometric: gravitational and dilatonic attraction cancels tensor-gauge repulsion exactly, so (I.31) is a static solution for any configuration of like-charged branes.

Reading off the charge radius. The coefficient in (I.30) is set by matching the source strength to N branes of physical tension T (measured in the frame of (I.26), where charge equals tension). Integrating $\nabla_\perp^2 H = -(\text{source})$ over the transverse space gives one master relation for all the branes below:

$$\boxed{\; r_h^{\tilde d} \;=\; \frac{2\kappa_D^2\, T\, N}{\tilde d\, \Omega_{\tilde d+1}} \;} \tag{I.32}$$

This is the supergravity avatar of the tension: the depth of the well a brane digs is its ADM tension divided by the transverse solid angle. Every exact coefficient in Table I.1 is (I.32) evaluated on the tension of Section 5.6, using $2\kappa_{10}^2 = (2\pi)^7 \alpha'^4 g_s^2$ and $2\kappa_{11}^2 = (2\pi)^8 \ell_p^9$ from Appendix I.1.

Preserved supersymmetry. Each solution is half-BPS: the Killing-spinor equations of the relevant supergravity (Appendix C) admit sixteen solutions, the constant spinors surviving a single projector. For a brane along $x^0 \cdots x^p$ the projector reads, schematically,

$$\big(\mathbf{1} \mp \Gamma^{01\cdots p}\,(\cdots)\big)\,\varepsilon = 0, \tag{I.33}$$

where the omitted factor is $\mathbf{1}$ for the M-branes, and carries a chirality operator Γ_* or a factor of i for the Dp and NS5 (the precise dressing, spelled out in Appendix C, encodes which supergravity and which frame). The Killing spinor takes the transverse profile $\varepsilon = e^{A/2}\varepsilon_0 = H^{-\tilde d/[4(D-2)]}\varepsilon_0$, the square root of the worldvolume warp factor $e^{2A} = H^{-\tilde d/(D-2)}$, with ε_0 constant. This gives $H^{-1/8}$ for the D3, $H^{-1/6}$ for the M2 and $H^{-1/12}$ for the M5. Two orthogonal branes impose two commuting projectors and preserve $\frac{1}{4}$ of the supersymmetry, and so on down; this counting drives the intersecting-brane bound states of Chapter 12.

I.2.2 The NS-NS branes: F1 and NS5

The Kalb-Ramond field B_2 has an electric source, the fundamental string F1, and a magnetic source, the solitonic five-brane NS5. Both are written most naturally in the string frame, related to the Einstein frame of (I.28) by $g^{\rm str}_{\mu\nu} = e^{\phi/2} g^{\rm E}_{\mu\nu}$.

F1 (electric B_2). The fundamental string solution of Dabholkar, Gibbons, Harvey and Ruiz Ruback (1990) has, in string frame,

$$\mathrm{d}s^2 = H_1^{-1}\big(-\mathrm{d}t^2 + (\mathrm{d}x^1)^2\big) + \sum_{i=1}^{8} (\mathrm{d}y^i)^2, \qquad e^{2\phi} = g_s^2\, H_1^{-1}, \qquad B_{01} = H_1^{-1} - 1, \tag{I.34}$$

with $H_1 = 1 + r_1^6/r^6$ harmonic in the eight transverse dimensions ($d = 2$, $\tilde d = 6$). The string-frame transverse space is flat: the warp factor sits entirely on the worldsheet directions. The dilaton falls to zero at the core, so the string is genuinely weakly coupled where it sits, consistent with

F1 being the elementary quantum of the theory rather than a soliton, and with the source being a delta function from the worldsheet action. Its charge is the number of strings, measured electrically by the Page integral over the transverse sphere,

$$N_1 = \frac{1}{(2\pi)^6\, g_s^2\, \ell_s^6} \oint_{S^7} e^{-2\phi} \star H_3, \qquad r_1^6 = 32\pi^2\, g_s^2\, N_1\, \ell_s^6, \tag{I.35}$$

the electric mate of the NS5 flux integral below. Its tension is the one every other tension is measured against,

$$T_{\mathrm{F1}} = \frac{1}{2\pi\alpha'} = \frac{1}{2\pi\ell_s^2} \;\sim\; g_s^0. \tag{I.36}$$

NS5 (magnetic B_2). The dual object is the Callan-Harvey-Strominger (1991) five-brane,

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \sum_{a=1}^{5} (\mathrm{d}x^a)^2 + H_5 \sum_{i=1}^{4} (\mathrm{d}y^i)^2, \qquad e^{2\phi} = g_s^2\, H_5, \qquad H_{ijk} = -\,\epsilon_{ijkl}\, \partial_l H_5, \tag{I.37}$$

with $H_5 = 1 + r_5^2/r^2$ harmonic in the *four* transverse dimensions ($d = 6$, $\tilde{d} = 2$). Now the worldvolume is flat and the transverse space is warped, the mirror image of the F1. The dilaton *grows* towards the core (an honest soliton, strongly coupled at $r \to 0$), and the H-flux through the surrounding three-sphere counts the fivebranes,

$$\frac{1}{(2\pi)^2\alpha'} \oint_{S^3} H_3 = N_5, \qquad r_5^2 = N_5\, \ell_s^2, \tag{I.38}$$

so the transverse throat radius is $\sqrt{N_5}\,\ell_s$, independent of g_s. The tension carries the two inverse powers of the coupling that mark a fundamental-string soliton,

$$T_{\mathrm{NS5}} = \frac{1}{(2\pi)^5\, g_s^2\, \ell_s^6} \;\sim\; g_s^{-2}. \tag{I.39}$$

I.2.3 The RR branes: Dp

Each RR potential C_{p+1} is sourced electrically by a Dp-brane, with p even in IIA and odd in IIB. The string-frame solution, sourced by N coincident branes, is

$$\boxed{\begin{aligned} \mathrm{d}s^2 &= H_p^{-1/2}\, \eta_{ab}\, \mathrm{d}x^a \mathrm{d}x^b + H_p^{1/2} (\mathrm{d}r^2 + r^2\, \mathrm{d}\Omega_{8-p}^2), \\ e^{\phi} &= g_s\, H_p^{(3-p)/4}, \qquad C_{01\cdots p} = H_p^{-1} - 1, \end{aligned}} \tag{I.40}$$

with $H_p = 1 + r_p^{7-p}/r^{\,7-p}$ harmonic in the $9-p$ transverse dimensions ($d = p+1$, $\tilde{d} = 7-p$). The dilaton coupling $a = (3-p)/2$ is what makes e^{ϕ} run for $p \neq 3$ and stay constant for $p = 3$. Evaluating (I.32) with the physical Dp tension of Eq. (6.35) gives the standard charge radius

$$r_p^{7-p} = d_p\, g_s\, N\, \ell_s^{7-p}, \qquad d_p = (2\sqrt{\pi})^{\,5-p}\, \Gamma\!\left(\tfrac{7-p}{2}\right), \tag{I.41}$$

and the tension and RR charge saturate the BPS bound, $\tau_p = \mu_p$,

$$T_{\mathrm{D}p} = \frac{1}{(2\pi)^p\, g_s\, \ell_s^{p+1}} \;\sim\; g_s^{-1}. \tag{I.42}$$

The single power $1/g_s$ places the D-branes between the F1 (g_s^0) and the NS5 (g_s^{-2}), and is the defining signature of a D-brane. The distinguished case is $p = 3$: the dilaton is constant and, writing $L^4 \coloneqq 4\pi g_s N \ell_s^4$, the charge radius is $r_3^4 = L^4$, so (I.40) is the extremal three-brane whose near-horizon throat is the $\mathrm{AdS}_5 \times S^5$ of Section 13.1. The D6 is special in the opposite direction: it is the magnetic dual of the D0, and it lifts to pure geometry, the Kaluza-Klein monopole (Taub-NUT) of the M-theory circle (Section 5.3).

I.2.4 The M-branes: M2 and M5

Eleven-dimensional supergravity has no dilaton, so its branes are the cleanest of all: pure geometry plus C_3-flux, with tensions set by ℓ_p alone. The electric M2 (Duff and Stelle, 1991) and the magnetic M5 (Güven, 1992) are

$$\text{M2:}\quad \mathrm{d}s^2 = H_2^{-2/3}\,\eta_{ab}\,\mathrm{d}x^a\mathrm{d}x^b + H_2^{1/3}\,\delta_{ij}\,\mathrm{d}y^i\mathrm{d}y^j,\quad C_{012} = H_2^{-1} - 1,\qquad H_2 = 1 + \frac{r_2^6}{r^6},\tag{I.43}$$

$$\text{M5:}\quad \mathrm{d}s^2 = H_5^{-1/3}\,\eta_{ab}\,\mathrm{d}x^a\mathrm{d}x^b + H_5^{2/3}\,\delta_{ij}\,\mathrm{d}y^i\mathrm{d}y^j,\quad F_{i_1\cdots i_4} = \epsilon_{i_1\cdots i_4}{}^{j}\partial_j H_5,\quad H_5 = 1 + \frac{r_5^3}{r^3},\tag{I.44}$$

with the M2 harmonic in 8 transverse dimensions ($d=3$, $\tilde d = 6$) and the M5 in 5 ($d=6$, $\tilde d = 3$). Their charge radii follow from (I.32) with the tensions of Eq. (5.33),

$$r_2^6 = 32\pi^2\,N\,\ell_p^6,\qquad r_5^3 = \pi\,N\,\ell_p^3,\qquad T_{\mathrm{M2}} = \frac{1}{(2\pi)^2\ell_p^3},\qquad T_{\mathrm{M5}} = \frac{1}{(2\pi)^5\ell_p^6}.\tag{I.45}$$

The two are Dirac dual: $2\kappa_{11}^2\,T_{\mathrm{M2}}\,T_{\mathrm{M5}} = 2\pi$, the minimal quantum (Exercise 5.8). Reduction on the M-theory circle turns (I.43) and (I.44) into the ten-dimensional branes: a wrapped M2 is the F1 and an unwrapped M2 is the D2; a wrapped M5 is the D4 and an unwrapped M5 is the NS5 (Section 5.3).

I.2.5 Tension and charge tables

Table I.1 gathers the catalogue in one place. Every tension is exact in the conventions of Appendix A, and every charge radius is (I.32) evaluated on that tension.

Brane	D	$d = p{+}1$	$\tilde d$	Tension T	Charge radius $r_h^{\tilde d}$
F1	10	2	6	$\dfrac{1}{2\pi\ell_s^2} \sim g_s^0$	$32\pi^2\,g_s^2 N\,\ell_s^6$
Dp	10	$p{+}1$	$7{-}p$	$\dfrac{1}{(2\pi)^p g_s\,\ell_s^{p+1}} \sim g_s^{-1}$	$d_p\,g_s N\,\ell_s^{7-p}$
NS5	10	6	2	$\dfrac{1}{(2\pi)^5 g_s^2\ell_s^6} \sim g_s^{-2}$	$N\,\ell_s^2$
M2	11	3	6	$\dfrac{1}{(2\pi)^2\ell_p^3}$	$32\pi^2\,N\,\ell_p^6$
M5	11	6	3	$\dfrac{1}{(2\pi)^5\ell_p^6}$	$\pi\,N\,\ell_p^3$

Table I.1: The extremal single-charge p-brane catalogue. Here $d_p = (2\sqrt{\pi})^{5-p}\Gamma(\frac{7-p}{2})$, so $d_3 = 4\pi$ and $r_3^4 = L^4 = 4\pi g_s N\ell_s^4$. All entries satisfy $\Delta = 4$; the tensions are those of Table 5.2 with the exact coefficients restored.

Remark I.3 (Beyond the lectures: black (non-extremal) branes). The extremal solutions are the zero-temperature limit of a two-parameter black-brane family. Introducing a blackening factor $f(r) = 1 - (r_0/r)^{\tilde d}$, the non-extremal D$p$ metric reads

$$\mathrm{d}s^2 = H^{-1/2}\big(-f\,\mathrm{d}t^2 + \mathrm{d}x_a^2\big) + H^{1/2}\big(f^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega_{8-p}^2\big),\tag{I.46}$$

with $H = 1 + r_h^{\tilde d}/r^{\tilde d}$. The charge radius r_h and the horizon radius r_0 are independent parameters of the black brane, related by $r_h^{\tilde d} = r_0^{\tilde d}\sinh^2\!\alpha$. The horizon at $r = r_0$ carries a Hawking temperature and a Bekenstein-Hawking entropy $S = A/4G$; extremality is $r_0 \to 0$ at fixed charge, which sends $T \to 0$ and recovers (I.28). Near-extremal D-branes are the gravitational side of finite-temperature holography and of the black-hole entropy counting of Chapter 12, where the microscopic degeneracy of the brane bound state reproduces $A/4G$.

Remark I.4 (Beyond the lectures: intersecting branes and the harmonic superposition rule). Bound states of different branes are built by a harmonic-function rule (Tseytlin 1996): each constituent contributes its own harmonic function, and the metric is assembled by giving each coordinate the product of the warp factors of the branes that are, or are not, extended along it. For the D1-D5-P system on $T^4 \times S^1$ used in Chapter 12, three harmonic functions H_1, H_5, H_p combine so that the near-horizon geometry is $\mathrm{AdS}_3 \times S^3 \times T^4$, the throat whose central charge counts the microstates. The rule works precisely for the supersymmetric intersections, those that preserve a common set of Killing spinors, one projector (I.33) per constituent, so the bound state is $\frac{1}{4}$- or $\frac{1}{8}$-BPS.

I.2.6 Near-horizon limits

Dropping the 1 in H isolates the throat, $r \ll r_h$, where the brane's own field dominates and the asymptotically flat region has been discarded. Whether the throat is a clean anti-de Sitter geometry depends entirely on the dilaton: the three cases with constant or absent dilaton (D3, M2, M5) give an exact $\mathrm{AdS}_{p+2} \times S^{\tilde{d}+1}$ with two independent radii, and these are exactly the branes whose worldvolume theories are conformal. Writing $H \to r_h^{\tilde{d}}/r^{\tilde{d}}$ in (I.40), (I.43) and (I.44):

$$\begin{aligned}
&\text{D3:} \quad \mathrm{AdS}_5 \times S^5, && L_{\mathrm{AdS}} = L_{S^5} = L, && L^4 = 4\pi g_s N\, \ell_s^4, \\
&\text{M2:} \quad \mathrm{AdS}_4 \times S^7, && L_{S^7} = R, \quad L_{\mathrm{AdS}} = \tfrac{1}{2}R, && R^6 = 32\pi^2 N\, \ell_p^6, \\
&\text{M5:} \quad \mathrm{AdS}_7 \times S^4, && L_{S^4} = R, \quad L_{\mathrm{AdS}} = 2R, && R^3 = \pi N\, \ell_p^3.
\end{aligned} \tag{I.47}$$

The sphere radius equals the transverse charge radius r_h; the ratio $L_{\mathrm{AdS}}/L_{\mathrm{sph}}$ is 1, $\frac{1}{2}$ and 2 for D3, M2 and M5 respectively. These three throats are maximally supersymmetric (the near-horizon limit doubles the sixteen preserved supercharges back to thirty-two, the extra sixteen being superconformal), and their isometries are the conformal group and R-symmetry of the dual $\mathcal{N} = 4$ super Yang-Mills, the $(2,0)$ theory, and the $(1,0)$ theory of Chapter 13. The relation $L^4 = 4\pi g_s N \ell_s^4$ for D3 is the master relation of holography (Maldacena 1998); the M-brane throats were identified in the same period.

Example I.5 (The D3 throat as $\mathrm{AdS}_5 \times S^5$)**.** Set $p = 3$ in (I.40). The dilaton coupling $a = 0$ makes $e^{\phi} = g_s$ constant, so string and Einstein frames coincide up to the constant g_s, and no running spoils the throat. With $H_3 = 1 + L^4/r^4$ and $L^4 \coloneqq 4\pi g_s N \ell_s^4$ from (I.41), drop the 1 in the throat $r \ll L$, so $H_3 \to L^4/r^4$ and $H_3^{-1/2} = r^2/L^2$, $H_3^{1/2} = L^2/r^2$. The metric becomes

$$\mathrm{d}s^2 = \frac{r^2}{L^2}\,\eta_{ab}\,\mathrm{d}x^a \mathrm{d}x^b + \frac{L^2}{r^2}\,\mathrm{d}r^2 + L^2\,\mathrm{d}\Omega_5^2. \tag{I.48}$$

The first two terms are AdS_5 in Poincaré coordinates of radius L (put $z = L^2/r$ to reach the standard form $L^2 z^{-2}(\eta_{ab}\mathrm{d}x^a\mathrm{d}x^b + \mathrm{d}z^2)$); the last is a round S^5 of the same radius L. Both radii equal the charge radius r_3, so $L_{\mathrm{AdS}} = L_{S^5} = L$ with $L^4 = 4\pi g_s N \ell_s^4$, the first line of (I.47).

The other branes do not give exact anti-de Sitter geometries, and the obstruction is always the running dilaton.

- **Dp, $p \neq 3$.** The near-horizon metric is only *conformal* to $\mathrm{AdS}_{p+2} \times S^{8-p}$, with $e^{\phi} \propto r^{(p-3)(7-p)/4}$ running across the throat. The dual $(p+1)$-dimensional super Yang-Mills has a dimensionful, running coupling and is not conformal. Passing to the dual frame $g_{\mathrm{dual}} = e^{c\phi} g$ (Boonstra, Skenderis and Townsend 1998) restores an exact AdS_{p+2} factor and makes the domain-wall/QFT correspondence precise. See Exercise 13.5.
- **NS5.** The transverse throat is $\mathbb{R}_{\phi} \times S^3$: a three-sphere of fixed radius $\sqrt{N_5}\,\ell_s$ times a real line along which the dilaton is *linear*, $\phi \sim -r_{\perp}/(\sqrt{N_5}\,\ell_s)$ in the proper transverse coordinate. This is the Callan-Harvey-Strominger throat. There is no AdS factor and no

mass gap from the geometry; the string coupling grows without bound down the throat. The decoupled worldvolume theory is not a local field theory but the six-dimensional *little string theory*, a nonlocal theory with Hagedorn growth, one of the genuinely exotic corners of the subject.

- **F1.** The string-frame throat (I.34) degenerates ($e^{2\phi} \to 0$, no independent AdS factor): the fundamental string is a singular source, not a smooth soliton. A regular $\mathrm{AdS}_3 \times S^3$ appears only in the bound state with the NS5 (or, S-dually, the D1-D5 system), where the two harmonic functions of Theorem I.4 together support a throat.

An exact $\mathrm{AdS} \times S$ near-horizon geometry with a genuine conformal dual is the exception, tied to the constant-dilaton branes D3, M2 and M5, not the rule.

I.2.7 Orientifold planes

An orientifold p-plane Op is the fixed locus of a combined worldsheet-parity and spacetime-reflection quotient (the T-dual of the worldsheet orientation projection Ω, constructed in Chapter 6). Unlike a D-brane it is not dynamical, it carries no worldvolume gauge field, and it can have *negative* tension and RR charge. For the standard Op^- plane both are fixed multiples of the Dp values,

$$T_{\mathrm{O}p^-} = -\,2^{p-5}\,T_{\mathrm{D}p}, \qquad \mu_{\mathrm{O}p^-} = -\,2^{p-5}\,\mu_{\mathrm{D}p}, \tag{I.49}$$

so an O3$^-$ carries $-\frac{1}{4}$ of a unit of D3 charge, an O9$^-$ carries -16 (the -32 half-planes on the double cover whose charge the SO(32) gauge stack cancels). Because the transverse space of an Op is the orbifold $\mathbb{R}^{9-p}/\mathbb{Z}_2$, the surrounding sphere is halved and the plane sources the harmonic profile (I.40) with charge radius -2^{p-5} times that of a single Dp: the negative coefficient is why the O-plane repels flux rather than attracting it. There are three further variants (Op^+, $\widetilde{\mathrm{O}p}^-$, $\widetilde{\mathrm{O}p}^+$), distinguished by discrete flux and yielding SO versus Sp gauge groups on coincident D-branes; the classification and its role in tadpole cancellation are in Chapter 6 and are used in Chapters 9 and 11. By Gauss's law a negative-tension, negative-charge plane is the sink that lets RR flux cancel on a compact space, so consistent compactifications with D-branes require orientifolds to soak up the net charge.

Exercise I.2 (The universal $\Delta = 4$ and the metric exponents)**.** *(Short.)* For each brane in Table I.1, read off $(a, d, \tilde{d}, D)$. Here a is the coupling to feed straight into the electric-formula dilaton $e^{\phi} = H^{a/2}$: it is $a = -1$ for the electric F1, $a = +1$ for the magnetic NS5 (the sign-flipped $a \to -a$ of the master magnetic rule applied to the H_3 coupling $a_{H_3} = -1$, so that $e^{\phi} = H^{+1/2}$ as in (I.37)), $a = (3-p)/2$ for the Dp, and $a = 0$ for the M-branes. Verify from (I.27) that $\Delta = a^2 + 2d\tilde{d}/(D-2) = 4$ in every case (the sign of a does not matter, since only a^2 enters). Then confirm that the metric exponents $-\tilde{d}/(D-2)$ and $d/(D-2)$ of (I.28), after the Weyl rescaling $g^{\mathrm{str}} = e^{\phi/2} g^{\mathrm{E}}$ with $e^{\phi} = H^{a/2}$, reproduce the string-frame powers quoted in (I.34), (I.37) and (I.40): worldvolume H^{-1} and transverse H^0 for the F1, worldvolume H^0 and transverse H^1 for the NS5, and $H^{\mp 1/2}$ for the Dp. *(Moral: one number, $\Delta = 4$, is the statement that all of these branes are BPS.)*

I.3 The duality dictionary

This section is the Rosetta stone for the web of Chapter 7. It collects, in one place and one convention, the maps that relate the five ten-dimensional strings and eleven-dimensional M-theory: T-duality on circles, the S-dualities that invert the coupling, and the U-duality groups that fuse them under toroidal reduction. It then tabulates how the branes and their charge lattices transport across those maps, and closes with a compact AdS/CFT dictionary card. Nothing here is derived from scratch; the physics lives in Chapters 6, 7 and 13, the

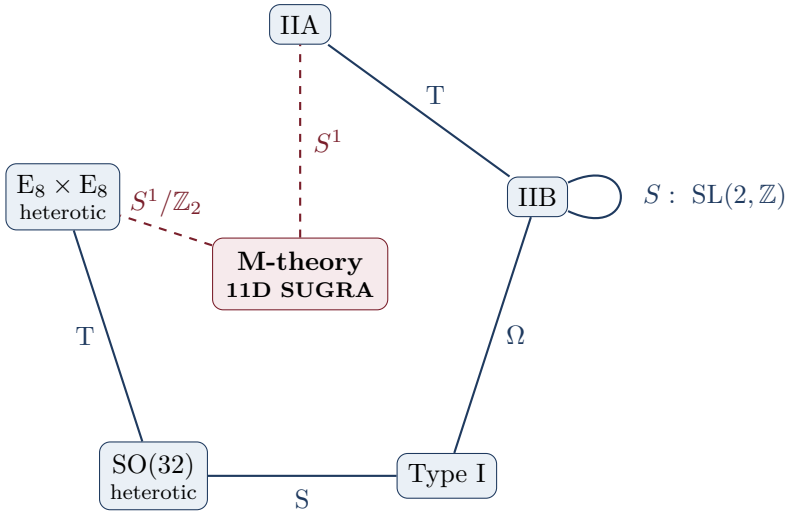


Figure I.1: The web of dualities. Solid links are exact equivalences: T-duality on a circle, the S-dualities that invert g_s, and the orientifold projection Ω (worldsheet parity) that defines Type I as IIB/Ω. Dashed spokes decompactify to eleven dimensions. This is the reference copy of Fig. 7.1.

effective actions and multiplet content in Appendix I.1, the brane solutions and tensions in Appendix I.2, the supersymmetry algebra and BPS bound in Appendix C, and the lattices in Appendix G. Conventions are those of Appendix A: $\ell_s = \sqrt{\alpha'}$, $g_s = e^{\langle\phi\rangle}$, $T_{\rm F1} = 1/(2\pi\alpha')$, and $2\kappa_{10}^2 = (2\pi)^7 \alpha'^4 g_s^2$.

I.3.1 The web at a glance

The five weakly coupled superstrings are cusps of one moduli space whose interior is M-theory. Figure I.1 draws the links; Table I.2 lists the corners with the data a reader most often wants at hand.

Theory	10D SUSY	Gauge group	Distinctive sector	Strong coupling
IIA	$\mathcal{N} = (1,1)$	U(1) (from C_1)	non-chiral; RR C_1, C_3	M-theory on S^1
IIB	$\mathcal{N} = (2,0)$	none	chiral; self-dual F_5	self-dual, SL(2, $\mathbb{Z}$)
Type I	$\mathcal{N} = (1,0)$	SO(32)	unoriented; C_2 only	SO(32) heterotic
het SO(32)	$\mathcal{N} = (1,0)$	SO(32)	chiral; B_2 only	Type I
het $E_8 \times E_8$	$\mathcal{N} = (1,0)$	$E_8 \times E_8$	chiral; B_2 only	M-theory on $S^1/\mathbb{Z}_2$

Table I.2: the five ten-dimensional string theories. All have 32 supercharges in the type II case and 16 in the type I/heterotic case. Field content and actions are in Appendix I.1.

I.3.2 T-duality

T-duality is the perturbative link: it is exact order by order in g_s, holds exactly in α', and never leaves the weakly coupled regime. On a single circle of radius R it acts as

$$R \longmapsto \tilde{R} = \frac{\alpha'}{R} = \frac{\ell_s^2}{R}, \qquad n \leftrightarrow w, \qquad g_s \longmapsto \tilde{g}_s = \frac{\ell_s}{R}\, g_s = \frac{\tilde{R}}{\ell_s}\, g_s, \tag{I.50}$$

exchanging Kaluza-Klein momentum n with winding w. The coupling rule keeps the lower-dimensional Newton constant fixed and is the zero-mode piece of the Buscher dilaton shift. In a general background with a compact shift isometry θ ($\theta \sim \theta + 2\pi$, no field depending on θ), the map is the set of Buscher rules (derived in Section 6.1.4), with metric and B-field components

in units of α':

$$\tilde{G}_{\theta\theta} = \frac{1}{G_{\theta\theta}}, \quad \tilde{G}_{\theta i} = \frac{B_{\theta i}}{G_{\theta\theta}}, \quad \tilde{B}_{\theta i} = \frac{G_{\theta i}}{G_{\theta\theta}}, \quad \tilde{\phi} = \phi - \tfrac{1}{2}\ln G_{\theta\theta}, \tag{I.51}$$

together with $\tilde{G}_{ij} = G_{ij} - (G_{\theta i}G_{\theta j} - B_{\theta i}B_{\theta j})/G_{\theta\theta}$ and $\tilde{B}_{ij} = B_{ij} - (G_{\theta i}B_{\theta j} - B_{\theta i}G_{\theta j})/G_{\theta\theta}$. The mixed rule exchanges the graviphoton $G_{\theta i}$ (momentum charge) with the winding photon $B_{\theta i}$, the general-background face of $n \leftrightarrow w$.

Applied to the Ramond-Ramond sector, a T-duality along θ adds or removes a leg $\mathrm{d}x^\theta$ on every potential, so it shifts each C_p by one form degree and shuttles between the even (IIA) and odd (IIB) RR ladders. This is why T-duality on a circle relates the two type II theories, and the two heterotic theories, as in Table I.3.

On a circle	becomes	Notes
IIA at radius R	IIB at radius α'/R	even $\leftrightarrow$ odd RR forms
het SO(32) at R	het $\mathrm{E}_8\times\mathrm{E}_8$ at α'/R	with a Wilson line breaking each factor
Type I at R	Type I′ (IIA/$\Omega\mathcal{I}$) at α'/R	orientifold with two O8-planes

Table I.3: T-duality across the circle. The perturbative duality group on T^d is the lattice automorphism group $O(d,d;\mathbb{Z})$ for type II and $O(d,d+16;\mathbb{Z})$ for heterotic, acting on the Narain lattice $\Gamma_{d,d}$ (resp. $\Gamma_{d,d+16}$) of Appendix G.

How the extended objects transport under a single T-duality along a direction x is fixed by whether their worldvolume wraps x. Table I.4 is the working table.

Object	x along worldvolume	x transverse
Dp	D$(p-1)$	D$(p+1)$
F1	momentum (pp-wave) along x	F1 (unchanged)
momentum along x	F1 wound on x	momentum (unchanged)
NS5	NS5	KK monopole (NUT circle x)
KK monopole (NUT circle x)	KK monopole	NS5

Table I.4: a single T-duality T_x acting on branes. A Dp loses (gains) a worldvolume leg and changes dimension by one, matching the RR-degree shift. The NS5/KK-monopole exchange is the NS-sector analogue; the KK monopole is the object whose transverse Taub-NUT circle is x. The F1 tension $1/(2\pi\alpha')$ is T-invariant.

Exercise I.3 (Buscher rules and $n \leftrightarrow w$)**.** For a circle background $G_{\theta\theta} = R^2/\alpha'$ with constant dilaton and $B = 0$, apply the Buscher rules Eq. (I.51) to obtain $\tilde{G}_{\theta\theta} = \alpha'/R^2$ and $\tilde{\phi} = \phi - \ln(R/\ell_s)$, and hence recover the circle map Eq. (I.50), including $\tilde{g}_s = (\ell_s/R)\, g_s$. Now switch on a constant graviphoton $G_{\theta i} \neq 0$ and show that the mixed rule sends it to a pure winding photon $\tilde{B}_{\theta i} = G_{\theta i}/G_{\theta\theta}$ (with $\tilde{G}_{\theta i} = 0$), the general-background face of the momentum-winding exchange $n \leftrightarrow w$.

I.3.3 S-duality and the ten-dimensional strong-coupling limits

S-dualities invert g_s and so trade the perturbative expansion for one with no perturbative meaning. Their evidence is BPS matching (Section 7.4). There are three inequivalent strong-coupling statements in ten (and eleven) dimensions, collected in Table I.5.

Theory	Strong coupling ($g_s \to \infty$)	Dictionary
IIB	IIB itself, $S : g_s \to 1/g_s$	$\mathrm{SL}(2,\mathbb{Z})$ on $\tau = C_0 + i/g_s$
IIA	M-theory on $S^1_{R_{11}}$	$R_{11} = g_s \ell_s$, $\ell_p = g_s^{1/3} \ell_s$
het $\mathrm{E}_8 \times \mathrm{E}_8$	M-theory on $S^1/\mathbb{Z}_2$	E_8 on each wall (Hořava-Witten)
het SO(32)	Type I	$g_s^{\mathrm{het}} = 1/g_s^{\mathrm{I}}$
Type I	het SO(32)	$\alpha'_{\mathrm{I}} = \alpha'_{\mathrm{het}}\, g_s^{\mathrm{het}}$

Table I.5: the strong-coupling faces of the five strings, and the two decompactification spokes to M-theory. Frame relations for the type I/heterotic pair: $\Phi_{\mathrm{I}} = -\Phi_{\mathrm{het}}$ and $G^{\mathrm{I}}_{\mu\nu} = e^{-\Phi_{\mathrm{het}}} G^{\mathrm{het}}_{\mu\nu}$ in string frame.

The IIB card. The RR scalar and dilaton form $\tau = C_0 + ie^{-\phi} = C_0 + i/g_s$ in the upper half-plane $\mathbb{H}$, and B_2, C_2 form a doublet $\mathcal{B} = (C_2, B_2)^{\mathsf{T}}$. The classical $\mathrm{SL}(2,\mathbb{R})$ of the Einstein-frame action is broken by charge quantisation to

$$\mathrm{SL}(2,\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : \; ad - bc = 1, \; a,b,c,d \in \mathbb{Z} \right\}, \quad \tau \mapsto \frac{a\tau + b}{c\tau + d}, \quad \mathcal{B} \mapsto \Lambda \mathcal{B}, \tag{I.52}$$

generated by $T : \tau \mapsto \tau + 1$ (the perturbative axion shift $C_0 \to C_0 + 1$) and $S : \tau \mapsto -1/\tau$ (the strong-weak map $g_s \to 1/g_s$ at $C_0 = 0$), with $S^2 = (ST)^3 = -\mathbf{1}$. A string carries charge $(p,q) =$ (F1 number, D1 number), transforming as

$$S : \begin{pmatrix} p \\ q \end{pmatrix} \mapsto \begin{pmatrix} -q \\ p \end{pmatrix}, \qquad T : \begin{pmatrix} p \\ q \end{pmatrix} \mapsto \begin{pmatrix} p+q \\ q \end{pmatrix}, \tag{I.53}$$

and the $\frac{1}{2}$-BPS (p,q)-string exists as a stable bound state exactly when $\gcd(p,q) = 1$. Its string-frame tension is

$$T^{\mathrm{str}}_{(p,q)} = \frac{1}{2\pi\alpha'} \, |p - q\tau| = \frac{1}{2\pi\alpha'} \sqrt{(p - qC_0)^2 + \frac{q^2}{g_s^2}} \qquad \text{(string frame)}, \tag{I.54}$$

while the Einstein-frame tension is the coset norm of the charge doublet, differing from Eq. (I.54) by the universal string conversion factor $g_s^{1/2}$,

$$T^{\mathrm{E}}_{(p,q)} = g_s^{1/2}\, T^{\mathrm{str}}_{(p,q)} = \frac{1}{2\pi\alpha'} \, \frac{|p - q\tau|}{\sqrt{\mathrm{Im}\,\tau}} = \frac{1}{2\pi\alpha'} \sqrt{(p \;\, q)\, \mathcal{M}(\tau)\, (p \;\, q)^{\mathsf{T}}} \qquad \text{(Einstein frame)}, \tag{I.55}$$

with the coset metric $\mathcal{M}(\tau) = (\mathrm{Im}\,\tau)^{-1} \left(\begin{smallmatrix} 1 & -\mathrm{Re}\,\tau \\ -\mathrm{Re}\,\tau & |\tau|^2 \end{smallmatrix} \right)$ (recall $\mathrm{Im}\,\tau = 1/g_s$, so the frame factor is $1/\sqrt{\mathrm{Im}\,\tau} = g_s^{1/2}$). The whole IIB brane spectrum organises under $\mathrm{SL}(2,\mathbb{Z})$ as in Table I.6.

Multiplet	Members	$\mathrm{SL}(2,\mathbb{Z})$ structure
strings	F1 $(1,0)$, D1 $(0,1)$	doublet, coprime (p,q) orbit
five-branes	NS5, D5	doublet, coprime (p,q) orbit
three-brane	D3	singlet (self-dual, fixed by S)
seven-branes	D7 and images	triplet; monodromies in $\mathrm{SL}(2,\mathbb{Z})$
pp-wave, KK monopole	momentum, KK5	singlets (geometric)

Table I.6: $\mathrm{SL}(2,\mathbb{Z})$ action on IIB branes. S exchanges F1 $\leftrightarrow$ D1 and NS5 $\leftrightarrow$ D5; the self-dual D3 makes the bulk duality descend to Montonen-Olive duality of $\mathcal{N} = 4$ SYM (Chapter 7); the seven-brane monodromies are the starting point of F-theory (Chapter 11).

Example I.6 (Reading a tension off the doublet)**.** At $C_0 = 0$, the string-frame formula Eq. (I.54) gives the F1 tension $T^{\mathrm{str}}_{(1,0)} = 1/2\pi\alpha'$ and the D1 tension $T^{\mathrm{str}}_{(0,1)} = 1/2\pi\alpha' g_s$, exchanged by $g_s \to 1/g_s$. The coprime state $(1,1)$ at $g_s = 1$ has $T^{\mathrm{str}} = \sqrt{2}/2\pi\alpha'$, strictly below the threshold

$2/2\pi\alpha'$ of a free F1 and D1, so it is a genuine bound state; the non-coprime $(2,0)$ sits exactly at threshold and is two marginal fundamental strings. The Einstein-frame values follow by the common factor $g_s^{1/2}$ of Eq. (I.55) and do not affect these comparisons. This is the arithmetic worked in Exercise 7.1.

I.3.4 U-duality and BPS multiplets

You do not need the exceptional-group machinery to use Table I.7: read the representation R_1 as the list of point-particle charges (momenta, windings, wrapped branes) and the mass formula Eq. (I.56) as the SL(2) tension Eq. (I.55) promoted to a larger duality group. The rest of the subsection makes that precise.

Under toroidal reduction the perturbative T-duality and the non-perturbative S-duality stop commuting and generate a larger arithmetic group, the U-duality group. Reducing eleven-dimensional supergravity on T^n (equivalently the type II string on T^{n-1}) to $D = 11 - n$ dimensions, the two-derivative action acquires the Cremmer-Julia symmetry $\mathrm{E}_{n(n)}(\mathbb{R})$ (split real form) acting on the scalar coset $\mathrm{E}_{n(n)}/K$, K its maximal compact subgroup. Charge quantisation cuts this to $\mathrm{E}_{n(n)}(\mathbb{Z})$, which contains the T-duality $O(n{-}1,n{-}1;\mathbb{Z})$ and the S-duality $\mathrm{SL}(2,\mathbb{Z})$ and fuses them (Section 7.3.3). Table I.7 is the maximal case.

D	n	$\mathrm{E}_{n(n)}$	U-duality $\mathrm{E}_{n(n)}(\mathbb{Z})$	particle rep R_1
9	2	$\mathrm{SL}(2)\times\mathbb{R}^+$	$\mathrm{SL}(2,\mathbb{Z})$	·
8	3	$\mathrm{SL}(3)\times\mathrm{SL}(2)$	$\mathrm{SL}(3,\mathbb{Z})\times\mathrm{SL}(2,\mathbb{Z})$	$(\mathbf{3},\mathbf{2})$, dim 6
7	4	$\mathrm{SL}(5)$	$\mathrm{SL}(5,\mathbb{Z})$	**10**
6	5	$\mathrm{SO}(5,5)$	$\mathrm{SO}(5,5;\mathbb{Z})$	$\mathbf{16}_s$
5	6	$\mathrm{E}_{6(6)}$	$\mathrm{E}_{6(6)}(\mathbb{Z})$	**27**
4	7	$\mathrm{E}_{7(7)}$	$\mathrm{E}_{7(7)}(\mathbb{Z})$	**56**
3	8	$\mathrm{E}_{8(8)}$	$\mathrm{E}_{8(8)}(\mathbb{Z})$	**248**

Table I.7: U-duality of M-theory on T^n ($D = 11 - n$), the maximal (32-supercharge) case. The low-rank names $\mathrm{SL}(5) = \mathrm{E}_{4(4)}$ and $\mathrm{SO}(5,5) = \mathrm{E}_{5(5)}$ identify members of the exceptional series. The point-particle (electric) charges fill the representation R_1; the derivation is quoted from Cremmer-Julia and Hull-Townsend (Section 7.3.3).

The point-like $\frac{1}{2}$-BPS states, momentum modes, winding strings, and wrapped branes alike, are the weights of the single representation R_1. Writing $q \in R_1$ for the integer charge vector and $V(\phi) \in \mathrm{E}_{n(n)}/K$ for the coset representative built from the moduli, the BPS mass is the U-invariant

$$M^2_{\frac{1}{2}\text{-BPS}} = q^{\mathsf{T}}\,\mathcal{M}(\phi)\,q, \qquad \mathcal{M} = V^{\mathsf{T}}V, \tag{I.56}$$

invariant under $q \to Uq$, $\mathcal{M} \to (U^{-1})^{\mathsf{T}}\mathcal{M}\,U^{-1}$. This is the direct descendant of the SL(2) doublet formula Eq. (I.55). The representation splits into U-orbits labelled by the preserved fraction $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}$, distinguished by U-invariant polynomials in q: a cubic $I_3(q)$ for $\mathrm{E}_{6(6)}$ in $D = 5$ and the quartic Cartan invariant $I_4(q)$ for $\mathrm{E}_{7(7)}$ in $D = 4$. The $\frac{1}{8}$-BPS black-hole entropies are the invariants themselves,

$$S_{D=5} \propto \sqrt{|I_3(q)|}, \qquad S_{D=4} \propto \sqrt{|I_4(q)|}, \tag{I.57}$$

matched microscopically in Chapter 12. The full statement, including the higher-fraction cases governed by the higher central-charge eigenvalues, is in Obers and Pioline (1999).

Remark I.7 (Beyond the lectures: which generators need branes). Inside $\mathrm{E}_{n(n)}(\mathbb{Z})$ the perturbative T-duality $O(n{-}1,n{-}1;\mathbb{Z})$ acts on momentum and winding, visible order by order on the worldsheet. Everything else, in particular the $\mathrm{SL}(2,\mathbb{Z})$ that rotates g_s and any generator mixing

a wrapped D-brane charge with a momentum charge, is non-perturbative: it can only be seen once the D-branes are included as dynamical states. The over-determination is the point: dozens of independently computed brane tensions assemble into $\mathrm{E}_{n(n)}$ multiplets with the masses Eq. (I.56) demands, and the closure is far tighter than any single coincidence.

I.3.5 Brane-charge lattices and how branes map

The supersymmetry algebra carries the brane charges as p-form central charges $Z_{M_1\cdots M_p}$ (Appendix C.1.6). In eleven dimensions the 528 generators on the right of $\{Q,Q\}$ split as 11 momenta, the 55 of the M2 charge Z_{MN}, and the 462 of the M5 charge Z_{MNPQR}; reducing on the M-theory circle turns these into the type IIA momentum, string, and brane charges. The reduction of the branes themselves is the master map of the web, Table I.8.

IIA object	tension	g_s-scaling	M-theory origin
D0	$\lvert n\rvert/(g_s\ell_s)$	g_s^{-1}	KK momentum n on S^1
F1	$1/(2\pi\alpha')$	g_s^{0}	M2 wrapping S^1
D2	$1/((2\pi)^2 g_s\ell_s^3)$	g_s^{-1}	M2 transverse to S^1
D4	$1/((2\pi)^4 g_s\ell_s^5)$	g_s^{-1}	M5 wrapping S^1
NS5	$1/((2\pi)^5 g_s^2\ell_s^6)$	g_s^{-2}	M5 transverse to S^1
D6	$1/((2\pi)^6 g_s\ell_s^7)$	g_s^{-1}	KK monopole (Taub-NUT)
D8	$1/((2\pi)^8 g_s\ell_s^9)$	g_s^{-1}	(massive IIA; no clean lift)

Table I.8: the type IIA brane spectrum from M-theory, using $R_{11}=g_s\ell_s$, $\ell_p=g_s^{1/3}\ell_s$, $T_{\mathrm{M2}}=1/((2\pi)^2\ell_p^3)$, $T_{\mathrm{M5}}=1/((2\pi)^5\ell_p^6)$. Read the scaling: the F1 is the only g_s-independent tension (it defines perturbation theory), every Dp carries g_s^{-1} (RR soliton), and the NS5 carries g_s^{-2} (magnetic NS object). The D8 couples to the Romans mass $F_0=m$ and has no simple eleven-dimensional origin; the massive IIA deformation it sources is discussed in Appendix I.1. General $T_{\mathrm{D}p}=1/((2\pi)^p g_s\ell_s^{p+1})$.

Exercise I.4 (M-theory reduction of tensions)**.** Using $R_{11}=g_s\ell_s$, $\ell_p=g_s^{1/3}\ell_s$, $T_{\mathrm{M2}}=1/((2\pi)^2\ell_p^3)$ and $T_{\mathrm{M5}}=1/((2\pi)^5\ell_p^6)$, reproduce the IIA tensions of Table I.8. Show that an M2 wrapping the circle gives the effective tension $T_{\mathrm{M2}}\,(2\pi R_{11})=1/(2\pi\alpha')$, the g_s^0 F1; the transverse M2 gives the D2 tension $\propto g_s^{-1}$; the wrapped M5 gives the D4 (g_s^{-1}); and the transverse M5 gives the NS5 (g_s^{-2}). Read off from the powers of g_s which objects are RR solitons and which is the magnetic NS object.

The general rule that fixes the g_s-power of a brane tension is its origin sector: NS-NS electric (F1) is g_s^0, NS-NS magnetic (NS5, KK monopole for the B-field dual) is g_s^{-2}, and every RR-charged Dp is g_s^{-1}. These are the scalings the U-duality multiplets Eq. (I.56) must and do reproduce. The charge lattices themselves are:

- *Perturbative (T-duality).* On T^d, momentum and winding fill the even self-dual Narain lattice $\Gamma_{d,d}$ (type II) or $\Gamma_{d,d+16}$ (heterotic), acted on by $O(d,d;\mathbb{Z})$ resp. $O(d,d+16;\mathbb{Z})$; see Appendix G.
- IIB *two-forms.* The F1/D1 charges and the NS5/D5 charges each fill a rank-two lattice $\mathbb{Z}^2$ carrying the $\mathrm{SL}(2,\mathbb{Z})$ doublet of Eq. (I.53); physical single-brane states are the primitive (coprime) vectors.
- *Non-perturbative (U-duality).* All point-particle charges fill R_1 of Table I.7, the wrapped-brane charge lattice on which $\mathrm{E}_{n(n)}(\mathbb{Z})$ acts.

A worked instance of a full multiplet is Exercise 7.9: on T^6 to $D=4$ the 6+6+1+15+15+1+6+6 = 56 point-like $\frac{1}{2}$-BPS charges (momenta, windings, wrapped D0/D2/D4/D6, wrapped NS5, KK monopoles) fill the **56** of $\mathrm{E}_{7(7)}$.

The two six-dimensional string-string dualities complete the picture of how a fundamental string can become a soliton: heterotic on T^4 equals IIA on K3, with $\phi_{\mathrm{IIA}} = -\phi_{\mathrm{het}}$ and the heterotic F1 realised as the IIA NS5 wrapped on K3; Type I equals SO(32) heterotic, with the heterotic F1 realised as the Type I D1 carrying 32 left-moving fermions. Both are BPS matchings (Section 7.4); the shared Narain lattice $\Gamma_{4,20} \cong H^{\mathrm{even}}(\mathrm{K3},\mathbb{Z})$ is the charge dictionary for the former.

I.3.6 The AdS/CFT dictionary card

The near-horizon throat of a $\frac{1}{2}$-BPS brane stack is an anti-de Sitter geometry, and closed strings on it are dual to the worldvolume theory of the branes (Chapter 13). The canonical entry is N D3-branes.

Parameter map ($\mathrm{AdS}_5 \times S^5$ / $\mathcal{N}=4$ SYM). With the D-brane normalisation $g_{\mathrm{YM}}^2 = 2\pi g_s$ of Chapter 6 and $\lambda \coloneqq g_{\mathrm{YM}}^2 N$,

$$\frac{L^4}{\alpha'^2} = 4\pi g_s N = 2\lambda, \qquad g_s = \frac{g_{\mathrm{YM}}^2}{2\pi} = \frac{\lambda}{2\pi N} \xrightarrow[\lambda \text{ fixed}]{} \frac{1}{N}. \tag{I.58}$$

Classical supergravity is trustworthy for $L \gg \ell_s$ ($\lambda \gg 1$) with small string loops ($g_s \ll 1$), the window $1 \ll \lambda \ll N$. The 't Hooft limit $N \to \infty$ at fixed λ is the classical ($g_s \to 0$) string limit, so the gauge-theory $1/N$ expansion is the closed-string genus expansion. (Some references set $g_{\mathrm{YM}}^2 = 4\pi g_s$, giving $L^4/\alpha'^2 = \lambda$; the geometry is the same.)

Symmetries. The isometry supergroup of the background equals the superconformal group of the dual:

$$\underbrace{\mathrm{SO}(4,2)}_{\text{isometry AdS}_5} \times \underbrace{\mathrm{SO}(6)}_{\text{isometry } S^5} = \underbrace{\mathrm{SO}(4,2)}_{\text{conformal}} \times \underbrace{\mathrm{SU}(4)_R}_{\text{R-symmetry}} \subset \mathrm{PSU}(2,2|4), \tag{I.59}$$

the bosonic part of the supergroup $\mathrm{PSU}(2,2|4)$, whose 32 fermionic generators (16 Poincaré Q plus 16 superconformal S) complete it.

Fields and operators. Each bulk field of mass m and spin s is dual to one gauge-invariant operator of dimension Δ set, for a scalar in AdS_{d+1}, by

$$\Delta(\Delta - d) = m^2 L^2, \qquad \Delta_\pm = \frac{d}{2} \pm \sqrt{\frac{d^2}{4} + m^2 L^2}, \tag{I.60}$$

stable down to the Breitenlohner-Freedman bound $m^2 L^2 \geq -d^2/4$. Masslessness in the bulk is a conserved current on the boundary; Table I.9 lists the entries fixed by symmetry. The source-to-operator relation is the Gubser-Klebanov-Polyakov-Witten formula, with ϕ_0 the boundary value of the bulk field ϕ,

$$Z_{\mathrm{string}}[\phi \to \phi_0] = \Big\langle \exp\Big(\int \mathrm{d}^d x \; \phi_0 \, \mathcal{O} \Big) \Big\rangle_{\mathrm{CFT}}. \tag{I.61}$$

Bulk field in AdS_5	Boundary operator	Δ	$\mathrm{SU}(4)_R$
graviton $g_{\mu\nu}$	stress tensor $T_{\mu\nu}$	4	**1**
SO(6) gauge field	R-current $J^\mu_{[ij]}$	3	**15**
axio-dilaton τ_{IIB}	marginal coupling $\mathrm{Tr}\, F^2 + \cdots$	4	**1**
KK scalars (S^5 harmonic k)	chiral primaries $\mathrm{Tr}\, X^{(i_1} \cdots X^{i_k)}$	k	$[0,k,0]$
massive string modes	Konishi, long operators	$\sim \lambda^{1/4}$	various

Table I.9: field-operator dictionary for $\mathrm{AdS}_5 \times S^5/\mathcal{N}=4$ SYM. The chiral primaries are $\frac{1}{2}$-BPS, with $\Delta = k$ protected at all coupling; the $\mathrm{SL}(2,\mathbb{Z})$ on τ_{IIB} is the Montonen-Olive duality of the gauge coupling.

Other brane throats. The same construction runs on the M-branes and on the D1-D5 system, Table I.10. The three maximally symmetric cases show the celebrated degree-of-freedom scalings N^2, $N^{3/2}$, N^3.

Branes	Near-horizon	Dual CFT	dim/$\mathcal{N}$	dof
N D3	$\mathrm{AdS}_5 \times S^5$	$\mathcal{N}=4$ SYM SU(N)	4D	N^2
N M2	$\mathrm{AdS}_4 \times S^7$	ABJM	3D $\mathcal{N}=8$	$N^{3/2}$
N M5	$\mathrm{AdS}_7 \times S^4$	$(2,0)$ theory	6D	N^3
D1-D5	$\mathrm{AdS}_3 \times S^3 \times M_4$	symmetric-product CFT	2D $(4,4)$	$Q_1 Q_5$
N Dp, $p \neq 3$	(dual-frame AdS_{p+2})	(p+1)D SYM	non-conformal	·

Table I.10: brane throats and their holographic duals. Radii: $L^4_{\mathrm{D3}} = 4\pi g_s N \ell_s^4$; for the M2, $R^6_{S^7} = 32\pi^2 N \ell_p^6$ with $R_{\mathrm{AdS}_4} = R_{S^7}/2$; for the M5, $R^3_{S^4} = \pi N \ell_p^3$ with $R_{\mathrm{AdS}_7} = 2R_{S^4}$. The D1-D5 count uses the two charges Q_1, Q_5 and underlies the black-hole entropy of Chapter 12.

Exercise I.5 (D3 radius and the Breitenlohner-Freedman window)**.** Using the D3 near-horizon radius $L^4 = 4\pi g_s N \ell_s^4$ of Table I.10 (derived from the master charge-radius formula of Appendix I.2), confirm the parameter map Eq. (I.58) and the validity window $1 \ll \lambda \ll N$. Then take a chiral primary $\mathrm{Tr}\, X^{(i_1} \cdots X^{i_k)}$ ($k \geq 2$), dual to the S^5 scalar harmonic with $m^2 L^2 = k(k-4)$: use Eq. (I.60) to show $\Delta_+ = k$, and check that every such mode lies at or above the Breitenlohner-Freedman bound $m^2 L^2 \geq -4$, saturated by the $k = 2$ (stress-tensor) multiplet.

Remark I.8 (Beyond the lectures: why $p = 3$ is special)*.* For N Dp-branes the worldvolume coupling $g^2_{\mathrm{YM}} \sim g_s \ell_s^{p-3}$ is dimensionful unless $p = 3$, so only the D3 theory is exactly conformal and only its throat is pure AdS with a constant dilaton. For $p \neq 3$ the dilaton runs and the near-horizon geometry is conformal to $\mathrm{AdS}_{p+2} \times S^{8-p}$ only in a Weyl-rescaled "dual frame"; the duality is then to a non-conformal super-Yang-Mills theory with a running effective coupling, useful over a window of energies rather than at a fixed point. The exercise on the general Dp throat is Exercise 13.5.